Supplement to Eclipses by Appointment: evidence for 273 physical eclipsing systems

Evidence register updated October 30, 2026

Unified evidence register

Current physical sample: **273 systems (261 confirmed, 12 probable)**. The 199 systems without a prior EB listing form a novelty subset of this physical sample. Seven plausible systems remain outside the count. Physical classification, prior catalogue status and completeness of review are distinct judgements.

The original 106 A entries and 120 figures are retained below. All A1-A204 objects are indexed; where the older supplement supplied no full entry, the archived report is reproduced when available, otherwise the existing objects.csv ledger is transcribed with an explicit coverage limit. Seventy R reports and seven returned batch reports are archived in full. Scientific report prose and tables appear inline; long ancillary console blocks are linked as text evidence to keep this reading copy navigable. A4 is the returned blind control; TIC 195865529 is a known catalogue recovery. Neither adds to the count. The other ten returned confirmations add ten unique objects.

Returned TIC 416285595 retains the reported confirmed classification with its depth-mismatch caveat. TIC 319185391 is plausible and outside the count, following the batch verdict over the report's more optimistic prose. Catalogue non-matches are limited to the stated searches.

A ledger and recovered evidence | R evidence | Returned per-object adjudications | Returned full reports

A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16 A17 A18 A19
A20 A21 A22 A23 A24 A25 A26 A27 A28 A29 A30 A31 A32 A33 A34 A35
A36 A37 A38 A39 A40 A41 A42 A43 A44 A45 A46 A47 A48 A49 A50 A51
A52 A53 A54 A55 A56 A57 A58 A59 A60 A61 A62 A63 A64 A65 A66 A67
A68 A69 A70 A71 A72 A73 A74 A75 A76 A77 A78 A79 A80 A81 A82 A83
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A198 A199 A200 A201 A202 A203 A204

R001 R002 R003 R004 R005 R006 R007 R008 R009 R010 R011 R012 R013
R014 R015 R016 R017 R018 R019 R020 R021 R022 R023 R024 R025 R026 R027
R028 R029 R030 R031 R032 R033 R034 R035 R036 R037 R038 R039 R040

R041 R042 R043 R044 R045 R046 R047 R048 R049 R050 R051 R052 R053 R054 R055 R056 R057 R058 R059 R060 R061 R062 R063 R064 R065 R066 R067 R068 R069 R070

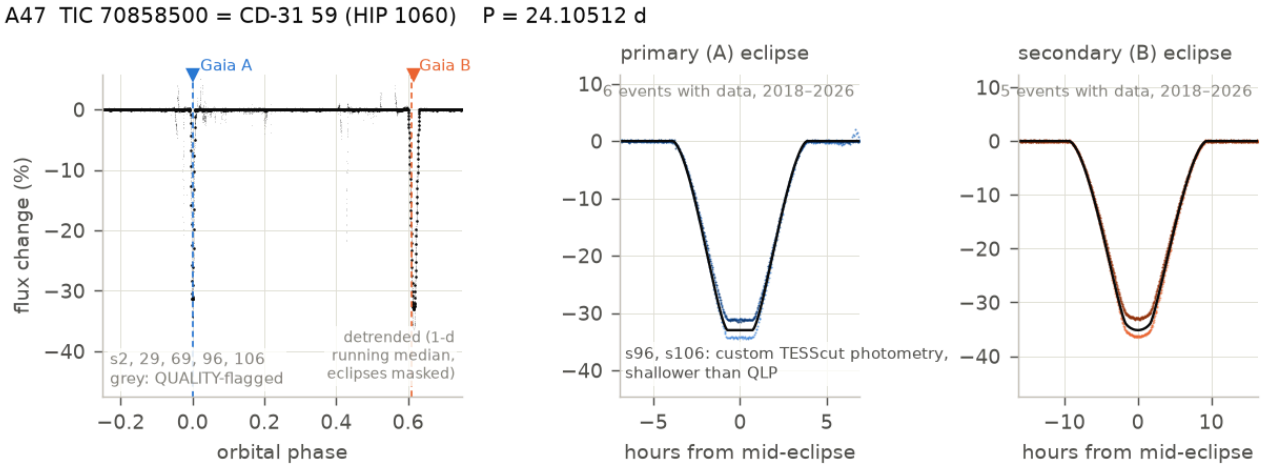
TIC 195865529 TIC 468181994 TIC 470235665 TIC 379251709 TIC 70665829 TIC 331293467 TIC 283532878 TIC 407877153 TIC 15855794 TIC 92881948 TIC 468827621 TIC 416285595 TIC 372683900 TIC 13106408 TIC 372683622 TIC 319185391 TIC 62328710 TIC 300903537 TIC 23112764 TIC 373062090 TIC 465073693 TIC 316405449 TIC 465263741 TIC 441741446

About this supplement

This supplement holds the per-object entries of the paper, one for each eclipsing system, heartbeat star, planet candidate and withdrawn object, with their figures. They are grouped under the section of the paper that discusses them (S3.1.2 holds the entries of the paper's Section 3.1.2, and so on) and are in the paper's order. Each entry gives the identifiers, the eclipse measurements, the ephemeris, the Gaia orbit, the independence checks, the catalogue status and the status word of the paper's Section 1 (Conventions), which also defines the terms used. Figure numbers follow the entry labels (Figure A47 belongs to entry A47). Depths are fractional losses of the system's light in the named pipeline's aperture. BTJD = BJD_{TDB} − 2457000. The analysis file behind each entry is named in the paper's Appendix B deposit (`tables/objects.csv`, column `source_path`).

S3.1.2 Confirmed and probable eclipsing binaries

A47. CD-31 59 = HIP 1060 (TIC 70858500)



A47. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 24.10512 d, Gaia's predicted conjunctions dashed. Middle: the six 34%, 6.9-h total eclipses at A, flat-bottomed during totality (sectors 2, 29, 69, 96, 106 ×2; the sector-96 event, in flagged cadences, is from custom TESScut photometry and shallower). Right: the five 36%, 16-h annular transits at B, with limb-darkened rounded bottoms.

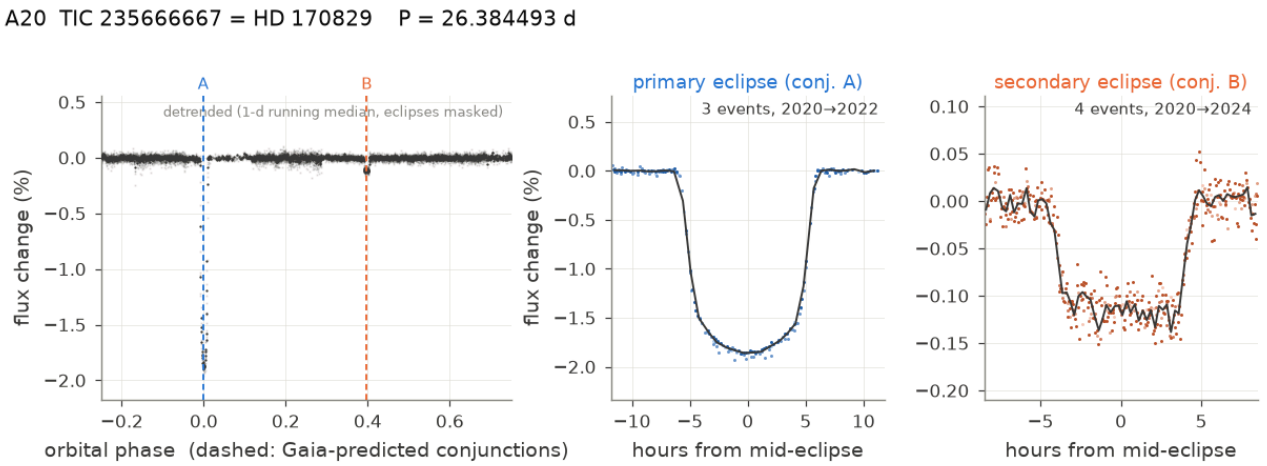
Identifiers	TIC 70858500; Gaia DR3 2320220880389321856; CD-31 59 = HIP 1060 = HE 0010-3037 = SAO 192405 = TYC 6990-585-1 = CPD-31 25 = PPM 275814; WDS 00132-3021 (HDS 31)
Position (J2000)	00:13:12.0 −30:20:59.5
Brightness	*G* 10.39; Tmag 10.01
Stars	Gaia SB2: star 1 (Gaia's “primary”, K_1 = 57.1 km/s) M_1 = 1.10 $M_\odot$, R_1 = 1.26 $R_\odot$, log *g* 4.28; star 2 M_2 = 1.22 $M_\odot$, R_2 = 1.83 $R_\odot$, log *g* 4.00, 67% of the TESS light (more evolved). SIMBAD type G2. Total luminosity 6.1 $L_\odot$, agreeing with Gaia's absolute magnitude. The WDS B component (*G* 13.24) lies 11.6″ away
Period	24.10512 ± 0.00001 d (Gaia 24.11059 ± 0.00424 d; −1.3σ)
Epochs	T_0(A) = BTJD 1367.0102 ± 0.0005 = BJD_{TDB} 2458367.0102; T_0(B) = BTJD 1357.7451 ± 0.0008 = BJD_{TDB} 2458357.7451. The two eclipses are nearly equal in depth (B deeper by 1.7 points). Phase 0 is set at A, the total eclipse, in the figure and the timing analysis; the VSX submission and Appendix A use B, the deeper, as the epoch, following VSX's convention
Eclipse at A	34.2% deep (PDCSAP 33.0%), 0.289 d (6.9 h) with a 0.069-d (1.7-h) flat bottom: total eclipse of star 1 behind star 2, near periastron; six events (s2, s29, s69, s96, s106 ×2) (the s96 A event lies entirely in QLP-flagged cadences and is measured in TARS and in custom photometry, this work)

Eclipse at B	35.9% deep (PDCSAP 34.7%; s096 diluted to 28.8% in QLP, 34.0% in TARS), 0.681 d (16 h) with a 0.128-d bottom: annular transit of star 1 across star 2, near apastron; five events (s2, s29, s69, s96, s106). B at phase 0.61564 after A
Orbit	Photometric $e \cos \omega = 0.1663$, $e \sin \omega = 0.4142$: $e = 0.446$, $\omega = 68.1°$; Gaia 0.4417 ± 0.0127, 69.2 ± 1.7° (within 1σ). Model: $i = 89.87°$, $k = 1.455$, χ^2 914/920
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s2, s29, s69); TASOC (s2); TGLC (s2); TARS (all sectors); custom TESScut photometry of s96 and s106 (this work)
Pixel attribution	TESScut difference images of all eclipses: flux-lost centroid 0.03–0.21 px from the target; the flux-lost image is 0.25× the direct image with 2% residuals; the *G* 13.24 neighbour 0.5 px away (would need about 470%) is excluded
Catalogue status	Not listed in: VSX, GCVS, NSV, ASAS-3, ASAS-SN, ATLAS, ZTF, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4 (including the unvetted table 2), Justesen & Albrecht 2021, Fetherolf et al. 2023, Gaia DR3 variability tables, SB9, ExoFOP, general web search; SIMBAD's 13 references include none on variability. Hipparcos: scatter 0.032 mag, no variability flag. A TESS-SPOC data-validation report for sector 69 carries a threshold-crossing event at 9.266 d, the B-to-A spacing, at very high signal-to-noise, never promoted to a TOI (Table A1)
Open issues	Fitting A and B separately shrinks the A-to-B separation by about 8.6 min over 120 cycles (3.5σ), the same sign as relativistic apsidal motion (which would give about 2.7 min); reported as tentative only
Next eclipses	A: BTJD 4331.940 (2026-10-18 10:26 UTC), 4356.045, 4380.150 (allow ±5 min); B: BTJD 4346.780 (2026-11-02 06:37 UTC), 4370.885
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-31 59 (HIP 1060) is a Hipparcos star with 35% eclipses that is not catalogued as eclipsing in any archive we searched. It is a double-lined pair in an eccentric ($e = 0.446$) orbit seen within 0.13° of edge-on. Its components are a 1.10 $M_\odot$ star and a slightly heavier, more evolved 1.22 $M_\odot$ companion that is 1.45 times larger and gives two-thirds of the light. Near periastron, where the stars are 26.5 $R_\odot$ apart and move at 154 km/s across the line of sight, the smaller star passes wholly behind the larger one. The result is a 6.9-h total eclipse whose flat bottom is totality and whose depth is the smaller star's share of the light. Near apastron (63.7 $R_\odot$, 64 km/s) the smaller star crosses the larger one, giving an annular transit 2.4 times longer, with a rounded bottom from limb darkening. One physical model reproduces the depths, shapes, durations, Gaia's e and ω, and the SB2 masses, and gives both radii.

The ephemeris was tested out of sample: three eclipses in sector 106, predicted before those data were examined, fall at O−C = +1.4, −1.5 and +1.4 min. Ground-based photometry of one eclipse, a joint fit with Gaia DR4 epoch radial velocities (masses and radii to about 1%), and continued timing to test the apsidal-motion hint are warranted. The system is the first of these eclipsing binaries to be submitted to the AAVSO VSX, and the submission is pending moderation.

A20. HD 170829 = HR 6950 = NSV 24480 (TIC 235666667)



A20. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 26.384493 d, Gaia's predicted conjunctions dashed. Middle: the three 1.65%, 11.8-h primary transits of the M dwarf across the subgiant (the fourth, in sector 80, lies in flagged cadences and is not drawn; see Open issues). Right: the four 0.106%, flat-bottomed secondary occultations.

Identifiers	TIC 235666667; Gaia DR3 4525943082344640256; HD 170829 = HR 6950 = HIP 90729 = NSV 24480
Position (J2000)	18:30:41.7 +20:48:54.2
Brightness	G 6.29; Tmag 5.79
Stars	G8 IV subgiant with an asteroseismic mass (1.11 $M_\odot$, Malla et al. 2024), R_1 = 1.76–1.86 $R_\odot$ + M dwarf of 0.19–0.21 $M_\odot$, R_2 = 0.21–0.23 $R_\odot$, about 3050 K; mass function 0.00534 $M_\odot$. SB9 orbit #1066; Gaia DR3 astrometric orbit
Period	26.384493 ± 0.000040 d (SB9 26.3847 ± 0.0001 d, −1.9σ; Gaia 26.384663 d, −0.2σ)
Epoch of primary	BTJD 2403.9798 = BJD_{TDB} 2459403.9798 (the epoch proposed for VSX)
Primary eclipse	1.65% (SPOC) / 1.76% (QLP) deep, 11.8 h, rounded U (full transit of the M dwarf across the subgiant); three events, sectors 26, 40, 53
Secondary eclipse	0.106% deep, 8.5 h, flat-bottomed (the M dwarf fully hidden); four events, sectors 26, 40, 53, 80; at phase 0.39677 after the primary
Orbit	Eclipse spacing gives e cos ω = −0.161 (SB9 −0.1607); duration ratio 1.39 gives e sin ω = −0.145 to −0.16 (SB9 −0.154). SB9: e 0.2223 ± 0.0002, ω 223.69 ± 0.06°. Gaia astrometric orbit: i = 89.7°; eclipse solution: impact parameter about 0, $i \approx 90°$
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP; TASOC raw (times within 3 min, depths within about 6%). TARS s053 fails on the saturated star
Pixel attribution	Centroids uninformative (saturated star, bleed columns). The brightest Gaia neighbour within 60″ (G 14.82) would need a 4,500% primary or a 290% secondary; no visual companion in WDS or CCDM (SIMBAD's "**" is the Gaia astrometric orbit). A Gaia E-type eclipsing binary 12″ away (P 0.92 d) contaminates the SPOC aperture (TCE_3, 75 ppm); it cannot produce the 1.65% or 0.106% eclipses
Catalogue status	**VSX type revision.** VSX: NSV 24480, type "VAR:", amplitude 0.08 V, no period. ASAS-SN "VAR", no period (saturated). Nothing in Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables; Hipparcos "constant" (the eclipses cover about 1.9% of the time); no TOI or CTOI; no eclipse in any of the 21 SIMBAD references or in a general web search. SPOC DV runs s14–s55 and s14–s86 carry both eclipses as TCEs
Open issues	Every event is 0.7–1.0 h early against the SB9 orbit. A full limb-darkened model joined to the SB9 orbit and the asteroseismic mass has not yet been made. A fourth primary falls in sector 80, in flagged cadences of QLP, SPOC and TESS-SPOC: a 1.7–1.9% dip centred near BTJD 3485.74 in flagged QLP cadences, with the data gap beginning at 3486.46, after egress
Next eclipses	Primary: BTJD 4330.048 (2026-10-16 13:07 UT), 4356.432, 4382.817; secondary: 4340.516, 4366.901 (about ±5 min formal; allow ±1 h given the SB9 tension)
Discovery	

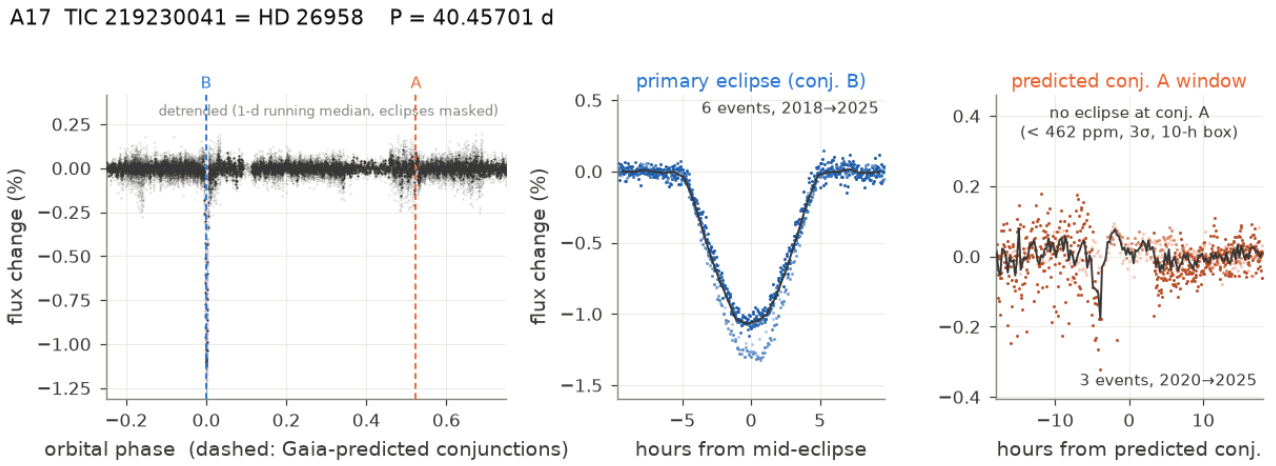
	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision, not a new star. HD 170829 is in VSX as NSV 24480 with type “VAR:” and no period; the eclipses are reported here for the first time. We propose revising it to type EA, P = 26.38449 d, epoch 2459403.9798.

HD 170829 is a bright G8 subgiant with an eclipsing M dwarf. At the primary eclipse the M dwarf crosses the subgiant, giving a limb-darkened U. At the secondary it is fully hidden behind the subgiant, giving a flat bottom. One geometry fits both. Taking the radius ratio from the primary depth and solving both durations gives R_2 = 0.21–0.23 $R_\odot$, as expected for a 0.2 $M_\odot$ dwarf, and an inclination that matches Gaia's independent astrometric orbit. The secondary depth implies a companion of about 3,050 K. Every eclipse falls within 1 h of the published SB9 spectroscopic orbit. A bright subgiant with asteroseismology, an eclipsing M dwarf and an astrometric orbit make an unusually complete combination. However, every event is 0.7–1.0 h early against the SB9 orbit, and no full model yet exists.

Both eclipses are observed in sector 26, and the sector-40 primary is complete in the raw SAP flux.

A17. HD 26958 (TIC 219230041)



A17. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 40.45701 d, Gaia's predicted conjunctions dashed. Middle: the six events at conjunction B, 2018–2025: five full 1.2%, 9.7-h eclipses and the partial egress of sector 5. Right: the data at conjunction A, where no eclipse is expected or seen.

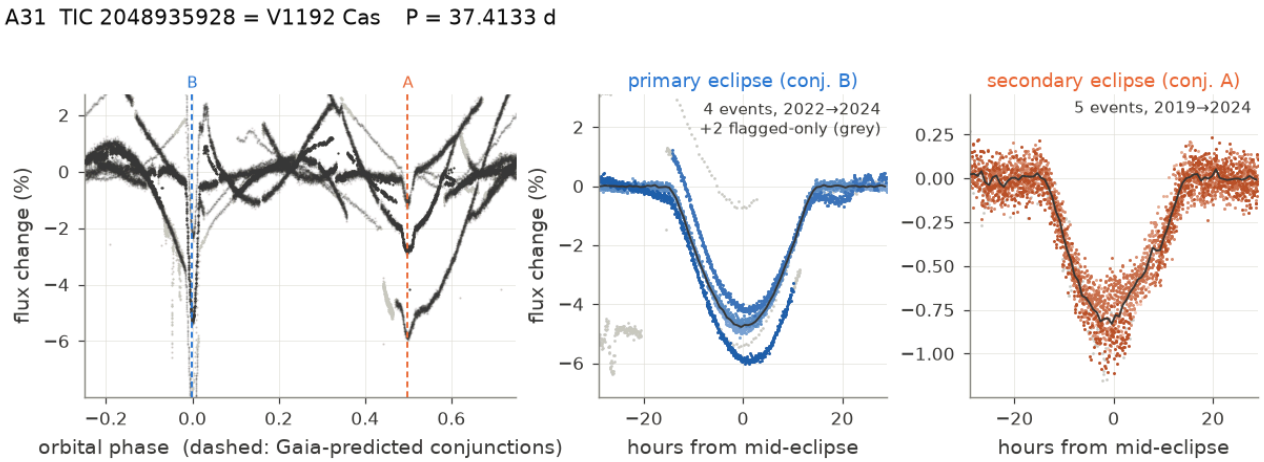
Identifiers	TIC 219230041; Gaia DR3 4830714373974009600; HD 26958
Position (J2000)	04:13:28.4 −49:07:18.6
Brightness	G 8.15; Tmag 7.51
Stars	K1 III giant (SIMBAD) + companion of at least 0.90–1.30 $M_\odot$ (minimum mass for a 1.0–2.0 $M_\odot$ giant), plausibly a ~1 $M_\odot$ dwarf or a fainter object of that mass such as an inner pair; mass function 0.2005 $M_\odot$; 163 pc. Gaia SB1, significance 617, all 18 RV epochs used
Period	40.45701 ± 0.00002 d (Gaia 40.454393 ± 0.001757 d; the eclipse period is longer by 1.49σ)
Epoch of eclipse	Ephemeris BTJD 386.0846 + 40.45701 E (BJD_{TDB} 2457386.0846); observed e.g. BTJD 3946.3012 ± 0.0009 (E88) = BJD_{TDB} 2460946.3012
Eclipse (B)	0.99–1.28% deep in QLP SAP (1.19–1.32% in the large-aperture flux), 9.4–9.8 h, rounded V (a pure V is worse by $\Delta\chi^2$ 21–175); five full events, sectors 3, 30, 31 and 97 (two), residuals −0.3 to +1.0 min; a partial egress in sector 5 also on the ephemeris
Other eclipse (A)	None: 113 ± 116 ppm in s031 (3σ limit 462 ppm for a 10-h box), below 561 ppm in s032; less than 4% of the B depth. The geometry predicts none (impact parameter at A exceeds 1 + k)
Orbit	Gaia e 0.1913, ω 258.54° (Gaia NSS solution); not refitted from the eclipses (only one kind seen). An inclination near 85° is estimated

Pipelines agreeing	QLP SAP and large-aperture flux; TGLC PSF and aperture, TASOC raw, eleanor-lite, T16 (s003); TARS (s003, s030, s031). No SPOC or TESS-SPOC light curve exists
Pixel attribution	No difference image; no centroid shift at mid-eclipse (TASOC, eleanor); no Gaia source within 42″, the two within 60″ would need 5944% and 67542% eclipses. TGLC's PSF photometry, which models this Gaia source and its neighbours separately, shows the same dip
Catalogue status	Not listed in: VSX, ASAS-SN, Kostov et al. 2025 tables 2–4, IJspeert et al. 2024, Prša et al. 2022, Gaia DR3 variability tables (`phot_variable_flag` NOT_AVAILABLE), Hipparcos (no variability flag), SIMBAD (SB*, K1III, 13 references, none on variability), general web search
Open issues	Radius ratio and impact parameter are degenerate (k 0.10–0.20 at impact 0.98–1.08); the companion's nature is inferred, not observed
Next eclipse	BTJD 4350.871 ± ~1.5 min (E = 98)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 26958 is a K giant at 163 pc with one eclipse per 40.457-d orbit. Five full eclipses from 2018 to 2025 fit a single linear ephemeris within 1.0 min, and five independent pipelines show the same dip at the same time and depth. The eclipse falls at the B conjunction. Under the standard radial-velocity convention the giant is the near body there, so the eclipse is most likely a grazing occultation of the companion rather than a transit of the companion across the giant. A total occultation is ruled out ($\Delta\chi^2 \approx 2100$ against a flat bottom), and the observed 9.7 h is half the central duration, so the eclipse must be grazing. The companion would then supply about 2–6% of the TESS light. A 1.0–1.16 $M_\odot$ main-sequence star would supply 7–10% in a blackbody estimate, within a factor of about 1.5 of the fitted value. K and M dwarfs fall below the minimum mass and are excluded.

The astrometric excess (RUWE 1.77 in DR3) is explained by the 40-d orbit's own photocentre motion, and no third body is required. Further constraints require spectroscopy to find the companion's lines, multi-colour photometry of the next eclipse (an occultation of a hotter dwarf should be deeper in the blue), and a joint RV and light-curve model.

A31. V1192 Cas = NSVS 3545309 (TIC 2048935928 = TIC 271237873)



A31. Left: QLP SAP flux folded on P = 37.4133 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 4.7%, 29-h grazing primary eclipses (the hotter companion occulted by the giant). Right: the 0.9%, 29-h secondary eclipses (the companion transiting the giant).

Identifiers	TIC 2048935928 (duplicate entry TIC 271237873, under which the TESS-SPOC files exist); Gaia DR3 2016683927990598784; V1192 Cas; NSVS 3545309
Position (J2000)	23:36:30.5 +63:27:28.7
Brightness	G 10.66; Tmag 9.79; VSX range 10.8–11.0
Stars	

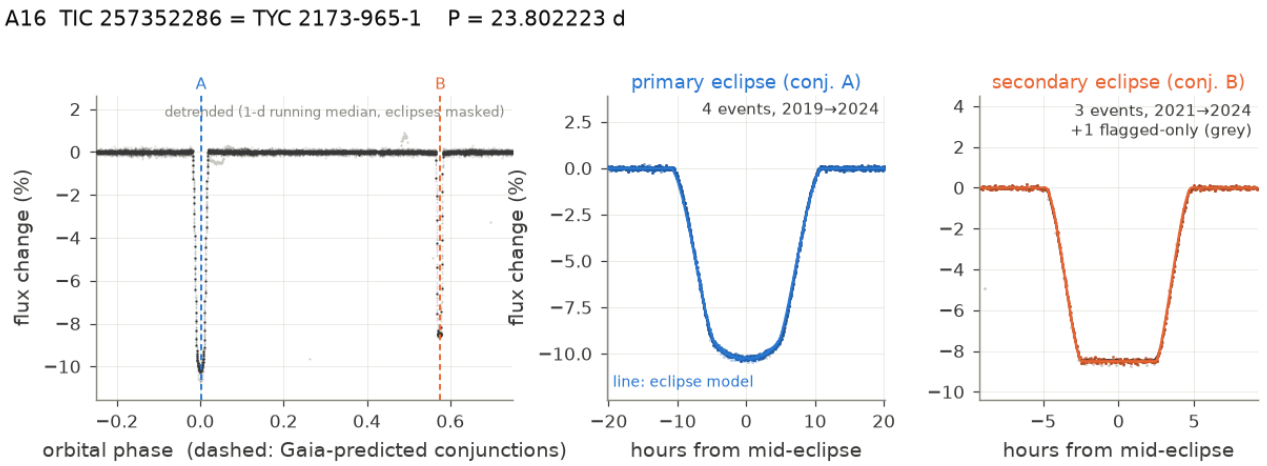
	K giant (Gaia: T_{eff} 5440 K, log *g* 2.74, R = 11.2 $R_\odot$, L = 98 $L_\odot$, FLAME mass 3.1 $M_\odot$) + hotter main-sequence companion of 1.34–1.97 $M_\odot$ (A/F type, about 5–10% of the TESS light); mass function 0.296 $M_\odot$; Gaia SB1. The composite light explains the hotter TIC T_{eff} (6850 K)
Period	37.4133 ± 0.0005 d from both eclipse kinds independently (Gaia 37.4195 ± 0.0041 d; −1.5σ)
Epoch of primary	T_0 = BTJD 1764.906 ± 0.018 = BJD_{TDB} 2458764.906 (*P* = 37.41329 ± 0.00047); secondary T_0 = BTJD 1783.585 ± 0.011
Primary eclipse (B)	4.3–4.9% deep, 28.4–29.2 h, V-shaped: grazing occultation of the hotter companion by the giant (Gaia's B conjunction, giant in front); six events, 2019–2025 (two in QLP-flagged cadences, clean in TEQUILA; two partial)
Secondary eclipse (A)	0.7–1.1% deep, 28–30 h: grazing transit of the companion across the giant; five events (the s085 one inflated by about 1 ppt of common mode)
Orbit	Secondary at phase 0.4993 ($e \cos \omega \approx -0.001$), duration ratio 1.0 ($e \sin \omega \approx 0$), in agreement with Gaia's e = 0.005 ± 0.006: circular. A central eclipse would last 46–56 h, so the geometry is grazing
Pipelines agreeing	QLP SAP; TESS-SPOC (as TIC 271237873, every post-s056 event); TEQUILA (s017, s018)
Pixel attribution	TESScut s058 difference images: largest flux-loss pixel is the target pixel for both kinds, offsets 1.8″ and 1.3″. Brightest star within 60″ (*G* 15.9 at 44″) would need 630%; the *G* 12.5 star at 109″ lies outside the difference signal
Catalogue status	**VSX type revision SRA → EA.** VSX/GCVS: V1192 Cas, SRA, *P* 37.4 d, epoch JD 2451512 (discovery: Kuzmin 2008, “New semiregular pulsating stars”); SIMBAD Mi*; ASAS-SN ROT, *P* 2.0518 d (probably spurious); Kostov table 2. Not listed in: Gaia DR3 variability tables, SB9, WDS, the TESS-era EB catalogues, general web search. ExoFOP was not queried
Open issues	Spectra for the companion's lines; multi-colour depths (the occultation should be deeper in the blue). Low-level intrinsic variability of the giant is possible but not a 37.4-d pulsation
Next eclipses	Primary: BTJD 4346.42, 4383.84 (±0.04); secondary: BTJD 4327.69, 4365.10 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision, not a new star. V1192 Cas is catalogued as a semi-regular pulsator (SRA) with *P* 37.4 d; that period is the orbital period found from the eclipses. We propose reclassification of the SRA (VSX) and Mi* (SIMBAD) types as EA.

V1192 Cas is a K giant of about 11 $R_\odot$ with a hotter main-sequence companion on a circular 37.4-d orbit, eclipsing at grazing incidence. The deeper eclipse falls where the giant is in front, so it is the occultation of the hotter companion. The shallow one is the companion's transit, which at a grazing chord reaches about 1%. Both last about 29 h, against 46–56 h for a central eclipse.

The catalogued pulsation was tested directly. ASAS-SN V photometry from 2014–2017 has a periodogram peak at 35.9 d, within the frequency resolution of 37.4 d. Folded on the TESS ephemeris extrapolated back about 25 cycles, it shows the primary eclipse at phase 0.00. Out of eclipse the rms is only 0.027 mag, not a 0.2-mag semi-regular range. The 3–5% slow variation in TESS is not locked to orbital phase and is partly QLP level jumps. Using Gaia's full parameter covariance (corr(ω, T_p) = 0.99999 for this near-circular orbit) shrank the conjunction-time error to 0.15–0.38 d. This makes the phase agreement a significant test, and the eclipses pass it.

A16. TYC 2173-965-1 (TIC 257352286)



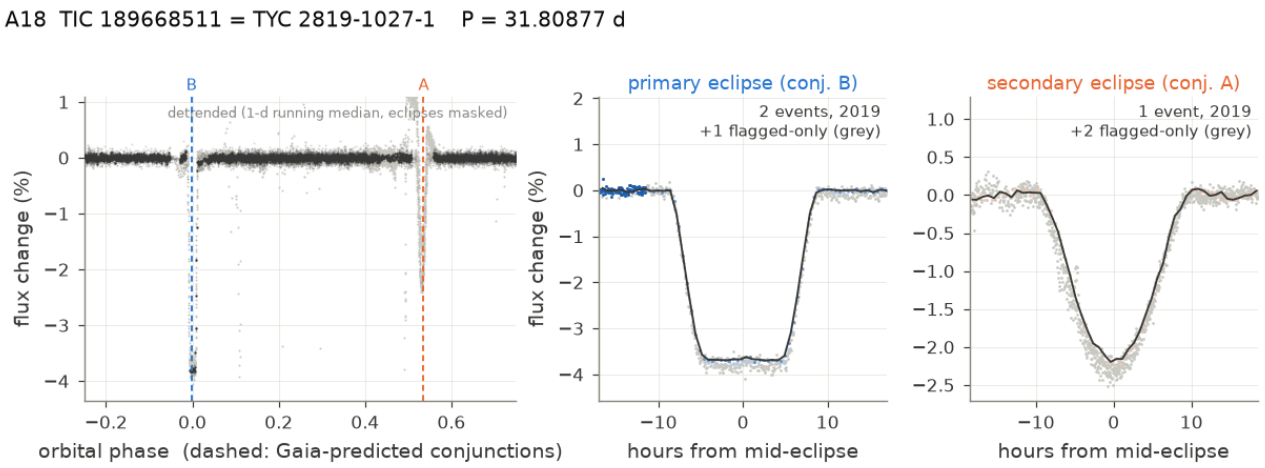
A16. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.802223 d, Gaia's predicted conjunctions dashed. Middle: the four round-bottomed 10% eclipses at A (sectors 15, 41, 55, 82). Right: the three flat-bottomed 8.45% eclipses at B (sectors 41, 55, 82).

Identifiers	TIC 257352286; Gaia DR3 1840217568772936448; TYC 2173-965-1
Position (J2000)	21:09:36.6 +23:45:30.2
Brightness	G 9.87; Tmag 9.42
Stars	Gaia SB1 (17 RV epochs, no secondary RVs). About 1.5 $M_\odot$ F/G subgiant (91.5% of the light) + about 1 $M_\odot$ G dwarf ($R \approx 0.94\ R_\odot$, about 5900 K); mass function 0.1473 $M_\odot$
Period	23.802223 ± 0.000022 d (Gaia 23.800734 ± 0.001153 d; difference +1.29σ)
Epoch of primary	T_0(A) = BTJD 2804.28519 ± 0.00089 = BJD_{TDB} 2459804.28519
Primary eclipse (A)	9.90–10.23% deep, 20.2 h, round-bottomed (limb-darkened transit of the subgiant by the companion), ingress 5.5 h; four events, sectors 15, 41, 55, 82
Secondary eclipse (B)	8.40–8.51% deep, 9.2 h, flat bottom 5.3 h (total occultation of the companion); three events, sectors 41, 55, 82; B − A = 13.65454 ± 0.00111 d (phase 0.57367)
Orbit	Eclipse model (sector 82): e = 0.4198, ω = 284.62°; Gaia 0.4239 ± 0.0073 (−0.56σ), 284.57 ± 0.87° (+0.05σ). i = 88.96°, $k = R_2/R_1$ = 0.317
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s41, s55); TESS-SPOC (s82); TARS (s41)
Pixel attribution	Difference images (SPOC s55 target pixel file; TESScut s41) have the target's PSF shape, centroid within 0.1 px of the target; no neighbour brighter than G 15 within 60″ could produce the dip (each would need a drop of more than 100%)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, ExoFOP (no TOI or CTOI), general web search; SIMBAD lists it as SB* with one reference (the Gaia DR3 multiplicity paper). Prior partial detection: a SPOC s14–s55 TCE at P = 13.5987 d, an alias folding both eclipses together, which never became a TOI
Open issues	None; spectroscopy at quadrature would give absolute masses and radii
Next eclipses	A: BTJD 4327.6275, 4351.4297; B: BTJD 4341.2820, 4365.0842 (1σ about 2.5–3 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 2173-965-1 is a detached, eccentric, totally eclipsing binary. At conjunction A the smaller companion transits the subgiant that Gaia's radial velocities track, giving the 20-h, round-bottomed eclipse. At B the subgiant totally hides the companion, giving the flat 8.45% eclipse, whose depth equals the companion's share of the light. The A eclipse lasts 2.2 times as long because it happens at a separation of $1.39a$ against $0.58a$ at B. One Keplerian eclipse model, fitted to sector 82 alone, reproduces all seven eclipses with residual rms 0.5–0.9 ppt, and its e and ω agree with Gaia's independent radial-velocity orbit. With the primary radius taken from the TIC and corrected for the companion's light, the solution is a self-consistent pair of a 1.5 $M_\odot$ F/G subgiant and a 1 $M_\odot$ G dwarf.

Seven eclipses are observed, including the sector-15 primary. The secondary's 5.3-h flat bottom is distinct from its 9.2-h total duration. The SPOC TCE at the alias period and the Gaia SB1 solution are prior partial detections of the system. QLP's detrended flux removes the 20-h eclipse in sectors 15 and 41; only SAP keeps it.

A18. TYC 2819-1027-1 (TIC 189668511)



A18. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 31.80877 d, Gaia's predicted conjunctions dashed. Middle: the three flat-bottomed 3.75% total occultations at B (sectors 17, 18, 85). Right: the three rounded 2.2% transits at A (sectors 17, 58, 85).

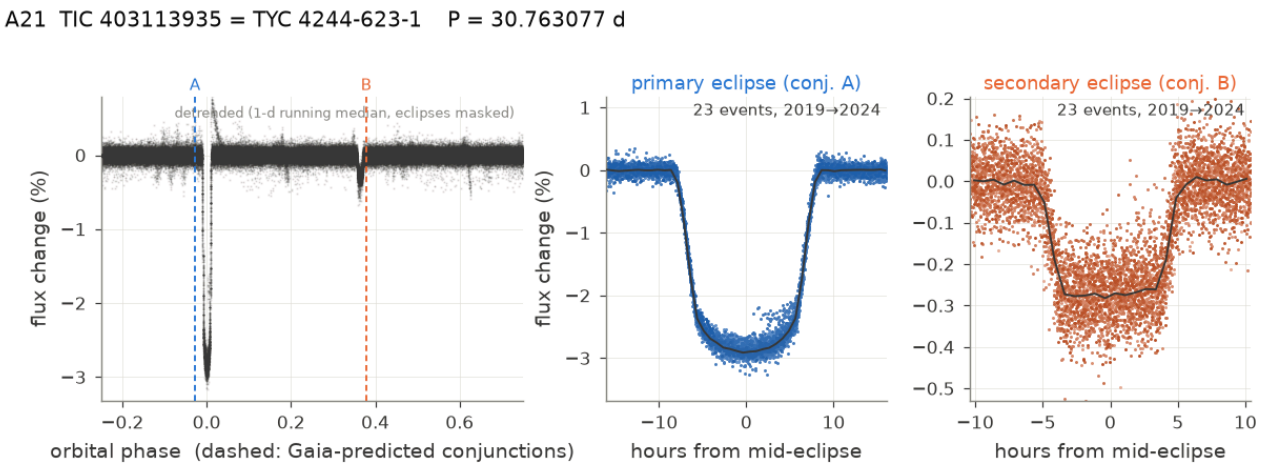
Identifiers	TIC 189668511; Gaia DR3 3468957854207709 44; TYC 2819-1027-1
Position (J2000)	01:42:18.1 +39:53:21.1
Brightness	G 9.84; Tmag 9.22
Stars	K giant of about 1.3–1.6 $M_\odot$ (R_1 fixed at 4.78 $R_\odot$) + roughly solar-type dwarf, 0.86–0.96 $M_\odot$, 0.67–0.75 $R_\odot$, 3.75% of the light; mass function 0.1364 $M_\odot$; Gaia SB1
Period	31.80877 ± 0.00001 d from B; 31.80862 ± 0.00003 d from A (Gaia 31.799234 ± 0.003676 d; the eclipse period is longer by 2.6σ)
Epoch of primary	T_0(B) = BTJD 1766.5861 = BJD_{TDB} 2458766.5861; each A follows its B by 16.894 d
Primary eclipse (B)	3.73–3.81% deep, 16.75–16.99 h, flat-bottomed (flat part 10.22 h): total occultation of the dwarf by the giant; sectors 17, 18, 85; O–C −0.9, +2.3, 0.0 min
Secondary eclipse (A)	2.12–2.25% deep, 18.0–18.6 h, short rounded bottom: transit of the dwarf across the limb-darkened giant; sectors 17, 58, 85; O–C +1.0, −0.7, +1.1 min. Predicted from the B geometry (19.3 h, about 2.2%) before being measured
Orbit	A follows B by phase 0.5311 (Gaia's elements predict 0.5379); e cos ω = −0.048 against Gaia's −0.059 ± ~0.009 (about 1.2σ). i ≈ 86.1–86.65°, off-centre chord
Pipelines agreeing	QLP SAP; TARS (s017, s018, s058, s085); TESS-SPOC SAP (s085, s058). No SPOC or TESS-SPOC product exists for s017–s018
Pixel attribution	TESScut difference images of the B eclipses in s017, s018 and s085 and the A eclipses in s017 and s058 (made after the pre-submission review; `appointments/vet2pix/`): in every clean event the deficit centroid lies within 0.16 px (3.4″) of the catalogue position and 0.2–2.8″ from the light centroid, and the deficit is a scaled copy of the target's image (residual 0.08–0.19); s058 A needs a 0.02-px pointing shift fitted out, and s085 A can be tested only on stray-light-flagged cadences, where it agrees too. The B depth is 3.7% in 1×1, 3×3 and 5×5 apertures. The only Gaia source within 60″ brighter than G 15 (G 13.31 at 58.7″, a rotating variable) would need a 95% eclipse, lies 2.9 px from the target where the loss is 0.2–12 ppt against 37 ppt on the target, and varies by 0.03 mag in Gaia; all others would need more than 1000%. QLP and TESS-SPOC give the same depths; TARS runs shallower
Catalogue status	Not listed in: VSX (only the rotating variable 58″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SIMBAD (SB* only), general web search
Open issues	

	Three of the six events lie in QLP-flagged scattered-light stretches (their depths, shapes and times match the clean ones, and TESS-SPOC SAP reproduces them); radial velocities or a spectrum would fix the giant's mass and the companion's light
Next eclipses	B at 1766.5861 + n × 31.80877; A 16.894 d after each B
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed: six eclipses of two kinds in four sectors at Gaia's period, the deficit on the target in difference images of five of them (added after the pre-submission review), and depth agreement between QLP and TESS-SPOC (not with TARS (s085 B 2.71% against QLP's 3.74%; s058 A 1.79% against 2.25%); the B and A periods, 31.80877 ± 0.00001 and 31.80862 ± 0.00003 d, differ by 4.7σ formally.

TYC 2819-1027-1 is a K giant with a roughly solar-type dwarf companion on a 31.8-d orbit. At B the giant is in front and totally hides the dwarf, giving flat-bottomed occultations. Their depth makes the companion 3.75% of the light, about 1.5 times its area ratio, so in the TESS band the companion is brighter per unit area than the giant, as a G dwarf of about 0.9–1.0 $M_\odot$ would be. With the giant's radius fixed, the B eclipse yields the companion's size and the impact geometry, and these predict the A transit (19.3 h, about 2.2% after limb darkening). The transit is observed three times at 18.0–18.6 h and 2.06–2.25%. Both kinds of eclipse give nearly the same period (4.7σ apart formally).

No other dip of 1% or more occurs in the four sectors.

A21. TYC 4244-623-1 (TIC 403113935)



A21. Left: QLP SAP flux over 28 sectors, detrended with a 1-d running median with the eclipses masked, folded on P = 30.763077 d, Gaia's predicted conjunctions dashed. Middle: the 2.77%, 16.1-h primary transits. Right: the 0.27%, 10.4-h flat secondary occultations, one of which was reported as a planet candidate.

Identifiers	TIC 403113935; Gaia DR3 2245031052604901120; TYC 4244-623-1; CTOI 403113935.01
Position (J2000)	20:14:39.7 +66:08:00.4 (TESS northern continuous viewing zone)
Brightness	G 9.84; Tmag 9.41
Stars	F/G subgiant, R_1 = 2.50–2.66 $R_\odot$ (assuming 1.2–1.4 $M_\odot$) + early-M dwarf of 0.39–0.43 $M_\odot$, 0.39–0.41 $R_\odot$, about 3530 K; mass function 0.0239 $M_\odot$. Gaia SB1, significance 53, RUWE 34.9
Period	30.763077 ± 0.000013 d (Gaia 30.785932 ± 0.010666 d; the eclipse period is 2.1σ shorter)
Epoch of primary	BTJD 2849.89772 = BJD_{TDB} 2459849.89772; secondary BTJD 2830.31820
Primary eclipse	2.77 ± 0.09% deep, 16.1 h, rounded U (transit); 20 clean events (23 fitted), rms 1.3 min, no curvature
Secondary eclipse	0.266 ± 0.014% deep, 10.4 h, flat-bottomed (occultation); 20 clean events (23 fitted), rms 3.6 min
Orbit	

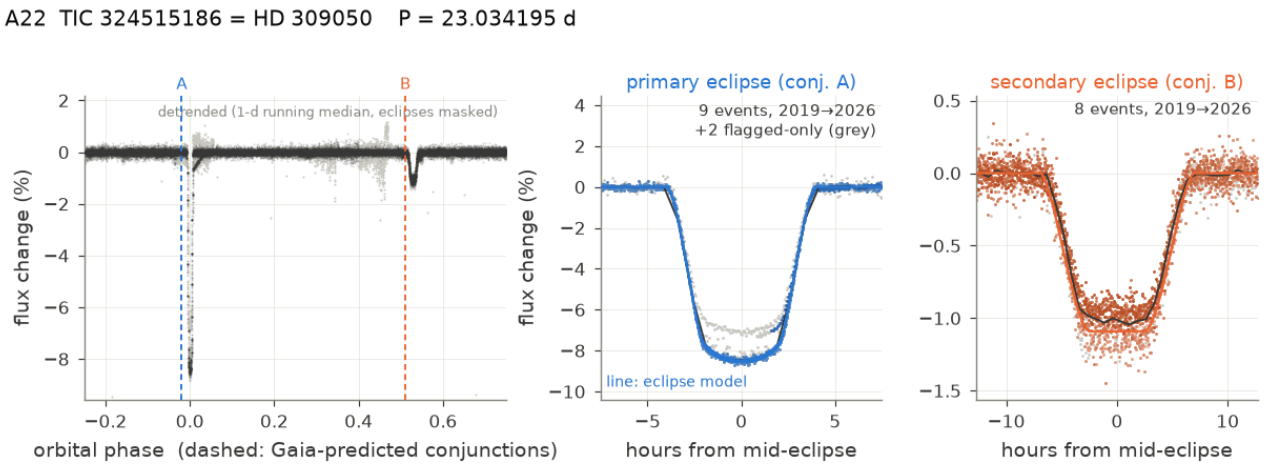
	Eclipse spacing 0.36354 of an orbit (Gaia predicts 0.40655): $e \cos \omega = -0.209$ against Gaia's -0.143 ± 0.018 (3.7σ); duration ratio gives $e \sin \omega = -0.23$ (Gaia -0.248). Eclipse-based $e \approx 0.32$, $\omega \approx 230°$ (Gaia 0.286, 240.06°). $i = 88.7°$, $k = 0.155$. No apsidal motion (primary and secondary periods agree to $(0.8 \pm 2.8) \times 10^{-5}$ d)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s056, s075); TASOC (s015); times within 6 min
Pixel attribution	No centroid shift in TESS-SPOC s056 during the primary. The only Gaia neighbour brighter than G 15 (G 13.93 at 49.9″) would need a 119% primary. RUWE 35 implies a probable star within about 1″
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, general web search; Kostov table 2 flag only. **A previous report misidentifies the signal:** ExoFOP CTOI 403113935.01 (submitted 2025-07-01 as a “single transit event”, epoch BJD 2459891.8453, 2,483 ppm, 9.9 h, 14.9 $R_\oplus$) is this star's secondary eclipse in sector 58. In the TESS-SPOC DV report (s56–s69), TCE 2 corresponds to the secondary eclipses
Open issues	Gaia's period and $e \cos \omega$ are inconsistent with the eclipse timing; an RV refit would constrain the true e and ω. High-resolution imaging for the RUWE-35 companion (a possible triple)
Next eclipses	Primary: BTJD 4357.2885 (2026-11-12 18:52 UTC), 4388.0516, 4418.8147; secondary: 4337.7090 (2026-10-24 04:57 UTC), 4368.4721 (about ±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with caveats. No difference image (only a null centroid shift in TESS-SPOC s056); RUWE 35 implies a star within about 1″; Gaia's period and $e \cos \omega$ are 2.1σ and 3.7σ off, so the eclipses follow Gaia's orbit only approximately.

Note: the star has an ExoFOP community planet candidate, CTOI 403113935.01, which is a misreading of the secondary eclipse of this 30.763-d eccentric eclipsing binary as a single planetary transit.

TYC 4244-623-1 is an F/G subgiant with an early-M companion on an eccentric 30.76-d orbit, observed in 28 TESS sectors. Twenty clean primaries and twenty clean secondaries give a period with 1.3-min timing rms. The primary depth is $1.14 \times k^2$, as limb darkening predicts. The ~1-d offset from Gaia's predicted times is a linear drift of −0.0229 d per orbit for both kinds of eclipse, which means that Gaia's period is too long. Extrapolated back to Gaia's epoch, the primaries and secondaries are offset in opposite directions, so Gaia's $e \cos \omega$ is also biased. Because TESS rules out apsidal motion, we attribute the discrepancy to Gaia's spectroscopic solution (RUWE 35, low significance).

RUWE 35 cannot come from this orbit, so a further star probably lies within about 1″, making the system a possible triple. The eclipse period is within 2.1σ of Gaia's SB1 period for the target, and the eclipses fall near its conjunctions. The secondary is phase-locked to the primary.

A22. HD 309050 (TIC 324515186)



A22. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.034195 d, Gaia's predicted conjunctions dashed. Middle: the 8.3%, 7.5-h flat-bottomed transits at A. Right: the 1.1%, 12.7-h total occultations at B.

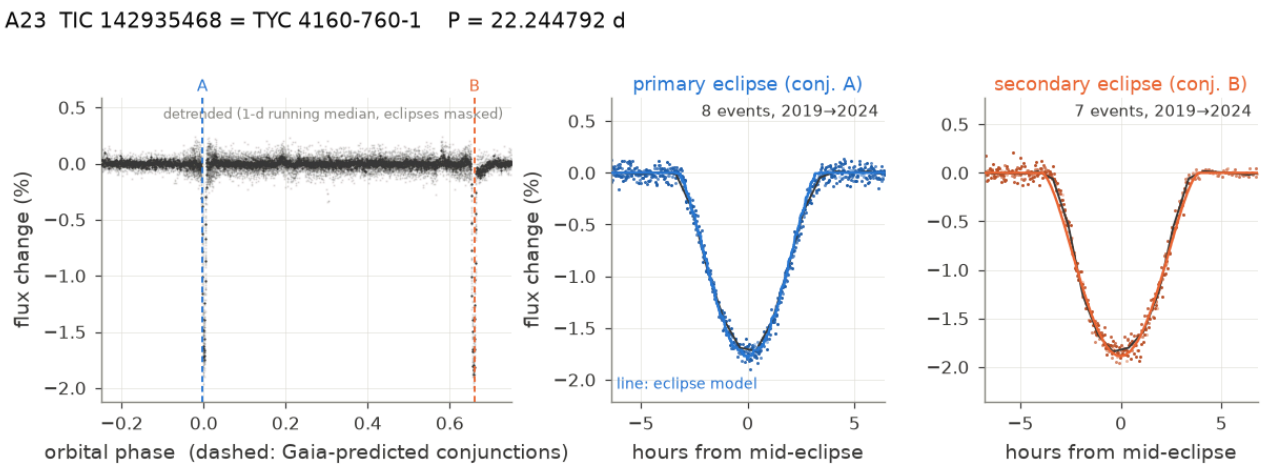
Identifiers	TIC 324515186; Gaia DR3 5333119169896513536; HD 309050
Position (J2000)	11:49:05.6 −63:46:58.3
Brightness	G 9.85; Tmag 9.44; V 9.97 (SIMBAD)
Stars	F2 subgiant, R_1 = 2.12 $R_\odot$ (TIC 2.21), M_1 = 1.39 $M_\odot$ (TIC) + M0 dwarf, m_2 = 0.572 $M_\odot$, R_2 = 0.59 $R_\odot$, about 3900 K, 1.09% of the light; mass function 0.0487 $M_\odot$; Gaia SB1
Period	23.034195 ± 0.000010 d (Gaia 23.0619 ± 0.0129 d; the eclipse period is 2.15σ shorter)
Epoch of primary	T_0(A) = BTJD 2657.89250 ± 0.00031 = BJD_{TDB} 2459657.89250
Primary eclipse (A)	7.9–8.33% deep, 7.5 h (flat part about 3.9 h), nearly flat-bottomed transit of the companion across the F star (b = 0.32); nine events, sectors 10–99, two in QLP-flagged cadences measured in other pipelines; O−C within ±0.3 min over 1500 d
Secondary eclipse (B)	0.92–1.14% deep, 12.4–13.8 h (flat part about 6 h), total occultation of the companion (b = 0.58); eight events; B − A = 12.17047 ± 0.00079 d (phase 0.52837)
Orbit	Joint physical model, e and ω free: e = 0.316, ω = 82.3° against Gaia 0.300 ± 0.017, 81.5 ± 3.7° (+0.9σ, +0.2σ); timing e cos ω = 0.0425 (Gaia 0.0443 ± 0.019). i = 88.72°, k = 0.277
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TGLC PSF; eleanor-lite; TARS (mid-times within 0–3 min for A, 2–11 min for B)
Pixel attribution	TESScut difference images (s37, s64): largest loss at the target pixel for both kinds, centroid of the loss within 0.04–0.1 px of the target; no loss at the G 14.1 star 18″ or the G 15.5 star 15″ away
Catalogue status	Not listed in: VSX (in a 120″ search the nearest entry, at 72″, is unrelated), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, SB9, WDS, Hipparcos, Gaia DR3 variability tables, CCDM, ExoFOP (no TOI/CTOI), 5 SIMBAD references, general web search. Kostov table 2 flags 16 faint stars 30–67″ away but not the star (probably contamination by the target's eclipses). A TESS-SPOC multi-sector DV report (s56–s69) detected the A eclipse at SNR 659; it never became a TOI
Open issues	None for the binary; two RV spectra would give the companion's absolute mass, an infrared spectral type would test T_2 ≈ 3900 K
Next eclipses	A: BTJD 4339.389, 4362.423; B: BTJD 4328.525, 4351.559 (±1.2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 309050 is an F2 subgiant with an early-M companion. At A the faint companion crosses the F star whose radial velocities Gaia measured. At B it is completely hidden, so the 1.09% depth is the companion's light directly. One geometry (k = 0.277, surface-brightness ratio 0.144, i = 88.7°) fits all six

modelled eclipses to the noise. Forcing 8% third light (SPOC's crowding) only moves k to 0.288. The light-curve e and ω, fitted freely, match Gaia's within 1σ. There is no sign of apsidal motion or timing variations.

The ~2-d offset from Gaia's predicted times is not constant. It grows from about −1 d to −4 d because Gaia's period is too long, and extrapolated back to Gaia's epoch the eclipses agree with Gaia's conjunctions (+1.3σ and +0.8σ). A Monte-Carlo conjunction predictor can jump a whole cycle when a sampled ω crosses 90° (conjunction A falls just after periastron here), inflating the window. The search's predictor re-centres the draws (Section 2.2) and was not affected.

A23. TYC 4160-760-1 (TIC 142935468)



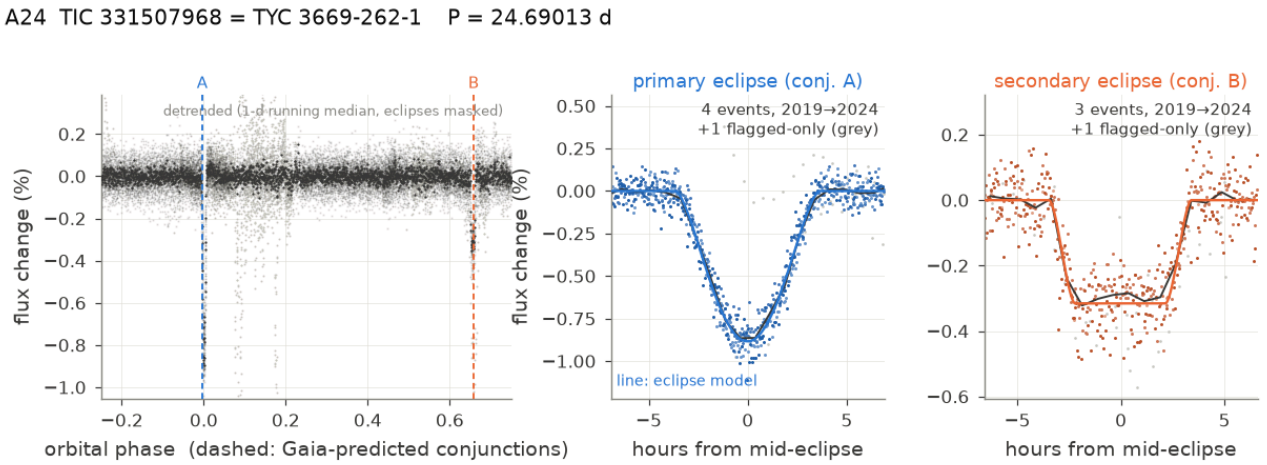
A23. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.244792 d, Gaia's predicted conjunctions dashed. Middle: the eight 1.70%, 6.4-h grazing transits at A. Right: the 1.8%, 6.8-h near-total occultations at B.

Identifiers	TIC 142935468; Gaia DR3 864679171266740736; TYC 4160-760-1
Position (J2000)	11:54:54.3 +66:03:48.2
Brightness	G 10.00; Tmag 9.45
Stars	Subgiant, R_1 = 1.97–2.15 $R_\odot$ (TIC 2.06), M_1 = 1.1–1.5 $M_\odot$ + M dwarf, m_2 = 0.57–0.68 $M_\odot$, R_2 = 0.59–0.65 $R_\odot$, about 3600 K, 1.9% of the light; mass function 0.0651 $M_\odot$; Gaia SB1
Period	22.244792 ± 0.000017 d (Gaia 22.247626 ± 0.00183 d; −1.55σ)
Epoch of primary	T_0(A) = BTJD 2228.68805 ± 0.00056 = BJD_{TDB} 2459228.68805
Primary eclipse (A)	1.66–1.76% deep, 6.0–6.7 h (about 6.4 h), rounded V/U: grazing transit (b = 1.10); eight events, sectors 14–75, O−C within ±3.5 min
Secondary eclipse (B)	1.73–1.91% deep, 6.6–7.0 h (about 6.8 h): near-total occultation of the companion (b = 0.75); five full and two partial events; B − A = 14.62903 ± 0.00091 d (phase 0.65764)
Orbit	Timing e cos ω = 0.2455 against Gaia's 0.2546 ± 0.0065 (−1.4σ); a model grid prefers Gaia's own e sin ω (−0.193). i = 87.17°, k = 0.30 (grazing eclipses let k, i and e sin ω trade off)
Pipelines agreeing	QLP SAP; SPOC 2-min s75; TESS-SPOC s75; TASOC s14/s15; TARS (13 events, within ±10 min)
Pixel attribution	SPOC s75 pixel file and TESScut s41/s48: the loss has the star's image shape, centroid within 0.1 px of the target; no Gaia source brighter than G 16 within 100″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 (all tables), WDS, Hipparcos, SB9, Gaia DR3 variability tables, CCDM, ExoFOP, 2 SIMBAD references (SB* from Gaia DR3), general web search. No SPOC DV products
Open issues	The k–i–e sin ω trade-off; radial velocities, multi-colour photometry of B and higher-cadence ingress/egress would break it
Next eclipses	A: BTJD 4341.943, 4364.188; B: BTJD 4334.328, 4356.572 (±2.5 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4160-760-1 is a subgiant with a cool dwarf companion of about 0.6 $M_\odot$. Its two eclipses are nearly equal in depth. At A the companion just clips the primary's limb and blocks about 1.7% of the light. At B, the closer conjunction, most of the companion is hidden, and the depth of about 1.8% is roughly the companion's whole light. The near-equal depths are therefore a coincidence of geometry. A companion carrying 2% of the light would not show in Gaia's spectra, so a single-lined solution is what one expects.

The eclipse durations are about 6.4 h and 6.8 h, and the offset from Gaia's predicted times is again a slow drift caused by the error in Gaia's period. Because both eclipses graze, the duration ratio does not constrain e sin ω well. The best grid point is Gaia's own value, with k = 0.30 (rms 410 ppm over eight modelled eclipses).

A24. TYC 3669-262-1 (TIC 331507968)



A24. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.69013 d, Gaia's predicted conjunctions dashed. Middle: the 0.86%, 6.9-h near-grazing V-shaped transits at A. Right: the 0.32%, 6.6-h flat-bottomed total occultations at B.

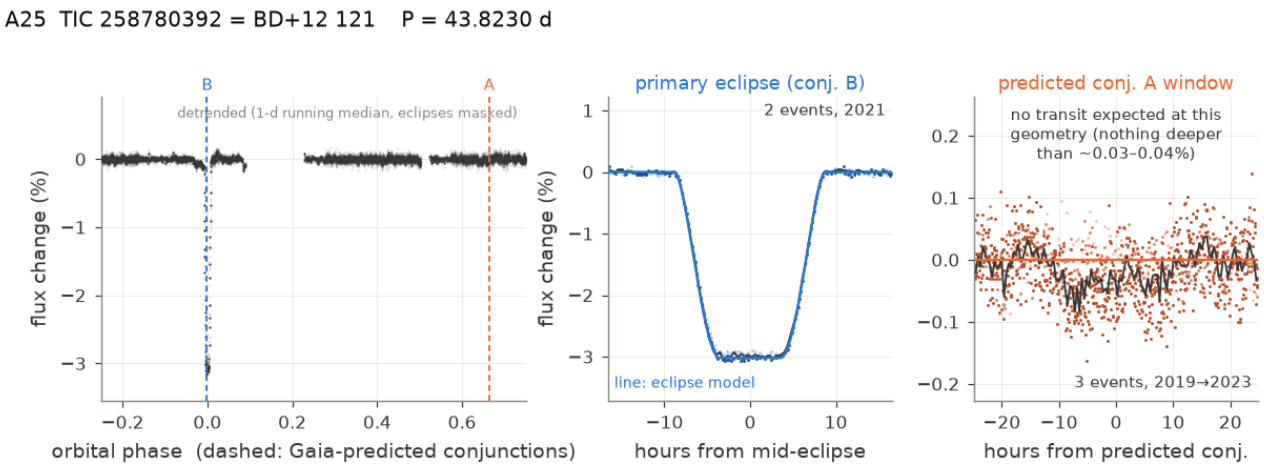
Identifiers	TIC 331507968; Gaia DR3 411152889920798336; TYC 3669-262-1
Position (J2000)	01:11:25.8 +54:21:26.3
Brightness	G 9.83; Tmag 9.28
Stars	G/K subgiant, R_1 = 1.89–2.11 $R_\odot$ (TIC 2.06), log g ≈ 3.85–3.9 + mid-M dwarf, M_2 = 0.29–0.36 $M_\odot$, R_2 = 0.29–0.33 $R_\odot$, roughly 3300 K, 0.32% of the light; mass function 0.01654 $M_\odot$; Gaia SB1
Period	24.69013 ± 0.00003 d (Gaia 24.68806 ± 0.00454 d; 0.46σ)
Epoch of primary	T_0(A) = BTJD 1797.4225 ± 0.0015 = BJD_{TDB} 2458797.4225; T_0(B) = BTJD 1813.6099 ± 0.0022
Primary eclipse (A)	About 0.86% deep, 6.9 h, nearly V-shaped (flat part 1.6 h): near-grazing transit of the subgiant (impact 0.997 R_1); four complete events (s18, s58 ×2, s85) and a partial in QLP-flagged s17 cadences; O−C −3.3 to +2.3 min
Secondary eclipse (B)	About 0.32% deep, 6.6 h, flat-bottomed (T_{23} ≈ 4.6 h): total occultation of the M dwarf (impact 0.535 R_1); three events (s18, s58, s85); phase 0.65562 ± 0.00011 after A
Orbit	One two-sphere Keplerian model fits all seven complete eclipses at χ^2/dof 1.055: e = 0.3824, ω = 307.96°, i = 87.67°, k = 0.155. Gaia: e 0.4019 ± 0.0059 (3.3σ apart), ω 307.30 ± 1.21° (within 0.5σ); e cos ω 0.2340 against 0.2435 ± 0.0076 (1.2σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP; TARS (mid-times within 0.002 d). No SPOC 2-min data
Pixel attribution	TESScut difference images (s58, s85 A): same shape as the target's PSF, peak in the target's own pixel, centroid offset 0.2–0.55 px (within the noise of a 0.8% signal). The VSX “RS” star 23″ away (G 17.89) would need a 1440% A dip; every other source within 60″ is G ≥ 17.07. No common mode (median of 110–150 stars −73 to +261 ppm)
Catalogue status	

	Not listed in: VSX (the “RS” entry 23″ away is a separate *G* 17.9 background star with no period), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI/CTOI), SIMBAD (SB*, one reference), general web search
Open issues	The photometric e is 3.3σ below Gaia's by Gaia's formal error; the tension is mild, since Gaia SB1 errors are commonly underestimated by a factor of about 1.5–2. A few RVs would refine e
Next eclipses	A: BTJD 4340.506, 4365.196 (±5 min); B: BTJD 4332.003, 4356.693 (±6 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3669-262-1 is a G/K subgiant of about 2.0 $R_\odot$ with an M dwarf of about 0.3 $M_\odot$. The A eclipse is a grazing transit of the subgiant, hence the V shape and a depth well below k^2. The B eclipse is a total occultation of the M dwarf, hence the flat bottom. The separation at A is 1.22a against 0.66a at B, so one inclination gives an impact parameter 1.9 times larger at A. The model's durations (7.3 and 6.8 h) match those measured. The occultation depth puts the companion at an absolute TESS magnitude of about 9.4, as expected for a dwarf of about 0.3 $M_\odot$.

The eclipse timing constrains e cos ω to about 10^{-4}. Fixing e and ω at Gaia's values raises χ^2 by more than 1600, so we adopt the photometric e and ω as the better determined. A further A eclipse, in fully flagged sector-17 cadences, is present in TESS-SPOC and TARS.

A25. BD+12 121 = ASAS J005930+1304.9 (TIC 258780392)



A25. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 43.8230 d, Gaia's predicted conjunctions dashed. Middle: the B windows with data; two identical 2.9%, 16.5-h flat-bottomed occultations (sectors 42 and 43) and a third window covering only the edge of an event. Right: the data at conjunction A, where the fitted geometry predicts no eclipse and none is seen (< 0.04%).

Identifiers	TIC 258780392; Gaia DR3 2776639323993879680; BD+12 121; ASAS J005930+1304.9
Position (J2000)	00:59:30.3 +13:04:56.1
Brightness	G 9.16; Tmag 8.52
Stars	G5/K giant (SIMBAD Sp G5), R_1 = 4.9–6.1 $R_\odot$ (TIC 6.07) + K0–K2 dwarf, $M_2 \approx 0.8$ $M_\odot$ (0.73–1.12 allowed), R_2 = 0.89–1.11 $R_\odot$, 3.0% of the light; mass function 0.1270 $M_\odot$; Gaia SB1
Period	43.8230 ± 0.0025 d from two eclipses one cycle apart; cycle count fixed by Gaia (Gaia 43.8190 ± 0.0114 d; 0.33σ)
Epoch of eclipse	T_0(B) = BTJD 2451.287 ± 0.0013 = BJD_{TDB} 2459451.287
Eclipse (B)	2.89–2.92% deep, 16.5 h, flat-bottomed (T_{23} 8.8–8.9 h, ingress/egress about 3.9 h): total occultation of the dwarf by the giant; two events (s42, s43) matching to 0.03% in depth and 0.1 h in duration; both 0.24 d (0.4σ) after Gaia's predicted B times
Other eclipse (A)	None, as predicted: at the fitted i the impact parameter at A is 1.29–1.34 R_1, beyond 1 + k = 1.18 in all 11 Gaia draws. Box limits at A: 120–240 ppm (s043), 280–430 ppm (s070)

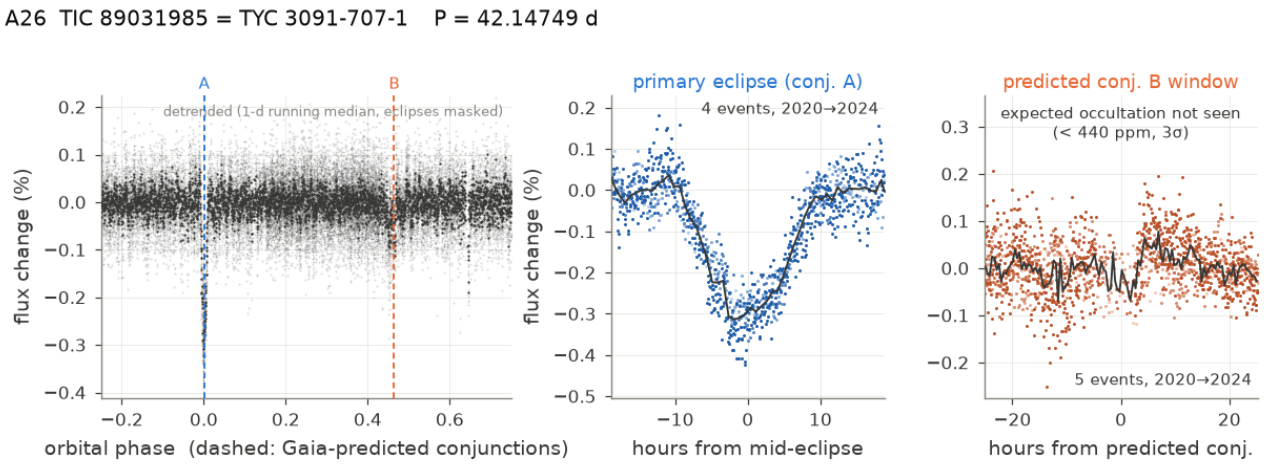
Orbit	e and ω held at Gaia draws (e 0.3841, ω 229.40°); one eclipse kind gives no independent $e \cos \omega$. i = 85.0–85.2°, k = 0.182
Pipelines agreeing	QLP SAP (quality-0 and all cadences); TARS (times within 0.003 d; TARS s43 depth 2.44%). TESS-SPOC exists only for s070
Pixel attribution	TESScut difference images (s42, s43): same shape as the target's PSF, offsets 1.0″ and 3.2″; the raw per-pixel depth is a flat ~2.8% across the whole PSF. Nearest Gaia neighbour G 19.58 at 17.7″. No common mode
Catalogue status	**VSX type revision.** VSX: ASAS J005930+1304.9 at the target, type MISC, P 91.5 d, amplitude 0.11 mag V; ASAS-SN "ROT:", no period. No eclipsing-binary catalogue, no Gaia DR3 variability row, nothing in SB9 or WDS, no TOI/CTOI, SIMBAD (SB*, 4 references); a general web search found nothing
Open issues	Only two eclipses; a third B eclipse and a spectrum (for M_1, R_1 and the companion's type) would improve the solution. An A transit, if ever seen, would force $i \geq 86°$
Next eclipses	B: BTJD 4335.68 ± 0.11, 4379.50 ± 0.11; A conjunction (no eclipse expected): 4364.83 ± 0.20
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision. The VSX entry ASAS J005930+1304.9 (MISC, 91.5 d) is this star, but it describes slow variability of the giant, not the eclipses. The ASAS-3 periodogram has power 0.35 at 91.2 d and 0.02 at twice the orbital period, and 0.03-mag eclipses filling 1.6% of the orbit cannot produce a 0.1–0.2-mag, 91-d signal. We propose reclassification from MISC/ROT to EA + ROT.

BD+12 121 is a G5/K giant with a K-dwarf companion. TESS observed two identical total eclipses one orbit apart, in which the giant hides the dwarf. Their depth, 3.0%, is the companion's share of the light. The same geometry predicts that the dwarf passes clear of the giant at the other conjunction. The two covered A windows are flat to 0.03–0.04%, so the data require the absence of an A eclipse, as Gaia's geometry predicts. The companion's absolute TESS magnitude (about 5.0) points to a K0–K2 dwarf of about 0.8 $M_\odot$. The light, mass function and radii are consistent.

The 91-d ASAS signal is attributed to slow, season-dependent modulation of the giant, most likely spots or activity, in agreement with ASAS-SN's independent "ROT:". TESS shows drifts of at most about 1.5% per sector and neither confirms nor contradicts that period.

A26. TYC 3091-707-1 (TIC 89031985)



A26. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 42.14749 d, Gaia's predicted conjunctions dashed. Middle: the four 0.29%, 18.8-h limb-darkened transits at A, 2019–2024. Right: the B window, where an occultation of about 0.2% was expected and none is seen (< 440 ppm, 3σ).

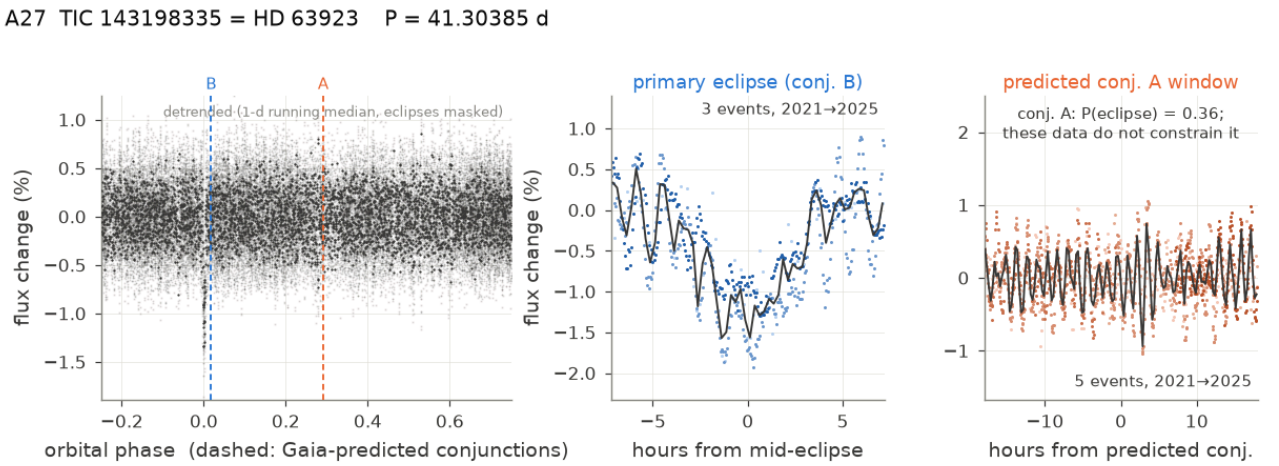
Identifiers	TIC 89031985; Gaia DR3 1344257629016319744; TYC 3091-707-1
Position (J2000)	17:36:17.8 +40:49:02.9

Brightness	*G* 9.46; Tmag 8.79
Stars	K giant, $R_1 \approx 7.5\ R_\odot$ (7.53 used) + a body of $R_2 \approx 0.54\ R_\odot$, minimum mass 0.50–0.67 $M_\odot$ for M_1 = 1.2–2.5 $M_\odot$; mass function 0.0341 $M_\odot$; systemic RV −57 km/s (SIMBAD); Gaia SB1
Period	42.14749 ± 0.00117 d (Gaia 42.144303 ± 0.003614 d; +0.8σ)
Epoch of eclipse	T_0(A) = BTJD 2645.4034 ± 0.0122 = BJD_{TDB} 2459645.4034 (cycle 15 of the fit)
Eclipse (A)	Stack: 2880 ppm deep (2535–3481 ppm per event), T_{14} = 18.8 h (17.0–21.7 h), flat fraction 0.2, limb-darkened U: a near-limb transit of the giant (impact ≈ 0.92 R_1); four events, sectors 26, 40, 52, 80; O−C −55 to +22 min (the outlier is the 30-min sector 26)
Other eclipse (B)	**Absent, though expected:** combined depth +29 ± 137 ppm over five windows, < 440 ppm at 3σ (< 917 ppm at any *e* cos ω); the physical Monte Carlo model predicts B with probability 0.999 and depth 2100 ppm [900, 4300]; only 4.6% of draws fall below 440 ppm
Orbit	With the TESS period and Gaia's t_{peri}, *e* (0.0631), ω (154.80°), the eclipses sit 0.0–0.1σ from Gaia's A time (±0.53 d) and about 35σ from B. *e*, ω not derivable from one eclipse kind
Pipelines agreeing	QLP SAP; TARS (all four); TASOC FLUX_RAW (s026); TESS-SPOC SAP and PDCSAP (s080)
Pixel attribution	TESScut difference images peak on the target pixel in all four sectors (offsets 0.28–0.35 px in the clean sectors); the *G* 15.65 star at 19.8″ would need 83% and is 10–24σ off; the *G* 11.56 star at 79″ is excluded by its own pixel. No common mode (target ranks 2nd–5th of 87–232 stars)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos (main, new reduction, variability annexes), SB9, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, one reference), general web search; not in the PTPS/JOTA SB1 sample (Jaros et al. 2026)
Open issues	The missing B eclipse: the companion's TESS-band surface brightness must be ≲ 0.13 of the giant's ($T_2 \lesssim 3050$ K), which contradicts a single main-sequence star of ≥ 0.5 $M_\odot$. The A depth is slightly shallow against the 4000 ppm predicted
Next eclipses	A: BTJD 4331.303 ± 0.048 (2026-10-17 19:15 UT), 4373.451 ± 0.049 (allow ±1 h); B predicted, not seen: 4350.85 ± 0.11
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3091-707-1 is a K giant of about 7.5 $R_\odot$ crossed every 42.15 d by a body of about 0.54 $R_\odot$ near the giant's limb, consistent with the mass function for a single K7–M0 dwarf. The four transits are on the target, reproduced by three other pipelines, and fall at Gaia's A conjunction.

The absence of an occultation at B is unexplained. The separation grows only 5.5% at B, so the companion should pass behind the giant and an occultation of about 0.2% was expected. The data limit it to below 440 ppm (injected 1000 and 2000 ppm signals were recovered exactly). To hide it, the companion must be fainter per unit area than a single main-sequence star of its mass. Two explanations remain viable. The companion may be a close M+M pair: two 0.3 $M_\odot$ stars at about 3400 K predict about 500 ppm, close to the limit. Alternatively, Gaia's *e* or ω may be substantially in error. A hot compact companion is excluded by GALEX's far-ultraviolet magnitude and by the radius. The missing occultation does not affect the detection. A spectrum, multi-band photometry of a transit, a TESS sector covering B, and an RV orbit would resolve it.

A27. HD 63923 (TIC 143198335)



A27. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 41.30385 d, Gaia's predicted conjunctions dashed. Middle: the three 1.07%, 7.2-h eclipses at conjunction B (sectors 35, 62, 88). Right: the data near conjunction A, which do not constrain an eclipse there.

Identifiers	TIC 143198335; Gaia DR3 5518561289173094400; HD 63923
Position (J2000)	07:49:14.9 −47:07:24.1
Brightness	G 9.94; Tmag 9.68
Stars	F0V (Michigan Spectral Survey), $R_1 \approx 2.7\ R_\odot$, with δ Sct-like variability; companion of m_2 = 0.63–0.71 $M_\odot$ for M_1 = 1.5–2.0 $M_\odot$ (a K dwarf on the main sequence); mass function 0.051 $M_\odot$. Gaia SB1, significance 20
Period	41.30385 ± 0.00030 d (Gaia 41.295873 ± 0.068698 d; +0.12σ)
Epoch of eclipse	T_0(B) = BTJD 2963.0484 ± 0.0044 = BJD_{TDB} 2459963.0484
Eclipse (B)	1.07% deep (1.03–1.17% per event), T_{14} = 7.1–7.4 h, flat fraction 0.3–0.4 (a V shape is worse by $\Delta\chi^2$ = 13, weak given the red noise); three events, sectors 35, 62, 88; O−C −5 to +9 min
Other eclipse (A)	Not constrained: P(eclipse) only 0.36 at A; the predicted centre falls on flagged data or a gap in s035 and s088
Orbit	With the TESS period and Gaia's t_{peri}, e, ω, the eclipses sit −0.6σ (±1.2 d) from B and 7.7σ from A; chance coincidence of an unrelated eclipser about 10^{-4}. e sin ω = −0.14 (Gaia), so B is near closest approach. Not derivable from one eclipse kind
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s035, s062); TARS (s035, s062, s088); CDIPS (s035); t_0 within 3–9 min
Pixel attribution	Difference images peak on the target in all three sectors; the fitted offset of the eclipse light excludes the 6.2″ G 14.38 neighbour at 8.7–16σ and the 35–80″ stars at 63–80σ (the target hypothesis is itself off by 2.2, 4.9 and 4.5σ). Only an unresolved source within about 2″ remains possible; TIC contamination ratio 0.243. No common mode
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP (contamination ratio 0.243), SIMBAD (SB*, nine references, none on eclipses), general web search; no GALEX match
Open issues	**Physical model unresolved.** An occultation of the mass-function companion cannot easily be this deep (at most 0.84%, range 0.62–1.13%, even if total) or this short (a total occultation would last ≳ 10 h); a ~0.3 $R_\odot$ transiting body fits the shape but not the mass
Next eclipses	B: BTJD 4367.379 ± 0.011 (2026-11-22 21:04 UTC), 4408.683 ± 0.011; A possible: 4337.45 ± 1.13
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	

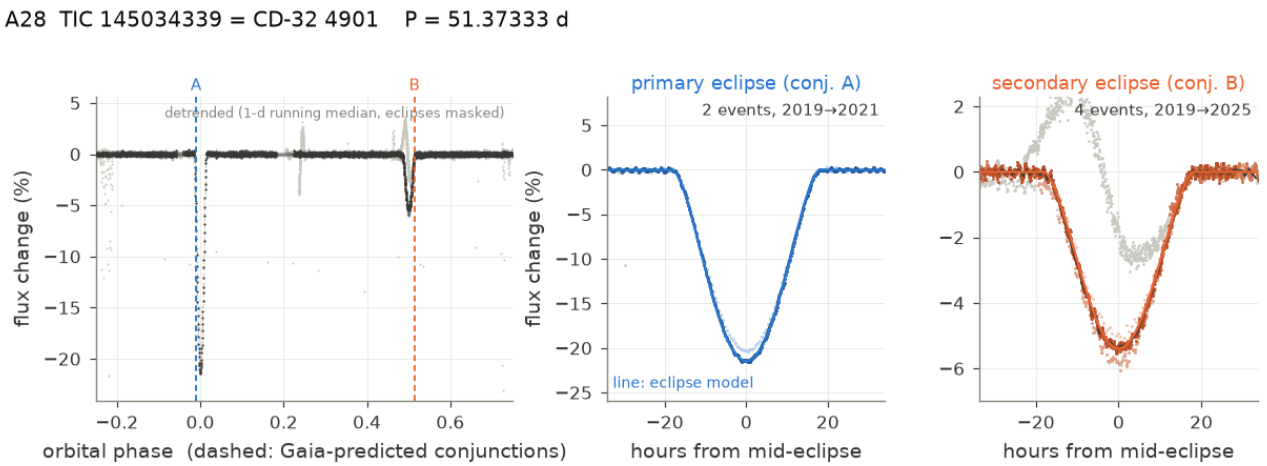
Confirmed, with caveats. The difference-image fit places the eclipse light on HD 63923 against the 6.2″ neighbour at 8.7–16σ, but the target hypothesis itself is off by 2.2, 4.9 and 4.5σ in the three sectors, so an unresolved source within about 2″ is not excluded; the 1.07% depth lies at the top of the 0.62–1.13% range allowed for an occultation of the mass-function companion (central value 0.84%); the physical model is unresolved.

Note: confirmed as an eclipsing system on the target (or on an unresolved source within about 2″), with its physical model unresolved. Either Gaia's orbit is in error (SB1 significance only 20, goodness-of-fit −1.5), or the companion is not a single main-sequence star.

HD 63923 is an F0 dwarf with three 1.07%, 7.2-h eclipses at Gaia's B conjunction. Three other pipelines reproduce the eclipses, and the difference images put the light on HD 63923 and not on its 6.2″ neighbour. The sector-62 eclipse (BTJD 3004.36) is distinct from a sector-wide systematic feature at BTJD 3007.

What the system is remains open. Under the convention validated on TIC 403113935 (A21), B means the companion passes behind the F star. However, an occultation of a 0.64 $R_\odot$, ~4300 K dwarf by a 2.7 $R_\odot$ F0 star can be at most 0.84% deep. The duration and flat fraction also imply R_1R_2 = 0.77–0.87 $R_\odot^2$ where about 1.7 is needed. A transit of the F star by a ~0.3 $R_\odot$ body ($k^2 \approx 1.2\%$) fits depth and shape. That reading, though, is the opposite of the convention at this conjunction, and no 0.67 $M_\odot$ object has that radius. An independent RV orbit is the decisive measurement. Further constraints would come from multi-colour eclipse depths (an occultation of a cool companion deepens to the red; a transit is grey), from high-resolution imaging within 2″, and from uncontaminated TESS coverage of an A window.

A28. CD-32 4901 (TIC 145034339)



A28. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 51.37333 d, Gaia's predicted conjunctions dashed; the secondary sits at phase 0.5000. Middle: the two 20–21%, 1.42-d primary eclipses (sectors 7 and 34). Right: the 5%, 1.4-d secondary eclipses.

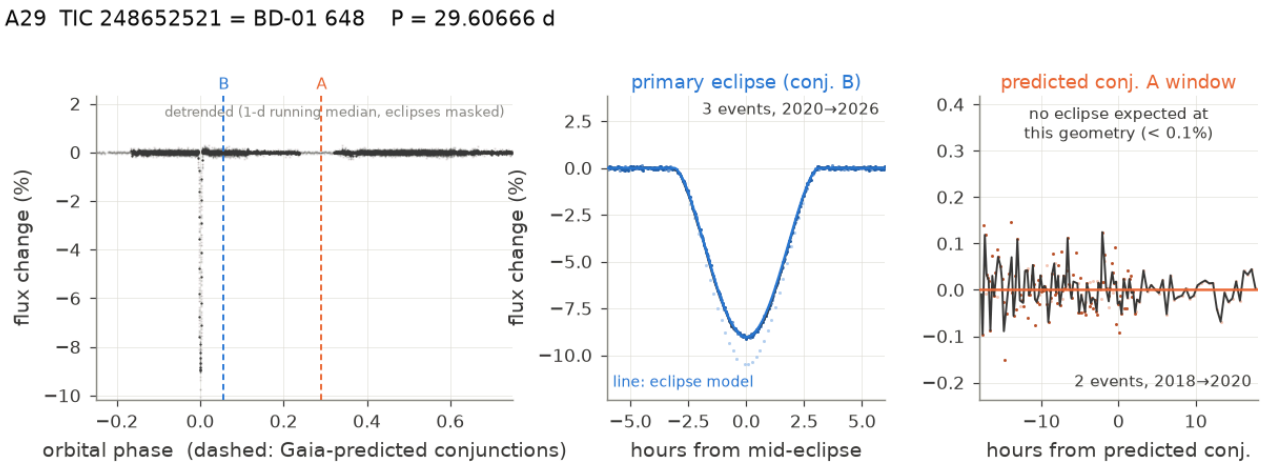
Identifiers	TIC 145034339; Gaia DR3 5547641825546896128; CD-32 4901 = CPD-32 2009 = TYC 7125-1800-1
Position (J2000)	08:08:46.9 −32:41:34.1
Brightness	*G* 9.74; Tmag 9.25
Stars	Gaia SB2 pair of large stars: $M_1 \sin^3 i$ = 1.147 $M_\odot$, $M_2 \sin^3 i$ = 0.577 $M_\odot$ (mass ratio 0.50 from Gaia's amplitudes); radii sum 8.5 $R_\odot$; companion 66% of the TESS light; TIC radius 8 $R_\odot$. X-ray source 1RXS J080846.6-324131 at 5.2″ and EUVE J0808-32.6 at 10″
Period	51.37333 ± 0.0002 d (Gaia 51.310621 ± 0.013853 d; the eclipse period is 0.063 d longer, 4.5σ)
Epoch of primary	T_0(A) = BTJD 1512.2307 ± 0.002 = BJD_{TDB} 2458512.2307

Primary eclipse (A)	20.0–21.2% deep, 1.42 d, partial V; two events (s7, s34)
Secondary eclipse (B)	4.9–5.5% deep, 1.38–1.42 d; four events (s8, s35, s61 on a scattered-light bump measured in TESS-SPOC, s88); O−C ≤ 0.0018 d; B − A = 25.6878 d (phase 0.50002)
Orbit	Circular: $e \cos \omega = 0.00004 \pm 0.0001$ against Gaia's 0.038 ± 0.007 (5.6σ); equal durations give $\lvert e \sin \omega \rvert \lesssim 0.01$ (Gaia -0.017 ± 0.006, not ruled out). Gaia $e = 0.042$, goodness-of-fit 14.9. Two-sphere fit: $i = 85.4°$
Pipelines agreeing	QLP SAP; TGLC PSF and aperture; GSFC-eleanor-lite; T16 difference imaging; TESS-SPOC (s61)
Pixel attribution	TESScut: the missing flux has the target's PSF shape, difference-centroid offset 0.026 px (A); neighbours within 60″ (G 14.75, 14.92) would need about 2100% and 2500%; RUWE 0.83
Catalogue status	Not listed under any name in: VSX, ASAS-SN, ASAS-3, GCVS, SB9, WDS, Hipparcos, Gaia DR3 variability tables (never tested by Gaia), Prša et al. 2022, Justesen & Albrecht 2021, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4 (three neighbours are in table 2, the target is not), ExoFOP, SIMBAD (SB*, one paper), general web search
Open issues	Gaia's e is probably spurious; Gaia's mass ratio would make the 0.58 $M_\odot$ star evolved, probably stripped. The weakest constraint is Gaia's mass ratio, not the eclipses. Partial eclipses leave the radius ratio free (the fit chose 2.3 and 6.2 $R_\odot$)
Next eclipses	A: BTJD 4337.764, 4389.137 (±0.011); B: BTJD 4363.450, 4414.823 (±0.005)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-32 4901 is a double-lined binary of two large stars on a 51.4-d orbit, with eclipses lasting 1.4 d. A 1.42-d eclipse needs the two radii to sum to at least 6 $R_\odot$. A circular two-sphere fit gives 8.5 $R_\odot$, partial V-shaped eclipses as observed, and a companion supplying 66% of the light. The depths need both stars large and bright, consistent with an SB2 and with the TIC radius. The nearby X-ray and EUV sources are consistent with an active giant binary. The photometric period is 4.5σ longer than Gaia's, but Gaia's conjunctions carried forward with it land within 0.3–0.5σ of the eclipses, and the deep eclipse falls at Gaia's A.

The secondary sits at phase 0.50002, so the orbit is circular, as expected for a giant at P = 51 d. Gaia's e = 0.042 disagrees by 5.6σ and is probably spurious given its goodness-of-fit of 15. If Gaia's velocity amplitudes are right, the less massive star must be evolved and probably stripped, as in a giant binary after mass transfer. Ground-based radial velocities and a spectrum of the cool component would resolve it.

A29. BD-01 648 (TIC 248652521)



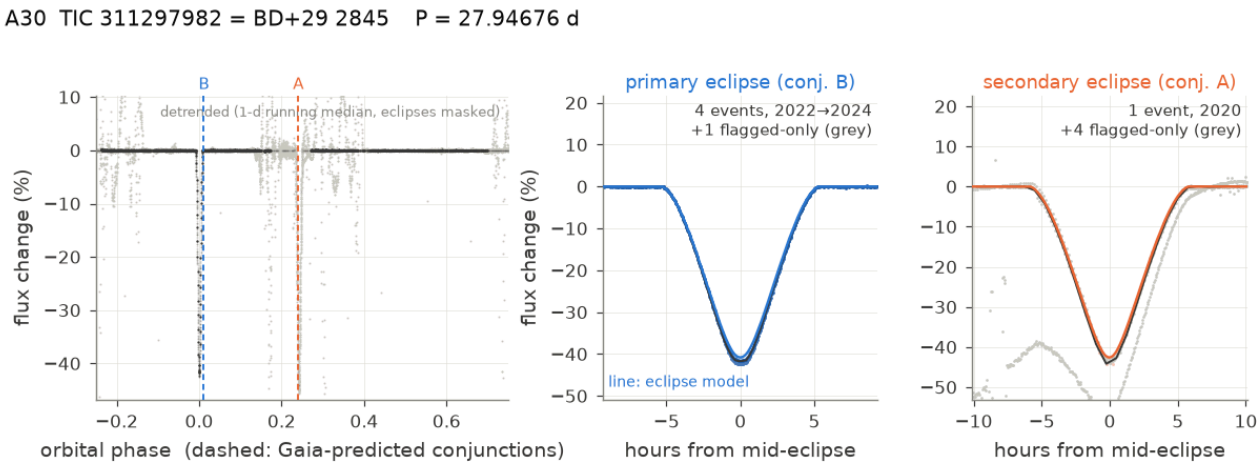
A29. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.60666 d, Gaia's predicted conjunctions dashed (the eclipse falls 1.67 d before Gaia's B). Middle: the three 9.5%, 5.8-h partial eclipses (sector 32, and two in sector 98; the sector-61 window, on a flagged scattered-light feature, is shown in a lighter tone and not used). Right: the A windows, flat to 0.1%, as the geometry predicts.

Identifiers	TIC 248652521; Gaia DR3 3206318533868790016; BD-01 648 = TYC 4727-450-1
Position (J2000)	04:28:42.2 −01:22:16.5
Brightness	*G* 9.99; Tmag 9.57
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 1.15 $M_\odot$, $M_2 \sin^3 i$ = 1.20 $M_\odot$ (mass ratio 1.04); twin model R = 1.69 $R_\odot$ each (radii free: 1.89 and 1.59), matching the luminosity (about 1.5 $R_\odot$ each at 6000 K)
Period	29.60666 ± 0.00002 d (Gaia 29.588805 ± 0.005297 d; the eclipse period is 0.0178 d longer, 3.4σ)
Epoch of eclipse	T_0 = BTJD 2177.8384 ± 0.0005 = BJD_{TDB} 2459177.8384
Eclipse (B, periastron side)	9.47–9.61% deep in a fixed aperture (QLP 8.8–10.2%), 0.24 d (5.8 h), rounded V (partial); three events (s32, s98 ×2); all O−C below 2×10^{-5} d (under 3 s)
Other eclipse (A)	Absent, not in a gap: all four A windows have data and are flat to 0.1%; the quality-0 data cover 99.7% of the orbit with no second eclipse deeper than 0.1% at any phase. At A the stars miss each other by 1.53 times the contact distance
Orbit	Gaia *e* 0.5486, ω 316.76° held fixed; *i* = 85.07°, B at 0.70 of contact. B is the conjunction nearest periastron (separation 0.51*a*, sky speed 150 km/s; A at 1.12*a*, 68 km/s)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s32); TARS (s32)
Pixel attribution	TESScut difference images: PSF-shaped deficit, centroid offset 0.03 px in all three events; the *G* 12.20 star at 90″ visibly does not dim; the *G* 14.30 star at 38″ would need 530%; RUWE 0.91
Catalogue status	Not listed under any name in: VSX, ASAS-SN, ASAS-3, GCVS, SB9, WDS, Hipparcos, Gaia DR3 variability tables, Prša et al. 2022, Justesen & Albrecht 2021, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP, SIMBAD (SB*, one paper), general web search
Open issues	The eclipse falls 1.669 d (0.056 in phase) before Gaia's B conjunction at Gaia's epoch: formally 25σ. Most likely Gaia's SB2 solution is distorted (goodness-of-fit 20); the alternative, that the eclipsing pair is not the Gaia pair, is much less likely but not excluded
Next eclipses	B: BTJD 4339.125, 4368.731 (±0.002); A (expected absent): 4346.09 ± 0.15
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with a caveat. The eclipse is 1.67 d (formally 25σ) from Gaia's B conjunction at Gaia's own epoch; we judge a distorted Gaia SB2 solution the likelier cause, but that the eclipsing pair is not the Gaia pair remains open.

BD-01 648 is a pair of near-twin stars of about 1.2 $M_\odot$ and 1.7 $R_\odot$ in an eccentric ($e \approx 0.55$) 29.6-d orbit. The stars partially eclipse each other near periastron, where they are close and fast. At the other conjunction they are more than twice as far apart and miss each other by 1.53 contact distances. The model predicts that A would eclipse only above about 87.5°, and B would then be at least 30% deep. The data show no second eclipse anywhere in the orbit. Three eclipses spanning five years agree to under three seconds.

The eclipse phase remains unresolved. Gaia's periastron time sits in the middle of its data span, so Gaia's period error cannot move the phase there. Yet the eclipse falls 1.67 d early. Shifting Gaia's orbit by that much would change the primary's velocity by 18 km/s rms, which Gaia's own velocities would show. The likelier explanation is that Gaia's SB2 solution is distorted (goodness-of-fit 20). Swapping the two components between epochs is the usual way near-twin SB2 fits go wrong. The much less likely alternative is that the eclipsing pair is not the Gaia pair. A few ground-based radial velocities near BTJD 4339 and around the following periastron would resolve it.

A30. BD+29 2845 (TIC 311297982)



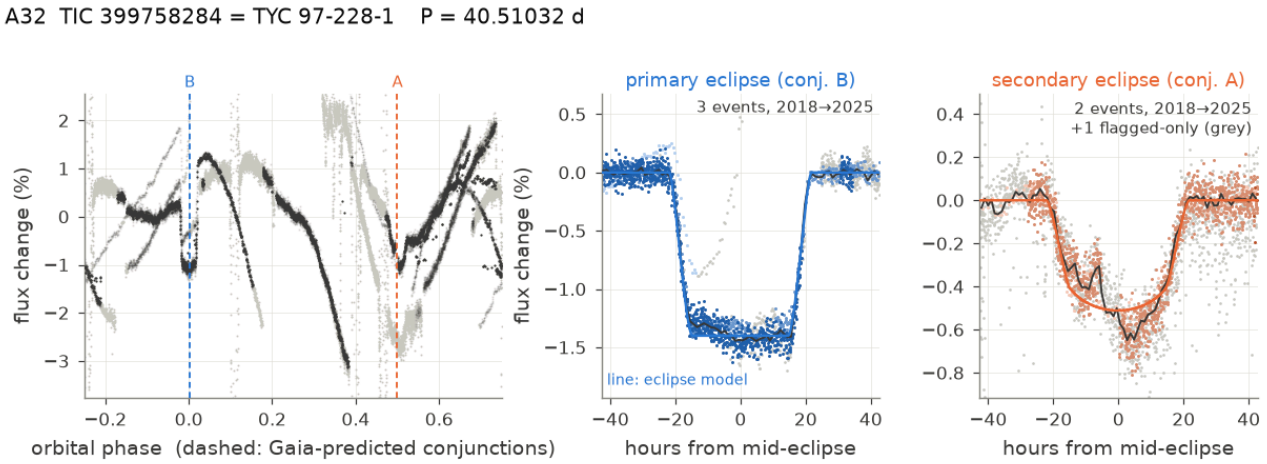
A30. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 27.94676 d, Gaia's predicted conjunctions dashed; flagged cadences, which hold half the eclipses, in a lighter tone. Middle: the five 40–42%, 9.3-h eclipses at B. Right: the five 42–45%, 10-h eclipses at A.

Identifiers	TIC 311297982; Gaia DR3 1305751357421836416; BD+29 2845 = TYC 2056-380-1 = SAO 84430 = PPM 104960 = AG+29 1593 = 2MASS J16314244+2901378
Position (J2000)	16:31:42.5 +29:01:37.9
Brightness	G 10.00; Tmag 9.60
Stars	Two slightly evolved F/G stars, Gaia SB2 (and APOGEE SB2, Kounkel et al. 2021): M $\sin^3 i$ = 1.12 + 1.09 $M_\odot$ with the TESS e (essentially the true masses at i = 89.5°); R_1 = 1.58, R_2 = 1.34 $R_\odot$; $L_2/L_1 \approx 0.71$; Gaia component temperatures 6110 and 5760 K; a = 50.4 $R_\odot$
Period	27.94676 ± 0.00001 d (Gaia 27.94708 ± 0.0013 d; −0.25σ)
Epoch of primary	T_0(B) = BTJD 1987.74538 ± 0.00042 = BJD_{TDB} 2458987.74538; T_0(A) = BTJD 1994.50441 ± 0.0018
Eclipse at A	42–45% deep, 9.9–10.3 h, near-V (nearly central, impact about 0.12–0.13 of R_1+R_2): star 1 (Gaia's primary) eclipsed by star 2; five events (s25, s51, s52, s78, s79), four in flagged cadences; O−C ±3 min
Eclipse at B	40–42% deep, 9.3–9.4 h: star 2 eclipsed; five events (s25 flagged in QLP, s51, s52, s78, s79); O−C ≤ 0.1 min. A→B phase 0.75815 (Gaia 0.77114 ± 0.00232)
Orbit	Physical fit to all ten eclipses: e = 0.420, ω = 353.6° against Gaia 0.4476 ± 0.0042, 348.67 ± 0.74° (roughly 6σ by Gaia's quoted errors; Gaia goodness-of-fit 23.7); i = 89.5°, k = 0.85, surface-brightness ratio 0.99. χ^2/n is 9.7 (red noise), so these values are indicative
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (all 10); TARS (6)
Pixel attribution	TESScut s079 B: largest flux-loss pixel is the target pixel, centroid offset 0.2″; nothing brighter than G 17.5 within 60″ (a 42% dip would need more than 1000%)
Catalogue status	Not listed under any name in: VSX, ASAS-SN, GCVS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 (all tables), WDS, Hipparcos and its variability annex, SB9, CCDM, Gaia DR3 variability tables, general web search; SIMBAD's one binary reference is the APOGEE SB2 list. ExoFOP was checked separately
Open issues	A joint fit of the APOGEE and Gaia velocities with the eclipses would give absolute radii and constrain e and ω
Next eclipses	B: BTJD 4335.273, 4363.220 (±0.001); A: BTJD 4342.028, 4369.975 (±0.004)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+29 2845 is a detached binary of two slightly evolved F/G stars of about 1.1 $M_\odot$ each, seen almost exactly edge-on, so the smaller star crosses the larger one nearly centrally and both eclipses are about 42% deep. Ten eclipses from 2020–2024 fit a single period. The radii (1.58 and 1.34 $R_\odot$, which add in quadrature to 2.07 $R_\odot$) match the TIC and Gaia single-star radius, and the Gaia multi-star temperatures are consistent.

Five of the ten eclipses lie in cadences flagged for scattered light, which may explain why the system has not been reported. Using Gaia's full covariance, the A eclipses fall within 0.23σ of prediction while the B eclipses are a constant 0.34–0.36 d early. Because the offset does not grow, it reflects the orbit shape rather than the period. The eclipses prefer $e \approx 0.42$, $\omega \approx 354°$, which amounts to a correction to Gaia's orbit rather than a problem with the eclipses. Gaia's errors are very probably understated. There is no apsidal motion within TESS.

A32. TYC 97-228-1 = ASASSN-V J045420.57+062810.8 (TIC 399758284)



A32. Left: QLP SAP flux folded on P = 40.51032 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 1.3–1.4%, 42-h flat total occultations of the G dwarf (sectors 32 and 98). Right: the 0.45% flat transits of the G dwarf across the giant (sectors 5 and 98).

Identifiers	TIC 399758284; Gaia DR3 3288089217923738112; TYC 97-228-1; ASASSN-V J045420.57+062810.8
Position (J2000)	04:54:20.6 +06:28:10.9
Brightness	G 10.66; Tmag 9.96
Stars	Chromospherically active K giant (T_1 = 4574 K), R_1 = 11.4–12.8 $R_\odot$ (TIC 10.8) + G dwarf, m_2 = 0.79–0.98 $M_\odot$, R_2 = 0.84–0.93 $R_\odot$, $T_2 \approx$ 5900–6000 K, 1.40% of the TESS light (for M_1 = 1.0–1.5 $M_\odot$); mass function 0.148 $M_\odot$; 886 pc; Gaia SB1
Period	40.51032 ± 0.00011 d (Gaia 40.501287 ± 0.007798 d; +1.16σ)
Epoch of primary	T_0(A) = BTJD 1443.6733 ± 0.0170 = BJD_{TDB} 2458443.6733; B at phase 0.49988 ± 0.00043
Primary eclipse (B)	1.32–1.44% deep, T_{14} 42.0–43.4 h, flat bottom 31.2–31.3 h: total occultation of the hotter G dwarf; two full events (s32, s98) and an ingress at the end of s5
Secondary eclipse (A)	0.45–0.46% deep (0.42–0.70% across pipelines), about 40 h, flat: full transit of the dwarf across the giant; two events (s5; s98, ingress lost, the weakest event)
Orbit	Circular: $e \cos \omega = -0.0002 \pm 0.0007$ (Gaia +0.003 ± 0.006; Gaia's e = 0.015 ± 0.007 is a 2σ detection). Two-sphere fit: a/R_1 = 5.27, b = 0.78, i = 81.4°, k = 0.072–0.073
Pipelines agreeing	QLP SAP; ELEANOR-lite, T16, TASOC, TGLC PSF and aperture (s5 A: five pipelines)
Pixel attribution	TESScut s32 B difference image peaks on the target pixel (noisier); TGLC PSF photometry puts the signal on the target; nothing brighter than G 17.95 within 60″. Common mode: target ranks first or second of 219–284 stars
Catalogue status	**VSX type revision ROT → EA+ROT.** VSX: ASASSN-V J045420.57+062810.8, ROT, P 39.81 d, amplitude 0.25 mag in g. Not in ASAS-SN II/366, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables or any eclipsing-binary catalogue; no TOI/CTOI; SIMBAD (SB*, one paper); a general web search found nothing
Open issues	The s98 A eclipse is weakly measured and the A durations carry spot red noise; a B eclipse with both edges in clean data and a spectrum for the faint companion would constrain the system further. Masses depend on the assumed M_1
Next eclipses	B: BTJD 4340.157, 4380.667; A: BTJD 4360.417, 4400.927 (±0.02 d; T_{14} about 1.78 d)
Discovery	

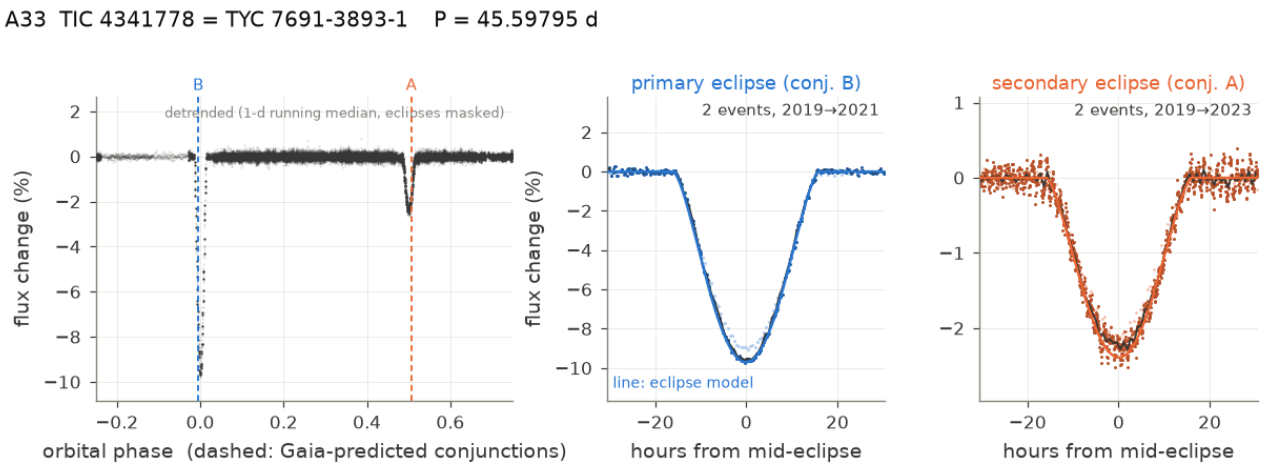
Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision. The star is in VSX as a rotating variable (ROT, 39.81 d) from ASAS-SN. That signal is real but due to spots, not eclipses; the eclipses are what is new, and we propose EA with P = 40.5103 d, keeping ROT (EA+ROT/RS).

TYC 97-228-1 is a detached eclipsing binary of a K giant of about 12 $R_\odot$ and a G dwarf on a circular 40.5-d orbit. Both eclipses are flat-bottomed. At A the dwarf crosses the giant's disk, and at B the giant hides the dwarf completely. B is the deeper because the dwarf is hotter per unit area. The central duration would be 63 h; the observed 42–43 h follows from an impact parameter of 0.78. The dwarf's radius, temperature, light share and mass all agree with the mass function.

The ASAS-SN rotation signal has an amplitude of 0.25 mag in g, far more than the eclipses or an ellipsoidal effect could produce. The star is spectroscopically active and TESS shows 2–3% spot-like drifts in every sector. Its 39.81-d period is 0.983 of the orbital period, which indicates near-synchronous rotation typical of an RS CVn-type giant. The rotation period could not be measured independently.

A33. TYC 7691-3893-1 (TIC 4341778)



A33. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 45.59795 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the two 8.8–9.5%, 30.5-h occultations of the A star (sectors 8 and 35). Right: the two 2.1–2.3%, 30.5-h near-V transits of the A star across the giant (sectors 9 and 62).

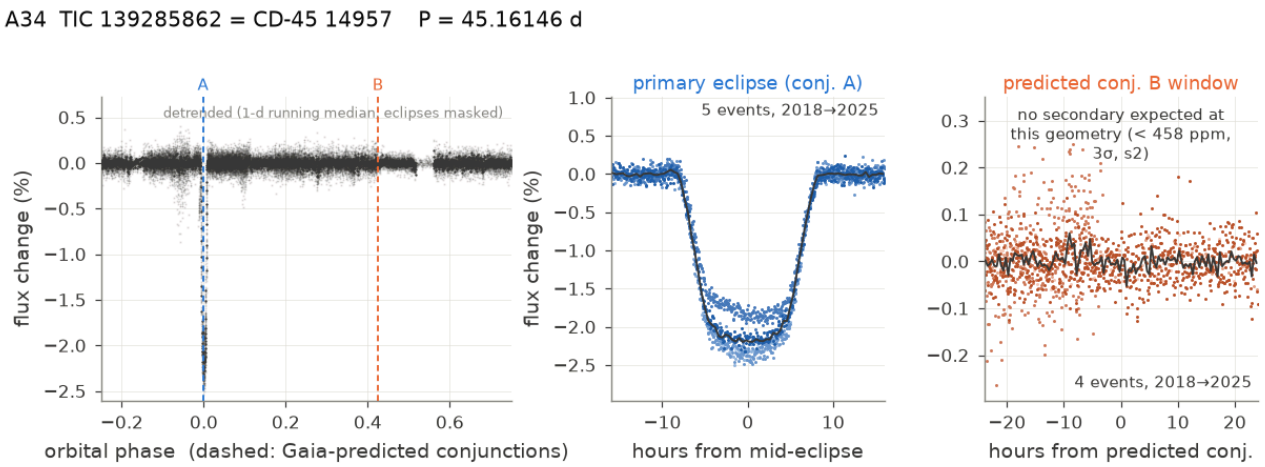
Identifiers	TIC 4341778; Gaia DR3 5430897742488880640; TYC 7691-3893-1
Position (J2000)	09:23:06.0 −37:32:12.4
Brightness	G 10.30; Tmag 9.83
Stars	Giant, R_1 = 10.2–10.7 $R_\odot$ (TIC 9.0) + A-type star, m_2 = 2.0–2.2 $M_\odot$, R_2 = 3.0–3.1 $R_\odot$, T_2 = 7200–9400 K, 23–24% of the TESS light (for M_1 = 2.5–3.0 $M_\odot$); mass function 0.373 $M_\odot$; Gaia SB1, spectral type F (Gaia); the TIC's 5957 K is a blend
Period	45.59795 ± 0.00008 d (Gaia 45.588324 ± 0.005264 d; +1.83σ)
Epoch of primary	T_0(A) = BTJD 1550.8760 ± 0.0030 = BJD_{TDB} 2458550.8760; B at phase 0.50010 ± 0.00006
Primary eclipse (B)	8.76–9.49% deep, T_{14} 30.5 h, 7-h bottom, ingress 11.7 h, near-V: partial occultation of the hot companion by the giant; two full events (s8, s35)
Secondary eclipse (A)	2.05–2.29% deep, 30.4–30.9 h, same near-V shape: partial transit across the giant's limb; two full events (s9, s62) plus an ingress (s35) and an egress (s89, after a gap, weaker)
Orbit	Circular: e cos ω = +0.00015 ± 0.00010, equal durations (e sin ω ≈ 0); Gaia's e cos ω = −0.016 ± 0.007 disagrees by 2.3σ, and Gaia's e (0.0174) is itself a 2.3σ detection and is biased upward. Joint fit: a/R_1 = 8.65, b = 1.02–1.03, k = 0.29, i = 83.2°
Pipelines agreeing	QLP SAP; ELEANOR, T16, TGLC PSF (s8 B); ELEANOR, T16, TEQUILA, TGLC PSF (s9 A); TESS-SPOC SAP and PDCSAP (s62 A)

Pixel attribution	TESScut s35 B: per-pixel depth peaks at 8.76% on the target pixel (5.0% at the *G* 13.3 star's pixel, the target's wings); the *G* 13.30 star at 45.7″ would need a 33% A dip and over 100% for B; TGLC PSF puts both signals on the target
Catalogue status	Not listed in: VSX (only a ROT star 51″ away, *G* 15.9), ASAS-SN II/366, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*; Turner 5 membership probability 0.00), general web search; Kostov table 2 only
Open issues	A correlated a/R_1–b–k valley in the fit; depths about 8% lower in s8/s9 (absorbed as extra dilution); masses depend on the assumed M_1. RVs or a spectrum to detect the A star, and multi-colour depths
Next eclipses	A: BTJD 4332.351, 4377.949; B: BTJD 4355.155, 4400.753 (±0.006 d; T_{14} about 1.27 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 7691-3893-1 is a detached, near-grazing eclipsing binary of a giant of about 10 $R_\odot$ and an A-type star of about 2 $M_\odot$ on a circular 45.6-d orbit. Both eclipses have the same duration and near-V shape, as a circular orbit requires, and cover the same overlap area. The occultation is about 4.3 times deeper because it hides the hot companion, which is 3.3–3.5 times brighter per unit area, while the transit covers the giant's darker limb. One geometry fits all four full eclipses to the noise, and the 2 $M_\odot$ A-type companion it requires is what the mass function demands.

All events fall within 0.17σ of Gaia's conjunctions.

A34. CD-45 14957 (TIC 139285862)



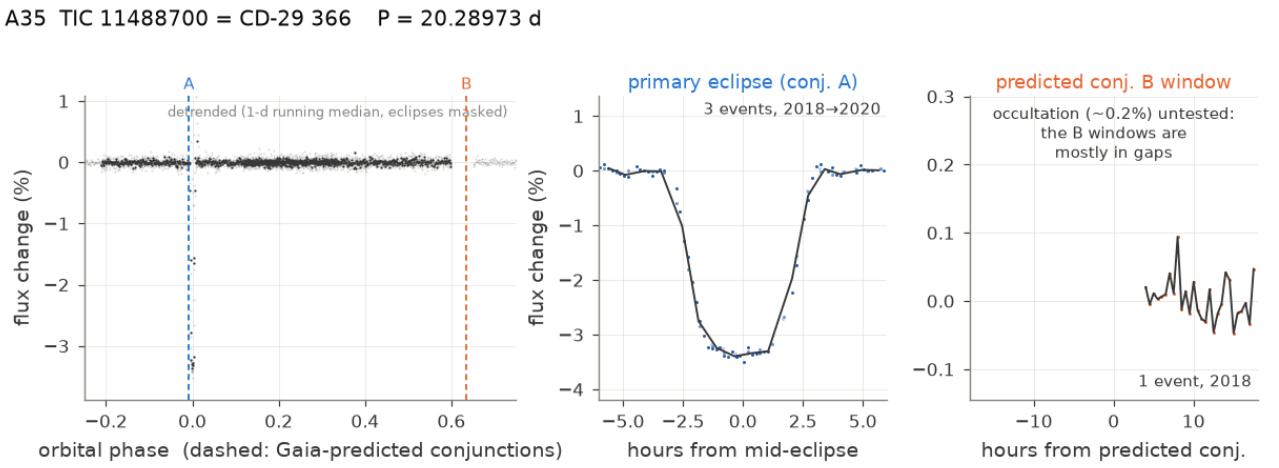
A34. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 45.16146 d, Gaia's predicted conjunctions dashed. Middle: the 2.3%, 16-h transits of the K dwarf across the K giant, 2018–2025. Right: the B window, where the geometry predicts no eclipse and none is seen.

Identifiers	TIC 139285862; Gaia DR3 6541487327863944960; CD-45 14957
Position (J2000)	23:12:03.2 −44:45:45.1
Brightness	*G* 10.51; Tmag 9.93
Stars	K giant, R_1 = 4.4–5.0 $R_\odot$ (Gaia FLAME 4.92 $R_\odot$, 1.63 $M_\odot$, log *g* 3.06) + K dwarf, R_2 = 0.67–0.76 $R_\odot$, $M_2(i = 90°)$ = 0.72/0.82/0.92 $M_\odot$ for M_1 = 1.3/1.63/2.0; mass function 0.0917 $M_\odot$; Gaia SB1
Period	45.16146 ± 0.00011 d (Gaia 45.1578 ± 0.0065 d; +0.6σ)
Epoch of eclipse	T_0(A) = BTJD 1354.3033 = BJD_{TDB} 2458354.3033
Eclipse (A)	2.11–2.31% deep (1.73% in s95, on a ramp), T_{14} 15.4–16.1 h, T_{23} 7.3–8.8 h, impact b = 0.67: transit of the K dwarf across the giant; five events (s2 egress only, s28, s68, s95, s96); O−C −5.5 to +16 min. A sixth, predicted at BTJD 4244.6367 ± 0.008 in s105 FFIs before those data were examined, was observed at 4244.6376 ± 0.001 (O−C +1.3 min) in custom TESScut photometry (this work)
Other eclipse (B)	

	None expected: the separation at B is 1.75 times larger, so $b_B = 1.17 > 1 + k$. Measured 3σ limits 570 and 458 ppm (s1, s2), < 1000 ppm (s95)
Orbit	Gaia e 0.294, ω 112.3°; e cos ω and e sin ω not measurable without B. Events +0.19 to +0.39 d after Gaia's A times (+0.73 to +0.83σ). $a/R_1 = 15.2$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s68); TGLC raw aperture (s2)
Pixel attribution	s28 difference image: loss on the target (offset +0.08, 0.00 px), core pixels lose 2.0–2.3% uniformly; the only Gaia neighbour within 60″ is G 19.9
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, general web search. SIMBAD's RAVE papers include it among RAVE + Gaia DR2 SB1 candidates (Birko et al. 2019); no eclipse
Open issues	RVs of the companion
Next eclipses	A: BTJD 4334.960, 4380.121, 4425.283 (±0.009)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-45 14957 is a K giant of about 5 $R_\odot$ crossed every 45.16 d by a K dwarf. Five transits from 2018 to 2025 define the period. A sixth transit, predicted for sector 105 (which has full-frame images but no QLP light curve), was recovered in TESScut photometry (this work) 1.3 min from the prediction, with the same depth and duration. Only one eclipse kind occurs: the transit impact parameter of 0.67 grows to 1.17 at the other conjunction, where the stars are 1.75 times farther apart, so the giant cannot occult the dwarf.

A35. CD-29 366 = Bok I 16B (TIC 11488700)



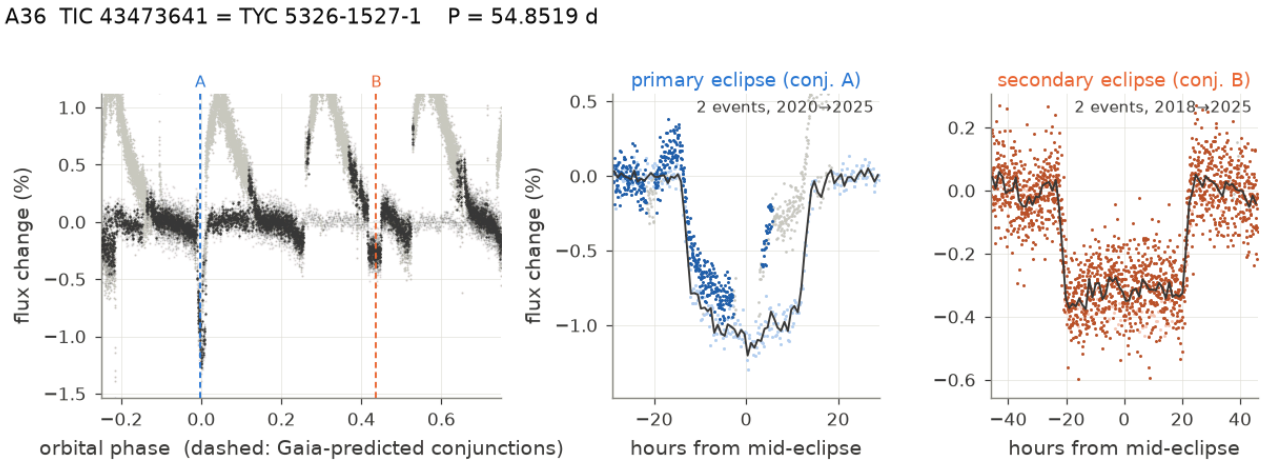
A35. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 20.28973 d, Gaia's predicted conjunctions dashed. Middle: the three 3.3%, 5.9-h U-shaped transits of the M dwarf across the F star (2018–2020). Right: the B window, mostly in data gaps; the expected 0.2% occultation is untested.

Identifiers	TIC 11488700; Gaia DR3 5032956687360117376; CD-29 366; Bok I 16B
Position (J2000)	01:11:28.1 −28:41:49.5
Brightness	G 10.64; Tmag 10.30
Stars	F star, R_1 = 1.41–1.48 $R_\odot$ (FLAME 1.65) + M dwarf, M_2 = 0.27–0.30 $M_\odot$, R_2 = 0.26 $R_\odot$; mass function 0.0090 $M_\odot$; Gaia SB1. A co-moving G 14.02 companion lies 1.65″ away (parallax 3.32 vs 3.29 mas, same proper motion)
Period	20.28973 ± 0.00016 d (Gaia 20.2758 ± 0.0109 d; +1.3σ)
Epoch of eclipse	BTJD 1404.1283 ± 0.0003 = BJD_{TDB} 2458404.1283
Eclipse (A)	3.28–3.35% deep, T_{14} 5.9 h, T_{23} 3.1 h, U-shaped, impact b = 0.62: transit of the M dwarf across the F star; three events (1383.8453 partial, 1404.1283, 2134.5620); O−C −4.9 to +4.8 min
Other eclipse (B)	Untested: expected occultation about 0.2%; only 16–21% of the predicted B windows have good data. B is therefore undetected rather than excluded

Orbit	Gaia e 0.228, ω 4°; events +0.85 to +1.37 d after Gaia's A times (+0.8 to +1.0σ). a/R_1 = 25
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s3, s30); TARS (s30); TGLC (s3); depths 3.1–3.7%, mid-times within 1 min
Pixel attribution	s30 difference image: loss lands on the pair (−0.02, +0.06 px); no pixel product can separate the two stars 1.65″ apart. The companion would need a 76% eclipse; the eclipse repeats at the period and phase of the F star's Gaia orbit
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS (nothing within 180″), Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Detecting B at about 0.2%; high-resolution imaging or RVs to rule out the 1.65″ companion completely. The s106 cutout is empty (target off-detector)
Next eclipses	A: BTJD 4346.142, 4366.432, 4386.722 (±0.024); B (about 0.2%): 4338.84 ± 0.54
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with a caveat. The co-moving G 14.02 companion 1.65″ away cannot be separated in any TESS product; the attribution rests on the period and phase of the F star's Gaia orbit (three events, one partial, 0.8–1.0σ from Gaia's A times) and on the physics.

CD-29 366 is an F star of about 1.45 $R_\odot$ transited every 20.29 d by an M dwarf of about 0.26 $R_\odot$. Two of the three dips lie in continuous clean data with flat background and centroids. The third is partly covered, after the flagged start of sector 3. No TESS pixel product can separate the co-moving companion 1.65″ away, but that companion would need a 76% eclipse to produce the dip. The eclipse also repeats at the period and phase of the F star's own Gaia orbit, and the physics is fully consistent with an M dwarf crossing the F star. The B windows are mostly in gaps, and the expected occultation is only about 0.2% deep.

A36. TYC 5326-1527-1 (TIC 43473641)



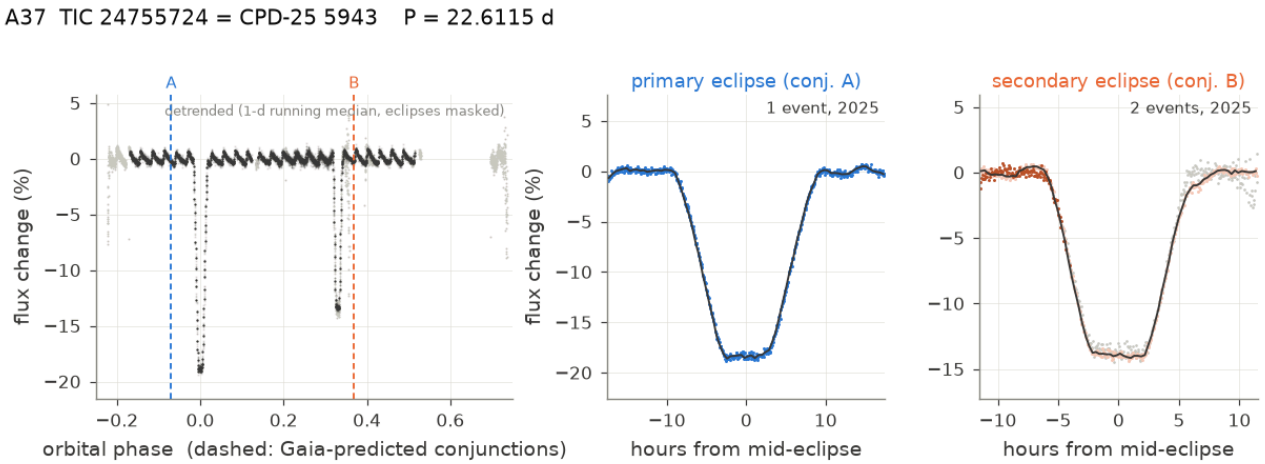
A36. Left: QLP SAP flux folded on P = 54.8519 d, Gaia's predicted conjunctions dashed. Middle: the 1.0%, 29-h flat transits at A (sectors 32 and 98). Right: the 0.39%, 46-h flat occultations at B (sectors 5 and 98).

Identifiers	TIC 43473641; Gaia DR3 3180643154951048064; TYC 5326-1527-1
Position (J2000)	04:59:10.9 −11:10:02.0
Brightness	G 11.00; Tmag 10.34
Stars	Giant, R_1 = 6.6/7.5/8.7 $R_\odot$ for M_1 = 1.0/1.5/2.5 $M_\odot$ (TIC 7.0, matching $M_1 \approx 1.1$) + K/M dwarf, R_2 = 0.67–0.88 $R_\odot$, about 3900 K, consistent with $M_2 \approx$ 0.55–0.65 $M_\odot$; mass function 0.0636 $M_\odot$; Gaia SB1
Period	54.8519 ± 0.0002 d (P_B = 54.85193 ± 0.00019, P_A = 54.8520 ± 0.0009; Gaia 54.8533 ± 0.0111 d, −0.1σ)
Epoch of primary	A: BTJD 2192.7826 ± 0.005 = BJD_{TDB} 2459192.7826; B: BTJD 1448.5445 ± 0.004

Primary eclipse (A)	1.03% deep, T_{14} 29.3 h, T_{23} 21.7 h, flat: transit of the dwarf across the giant (b = 0.56); two events (s32; s98, egress in flagged scattered light)
Secondary eclipse (B)	0.375–0.391% deep, T_{14} 45.8–47.2 h, T_{23} about 40 h, flat: total occultation of the dwarf at 1.64 times the A separation; two events (s5, s98)
Orbit	A→B phase 0.43189 and duration ratio 1.611 give e cos ω = −0.104, e sin ω = +0.241, e = 0.262, ω = 113°; Gaia −0.092 ± 0.008 and +0.241 ± 0.008 (1.6σ and 0.0σ; Gaia e 0.2579, ω 110.93°). a/R_1 = 10.6
Pipelines agreeing	QLP SAP; raw eleanor (s5 B 0.36%); raw TGLC aperture 0.46% and PSF 0.51%. Calibrated TGLC fluxes erase the two-day eclipse with a one-day filter
Pixel attribution	s32 difference image: A loss on the target pixel (+0.12, +0.18 px; 2.3 px from the G 11.85 neighbour at 58.5″, which would need 2.2%/0.9% eclipses); target pixel loses 1.3%, pixels toward the neighbour ≤ 0.3%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	B's ingress (2.9–3.7 h) is shorter than A's scaled by the velocity ratio, attributed to limb darkening biasing the A fit; this is not a refutation. A fully covered A eclipse at 200-s cadence fitted with limb darkening would resolve it
Next eclipses	A: BTJD 4332.01 (±0.04; 4332.008 ± 0.009 with P_B), 4386.86; B: BTJD 4355.703, 4410.555 (±0.05)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 5326-1527-1 is a giant of about 7 $R_\odot$ with a K/M dwarf companion on an eccentric 54.85-d orbit. The dwarf's transit lasts 29 h. Its occultation happens at 1.64 times the separation and so lasts almost two days. With both kinds of eclipse seen, the eclipse timing and durations give e cos ω and e sin ω independently of Gaia's radial velocities, and they reproduce Gaia's to 1.6σ and 0.0σ. The occultation depth (0.39% of the light) needs a companion about 0.4 times as bright per unit area as the giant, about 3900 K, consistent with the mass function.

A37. CPD-25 5943 (TIC 24755724)



A37. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.6115 d, Gaia's predicted conjunctions dashed. Middle: the single 18.7%, 17.5-h flat transit at A in sector 91. Right: the two 13.4%, 11.5-h flat total occultations at B (the second in flagged cadences).

Identifiers	TIC 24755724; Gaia DR3 4109331632663213440; CPD-25 5943
Position (J2000)	17:10:32.5 −26:00:49.1
Brightness	G 10.55; Tmag 10.21
Stars	A5 star (Wallenquist 1939), R_1 = 2.7–2.9 $R_\odot$ + companion, R_2 = 1.24–1.36 $R_\odot$, 13.4% of the light (about 6400 K and 1.3 $M_\odot$ needed); mass function 0.0926 $M_\odot$ gives $M_2(i = 90°)$ = 0.81–0.98 $M_\odot$ for M_1 = 1.6–2.2; Gaia SB1 (significance 11)

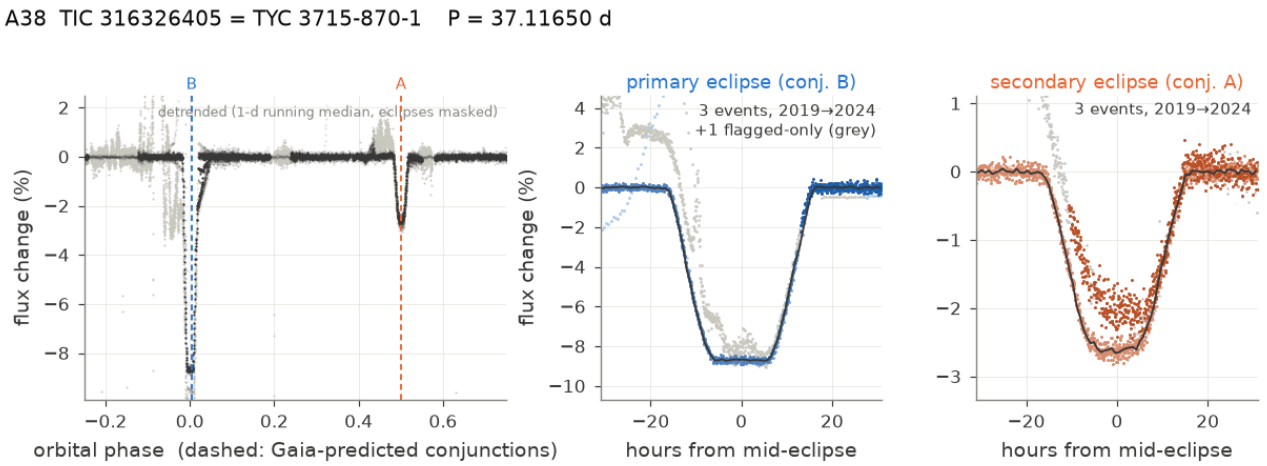
Period	P_B = 22.6115 ± 0.0029 d from two B eclipses (Gaia 22.5960 ± 0.0215 d; +0.7σ). The Harvard plates refine it (Section 5.6): the eclipse appears on 47 plates from 1890–1990 at P = 22.6131 ± 0.0002 d, inside the TESS error and twenty times tighter
Epoch of primary	A: BTJD 3793.2998 ± 0.0014 = BJD_{TDB} 2460793.2998
Primary eclipse (A)	18.73% deep, T_{14} 17.5 h, T_{23} 5.9 h, flat (b_A = 0.22); one event, sector 91
Secondary eclipse (B)	13.43–13.46% deep, 11.4–11.5 h, T_{23} 4.7–4.8 h, flat: total occultation; two events (3778.1317; 3800.7432 in flagged cadences)
Orbit	A→B phase 0.3292 and duration ratio 0.658 give e cos ω = −0.266, e sin ω = −0.204, e = 0.335, ω = 217°; Gaia −0.092 ± 0.058 (3.0σ off) and −0.220 ± 0.041 (0.4σ); Gaia e 0.246 ± 0.037, ω 247 ± 14°. a/R_1 = 17
Pipelines agreeing	QLP SAP (one sector, s91); no second pipeline was listed for this star
Pixel attribution	TESScut s91 difference images: loss centroid (−0.05, +0.16) px from the target at A and (−0.01, +0.04) px at B; peak-loss pixel is the target pixel (16% A, 11% B). The VSX "S" source 8.3″ away (G 18.47, probably a spurious detection in the target's halo) would need more than 26,000%; every Gaia star with G < 16 within 150″ would need more than 100%
Catalogue status	Not listed in: VSX (an "S" entry 8.3″ away, from the Gaia DR3 classifier, no period; an "RS" source at 53″, G 17.1), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Two tensions with Gaia: e cos ω 3.0σ off (plausibly an underestimated ω error in a weak orbit), and the 13.4% occultation needs a brighter companion than the mass function allows (Gaia's K1 is probably diluted). Neither moves the eclipse off the target. Needs a second A eclipse and ground spectroscopy. The A eclipse falls +3.88 d (+1.1σ) after Gaia's predicted A time. The Harvard plates (Section 5.6) supply a second, independent century of eclipses at this ephemeris (z = 11.9, false-alarm probability below 0.5%), so the period no longer rests on one sector and Gaia; the status is kept pending a second TESS epoch
Next eclipses	A: BTJD 4335.98, 4358.59; B: BTJD 4343.42, 4366.03 (±0.07–0.08)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable, see below
Status	Probable. One sector (s91) only: one transit and two occultations, the second in flagged cadences; no second pipeline; e cos ω 3.0σ from Gaia; the 13.4% total occultation needs a companion of about 1.3 $M_\odot$ against about 0.85 $M_\odot$ from the mass function. The A eclipse falls +3.88 d (+1.1σ) after Gaia's predicted A time. A second A eclipse measured in an independent pipeline would confirm it.

Note: probable, not confirmed. Both eclipse kinds were seen in a single sector (s91): one transit and two occultations, the second in cadences QLP flagged for scattered light, and no second pipeline was listed. The eclipse spacing puts e cos ω 3.0σ from Gaia's. The total occultation needs a companion of about 1.3 $M_\odot$ where the mass function gives about 0.85 $M_\odot$. The A eclipse falls 3.88 d (+1.1σ) after Gaia's predicted A time. A second A eclipse measured in an independent pipeline would confirm it.

CPD-25 5943 is an A5 star of about 2.8 $R_\odot$ with a companion of about 1.3 $R_\odot$, observed in a single TESS sector that contains one deep flat transit and two identical flat total occultations one orbit apart. The signal is confirmed on the target. The difference images put both losses on the A star, and the VSX short-timescale source 8.3″ away is faint, astrometrically poor and probably spurious. The ~1-day, ~1% sawtooth in the light curve tracks pointing resets and is not stellar.

Two results disagree with Gaia's orbit. The eclipse spacing gives e cos ω 3.0σ from Gaia's, while the duration ratio matches. Because B is a total occultation, its depth shows that the companion supplies 13.4% of the light. That needs a star of about 1.3 $M_\odot$, where the mass function gives about 0.85; a 0.85 $M_\odot$ dwarf would give about 7%. The likely explanation is that the companion's lines blend into the A star's spectrum and dilute Gaia's velocity amplitude. The alternative is a much more massive primary.

A38. TYC 3715-870-1 (TIC 316326405)



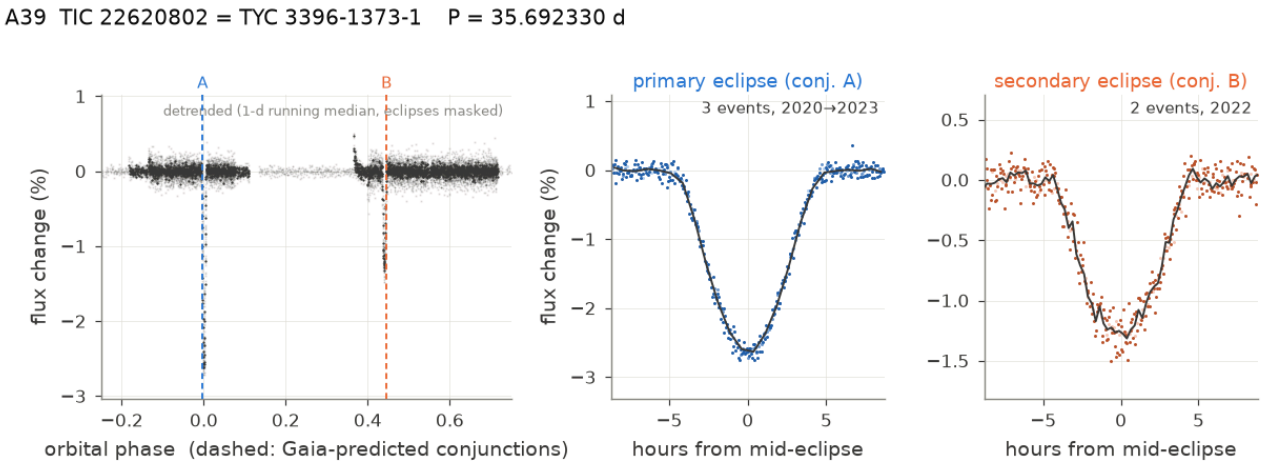
A38. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 37.11650 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 8.7%, 31-h flat total occultations of the hot companion at B. Right: the 2.5%, 31-h flat transits at A.

Identifiers	TIC 316326405; Gaia DR3 450207237998808064; TYC 3715-870-1
Position (J2000)	03:28:28.7 +59:17:36.2 (2° from the Galactic plane, 845 pc)
Brightness	G 10.63; Tmag 9.93
Stars	Giant, R_1 = 9.08 $R_\odot$ (TIC), probably near 5000 K + hot dwarf, $M_2(i = 90°)$ = 1.38/1.56/1.72 $M_\odot$ for M_1 = 2.0/2.5/3.0, R_2 ≈ 1.4–1.5 $R_\odot$, 8.7% of the light; mass function 0.229 $M_\odot$; Gaia SB1 (K_1 39.06 km/s)
Period	37.116502 ± 0.000308 d (Gaia 37.1120 ± 0.0055 d; +0.81σ)
Epoch of primary	T_0(B) = BTJD 1839.8511 ± 0.009 = BJD_{TDB} 2458839.8511; T_0(A) = BTJD 1821.2932 ± 0.011
Primary eclipse (B)	8.65–8.82% deep, T_{14} 31.4–31.6 h, T_{23} 14.2–14.3 h, flat: total occultation of the hot companion; two measured (s19, s59) plus a flagged ingress in s86
Secondary eclipse (A)	2.49–2.56% deep, T_{14} 30.3–31.3 h, T_{23} about 12 h, flat: transit of the companion across the giant (k ≈ 0.15–0.17); three events (s19, s59, s86 on a systematic feature, timing only)
Orbit	Circular: B–A phase 0.49999 ± 0.0004, e cos ω = 0.0000 ± 0.0006 (about ±0.001 including systematics; Gaia +0.0057 ± 0.0048); duration ratio 1.025, e sin ω ≈ +0.012 (Gaia +0.0045 ± 0.0049). b ≈ 0.74, i ≈ 85°
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s59); TEQUILA (s19)
Pixel attribution	TESScut s59: both deficits centred on the target pixel, centroid within 0.35 px; core depths A 2.58%, B 8.94%; a brighter star about 95″ away does not dip; brightest Gaia neighbour within 60″ is G 18.0
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	The surface-brightness ratio J ≈ 3.4–4.3 needs a 9000–9900 K companion with the TIC T_{eff} (5728 K), too hot for 1.5 $M_\odot$, but 7400–8100 K if the giant is near 5000 K; the TIC T_{eff} is from blended light. A spectrum is needed; the binary interpretation is unaffected
Next eclipses	B: BTJD 4363.773, 4400.890; A: BTJD 4345.215, 4382.332 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3715-870-1 is a giant of about 9 $R_\odot$ with a hot main-sequence companion of 1.4–1.7 $M_\odot$ on a circular 37.1-d orbit. Both eclipses are flat-bottomed and about 31 h long. At B the giant hides the companion completely, so the 8.7% depth is the companion's share of the light. At A the companion crosses the giant's disk. A central eclipse would last 38–42 h; the observed 31 h implies an impact parameter of about 0.74 and predicts the 14-h flat part seen. One geometry fits both kinds of eclipse. A

companion carrying 9% of the light makes Gaia's single-lined classification expected. The only tension is the giant's temperature, which the TIC took from blended light; a spectrum would show the hot companion's lines in the blue.

A39. TYC 3396-1373-1 (TIC 22620802)

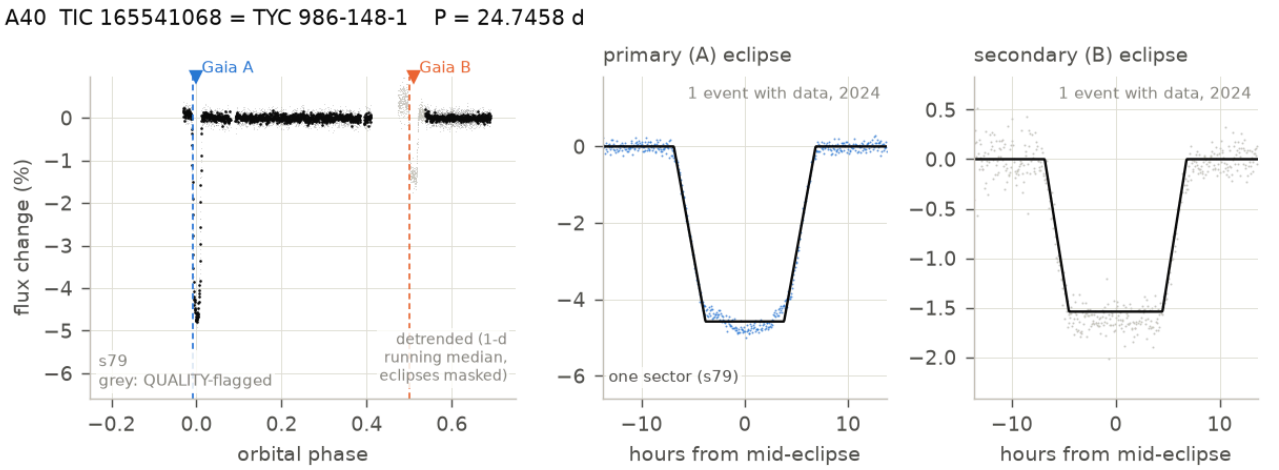


A39. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 35.692330 d, Gaia's predicted conjunctions dashed. Middle: the 2.6%, 8.7-h grazing eclipses at A. Right: the 1.25%, 8.7-h grazing eclipses at B.

Identifiers	TIC 22620802; Gaia DR3 977538409156198272; TYC 3396-1373-1
Position (J2000)	07:11:50.2 +46:52:45.4
Brightness	*G* 10.83; Tmag 10.21
Stars	Subgiant (T_1 = 4908 K) + K dwarf, M_2 = 0.64–0.76 $M_\odot$ (for M_1 = 1.1–1.5), T_2 ≈ 4100 K; mass function 0.0862 $M_\odot$; Gaia SB1 (K_1 28.66 km/s)
Period	35.692330 ± 0.000062 d (Gaia 35.6858 ± 0.0072 d; +0.90σ)
Epoch of primary	T_0(A) = BTJD 2602.95779 ± 0.0007 = BJD_{TDB} 2459602.95779; T_0(B) = BTJD 2582.9649 ± 0.0011
Primary eclipse (A)	2.56–2.58% deep, 8.74–8.76 h, V-shaped (T_{23} only about 2 h), partial: grazing transit (b ≈ 1.07 R_1); three events (s47, s60; s20 complete in TARS, 21 cycles earlier)
Secondary eclipse (B)	1.23–1.26% deep, 8.5–9.0 h, V-shaped, partial; two events (s47, s60)
Orbit	B–A phase 0.43985 ± 0.00004 (Gaia 0.4485 ± 0.0104): e cos ω = −0.0946 against Gaia −0.0810 ± 0.0164 (−0.8σ); equal durations, e sin ω ≈ +0.001 (Gaia +0.010 ± 0.014). O–C within ±1.4 min at every event
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s60); TARS (s20, s47, s60); times within 5 min
Pixel attribution	TESScut s60: deficits fit the out-of-eclipse PSF (A core 2.56 ± 0.06%, B 1.23 ± 0.10%), centroid within 0.15 px; brightest Gaia neighbour within 60″ is *G* 18.5
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	None for the binary; a limb-darkened model would constrain b and k
Next eclipses	A: BTJD 4351.882, 4387.574; B: BTJD 4331.889, 4367.581 (±0.003)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3396-1373-1 is a subgiant grazed by a K dwarf of about 0.7 $M_\odot$ on a nearly circular 35.7-d orbit. A central eclipse would last about 16 h. The observed 8.7 h requires an impact parameter of about 1.07 primary radii, so both eclipses are partial and V-shaped. The depth ratio of the two eclipses, 0.48, approximates the surface-brightness ratio and gives the companion about 4100 K. All measurements are consistent with one geometry: neither eclipse is flat, B lasts 8.5–9 h, and the sector-20 eclipse is complete in TARS and fits the ephemeris to within a minute.

A40. TYC 986-148-1 (TIC 165541068)



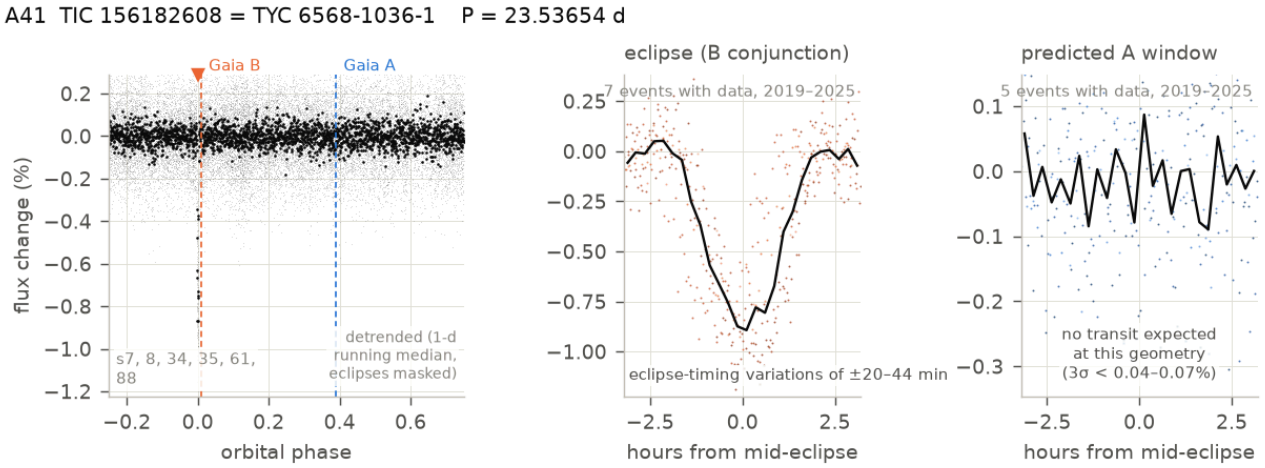
A40. Left: QLP SAP flux of sector 79, the only sector, detrended with a 1-d running median with the eclipses masked, folded on Gaia's P = 24.7458 d, Gaia's predicted conjunctions dashed. Middle: the 4.6%, 13.7-h transit at A on clean data. Right: the 1.5%, 13.7-h flat occultation at B, entirely in flagged cadences (shown in a lighter tone).

Identifiers	TIC 165541068; Gaia DR3 4540690625639510528; TYC 986-148-1
Position (J2000)	17:19:11.3 +12:29:51.2
Brightness	G 10.80; Tmag 10.16
Stars	Subgiant + dwarf, M_2 = 0.61–0.73 $M_\odot$ (for M_1 = 1.1–1.5), $R_2 \approx$ 0.64–0.71 $R_\odot$, 1.54% of the light; mass function 0.0772 $M_\odot$; Gaia SB1 (K_1 31.11 km/s)
Period	Gaia 24.7458 ± 0.0025 d (one event of each kind; no photometric period)
Epoch of primary	A: BTJD 3463.5943 ± 0.0010 = BJD_{TDB} 2460463.5943
Primary eclipse (A)	4.58% deep, T_{14} 13.73 h, T_{23} 7.6 h: transit (b ≈ 0.72, $k \approx$ 0.19–0.22); one event, sector 79, clean data
Secondary eclipse (B)	1.54% deep (TESS-SPOC 1.51%), T_{14} 13.65 h, T_{23} 9.0 h, flat: total occultation; one event, BTJD 3476.1984 ± 0.0031, all 389 cadences within ±0.45 d flagged (bits 2048/4096)
Orbit	A–B gap 12.604 d (P/2 = 12.373): e cos ω = +0.0147 ± 0.0002 against Gaia's +0.0176 ± 0.0057 (−0.5σ); durations give e sin ω ≈ −0.003 (Gaia +0.004 ± 0.007). Gaia's conjunctions within 0.11σ of both events. $i \approx 87°$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (the B event; A in QLP and TARS); TARS (A only; it has a gap over B)
Pixel attribution	TESScut s79: deficit follows the target PSF for both (A core 4.51 ± 0.16%, B 1.62 ± 0.23%), centroid within 0.2 px. None of 23 other QLP stars on the same CCD dips more than 1% in the B window. Gaia neighbours within 60″ are $G \geq$ 17.76 (B would need more than 900%)
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	**One sector.** The period rests on Gaia; a second epoch of A (4.6%, 13.7 h, readily observable from the ground) would fix it. Mild tension: the companion's implied temperature (3700–3860 K) is below the 4100–4300 K expected, which is not evidence against the eclipses
Next eclipses	A: BTJD 4329.696, 4354.442; B: BTJD 4342.300, 4367.046 (±0.09–0.10, from Gaia's period error)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Probable. One transit and one occultation in a single sector, the occultation wholly in flagged cadences, and the period is Gaia's, not photometric (Appendix A marks it so): the eclipses are on the target and self-consistent, but the protocol's period-from-the-eclipses check is missing until a second epoch. TESS-SPOC confirms the flagged event.

Note: one sector. Both eclipses were seen once, in sector 79, and the B eclipse lies wholly in cadences QLP flagged for scattered light. The eclipses are on the target; the photometric period awaits a second epoch, and until then the status is probable.

TYC 986-148-1 is a subgiant with a dwarf companion of about 0.67 $M_\odot$ on a nearly circular 24.7-d orbit. The flagged B event was tested in detail. It has a sharp ingress, a flat bottom and a sharp egress, while the scattered-light background is a smooth bell that peaks before the event and is back at baseline by egress. An equal bell at 3461.5 produces no dip. The centroids are smooth. The same depth, time and duration are obtained from TESS-SPOC, with its own aperture and background. The pixel deficit follows the target's PSF, and no dip is seen in any other star on the CCD. The occultation's total duration equals the transit's to 0.1 h, as it must for a near-circular orbit, and its phase gives an e cos ω that matches Gaia's independent value. The B eclipse is therefore confirmed despite the quality flags.

A41. TYC 6568-1036-1 (TIC 156182608)



A41. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.53654 d, Gaia's predicted conjunctions dashed. Middle: the seven 0.7–1.0%, 3-h grazing occultations at B, 2018–2025, in time order (the 2025 events are deeper and longer). Right: the A window, flat as the geometry predicts.

Identifiers	TIC 156182608; Gaia DR3 5692749838260791168; TYC 6568-1036-1
Position (J2000)	08:18:03.3 −28:34:59.7
Brightness	G 10.88; Tmag 10.52
Stars	F star + K dwarf, m_2 = 0.64 $M_\odot$ at i = 90°; mass function 0.067 $M_\odot$; a = 43.3 $R_\odot$; Gaia SB1, RUWE 2.04
Period	23.53654 ± 0.00017 d, linear fit (Gaia 23.539603 ± 0.002965 d; 1.0σ)
Epoch of eclipse	T_0(B) = BTJD 1501.940 ± 0.011 = BJD_{TDB} 2458501.940
Eclipse (B)	0.69–0.99% deep (QLP), T_{14} 2.9–3.5 h, T_{23} about 0.8–1.1 h: grazing occultation of the K dwarf by the F star, near periastron; seven events, 2018–2025 (s7, s8, s34, s35, s61, s88 ×2)
Other eclipse (A)	None expected (impact distance at A 2.04 times larger, 3.9 $R_\odot$ against R_1+R_2 = 2.18) and none seen: 100% coverage of the A window, 3σ limits 386–668 ppm
Orbit	At Gaia's epoch the eclipses sit −0.21 ± 0.11 d (−1.9σ) from B and −34σ from A. Separation 0.635a at B, 1.29a at A. Impact d/(R_1+R_2) = 0.88 (s61), 0.83 (s88)
Pipelines agreeing	QLP SAP; TESS-SPOC (s7); eleanor (s7); SPOC 2-min (s34, s35); times agree to about 2 min, so the timing variations are reproduced in three pipelines
Pixel attribution	Difference-image centroids 0.7–4.5″ from the target (s34, s61, s88); the G 13.41 neighbour at 17.8″ would need 7.7% and is excluded by the centroids; the G 12.22 star at 43.7″ would need 2.6% and is about 2 px away; depth falls as the aperture grows
Catalogue status	

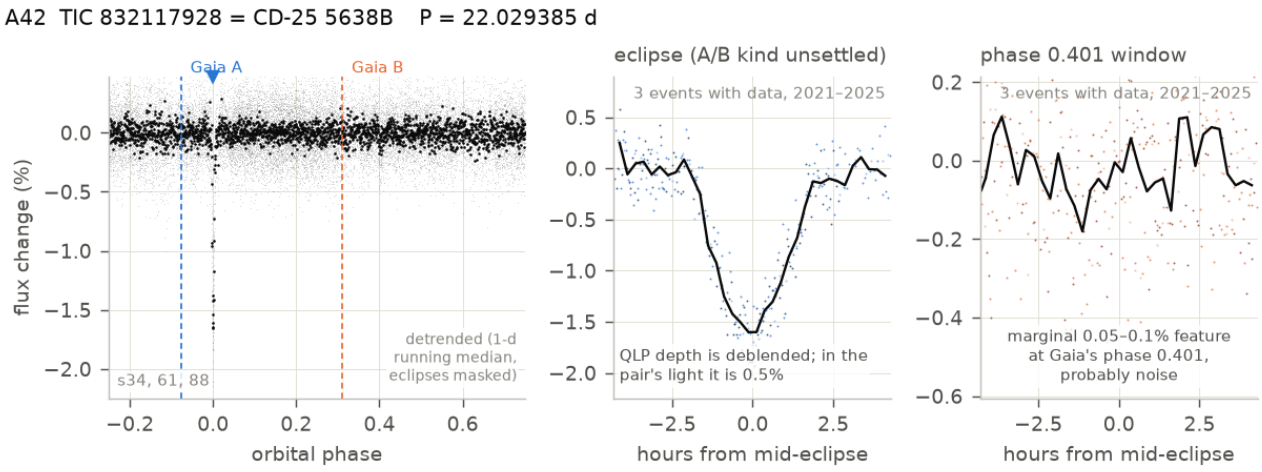
	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP (no TOI/CTOI); Kostov table 2 only. SIMBAD's reference list was not checked. A prior partial detection: an unvetted SPOC s1–s36 TCE at *P* 47.069 d (= 2P)
Open issues	**Eclipse-timing variations:** O–C +21, +20, −6, −11, −44, +17, +11 min, χ^2 = 783 for 6 dof. **Depth change:** the absolute lost flux grew from 76–77 to 106 e/s (1.39×) and T_{14} from 2.9–3.0 to 3.48 h in s88. More eclipse times, RVs for a long-period trend, Gaia DR4 epoch astrometry
Next eclipses	B: BTJD 4349.861, 4373.398, 4396.935 (±0.05 d including the timing variations)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: confirmed, with eclipse-timing variations. The eclipse times vary by tens of minutes in a coherent way, and the 2025 eclipses are deeper and longer than earlier ones. The most likely interpretation, perturbation of the orbit by a third body, is not yet supported by a measured third orbit.

TYC 6568-1036-1 is an F star whose K-dwarf companion is partly hidden behind it once per 23.54-d orbit, near periastron. The central duration would be 6.2 h; the observed 3.0–3.5 h is a graze. At the other conjunction the stars miss by twice the distance, so no transit is expected, and none is seen.

The timing residuals are not scatter, since three other pipelines reproduce them to about 2 min. A light-travel-time orbit is excluded as their cause: it would need a semi-amplitude of at least 3.8 AU with an outer period under about 6 yr, and hence a tertiary of at least 2 $M_\odot$ that would dominate the light. A dynamical perturbation with an outer period of a few years fits, and Gaia's RUWE of 2.04 supports an outer orbit. In fixed apertures the 2025 eclipses lose 1.39 times more flux and last longer, as a grazing eclipse with a shrinking impact parameter does. We interpret this as an inclination change driven by the same third body. Predicted depths for a companion supplying 3.4% of the light are about half those observed: the change is in the observed direction and within a factor of 2.

A42. CD-25 5638B (TIC 832117928)



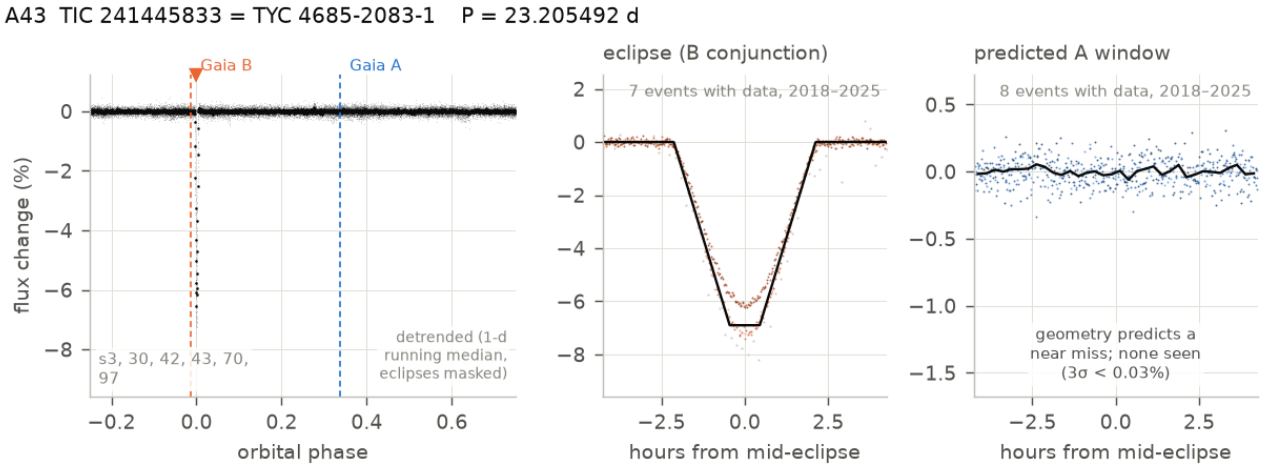
A42. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 22.029385 d, Gaia's predicted conjunctions dashed. Middle: the V-shaped eclipses in QLP's deblended flux (0.5% of the 1.3″ pair's combined light), 2019–2025; the sector-7 event, first found in eleanor-lite, is included where QLP has data. Right: the data near phase 0.40, where a marginal feature is probably noise.

Identifiers	TIC 832117928; Gaia DR3 5694223905397676416; CD-25 5638B (component B of a 1.3″ visual pair)
Position (J2000)	08:09:03.3 −26:17:15.5
Brightness	*G* 10.82; Tmag 10.50

Stars	If on B (Gaia FLAME R 2.24 $R_\odot$, M 1.50 $M_\odot$): companion $m_2 \geq 0.41\ M_\odot$; mass function 0.0194 $M_\odot$. Gaia SB1 on component B
Period	22.029385 ± 0.000018 d (Gaia 22.004 ± 0.060 d; 0.42σ)
Epoch of eclipse	T_0 = BTJD 1508.34974 ± 0.00098 = BJD_{TDB} 2458508.34974
Eclipse	0.56% (eleanor) and 0.40–0.55% (TESScut apertures) of the pair's combined light; QLP's 1.5–1.9% is its deblended value. V-shaped, T_{14} 3.8–4.8 h. Four events: s7 (new, eleanor-lite, predicted to 0.002 d), s34, s61, s88; O–C ≤ 1.4 min
Other eclipse	A marginal feature at phase 0.401 (515–1014 ppm per sector, 3.55 robust σ) at Gaia's predicted separation; the predicted M-dwarf occultation is ≤ 0.045% blended and Gaia's geometry predicts a miss, so it is probably noise
Orbit	Gaia's orbit too poorly constrained to discriminate (*e* 0.28 ± 0.13, ω ±22°, t_p ±1.4 d); at Gaia's epoch the eclipses sit +1.0σ from A and −1.5σ from B. If on B, central duration at A 9.9 h, so the observed ~4 h is a graze
Pipelines agreeing	QLP SAP; eleanor-lite (s7); TESScut apertures
Pixel attribution	The two stars of the pair cannot be separated in TESS pixels; the s61 difference image is centred on the pair (4.7″). QLP centroid shift +0.25 ± 0.21 mpx against +0.26 expected if B is eclipsed and −0.17 if A: favours B at about 2σ. Every other star within 90″ would need ≥ 22%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP; Kostov table 2 only. SIMBAD's reference list was not checked
Open issues	Which component eclipses: the period match favours B strongly, the centroid weakly (2σ); Gaia's RVS spectra of a 1.3″ pair may be blended. Speckle or AO-resolved photometry during an eclipse, or resolved RVs, would resolve it
Next eclipses	BTJD 4328.111, 4350.140, 4372.170 (±0.003 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed in the pair's combined light, with the host component probably B (the period match is strong, the centroid shift favours B at only about 2σ); no single eclipse is measured in two independent pipelines (s7 in eleanor-lite only, the rest in QLP and custom TESScut photometry).

CD-25 5638 is a 1.3″ visual pair. The eclipses are confirmed in the pair's combined light, at a period that matches, to 0.42σ, the radial-velocity orbit Gaia measured on component B. Three lines of evidence bear on which star eclipses. The period match is strong, because a chance eclipsing binary on A within ±0.06 d of that period is very improbable. The centroid shift is weak, favouring B at about 2σ. Gaia's *e* and ω are not decisive. We therefore interpret the system as probably a companion of at least 0.41 $M_\odot$ grazing B. The predicted transit depth (0.72% of B's light) is within a factor of 2 of the observed (about 1.5% of B).

A43. TYC 4685-2083-1 (TIC 241445833)

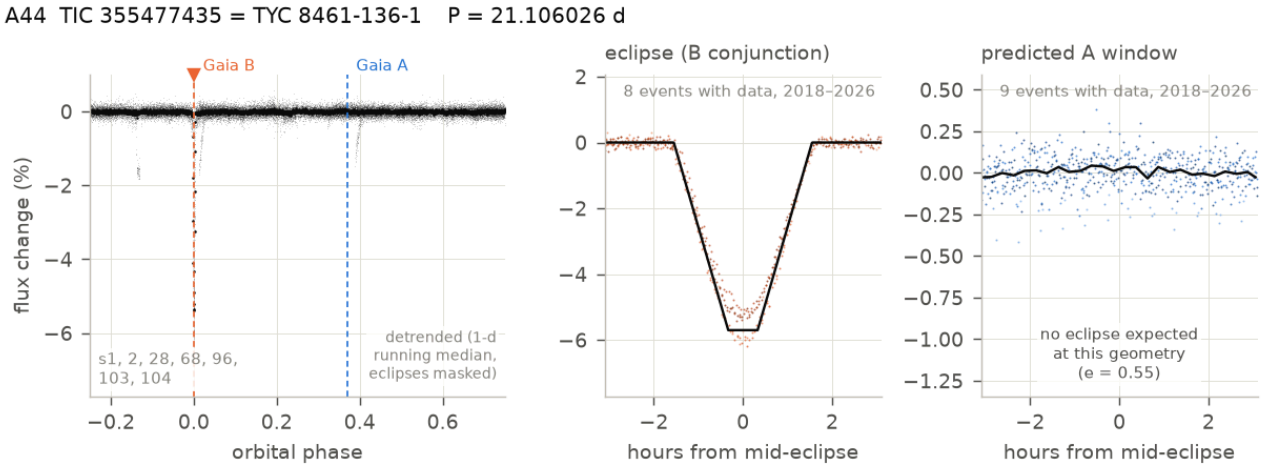


A43. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.205492 d, Gaia's predicted conjunctions dashed. Middle: the seven 6.9%, 4.25-h grazing eclipses at B, 2018–2025. Right: the A window, where the near-twin geometry just misses eclipsing and the data are flat (< 300 ppm).

Identifiers	TIC 241445833; Gaia DR3 2508607251322270464; TYC 4685-2083-1
Position (J2000)	01:39:18.9 −01:39:56.3
Brightness	G 10.81; Tmag 10.38
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 0.935, $M_2 \sin^3 i$ = 0.993 $M_\odot$; $a \sin i$ = 42.6 $R_\odot$; radii about 0.95 $R_\odot$ each fit both conjunctions
Period	23.205492 ± 0.000007 d (Gaia 23.19576 ± 0.00395 d; 2.47σ short)
Epoch of eclipse	T_0(B) = BTJD 1389.70430 ± 0.00052 = BJD_{TDB} 2458389.70430
Eclipse (B, periastron side)	6.8–7.0% deep (SPOC 2-min s30, s43 6.82%; TESS-SPOC s70 6.90%), T_{14} 4.25 h, T_{23} 0.9 h, grazing (d/S = 0.74–0.85); seven events, 2018–2025, including s3 and a flagged s42 event; O−C ≤ 1.5 min
Other eclipse (A)	None: 100% coverage, per-epoch −350 to +250 ppm, 3σ limit 280–345 ppm. The geometry predicts no A eclipse, but only narrowly
Orbit	Projected to Gaia's epoch, the eclipse sits +0.283 ± 0.161 d (+1.8σ) from Gaia's B conjunction; the TESS-era 3σ offset is mostly Gaia's period error. Gaia e 0.2711, ω 330.17°. r_A/r_B = 1.31
Pipelines agreeing	QLP SAP; SPOC 2-min (s30, s43); TESS-SPOC (s3, s70); eleanor (s3)
Pixel attribution	Difference-image centroid on the target (0.2–0.4 px); SPOC centroid shifts under 1 mpx at 7% depth; the only stars within 60″ are $G \approx 21$
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP. A prior partial detection: an unvetted SPOC s1–s46 TCE at P 348.08 d (= 15P)
Open issues	A mild t_p/ω tension with Gaia, which is not significant. The A conjunction is a sensitive radius test, where a shallow graze is just possible
Next eclipses	B: BTJD 4336.802, 4360.007, 4383.213 (±0.001 d); A windows +8.13 ± 0.19 d later (4344.93, ...)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4685-2083-1 is a pair of near-twin stars of about 1 $M_\odot$ in a 23.2-d orbit that graze each other near periastron. The central duration there would be 7.3 h; the observed 4.25 h is a graze. At the other conjunction the stars are 1.31 times farther apart. Twins of 1.03 $R_\odot$ would make a 4.2% eclipse with A missing. Twins of 0.9 $R_\odot$ would make a 7.0% eclipse with a 0.16% graze at A, which the data exclude. At about 0.95 $R_\odot$ both observations fit. QLP's 6.0% depth in s97 is a normalisation effect.

A44. TYC 8461-136-1 (TIC 355477435)

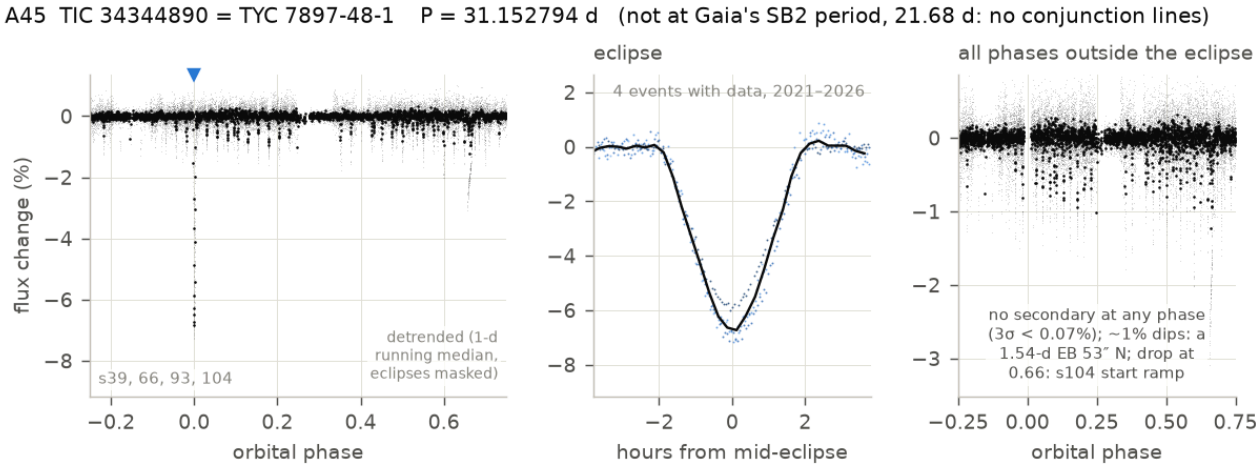


A44. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.106026 d, Gaia's predicted conjunctions dashed. Middle: the eight 5.7%, 3.1-h grazing eclipses at B, 2018–2026 (the last in the January 2026 sectors). Right: the A window; at e = 0.55 no eclipse is possible there.

Identifiers	TIC 355477435; Gaia DR3 6501783310150885376; TYC 8461-136-1
Position (J2000)	23:26:45.3 −50:44:19.7
Brightness	G 10.92; Tmag 10.50
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 1.138, $M_2 \sin^3 i$ = 1.034 $M_\odot$; $a \sin i$ = 41.6 $R_\odot$; radii about 1.2 and 1.0 $R_\odot$ give the observed depth
Period	21.106026 ± 0.000006 d (Gaia 21.100720 ± 0.003288 d; 1.6σ)
Epoch of eclipse	T_0(B) = BTJD 1325.49469 ± 0.00051 = BJD_{TDB} 2458325.49469
Eclipse (B, periastron side)	5.5–5.9%, constant (QLP's 5.0–5.2% in s103/s104 is a pipeline artefact), T_{14} 3.1 h, T_{23} 0.66 h, grazing (d/S ≈ 0.75); eight events, 2018–2025, including s1 ×2 and s68; O−C ≤ 2.4 min
Other eclipse (A)	Impossible: impact distance at A is 5.1 $R_\odot$ against R_1+R_2 = 2.2 $R_\odot$ (r_A/r_B = 3.13). All epochs flat to ±200 ppm apart from two single-epoch artefacts
Orbit	e = 0.546 (Gaia). At Gaia's epoch the offset from B is −0.016 ± 0.047 d (−0.3σ); the +0.4 to +0.9 d TESS-era offsets are entirely Gaia's period error
Pipelines agreeing	QLP SAP; eleanor (s1); TESScut apertures (lost flux 587 and 579 e/s in s28 and s103)
Pixel attribution	Difference image on the target; the only neighbour within 90″ is G 19.4 at 13.6″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP
Open issues	None
Next eclipses	B: BTJD 4343.656, 4364.762, 4385.868 (±0.001 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 8461-136-1 is a near-twin pair of about 1.1 and 1.0 $M_\odot$ in a highly eccentric (e = 0.55) 21.1-d orbit. The stars graze each other at the periastron-side conjunction; at the other they are more than three times farther apart, so no A eclipse can occur. Eight eclipses over seven years keep a constant depth, and once Gaia's period error is removed the eclipses fall where Gaia's orbit puts conjunction B, to 0.3σ.

A45. TYC 7897-48-1 (TIC 34344890)



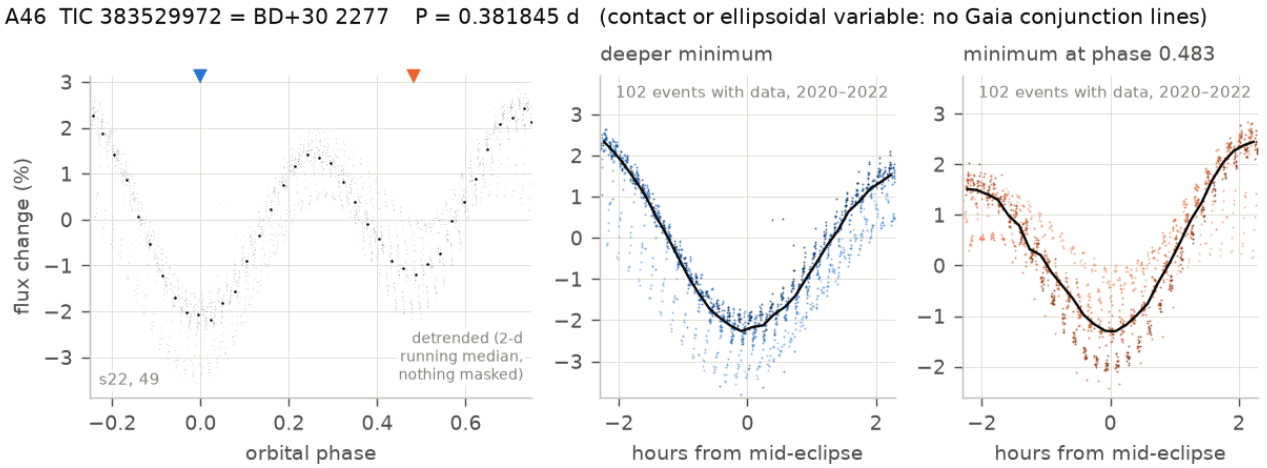
A45. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 31.152794 d. Middle: the four 6–7%, 3.8-h grazing eclipses (sectors 39, 66, 93 and 104, the last in January 2026). Right: all phases outside the eclipse, including phase 0.5; no secondary is seen (< 0.07%).

Identifiers	TIC 34344890; Gaia DR3 5956391835910597888; TYC 7897-48-1
Position (J2000)	17:49:36.3 −43:12:27.7
Brightness	G 10.93; Tmag 10.50
Stars	Gaia SB2 of near-twin stars seen almost edge-on (K_1 56.7, K_2 59.7 km/s, e 0.58, M $\sin^3 i$ 0.98 and 0.93 $M_\odot$) at P = 21.684 d; the eclipsing pair's relation to it is unresolved
Period	31.152794 ± 0.000072 d, the only period the coverage allows (P/2 ruled out). Gaia's SB2 period is 21.683770 ± 0.002520 d
Epoch of eclipse	T_0 = BTJD 2382.7596 ± 0.0037 = BJD_{TDB} 2459382.7596
Eclipse	5.70–6.80% deep, T_{14} 3.7–3.9 h, flat bottom about 0.8 h, close to V (likely grazing); four events, one per sector (s39, s66, s93, s104), 2021–2025. O−C −3.6 to +4.6 min, χ^2 131 for 2 dof (errors underestimated, or real timing variations)
Secondary eclipse	None secure: phase coverage 99%, 3σ limit about 0.07%; one unconfirmed 0.5% dip at phase 0.096
Orbit	The eclipses do not fall at the conjunctions of Gaia's 21.684-d SB2 orbit. 31.15/21.68 = 1.44 cannot be the inner and outer orbits of a stable hierarchy, and the beat period (71.3 d) is not a known alias
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s39)
Pixel attribution	Difference-image centroid within 0.1 px (~2″) of the target in s66 and s93; the G 13.19, 12.66 and 11.73 stars at 57–65″ are excluded; Gaia source 5956391801551111552 at 3.3″, with no G magnitude, cannot be excluded
Catalogue status	Not listed in: VSX, ZTF (Chen et al. 2020), ATLAS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, WDS, CCDM, Hipparcos, SB9, ExoFOP, general web search. The VSX "E" star 53″ north is a separate star (its true period is 1.5432 d, not VSX's 12.3202 d)
Open issues	**Orbit unresolved.** Either Gaia's period is wrong and the 31.15-d pair is the SB2 itself, or the system is a 2+2 quadruple, or the eclipses come from a blend within about 2″. RVs phased at 31.15 d against 21.68 d would distinguish between these
Next eclipses	BTJD 4345.386 ± 0.006
Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)
Status	Confirmed as an eclipsing system on, or within about 2″ of, TYC 7897-48-1, found as by-catch. Its relation to the Gaia SB2 orbit is open (Gaia's period wrong, a 2+2 quadruple, or a blend within 2″), so Gaia's orbit is not counted as contradicted; the Gaia source 3.3″ away without a G magnitude cannot be excluded.

Note: a confirmed eclipser on, or within about 2″ of, the target, with an unresolved orbit. The eclipses are real and on or near TYC 7897-48-1, but they do not follow the Gaia orbit that brought the star into the sample, and the two periods cannot belong to a stable triple; Gaia's orbit is not counted as contradicted.

TYC 7897-48-1 shows one grazing eclipse of 6–7% per TESS sector, four in all, on a single period of 31.15 d. A 3.894-d period is one-eighth of this value. The 0.77-d dips also present are a separate eclipsing binary 53″ north, whose alternating depths give a true period of 1.5432 d. A 3.8-h eclipse fits an $e \approx 0.58$ pair of about 1 $M_\odot$ at either period, so three interpretations remain: Gaia's period is wrong and the eclipsing pair is the Gaia SB2 itself; a 2+2 quadruple; or a blend within about 2″. A Gaia source 3.3″ away without a magnitude cannot be excluded by the pixels.

A46. BD+30 2277 = WDS ES 436 (TIC 383529972)



A46. QLP SAP flux of the 1.87″ pair, detrended with a 2-d running median (nothing masked), folded on P = 0.381845 d (sectors 22 and 49), showing a contact-binary or ellipsoidal light curve with unequal minima (4.6% and 3.4%) and unequal maxima (O'Connell effect). The Gaia 46.3-d orbit of component A is not the period of this variability.

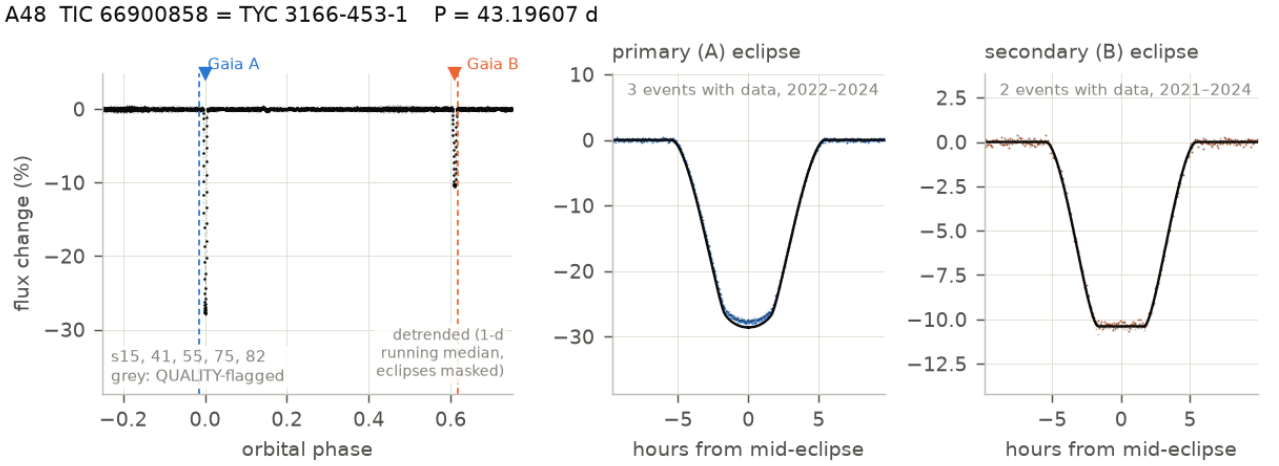
Identifiers	TIC 383529972; Gaia DR3 4011717574379114240 (component A); component B = Gaia DR3 4011717574379114112; BD+30 2277; WDS ES 436
Position (J2000)	12:28:27.5 +29:52:42.3
Brightness	G 10.63 (A), 10.73 (B); Tmag 10.12
Stars	A 1.87″ physical pair (parallaxes 4.02 and 4.15 mas). Component A is a Gaia SB1 at P 46.340 d (K_1 = 3.0 km/s, RUWE 4.6). The variable is probably component B
Period	0.381845 ± ~0.000002 d (Fourier fit 0.3818455; 99 minimum timings 0.3818442 ± 0.0000005); not an alias of the 46.3-d orbit
Epoch of deeper minimum	BTJD 2645.0111 ± 0.0007 = BJD_{TDB} 2459645.0111
Minima	4.6% and 3.4% jointly (4.33/1.97% in s22, 4.56/4.00% in s49), second minimum at phase 0.483; peak-to-peak 4.4–4.7%; O'Connell effect −0.7 to −0.9%; shape changes between sectors. In the variable star's own light, about 9.6% if B, 8.8% if A
Type	A spotted contact binary (EW) or an ellipsoidal variable
Pipelines agreeing	QLP SAP (s22, s49)
Pixel attribution	On the pair: difference image within 0.1 px, excluding the G 10.47 star 76″ away. Which component: the fitted centroid lies 0.03–0.05 px closer to B in all six tests, but systematics are about 0.1 px, so the test is inconclusive. Gaia RVs: A's 17 RVS transits have a robust RV amplitude of only 7.6 km/s, so A is unlikely to be a contact binary; B has no Gaia RV
Catalogue status	Not listed in: VSX, ZTF (Chen et al. 2020), ATLAS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, CCDM, Hipparcos, SB9, ExoFOP, general web search; WDS has the pair only
Open issues	**Host component unresolved.** Speckle or adaptive-optics photometry separating A and B would resolve it
Next minima	Deeper minima: BTJD 4325.511, 4325.893, 4326.275 (+ n × 0.381845)

Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)
Status	By-catch variable, not counted among the eclipsing systems of this paper. A contact-binary or ellipsoidal light curve on an unresolved 1.87″ pair, in one pipeline, with the host component undetermined: a new variable, but not shown to be an eclipsing system.

Note: a new variable on the BD+30 2277 pair, with the host component unresolved. It is a short-period (9.2-h) contact or ellipsoidal variable, not a long-period eclipser, found incidentally rather than at a predicted conjunction, and it is not counted among the 104 eclipsing systems.

BD+30 2277 is a 1.87″ visual pair of two stars of nearly equal brightness at the same distance; component A carries a Gaia single-lined orbit of 46.3 d. TESS shows a 0.3818-d light curve with unequal minima and maxima whose shape changes between 2020 and 2022, typical of a spotted contact binary or an ellipsoidal variable. The pixels cannot tell which star varies. Gaia's radial velocities of A scatter by only 7.6 km/s, too little for a contact binary; B is therefore the more likely host. If B is the variable, the system is a bound 2+2 quadruple, consisting of the Gaia SB1 binary A and the contact binary B.

A48. TYC 3166-453-1 (TIC 66900858)



A48. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 43.19607 d, Gaia's predicted conjunctions dashed; Gaia's A falls about 0.9 d early (Gaia's eccentricity exceeds the photometric value). Middle: the three 27%, 9.7-h annular transits at A. Right: the two 10.4%, 9.9-h flat total occultations at B.

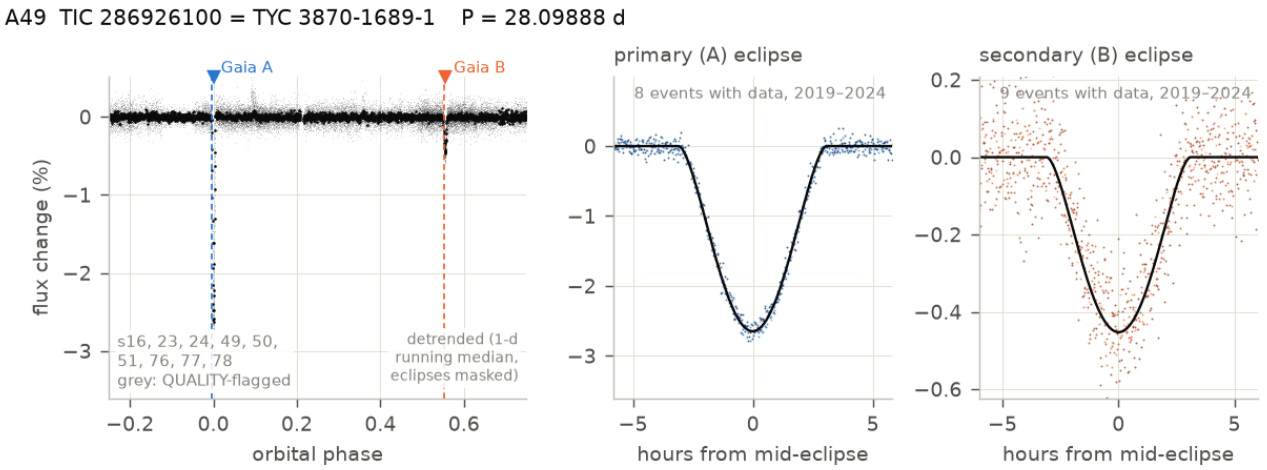
Identifiers	TIC 66900858; Gaia DR3 1871082303289948544; TYC 3166-453-1
Position (J2000)	20:46:48.0 +37:33:55.9
Brightness	G 10.98; Tmag 10.53
Stars	Slightly evolved primary, $M_1 \approx 1.3\ M_\odot$ (Gaia binary_masses 1.06; the parallax luminosity favours 1.3), R_1 = 1.37–1.45 $R_\odot$ + K dwarf, M_2 = 0.73–0.82 $M_\odot$, R_2 = 0.73–0.77 $R_\odot$, about 4600 K, 10.4% of the light; mass function 0.122 $M_\odot$; Gaia SB1
Period	43.19607 ± 0.00003 d (Gaia 43.19275 ± 0.01065 d; +0.3σ)
Epoch of primary	T_0(A) = BTJD 2804.3848 ± 0.0004 = BJD_{TDB} 2459804.3848; T_0(B) = BTJD 2441.9431 ± 0.0008
Primary eclipse (A)	27.0–27.4% deep, 0.403 d (9.7 h), flat part 0.126 d: annular transit of the K dwarf across the primary; three events (2804.38, 3365.93, 3538.72); O–C under 0.1 min
Secondary eclipse (B)	10.3–10.5% deep, 0.414 d (9.9 h), flat part 0.159 d: total occultation of the K dwarf; two events (s41, every QLP cadence flagged but clean in TESS-SPOC; s55). B at phase 0.60938 after A (Gaia 0.6334 ± 0.0085, −2.8σ)
Orbit	Photometric e cos ω = 0.1727, e sin ω = 0.002: e = 0.173, ω = 0.7°; Gaia e cos ω 0.211 ± 0.014, e sin ω −0.005 (Gaia e 0.2111, ω 358.72°). Model: k = 0.528, $i \approx 90°$, χ^2 460/460
Pipelines agreeing	QLP SAP; TESS-SPOC (s41, s55); TARS (all sectors)

Pixel attribution	TESScut difference images: flux lost peaks on the target pixel, centroid 0.03–0.21 px; the QLP centroid shifts at A and B (0.075 and 0.02 px, ratio 3.7) match the depth ratio (3.5), so both eclipses come from the same source; the *G* 12.67 neighbour at 39″ (would need about 130%) is excluded
Catalogue status	Not listed in: VSX, GCVS, NSV, ASAS-3, ASAS-SN, ATLAS, ZTF, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Justesen & Albrecht 2021, Fetherolf et al. 2023, Gaia DR3 variability tables, SB9, ExoFOP, SIMBAD (SB*, one reference), general web search
Open issues	**Gaia eccentricity revised:** Gaia's *e* exceeds the photometric value by about 0.04 (2.8σ). RVs to fix K_1 against the photometric *e* and ω; spectroscopic T_{eff} and log *g*. The Harvard plates (Section 5.6) show the eclipse at this ephemeris (z = 9.4 on 15 plates) but more strongly at P = 43.19588 d, 6σ below the TESS period, a 3.8-h shift of the eclipse centre over 830 cycles: an underestimated period error or apsidal motion
Next eclipses	B: BTJD 4342.570 (2026-10-29 01:37 UTC), 4385.766; A: BTJD 4359.443 (2026-11-14 22:36 UTC), 4402.639
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: the eclipses contradict part of Gaia's orbit. The eclipse spacing gives *e* = 0.173 where Gaia's SB1 has *e* = 0.211 ± 0.014 (2.8σ), which is why Gaia's predicted A times run about 0.9 d early. The timing measurement is model-independent, and light-travel time across the orbit (about a minute) cannot explain a one-day shift.

TYC 3166-453-1 is a slightly evolved star of about 1.3 $M_\odot$ with a K-dwarf companion of about 0.75 $M_\odot$, seen edge-on. At A the dwarf crosses the primary's disk entirely (an annular transit, 27% deep); at B it is hidden completely, and the flat 10.4% depth is its share of the light. Five eclipses fit a single linear ephemeris to within 0.1 min; one of them, in sector 41, is wholly in QLP-flagged cadences but present with quality zero in TESS-SPOC. A companion supplying 10% of the light is consistent with Gaia's single-lined solution.

A49. TYC 3870-1689-1 (TIC 286926100)



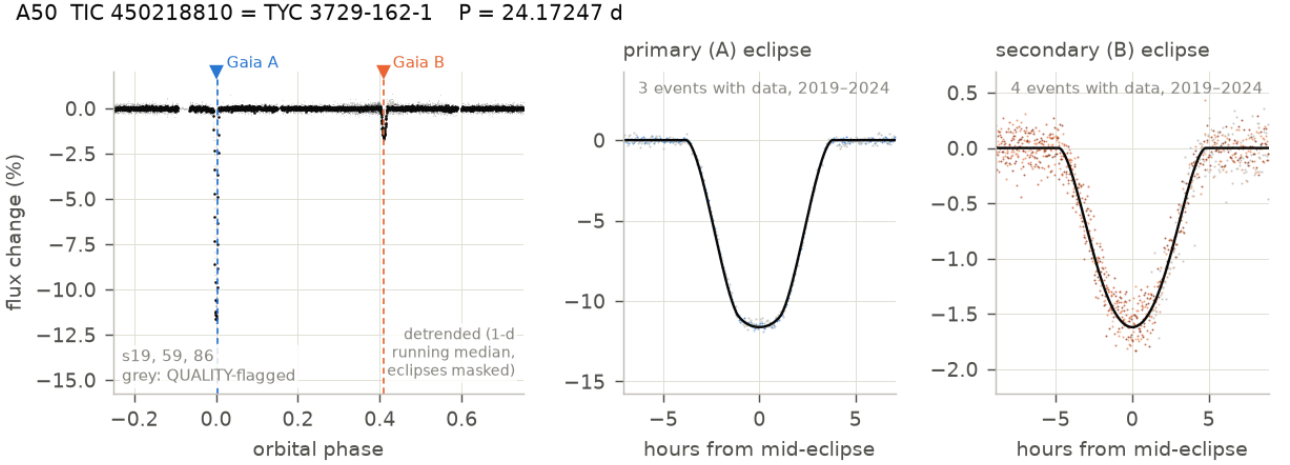
A49. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 28.09888 d, Gaia's predicted conjunctions dashed. Middle: the eight 2.6%, 5.8-h grazing eclipses at A. Right: the nine 0.45%, 6-h grazing eclipses at B.

Identifiers	TIC 286926100; Gaia DR3 1404550929673449216; TYC 3870-1689-1
Position (J2000)	15:49:15.4 +52:32:40.6
Brightness	*G* 10.40; Tmag 10.01
Stars	Primary R_1 = 1.64–1.75 $R_\odot$ (TIC 1.80), M_1 = 1.0–1.25 $M_\odot$ assumed + M dwarf, m_2 = 0.46–0.53 $M_\odot$, R_2 = 0.46–0.49 $R_\odot$, T_2 ≈ 3670 K; mass function 0.0465 $M_\odot$; Gaia SB1; APOGEE entries in SIMBAD

Period	28.09888 ± 0.00001 d (P_A 28.09887 ± 0.00001, P_B 28.09898 ± 0.00004; Gaia 28.1076 ± 0.0067 d, −1.3σ)
Epoch of primary	T_A = BTJD 2645.0156 ± 0.0003 = BJD_{TDB} 2459645.0156; T_B = BTJD 2688.6996 ± 0.0008
Primary eclipse (A)	2.53–2.69% deep, T_{14} 5.8 h, ingress 2.3 h, V-like (flat about 1.2 h): grazing transit of the M dwarf; eight events 2019–2024 (one, s023, wholly in flagged cadences); O−C ≤ 1 min
Secondary eclipse (B)	0.41–0.48% deep, T_{14} mostly 5.8–6.0 h: grazing occultation; nine events; O−C ≤ 5 min. B at phase 0.55468 ± 0.00003 after A (Gaia 0.5567 ± 0.0083)
Orbit	e cos ω = 0.0860 against Gaia's 0.0892 ± 0.013; e sin ω ≈ −0.005 (Gaia −0.029 ± 0.0135, +1.7σ); i = 87.8°, b/(1+k) ≈ 0.80 at both conjunctions
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s50, s76); TARS (s50, s76); within 0–3 min
Pixel attribution	TESScut s50 and s76 difference images (A) peak on the target pixel at 64σ and 79σ, centroid 0.7″ from the out-of-eclipse centroid; the only neighbour (G 15.32 at 53″) would need 247%. Background excursions change sign from event to event while the depth stays constant
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (Gaia multiplicity and APOGEE entries only), general web search
Open issues	P_A and P_B differ by 2.7σ (possibly slow apsidal motion; not significant); the radius-ratio degeneracy of a grazing fit (multi-colour photometry would break it); APOGEE RVs not yet analysed
Next eclipses	A: BTJD 4330.949, 4359.047; B: BTJD 4346.534, 4374.633 (formal 1–2 min; allow about 10 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3870-1689-1 is a star of about 1.7 $R_\odot$ with an M-dwarf companion of about 0.5 $M_\odot$, grazing at both conjunctions of a mildly eccentric 28.1-d orbit. Seventeen eclipses over five years give a period to 0.00001 d, and the eclipse spacing reproduces Gaia's e cos ω. The companion's fitted radius and temperature lie where main-sequence stars of its mass do; the alternative fit with a large companion would need 1.2–1.3 $R_\odot$ for 0.5 $M_\odot$, which is unphysical.

A50. TYC 3729-162-1 (TIC 450218810)



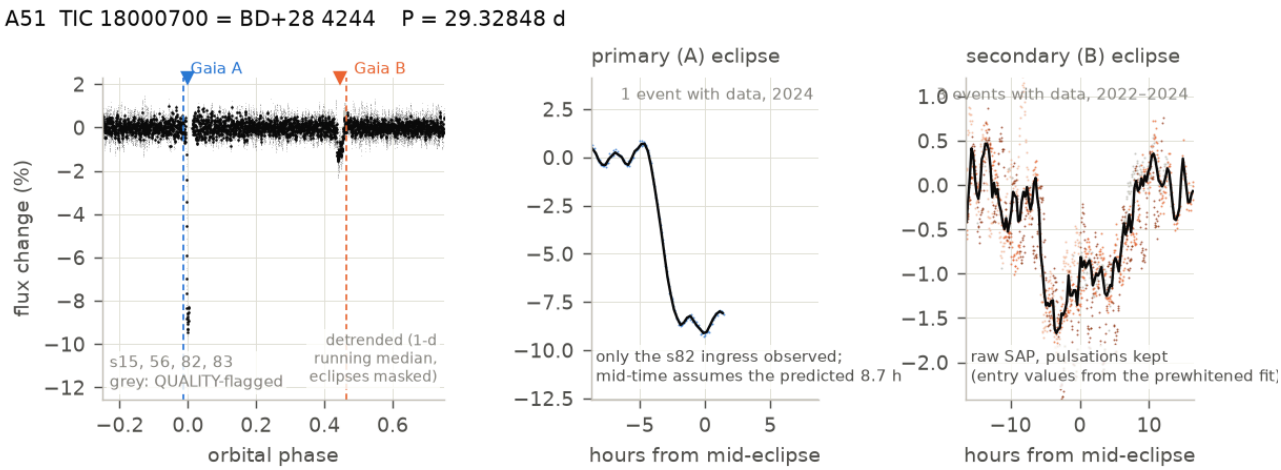
A50. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.17247 d, Gaia's predicted conjunctions dashed. Middle: the A windows with data: three 11.5%, 7.1-h flat-bottomed transits and a fourth window covering only part of an event. Right: the three 1.6%, 8.9-h occultations at B.

Identifiers	TIC 450218810; Gaia DR3 473363571357621760; TYC 3729-162-1
Position (J2000)	03:55:22.1 +59:49:27.7
Brightness	G 10.95; Tmag 10.40
Stars	Primary R_1 = 1.69–1.88 $R_\odot$ (TIC 1.77), M_1 = 0.95–1.2 $M_\odot$ assumed + K/M dwarf, m_2 = 0.59–0.68 $M_\odot$, R_2 = 0.56–0.63 $R_\odot$, T_2 ≈ 3850–4015 K, L_2/L_1 ≈ 0.02–0.025; mass function 0.0879 $M_\odot$; Gaia SB1

Period	24.17247 ± 0.00002 d (Gaia 24.176370 ± 0.007485 d; −0.5σ)
Epoch of primary	T_A = BTJD 2926.6627 ± 0.0004 = BJD_{TDB} 2459926.6627; T_B = BTJD 3274.9793 ± 0.0011
Primary eclipse (A)	11.3–11.9% deep, T_{14} 7.0–7.2 h, flat bottom 2.1 h, ingress 2.5 h: transit of the companion (b_A = 0.57); three events (s19 partly flagged, s59, s86 all flagged); O–C ≤ 0.4 min
Secondary eclipse (B)	1.46–1.72% deep, T_{14} about 8.9 h: occultation of the companion (b_B = 0.83–0.90); three events (s19, s59, s86). B at phase 0.40965 ± 0.00004 after A (Gaia 0.4063 ± 0.0111)
Orbit	e cos ω = −0.1399 against Gaia −0.1434 ± 0.017; e sin ω from the shapes 0.184 (e = 0.231) against Gaia 0.238 ± 0.013; i ≈ 88.2°. B lasts longer because the separation is larger there
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s19, s59); TARS (s19, s59, s86)
Pixel attribution	TESScut s059 A difference image peaks on the target at 224σ, centroid 0.9″ from the out-of-eclipse centroid; B in s019 and s059 also on target; G 15.4 neighbours at 20″ and 56″ would need 680% and 506%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (Gaia multiplicity only), general web search; Kostov table 2 only
Open issues	A mild tension (2.5–3σ) between the shape-based e sin ω and Gaia's (forcing Gaia's e, ω costs $\Delta\chi^2$ ≈ 11), with optimistic errors and partial or flagged eclipses that can bias it; one clean A and B pair in the same sector would test Gaia's e of 0.278
Next eclipses	A: BTJD 4328.666, 4352.839; B: BTJD 4338.568, 4362.741 (a few minutes)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3729-162-1 is a star of about 1.8 $R_\odot$ with a K/M-dwarf companion of about 0.6 $M_\odot$ on an eccentric 24.17-d orbit. At A the companion crosses the primary's disk (11.5%, flat-bottomed); at B it is hidden, and the 1.6% depth is its share of the light. The eclipse timing reproduces Gaia's e cos ω within 0.3σ. Box-fit estimates of the A eclipse (9.7%, 4 h) underestimate the model-fit depth and duration given above.

A51. BD+28 4244 (TIC 18000700)



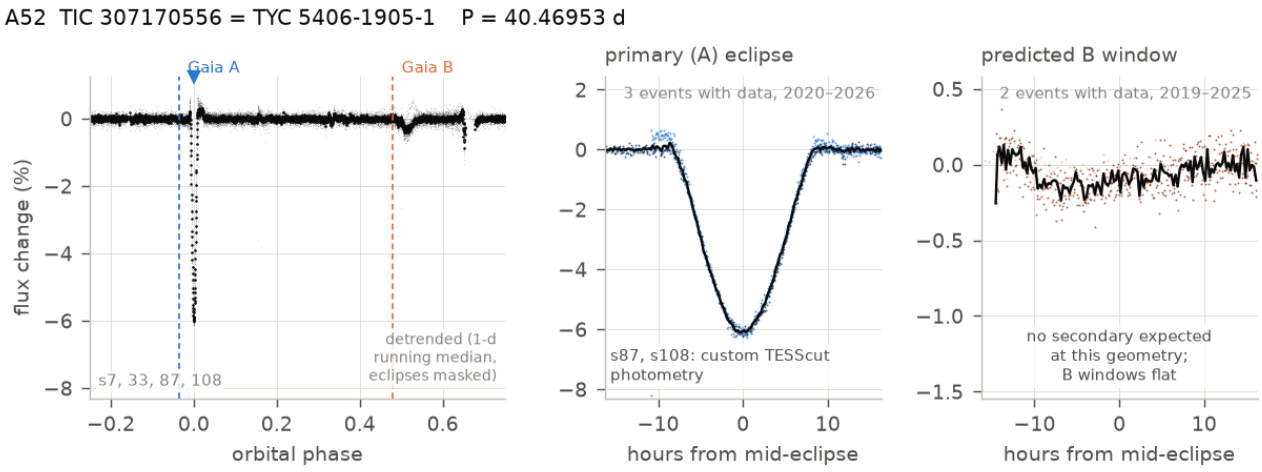
A51. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.32848 d, Gaia's predicted conjunctions dashed; the star pulsates throughout. Middle: the ≥ 8.6% transit ingress at A at the end of sector 82. Right: the four B windows with data (the entry's fit uses all four; one is partly flagged), the 1.1–1.3%, 16.4-h occultations shown in raw SAP flux with the pulsations present (depth and duration from the prewhitened fit; see text).

Identifiers	TIC 18000700; Gaia DR3 1896494009856522240; BD+28 4244
Position (J2000)	21:57:03.2 +29:18:20.1
Brightness	G 10.46; Tmag 10.19
Stars	F0 primary of 2.2 $R_\odot$ (M_1 ≈ 1.6 $M_\odot$ assumed), a hybrid δ Sct/γ Dor pulsator (12.08, 12.37, 4.08, 3.16, 2.70 c/d at ≤ 2.5 ppt) + K dwarf of about 0.7 $R_\odot$, $m_2(i = 90°)$ ≈ 0.73 $M_\odot$, 4500–4800 K; mass function ≈ 0.071 $M_\odot$; a ≈ 53 $R_\odot$; Gaia SB1
Period	P_B = 29.32848 ± 0.00033 d (Gaia 29.3189 ± 0.0156 d; +0.6σ)

Epoch of eclipse	T_0(B) = BTJD 3571.1856 ± 0.0044 = BJD_{TDB} 2460571.1856; A first contact BTJD 3557.9714 ± 0.0012
Eclipse at B	1.12–1.34% deep after prewhitening, T_{14} 15.6–16.5 h, T_{23} about 10 h, flat: total occultation of the K dwarf (its light share), near apastron; four events (s15 all flagged, 1.8%; s56 partly flagged; s82; s83); all O–C under 0.02 d
Eclipse at A	Only an ingress: first contact BTJD 3557.9714, ingress 1.86 h, depth ≥ 8.6% before the data end at the close of s82; transit of the dwarf across the F star (R_2/R_1 ≈ 0.3), predicted full duration 8.7 h near periastron. In all other sectors the A conjunction fell in gaps at sector boundaries
Orbit	A about 16.3 d after B (Gaia predicts 15.26 d); implied $\|e \cos \omega\|$ ≈ 0.09 against Gaia −0.03 ± 0.09; predicted central durations 17.2 h (B) and 8.7 h (A). Every B within 0.3 d of Gaia's prediction
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and TARS (all four measured events, including the A ingress: 8.3% and 9.1%)
Pixel attribution	TESScut: every event on the target (e.g. the A ingress: target loses 9.69%, χ^2/pix 0.80 against 94 for the next source); all neighbours $G \geq 14.4$ (one would need ≥ 75%). No common-mode signal at any event (166–297 comparison stars)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP (no TOI/CTOI), SIMBAD (SB*, Gaia DR3 orbit reference only)
Open issues	A full A eclipse (predicted 8.7 h) has not yet been observed; it would give e sin ω. RVs for the K_1 phase
Next eclipses	B: BTJD 4333.73, 4363.05 (±0.01); A: BTJD 4350.02, 4379.35 (±0.06; assumes the predicted duration)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+28 4244 is a pulsating F0 star in an eclipsing binary with a K-dwarf companion. Its δ Sct and γ Dor pulsations, at up to 2.5 ppt, distorted the eclipse profiles. Prewhitening twelve pulsation frequencies reduced the scatter from about 3.9 to 1.5 ppt and left flat-bottomed 16-h occultations. The 9% drop at the end of sector 82 is the start of the transit at the periastron-side conjunction, with a 1.9-h ingress that fits the predicted 8.7-h transit. Every event is on the target in the pixels, reproduced by two other pipelines, and absent from the common mode.

A52. TYC 5406-1905-1 = ASAS J071301-1406.7 (TIC 307170556)



A52. Left: SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 40.46953 d, Gaia's predicted conjunctions dashed (computed with Gaia's 40.543-d period, which the eclipses contradict). Middle: the three 6%, 16-h grazing eclipses at A (sectors 33, 87 under scattered light, and 108). Right: the B windows, where no eclipse is expected at e ≈ 0.31 and none is seen.

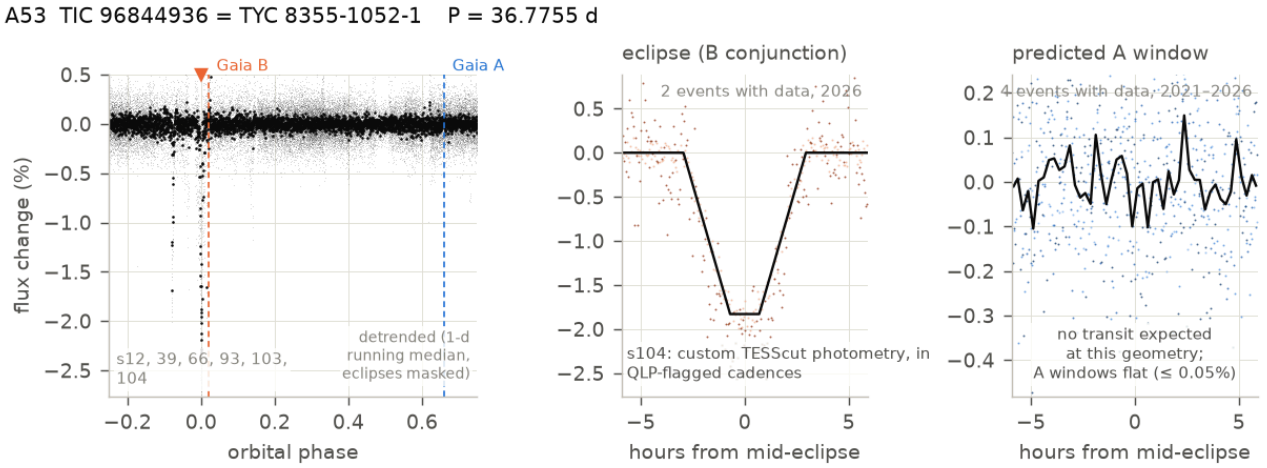
Identifiers	TIC 307170556; Gaia DR3 3032498188045355136; TYC 5406-1905-1; ASAS J071301-1406.7
Position (J2000)	07:13:01.5 −14:06:42.9

Brightness	*G* 10.49; Tmag 9.82
Stars	Gaia SB2 equal-mass pair: K_1 41.0, K_2 40.3 km/s, M $\sin^3 i$ = 0.961 and 0.977 $M_\odot$, $a \sin i$ = 61.8 $R_\odot$. Gaia FLAME (as a single star) R 5.6 $R_\odot$, L 23.5 $L_\odot$: probably two ~1 $M_\odot$ stars on the lower red-giant branch
Period	40.46953 ± 0.00005 d (formal; allowing for red noise the uncertainty is about three times larger). Gaia 40.543005 ± 0.014906 d: −0.0735 d, **−4.9σ**
Epoch of eclipse	T_0 = BTJD 2206.5875 ± 0.0019 = BJD_{TDB} 2459206.5875
Eclipse (A, periastron)	5.92–6.16% deep, T_{14} 16.10–16.36 h, T_{23} 3.4–3.5 h, ingress about 6.4 h: grazing, V/U-shaped; three events: s33, s87 (under a scattered-light surge in flagged cadences, recovered with local-background photometry), s108 (within about 6 min of the time predicted from s33 and s87); O−C after refitting all three +1.0, −3.3, +2.3 min
Other eclipse (B)	None expected (impact parameter at B 1.9× larger) and none seen: the s7 and s87 B windows are flat
Orbit	Gaia *e* 0.3085, ω 85.93°. At Gaia's epoch the eclipse and Gaia's A conjunction agree (+1.7σ); the disagreement is entirely in the period. Gaia goodness-of-fit 11.5
Pipelines agreeing	QLP SAP (s33); custom TESScut aperture photometry (this work) for all three. Gaia epoch photometry: the faintest of 51 G transits falls 3.0 h from mid-eclipse on the TESS ephemeris, 2.84 d out of eclipse on Gaia's period (chance about 0.6%); a single point, +0.12 mag against a scatter of 0.025 mag, about twice what the eclipse can produce at that phase, so supporting rather than independent evidence
Pixel attribution	Difference images of all three events: uniform 5.2–5.3% fractional depth in every core pixel, about 0 off target; no Gaia neighbour within 60″ brighter than *G* 14.5 (would need 245%)
Catalogue status	**VSX type revision MISC → EA.** VSX: ASAS J071301-1406.7, MISC, *P* 79.3 d, at 1″; ASAS-SN "ROT:", *P* 441.8 d; Gaia flags it VARIABLE with classifier RS (score 0.35). Nothing in Gaia's eclipsing-binary table, SB9, WDS or ExoFOP
Open issues	**Gaia period corrected** by 4.9σ; an RV refit at 40.4695 d would test Gaia's solution. The B window in s33 and s108 was not usable
Next eclipses	A: BTJD 4351.473, 4391.942, 4432.412 (±0.01); a B eclipse, if any, near 4372.28 ± 0.93
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: (1) a VSX type revision: the star is in VSX as ASAS J071301-1406.7, type MISC, *P* 79.3 d, and the eclipses are what is new; (2) the eclipses contradict Gaia's period, which exceeds the eclipse period by 4.9σ, while agreeing with Gaia's conjunction phase at Gaia's own epoch.

TYC 5406-1905-1 is a pair of equal-mass stars, likely low-red-giant-branch giants of about 1 $M_\odot$ each, on an eccentric 40.47-d orbit. They graze each other at periastron, where the 16.3-h chord (about 9 $R_\odot$) is consistent with the separation and speed. At the far conjunction they miss. The period rests on three eclipses. One of them lies under a scattered-light surge where QLP's background subtraction fails (12 of 247 stars in that sector show spurious dips there). The eclipse in sector 108 falls exactly at cycle 52 of the TESS period, about 3.8 d from where Gaia's period would put it. One Gaia epoch-photometry point is consistent with the TESS period and not with Gaia's, which supports but does not independently confirm it.

A53. TYC 8355-1052-1 (TIC 96844936)



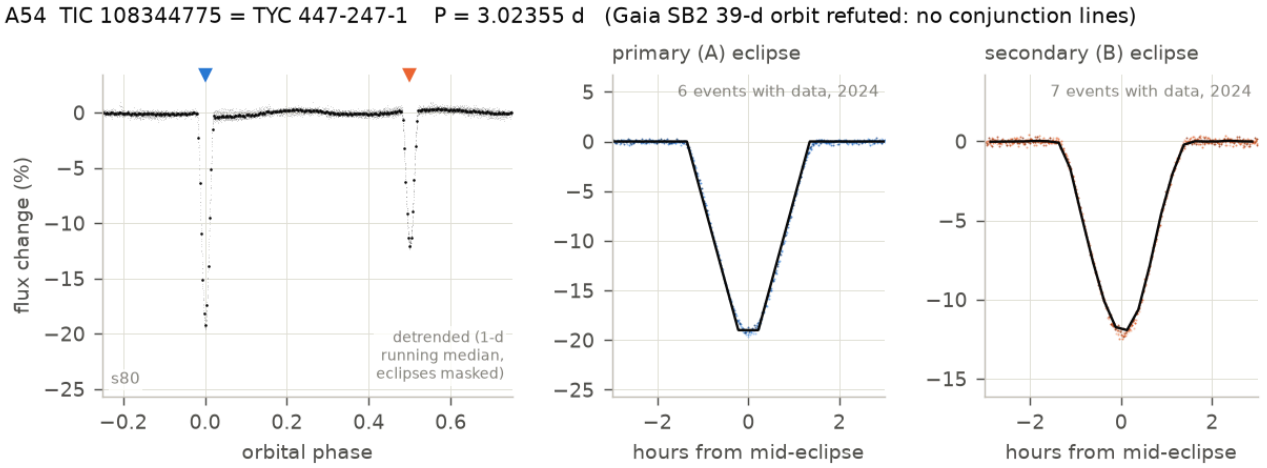
A53. Left: SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 36.7755 d, Gaia's predicted conjunctions dashed. Middle: the two 1.8–2.4%, 5.9-h occultations at B (sector 103 clean; sector 104 in QLP-flagged cadences, shown in a lighter tone). Right: the A windows, flat as the geometry predicts.

Identifiers	TIC 96844936; Gaia DR3 5921931286046605952; TYC 8355-1052-1
Position (J2000)	17:40:45.0 −52:25:24.0
Brightness	*G* 10.94; Tmag 10.46
Stars	Primary of 3.0 $R_\odot$, 1.62 $M_\odot$ (Gaia FLAME) + companion of about 0.95 $M_\odot$, about 6.5% of the TESS light; mass function 0.130 $M_\odot$; Gaia SB1; SIMBAD: GALAH and open-cluster papers
Period	36.7755 ± 0.006 d, the interval between the two events (4192.621 – 4155.8455; the rounded value 36.772 does not reproduce the next-eclipse times given below) (Gaia 36.7818 ± 0.0092 d; −0.7σ)
Epoch of eclipse	BTJD 4155.8455 ± 0.0007 (QLP) = BJD_{TDB} 2461155.8455; our TESScut photometry 4155.8488 ± 0.0008
Eclipse (B)	1.83% (QLP) / 2.44% (our TESScut photometry) deep, T_{14} 5.88 h, T_{23} 1.40 h: partial occultation of the companion (35–39% of its disc covered, b ≈ 0.8, i ≈ 84°). Event 1: s103, clean. Event 2: BTJD 4192.621 ± 0.006 in s104, in QLP-flagged scattered-light cadences, 1.8–2.5% across 12 aperture and background choices in our TESScut photometry
Other eclipse (A)	None expected (projected separation 2.87× larger) and none seen: four A windows (s39, s66, s93, s104) flat to ≤ 0.05 ± 0.03%
Orbit	Gaia e 0.5207, ω 248.07°; central duration at B would be 9.9 h against 5.9 h observed
Pipelines agreeing	QLP SAP (s103); custom TESScut photometry (this work) for both events
Pixel attribution	s103 difference image on target (1.85–2.25% in each core pixel, 0.00% off); s104 difference image noisier (pointing jitter) but strongest in the core. No common mode (228 stars, median −17 ppm). Neighbours: a 3.7″ source with no Gaia photometry, and stars G ≥ 16.1 (would need more than 100%)
Catalogue status	Not listed in: VSX, ASAS-SN, Gaia DR3 eclipsing-binary table, SB9, WDS, ExoFOP (no TOI/CTOI)
Open issues	**One event in flagged data.** Fainter stars in the s104 cutout scatter by ±1–4% in that stretch. One further clean B event would secure the confirmation
Next eclipses	B: BTJD 4339.723, 4376.498 (±0.01–0.03)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable (see below)
Status	Probable. The second event (sector 104) lies in QLP-flagged scattered-light cadences and was recovered only in custom TESScut photometry, where the depth is reduction-dependent (1.8–2.5% across 12 aperture and background choices; an independent re-reduction made for the referee review found 2.0–3.3%, with four of ten reductions degenerate); the period is a single interval, 36.7755 d, with the cycle count fixed by Gaia (two cycles, 18.39 d, is untested); a 3.7″ source without Gaia photometry shares the pixel; a flagged 2.2–2.6% drop at the end of sector 66, inside a B window, is inconclusive. One further clean event would confirm it.

Note: probable, not confirmed. The first occultation (sector 103) is clean and on the target. The second (sector 104) sits in cadences QLP flagged for scattered light and was recovered only in custom TESScut photometry (this work), where its depth depends on the reduction. The period is a single interval, with the cycle count taken from Gaia. The confirmation therefore depends partly on flagged data, and one further clean event is needed.

TYC 8355-1052-1 is a star of about 3 $R_\odot$ and 1.6 $M_\odot$ with a companion of about 0.95 $M_\odot$ on an eccentric 36.8-d orbit. The eclipses fall where Gaia's primary is in front, so they are occultations of the companion. At an impact parameter of about 0.8 the primary covers 35–39% of the companion's disc. For a companion carrying about 6.5% of the light, that reproduces the 2.3–2.5% depth. At the other conjunction the stars are 2.87 times farther apart in projection, and the four covered windows are flat. The second event improves the fit by $\Delta\chi^2 \approx 1150$ over no eclipse, and the brightest other star in the cutout is flat at that time. The 36.7755-d interval agrees with Gaia's period (36.7818 ± 0.0092 d) to 0.7σ.

A54. TYC 447-247-1 (TIC 108344775)



A54. Left: QLP SAP flux of sector 80, detrended with a 1-d running median with the eclipses masked, folded on P = 3.02355 d; the out-of-eclipse wave is a near-synchronous spot modulation. Middle: the 18.8%, 2.7-h primary eclipses (five measured; a sixth window holds partial data). Right: the seven 11.7% secondary eclipses at phase 0.5. Gaia's 39-d orbit is not shown (see text).

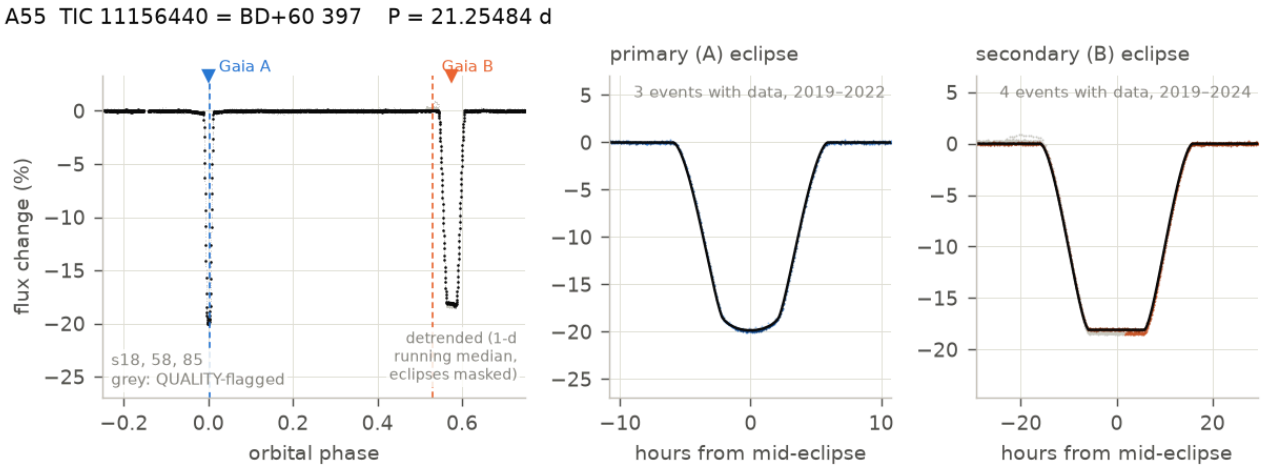
Identifiers	TIC 108344775; Gaia DR3 4279599735525131776; TYC 447-247-1
Position (J2000)	18:39:53.6 +01:43:01.4
Brightness	G 10.95; Tmag 10.29
Stars	K-dwarf pair, about 4800 K (AMBRE: T_{eff} 4837, log g 4.89, Vbroad 13.74 ± 0.29 km/s), radii about 0.8 and 0.7 $R_\odot$; 111 pc, absolute G 5.7. Gaia RUWE 10.58, non_single_star = 3, plus an Acceleration7 astrometric solution (significance 90.7)
Period	3.02348 ± 0.00001 d, from the ASAS-SN V photometry of 2015–2018 (3.023466; bootstrap 16–84%: 3.023454–3.023478) and the Harvard plates (3.023482 ± 0.000003; Section 5.6), with the epoch from sector 80. Sector 80 alone gives 3.02351–3.02358 depending on the reduction and eclipse kind. Gaia's SB2 solution gives 38.956902 ± 0.013672 d, which we show to be spurious
Epoch of primary	T_0 = BTJD 3482.0379 ± 0.0001 = BJD_{TDB} 2460482.0379
Primary eclipse	18.80% (our photometry) / 19.00% (QLP) deep, T_{14} 2.70 h, T_{23} 0.44 h; five events in s80
Secondary eclipse	11.65% / 11.83% deep, T_{14} 2.59 h; seven events in s80; at phase 0.5 within −0.3 ± 0.14 min
Orbit	Circular: $e \cos\omega \approx -1\times10^{-4}$, duration ratio gives $e \sin\omega \approx -0.02$. Gaia's 39-d SB2 ($K_1$+$K_2$ = 190 km/s) would need M $\sin^3 i$ = 14.0 and 13.5 $M_\odot$; the same amplitudes at 3.0235 d give 2.15 $M_\odot$ total against about 1.5 expected
	QLP SAP; TESS-SPOC SAP (18.2% and 11.4%, same times); our TESScut photometry

Pipelines agreeing	Difference images: uniform 17.2–18.0% (primary) and 10.0–11.0% (secondary) in every core pixel, about 0 off target; nearest bright neighbour (*G* 15.9 at 13″) would need more than 100%
Pixel attribution	
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022 (covers sectors 1–26 only), IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 eclipsing-binary table, SIMBAD (not as an EB), ExoFOP, general web search
Open issues	**Gaia orbit refuted.** The 39-d period is not a simple sampling alias of 3.02 d (tested with the epochs of three neighbouring Gaia sources; the RVS epochs themselves were not obtainable). A probable triple: an outer companion with a period much longer than Gaia's 34-month baseline. A few RVs and Gaia DR4 epoch RVs would check the 3.02-d SB2. Only one TESS sector
Next eclipses	Primary: $T_0 + n \times 3.02348$, e.g. BTJD 4328.612 (2026-10-15 02:41 UTC; n = 280, ±0.003)
Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)

Note: the eclipsing binary is confirmed, and the Gaia orbit that brought it into the sample is refuted. The 3.02-d period differs from Gaia's, so the system was found incidentally rather than at the Gaia period.

TYC 447-247-1 is a detached pair of K dwarfs on a circular 3.02-d orbit, observed in a single TESS sector with twelve eclipses. Three independent facts tie the stars to the short period. The first is the eclipse signal itself. The second is a spot wave of about 2.9 d, from near-synchronous rotation. The third is a spectroscopic line broadening of 13.7 km/s, which equals the rotation speed of a 0.8 $R_\odot$ star at 3.02 d; at 39 d it would be about 1 km/s. Gaia's double-lined solution at 39 d would make these K dwarfs 14 $M_\odot$ stars. At 3.02 d the same velocity amplitudes agree with expectation (about 13% high). The Gaia SB2 solution is therefore spurious: its period is misidentified, although its velocity scale is approximately correct. Gaia's large RUWE and acceleration solution point to a wide third star, though not at a 39-d period.

A55. BD+60 397 (TIC 11156440)



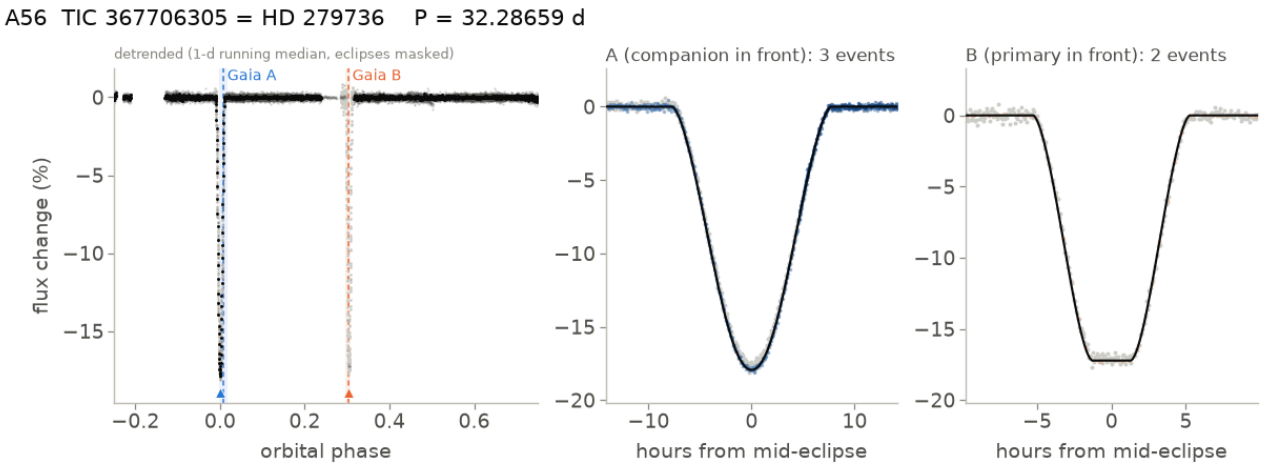
A55. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.25484 d, Gaia's predicted conjunctions dashed. Middle: the 19.5%, 11-h central transits at A. Right: the 18.1%, 29-h flat total occultations at B; the 2.7:1 duration ratio comes from the eccentric orbit.

Identifiers	TIC 11156440; Gaia DR3 508026396981317504; BD+60 397
Position (J2000)	01:58:12.9 +60:43:12.8
Brightness	*G* 10.00; Tmag 9.53
Stars	A-type subgiant, $M_1 \approx 2.6\ M_\odot$, R_1 = 4.1–4.5 $R_\odot$ (4.32 from the split light at 545 pc) + early-F/A companion, M_2 = 1.5–1.8 $M_\odot$, R_2 = 1.9–2.1 $R_\odot$, at nearly the same temperature (L_2/L_1 = 0.221); mass function 0.256 $M_\odot$; SIMBAD A7; Gaia SB1, RUWE 2.93
Period	21.25484 ± 0.00001 d (Gaia 21.257446 ± 0.002133 d; –1.2σ)
Epoch of primary	A: BTJD 1801.3737 ± 0.0005 = BJD_{TDB} 2458801.3737

Primary eclipse (A)	19.23–19.60% deep, T_{14} 10.78–10.84 h, T_{23} 3.95 h, rounded bottom: central transit of the companion (k = 0.466); four events (s18, s58 ×2; s85 flagged, timing only)
Secondary eclipse (B)	18.03–18.13% deep, T_{14} 29.37–29.42 h, T_{23} 12.73–12.75 h, flat: central total occultation of the companion (its light share); four events (s18 start, new; 1813.60; 2897.59; 3620.26). O−C ≤ 2.9 min
Orbit	Duration ratio 2.7 = (1+e sin ω)/(1−e sin ω) with e sin ω = 0.458, against Gaia +0.457 ± 0.039 (0.0σ). B−A phase 0.57510 against 0.527 ± 0.017: e cos ω = +0.105 against Gaia +0.037 ± 0.023 (**+2.9σ**); ω 77° against Gaia 85.3 ± 2.9°. i ≈ 90°, both impact parameters about 0
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s58 A ×2, s58 B, s85 B; t_0 within 0.5 min)
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target's Gaia position; no neighbour within 60″ brighter than G 14.8 (would need more than 100%). Common mode clean
Catalogue status	Not listed in: VSX, ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS, KELT (target list only), WISE variables, Justesen & Albrecht 2021, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, CCDM, Hipparcos, ExoFOP, general web search. The 25″ G 16.0 neighbour's Kostov table 2 flag is almost certainly due to contamination of its aperture by the target's eclipses
Open issues	e cos ω is 2.9σ from Gaia; a bias in the Gaia SB1 solution is plausible because the companion's lines, at nearly the same temperature, blend. A few spectra for K_2 and ω
Next eclipses	A: BTJD 4330.701 (2026-10-17 04:43 UTC), 4351.956; B: BTJD 4342.925 (2026-10-29 10:05 UTC), 4364.179 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+60 397 is an A-type subgiant of about 4.3 $R_\odot$ with an early-F or A-type companion of about 2 $R_\odot$ at nearly the same temperature, seen edge-on in an eccentric 21.25-d orbit. Both eclipses are central. At A the companion crosses the primary's disc (19.5%). At B it is hidden behind the primary for 12.7 h of a 29-h eclipse (18.1%, its share of the light). The occultation lasts 2.7 times as long as the transit because the stars are 0.53a apart at A and 1.44a at B, and this duration ratio reproduces Gaia's e sin ω. The only significant disagreement with Gaia is in e cos ω, and it does not call the eclipses themselves into question.

A56. HD 279736 (TIC 367706305)



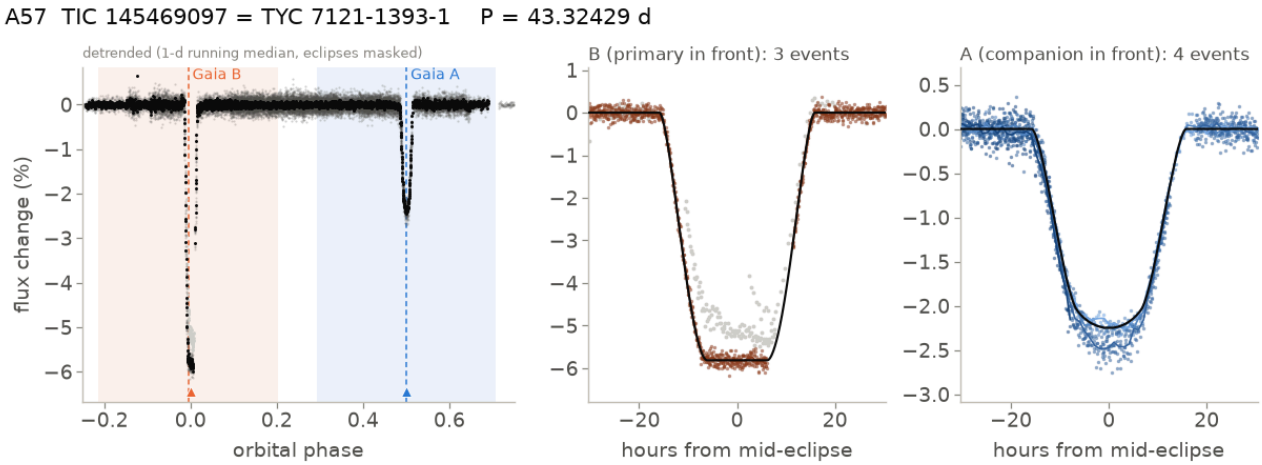
A56. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 32.28659 d, Gaia's predicted conjunctions dashed. Middle: the 17.5%, 14-h partial transits at A. Right: the 17.2%, 9.8-h total occultations at B.

Identifiers	TIC 367706305; Gaia DR3 1774751310738398 72; HD 279736
Position (J2000)	04:21:26.9 +36:36:25.7
Brightness	G 10.86; Tmag 10.35
Stars	

	G0 star (SIMBAD; APOGEE T_{eff} 6322 K, log *g* 4.40, [Fe/H] −0.59), M_1 = 1.0–1.3 $M_\odot$, R_1 = 2.05–2.20 $R_\odot$ + K companion, M_2 = 0.8–0.9 $M_\odot$, R_2 ≈ 1.0 $R_\odot$, L_2/L_1 = 0.21; mass function 0.160 $M_\odot$; Gaia SB1
Period	32.28659 ± 0.00001 d (Gaia 32.290060 ± 0.013184 d; −0.3σ)
Epoch of primary	A: BTJD 2914.8973 ± 0.0005 = BJD_{TDB} 2459914.8973
Primary eclipse (A)	17.53% deep, T_{14} 14.17 h, T_{23} 2.85 h, V-like: partial transit (b/R_1 = 0.61 > 1−*k*); three events (s19 flagged, s59, s86 flagged)
Secondary eclipse (B)	17.24% deep, T_{14} 9.82 h, T_{23} 3.26 h, flat: total occultation (b/R_1 = 0.40); two events (s19; s59 flagged). O−C ≤ 0.5 min
Orbit	B−A phase 0.30391 (Gaia 0.2954 ± 0.0073): $e \cos \omega$ = −0.306 against Gaia −0.320 ± 0.012 (+1.2σ); $e \sin \omega$ = −0.217 (fit) / −0.183 (durations) against −0.206 ± 0.012; Gaia *e* 0.3811. *i* = 88.74°, *k* = 0.474
Pipelines agreeing	QLP SAP; TESS-SPOC (s19 A/B, s59 A); TARS (s59 A/B)
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target; no neighbour within 60″ brighter than *G* 14.8. Common mode clean
Catalogue status	Not listed in: VSX (the ROT entry at 40″ is the *G* 14.8 neighbour Gaia DR3 177475062354364544), ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS (a row at 28″ is a different star), KELT, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, Gaia DR3 variability tables, SB9, WDS, Hipparcos, ExoFOP, general web search
Open issues	APOGEE's single visit (RV 77.42 km/s) is about 10 km/s short of the predicted ~88 km/s; this is not decisive given the uncertain γ and line blending
Next eclipses	A: BTJD 4335.507 (2026-10-22 00:02 UTC), 4367.794; B: BTJD 4345.319 (2026-10-31 19:31 UTC), 4377.606 (±1–2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 279736 is a metal-poor G0 star of about 2.1 $R_\odot$ with a companion of about 0.85 $M_\odot$ and 1 $R_\odot$ on an eccentric 32.3-d orbit. The two eclipses are nearly equal in depth but different in kind: at A the companion crosses only part of the primary's disc (a partial, V-like transit), and at B the primary hides the companion completely (a flat total occultation). The eclipses give *e*, ω and *P* all within 1.2σ of Gaia's independent orbit.

A57. TYC 7121-1393-1 (TIC 145469097)



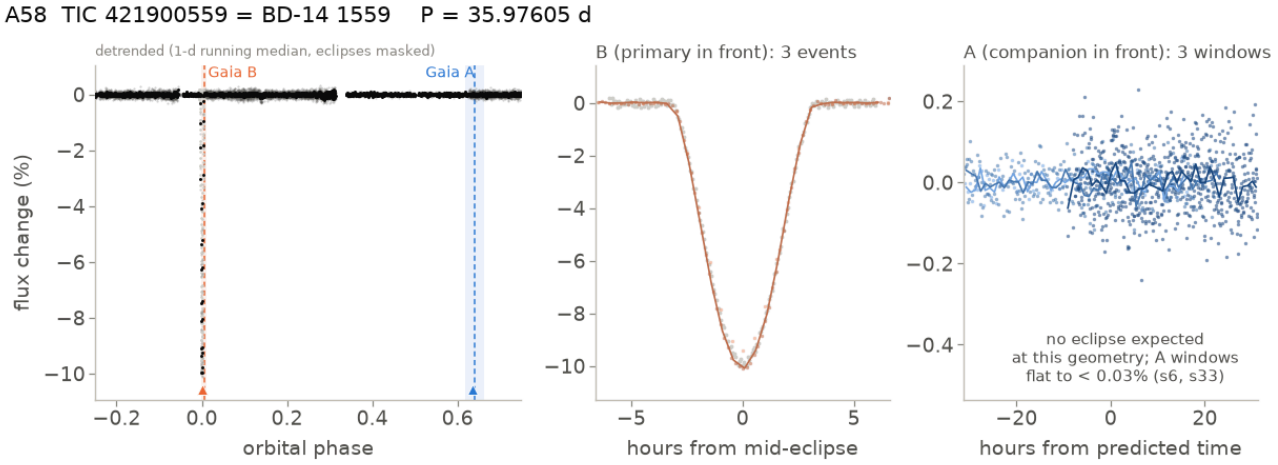
A57. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 43.32429 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 5.8%, 30-h flat total occultation at B in sector 88, with flagged partials from sectors 8 and 35. Right: the 2.2%, 30-h annular transits of the F dwarf across the giant at A.

Identifiers	TIC 145469097; Gaia DR3 5548179074417798400; TYC 7121-1393-1
Position (J2000)	08:11:16.4 −31:37:04.1
Brightness	*G* 10.62; Tmag 10.02
Stars	

	K giant, M_1 = 1.4–2.6 $M_\odot$, R_1 = 7.7–9.1 $R_\odot$ (TIC 8.0) + hotter F dwarf (about 6800 K), M_2 = 1.0–1.5 $M_\odot$, R_2 = 1.2–1.4 $R_\odot$, L_2/L_1 = 0.062; mass function 0.186 $M_\odot$; Gaia SB1
Period	43.32429 ± 0.00007 d (Gaia 43.336415 ± 0.007917 d; −1.5σ)
Epoch of eclipse	B: BTJD 3696.526 ± 0.001 = BJD_{TDB} 2460696.526; A: BTJD 1508.648 ± 0.002
Deeper eclipse (B)	5.78% deep, T_{14} 30.4 h, T_{23} 15.4 h, flat: total occultation of the F dwarf; one full event (s88, egress across a data gap) plus partials in flagged cadences (s8 egress, about 4.4%; s35 ingress, about 4.7%), all at phase 0.5000
Shallower eclipse (A)	2.15–2.40% deep (s61's 2.40% includes a known common-mode systematic), T_{14} 30.1–30.8 h: annular transit across the giant (b/R_1 = 0.76 < 1–k); three events (s7, s34, s61) plus an ingress at the end of s88, predicted before inspection of the data (A = 3718.187) and matched to 0.07% in every bin. O–C ≤ 1.3 min
Orbit	Circular: B–A phase 0.50004 ± 0.00009 (Gaia 0.4966 ± 0.0039), e cos ω = 0.000 (Gaia −0.006 ± 0.006), equal 30-h durations (e sin ω +0.011/0.000 against +0.002 ± 0.007). i = 85.2°, k = 0.155
Pipelines agreeing	QLP SAP; eleanor-lite, T16, TGLC aperture (s7 A); TESS-SPOC (s61 A). No SPOC-family light curve for s88, so the full B has no independent check
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target; no neighbour within 60″ brighter than G 14.8. Same-camera stars flat during the s88 ingress (median −0.04%)
Catalogue status	Not listed in: VSX, ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS, KELT, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, Hipparcos, ExoFOP, general web search
Open issues	The one full B has no second-pipeline check; ground photometry of the next B would cover the egress, which fell in the sector-88 data gap
Next eclipses	B (deeper): BTJD 4346.391 (2026-11-01 21:22 UTC), 4389.715; A: BTJD 4368.051, 4411.375 (±7–9 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 7121-1393-1 is a K giant of about 8 $R_\odot$ with a hotter F-dwarf companion of about 1.3 $R_\odot$ on a circular 43.3-d orbit. Both eclipses last 30 h. At A the dwarf crosses the giant's face entirely, an annular transit 2.2% deep. At B the giant hides the dwarf completely. This occultation is deeper than the transit, at 5.8%, because the dwarf is about 2.5 times brighter per unit area in the TESS band. The ingress of an A eclipse at the end of sector 88 was predicted before those data were examined, and it is present. Two further partial B eclipses lie in flagged cadences at phase 0.5000.

A58. BD-14 1559 (TIC 421900559)

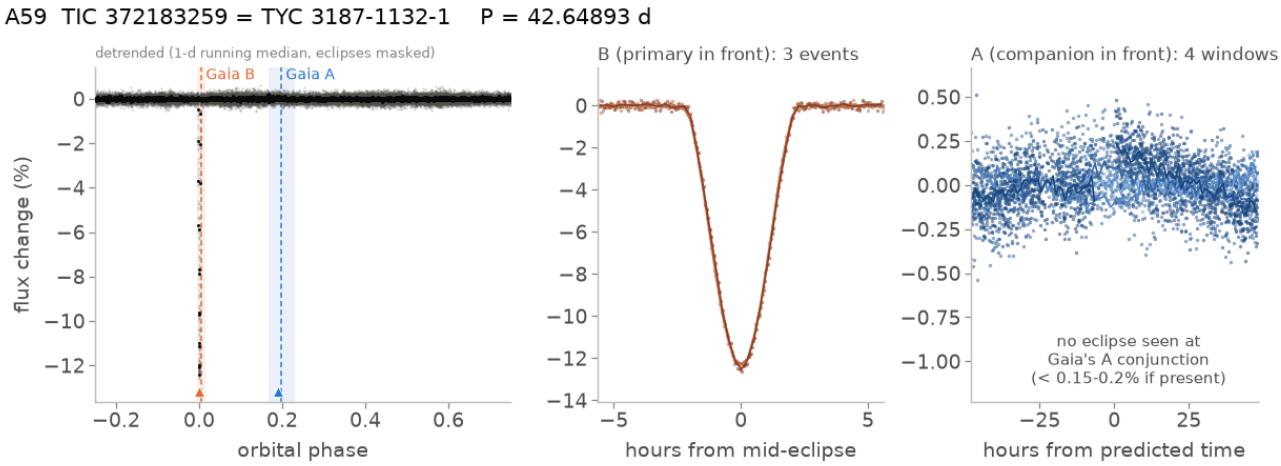


A58. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 35.97605 d, Gaia's predicted conjunctions dashed. Middle: the three 9.8%, 6.0-h partial occultations at Gaia's B conjunction (sector 87 in QLP-flagged cadences). Right: the covered A windows (three with usable data of the four the entry lists), flat as the geometry predicts.

Identifiers	TIC 421900559; Gaia DR3 2947549198126297984; BD-14 1559
Position (J2000)	06:43:55.3 –15:07:36.5
Brightness	*G* 10.03; Tmag 9.56
Stars	G star (GSP-Phot 5830 K; Gaia FLAME R 2.13 $R_\odot$, M 1.25 $M_\odot$); the eclipse fit gives R_1 = 1.56 $R_\odot$, R_2 = 0.83 $R_\odot$, companion 12.1% of the light; companion $m_2(i = 90°) \approx 0.76\ M_\odot$, a K dwarf; mass function 0.108 $M_\odot$; Gaia SB1, K_1 34.1 km/s, RUWE 1.21
Period	35.97605 ± 0.00002 d (Gaia 35.9893 ± 0.0083 d; –1.6σ)
Epoch of eclipse	T_0 = BTJD 2203.38951 = BJD_{TDB} 2459203.38951
Eclipse (B, near periastron)	9.2–10.1% deep, T_{14} 5.99–6.21 h, T_{12} 2.38–2.39 h: partial occultation of the companion, the primary in front (Gaia's B conjunction) (i = 88.1°); three events (s006, s033, s087 with all but 7 of 260 cadences flagged); O–C ≤ 0.5 min. Gaia's B is ν = 27°, about 1 d after periastron
Other eclipse (A)	None possible (sky separation 2.5–2.8 $R_\odot > R_1 + R_2$) and none seen: s006 ≤ 290 ppm, s033 ≤ 190 ppm, s087 about 800 ppm at a scattered-light edge, s108 flat
Orbit	Gaia e 0.438 ± 0.058, ω 243.1 ± 2.6°. With Gaia's period the eclipses sit at B at –1.7σ; carrying Gaia's periastron forward with the eclipse period puts them –0.18 d (–0.77σ) from B and 16σ from A. One eclipse kind only, so e cos ω and e sin ω cannot be derived from timing
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s006, s033); TARS (s006; s087 at 10.09%)
Pixel attribution	TESScut s87 difference image: centroid 0.3″ from the target, 3×3 aperture loses 9.5%; brightest neighbour within 60″ is *G* 17.15 at 34.5″ (would need about 70×)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia NSS reference only), general web search
Open issues	Radius conflict: the fitted R_1 runs 15–25% below Gaia's (about 1.8 $R_\odot$ after the companion's light); with R_1 fixed at 2.13, χ^2 rises from 774 to 1063. An SB2 spectrum would fix which star is in front and R_1
Next eclipses	B: BTJD 4361.952, 4397.928 (±0.001); A (no eclipse expected) 4384.8 ± 0.9
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD-14 1559 is a G star with a K-dwarf companion of about 0.76 $M_\odot$ on an eccentric 36-d orbit. Three identically shaped 9.8%, 6-h eclipses over seven years fall at the conjunction just after periastron. At the far conjunction the stars cannot overlap, and the four covered windows there are flat. The apparent –1.8σ offset from Gaia's prediction is Gaia's period error accumulated over about 50 cycles. Carried forward with the eclipse period, Gaia's periastron puts the eclipses within 0.8σ of B. The sector-87 eclipse sits almost entirely in QLP-flagged cadences and is recovered in TARS and in the pixels. A model with the companion in front would need it to carry 30% of the light, which is implausible.

A59. TYC 3187-1132-1 (TIC 372183259)



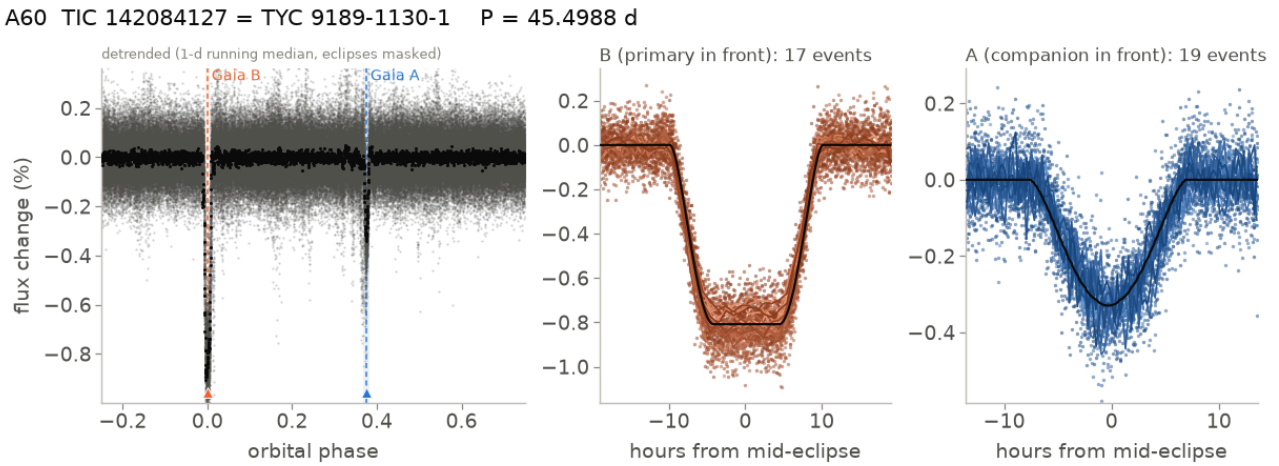
A59. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 42.64893 d, Gaia's predicted conjunctions dashed. Middle: the three 12.3%, 4.05-h partial occultations at Gaia's B conjunction near periastron (sector 16 in QLP-flagged cadences). Right: the A windows, flat to about 0.2%.

Identifiers	TIC 372183259; Gaia DR3 1966468304243902848; TYC 3187-1132-1
Position (J2000)	21:38:30.7 +40:58:26.7
Brightness	G 10.78; Tmag 10.39
Stars	Primary of TIC 6213 K, R 1.65 $R_\odot$, M 1.20 $M_\odot$ at 307 pc (Gaia about 1.64 $R_\odot$) + companion $m_2(i = 90°) \approx 0.69\ M_\odot$; the eclipse fit gives $R_1 = 1.14\ R_\odot$, $R_2 = 0.73\ R_\odot$, companion 12.8% of the light; mass function 0.0914 $M_\odot$; Gaia SB1, K_1 37.1 km/s
Period	42.64893 ± 0.00006 d (Gaia 42.5594 ± 0.0391 d; +2.3σ)
Epoch of eclipse	T_0 = BTJD 3537.57169 = BJD_{TDB} 2460537.57169
Eclipse (B, near periastron)	11.9–12.7% deep, T_{14} 4.03–4.09 h, T_{12} 1.64–1.69 h (b ≈ 0.7): partial occultation of the companion, the primary in front (Gaia's B conjunction) (i = 88.6°); three events (s016, new, every in-eclipse QLP cadence flagged; s082; s083); O−C ≤ 1.1 min. Gaia's B is ν = 319°, 0.75 d before periastron
Other eclipse (A)	Not seen: four A windows (s076, s016, s082, s015) limit any A eclipse to ≲ 0.15–0.2%, which rules out the smallest-R_1 solutions
Orbit	Gaia e 0.672 ± 0.033, ω 310.7 ± 5.4°. With Gaia's period the eclipse lands at A (−0.6σ); with the eclipse period it sits at **B**, −0.35 d (−0.83σ), and A is excluded at 6.4σ. The durations agree: at A (about 50 km/s) a 12% eclipse would last 14–18 h; at B (about 154 km/s) about 6 h central. One eclipse kind only
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP and TARS (s082, s083); TESS-SPOC SAP leaves the s016 eclipse unflagged
Pixel attribution	TESScut difference images put all three on the target (centroid offsets 1.3″, 3.1″, 1.0″; 3×3 losses 9.4–9.9%); brightest neighbour within 60″ is G 16.09 at 19.6″ (would need about 16×)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia NSS reference only), general web search
Open issues	Unresolved radius conflict: the eclipse shape and duration require $R_1 + R_2 \approx 1.9\ R_\odot$ against about 2.2–2.3 expected (fitted R_1 1.1–1.3 against 1.5–1.65; fixing R_1 at 1.53–1.65 raises χ^2 by 460–1080). Possible causes: Gaia's e or ω off, third light, a non-main-sequence companion, the simple disc model. Spectroscopy and an independent RV orbit would resolve it
Next eclipses	B: BTJD 4347.901, 4390.550 (±0.001); A (≲ 0.2% if present) 4356.0 ± 1.3
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3187-1132-1 is a star of about 1.6 $R_\odot$ with a companion of about 0.7 $M_\odot$ on a highly eccentric ($e \approx 0.67$) 42.6-d orbit. Of its three eclipses, the one in sector 16 lies entirely in QLP-flagged cadences. The period is 42.64893 d (a single-cycle interval gives 42.647). Gaia's period is 2.3σ short. Over 74

cycles that moves the predicted conjunction by about 6 d and would put the eclipses at A. With the eclipse period they fall at B, just before periastron, where the stars move fast enough to make a 4-h eclipse. The eclipses are not in doubt, but the stellar radii they imply are smaller than expected, and this discrepancy remains unresolved.

A60. TYC 9189-1130-1 (TIC 142084127)



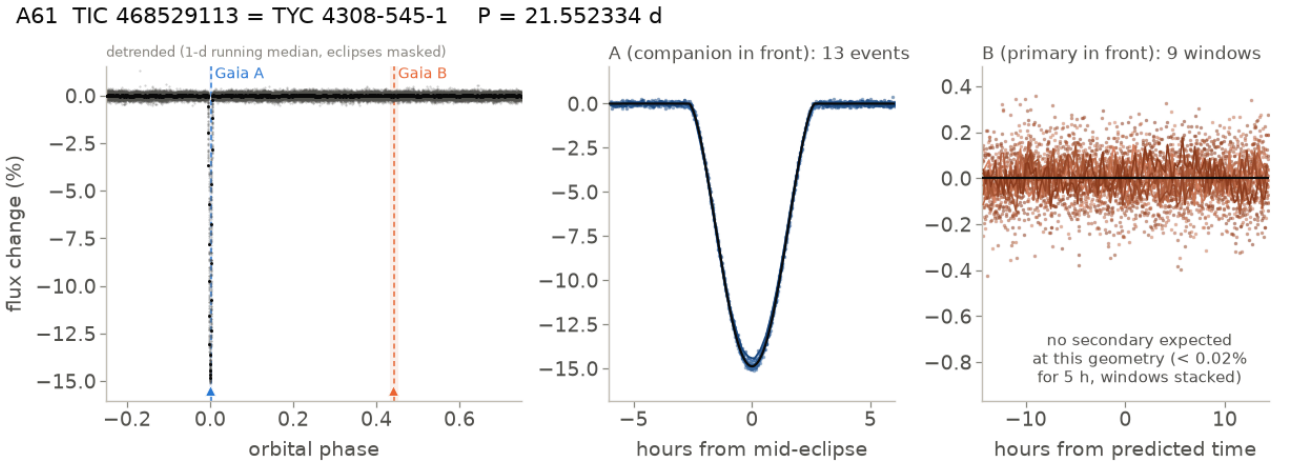
A60. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 45.4988 d, Gaia's predicted conjunctions dashed. Middle: the seventeen 0.8%, 19-h flat total occultations at B. Right: the nineteen 0.3%, 13.6-h V-shaped grazing transits of the K dwarf across the giant at A.

Identifiers	TIC 142084127; Gaia DR3 5261799966596528640; TYC 9189-1130-1
Position (J2000)	06:32:46.2 −74:36:28.9 (southern TESS continuous viewing zone)
Brightness	*G* 10.67; Tmag 10.02
Stars	Oscillating red giant, about 4800 K, R_1 = 5.7–6.3 $R_\odot$ (Gaia L 15.2 $L_\odot$ implies about 5.7) + K dwarf, m_2 = 0.7–0.9 $M_\odot$, R_2 = 0.67–0.78 $R_\odot$, T_2 ≈ 4200 K, 0.81% of the light; mass function 0.0938 $M_\odot$; Gaia SB1. Seismology: Beck et al. 2024 ν_{max} 67, Δν 4.7; Mackereth et al. 2021 pipelines disagree (BHM ν_{max} 125.8, R 7.5; A2Z ν_{max} 7.8)
Period	B: 45.498805 ± 0.000043 d; A: 45.498766 ± 0.000086 d (they agree; no apsidal motion detected). Gaia 45.50477 ± 0.0073 d (−0.8σ)
Epoch of eclipse	B: BTJD 2553.26450 ± 0.00083 = BJD_{TDB} 2459553.26450; A: BTJD 2615.83808 ± 0.00145
Deeper eclipse (B)	0.72–0.86% deep, T_{14} 19.1 h, T_{23} 10.7 h, flat: total occultation of the K dwarf; seventeen events; O−C rms about 5 min
Shallower eclipse (A)	0.28–0.35% deep, T_{14} 13.6 h, T_{23} 3.5 h, V-shaped: partial (grazing) transit of the K dwarf across the giant; nineteen events
Orbit	A−B separation 17.0748 d against Gaia 17.0746 ± 0.22 (0.00σ); e cos ω +0.1957 ± 0.0001 against Gaia +0.1958 ± 0.008. Physical fit to all 36 events: e = 0.2261, ω = 329.7° (Gaia 0.2308 ± 0.008, 328.07 ± 1.74°), e sin ω = −0.114 (Gaia −0.122 ± 0.007), i = 85.30°. The duration ratio cannot give e sin ω here because A is grazing and B total
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s066, both kinds); TGLC PSF flux (s004, same times, diluted)
Pixel attribution	TESS-SPOC s066 TPF difference images: B centroid 1.8″ from the target (S/N 60), A 7.3″ (0.35 px, S/N 8); nearest star brighter than *G* 15 is 117″ away; brightest neighbour within 60″ (*G* 16.75 at 57″) would need 215%
Catalogue status	No eclipse catalogued: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, general web search. Beck et al. 2024 lists the SB1 and its seismology with the inclination blank and no eclipse
Open issues	An RV of the K dwarf, or a seismic mass, would give the red giant a well-measured mass
Next eclipses	B: BTJD 4327.718 (2026-10-14 05:13 UTC), 4373.217 (±3 min); A: BTJD 4344.791, 4390.290 (±5 min)
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 9189-1130-1 is an oscillating red giant of about 6 $R_\odot$ with a K-dwarf companion on a 45.5-d orbit. It lies in the southern continuous viewing zone, so 36 eclipses across 2018–2025 could be measured. The giant hides the dwarf completely at B (0.8%, its share of the light) and the dwarf grazes the giant's limb at A (0.3%). A physical model fitted to all 36 events reproduces Gaia's eccentricity and periastron angle. The star is already in the seismology literature as a binary giant, where pipelines disagree on its oscillation frequencies and so on its radius. For an assumed mass of 1.2–2.0 $M_\odot$, the eclipses fix R_1 at 5.7–6.3 $R_\odot$. That radius matches the Gaia luminosity and argues against the 7.5 $R_\odot$ solution.

A61. TYC 4308-545-1 (TIC 468529113)



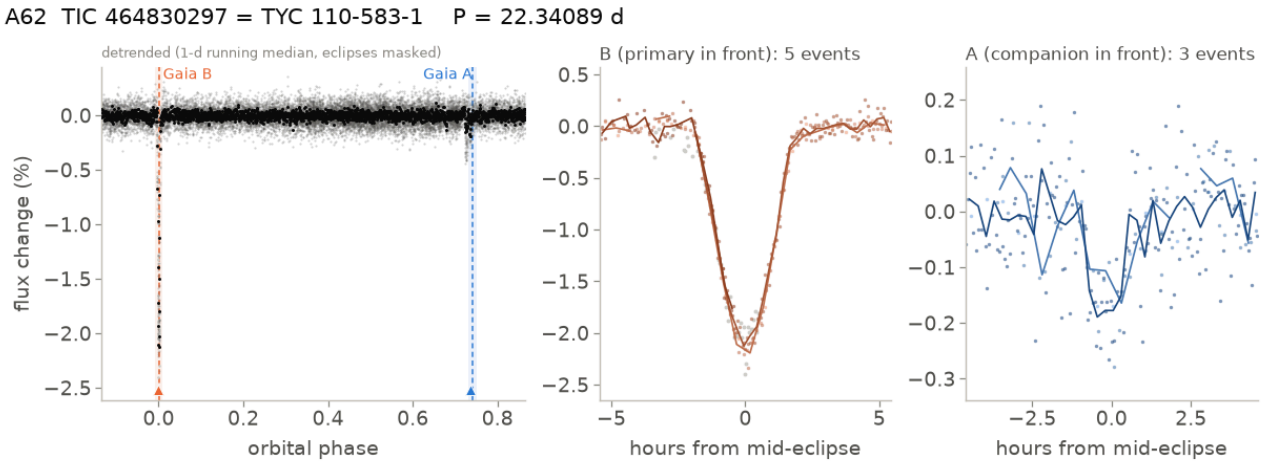
A61. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.552334 d, Gaia's predicted conjunctions dashed. Middle: the thirteen 14.6%, 4.9-h transits at A. Right: the covered B windows (nine drawn; the entry's limits use ten), flat as the geometry predicts.

Identifiers	TIC 468529113; Gaia DR3 538862715972209920; TYC 4308-545-1
Position (J2000)	00:58:32.4 +74:58:03.2
Brightness	G 10.40; Tmag 9.98
Stars	F star, $R_1 \approx 1.9\ R_\odot$ ($M_1 = 1.4\ M_\odot$ assumed) + K dwarf, m_2 0.84 $M_\odot$, $R_2 \approx 0.79\ R_\odot$ (k 0.41); a 42.6 $R_\odot$; mass function 0.1177 $M_\odot$; Gaia SB1
Period	21.552334 ± 0.000007 d (Gaia 21.55579 ± 0.0036 d; −0.95σ)
Epoch of primary	T_0 = BTJD 2481.89522 ± 0.00022 = BJD_{TDB} 2459481.89522
Eclipse (A)	14.2–15.0% deep, T_{14} 4.9 h, T_{23} 1.0 h: transit of the K dwarf; thirteen events s018–s086 (s079 partial); χ^2/dof 1.1/11, all O−C under 1 min
Other eclipse (B)	None possible: sky separation at B is 1.43 (R_1+R_2) (1.37–1.47 in every refit); a B eclipse would need i > 87.5°. Ten covered windows give stacked 3σ limits of 0.03% (3 h), 0.02% (5 h), 0.013% (8 h); a total occultation, had it happened, would be about 7%
Orbit	Fit with Gaia's e = 0.497, ω = 99.4° fixed: i = 86.5°; r_A = 0.506a, r_B = 1.477a. Observed minus Gaia −0.25 to −0.55 d, a constant −1.0σ, all from the period
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (s059, depth 14.4% after crowding correction); TARS (same times)
Pixel attribution	Difference-image centroid 2.8–3.5″ from the target and 49″ from the only neighbour bright enough to produce the dip (G 11.59 at 45.3″, which would need a 45% dip), at S/N 200–300
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, VizieR all-catalogue cone, SIMBAD ([SB*] only), general web search
Open issues	No photometric question remains open; spectroscopic detection of the companion
Next eclipses	A: BTJD 4335.396 (2026-10-21 21:25 UTC), 4356.948 (2026-11-12 10:44 UTC), 4378.501 (2026-12-04 00:00 UTC) (±1 min). No B expected
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4308-545-1 is an F star of about 1.9 $R_\odot$ with a K-dwarf companion of about 0.84 $M_\odot$ on an eccentric ($e \approx 0.50$) 21.55-d orbit. All thirteen transits from 2019 to 2025 are measured, with every O–C under a minute. The other conjunction falls near apastron, where the stars pass 1.43 times their summed radii apart on the sky, so no occultation is expected. Ten covered windows show none, to a limit of 0.02%; a total occultation of a K dwarf would have been about 7% deep.

A62. TYC 110-583-1 (TIC 464830297)



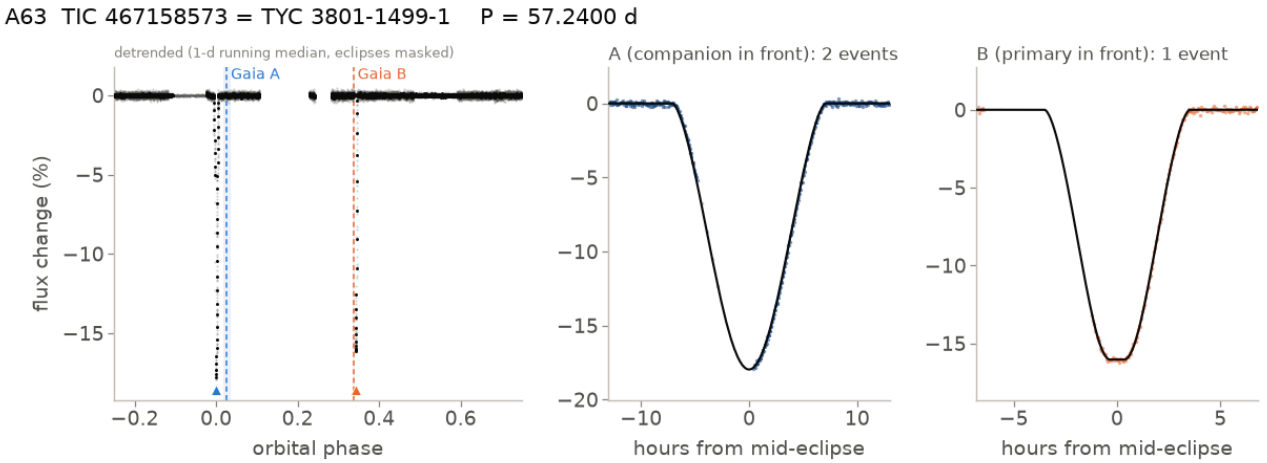
A62. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.34089 d, Gaia's predicted conjunctions dashed. Middle: the five 2.1%, 3.7-h grazing occultations at B. Right: the 0.15%, 2-h graze at A in the three covered windows.

Identifiers	TIC 464830297; Gaia DR3 3240092992751580416; TYC 110-583-1
Position (J2000)	05:00:47.7 +05:55:39.4
Brightness	G 10.76; Tmag 10.35
Stars	Primary (TIC radius) + companion of minimum mass 0.78 $M_\odot$; the depths need $k \approx 0.45$, $R_2 \approx 0.61\ R_\odot$, about 4650 K; mass function 0.1255 $M_\odot$; Gaia SB1
Period	22.34089 ± 0.00001 d (Gaia 22.329766 ± 0.005097 d; +2.2σ)
Epoch of eclipse	B: T_0 = BTJD 1440.0266 ± 0.0004 = BJD_{TDB} 2458440.0266
Deeper eclipse (B)	1.99–2.19% deep, near-V, 3.7 h: grazing occultation of the companion (primary in front at B); four full events (1440.03; 1462.37; 2177.28; 4009.23) and a partial near 4031.57 in flagged data, on the ephemeris; O–C under 1 min
Shallower eclipse (A)	About 0.15% deep, about 2 h: grazing transit ($b_A = 1.27\ b_B$); detected in all three covered A windows at the same offset: 1456.452 (1450 ± 216 ppm), 2193.701 (1606 ± 204), 4025.654 (1582 ± 184), where the largest dip elsewhere in those sectors is 839–1123 ppm
Orbit	A–B = 16.425 d gives e cos ω = −0.376 against Gaia −0.386 ± 0.015 (+0.7σ); Gaia e 0.4042, ω 197.15°. Propagated back 47 cycles, the B ephemeris lands −0.07σ from Gaia's B conjunction at Gaia's epoch
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS; eleanor-lite; TGLC aperture and PSF (s5 eclipses at 1.9–2.35%)
Pixel attribution	Difference-image centroids within 0.03–0.05 px of the out-of-eclipse centroid; the G 16.80 star at 10.5″ would need 434%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	A mild tension: a main-sequence 0.75 $R_\odot$ companion would eliminate the A eclipse, so the companion is slightly small or the primary radius is in error (FFI smearing not modelled)
Next eclipses	B: BTJD 4344.342, 4366.683 (±0.001); A: BTJD 4338.426, 4360.767 (±0.005)
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 110-583-1 has a companion of at least 0.78 $M_\odot$ on an eccentric 22.34-d orbit, and the two stars only graze at both conjunctions. The 2% eclipses fall where Gaia's primary is in front (occultations of the companion). A 0.15% graze appears at the other conjunction in all three covered windows. Each is at the same offset, and each is larger than any other dip in its sector. Together with the main eclipses, the graze gives e cos ω within 0.7σ of Gaia. A further full occultation is present in 2018.

A63. TYC 3801-1499-1 (TIC 467158573)



A63. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 57.2400 d, Gaia's predicted conjunctions dashed. Middle: the 18%, 13-h V-shaped transit at A in sector 60, with the ingress at the end of sector 20. Right: the 16%, 6.4-h flat occultation at B in sector 74.

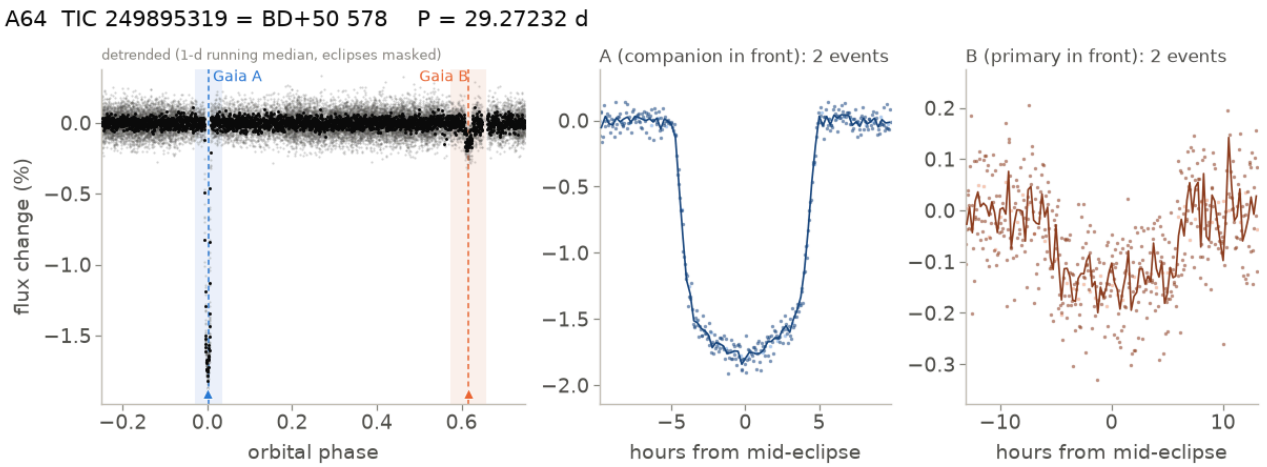
Identifiers	TIC 467158573; Gaia DR3 1037239725881771008; TYC 3801-1499-1
Position (J2000)	08:47:33.6 +56:45:12.2
Brightness	G 10.95; Tmag 10.50
Stars	Primary $R_1 \approx 1.22\ R_\odot$ + K2–3 dwarf of about 0.75 $M_\odot$ (mass-function minimum 0.69), $R_2 \approx 0.73\ R_\odot$, about 4900 K, 19% of the light; Gaia SB1
Period	57.2400 ± 0.0003 d over 19 cycles (Gaia 57.249210 ± 0.016133 d; −0.57σ)
Epoch of primary	A ingress fitted with the s60 template: BTJD 1868.8865 ± 0.005 = BJD_{TDB} 2458868.8865; s60 A centre about 2956.45, the midpoint of ingress and egress (the minimum falls in a data gap)
Primary eclipse (A)	≥ 17.4% observed (18.0% modelled), V-shaped, $T_{14} \approx$ 12.7–13 h: transit; s60 (ingress 2956.19, egress ends 2956.72) and an ingress at the end of s20 (QUALITY 0, identical in TESS-SPOC SAP and PDCSAP)
Secondary eclipse (B)	16.0% deep, flat bottom, $T_{14} \approx$ 6.4 h assuming symmetry: occultation of the companion; one event (s74, BTJD 3319.528, ingress in a gap)
Orbit	One geometry fits both kinds: e 0.457, ω 240.0°, i = 89.40°, k = 0.60; Gaia e 0.497 ± 0.012 (−3.3σ), ω 236.6 ± 1.3° (+2.6σ); e cos ω −0.229 against −0.274 (about 3.8σ); e sin ω −0.396 against −0.415 (about 1.5σ). Fixing e, ω at Gaia's values puts B 1.8 d early. Gaia goodness-of-fit 2.84
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (16.9–18.4% in s60; 15.3–16.0% in s74)
Pixel attribution	Difference-image centroid offsets 0.04 px (s60), 0.05 px (s20), 0.17 px (s74, post-gap settling); no neighbour could produce 16%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	

	Gaia orbit tension: e cos ω about 3.8σ from Gaia's, whose errors may be underestimated. Three eclipses in all; no other conjunction in covered data (s20, s47, s60, s74). **Not seen on the Harvard plates** (Section 5.6): the nine plates inside the A window, 1893–1989, give 0.066 ± 0.036 mag against the 0.21 mag the TESS depth predicts, 4σ short, with injected eclipses recovered 10 times in 10 and no signal within ±40 period errors; the eclipse was shallower or absent before TESS, or the period is off by more than 0.012 d
Next eclipses	A: BTJD 4330.207, 4387.447 (±0.008); B: BTJD 4349.848, 4407.088 (±0.007)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with caveats. No eclipse is completely observed (the s60 A minimum falls in a gap, s20 is an ingress only, the s74 B ingress is in a gap); one B event; the cycle count of 19 between the s20 ingress and the s60 transit is fixed by Gaia's period (18 or 20 cycles would give 60.42 or 54.38 d); three pipelines agree on what is seen; e cos ω is 3.8σ from Gaia's.

Note: confirmed with caveats, in tension with Gaia's orbit. One eclipse geometry fits both the transit and the occultation, with e = 0.457 against Gaia's 0.497 ± 0.012. Forcing Gaia's e and ω misplaces the occultation by 1.8 d. No eclipse is completely observed, and the cycle count between sectors 20 and 60 is fixed by Gaia's period.

TYC 3801-1499-1 is a star of about 1.2 $R_\odot$ with a K-dwarf companion of about 0.75 $M_\odot$ on an eccentric 57.24-d orbit, the longest period in this set. Three eclipses are seen: a deep V-shaped transit in sector 60, the ingress of an earlier one at the end of sector 20, and a 16% flat occultation in sector 74. The occultation is nearly as deep as the transit, so the companion carries about 19% of the light, as a K2–3 dwarf would beside this primary. The period rests on 19 cycles between the sector-20 ingress and the sector-60 transit.

A64. BD+50 578 (TIC 249895319)



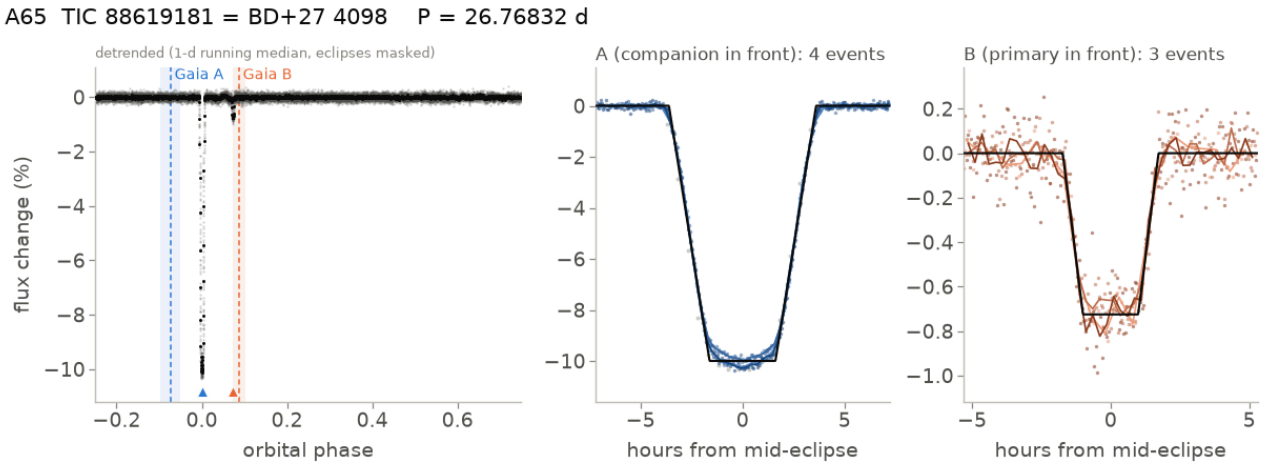
A64. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.27232 d, Gaia's predicted conjunctions dashed. Middle: the two 1.67%, 9.9-h flat-bottomed transits of the M dwarf at A. Right: the 0.15%, 13-h occultation of the M dwarf at B, stacked from both windows.

Identifiers	TIC 249895319; Gaia DR3 4513096358441890 56; BD+50 578
Position (J2000)	02:33:53.5 +51:36:39.1
Brightness	G 10.37; Tmag 9.77
Stars	Giant of 2.34 $R_\odot$ + M dwarf of about 0.26 $M_\odot$, R_2 = 0.29 $R_\odot$ (k = 0.123), about 3000 K from the occultation depth; mass function 0.0081 $M_\odot$; Gaia SB1
Period	29.27232 ± 0.00005 d over 38 cycles (the 37- and 39-cycle aliases, 30.06 and 28.52 d, excluded by Gaia); Gaia 29.285986 ± 0.019640 d (−0.70σ)
Epoch of primary	A: BTJD 2906.2943 ± 0.0005 = BJD_{TDB} 2459906.2943; also 1793.9463 ± 0.0007

Primary eclipse (A)	1.665–1.680% deep, T_{14} 9.89–9.90 h, flat 6.89–6.90 h: transit of the M dwarf (b ≈ 0.4); two events, identical in shape (s18, s58)
Secondary eclipse (B)	Stacked 1476 ± 70 ppm, T_{14} 13.1 ± 0.6 h (13.5 predicted), flat 9.8 h: occultation of the M dwarf; same offset in both B windows (1811.947: 1408 ± 75 ppm; 2895.063: 1429 ± 68, TESS-SPOC 1542 ± 65). Relative to the sector rms, each dip is about 3σ, together about 4σ
Orbit	B–A = 18.027 d gives e cos ω = +0.1805 (Gaia +0.177 ± 0.039); duration ratio 1.32 gives e sin ω ≈ +0.14 (Gaia +0.160). −0.11σ from Gaia's conjunction at Gaia's epoch
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (1.75–2.04%)
Pixel attribution	Centroids within 0.01 px (s58) and 0.13 px (s18); the G 15.03 star at 53″ would need 127%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	Two transits and two occultation windows; the s85 transit was not covered and its B fell in flagged scattered light
Next eclipses	A: BTJD 4340.638, 4369.910 (±0.005); B: BTJD 4329.393, 4358.665 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+50 578 is a giant of about 2.3 $R_\odot$ with an M-dwarf companion of about 0.26 $M_\odot$ on a 29.27-d orbit. Two transits of identical shape, four years apart, fix the period at 38 cycles. A 0.15% occultation of the M dwarf appears at the same phase in both windows that cover it, and also in TESS-SPOC. The occultation's timing and duration reproduce Gaia's e cos ω and e sin ω. The offset of about 1 d between the observed transits and the times predicted from Gaia's elements is the accumulated error of Gaia's period.

A65. BD+27 4098 (TIC 88619181)



A65. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 26.76832 d, Gaia's predicted conjunctions dashed (Gaia's e and ω put B 4.3 d after A; it falls 1.95 d after). Middle: the four 10%, 7.2-h flat transits of the M dwarf at A. Right: the three 0.72%, 3.4-h flat occultations at B.

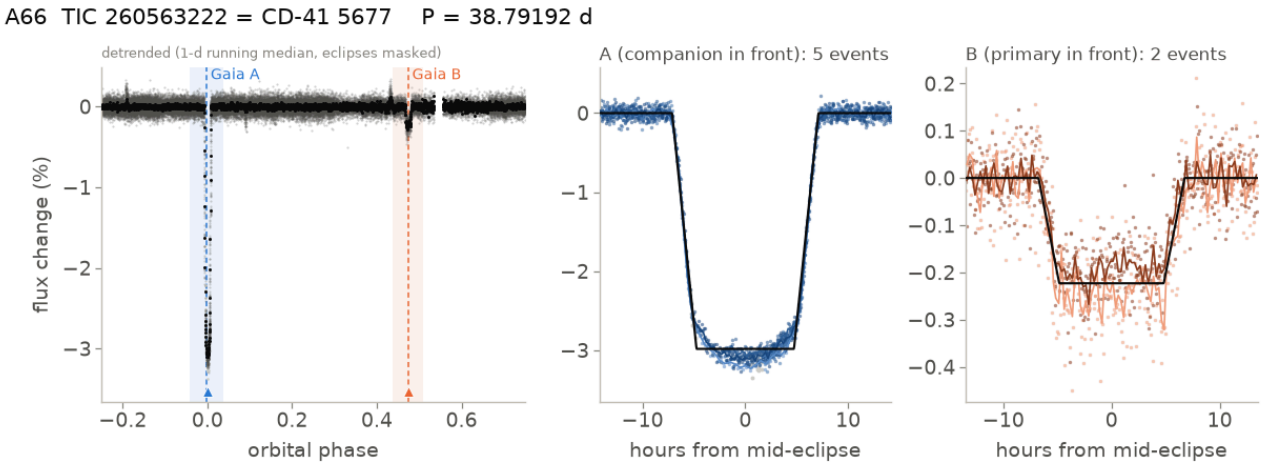
Identifiers	TIC 88619181; Gaia DR3 1848755826318803968; BD+27 4098
Position (J2000)	21:33:50.8 +27:43:45.3
Brightness	G 10.45; Tmag 10.02
Stars	F0 star (SIMBAD; Gaia GSP-Spec 6333 K, log g 3.79), R_1 ≈ 1.37 $R_\odot$ (TIC 1.54) + M dwarf, m_2 ≈ 0.29–0.33 $M_\odot$, R_2 ≈ 0.41 $R_\odot$ (k = 0.302), about 3300–3400 K, L_2/L_1 = 0.0070; mass function 0.0107 $M_\odot$; Gaia SB1, RUWE 7.8
Period	26.76832 ± 0.00001 d (s55–s82; $P_A = P_B$); with the s15 eclipse 26.76842 ± 0.000005 d. Gaia 26.76904 ± 0.023 d (−0.03σ)
Epoch of primary	A: BTJD 2819.5596 ± 0.0002 = BJD_{TDB} 2459819.5596
Primary eclipse (A)	9.71–10.18% deep, T_{14} 7.21–7.23 h, T_{23} 3.2 h, flat (b ≈ 0.4): transit of the M dwarf across the F star; four events (1722.0515, in s15 flagged cadences; 2819.56; 2846.33; 3542.30)

Secondary eclipse (B)	0.70–0.73% deep, T_{14} 3.41–3.46 h, T_{23} 2.0 h, flat: total occultation of the M dwarf; three events; B–A constant at 1.9527–1.9534 d (phase 0.07296)
Orbit	Joint fit: e cos ω = −0.691, e sin ω = −0.383, ***e* = 0.790, ω = 209°**, i = 88.7° (the joint fit was stable only for e sin ω between −0.40 and −0.35; no formal errors on e and ω); Gaia e = 0.688 ± 0.017 (+6σ), ω = 223.8 ± 4.3° (−3.4σ). Gaia's e, ω predict phase 0.161 ± 0.017 (4.30 d), −5.1σ from observed, and with them fixed the model cannot place B (χ^2 6948 against 2008). Periastron 2821.12 against Gaia 2821.44 ± 0.40 (−0.8σ): the disagreement is confined to e and ω
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s56, s82: 9.7–9.8%, 0.69–0.71%)
Pixel attribution	TESScut difference images on target for both kinds (A 1.7–3.2″ off at S/N 400–600, within the ~3″ floor; B 5.1″ at S/N 74, 20–60″ from every Gaia neighbour); the only feasible neighbour (G 12.33 at 58″, would need 57%) is 59″ from the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity and JSDC only), general web search
Open issues	**Gaia orbit shape corrected.** The four A times do not fit a line (χ^2 156 for 2 dof; O–C −2.4, +1.8, +1.6, −2.0 min): possible eclipse-timing variations, consistent with a third body (RUWE 7.8). R_2 is 10–15% above the main sequence for 0.3 $M_\odot$. RVs measuring e directly (predicted about 0.79); high-resolution imaging or Gaia DR4 astrometry for the third body
Next eclipses	A: BTJD 4345.354, 4372.122 (±0.001; ETVs of ±2–10 min possible); B: BTJD 4347.307, 4374.075
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: confirmed at Gaia's period, with Gaia's orbit shape contradicted. The 1.95-d spacing between transit and occultation, together with the two durations, needs e = 0.790 and ω = 209°, against Gaia's 0.688 ± 0.017 and 224 ± 4°. With Gaia's values the occultation cannot be placed at all. The transit times also deviate from a linear ephemeris by about ±2 min, a hint of a third body. The e cos ω disagreement comes from the eclipse timing and is secure; the 6σ figure for e rests on a fit without formal errors.

BD+27 4098 is an F0 star with an M dwarf of about 0.3 $M_\odot$ on a very eccentric 26.77-d orbit. The transit and the occultation are only 1.95 d apart because periastron falls between them, 1.6 d after the transit and 0.4 d before the occultation. The durations test the fitted geometry. With Gaia's e and ω a central transit would last 12–12.5 h. The observed flat transit lasts 7.2 h ($b \approx 0.4$), which matches the fitted orbit. The fitted orbit, however, predicts a 3.9–4.1-h occultation, against the 3.4–3.5 h observed. Both eclipses are flat-bottomed, and the occultation depth gives the M dwarf's share of the light directly; that share is in line with the main sequence. The fourth transit, in 2019, lies in QLP-flagged cadences.

A66. CD-41 5677 (TIC 260563222)

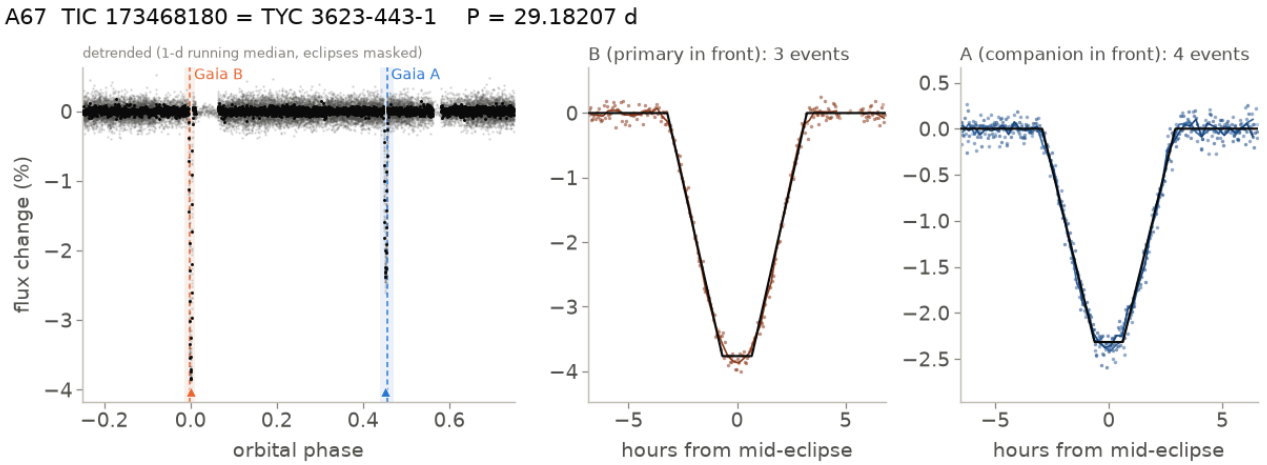


A66. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 38.79192 d, Gaia's predicted conjunctions dashed. Middle: the 3.0%, 14-h flat transits of the M dwarf at A. Right: the two 0.22%, 13.5-h flat occultations at B.

Identifiers	TIC 260563222; Gaia DR3 5416799976193456256; CD-41 5677
Position (J2000)	10:12:10.2 −42:30:11.4
Brightness	G 10.06; Tmag 9.65
Stars	Primary $R_1 \approx 2.3\ R_\odot$ (TIC 2.66) + M dwarf, $m_2 \approx 0.37$–$0.43\ M_\odot$, $R_2 \approx 0.39\ R_\odot$ (k = 0.165, main sequence), L_2/L_1 = 0.0022; mass function 0.0235 $M_\odot$; Gaia SB1; SIMBAD: GALAH DR2/DR3
Period	38.79192 ± 0.00002 d (Gaia 38.795818 ± 0.007172 d; −0.54σ)
Epoch of primary	A: BTJD 3027.5910 ± 0.0006 = BJD_{TDB} 2460027.5910
Primary eclipse (A)	2.95–3.08% deep, T_{14} 14.2 h, T_{23} 9.5 h: transit of the M dwarf; four full events (1553.50, 2290.55, 3027.59, 3764.64) and a partial (3725.874); s63 event 39 of 172 cadences flagged, TESS-SPOC time within 0.7 min; O−C within ±1 min
Secondary eclipse (B)	0.204–0.242% deep, T_{14} 13.4–13.6 h, T_{23} 9.5–9.8 h, flat: total occultation of the M dwarf; two events (3744.2316, 4054.5667)
Orbit	Phase separation 0.47395 (Gaia 0.4765 ± 0.0067, −0.4σ); $e \cos \omega = -0.0409$, $e \sin \omega = -0.044$ (±0.003 statistical, about ±0.01 with red noise): e = 0.060, ω = 227° against Gaia 0.061 ± 0.011, 232 ± 10°. i = 89.7°
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (B in s89: 0.32–0.36%, against 0.20–0.24% in QLP and TESS-SPOC, so TARS disagrees on the depth)
Pixel attribution	TESScut difference images on target for both kinds (B 2.6″ and 4.3″; neighbours beyond 52″); the only feasible neighbour (G 13.68 at 59″, would need 84%) is 57–62″ from the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	The occultation gives L_2/L_1 = 0.22% directly, implying $T_2 \approx 3300$ K against about 3800 K from the main-sequence relation: a mild tension (a factor of 2 in L_2). One RV epoch, or space-quality photometry of the next B; whether the T_2 tension holds depends on which B depth is right
Next eclipses	B: BTJD 4364.902 (2026-11-20 09:40 UTC), 4403.694; A: BTJD 4346.517, 4385.309 (±0.002)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-41 5677 is a star of about 2.3 $R_\odot$ with an M dwarf of about 0.4 $M_\odot$ on a nearly circular 38.8-d orbit. Both eclipses are flat-bottomed and long: a 3% transit of the M dwarf and a 0.22% total occultation, whose depth is the dwarf's share of the light. The eclipse timing and shapes give e = 0.060 and ω = 227°, matching Gaia's independent orbit within 1σ.

A67. TYC 3623-443-1 (TIC 173468180)

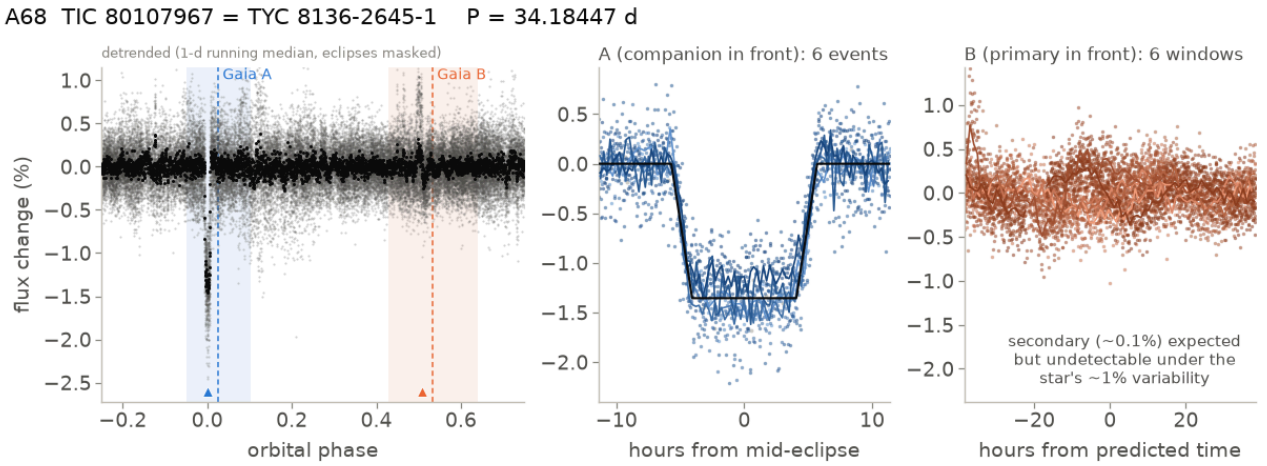


A67. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.18207 d, Gaia's predicted conjunctions dashed. Middle: the three 3.8%, 6.5-h near-total occultations at B, deeper than the transits. Right: the four 2.3%, 5.9-h grazing transits at A.

Identifiers	TIC 173468180; Gaia DR3 1941146757609930240; TYC 3623-443-1
Position (J2000)	23:12:27.6 +45:52:28.9
Brightness	*G* 10.82; Tmag 10.30
Stars	Primary $R_1 \approx 1.6\ R_\odot$ (FLAME 1.73) + K dwarf, $m_2 \approx 0.67$–$0.76\ M_\odot$, $R \approx 0.75\ R_\odot$, 4700–5000 K, $L_2/L_1 \approx 0.07$–0.10; mass function $0.115\ M_\odot$; Gaia SB1
Period	29.18207 ± 0.00001 d (Gaia 29.179837 ± 0.002830 d; +0.8σ)
Epoch of primary	A: BTJD 2878.8658 ± 0.0005 = BJD_{TDB} 2459878.8658
Shallower eclipse (A)	2.25–2.36% deep, T_{14} 5.8–5.9 h, T_{23} 1.2–1.4 h, nearly V-shaped: grazing transit of the primary's darkened limb ($b \approx 1.2\ R_1$); four events (1740.76, 1769.95, 2878.87, 3608.42); O−C ≤ 0.4 min
Deeper eclipse (B)	3.75–3.77% deep, T_{14} 6.4–6.5 h, T_{23} 1.3–1.4 h: near-total occultation of the companion ($b \approx 0.95\ R_1$), removing roughly half its light; three events (1756.76, 1785.94, 3595.23)
Orbit	Phase separation 0.54816 (Gaia 0.5433 ± 0.0049, +1.0σ); $e \cos \omega$ = 0.0745–0.0756 against Gaia 0.0676 ± 0.0077 (+1σ); $e \sin \omega$ poorly constrained (−0.20 to −0.05), Gaia's −0.119 in the middle. The duration-ratio estimate (+0.05) is wrong because the impact parameters differ. $i \approx 88°$; stars $1.11a$ apart at A, $0.88a$ at B
Pipelines agreeing	QLP SAP; TESS-SPOC (B in s16, s17, s84: 3.66–3.83%)
Pixel attribution	TESScut difference images: A and B 0.1–2.3″ from the target; every neighbour would need more than 180%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	A spectrum for the K-dwarf lines ($K_2 \approx 51$ km/s for $M_1 = 1.05$); the system should be detectable as SB2
Next eclipses	A: BTJD 4337.969, 4367.151 (±0.001); B: BTJD 4353.966, 4383.148
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3623-443-1 is a star of about 1.6 $R_\odot$ with a K dwarf of about 0.7 $M_\odot$ on a mildly eccentric 29.18-d orbit. The occultation is deeper than the transit. Gaia's orbit puts the stars farther apart at A ($1.11a$) than at B ($0.88a$), so at A the companion only clips the primary's limb-darkened edge, giving a 2.3% eclipse. At B the primary covers about half of the companion, which carries 7–9% of the light, giving a 3.8% eclipse. Both chords are near grazing, so both eclipses last about 6 h against 11–13 h for central ones.

A68. TYC 8136-2645-1 (TIC 80107967)



A68. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 34.18447 d, Gaia's predicted conjunctions dashed; the star varies at about 1%. Middle: the six 1.3%, 11.4-h flat transits at A. Right: the B windows, where the expected 0.1% occultation is below the star's variability.

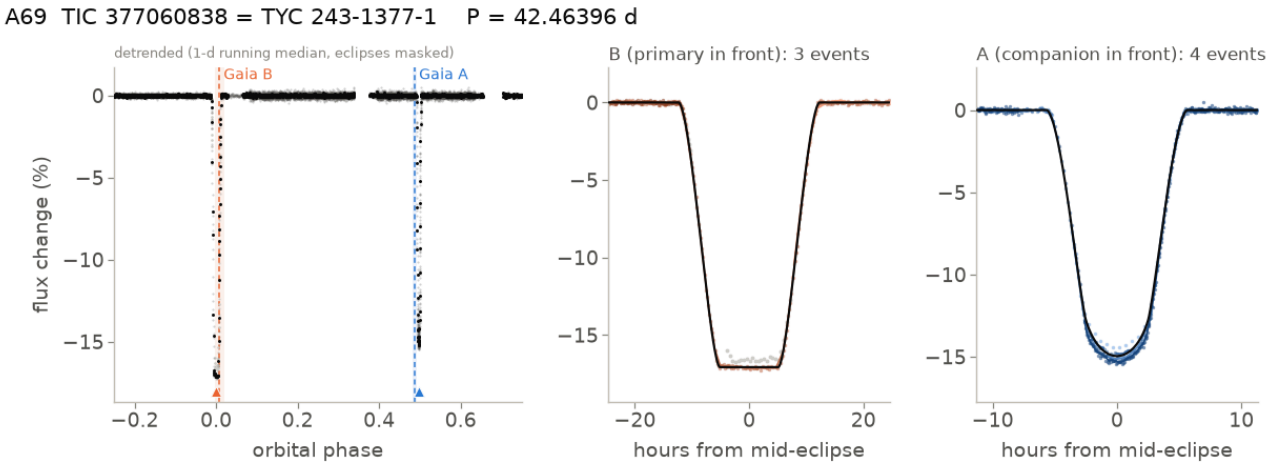
Identifiers	TIC 80107967; Gaia DR3 5520398813627036288; TYC 8136-2645-1
Position (J2000)	08:04:48.2 −45:32:43.4
Brightness	*G* 11.00; Tmag 10.58
Stars	Primary $R_1 \approx 2.6\ R_\odot$ (TIC 2.45) + M dwarf, $m_2 \approx 0.25$–$0.28\ M_\odot$, $R_2 \approx 0.27$–$0.29\ R_\odot$ (k = 0.111); mass function 0.0066 $M_\odot$; Gaia SB1. A *G* 14.46 star (TIC 818814154) lies 4.4″ away
Period	34.18447 ± 0.00008 d (Gaia 34.164065 ± 0.036796 d; +0.55σ)
Epoch of primary	T_0 = BTJD 1521.3074 = BJD_{TDB} 2458521.3074
Eclipse (A)	1.20–1.49% deep, T_{14} 11.1–11.7 h, T_{23} 8.0–8.5 h, flat: transit of the M dwarf near periastron (predicted central duration 11.2–11.7 h); six events 2018–2025; χ^2_{red} 3.2, O−C up to +8.5 min (s35, partly flagged; red noise from the star's variability); phases ≤ 0.27σ from Gaia's A
Other eclipse (B)	Not detected, and not expected to be: geometry allows a 15–17-h total occultation, but $L_2/L_1 \approx$ 0.10–0.15% sits under the star's red noise (spurious box-fit dips up to 0.25–0.67%)
Orbit	No B measured, so e cos ω is unmeasurable; a depth/duration fit with Gaia's e, ω fixed is as good as free (χ^2 2421.1 against 2421.0). $i \approx 89°$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s8, s34, s35: 1.20%, 1.39–1.43%); TARS (single-event times within ±5–8 min of QLP)
Pixel attribution	**Not decisive.** The 4.4″ neighbour would need a 33% eclipse, which is physically possible. TGLC's aperture light curve of the neighbour shows exactly the 32.8% contamination expected if the target dims 1.3%; TGLC PSF photometry splits the dip (target 0.9–1.3%, neighbour 13–18%); difference-image centroids favour the target (0.5″ against 4.6″ in s34; 1.5″ against 3.0″ in s62), within the ~3″ systematic floor. The *G* 14.78 star at 35″ is excluded by the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Host attribution rests on the period and phase match (a chance match of a background eclipsing binary in both period and phase, not quantified) and self-consistent physics, not pixels. AO or speckle imaging of the 4.4″ neighbour plus ground photometry with a ~1.5″ aperture during a transit would resolve the attribution: 1.3% on the target, about 33% on the neighbour
Next eclipses	A: BTJD 4358.619, 4392.803 (±0.007); B (about 0.1%, undetectable from TESS): 4341.79, 4375.98 (±1.46)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable (see below)
Status	

Probable. The host is not pixel-proven: the *G* 14.46 star 4.4″ away could host a 33% eclipse, TGLC PSF photometry splits the dip between the two stars, and the difference-image centroids favour the target only within the ~3″ systematic floor. The attribution rests on the match of period and phase to Gaia's orbit and on self-consistent physics; the chance of a background eclipsing binary matching both has not been quantified. Resolved imaging, or photometry through a ~1.5″ aperture during a transit, would resolve it.

Note: probable, not confirmed. The system eclipses at Gaia's period, but the host is not pixel-proven. A *G* 14.5 star 4.4″ from the target could host a 33% eclipse, and TESS pixels (21″ across) cannot separate the two. The target is adopted as the host on physical consistency and on the period and phase match, not on pixel data.

TYC 8136-2645-1 is a star of about 2.6 $R_\odot$ with an M dwarf of about 0.26 $M_\odot$ on a 34.18-d orbit. Six flat transits from 2018 to 2025 fall within 0.27σ of Gaia's predicted conjunction, with a period 0.55σ from Gaia's. A background eclipsing binary would have to match both the period and the conjunction phase by chance. That is unlikely, but the probability has not been quantified. The 11.4-h duration, the flat bottom, the 1.3% depth and the companion mass from the mass function all fit an M dwarf crossing this star near periastron. Its occultation, about 0.1%, is lost under the star's own 1% variability, as expected.

A69. TYC 243-1377-1 (TIC 377060838)



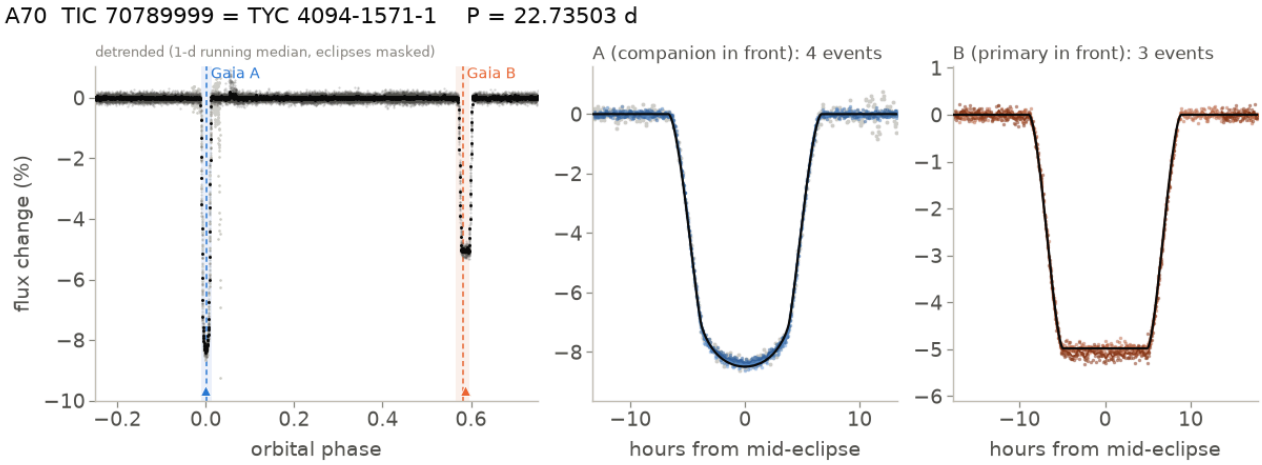
A69. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 42.46396 d, Gaia's predicted conjunctions dashed. Middle: the 17%, 24.6-h flat total occultations at B, deeper and longer than the transits. Right: the four 14.6%, 11.3-h full transits at A.

Identifiers	TIC 377060838; Gaia DR3 3853435449561968000; TYC 243-1377-1
Position (J2000)	09:51:56.8 +06:29:19.4
Brightness	*G* 10.63; Tmag 10.16
Stars	G subgiant, R_1 ≈ 2.6–2.7 $R_\odot$, M_1 ≈ 1.4–1.6 $M_\odot$, T_1 ≈ 5400 K + smaller, hotter dwarf, m_2 0.94–1.0 $M_\odot$, R_2 = 0.92–1.04 $R_\odot$, about 6000 K, surface-brightness ratio S_2/S_1 = 1.40; total about 7 $L_\odot$ (Gaia FLAME 7.2 $L_\odot$, 1.42 $M_\odot$); mass function 0.150 $M_\odot$; Gaia SB1
Period	42.46396 ± 0.00002 d (Gaia 42.46599 ± 0.01286 d; −0.2σ); B alone 42.46396 ± 0.00007
Epoch of primary	$T_{0,A}$ = BTJD 1538.63354 = BJD_{TDB} 2458538.63354; $T_{0,B}$ = BTJD 1517.53332
Primary eclipse (A)	14.1–15.0% deep, T_{14} 11.3 h, nearly flat bottom 5.0 h, U-shaped from limb darkening: full transit of the companion (b_A ≈ 0.0, *k* = 0.384); four events (s008, s035, s046, s072); every O−C under 0.2 min
Secondary eclipse (B)	17.0% deep, T_{14} 24.6 h, flat 10.3 h: total occultation of the companion near apastron (b_B = 0.24); two full events (s045; s008 in flagged cadences) plus an ingress in flagged cadences at the end of s046
Orbit	

	B–A phase 0.50310 ± 0.00002 (Gaia 0.5205 ± 0.0090, –1.9σ); *e* cos ω +0.0045 (Gaia +0.0295); *e* sin ω +0.378 (Gaia +0.403); *e* 0.378 against 0.404 ± 0.0145 (1.8σ); ω 89.3° against 85.8 ± 1.8°. The fit tied to Gaia's duration ratio fits B's ingress and egress noticeably worse
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (A s35); TESS-SPOC (A s35, s72); eleanor-lite (A s8); TARS (A ×4, B s45); every refit within 0–3 min
Pixel attribution	SPOC TPF and TESScut scene fits: lost flux on the target in all four eclipses tested (13.8–16.3%); the only bright neighbour (*G* 11.37 at 45″, would need 30–34%) gets +2.0, −0.8, +1.4, −0.1%; $\Delta\chi^2$ 3.5×10^5 to 2×10^6 in favour of the target
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	A mild tension: a 1.0 $M_\odot$ star at 6000 K is slightly hot for its mass (possibly low metallicity), easing if the primary is cooler (Gaia 5937 K against TIC 5446 K). No formal error on *e* sin ω (the free-ρ fit failed). Spectra for an SB2 detection and the companion's temperature
Next eclipses	A: BTJD 4341.255, 4383.719; B: BTJD 4362.619, 4405.083 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 243-1377-1 is a G subgiant of about 2.6 $R_\odot$ with a dwarf companion of about 1 $M_\odot$ and 1 $R_\odot$ on an eccentric 42.46-d orbit. The occultation is deeper than the transit, even though the transit is full. The companion is smaller and hotter, with a surface about 1.4 times brighter than the primary's. Hiding it entirely therefore removes more light (17%) than covering part of the primary with it (14.6%). The occultation also lasts more than twice as long because it falls near apastron, where the stars move 2.2 times slower. One geometry fits all five eclipses, with white residuals.

A70. TYC 4094-1571-1 (TIC 70789999)



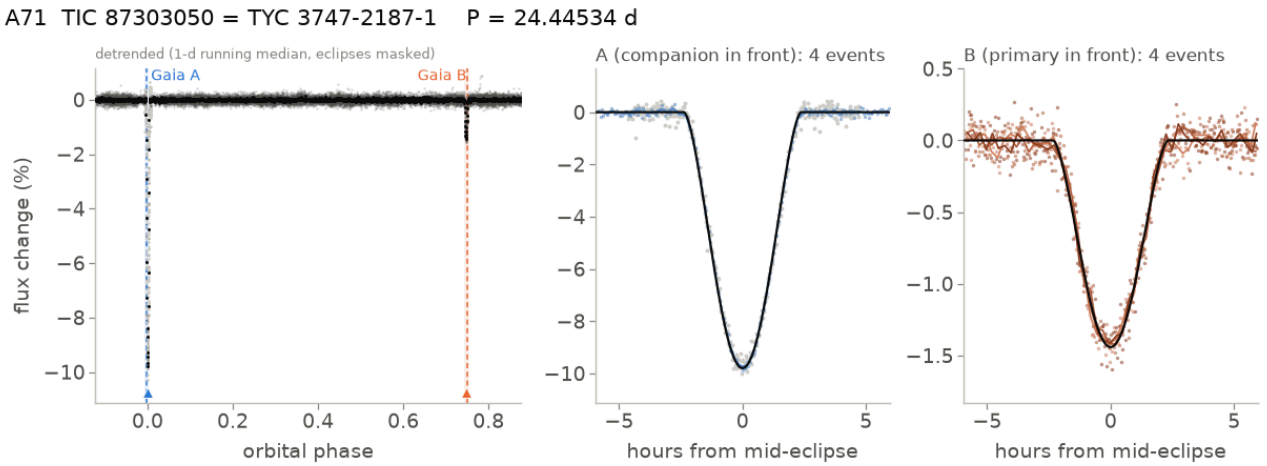
A70. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 22.73503 d, Gaia's predicted conjunctions dashed. Middle: the 8.2%, 13.3-h flat transits at A. Right: the three 5.1%, 17.8-h flat total occultations at B.

Identifiers	TIC 70789999; Gaia DR3 2822584818250071O4; TYC 4094-1571-1
Position (J2000)	05:53:58.9 +60:19:36.1
Brightness	*G* 10.17; Tmag 9.76
Stars	Subgiant, R_1 = 3.0–3.4 $R_\odot$ (TIC 3.60; Gaia 3.21–3.57, FLAME 1.83 $M_\odot$), T_1 6300–6600 K + G dwarf, m_2 0.9–1.0 $M_\odot$, R_2 = 0.83–0.94 $R_\odot$, T_2 = 5600–5900 K, L_2 = 5% of the light; mass function 0.120 $M_\odot$; Gaia SB1; SIMBAD: 2019 TESS asteroseismic target list
Period	22.73503 ± 0.00001 d (Gaia 22.73174 ± 0.00324 d; +1.0σ)
Epoch of primary	$T_{0,A}$ = BTJD 1823.53837 = BJD_{TDB} 2458823.53837; $T_{0,B}$ = BTJD 1836.88172
Primary eclipse (A)	

	8.09–8.35% deep, T_{14} 13.3 h, 7.6 h flat, U-shaped: full transit ($b_A \approx 0$, $k = 0.275$); four events (s019, s059, s060; s073 in flagged cadences, clean); plus a further A eclipse at 2937.555 on a scattered-light feature, unusable for timing; O–C ≤ 0.1 min for the events in the fit; the flagged s073 event is −3 min
Secondary eclipse (B)	5.00–5.13% deep, T_{14} 17.8 h, 9.9 h flat: total occultation ($b_B = 0.19$); three events (s019, s059, s073)
Orbit	B–A phase 0.58691 ± 0.00003 (Gaia 0.5794 ± 0.0099, +0.8σ); e cos ω +0.135 (Gaia +0.124); e sin ω +0.148 (+0.140); e = 0.201 against 0.187 ± 0.020; ω = 47.5° against 48.4 ± 3.4°. The Gaia-tied fit is as good as the free one
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (A s19, s59, s60; B s19, s59, s73; 0–1 min); TARS (all six within 1 min, B s73 within 6 min in noisy data)
Pixel attribution	TESScut s59 scene fit, A and B: lost flux on the target (8.2%, 5.1%); $\Delta\chi^2 \approx 3\times10^5$ for target over neighbour; centroid within 0.08 px; neighbours within 60″ (G 15.6–15.7) would need more than 100%
Catalogue status	Not a catalogued EB: nothing in VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, WDS, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP or a general web search. Flagged by Kostov et al. 2025 table 2 (unvetted) and by a TESS-SPOC multi-sector TCE at P = 11.3675 d (half the true period), depth 8.4%, never followed up
Open issues	Spectroscopy for SB2 and the companion's temperature
Next eclipses	B: BTJD 4337.735, 4360.470; A: BTJD 4347.126, 4369.862 (±1 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4094-1571-1 is a subgiant of about 3.2 $R_\odot$ with a G dwarf of about 1 $M_\odot$ on a mildly eccentric 22.74-d orbit. Both eclipses are central and flat-bottomed: a full transit of the dwarf (8.2%) and its total occultation (5.1%, the dwarf's share of the light). The eclipses reproduce Gaia's e and ω within 1σ. A TESS-SPOC multi-sector search had already detected the transit as a threshold-crossing event, at half the true period. It was not followed up, and we do not treat it as a prior catalogue entry.

A71. TYC 3747-2187-1 (TIC 87303050)



A71. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.44534 d, Gaia's predicted conjunctions dashed. Middle: the four 9.7%, 4.5-h V-shaped grazing eclipses at A. Right: the four 1.4% grazing eclipses at B.

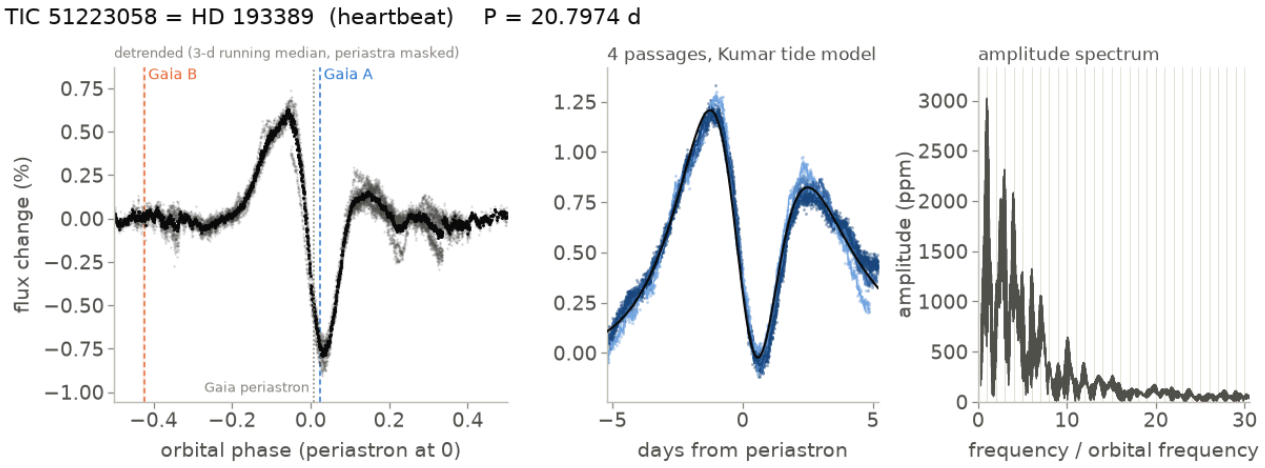
Identifiers	TIC 87303050; Gaia DR3 2809691025802535 68; TYC 3747-2187-1
Position (J2000)	05:15:50.8 +58:13:29.8
Brightness	G 10.62; Tmag 10.20
Stars	G0 star (LAMOST; RV −25.2 km/s), R_1 from the crossing speed 1.0–1.06 $R_\odot$, depending on b (TIC 1.18; Gaia 1.24–1.25) + K dwarf, m_2 = 0.62–0.68 $M_\odot$; a = 42–44 $R_\odot$; mass function 0.089 $M_\odot$; Gaia SB1
Period	24.44534 ± 0.00001 d (Gaia 24.44347 ± 0.00331 d; +0.6σ)

Epoch of primary	$T_{0,A}$ = BTJD 1833.11980 = BJD_{TDB} 2458833.11980; $T_{0,B}$ = BTJD 1826.93956
Primary eclipse (A)	9.6–9.8% deep, V-shaped, 4.3–4.7 h: grazing transit of the companion; four events (s019, s059; s073 and s086 in flagged cadences, clean)
Secondary eclipse (B)	1.36–1.47% deep, V-shaped, grazing; four events (s019, s059, s073, s086); O–C ≤ 2 min
Orbit	B–A phase 0.74718 ± 0.00003 (Gaia 0.7514 ± 0.0044, −1.0σ); e cos ω = +0.399 (Gaia +0.406); e sin ω unconstrained by the durations; e = 0.399–0.41 against Gaia 0.409 ± 0.0075; i ≈ 87.6–87.8°
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s19: A 9.5%, B 1.33–1.38%); TESS-SPOC (A s19, s59; B s19, s59, s73); TARS (all six within 0–3 min)
Pixel attribution	SPOC TPF s19 and TESScut s59: lost flux on the target (A 8.7%, 9.1%; B 1.2%, 1.4%); centroid shift ≤ 0.03 px (A), ≤ 0.1 px (B); nearest bright neighbour (G 16.07 at 41″) would need more than 100%
Catalogue status	Not a catalogued EB: nothing in VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, WDS, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP or a general web search; Kostov et al. 2025 table 2 only (unvetted)
Open issues	Both chords graze, so k, b and third light are not separable: the free fit (k = 0.83, b = 1.37) and the Gaia-tied fit (k = 0.66) are statistically identical; the free-fit companion (R_2 ≈ 0.85 $R_\odot$, about 3700 K) is implausible for 0.65 $M_\odot$, and the expected k ≈ 0.55 presumably lies within the degeneracy (the k-fixed fits failed). Multi-band photometry or an SB2 detection would break it
Next eclipses	A: BTJD 4350.990 (2026-11-06 11:38 UTC), 4375.435; B: BTJD 4344.809, 4369.255 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3747-2187-1 is a G0 star with a K dwarf of about 0.65 $M_\odot$ on an eccentric (e ≈ 0.40) 24.45-d orbit. The two stars graze at both conjunctions: both eclipses are V-shaped and last 4.5 h, against 7–8 h for central ones. Eight eclipses over six years, two of them recovered from QLP-flagged cadences, fit a linear ephemeris to within 2 min. Their spacing gives Gaia's e cos ω. The 1.4:9.7 depth ratio fits a K dwarf behind a G0 star; the grazing geometry leaves the radius ratio open.

S3.1.3 Heartbeat stars

HD 193389 (TIC 51223058)



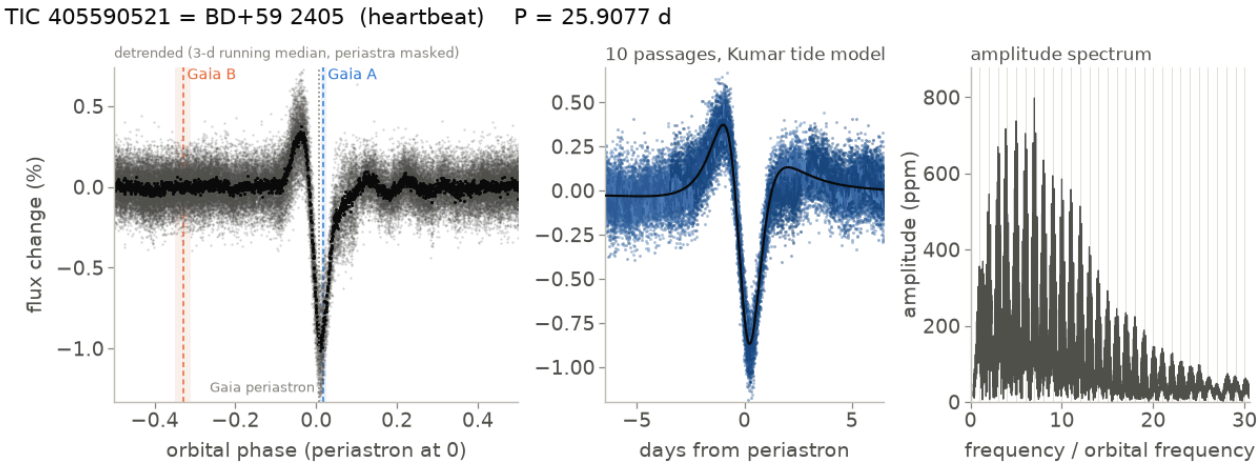
HD 193389. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 20.797 d, periastron and Gaia's conjunctions marked. Middle: the 1.2% peak-to-trough heartbeat at each covered periastron (four passages overlaid) with the Kumar et al. (1995) model (i ≈ 35°). Right: the amplitude spectrum of the light curve, against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 51223058; Gaia DR3 1808373856612878464; HD 193389; WDS HDS 2905
Position (J2000)	20:19:31.5 +15:32:42.2
Brightness	G 7.45; Tmag 7.06

Stars	F5.5 II-III bright giant (SIMBAD), L ≈ 210 $L_\odot$, R_1 ≈ 11 $R_\odot$ (8–16 from the Hipparcos parallax range; TIC 13.3), M_1 ≈ 3.5 $M_\odot$ assumed + companion m_2 ≈ 2.3 $M_\odot$ at i = 35° (about 10% of the light); mass function 0.0655 $M_\odot$; Gaia SB1, K_1 33.4 km/s, RUWE 2.06. A 0.3″ visual companion (HDS 2905, ΔHp ≈ 3.7) probably explains the RUWE
Period	20.7974 ± 0.0015 d from the troughs (Gaia 20.79663 ± 0.00058 d; 0.5σ)
Epoch	Trough BTJD 2778.057 = BJD_{TDB} 2459778.057 (troughs 2778.057, 2798.834, 2819.718, 3526.775; O−C ≤ 0.05 d); the trough falls about 0.6 d after periastron
Periastron signature	A 4–5-d rise to a peak 1.05–1.23 d before periastron, a steep fall, a trough 0.54–0.63 d after periastron, then a lower secondary maximum and slow decline; peak to trough 12,160–12,410 ppm in all three fully covered passages (s54, s55, s81)
Tidal model *e*, ω vs Gaia	Orbit free: e = 0.371, ω = 77.6°, i = 35.5° against Gaia e 0.3555 ± 0.0044, ω 77.26 ± 0.86°; from the light curve alone, ω agrees with Gaia's to 0.3° and e to 0.016 (3.2σ by Gaia's quoted error). Periastron shifted +0.28 d (about 3σ), attributed to the simplified model, with apsidal motion also possible. Sharp χ^2 minimum at i ≈ 35°; fitted S = 5.4×10^{-3} against 4.3×10^{-3} predicted for an equilibrium tide at that inclination
Eclipse	None: the heartbeat model beats a trapezoid eclipse by ΔBIC ≈ 67,400; an eclipse at A would need $i \gtrsim 70°$; any eclipse on top of the heartbeat is ≲ 0.05% against the 1.2% depth of the candidate eclipse; B windows flat
Oscillations	Not searched; a search for tidally excited oscillations at orbital harmonics remains to be made
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s81: 11,136 and 11,514 ppm against QLP's 11,621)
Pixel attribution	TESScut difference images (peak minus trough): the 5×5 core loses 10,300–11,100 ppm, centred on the target; the brightest Gaia neighbour (G 12.54 at 16.6″) would need 132%. HDS 2905 B (0.3″) would need about 37% and cannot be separated in pixels; the signal is locked to HD 193389's own RV periastron and reproduces that orbit
Catalogue status	Not listed as a heartbeat or eclipsing binary in: VSX (only an unrelated G 18.2 variable at 24″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, SB9, Gaia DR3 eclipsing-binary table (Gaia's variability classification: ACV|CP|MCP|ROAM|ROAP|SXARI), ExoFOP, SIMBAD's 22 references, general web search
Open issues	A heartbeat model fitted jointly with the Gaia RVs; spectra for the about 2.3 $M_\odot$ companion. Only three TESS sectors
Next periastra	BTJD 4337.24, 4358.04, 4378.83 (A conjunction 4337.57 ± 0.12, 4358.36, 4379.16)
Discovery	Conjunction-window search (Section 2.5), as a candidate eclipse; reclassified as a heartbeat by independent re-analysis with separately written code (Section 2.5)

HD 193389 is an F-type bright giant of about 11 $R_\odot$ on a 20.8-d orbit with e = 0.36. The 1.2% dip after periastron is not an eclipse at the A conjunction. The same asymmetric peak-and-trough pattern recurs at every periastron, with the trough 0.2–0.3 d after the conjunction rather than on it. A tidal model fitted to the light curve alone reproduces Gaia's orbit. At periastron the giant's radius is about 0.3 of the separation, so a heartbeat of about this size is expected. The eclipse interpretation is refuted, and the heartbeat has not been reported before.

BD+59 2405 (TIC 405590521)



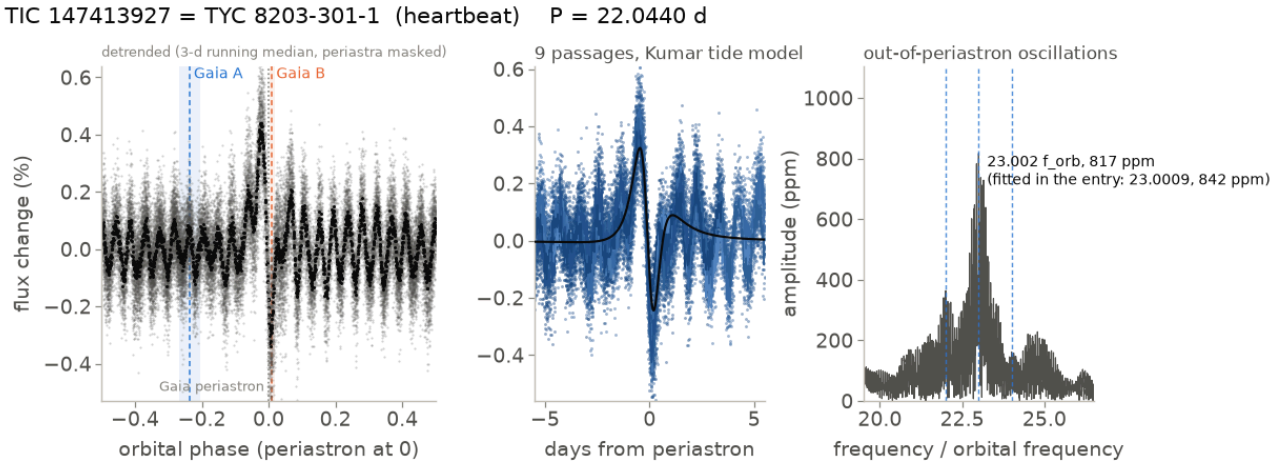
BD+59 2405. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 25.9077 d, periastron and Gaia's conjunctions marked. Middle: the +0.3–0.4% brightening and 1.2% dip at each periastron (ten passages overlaid; seven were fitted) with the Kumar et al. (1995) model ($i \approx 60$–$65°$). Right: the amplitude spectrum of the light curve, against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 405590521; Gaia DR3 2203762361132796800; BD+59 2405
Position (J2000)	21:37:16.0 +60:29:23.1
Brightness	*G* 10.40; Tmag 9.89
Stars	A2 star (SIMBAD), member of Trumpler 37, $R_1 \approx 7\ R_\odot$ (TIC 6.96), $M_1 \approx 2.5\ M_\odot$ assumed + companion $m_2 \approx 1.5$–$1.85\ M_\odot$; mass function 0.2225 $M_\odot$; Gaia SB1
Period	25.9077 ± 0.0004 d from the tidal fit (Gaia 25.8987 ± 0.0077 d; +1.2σ)
Epoch	Periastron t_p = BTJD 3562.21 ± 0.07 = BJD_{TDB} 2460562.21 (Gaia 3561.28 ± ~0.95, +1σ)
Periastron signature	A +0.3–0.4% brightening and about 1.2% dip in every orbit, in seven fitted periastron windows (s17, 56, 57, 76, 77, 83, 84); the model's flux minimum falls at the A conjunction (2837.14, 3562.56, 3588.45); per-window timing shifts under 0.03 d
Tidal model *e*, ω vs Gaia	e = 0.5846 ± 0.0033 against Gaia 0.587 ± 0.025; ω = 69.4 ± 4.7° against 70.0 ± 2.1°; i = 60.5 ± 5.4° (65 ± 7° without s17); S = 465–515 ppm; χ^2 8620 for 3046 points. With Gaia's orbit fixed the fit is poor (i = 37°, χ^2 155k). Amplitude check: $(m_2/M_1)(R_1/d_{peri})^3 \approx 1.6\%$ for $R_1 \approx 7\ R_\odot$
Eclipse	None: residuals in a 0.3-d box at each A conjunction are at the red-noise level, so any A eclipse is shallower than about 0.1%; A needs $i \gtrsim 69°$ and B $i > 84°$, and the fitted 60–65° predicts none
Oscillations	None reported; residual rms 460–880 ppm against about 350 white. An RV test for tidally excited oscillations remains to be made
Pipelines agreeing	QLP SAP; TESS-SPOC s83 (SAP dip 1.03% at 3562.48, brightening +0.21%; PDCSAP 0.88% / +0.27%)
Pixel attribution	Not tested separately; the signal is locked to the star's own RV periastron and reproduces that orbit's e and ω
Catalogue status	No prior heartbeat claim: VSX, ASAS-SN, Solanki et al. 2025 (no source within 30″), Callahan et al. 2025 (112 heartbeat systems from Gaia SBs + TESS; not included); the Kirk et al. 2016, Shporer et al. 2016 and Wrona et al. 2022 tables cover Kepler and OGLE fields. No TOI/CTOI; a general web search found nothing
Open issues	A joint RV and light-curve fit; an RV test for tidally excited oscillations
Next periastra	Heartbeat minima BTJD 4339.79, 4365.70, 4391.60 (±0.08)
Discovery	Conjunction-window search (Section 2.5), flagged as a heartbeat candidate; confirmed by independent re-analysis with separately written code (Section 2.5)

BD+59 2405 is an A2 star of about 7 $R_\odot$ in the young cluster Trumpler 37, with a companion of about 1.6 $M_\odot$ on a 25.9-d orbit with e = 0.58. Its light rises by about 0.3% and falls by about 1.2% at every periastron. A tidal model fitted to the light curve alone returns Gaia's eccentricity and periastron angle

to within 0.3σ at an inclination of 60–65°, where no eclipse is possible. The residuals at the conjunction limit any eclipse to about 0.1%. The star is not in the heartbeat catalogues we checked (Callahan et al. 2025; Solanki et al. 2025).

TYC 8203-301-1 (TIC 147413927)



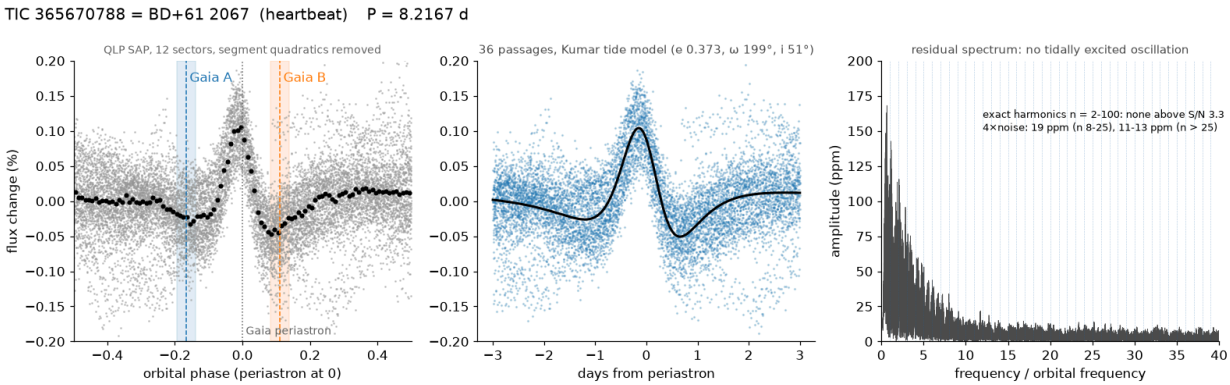
TYC 8203-301-1. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 22.0440 d, periastron and Gaia's conjunctions marked. Middle: the heartbeat at each covered periastron (nine passages overlaid) with the Kumar et al. (1995) model ($i \approx 45°$); the oscillation is present in the plotted data. Right: the amplitude spectrum of the out-of-periastron data, with 22, 23 and 24 times the orbital frequency marked; the peak at 23 times the orbital frequency is 817 ppm in this panel, against the fitted 842 ppm given in the entry.

Identifiers	TIC 147413927; Gaia DR3 5362949222888372608; TYC 8203-301-1
Position (J2000)	10:52:52.3 −47:42:04.8
Brightness	G 10.44; Tmag 10.13
Stars	Late-main-sequence star in the γ Dor instability strip: Gaia FLAME R 3.12 $R_\odot$, M 2.07 $M_\odot$, L 28 $L_\odot$; GSP-Phot 7530 K, log g 3.67 (TIC 7255 K, 3.19 $R_\odot$, 1.63 $M_\odot$) + companion m_2 = 0.94–1.08 $M_\odot$ at i = 44.6°; a = 45–48.5 $R_\odot$; mass function 0.0440 $M_\odot$; Gaia SB1, K_1 36.6 km/s
Period	22.0440 ± 0.0005 d from the tidal fit (Gaia 22.0393 ± 0.0080 d; 0.6σ); the three oscillation harmonics imply 22.0432, 22.0438, 22.0435 d
Epoch	Periastron t_p (N = 0) = BTJD 399.972 ± 0.078 = BJD_{TDB} 2458399.972 (Gaia 399.948 ± 0.136; 0.15σ)
Periastron signature	Peak 0.47 d before periastron, trough 0.07 d after; peak to trough 7.0–7.8 ppt in QLP; five passages fully covered (s10, s63, s90 twice, s100)
Tidal model *e*, ω vs Gaia	e = 0.670 ± 0.007 against Gaia 0.680 ± 0.029 (0.3σ); ω = 244.5 ± 4.5° against 241.1 ± 4.2° (0.55σ); i = 44.6 ± 4.6°; S = 2.05×10^{-4} against 1.3–1.9×10^{-4} predicted for an equilibrium tide; reduced χ^2 2.9, jackknife errors over the five passages
Oscillations	Main peak 1.0434063 ± 0.0000136 c/d (842 ppm), coherent over 2550 d, at **23.0009 ± 0.0006** times the orbital frequency (+1.4σ from 23); weaker peaks at 22.0002 ± 0.0006 (about 170 ppm) and 24.0006 ± 0.0008 (about 130 ppm): tidally excited oscillations at n = 22 and 24, or orbital sidebands of n = 23; in either case they are locked to the orbit. Other peaks not at harmonics: 0.15–0.19 c/d (γ Dor-like) and 0.644 c/d
Eclipse	None possible: at i = 44.6° an eclipse needs i > 75° (near-periastron conjunction) or > 86° (far); residual boxes at both conjunctions are at the noise level; no eclipse ≳ 0.05%
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s10, s63: peak to trough 6.9–7.7 ppt)
Pixel attribution	TESScut s63 difference image (peak minus trough) is PSF-shaped and on the target, 6.4 ppt in the 3×3 aperture (centroid 0.4 px from the reference, read as jitter on a sub-percent signal); neighbours within 60″ (G 15.76–16.85) would need about 100–270% variation
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, one reference), general web search; not in Callahan et al. 2025, the APOGEE heartbeat search (Callahan et al. 2026), Li et al. 2024a,b or Solanki et al. 2025, whose searches cut at $P \lesssim 12$ d

Open issues	The s99/s100 2-min data; a PHOEBE fit with the Gaia RVs; spectroscopy
Next periastra	BTJD 4345.85 (peak 4345.38, trough 4345.92), 4367.90, 4389.94 (±0.12)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 8203-301-1 is a star of about 7300–7500 K and 3.1 $R_\odot$ on a 22.04-d orbit with e = 0.68, seen at about 45°. The heartbeat reproduces Gaia's orbit to within 0.6σ. The light curve also carries a coherent 842-ppm oscillation at 23.0009 times the orbital frequency, with weaker peaks at exactly 22 and 24 times it. The three harmonics each imply the orbital period independently, and all three agree with the heartbeat's. An oscillation held at an exact harmonic of the orbit over 2550 d is the signature of a tidally excited oscillation, and that is how we read it. The published TESS heartbeat searches stop at periods of about 12–13 d, which is why a 22-d system would not be in them.

BD+61 2067 (TIC 365670788)



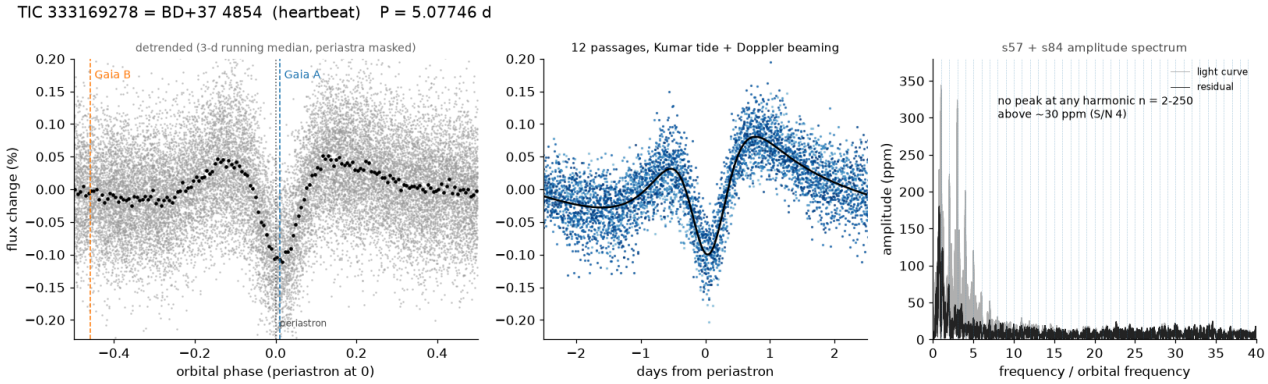
BD+61 2067. Left: QLP SAP flux, 12 sectors, with a quadratic removed per segment, folded on Gaia's P = 8.2167 d, Gaia's periastron and conjunctions marked. Middle: 36 periastron passages overlaid with the Kumar et al. (1995) model (e 0.373, ω 199°, i 51°). Right: the residual amplitude spectrum against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 365670788; Gaia DR3 2195886868574308608; BD+61 2067; TYC 4251-1269-1; SAO 19082
Position (Gaia DR3, J2016)	20:53:08.12 +62:10:15.7; 226 pc
Brightness	G 9.01; Tmag 8.65
Stars	F8 (SIMBAD), T_{eff} 6170–6400 K, R_1 2.55–2.65 $R_\odot$, M_1 1.3–1.6 $M_\odot$ + M dwarf, $m_2 \approx$ 0.23–0.26 $M_\odot$ at i = 51°; mass function 0.00244 $M_\odot$; Gaia SB1. A co-moving G 15.38 star 25.3″ away (about 5,700 au) makes it a probable hierarchical triple
Period	8.216284 ± 0.000094 d from twelve per-sector periastron times (Gaia 8.216686 ± 0.000589 d; −0.7σ)
Epoch	Periastron BTJD 2953.1308 ± 0.0083 = BJD_{TDB} 2459953.1308
Periastron signature	A +1.0-ppt hump peaking 0.15 d before periastron and a −0.55-ppt dip 0.66 d after; 1.45–2.14 ppt peak to trough in each of 12 sectors (2019–2024), 36 passages
Tidal model *e*, ω vs Gaia	From the light curve alone, P fixed: e = 0.373 ± 0.017 against Gaia 0.364 ± 0.063 (0.1σ); ω = 198.7 ± 6.5° against 192.3 ± 4.0° (0.9σ, compared mod 180°); i = 51 ± 6°; periastron +0.04 ± 0.05 d from Gaia's. S = 2.65×10^{-4} against 3.1–3.7 × 10^{-4} expected for an equilibrium tide. Errors from a bootstrap over sectors
Eclipse	None: an eclipse needs i > 80°; any eclipse is shallower than 130–150 ppm (3σ) at either conjunction
Oscillations	No tidally excited oscillation: no exact harmonic n = 2–100 above S/N 3.5; limits 19 ppm (n = 8–25) and 11–13 ppm (n > 25)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (12 sectors: e 0.357, ω 198.0°); SPOC 2-min (s57, s58); own TESScut photometry
Pixel attribution	

	The per-pixel heartbeat image is star-shaped and centred on the target to ≤ 0.5″ in s57, s76 and s84. The brightest neighbour within 60″ (the co-moving *G* 15.38 star) would need 58% variability; GJ 809 at 109″ shows none
Catalogue status	Not in VSX, ASAS-SN, SB9, the Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Kirk et al. 2016, Shporer et al. 2016 or Wrona et al. 2022; SIMBAD SB*, 13 references, none on variability
Open issues	ω compared mod 180° only (a beaming term does not choose the branch here); a joint RV and light-curve fit; RVs of the wide companion
Next periastra	BTJD 4333.467, 4341.683, 4349.899 (±0.018); no eclipse expected
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

BD+61 2067 is an F8 star of about 2.6 $R_\odot$ with an M-dwarf companion on an 8.2-d orbit with e = 0.37. At every periastron its light rises by 0.1% and falls by half that, in all twelve TESS sectors from 2019 to 2024. A tidal model fitted to the light curve alone returns Gaia's eccentricity and periastron angle within 1σ, at an inclination where no eclipse is possible. It is the clearest of the four.

BD+37 4854 (TIC 333169278)



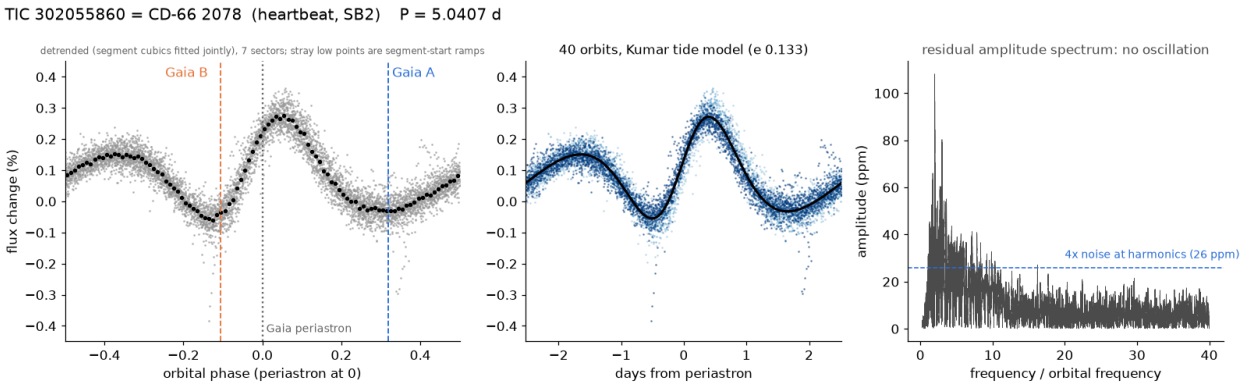
BD+37 4854. Left: the detrended QLP light curve folded on Gaia's P = 5.0776 d, periastron and Gaia's conjunctions marked. Middle: 12 periastron passages with the tide-plus-beaming model. Right: the residual amplitude spectrum (s57 and s84) in units of the orbital frequency.

Identifiers	TIC 333169278; Gaia DR3 1919980437220007296; BD+37 4854; TYC 3231-1540-1; SAO 73302
Position (J2000)	23:30:10.38 +38:24:56.0; 253 pc
Brightness	*G* 9.68; Tmag 9.38
Stars	F2 (SIMBAD), T_{eff} 6670–7280 K, R_1 1.8–1.95 $R_\odot$, M_1 1.5–1.6 $M_\odot$ + M dwarf, $m_2 \approx$ 0.48–0.51 $M_\odot$ at i = 51.5°; mass function 0.0140 $M_\odot$; Gaia SB1, K_1 31.5 km s^{-1}. RUWE 4.2 is far more than this orbit can cause, so a third body is possible
Period	5.077463 ± 0.000021 d from Gaia's periastron and three TESS sectors (Gaia 5.077644 ± 0.000140 d; −1.3σ)
Epoch	Periastron BTJD 2870.807 ± 0.008 = BJD_{TDB} 2459870.807
Periastron signature	An “M”: a hump 0.53 d before periastron, a dip 0.04 d after it on Gaia's A conjunction, and a higher hump 0.77 d after; peak to trough 1540–1720 ppm in QLP s16, s57, s84 and 1570–1780 ppm in TESS-SPOC; present at every covered periastron
Tidal model *e*, ω vs Gaia	Tide plus Doppler beaming, from the light curve alone: e = 0.304 ± 0.009 against Gaia 0.315 ± 0.010 (−0.8σ); ω = 79.5 ± 2.7° against 81.7 ± 2.2° (−0.6σ, not mod 180°); i = 51.5 ± 6.4°; S = 3.9 ± 0.8 × 10^{-4} (3.9–5.8 expected); beaming 3.4 ± 0.5 × 10^{-4} (2.9 expected); χ^2/dof 2.1. The pure tide returns the mirror angle 100 ± 4°; adding beaming lowers χ^2 by 1807 and an injection reproduces the mirror. With the shape frozen, the beaming amplitude refitted sector by sector is 4.2, 3.2, 3.4 × 10^{-4} in QLP (constancy χ^2 11.5 for 2 dof) and 3.8, 3.5, 3.8 in TESS-SPOC (2.0 for 2 dof), and every one of the 12 passages alone returns it positive at more than 2σ; the tide amplitude is no more constant (6.7 and 5.7 for 2 dof)

Eclipse	None: an eclipse needs $i > 77°$; any eclipse is ≤ 160 ppm at A and ≤ 100 ppm at B (3σ)
Oscillations	No tidally excited oscillation (limit about 30 ppm at harmonics n = 2–250); no δ Sct or γ Dor pulsation
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (e 0.307, ω 79.7°) and PDCSAP; own TESScut photometry
Pixel attribution	TESScut hump-minus-dip images: the 3×3 core changes by 1380–1470 ppm in s16, s57, s84 against 1395 expected, PSF-shaped, centroid ≤ 0.16 px; the brightest neighbour (G 13.93, 31.5″) would need 7% and its pixels do not vary
Catalogue status	Not in VSX, ASAS-SN, SB9, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Wrona et al. 2022, Kirk et al. 2016, Shporer et al. 2016 or the Gaia DR3 variability tables; SIMBAD SB*, 4 references
Open issues	A joint RV and light-curve fit; the cause of RUWE 4.2; inclination loosely set; four sectors, one dominated by scattered light
Next periastra	BTJD 4328.041, 4333.118, 4338.196 (±0.008)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

BD+37 4854 is an F2 star with an M-dwarf companion on a 5.08-d orbit with e = 0.31. Its periastron pulse is an M-shape of 0.17%. Near ω = 90° a pure tidal model cannot tell the true periastron angle from its mirror image and picks the mirror. Doppler beaming of the F star, whose size the Gaia velocity fixes, is antisymmetric about periastron and settles it: with that term the light curve returns Gaia's ω to 0.6σ and its e to 0.8σ. The term is not a detrending artefact: refitted with the orbit shape frozen, its sign comes back from every periastron passage on its own and its size holds across three sectors and two pipelines.

CD-66 2078 (TIC 302055860)



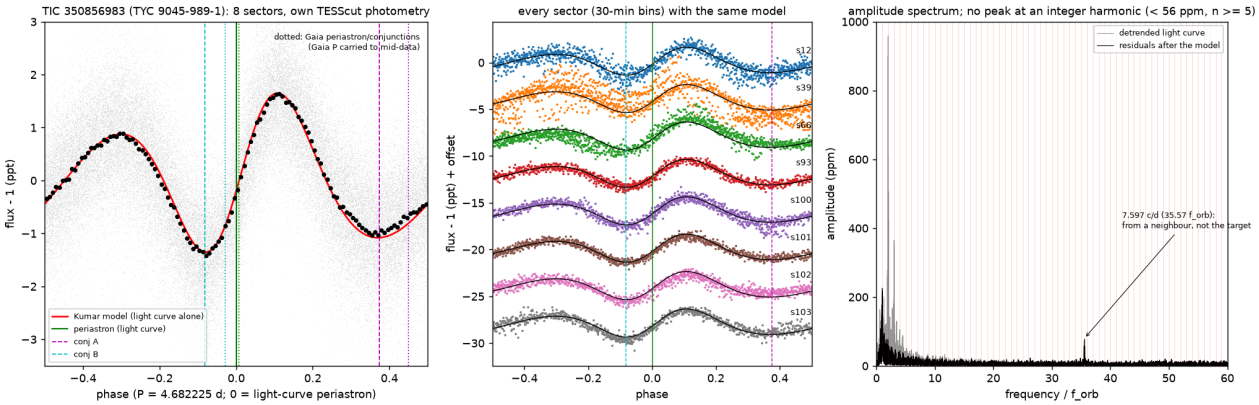
CD-66 2078. Left: QLP SAP flux with baselines removed, folded on Gaia's P = 5.0407 d, Gaia's periastron and conjunctions marked. Middle: 40 orbits against time from periastron, with the model. Right: the residual amplitude spectrum in units of the orbital frequency, with the 26-ppm limit.

Identifiers	TIC 302055860; Gaia DR3 5813326715771172992; CD-66 2078; CPD-66 3086; TYC 9064-655-1
Position (Gaia DR3, J2016)	17:20:15.59 −66:45:26.2; 364 pc
Brightness	G 9.90; Tmag 9.55
Stars	Two near-twin F stars (Gaia two-star fit 6338 and 6291 K), each about 1.9 $R_\odot$ and 1.4–1.5 $M_\odot$; Gaia **SB2**, K_1 80.2, K_2 73.5 km s^{-1}, M $\sin^3 i$ 0.88 and 0.96 $M_\odot$, so $i \approx 56–63°$
Period	5.040480 ± 0.000030 d (TESS; Gaia 5.040691 ± 0.000063 d, −3.0σ, or −1.3σ with Gaia's error scaled by its χ^2/dof)
Epoch	Periastron BTJD 3412.468 ± 0.021 = BJD_{TDB} 2460412.468
Periastron signature	A hump 0.21 d after periastron and dips at both conjunctions, the deeper at B; 3.05–3.76 ppt peak to trough in each of 7 sectors
Tidal model e, ω vs Gaia	

	e = 0.134 ± 0.005 (stat) ± 0.004 (sys) against Gaia 0.152 ± 0.003 (−2.6σ; −1.8σ with Gaia's error scaled by its χ^2/dof of 6.2); ω = 137.8 ± 2.9° (mod 180°) against 140.7 ± 1.8° (0.9σ). The pure tide wants i = 40° against 56–63° from the masses; adding irradiation of each twin by the other fits at i = 59° with the expected tidal amplitude and e 0.130
Eclipse	None: an eclipse needs i > 75°; any eclipse is shallower than about 75 ppm
Oscillations	No tidally excited oscillation above 26 ppm (harmonics n = 9–60); nothing above S/N 4 to 80 c/d
Pipelines agreeing	QLP SAP (e 0.134); SPOC 2-min s12–13 (0.142 SAP, 0.132 PDCSAP); TESS-SPOC (0.139); own TESScut regression
Pixel attribution	TESScut template regression in s13, s39, s101: PSF-shaped, centred 0.9–1.7″ from the target in the 200-s and 10-min sectors; the brightest neighbour within 60″ (G 16.11) would need 98%
Catalogue status	Not in VSX, ASAS-SN, SB9, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, the Gaia DR3 variability tables, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Wrona et al. 2022, Kirk et al. 2016 or Shporer et al. 2016; SIMBAD SB*, 1 reference
Open issues	The e tension with Gaia, whose SB2 solution has χ^2/dof 6.2 and a period 3σ off; independent RVs or Gaia DR4 epoch velocities would settle it
Next periastra	BTJD 4329.835, 4334.876, 4339.916 (±0.022)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

CD-66 2078 is a pair of near-twin F stars on a 5.04-d orbit with e = 0.15, which Gaia resolved as a double-lined binary. Its 0.3% periastron pulse is in all seven TESS sectors and on the target. The light curve returns Gaia's periastron angle within 1σ and an eccentricity 0.018 lower, a gap typical of heartbeat fits against Gaia orbits (Callahan et al. 2025) and within twice Gaia's error once that is scaled to its own fit quality.

TYC 9045-989-1 (TIC 350856983)



TYC 9045-989-1. Left: own TESScut photometry folded on the photometric period 4.6822 d with the model, light-curve and Gaia periastron and conjunctions marked. Middle: all eight sectors with the same model. Right: the amplitude spectrum in units of the orbital frequency.

Identifiers	TIC 350856983; Gaia DR3 5828538665312585856; TYC 9045-989-1; 2MASS J16281728-6345236
Position (Gaia DR3, J2016)	16:28:17.26 −63:45:24.1; 395 pc
Brightness	G 9.72; Tmag 9.37
Stars	Two near-twin F stars (q = 0.95), T_{eff} ≈ 6400–6600 K, each about 2.3–2.5 $R_\odot$; Gaia **SB2**, M $\sin^3 i$ 0.82 and 0.78 $M_\odot$
Period	4.682234 ± 0.000021 d (photometric, eight sectors; Gaia 4.682539 ± 0.000076 d, 4σ long)
Epoch	Light minima at conjunction A BTJD 3800.117 and B 3802.654 (±0.003); periastron 3798.36 ± 0.10
Periastron signature	

	A dip of 1.4 ppt at conjunction B, a hump of 1.65 ppt 0.52 d after periastron and a second dip at A; 2.8–3.6 ppt peak to trough in every sector; the second orbital harmonic (ellipsoidal) is 1.0–1.2 ppt
Tidal model *e*, ω vs Gaia	e = 0.109 ± 0.005 against Gaia 0.119 ± 0.006 (–1.3σ); ω = 127 ± 8° (mod 180°) against 106.6 ± 3.3°: **+20°, 2.4σ**, the same in three independent data sets and with the period free; not the mirror branch, beaming, reflection or apsidal motion. Carried back with the photometric period, the B conjunction lands on Gaia's to 0.3 h
Eclipse	None: an eclipse needs i > 68° against the fitted 40–44°; limit about 200 ppm
Oscillations	None at the orbital harmonics (limit 56 ppm for n ≥ 5); a 7.597 c/d signal belongs to a neighbour, since it grows with the aperture
Pipelines agreeing	QLP SAP, TESS-SPOC SAP and PDCSAP, TARS, eleanor-lite, own TESScut photometry
Pixel attribution	TESScut difference images in five sectors have the target's PSF shape; the amplitude does not depend on aperture size; the brightest neighbour within 60″ (G 15.28) would need 49%
Catalogue status	In no variable-star, eclipsing-binary or heartbeat catalogue checked (as for H6). Callahan et al. (2025) inspected every Gaia DR3 SB2 and did not list it
Open issues	ω 20° from Gaia's; at e = 0.12 the signal is largely ellipsoidal; the inclination from the amplitude implies masses heavy for the luminosity; an RV orbit would settle all three
Next periastra	BTJD 4327.45, 4332.14, 4336.82 (±0.10); conjunction B minima 4331.747, 4336.429 (±0.01)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; kept as a candidate by independent re-analysis with separately written code
Status	Candidate. The pulse is on the target, in every sector and locked to Gaia's conjunctions, and e agrees, but ω does not

TYC 9045-989-1 is a pair of near-twin F stars on a 4.68-d orbit with e = 0.12. Its light curve is mostly the ellipsoidal variation of the distorted stars, with a periastron asymmetry of a few tenths of a part per thousand on top. The asymmetry is locked to Gaia's orbit and returns its eccentricity, but places periastron 20° from where Gaia does. Gaia's solution has a poor fit and a period 4σ off, so the discrepancy may be Gaia's; until an RV orbit settles it, the star is a candidate.

S3.2 Eclipses on edge-on Gaia astrometric orbits (A19, A72–A74)

A19 TIC 219766411 = Gaia DR3 1440245818754478976

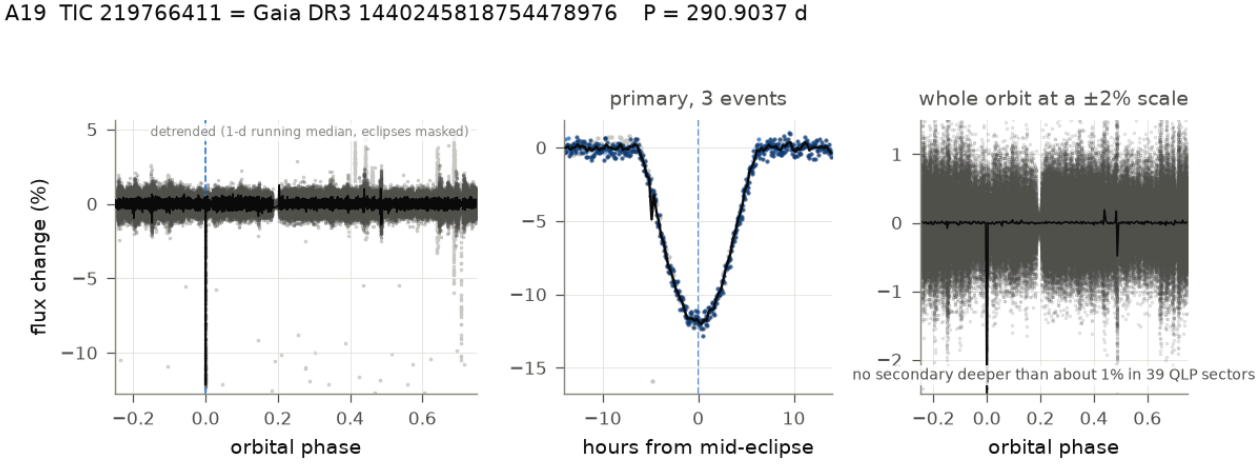


Figure A19. TIC 219766411: the three ~125-ppt, ~7-h eclipses at BTJD 1936.36 (sector 23), 2809.07 (sector 55, in flagged cadences) and 3390.88 (sector 76), on P = 290.9037 d. Left: the flux, detrended with a 1-d running median with the eclipses masked, folded on P. Middle: the primary eclipse (3 events) against hours from mid-eclipse. Right: the whole orbit at a ±2% scale; no secondary deeper than about 1% in 39 QLP sectors.

Identifiers	TIC 219766411; Gaia DR3 1440245818754478976
Brightness	Tmag 12.39 (G 12.8)

Stars	TIC: T_{eff} 6190 K, 1.26 $R_{\odot}$; a faint companion (secondary eclipse < ~1%)
Period	290.9037 ± 0.0006 d (formal, three events) (Gaia astrometric: 292.7769 ± 2.7548 d; agreement 0.7σ)
Epoch	BTJD 1936.3573 ± 0.0028 = BJD_{TDB} 2458936.3573
Primary eclipse	124, 127 and 125 ppt (Gaussian fits, raw QLP SAP), FWHM ~7 h; O–C −21 ± 9, +2 ± 2, −1 ± 1 min (sector 23 is 30-min cadence)
Secondary eclipse	None deeper than about 1% in hourly bins over all 39 QLP sectors, flagged cadences included
Orbit	Gaia: i = 91.190° (derived from Thiele–Innes), e = 0.418, ω = 15° (mod 180°). The eclipses fall 26 d (68%: 16–36 d) before the nearer predicted conjunction; 0.6% of Monte Carlo draws reach zero, about 2.5σ
Pipelines agreeing	QLP SAP (three events); TESS-SPOC recovers the sector-23 event at 85 ppt (no TESS-SPOC products for the other two sectors)
Pixel attribution	Sector-23 difference image: deficit 0.19 TESS pixel from the target; no other Gaia source within 30″
Catalogue status	Not listed in: VSX, Prša et al. (2022), IJspeert et al. (2024), Kostov tables 3 and 4, or the eccentric-EB catalogue of Tang et al. (2025; J/AJ/170/356), with no row within 30″; no Gaia variability or EB row; SIMBAD "*". Kostov table 2 only
Open issues	The 2.5σ conjunction offset, recorded as a caveat on Gaia's error model rather than as confirmation. Spectroscopy (an SB1 at Gaia's period with a consistent K) would be decisive
Next eclipse	BTJD 4554.491 ± 0.005 (2027 May 28), computed here from the three-event ephemeris (formal error; the three-event fit has $\chi^2 \approx 7.4$ for one degree of freedom, so the true error is about 2.7 times larger)
Method	Edge-on astrometric-orbit search (Section 2.6); third event and Gaia-covariance check by independent re-analysis (Section 2.5)
Status	Probable. The period agrees with Gaia's and the events repeat, but the conjunction is 2.5σ off, the sector-55 event exists only in flagged cadences and the sector-76 event only in QLP, the one independent depth (TESS-SPOC, sector 23) is 30% shallower than QLP's, the neighbour check reaches 30″ only, and the independent re-analysis itself names spectroscopy as the decisive test. The same protocol passed TIC 316045876, since withdrawn (Section 3.8)

The third event was in sector 55, where every in-eclipse cadence carries QLP quality flag 4096. A search restricted to QUALITY = 0 shows only two events. Gaia's period agrees to 0.7σ and the eclipses repeat three times. We therefore interpret the 26-d offset as a measurement of Gaia's error model for a G = 12.8 star over 34 months rather than as evidence of a different orbit. The timing should not be cited as confirmation.

A72 TIC 279636729 = Gaia DR3 4720373709202359808

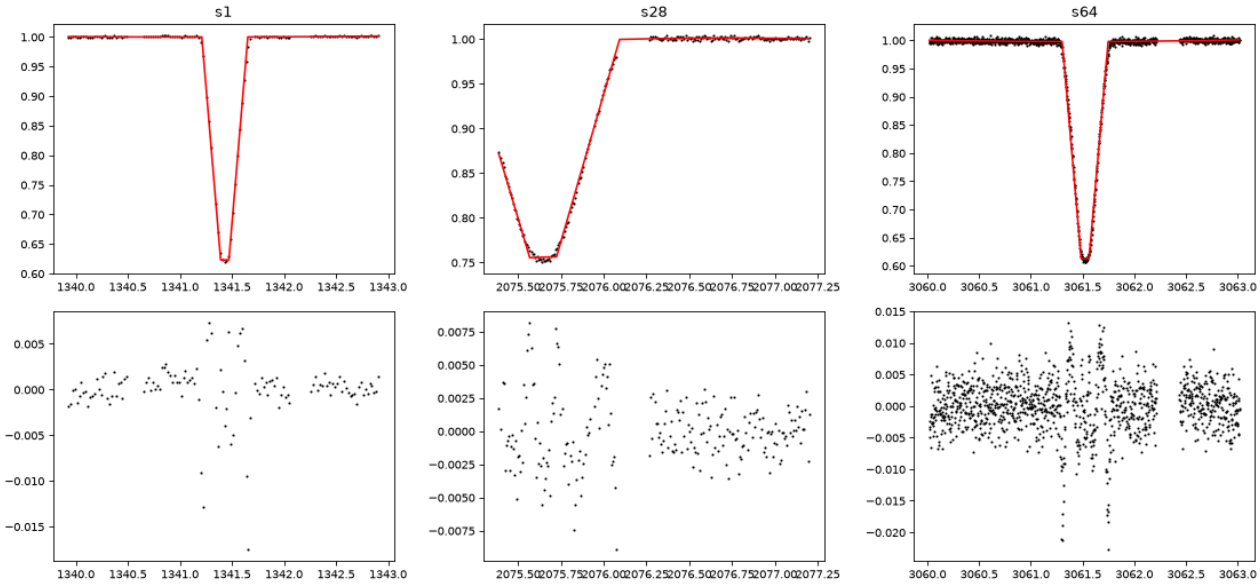


Figure A72. TIC 279636729, raw QLP SAP flux with trapezoid fits (red; top) and residuals (bottom) for sectors 1, 28 and 64 (left to right): the 38% primaries in sectors 1 and 64 and the 24%, 21-h secondary in sector 28, whose data begin part-way through ingress, just after a mid-sector gap. The third primary, in sector 96 under scattered-light flags, is not shown.

Identifiers	TIC 279636729; Gaia DR3 4720373709202359808
Position (J2000)	48.475926, −64.367416 deg (TIC)
Brightness	Tmag 12.01; G 12.43
Stars	F-type (TIC: T_{eff} 6081–6213 K, 1.3–1.4 $R_\odot$, ~1.2 $M_\odot$) at 560 pc with a companion of comparable size and temperature supplying 20–25% of the light (from the depths and the Gaia photocentre orbit, a_0 = 0.351 mas); total mass ~2.2 $M_\odot$, a ≈ 0.91 AU
Period	215.0116 ± 0.0002 d from three primaries over 12 cycles (χ^2 0.01); Gaia astrometric `Orbital` solution 217.22 ± 2.04 d, agreement at −1.1σ. P/2 and P/4 are excluded by flat data at their predicted times; P/3 is excluded only by Gaia's period
Epochs	Primary T_0 = BTJD 1341.4321 ± 0.0009 = BJD_{TDB} 2458341.4321; primaries at 1341.4321, 3061.5252 and 3921.565–3921.576 (O−C 0.0, +0.1, −0.9 min); secondary at 2075.6456 ± 0.0012, phase 0.4148
Primary eclipse	37.7–38.8%, total duration 10.6 h, ingress 4.4 h, flat bottom 1.9 h
Secondary eclipse	24.2%, 21.4 h, at phase 0.4148: e cos ω between −0.14 and −0.06; the duration ratio of 2.0 gives e sin ω ≈ 0.34, so e ≈ 0.36, ω ≈ 111°, against Gaia's e = 0.40 ± 0.11 and $\lvert e \cos \omega \rvert$ at Gaia's 84th percentile: agreement at about 1σ
Predicted event	Sectors 1 and 64 alone predict the sector-96 primary at 3921.5717; every QLP cadence there is flagged on the tail of a scattered-light episode, and the raw flux holds the eclipse bottom and a clean egress at 3921.565–3921.576 with the sector-64 shape. The seven other scattered-light episodes of that sector, aligned on their end, show the flux already recovered where this one sits at 0.6; custom cutout photometry (this work) with a separately estimated background reproduces the eclipse
Pixel attribution	Difference images from TESScut cutouts for all four events: deficit centroid within 0.02–0.2 px of the Gaia position each time, while the out-of-eclipse light centroid is pulled 0.2–0.6 px toward the G 13.09 neighbour at 41″, whose own pixel loses 0.4–4.8% (it would need 71%). The only other star within 60″ (G 17.07) would need 2768%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability and eclipsing-binary tables, SB9, WDS, ExoFOP, SIMBAD, general web search; ADS full text (TIC and Gaia identifiers) none. Kostov table 2 only
Gaia conjunctions	The primaries fall 2–29 d before Gaia's predicted conjunction, the spread growing with cycle because Gaia's period is 2.2 d longer than the eclipse period; consistent within the Monte Carlo ranges. The test is weak

Next eclipses	Primary BTJD 4351.595 ± 0.002 (2026-11-07), then 4566.607, 4781.618; secondary 4440.773 ± 0.003 (rests on one event)
Method	Short-period search of the edge-on astrometric sample (Section 2.6); confirmed by independent re-analysis (Section 2.5)

Both stars are luminous: a 38% primary needs the eclipsed star to supply at least 39% of the light, and one impact parameter of about 0.7 fits both the 10.6-h primary and the 21-h secondary once the eccentric geometry is allowed for. The star should be double-lined, and with the astrometric orbit a spectroscopic one would give the masses directly. The next secondary, in late February 2027, would fix $e \cos \omega$. In the period-window search of Section 2.6 the system scored $z = 2.6$, below the detection threshold.

A73 TYC 4590-364-1 = TIC 315721826

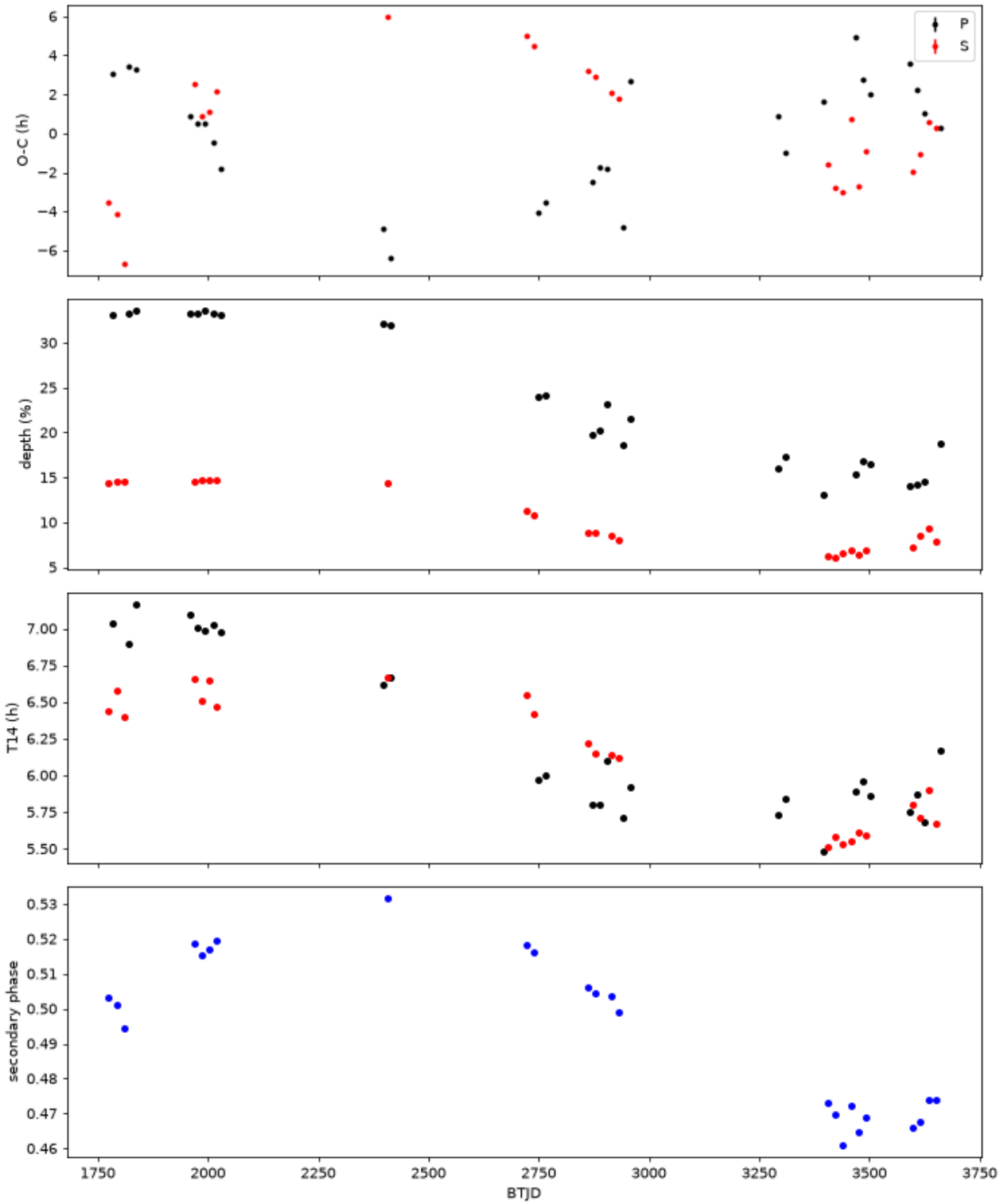


Figure A73. TYC 4590-364-1, from fits to 27 primaries (black) and 24 secondaries (red) in 21 QLP sectors, 2019–2024, from top to bottom: timing residuals against a linear 17.54-d ephemeris, eclipse depths, total durations and the phase of the secondary (blue). The eclipses fade from 33% to 13% and narrow, the timings vary by hours, and the secondary's phase rises then falls.

Identifiers	TIC 315721826; TYC 4590-364-1; Gaia DR3 (RUWE 1.93)
Position (J2000)	310.554255, +78.340072 deg (TIC)
Brightness	Tmag 10.81; G 11.28; V ≈ 11.6
Stars	

	TIC: T_{eff} 5932 K, 1.46 $R_\odot$, at 282 pc (parallax 3.55 mas), BP−RP 0.90; inner pair of about 2 $M_\odot$ in total, $a_{in} \approx 0.167$ AU. Gaia astrometric `Orbital` solution: $P_{out} = 124.72 \pm 0.24$ d, $e_{out} = 0.41 \pm 0.08$, $a_0 = 0.719$ mas, implying a third body of about 0.9 $M_\odot$ if dark
Period	17.538 d (mean of the primaries; the secondaries give 17.531 d because $e \cos \omega$ is changing); single-cycle intervals range from 17.386 to 17.851 d. Not Gaia's period: the eclipsing pair is the inner binary of a triple whose outer orbit Gaia measured
Epochs	27 primaries and 24 secondaries fitted in sectors 17–86, median timing error 0.8 min; against a linear ephemeris the primaries' O−C spans −6.4 to +5.0 h (rms 2.9 h), the secondaries' −6.7 to +6.0 h. The largest single jump, 17.851 d between the clean eclipses at BTJD 2940.800 and 2958.651, is real
Primary eclipse	Fading: 33.1–33.6% (2019–20), 32.0% (2021), 24.0% (2022), 18.6–20.2% (late 2022), 12.7–17% (sectors 73–80, December 2023 to July 2024), 14.1–14.9% (sectors 84–86, late 2024); total duration 7.0 h falling to 5.5–5.9 h
Secondary eclipse	14.4–14.8% falling to 6–9%; its phase 0.500 (2019), 0.515–0.520 (2020), 0.517 (2022), 0.504–0.509 (late 2022), 0.465–0.494 (sectors 77–80), 0.466–0.474 (sectors 84–86): $e \cos \omega$ rises then falls by about 0.05
Attribution of the fading	Custom cutout photometry (this work) at an early, a middle and a late epoch gives primary/secondary depths of 31.7/14.1% (sector 19), 18.8/8.6% (58) and 12.3/6.3% (79), the same within 0.5% for 1×1 to 7×7-pixel apertures; the star's flux against a fixed reference star stays constant (ratios 1.856, 1.954, 1.843), so no added light dilutes it, and the eclipses narrow, which dilution would not do
Pixel attribution	Difference images at sectors 19, 58 and 79: deficit centroid within 0.6″ of the light centroid, depths at the target pixel 31.5%, 18.3%, 12.2%. The nearest Gaia source (G 17.95 at 8.5″) would need an eclipse 150 times deeper than its whole light; nothing brighter than G 17.8 within 60″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability and eclipsing-binary tables, SB9, WDS, ExoFOP, general web search; SIMBAD has no references; ADS full text (TIC and Tycho identifiers) none. Kostov table 2 only
Third body	Required: timing variations of hours that change cycle to cycle, $e \cos \omega$ changing by 0.03–0.05 in a year, and the depth and width falling together with no change in total flux (an inclination or conjunction-geometry change); relativity and tides are orders of magnitude too slow. Gaia's 124.7-d companion fits the scale (expected dynamical timing amplitude ~4 h against the observed rms of 3 h; light-travel time only ~2 min; nodal precession on ~8 yr/cos i_{mut}) but is in tension with stability: $a_{out}/a_{in} \approx 4.2$ against the stability limit of Mardling & Aarseth (2001), 6–7 at e_{out} 0.41. The perturbations do not localise to Gaia's periastron. A third body is required; its identification with Gaia's companion is not established
Next eclipses	Primaries BTJD 4344.09, 4361.62, 4379.16; secondaries 4334.60, 4352.13 (±0.3 d because of the timing variations)
Method	Short-period search of the edge-on astrometric sample (Section 2.6); confirmed by independent re-analysis (Section 2.5)

A detached 17.5-d pair whose eclipses are disappearing within the TESS era is rare in itself. Two things would resolve its nature. The first is a photodynamical fit of the times, depths and widths, with Gaia's outer orbit first as a prior and then left free; the fit must reproduce the jump at BTJD 2958 and remain stable. The second is spectroscopy of the systemic velocity over about 125 d. Before TESS, ASAS-SN V photometry (527 epochs, 2014–2018, one or two points per eclipse) catches the primary eclipse in every season, 11–20 h after the TESS-era linear ephemeris predicts and at least 33% deep in June 2018, so the fade began after mid-2018. Whether the eclipses were deeper or shallower in 2014–2016, and so whether the fade is periodic, those data cannot say. Harvard plates (DASCH DR7, 1894–1990; 1,939 after cleaning) show no eclipse signal at any period; eclipses of the 2019 depth injected into the plates are recovered 60–93% of the time and those of the 2023 depth never, so the plates disfavour a century at the 2019 depth and say nothing about the shallower state. The late-2024 depths may show the start of a rebound. The period-window search of Section 2.6 detected the system at an alias period of 125.27 d.

A74 TIC 427345302 = Gaia DR3 2226943914530555776

A74 TIC 427345302 = Gaia DR3 2226943914530555776 P = 173.8755 d

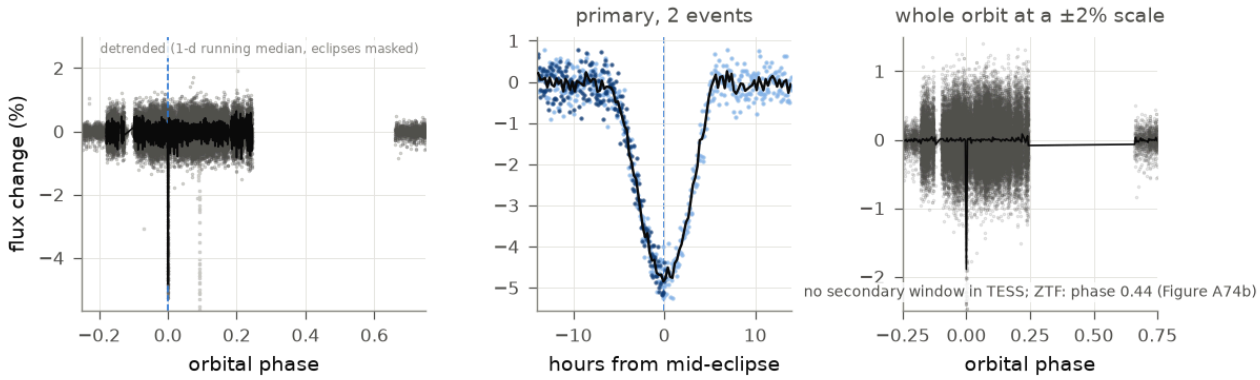


Figure A74. TIC 427345302: the two 4.6%, 10.8-h grazing eclipses at BTJD 2884.986 (sector 58) and 3406.613 (sector 77), three cycles of 173.8755 d apart. Left: the QLP SAP flux of nine sectors, detrended with a 1-d running median with the eclipses masked, folded on *P*. Middle: the two events against hours from mid-eclipse. Right: the whole orbit; no TESS sector covers the secondary window.

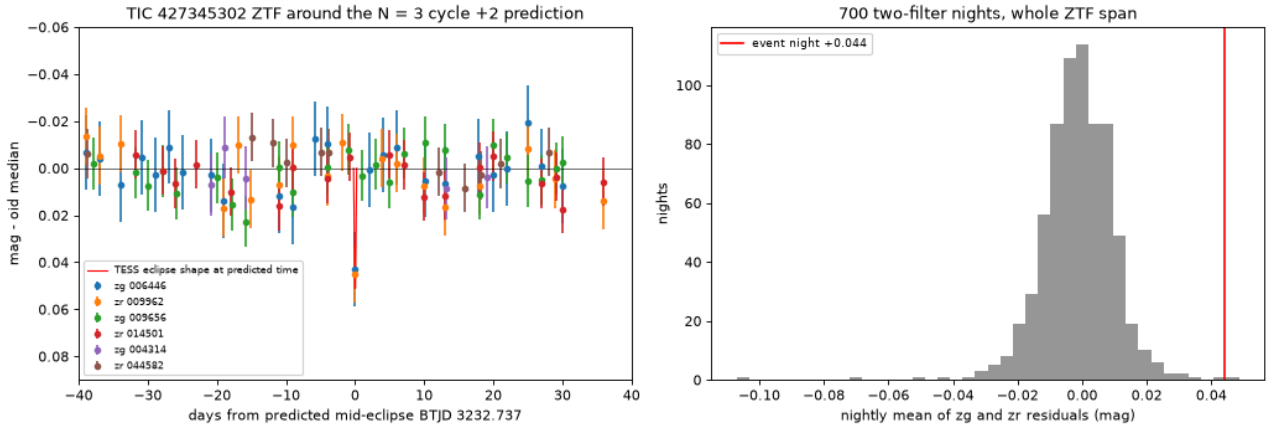


Figure A74b. Left: ZTF g and r photometry of TIC 427345302 within 40 d of the predicted mid-time of the eclipse between the two TESS events, BTJD 3232.737 (2023 October 15), each point a ZTF epoch relative to its object's median, with the TESS eclipse shape at the predicted time. Right: the distribution of the nightly mean of the g and r residuals over the 700 ZTF nights with both filters; the event night is the red line.

Identifiers	TIC 427345302; Gaia DR3 2226943914530555776 (RUWE 1.86); no other catalogue name
Position (J2000)	344.194757, +71.598251 deg (TIC)
Brightness	Tmag 12.10; G 12.55
Stars	TIC: T_{eff} 6168 K, 1.40 $R_\odot$, 1.18 $M_\odot$, at 540 pc (parallax 1.85 mas). Gaia astrometric `Orbital` solution: P = 173.19 ± 0.87 d, e = 0.062 ± 0.078, a_0 = 0.401 mas (0.217 AU), significance 25
Period	173.8755 ± 0.0005 d: the interval between the two TESS eclipses divided by three. TESS alone allowed 1, 2, 3, 5 or 6 cycles between them (the other divisors put an eclipse in flat observed data); ZTF excludes 5 and 6 (flat on predicted nights) and shows the eclipse at the 3-cycle prediction, while 1 and 2 cycles predict nothing on that night. Gaia 173.19 ± 0.87 d (+0.8σ)
Epochs	BTJD 2884.9860 ± 0.0013 = BJD_{TDB} 2459884.9860 (sector 58); 3406.6126 ± 0.0006 (sector 77)
Eclipse	4.62% (sector 58) and 4.6–5.3% (sector 77) deep, T_{14} 10.8 h, ingress about 4.2 h (T_{12}/T_{14} ≈ 0.4), V-shaped: grazing. Both events fully covered; no flagged cadences at the first; the second ends 0.3 d before a 10-d gap. No other dip deeper than 3% in nine sectors
Third eclipse (ZTF)	

	On the night BTJD 3232.68–3232.72 (2023 October 15), with mid-eclipse predicted at 3232.737 ± 0.002, ZTF reads the star 0.043 ± 0.016 mag faint in g and 0.045 ± 0.012 mag faint in r (0.051 and 0.048 expected from the TESS shape); the nights 21 h before and 23 h after are at baseline. Among the 700 ZTF nights with both filters, one other is as faint, so the chance that the one two-filter night inside a predicted window is this faint by accident is 0.1–0.3%. ZTF resolves the star from every neighbour (nearest G 15.4 at 40″). Comparison stars from the same exposures were not checked (the IRSA service failed during the check); that is the item that keeps the status at probable
Secondary (ZTF)	The only other faint ZTF nights, BTJD 1743.9 (five epochs, 0.04–0.06 mag, two fields and both filters) and 2091.9 (four epochs, 0.03–0.07 mag), are 2.00 cycles apart at phase 0.438, inside Gaia's predicted secondary window (phase 0.38–0.50 at 95%); the nights around them are at baseline. A secondary as deep as the primary, about 5% in g. Not a tested prediction: it was found in the same archival check
Orbit	The secondary phase gives $e \cos \omega \approx -0.10$ ($e \geq 0.10$; Gaia 0.062 ± 0.078). The TESS eclipses fall at Gaia's second conjunction (offsets +13 and +15 d against an uncertainty of −4 to +31 d)
Pixel attribution	TESS difference images of both events: 4.7% and 4.5% at the target pixel, deficit centroid within 0.15 px of the target; the neighbours that could host it (G 14.22 at 59″, G 15.41 at 40″) lose 0.1–0.7% where they would need 21% and 64%. The ZTF dimming is of the resolved target
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP, general web search; SIMBAD has no entry; ADS full text (TIC and Gaia identifiers) none. Kostov table 2 only. A TESS-SPOC data-validation report for sector 52 carries a threshold-crossing event at 1.143 d, 0.3% deep, unrelated to the eclipses (Table A1)
Open issues	The companion. For a dark companion the photocentre orbit gives $m_2 \approx 0.5\ M_\odot$, but a secondary as deep as the primary needs a companion of comparable surface brightness, so the photocentre understates the orbit and the companion is more massive; a near-equal pair grazing each other at both conjunctions would give the similar depths seen. Not modelled here
Next eclipses	Primaries BTJD 4449.87 (2027-02-13), 4623.74 (2027-08-06); secondaries 4352.11 (2026-11-07), 4525.98 (2027-04-30); ±0.05 d. No TESS sector covers the star before 2027; at Dec +71.6° the November 2026 secondary is observable from northern sites
Method	Short-period search of the edge-on astrometric sample (Section 2.6; the inspection of the deepest single events); checked by independent re-analysis (Section 2.5); third eclipse from ZTF archival photometry

Note: probable. The third eclipse rests on one ZTF night whose comparison stars have not been checked, and the secondary identification comes from the same archival check rather than a tested prediction. One clean eclipse of either kind from the ground would make the system confirmed.

An F-type star with a 173-d Gaia astrometric orbit seen nearly edge-on shows two 4.6% grazing eclipses in TESS, three cycles apart. ZTF, read at the predicted time of the cycle between them, shows the star faint by the same amount for one night. The two TESS events alone allowed five spacings: one, two, three, five or six cycles between them. The ZTF record is flat on the nights that five or six cycles predict, which removes those two. The ZTF eclipse falls on the night that three cycles predict; one or two cycles predict nothing on that night. The same ZTF data hold two nights of a secondary as deep as the primary, at one phase, 0.438. A secondary that deep means the companion is a star of comparable surface brightness, not the faint dwarf that a dark-companion reading of the photocentre would give. The system is 12th magnitude with 5% eclipses lasting 11 h, in reach of a small telescope. The next secondary falls on 2026-11-07 and the next primary on 2027-02-13.

S3.3 Eclipsing systems from the Gaia-orbit period folds (A5–A12,

A75–A109)

A5 HD 106770 = TIC 22388285

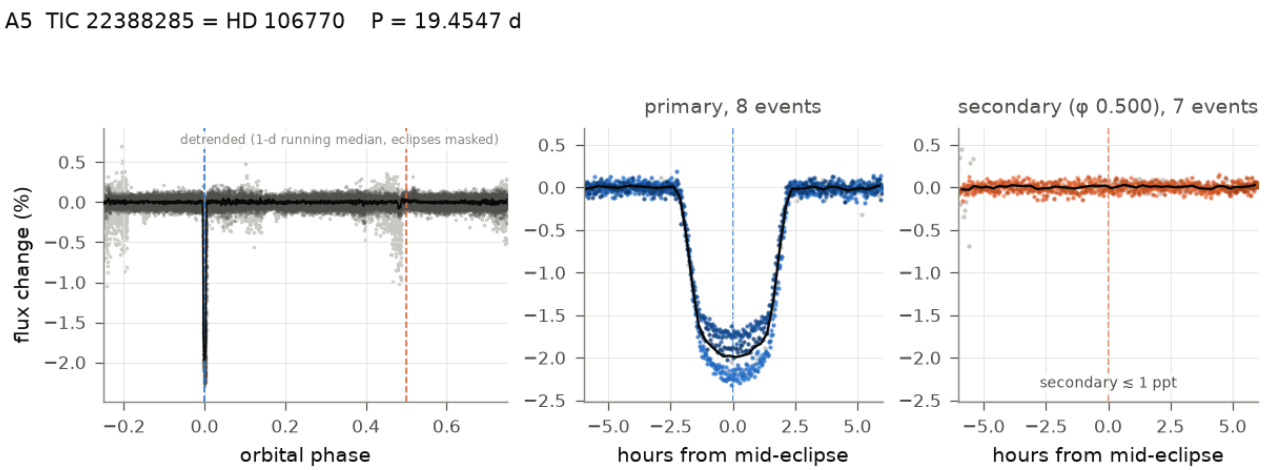


Figure A5. HD 106770, QLP SAP flux folded on the search period P = 19.4547 d (left panel detrended with a 1-d running median, eclipses masked): a ~20-ppt eclipse (eight events in the fold; the search's box gave 19 ppt and 2.9 h; the raw-flux timing fit to the seven fully covered events gives 20.3 ppt and 4.4 h); any secondary is ≲ 1 ppt.

Identifiers	TIC 22388285; HD 106770; TYC 8234-1406-1; Gaia DR3 6130370305216737408
Position (J2000)	184.19851, −47.74244 deg (TIC)
Brightness	Tmag 8.98; G 9.43
Stars	G5V (TIC: T_{eff} 5651 K, 1.14 $R_{\odot}$) at 97 pc, with a spectroscopic companion
Period	19.454722 ± 0.000010 d, timing fit of seven events (search fold 19.4547 d; Gaia SB1C: 19.4564 ± 0.0031 d)
Epoch	BTJD 3494.45294 ± 0.00039 = BJD_{TDB} 2460494.45294 (timing fit; the search's box epoch was 1587.8864)
Primary eclipse	20.3 ppt, 4.4 h in raw QLP SAP flux (timing fit; the search's box gave 19 ppt, 2.9 h)
Secondary eclipse	≲ 1 ppt
Orbit	Gaia's solution is circular (SB1C); no conjunction-phase test is possible
Catalogue status	In no variable catalogue. SIMBAD "**", known SB, zero references. Not in any catalogue searched (Section 2.7); Algol was recovered as a positive control
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. The period agreement with Gaia is by construction of the fold; the orbit is circular, so no conjunction-phase test is possible; one pipeline; no pixel test. A difference image or a second pipeline would confirm it

A G dwarf at 97 pc whose spectroscopic orbit Gaia published, and whose 19-ppt eclipses are uncatalogued. The contamination ratio is 0.003, so the aperture holds little light from anything else, and the period agrees with Gaia's to within its quoted error. Status: probable (see the Status row).

A6 BD−13 1046 = TIC 348773952

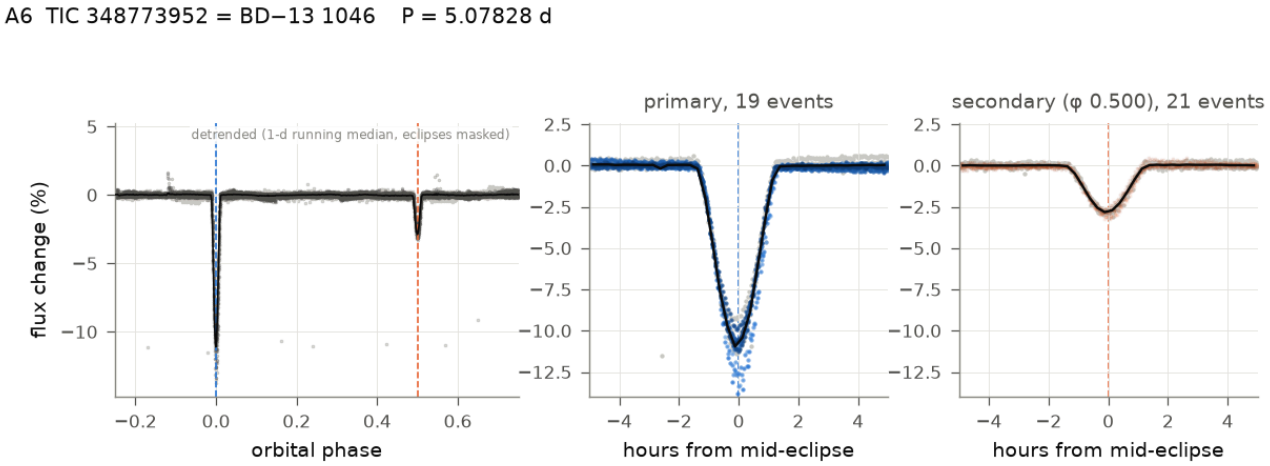


Figure A6. BD−13 1046, QLP SAP flux folded on the search period P = 5.07828 d (left panel detrended with a 1-d running median, eclipses masked): a ~110-ppt primary and a ~28-ppt secondary at phase 0.5.

Identifiers	TIC 348773952; BD−13 1046; TYC 5329-1207-1; Gaia DR3 2987862104761137280
Position (J2000)	75.15913, −13.71899 deg (TIC)
Brightness	Tmag 9.60; G 10.18
Stars	K dwarf (TIC: T_{eff} 4912 K, 0.74 $R_{\odot}$) at 63 pc, with a spectroscopic companion
Period	5.0782783 ± 0.0000010 d, timing fit of 13 primaries (search fold 5.07828 d; Gaia SB1: 5.078108 d)
Epochs	Primary BTJD 3609.56921 ± 0.00016 = BJD_{TDB} 2460609.56921; secondary BTJD 3657.81301 ± 0.00016 (timing fits; the search's box epoch was 1441.1441)
Primary eclipse	110 ppt, 2.55 h in raw QLP SAP flux (timing fit; the search's box gave 114 ppt in a 1.9-h box)
Secondary eclipse	28 ppt, 2.55 h, at phase 0.5
Orbit	Eclipse falls on Gaia's predicted conjunction: Δφ = +0.006
Catalogue status	No eclipsing classification anywhere; RAVE lists solar-like activity and Schmitt et al. (2026) a period only. Known SB. Not in any catalogue searched (Section 2.7)
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. The eclipse falls on Gaia's predicted conjunction, but the period agreement is by construction of the fold, and there is one pipeline and no pixel test

The deepest of A5–A7, a K dwarf 63 pc away eclipsing by about 11%, with a secondary eclipse at phase 0.5. It had been catalogued only for activity and a period, not as an eclipsing binary.

A7 UCAC4 130-138342 = TIC 271321097

A7 TIC 271321097 = UCAC4 130-138342 P = 13.9107 d

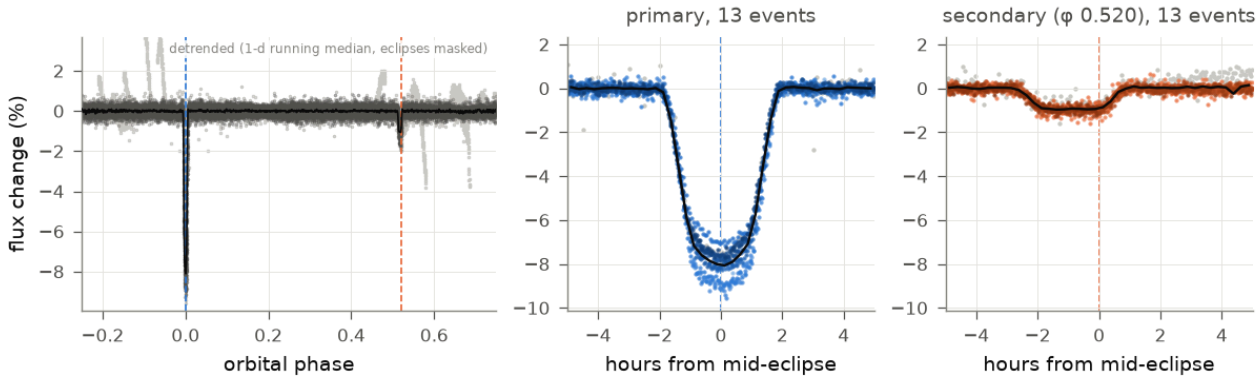


Figure A7. UCAC4 130-138342, QLP SAP flux folded on the search period P = 13.9107 d (left panel detrended with a 1-d running median, eclipses masked): a primary of ~70–80 ppt (in a contaminated aperture; the search's box gave 68 ppt, the raw-flux fit 80 ppt) and a ~10-ppt secondary at phase 0.52.

Identifiers	TIC 271321097; UCAC4 130-138342; Gaia DR3 5826591602034536320
Position (J2000)	233.04765, −64.14559 deg (TIC)
Brightness	Tmag 10.93; G 11.69
Stars	K dwarf (TIC: T_{eff} 4223 K, 0.71 $R_{\odot}$) at 80 pc, with a spectroscopic companion
Period	13.9106635 ± 0.0000071 d, timing fit of 12 primaries (search fold 13.9107 d; Gaia SB1: 13.915147 d)
Epochs	Primary BTJD 3685.73790 ± 0.00034 = BJD_{TDB} 2460685.73790; secondary BTJD 3887.68375 ± 0.0007 (timing fits; the search's box epoch was 1626.9584)
Primary eclipse	80 ppt, 3.8 h in raw QLP SAP flux (timing fit; the search's box gave 68 ppt; TIC contamination ratio 0.41, so diluted)
Secondary eclipse	10 ppt, 3.4 h, at phase 0.52
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = −0.017
Catalogue status	Only a Gaia DR2 large-amplitude-variable flag, untyped. The VSX eclipsing stars 107″ away have other periods. Not in any catalogue searched (Section 2.7). **Prior report (ADS full text):** Bryant et al. (2023), vetting giant-planet candidates around low-mass stars, identified TIC 271321097 as an eclipsing binary from the extended-mission 10-min light curve (the primary-mission 30-min data had shown only the secondary eclipses, the primaries falling in gaps) and excluded it; no period, epoch or catalogue entry was published. This is the only one of the 104 with a prior mention of its eclipses
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. 41% of the aperture light is foreign; the eclipse falls on Gaia's conjunction, but the period agreement is by construction of the fold, and there is one pipeline and no pixel test. Bryant et al. (2023) saw the eclipses in TESS-SPOC data, which supports their reality but not the attribution

The aperture is crowded (41% of its light is attributed to neighbours), so the depth is a lower limit on the eclipse of the star itself. The attribution rests on the period agreeing with Gaia's orbit and the eclipse falling on Gaia's conjunction; no pixel-level attribution test is reported.

A8 HD 131614 = TIC 402800288

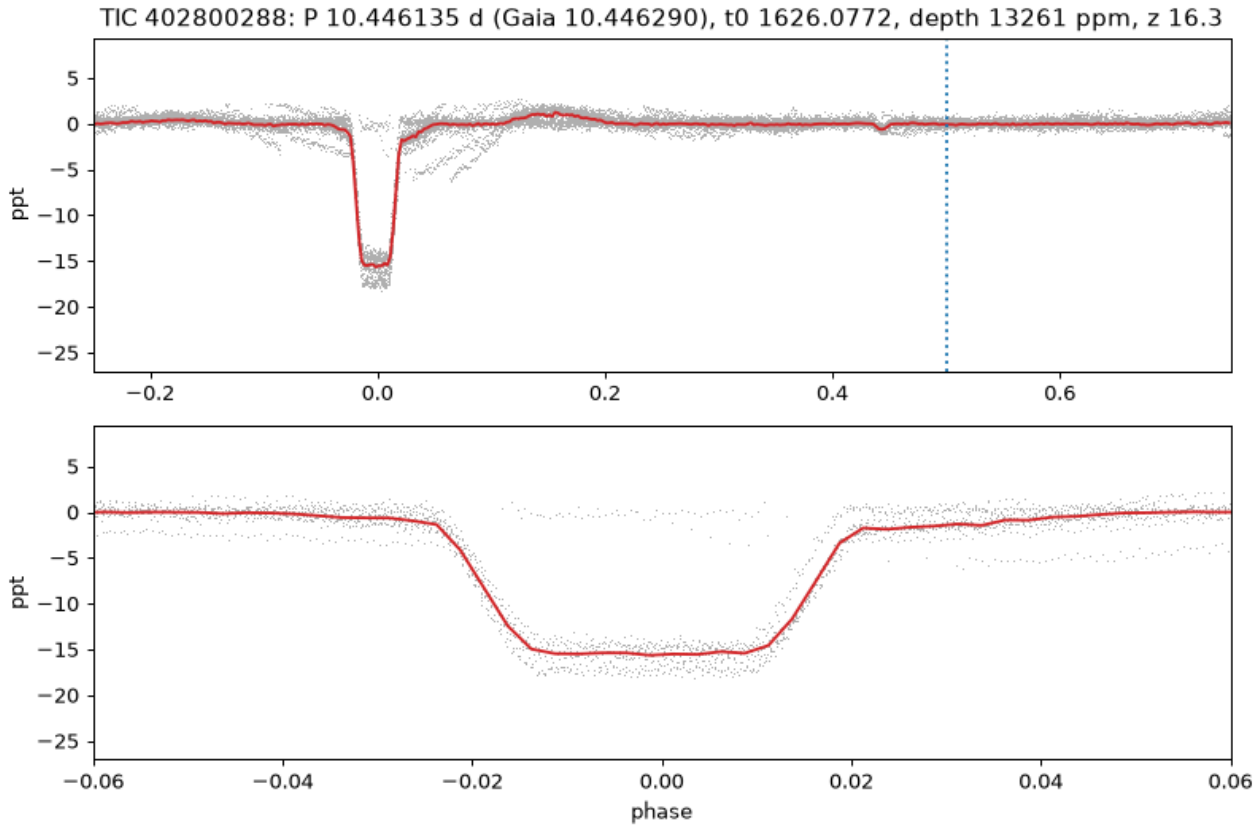


Figure A8. HD 131614, the discovery fold on P = 10.4461 d in the search's detrended flux, in which the eclipse is suppressed toward zero in the sectors after 56 (the points near 0 ppt inside the eclipse); in raw SAP flux with per-event baselines it is 18.2 ppt and about 11 h (Section 3.3), stable across all ten sectors.

Identifiers	TIC 402800288; HD 131614; TYC 9266-1012-1; Gaia DR3 5793878347940289152
Position (J2000)	224.85938, −72.53302 deg (TIC)
Brightness	Tmag 8.47; G 8.91
Stars	Evolved primary (TIC: T_{eff} 5972 K, 5.8 $R_{\odot}$, giant) at 405 pc, with a spectroscopic companion
Period	10.4462098 ± 0.0000098 d, timing fit of 21 events (search fold 10.4461 d; Gaia: 10.446290 d)
Epoch	BTJD 3391.45705 ± 0.00071 = BJD_{TDB} 2460391.45705 (timing fit; the search's box epoch, 1626.0772, was 0.12 d off the fitted line)
Primary eclipse	18.2 ppt against flux 0.75–1.75 durations away (15.1–15.7 ppt with a local quadratic baseline: the flux curves into the eclipse), about 11 h (trapezoid fits in QLP, TESS-SPOC and TARS give 10.9–11.2 h; the 13.2 h first quoted from per-event boxes is not reproduced), stable across all 10 sectors (13.3 ppt in the search's fold)
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = +0.004 (0.9σ, Monte Carlo through Gaia's errors)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s65, s66: 15.1 ppt, 10.9 h, mid-times within 0.1–0.3 min of QLP's); TARS; SPOC's own multi-sector search detected the eclipse (MES 91) with a centroid 1.4 ± 2.8″ from the catalogue position. The QLP detrended flux reduces the eclipse depth from 14.7 to 0.8 ppt in some sectors
Pixel attribution	TESScut difference images of two eclipses each in s65 and s38: the deficit has the star's own image shape (residual 3–4%), its centroid 0.2″ and 0.7″ from the light centroid, and the same depth in 1×1, 3×3 and 5×5 pixel apertures. No Gaia source within 60″ is brighter than G 14.62; the brightest would need a 293% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not in any catalogue searched (Section 2.7): VSX, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025), tables 3 and 4; confirmed by independent cone searches. Known SB
Method	Gaia-orbit period fold (Section 2.4); checked by independent re-analysis (Section 2.5)
Status	

	Confirmed: 21 events in ten sectors at Gaia's period on the eccentric orbit's conjunction phase (0.9σ), the deficit on the target in difference images, and TESS-SPOC agreeing with QLP (both added after the pre-submission review)

A8 is the clearest case of the detrending artefact of Section 2.8. In QLP detrended flux the eclipse depth collapses in some sectors; in raw SAP flux with a local baseline around each event it is ~18 ppt in all ten sectors. The same raw-flux re-measurement rejects the three shallowest candidates of the fold (Section 3.8).

A9 HD 7601 = TIC 399009076

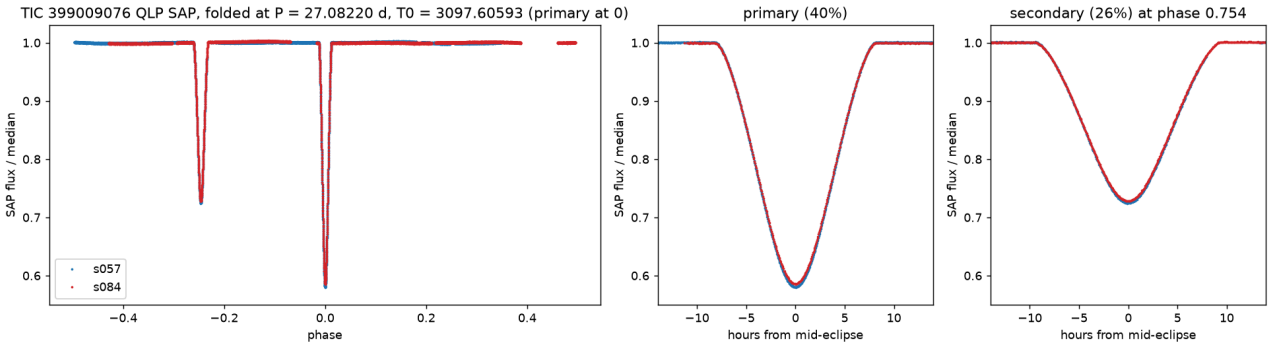


Figure A9. HD 7601, raw QLP SAP flux of sectors 57 and 84 folded on P = 27.0822 d: a 40%, 14-h V-shaped primary and a 26%, 16-h secondary at phase 0.754. The 13.4166-d period of the search's two-sector fold is an alias.

Identifiers	TIC 399009076; HD 7601; TYC 1747-1864-1; Gaia DR3 293150213945293824
Position (J2000)	19.09573, +23.78509 deg (TIC)
Brightness	Tmag 8.56; G 8.97
Stars	F-type (TIC: T_{eff} 6152 K, 3.5 $R_\odot$) at 274 pc; Gaia DR3 lists a double-lined (SB2) orbit
Period	27.082201 ± 0.000018 d from the five eclipses in sectors 57 and 84 (timing fit); the search's 13.4166 d (Gaia's 13.4234 d) does not phase the eclipses: the 13.42-d prediction at BTJD 2867.45 falls on 48 good cadences at flux 1.000
Epochs	T_0(primary) = BTJD 3097.60593 ± 0.00023 = BJD_{TDB} 2460097.60593; T_0(secondary) = BTJD 3253.43045 ± 0.00028, phase 0.754
Primary eclipse	39.8% (trapezoid; ~41% at minimum), 14.3 h, V-shaped (partial); three events (2853.866, 2880.948, 3585.086)
Secondary eclipse	26.1% (~27% at minimum), 16.5 h, at phase 0.754; two events (2874.280, 3605.499). From the phase and durations $e \approx 0.41$, $\omega \approx 10°$
Long baseline	ASAS-3 V (2003–2009, ~200 cycles earlier) folded on the TESS ephemeris: its three faintest points (0.62, 0.51, 0.34 mag; 0.62 mag = 43% = the TESS primary depth) fall at phases −0.001 to −0.005 and seven more at 0.748–0.758; at either 13.42-d period the same points scatter at random
Gaia orbit	Not a confirming match. Gaia's SB2 solution (P 13.4234 ± 0.0012 d, e 0.073, K_1 37.8, K_2 33.0 km/s, 14 transits, F2 = 12.8) fits the eclipses at no period: $2P_{Gaia}$ = 26.847 d differs from 27.082 by 100σ, the eclipse $e \cos \omega$ of 0.41 contradicts Gaia's 0.035, and $(M_1+M_2) \sin^3 i$ = 0.49 $M_\odot$ is far too small for an eclipsing G = 9.0 pair at 274 pc. Gaia's period is erroneous; the exact alias is not identified (DR3 publishes no epoch velocities for this star)
Pixel attribution	Difference images of all five events (sectors 57 and 84): the lost flux is a copy of the target's PSF with its centroid 0.09–0.44 px from the out-of-eclipse centroid, pointing away from the G 11.55 neighbour 6 px west; nothing within 36″ brighter than G 19.8. Only an unresolved companion inside the PSF cannot be excluded
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches; ADS full text). Known SB (Gaia)
Next eclipses	Primary: BTJD 4343.387 (2026-10-29 21:16 UT), 4370.469; secondary: 4336.718, 4363.800
Method	Gaia-orbit period fold (Section 2.4), retained under the supplementary criterion for deep signals (Section 2.4); period from the eclipse-timing fit (Appendix A)

A9 has the deepest eclipses found by the fold, but the fold assigned it an incorrect period. Its eclipses are so deep that they inflate the star's own random-period null. Its z score (3.6) therefore fell below the cut, and it was retained under the supplementary criterion for deep, high-S/N signals (Section 2.4). The fold at Gaia's period produced one smeared 20% dip. The conjunction-phase test, which compares the events with Gaia's conjunction *at Gaia's period*, passed spuriously, because the fold merged two different eclipses. An eclipse-timing fit (Appendix A) gives the 27.08-d ephemeris, which the raw sectors, ASAS-3 photometry two decades earlier and the pixels confirm. A9 is kept as a confirmed eclipsing binary at 27.08 d, with Gaia's orbit recorded as a discrepancy rather than a confirmation.

A10 HD 99010 = TIC 300395201

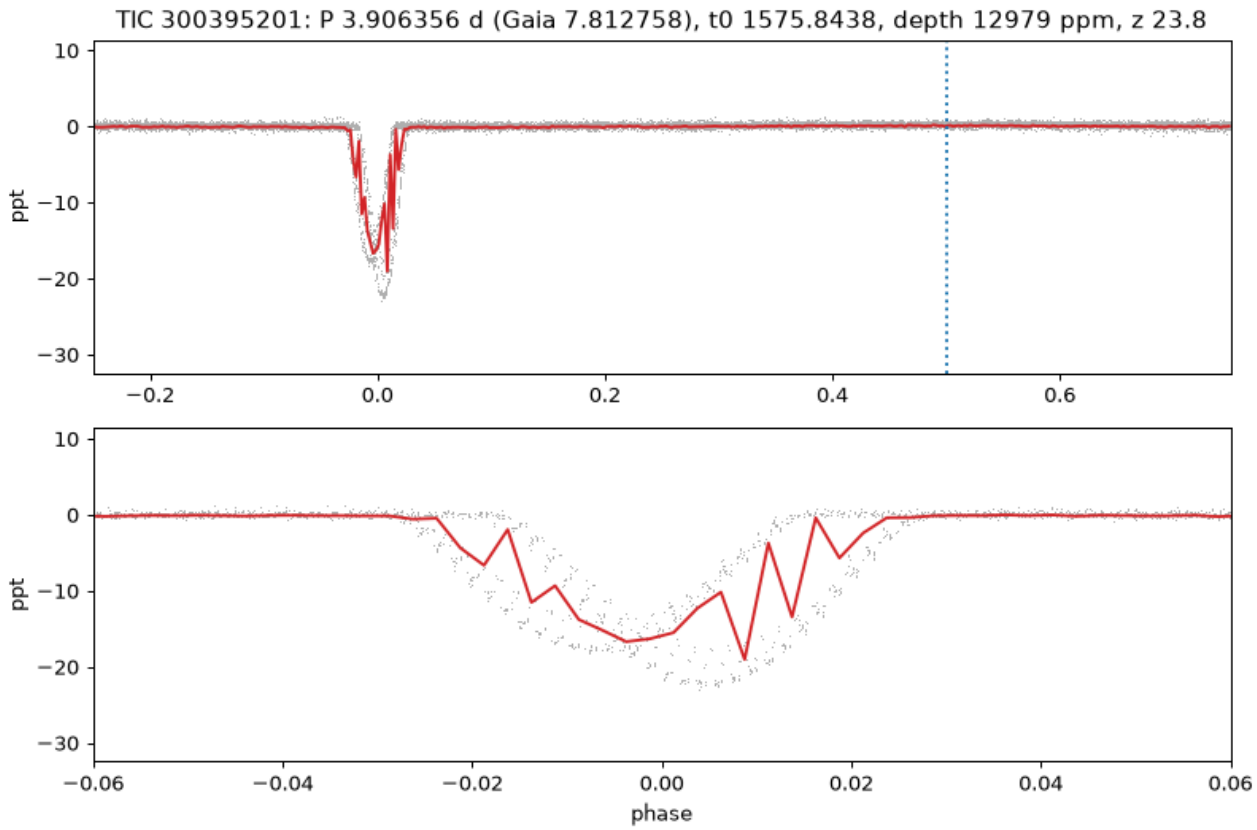


Figure A10. HD 99010, the discovery fold at P/2 = 3.906356 d, half the 7.8128-d orbital period, which stacks the two eclipses into one dip; the raw-flux timing fit gives 22.0 and 17.3 ppt for the two eclipses.

Identifiers	TIC 300395201; HD 99010; TYC 8617-227-1; Gaia DR3 5348342423752784384
Position (J2000)	170.78242, −53.10540 deg (TIC)
Brightness	Tmag 8.45; G 8.81
Stars	F-type primary (TIC: T_{eff} 6372 K, 2.2 $R_{\odot}$) at 175 pc; Gaia double-lined circular orbit (SB2C)
Period	7.8127421 ± 0.0000013 d, timing fit of 17 primaries and 20 secondaries (search fold 7.8128 d; Gaia: 7.812758 d)
Epochs	Primary BTJD 3384.50567 ± 0.0001 = BJD_{TDB} 2460384.50567; secondary BTJD 3427.43069 ± 0.00016 (timing fits; the search's box epoch was 1579.7618)
Eclipses	22.0 and 17.3 ppt, 4.0 and 3.8 h, alternating, in raw SAP flux (timing fit; the 18.3 and 9.3 ppt of the independent re-analysis came from detrended flux)
Orbit	Circular SB2C: no conjunction-phase test is possible
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches). Known SB
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. With a circular orbit no conjunction-phase test is possible, and the period agreement, the only attribution, is guaranteed by construction of the fold; one pipeline, no pixel test

Two eclipses per orbit at Gaia's double-lined period. With a circular orbit the attribution rests on the period agreement alone, which is why it is listed as probable.

A11 BD+40 3254 = TIC 159892657

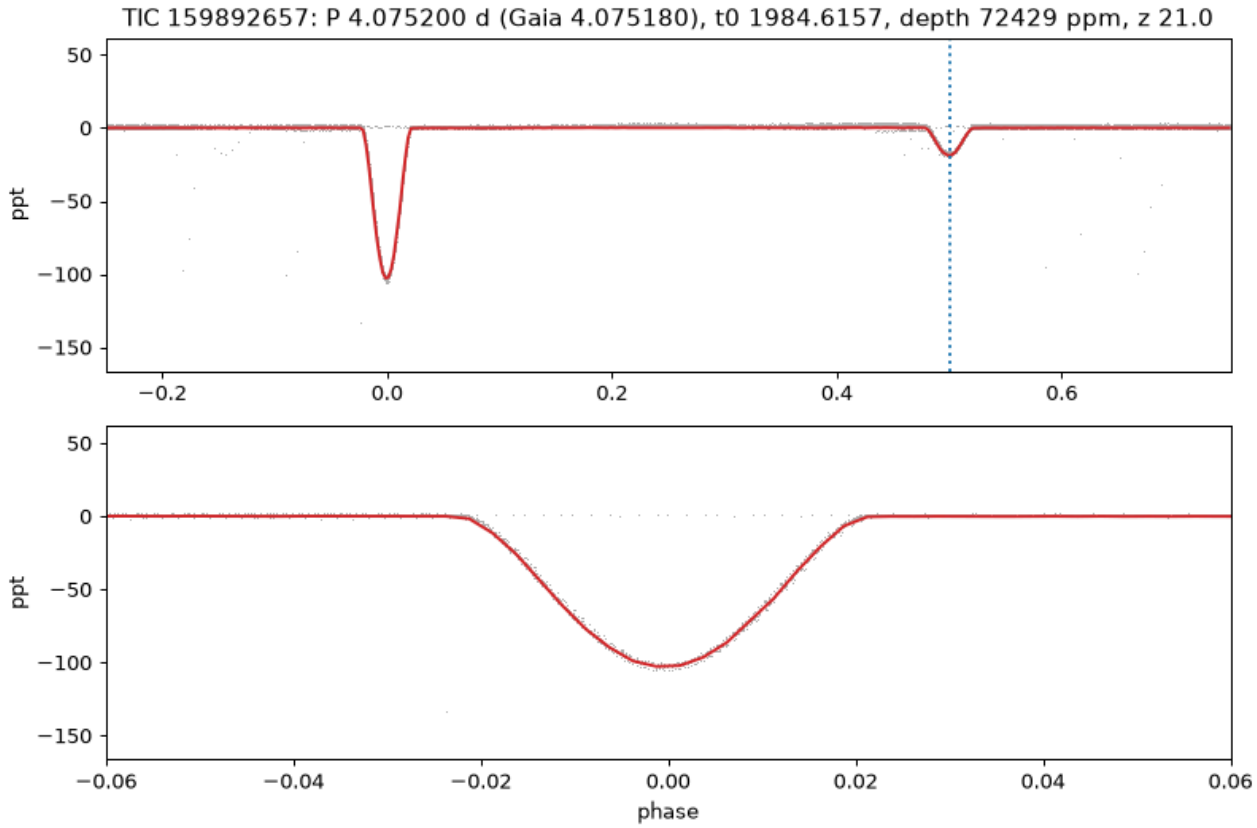


Figure A11. BD+40 3254, the discovery fold on P = 4.0752 d: a ~9% primary (the panel's own box depth reads 7.2%) and a 2% secondary at phase 0.498 (detrended search flux; the raw-flux fit gives 101 and 19 ppt).

Identifiers	TIC 159892657; BD+40 3254; HIP 87971 (TIC cross-match); TYC 3093-922-1; Gaia DR3 4611458011500418176
Position (J2000)	269.54374, +40.11201 deg (TIC)
Brightness	Tmag 8.42; G 8.79
Stars	F-type primary (TIC: T_{eff} 6192 K, 1.7 $R_{\odot}$) at 129 pc, with a spectroscopic companion
Period	4.0751904 ± 0.0000009 d, timing fit of 46 primaries and 40 secondaries (search fold 4.0752 d; Gaia: 4.075180 d)
Epochs	Primary BTJD 3133.82003 ± 0.00011 = BJD_{TDB} 2460133.82003; secondary BTJD 3184.76020 ± 0.00014 (timing fits; the search's box epoch was 1984.6157)
Primary eclipse	101 ppt, 4.0 h in raw SAP flux (timing fit); ~9% in 7 of 8 sectors in the independent re-analysis (sector 26's core diluted by the detrend; its minimum still 3.7%); 7.2% in the search's fold
Secondary eclipse	19 ppt, 3.8 h, at phase 0.498
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = +0.003
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP and TESS-SPOC (s74, s79, s80: primary 97.8–98.3 ppt, secondary 17.7–17.9 ppt, 3.94 h, secondary at phase 0.4998, mid-times within 0.1 min of QLP's in s74 and s79); SPOC's own search detected both eclipses (MES 4630). Against the SPOC products the 2024 primaries fall 0.9 ± 0.15 min earlier than this entry's ephemeris and the period differs by 3×10^{-6} d (3.3σ): this star's QLP sector 80 light curve runs 2.7 min late against SPOC 2-min and TESS-SPOC on the same seven events (one star, one sector: not a statement about the pipeline), and it entered the timing fit, so the errors quoted here are understated by about a minute at 2024 epochs
Pixel attribution	

	TESScut difference images of seven primaries in s74 and six in s40: the deficit has the star's own image shape (residual 2%), its centroid 0.1″ from the light centroid and within 0.2 px of the catalogue position. All 12 Gaia sources within 60″ are G ≥ 17.3 and would need eclipses of 25,000% or more. A companion within about 1″ is not excluded
Catalogue status	Listed only in Kostov et al. (2025), table 2, as an unvetted machine-learning candidate. Known SB
Method	Gaia-orbit period fold (Section 2.4)
Status	Confirmed: 86 eclipses in eight sectors at Gaia's period on the conjunction phase, the deficit on the target in difference images, and SPOC 2-min and TESS-SPOC agreeing with QLP (both added after the pre-submission review)

A 9% eclipse on an eighth-magnitude star, listed only as an unvetted machine-learning candidate (Kostov et al. 2025). Both the conjunction phase and the near-0.5 secondary agree with Gaia's nearly circular orbit.

A12 HD 217492 = TIC 436758044

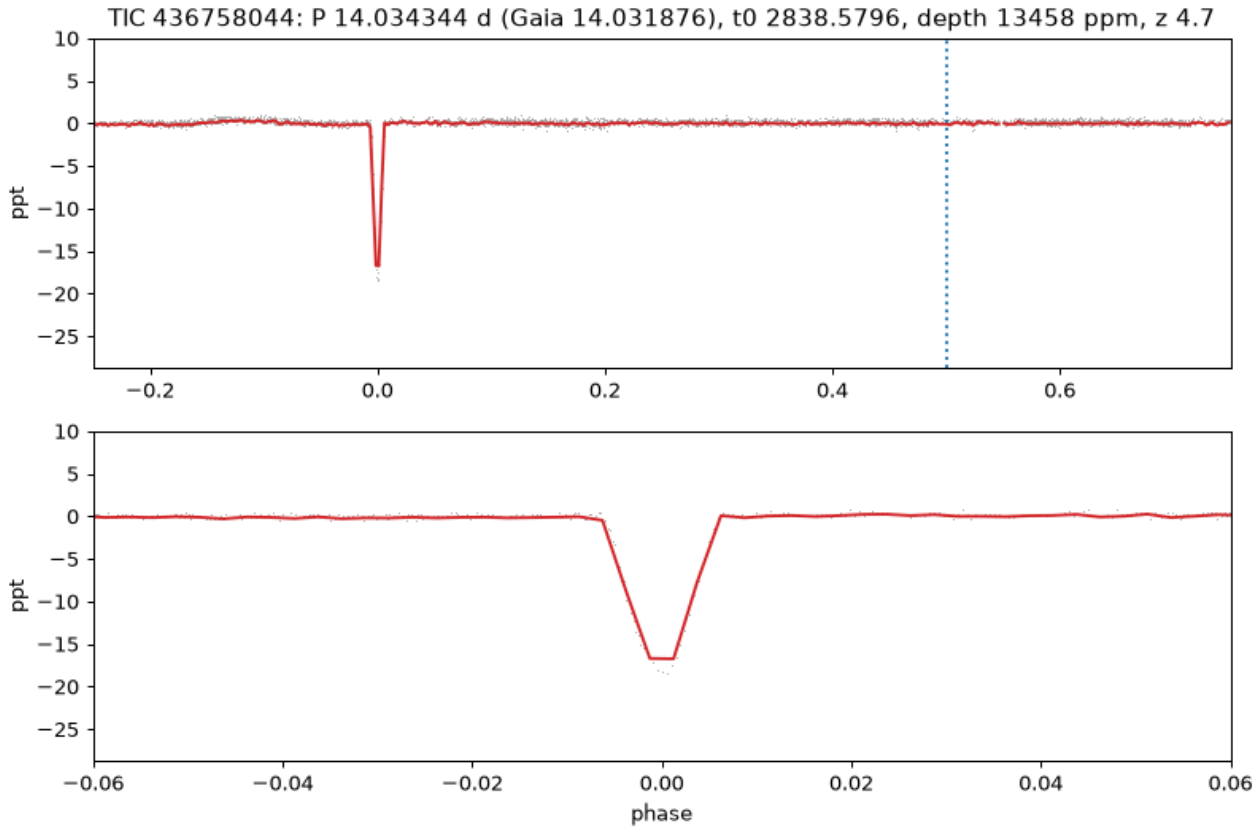


Figure A12. HD 217492, the discovery fold on P = 14.0343 d (detrended QLP search flux; the panel title gives z 4.7): a narrow eclipse whose box depth is 13.5 ppt and whose deepest bins reach about 17 ppt; the raw-flux fit gives 18.2 ppt, and TESS-SPOC confirms the events.

Identifiers	TIC 436758044; HD 217492; HIP 113630 (TIC cross-match); TYC 2242-1176-1; Gaia DR3 2843881366341547520
Position (J2000)	345.19954, +26.50218 deg (TIC)
Brightness	Tmag 8.33; G 8.66
Stars	F-type primary (TIC: T_{eff} 6526 K, 2.5 $R_{\odot}$) at 193 pc; Gaia double-lined orbit (SB2)
Period	14.0343984 ± 0.0000110 d, timing fit of the two QLP events with the cycle count (51) from the search period (search fold 14.0343 d; Gaia: 14.031876 d; 4σ apart)
Epoch	BTJD 3203.47040 ± 0.00028 = BJD_{TDB} 2460203.47040 (timing fit; the search's box epoch was 2838.5796). TESS-SPOC events at BTJD 2852.61, 3568.36 and 3582.40
Primary eclipse	

	18.2 ppt, 4.0 h in raw QLP SAP flux (trapezoid fits to the two events); 13.5 ppt in the search's detrended fold (minimum bin 14.1 ppt at phase −0.003); 8.2, 8.9 and 9.1 ppt in a ±0.10-d core in TESS-SPOC, ~16 ppt undiluted (fold minimum 15.98 ppt)
Orbit	Conjunction offset +0.043 ± 0.008: a **5.5σ tension** with Gaia's SB2 elements
Pipelines agreeing	QLP, TESS-SPOC
Pixel attribution	In the SPOC target-pixel files all three events dip the summed flux 13.3–14.2 ppt while the flux-weighted centroid moves ≤ 0.0015 px: the eclipsing source is within ~0.1 px (~2″) of HD 217492
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches). Known SB
Open issues	The period and conjunction tensions are with Gaia's SB2 elements, or indicate an unresolved third body. The Gaia epoch RVs that would allow re-deriving the orbit are not public: both Gaia epoch-RV tables hold no rows for this source, so resolution requires the Gaia DR4 epoch RVs
Method	Gaia-orbit period fold (Section 2.4), retained at $z = 4.7$ under the supplementary criterion for deep signals (Section 2.4); confirmed by independent re-analysis with the orbit caveat, after an epoch-RV check (Section 2.5)

The eclipse is real in two pipelines and falls on the target to within ~2″, so it is attributed to HD 217492. Either Gaia's double-lined elements for this star are biased, or a third star in the system perturbs the orbit or the line profiles. The 5.5σ conjunction offset appears only when the uncertainties of all of Gaia's elements are propagated into the conjunction phase (Section 2.2).

A75 TYC 3255-1824-1 = TIC 202282515

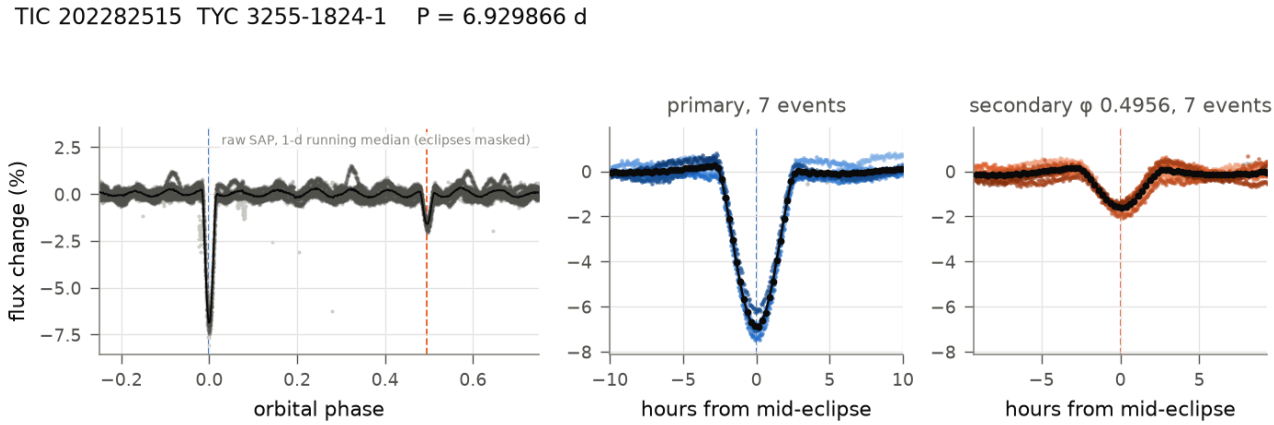


Figure A75. TYC 3255-1824-1, raw QLP SAP flux of sectors 57 and 84, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on $P = 6.929866$ d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.4956 (7 events of each kind plotted; the 6 fully covered primaries are those in the timing fit): a 68-ppt, 5.0-h V-shaped primary and a 16-ppt, 4.6-h secondary.

Identifiers	TIC 202282515; TYC 3255-1824-1; Gaia DR3 394192545103488768
Position (J2000)	5.13060, +49.24466 deg (TIC)
Brightness	Tmag 9.66; G 9.95
Stars	TIC: T_{eff} 6840 K, 2.12 $R_\odot$ (Gaia GSP-Phot 6978 K) at 325 pc; Gaia single-lined orbit (SB1, e = 0.012 ± 0.006, K_1 46.3 km/s). The mass function, 0.071 $M_\odot$, gives $M_2 \geq 0.70$ $M_\odot$ for $M_1 = 1.48$ $M_\odot$; the depth ratio, 0.23, indicates a K-dwarf companion of roughly 4500–5000 K
Period	6.929866 ± 0.000015 d, timing fit of 6 primaries; 7 secondaries fitted at that period (Gaia: 6.929767 ± 0.000185 d; +0.5σ)
Epochs	Primary BTJD 3111.42326 ± 0.00065 = BJD_{TDB} 2460111.42326; secondary BTJD 3114.85767 ± 0.00293 (phase 0.49560 ± 0.00042)
Primary eclipse	67.6 ppt, 5.00 h (ingress 1.95 h), V-shaped, in raw QLP SAP flux (trapezoid fit to the fold of 6 events). The sector-84 events are 8% shallower than those of sector 57 (medians 64.4 and 70.4 ppt) in every pipeline and in TESScut photometry, a sector-dependent dilution; alternate events within sector 57 agree to within the event scatter (about 3 ppt)

Secondary eclipse	15.5 ppt, 4.6 h, at phase 0.4956 (7 events). The secondary mid-times scatter by about ±10 min (O−C rms 9.3 min against formal errors of 1–2 min), event by event identically in QLP, TESS-SPOC and TARS
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at +0.0049 ± 0.0028 (+1.7σ), secondary on conjunction B at −0.0054 ± 0.0023 (−2.3σ); carried to the TESS epochs with Gaia's period, +0.0103 ± 0.0103 (+1.0σ) and −0.0002 ± 0.0117 (0.0σ). The mean of the two offsets is on Gaia's conjunctions to 0.0003 in phase. $e \cos \omega = -0.0069 \pm 0.0007$ from the eclipse spacing, against Gaia's +0.0087 (+0.0071/−0.0065): 2.3σ apart, which is the opposite-signed split of the two offsets
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s57, s84: 68.6 and 16.7 ppt against QLP's 69.1 and 16.7 ppt; matched mid-times +0.03 ± 0.13 min over 6 primaries and +0.15 ± 0.49 min over 7 secondaries); SPOC 2-min SAP and PDCSAP (s57: −0.01 ± 0.14 min over 4 primaries); TARS. Depths agree within 1.5% across pipelines. SPOC's s57 search detected the primary as a TCE at 6.92985 ± 0.00014 d (MES 159), 0.09 ± 2.50″ from the out-of-transit centroid, and the secondary as a second TCE (MES 39); the TESS-SPOC s84 search detected the primary (MES 95). None was flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 4 primaries in s57 and 2 in s84: deficit centroid 0.1″ from the light centroid in both, a scaled copy of the star's image (residual 4%), depth 61 ppt (s57) and 55 ppt (s84) alike in 1×1, 3×3 and 5×5 pixel apertures (the eclipse-core median of a V-shaped event). The brightest Gaia source within 60″, G 15.62 at 57″, would need a 1250% eclipse
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	The 2.3σ difference between the photometric and Gaia values of $e \cos \omega$, and the ±10-min scatter of the secondary mid-times, which the predicted secondary times below do not include
Next eclipses	Primary: BTJD 4331.080 (2026-10-17 13:47 UTC), 4338.010, 4344.939; secondary: BTJD 4327.584 (2026-10-14 01:54 UTC), 4334.514 (allow ±12 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A detached F-type star with a K-dwarf companion on a nearly circular 6.93-d orbit, seen at a moderately high impact parameter (b ≈ 0.75 of the sum of the radii), so that the 68-ppt primary is partial and V-shaped. Six primaries and seven secondaries in two sectors fall on Gaia's two conjunctions, the deficit is on the target in difference images, and TESS-SPOC and SPOC 2-min reproduce QLP's depths and mid-times to 0.1 min. The photometric $e \cos \omega$ differs from Gaia's by 2.3σ, but both give $e < 0.02$, and the difference appears only as an opposite-signed split of the two conjunction offsets. The secondary times scatter by ±10 min in every pipeline, so the scatter lies in the star or in the local baseline of the shallow eclipse, not in QLP; further sectors would show whether it is periodic.

A76 TYC 3256-1533-1 = TIC 202710940

TYC 3256-1533-1 = TIC 202710940 P = 3.306372 d

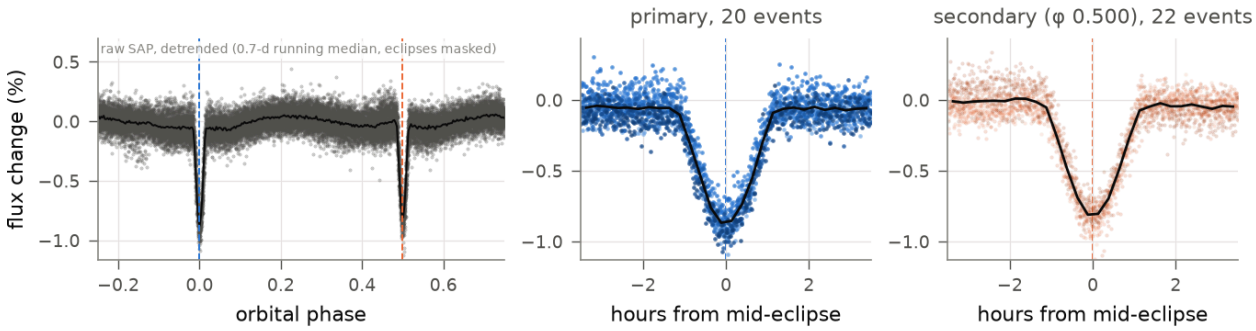


Figure A76. TYC 3256-1533-1, raw QLP SAP flux of sectors 57, 84 and 85, sector medians removed and detrended by a 0.7-d running median with the eclipses masked, folded on P = 3.3063718 d (left), with zooms on the primary and on the secondary at phase 0.500, each event normalised to its own baseline (20 primaries and 22 secondaries plotted; the 18 and 20 fully covered events are those in the timing fit): two nearly equal grazing eclipses, 8.1 and 7.8 ppt deep and both 2.25 h long. The out-of-eclipse flux carries 2–3 ppt of spot modulation.

Identifiers	TIC 202710940; TYC 3256-1533-1; Gaia DR3 391596224492510592
Position (J2000)	7.91761, +49.89143 deg (TIC)
Brightness	Tmag 9.48; G 9.92
Stars	TIC: T_{eff} 5923 K, 2.50 $R_\odot$ (the combined light of both stars) at 277 pc; Gaia double-lined orbit (SB2, e = 0.010 ± 0.004, K_1 95.5, K_2 96.8 km/s): a near-twin pair (q = 0.99) with M $\sin^3 i$ = 1.23 and 1.21 $M_\odot$, two slightly evolved G0-type stars of 1.6–1.8 $R_\odot$ each; at the inclination implied by the grazing eclipses, $i \approx 75$–77°, the masses are about 1.3–1.4 $M_\odot$. Spot modulation of 2–3 ppt semi-amplitude with a period of 3.2–3.7 d per sector, consistent with synchronous rotation
Period	3.3063718 ± 0.0000018 d, timing fit of 18 primaries; 20 secondaries give 3.3063711 ± 0.0000018 d (Gaia: 3.306368 ± 0.000030 d; +0.13σ). Twin eclipses: the period is twice the fold period, 1.653192 d. The two eclipses have the same shape and duration and differ in depth by only 4%, so the photometry alone does not separate P from P/2; Gaia's double-lined orbit at 3.306 d does
Epochs	Primary BTJD 3299.60466 ± 0.00020 = BJD_{TDB} 2460299.60466; secondary BTJD 3294.64483 ± 0.00020 (phase 0.49992)
Primary eclipse	8.11 ± 0.33 ppt, 2.25 h, V-shaped (grazing), in raw QLP SAP flux (trapezoid fits to 18 events)
Secondary eclipse	7.82 ± 0.44 ppt, 2.25 h, at phase 0.49992 (20 events). The depth difference, 0.29 ± 0.13 ppt (2.2σ), is 0.5–2.5% in the other pipelines, so which eclipse is called primary is a 2σ choice
Orbit	At Gaia's epoch with the TESS period (880 cycles back): primary on conjunction B at +0.0030 ± 0.0013 (2.3σ), secondary on conjunction A at +0.0023 ± 0.0013 (1.7σ); carried to the TESS epochs with Gaia's period, +0.0041 ± 0.0080 (0.5σ) and +0.0034 ± 0.0077 (0.4σ). Both eclipses follow Gaia's conjunctions by a common 0.0027 in phase (13 min). The eclipse spacing, 0.4999, matches Gaia's predicted 0.4995 ± 0.0022 (e cos ω = −0.0001), so the common shift is not an e cos ω effect. With q = 0.99 the two stars are interchangeable
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s57, 7 events of each kind: median mid-time differences 0.03–0.21 min, depths within 5%); TESS-SPOC SAP and PDCSAP (s57, s84, s85, 17 primaries and 19 secondaries: within 0.21 min, depths within 4.3%); TARS. SPOC's s57 search detected the eclipses as a TCE at the half period, 1.653144 ± 0.000031 d (MES 123.5, 7986 ppm, 2.36 h, impact parameter 0.98), 0.52 ± 2.50″ from the out-of-transit centroid; with both eclipses folded together it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images in s57 and s84, 14 eclipses of both kinds stacked in each: deficit centroid 1.0″ and 0.9″ from the light centroid, within 0.10 px of the catalogue position, a scaled copy of the target's image (residual 8–9%), depth 6.6–7.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 32 Gaia sources within 60″, only G 14.12 at 23.5″ (would need a 39% eclipse) and G 14.33 at 45.4″ (47%) are bright enough; either would pull the deficit centroid by more than 20″, and both are excluded. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 table 3, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; north of ASAS-3 coverage. Kostov et al. 2025 table 2 lists only the neighbour TIC 202710934 (Tmag 13.4, 24″); no period is compared there, so this may, but need not, be this star's eclipses in the neighbour's aperture. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	The common 13-min (2σ) delay of both eclipses relative to Gaia's conjunctions. A light-travel-time effect from a third body cannot be tested against Gaia's single epoch
Next eclipses	Primary: BTJD 4327.886 (2026-10-14 09:09 UTC), 4331.193, 4334.499; secondary: BTJD 4329.539 (2026-10-16 00:49 UTC), 4332.846 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A near-twin pair of slightly evolved G stars (q = 0.99) on a circular 3.31-d orbit, seen at an impact parameter of about 0.93–0.95 of the sum of the radii ($i \approx 75$–$77°$ for radii of 1.6–1.8 $R_\odot$): the eclipses are grazing, which is why two equal stars give only 8-ppt dips. Because the two eclipses are equal to within 4% in every pipeline, the photometry alone cannot distinguish the period from its half; the double-lined Gaia orbit at 3.306 d, which a 1.65-d binary could not produce, fixes it, and the TESS period agrees with Gaia's to 0.13σ. Both eclipses fall on Gaia's conjunctions with a common 13-min delay at the 2σ level, which does not affect the attribution. The spot modulation near the orbital period is that expected of a tidally synchronised, active pair.

A77 BD+44 219 = TIC 196475343

TIC 196475343 BD+44 219 P = 6.370415 d

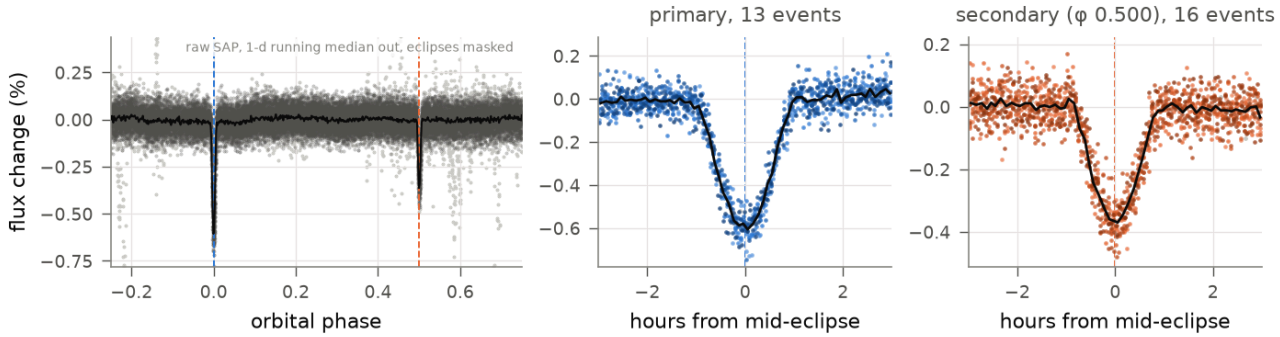


Figure A77. BD+44 219, raw QLP SAP flux of sectors 17, 57, 84 and 85, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on P = 6.370415 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.500, each event normalised to its own baseline (13 primaries and 16 secondaries plotted, including sector 17; the 11 and 12 fully covered events are those in the timing fit): a 6.1-ppt, 1.9-h primary and a 3.7-ppt, 1.7-h secondary, both grazing.

Identifiers	TIC 196475343; BD+44 219; TYC 3263-811-1; Gaia DR3 377002711596378752
Position (J2000)	15.62474, +45.18042 deg (TIC)
Brightness	Tmag 9.12; G 9.51
Stars	G0 (SIMBAD; TIC: T_{eff} 6196 K, 1.79 $R_\odot$) at 185 pc; Gaia double-lined orbit (SB2, e = 0.011 ± 0.005, K_1 75.4, K_2 75.0 km/s): twins with q = 1.004 and M sin³i = 1.120 and 1.125 $M_\odot$, two G0 dwarfs of about 1.15 $M_\odot$ and 1.2 $R_\odot$
Period	6.370415 ± 0.000005 d, timing fit of 11 primaries; 12 secondaries give 6.370416 ± 0.000003 d (Gaia: 6.370535 ± 0.000148 d; −0.8σ). The fold ran at half the orbital period, 3.1851971 d; the alternate eclipses differ in depth (6.07 ± 0.07 against 3.74 ± 0.10 ppt, errors of the means: 20σ) and in duration
Epochs	Primary BTJD 3407.33044 ± 0.00027 = BJD_{TDB} 2460407.33044; secondary BTJD 3321.33200 ± 0.00016 (phase 0.50034 ± 0.00005)
Primary eclipse	

	6.07 ppt (event-to-event scatter 0.22 ppt), 1.9–2.0 h, in raw QLP SAP flux (trapezoid fits to 11 events in s57, s84 and s85; the 30-min cadence of s17 does not resolve the eclipse well enough for the timing fit)
Secondary eclipse	3.74 ppt (event-to-event scatter 0.34 ppt), 1.7 h, at phase 0.50034 (12 events)
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at −0.0009 ± 0.0019 (0.5σ), secondary on conjunction B at +0.0013 ± 0.0022 (0.6σ). Eclipse spacing 0.50034 against Gaia's predicted 0.4982 ± 0.0035 (+0.6σ); $e \cos \omega$ = +0.0005 against Gaia's −0.003 ± 0.005. The deeper, longer eclipse falls at conjunction A, the nearer one for Gaia's $e \sin \omega$ = +0.009: for two equal stars in grazing geometry the overlap areas at the two conjunctions are in the ratio 0.607, against an observed depth ratio of 0.616
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s57, s84, s85, 22 matched events: mean mid-time differences 0.0–0.9 min per sector, depths 5.82 and 3.66 ppt against QLP's 5.72 and 3.63 ppt with the same quadratic baseline); TARS (5.98 and 3.25 ppt). There are no SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of 3 primaries in s57 and of 4 primaries and 4 secondaries in s84: deficit centroid 0.4–0.8″ from the light centroid, with the target's image shape (residual 6–10%), depth 4.7, 4.7, 4.8 ppt (s57) and 5.1, 5.2, 5.2 ppt (s84) in 1×1, 3×3 and 5×5 pixel apertures, and 3.2 ppt for the secondaries. The brightest Gaia source within 60″, G 16.89 at 31″, would need a 547% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, the Gaia DR3 eclipsing-binary table, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. The Gaia DR3 variability summary has a row (39 epochs, G range 0.022 mag) with no classification. SIMBAD's three references (two stellar-diameter catalogues and the Gaia DR3 non-single-star catalogue) include none on variability. Known SB (Gaia DR3)
Next eclipses	Primary: BTJD 4331.041 (2026-10-17 12:50 UTC), 4337.411, 4343.781; secondary: BTJD 4327.858 (2026-10-14 08:27 UTC), 4334.228 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

Two equal G0 dwarfs (q = 1.004) in a nearly circular 6.37-d orbit, seen so close to the grazing limit (b ≈ 0.95 of the sum of the radii) that the eclipses last under 2 h and remove only a few parts per thousand. The two eclipses are nevertheless unequal, 6.1 against 3.7 ppt. Gaia's small $e \sin \omega$ of +0.009 makes the stars 1.9% closer at conjunction A than at B, and in a grazing geometry this changes the overlap area enough to reproduce both the depth ratio (0.616 observed, 0.607 predicted) and the duration ratio for a single inclination. The unequal eclipses are thus consistent with the spectroscopic $e \sin \omega \approx 0.009$, a check on the orbit rather than an independent measurement of it. Both eclipses fall on Gaia's conjunctions within 0.6σ. Radial velocities at higher precision and a joint fit would give the inclination and both radii of a grazing twin.

A78 HD 6217 = TIC 268587594

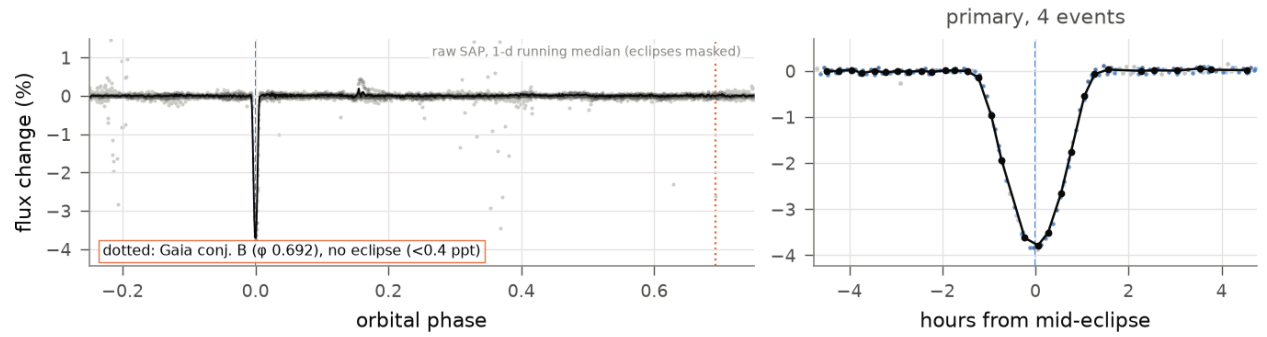


Figure A78. HD 6217, raw QLP SAP flux of sectors 3 and 30, sector medians removed and detrended by a 1-d running median with the eclipse masked, folded on P = 9.140474 d (left; flagged cadences grey), with a zoom on the primary, each event normalised to its own baseline: a 37-ppt, 2.4-h V-shaped eclipse at Gaia's conjunction A. The dotted line marks Gaia's conjunction B at phase 0.692, where no eclipse is seen (< 0.4 ppt).

Identifiers	TIC 268587594; HD 6217; CD-24 458; CPD-24 106; TYC 6422-1324-1; Gaia DR3 2345174743458805760
Position (J2000)	15.73469, −23.88401 deg (TIC)
Brightness	Tmag 8.84; G 9.27
Stars	G2/3V (SIMBAD; TIC: T_{eff} 5887 K, 1.36 $R_\odot$) at 115 pc; Gaia double-lined orbit (SB2, e = 0.4017 ± 0.0045, ω = 42.7 ± 0.7°, K_1 73.6, K_2 64.8 km/s): M $\sin^3 i$ = 0.90 and 1.03 $M_\odot$, two G dwarfs of about 0.9 and 1.0 $M_\odot$
Period	9.140474 ± 0.000004 d, timing fit of 4 primaries in s3 and s30 (with a fifth, in flagged cadences, 9.140477 ± 0.000004 d) (Gaia: 9.140352 ± 0.000190 d; +0.6σ)
Epoch	Primary BTJD 1764.02178 ± 0.00018 = BJD_{TDB} 2458764.02178
Primary eclipse	36.5 ppt, 2.38 h (ingress 0.88 h), V-shaped, in raw QLP SAP flux (trapezoid fit to the fold of 4 events); 37.7 ppt in s30; 37.3 ppt in SPOC 2-min
Secondary eclipse	Not detected (< 0.4 ppt, 3σ) at Gaia's conjunction B, predicted at phase 0.6922 ± 0.0030 and covered in four cycles; no dip above S/N 3.5 at any phase. In the e = 0.40 orbit the projected separation at B (1.152 a) is 1.75 times that at A (0.659 a), so the impact parameter of about 0.84 (in units of the sum of the radii) implied by the primary's duration becomes about 1.47 at B: the stars do not eclipse there
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at −0.0012 ± 0.0007 (−1.7σ; −0.27 ± 0.16 h); carried to the TESS epochs with Gaia's period, +0.0008 ± 0.0035 (+0.2σ). Conjunction B is 80σ away. With one eclipse per orbit, $e \cos \omega$ is not tested by the eclipse spacing
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s3: 37.3 ppt; mid-times within 0.6 min on two events at QLP's 30-min cadence); TESS-SPOC SAP and PDCSAP (s3, s30: 36.6–36.8 ppt; QLP − TESS-SPOC +0.05 ± 0.08 min on 3 events); TARS; TESScut photometry of the 200-s images of sector 107 (this work): two primaries at O−C +0.74 ± 0.15 and +1.22 ± 0.14 min, within the ephemeris's propagated error (1.6 min) about 2100 d after the last QLP sector, 37.6 ppt. SPOC's s3 search detected the eclipse as a TCE at 9.14050 ± 0.00013 d (MES 395), 0.41 ± 2.50″ from the catalogue position; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 1 primary in s3 (30-min cadence) and 2 in s30: deficit centroid 0.7″ and 0.4″ from the light centroid, a scaled copy of the star's image (residual 3.6–3.7%), depth 31.8, 31.7 and 31.8 ppt in 1×1, 3×3 and 5×5 pixel apertures (s30; the eclipse-core median of a V-shaped event). The only Gaia source within 60″ is G 20.18 at 53″ (would need 84,000%); the brightest within 90″, G 16.10 at 66″, would need about 2000%
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. SIMBAD's eight references include none on variability. Known SB (Gaia DR3)

Next eclipses	Primary: BTJD 4332.495 (2026-10-18 23:44 UTC), 4341.635, 4350.776 (allow ±5 min); no secondary eclipse
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A pair of G dwarfs in an e = 0.40 double-lined orbit that eclipse at only one conjunction. At conjunction A, on the periastron side, the stars are 0.66 a apart in projection and overlap in a grazing eclipse, at an impact parameter of about 0.84 of the sum of the radii from its duration. At conjunction B they are 1.15 a apart and miss each other, and the light curve is flat there to 0.4 ppt. The single eclipse is attributed to the Gaia orbit by its period (+0.6σ) and its phase (1.7σ at Gaia's epoch), and the 2018–2020 baseline gives the period to 4×10^{-6} d. The ephemeris was tested out of sample: two eclipses in sector 107 fall within its propagated error. Because only one eclipse per orbit occurs, the photometry alone constrains neither e nor ω; the radii and inclination require a joint fit with radial velocities.

A79 BD+64 520A = TIC 416277351

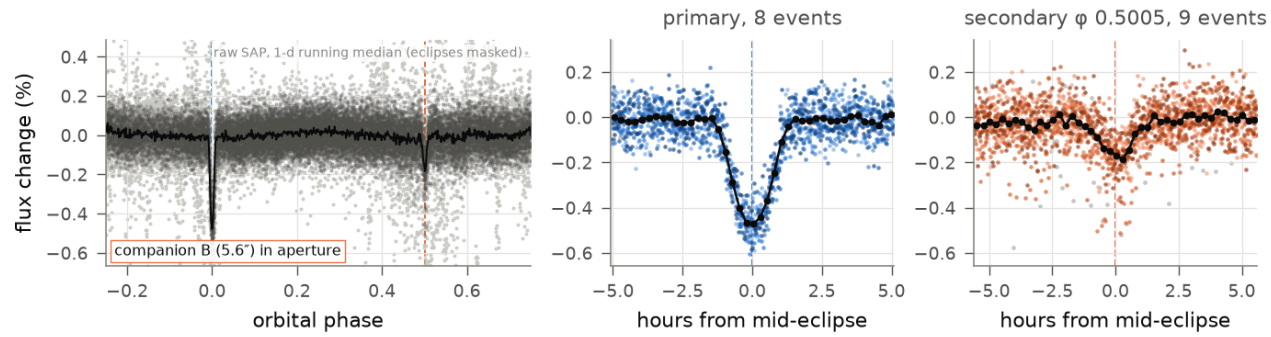


Figure A79. BD+64 520A, raw QLP SAP flux of sectors 59, 73 and 86, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on P = 6.607651 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.5005, each event normalised to its own baseline: a 4.5-ppt, 2.5-h V-shaped primary and a 1.7-ppt, 2.8-h secondary. The aperture includes the visual companion BD+64 520B, 5.6″ away; QLP's SAP depths here are already corrected for its light.

Identifiers	TIC 416277351; BD+64 520A; HIP 25046; TYC 4088-1339-1; Gaia DR3 479996547047333120; WDS J05218+6454A (STF 659A)
Position (J2000)	80.44749, +64.90443 deg (TIC)
Brightness	Tmag 9.36; G 9.75
Stars	F2 (SIMBAD; TIC: T_{eff} 6506 K, 2.61 $R_\odot$) at 308 pc; Gaia single-lined orbit (SB1, e = 0.009 ± 0.005, K_1 49.4 km/s). The mass function, 0.082 $M_\odot$, gives $M_2 \geq 0.70$ $M_\odot$; the secondary-to-primary depth ratio, 0.35, indicates a companion of roughly 5000 K, a K dwarf of about 0.8 $M_\odot$. The visual companion BD+64 520B (TIC 416277353; G 10.62, 5.6″) has the same parallax and colour and 0.445 of A's G-band flux; Gaia DR3 sees its radial velocity as constant (robust amplitude 6.2 km/s over 21 transits)
Period	6.607651 ± 0.000012 d, timing fit of 8 primaries; 9 secondaries give 6.607705 ± 0.000030 d (1.7σ) (Gaia: 6.607711 ± 0.000099 d; −0.6σ)
Epochs	Primary BTJD 3149.97660 ± 0.00045 = BJD_{TDB} 2460149.97660; secondary BTJD 3153.28375 ± 0.00125 (phase 0.50050 ± 0.00019)
Primary eclipse	4.54 ppt, 2.54 h (ingress 0.93 h), V-shaped (grazing), in raw QLP SAP flux (trapezoid fit to the fold of 8 events). QLP's SAP depth is corrected for B's light (it exceeds SPOC's uncorrected SAP depth by a factor of 1.52 on every matched event) and is the depth on A; in the combined light of A and B the eclipse is about 3.0 ppt (SPOC 2-min SAP 2.95 ppt, crowding 0.69; PDCSAP 4.26 ppt)
Secondary eclipse	1.69 ppt, 2.80 h, at phase 0.50050 (9 events of 1.3–2.8 ppt); 1.48 ppt in SPOC 2-min PDCSAP

Orbit	At Gaia's epoch with the TESS period: primary on conjunction A (A eclipsed) at −0.0024 ± 0.0020 (−1.2σ), secondary on conjunction B at +0.0037 ± 0.0019 (+1.9σ); carried to the TESS epochs with Gaia's period, −0.0061 ± 0.0065 (−1.0σ) and +0.0002 ± 0.0070 (0.0σ). $e \cos \omega$ = +0.0008 ± 0.0003 from the eclipse spacing, against Gaia's −0.0082 (+0.0050/−0.0055): 1.7σ apart, both circular to about 0.01
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s59, s73: matched mid-times −0.20 ± 0.70 min over 6 primaries and +0.07 ± 2.31 min over 6 secondaries); TESS-SPOC SAP and PDCSAP (s59, s73: 4.33 ppt in PDCSAP); TARS. SPOC's multi-sector search (s14–s78) detected the primary as a TCE at 6.607629 ± 0.000018 d (MES 56, 4458 ppm), 1.48 ± 2.51″ from the out-of-transit centroid, and the secondary as a second TCE (MES 15.6); neither was flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 4, 3 and 1 primaries in s59, s73 and s86: deficit centroid 1.7″, 1.8″ and 0.4″ from the light centroid and within 0.11 px of the catalogue position, a scaled copy of the star's image (residual 17–23%, from the faintness of a 3-ppt signal), depth 2.7–2.8 ppt in 3×3 and 5×5 pixel apertures; the 1×1 depths are not stable (3.42, 3.04 and 2.15 ppt). The G 15.60 star at 38″ would need an eclipse of about 95% of its light and would move the deficit centroid by more than a pixel; only B (would need 9.7 ppt) is bright and close enough. Along the A–B axis the deficit-minus-light centroid shift, −0.78 ± 0.57″ over three sectors, lies 1.7σ from the −1.7″ expected for an eclipse on A and more than about 5σ from the +3.9″ expected for one on B (null distribution from six stacks per sector at off-eclipse phases, so the error of the null is itself uncertain by about 30%)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. WDS lists the visual pair STF 659 AB (5.6″). SIMBAD's 13 references include none on variability. Known SB (Gaia DR3)
Open issues	The pixel separation of A from B rests on a 0.27-px centroid shift; the attribution to A rests equally on the eclipses' agreement with A's Gaia orbit and on B's constant radial velocity
Next eclipses	Primary: BTJD 4332.746 (2026-10-19 05:50 UTC), 4339.354, 4345.961; secondary: BTJD 4329.446 (2026-10-15 22:37 UTC), 4336.053 (allow ±10 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+64 520A, an F2 star and Hipparcos target, is the A component of the visual pair STF 659, with a companion of 0.45 times its brightness 5.6″ away, inside every TESS aperture. The eclipses are shallow and grazing (b ≈ 0.96 of the sum of the radii): 4.5 ppt on A and 1.7 ppt at phase 0.5005, from a K-dwarf companion on a circular 6.61-d orbit. Three independent lines place them on A rather than B. They have the period of A's Gaia orbit and fall on both of its conjunctions; B is radial-velocity constant in Gaia to a few km/s, whereas a companion able to produce a 9.7-ppt eclipse on B would move it by tens of km/s; and the difference-image centroid along the A–B axis falls where an eclipse on A puts it. The entry's depths are those of A alone; in the combined light the primary is about 3.0 ppt. Adaptive-optics or speckle-resolved photometry of one eclipse would separate A and B directly.

A8o TYC 1310-1984-1 = TIC 19702288

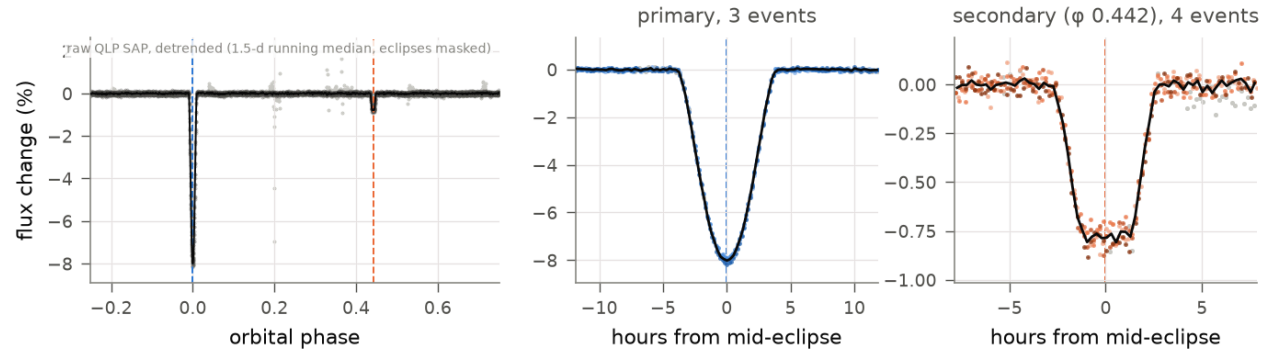


Figure A80. TYC 1310-1984-1, raw QLP SAP flux of sectors 43–45, sector medians removed and detrended by a 1.5-d running median with the eclipses masked, folded on P = 17.02161 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.442, each event normalised to its own baseline: an 80-ppt, 7.4-h primary with a rounded, limb-darkened bottom (transit of the companion) and a 7.7-ppt, 4.9-h flat-bottomed secondary (total occultation).

Identifiers	TIC 19702288; TYC 1310-1984-1; Gaia DR3 3404186685061564288
Position (J2000)	84.34853, +22.45937 deg (TIC)
Brightness	Tmag 9.48; G 9.86
Stars	TIC: T_{eff} 6225 K, 1.62 $R_\odot$ at 200 pc; Gaia single-lined orbit (SB1, e = 0.267 ± 0.011, ω = 245.6 ± 3.2°, K_1 27.4 km/s). The companion is an M dwarf: the mass function, 0.032 $M_\odot$, gives $M_2 \geq$ 0.45 $M_\odot$ for M_1 = 1.20 $M_\odot$; the primary depth gives $R_2/R_1 \approx 0.28$ ($R_2 \approx 0.46$ $R_\odot$); the occultation depth corresponds to a TESS-band surface-brightness ratio of about 0.1 (roughly 3600 K)
Period	17.02161 ± 0.00032 d, timing fit of 3 primaries; 4 secondaries give 17.02189 ± 0.00045 d (Gaia: 17.02361 ± 0.00114 d; –1.7σ). ASAS-3 V photometry of 2004–2009, about 353 cycles earlier, shows the primary above 5σ only for 17.0215 < P < 17.0221 d (peak 7.9σ at 17.02195 d) and nothing at Gaia's period
Epochs	Primary BTJD 2516.54908 ± 0.00042 = BJD_{TDB} 2459516.54908; secondary BTJD 2524.0685 (phase 0.44176 ± 0.00004)
Primary eclipse	79.8 ppt (78.7–80.7 ppt per event), 7.4 h, rounded and limb-darkened ($T_{23}/T_{14} \approx 0.2$), in raw QLP SAP flux (trapezoid fits to 3 events)
Secondary eclipse	7.7 ppt (7.3–8.0 ppt per event), 4.9 h, flat-bottomed ($T_{23}/T_{14} \approx 0.6$), a total occultation, at phase 0.44176 (4 events). It falls before phase 0.5 because e cos ω < 0, and is shorter than the primary because conjunction B is the nearer one in projection (0.75 a against 1.23 a at A)
Orbit	At Gaia's epoch (the TESS ephemeris carried 125 cycles back, where Gaia's period and the TESS period give the same result): primary on conjunction A at −0.0082 ± 0.0070 (1.2σ), secondary on conjunction B at +0.0057 ± 0.0034 (1.7σ); conjunction B is excluded for the primary (130σ). The eclipse spacing, 0.44176, gives e cos ω = −0.0915 against Gaia's −0.110 ± 0.013 (1.5σ); the duration ratio, 0.66, compares with about 0.61 predicted for equal impact parameters from Gaia's e sin ω = −0.243
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s43–s45: 3 primaries at 76.2 and 79.3 ppt, 4 secondaries at 7.4 and 7.7 ppt; mean mid-time differences from QLP +0.19 ± 0.55 and +0.42 ± 0.84 min); TARS (79.8 and 8.1 ppt); aperture photometry on TESScut images (this work; s44, s45: 77.9 ppt); ASAS-3 (the primary at +72 mmag, 7.0σ, at the TESS period). The TESS-SPOC s45 search detected a single primary as a TCE at 17.0347 d (79.8 ppt, 7.79 h, MES 431), 0.45 ± 2.50″ from the out-of-transit centroid; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of the single primaries in s44 and s45: deficit centroid 0.8″ and 0.5″ from the light centroid, a scaled copy of the star's image (residual 9–11%), depth 67.8, 68.4 and 65.9 ppt in 1×1, 3×3 and 5×5 pixel apertures (s44; the median over the rounded bottom). The two s44 secondaries, stacked, are also on the target (7.3 ppt, 2.4″ from the light centroid). No Gaia source within 60″ is bright enough: the brightest, G 13.73 at 43″, would need a 289% eclipse. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), the SuperWASP false-positive catalogue (Schanche et al. 2019), general web search. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	TESS has observed the star in one season only (2021): in sectors 71 and 72 it falls off the science pixels. The period is therefore known to ±0.00032 d, and the predicted times carry ±49 min (1σ). A joint fit of the TESS and ASAS-3 timings would reduce the period error
Next eclipses	Primary: BTJD 4337.861 (2026-10-24 08:34 UTC), 4354.883; secondary: BTJD 4328.359 (2026-10-14 20:32 UTC), 4345.381 (allow ±148 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

An F-type star (6200 K) with an M-dwarf companion of about 0.45 $M_\odot$ and 0.46 $R_\odot$ on an eccentric (e = 0.27) 17.02-d orbit. The 80-ppt primary is the M dwarf crossing the F star, with a rounded, limb-darkened bottom; the 7.7-ppt secondary is its total occultation, flat-bottomed. Both eclipses fall on Gaia's conjunctions within 1.7σ at Gaia's own epoch, and one geometry with Gaia's e and ω accounts for the secondary's phase (0.442) and its shorter duration at the nearer conjunction; the depth ratio is set by the surface-brightness ratio of the two stars. ASAS-3 recovers the primary more than a decade before TESS and restricts the period to 17.0215–17.0221 d, in agreement with the TESS period, which differs from Gaia's 17.02361 d by 1.7σ. With only three primaries from one TESS season, the next eclipses are uncertain by about 50 min (1σ); a single timed eclipse from the ground would reduce this to minutes, and the total occultation makes the system suitable for a direct measurement of the M dwarf's temperature.

A81 CD−24 5530 = TIC 109061795

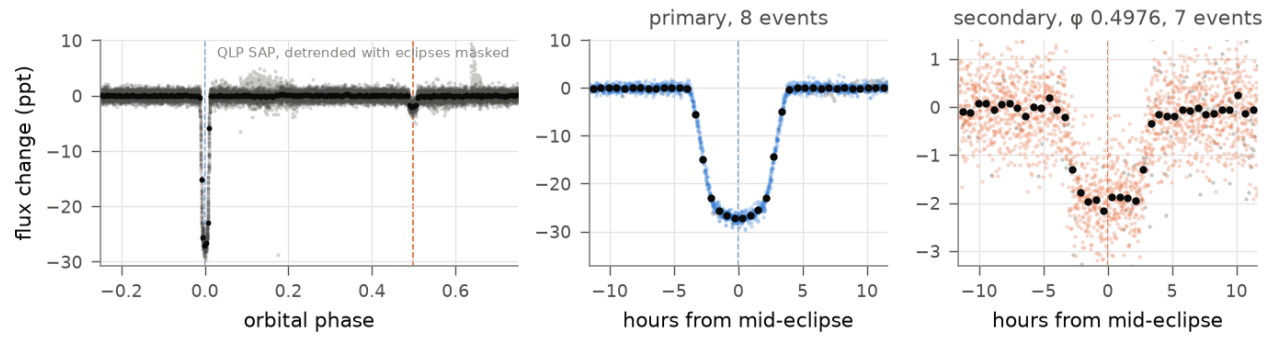


Figure A81. CD-24 5530, raw QLP SAP flux of sectors 7, 34, 61 and 88, sector medians removed and detrended per sector by 1-d bin medians with the eclipses masked, folded on P = 13.2271715 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.4976 (binned medians in black): a flat-bottomed 25.6-ppt, 7.25-h primary and a 1.9-ppt secondary of the same duration.

Identifiers	TIC 109061795; CD-24 5530; CPD-24 2516; TYC 6542-2411-1; Gaia DR3 5615242068166737280
Position (J2000)	112.91182, −24.52605 deg (TIC)
Brightness	Tmag 9.34; G 9.77
Stars	TIC: T_{eff} 5899 K, 2.32 $R_\odot$ (Gaia GSP-Phot 5925 K) at 247 pc; Gaia single-lined orbit (SB1, e = 0.039 ± 0.011, K_1 23.1 km/s). The companion is an M dwarf: the mass function, 0.0168 $M_\odot$, gives $M_2 \geq 0.32$ $M_\odot$ for M_1 = 1.07 $M_\odot$; the flat-bottomed primary gives $R_2/R_1 \approx 0.16$ ($R_2 \approx 0.37$ $R_\odot$, about 0.33–0.35 $M_\odot$); the depth ratio, 0.074, gives about 3200 K. $i \approx 86$–$88°$
Period	13.2271715 ± 0.0000023 d, timing fit of 8 primaries; 7 secondaries give 13.227137 ± 0.000023 d (−1.5σ) (Gaia: 13.22633 ± 0.00115 d; +0.7σ)
Epochs	

	Primary BTJD 3040.42781 ± 0.00013 = BJD_{TDB} 2460040.42781; secondary BTJD 3047.00989 (phase 0.49762 ± 0.00011)
Primary eclipse	25.64 ± 0.14 ppt, 7.25 h, flat-bottomed ($T_{23}/T_{14} \approx 0.6$), in raw QLP SAP flux (trapezoid fit to the fold of 8 events); alternate events 26.03 and 25.76 ppt. The QLP depth per sector falls from 27.8 ppt (s7) to 25.4 ppt (s88) at constant duration and depth ratio, while TESS-SPOC SAP is constant at 24.4–25.0 ppt and PDCSAP at 26.7–27.2 ppt: the change is in the dilution of QLP's aperture by the G 13.94 neighbour, not in the binary
Secondary eclipse	1.89 ± 0.04 ppt, 7.25 h, at phase 0.49762 (7 events of 1.7–2.2 ppt; 1.85–2.19 ppt in every pipeline)
Orbit	At Gaia's epoch with the TESS period (200 cycles back): primary on conjunction A at −0.0014 ± 0.0025 (0.6σ), secondary on conjunction B at −0.0036 ± 0.0030 (1.2σ); carried to the TESS epochs with Gaia's period, +0.0110 ± 0.0165 (0.7σ) and +0.0085 ± 0.0170 (0.5σ). $e \cos \omega$ = −0.0037 ± 0.0002 from the eclipse spacing, against Gaia's −0.0006 ± 0.0071 (0.4σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s7, s34, s61, 6 primaries: matched mid-times −0.21 ± 0.24 min from QLP's, 24.9 ppt in SAP and 26.9 ppt in PDCSAP); TARS (8 primaries, 25.9 ppt); TGLC aperture photometry (s7: 27.9 ppt, −0.51 ± 0.19 min); ASAS-3 V (2001–2009, about 497 cycles earlier): +21.0 ± 5.0 mmag in the eclipse core on the TESS ephemeris against 28.2 mmag expected, consistent with the eclipse but only about 2.5σ against the scatter of trial periods (sd 7.8 mmag). The TESS-SPOC s61 search detected the eclipses as a TCE at P/2, 6.61375 d (primaries and secondaries folded together; 27.0 ppt, MES 95), 0.12 ± 2.5″ from the catalogue position; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 2 primaries each in s34, s61 and s88: the deficit centroid is 0.6–2.0″ from the catalogue position, while the out-of-eclipse light centroid is pulled up to 0.20 px toward the G 13.94 neighbour 16.9″ (0.8 px) away. The depth is largest on the target's pixel (27.3, 25.1 and 23.8 ppt in 1×1, 3×3 and 5×5 pixel apertures in s34) and falls in larger apertures, which take in the neighbour's constant light. That neighbour would need a 122% eclipse; none of the 75 Gaia sources within 60″ can produce the dip. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search. Listed in Kostov et al. (2025), table 2, as an unvetted machine-learning candidate. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Next eclipses	Primary: BTJD 4336.691 (2026-10-23 04:33 UTC), 4349.918; secondary: BTJD 4330.046 (2026-10-16 13:05 UTC), 4343.273 (allow ±2 min for primaries and ±6 min for secondaries, whose phase is less well fixed; the secondaries also scatter by 4.2 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A slightly evolved G star (2.3 $R_\odot$) with an M-dwarf companion of about 0.35 $M_\odot$ on a near-circular 13.23-d orbit. The primary is a flat-bottomed transit of the M dwarf, and the 1.9-ppt secondary at phase 0.4976 is its occultation, implying a temperature near 3200 K. Both eclipses fall on Gaia's conjunctions within 1.2σ, and the TESS period agrees with Gaia's to 0.7σ. A G 13.94 star 17″ away shares the TESS pixels: the difference images place the deficit on CD-24 5530 while the light centroid is displaced toward the neighbour, and the slow decrease of the QLP depth from sector to sector, absent from TESS-SPOC, is a change in that neighbour's contribution to QLP's aperture. ASAS-3 photometry two decades earlier is consistent with the eclipse where the TESS ephemeris carried back 497 orbits places it.

A82 CD−83 126 = TIC 317250623

TIC 317250623 CD-83 126 P = 19.60528 d

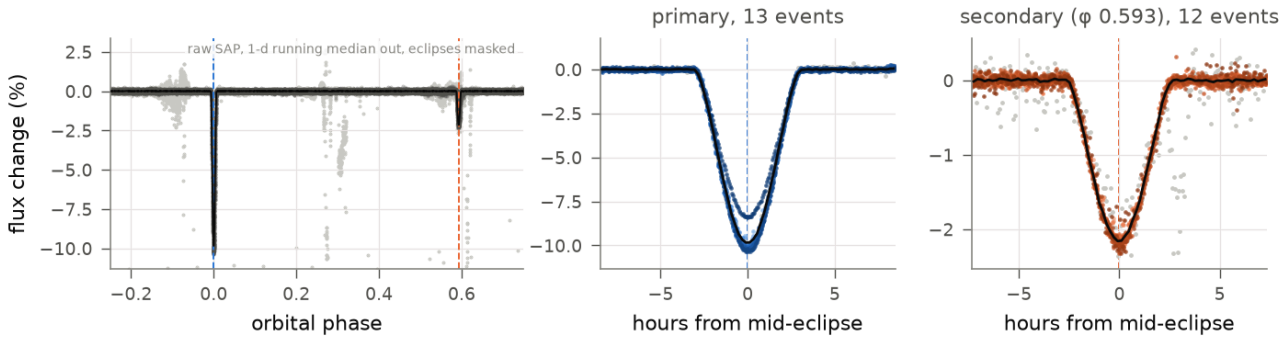


Figure A82. CD-83 126, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 19.605278 d (flagged cadences grey). Left: the 98-ppt primary at phase 0 and the 20.6-ppt secondary at phase 0.593. Middle: the 5.65-h primary, each event normalised to its own out-of-eclipse flux; the shallower track is the sector-94 event in QLP flux, which TARS gives at full depth. Right: the 4.9-h secondary, aligned on its own linear ephemeris. The panels plot 13 primaries and 12 secondaries, including partly covered events; the timing fits use the 10 primaries and 11 secondaries that are fully covered.

Identifiers	TIC 317250623; CD-83 126; CPD-83 220; TYC 9506-410-1; Gaia DR3 5193353340543454848
Position (J2000)	122.99276, −84.23232 deg (TIC)
Brightness	Tmag 8.98; G 9.41
Stars	G0 (SIMBAD; TIC: T_{eff} 5301 K, 2.34 $R_\odot$ for the combined light) at 159 pc; Gaia SB2 orbit (e = 0.1862 ± 0.0053, ω = 36.0 ± 2.5°, K_1 = 49.57 and K_2 = 48.74 km/s): a near-twin pair of about 0.92 $M_\odot$ each at $i \approx 86°$, a = 37.4 $R_\odot$. Gaia RUWE 6.7
Period	19.605278 ± 0.000014 d, timing fit of 10 primaries (Gaia: 19.60653 ± 0.00145 d; −0.9σ). The secondaries alone give 19.604896 ± 0.000013 d (see Open issues)
Epochs	Primary BTJD 3060.10199 ± 0.00051 = BJD_{TDB} 2460060.10199; secondary BTJD 3071.72505 ± 0.00040 (phase 0.5929; 0.5945 in sector 11, 0.5928 in sector 65)
Primary eclipse	98 ppt, 5.65 h in raw QLP SAP flux (trapezoid fits to 10 events); the sector-94 event is 81.9 ppt in QLP and 96.1 ppt in TARS
Secondary eclipse	20.6 ± 1.1 ppt, 4.9 h, at phase 0.593 (11 events). It is a grazing eclipse (b ≈ 0.9) at conjunction B, where the stars are 40.6 $R_\odot$ apart against 32.6 $R_\odot$ at conjunction A; this, not a difference between the stars, makes it much shallower than the primary
Orbit	Primary on Gaia's conjunction A: Δφ = −0.0022 ± 0.0021 (1.1σ); secondary on conjunction B: −0.0054 ± 0.0023 (2.3σ; an upper bound, since it carries the secondary's non-constant period back 125 cycles). Eclipse separation 0.59285 against Gaia's 0.5960 ± 0.0041 (0.8σ): $e \cos \omega$ = +0.146 against Gaia's +0.1505 ± 0.0065. The observed drift carried back to Gaia's epoch gives a separation of 0.5957
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s65–s67: mid-times within 0.0–0.7 min of QLP's, depths within 2–6%); eleanor-lite (s11–s13) and TGLC aperture photometry (s11) within 0.7 min (TGLC PSF 1.7 min); TARS (all sectors, 20 events, −1.3 to +1.5 min). The TESS-SPOC multi-sector search (s56–s69) produced TCEs on both eclipses, at 19.6042 d (MES 1472, 98.9 ppt) and 19.6053 d (MES 302, 21.0 ppt), with centroids 0.53 ± 2.54″ and 0.19 ± 2.54″ from the TIC position; neither is flagged as an eclipsing binary. ASAS-3 (2000–2009) shows the primary at the TESS period (+94 mmag, SNR 9.7), so the primary ephemeris holds from 2000 to 2023 to about $\pm 3 \times 10^{-5}$ d
Pixel attribution	TESScut difference images in s27 (two primaries) and s66 (one primary, one secondary): the peak deficit is in the target's pixel, the deficit centroid 3.4–3.6″ (0.16 px) from the light centroid; the deficit has the star's image shape (residual 19–23%, from uneclipsed diffuse light in the cutouts) and primary depths of 84/83/78 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest of the nine Gaia sources within 60″ is G 18.15 at 42.8″; every neighbour would need an eclipse of more than 30,000%. A companion within about 1″ is not excluded
Catalogue status	

	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, G0); its SIMBAD references (RAVE, APASS, diameter catalogues, Gaia DR3) do not mention variability. TESS-SPOC TCEs at both eclipses, not promoted to TOIs (Table A1)
Open issues	The secondary drifts against the primary. The two linear periods differ by 0.00038 ± 0.00002 d (18σ); measured within single sectors, the secondary-minus-primary interval is 11.65577 d in sector 11 and 11.62176 d in sector 65, a change of −49.0 min in 75 cycles, also seen in TARS and in eleanor-lite and TGLC against TESS-SPOC. The secondary O−C is curved. As apsidal motion this would need dω ≈ 0.020° per cycle (1.47° in 75 cycles at 33.4 min per degree for e = 0.186, ω = 36°; U ≈ 990 yr), more than 100 times the general-relativistic rate, and the tidal term is negligible (R/a ≈ 0.035); a third body perturbing the orbit is the likely cause, consistent with RUWE 6.7 for a near-twin. QLP depths rise from Year 1 to Year 5 (secondary 19.0 to 21.3 ppt), but TARS shows a much smaller rise and none in the primary, so a depth change is not established
Next eclipses	Primary: BTJD 4334.445 (2026-10-20 22:41 UTC), 4354.050 (allow ±5 min); secondary, from its own linear ephemeris: BTJD 4346.043 (2026-11-01 13:02 UTC), 4365.648 (allow ±15 min, given the drift)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-83 126 is an eccentric (e = 0.19) pair of nearly equal G stars of about 0.92 $M_\odot$, seen at i ≈ 86°. The 98-ppt primary occurs at the closer conjunction; at the wider one the chord is nearly grazing, so the secondary is only 21 ppt deep. Both eclipses fall on Gaia's double-lined conjunctions, their separation reproduces Gaia's e cos ω, the deficit is on the target, and SPOC's own search detected both eclipses. Between 2019 and 2023 the secondary moved 49 min earlier relative to the primary, with a curved O−C. As apsidal motion that would be more than a hundred times faster than general relativity allows for this orbit, and together with a Gaia RUWE of 6.7 for a near-twin it points to a third body. Continued timing of both eclipses and the Gaia DR4 epoch radial velocities can test this; because the secondary is near grazing, its depth is also a sensitive monitor of any change in inclination.

A83 BD+13 1919 = TIC 7060775

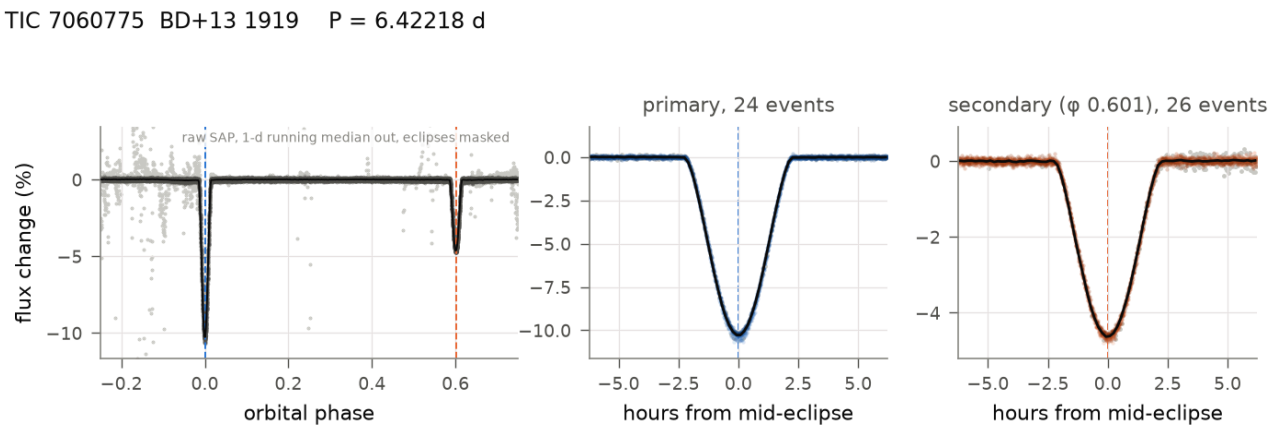


Figure A83. BD+13 1919, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 6.422180 d (flagged cadences grey). Left: the 101-ppt primary at phase 0 and the 45.9-ppt secondary at phase 0.601. Middle and right: the two eclipses, each event normalised to its own out-of-eclipse flux; both last 4.15 h. The panels plot 24 primaries and 26 secondaries, including partly covered events; the timing fits use the 23 and 24 that are fully covered.

Identifiers	TIC 7060775; BD+13 1919; TYC 804-821-1; Gaia DR3 650833814750768768
Position (J2000)	126.92111, +12.76900 deg (TIC)
Brightness	Tmag 9.16; G 9.55
Stars	

	F7 (SIMBAD; TIC: T_{eff} 6133 K, 1.89 $R_\odot$ for the combined light) at 198 pc; Gaia SB2 orbit (e = 0.234 ± 0.019, ω = 50.0 ± 1.9°; K_1/K_2 = 1.19, so Gaia's secondary is the heavier star): about 0.98 and 1.16 $M_\odot$ at i ≈ 85°, a ≈ 18.7 $R_\odot$. The two eclipse durations give $R_1 + R_2 \approx 2.4\ R_\odot$, two late-F dwarfs. RUWE 1.06
Period	6.422180 ± 0.000001 d, timing fit of 23 primaries; 24 secondaries give 6.422175 ± 0.000001 d (Gaia: 6.421481 ± 0.000299 d; +2.3σ)
Epochs	Primary BTJD 2741.55165 ± 0.00012 = BJD_{TDB} 2459741.55165; secondary BTJD 2841.74306 ± 0.00011 (phase 0.60080). O–C rms 0.5 min for each kind
Primary eclipse	101 ± 2 ppt, 4.15 h in raw QLP SAP flux (trapezoid fits to 23 events); SPOC 2-min SAP 96.6 ppt, PDCSAP 98.2 ppt
Secondary eclipse	45.9 ± 0.7 ppt, 4.15 h, at phase 0.60080 (24 events); SPOC 2-min SAP 43.2 ppt, PDCSAP 44.2 ppt. Impact parameters b ≈ 0.57 at conjunction A (separation 15.0 $R_\odot$) and b ≈ 0.82 at conjunction B (21.5 $R_\odot$) reproduce the equal durations and the depth ratio
Orbit	Primary on Gaia's conjunction A: Δφ = +0.0023 ± 0.0030 (0.8σ); secondary on conjunction B: +0.0061 ± 0.0060 (1.0σ). Eclipse separation 0.60080 against Gaia's 0.5970 ± 0.0085 (+0.4σ): e cos ω = +0.158 against Gaia's +0.150 ± 0.013
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s34, s44; 13 events, mid-times within 0.2 min of QLP's); TESS-SPOC SAP (s7, s34, s44, s71; 25 events, within 0.1 min); eleanor-lite (s7, 0.1 min); TARS. QLP depths are 2–6% deeper than the other pipelines'. The SPOC multi-sector search (s1–s46) produced TCEs on both eclipses, at 6.4221771 d (MES 2600, 101.0 ppt) and 6.422174 d (MES 1017, 45.3 ppt), with centroids 0.48 ± 2.54″ and 0.55 ± 2.52″ from the TIC position; neither is flagged as an eclipsing binary. ASAS-3 (2000–2009) is consistent with the TESS ephemeris at SNR ≈ 3
Pixel attribution	TESScut difference images in s45 (four primaries, three secondaries) and s71 (three primaries): the deficit centroid is 0.2–0.4″ from the light centroid (both share a 0.2–0.3 px offset from the catalogue position, which lies in the cutouts' astrometry); the deficit is PSF-shaped (residual 9–23%), with primary depths of 87.1/86.7/86.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest of six Gaia sources within 60″, G 14.60 at 16.0″, would need a 1067% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, F7). SIMBAD links the star to Whitney (1958, AJ 63, 243), a study of the eclipsing binary TW Cnc 34′ away, most likely as a comparison star; that paper itself was not examined. SPOC TCEs at both eclipses, not promoted to TOIs (Table A1)
Open issues	The secondaries give a period shorter than the primaries' by (5 ± 1.4) × 10^{-6} d, the size expected from apsidal motion in this orbit: general relativity gives about 4.9 × 10^{-6} rad per cycle and the tidal term about 1.6 × 10^{-6} rad per cycle, together about 4.7 × 10^{-6} d per cycle in the eclipse spacing. At about 3σ, with mid-times quantised at 1 min, this is a hint of apsidal motion only
Next eclipses	Primary: BTJD 4327.830 (2026-10-14 07:56 UTC), 4334.252, 4340.674 (allow ±2 min); secondary: BTJD 4331.689 (2026-10-18 04:32 UTC), 4338.111 (allow ±4 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+13 1919 is an eccentric (e = 0.23) double-lined pair of late-F dwarfs of about 1.0 and 1.2 $M_\odot$, with 10% and 4.6% eclipses of equal duration. The deeper eclipse falls at the closer conjunction, where the chord is nearer the centre; the shallower one, at the wider conjunction, is close to grazing. Both fall on Gaia's conjunctions within 1σ, the eclipse spacing reproduces Gaia's e cos ω, the 47 eclipses fit linear ephemerides with 0.5-min residuals, and SPOC's own search detected both eclipses at very high significance without either becoming a TOI. The TESS period is 2.3σ longer than Gaia's, a difference of the size seen in several other Gaia orbits tested here. The two eclipse kinds give a hint of apsidal motion, to which general relativity and tides both contribute; further seasons of TESS timing can test it.

A84 TYC 3805-1342-1 = TIC 456339196

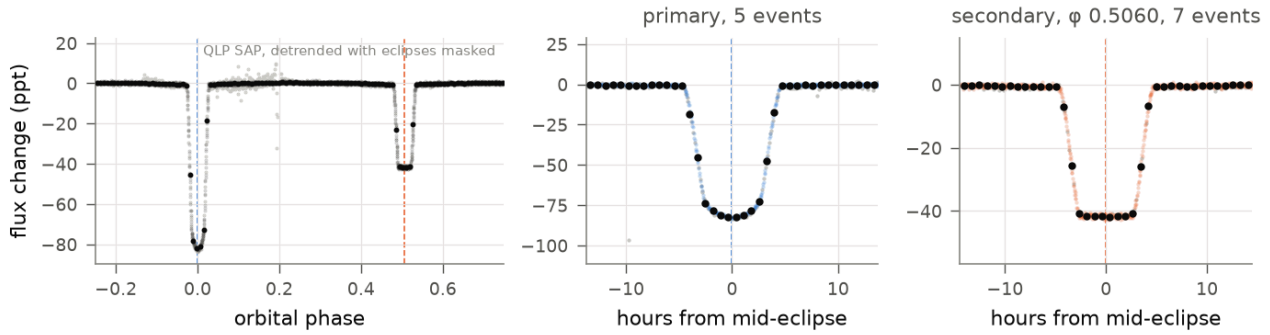


Figure A84. TYC 3805-1342-1, QLP SAP flux detrended with the eclipses masked and folded on P = 7.106115 d (binned points black). Left: the 78.6-ppt primary at phase 0 and the 41.3-ppt secondary at phase 0.5060. Middle: the 8.6-h primary, with the rounded bottom of a transit across a limb-darkened star. Right: the 9.0-h secondary, flat-bottomed: a total occultation of the companion.

Identifiers	TIC 456339196; TYC 3805-1342-1; Gaia DR3 1017462707153422976
Position (J2000)	135.49647, +52.53628 deg (TIC)
Brightness	Tmag 9.57; G 9.92
Stars	Slightly evolved F-type primary (TIC: T_{eff} 6384 K, 2.89 $R_\odot$) at 388 pc; Gaia SB1 orbit (e = 0.008 ± 0.005, K_1 = 52.66 km/s, f(M) = 0.107 $M_\odot$). The eclipses make the companion a G9–K0 dwarf of T_{eff} ≈ 5300 K, giving 4.1% of the TESS light, and of about 0.78 $M_\odot$ (the mass-function minimum is 0.77 $M_\odot$). The radii await a light-curve model: the long flat bottom of the occultation requires a nearly central eclipse and a radius ratio of about 0.25 or less
Period	7.1061147 ± 0.0000043 d, timing fit of 5 primaries; 7 secondaries give 7.106109 ± 0.000002 d (Gaia: 7.105995 ± 0.000157 d; +0.8σ). The fold ran at half the orbital period, 3.5531 d. The alternate eclipses differ in depth (78.6 and 41.3 ppt) and are 0.506, not 0.5, apart in phase
Epochs	Primary BTJD 2530.89350 ± 0.00012 = BJD_{TDB} 2459530.89350; secondary BTJD 2534.48917 (phase 0.50600 ± 0.00005)
Primary eclipse	78.6 ± 0.3 ppt, 8.6 h in raw QLP SAP flux (trapezoid fits to 5 events), rounded bottom: the transit of the companion across the F star
Secondary eclipse	41.3 ± 0.1 ppt, 9.0 h, at phase 0.50600 (7 events), flat-bottomed: the total occultation of the companion. The 0.006 offset from phase 0.5 (0.0426 d, 61 min) is the orbit's small eccentricity
Orbit	With the TESS period carrying the eclipses back 301 cycles to Gaia's epoch: primary on Gaia's conjunction A, Δφ = −0.0022 ± 0.0018 (1.2σ); secondary on conjunction B, +0.0002 ± 0.0018 (0.1σ). $e \cos \omega$ = +0.0094 ± 0.0001 from the secondary's phase, against Gaia's +0.0069 ± 0.0049 (0.5σ). The deeper eclipse is at conjunction A, where the radial-velocity star is behind, as expected when the visible spectrum is the larger, hotter star's
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s21, s47: primary 78.4–78.6 ppt, secondary 40.6–40.7 ppt; matched mid-times −0.07 ± 0.16 and +0.39 ± 0.64 min from QLP's); TARS (81.3 and 41.8 ppt). The TESS-SPOC search in s47 produced one TCE at 3.55304 d, half the orbital period (MES 929, mean depth 42.2 ppt), not flagged as an eclipsing binary, with its centroid 0.10 ± 2.5″ from the TIC position
Pixel attribution	TESScut difference images in s21 (two primaries, four secondaries) and s47 (three primaries, four secondaries): both eclipse kinds have deficit centroids 0.1–0.7″ from the light centroid, as scaled copies of the star's image (residual 1.3–3.1%), with depths flat across apertures (primary 79.1/79.0/79.2 ppt in 1×1, 3×3 and 5×5 pixels in s21). Only two Gaia sources lie within 60″: G 16.76 at 54.8″ and G 17.27 at 58.7″, which would need eclipses of 4306% and 6886%. A companion within about 1″ is not excluded
Catalogue status	

	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, from Gaia DR3 only, no spectral type). No ASAS-3 coverage. TESS-SPOC TCE at half the period, not promoted to a TOI (Table A1)
Next eclipses	Primary: BTJD 4328.741 (2026-10-15 05:46 UTC), 4335.847, 4342.953; secondary: BTJD 4332.336 (2026-10-18 20:03 UTC), 4339.442 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3805-1342-1 is a slightly evolved F star with a G9–K0 dwarf companion in a 7.1-d, nearly circular orbit. The two eclipses are both deep but unequal: a 78.6-ppt transit with a limb-darkened bottom and a 41.3-ppt flat-bottomed occultation whose depth is the companion's 4.1% share of the light, giving a surface-brightness ratio of 0.53 and T_{eff} ≈ 5300 K against 6400 K. The secondary sits at phase 0.5060, so no half-period fold can align the two, and its offset gives $e \cos \omega = 0.0094$, within 0.5σ of Gaia's. Both eclipses lie on Gaia's conjunctions, the deficit is on the target, and TESS-SPOC reproduces the depths to 1%. Radial velocities of the companion's lines, if detectable at 4% of the light, would give both masses.

A85 CD−27 6336 = TIC 38388435

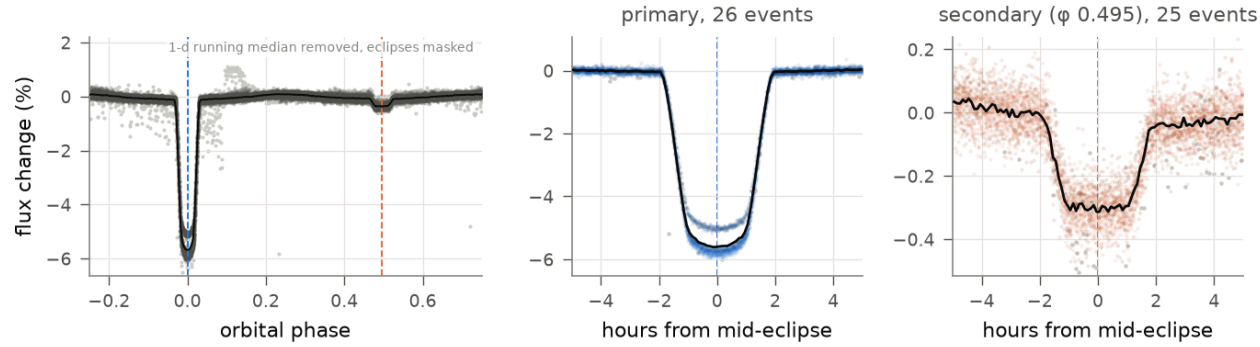


Figure A85. CD-27 6336, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 2.6164745 d (flagged cadences grey). Left: the 55-ppt primary at phase 0 and the 2.8-ppt secondary at phase 0.495. Middle: the 3.6-h primary, each event normalised to its own out-of-eclipse flux; the shallower track is sector 99, where the QLP flux is diluted (about 50 ppt). Right: the flat-bottomed secondary, 3.7 h. The panels plot 26 events of each kind, including partly covered events; the timing fits use the 25 that are fully covered.

Identifiers	TIC 38388435; CD-27 6336; CPD-27 3674; SAO 177090; TYC 6598-160-1; Gaia DR3 5636144020418779008
Position (J2000)	137.87691, −28.14134 deg (TIC)
Brightness	Tmag 9.14; G 9.46
Stars	F-type primary (TIC: T_{eff} 6772 K, 1.72 $R_\odot$, 1.45 $M_\odot$) at 202 pc; Gaia SB1 orbit (e = 0.025 ± 0.011, ω = 167 ± 21°, K_1 = 40.36 km/s, f(M) = 0.0178 $M_\odot$). The companion is an M dwarf of about 0.39 $M_\odot$ (0.32–0.46 for the range of primary masses), R ≥ 0.40 $R_\odot$, T_{eff} ≈ 3300–3500 K. A G 14.30 Gaia source 2.2″ away contributes about 1.1% of the light. RUWE 0.90
Period	2.6164745 ± 0.0000005 d, timing fit of 25 primaries; 25 secondaries give 2.6164734 ± 0.0000019 d (Gaia: 2.616527 ± 0.000044 d; −1.2σ)
Epochs	Primary BTJD 3518.67725 ± 0.00008 = BJD_{TDB} 2460518.67725; secondary BTJD 3519.97302 (phase 0.49524 ± 0.00008). Primary O−C rms 0.21 min
Primary eclipse	55 ppt, adopted from the difference images (55.1–55.4 ppt in s62 and s99) and from SPOC 2-min and TESS-SPOC (54.5–56.0 ppt); 3.6 h (trapezoid fits to 25 events in raw QLP SAP flux; 3.5–3.7 h by fitting method). QLP trapezoid fits give 56.6, 55.3 and 48.7 ppt in sectors 62, 89 and 99; the sector-99 QLP flux is diluted, since the sector-99 difference image gives 55.4 ppt

Secondary eclipse	2.8 ppt (2.7 ppt in QLP, 2.8–2.9 ppt in SPOC), 3.7 h, flat-bottomed, at phase 0.49524 (25 events): the total occultation of the M dwarf
Orbit	With the TESS period carrying the eclipses to Gaia's epoch: primary on Gaia's conjunction A, $\Delta\varphi = -0.0045 \pm 0.0037$ (1.2σ); secondary on conjunction B, $+0.0054 \pm 0.0034$ (1.6σ); conditioning Gaia's elements on the TESS period gives 1.8σ and 1.9σ. $e \cos \omega = -0.0075 \pm 0.0001$ from the secondary's phase, against Gaia's -0.023 ± 0.010 (1.5σ)
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP and TESS-SPOC SAP and PDCSAP (s62; 10 shared primaries, median mid-time differences 0.00–0.04 min, rms ≤ 0.11 min); TARS (s62, s89; +0.5 min). The SPOC 2-min search in s62 produced a TCE at 2.616486 d (MES 504, 57.2 ppt), its centroid 0.79 ± 2.5″ from the TIC position, and a second TCE on the secondary (3.0 ppt, MES 29); TESS-SPOC gives MES 759 and 37. Neither is flagged as an eclipsing binary. ASAS-3 V photometry (2000s, about 2700 cycles before the TESS epoch) folded on the TESS ephemeris shows the eclipse: +65 ± 4 mmag in 21 points in the eclipse core, against 61 mmag expected, and the TESS period gives the deepest signal among trial periods within ±4 eclipse durations of drift
Pixel attribution	TESScut difference images in s62 (10 events) and s99 (7 events): the deficit centroid is 0.2″ from the light centroid, with the peak deficit in the target's pixel; PSF-fit residuals 2.3% and 3.4%; depths 55.4/55.1/55.3 and 55.9/55.4/55.4 ppt in 1×1, 3×3 and 5×5 pixel apertures. None of the 22 Gaia sources within 60″ can make the eclipse: the G 14.30 star at 2.2″ would need 473%, and G 15.99 at 28.6″ 2225%. A source brighter than G 12.6 within about 1″ is not excluded; Gaia shows none
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, no spectral type, three references). SPOC and TESS-SPOC TCEs in s62, not promoted to TOIs (Table A1)
Next eclipses	Primary: BTJD 4329.784 (2026-10-16 06:52 UTC), 4332.401, 4335.017; secondary: BTJD 4328.464 (2026-10-14 23:10 UTC), 4331.080 (allow ±2 min; the secondaries scatter by 2.4 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-27 6336 is an F star eclipsed every 2.62 d by an M dwarf of about 0.4 $M_\odot$. The 55-ppt primary is a transit at impact parameter about 0.5–0.65, and the flat-bottomed 2.8-ppt secondary is the dwarf's total occultation. The orbit is nearly circular: the secondary at phase 0.4952 gives $e \cos \omega = -0.0075$, 1.5σ from Gaia's value, and the two eclipses fall on Gaia's conjunctions within 1.6σ. The deficit is on the target in two sectors, SPOC 2-min and TESS-SPOC timings agree with QLP's to 0.1 min, and ASAS-3 data from the 2000s show the eclipse on the TESS ephemeris. With a Tmag 9.1 primary and a deep transit by a low-mass star, the system is suited to radial-velocity work on the companion's mass and radius.

A86 BD−01 2238 = TIC 170891257

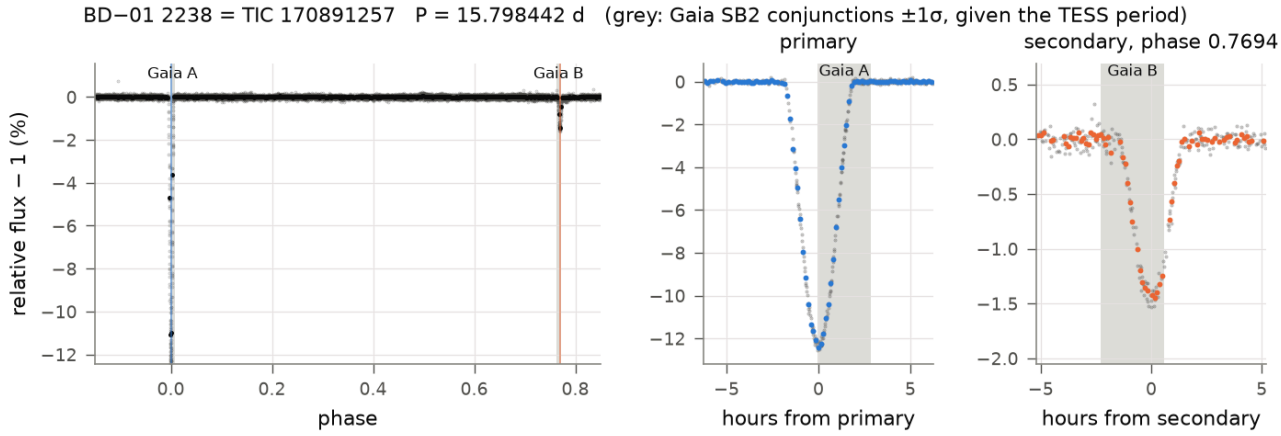


Figure A86. BD-01 2238, QLP SAP flux with sector medians removed, detrended by a running median with the eclipses masked and folded on P = 15.798442 d; individual cadences grey, binned points coloured. Grey bands mark Gaia's predicted conjunctions A and B (±1σ) carried with the TESS period. Left: the 120-ppt primary at phase 0 and the 14-ppt secondary at phase 0.7694. Middle: the V-shaped 3.5-h primary. Right: the 2.8-h grazing secondary.

Identifiers	TIC 170891257; BD-01 2238; TYC 4884-610-1; Gaia DR3 3838406607374037632
Position (J2000)	139.65855, −2.55169 deg (TIC)
Brightness	Tmag 9.41; G 9.87
Stars	F9 (SIMBAD; TIC: T_{eff} 5732 K, 1.28 $R_\odot$ for the combined light) at 135 pc; Gaia SB2 orbit (e = 0.4313 ± 0.0121, ω = 23.89 ± 1.45°, K_1 = 57.8 and K_2 = 57.6 km/s, so q = 1.00): two equal dwarfs of about 0.93 $M_\odot$ at $i \approx 87°$ (the F9 type and the TIC temperature describe their combined light), a sin i = 32.5 $R_\odot$. RUWE 1.10
Period	15.798442 ± 0.000003 d, timing fit of 4 secondaries spanning 2543 d; 4 primaries give 15.798447 ± 0.000013 d, in agreement (Gaia: 15.789859 ± 0.002086 d; +4.1σ)
Epochs	Primary BTJD 2811.62273 ± 0.00024 = BJD_{TDB} 2459811.62273; secondary BTJD 2744.78595 (phase 0.76941)
Primary eclipse	120 ppt (118.8–122.6), 3.5 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 4 events in s35 and s61); TESS-SPOC 115–117 ppt
Secondary eclipse	14 ppt, 2.8 h, at phase 0.76941 (4 events: two in s61, one in s99 in flagged but clean cadences, and one in s08). The stars are nearly equal; the secondary is shallow because it is grazing (b ≈ 0.90) at conjunction B, where the separation is 0.986a against 0.693a at conjunction A (b ≈ 0.63)
Orbit	Primary near conjunction A, secondary near conjunction B. Conditioned on the TESS period, which lies 4.1σ from Gaia's, through Gaia's full covariance: −0.0037 ± 0.0039 (0.9σ) and +0.0022 ± 0.0036 (0.6σ); with the correlations ignored, −0.0214 ± 0.0064 (3.3σ) and +0.0001 ± 0.0082 (0.0σ); with Gaia's own period carried to the TESS epochs (72–153 cycles), +0.062 ± 0.016 (3.8σ) and +0.081 ± 0.020 (4.1σ). The eclipse spacing, 0.7694, against Gaia's predicted 0.748 ± 0.008, differs by 2.7σ; it gives e cos ω = +0.4308 at Gaia's e sin ω, against Gaia's +0.394 ± 0.014 (2.5σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s35, s61: 3 primaries and 2 secondaries, same-event mid-times within 0.3 min of QLP's, depths 115–117 and 13.7–13.8 ppt); SPOC 2-min (the s08 secondary, 13.8 ppt); TARS (117.0 and 14.1 ppt). The SPOC search in s61 produced TCEs on both eclipses: the primary at 15.79788 d (MES 632, 120.4 ppt), its centroid 0.44 ± 2.50″ from the TIC position, and the secondary (MES 71, 13.2 ppt); the s08 2-min search produced a TCE on the s08 secondary. None is flagged as an eclipsing binary. ASAS-3 shows the primary on the TESS ephemeris 387 cycles earlier (5 points at 103 mmag, SNR 11)
Pixel attribution	TESScut difference images of the primary in s61 (one event) and s35 (two) and of the secondary in s61 (two): the deficit centroid is 0.1–0.6″ from the light centroid, with the peak deficit in the target's pixel each time; PSF-fit residuals 1.4–1.5% (primary) and 5.7% (secondary); primary depths 103.1/103.5/103.2 ppt in 1×1, 3×3 and 5×5 pixel apertures in s61. The brightest Gaia source within 90″, G 15.87 at 41.5″, would need eclipses of 3033% (primary) and 367% (secondary); the other eight are G ≥ 17.37. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 variability tables (target and the four sources within 60″), SB9, WDS (120″), Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, F9; also typed Em*); its references (RAVE, APASS, LAMOST, Gaia DR3) report no eclipses. SPOC TCEs in s08 and s61, not promoted to TOIs (Table A1)
Open issues	Both the period (4.1σ) and the eclipse spacing (2.7σ) are in tension with Gaia's orbit, and the agreement of the conjunctions depends on conditioning Gaia's covariance on the TESS period; a light-curve model fitting e and ω from TESS alone is not yet made. $R_1 + R_2 = 1.58\ R_\odot$ from the trapezoid durations is a little below the ~1.75 $R_\odot$ expected for two 0.93-$M_\odot$ dwarfs
Next eclipses	Primary: BTJD 4328.273 (2026-10-14 18:36 UTC), 4344.072; secondary: BTJD 4340.429 (2026-10-26 22:18 UTC), 4356.227 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD-01 2238 is a pair of equal-mass dwarfs of about 0.93 $M_\odot$ (q = 1.00) in a 15.8-d orbit with e = 0.43, seen at $i \approx 87°$. The orbit alone sets the difference between the 120-ppt and 14-ppt eclipses of two equal stars: the deep one is at the close conjunction and the grazing one at the far conjunction, at phase 0.7694. The TESS period is 0.0086 d (4.1σ) longer than Gaia's. Conditioned on the TESS period through Gaia's full covariance, both eclipse kinds fall on Gaia's conjunctions within 1σ, but with the correlations ignored the primary is 3.3σ off, and the eclipse spacing differs from Gaia's prediction by 2.7σ; the period and the spacing are both in tension with the Gaia orbit. The deficit is on the target for both eclipses, TESS-SPOC and SPOC 2-min agree with QLP, and ASAS-3 shows the primary on the same ephemeris two decades earlier. A light-curve and radial-velocity model would resolve the 2.5σ difference in $e \cos \omega$ and give the radii of two nearly identical stars.

A87 HD 89725 = TIC 463896609

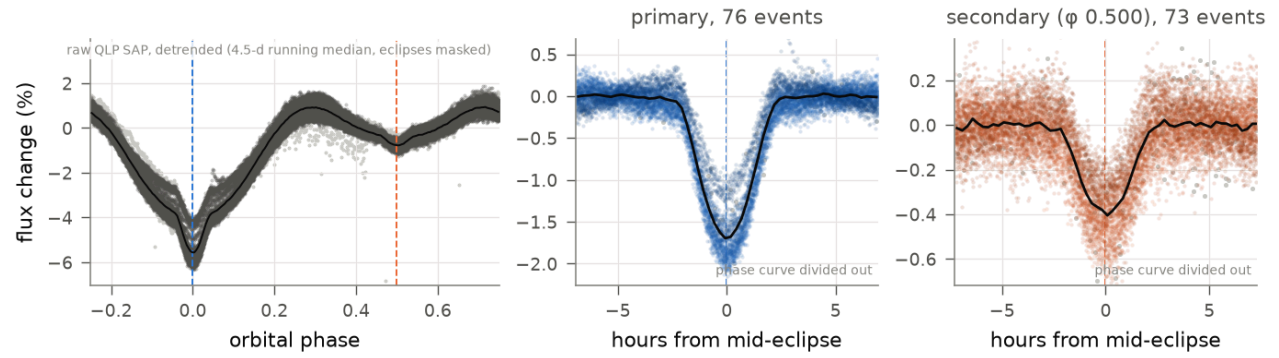


Figure A87. HD 89725, QLP SAP flux with sector medians removed, detrended by a 4.5-d running median with the eclipses masked and folded on P = 2.2366733 d (flagged cadences grey). Left: the full orbit, with the 6% peak-to-peak phase curve and the eclipses at phases 0 and 0.500. Middle and right: the V-shaped 4.3-h primary (16 ppt) and the 3.6-ppt secondary, each with the phase curve divided out. The panels plot 76 primaries and 73 secondaries, including partly covered events; the timing fit uses the 70 fully covered primaries.

Identifiers	TIC 463896609; HD 89725; CD-57 3153; CPD-57 2988; SAO 237919; TYC 8608-545-1; Gaia DR3 5258656806462084096
Position (J2000)	154.91715, −58.09636 deg (TIC)
Brightness	Tmag 8.89; G 9.13
Stars	

	F0/3IV/V (SIMBAD; Gaia GSP-Phot T_{eff} 7159 K; the TIC's 11,639 K is inconsistent with BP−RP 0.55 and the spectral type) at 371 pc, in a crowded Carina field; Gaia SB1 orbit (e = 0.050 ± 0.018, ω = 134 ± 24°, K_1 = 46.75 km/s, f(M) = 0.0236 $M_\odot$: $m_2 \geq 0.50\ M_\odot$ for $M_1 \approx 1.8\ M_\odot$), RUWE 1.68. The out-of-eclipse light curve has a reflection term (cos 2πφ, −18.3 ppt) and an ellipsoidal term (cos 4πφ, −14.6 ppt). The ellipsoidal term is of the size the F star's own tidal distortion gives in a detached F + M pair (about 11–15 ppt); the reflection term is too large for a detached main-sequence companion and suggests, as a hypothesis, a semi-detached, Algol-like system with an irradiated, Roche-lobe-filling companion
Period	2.2366733 ± 0.0000008 d, timing fit of 70 primaries, with errors scaled by 4.0 because the individual times scatter on the steep phase curve (O−C rms 3.1 min, no trend) (Gaia: 2.236642 ± 0.000026 d; +1.2σ)
Epochs	Primary BTJD 3195.87831 ± 0.00025 = BJD_{TDB} 2460195.87831; secondary BTJD 3196.9962 (phase 0.49982 ± 0.00036)
Primary eclipse	16.4 ppt (median of 70 events; sd 1.6), 4.3 h, V-shaped (grazing; SPOC fits give b = 0.99), in raw QLP SAP flux against a Fourier phase-curve model. The QLP depth varies by sector with dilution: 12.8 ppt in s99, 14.6–15.1 ppt in s9–s10, 17–18 ppt in s36–s64. TESS-SPOC SAP 14.5 ppt, PDCSAP 16.8 ppt
Secondary eclipse	3.6 ± 0.1 ppt (65σ) in a box at phase 0.5000 in the folded light curve against the phase-curve model; a trapezoid fit gives 4.0 ppt, 4.6 h, at phase 0.4998: the occultation of the companion
Orbit	With the TESS ephemeris carried 1255 cycles back to Gaia's epoch: secondary on Gaia's conjunction B, Δφ = −0.0016 ± 0.0064 (0.2σ); primary −0.0210 ± 0.0049 from conjunction A (4.3σ; 68 min, inside the eclipse half-width of 0.040 in phase). e cos ω = −0.0003 ± 0.0006 from the eclipse separation, against Gaia's −0.030 ± 0.015 (2.0σ); Gaia places B 0.481 ± 0.009 after A, against 0.4998 observed
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s9–s64, 62 events; 51 matched mid-times differ from QLP's by +0.15 ± 0.74 min); TARS (75 events, 15.7 ppt); TGLC (18.6 ppt aperture, 16.0 ppt PSF) and eleanor-lite (14.2–14.5 ppt) in s9–s10. TESS-SPOC searches produced TCEs in s36, s37, s63, s64 and s56–s69 (2.23660 d, 16.2 ppt, MES 26–43), centroids 0.9–1.3 ± 2.5″ from the TIC position, none flagged as an eclipsing binary. ASAS-3 V photometry folded on the TESS ephemeris (about 3000 cycles back) is consistent with the eclipse: +32 ± 5 mmag in 27 points in the eclipse core, a figure that includes the phase-curve minimum at the primary; the broad minimum does not single out the TESS period among trial periods
Pixel attribution	TESScut difference images in s63 and s90 (11 events each): the deficit centroid is 1.4″ and 2.6″ (0.06 and 0.12 px) from the catalogue position, with the peak deficit in the target's pixel; the light centroid is pulled 3–4″ away by neighbours (152 Gaia sources within 60″). Depths fall with aperture (24.3/23.8/22.5 ppt in 1×1, 3×3 and 5×5 pixels in s63; larger than the light-curve depth because the short windows include phase-curve curvature), as dilution predicts when the target is the source. The PSF-fit residual is 36–37%, from the crowding and the phase curve. Two neighbours are bright enough in principle, G 13.06 at 34″ (would need 63%) and G 13.28 at 57″ (77%), but their pixels lose only 18 and 4–5 ppt, the target's wing, and the deficit centroid excludes them. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (only a neighbour 47″ away, type L), the ASAS-SN variable-star catalogue, Gaia DR3 variability tables (only the same neighbour, as a long-period variable), SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP TOIs and CTOIs, the SuperWASP false-positive catalogue (Schanche et al. 2019), or a general web search. Known SB (SIMBAD SB*, F0/3IV/V; references are photometric catalogues only). TESS-SPOC TCEs, not promoted to TOIs (Table A1)
Open issues	The primary falls 4.3σ before Gaia's conjunction A while the secondary is on conjunction B. Gaia's eccentricity (2.8σ from zero for a 2.24-d binary that is expected to be circular) compresses the A-to-B interval and accounts for part of this; the mean offset of the pair, −0.011 ± 0.004, leaves about 2.8σ unexplained, possibly distortion of the radial velocities by the irradiated companion (RUWE 1.68)
Next eclipses	Primary: BTJD 4327.635 (2026-10-14 03:17 UTC), 4329.872, 4332.108; secondary: BTJD 4328.753 (2026-10-15 06:07 UTC), 4330.990 (allow ±2 min for primaries and ±4 min for secondaries; the primaries scatter by 3.1 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 89725 is an F0 subgiant with a 2.24-d companion whose grazing 16-ppt eclipse and 3.6-ppt occultation are superimposed on a 6% peak-to-peak phase curve. The phase curve is the notable feature. Its 15-ppt ellipsoidal term is about what the F star's own distortion gives, but its 18-ppt reflection term, with its minimum at the primary eclipse, is too large for a detached F + M pair; a semi-detached, Algol-like system in which the companion fills its Roche lobe and is strongly irradiated is the hypothesis that would explain it. The eclipses are on the target in two sectors despite the crowded field, and TESS-SPOC reproduces the timings. The secondary falls on Gaia's conjunction B, while the primary is 0.021 in phase early relative to conjunction A, partly because Gaia's small nonzero eccentricity compresses the predicted spacing; the mean offset of the two eclipses still leaves about 2.8σ unexplained. Spectroscopy to detect the companion's lines and a combined light-curve model would establish whether mass transfer is under way.

A88 TYC 3827-170-1 = TIC 302351416

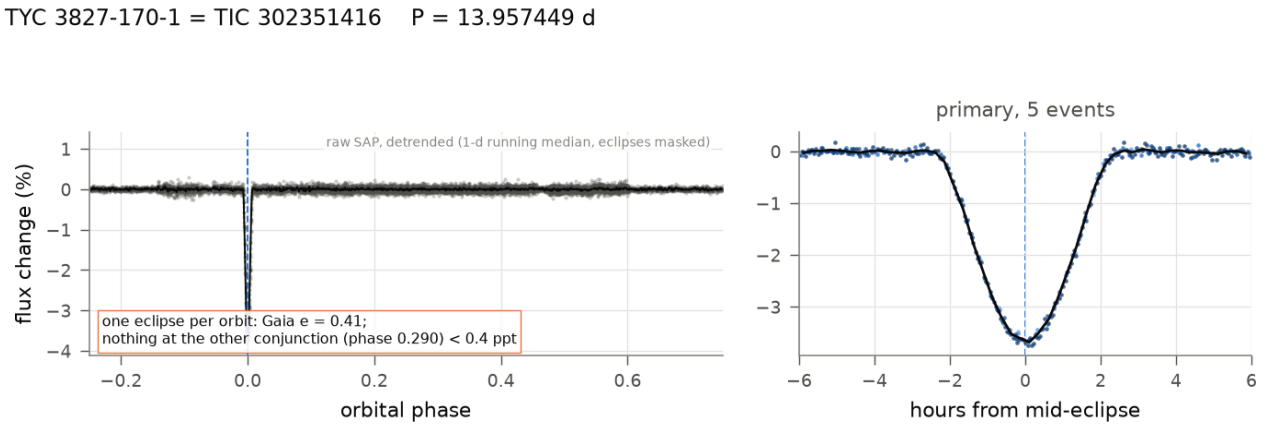


Figure A88. TYC 3827-170-1, QLP SAP flux (unflagged cadences), each sector divided by its median, detrended by a 1-d running median with the eclipse masked and folded on P = 13.9574485 d. Left: the full orbit, with one eclipse per orbit; Gaia's other conjunction, at phase 0.29 (not marked), shows no eclipse (< 0.4 ppt). Right: the five events of the 35.7-ppt, 4.5-h V-shaped eclipse.

Identifiers	TIC 302351416; TYC 3827-170-1; Gaia DR3 856255301585891840
Position (J2000)	168.61584, +56.20059 deg (TIC)
Brightness	Tmag 9.42; G 9.81
Stars	Slightly evolved F9–G0 primary (TIC: T_{eff} 6112 K, 2.03 $R_\odot$) at 240 pc; Gaia SB1 orbit (e = 0.4118 ± 0.0030, ω = 318.60 ± 0.81°, K_1 = 45.57 km/s, f(M) = 0.104 $M_\odot$: $m_2 \geq 0.71\ M_\odot$ for M_1 = 1.15 $M_\odot$, a ≈ 30 $R_\odot$)
Period	13.9574485 ± 0.0000066 d, timing fit of 5 eclipses, errors scaled by 2.7 (Gaia: 13.956758 ± 0.000425 d; +1.6σ). Odd and even eclipses agree (35.82 and 35.50 ppt) and nothing is seen at phase 0.5, so the period is P
Epochs	Primary BTJD 2928.45698 ± 0.00023 = BJD_{TDB} 2459928.45698
Primary eclipse	35.7 ± 0.4 ppt, 4.5 h, V-shaped (flat fraction 0.2), in raw QLP SAP flux (trapezoid fits to 5 events in s21, s48 and s75); TESS-SPOC 34.7 ppt
Secondary eclipse	Not detected (< 0.4 ppt, 3σ, at Gaia's conjunction A, phase 0.29 after the eclipse; nothing elsewhere in phase above 0.32 ± 0.46 ppt). At conjunction A the stars are 1.75 times farther apart (1.141a against 0.653a), so for the impact parameter of the observed eclipse (b ≈ 0.78) the other conjunction has b ≈ 1.4 and no eclipse
Orbit	The eclipse is on Gaia's conjunction B (ν = 270° − ω): Δφ = +0.0000 ± 0.0014 (0.0σ) with the conjunction at Gaia's epoch carried by the TESS period; +0.009 ± 0.006 (1.5σ) with Gaia's own period carried 181 cycles. Conjunction A is excluded (−0.299 ± 0.002). With the standard radial-velocity sign convention the RV star is in front at conjunction B, which would make this eclipse an occultation of the companion. A grazing occultation of a companion that supplies about 5% of the light could hardly remove 36 ppt, which mildly favours the reading as a transit of the radial-velocity star. Single-eclipse system: no *e* cos ω test

Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s21, s48, s75; 5 events: 34.7–34.8 ppt, 4.40 h, mid-times within 0.0–0.6 min of QLP's event by event); TARS (36.0 ppt). No SPOC 2-min data and no SPOC TCE
Pixel attribution	TESScut difference images in s21, s48 and s75 (2, 2 and 1 events): the deficit centroid is 0.2–0.7″ from the light centroid and 0.04–0.15 px from the catalogue position, with the peak deficit in the target's pixel and the target's image shape (residual 2.2–7.8%); depths 29.7–33.7 ppt in 1×1, 3×3 and 5×5 pixel apertures (the core window of a V-shaped eclipse is shallower than the fitted depth). Gaia DR3 has no source within 60″; the nearest within 90″ is G 19.98 at 65.3″, and only a star brighter than G 13.42 could make the eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary` (target and neighbours within 90″), SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2 and 3, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit; three catalogue references). No ASAS-3 data
Open issues	The five mid-times scatter about the linear ephemeris by more than their errors (O–C −0.8 to +1.5 min, χ^2 22.6 for 3 degrees of freedom); the sector-21 offsets of +1.0 and +1.5 min are reproduced by TESS-SPOC. Which star is eclipsed rests on the radial-velocity sign convention (the depth mildly favours a transit of the radial-velocity star), and the 36-ppt depth of a grazing eclipse constrains the companion only weakly
Next eclipses	Primary: BTJD 4338.159 (2026-10-24 15:50 UTC), 4352.117 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3827-170-1 is a slightly evolved F9–G0 star with a companion of at least 0.71 $M_\odot$ in a 14.0-d orbit with e = 0.41. Only one eclipse occurs per orbit: a 35.7-ppt, 4.5-h V-shaped grazing eclipse at conjunction B, where the stars are closest, while at conjunction A they are 1.75 times farther apart and miss each other, consistent with the < 0.4-ppt limit there. The eclipse falls on Gaia's conjunction B, at +0.0000 ± 0.0014 in phase, when Gaia's epoch is carried by the TESS period, the deficit is on the target in three sectors in an otherwise empty field, and TESS-SPOC reproduces the depth and timings. The system shows how an eccentric orbit can hide one of its two eclipses entirely; radial velocities that resolve the companion's lines would determine which star is eclipsed and how massive it is.

A89 CPD−85 289 = TIC 388287047

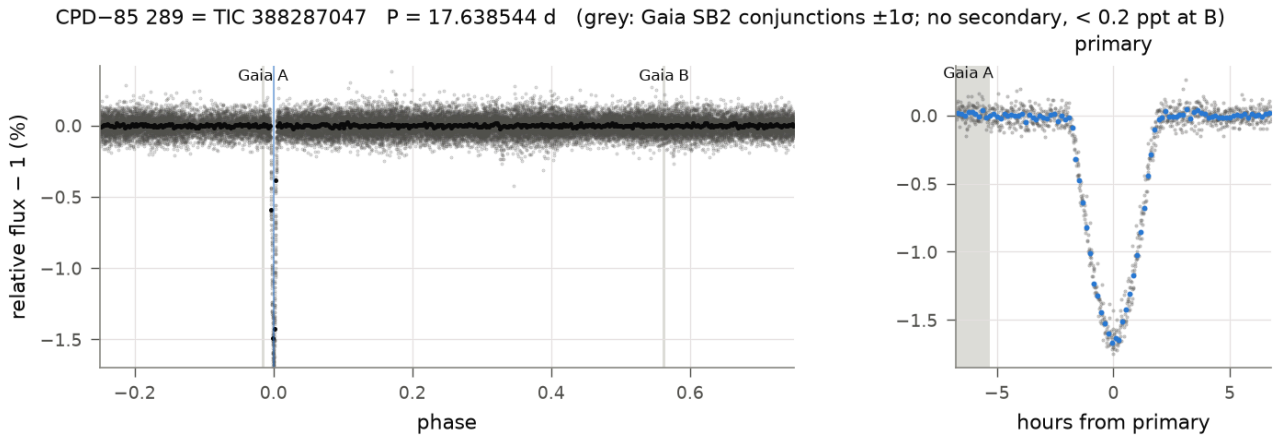


Figure A89. CPD-85 289, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 17.638544 d; grey bands are Gaia's predicted conjunctions A and B (±1σ). Left: the full orbit, with no secondary eclipse (< 0.2 ppt at B). Right: the 16-ppt, 3.9-h V-shaped primary, which falls about 6 h after the predicted conjunction A (grey band at the left edge).

Identifiers	TIC 388287047; CPD-85 289; TYC 9511-2297-1; Gaia DR3 5190617098481961984
Position (J2000)	169.88291, −85.71758 deg (TIC)
Brightness	Tmag 9.32; G 9.73

Stars	G0 (SIMBAD; TIC: T_{eff} 6120 K, 2.24 $R_\odot$, combined light of both stars) at 244 pc; Gaia SB2 orbit (e = 0.2082 ± 0.0052, ω = 54.6 ± 1.9°, goodness of fit 4.5). $M_1 \sin^3 i$ = 1.32 $M_\odot$ and $M_2 \sin^3 i$ = 1.10 $M_\odot$ (q = 0.83), so about 1.33 + 1.11 $M_\odot$ at i ≈ 85°, the companion giving roughly a third of the light; a sin i = 38.3 $R_\odot$
Period	17.638544 ± 0.000019 d, timing fit of 7 primaries in five sectors (Gaia: 17.63675 ± 0.00102 d; +1.8σ). Odd and even depths agree (15.85 ± 0.77 and 15.61 ± 0.51 ppt)
Epochs	Primary BTJD 3032.65988 ± 0.00069 = BJD_{TDB} 2460032.65988
Primary eclipse	16.1 ± 0.2 ppt at 200-s cadence (15.2 ppt at 10 min and 14.0 ppt at 30 min, smeared), 3.8–3.9 h, V-shaped (flat fraction about 0.2), in raw QLP SAP flux (trapezoid fits to 7 events); a grazing eclipse (impact parameter about 0.9)
Secondary eclipse	Not detected: 0.09 ± 0.04 ppt at Gaia's conjunction B (< 0.2 ppt, 3σ) and −0.04 ± 0.04 ppt at phase 0.5. None is expected: at B the stars are 1.152a apart against 0.818a at A, and the inclination that gives the observed grazing eclipse at A puts the impact parameter at B at 1.25
Orbit	Primary 0.0147 ± 0.0022 in phase (6.2 ± 0.9 h) after Gaia's predicted conjunction A, conditioned on the TESS period: 6.7σ with Gaia's full covariance, 3.3σ with Gaia's errors doubled, 2.2σ with the correlations ignored. Closing the gap needs ω = 47.4° at fixed T_p (−3.8σ) or T_p later by 0.25 d at fixed ω (2.7σ). The eclipse cannot be at B: an eclipse there would force one at A (impact parameter 0.67). No secondary, so no e cos ω test
Pipelines agreeing	QLP SAP (7 events, 15.98 ppt, 3.78 h); TESS-SPOC SAP and PDCSAP (5 events, 15.19 and 15.50 ppt, 3.79 h, mid-times within 0.1 min of QLP's on the same events); SPOC 2-min (s38: 14.51 ppt, 3.71 h, mid-time within 0.1 min of QLP's); TARS (7 events). SPOC multi-sector TCE (s56–s69): P 17.63837 ± 0.00023 d, depth 15.87 ppt, 3.97 h, impact parameter 0.97, MES 232, not flagged as an eclipsing binary, centroid 0.13 ± 2.57″ from the TIC position; single-sector TCEs in s66 (MES 183) and s38 (MES 92)
Pixel attribution	TESScut difference images in sectors 66 (2 events), 13 (2) and 38 (1): deficit centroid 0.3″, 0.5″ and 0.1″ from the light centroid, PSF-shaped (residual 6%), the peak deficit on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (14.0, 13.5, 13.3 ppt in s66). The brightest of 27 Gaia sources within 60″ (G 16.01 at 48.7″) would need 522%; the G 13.10 star at 80.5″ would need 37% and loses only 0.2–3.0 ppt at its own pixel. A companion within about 1″ is not excluded (RUWE 1.00)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and all 27 sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB: SIMBAD type SB*, in the APOGEE double-lined list of Kounkel et al. (2021); no eclipses reported. SPOC TCEs as above, not promoted to a TOI
Open issues	The conjunction offset, whose significance rests on Gaia's ω–T_p correlation (0.968) and a poorly fitting SB2 solution. The primary O–C runs about +3 min (2019), −7 min (2021; the s38 event is 6.8 min early in QLP, TESS-SPOC and SPOC 2-min alike) and about 0 (2023–25), rms 3.1 min against fitted errors about four times smaller; a light-travel-time signal from a third body is possible but untested with seven events
Next eclipses	Primary: BTJD 4337.912 (2026-10-24 09:54 UTC), 4355.551 (allow ±7 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CPD-85 289 is a known double-lined binary of two stars of about 1.33 and 1.11 $M_\odot$ in a 17.6-d orbit with e = 0.21, and the eclipses are new. Only conjunction A, where the stars are closest, is eclipsed, in a 16-ppt grazing eclipse; at B the projected separation is too large for any eclipse, which accounts for the missing secondary. The eclipses are on the target in three sectors of difference images and are reproduced by TESS-SPOC, SPOC 2-min and SPOC's own transit search, and the period agrees with Gaia's to 1.8σ. The primary falls 6.2 h after Gaia's predicted conjunction; the formal 6.7σ offset depends on the strong ω–T_p correlation of a poorly fitting SB2 solution, and is most likely an underestimated error in Gaia's ω rather than a misattribution. Gaia DR4 epoch radial velocities would test that directly, and further TESS sectors would show whether the timing residuals are a light-travel-time effect.

A90 CD−73 685 = TIC 454747287

CD−73 685 = TIC 454747287 P = 5.634764 d

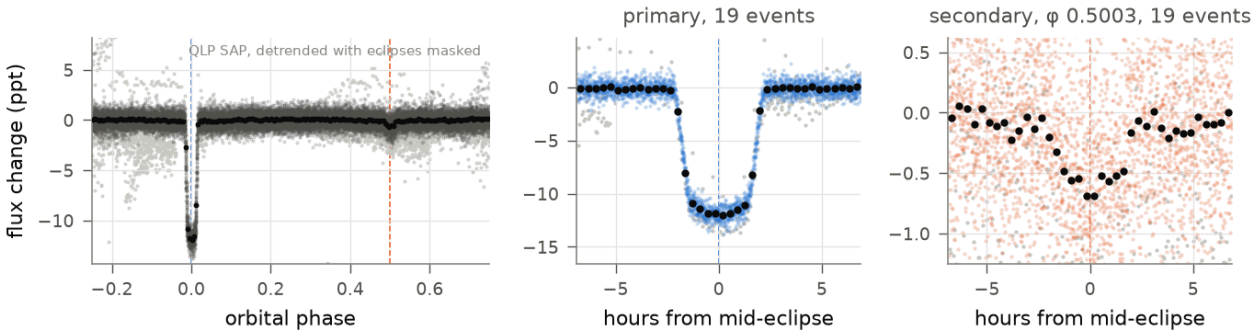


Figure A90. CD-73 685, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 5.634764 d (flagged cadences in light grey). Left: the full orbit. Middle: the 19 flat-bottomed 11.7-ppt, 4.3-h primaries. Right: the 0.49-ppt, 3.9-h secondary at phase 0.5003 (19 events).

Identifiers	TIC 454747287; CD-73 685; CPD-73 881; TYC 9238-667-1; Gaia DR3 5226756297640451840
Position (J2000)	175.63397, −73.86180 deg (TIC)
Brightness	Tmag 9.35; G 9.89
Stars	TIC: T_{eff} 5480 K, 2.10 $R_\odot$ (Gaia GSP-Phot 6270 K), a slightly evolved F/G star, at 183 pc; Gaia SB1 orbit (e = 0.019 ± 0.020, K_1 = 14.13 km/s, f(M) = 0.00165 $M_\odot$). The companion is an M dwarf: m_2 ≈ 0.13–0.14 $M_\odot$, with a radius ratio of at least 0.108. The flat bottom (T_{23}/T_{14} ≈ 0.6) and the 4.3-h duration together imply an impact parameter of about 0.75 and a primary radius of about 1.7 $R_\odot$, below the TIC's 2.1 $R_\odot$, so R_2 ≈ 0.18–0.23 $R_\odot$; and the secondary-to-primary depth ratio (0.042) matches a 2800–3000 K companion
Period	5.6347635 ± 0.0000015 d, timing fit of 19 primaries (Gaia: 5.634718 ± 0.000250 d; +0.18σ). Odd and even depths agree (11.67 and 11.62 ppt)
Epochs	Primary BTJD 2691.10727 ± 0.00013 = BJD_{TDB} 2459691.10727; secondary BTJD 2693.92630 (phase 0.50029 ± 0.00024)
Primary eclipse	11.70 ± 0.06 ppt, 4.30 h, flat-bottomed (T_{23}/T_{14} ≈ 0.6), in raw QLP SAP flux (trapezoid fits to 19 events; 11.0–12.0 ppt per event); the transit of the M dwarf across the primary
Secondary eclipse	0.49 ± 0.03 ppt, 3.9 h, at phase 0.50029 ± 0.00024 (fold fit to 19 events, each about 3σ); the occultation of the M dwarf
Orbit	Primary on Gaia's predicted conjunction A: +0.0041 ± 0.0058 in phase (0.7σ), secondary on conjunction B: +0.0032 ± 0.0057 (0.6σ), both at Gaia's epoch with the TESS period. e cos ω = +0.0005 ± 0.0004 from the eclipse spacing, Gaia +0.0005 ± 0.017 (0.0σ): circular
Pipelines agreeing	QLP SAP (19 events, 11.47 ppt, 4.32 h); TESS-SPOC SAP and PDCSAP (18 events, 11.13 and 11.46 ppt, mid-times −0.07 ± 0.23 min from QLP's on 14 matched events); SPOC 2-min SAP and PDCSAP (s39, 11.08 and 11.24 ppt, mid-times −0.14 ± 0.17 min); TGLC aperture flux (s11, 12.75 ppt). SPOC multi-sector TCE (s56–s69): P 5.63481 d, 11.84 ppt, 4.42 h, MES 217, not flagged as an eclipsing binary, centroid 0.11 ± 2.51″ from the TIC position; a second TCE is the secondary (P 5.6345 d, 531 ppm, MES 9.9, at phase 0.500)
Pixel attribution	TESScut difference images in sectors 39 and 65 (four events each): deficit centroid 0.2″ and 0.3″ from the light centroid, PSF-shaped (residual 7–8%), the peak deficit on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (11.8, 11.6, 11.4 ppt in s39). The brightest of 46 Gaia sources within 60″ (G 14.98 at 36.4″) would need 128%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit). ASAS-3 photometry (15–18 mmag scatter, 12 points in the eclipse core) is too noisy to detect the eclipse. SPOC TCEs as above, not promoted to a TOI
Next eclipses	

	Primary: BTJD 4330.823 (2026-10-17 07:47 UTC), 4336.458, 4342.093; secondary: BTJD 4328.008 (2026-10-14 12:13 UTC), 4333.642 (allow ±2 min for primaries and ±6 min for secondaries; the individual secondaries, each about 3σ, scatter by 30 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-73 685 is a slightly evolved F/G star eclipsed by an M dwarf of about 0.13 $M_\odot$ in a circular 5.63-d orbit. The 11.7-ppt flat-bottomed transit and the 0.49-ppt occultation at phase 0.5003 fall on Gaia's two predicted conjunctions, the period agrees with Gaia's to 0.2σ, and the eclipse spacing gives the same e cos ω as Gaia's orbit. Difference images in two sectors, three further pipelines and SPOC's own transit search all place the signal on the target. The deep eclipse is at the conjunction where the radial-velocity star is behind its companion, as expected for a single-lined system. With radial velocities and a better primary temperature (TIC and Gaia differ by about 800 K), the system would give a mass and radius for a very-low-mass star.

A91 HD 311112 = TIC 311635507

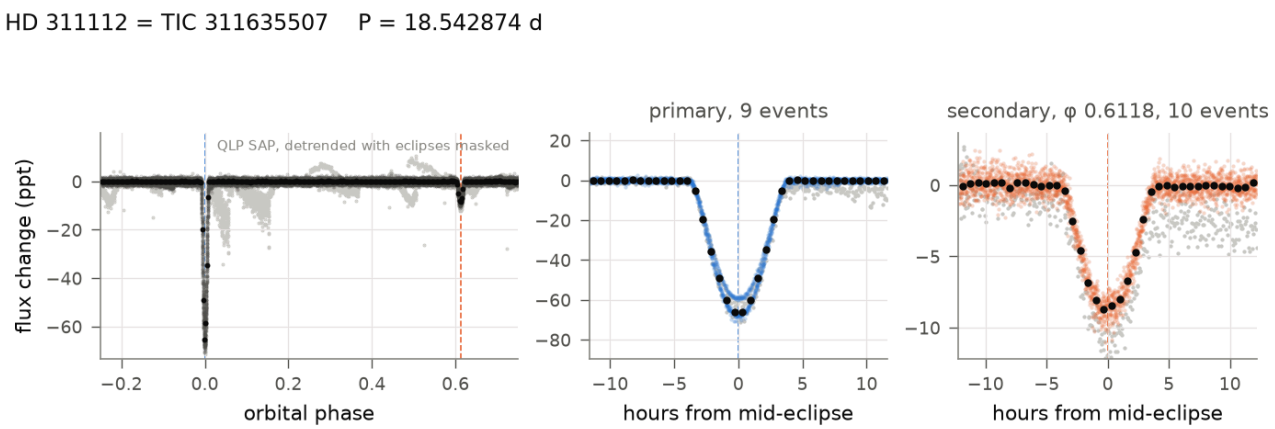


Figure A91. HD 311112, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 18.542874 d (flagged cadences in light grey). Left: the full orbit, with the secondary at phase 0.6118. Middle: the nine 67-ppt, 7.3-h near-V primaries; the two depth levels are the sectors before and after the depth step of sectors 99–100 (59 ppt). Right: the ten 8.7-ppt, 7.6-h secondaries.

Identifiers	TIC 311635507; HD 311112; CD-65 1201; CPD-65 1787; TYC 8986-1004-1; Gaia DR3 5860649009076840320
Position (J2000)	181.56340, −66.42961 deg (TIC)
Brightness	Tmag 9.38; G 9.84
Stars	F5 (SIMBAD; TIC: T_{eff} 5919 K, 2.33 $R_\odot$, slightly evolved) at 239 pc; Gaia SB1 orbit (e = 0.184 ± 0.014, ω = 6.2 ± 6.2°, K_1 = 29.32 ± 0.36 km/s). f(M) = 0.046 $M_\odot$, so $m_2 \geq 0.48\ M_\odot$ for M_1 = 1.08 $M_\odot$; radius ratio ≥ 0.26 ($R_2 \geq 0.6\ R_\odot$) and the secondary-to-primary depth ratio (0.13) indicate a K/M dwarf of about 3600 K. Crowded field (322 Gaia sources within 60″; TIC contamination ratio 0.10)
Period	18.5428744 ± 0.0000049 d, timing fit of 9 primaries; the 10 secondaries give 18.542873 ± 0.000009 d (Gaia: 18.54250 ± 0.00509 d; +0.07σ)
Epochs	Primary BTJD 3479.18152 ± 0.00017 = BJD_{TDB} 2460479.18152; secondary BTJD 3490.52516 (phase 0.61175 ± 0.00003)
Primary eclipse	66.74 ± 0.19 ppt, 7.27 h, nearly V-shaped ($T_{23}/T_{14} \approx 0.2$), in raw QLP SAP flux (trapezoid fits to 9 events). The QLP depth is 67.3–68.4 ppt in sectors 37–65 and 58.9–59.5 ppt in sectors 99–100, with unchanged duration and the secondary scaled in proportion: a change of dilution or background in this crowded field, not of geometry. A fixed 3×3 TESScut aperture shows a smaller step (59.1 ppt in s65, 55.6 ppt in s99), so the change is not confined to QLP's aperture. Crowding-corrected TESS-SPOC PDCSAP gives 69.5 ppt
Secondary eclipse	8.70 ± 0.06 ppt, 7.63 h, at phase 0.61175 ± 0.00003 (10 events); nothing at phase 0.5 (−0.01 ± 0.04 ppt). The secondary is displaced from phase 0.5 by the eccentric orbit

Orbit	Primary on Gaia's predicted conjunction A: −0.0007 ± 0.0056 in phase (0.1σ); secondary on conjunction B: −0.0042 ± 0.0049 (0.9σ), both at Gaia's epoch with the TESS period. Eclipse spacing 0.61175 against Gaia's predicted 0.6154 ± 0.0092 (0.5σ). e cos ω = 0.1764 ± 0.0001 (Kepler's equation, Gaia's e sin ω held), Gaia 0.183 ± 0.015, implying e = 0.178, ω = 6.4°; the duration ratio gives e sin ω ≈ +0.024 (Gaia +0.020 ± 0.019)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (7 primaries and 7 secondaries: 61.6 and 8.58 ppt; primary mid-times +0.06 ± 0.05 min from QLP's on 4 matched events, secondaries −0.5 ± 1.4 min) and PDCSAP (69.5 and 9.67 ppt, CROWDSAP 0.87–0.89); TARS (65.9 and 9.51 ppt); TGLC (s11). TESS-SPOC TCEs: the primary in s37 (P 18.54296 d, 72.2 ppt, 7.6 h, MES 750, centroid 0.48 ± 2.5″ from the TIC position) and s65 (MES 349, 0.84″), the secondary in s38 (10.4 ppt, MES 80); never flagged as an eclipsing binary
Pixel attribution	TESScut difference images in sectors 37, 65 and 99 (two primaries each): deficit centroid 1.4″, 1.4″ and 1.6″ from the light centroid, the peak deficit on the target pixel, PSF-fit residual 9–22% (crowded field); depth stable across 1×1, 3×3 and 5×5 pixel apertures (61.4, 59.1, 58.5 ppt in s65). A source would need G < 12.78 to make the dip; the brightest within 90″ is G 13.42 at 85.5″, and within 60″ G 13.78 at 28.5″ would need 257%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (one unrelated G 18.5 variable 55″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit). The primary eclipse is present in ASAS-3 photometry (2001–2009) at the TESS period, 374 cycles before TESS: +74.0 ± 8.3 mmag in the eclipse core against 75.0 expected, with a trial-period scan peaking at 18.542845 d. TESS-SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4332.154 (2026-10-18 15:44 UTC), 4350.697; secondary: BTJD 4343.497 (2026-10-29 23:59 UTC), 4362.040 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 311112 is a slightly evolved F5 star with a K or early-M dwarf companion of at least 0.48 $M_\odot$ in an 18.54-d orbit with e = 0.18. A near-grazing 67-ppt primary and an 8.7-ppt secondary at phase 0.6118 fall on Gaia's two predicted conjunctions to within 1σ, and the eclipse spacing gives e cos ω = 0.176, within 0.5σ of Gaia's. The deficit lies on the target in three sectors of difference images in a crowded field where no neighbour is bright enough to produce it, and TESS-SPOC and TARS reproduce both eclipses. The eclipse is also recovered at full depth in ASAS-3 photometry two decades old, on the TESS ephemeris carried back 374 cycles; the ASAS-3 data are not part of the ephemeris fit. The QLP depth falls by 12% from sectors 37–65 to 99–100 with the secondary in proportion and the duration unchanged, a change of dilution that a fixed TESScut aperture also shows, though smaller; depths for this star should be quoted with their pipeline.

A92 BD+69 653 = TIC 148877735

TIC 148877735 BD+69 653 P = 3.607450 d

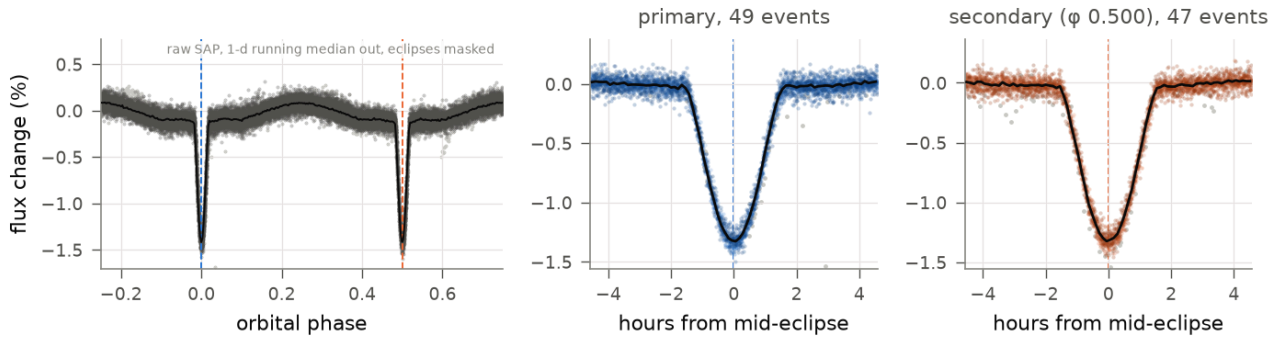


Figure A92. BD+69 653, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 3.607450 d, twice the fold period. Left: the full orbit, with two equal eclipses at phases 0 and 0.500 and an ellipsoidal variation of at least 1–2 ppt peaking at quadrature (the running median removes part of it). Middle and right: the 13.2-ppt, 3.05-h V-shaped primary and the identical secondary (49 and 47 events plotted; 39 and 36 fully covered events fitted individually, 35 and 31 of them in the timing fit).

Identifiers	TIC 148877735; BD+69 653; TYC 4393-1419-1; Gaia DR3 1683351069470284032
Position (J2000)	183.42497, +69.15763 deg (TIC)
Brightness	Tmag 9.33; G 9.66
Stars	F5 (SIMBAD; TIC: T_{eff} 6516 K, 2.54 $R_\odot$ as a single star) at 316 pc; Gaia SB2 orbit (e = 0.002 ± 0.007, K_1 = 89.72 ± 0.61, K_2 = 89.46 ± 0.63 km/s). Twins: q = 1.003, M $\sin^3$i = 1.074 and 1.077 $M_\odot$, about 1.19 $M_\odot$ each at i ≈ 75°, and about 1.8 $R_\odot$ each if they share the light equally: slightly evolved F5 stars. a sin i = 12.8 $R_\odot$
Period	3.6074499 ± 0.0000008 d, timing fit of 35 primaries (31 secondaries give 3.6074495 ± 0.0000009 d) (Gaia: 3.607477 ± 0.000048 d; −0.6σ). Twin eclipses: the period is twice the fold period 1.8037182 d. The two eclipses are identical in depth and duration, so the photometry alone cannot separate P from P/2; the Gaia SB2 orbit, with two equal, anti-phased velocity curves, fixes P
Epochs	Primary BTJD 3110.82736 ± 0.00008 = BJD_{TDB} 2460110.82736; secondary BTJD 3101.80861 ± 0.00009 (phase 0.49997 ± 0.00003). For twins the labels are arbitrary: the primary here is the eclipse at Gaia's conjunction B, of Gaia's secondary component
Primary eclipse	13.21 ± 0.26 ppt, 3.05 ± 0.05 h, V-shaped (flat fraction about 0.2), in raw QLP SAP flux (trapezoid fits to 39 events); 12.4–12.7 ppt with a quadratic baseline in all pipelines
Secondary eclipse	13.16 ± 0.30 ppt, 3.05 h, at phase 0.49997 (36 events)
Orbit	Primary on Gaia's predicted conjunction B: +0.0009 ± 0.0020 in phase (0.5σ); secondary on conjunction A: +0.0011 ± 0.0019 (0.5σ), at Gaia's epoch with the TESS period; Gaia predicts the separation 0.5001 ± 0.0033. e cos ω = −0.0001 ± 0.0001 from the eclipse spacing (Gaia +0.0001 ± 0.0052); equal durations give e sin ω ≈ 0: circular
Pipelines agreeing	QLP SAP; SPOC 2-min SAP (s41, 13 events: mid-times +0.08 ± 0.30 min, depth ratio 1.03); TESS-SPOC SAP (s41 and s74, mid-times within 0.06 min); TASOC and TARS (s14, s15). In the 30-min sectors 14 and 15 the QLP mid-times are 5.3–6.5 min later than TESS-SPOC, TASOC (including its 2-min pixel light curve) and TARS on the same events; those sectors are excluded from the timing fit. SPOC multi-sector TCE (s14–s50): P 1.8037244 ± 0.0000016 d (half the period), depth 12.76 ppt, 3.10 h, impact parameter 0.99, MES 314, 40 transits, not flagged as an eclipsing binary, centroid 0.45 ± 2.53″ from the TIC position
Pixel attribution	TESScut difference images in sectors 47 (8 primaries) and 74 (7): deficit centroid 0.3″ and 0.6″ from the light centroid, PSF-shaped (residual 5.4% and 2.5%); depth stable across 1×1, 3×3 and 5×5 pixel apertures (11.5, 11.3, 11.2 ppt in s47). The only Gaia sources within 60″ (G 18.56 and 19.74, both at 52″) would need 4787% and 14300%. A companion within about 1″ is not excluded (RUWE 0.99)
Catalogue status	

	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search; no ASAS-3 data at this declination. Known SB (SIMBAD SB*, F5, from the Gaia DR3 orbit). SPOC TCE at the half period as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4330.145 (2026-10-16 15:29 UTC), 4333.753, 4337.360; secondary: BTJD 4328.342 (2026-10-14 20:12 UTC), 4331.949 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+69 653 is a pair of nearly identical, slightly evolved F5 stars (q = 1.003, about 1.2 $M_\odot$ each) in a circular 3.61-d orbit seen at about 75°, so that each star grazes the other at conjunction and the two V-shaped eclipses are equal at 13.2 ppt. The photometry alone admits half the period; Gaia's double-lined orbit fixes the true one, and the two eclipses fall on its two conjunctions to 0.5σ. The deficit lies on the target in difference images, and SPOC's own transit search detected the eclipses at the half period with a centroid on the star (MES 314); because the alternate events are equal at that period, it was not flagged as an eclipsing binary. Ellipsoidal variation of 1–2 ppt is visible out of eclipse; it is measured after a 1-d running median on a 3.6-d orbit, which suppresses part of it, so it is a lower bound. In this star's two 30-min sectors the QLP mid-times are about 6 min later than those of three other pipelines on the same events, so the offset lies in the QLP light curve rather than in the star; whether other stars in those sectors share it has not been tested.

A93 HD 112601 = TIC 175734936

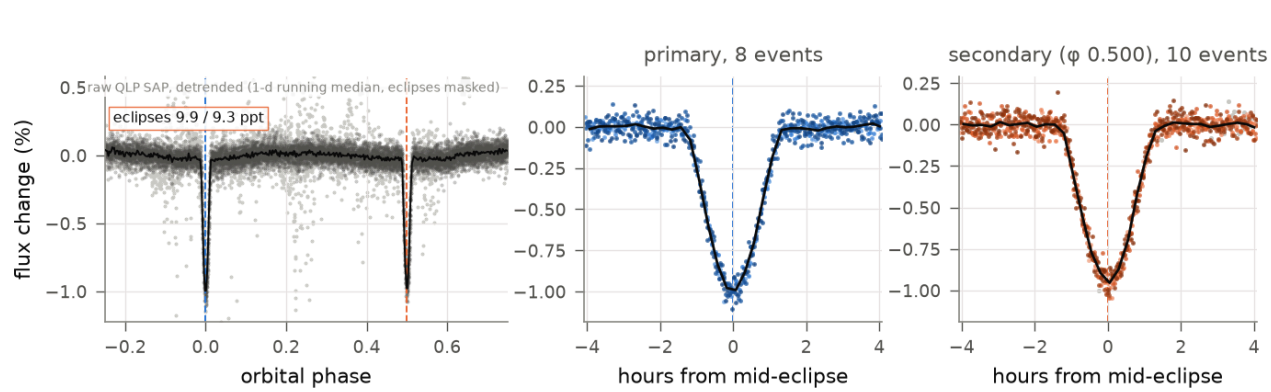


Figure A93. HD 112601, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 4.337953 d, twice the fold period (flagged cadences in light grey). Left: the full orbit, with eclipses at phases 0 and 0.500. Middle: the 9.9-ppt, 2.55-h V-shaped primary (8 events). Right: the 9.3-ppt secondary of the same duration (10 events).

Identifiers	TIC 175734936; HD 112601; BD-01 2746; HIP 63269; TYC 4950-1175-1; 1SWASP J125754.13-020310.2; Gaia DR3 3685481410582132864
Position (J2000)	194.47556, −2.05285 deg (TIC)
Brightness	Tmag 8.95; G 9.34
Stars	F5V (SIMBAD; Gaia GSP-Phot T_{eff} 6194 K, blended light) at 232 pc from the Gaia parallax (TIC 118 pc; RUWE 4.64); Gaia SB2C orbit (circular; K_1 = 80.54 ± 0.64, K_2 = 85.97 km/s, q = 0.94, goodness of fit 5.6). $M_1 \sin^3 i$ = 1.07 and $M_2 \sin^3 i$ = 1.00 $M_\odot$: two F stars of about 1.1 $M_\odot$, a = 14.8 $R_\odot$. A third star is present: one CORALIE spectrum shows three line systems (Schanche et al. 2019), consistent with the high RUWE and the discrepant distances; it dilutes the eclipses
Period	4.337953 ± 0.000001 d, timing fit of 8 primaries and 10 secondaries (Gaia: 4.337861 ± 0.000105 d; +0.9σ). Twin eclipses: the period is twice the fold period 2.168977 d; at the fold period alternate eclipses differ in depth by 0.46 ± 0.08 ppt (5.5σ)
Epochs	Primary BTJD 3380.72231 ± 0.00018 = BJD_{TDB} 2460380.72231; secondary BTJD 3382.89142 (phase 0.50003 ± 0.00006)
Primary eclipse	

	9.89 ± 0.06 ppt, 2.55 h, V-shaped (flat fraction 0.2), in raw QLP SAP flux (trapezoid fits to 8 events); grazing (impact parameter about 0.9)
Secondary eclipse	9.33 ± 0.05 ppt, 2.55 h, at phase 0.50003 ± 0.00006 (10 events); 0.56 ppt shallower than the primary (7σ)
Orbit	Circular orbit (SB2C): no conjunction-phase test. Gaia's reference epoch falls 0.2469 ± 0.0012 in phase (19 min from quadrature) after a TESS primary, as expected if it marks a velocity extremum; the secondary at phase 0.50003 agrees with a circular orbit
Pipelines agreeing	QLP SAP (s46, s91); SPOC 2-min SAP (s46: 9.69 and 9.28 ppt, event-to-event scatter 0.19 and 0.21 ppt, standard errors 0.08 and 0.09 ppt, so the difference is 3.4σ) and PDCSAP (9.65 and 9.25 ppt); QLP mid-times later than SPOC's by +0.39 ± 0.22 min (5 primaries) and +0.19 ± 0.17 min (6 secondaries; standard errors); TESS-SPOC SAP (s46: 9.60 and 9.20 ppt). SPOC and TESS-SPOC TCEs at half the period (2.16895 and 2.16896 d, 9.4 ppt, 2.61–2.63 h, impact parameter 0.98, MES 181 and 199), not flagged as an eclipsing binary, centroids 0.27–0.62 ± 2.50″ from the TIC position
Pixel attribution	TESScut difference images in sectors 46 and 91, primaries and secondaries separately (3–6 events each): deficit centroid 0.4–0.7″ from the light centroid, PSF-shaped (residual 7–9%); depth stable across 1×1, 3×3 and 5×5 pixel apertures (8.1, 8.3, 8.3 ppt for the s46 primaries, a core average). No Gaia source within 60″; the G 12.05 star at 80.5″ would need 12% and would displace the centroid by pixels. A source within about 1″ is not excluded, and the spectroscopic third star is one
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI). Known SB (SIMBAD SB*, F5V). Previously catalogued: SIMBAD lists the type EB* among its object types, from the SuperWASP disposition catalogue (Schanche et al. 2019), which lists 1SWASP J125754.13-020310.2 as an eclipsing binary with follow-up flag “blended eclipsing binary” and three line systems in one CORALIE spectrum, with no period. ASAS-3 shows the eclipse marginally at the TESS ephemeris (+9 mmag, 2.7σ, against about 10 mmag expected). SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4330.734 (2026-10-17 05:43 UTC), 4335.072, 4339.410; secondary: BTJD 4328.565 (2026-10-15 01:40 UTC), 4332.903 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 112601 is a hierarchical triple whose inner pair, two F stars of about 1.1 $M_\odot$ with q = 0.94, eclipses every 4.338 d in a circular orbit. The two eclipses are grazing and nearly equal (9.9 and 9.3 ppt), and their 7σ depth difference is reproduced in SPOC 2-min photometry (3.4σ in one sector); Gaia's double-lined period agrees to 0.9σ. Both eclipse kinds lie on the target in difference images. The SuperWASP disposition catalogue lists it as an eclipsing binary, with a follow-up disposition of blended eclipsing binary and a note that one spectrum showed three line systems; no period or eclipse parameters were published. The ephemeris, depths and near-twin nature given here are new. Because SIMBAD carries that flag as an eclipsing-binary type, the system counts as previously catalogued under the definition of Section 1 and is not among the new systems counted in this paper. The third star dilutes the eclipses and accounts for the large RUWE and the discrepant distances; spectra resolving all three components would give its light fraction and, with the eclipses, masses for the inner pair.

A94 HD 114997 = TIC 245194662

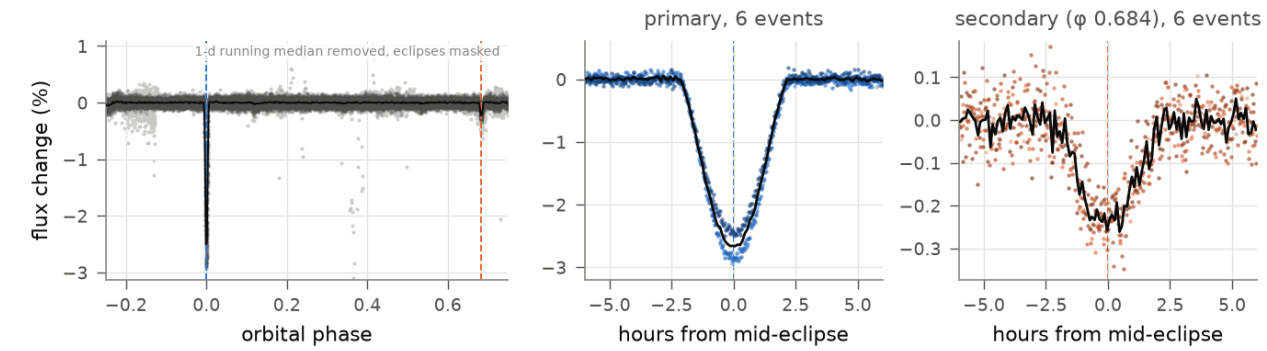


Figure A94. HD 114997, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 17.279547 d (flagged cadences in light grey). Left: the full orbit, with the secondary at phase 0.684. Middle: the six 28-ppt, 4.2-h V-shaped primaries (the shallower points are the diluted sector-100 events). Right: the six 2.5-ppt, 4.0-h secondaries.

Identifiers	TIC 245194662; HD 114997; CPD-53 5525; TYC 8649-1616-1; Gaia DR3 6068115377041311360
Position (J2000)	198.79864, −53.67158 deg (TIC)
Brightness	Tmag 8.92; G 9.35
Stars	G3V (SIMBAD; TIC: T_{eff} 6096 K, 1.38 $R_\odot$, 1.14 $M_\odot$) at 121 pc; Gaia SB1 orbit (e = 0.300 ± 0.009, ω = 358.1 ± 1.6°, K_1 = 23.87 ± 0.20 km/s). f(M) = 0.0211 $M_\odot$, so m_2 ≈ 0.36 $M_\odot$ at i = 90° (0.29–0.43 $M_\odot$): an M dwarf, radius ratio ≥ 0.17; a = 32.2 $R_\odot$. Crowded field (78 Gaia sources within 60″; TIC contamination ratio 0.13)
Period	17.2795467 ± 0.0000052 d, timing fit of 6 primaries; the 6 secondaries give 17.279539 ± 0.000033 d (Gaia: 17.28057 ± 0.00138 d; −0.7σ)
Epochs	Primary BTJD 3353.15051 ± 0.00022 = BJD_{TDB} 2460353.15051; secondary BTJD 3364.96274 (phase 0.68360 ± 0.0001)
Primary eclipse	28 ppt, 4.1–4.2 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 6 events): 27.7–28.5 ppt in sectors 11, 37 and 64 and 24.0 ppt in sector 100, most likely diluted by the red semi-regular variable 50″ away; TESS-SPOC PDCSAP 28.4–28.7 ppt. A grazing eclipse (impact parameter about 0.84; SPOC 0.96)
Secondary eclipse	2.5 ppt (sd 0.3), about 4.0 h, at phase 0.68360 (6 events); recovered at 2.1–3.6 ppt in every pipeline. Displaced from phase 0.5 by the eccentric orbit; equal durations give e sin ω ≈ 0 (Gaia −0.010)
Orbit	Primary on Gaia's predicted conjunction A: +0.0039 ± 0.0028 in phase (1.4σ); secondary on conjunction B: −0.0006 ± 0.0037 (0.2σ), at Gaia's epoch with the TESS period (1.3σ and 0.7σ when conditioned on the TESS period through Gaia's full covariance). Eclipse spacing 0.6836 against Gaia's predicted 0.6880 (0.6823–0.6938). e cos ω = +0.2926 ± 0.0001 from the eclipse spacing, Gaia +0.2999 ± 0.0094 (0.8σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s37, s64: 3 primaries, 26.2 and 28.6 ppt, mid-times −0.3 min from QLP's); TARS (4 primaries, 27.6 ppt, +0.01 min); eleanor-lite (s11, 26.9 ppt, +0.8 min). TGLC (s11) agrees in time (+0.7 min) but not in depth (39.5 ppt). TESS-SPOC TCE in s64: P 17.27980 ± 0.00027 d, 29.1 ppt, 4.40 h, impact parameter 0.96, MES 320, centroid 0.18 ± 2.5″ from the TIC position; a TCE in s37 (MES 105) combined primary and secondary at 11.81 d
Pixel attribution	TESScut difference images in sectors 64 and 100 (two primaries each): deficit centroid 2.8″ and 2.1″ from the catalogue position, with the peak deficit on the target pixel; the light centroid is pulled 10–11″ east by neighbours. On the target's 3×3 pixels the deficit is PSF-shaped to 14–19%; the depth falls in larger apertures (23.3, 22.4, 18.6 ppt in 1×1, 3×3, 5×5 in s64) as neighbours' light enters. Two Gaia sources within 60″ are bright enough: the red semi-regular variable (G 11.83, BP−RP 4.7, 50.3″; about 13% needed in the TESS band) and G 12.13 at 52.3″ (39%); both lie 2.4 px from the deficit centroid and lose only 0.8–8.4 ppt at their own pixels, the target's wing, against about 300 ppt if either were the source. All others would need more than 200%

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, G3V). Listed in Kostov et al. (2025) table 2. The only catalogued variable within 60″ is the G 11.83 neighbour at 48.7″, ASASSN-V J131506.19-534018.9 (VSX type SR, P = 513 d; a Gaia DR3 long-period variable); a long-period variable cannot produce a 4-h eclipse. The primary is present in ASAS-3 photometry (2000–2009) at the TESS ephemeris: +26.5 mmag (4.9σ) in the eclipse core, holding across the TESS period ± 8σ and absent at Gaia's period. TESS-SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4338.085 (2026-10-24 14:06 UTC), 4355.364; secondary: BTJD 4332.617 (2026-10-19 02:53 UTC), 4349.897 (allow ±2 min; single secondaries scatter by 4.6 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 114997 is a G3 dwarf with an M-dwarf companion of about 0.36 $M_\odot$ in a 17.28-d orbit with e = 0.30, seen nearly edge-on so that both conjunctions are grazed. The 28-ppt primary and the 2.5-ppt secondary fall on Gaia's two predicted conjunctions, and the secondary's displacement to phase 0.684 gives e cos ω = 0.293, within 0.8σ of the spectroscopic value. That offset is fixed by the orbit and could not be reproduced by a blended neighbour. In this crowded field the difference images place the deficit on the target pixel, away from the two bright neighbours, five pipelines recover the eclipses at the same times (TGLC with a deeper 39.5 ppt), and the TESS-SPOC transit search detects them with a centroid on the target (MES 320). The primary is also present in ASAS-3 photometry at the TESS ephemeris (six in-eclipse points, 4.9σ), which confirms that the ephemeris holds about 20 years earlier; the ASAS-3 data are not part of the timing fit. The QLP depth in sector 100 is diluted, so depths should be quoted with their pipeline and sector.

A95 HD 234179 = TIC 154346187

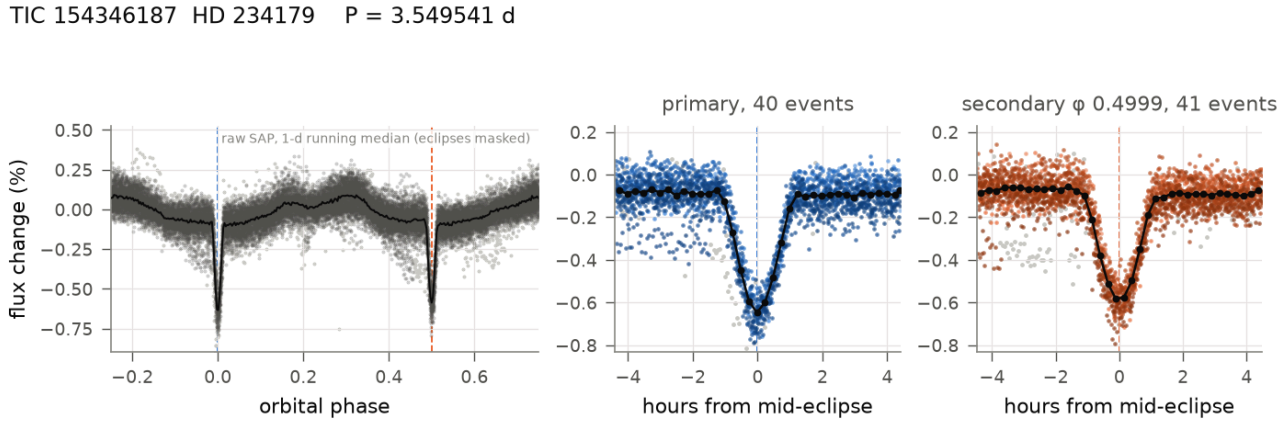


Figure A95. HD 234179, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 3.549541 d, twice the fold period. Left: the full orbit, with eclipses at phases 0 and 0.4999 and a 1–2 ppt non-sinusoidal modulation out of eclipse. Middle: the 5.6-ppt, 2.2-h V-shaped primary. Right: the 5.1-ppt secondary (40 and 41 events plotted; 36 and 31 fully covered events in the timing fit).

Identifiers	TIC 154346187; HD 234179; BD+53 1736; TYC 3860-1655-1; Gaia DR3 1605802239486264448
Position (J2000)	222.01636, +53.16917 deg (TIC)
Brightness	Tmag 8.98; G 9.38
Stars	G0 (SIMBAD; TIC: T_{eff} 6119 K, 2.65 $R_\odot$ from the combined light) at 256 pc; RUWE 5.5. Gaia DR3 gives two orbits for this source: an SB2C (circular; P = 3.549504 ± 0.000043 d, K_1 = 104.83 ± 0.56, K_2 = 77.41 km/s, goodness of fit 23.1) and an astrometric orbit (P = 541.2 ± 2.2 d, e = 0.113 ± 0.022, photocentre semi-major axis 1.66 mas = 0.40 AU). Inner pair: M $\sin^3 i$ = 0.95 and 1.28 $M_\odot$, about 1.05 and 1.4 $M_\odot$ at i ≈ 75–78°, a sin i = 12.8 $R_\odot$. Hierarchical triple: Gaia's astrometric orbit is the outer orbit of the eclipsing pair about an unresolved third body; the timing amplitude requires $M_3 \gtrsim 0.66\ M_\odot$

Period	3.549541 ± 0.000002 d, timing fit of 36 primaries (Gaia SB2C: 3.549504 ± 0.000043 d; +0.9σ). Twin eclipses: the period is twice the fold period 1.77478 d; at the fold period alternate eclipses differ in depth (5.85 ± 0.09 and 5.18 ± 0.08 ppt, 5.6σ)
Epochs	Primary BTJD 2489.65835 ± 0.00031 = BJD_{TDB} 2459489.65835; secondary BTJD 2491.43279 ± 0.00035 (phase 0.49991 ± 0.00010)
Primary eclipse	5.59 ppt, 2.20 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 36 events); 5.5–5.9 ppt in every sector; grazing (impact parameter about 0.9)
Secondary eclipse	5.09 ppt, 2.24 h, at phase 0.49991 ± 0.00010 (31 events); about 10% shallower than the primary in every pipeline
Orbit	Circular orbit (SB2C): no formal conjunction-phase test. If DR3's circular solutions fix ω = 0 so that the periastron epoch is the ascending node, the primary falls −0.0007 ± 0.0009 in phase (−3.8 ± 4.8 min, −0.8σ) from conjunction A. Outer orbit: the 67 eclipse times (four seasons) are consistent with the light-travel-time curve whose shape is fixed by Gaia's astrometric orbit. With the period, eccentricity and phase fixed and the amplitude, its sign and a linear ephemeris free, K = −3.39 ± 0.32 min (error scaled by √1.6 for χ^2/dof = 102.4/63, down from 391.5/64 without the term; O−C rms 1.39 min, from 2.71), against a minimum of 3.25 min predicted from Gaia's photocentre orbit. About 11% of randomly chosen outer phases fit as well as Gaia's, so the chance of such a match is about 0.1. A free sinusoid search (50–3000 d) finds 528 d, but 318 and 1463 d fit almost as well ($\Delta\chi^2 \approx 5$)
Pipelines agreeing	QLP SAP (s49, s50, s76, s77: 5.75 and 5.01 ppt); TESS-SPOC SAP and PDCSAP (5.44 and 4.93 ppt), mid-times −0.15 ± 0.25 min (17 primaries) and −0.11 ± 0.29 min (17 secondaries) from QLP's; TARS (5.50 and 4.92 ppt). TESS-SPOC single-sector TCEs in s49, s50, s76 and s77 at half the period (1.7747–1.7749 d, 5.1–5.3 ppt, 2.2 h, MES 71–91), not flagged as an eclipsing binary, centroids 0.49–0.99″ from the TIC position
Pixel attribution	TESScut difference images in sectors 49 (6 primaries) and 76 (5): deficit centroid 0.9″ and 1.3″ from the light centroid, PSF-shaped (residual 6–9%), on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (4.35, 4.39, 4.41 ppt in s49, a core average). The only Gaia sources within 60″ (G 17.17 at 42.4″, G 18.69 at 44.0″) would need 740% and 3000%. The third body (about 8 mas) is unresolved and its light is part of the target's
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search; no ASAS-3 data. Known SB (SIMBAD SB*, G0); SIMBAD references are catalogues only. TESS-SPOC TCEs at half the period as above, not promoted to a TOI
Open issues	The outer period is sampled sparsely by seven sectors in four seasons, so the timing does not by itself fix it. The predicted minimum amplitude assumes that the third body's fraction of the light is below its fraction of the mass. The mass budget does not close: K = 3.39 min needs $M_3 \approx 0.69\ M_\odot$, and a main-sequence star of that mass would shrink the photocentre orbit to about 0.34 AU, against Gaia's 0.40 AU, so the third body would have to be under-luminous for its mass. The out-of-eclipse modulation of 1–2 ppt, probably spots on a synchronised star, is not analysed. Eclipse times may depart from the linear ephemeris by up to ±3.4 min of light-travel time
Next eclipses	Primary: BTJD 4328.321 (2026-10-14 19:44 UTC), 4331.870, 4335.420; secondary: BTJD 4330.095 (2026-10-16 14:19 UTC), 4333.645 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 234179 is a hierarchical triple. Its inner pair, of about 1.05 and 1.4 $M_\odot$ in Gaia's circular 3.55-d double-lined orbit, shows two grazing eclipses of 5.6 and 5.1 ppt, unequal in every pipeline, so the true period is twice the fold period. Gaia's separate 541-d astrometric orbit for this source, which is the outer orbit of the eclipsing pair about a third body, predicts a light-travel-time variation of at least 3.25 min if the third body supplies a smaller fraction of the light than of the mass. The eclipse times are consistent with it: a light-travel-time variation is present (QLP and TESS-SPOC times agree to under 0.4 min, so it is not a pipeline effect), and fitting only its amplitude, with the period, eccentricity and phase fixed by Gaia, gives 3.39 ± 0.32 min and removes three-quarters of the timing χ^2. The test is not strong: 67 timings in four seasons, three free parameters plus the sign of K, a chance coincidence of about 0.1, and outer periods of 318 and 1463 d not excluded. The amplitude also implies a third body of about 0.69 $M_\odot$

that is fainter than a main-sequence star of that mass, which is unexplained. The deficit is on the target in difference images and TESS-SPOC reproduces both eclipses. Eclipse timing over a full outer orbit, and radial velocities of the third star, would test the outer orbit and give the outer mass ratio and the mutual inclination of the two orbits.

A96 HD 140021 = TIC 426877400

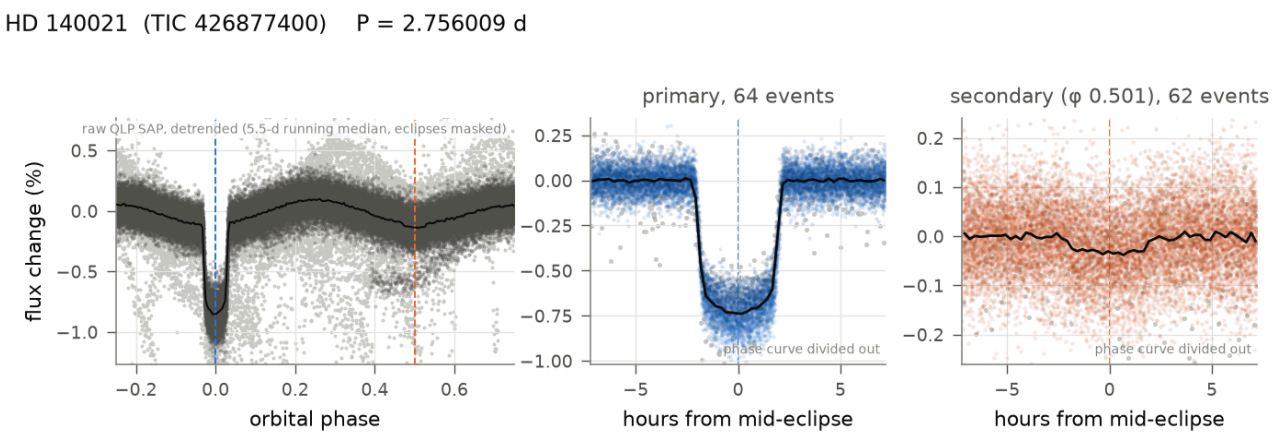


Figure A96. HD 140021, raw QLP SAP flux (sector medians removed, flagged cadences light grey), detrended by a 5.5-d running median with the eclipses masked and folded on P = 2.7560089 d. Left: the full orbit, with the 0.9-ppt ellipsoidal variation; the points near −0.5% at phases 0.38–0.48 are the sector-99 start ramp. Middle: the 7.6-ppt, 4.5-h flat-bottomed primary; right: the 0.31-ppt secondary near phase 0.50 (both with the phase curve divided out).

Identifiers	TIC 426877400; HD 140021; CD−71 1198; CPD−71 1917; TYC 9268-1801-1; Gaia DR3 5794909926060103424
Position (J2000)	236.72191, −71.67431 deg (TIC)
Brightness	Tmag 9.62; G 10.00
Stars	F6/7IV (SIMBAD; TIC: T_{eff} 6215 K, 2.12 $R_\odot$) at 271 pc; Gaia SB1 orbit (e = 0.079 ± 0.030, ω = 8 ± 15°, K_1 16.8 km/s). The mass function (0.00134 $M_\odot$) gives $m_2 \geq 0.13\ M_\odot$ and the transit depth $R_2 \approx 0.18\ R_\odot$: an M-dwarf companion. Ellipsoidal variation of 0.9 ppt (cos 4πφ amplitude) and a 0.25-ppt sin 2πφ term, possibly Doppler beaming
Period	2.7560089 ± 0.0000006 d, timing fit of 61 primaries (Gaia: 2.756044 ± 0.000090 d; −0.4σ). Odd and even primaries agree (7.42 ± 0.11 and 7.37 ± 0.11 ppt)
Epochs	Primary BTJD 3574.15160 ± 0.00014 = BJD_{TDB} 2460574.15160 (errors scaled by 2.1 for χ^2/dof = 4.4); secondary BTJD 3575.533 (phase 0.501 ± 0.010, from the fold; the error is the spread between detrending choices and sector subsets, which give 0.5085–0.521)
Primary eclipse	7.6 ppt (median of 61 events), 4.5 h, flat-bottomed, in raw QLP SAP flux (trapezoid fits with a local quadratic baseline); 7.7–8.1 ppt in sectors 12–93 and 6.4–7.1 ppt in sectors 99–102, a QLP aperture effect (the difference images give 7.9–8.1 ppt in both sectors 65 and 100); TESS-SPOC SAP 7.3 ppt, 4.3 h
Secondary eclipse	0.31 ± 0.02 ppt (17σ), 4.7 h, at phase 0.501 ± 0.010, measured in the fold against a model of the phase curve; single events are below the noise
Orbit	Primary 0.028 ± 0.010 in phase after Gaia's conjunction A (2.8σ), inside the eclipse half-width of 0.036; secondary −0.019 ± 0.013 from conjunction B (1.4σ); evaluated at the eclipse nearest Gaia's epoch, and the same with Gaia's or the TESS period. Conjunction B is excluded for the primary (55σ). e cos ω = +0.002 ± 0.015 from the secondary phase, against Gaia's +0.075 ± 0.028
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s12, s39, s65, s66; 27 events, 7.3 and 7.9 ppt, 4.3 h; mid-times of 25 matched events within −0.6 ± 1.9 min of QLP's); TARS (43 events, 7.6 ppt); eleanor-lite (s12, 7 events, 7.4 ppt). TESS-SPOC TCEs in s39 (P 2.75605 d, MES 110), s65 (MES 97), s66 (MES 62) and s56–s69 (P 2.755995 ± 0.000026 d, 8.15 ppt, MES 100), none flagged as an eclipsing binary; multi-sector centroid 0.45 ± 2.54″ from the out-of-transit centroid
Pixel attribution	

	TESScut difference images in s65 (9 events) and s100 (8 events): deficit centroid 1.2″ and 2.2″ from the light centroid, on the target's pixel, PSF-shaped (residual 10–14%); depth 8.1/8.1/7.8 and 7.5/7.9/7.3 ppt in 1×1/3×3/5×5 pixel apertures. Three Gaia sources within 60″ are bright enough in principle: G 13.55 at 54.7″ (would need 20%), G 14.94 at 35.7″ (69%) and G 15.03 at 26.3″ (75%); each would move the deficit centroid by more than 1 px, and the G 13.55 star's own pixel loses only 0.5–3.3 ppt. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (its only entry within 60″ is a rotational variable 52″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists, and the SuperWASP false-positive catalogue (Schanche et al. 2019); no SIMBAD reference or ADS full-text record mentions variability. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	The secondary phase depends on the detrending at the 0.01 level, so the eclipses do not yet test Gaia's eccentricity; a radial-velocity orbit would. The primary O−C shows structure of ±1–2 min between sectors (rms 1.7 min)
Next eclipses	Primary: BTJD 4329.298 (2026-10-15 19:11 UTC), 4332.054, 4334.810; secondary: BTJD 4327.923 (2026-10-14 10:10 UTC), 4330.679 (allow ±2 min, plus event-to-event scatter of 1.7 min rms, for the primary; the secondary times are uncertain by about 40 min, from the phase error)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 140021 is an F6/7 subgiant eclipsed every 2.756 d by a low-mass companion. The flat-bottomed 7.6-ppt transit, the 0.31-ppt occultation near phase 0.50 and the 0.9-ppt ellipsoidal variation fit one geometry: a companion of about 0.18 $R_\odot$ and at least 0.13 $M_\odot$, an M dwarf, whose occultation depth matches its expected TESS-band surface brightness. The primary falls 0.028 ± 0.010 in phase after Gaia's conjunction A (2.8σ), inside the eclipse, and the secondary 0.019 ± 0.013 before conjunction B. The shallow secondary's phase depends on the detrending at the 0.01 level, so the eclipses give only $e \cos\omega = 0.002 \pm 0.015$, consistent both with a circular orbit, as expected for a 2.76-d F+M binary, and with Gaia's $e = 0.079 \pm 0.030$. Radial velocities would measure the eccentricity and give the M dwarf's mass and, with the transit, its radius.

A97 TYC 1529-387-1 = TIC 351595900

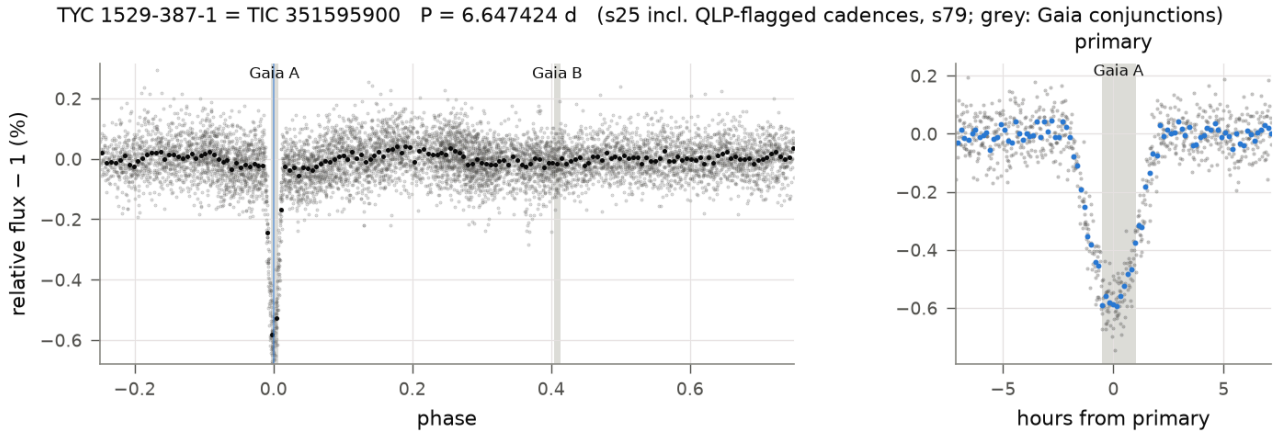


Figure A97. TYC 1529-387-1, raw QLP SAP flux of sectors 25 (including the QLP-flagged cadences, which cover three of the four sector-25 eclipses) and 79, sector medians removed, detrended by a 0.75-d running median with the eclipses masked and folded on $P = 6.647424$ d; grey bands mark Gaia's predicted conjunctions A and B. Left: the full orbit, with nothing at conjunction B (phase 0.41). Right: the 5.9-ppt, 4.1-h V-shaped primary, centred on conjunction A.

Identifiers	TIC 351595900; TYC 1529-387-1; Gaia DR3 4561305128630471168
Position (J2000)	253.94271, +19.00811 deg (TIC)
Brightness	Tmag 9.32; G 9.73
Stars	

	Evolved F star (TIC: T_{eff} 6198 K, 3.12 $R_\odot$, log g 3.53) at 339 pc; Gaia SB1 orbit (e = 0.147 ± 0.013, ω = 172.7 ± 3.5°, K_1 31.5 km/s). With the inclination from the eclipse ($i \approx 79°$) the companion is about 0.38 $M_\odot$, an M dwarf
Period	6.647424 ± 0.000005 d, timing fit of 7 primaries in sectors 25 and 79, flagged cadences included; an event is used when it is fully covered and its trapezoid fit converges inside the search window, which excludes the sector-79 event at BTJD 3457.65, whose cadences are all flagged and carry artefacts of 26–32 ppt; the 219 cycles between the sectors are unambiguous (Gaia: 6.647769 ± 0.000184 d; −1.9σ). Odd and even primaries agree (6.01 ± 0.10 and 6.03 ± 0.15 ppt)
Epochs	Primary BTJD 2826.14200 ± 0.00052 = BJD_{TDB} 2459826.14200
Primary eclipse	5.9 ± 0.2 ppt, 4.1 h, V-shaped (grazing), in raw QLP SAP flux (trapezoid fits to 7 events, O−C rms 1.8 min); TESS-SPOC SAP 5.7 ppt, 4.1 h
Secondary eclipse	Not detected: < 0.2 ppt (3σ) at Gaia's conjunction B (phase 0.409), −0.06 ± 0.05 ppt at phase 0.5. The occultation of an M dwarf by an F subgiant is expected at about 0.1 ppt
Orbit	Primary on Gaia's conjunction A: Δφ = −0.0015 ± 0.0048 (0.3σ, −0.25 h), with Gaia's orbit carried by the TESS period (−0.025 ± 0.013, 1.9σ, with Gaia's own period); conjunction B is 0.42 in phase away. No secondary, so no e cos ω test
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (all 7 primaries, 5.7 ppt, 4.1 h, median O−C −0.1 and −1.2 min; the sector-25 cadences flagged by QLP are unflagged in TESS-SPOC; one matched event 0.5 min apart); TARS (5 events, 6.1 ppt). TESS-SPOC TCE in s79: P 6.6486 ± 0.0008 d, 5.77 ppt, 4.14 h, b 0.99, MES 38, not flagged as an eclipsing binary; centroid 1.39 ± 2.51″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in s79 (3 events) and s25 (2 events): deficit centroid 1.5″ and 2.8″ from the light centroid, on the target's pixel, PSF-shaped (residual 13% and 23%, at low signal-to-noise); depth 5.4/5.2/5.1 ppt in 1×1/3×3/5×5 pixel apertures (s79). The brightest Gaia neighbour that could matter, G 15.46 at 25.7″, would need a 118% eclipse, and its own pixel loses only the target's wings; all 21 sources within 90″ are excluded. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all 11 sources within 60″), SB9, WDS, the ExoFOP TOI and CTOI lists; ADS full text has nothing on the star. Known SB (SIMBAD SB*, from Gaia DR3, its only reference)
Next eclipses	Primary: BTJD 4328.460 (2026-10-14 23:05 UTC), 4335.107, 4341.755 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A grazing eclipse of an evolved F star by an M-dwarf companion of about 0.38 $M_\odot$. The single eclipse kind is the companion's transit at conjunction A, on Gaia's predicted phase to 0.3σ; the occultation at conjunction B, which Gaia's eccentric orbit places at phase 0.41, is expected at about 0.1 ppt and lies below the 0.2-ppt limit. The 7 events span two sectors 219 cycles apart, and three of the four sector-25 eclipses fall entirely in cadences that QLP flags but TESS-SPOC does not (the fourth, at BTJD 2008.51, is unflagged); with the flagged cadences excluded, the period from sector 79 alone (6.64776 ± 0.00075 d) agrees. Radial velocities over one orbit, with the transit, would give the companion's mass and radius; more TESS sectors would tighten the period, now 1.9σ below Gaia's.

A98 BD+45 2574 A = TIC 157469799

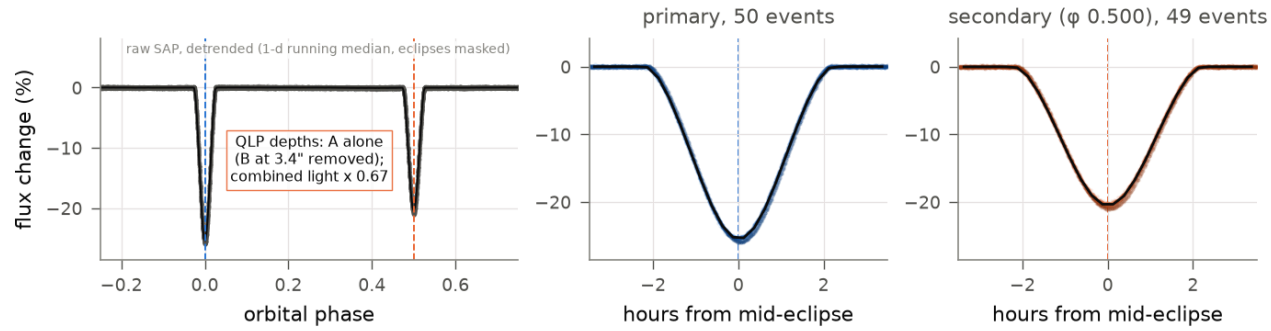


Figure A98. BD+45 2574 A, raw QLP SAP flux (sector medians removed), detrended by a 1-d running median with the eclipses masked and folded on P = 3.5063244 d, twice the fold period. Left: the full orbit, with two unequal eclipses half an orbit apart. Middle: the 245-ppt, 3.85-h primary; right: the 196-ppt, 3.85-h secondary at phase 0.500 (50 and 49 events plotted; 41 and 44 fully covered events in the timing fit). QLP's flux is corrected for the light of the 3.4″ companion B, so these are depths in A alone; in the combined light they are 0.67 times as deep.

Identifiers	TIC 157469799; BD+45 2574 A; TYC 3095-174-1; Gaia DR3 1345750246115251840; WDS J17399+4119 A (A 695 A)
Position (J2000)	264.96617, +41.31447 deg (TIC)
Brightness	Tmag 8.95; G 9.33
Stars	TIC: T_{eff} 6171 K, 1.52 $R_\odot$, at 147 pc; Gaia double-lined circular orbit (SB2C; K_1 83.5, K_2 91.2 km/s, q = 0.92), $M_1 \sin^3 i$ = 1.01 and $M_2 \sin^3 i$ = 0.93 $M_\odot$: two late-F/G dwarfs, $i \approx 87°$. Component A of the visual pair A 695 (WDS 17399+4119, 3.3–3.4″, first measured 1893); B (Gaia DR3 1345750216051287040, G 10.07) shares A's parallax (6.73 and 6.76 mas) and proper motion, and Gaia measured its velocity as constant to 3.6 km/s
Period	3.5063244 ± 0.0000002 d, timing fit of 41 primaries (and 3.5063237 ± 0.0000002 d from 44 secondaries) (Gaia: 3.506308 ± 0.000145 d; +0.1σ). The two eclipses differ in depth by 25% (about 60σ): the period is twice the fold period, 1.753154 d
Epochs	Primary BTJD 3172.71372 ± 0.00002 = BJD_{TDB} 2460172.71372; secondary BTJD 3016.68227 ± 0.00003 (phase 0.49999). Both fits with the QLP sector-80 times shifted by −2.70 min (see Pipelines agreeing); O−C rms 0.24 and 0.45 min
Primary eclipse	245.1 ± 2.6 ppt in QLP SAP flux, which carries the TIC dilution correction (contamination ratio 0.495), i.e. the depth in A alone; 166 ppt in the combined light of A and B; 3.85 h (trapezoid fits to 41 events); odd and even 245.8 and 245.5 ppt
Secondary eclipse	196.4 ± 4.3 ppt in A alone (131 ppt in combined light), 3.85 h, at phase 0.49999 (44 events)
Orbit	Circular orbit (SB2C): no conjunction-phase test. If Gaia's reference epoch for a circular orbit is the time of the primary's maximum velocity, the deeper eclipse falls 0.0025 ± 0.0027 in phase from the predicted conjunction (0.9σ) and is the eclipse of the more massive star; this depends on that convention and is not used for the attribution
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s25, s26; mid-times −0.05 and +0.27 min from QLP's; PDCSAP 253 and 199 ppt, SAP 167 and 131 ppt); TESS-SPOC (all seven sectors, mid-times within ±0.5 min of QLP's in s25–s79); TARS. In sector 80 the QLP times are 2.7 min late against TESS-SPOC (−2.69 ± 0.05 and −2.72 ± 0.06 min for the two eclipses) and TARS, the same offset as for A11. SPOC TCEs at the half period in s25, s26 and s14–s26 (1.7531525 d, MES 4313), flagged as suspected eclipsing binaries; difference-image centroid 1.23 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in s40, s53, s79 and s80 (6–8 events per stack, both eclipses): the deficit is a copy of the blended image (residual 6–7%), 1.0–1.2″ from the light centroid, with depth stable across 1×1/3×3/5×5 pixel apertures (133–142 ppt). Companion B, 3.4″ away, is unresolved, but the deficit centroid moves −1.03 ± 0.03″ along the A→B line in all eight stacks, against −0.94″ predicted for eclipses on A and +1.85″ for eclipses on B; B is excluded. No other Gaia source within 60″ is brighter than G 17.17

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2 and 3, Gaia DR3 variability tables, SB9, the ExoFOP TOI and CTOI lists (no ASAS-3 coverage); SIMBAD's references for the pair are visual double-star measurements, and ADS full text has nothing on eclipses. Known SB (SIMBAD SB* for A, from Gaia DR3)
Next eclipses	Primary: BTJD 4329.801 (2026-10-16 07:13 UTC), 4333.307, 4336.813; secondary: BTJD 4328.048 (2026-10-14 13:09 UTC), 4331.554 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+45 2574 A is a double-lined pair of late-F/G dwarfs (q = 0.92) in a 3.5-d circular orbit, the brighter member of the visual binary A 695. The two eclipses, 25% and 20% of A's light, are what a slightly unequal twin produces at $i \approx 87°$. Because the 3.4″ companion B is unresolved in TESS, the attribution rests on the motion of the deficit centroid along the A–B line, which matches an eclipse on A and excludes B, and on B's constant Gaia velocity; the period agrees with A's double-lined orbit to 0.1σ. QLP's flux for this star is already corrected for B's light, so its depths exceed those in SAP flux from the other pipelines by the factor 1.495. B shares A's parallax and proper motion, so the system is a triple, with B at a projected 500 au. A joint fit of the eclipses with Gaia DR4 epoch velocities, or new spectra, would give both eclipsing stars' masses and radii.

A99 TYC 3925-1852-1 = TIC 352236055

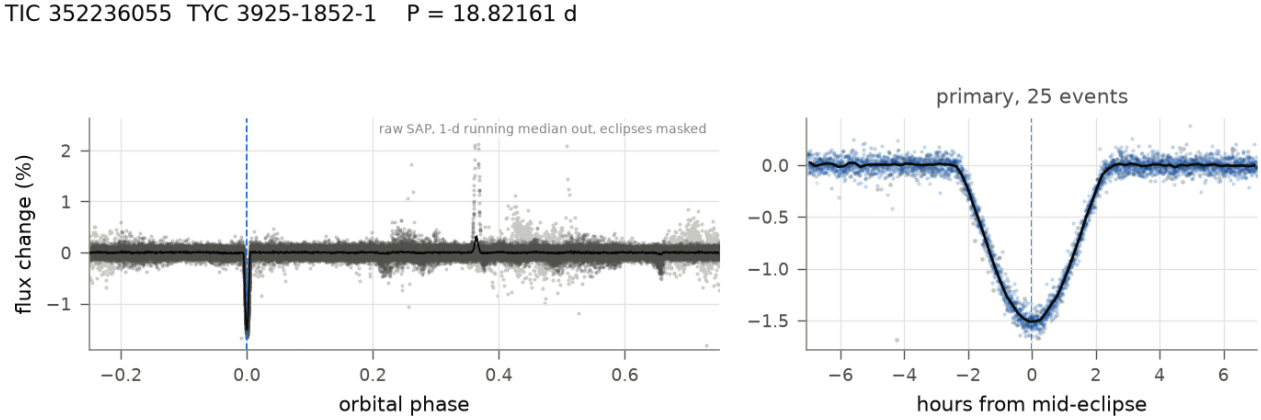


Figure A99. TYC 3925-1852-1, raw QLP SAP flux (sector medians removed, flagged cadences light grey), detrended by a 1-d running median with the eclipses masked and folded on P = 18.821605 d. Left: the full orbit, with one eclipse kind; the brightening at phase 0.365 is a single sector-82 QLP event absent from TESS-SPOC and TARS, and the shallow dip near 0.655 comes from one sector-73 orbit. Right: the 14.9-ppt, 4.7-h V-shaped eclipse, every event normalised by its own out-of-eclipse median.

Identifiers	TIC 352236055; TYC 3925-1852-1; Gaia DR3 2141529070981241088
Position (J2000)	292.15643, +55.31563 deg (TIC)
Brightness	Tmag 9.28; G 9.70
Stars	Subgiant (TIC: T_{eff} 6047 K, 2.53 $R_\odot$) at 271 pc; Gaia SB1 orbit (e = 0.398 ± 0.004, ω = 230.8 ± 0.6°, K_1 40.8 km/s). The mass function (0.102 $M_\odot$) gives $m_2 \geq 0.70\ M_\odot$, a K dwarf; $i \approx 83°$ from the eclipse duration
Period	18.821605 ± 0.000006 d, timing fit of 23 events, with the two sector-80 times (O–C +2.7 min each) corrected by −2.70 min for the QLP sector-80 timestamp offset measured in A98; dropping them instead gives 18.821603 d (Gaia: 18.821345 ± 0.00093 d; +0.3σ). Odd and even events agree (15.00 and 14.83 ppt), and the 2P fold shows equal depths at phases 0 and 0.5
Epochs	Primary BTJD 3160.83009 ± 0.00017 = BJD_{TDB} 2460160.83009 (errors scaled by 1.7; O–C rms 1.8 min)
Primary eclipse	

	14.9 ± 0.4 ppt, 4.7 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 23 fully covered events in 16 of the 18 QLP sectors); TESS-SPOC SAP 13.6 ppt and PDCSAP 14.4 ppt on the same five events. It falls at conjunction B, where the visible star is in front: a grazing occultation of the companion
Secondary eclipse	Not detected: < 0.13 ppt (3σ) at Gaia's conjunction A (phase 0.6665 ± 0.008), and nothing above S/N 6.3 at any phase. At conjunction A the stars are 44 $R_\odot$ apart against 23 $R_\odot$ at B, and the inclination that gives the observed chord at B gives an impact parameter of about 1.6 at A, so no eclipse is expected there
Orbit	Eclipse on Gaia's conjunction B: $\Delta\varphi = +0.0010 \pm 0.0008$ (1.2σ, +0.4 h), with the TESS ephemeris carried to the orbit nearest Gaia's periastron epoch (147 cycles) and Gaia's elements drawn with their full covariance; conjunction A is excluded (+0.3345, 148σ). No secondary, so no $e \cos \omega$ test (Gaia: -0.252 ± 0.004)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s14, s41, s82; 5 events, 13.6 and 14.4 ppt, 4.6 h; mid-times within 0.0–0.5 min of QLP's at 200-s and 600-s cadence, 3.6 min in the 30-min s14); TARS (5 events, 14.5 ppt); TASOC (s14, 14.0 ppt). No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images in s41 and s82 (2 events each): deficit centroid 1.1″ and 0.4″ from the light centroid, PSF-shaped (residual 7–10%), depth 12.4/12.7/12.2 and 12.4/12.3/11.7 ppt in 1×1/3×3/5×5 pixel apertures. Only two Gaia sources within 60″ are bright enough: G 13.18 at 42.5″ (would need 38%) and G 14.05 at 42.9″ (84%); either would put the deficit about 2 px away, and their own pixels lose less than the target's. A companion within about 1″ is not excluded (RUWE 1.10)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists (no ASAS-3 coverage); ADS full text has nothing on the star. Known SB (SIMBAD SB*, from Gaia DR3, its only reference)
Next eclipses	Primary: BTJD 4327.770 (2026-10-14 06:25 UTC), 4346.591 (allow ±2 min, plus event-to-event scatter of 1.8 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3925-1852-1 is a subgiant with a K-dwarf companion in an 18.8-d orbit of $e = 0.40$. The one eclipse it shows falls on Gaia's conjunction B, where the stars are 23 $R_\odot$ apart and the visible star passes in front: it is a grazing occultation of the companion, and its depth measures the companion's light. At conjunction A, near apastron, the separation is nearly twice as large and the same inclination misses, which accounts for the empty conjunction A (< 0.13 ppt). A 15-ppt occultation requires most of the companion's disc to be covered, somewhat more than the impact parameter from the duration allows with the TIC radius, but within the uncertainties of that radius and of the primary's mass. With 23 events over five years, it is a well-sampled eccentric binary that eclipses at only one conjunction. Radial velocities of the companion, or a spectrum showing its lines, would fix its mass and the inclination.

A100 HD 353655 = TIC 425304484

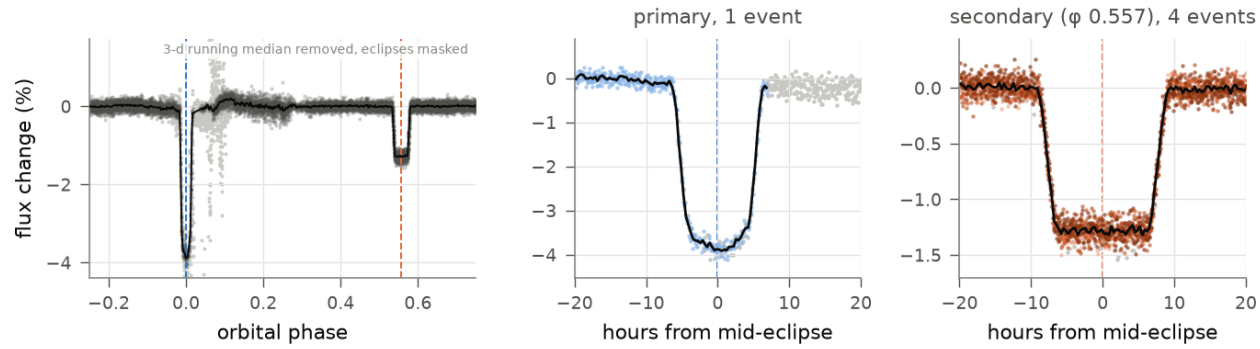


Figure A100. HD 353655, raw QLP SAP flux of sectors 54 and 81 (sector medians removed, flagged cadences light grey), detrended by a 3-d running median with the eclipses masked and folded on P = 16.260285 d. Left: the full orbit, with the deep eclipse at phase 0 and the shallow one at phase 0.557. Middle: the single 35–38-ppt, 12-h primary (sector 81; the grey points after egress are flagged scattered-light cadences). Right: the four 13-ppt, 18-h flat-bottomed secondaries. Zoomed events are normalised by their own out-of-eclipse median.

Identifiers	TIC 425304484; HD 353655; TYC 1615-291-1; Gaia DR3 4317746535565107840; WDS J19475+1512AB (TDT 1753)
Position (J2000)	296.88746, +15.20305 deg (TIC)
Brightness	Tmag 9.46; G 9.94
Stars	G0 (SIMBAD; TIC: T_{eff} 5955 K, 4.14 $R_\odot$, log g 3.24, a subgiant) at 440 pc; Gaia SB1 orbit (e = 0.208 ± 0.009, ω = 64.4 ± 1.7°, K_1 35.4 km/s). The mass function (0.0698 $M_\odot$) gives m_2 = 0.58–0.67 $M_\odot$ for M_1 = 1.1–1.4 $M_\odot$, and the occultation depth a companion near 4500 K: a K dwarf. The WDS companion at 2.6″ is G 19.0
Period	16.260285 ± 0.000024 d, timing fit of 4 secondaries in sectors 54 and 81, 45 cycles apart; Gaia's period excludes 44 or 46 cycles (Gaia: 16.25919 ± 0.00100 d; +1.1σ). The two eclipses differ in depth and duration, so P is the orbital period
Epochs	Primary BTJD 3531.06071 ± 0.00039 = BJD_{TDB} 2460531.06071 (one event); secondary BTJD 3231.16518 ± 0.00053 (phase 0.55656 ± 0.00004)
Primary eclipse	35–38 ppt, 12.1 h, flat-bottomed, from the one sector-81 event at BTJD 3531.06 (the other predicted events fall in data gaps or flagged cadences): 35.0 ppt in raw QLP SAP flux (trapezoid fit) and 35.1 ppt in TESS-SPOC SAP, against 38.0 ppt in TESS-SPOC PDCSAP, 37.5 ppt in TARS and 37.9 ppt in the 1×1-pixel difference image; the end of egress and the baseline after it fall in flagged scattered-light cadences, which limits the baseline fit. ASAS-3 V (2000–2009) folded on this ephemeris shows it at +48.5 mmag (11.5σ; eight of nine in-eclipse points at +44 to +53 mmag); the signal vanishes at Gaia's period
Secondary eclipse	12.9–13.5 ppt (13.0), 18.2 h, flat-bottomed (total), at phase 0.55656 (4 events); ASAS-3 +15 mmag (3.9σ). It falls 0.557 in phase after the primary because the orbit is eccentric; the duration ratio, 18.2/12.1, gives e sin ω ≈ +0.20
Orbit	Primary 0.0002 ± 0.0021 in phase from Gaia's conjunction A (0.1σ), secondary −0.0014 ± 0.0035 from conjunction B (0.4σ), with the TESS period and Gaia's (e, ω, T_p) and their covariance at Gaia's epoch. e cos ω = 0.08897 ± 0.00006 from the eclipse spacing against Gaia's 0.0897 ± 0.0083 (−0.1σ); e sin ω ≈ 0.20 from the durations against Gaia's 0.187
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (primary at BTJD 3531.06093, 0.3 min from QLP's; secondaries 12.6–13.7 ppt, within 1–3 min of QLP's); TARS (primary +0.8 min, 37.5 ppt; two secondaries, 13.7–14.3 ppt). TESS-SPOC TCE in s54: P 16.2598 ± 0.0015 d, 14.5 ppt, 18.3 h, MES 181, centroid 1.3 ± 2.5″ from the out-of-transit centroid; in s81 a TCE at 23.50 d (MES 231) pairs a secondary with the primary. ASAS-3 detects both eclipses
Pixel attribution	

	TESScut difference images: for the primary (s81, one event) the deficit centroid is 2.5″ from the light centroid, PSF-shaped (residual 2.7% in 3×3), depth 37.9/36.9/34.9 ppt in 1×1/3×3/5×5 pixel apertures; for the secondary (s54 and s81, two events each) 2.0″ and 2.7″ from the light centroid (residual 12–23%, at lower signal-to-noise under scattered light). In this crowded field (260 Gaia sources within 60″) none is bright enough to make the primary: the brightest, G 14.15 at 54.5″, would need 174% (65% for the secondary), and its own pixel loses only 0.8–5.9 ppt. A companion within about 1″ is not excluded; the 2.6″ WDS companion is 9 mag too faint
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 3 and 4, Gaia DR3 variability tables, SB9, the ExoFOP TOI and CTOI lists; ADS full text has nothing on the star. Listed in Kostov et al. (2025) table 2. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	TESS has observed the primary once; its timing rests on that event and on ASAS-3, and its depth is uncertain at the 35–38 ppt level. The companion radius implied by the transit is large for its mass unless the subgiant's radius is about 20% below the TIC value
Next eclipses	Primary: BTJD 4327.815 (2026-10-14 07:30 UTC), 4344.075; secondary: BTJD 4336.865 (2026-10-23 08:43 UTC), 4353.125 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 353655 is an evolved G star with a K-dwarf companion in a 16.26-d orbit of $e = 0.21$. At conjunction A, near periastron (separation $0.81a$), the companion transits the subgiant in a 35–38-ppt, 12-h eclipse; at conjunction B ($1.18a$) the subgiant totally occults the companion in a flat-bottomed 13-ppt eclipse half again as long. Both eclipses fall on Gaia's predicted conjunctions, and the two orbital quantities the eclipses measure, $e\cos\omega$ from their spacing and $e\sin\omega$ from their duration ratio, both match the spectroscopic orbit. TESS observed the deeper eclipse only once, but ASAS-3 photometry two decades earlier recovers it at 11.5σ on the TESS ephemeris. The transit depth requires a radius ratio of 0.19–0.20 and so, with the TIC radius of the subgiant, a companion of at least 0.77–0.81 $R_\odot$, larger than the 0.6–0.7 $R_\odot$ expected for a 0.6–0.7 $M_\odot$ dwarf; the two agree only if the subgiant is about 20% smaller than the TIC gives. A radial-velocity orbit and ground-based photometry of a primary eclipse would determine the radii.

A101 HD 192721 = TIC 14596588

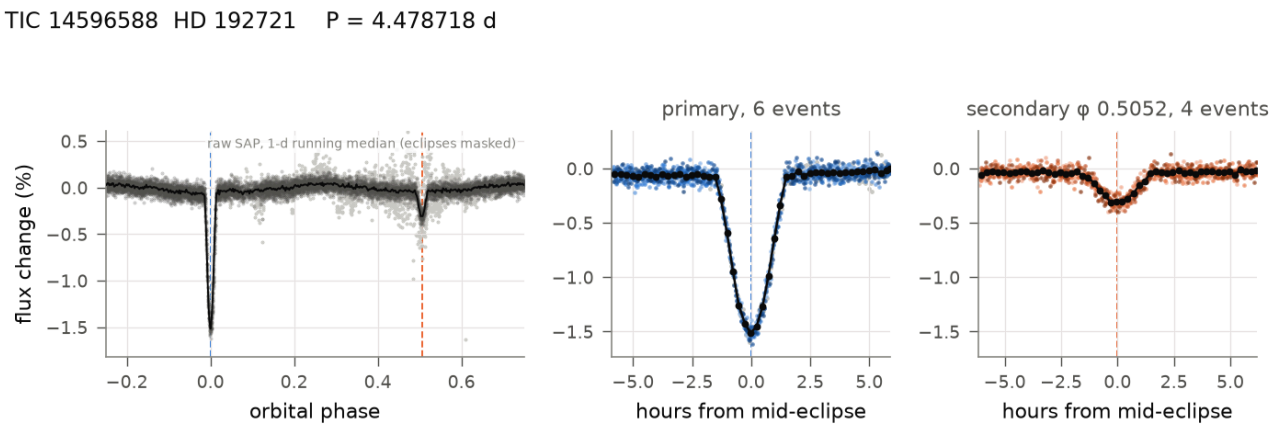


Figure A101. HD 192721, raw QLP SAP flux of sector 92 (the only sector; sector median removed, flagged cadences light grey), detrended by a 1-d running median with the eclipses masked and folded on $P = 4.478718$ d. Left: the full orbit. Middle: the 14.4-ppt, 3.0-h primary; right: the 2.8-ppt, 3.1-h secondary at phase 0.5052, displaced from 0.5 by a small eccentricity.

Identifiers	TIC 14596588; HD 192721; BD–14 5697; TYC 5752-100-1; Gaia DR3 6876279994232577536
Position (J2000)	304.21183, –14.19728 deg (TIC)
Brightness	Tmag 8.89; G 9.12
Stars	

	A1mA8-F2, an Am star (SIMBAD; Renson 53787; TIC: T_{eff} 7530 K, 2.31 $R_\odot$) at 276 pc; Gaia SB1 orbit (e = 0.009 ± 0.011, K_1 51.4 km/s). The mass function (0.063 $M_\odot$) gives $m_2 \geq 0.73\ M_\odot$, and the secondary-to-primary depth ratio (0.19) a companion near 4500 K: a K dwarf. A co-moving G 14.54 star lies 3.9″ away (parallax 3.63 against 3.57 mas)
Period	4.478718 ± 0.000065 d, timing fit of 5 primaries in one sector (Gaia: 4.478818 ± 0.000142 d; −0.6σ). Odd and even primaries agree (14.40 and 14.37 ppt in the 2P fold)
Epochs	Primary BTJD 3818.15412 ± 0.00013 = BJD_{TDB} 2460818.15412; secondary BTJD 3820.41690 ± 0.00060 (phase 0.50523 ± 0.00013)
Primary eclipse	14.4 ppt, 2.95 h (ingress 1.16 h), V-shaped (grazing, b ≈ 0.9), in raw QLP SAP flux (trapezoid fits to 5 events, O−C rms 0.24 min); TARS 13.6 ppt; photometry from SPOC-calibrated TESScut images 14.3 ppt
Secondary eclipse	2.8 ppt, 3.1 h, at phase 0.50523 ± 0.00013 (4 events); the offset from 0.5 (34 min) is a small eccentricity, the light-travel delay accounting for under 0.1 min of it
Orbit	Primary −0.0034 ± 0.0168 in phase from Gaia's conjunction A (−0.20σ), secondary +0.0028 ± 0.0168 from conjunction B (+0.17σ), with the TESS period and Gaia's (e, ω, T_p) and their covariance at Gaia's epoch (−0.70σ and −0.55σ with Gaia's period carried 766 cycles to the TESS epochs). $e \cos \omega$ = +0.0082 ± 0.0002 from the secondary phase against Gaia's −0.0009 (+0.0061/−0.0101), +1.1σ
Pipelines agreeing	QLP SAP; TARS (5 primaries and 4 secondaries, 13.6 and 2.7 ppt; primary mid-times −0.25 ± 0.36 min from QLP's); aperture photometry on SPOC-calibrated TESScut images (14.3 and 2.8 ppt; primary mid-times −0.01 ± 0.20 min from the QLP ephemeris). No SPOC 2-min or TESS-SPOC light curve exists for this sector, and so no TCE
Pixel attribution	TESScut difference images in s92: the primary deficit (5 events) is 0.3″ from the light centroid, PSF-shaped (residual 3.9%), depth 12.8/12.3/12.1 ppt in 1×1/3×3/5×5 pixel apertures; the secondary deficit (4 events) is 1.3″ from the light centroid. The 3.9″ companion (G 14.54) would need a 214% primary, and the deficit-centroid shift along the line to it excludes it at 11.8 standard deviations for the primary and 3.0 for the secondary (where it would need 41%). The brightest other Gaia source within 60″, G 14.44 at 50.8″, would need 196%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists; Paunzen et al. (2013) list it as constant or undetermined in STEREO photometry of chemically peculiar stars; ADS full text has nothing on eclipses. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	One sector only, so the period is known to 6.5×10^{-5} d. ASAS-3 (463 points, 2000–2009) shows no dip at this period carried 1622 cycles back (+1.6 ± 6.0 mmag); a scan of trial periods gives a marginal dip of the expected size (+16.4 ± 4.1 mmag, S/N 4.0) at 4.479112 d, 6σ from the TESS period among about 100 independent trials, which is not adopted
Next eclipses	Primary: BTJD 4328.728 (2026-10-15 05:25 UTC), 4333.207, 4337.685; secondary: BTJD 4330.991 (2026-10-17 11:44 UTC), 4335.469 (allow ±32 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 192721 is a metallic-line A star with a K-dwarf companion in a 4.48-d orbit, seen in grazing eclipse. Its Am classification fits a short-period binary, in which tidal synchronisation slows the star's rotation. Both eclipses fall on Gaia's predicted conjunctions, and the secondary's displacement to phase 0.5052 measures a small eccentricity ($e \cos \omega$ = 0.008) that Gaia's orbit, consistent with zero, does not resolve. The data are a single sector, with no SPOC product; the period comes from five primaries and its uncertainty grows to about half an hour by the next eclipses. Ground-based timing of one or two eclipses would fix the period and test the marginal ASAS-3 alternative at 4.47911 d.

A102 HD 194269 = TIC 72160748

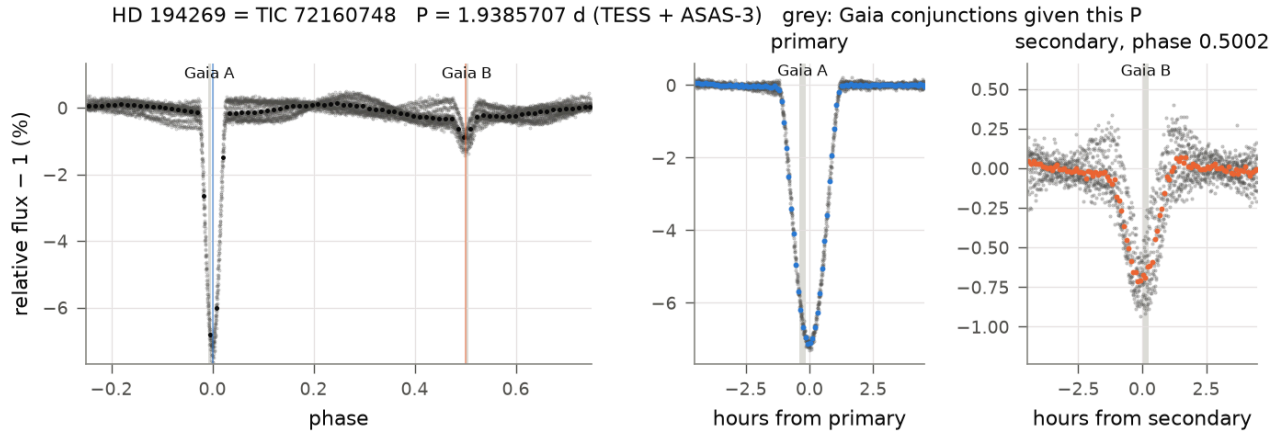


Figure A102. HD 194269, raw QLP SAP flux of sector 92 (the only sector; sector median removed), detrended by a 0.75-d running median with the eclipses masked and folded on P = 1.9385707 d, the period from TESS and ASAS-3; grey bands mark Gaia's conjunctions A and B recomputed with this period. Left: the full orbit; the residual out-of-eclipse structure is spot modulation not at the orbital period. Middle: the 69-ppt, 2.3-h V-shaped primary; right: the 7.1-ppt secondary at phase 0.5002.

Identifiers	TIC 72160748; HD 194269; BD−21 5717; CPD−21 7674; TYC 6341-1343-1; Gaia DR3 6859061882657414144
Position (J2000)	306.37372, −21.48726 deg (TIC)
Brightness	Tmag 8.65; G 9.04
Stars	F8V (SIMBAD; TIC: T_{eff} 6174 K, 1.26 $R_\odot$) at 106 pc; Gaia SB1 orbit (e = 0.070 ± 0.012, ω = 318 ± 5°, K_1 58.9 km/s). The mass function (0.0408 $M_\odot$) gives $m_2 \approx 0.49\ M_\odot$ at $i \approx 81.5°$, and the depth ratio (0.10) a companion near 3800 K: an early M dwarf. The primary is active: about 1% spot modulation within the sector and a ROSAT X-ray association
Period	1.9385707 ± 0.0000005 d from ASAS-3 V (520 points, 2000–2009) folded on the TESS epoch over a scan of trial periods (eclipse 64 mmag deep, S/N 14.8; next-best period S/N 3.9); TESS alone: 1.938562 ± 0.000029 d, timing fit of 11 primaries. The TESS period carried back 3700 cycles is uncertain by 0.11 d, 0.055 in phase, so the ASAS-3 cycle count is unambiguous (Gaia: 1.938490 ± 0.000016 d; +5.0σ)
Epochs	Primary BTJD 3815.72949 ± 0.00012 = BJD_{TDB} 2460815.72949; secondary BTJD 3814.76054 (phase 0.50017 ± 0.00016)
Primary eclipse	69.0 ppt, 2.3 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 11 events with a quadratic baseline; 70.2 ± 1.5 ppt per event with linear baselines, drifting from 68.8 to 73.5 ppt in the second orbit as the scatter rises); SPOC 2-min SAP 69.7 ppt. Odd and even primaries agree (69.8 ± 0.8 and 70.5 ± 0.6 ppt)
Secondary eclipse	7.1 ppt, 2.3 h, at phase 0.50017 ± 0.00016 (11 events); SPOC 7.09 ppt
Orbit	With Gaia's published orbit the primary falls +0.096 ± 0.019 in phase from conjunction A (5.2σ) and the secondary +0.063 ± 0.013 from conjunction B (4.7σ). Gaia's period is 5.0σ short and correlated with its eccentricity (ρ = −0.84); conditioning Gaia's elements on the measured period gives e = 0.021 ± 0.006 (still 3.5σ from circular) and offsets of +0.0057 ± 0.0025 (2.3σ, +0.26 h) for the primary and −0.0032 ± 0.0012 (2.7σ) for the secondary. $e \cos \omega$ = +0.0003 ± 0.0003 from the secondary phase against Gaia's +0.052 ± 0.011 (−4.9σ; +0.014 ± 0.005 after the conditioning, 2.9σ)
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s92; 11 primaries, 69.7 and 70.8 ppt, 2.27 h, median O−C +0.05 min against QLP's +0.09 min); TARS (69.1 ppt); ASAS-3 V (64 mmag). SPOC TCEs: the primary (P 1.9385709 ± 0.0000051 d, 72.8 ppt, 2.42 h, b 0.90, MES 721) and the secondary (7.09 ppt, 2.33 h, phase 0.49999, MES 77), neither flagged as an eclipsing binary; primary centroid 0.12 ± 2.50″ from the out-of-transit centroid
Pixel attribution	

	Difference images of 11 primaries in s92 in both the TESScut cutout and the SPOC target-pixel file: deficit centroid 0.2″ from the light centroid, PSF-shaped (residual 2.0% and 1.9%), depth 62.8/62.5/62.1 ppt in 1×1/3×3/5×5 pixel apertures; the secondary deficit (12 events) is 2.3″ (0.1 px) from the light centroid. The brightest Gaia source within 90″, G 13.89 at 49.4″, would need a 616% primary (63% for the secondary, while its own pixel loses only the target's wings). A companion within about 1″ is not excluded (RUWE 0.92)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all 22 sources within 60″), SB9, WDS, the ExoFOP TOI and CTOI lists; no SIMBAD reference or ADS full-text record mentions variability. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	Even with the measured period, Gaia's orbit differs from the eclipses at 2.3–3.5σ (conjunction offsets 2.3 and 2.7σ, e cos ω 2.9σ, *e* 3.5σ from circular). The update is linear, 5σ from Gaia's published values, not a refit of the epoch velocities, which are not public
Next eclipses	Primary: BTJD 4327.512 (2026-10-14 00:14 UTC), 4329.451, 4331.389; secondary: BTJD 4328.482 (2026-10-14 23:30 UTC), 4330.420 (allow ±2 min; single secondaries scatter by 1.6 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 194269 is an active F8 dwarf with an early-M companion of about 0.5 $M_\odot$ in a 1.94-d orbit. The secondary at phase 0.5002 shows the orbit to be circular, as expected at this period, and the spot modulation and X-ray emission fit a primary kept magnetically active by the close orbit. With Gaia's published elements the eclipses fall 5σ from the predicted conjunctions. ASAS-3 photometry 3700 cycles before TESS fixes the period at 1.9385707 d: the single-sector TESS period, carried back, is uncertain by only 0.055 in phase, so the cycle count is secure. Gaia's period is 5.0σ short of this. Because Gaia's period and eccentricity are strongly anti-correlated, correcting the period reduces the tension in Gaia's e = 0.07 without removing it: the conditioned orbit has e = 0.021 ± 0.006, still 3.5σ from circular, e cos ω 2.9σ from the eclipses' value, and conjunctions 2.3 and 2.7σ from the eclipses. The system shows how a Gaia period error of a few σ can propagate into both the conjunction phase and the eccentricity, and Gaia DR4 epoch velocities would test this directly.

A103 BD+41 3794 = TIC 188258837

BD+41 3794 = TIC 188258837 P = 5.985501 d

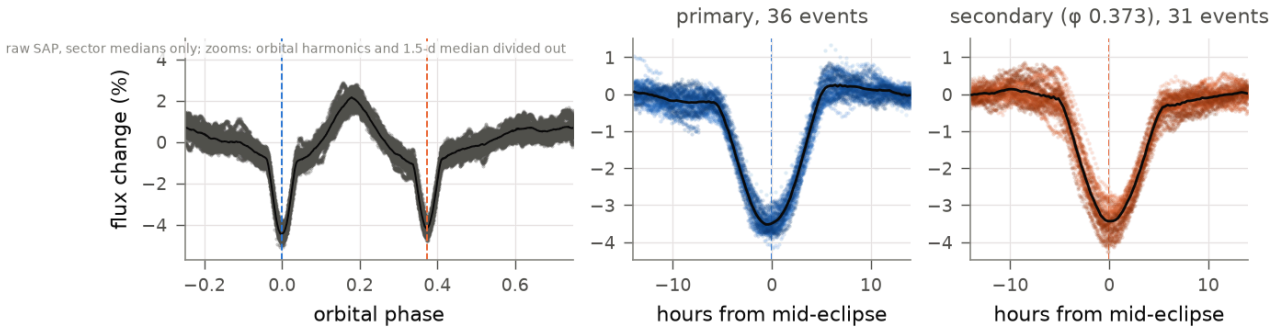


Figure A103. BD+41 3794, raw QLP SAP flux with only the sector medians removed, folded on the fitted P = 5.985501 d. Left: the full orbit, with the two eclipses (dashed lines at phases 0 and 0.373) and the 2.5% periastron brightening peaking at phase 0.18. Middle and right: the primary (36 ppt, 11.0 h) and the secondary (32 ppt, 10.45 h), with the orbital harmonics and a 1.5-d running median divided out (36 and 31 events plotted; 31 and 29 fully covered events in the timing fit).

Identifiers	TIC 188258837; BD+41 3794; ALS 11405; LS III +42 16; TYC 3161-510-1; Gaia DR3 2068151478876679168
Position (J2000)	308.00918, +42.20725 deg (TIC)

Brightness	Tmag 8.92; G 9.75
Stars	B0.2 III (SIMBAD) at 1390 pc, in the Cygnus OB region at b = +1.5°; the TIC T_{eff} of 9072 K is an effect of reddening (E(B−V) = 1.24). Gaia SB1 orbit (e = 0.198 ± 0.048, ω = 345.0 ± 15.6°, K_1 = 135.5 km/s). The mass function, 1.45 $M_\odot$, gives a companion of at least 10.6 $M_\odot$ (a B1–B2 dwarf) for an 18 $M_\odot$ primary; a ≈ 42 $R_\odot$. Stochastic low-frequency variability of 2–3 ppt rms, and a 2.5% periastron brightening at phase 0.18, the same in all eight sectors
Period	5.9855010 ± 0.0000159 d, timing fit of 31 primaries; the 29 secondaries give 5.9854465 ± 0.0000256 d (1.8σ apart). Gaia: 5.986001 ± 0.001368 d (−0.4σ)
Epochs	Primary BTJD 3144.94717 ± 0.00108 = BJD_{TDB} 2460144.94717; secondary BTJD 3135.20653 ± 0.00175 (phase 0.3726)
Primary eclipse	36.2 ± 2.3 ppt, 11.0 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 31 events); 32.5 ppt in TESS-SPOC SAP, 34.0 ppt in PDCSAP. O−C rms 8.4 min
Secondary eclipse	31.8 ± 3.7 ppt, 10.45 h (29 events), at phase 0.3726 after the primary, displaced from 0.5 by the eccentric orbit. O−C rms 14.3 min. The separation is constant to ±0.0008 in phase across eight sectors over five years: no apsidal motion is detected (< 0.0003 per year)
Orbit	Primary at Gaia's conjunction B: −0.0031 ± 0.0166 in phase (0.2σ); secondary at conjunction A: −0.0133 ± 0.0193 (0.7σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The swapped assignment is off by 0.386 ± 0.019. The eclipse spacing gives e cos ω = +0.201 (Gaia +0.184 ± 0.049, 0.35σ); the duration ratio, 1.05 ± 0.05, gives e sin ω ≈ −0.02 ± 0.03 (Gaia −0.048 ± 0.052). Gaia's periastron falls at phase 0.15. The primary is the eclipse of the companion by the B0.2 giant, at the conjunction where the stars are closer (0.914a against 1.013a), which makes it the deeper of two grazing eclipses (b ≈ 0.75–0.8)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s56, s75, s76, s82): on 17 matched primaries and 17 secondaries the mid-times differ from QLP's by +0.15 and +0.17 min (median), 0.6–0.8 min rms, and the depth ratios are 0.97 (SAP) and 1.00 (PDCSAP). The 8–14 min O−C scatter is therefore intrinsic to the star. No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of five primaries each in s56 and s82 and four secondaries in s56: deficit centroid 0.5–0.9″ from the light centroid, peaked on the target pixel and a scaled copy of the target's image over the central 5×5 pixels (residual 8–18%); depth 36–39 ppt in 1×1 and 3×3 pixel apertures, slightly diluted in 5×5 by a neighbour. The brightest Gaia source within 60″ (G 14.41, 47″) would need a 266% eclipse. Three brighter stars at 64–89″ (G 9.98, 11.85, 11.92) would need 8–30% eclipses and lose ≤ 5 ppt at their own pixels. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI); no ASAS-3 data. SIMBAD type SB*. Its eleven SIMBAD references are OB-star catalogues, photometry and Galactic-structure papers; none reports eclipses or an orbit. Two neighbours 47″ away appear in Kostov et al. (2025) table 2, most likely through this star's eclipses in their apertures. Known SB
Open issues	The eclipse times scatter by 8–14 min about the linear ephemeris because of the star's own red noise, which limits the timing precision
Next eclipses	Primary: BTJD 4330.076 (2026-10-16 13:46 UTC), 4336.062, 4342.047; secondary: BTJD 4332.307 (2026-10-18 19:18 UTC), 4338.292 (allow ±14 min, plus event-to-event scatter of 8 min rms for the primary and 14 min for the secondary)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+41 3794 (ALS 11405) is an early-B giant in the Cygnus OB region in an eccentric (e ≈ 0.2) 5.99-d orbit with a companion of at least 10.6 $M_\odot$. Both eclipses are grazing and of similar depth, and they fall on Gaia's two conjunctions with the TESS period; the spacing gives e cos ω in agreement with Gaia's. A 2.5% brightening peaks near Gaia's periastron phase and repeats unchanged over five years; it is the tidal and irradiation signature of periastron passage in a massive eccentric pair. The giant's stochastic low-frequency variability shifts individual eclipse times by several minutes, which TESS-SPOC reproduces event by event. A double-lined spectroscopic orbit would give the masses of a young massive binary directly, and continued timing would measure apsidal motion, which is not yet detected.

A104 BD+42 3880 = TIC 353453805

BD+42 3880 = TIC 353453805 P = 14.350949 d

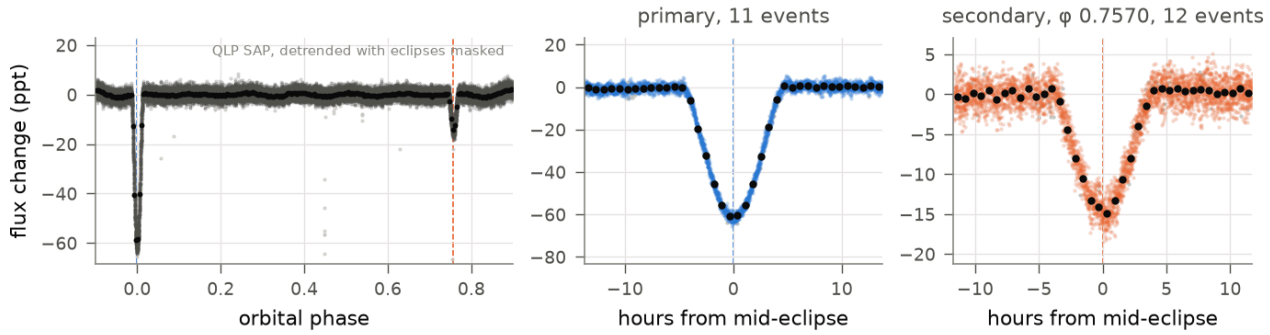


Figure A104. BD+42 3880, QLP SAP flux with sector medians removed, detrended with the eclipses masked, folded on the fitted P = 14.350949 d. Left: the full orbit, with the primary at phase 0 and the secondary of the eccentric orbit at phase 0.757. Middle: the 11 primaries (61 ppt, 8.6 h, nearly V-shaped). Right: the 12 secondaries (14.6 ppt, 7.3 h).

Identifiers	TIC 353453805; BD+42 3880; TYC 3174-31-1; Gaia DR3 2066657521752860544
Position (J2000)	312.47466, +42.83717 deg (TIC)
Brightness	Tmag 9.50; G 9.75
Stars	A2 (SIMBAD; TIC: T_{eff} 9825 K, 4.3 $R_\odot$ for the combined light) at 658 pc; Gaia SB2 orbit (e = 0.4754 ± 0.0084, ω = 24.06 ± 0.91°, K_1 = 86.6 and K_2 = 75.6 km/s, goodness of fit 17.1). With the inclination from the eclipse durations (82.9°) the masses are 2.06 and 2.36 $M_\odot$, two late-B or early-A stars with $R_1 + R_2$ = 5.4 $R_\odot$; a ≈ 40.7 $R_\odot$
Period	14.3509487 ± 0.0000051 d, timing fit of 11 primaries (O−C rms 0.23 min); the 12 secondaries give 14.350951 ± 0.000017 d. Gaia: 14.352119 ± 0.000459 d (−2.55σ)
Epochs	Primary BTJD 3309.02608 ± 0.00010 = BJD_{TDB} 2460309.02608; secondary BTJD 3319.88908 (phase 0.75695 ± 0.00005)
Primary eclipse	60.9 ppt, 8.6 h in raw QLP SAP flux (trapezoid fits to 11 events; 59.7–61.6 ppt per event), nearly V-shaped; 55.5 ppt in TESS-SPOC SAP, 56.5 ppt in PDCSAP
Secondary eclipse	14.6 ppt, 7.3 h (12 events), at phase 0.757: the orbit is eccentric, and no dip is present at phase 0.5. The secondary is shallower mainly because it is more grazing (impact parameter 0.89 against 0.60 at the primary)
Orbit	Primary at Gaia's conjunction A, secondary at conjunction B (the swapped assignment is off by 0.23 in phase). With Gaia's conjunctions carried to the TESS epochs with the TESS period: secondary +0.0027 ± 0.0034 (0.8σ), primary +0.0180 ± 0.0022 (**8.2σ**). The eclipses are 0.75695 apart in phase against Gaia's predicted 0.7722 ± 0.0054 (2.8σ); at Gaia's e sin ω = 0.194 the spacing requires e cos ω = 0.408 (Gaia 0.434 ± 0.009), that is e = 0.452 and ω = 25.4°. The longer eclipse falls at the closer conjunction (0.648a against 0.960a), where the chord is more central
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (all seven sectors): matched mid-times differ from QLP's by −0.02 ± 0.13 min (11 primaries) and −0.18 ± 1.04 min (secondaries); depths 8% (SAP) and 6% (PDCSAP) shallower for both eclipses alike, a common scale difference between the pipelines. TESS-SPOC TCEs in every sector: primary P 14.350–14.352 d, MES 172–304; the secondary as a second TCE (MES 17–75); centroid offsets 0.2–2.0 ± 2.5″ from the TIC position
Pixel attribution	TESScut difference images of two primaries each in s56 and s75: deficit centroid 0.2″ and 1.5″ from the light centroid, a scaled copy of the star's image (residual 3.6–4.7%), core depth 48–49 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 49 Gaia sources within 60″, the brightest (G 15.95, 58″) would need an 1850% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP (no TOI or CTOI). SIMBAD type SB*; its references are Vilnius photometry, extinction studies and diameter catalogues. Known SB
Open issues	

	The primary lies 0.018 in phase (6 h) from Gaia's conjunction A, 8σ on Gaia's formal errors. This is the same discrepancy as the eclipse spacing (2.8σ) and the period (2.55σ), and Gaia's goodness of fit of 17 suggests that its formal errors are underestimated. A ~2-ppt brightening near Gaia's periastron phase (−0.07) is suggested but not established
Next eclipses	Primary: BTJD 4327.943 (2026-10-14 10:34 UTC), 4342.294; secondary: BTJD 4338.806 (2026-10-25 07:17 UTC), 4353.157 (allow ±2 min, plus event-to-event scatter of 1.9 min rms for the secondary)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+42 3880 is a pair of late-B or early-A stars of about 2.1 and 2.4 $M_\odot$ in a 14.35-d orbit with e ≈ 0.45, seen at about 83°. Both eclipses are partial; the deeper and longer one falls near periastron, where the stars are closest and the chord is most central, and the secondary falls at phase 0.757. The attribution rests on the pixels and on TESS-SPOC, which reproduces both eclipses with the same timings and its own threshold-crossing events. The eclipse spacing, e cos ω = 0.408, differs from Gaia's 0.434 ± 0.009 by 2.8σ, and Gaia's period is 2.55σ from the TESS period. Because both tensions come from a solution with a goodness of fit of 17, we attribute them to underestimated Gaia errors; the TESS values should supersede them. Gaia DR4 epoch radial velocities fitted jointly with the eclipse timings would give both masses and radii.

A105 BD+67 1313 = TIC 321718006

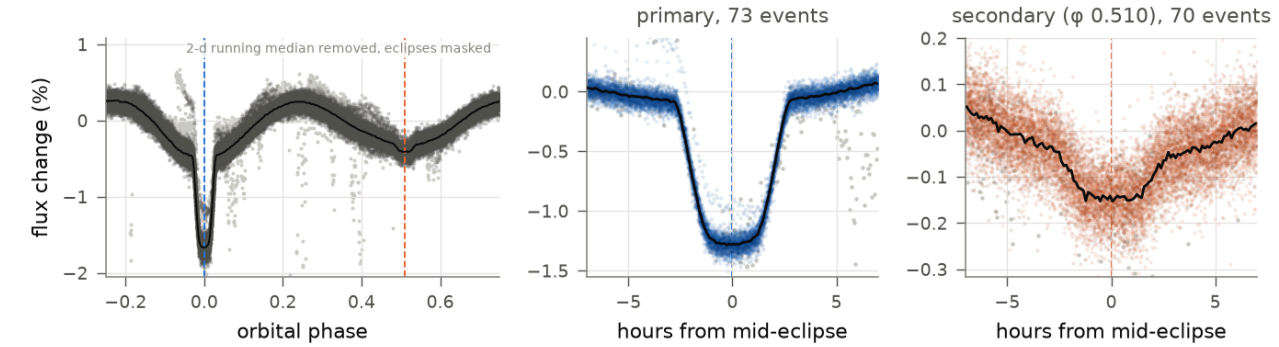


Figure A105. BD+67 1313, QLP SAP flux with sector medians removed and a 2-d running median removed with the eclipses masked, folded on the fitted P = 3.925329 d; flagged cadences in grey. Left: the full orbit, with the ellipsoidal modulation (about 0.6% peak to peak) and the eclipses at phases 0 and 0.510. Middle: the grazing primary (11.8 ppt, 5.6 h). Right: the secondary (0.9 ppt, 5.3 h); each event is normalised by its own out-of-eclipse median, so the zoom keeps part of the ellipsoidal curvature.

Identifiers	TIC 321718006; BD+67 1313; SAO 19407; TYC 4461-1835-1; Gaia DR3 2222179382751488384
Position (J2000)	321.50404, +67.71901 deg (TIC)
Brightness	Tmag 8.84; G 9.21
Stars	F8 (SIMBAD; TIC: T_{eff} 6455 K, 3.56 $R_\odot$, log *g* 3.45) at 336 pc; Gaia SB1 orbit (e = 0.015 ± 0.011, K_1 = 29.03 ± 0.47 km/s). The mass function, 0.00995 $M_\odot$, gives a companion of about 0.30 $M_\odot$ (0.24–0.35) for the TIC mass of 1.31 $M_\odot$: an M dwarf grazing an evolved F star. Ellipsoidal variation of about 0.6% peak to peak
Period	3.9253291 ± 0.0000007 d, timing fit of 68 primaries (O−C rms 0.98 min). Gaia: 3.925365 ± 0.000079 d (−0.45σ)
Epochs	Primary BTJD 3144.07847 ± 0.00008 = BJD_{TDB} 2460144.07847; secondary BTJD 3146.08177 (phase 0.51035 ± 0.00015)
Primary eclipse	11.8 ppt, 5.6 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 68 events; 11.7–11.9 ppt per event); 11.2–12.9 ppt depending on the baseline and pipeline. Odd and even depths agree to 1.0σ

Secondary eclipse	0.9 ppt, 5.3 h (63 events), at phase 0.51035. The depth is uncertain by a factor of about two (0.7–1.7 ppt) because it depends on the baseline model of the ellipsoidal variation. The depth ratio, 0.075, matches the TESS-band surface-brightness ratio of a 3300–3500 K dwarf and a 6300 K primary
Orbit	Primary at Gaia's conjunction A: +0.0003 ± 0.0034 in phase (0.1σ); secondary at conjunction B: +0.0037 ± 0.0033 (1.1σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipse spacing gives e cos ω = +0.0163 ± 0.0002 (Gaia +0.0104, +0.0105/−0.0084; 0.6σ). At conjunction A the F star is behind, so the deeper eclipse is of the F star
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP: 19 primaries, mid-times within +0.00 and +0.06 min of QLP's (median), 0.22 and 0.33 min rms; depth 11.2–11.3 ppt against 11.7 ppt. TESS-SPOC multi-sector TCE (s56–s69) at 3.925314 ± 0.000022 d, MES 186, 11.5 ppt, 5.57 h, impact parameter 0.84, centroid 0.45 ± 2.5″ from the TIC position; the secondary as a second TCE (MES 12.4, 0.90 ppt)
Pixel attribution	TESScut difference images of six primaries each in s56 and s83: deficit centroid 0.4″ and 0.6″ from the light centroid, PSF-shaped (residual 2.2–3.4%), depth 12.1–12.6 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 22 Gaia sources within 60″, the brightest (G 16.31, 44.5″) would need an 822% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3 and 4, ExoFOP (no TOI or CTOI). Listed in Kostov et al. (2025) table 2. SIMBAD type SB*, with four references (diameter catalogues, the TESS asteroseismic target list and Gaia DR3). Known SB
Next eclipses	Primary: BTJD 4329.528 (2026-10-16 00:36 UTC), 4333.453, 4337.379; secondary: BTJD 4327.606 (2026-10-14 02:28 UTC), 4331.531 (allow ±2 min; single secondaries scatter by 12 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+67 1313 is a slightly evolved F8 star (log g 3.45) with an M-dwarf companion of about 0.3 $M_\odot$ in a 3.93-d, nearly circular orbit. The 11.8-ppt primary is a grazing eclipse (impact parameter about 0.82), and the 0.9-ppt secondary at phase 0.510 is the occultation of the M dwarf. Both lie on Gaia's conjunctions to within 1.1σ, and the secondary's phase gives a small, significant eccentricity (e cos ω = 0.016) consistent with Gaia's. The eclipses sit in the minima of a 0.6% ellipsoidal modulation, which leaves the secondary depth uncertain by a factor of two. The system was listed only as an unvetted machine-learning candidate and as a TESS-SPOC threshold-crossing event that did not become a TOI. A spectroscopic mass and radius for the F star, combined with a fit to the grazing eclipse, would fix the radius of the M dwarf.

A106 HD 205273 = TIC 352935659

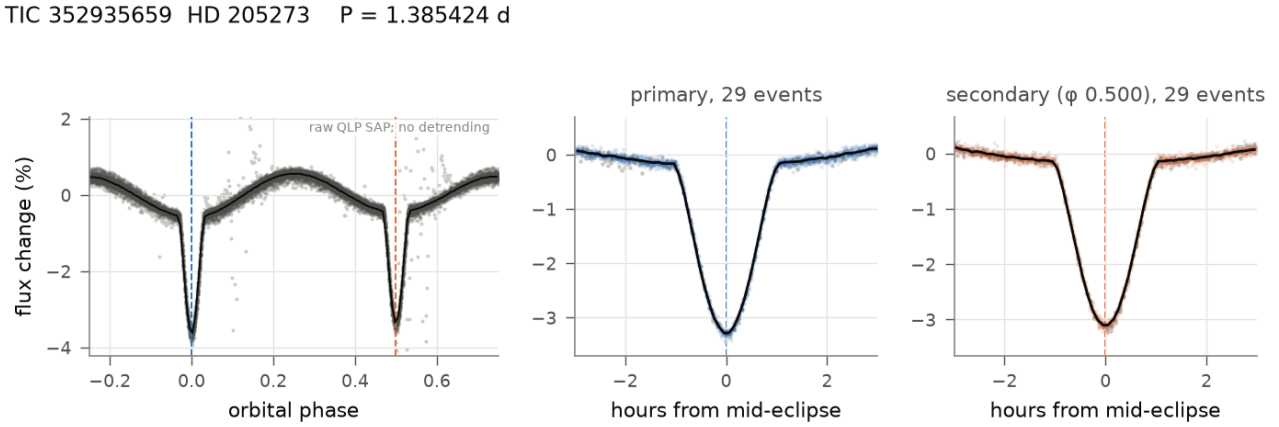


Figure A106. HD 205273, raw QLP SAP flux with sector medians removed and no detrending, folded on the fitted P = 1.385424 d; flagged cadences in grey. Left: the full orbit, with the ellipsoidal modulation of about 1.1% peak to peak and two grazing, V-shaped eclipses at phases 0 and 0.500. Middle and right: the primary (30.6 ppt) and the secondary (29.2 ppt), both 1.95 h long, each event normalised by its own out-of-eclipse median (29 events of each kind plotted; 27 and 26 fully covered events in the timing fit).

Identifiers	TIC 352935659; HD 205273; BD+04 4700; SAO 126895; TYC 550-1303-1; Gaia DR3 2699348496989329536
Position (J2000)	323.51547, +4.92603 deg (TIC)
Brightness	Tmag 8.71; G 9.05
Stars	F5 V (SIMBAD; TIC: T_{eff} 6456 K, 1.80 $R_\odot$ for the combined light) at 165 pc; Gaia SB2 orbit (e = 0.011 ± 0.004, K_1 = 122.77 ± 0.58 and K_2 = 127.64 km/s, q = 0.962). With the inclination from the eclipse durations (about 71°) the masses are about 1.37 and 1.32 $M_\odot$: two F5 dwarfs of about 1.3 $R_\odot$ each; a sin i = 6.9 $R_\odot$, so a ≈ 7.3 $R_\odot$. The ROSAT source 1RXS J213405.3+045548, 27.6″ away and within its positional error, is probably this binary
Period	1.3854241 ± 0.0000004 d, timing fit of 27 primaries (O–C rms 0.32 min) and 26 secondaries (0.24 min). Twin eclipses: the period is twice the fold period 0.692713 d. Gaia: 1.3854194 ± 0.0000044 d (+1.1σ)
Epochs	Primary BTJD 3402.27646 ± 0.00009 = BJD_{TDB} 2460402.27646; secondary BTJD 3402.96917 (phase 0.50000 ± 0.00009)
Primary eclipse	30.6 ppt, 1.95 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 27 events); 28.9 ppt in TESS-SPOC SAP
Secondary eclipse	29.2 ppt (sd 0.46), 1.95 h (26 events), at phase 0.50000; 27.6 ppt in TESS-SPOC SAP. The 1.3-ppt (4%) difference between the two eclipses is the same in QLP and TESS-SPOC and is significant at 11.6σ in QLP
Orbit	Primary at Gaia's conjunction A: +0.0021 ± 0.0014 in phase (1.5σ); secondary at conjunction B: −0.0048 ± 0.0018 (2.6σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipses are exactly half a period apart, e cos ω = 0.0000 ± 0.0002, against Gaia's +0.0107 ± 0.0040 (2.7σ); the secondary's offset is this difference. The deeper eclipse, at A, is of Gaia's star 1, the slightly more massive star
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP and TARS (s55, s82): matched mid-times within −0.01 and −0.08 min of QLP's for primaries and secondaries (median; rms 0.05 and 0.15 min), with the same unequal depths; TARS within +0.36 and −0.15 min. TESS-SPOC TCEs at half the period, 0.6927155 d (MES 308 in s55, 626 in s82, 29.0 ppt, impact parameter 0.97), centroids 2.1 and 1.1 ± 2.5″ from the TIC position. ASAS-3 photometry shows the eclipses about 8,500 cycles before TESS: the 20 points in the predicted eclipse windows are 34 mmag fainter than the rest (10σ)
Pixel attribution	TESScut difference images stacking both eclipses, 27 events in s55 and 31 in s82: deficit centroid 0.6″ and 0.1″ from the light centroid, PSF-shaped (3×3 residual 1.1–1.5%), depth 26.3–26.9 ppt in 1×1, 3×3 and 5×5 pixel apertures; the per-event pixel depths alternate between the two eclipses. The brightest of the 7 Gaia sources within 60″ (G 18.62, 22″) would need a 20,000% eclipse. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI). SIMBAD type SB*, with six references (spectral classification, photometric and diameter catalogues, a study of young stars); none reports variability. Known SB
Next eclipses	Primary: BTJD 4327.740 (2026-10-14 05:39 UTC), 4329.125, 4330.511; secondary: BTJD 4328.432 (2026-10-14 22:16 UTC), 4329.818 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 205273 is a detached pair of nearly equal F5 dwarfs (q = 0.96) in a 1.385-d orbit, seen at about 71°, so that both eclipses are grazing and V-shaped. The two eclipses differ in depth by 4%, the same in two pipelines and in the alternating depths of the difference images, which identifies the true period as Gaia's SB2 period. The eclipses are exactly half a period apart, as expected for a tidally circularised orbit of this period; Gaia's small eccentricity (0.011 ± 0.004) is most likely the bias that affects fitted eccentricities of orbits with tidally distorted components, and it accounts for the 2.6σ offset of the secondary. The eclipses are also present in ASAS-3 photometry, so the ephemeris holds across about 8,500 cycles. A light-curve and radial-velocity solution would give masses and radii for two main-sequence F stars.

A107 BD+32 4452 = TIC 397680054

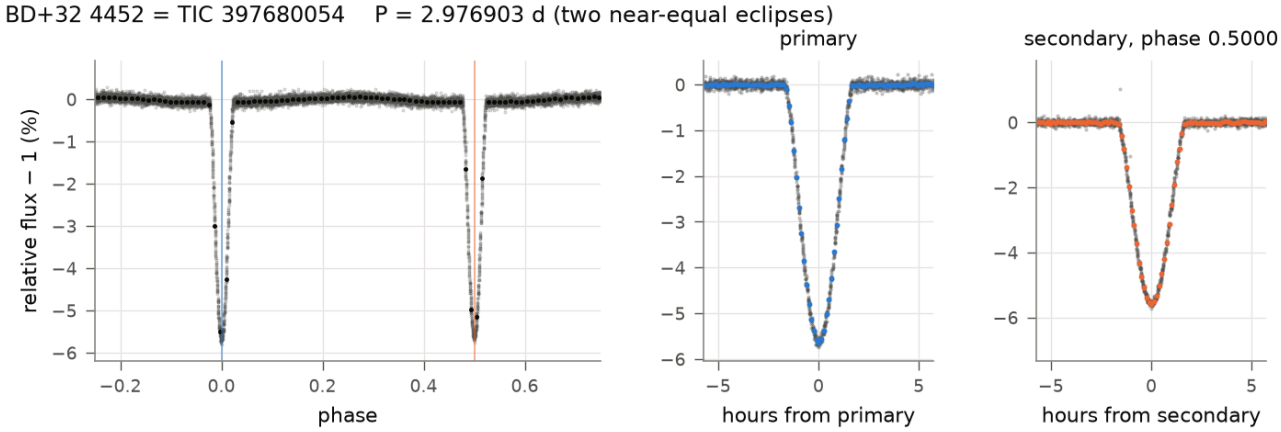


Figure A107. BD+32 4452, QLP SAP flux with sector medians removed and a running median removed with the eclipses masked, folded on the fitted P = 2.976903 d, twice the fold period. Left: the full orbit, with two near-equal eclipses at phases 0 and 0.5000. Middle and right: the primary (55.7 ppt) and the secondary (55.2 ppt), both V-shaped and 3.15 h long.

Identifiers	TIC 397680054; BD+32 4452; SAO 72451; TYC 2739-1543-1; Gaia DR3 1901472839026009344
Position (J2000)	338.24512, +33.00083 deg (TIC)
Brightness	Tmag 9.49; G 9.82
Stars	F2 (SIMBAD; TIC: T_{eff} 6629 K, 2.04 $R_\odot$ for the combined light) at 276 pc; Gaia SB2C orbit (circular; K_1 = 103.1 ± 0.7 and K_2 = 100.7 ± 29.4 km/s). M $\sin^3 i$ = 1.29 and 1.32 $M_\odot$, q ≈ 1; with the inclination from the eclipse duration (about 79°), about 1.37 $M_\odot$ each, uncertain by more than ±50% because K_2 is uncertain by ±29%: twin F stars, a sin i = 12.0 $R_\odot$
Period	2.976903 ± 0.000001 d, timing fit of 14 primaries (O–C rms 0.49 min); the 12 secondaries give 2.976902 ± 0.000001 d. Twin eclipses: the period is twice the fold period 1.48845 d. Gaia: 2.976915 ± 0.000039 d (−0.3σ)
Epochs	Primary BTJD 2681.70466 ± 0.00012 = BJD_{TDB} 2459681.70466; secondary BTJD 2739.75430 (phase 0.50001 ± 0.00006)
Primary eclipse	55.7 ppt, 3.15 h in raw QLP SAP flux (trapezoid fits to 14 events), V-shaped (flat fraction 0.2); 53.9 ppt in TESS-SPOC SAP, 55.2 ppt in PDCSAP

Secondary eclipse	55.2 ppt, 3.15 h (12 events), at phase 0.50001. The primary is deeper by 0.46 ± 0.13 ppt (3.6σ; 5.4σ in s56 alone), and TESS-SPOC SAP and PDCSAP show the same 0.45–0.55-ppt difference. Odd and even depths within each eclipse agree
Orbit	Circular orbit (SB2C): no conjunction-phase test. The secondary at phase 0.50001 ± 0.00006 gives e cos ω = 0.0000 ± 0.0001
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s16, s56; 15 primaries and 15 secondaries) and TARS: the same T14 (3.16–3.17 h), median O−C within 0.2 min of the ephemeris in every product, depths 53.5–55.2 ppt against QLP's 54.3–55.0 in the same per-event fits. TESS-SPOC TCE in s56 at half the period, 1.4884527 d, MES 1077, 56.0 ppt, impact parameter 0.95; centroid 0.59 ± 2.50″ from the TIC position and 0.09 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images of nine primaries and nine secondaries in s56 and six primaries in s16: deficit centroid 0.1–0.2″ from the light centroid, on the target pixel, PSF-shaped (residual 1.6–2.2%), depth 46.7–48.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest Gaia source within 60″ (G 15.37, 28.9″) would need a 929% eclipse; all 43 sources within 90″ are excluded. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables (the target and all 24 sources within 60″), Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI); no ASAS-3 data. SIMBAD type SB*, with three references (diameter catalogues and Gaia DR3). Known SB
Next eclipses	Primary: BTJD 4327.932 (2026-10-14 10:15 UTC), 4330.909, 4333.886; secondary: BTJD 4329.421 (2026-10-15 21:58 UTC), 4332.397 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+32 4452 is a pair of nearly identical F2 stars, of about 1.4 $M_\odot$ each (uncertain by more than ±50%, since Gaia's K_2 is uncertain by ±29%), in a circular 2.98-d orbit, seen at about 79°, so that each star grazes the other and the two V-shaped eclipses are 56 ppt deep. The period is Gaia's double-lined circular period; the 0.5-ppt depth difference between the eclipses, seen in three products, shows that the stars differ slightly in surface brightness and fixes which eclipse is which. A circular orbit has no predicted conjunction phase to test, so the attribution rests on the pixels, where both eclipses are centred on the target in two sectors, and on TESS-SPOC and its own threshold-crossing event. A double-lined radial-velocity curve with a better K_2 than Gaia's (±29%) would give precise masses for two F stars.

A108 TYC 2747-436-1 = TIC 239649487

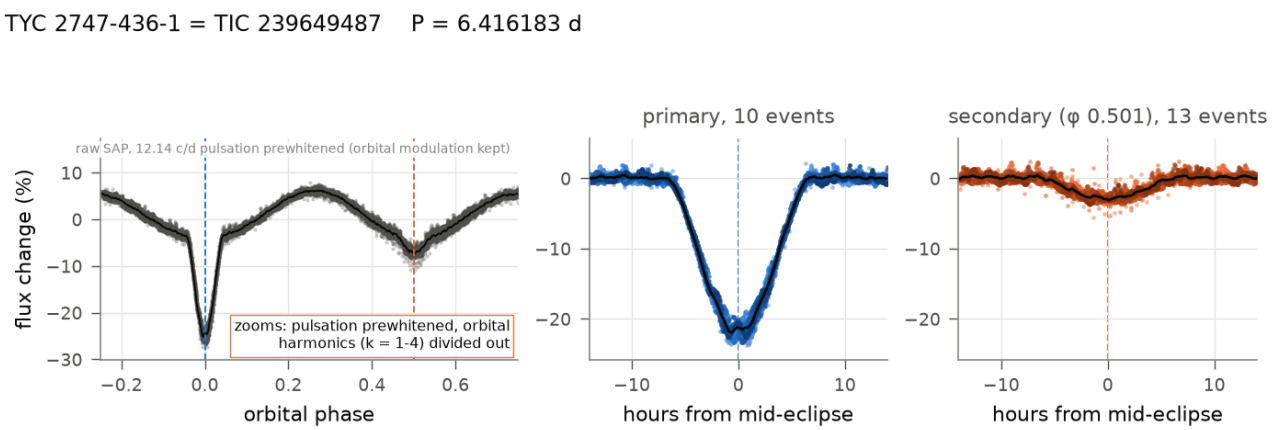


Figure A108. TYC 2747-436-1, raw QLP SAP flux folded on the fitted P = 6.416183 d. Left: with the 12.14 c/d δ Scuti pulsation prewhitened and the orbital modulation kept, showing the ellipsoidal variation with the primary at phase 0 and the secondary at phase 0.501 in the deeper ellipsoidal minimum. Middle and right: with the pulsation prewhitened and orbital harmonics (k = 1–4) divided out, the primary (214 ppt, 12.6 h) and the secondary (about 28 ppt, about 11.7 h; 13 secondaries plotted, 11 fully covered and fitted).

Identifiers	TIC 239649487; TYC 2747-436-1; Gaia DR3 1904322292128388096

Position (J2000)	338.96658, +35.84712 deg (TIC)
Brightness	Tmag 9.13; G 9.53
Stars	T_{eff} 7236 K (Gaia DR3; TIC: T_{eff} 6182 K, 3.82 $R_\odot$, a single-star estimate for the blended light) at 389 pc; Gaia SB2 orbit (e = 0.0096 ± 0.0095, K_1 = 30.91 and K_2 = 111.34 km/s, q = 0.278). Masses about 1.55–1.65 and 0.43–0.46 $M_\odot$ for i ≥ 75°; a ≈ 18 $R_\odot$. The more massive star is a high-amplitude δ Scuti pulsator: 12.1381 c/d (1.977 h, semi-amplitude 50 ppt) with harmonics at 24.276 and 36.414 c/d, an independent mode at 15.6913 c/d (17 ppt) and combination frequencies. An ellipsoidal modulation of semi-amplitude 48–53 ppt, phase-coherent over five years, suggests that the low-mass secondary is close to filling its Roche lobe (about 5.0 $R_\odot$); a semi-detached Algol with a pulsating mass gainer is the working hypothesis, pending a light-curve model
Period	6.4161826 ± 0.0000024 d, timing fit of 10 primaries (O–C rms 1.44 min); the 11 secondaries give 6.416179 ± 0.000040 d. Gaia: 6.416343 ± 0.000171 d (−0.94σ)
Epochs	Primary BTJD 3233.51625 ± 0.00014 = BJD_{TDB} 2460233.51625; secondary BTJD 3236.72891 (phase 0.50071 ± 0.00037)
Primary eclipse	214 ± 2 ppt, 12.6 h in QLP SAP flux with the pulsation prewhitened and the orbital harmonics divided out (trapezoid fits to 10 events), flat fraction 0.2; in the raw pulsating flux, 214 ppt in QLP against 203 ppt in TESS-SPOC SAP, 206 ppt in PDCSAP and 202 ppt in TARS. The pulsation amplitude, in units of the out-of-eclipse flux, falls in the primary-eclipse cores (39.4 against 50.5 ppt) and not at the secondary, so the pulsator is the star eclipsed at primary
Secondary eclipse	About 28 ppt (26–37 ppt per event), about 11.7 h (11 events), at phase 0.50071 ± 0.00037. Its core is W-shaped in s56 and s83, probably from residual pulsation or an imperfect ellipsoidal model; the pulsation amplitude does not drop in this eclipse
Orbit	Primary at Gaia's conjunction A: −0.0018 ± 0.0028 in phase (0.6σ); secondary at conjunction B: −0.0062 ± 0.0023 (2.7σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipse spacing gives e cos ω = +0.0011 ± 0.0006 (Gaia +0.0078 ± 0.0074, 0.9σ): the orbit is circular to 0.001, and Gaia's small, upward-biased eccentricity moves its conjunction B to phase 0.505, which accounts for the secondary's offset. The primary is the eclipse of the more massive star (K_1 = 30.9 km/s)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (s16, s56, s83): the ten primary mid-times agree with QLP's to −0.10 min median, 0.61 min rms, depth ratio 0.94; TARS +1.7 min median, 5.0 min rms, depth ratio 0.92. The secondary is too shallow against the ±5% pulsation for an event-by-event comparison in raw flux. No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of four primaries each in s56 and s83: deficit centroid 1.0″ and 0.4″ from the light centroid, on the target pixel, a scaled copy of the target's image (residual 3.7–4.0%), core depth 187–193 ppt in 1×1, 3×3 and 5×5 pixel apertures. All 16 Gaia sources within 60″ are G ≥ 15.15; the brightest (56.6″) would need a 3800% eclipse, and none could produce the 10% peak-to-peak pulsation. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing or pulsating in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2 and 3, ExoFOP (no TOI or CTOI); no ASAS-3 data. The only VSX object within 60″ is the neighbour Gaia DR3 1904322257768650624 (rotational variable, G 15.15, 56.6″). SIMBAD type SB*; its only reference is Gaia DR3. Known SB
Open issues	The light shares are not consistent: the 22% drop in pulsation amplitude would imply that the pulsator supplies about 95% of the light, while a 48–53 ppt ellipsoidal semi-amplitude from a lobe-filling 0.45 $M_\odot$ star needs that star to supply about half; a light-curve model is needed to resolve this and to test the semi-detached configuration. The W-shaped core of the secondary is unexplained, and its event times scatter by 17 min rms. QLP's primary is about 6% deeper than TESS-SPOC's
Next eclipses	Primary: BTJD 4330.683 (2026-10-17 04:17 UTC), 4337.100, 4343.516; secondary: BTJD 4333.896 (2026-10-20 09:23 UTC), 4340.312 (allow ±2 min; single secondaries scatter by 17 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 2747-436-1 is a double-lined eclipsing binary of about 1.6 and 0.45 $M_\odot$ in a circular 6.42-d orbit, whose more massive star is a high-amplitude δ Scuti pulsator with a dominant period of 1.98 h. The 21% primary eclipse is of the pulsator: the pulsation amplitude falls during the primary eclipse and not

during the secondary. The large ellipsoidal modulation suggests that the 0.45 $M_\odot$ companion is close to filling its Roche lobe, which would make this an oEA system, an Algol with a pulsating mass gainer; this remains a hypothesis, because the size of the amplitude drop and the ellipsoidal amplitude imply different light shares for the two stars, and only a light-curve model can reconcile them. Both eclipses lie on Gaia's conjunctions; the secondary's 2.7σ offset follows from Gaia's small and probably spurious eccentricity. Neither the pulsation nor the eclipses was previously catalogued. Pulsation timing around the orbit would measure the pulsator's light-travel-time orbit independently of Gaia, and eclipse mapping of the pulsation would constrain its mode geometry.

A109 BD+43 1585 = TIC 274331697

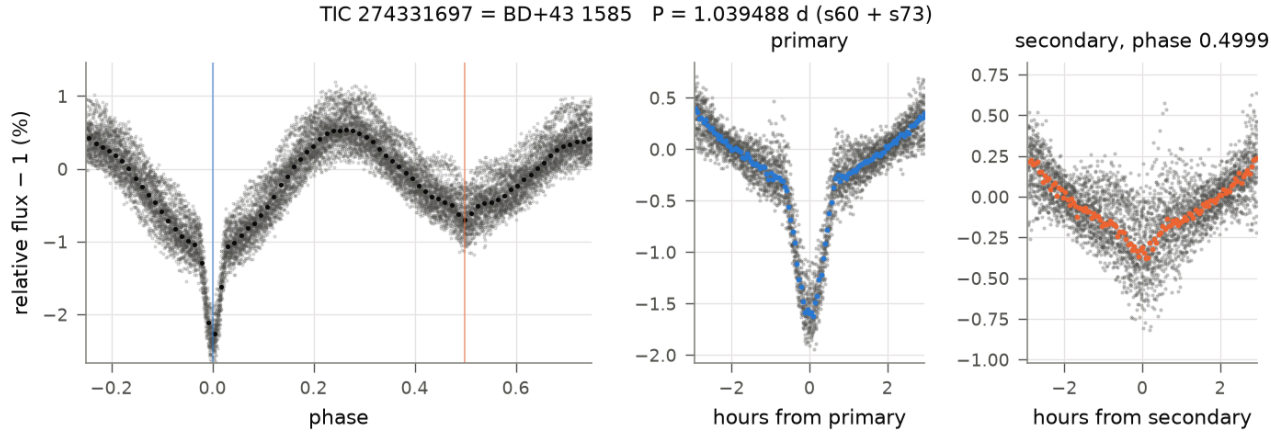


Figure A109. BD+43 1585, raw QLP SAP flux (sector medians removed; a running median with the eclipses masked, long enough to keep the orbital wave) folded on the sector 60/73 ephemeris, P = 1.0394878 d; the 2.5-d^{-1} pulsation is not removed and accounts for most of the scatter. Left: the full orbit, with the 6-ppt ellipsoidal double wave and the measured eclipse phases 0 and 0.4999 marked. Middle: the 1.3-h V-shaped primary. Right: the 1.4–1.6-ppt secondary. In both zooms the eclipses sit on the curvature of the wave, so the apparent depths there exceed the fitted ones.

Identifiers	TIC 274331697; BD+43 1585; TYC 2949-1328-1; Gaia DR3 963883711690164992
Position (J2000)	100.50093, +42.89616 deg (TIC)
Brightness	Tmag 9.62; G 9.95
Stars	F2 (SIMBAD; TIC: T_{eff} 6812 K, 1.59 $R_\odot$) at 246 pc; Gaia SB1 orbit (e = 0.019 ± 0.018, ω = 148 ± 36°, K_1 = 72.8 ± 1.4 km/s); RUWE 3.11 with no sign of a resolved double (ipd_frac_multi_peak 0). The mass function (0.0416 $M_\odot$) gives $m_2 \approx 0.6\ M_\odot$ at the inclination implied by the eclipse duration ($i \approx 69$–$70°$, impact parameter about 0.9), and the depth ratio (0.12) a companion near 3800–3900 K: a late-K or early-M dwarf. A pulsation at 2.503 d^{-1} (with its harmonic and close peaks at 2.48–2.58 d^{-1}), 1.2–3.2 ppt and changing from sector to sector, is in this star's light: a γ Doradus-type pulsation of the F2 primary or variability of an unresolved third star
Period	1.0394878 ± 0.0000005 d, timing fit of 34 primaries in sectors 60 and 73 (Gaia: 1.0394549 ± 0.0000076 d; +4.3σ). The period is not constant: the sector 20 and 60 epochs give 1.0395047 d (+6.5σ from Gaia's), 16 ppm longer, and no linear ephemeris fits the three sector epochs (χ^2 3678 for 1 degree of freedom)
Epochs	Primary BTJD 3134.11958 ± 0.00009 = BJD_{TDB} 2460134.11958; secondary BTJD 3073.30944 ± 0.00052 (phase 0.49991 ± 0.00050). Sector 20 epoch BTJD 1854.49221 ± 0.00013, 25.8 min early on this ephemeris
Primary eclipse	12.6 ppt (per-event scatter 0.4 ppt), 1.28 h, V-shaped (flat fraction 0.2), in QLP SAP flux after a model of the orbital harmonics and the pulsation (fitted with the eclipses masked) is divided out (trapezoid fits to 34 events); 12.3 and 12.7 ppt in sectors 60 and 73 in raw-flux fits. Odd and even primaries agree (12.55 ± 0.08 and 12.46 ± 0.11 ppt). Grazing
Secondary eclipse	1.44 ± 0.03 ppt in the fold (1.6 ppt median per event), 1.3 h, at phase 0.49991 ± 0.00050 (34 events)
Orbit	

	The result depends on how the epoch is carried to Gaia's, because the period changes. From the sector 20 epoch, 1409 cycles after Gaia's reference epoch, with the sector 20–60 period: primary −0.0105 ± 0.0067 in phase from conjunction A (1.6σ), or −0.011 ± 0.024 (0.5σ) when the observed 16-ppm period change is carried as a systematic error; secondary +0.041 ± 0.042 from conjunction B (1.0σ). From the sector 60/73 epoch with the sector 60–73 period: +0.027 ± 0.006 (4.2σ) on the formal error, +0.027 ± 0.043 (0.6σ) with the systematic. With Gaia's own period: +0.048 ± 0.010 (4.7σ) from sector 20 and +0.104 ± 0.019 (5.6σ) from sectors 60/73. The period that would put the sector 20 primary exactly on conjunction A, 1.039490 ± 0.000008 d, lies within the range of the TESS periods. $e \cos \omega = -0.0001 \pm 0.0008$ from the secondary phase against Gaia's −0.015 ± 0.014 (1.1σ): circular
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (11.7–12.7 ppt, median O−C −0.26 to +0.35 min against the QLP ephemeris in sectors 60 and 73); TARS (11.9 and 12.3 ppt, +0.21 and −0.39 min). All four products put the sector 20 eclipses 25–30 min early, and the sector 20 secondaries shift with the primaries (−22 to −28 min). TESS-SPOC TCEs in sector 60 (P 1.0394974 ± 0.0000150 d, 12.09 ppt, 1.33 h, b 0.99, MES 76) and sector 73 (P 1.0395090 ± 0.0000134 d, 12.35 ppt, MES 65), with the secondary as a second TCE in each (1.71 and 1.15 ppt, phase 0.5007); none flagged as an eclipsing binary; centroids 0.37 and 0.28 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in sectors 20, 60 and 73 (11, 16 and 19 primaries): deficit centroid 0.8″, 0.9″ and 0.6″ from the light centroid, PSF-shaped (residual 5.5%, 3.6%, 3.7%), on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (12.24, 12.29, 12.17 ppt in sector 60). The secondary stacks are on the target as well. The only neighbour bright enough, Gaia DR3 963883608610951552 (G 13.67 at 40.1″, classified δ Scuti/γ Doradus/SX Phoenicis by Gaia and in VSX), would need a 40% eclipse, and its own pixel loses only the target's wings. Per-pixel amplitude maps place both the ellipsoidal wave and the 2.503-d^{-1} signal on the target (0.06–0.20 px), not on the neighbour (1.7–1.9 px). A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing or variable in: VSX (only the neighbour above), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all sources within 60″ except the neighbour), SB9, WDS, ExoFOP (no TOI or CTOI), web search; SIMBAD references are catalogues only. Known SB (SIMBAD SB*, from Gaia DR3). TESS-SPOC TCEs as above, not promoted to a TOI
Open issues	The eclipse period differs by 16 ppm between the sector 20–60 and 60–73 baselines, and both eclipses shift together, as light-travel time in a wider orbit would produce; three epochs cannot fix the outer orbit. Gaia's period is 43–48 ppm shorter than every TESS value (4.3–6.5σ on the formal errors); with the observed 16-ppm change added as a systematic error the difference is 1.8–2.7σ, so it may reflect the same variation over Gaia's 2014–2017 baseline. The ellipsoidal semi-amplitude (6.3 ppt) is below the roughly 11 ppt expected, within the factor-of-two uncertainty of the stellar radius and the gravity-darkening coefficient. Whether the pulsation belongs to the primary or to a third star is not determined
Next eclipses	Primary: BTJD 4328.491 (2026-10-14 23:44 UTC), 4329.531, 4330.570; secondary: BTJD 4327.971 (2026-10-14 11:15 UTC), 4329.011 (allow ±30 min: the formal ephemeris error is 0.8 min, but the sector 20 epoch lies 25.8 min from this ephemeris, and a timing variation of that size may be present; single secondaries scatter by 4.2 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+43 1585 is an F2 star with a late-K or early-M companion of about 0.6 $M_\odot$ in a circular 1.04-d orbit, seen at about 70°, so that both eclipses are grazing: a 12.6-ppt primary and a 1.4–1.6-ppt secondary on a 6-ppt ellipsoidal wave. The deficit is on the target in difference images in three sectors, TESS-SPOC and TARS reproduce both eclipses, and the δ Scuti-type neighbour 40″ away is excluded both as the eclipse source and as the source of the 2.5-d^{-1} pulsation, which lies in the target's own light. The eclipse times are not on one linear ephemeris: sector 20 is 25.8 min early against sectors 60 and 73 in every pipeline, and the secondary shifts with the primary, as light-travel time would produce and apsidal motion in a circular orbit would not. With RUWE 3.1, this points to a third body. The changing period also governs the comparison with Gaia. Carried with the TESS period from the epoch nearest Gaia's, the primary is consistent with conjunction A (1.6σ on the formal error), whereas Gaia's own period, 43–48 ppm

shorter, places it 4.7σ away, so the comparison has no single significance. Eclipse timing over several seasons would give the outer orbit, and Gaia DR4 epoch astrometry and velocities would test whether Gaia's short period reflects the same variation.

S3.4 Eclipsing systems from the M-dwarf transit searches (A1–A4)

A1 TIC 311958594 = UCAC4 083-054218

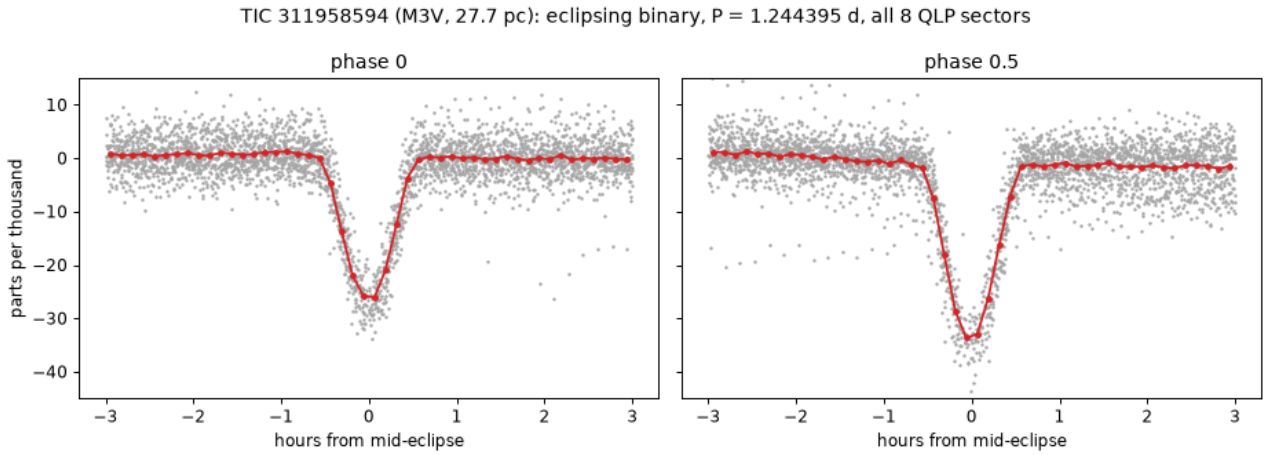


Figure A1. TIC 311958594, all 8 QLP sectors folded on P = 1.2443947 d: the two unequal V-shaped eclipses, 18.5 and 22.8 ppt (1.85% and 2.28%) deep in the QLP aperture, half an orbit apart, at phase 0 (left) and phase 0.5 (right).

Identifiers	TIC 311958594; Gaia DR3 5803288101584023040; 2MASS J17342735-7326519; UCAC4 083-054218
Position (J2000)	17:34:27.3370 −73:26:51.896 (263.613904, −73.447749 deg; TIC/Gaia-propagated)
Brightness	Tmag 9.82; G 11.0004; V 12.235
Stars	M3V (0.56 $R_\odot$ in TIC) + M dwarf, near-equal, unresolved; at 27.7 pc. Co-moving with the K7 dwarf TIC 373015604, 98″ away (~2,700 AU): a hierarchical triple
Period	1.2443947 ± 0.0000016 d (no Gaia orbit)
Epoch	BTJD 1629.45849 = BJD_{TDB} 2458629.45849 (the shallower eclipse)
Eclipses	2.28% and 1.85% (QLP box depths), V-shaped, ~0.7 h, half an orbit apart; grazing (a central eclipse of two 0.5 $R_\odot$ stars at this period would last ~2 h). Depth ratio B/A = 1.20–1.33 in the six sectors at 10-min cadence or finer, each at +4.7 to +8.9σ
Out of eclipse	Brightening at quadrature (ellipsoidal) and a ±5–10 ppt spot wave locked to the orbit, whose phase drifts from sector to sector
Pixel attribution	Deficit centroid 3–5″ from this star and 13–15″ from the giant HD 158062 (TIC 311958593) in S093 and S100, two field orientations
Catalogue status	Not in any catalogue searched (Section 2.7) except Kostov et al. (2025), table 2. SIMBAD lists it as a proper-motion star. Not in Gaia DR3 `vari_eclipsing_binary` or NSS; RUWE 1.09. Gaia photometry puts it ~1 mag above a single M3V (M_G 8.79 at BP−RP 2.48), as a near-equal unresolved pair should be
Open issues	Type EW or EB uncertain; the detached geometry has not been fitted
Method	M-dwarf transit search (Section 2.3); confirmed by independent re-analysis (Section 2.5)

The signal was first detected as a 0.622-d, 240-ppm transit-like signal on the K7 dwarf TIC 373015604. Its depth depended on the aperture (~450 ppm in sectors 39, 66 and 93, ~0 in sectors 100, 102 and 103), and the per-pixel deficit lies ~95″ away. In the M dwarf's own QLP light curve the signal is two unequal eclipses at 1.2443947 d. After a 1-d running-median detrending, the inequality appears insignificant in two sectors (S066 at +1.1σ, S039 at −2.0σ). The detrending is the cause: beneath the drifting spot wave, it tilts the baseline differently under each eclipse. Measured in a 43-min core against a local linear baseline, the eclipses are unequal in every sector at 10-min cadence or finer. In sectors 12

and 13, at 30-min cadence, the ratio is 1.11, because the cadence smears a 0.7-h eclipse. Parallaxes of 36.12 and 36.01 mas and proper motions of (24.1, −58.6) and (23.1, −54.3) mas yr^{-1} tie the M dwarf to TIC 373015604. A spotted, tidally locked M+M pair is common at this period.

A2 TIC 183494856 = UCAC4 070-000968

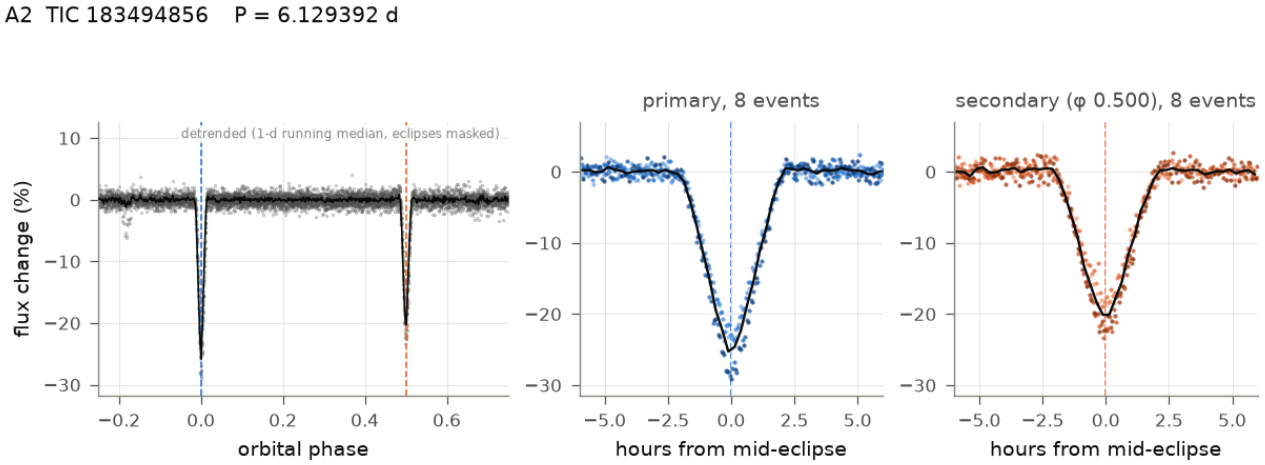


Figure A2. TIC 183494856 folded on P = 6.129392 d with the deeper eclipse at phase 0. The shallower eclipse falls at phase 0.50010. Absolute depths depend on the aperture; their ratio, 0.79, does not.

Identifiers	TIC 183494856; Gaia DR3 4636365763800590208; 2MASS J01134830-7602338; UCAC4 070-000968
Position (J2000)	01:13:48.3096 −76:02:33.825 (18.451290, −76.042729 deg)
Brightness	Tmag 14.81; G 15.2985; V 15.48
Stars	Solar-type primary (TIC: T_{eff} 5601 K, 0.979 $R_\odot$, 0.99 $M_\odot$, ρ 1.055 $\rho_\odot$) at 1,150 pc (parallax 0.840 ± 0.034 mas); RUWE 0.955
Period	6.129392 ± 0.000006 d (no Gaia orbit)
Epoch	BTJD 3.74608 = BJD_{TDB} 2457003.74608 (deeper eclipse)
Primary eclipse	21–29% (own 3×3-pixel aperture, by sector), 29.1% (GSFC-ELEANOR-LITE S001), 44.9% (TGLC S001); V-shaped, partial
Secondary eclipse	17–23% (own aperture), at phase 0.50010 (+0.9 min): consistent with a circular orbit. Shallow/deep ratio 0.79 in all four reductions
Timing	rms 2.1 min over 27 of 31 eclipse times, 757-d baseline, 124 cycles (sectors 1, 13, 27, 28)
Pipelines agreeing	Own aperture photometry, GSFC-ELEANOR-LITE (S001, S013), TGLC (S001). TASOC's corrected flux has 88% scatter and is unusable; too faint for QLP
Pixel attribution	Deficit centroid 1.1″ (S27) and 1.3″ (S28) from this star, 70.5″ and 71.2″ from the M dwarf on which it was found
Catalogue status	Not listed in any of: VSX, the ASAS-SN variable-star catalogue (II/366; Jayasinghe et al. 2018), Gaia DR3 `vari_eclipsing_binary`, `vclassre` and NSS, Prša et al. (2022), ATLAS variables, Kepler/K2 EBs, OGLE-IV, WISE variables (Chen et al. 2018), Gaia DR2 variables, SIMBAD (nearest entry a galaxy 10.7″ away). Listed only in Kostov et al. (2025), table 2. No TOI or CTOI
Open issues	Type EA or EB uncertain; no morphology fit. Sectors 94 and 95 put the star in the inter-CCD gap, so no longer baseline exists
Method	M-dwarf transit search (Section 2.3)

The signal was first detected as a 1.9 $R_\oplus$ planet candidate on the 0.217 $R_\odot$ M dwarf TIC 183494855 (P 3.06465 d, 6,400 ppm, 24.5σ). Its half-depth width of 1.77 h exceeded the 1.00-h maximum possible transit on that star, and the alternating events differed at 3.4σ when folded at twice the period. The centroid moved the signal 70″ to this star. Here the eclipses are 20–29% deep, and the 1.77-h duration fits a star of 1.06 $\rho_\odot$ at impact parameter ~0.7. At Tmag 14.81 the star is below QLP's ~13.5 limit, which explains its absence from full-frame-image pipeline products.

A3 HIP 83929 = TIC 85379511 = G 181-29

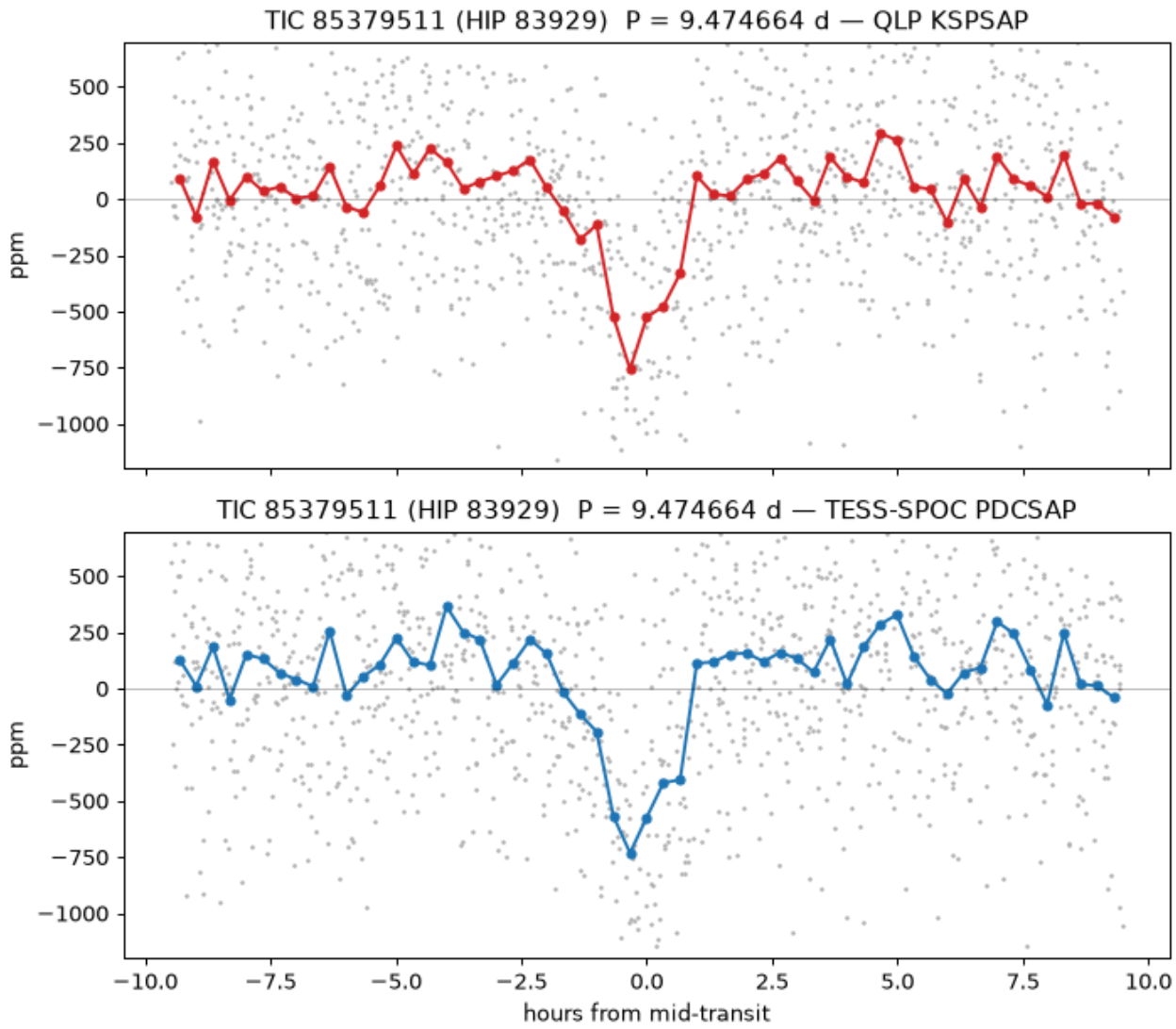


Figure A3. HIP 83929 folded on P = 9.474664 d: the 515–581 ppm, 1.58-h dip in QLP KSPSAP (top) and TESS-SPOC PDCSAP (bottom) flux. Phase 0.5, where no dip is found, is not shown.

Identifiers	HIP 83929; TIC 85379511; Gaia DR3 1354026652389939968; G 181-29; 2MASS J17091965+4126095
Position (J2000)	17:09:19.6205 +41:26:09.856 (257.331752, +41.436071 deg)
Brightness	Tmag 9.66; G 10.4472; V 11.02
Stars	K4V, 0.67 $R_\odot$, at 39.1 pc, with an unseen stellar companion (Gaia DR3 RUWE 3.87; robust RV amplitude 73.2 km s^{-1} over 24 epochs, χ^2 p-value 0.0). If the RV period equals the eclipse period, the companion is ~0.37 $M_\odot$ (0.365 $M_\odot$; an estimate, not a lower limit)
Period	9.474664 d (photometric; Gaia publishes no orbit for this star)
Epoch	BTJD 1993.379 = BJD_{TDB} 2458993.379
Primary eclipse	515 ± 40 ppm (QLP), 581 ± 39 ppm (TESS-SPOC); 1.58 h, against a 3.0-h maximum central duration: grazing
Secondary eclipse	None: 15 ± 42 ppm at phase 0.5 (predicted $\lesssim$ 25 ppm for an M-dwarf companion)
Pipelines agreeing	QLP (7 sectors, 12/13 events positive) and TESS-SPOC (7 sectors, 13/14 events positive). The QLP odd/even difference (600 vs 435 ppm) reverses in SPOC (500 vs 647), so it is not an alternating-depth signal
Pixel attribution	

	On target in five sectors (deficit centroid 0.5–3.2σ from the star); the only star inside one pixel, TIC 1270686103 (Tmag 16.37, 19″ N, would need a 25% eclipse), is rejected in three of the five sectors (at 5.4σ in S52) while S53 and S79 leave it level with the target; brighter neighbours excluded at ≥ 9.5σ. In a cyclic-shift null test, 0 of 100 trials reach the observed coherent SNR of 12.6
Catalogue status	Not listed as a variable of any eclipsing type. VSX: HAT-151-0000522, type “VAR”, P 12.895517 d, amplitude 0.012 V (a rotational signal 20× the eclipse depth at a different period; SIMBAD types the star Ro*); kept under Section 2.7, and the eclipses are new. In Hipparcos since 1991; 21 SIMBAD references, none about binarity; no TOI or CTOI; ExoFOP empty. A TESS-SPOC threshold-crossing event at 9.4748 d, the eclipse period (632 ppm, MES 8), exists in the multi-sector data-validation products and was never promoted (Section 2.7)
Open issues	Gaia DR3 publishes the RV amplitude, not the RV period: a planet is strongly disfavoured but not formally excluded. One night of spectroscopy or Gaia DR4 epoch RVs would resolve it. Sectors 78 and 79 were originally measured shallower in both pipelines (SPOC 219/233 ppm against ~700); an independent re-reduction does not reproduce this (TESS-SPOC 473 and 607 ppm, QLP not shallower), so no trend is claimed
Method	M-dwarf transit search (Section 2.3), coherent search; the sole survivor of the vetting of 40 peaks; arithmetic independently cross-checked
Status	Probable. The dip is real (three reductions, 6–7σ, 0 of 400 random-phase boxes reach it) and on or near the target, but the companion's radial-velocity period is unknown, so its mass and even its stellar nature rest on the assumption that the RV period equals the eclipse period; the one-pixel neighbour is rejected in only three of five sectors; no timing fit, so no next eclipse is predicted

The signal resembles a 1.7 $R_\oplus$ super-Earth with an equilibrium temperature of ~590 K on a bright K dwarf. Every photometric test passed: seven sectors over four years with no single sector carrying it, a duration well inside the star's maximum, pixels on target, a second pipeline, and 0 of 100 trials in a null test using the same grid and filter as the observation. Gaia DR3 radial velocities strongly disfavour a planet. An RV amplitude of 73.2 km s^{-1} indicates a stellar companion. At the photometric period the mass function gives ~0.37 $M_\odot$, and the primary's photocentre wobble would be ~0.71 mas at 39.1 pc, the right size for RUWE 3.87. A 9.47-d planet inside such a binary would also be dynamically difficult to retain. One alternative, a ~1 $M_\odot$ white dwarf on a 60–66 d orbit, is disfavoured because its wobble (~6 mas) would be an order of magnitude more than the RUWE reflects. The dip is most plausibly a grazing stellar eclipse. Its nature is undetermined until the companion's period is measured.

A4 TIC 70665829 = UCAC4 537-050325

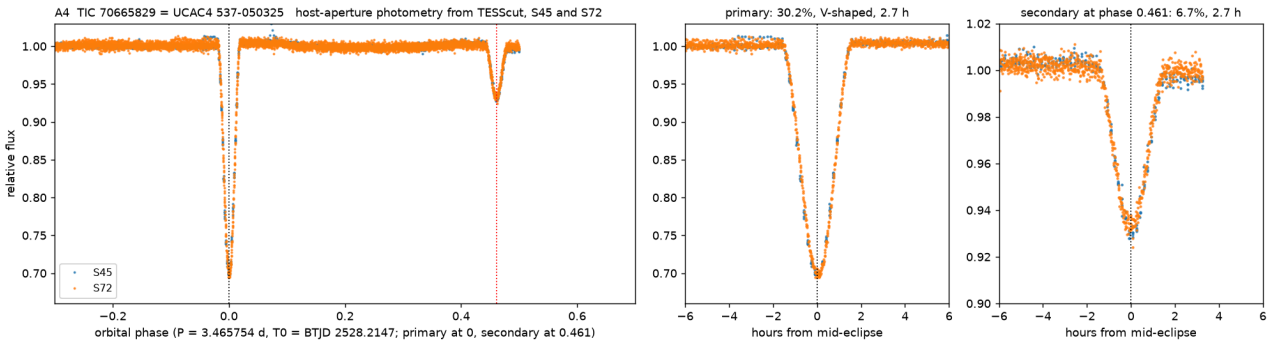


Figure A4. TIC 70665829, aperture photometry extracted at the host in the TESScut cutouts of sectors 45 and 72, folded on P = 3.465754 d with the 30% primary at phase 0 and the 6.7% secondary at phase 0.461.

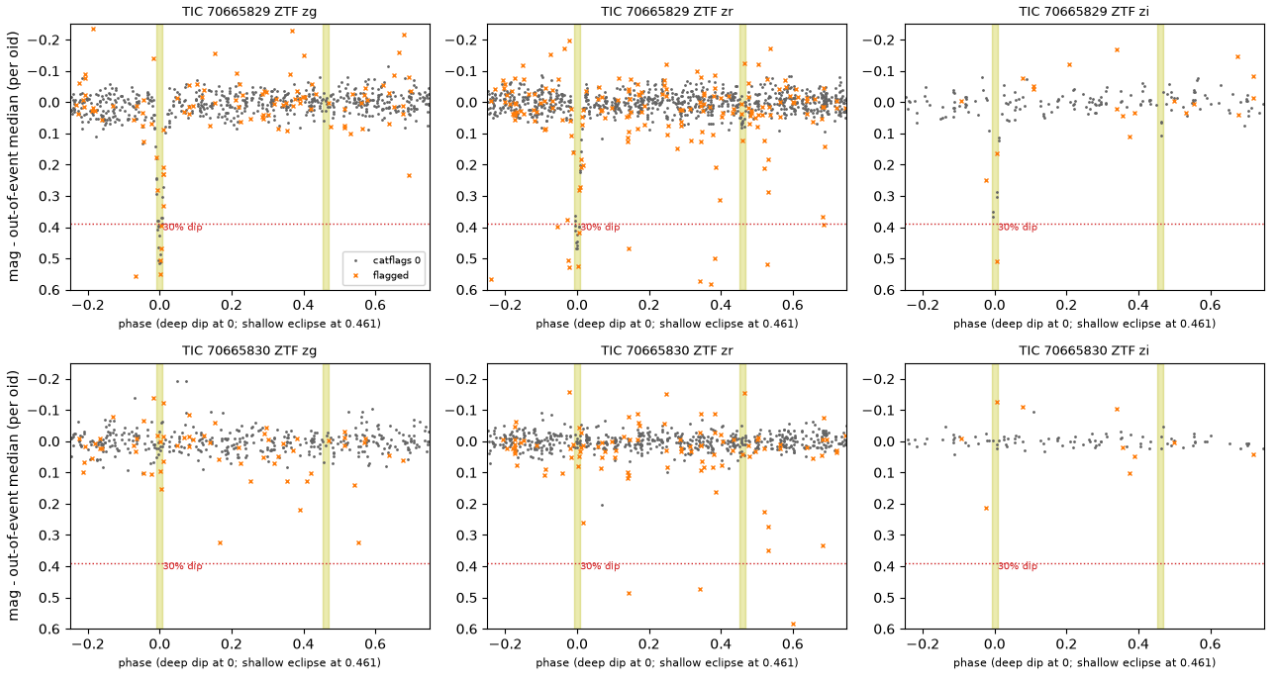


Figure A4b. ZTF *g*, *r* and *i* photometry (2018–2025) of TIC 70665829 (top) and of the M dwarf TIC 70665830 at 60″ (bottom), which ZTF resolves, folded on the TESS ephemeris with the primary at phase 0 (yellow bands: the predicted eclipse windows). Every ZTF epoch inside a primary window is faint on TIC 70665829, by 0.40 mag at the median; the same windows on the M dwarf show nothing.

Identifiers	TIC 70665829; Gaia DR3 3890379452571731456; UCAC4 537-050325
Position (J2000)	155.384788, +17.203515 deg
Brightness	Tmag 12.29; V 13.12; range at primary minimum T 12.294–12.684, at secondary minimum 12.370
Stars	K dwarf (TIC: T_{eff} 5080 K, 0.76 $R_\odot$, 0.85 $M_\odot$) at 240 pc + an M dwarf of ~0.54 $M_\odot$ (from the mass function; the two-sphere fit gives R_2 0.42–0.54 $R_\odot$, T_2 ≈ 3600–3700 K, L_2/L_1 0.07–0.11); *i* 87–89°, both eclipses partial (impact parameter 0.36–0.56 $R_\odot$)
Period	3.4657540 ± 0.0000093 d (joint timing fit of 11 primary and 10 secondary eclipses; O–C rms 0.3 min over 218 cycles). Gaia NSS SB1: 3.465888 ± 0.000285 d, agreement at 0.5σ; the primary and secondary periods agree to 1.1 ± 1.6 s
Epochs	Primary T_0 = BTJD 2528.2147 ± 0.002 = BJD_{TDB} 2459528.2147; last observed primary BTJD 3283.7496. Secondary at phase 0.4607 (BTJD 2529.8114)
Primary eclipse	30.2% (29.8–30.5% over the 11 events; 30.7–31.1% after removing the ~3% of foreign light in the aperture), V-shaped, full width 2.75 h, FWHM 1.56 h
Secondary eclipse	6.73 ± 0.06% at mid-eclipse, V-shaped, full width ~2.7 h, at phase 0.4607: $e \cos \omega = -0.062$, against Gaia's −0.077 ± 0.040 (0.4σ); equal durations give $e \sin \omega \approx 0$. The depth ratio 0.22 is the M/K surface-brightness ratio in the TESS band
Orbit	Gaia SB1: K_1 60.97 ± 1.75 km s^{-1}, *e* 0.077 ± 0.040, ω 175 ± 22°, mass function 0.081 $M_\odot$; RUWE 1.85. With the K dwarf (the RV star) eclipsed at the deep event, Gaia's ω predicts the secondary at phase 0.451 ± 0.025 (observed 0.461, 0.4σ); the other assignment predicts it at 0.451 and the deep eclipse is at 0.539 (3.5σ)
Pixel attribution	Same method for both eclipse kinds, in two field orientations: the 30% eclipse's deficit centroid is 2.7″ (S45) and 2.0″ (S72) from this star, the 6.7% one's 4.4″ and 3.2″, both peaking on the star's pixel; 58–60″ from the M dwarf TIC 70665830 on which the signal was first seen. No star within 90″ could supply the 30%: the only one brighter than Tmag 17, the M dwarf at 60″, puts 1.5–4% of its light in the aperture and would need an intrinsic depth above 1500%
Independent photometry	ZTF resolves the pair: 46 epochs inside predicted primary windows over 2018–2025 are all faint on TIC 70665829, median +0.396 mag (30.6%), while the same windows on TIC 70665830 give +0.006 mag. QLP, TESS-SPOC and TARS light curves of the star show the primary at 27–30% in sectors 45, 46, 48, 72 and a January 2026 sector; the M dwarf's own 2-min SPOC aperture sees it at 1–2%, as contamination
Catalogue status	

	Known SB (SIMBAD SB*, from the Gaia solution); nothing in VSX within 60″; not in the TESS-era eclipsing-binary tables; nothing at the position in the ASAS-SN variables
Next eclipses	Primary: BTJD 4330.4068 (2026-10-16 21:49 UTC), 4333.8725, 4337.3383, 4340.8040, 4344.2698, then every 3.46575 d; secondary 1.597 d after each primary
Method	M-dwarf transit search (Section 2.3), from one of four remaining eclipse-like candidates of the faint late-M search, as a 6.7% eclipse; the 30% primary was identified later in the same photometry and attributed to this star by the tests described below

The faint-M search found a 1.58% dip at 3.46577 d on the M dwarf TIC 70665830, clean on the fold but 1.5 times longer than that star allows. The pixels put it on TIC 70665829, the Tmag 12.29 star three TESS pixels (60″) away, where Gaia already had a single-lined orbit at the same period. The search's fold put the 6.7% eclipse at phase 0 and tested for a secondary only at phase 0.500. On that fold the 30% eclipse falls at phase 0.539, 3.3 h later, so the search detected only the 6.7% eclipse although the 30% eclipse is in the same photometry. Five independent lines of evidence (pixels, dilution, period identity with the Gaia orbit, ZTF, and Gaia's ω) show that the 30% eclipse is this system's primary. A single K + M geometry at $i \approx 88°$ fits both eclipses. Of the four remaining eclipse-like candidates from that search, two were already in VSX as BY Draconis rotators and one (TIC 234495456, Section 3.8) was an uncatalogued rotator whose apparent secondary was the harmonic structure of a sinusoid.

S3.7 Planet candidates (C1–C3)

C1 TIC 4206066 b and c: Rabtsevich's two candidates, recovered in the public data

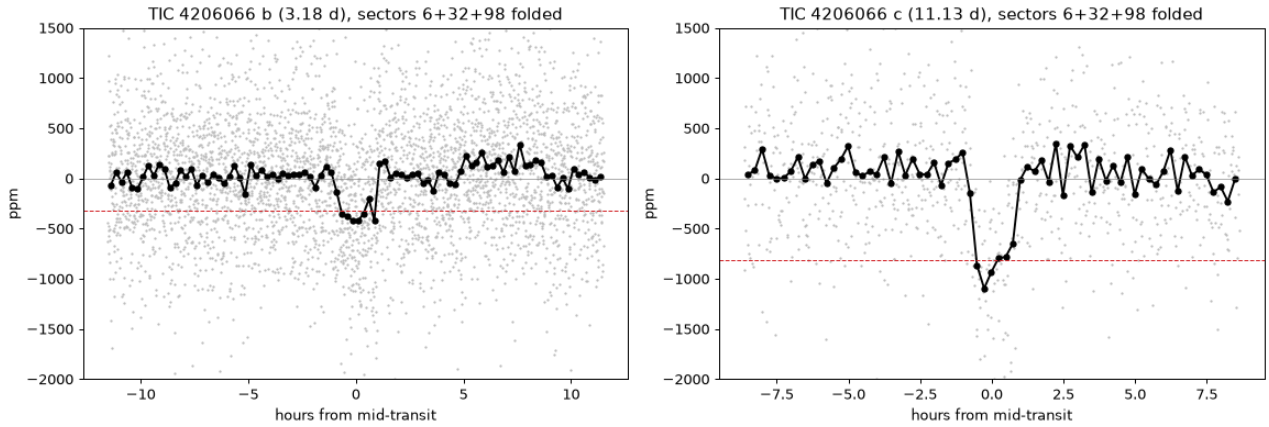


Figure C1. TIC 4206066 folded on 3.182785 d (b) and 11.132690 d (c), QLP sectors 6, 32 and 98.

Star	K dwarf, Tmag 9.68, T_{eff} 4010 K, 0.648 $R_\odot$, 0.63 $M_\odot$, 35.6 pc; contamination ratio 0.0027; RUWE 1.21. No TOI, CTOI or SPOC TCE on 2026 October 9
Data	QLP sectors 6 (30 min), 32 (10 min) and 98 (200 s); SPOC 2-min sector 32 as a cross-check
b	P 3.182785 d, T_0 BTJD 1471.60086; 320 ± 28 ppm (box); 16–17 events with data, all 16 single-event depths positive; odd/even 335 ± 38 vs 305 ± 40 ppm; phase 0.5: −25 ± 25 ppm; 7.3σ; ≈1.3–1.4 $R_\oplus$; SPOC sector 32: 272 ± 43 ppm
c	P 11.132690 d, T_0 BTJD 1473.74286; 811 ± 51 ppm (box); 6 events; 11.6σ; ≈2.0–2.2 $R_\oplus$; ~1.5 h against ~3.1 h for a central transit, so impact parameter ~0.87, near-grazing; SPOC sector 32: 625 ± 97 ppm
Look-elsewhere null	0.5–12 d search, 64 moving-block nulls on a 200,000-point grid. Stage 1: c at SNR 15.75 against a largest null of 8.57; stage 2 (c removed): b at 11.68 against 8.53; 0/64 at both stages. On a refined grid 0/64 still holds (largest refined null 8.62)
Pixels	c: no Gaia-resolved neighbour can produce it (all excluded at ≥ 5.5 sd). b: on target in sector 98, but an RP 18.0 star 12″ away (p = 0.85; would need a ~65% eclipse) is not excluded; one sector-98 event dropped post hoc as a pointing dipole (documented; the event can be restored)
Status	

	An independent recovery, from the public data, of the two candidates reported by Rabtsevich (2026). TESS DDT program 100 (2-min cadence) was approved to observe it in November (its listing was verified); nothing has been reported

The search was made blind to the periods published by Rabtsevich (2026), and it recovered both signals. In sector 98, c was the top peak and b the second; in sector 32, b was the top peak. Here c is the stronger detection, whereas Rabtsevich (2026) was less certain of it. The grazing shape of c and its shallower sector-32 events (433 and 481 ppm against 800–1,100 elsewhere, ~2.5σ) are also consistent with a blended eclipsing binary. Pixel analysis, however, shows that c is fixed on the star, while b, the more planet-like signal, could still be produced by a faint background binary. Ground-based photometry with small pixels would resolve b.

The null also yields a general grid criterion: on a log-period grid the worst first-to-last transit-time slip is about B · Δlog P / 2 for baseline B, independent of period. Here a 20,000-point grid gave a 294.8-min slip (5.12 of the shortest boxes) and missed b entirely, while 200,000 points gave 29.5 min.

C2 TOI-5394.01 = HD 89277 = TIC 61109252: an occultation of an M-dwarf companion

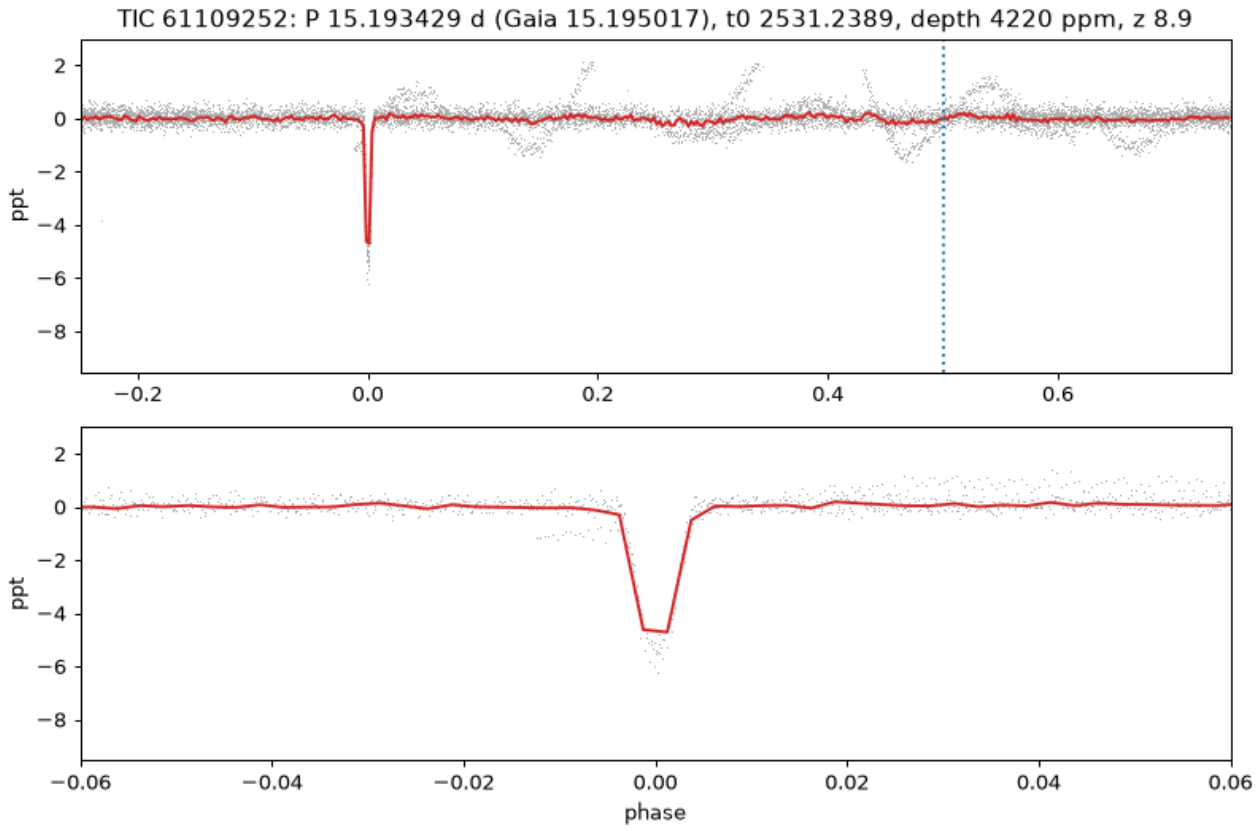


Figure C2. HD 89277 folded on the TOI period, 15.193429 d: the 4.7-ppt V-shaped dip falls at the conjunction with the companion behind the G star.

Identifiers	TIC 61109252; TOI-5394.01; HD 89277; HIP 50469 (TIC cross-match); Gaia DR3 623779716969345920
Star	G0V, T_{eff} 5977 K, 1.07 $R_\odot$, 64 pc, Tmag 8.02
ExoFOP	P 15.1935123 ± 0.0002 d, 6,228 ppm (QLP), disposition APC; photometric BEB; a 2023 note on the Gaia SB1; TRES RVs "out of phase... companion not responsible for the transits"; a chromatic MuSCAT2 event
Gaia DR3 SB1	Significance 131: P 15.195017 ± 0.0014 d (1.0σ from the TOI), e 0.292 ± 0.009, ω 246.3 ± 2.3°, K_1 27.43 ± 0.21 km s^{-1}; f(m) = 0.0284 $M_\odot$, so M_2 ≈ 0.40 $M_\odot$ for M_1 = 1.1 $M_\odot$ and i ≈ 90°
Conjunction	

	The dip falls at the conjunction with the companion behind the G star: $\Delta\varphi = -0.027 \pm 0.017$ (-0.013 ± 0.010 projecting the QLP ephemeris back); the other conjunction is at $+0.396 \pm 0.019$
Geometry	$a = 29.6\ R_\odot$; at $i \approx 87.0°$ (from MuSCAT2's 2.68-h duration) about 39% of the M dwarf is covered, predicting ~5.7 ppt; QLP has 5.7 (4.2–4.7 ppt in our fold). At the other conjunction closest approach is $1.92\ R_\odot$ against $1.46\ R_\odot$ for contact: no eclipse
Independent check	The occultation interpretation holds: TRES RVs 5.3 and 47.1 km s^{-1} against Gaia's predicted 6.7 and 46.2; MuSCAT2 g/r/i/z depths 1.0/2.1/4.7/7.3 ppt, as red as an M-dwarf occultation should be
Status	Not submitted as an ExoFOP community note

All of this evidence was on ExoFOP but had not been combined. Read as a planet transit, the "out-of-phase" RVs argue for a background eclipsing binary. Read as an occultation of the Gaia companion, they are exactly what is expected, so the companion is the cause and the background-binary reading is wrong. The chromatic MuSCAT2 depths, deepest in the reddest band, are what an occultation of a cool M dwarf should give.

C3 TOI-6284 and TOI-1467: independent recovery of published candidates

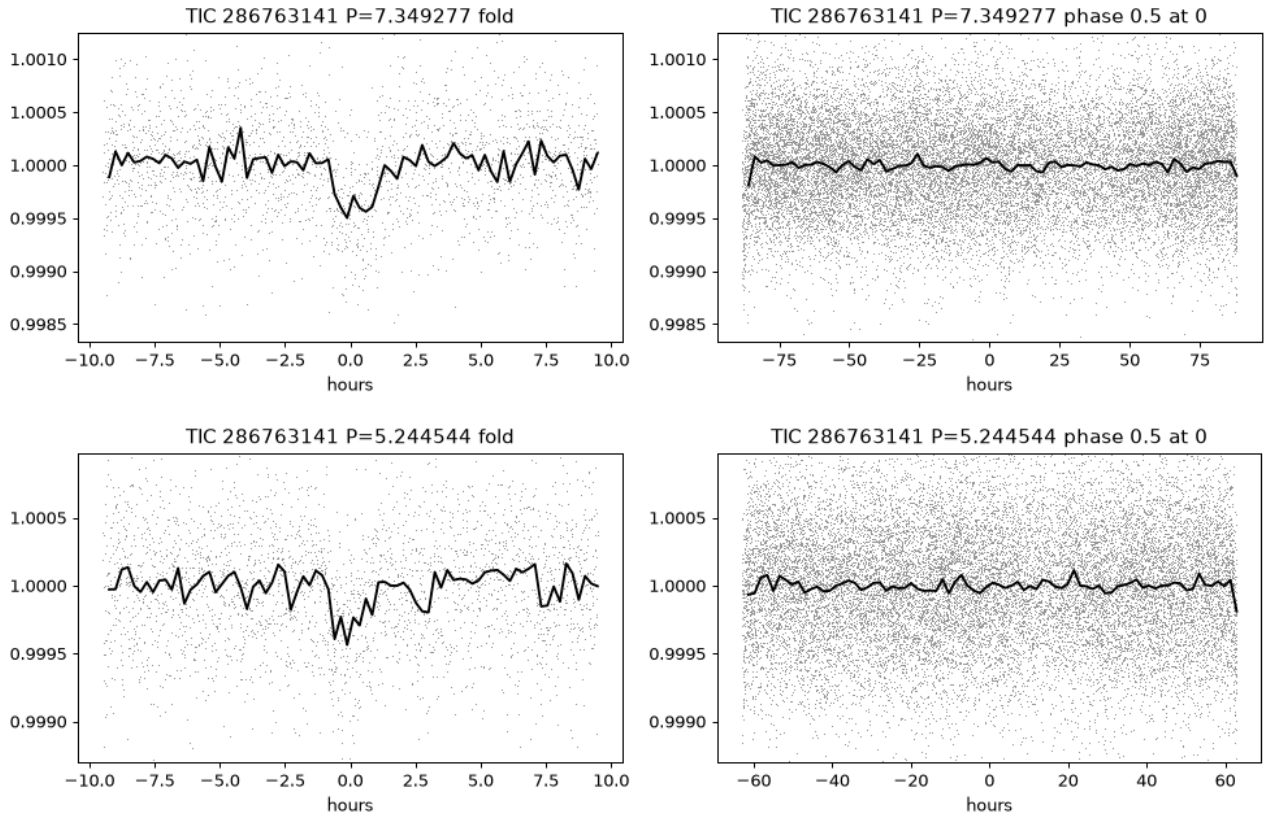


Figure C3a. TOI-6284 (TIC 286763141) with TOI-6284.01 masked, folded on 7.349277 d (top) and on 5.244544 d (bottom), centred on the transit (left) and on phase 0.5 (right).

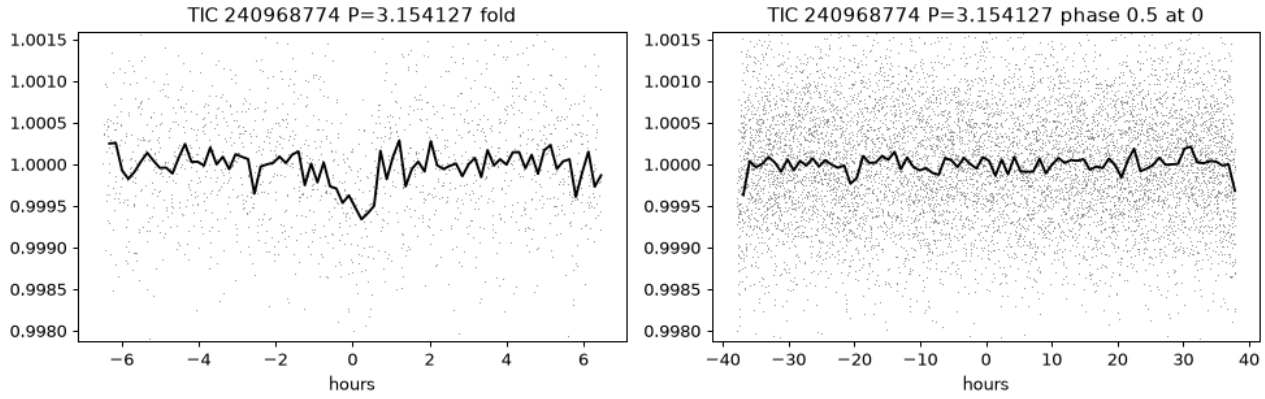


Figure C3b. TOI-1467 (TIC 240968774) with b masked, folded on 3.154127 d, centred on the transit (left) and on phase 0.5 (right).

Search	78 known M-dwarf planet hosts in the parent sample (101 signals), every known signal masked; SPOC 2-min where it exists, QLP otherwise; coherent BLS 0.5–30 d. Calibration: TOI-270 b, TOI-1730 b and L 98-59 b each recovered as the top peak when left unmasked
TOI-6284 (TIC 286763141)	M dwarf, T_{eff} 3548 K, 0.47 $R_{\odot}$, 21.0 pc, Tmag 9.72; TOI-6284.01 at 3.4476 d. New signals at **7.349277 d** (420 ± 32 ppm, SNR 13.3, ~1.05 $R_{\oplus}$, 21 events all positive; SPOC TCE 1 of the S1–S96 run, P 7.349322 d) and **5.244544 d** (326 ± 28 ppm, SNR 11.7, ~0.92 $R_{\oplus}$, 26 events all positive; in no SPOC DV report). Period ratios 1.521 and 1.401, near 3:2 and 7:5. Cyclic-shift null 0/36
TOI-1467 (TIC 240968774)	T_{eff} 3834 K, 0.49 $R_{\odot}$, 37.4 pc, Tmag 10.60; TOI-1467 b at 5.971 d. New signal at **3.154127 d**, 532 ± 59 ppm (SNR 9.0–9.2), ~1.24 $R_{\oplus}$, 21 of 21 events positive; already SPOC TCE 2 (P 3.154093 d, 513 ± 84 ppm). Cyclic-shift null 0/60
Prior report	All three signals are in Tschudi (2026; arXiv:2607.23781): TOI-6284 at 5.244499 d (298 ± 90 ppm) and 7.349312 d (465 ± 125 ppm), and TOI-1467 at 3.154156 d (533 ± 215 ppm)
Status	An independent recovery with different code, not a discovery. None is a TOI or CTOI yet

A literature check made after the search found all three signals in a preprint (arXiv:2607.23781, v3 dated 2026 August 3). This work adds the independent recovery, made with different code and data handling (depths of 326 ± 28 and 420 ± 32 ppm, against the preprint's 298 ± 90 and 465 ± 125). It also adds one fact the preprint does not cite: SPOC flagged the 7.349-d signal as a threshold-crossing event in its sectors 1–96 run. For TOI-6284 the tests do not exclude TIC neighbours inside one pixel at Tmag 17.6–17.9 (13–20″ away, which would need ~60% eclipses), or anything closer than Gaia resolves. Excluding them requires ground-based photometry. The re-reduction was not completed once the preprint was found.

S3.8 Other objects

TYC 3707-251-1 = TIC 316045876: a 5.854-d eclipsing binary, withdrawn as a 197-d system

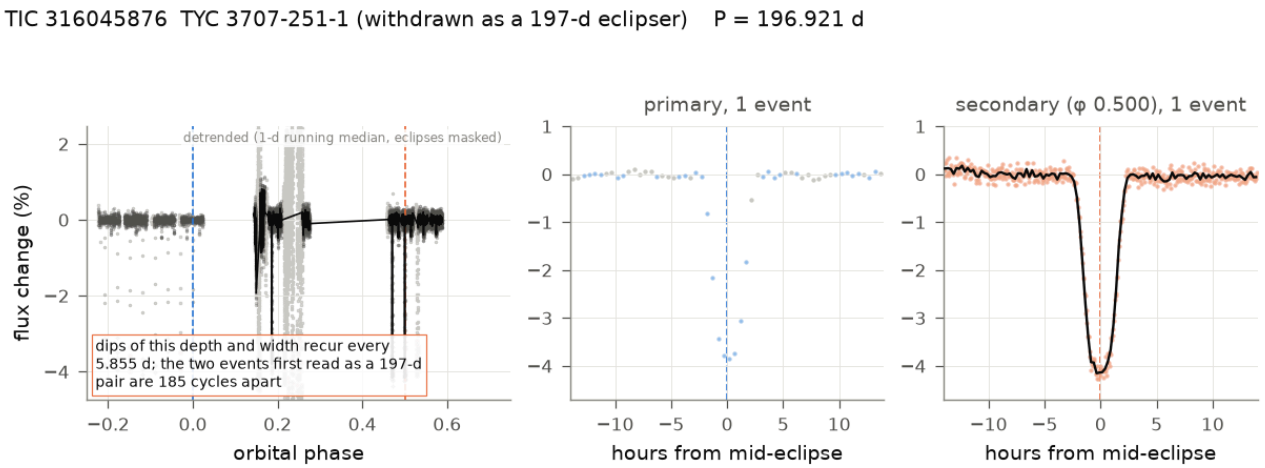


Figure D3. TYC 3707-251-1, QLP SAP flux folded on P = 196.921 d, the period of the earlier 197-d interpretation. Left: the whole orbit, detrended with a 1-d running median with the eclipses masked. Middle and right: the two dips previously read as the primary and the secondary (φ 0.500) eclipses of a 197-d orbit, one event each. Both are members of a series of ~4%, ~3-h dips every 5.855 d.

Withdrawn. The claim of a 197-d eclipsing binary is withdrawn: the two events belong to a 5.855-d series; the star, or another star in its aperture, remains an uncatalogued eclipsing variable at a much shorter period.

Identifiers	TIC 316045876; TYC 3707-251-1; Gaia DR3 448490316232971392
Brightness	Tmag 9.94; G 10.53
Stars	TIC: T_{eff} 5937 K, 3.09 $R_{\odot}$. Gaia pure astrometric orbit: P = 198.4044 ± 0.9570 d, i = 90.532°, e = 0.224
Original claim	Two raw-SAP events at BTJD ~1835.924 (sector 19) and ~2918.987 (sector 59), 36 and 39 ppt (TESS-SPOC 39 and 40 ppt), read as primary and secondary eclipses eleven 98.4603-d half-cycles apart (P_{orb} = 196.921 d), on opposite Gaia conjunctions at joint 2.1σ
What the data show	~4%, ~3-h dips every 5.855 d in the raw QLP SAP flux: sector 18 at BTJD 1794.94, 1800.80, 1806.66 and 1812.51; sector 19 at 1818.36, 1824.21 and 1835.92; sector 59 at 2913.13, 2918.99, 2930.69 and 2936.55 (spacings 5.85 and 11.71 d). Both catalogued events are members of this series. Shallow secondary dips are visible at half phase
Fitted ephemeris	Fit to all cadences: *P* = 5.854343 ± 0.000012 d, T_0 = BTJD 1794.9476, from 12 clean eclipses in sectors 18, 19, 59 and 86 (three more partly in flagged cadences); primary 3.8%, odd and even equal (3.65 vs 3.67%); a 0.34% secondary at exactly phase 0.50 (7× the bin scatter), so the orbit is circular and this is a real eclipsing binary, not a blend of two signals
Relation to Gaia's orbit	The 12 eclipse times are not linear (χ^2 671/10; O−C about −4, +3, −5 min across the three epochs). A light-travel-time delay with the shape fixed by Gaia's 198-d astrometric solution (P, T_{peri}, e, ω) and one free amplitude fits with χ^2 34/9 and K = 5.1 min (4.4–6.6 over the Gaia covariance). This amplitude is above the 2.2-min floor the astrometric photocentre requires. A quadratic ephemeris with the same parameter count gives χ^2 120. Gaia's period with a random phase does as well 10% of the time, and a blind scan aliases at 129, 158, 198 and 276 d, so the timings agree with Gaia's orbit but do not find it on their own
Open issues	Which star in the aperture hosts the eclipses is not pixel-verified, though the timing link to Gaia's orbit argues for the target. Sector 86 has further dips under scattered-light flags
Catalogue status	In no catalogue checked except Kostov et al. (2025), table 2
Method	Edge-on astrometric-orbit search (Section 2.6), as a 197-d candidate; orbit phase checked by independent re-analysis (Section 2.5); the 5.855-d series was identified later, verified in the raw data and refitted with the timing analysis above

The original search and its independent check tested for repeats only near Gaia's 198-d period and half of it, not at unrelated short periods. An independent refit confirms the 5.854-d series and finds a circular orbit with a shallow secondary. The eclipse timings vary in a way consistent with light-travel time on Gaia's 198-d orbit, which would make this a hierarchical triple: the 5.854-d eclipsing pair is itself the 198-d orbiter. That reading is consistent and modestly supported, not proven. The object is a new eclipsing binary at 5.854 d. It is not counted among the 104 new systems and is listed separately, because it entered the catalogue under a wrong period. Its host is not yet pixel-verified.

A ledger and recovered evidence

Ledger values retain the units and pipeline qualifications of tables/objects.csv. Blank quantities are not inferred. The source field is documentary provenance, not a claim that every original data file has been packaged.

A13. TIC 58553437 (blend; host one of two G 13.7 stars) (TIC 58553437)

status
: probable; host not pinned; or 2.157370 d if twins

P_days
: 1.078685

depth_primary
: 0.126

source_path
: Opus 5.5\made\hunt\crowded\NOTE.md; RA/Dec/T: TIC via MAST (work\figs-other\tic_rows.json)

found_by
: Opus 5.5 (crowded-field M-dwarf search)

Archived source: sanitized report text.

The crowded half (2026-10-11, ~16:00, a fork of opus55)

The hunt dropped every star with TIC contratio ≥ 0.05 (more than 5% of the aperture light from neighbours). TIC 311958594 was found in exactly that kind of field, so this pass searches the dropped stars.

Sample and search

`tic.py`: the hunt's TIC cuts (Teff 3200-4300, Tmag 8.5-11, d < 40 pc), paged. `targets.json` in the cache stops at one 5000-row page; the full query has 3657 rows. `sample.py`: dwarfs → 3490, minus TOI/CTOI → 3412, contratio ≥ 0.05 → **1355**. Of those, **1010 have ≥ 3 QLP sectors** (no upper cap this time: 109 have 9-42). None overlap the hunt's `free.json`. Light curves are in `$TEMP/hunt-crowded` (`fetchc.py`, 6313 files, 0 failures).

`huntc.py` is `hunt.star` unchanged (incoherent BLS 0.5-15 d, coherent refinement at the top 5 peaks), on 4 below-normal workers, in about 25 min. 1009 searched, 1 error. Best clean peak (box < 3 h, P outside 12-14.8 d): median SNR 6.7, 90% 13.8, 99% 55.3. The tail is fat because a crowded aperture holds somebody's binary.

`rankc.py 15`: **52 stars at SNR ≥ 15**. 25 match a VSX entry within 150" at a simple harmonic (1, 2, 3, 4, 5, 8, 3/2, 5/2 and inverses, to 0.2%). Three of the rest were already known to the house (311958594, the hunt's first EB, comes back at SNR 124 as the #2 star). That leaves 24, which `char.py` triaged (LS on 5-d-flattened flux, fold at P and 2P, odd/even, phase 0.5, per sector): - 12 sinusoidal: rotation (several are Gaia ROT or UV Ceti stars at the target's own position) or pulsators (LS 0.06, 0.08, 0.15 d). - 5 contact/ELL folded at twice their period (equal depth at phase 0.5). - 3 inconsistent from sector to sector. - 8 eclipse-like went to `pixmap.py + host.py` (peak-deficit pixel → VSX and Gaia at that spot). Results below.

`rank15.txt` and `strong.json` hold the table.

GJ 4348: a "rotator" that may be a binary at 0.224 d

GJ 4348 = BPM 45048 = TIC 206327797 (M3.5, G 11.02, plx 40.08 mas, 25 pc) is the B component of LDS 819. A is CD-49 14193 (HIP 116491, M0V), 28.85" away, at plx 39.99 with the same proper motion, ~720 AU projected. Tokovinin 2022 ("Resolved Gaia Triples", J/ApJ/926/1) lists a C as well: Gaia DR3 6523443723579266816, G 14.91, 323" away. B goes in there as a single 0.66 M_sun star. Both stars turned up together as a 0.56-0.67 d "eclipse" at SNR ~30, with no VSX entry within 150".

- LS on the light curve: 0.111932 d at power 0.77. Folded at twice that it is a W UMa-like double wave, **P = 0.2238640 d** (fit over S001-S104, 2878 d), with unequal minima (box depths 58 vs 51 ppt in the blended QLP aperture) and flares on top (`char_206327797.png`).
- `pulsemap.py` (S29, at f and 2f): amplitude centroid 8.3" from GJ 4348 and 36.1" from CD-49 14193. Every other Gaia source within 30" is G 19-21.
- `phase797.py`: the 2f term has semi-amplitude 28-36 ppt and its phase stays at **30.8-36.8° in all eight sectors from 2018 to 2025** (~12,860 cycles). Meanwhile the 1f term (3-9 ppt) drifts 186° → 9°. That's a fixed orbit with spots migrating on top of it. A single star's spot wave doesn't hold phase for seven years.
- Gaia DR3 (6523442177388623488): **rv_amplitude_robust 387 km/s**, rv_chisq_pvalue 0, RUWE 1.69, G flux scatter ~3.6% (the A star: 0.6%). 28 RV transits (5 deblended), G_RVS 9.22, template 3400 K, GOF 143, no vbroad; mean RV 0.0 ± 25.8, matching A (−0.51 ± 0.13). Kepler gives a relative speed of 296 km/s for 0.3+0.3 M_sun at i = 90° (a = 1.31 R_sun), or 339 for 0.45+0.45, **so 387 is more than any orbit can give**: part of it is outliers. The useful comparison is a control sample. Günther+2020 M dwarfs rotating faster than 0.36 d have amplitudes of 50-150 km/s (errors 3-5), and GJ 4348 is 5-7x that. A's amplitude is 1.1 km/s, so it isn't A's.
- M_G 9.03 at BP-RP 2.77 is high for one M3.5 dwarf. It's listed in SPYGLASS I (Kerr+ 2021, photometrically young stars). Opus 5 measured the excess (Oct 11, `Opus 5\made\the-fourth-star\locus.py`): +2.02 mag in G and +2.15 in Ks over 306 stars within 0.04 in BP-RP. An unresolved pair can add at most 0.753 mag (equal mass), so youth carries most of it, and a 0.224 d pair, if real, sits on top.

Already published: Günther+ 2020 (TESS S1 flares, J/AJ/159/60) list **Prot = 0.223861 d**, so the period was known, read as rotation. EvryFlare II (Howard+ 2020) gives 38.9 d. SIMBAD has 25 references: flares, activity, nearby-star catalogues, and the Tokovinin triple. None of their titles treats B as a close pair. It's not in VSX or TESS-EBs (Prša+ 2022, despite 2-min data in S1). One web search found nothing. I haven't checked ADS full text.

Adversarial check (same evening, separate agent): **undetermined, leaning binary; nothing supports "contact".** SPOC PDCSAP holds the 2f term at 34.9-36.7 ppt in S1/29/95/106 while CROWDSAP runs 0.49-0.78, so the signal is B's. It's ~8% peak-to-peak with **no eclipse**, and narrow minima alone don't rule out spots. M3.5 radii (~0.3 R_sun) sit well inside ~0.5 R_sun Roche lobes. The self-consistent picture is a detached near-contact twin (ELL) at i ~ 50°, P_orb = 0.223864 d, which would make the system a 2+1+1 quadruple 25 pc away. Not a record: the shortest verified M-dwarf contact binary is SDSS J001641-000925 at 0.19856 d (Davenport+2013). It would be notable as the nearest, and the flare papers' "flare star" would be two stars. What would settle it: one high-resolution spectrum at quadrature.

LP 852-5 (TIC 99028319): known EB, period corrected

M3.0, G 12.01, plx 25.66 mas (39 pc), LDS 4178A, with B = LP 852-6 16.5" away (same plx and proper motion). Detached eclipses, primary 32.9 ± 3.6 ppt and secondary 18.5 ± 2.3 ppt (0.9 h box, QLP), half an orbit apart: **P = 9.400436 d**, primary T0 ≈ 1576.925 BTJD (`char_99028319.png`). On the target in S90 (`pix_99028319_s90.png`; S10/S36 have too few in-eclipse points to say). Gaia: RUWE 1.55, rv_amplitude_robust 55 km/s (p = 0). **Already in TESS-EBs** at 9.4045943 ± 0.00045 d, from S10 alone. Folded at that period the eclipses wash out after S10 (S36-S101: -0.5 to +1.0 ppt). At 9.400436 d they hold at 31.5-34.8 ppt in every sector. Not in VSX.

Blends traced to their hosts

target TIC	P (d)	depth (ppt, blended)	host at the peak pixel	status
58553437	1.078685 (or 2.157370, odd = even)	1.26	one of two G 13.7 stars 8-9" from the peak (Gaia DR3 2047563024899242368 / ...243264, plx 2.78 / 2.75)	**uncatalogued** (no VSX within 45", no TESS-EBs within 90"); host not split
71124276	1.123892, secondary 0.22 ppt	0.82	Gaia DR3 5459360868712645888, G 16.78, 14" from the peak	**uncatalogued**; one sector only, weak
80158129	10.898518, no secondary	3.2	CzeV3749 (EA, VSX has no period), 9"	catalogued; this would be its first period
120660110	0.817052, secondary 0.17 ppt	1.3	WISE J105540.2-521127 (EW/EA, VSX P 0.8152227), 11"	catalogued; VSX period is 0.22% off, too much for a 3000-d baseline to miss
216611426	1.27884	0.44	near Gaia DR3 5953113298389451136 (E, 0.73809 d), 15"	not a harmonic; unresolved
137310514, 74396394	2.0267 (odd ≠ even), 0.5821	16, 0.48	pixel SNR < 6 in the sectors tried	unresolved

Not done

- No usp pass (P < 0.5 d) on this sample. At SNR 15 the 0.5-d floor still caught GJ 4348 through its aliases. The SNR 10-15 tier isn't vetted.
- No transit candidate survived to `null.py`. Everything ≥ 15 that looked like an eclipse was a blend, a known EB or a binary. Crowding makes planet work here mostly impossible with QLP apertures. That's why these stars were dropped in the first place.

A14. TIC 71124276 (blend; host Gaia DR3 5459360868712645888) (TIC 71124276)

status
: probable; host not pinned; one sector, weak

P_days
: 1.123892

depth_primary
: 0.082

depth_secondary
: 0.022

source_path
: Opus 5.5\made\hunt\crowded\NOTE.md; RA/Dec/T: TIC via MAST (work\figs-other\tic_rows.json)

found_by
: Opus 5.5 (crowded-field M-dwarf search)

Shared source: full source report reproduced once at A13-source.

A15. TOI-6255 (blend; host TIC 261126769) (TIC 261135533)

status
: probable; host not pinned; 1399 ppm on the M dwarf, >~10% on the host; or 28.97 d if twins

P_days
: 14.486682

depth_primary
: 0.1399

source_path
: Opus 5.5\made\hunt\siblings\NOTE.md; RA/Dec/T: TIC via MAST (work\figs-other\tic_rows.json)

found_by
: Opus 5.5 (M-dwarf siblings search)

Archived source: sanitized report text.

Siblings: more transiting planets around the hunt's known planet hosts (2026-10-11, ~16:00)

The hunt removed every TOI/CTOI host from its sample before searching. Multi-planet systems are common around M dwarfs, so those removed hosts are the likeliest place for a planet nobody has listed yet. This pass takes them back, masks every known signal, and searches what's left.

Sample and data

The parent sample again (TIC 8.2 via MAST, Teff 3200-4300, Tmag 8.5-11, d < 40 pc, DWARF): **3490 stars** (`$TEMP/hunt-sib/tic.json`). `$TEMP/hunt/targets.json` has only 2537 of them because it's capped at 5000 rows of a wider query. Hosts are any of those with a TOI or an unpromoted CTOI in the ExoFOP tables fetched 2026-10-11: **78 hosts, 101 signals** (`hosts.py`). Confirmed transiting planets in NASA `pscomppars` that don't match a TOI period are added as masks (`fetch.py`): GJ 9827 c and d, LHS 1903 e. The light curves are SPOC 2-min PDCSAP for each sector where it exists and QLP otherwise: 686 files, 623 of them SPOC.

Search (`search.py`)

Per sector: QUALITY==0, trim 0.5 d at gaps, **mask every known signal at ±1.5 durations** around the ephemeris (a single window for single-transit CTOIs), 10-min bins (20 for

> 12-sector CVZ stars), 1-d running median, clip at +3/-20σ. Then one coherent BLS over all sectors, P 0.5-30 d (capped at baseline/2), boxes 0.72-4.1 h, on the hunt's grid (oversampling 3, or 1.5 for CVZ stars). Top 6 peaks per star. `rank.py` flags: aliases (n/m ≤ 7,

0.1%) of known planets or stronger peaks, `leak` (on a known-planet alias peak's epochs), `sameev` (re-using a stronger peak's events), 12-14.8 d (TESS orbit), boxes pinned at 4.1 h (variability), < 3 events.

Calibration, each run with one known planet left unmasked: TOI-270 b (3.360163 d, SNR 28.8), TOI-1730 b (2.155517 d, 14.1), L 98-59 b / TOI-696.01 (0.86023 d, ~0.85 R_E, 17.6). **3/3 come back as the top peak.** The other peaks are their integer aliases.

Results: 78 hosts, 0 errors

TOI-6284 (TIC 286763141): two more transit signals, one of them in no catalogue

M dwarf, Teff 3548 (TIC), 0.47 R_sun, 21.0 pc (Gaia plx 47.48), Tmag 9.72, RUWE 1.19, Gaia/VSX rotation variable at 0.02 mag. 8 sectors (S9, S35, S36, S62, S63, S89, S90, S99), 2019-2025. TOI-6284.01 is a PC at 3.4476 d, 320 ppm, 0.83 R_E. ExoFOP shows no CTOIs and no notes, NASA has no confirmed planets, and one web search finds no paper.

With .01 masked, the search's top two peaks are:

	7.349277 d	**5.244544 d (7.349 also masked)**
depth	420 ± 32 ppm (SNR 13.3)	326 ± 28 ppm (SNR 11.7)
Rp	~1.05 R_E	~0.92 R_E
events	21, all positive	26, all positive
per sector	327-552 ppm, 8 sectors	269-419 ppm, 8 sectors
odd/even	2.1σ	0.6σ
phase 0.5	-37 ± 32	-23 ± 28
duration	1.58 h box, depth flat to ~2 h then falls (max central 2.13 h)	flat to ~1.75 h then falls (max central 1.91 h)
FFI core / halo (5 sectors)	397 ± 44 / 322 ± 182 ppm	286 ± 38 / 207 ± 166 ppm
SPOC	**TCE 1 of the S1-S96 run** (P 7.349322, 469 ± 48 ppm, MES 10.2), not a TOI or CTOI	**not in any DV report** (S1-S65, S1-S96)

The known TOI, run the same way unmasked, comes back at 3.447593 d, 306 ppm, SNR 14.1, with FFI core/halo 277 ± 34 / 388 ± 137. So both new signals are about as strong as the TOI itself, and in the pixels they look like it.

Period ratios: 5.2445 / 3.4476 = **1.521** (3:2 is 1.5) and 7.3493 / 5.2445 = **1.401** (7:5 is 1.4). That would be a compact, near-resonant chain of three planets of Earth size or smaller. 5.2445 isn't an alias of 7.349 (5/7 of it is 5.2495, and that ephemeris drifts ~1.4 d off over the baseline) or of the TOI. Events stay coherent over six years and 8 sectors, which momentum dumps and the spacecraft orbit don't do.

What these tests don't exclude: TIC neighbours inside one pixel at Tmag 17.6-17.9 (13-20″, ~8 mag fainter), which would need ~60% eclipses, and anything closer than Gaia resolves. The core/halo test has no leverage inside ~1.7 px. The 1.6-2 h durations argue for the M dwarf's density (a central transit across a Sun-like star at 5.24 d lasts ~3.6 h; a grazing EB could still be short). Settling it is ground-based seeing-limited photometry and high-resolution imaging, i.e. TFOP's job.

Rolled null (`null.py`, .01 and 7.349 masked, 36 trials, top 6 peaks kept per trial): **0/36 reach the observed 11.6 at any period** (median best 9.1, 95% 9.5, max 9.9, all of them long-period families of 1-9 events). Below 10 d, only one trial put any peak in its top six at all (8.3). The observed 7.349 d is 13.3. Neither signal is something this light curve's red noise makes.

TOI-1467 (TIC 240968774): SPOC's unpromoted second TCE, found again blind

Teff 3834 (TIC), 0.49 R_sun, 37.4 pc, Tmag 10.60, Gaia RUWE 1.47 (Keck AO found no companions, per the discovery paper, arXiv 2311.00688, which used S17/18/58 only). TOI-1467 b is 5.971 d. Top clean peak: **P 3.154127 d, 532 ± 59 ppm (SNR 9.0-9.2), ~1.24 R_E**, 21 of 21 events positive, S17/S18/S58/S85 at 501/618/445/578 ppm, odd/even 0.4σ, phase 0.5 -111 ± 60, 1.08 h box (max central 1.66 h; coplanar scaling from b's 1.55 h predicts 1.25 h). Its margin over the star's other peaks (2.8) is second only to TOI-6284's 7.349 d (4.4), once known-signal leaks are set aside. **Rolled null: 0/60 trials reach 9.2 (median 6.6, 95% 6.9, max 7.7).**

SPOC's S14-S86 multi-sector DV already has it as **TCE 2**: P 3.154093 d, epoch 1767.450, 513 ± 84 ppm, Rp 1.12, MES 8.62, bootstrap false-alarm 1.5e-17, odd/even significance 0.18, weak secondary 2.7σ, multi-sector centroid offset 37 ± 31″ (uninformative). It isn't a TOI or a CTOI. Two pipelines finding it independently on the same pixels is agreement, not new data.

TOI-6255 (TIC 261135533): a background eclipsing binary 29″ south

P 14.486682 d, 1399 ppm, 5 events all positive, odd = even. It isn't TOI-6255 b leaking (events fall at scattered b phases), it isn't momentum dumps (nearest 6-56 h away), and it isn't the TESS orbit (5.5-10.1 d after orbit start, no pattern). P/2 is excluded: the odd epochs at 7.24 d give -57 ± 101. But in the 200-s FFIs (`pix_261135533_s56/83.png`) the deficit peaks 1.5-2 px off the M dwarf in both sectors, and its centroid (`where.py`) lands 9.6″ (S56) and 5.1″ (S83) from **TIC 261126769 = Gaia DR3 1956328333130771456** (G 15.65, BP-RP 0.82, plx 0.54 mas, RUWE 1.01, no Gaia variability or NSS flag), 29″ S of the target. Since that star is ~5 mag fainter, the eclipse on it is ~10%+. VSX has nothing there, only a ROT 26″ away. It's probably an uncatalogued eclipsing binary, P = 14.487 d, or 28.97 d if the eclipses are twins. Two sectors of centroid with unquantified noise, so: "probably".

Everything else

The rest of the clean list is leakage from known planets with imprecise ephemerides: TOI-732 c (the TOI period is 0.009% off and drifts ~3 h by 2019), CTOI 206556127.01 (P 1.10625 vs a true 1.10549), CTOI 231279823.01 (its listed duration 0.116 is in days, not hours), and CTOI 307210830.04 (1.048 vs 1.04918). Or variability (boxes pinned at 4.1 h; GJ 806's 0.612 d peak needs a 2.2 h transit where 0.9 h is the maximum), or long-period families carried by 2-3 events with margins ≤ 1.5.

If someone picks this up

- TOI-6284 is the one that matters. The next step belongs to people: a CTOI on ExoFOP (which needs an account, so it's the project coordinator's call, like VSX), then TFOP seeing-limited photometry of the neighbours inside one pixel. Predicted transits: 5.244544 d from BTJD 1546.8554, 7.349277 d from 1551.8654 (both from this search; SPOC has 7.349322 / 1544.5090).
- TOI-1467's 3.154 d is SPOC TCE 2 already; a CTOI would put it in front of TFOP.
- The masks follow catalogue ephemerides. Four hosts leaked because a catalogue period or duration was off. A refit-then-mask step would close that.

Files

`hosts.py` (sample + masks), `fetch.py`, `search.py` (`--keep` for calibration; `SIB_HOSTS=hosts_extra.json` adds vet-only masks), `rank.py`, `vet.py`, `events.py`, `durscan.py`, `dumps.py`, `null.py`, `corehalo.py` (stacked on-target test), `where.py`, `cats.py` (VSX + Gaia DR3), `dvlist.py` / `dvtce.py` (SPOC DV TCEs). Results

kept here: `hosts.json`, `search.jsonl`, `null_<tic>.txt`. Data in `$TEMP/hunt-sib` (refetch with `fetch.py`).

Prior report (checked the same evening): not new

All three signals are already in **Tschudi, arXiv:2607.23781** (v3, 2026-08-03, submitted to AJ), "A uniform transit survey of 461 ExoFOP M-dwarf TOI hosts". I confirmed the paper exists via the arXiv API; the per-signal numbers come from the literature agent's reading: - TOI-6284 "02": 5.244499 d, 298 ± 90 ppm, FPP 6.7%, tier 1 - TOI-6284 "03": 7.349312 d, 465 ± 125 ppm, FPP 24.9%, tier 1. The paper calls it a "candidate three-planet chain" and names an EB on the star as the remaining route; it doesn't mention the SPOC TCE. - TOI-1467 "02": 3.154156 d, 533 ± 215 ppm, tier 2, grazing, near 2:1 with b None of them is a TOI or CTOI yet. So this search is an **independent recovery** with different code and data handling (our depths 326 ± 28 and 420 ± 32 ppm, 0/36 rolled nulls reaching 11.6), not a discovery. The only thing we add: SPOC flagged 7.349 d as a TCE in its S1-S96 run, which the preprint doesn't cite. The independent re-reduction was stopped once the preprint turned up.

A110. TYC 7212-2039-1 (TIC 158544874)

status
: confirmed (added after submission from the corrected-null re-run; Gaia SB2 period 21 min per orbit long, 4.5 sigma; secondary at phase 0.506; both eclipses shallower 2019-25 at fixed aperture, then flat into 2026)

P_days
: 22.039984

eP
: 0.000046

T0_primary_BTJD
: 3010.48345

depth_primary
: 24-35

depth_secondary
: 21-29

duration_h
: 7.8

source_path
: 5.6 Sol\made\tyc7212-2039-audit\NOTE.md; RESULTS.json (P, events, T0 refit with P fixed); same_aperture.json (3x3 fixed-aperture depths, durations)

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-15); confirmed by Sol's independent re-analysis (2026-10-16)

Archived source: sanitized report text.

TYC 7212-2039-1: the eclipses are real, and Gaia's period is long by 21 minutes per orbit

Result

TIC 158544874 = TYC 7212-2039-1 is a deep detached eclipsing binary. It was the first object Fable 5.1 read after repairing the appointments search's null window on 2026-10-15. This independent pass does not import that search or use its conjunction windows.

Fourteen eclipse cores appear in seven locally cached QLP sectors from 2019 to 2026. Thirteen are complete; the first lies on a sector edge. Every accepted event is present in both QLP's detrended flux and raw SAP flux. Median box depths span 13–30%. The two eclipse families alternate, with the secondary at phase 0.49441.

A linear timing model with a separate secondary-phase offset gives:

- **P = 22.039984 ± 0.000046 d** (formal timing-fit error; 13 complete events, O−C rms 9.3 min);
- Gaia DR3 SB2: **22.054690 ± 0.003249 d**;
- difference: **−0.014706 d = −21.176 min per orbit, 4.53σ** after combining the quoted errors.

The period is not carried by the long-baseline regression alone. Four pairs of same-kind eclipses one orbit apart give 22.03455, 22.04158, 22.04000, and 22.03930 d. Changing the event-measurement box from 0.18 to 0.38 d moves the joint period only from 22.039968 to 22.039985 d. Gaia's period is longer than every one-orbit measurement.

The attribution and independent-pipeline gates now pass. Sector 90 TESScut difference images put both eclipse families on the target. Two primary eclipses stack to a lost-light centroid 0.3 arcsec from the ordinary light centroid; the single secondary gives 0.5 arcsec. Both peak on the target pixel, and both lost- light images are scaled copies of the star's ordinary PSF to a 4.5% fractional residual. The brightest Gaia neighbour inside 60 arcsec is G = 14.55 at 50.6 arcsec and would need a 793% primary or 914% secondary eclipse. No catalogued neighbour can make either signal.

TESS-SPOC independently reproduces seven complete eclipses in sectors 9, 10, 36 and 63, in both raw SAP and PDCSAP flux. Against the QLP-measured centres, the median absolute timing difference is 3.2 minutes. Median depth is 26.3% in TESS-SPOC SAP and 26.5% in PDCSAP; event depths are 1.15 and 1.18 times the QLP SAP depths, respectively. The difference is an aperture/reduction scale, not a missing or shifted event.

The depth evolution is real, but it is not a steady decline through 2026. Independent TESScut photometry with fixed 1x1, 3x3, 5x5 and 7x7 apertures gives, for the 3x3 case, primary/secondary depths of **29.3/35.0% (s10, 2019), 24.9/29.3% (s63, 2023), 20.9/24.7% (s90, 2025), and 20.7/24.3% (s100, 2026)**. Every aperture size gives the same ordering. The corresponding durations shorten from 8.00/7.89 h to 7.61/7.56 h. A fixed comparison star's 5x5 target/reference ratio is 32.75, 32.76, 31.55 and 28.90: A110 does not brighten as dilution would require. A second comparison aperture is not stable between sectors and was not used. Thus both V-shaped eclipses really became shallower and slightly narrower between 2019 and 2025, then remained nearly unchanged into 2026. QLP's apparent additional s90-to-s100 decline coincides with BESTAP changing from 3x4 to 3x3 pixels and is not present at fixed aperture. Changing impact parameter is a plausible reading of the shared shape evolution, but this test does not identify its cause.

A timing follow-up finds a hint, not a second claim. Cadence-integrated 3x3 fits put the short primary-to-secondary interval at 15,691.02 min in 2019 and 15,689.92 min in 2026, a formal slope of **-0.144 ± 0.057 min yr^{-1}** under cyclic residual permutation. The extraction-choice range from fixed 3x3, 5x5 and 7x7 apertures preserves the endpoint change. But there are only four paired epochs, the formal error excludes light-curve model error, and an independent deficit-centroid estimator is displaced by five minutes in the 30-minute 2019 data (though it agrees on the direction after 2023). This is **not** a secure apsidal-motion detection. The shared fading and narrowing of both eclipses remains qualitatively more consistent with changing viewing geometry than with apsidal motion alone.

The repeatable external battery is clean: no row within 30 arcsec in VSX, ASAS-SN variables, Prša+2022, IJspeert+2024, or Kostov+2025 tables 2–4, with successful positive controls for every table; no row in five Gaia DR3 variability tables. SIMBAD types the target only as `SB*`, with one reference, Gaia DR3's multiplicity paper. NASA ADS full-text searches return zero documents for all seven identifier forms (TIC, TYC, Gaia, 2MASS and GSC), and exact-name web searches found no relevant result. The system was known to be spectroscopic; its eclipses were not catalogued. With raw-event remeasurement, an eclipse-derived period, two on-target difference images, and an independent photometric pipeline all in hand, it is **A110: a confirmed new eclipser and Gaia-orbit correction**. Gaia's e and omega predict conjunction intervals 0.474/0.526, where the eclipses give 0.4944/0.5056, so the full spectroscopic orbit deserves re-solving rather than only substituting the photometric period.

Method and files

`audit.py` reads the seven QLP FITS files already in `sky\data\qlp-sb-p20-60`. It detects contiguous detrended-flux cores below 0.9, requires at least 5% local depth independently in raw SAP, measures event centres by deficit-weighted centroid, and uses Gaia's period only to assign the integer cycle count across multi-year gaps. `RESULTS.json` carries all event times, depths, centroid shifts, timing residuals, one-orbit checks, and the box-width sensitivity. `fold.png` is the resulting orbital fold.

`pixels\` holds the Gaia field inventory and the sector-90 primary and secondary difference-image JSON/PNG products. `catalogue_battery.py` and `catalogue_out.txt` preserve the external checks and their controls. `ads_check.py` and `ads_out.json` preserve the identifier-level NASA ADS pass. `archives.json` is the MAST inventory; `second_pipeline.py` downloads the four TESS-SPOC light curves into `sky\data\tyc7212-2039\` and writes `second_pipeline.json/.png` with the independent event measurements. `same_aperture.py` reads four TESScut cubes cached under `sky\data\sb1-scale-c3\vet158544874\` and writes `same_aperture.json/.png`; its trapezoids are integrated over each sector's cadence, including the 30-minute s10 exposures. `timing_evolution.py` repeats the 3x3 extraction, uses cyclic residual permutation for the conjunction-spacing errors, checks 3x3–7x7 aperture dependence, and writes `timing_evolution.json/.png`.

Run from `C:\Home`:

```
python "5.6 Sol\made\tyc7212-2039-audit\audit.py"
```

— Sol, revised in the 2026-10-16 10:30 session

A111. CD-28 8698 (TIC 20379300)

status
: confirmed (added after submission from the corrected-null re-run; K subgiant + M dwarf, primary 4.9-5.1% near periastron, secondary a 0.5-0.6% total occultation 17-18 h; e, omega within 1 sigma of Gaia's)

P_days
: 32.60569

eP
: 0.00003

T0_primary_BTJD
: 3759.1644

depth_primary
: 4.9-5.1

depth_secondary
: 0.50-0.62

duration_h
: 7.9

source_path
: Fable 5.1\made\appointments\vet20379300\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 20379300 (CD-28 8698, TYC 6661-930-1), adversarial check, 2026-10-17

Verdict: CONFIRMED. It is an eccentric eclipsing binary on the target: a K-type subgiant with an M-dwarf companion, P = 32.60569 d. Both kinds of eclipse are present and on target, and both follow one ephemeris. Separately, the timing and the durations give the same e and ω as the Gaia SB1 orbit, within 1σ. It is not in any catalogue I checked.

Archives and web

- cats.py: ASAS-SN none; Prša+2022, IJspeert+2024 and Kostov+2025 t2/t3/t4 none; SIMBAD CD-28 8698 [SB*]. VSX returned 503 in cats.py, so I re-ran it through the VizieR cone (archive2.py; the query was checked on Algol): no VSX entry within 120". SB9, WDS and the TESS-EB catalogue (J/ApJS/258/16): none. Gaia DR3 vari_summary and vari_eclipsing_binary: none. ExoFOP: no TOI or CTOI, no spectroscopy, no imaging. A web search on the three names found nothing.
- Gaia DR3: RUWE 2.30, GSP-Phot Teff 5453 K, logg 4.0, R 2.28; FLAME R 2.39, M 1.30. No neighbour brighter than G 18.2 within 60". There is one G 11.92 star at 67.5". To fake the dips it would need a 19% dip (A) or a 2.3% dip (B).

Eclipses (raw QLP SAP; trapezoid × quadratic baseline; fits_20379300.json)

kind	sector	mid (BTJD)	depth	T14	T12	notes
A	s036	2291.9080 ± 0.0010	5.07 %	7.91 h	1.62 h	every QLP cadence flagged; TESS-SPOC SAP/PDCSAP has it unflagged at 4.9/5.0%, TARS at 5.3%
A	s090	3759.1651 ± 0.0010	4.89 %	7.91 h	1.63 h	TARS 5.1%
A	s100	4085.2209 ± 0.0010	4.32 %	7.91 h	1.56 h	egress cadences flagged; q==0-only fit gives 4.41%, 7.85 h
B	s009	1553.6547 ± 0.0020	0.54 %	17.1 h	2.8 h	30-min cadence, sits on a baseline step; lower weight
B	s036	2303.5856 ± 0.0032	0.60 %	18.1 h	3.5 h	TESS-SPOC 0.55%, TARS 0.67%
B	s063	3020.9093 ± 0.0021	0.62 %	17.7 h	3.4 h	egress falls in a short gap
B	s090	3770.8426 ± 0.0014	0.62 %	17.8 h	3.5 h	TARS 0.56%
B	s100	4096.8968 ± 0.0015	0.50 %	17.5 h	3.2 h	

- A dips are flat-bottomed with limb-darkening curvature. B dips are flat boxes about 11 h long, which means a total occultation.
- In s100 both kinds are 10–18% shallower. The A/B depth ratio stays at 7.9–8.8 in every sector, so the change is dilution and one source makes both.
- None of the times is on a listed sector-wide systematic, and none sits at a gap edge.
- Blind dip census (census.py): nothing else deeper than 0.2% except edge and steep-trend artefacts.
- Background is smooth through every event. The s090 A shows a −0.015 px shift in SAP_X/Y; the pixel-level map below shows the light is lost on the target.

Ephemeris (ephem.txt)

- A alone: P = 32.60570 ± 0.00005 d. B alone: P = 32.60568 ± 0.00003 d.
- Joint fit: T_A = 3759.1644 ± 0.0012, T_B = 3770.8411 ± 0.0011, P = 32.60569 ± 0.00003 d. $\chi^2 = 1.4$ for 5 dof; every O−C is under 2.5 min, including the flagged s036 primary (−0.7 min).
- P − P_Gaia = +0.0123 d, which is +0.93σ. Against the Gaia-predicted conjunctions, the A eclipses fall +0.95 to +1.63 d late (+1.1σ) and the B eclipses +0.3 to +1.2 d late (+0.3 to +0.7σ). Most of that drift is the period difference.
- Coverage: every predicted eclipse that falls inside the data is seen. The s009 A (1541.98) and the s063 A (3041.89) fall just outside their sectors.
- B − A separation = 11.677 d (phase 0.3581); Gaia's orbit predicts 12.07 ± 0.82 d.
- Duration ratio T14_B/T14_A = 2.25; Gaia's orbit implies 2.36 [2.10, 2.66].
- These give $e \sin\omega \approx 0.384$ and $e \cos\omega = -0.208$, so $e = 0.437$ and $\omega = 118°$. Gaia has $e = 0.446 \pm 0.053$ and $\omega = 114.9 \pm 3.8°$: −0.6σ in $e \cos\omega$, −0.4σ in $e \sin\omega$. Gaia's orbit is not contradicted.

Independence

- TESS-SPOC s036 (SAP and PDCSAP), TARS s036 and TARS s090 all show both kinds at the same depths (indep_20379300.png).
- Difference image, TESScut s090 15×15:

- A, plain difference (diff_s090A.png): 4.8% at the target pixel; deficit centroid 2" from the light centroid; −0.3% at the G 11.9 neighbour's pixel.
- The plain difference for B is dominated by a pointing-drift dipole, so I fitted each pixel separately (pixfit.py: quadratic + centroid-drift terms + a fixed-shape trapezoid).
- Per-pixel result: A depth 4.45/4.82/4.61% and B depth 0.58/0.59/0.56% in 1×1/3×3/5×5 apertures. Both eclipse maps have the shape of the target's PSF (pix_s090A.png, pix_s090B.png). The neighbour's pixel shows +0.05% at B.

Physics (physics.txt)

- Mass function f(m) = 0.0363 M⊙. For M1 = 0.95 / 1.1 / 1.25 / 1.4 M⊙ that gives m2 (i = 90°) = 0.41 / 0.44 / 0.48 / 0.51 M⊙.
- Fitting the durations (T14A, T14B, T23B, with e and ω from the timing) gives R1/a = 0.039–0.041, i = 89.5–90°, b_A ≤ 0.11 and b_B ≤ 0.26.
- The solution consistent with the depths has k = 0.21 and i ≈ 89.6°. That gives R1 = 1.9–2.1 R⊙ and R2 = 0.42–0.44 R⊙, about what a 0.45 M⊙ M dwarf should have, consistent with m2.
- Mild tension: Gaia's radius is 2.3–2.4 R⊙. A star that large would need b_A ≈ 0.6 to keep the 7.9-h primary, and then there could be no B eclipse at all. So the eclipses prefer R1 ≈ 2.0 R⊙.
- Which star is in front: A is at ν = 90° − ω, where the primary is farther from us, so the M dwarf transits the subgiant. This is near periastron, so it is the short event. B, near apastron, is the total occultation of the M dwarf.
- The B depth gives a TESS-band surface-brightness ratio J = 0.11, so T2 ≈ 3300–3500 K. Depths, durations and shapes all fit one geometry.

Next eclipses (BTJD, ±2 min)

- A: 4150.433, 4183.038, 4215.644, 4248.250, 4280.855, 4313.461, 4346.067, 4378.672
- B: 4162.109, 4194.715, 4227.321, 4259.926, 4292.532, 4325.138, 4357.743, 4390.349

What would settle it further

It is already settled as an EB. For masses and radii: - RV follow-up to tie Gaia's γ, K1 and phase to the eclipse ephemeris. Here K1 = 24.7 km/s and ω should come out near 118°. - A ground-based light curve of an A eclipse, in a bluer band, to get the limb-darkened k and b and resolve the R1 tension. - Spectroscopy to look for the M dwarf in the IR, which would make it SB2.

A112. TYC 9053-432-1 (TIC 305944592)

status
: confirmed (added after submission from the corrected-null re-run; F subgiant + 0.30 Rsun M dwarf, 0.86% transit constant at fixed aperture 2019-2026; secondary expected at 0.04-0.07%, not seen)

P_days
: 34.48723

eP
: 0.00005

T0_primary_BTJD
: 1626.1182

depth_primary
: 0.86

duration_h
: 18.7

source_path
: Fable 5.1\made\appointments\vet305944592\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 305944592 (TYC 9053-432-1, Gaia DR3 5910672149533277952): adversarial check, 2026-10-17

Verdict: CONFIRMED. It is a single-lined F subgiant with an M-dwarf companion that transits it once per orbit, at the Gaia A conjunction. P = 34.48723 ± 0.00005 d.

Archives (queried 2026-10-17)

- TAPVizieR returned HTTP 503 (so `cats.py` gave "VSX: error"). I repeated the queries as VizieR asu-tsv cones of 60": VSX found only Gaia DR3 5910672179592197504 [DSCT|GDOR], which is 10.7" away at G 17.69. That star cannot produce either the 0.86% dips or the 0.2% pulsations, so the VSX entry is not prior art. SB9, WDS, ASAS-SN II/366, Prša+2022, IJspeert+2024 OBAF-EB, Kostov+2025 t2/t3/t4 and Gaia I/358 veb all returned nothing.
- Gaia DR3 vari_summary, vari_eclipsing_binary and vari_classifier_result have no rows. SIMBAD has TYC 9053-432-1 [SB*] at 0.0", which is the Gaia orbit only. ExoFOP has no TOI, no CTOI and no spectra. I did not run a web search beyond these catalogues.
- Gaia AP: Teff 6608 K, logg 3.62, R_FLAME 3.25 Rsun, M_FLAME 1.85 Msun, L 18 Lsun. TIC gives R 3.27 Rsun and M 1.46 Msun. RUWE 0.95.

Eclipses

I fitted SAP_FLUX from QLP with a trapezoid times a quadratic baseline. The star pulsates coherently at 12.00, 13.22, 10.65, 9.75, 10.96 and 9.42 c/d, with the same set of modes in every sector from 2019 to 2026. I removed 12 sine terms per sector (prewhitening), which cuts the rms from 0.26% to about 0.1%. Errors come from a prayer-bead (cyclic residual-shift) analysis.

sector	t0 (BTJD)	depth (QLP pw)	own 3x3 TESScut depth	T14	T12
s012 (partial, egress only, all flagged)	1626.113 ± 0.008	(shape fixed)	–	–	–
s013	1660.6024 ± 0.0037	0.953%	0.866%	19.2 h	3.2 h
s039	2384.8401 ± 0.0026	0.872%	0.862%	17.7 h	0.9 h
s100 (partial, clean q=0 egress)	4074.714 ± 0.005	(shape fixed)	–	–	–
s101	4109.1966 ± 0.0027	0.789%	0.859%	18.4 h	2.3 h
s102	4143.6876 ± 0.0036	0.744%	0.859%	18.2 h	2.0 h

- **Two more events than the classifier reported:** partial eclipses at s012 1626.11 and s100 4074.71, each with egress at the predicted time.
- **Detrending and flags:** fits on raw q=0 data, on all cadences, and on prewhitened data agree to within 0.007 d in t0. In s013 there are 10 flagged cadences inside the event (9 with q=4096 and 1 with q=4132, a background spike). Dropping the spike and including the rest gives 0.918% and t0 1660.6014. The flags make no difference.
- **Depth is constant:** the fall seen in QLP (0.95 → 0.74%) comes from QLP's aperture and background handling, not the star. In my own fixed 3x3 aperture the four depths are 0.866, 0.862, 0.859 and 0.859%, constant to 1% between 2019 and 2026. Durations are 18.3–19.8 h, so there is no sign of the impact parameter changing.
- **Other pipelines agree:** TESS-SPOC s013 gives SAP 0.86% / PDCSAP 0.92%, T14 19 h. TESS-SPOC s039 gives 0.85% / 0.87%, 18.5 h. Eleanor-lite s013 gives 0.87–0.93%, and eleanor-lite s012 recovers the partial event at 1626.14.
- **Difference images** (TESScut, all four full events): the largest deficit is on the target pixel every time, and the deficit centroid sits 3–7" (under 0.35 px) from the light centroid. The only neighbour bright enough to matter is G 12.34 at 53"; it would need a ≥5.7% eclipse, and its own pixel shows −0.1% to +0.2%. All other neighbours are G ≥ 14.7 and would need eclipses deeper than about 75%.
- **Census:** with 6, 12 and 19-h boxes in every sector, the four eclipses are the only dips deeper than 0.3%.

Ephemeris

- T0 = BTJD 1626.1182 ± 0.0026, P = 34.48723 ± 0.00005 d. The fit uses all 6 events: $\chi^2 = 3.3$ for 4 degrees of freedom, every |O−C| under 7 min. The cycle count is unique because s101 and s102 hold consecutive eclipses.
- Compared with Gaia: P_Gaia = 34.5029 ± 0.0432 d, a difference of −0.36σ.
- Projected back to the Gaia epoch, the ephemeris puts an eclipse at 384.578. Gaia's A conjunction there is 384.40 ± 0.96 (+0.19σ). Gaia's B conjunction is 19 d away. In the 2019–2026 windows the events fall 0.17–0.29σ early.
- Only one eclipse kind is seen, so $e \cos \omega$ and $e \sin \omega$ cannot be measured from the light curve. Gaia gives $e \cos \omega = -0.081$ and $e \sin \omega = -0.246$.

Physics

- Mass function f(m) = 0.00525 Msun, so m2(i = 90°) = 0.248 Msun for M1 = 1.46, or 0.289 Msun for M1 = 1.85.
- a = 53–57 Rsun.
- From the depth, k = 0.093, which gives **R2 = 0.30 Rsun. That is the main-sequence radius of a 0.25–0.29 Msun M dwarf**, so the mass from the RVs and the radius from the eclipse agree.
- The A conjunction (companion in front) falls near apastron: separation 66–71 Rsun, sky speed 7.6–8.2 Rsun/d. The **predicted central duration is 20.8–22.6 h**. The observed 18.7 h implies b = 0.48–0.62 and i = 88.3–88.7°. The T12/T14 ratio gives b ≈ 0.36 independently, roughly consistent. The predicted 19-h duration fits.
- **The secondary (B) eclipse is geometrically required** (b_B ≈ 0.3, T14 ≈ 12–13 h) but should be only about 0.04–0.07% deep (my crude M-dwarf surface-brightness estimate). It falls at A + 15.54 ± 1.19 d. Nothing is detected. Stacking every sector in a 12.5-h box gives at most 0.034% (at A + 15.21 d), and noise peaks of the same height appear elsewhere, so the upper limit is about 0.035%. This is mild tension at most. The classifier's "no B at 0.3%" is fully expected.

Next eclipses (BTJD)

- A (0.86%, 18.7 h): 4350.609, 4385.096, 4419.584, 4454.071 (±0.005)
- B (≲0.05%, about 12.5 h): 4331.7, 4366.2, 4400.6, 4435.1 (±1.2)

What would settle it further

- A spectrum at quadrature, to check the K1 and γ of the RV orbit.
- A stacked TESS secondary at the 0.03% level, which would also measure e cos ω (from the timing) and the M-dwarf temperature.
- Ground photometry of the A eclipse on 4350.61, which is above the ~0.3% level that ground photometry reaches.

A113. TYC 4251-1144-1 (TIC 352914772)

status
: confirmed (added after submission from the corrected-null re-run; Gaia's 37.6646-d circular SB2 period is the 63-d scan alias of the true 23.637 d; eccentric, secondary at phase 0.681; curved O-C, T0 from cycles >= 70, allow 10 min)

P_days
: 23.63714

eP
: 0.00001

T0_primary_BTJD
: 1722.60576

depth_primary
: 14.5-15.4

depth_secondary
: 1.8-2.2

duration_h
: 8.4

source_path
: Fable 5.1\made\appointments\vet352914772\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 352914772 (TYC 4251-1144-1) -- adversarial check, 2026-10-17

Verdict: CONFIRMED. A detached, eccentric eclipsing binary at **P = 23.6371 d**, with both eclipse kinds on the target. Gaia's SB2C period of 37.665 d is wrong.

Archives

Source	Query	Result
`cats.py --r 60`	VSX, ASAS-SN, EB catalogues, SIMBAD	VSX TAP returned 503, so I redid it via VizieR asu-tsv: 0 rows within 60". ASAS-SN II/366: none. Prša+2022, IJspeert+2024, Kostov t2/t3/t4: none within 5". SIMBAD: TYC 4251-1144-1 [SB*].
VizieR	SB9, WDS (30")	none
Gaia DR3	vari_summary, vari_eclipsing_binary	no rows; phot_variable_flag NOT_AVAILABLE
ExoFOP	TOI / CTOI	none; TIC contamination ratio 0.094
Web	TYC and TIC names	nothing

It is not previously known as an eclipser. Its MAST light curves are QLP, TESS-SPOC FFI and TASOC (s15), over 13 sectors (s15–17, 24, 56–58, 76–78, 83–85).

Gaia NSS SB2C: - P 37.6646 ± 0.0154 d, circular - K1 38.42 ± 0.69 km/s, K2 42.37 ± 3.22 km/s - 20 RV transits; goodness_of_fit 25.6 (a poor fit); RUWE 1.35

Eclipses

I fitted raw QLP SAP_FLUX (q==0, no trimming) with a trapezoid times a quadratic baseline (`ecl.py`, `ecl.json`, `ecl_P.png`, `ecl_S.png`).

Primaries. 13 fitted, plus s15 E0 at 1722.59, which is entirely flagged but plainly present in the flagged cadences (`fold.png`). - Depth 14.5–15.4%; T14 8.38–8.41 h; T12 3.45 h; flat bottom 1.45–1.50 h. - Mid-time errors are 0.1–0.5 min. - Within s57 there are two primaries 23.637 d apart with equal depths (15.14% and 15.19%). This alone rules out a 37.665-d period.

Secondaries. 13 fitted, plus s58 E49 at 2896.92, which is flagged but present. - Depth 1.84–2.20%; T14 8.8–9.4 h.

Other checks. - Background and centroids show no event-like excursion beyond the ≤10-mpx shifts expected from a 15% dimming of the target with neighbours in the aperture. - Blind census of all 13 sectors (1.5-d running median, 1-h bins, threshold 0.3%; `census.py`): no other dips.

Ephemeris (`ephem.py`, `ephem.json`, `oc.png`)

	T0 (BTJD)	P (d)	χ^2 / dof
P-kind	1722.59629 (143)	23.637271 (22)	1999 / 11
S-kind	1738.69628 (62)	23.637135 (11)	35.7 / 11

- Mean secondary phase: 0.68113.

Gaia's period. Gaia's 37.6646 d is **14.03 d = 913 σ** away. 1/23.6373 − 1/37.6646 = 1/63.5 d, which matches Gaia's 63-d scan-precession alias. No n/m ratio rescues it: 3/2 × P = 35.456 d is still 144 σ off. Folding at 37.6646 d, or at ±3 σ from it, scatters the eclipses over about 10 phases (`fold.png`).

The O-C is curved. - Primaries: −6.8 min (2019) → +2.0 (2022) → 0 (2024) → −1.6 (2025). A quadratic fit gives χ^2 10.6. - The secondary-minus-primary separation within a cycle falls by 14 min over 6 years: 16.1040 → 16.0930 → 16.0896 → 16.090 d. - TESS-SPOC PDCSAP s16/17/24 reproduce the 2019 times to ≤1.1 min, so this is in the data, not a QLP artefact. - GR plus tidal apsidal motion would give ~0.01 min over this baseline. If the signal is real it points to a third body. This is a side finding and is not needed for the verdict.

Independence

Second pipeline. TESS-SPOC FFI (s16, 17, 24, 57, 84): - Mid-times agree with QLP to 0.0–0.1 min (200-s primaries) and ≤1.5 min (secondaries). - PDCSAP depths are 14.4–15.4% and 1.9–2.1%, with the same durations.

Difference images. TESScut s57, 15×15 (`diff_s57P`, `diff_s57P2`, `diff_s57S`), covering both s57 primaries and the s57 secondary: - The deficit centroid lies 2.2–2.4" from the out-of-eclipse light centroid, i.e. 0.1–0.2 px from the target's Gaia position. - The 3×3 box on the target holds 0.80–0.81 of the deficit, matching its 0.79–0.80 share of the light. - At the pixel of the G 13.11 neighbour (31") the fractional depth is 6% against 14% on the target (0.8% against 1.8% for the secondary).

Neighbour dilution (Gaia, G < 16 within 90"):

Neighbour	Separation	Primary would need	Secondary would need	Status
G 15.82	9.4"	143× its depth	315%	impossible
G 13.11	31.4"	180%	26%	primary impossible; secondary excluded by the difference image

Physics

My own EB model (`ebfit.py`): two limb-darkened spheres on a Kepler orbit, fitted to the 2022 200-s eclipses of both kinds. Residual rms 0.66 ppt, with no structure (`ebfit.png`).

Quantity	Value
e cos ω	0.2801 ± 0.0006
e sin ω	0.226 ± 0.009
e	0.360
i	86.73°
R1/a, R2/a	0.0512, 0.0258 (R2/R1 = 0.50)
Surface-brightness ratio J	0.63
L2/L1 (TESS band)	0.16
Periastron	BTJD 3611.986 + n × 23.63714

- **Primary eclipse:** the smaller, cooler star transits the larger star near periastron (r = 0.71a, b = 0.79 R1).
- **Secondary eclipse:** a grazing occultation of the small star near apastron (r = 1.12a).
- For a total mass of 2.2–3.2 Msun: a = 45–51 Rsun, R1 = 2.3–2.6 Rsun (an evolved primary; TIC gives R 3.5 and Gaia logg 3.54), R2 = 1.2–1.3 Rsun.
- **Gaia's circular solution is excluded:** e must be at least e cos ω = 0.280.

Gaia's K's at the true orbit (P 23.637 d, e 0.36): - M1 $\sin^3 i$ = 0.59 [0.51, 0.69] Msun, M2 $\sin^3 i$ = 0.54 [0.49, 0.59] Msun, so M_total ≈ 1.13 Msun at i = 86.7°. - That is impossible for a G 10.43 star at 509 pc (M_G ≈ +1.5 to +1.9). - Plausible masses (1.5 + 1.2–1.35 Msun) imply K1, K2 of about 52 and 58 km/s at 23.64 d. - So Gaia's K's are about 70–75% of the true values, which is what an RV orbit fitted at an alias period produces. The published SB2C orbit does not fit a 23.64-d eccentric orbit as it stands. That counts against the Gaia solution, not against the eclipses. - The mass ratio K1/K2 = 0.91 ± 0.07 and the light ratio of 0.16 (subgiant plus MS companion) are mutually plausible.

The orbit should be: P 23.63714 d, e 0.36, periastron BTJD 3611.986, i 86.7°.

Next eclipses (BTJD)

These come from a linear fit to E ≥ 70 only, because the O-C is curved. Allow ±10 min for the ETV.

Kind	Ephemeris	Next times
P-kind	1722.60576 + 23.637138 E	3660.8511, 3684.4882, 3708.1254, 3731.7625, 3755.3996, 3779.0368
S-kind	1738.69540 + 23.637137 E	3653.3035, 3676.9406, 3700.5778, 3724.2149, 3747.8520, 3771.4892

What would settle it further

- Ground-based RVs over one 23.6-d cycle would replace the alias SB2C solution and give true masses: with i known, K1 and K2 of roughly 52/58 km/s should come out.
- Continued eclipse timing would test the curved O-C and the shrinking primary–secondary separation (third body?).

A114. TYC 6528-895-1 (TIC 65660608)

status
: confirmed (added after submission from the corrected-null re-run; three identical grazing eclipses 2021-2025, impact 0.7 (R1+R2); no secondary possible at Gaia's e sin omega; Gaia's K1, K2 about 14% low for the inclination, a third light suspected, RUWE 38)

P_days
: 41.56002

eP
: 0.00006

T0_primary_BTJD
: 2230.72319

depth_primary
: 9.3-9.5

duration_h
: 4.2

source_path
: Fable 5.1\made\appointments\vet65660608\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 65660608 (TYC 6528-895-1), adversarial check, 2026-10-17

Verdict: CONFIRMED. It is an eclipsing binary on the target: - Three grazing eclipses with identical shapes, all on one exact linear ephemeris, P = 41.56002 d. - The eclipses sit at the A conjunction of Gaia's SB2 orbit. - The other kind of eclipse cannot occur at any stellar size, given Gaia's e and ω.

Open problem: with Gaia's K1 and K2, an eclipsing inclination makes the two stars 0.78 and 0.85 M⊙. The colour (about 6100 K) says roughly 1.15 M⊙ each. Gaia's K amplitudes would have to be about 14% low (about 3.5σ), possibly because a third star's light is mixed into the spectrum (RUWE 38). The catalogue checks below found no prior eclipsing-binary classification.

Archives and web

- cats.py: ASAS-SN none; EB tables none; SIMBAD TYC 6528-895-1 [SB*]. VSX returned 503 there; through the VizieR cone it has nothing at the target. Within 120" VSX has only four faint Gaia variables, at 104–118" and G 14.0–17.7 (two EAs, a DSCT|GDOR, an RS).
- SB9, WDS, TESS-EB catalogue, Gaia vari_summary and vari_eclipsing_binary: none. ExoFOP: no TOI or CTOI, no spectra, no imaging.
- A web search on TYC 6528-895-1, TIC 65660608 and 2MASS J07134548-2606443 found nothing.

- Gaia DR3: RUWE 38.4, parallax 7.28 ± 0.56 mas (probably corrupted), bp_rp 0.71, GSP-Phot Teff 6136 K, logg 4.37.
- A Gaia source 3.1" away has no G magnitude: RP 15.68, parallax 0.01 mas, so it is a background star. To fake a 9% dip it would need a >100% dip, as would every neighbour within 90" (G 13.8–16).

Eclipses (raw QLP SAP, q==0; fits_poly.json, graze_k1.0_J1.0.json)

sector	mid (BTJD)	QLP depth (quadratic / cubic baseline)	own TESScut 5×5 depth	trapezoid T14	T12
s034	2230.72321 ± 0.0010	9.22 / 9.37%	9.43 ± 0.07%	4.18 h	1.67 h
s061	2978.80354 ± 0.0010	9.09 / 9.02%	9.30 ± 0.03%	4.19 h	1.68 h
s087	3685.32395 ± 0.0010	8.62 / 8.55%	9.45 ± 0.02%	4.17 h	1.65 h

- Shape: nearly V, flat part 0.84 h. Contact-to-contact length in the physical model is 4.5 h.
- Depth changes: in my own fixed-aperture TESScut photometry the depth is constant, 9.3–9.45% (3×3: 9.45/9.23/9.38%). So the QLP decline (the classifier's 10.1 → 8.3%) is QLP dilution or background, not a real change.
- It is not the detrending against the pulsation: quadratic and cubic baselines agree to 0.15%, and the mid-times agree to 3e-5 d.
- Durations are constant to about 1 min. A real inclination drift big enough for the QLP depth change would have shortened T14 by about 4 min.
- TESS-SPOC s061 shows the eclipse at 8.6% (SAP) and 8.9% (PDCSAP), crude in/out estimate.
- None of the three times is on a listed systematic, and all are in clean data away from gaps.
- Blind dip census: only these three. The next-strongest dips, 0.6–0.7%, are pulsation minima or momentum-dump gaps (zoom_census.png).

Ephemeris (ephem.txt)

- T0 = 2230.72319, P = 41.56002 ± 0.00006 d. O–C is 0.0, 0.0, 0.0 min (cycles 0, 18, 35).
- The cycle count is unique: the other counts give 39.4 or 44.0 d, against Gaia's 41.69 ± 0.09. Because cycle 35 is odd, the true period cannot be 83.1 d.
- P − P_Gaia = −0.130 d, which is −1.43σ.
- Conjunction kind, with Gaia's T_peri, e and ω drawn and P fixed at 41.560:
- A (ν = 90° − ω = 21°, 0.8 d after periastron) is predicted at 2231.18 ± 1.02; the observed eclipse is −0.45σ from it.
- B is +15.3 d away, which is +4.3σ.
- So these are A eclipses: Gaia component 1 (K1 = 42.8 km/s, the lower-mass one) is behind, and component 2 is in front.

Independence

- TESS-SPOC s061 light curve: covered above.
- Per-pixel eclipse fits on TESScut s034, s061 and s087: in all three the eclipse-amplitude map has the shape of the target's PSF, and its centroid is 2–5" from the light centroid (pix_s0*E.png).
- The 1.2–2 d variability is also on the target. Its pixel map follows the target PSF (amplitude/light 1.04 / 1.00 / 0.97 in 1×1/3×3/5×5 in s087; var_s0*.png), so it is not the VSX GDOR star at 118".

Physics (graze.py; graze_k1.txt)

Orbit. The Gaia SB2 orbit (K1 = 42.8 ± 2.2, K2 = 39.2 ± 2.3 km/s, e = 0.471, ω = 68.9°) gives: - $M1 \sin^3 i$ = 0.78 [0.66, 0.90] M⊙ and $M2 \sin^3 i$ = 0.85 [0.73, 0.97] M⊙; q = M2/M1 = 1.09. - a sin i = 59.4 R⊙. The separation is 32.1 R⊙ at A and 82.4 R⊙ at B (ratio 2.57).

Grazing fit. Two equal limb-darkened disks (k = 1, J = 1) fit each event with an impact of 0.69–0.70 of (R1 + R2). With Gaia's a sin i this gives R1 = R2 = 0.95 R⊙ and i = 87.6°. An eclipse at A needs i > 86.6°. So a 4.5-h, 9% V-shaped graze fits two known-mass stars at this period.

Mass tension. At i ≈ 88° the masses are 0.78 and 0.85 M⊙, which is too low for 6100 K stars. Stars of about 1.15 M⊙ would need $\sin^3 i$ = 0.68 (i ≈ 61°), and then there could be no eclipse.

The consistent picture is Gaia's K1 + K2 being about 14% low. That would give masses of about 1.15/1.26 M⊙ and radii of about 1.08 R⊙, both plausible. A third star's light diluting the RVS lines (and the eclipses), suggested by RUWE 38, is the likely cause. It needs checking. It is not a reason to doubt the eclipse.

The missing other-kind eclipse. It is not shallow; it cannot happen, and the argument does not depend on the stars' sizes: - $b_B = b_A \times r_B/r_A = 0.69 \times 2.57 = 1.77$ (in units of R1 + R2), which is greater than 1. - A B eclipse would need $r_B/r_A < 1.45$, i.e. e sin ω < 0.18, against Gaia's 0.44 ± 0.045 (5.7σ).

The B windows are consistent with this. The data at 1508.35 (s007), 2962.95 (s061, sector start), 3669.47 (s087) and 3711.03 (s088), each ±1.65 d, show only the pulsation and momentum-dump gaps. The 2214.87 window (s033) straddles the orbit gap.

Next eclipses (BTJD)

- A, ±4–5 min: 4142.484, 4184.044, 4225.604, 4267.164, 4308.724, 4350.284, 4391.844, 4433.404, 4474.964, 4516.524.
- B: no eclipse expected. The conjunctions are at A + 25.70 ± 1.65 d: 4168.19, 4209.75, 4251.31, 4292.87, 4334.43.

What would settle it further

- High-resolution RV spectra, two epochs near quadrature, to re-measure K1 and K2. This resolves the mass tension and tests for a third set of lines (a tertiary, RUWE 38).
- A ground-based light curve of one A eclipse, for example at 4142.48 or 4184.04, to refine the impact parameter.
- Gaia DR4 epoch astrometry and RVs, which will show whether a tertiary drives the RUWE.

A115. TYC 9038-1390-1 (TIC 447307865)

status
: confirmed (added after submission from the corrected-null re-run; Gaia SB2 near twins, five grazing 1.3% eclipses 2019-2026 at one conjunction; the other conjunction's minimum separation exceeds R1+R2, no eclipse expected)

P_days
: 40.01697

eP
: 0.00002

T0_primary_BTJD
: 1628.6578

depth_primary
: 1.3

duration_h
: 5.4

source_path
: Fable 5.1\made\appointments\vet447307865\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 447307865 (TYC 9038-1390-1, Gaia DR3 5830628390235398272, SB2): adversarial check, 2026-10-17

Verdict: CONFIRMED. This is a grazing eclipse between the two near-twin stars of the Gaia SB2, seen at the A conjunction. P = 40.01697 ± 0.00002 d. There are **five** eclipses, not three: s012 and s039 were missed by the classifier.

Archives (queried 2026-10-17)

- TAPVizieR returned HTTP 503, so I used VizieR asu-tsv cones of 60". VSX has only Gaia DR3 5830628390235409792 [E, P 0.327 d] at 36.8". That star is G 18.78, so it cannot produce a 1.2% dip.
- SB9, WDS, ASAS-SN, Prša+2022, IJspeert+2024, Kostov+2025 t2/t3/t4: none. The target has no rows in Gaia's vari tables. ExoFOP has no TOI and no CTOI. SIMBAD has TYC 9038-1390-1 [SB*], which is the Gaia orbit only. I did not run a web search beyond these catalogues.
- Gaia AP, fitted as if single: Teff 5707 K, R 2.10 Rsun, L 4.24 Lsun.
- Neighbours: G 13.02 at 36.7" would need a ≥15% eclipse and G 13.13 at 52.6" ≥17%. The rest within 60" are G ≥ 14.2 (≥40%) or the G 16.9–17.4 pair at 7" (>100%).

Eclipses

I fitted QLP SAP_FLUX with a trapezoid times a quadratic baseline, with prayer-bead errors. This star does not pulsate.

sector	t0 (BTJD)	QLP depth	own 3x3 TESScut depth	T14	T12	notes
s012	1628.6571 ± 0.0015	1.23% (30-min)	1.34%	5.3 h	1.9 h	q=0, new
s039	2388.9805 ± 0.0009	1.33%	1.28%	5.3 h	1.8 h	every cadence q=4096; new
s066	3109.2857 ± 0.0012	1.38%	1.27%	5.3 h	2.0 h	dip cadences q=0
s101	4109.7093 ± 0.0005	1.24%	1.28%	5.4 h	2.1 h	clean
s102	4149.7270 ± 0.0007	1.05%	1.27%	5.4 h	2.1 h	clean

- **The s066 event is real.** I took 19 unrelated stars on the same camera 2 / CCD 4 at all cadences: their median shows a +0.001% deficit at 3109.28, and none dips by more than 0.3%. The cadences inside the dip are q=0; the flagged stretch (bit 30) starts around 3109.5, after egress. The fit is identical with or without the flagged cadences. TARS photometry recovers the event (1.32%), and so does my own TESScut photometry (1.27%).
- **The s039 event is real** even though every cadence is flagged: 15 same-CCD stars are clean at that time, and TARS gives 1.45%.
- **The s093 dip at 3849.85 is a sector systematic.** It is a 0.4% pre-gap drop, and 12 of 21 unrelated stars on the same CCD show it too (median 0.36%).
- **Depth is constant:** 1.27–1.28% in my own aperture in every 2021–2026 sector. The QLP s102 value of 1.05% is a QLP artefact. A grazing eclipse is very sensitive to inclination, so this rules out any measurable inclination drift.
- **Census:** the primaries are the only dips deeper than 0.3% in any sector.
- **Difference images** (TESScut s039/066/101/102): the largest deficit is on the target pixel every time, and the deficit centroid sits 1.3–4.4" from the light centroid. The G 13.02 star on the adjacent pixel shows 0.3–0.9%, which is the wing of the target's PSF; it would need 15%.

Ephemeris

- Five events give T0 = 1628.6578 ± 0.0009 and P = 40.01697 ± 0.00002 d, with $\chi^2 = 1.0$ for 3 degrees of freedom and every $|O-C| \leq 1.1$ min.
- **The two clean events alone fix the period:** s101 to s102 is one cycle, P = 40.0177 ± 0.0010. s066 then falls exactly 25.000 cycles before s101, so the cycle count is unambiguous.
- Compared with Gaia: P = 39.9995 ± 0.0261 d, a difference of +0.67σ. The events fall +0.75 to +0.80σ after the A conjunction. At the Gaia epoch the ephemeris gives 388.108 against Gaia's A conjunction at 387.74 ± 0.60 (+0.6σ). Gaia's B conjunction is 24 d away.
- Only one eclipse kind is seen, so e cos ω and e sin ω cannot be measured from the light curve. Gaia gives e sin ω = 0.065.

Physics

- M1 $\sin^3 i$ = 1.041 Msun and M2 $\sin^3 i$ = 1.062 Msun, so the stars are near twins (q = 0.98). a sin i = 63.1 Rsun.
- At A ($\nu = 90° - \omega$), component 1 is behind and is eclipsed by component 2. The separation is 56.9 Rsun and the sky speed 10.76 Rsun/d. **The central duration for R1 + R2 ≈ 3.0 Rsun would be 13.4 h; the observed 5.35 h therefore means the eclipse is grazing.**

- I fitted the s101 event with two limb-darkened disks, assuming equal stars and equal light and fixing the sky speed from the Gaia orbit. The best fit is **R = 1.6 Rsun each at b = 0.91** (b being the minimum separation as a fraction of R1 + R2), with residuals at the noise level. R = 1.5 is worse by $\Delta\chi^2$ 36 and R = 1.7 by $\Delta\chi^2$ 107. Two 1.5–1.6 Rsun stars at 5700 K reproduce Gaia's combined L = 4.2 and R_single = 2.1.
- **So 1.2–1.3% in 5 h fits a grazing eclipse of two similar stars.** A small companion transiting one of them fits worse ($\Delta\chi^2$ +87 to +107) and is ruled out anyway, because K1 ≈ K2 makes the companion a near-equal-mass star.
- **The other eclipse belongs at B**, at A + 24.75 ± 0.91 d. At B the stars are farther apart by $r_B/r_A = (1 + e \sin\omega)/(1 - e \sin\omega) = 1.135 \pm 0.056$. That puts the closest approach at 3.32 Rsun against R1 + R2 = 3.2 Rsun, so **for the nominal orbit B narrowly misses**. B would be eclipsed only if the ratio is below 1.10 (about 0.6σ), and even then very shallowly.
- **B is covered:** the window falls in s039, s093, s100, s102 and s103, and the stack covers the whole ±3σ range. Nothing is found above about 0.03% in a 4.5-h box. The only feature is the s102 4133.0 common-mode step, which lands at offset 23.17 d.

Next eclipses (BTJD)

- A (1.27%, 5.4 h, V-shaped): 4349.812, 4389.829, 4429.846, 4469.863 (±0.002)
- B (expected absent; if grazing, ≲0.03%): 4334.5, 4374.6, 4414.6 (±0.9)

What would settle it further

- High-resolution spectra that resolve both line sets, giving the light ratio and the two radii. Together with the duration and depth this pins b. A light ratio well away from 1 would force a refit with unequal stars.
- A ground-based light curve of the 4349.81 eclipse, which is 1.3% deep and 5 h long. It would also check the A-only geometry against e sin ω.

A116. TYC 3608-1307-1 (TIC 455491894)

status
: probable (added after submission from the corrected-null re-run; two 2.3% flat-bottomed eclipses in s076 and s083 exactly 3.000 Gaia periods apart, on target in difference images, 1 of 249 stars dipping at the s083 time; one eclipse kind and two events, so probable; the events are 1.57 d from Gaia's conditional conjunction phase)

P_days
: 57.3498

eP
: 0.0100

T0_primary_BTJD
: 3564.19008

depth_primary
: 2.3

duration_h
: 5.4

source_path
: Fable 5.1\made\appointments\vet455491894\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); re-analysed by an Opus-model subagent (2026-10-17)

Archived source: sanitized report text.

TIC 455491894 (TYC 3608-1307-1) -- adversarial check, 2026-10-17

Verdict: PLAUSIBLE. The two B-kind dips are real, on target, and exactly 3 Gaia periods apart, so they almost certainly belong to the SB2. Two things keep this short of CONFIRMED: - There are only two events, both of one kind. - Their phase is 1.57 d early. Once TESS pins the period, that is 5.6–13 σ off Gaia's prediction, not the classifier's −2.8 σ.

Archives

Source	Query	Result
`cats.py --r 60` and VizieR asu-tsv	VSX, ASAS-SN, SB9, WDS, Prša+2022, IJspeert+2024, Kostov t3/t4 (TAPVizieR was returning 503)	none
SIMBAD	position	TYC 3608-1307-1 [SB*]
Gaia DR3	vari_summary, vari_eclipsing_binary	no rows
ExoFOP	TOI / CTOI	none; contamination ratio 0.21
Web	TYC and TIC names	nothing
Kostov+2025 table 2 (unvetted ML list)	60" cone	flags the **neighbour** TIC 455491931 (34", Tmag 13.46, = Gaia G 14.44), not the target

Gaia NSS SB2: - P 57.3498 ± 0.0100 d, e 0.3464 ± 0.0057, ω 312.29 ± 1.14°, T_peri BTJD 414.716 ± 0.188 - K1 33.66 ± 0.33 km/s, K2 36.96 ± 0.27 km/s - 28 RV transits; goodness_of_fit 28.8 (a poor fit)

MAST light curves: QLP, TARS and TESS-SPOC FFI for s16, 56, 76, 83.

Eclipses

Raw QLP SAP_FLUX, q==0, no trimming (`ecl.py`):

Event	Mid-time (BTJD)	Depth	T14	T12	Flat bottom
s76 B	3392.13935 ± 0.7 min	2.31%	5.43 h	2.03 h	1.36 h
s83 B	3564.19008 ± 0.6 min	2.30%	5.52 h	2.21 h	1.10 h

- Neither event has a flagged cadence in eclipse. Background and centroid excursions are ≤ 1.5 mpx.
- s76 has no data gap within 2 d. The s83 gap (3564.84–3565.03) starts 0.53 d after egress, so both events have full pre- and post-baseline.
- The classifier's "2.6% V, 3.4 h" was a box estimate; the trapezoid has a short flat bottom.
- TESS-SPOC PDCSAP gives 3392.13793 (2.28%, 5.39 h) and 3564.18991 (2.36%, 5.47 h): the same events, within 2 min.

Common mode (`commonmode.py`, every star in the shared folder): - At 3564.19 the target dips 2.04% and ranks 1st of 249. The other stars' median is 0.015%; the 23 on the target's camera/CCD have median 0.04% and max 0.10%. - At the listed common-mode time 3564.45 the other stars' median is 0.05% and the target reads −0.04%, i.e. clean. - At s76 3392.14 the target ranks 1st of 227; the other stars' median is −0.006%. - So neither event is a sector systematic.

Blind census (>0.3%, 1-h bins): nothing else in s16, s56 or s83. s76 has 0.3–0.38% wiggles at 3371.4–3373.4 inside an SAP hump; there the target ranks 7th of 149, so this is common-mode-like and not eclipse-shaped.

Ephemeris

- The spacing is 172.05073 d = 3 × **57.35024 ± 0.0003 d**, which matches Gaia's P to 0.04 σ. P/2 and P/4 are not compatible with Gaia's P.
- With P fixed at the TESS value, Gaia predicts B at 3393.712 and 3565.763. Both observed events are **−1.573 d** early, by the same amount.
- **The classifier's −2.8 σ is out of date.** It included Gaia's period error × 51 cycles (0.51 d), which the TESS spacing removes. Conditioned on the TESS P, the predicted-B σ is 0.22 d with independent errors, 0.12 d with corr(T_peri, ω) = +0.926 (the only strong term in corr_vec), or 0.28 d with −0.926. That makes the offset −7.3, −13.4 or −5.6 σ.
- **A period shift within Gaia's errors cannot explain it.** It would need P = 57.319 d (−3 σ_Gaia), and the TESS spacing excludes that at about 100 σ.
- **Single-parameter fixes:** T_peri −1.573 d (−8.4 σ), or ω +17.7° (+15.5 σ, giving e cos ω 0.300 instead of 0.233), or e −0.172 (−30 σ).
- The most economical reading is that Gaia's T_peri (and perhaps ω) errors are underestimated about 3–7× (gof 28.8). Without an A eclipse the photometry cannot measure e cos ω independently.

A-kind

- Gaia's e and ω put A at B + 19.958 ± 0.233 d (phase 0.348).

- **s83 3584.147:** q==0 data, one-sided baseline; depth +0.02 ± 0.01% for T14 of 5.5–9 h.
- **s16 1748.94:** flagged cadences only; +0.04 ± 0.04%.
- **s56 2838.59:** falls in the data gap.
- At Gaia's nominal unshifted A times: s16 1750.51 is in a gap, s56 2840.17 is covered and flat, s83 3585.72 is after the sector ends.
- **Phase coverage:** 99.5% of the orbit is measurable with a 6-h box. The max box depth over phases 0.25–0.45 after B is 0.12%, and the only uncovered sliver is 0.345–0.350, at the end of s83.
- **Constraint:** no A eclipse deeper than about 0.05% if A − B ≤ 20.1 d. That covers about 74% of the Gaia A − B distribution; A − B > 20.1 d is unconstrained.
- **This absence is expected.** The conjunction separations are 0.700a at B (near periastron) and 1.183a at A, so b_A = 1.69 b_B.

Independence

Second pipeline: TESS-SPOC (above).

Difference images (TESScut s76 and s83; `diff_s76B`, `diff_s83B`): - The flux lost peaks on the target pixel. The deficit centroid is 0.16 px (s76) and 0.09 px (s83) from the target's Gaia position. - The deficit is more concentrated than the light: 0.82 of it sits in the 3×3 box, which holds only 0.64 of the light.

Neighbour dilution: fractional dip each neighbour would need for a 2.3% dip on the target.

Neighbour	Separation	Would need
G 12.87	51"	20%
G 13.73	29"	45%
G 13.81	39"	49%
G 14.40	29"	84%
G 14.44 (= the Kostov-flagged TIC 455491931)	35"	87%
everything fainter than G ≈ 14.7	—	>100%

The 20–87% cases are physically possible on dilution alone, but every one of those stars lies 1.4–2.4 px from the deficit centroid, and their pixels show lower fractional depth (0.6–1.9%) than the target pixel (2.1–2.2%). They are excluded.

Physics

Masses from Gaia's K's (P 57.350 d, e 0.346): - M1 $\sin^3 i$ = 0.904 ± 0.015 Msun, M2 $\sin^3 i$ = 0.823 ± 0.017 Msun, q = 0.910 - a sin i = 75.07 Rsun; f(M) = 0.187 Msun

Which star is in front at each conjunction: - At B ($\nu = 270° - \omega$) the Gaia primary is in front: the observed eclipse is a grazing occultation of the Gaia secondary. - At A the secondary would be in front, but the separation there is 1.69× larger.

My EB model (`ebfit.py`): Gaia e and ω fixed, star 2 = Gaia primary, four radius/light-ratio priors (R_prim/R_sec, L_prim/L_sec = 1/1, 1.15/1.5, 1.3/2, 0.9/0.7; `ebfit_con_*`). All four fit the two B eclipses equally well (rms 0.71 ppt): - i 87.4–87.6° - R1 + R2 = 0.035–0.036a = 2.6–2.7 Rsun, i.e. about 1.35 Rsun each - b_B = 0.86–0.87, so b_A = 1.45–1.48 (in units of R1 + R2): **no A eclipse is predicted under any of them.** This is consistent with what the data show.

Luminosity check: two stars of about 1.35 Rsun and 5700 K give M_G ≈ 3.5, against the observed 3.2–3.4 (parallax 3.626 mas). TIC's single-star R of 1.96 is √2 × 1.39. So at i ≈ 87.5° the masses of about 0.90 + 0.82 Msun are consistent with slightly evolved, old G stars.

Next eclipses (BTJD)

Kind	Times	Uncertainty
B	3621.5403, 3678.8906, 3736.2408, 3793.5910, 3850.9413 (3564.19008 + 57.35024 n)	±0.4–2.2 min
A (only if Gaia e, ω are right; none is expected geometrically)	3641.498, 3698.849, 3756.199, 3813.549, 3870.899	±0.23 d

What would settle it

- A third B eclipse at a predicted time (e.g. 3621.540, which ground photometry could catch: 2.3% deep, 5.5 h) would confirm the period and the phase.
- Ground RVs at a few phases would fix T_peri and ω and test the −1.57 d offset directly. The offset predicts periastron at BTJD 3567.41 + 57.35024 n (Gaia T_peri 414.716 − 1.573, carried forward 55 cycles), 3.22 d after the s83 B eclipse.

A117. TYC 629-1389-1 (TIC 404715740)

status
: confirmed (added after submission from the corrected-null re-run; Gaia SB1 period 3.5 sigma long, 87 min per orbit; 21.5% primaries, 3.3% secondary at phase 0.473; e cos omega and e sin omega agree with Gaia's)

P_days
: 36.54015

eP
: 0.00003

T0_primary_BTJD
: 2492.88004

depth_primary
: 21.5

depth_secondary
: 3.3

duration_h
: 6.9

source_path
: Fable 5.1\made\appointments\vet404715740\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by an Opus-model re-analysis (2026-10-17)

Archived source: sanitized report text.

TIC 404715740 (TYC 629-1389-1): adversarial check report

Opus 5.5 (subagent of Fable 5.1's session), 2026-10-17. Scripts, outputs and figures are in this folder. Downloads are in `C:\Home\sky\data\qlp-sb-p20-60\vet20261017\404715740\`.

Verdict: CONFIRMED (new eclipsing SB1; the Gaia period is 3.5 sigma too long)

Archives (what was queried and what came back)

- cats.py --r 60: the VSX TAP returned HTTP 503 twice, so VSX is unchecked. ASAS-SN: none. SIMBAD: TYC 629-1389-1 [SB*] only.
- VizieR cones (archives_out.txt):
- WDS 60": nothing
- Hipparcos main, new reduction, variability annexes: nothing
- SB9: nothing

- Gaia DR3 vari_classifier, vari_eclipsing_binary: nothing
- CCDM: nothing
- Gaia DR3 within 60": the target, plus two stars at G 18.7 and 18.8 (39-41").
- ExoFOP JSON: no TOI, no CTOI, no files, imaging or spectroscopy. The TOI list has no row.
- Web search for "TYC 629-1389-1": no pages about the star.
- MAST (mast_inventory.txt): QLP, TESS-SPOC and TARS for s42, s43, s70 and s71. SPOC also has a 2-min light curve and target pixel file for s42.

Eclipses (trapezoid + quadratic baseline, prayer-bead errors; trap.json, trap.png, zoom_events.png)

kind	sector / product	mid-time BTJD	depth	T14	flat bottom	ingress
A primary	s42 SPOC 2-min SAP (bit 4096 only)	2456.3406 +- 0.0007	20.97%	6.90 h	2.1 h	2.4 h
A primary	s43 QLP (TESS-SPOC)	2492.87984 +- 0.00013 (2492.87952)	21.36% (21.07%)	6.96 h	2.0 h	2.46 h
A primary	s70 QLP (TESS-SPOC)	3223.68303 +- 0.00010 (same)	21.55% (21.01%)	6.95 h	2.1 h	2.43 h
B secondary	s71 QLP (TESS-SPOC)	3240.97563 +- 0.00054 (3240.97524)	3.26% (3.18%)	9.23 h	2.55 h	3.34 h

- QLP flags the s42 primary (scattered light). SPOC 2-min and TESS-SPOC both record it cleanly, carrying only quality bit 4096. That makes it a third primary.
- Background and centroids are flat through all four events. TARS shows them too.
- Blind dip census (census.txt; QLP and TESS-SPOC q==0, plus SPOC 2-min): nothing else deeper than 0.2%, apart from two things that are not eclipses:
- TESS-SPOC s71 3253.09 (0.31%) is common-mode: 81% of 102 s71 stars dip by more than 1000 ppm there (s6_common.py, all cadences).
- The s42 2-min 4096-flagged scattered-light stretches.
- With this ephemeris, no other eclipse of either kind falls in the data:
- The secondary at 2473.63 falls in the s42/s43 gap.
- The secondary at 3204.43 falls before s70 starts.
- The primary at 3260.22 falls after s71 ends.

Ephemeris (ephem_out.txt)

- Primary: P = 36.54015 +- 0.00003 d, T0 = 2492.88004 +- 0.00038 BTJD (N = -1, 0, 20). O-C is +1.1 / -0.3 / 0.0 min.
- The consecutive s42 to s43 pair alone gives 36.539 d. Counting 19 or 21 cycles across the 731-d gap would give 38.46 or 34.80 d instead, so the cycle count is unambiguous.
- Gaia P = 36.6007 +- 0.0172 d, so the difference is -0.0605 d, or -3.5 sigma.
- Running Gaia's t_peri, e and w with the eclipse period predicts A 0.38 d late (0.5 sigma of t_peri) and B 0.08 d late.
- So the consistent 4-5 d earliness is entirely a Gaia period error, built up over 58-78 cycles from the Gaia epoch. It is not a T_peri error and not a wrong e cos w.
- Secondary phase is 0.47325, i.e. A to B is 17.293 d. Gaia predicts 17.02 +- 0.37 d, so the observed spacing is +0.7 sigma.

- Exact Kepler gives e cos w = -0.0416 (Gaia -0.053 +- 0.016).
- The duration ratio gives e sin w = 0.142 to first order (Gaia 0.140).

Independence and position

- TESS-SPOC, SPOC 2-min (s42) and TARS all show the eclipses at the same times and depths.
- TESScut difference images (diff_*s43/s70, difflin*_s71) put the maximum on the target pixel:
- primary 20.5% (s43) and 20.3% (s70) per-pixel depth at the target;
- secondary 3.25% at the target pixel (per-pixel linear baseline across the pre-ingress gap).
- Neighbours: only the G 18.7/18.8 stars at 40". Either would need to dim by 21% x 10^(0.4 x 8.9), far more than 100%, so they are excluded.

Physics (s8_model.py; model.png, model_out.txt)

- Joint Keplerian two-sphere model with limb darkening, fitted to the s43 and s70 primaries and the s71 secondary. P is fixed. chi2 = 600 for 1171 points.
- Fit results:
- r1 = R1/a = 0.02264, r2 = 0.01029, k = 0.455
- i = 89.39 deg; impact parameter b_A = 0.40, b_B = 0.54 (the occultation is just total)
- central surface-brightness ratio J = 0.174
- e cos w = -0.0416, e sin w = +0.148, so e = 0.153 and w = 105.7 deg (Gaia 0.150 +- 0.011, 110.7 +- 6.4)
- Fixing e sin w at Gaia's 0.140 costs only delta chi2 = 19 and changes nothing else.
- Mass function f(m) = 0.0605 Msun. With M1 = 1.25 Msun, m2 = 0.589 Msun (0.55-0.63 for M1 = 1.1-1.4) and a = 56.8 Rsun.
- That gives R1 = 1.29 Rsun (TIC 1.41), log g1 = 4.32, and R2 = 0.58 Rsun, a normal main-sequence radius for 0.59 Msun.
- J gives T2 of about 4070 K, a K7-M0 dwarf contributing 3.3% of the TESS light.
- Deep eclipse (A, nu = 90 - w): the dwarf transits the F star, U-shaped from limb darkening. Shallow eclipse (B): the F star occults the dwarf, flat-bottomed.
- Depths, durations, shapes, the timing-derived e and w, the RV mass function and the TIC radius all fit one geometry.

Next eclipses (BTJD; uncertainty under 0.1 h)

- Primary (A): 4356.428, 4392.968, 4429.508, 4466.048, 4502.588, 4539.128 (T0 2492.88004 + 36.54015 N)
- Secondary (B): 4337.180, 4373.720, 4410.260, 4446.801, 4483.341, 4519.881 (phase 0.47325; based on a single event, so about 0.01 d)

What would settle it further

- It is already settled as an EB.
- A second secondary would pin e cos w beyond one event.
- Ground photometry of a primary (21%, mid-time 4356.43 BTJD = JD 2461356.43) would check the period.

- RVs at quadrature would give an SB1 orbit with the corrected P. Gaia DR4 epoch RVs should show the 36.540-d period.
- Retry VSX when its TAP is up.

A118. TYC 6536-1088-1 (TIC 64696335)

status
: confirmed (added after submission from the corrected-null re-run; red-clump giant + K7 dwarf on Gaia's circular SB1 orbit, P +1.9 sigma from Gaia's; 0.56% transits and 0.24% occultations about 30 h each; seven events 2019-2026 on one ephemeris, the s108 occultation out of sample at +0.3 h)

P_days
: 41.86524

eP
: 0.00087

T0_primary_BTJD
: 2238.650

depth_primary
: 0.56

depth_secondary
: 0.24

duration_h
: 32

source_path
: Fable 5.1\made\appointments\vet64696335\REPORT.md

found_by
: Fable 5.1 (appointments search, corrected null, 2026-10-17); confirmed by Opus 5.5's re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 64696335 (TYC 6536-1088-1): adversarial check

Opus 5.5, 2026-10-18, 20:15 makeup session. The Oct 17 subagent that started this left scripts, plots, MAST downloads and its own TESScut photometry (`s8_cut.py` → `phot_*.npz`, including **s108 and s109**), but no results. I used its photometry and downloads and wrote my own analysis in `o55\`.

Verdict: CONFIRMED

This is a detached, circular EB: a red-clump giant (~2.1 M⊙, ~8.7 R⊙, 5150 K) with a K7-ish dwarf companion (~0.64 R⊙, ~4200 K) on Gaia's SB1 orbit, at a slightly longer period. It is new. The deciding facts:

- **Seven events on one linear ephemeris over 7.7 years**, both kinds, at phase 0.4993. The four clean events fit with χ^2 1.0/1 (with a 29-min floor for the giant's red noise), and every O−C is ≤ 0.52 h. The ephemeris then predicts the three events I hadn't used. It gets s87 B to +1.1 h and s88 A to −1.5 h. **s108 B, out of sample, comes in at +0.3 h.** s108 was in neither QLP nor the search; that event is only a 2σ fit, so it supports the result rather than carrying it.

- **Difference images (s34 A, s33 B, s61 B) put the deficit on the target**, with the deficit centroid 4.1″, 0.9″ and 1.4″ from the light centroid. For each neighbour, the deficit fits the target's own image better than the same PSF moved to the neighbour, by $\Delta\chi^2$ 4099 / 4264 / 5016 (s34), 595 / 531 / 669 (s33) and 916 / 1174 / 1169 (s61). The neighbours are the G 14.41 at 32.6″, the G 15.65 at 28.0″ (RUWE 4.8) and the G 16.10 at 42.9″. A neighbour would have needed a 13% / 40% / 61% A eclipse. The s87 difference image is dominated by scattered light and is not used.
- **Second pipelines.** For s7 B, eleanor-lite RAW/CORR/PCA give 2960 / 2166 / 2825 ppm, and TGLC aperture 3391 ppm. **TGLC PSF gives only 415 ± 398 ppm**, the one dissent; with nothing brighter than G 16.8 within 27″ in Gaia, I read it as a PSF-fit failure on a 30-min sector, not a host problem. For s61 B, TESS-SPOC SAP and PDCSAP give 2663 and 3748 ppm (6.8σ and 4.4σ against the same fit at 169 random off-conjunction centres in that sector). The errors come from fake-centre scatter, which carries the red noise.
- **The giant doesn't make these dips on its own.** A blind census of all six QLP sectors (`s5_census.py`, boxes 2–26 h, 3-d running median) finds long dips deeper than 1500 ppm only at the conjunction times (s7, s33, s34). The other candidates are 2-h boxes, sector edges, or the s88 background spike. A common-mode test of 26-h boxes on 246–333 other stars per sector is clean at each event: median −21 to −150 ppm, against the target's +2000 to +5200 ppm.

Measurements (own r = 2.5 px TESScut, quadratic baseline, trapezoid)

kind	sector	t0 (BTJD)	depth ppm	T14 h	note
B	7	1506.031 ± 0.008	2450 ± 82	28.2	clean
B	33	2217.770 ± 0.015	2493 ± 144	31.0	clean
A	34	2238.650 ± 0.005	5617 ± 74	32.5	clean, T23 18 h
B	61	2971.315 ± 0.008	2463 ± 74	28.3	clean
B	87	3683.078 ± 0.018	2288 ± 248	29.2	scattered light, 85% coverage
A	88	3703.869 ± 0.025	7028 ± 522	33.3	at a gap and background spike
B	108	4311.022 ± 0.141	1426 ± 663	26.4	out of sample, weak

These errors are residual-permutation errors and understate the red noise; the fake-centre nulls above are the honest ones.

Ephemeris (`o55\ephem.py`): P = **41.86524 ± 0.00087 d**, T_A = 2238.650 ± 0.021, T_B = 2217.748 ± 0.013 BTJD. Gaia's P is 41.84054 ± 0.0130, which puts ours +1.9σ (+35.6 min per orbit). The Gaia A conjunction at cycle 43 is predicted at 2237.70 ± 6.5, and the measured A is +0.15σ from it. The A–B phase of 0.4993 gives $e\cos\omega \approx -0.001$, and the equal durations give $e\sin\omega \approx 0$. Both match Gaia's e = 0.021 ± 0.012.

Physics (`o55\`, by hand; not a light-curve fit)

f(M) = 0.0322 M⊙ (K1 19.50 km/s). The A event is the transit of the giant by the companion (Gaia's A conjunction, deeper eclipse). The B event is the companion's occultation, so its depth (0.24%) is the companion's light share. That gives $k \approx 0.074$ and surface-brightness ratio $J \approx 0.44$, so T2 ≈ 4200 K and R2 ≈ 0.64 R⊙ for R1 = 8.7 R⊙ (TIC): a K7 dwarf. With m2 ≈ 0.63 M⊙, f(M) gives M1 ≈ 2.1 M⊙ and $a \approx 70$ R⊙. The 32-h T14 then requires $b \approx 0.72$ ($i \approx 85°$) and predicts a ~5-h ingress. The A fit's 7 h is about right for a limb-darkened transit; the B fits are too noisy to say. L1 ≈ 48 L⊙ is a red-clump giant, consistent with M_G ≈ 0.9 − A_G at Gaia's 1.05 kpc. These numbers agree with each other, but none of them is fitted jointly. A proper two-sphere fit is the next step if the paper wants radii.

Archives

None of these list the eclipses:

- VSX (VizieR B/vsx, 120″): the only entry is Gaia DR3 5609478320825593984, type E, P 0.235 d, G ~19 at 92″, which can't make 29-h dips.
- ExoFOP TOI and CTOI lists: none.
- ASAS-SN: none. SIMBAD: SB* only.
- Gaia DR3 vari / EB: none. SB9, WDS, Hipparcos: none.
- Zenodo: zero hits on either name. Web search: nothing on either name.

Next eclipses (BTJD, ±0.05 d; each lasts ~32 h)

- A (transit, 0.56%): 4331.912 (observed at 4331.901, see below), 4373.777, 4415.643, 4457.508
- B (occultation, 0.24%): 4352.876, 4394.741, 4436.606, 4478.472

The out-of-sample A transit (Opus 5.5, 2026-10-22 18:30)

The s109 TESScut was re-released by Oct 22 18:30 EDT, ending at **4331.69**: the predicted ingress (~4331.24) and about 11 h of the transit, no egress yet. The first-orbit-only files were kept as `*.first-orbit.*` in the data folder, and `s8_cut.py 109` was re-run.

- **Own-background photometry shows the ingress on time** (`o55\s109_orbit2.png`). In 2-h bins, flux is flat to −0.22% through 4331.18, then −0.37% at 4331.27, −0.69% at 4331.35, and −0.89 to −0.99% from 4331.43 on. The pre-ingress trend is included in those numbers.
- **Fit** (`o55\s109_fit.py`; s34 shape fixed at T14 32.5 h and T23 18 h; free tc, depth and quadratic baseline): **tc 4331.901 ± 0.009**, depth **6685 ± 212 ppm** (s34 5617, s88 7028). O−C = **−15 min, −0.2σ** against the ±0.05 prediction. The tc comes from ingress alone with the shape held fixed, so its ±12 min understates the real uncertainty.
- **On target, not common-mode** (`o55\s109_neigh.py`): the step is −0.83%, −0.84% and −0.80% in r = 1.5, 2.5 and 3.5 px. The bright neighbour's aperture sums change by −0.08%, −0.05% and +0.01%. A small pointing shift at the same time moves single off-target pixels by ±1% with opposite signs across rows (`o55\s109_common.py`), but it conserves flux in an aperture. The own background is flat across the window.
- **Null** (`o55\s109_ingress.py`): the same baseline-extrapolation statistic at 549 fake centres in s108 and s109 has std 1003 ppm. None reaches the observed 4188 ppm.
- **SPOC FLUX_BKG is ~0 in this release.** `spoc_*` columns carry the first-orbit scattered-light hump; use only `own_*` for s109.

So the ephemeris built from the four clean events (s7, s33, s34, s61; the last at 2971.3) predicted a transit 1360 d (32.5 orbits) past the last event it used. The transit came on time and at a depth within the range of the earlier ones. When the rest of s109's second orbit is released (it runs to about 4345), re-run `s8_cut.py 109` (move the cached cut first) and refit with the egress included.

A119. TYC 4320-529-1 (TIC 280129511)

status
: confirmed (per-object re-analysis 2026-10-18; Gaia SB2 near twins, grazing eclipses at both conjunctions, 3.4 and 2.8 ppt, ellipsoidal locked to them; T0 and P from the deeper eclipse; the batch's 36-event count included background-dominated s25/s78/s79 events)

P_days
: 6.348792

eP
: 0.000015

T0_primary_BTJD
: 4336.598

depth_primary
: 0.34

depth_secondary
: 0.28

duration_h
: 3.1

source_path
: Fable 5\made\sb1-scale\pervet\vet280129511\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 280129511: per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.93, P 6.34879 d, 2.7 ppt, 36 events, dphi +0.022.

Verdict: CONFIRMED, a grazing eclipsing binary of near-twin stars, with both eclipses present.

Deciding numbers 1. **Both eclipses, at both of Gaia's conjunctions.** In the clean QLP sectors (s18/19/52/58/59/86) the joint fits are:

Eclipse	Events	Depth	Duration	Δχ² vs no eclipse	Period (d)	O−C rms
I	18	2.82 ppt	2.88 h	1921	6.348801 ± 0.000016	7.5 min
II	16	3.38 ppt	3.12 h	1744	6.348792 ± 0.000015	7.8 min

The timings match Gaia's predicted conjunctions to 0.7σ and about 1σ. 2. **An independent pipeline sees both.** TESS-SPOC SAP and PDCSAP in s58/59/78/79 give eclipse I at 2.4–2.9 ppt and eclipse II at 3.2–3.8 ppt in every sector. That includes s78/s79, where QLP is wrecked by background. 3. **The geometry is self-consistent.** From the Gaia double-lined orbit, the total mass × $\sin^3 i$ is 2.855 $M_\odot$ (mass ratio 1.02) and a sin i is 20.47 $R_\odot$. - Twins of about 3.1–3.6 $R_\odot$ near i ≈ 70–75° (just at the

grazing limit) give about 3 ppt and about 3 h. - The ellipsoidal variation is 1.59 ± 0.02 ppt, with its minima locked to the eclipses. - Eclipse II falls at phase 0.49768 after eclipse I, giving e cos ω ≈ −0.004. Gaia predicts a separation of 0.5048, which agrees to about 1.5σ given Gaia's ±13.8° on ω.

Location: the batch difference image (s58) puts the flux loss 2.4″ (0.12 px) from the target, peaked on the target pixel. The three neighbours that could fake it (G 15.6, 16.1 and 16.9 at 30–42″, needing 21–67% eclipses) are excluded. Archives show nothing beyond SIMBAD's spectroscopic-binary flag.

Row notes (none disqualifying) - The 6.34879 d period is right. Equal-looking eclipses could suggest the search's half period (3.17 d), but the depth difference (2.8 vs 3.4 ppt, steady in every sector) and the off-half separation rule it out. - TESS P is 2.8σ from Gaia's 6.348166 ± 0.000223. Gaia's fit quality is poor (goodness of fit 23.5, RUWE 2.2), so its error bar is probably understated. - The batch's 36-event count includes QLP s25/s78/s79 events that are background-dominated (one in s79 comes out at −4.6 ppt). Only the six clean sectors were used for depth and timing.

Next predicted times (BTJD) - Eclipse I: 4333.442, 4339.791 (σ ≈ 0.015 d). - Eclipse II: 4336.598, 4342.947 (σ ≈ 0.01 d).

Would settle it further: a two-star light-curve model of eclipses plus ellipsoidal variation, with the Gaia velocities, for masses and radii. That is not needed for the confirmation itself.

A120. TYC 8982-3032-1 (TIC 382605794)

status
: confirmed (per-object re-analysis 2026-10-18; M dwarf grazing the F star at conjunction A, 3.1 ppt, 81 events; the bright 27 arcsec co-moving neighbour WDS TOB 243 excluded, likely a triple)

P_days
: 2.140795

eP
: 0.000001

T0_primary_BTJD
: 3055.59400

depth_primary
: 0.31

duration_h
: 1.7

source_path
: Fable 5\made\sb1-scale\pervet\vet382605794\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 382605794 (TYC 8982-3032-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.58, P 2.14079 d, 2.8 ppt, 71 events, dphi +0.012.

Verdict: CONFIRMED, grazing transit of the F star by its M-dwarf companion

Deciding numbers: 1. **Right conjunction:** the eclipse lands +0.50σ from Gaia conjunction A in s011 (the sector nearest Gaia's epoch), the conjunction where the companion is in front. B is excluded at 6–19σ. 2. **Right period:** TESS P = 2.140795 ± 0.000001 d against Gaia's 2.140700 ± 0.000082 (+1.2σ). All 81 events have positive depth; O–C rms is 6 min. 3. **Right star:** - The eclipse-to-ellipsoidal ratio is about 1.0 in every pipeline (QLP, TESS-SPOC PDCSAP and SAP, TARS, TGLC), so both signals come from one star. - The bright neighbour (G 10.25 at 27", TIC 382605852) shows 0.39 ± 0.25 ppt in its PSF-deblended TGLC light curve; the target shows 2.74 ± 0.24. - TESS-SPOC pixel difference images in s065 and s038 prefer the target over every neighbour by $\Delta\chi^2 \geq 285$.

Row fixes: - Depth is **3.1 ppt** (paper 2.8); PDCSAP, which corrects for crowding, gives 3.4–4.0. - Event count is **81** (paper 71).

Other checks: - **Archives:** no VSX, ASAS-SN or EB-table match; Gaia vari tables empty; SIMBAD lists it only as SB*; no TOI/CTOI; no literature found. - **WDS:** the target and the 27" neighbour are the visual pair TOB 243 (first observed 1923). The neighbour shares the parallax and proper motion, so this is probably a triple. - **Background:** the in-eclipse change is +9 ± 12 e-/s, against +41 e-/s needed to fake the dip. A regression test bounds any background-made dip at < 0.09 ppt per sector. - **Systematics and control phases:** no event falls near a listed sector-wide systematic time. All control phases are null. - **Physics:** - Companion m2 ≈ 0.32–0.34 Msun on a graze (i ≈ 73°, b ≈ 1.1). A limb-darkened model reproduces both the 3.1 ppt depth and the 1.7 h duration. - The ellipsoidal (3.2 ppt) runs 0.56–0.66× the simple prediction, which hints R1 is nearer 1.75 Rsun than the TIC's 2.07.

Caveats: - Depth varies across sectors more than noise allows (χ^2 123/80). The shallower sectors (s099–101) use QLP's newer 3×3 aperture, so read it as aperture systematics. - A tentative secondary sits at phase 0.500 (0.2–0.4 ppt, S/N 3.4). That depth fits the companion's occultation, and the phase implies e ≈ 0. That is 2.2σ from Gaia's e cos ω, but Gaia's e = 0.066 reads as the usual positive bias for near-circular orbits. - Doppler beaming (expected ~0.4 ppt) isn't detected (≤ 0.12 ppt). Likely FFI detrending eating a 2-day signal; it doesn't affect attribution.

Next eclipses (BTJD): - Primary: 4400.0135 ± 0.0007, 4402.1543, 4404.2951 (generally 3055.59400 + 2.140795·(E−680)). - Secondary, if real: 4401.0839, 4403.2247.

A121. TYC 9010-4187-1 (TIC 384249385)

status
: confirmed (per-object re-analysis 2026-10-18; occultation of the M dwarf at conjunction B, 3.3 ppt, 17 events; conjunction offset -0.017; the G 13.8 neighbour 13 arcsec away shows the dip only as PSF leakage and must not be catalogued as an eclipser itself)

P_days
: 7.805870

eP
: 0.000011

T0_primary_BTJD
: 3653.55840

depth_primary
: 0.33

duration_h
: 3.24

source_path
: Fable 5\made\sb1-scale\pervet\vet384249385\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 384249385 (TYC 9010-4187-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.26, P 7.80588 d, 2.9 ppt, 12 events, dphi +0.017.

Verdict: CONFIRMED, occultation of the M-dwarf companion by the primary

Deciding numbers: 1. **Right conjunction:** the eclipse is −0.68σ from Gaia conjunction B in s011, the conjunction where the primary is in front. A is excluded at 5.7–16σ. 2. **Right period:** TESS P = 7.805870 ± 0.000011 d against Gaia's 7.806941 ± 0.0013 (−0.83σ). There are 17 events, all at S/N > 3, with consistent depths (χ^2 17.5/16). 3. **Right star:** - The G 13.82 star 13" away (TIC 384244452) shows a 27–39 ppt dip in its own QLP and TGLC-PSF light curves. That is about what it would need (≈ 32 ppt) to be the source, so it was tested directly. - Difference-image positions along the target→neighbour axis exclude the neighbour as host at **8.6σ, 10.3σ, 18.6σ and 8.6σ** in s038, s100, s101 and s102. The implied loss on the target is 3.0–3.8 ppt. - The neighbour's dip is leakage from the target; TGLC's PSF fit can't separate stars 0.6 px apart. The neighbour should **not** be catalogued as an eclipser.

Row fixes: - Depth is **3.3 ppt** (paper 2.9). - Event count is **17** (paper 12; s011 and s012 add 5). - The conjunction offset is **−0.017 at conjunction B**; the CSV's +0.017 has the sign dropped. If the paper calls this a primary eclipse or transit, it should say occultation.

Other checks: - **Second pipelines:** PDCSAP 3.98, TARS 2.8–4.0 and TGLC 3.5 ppt all agree. - **Background:** at most ±0.26 ppt per sector, against a 1.7–2.6 ppt dip in the same windows. - **Systematics and control phases:** no momentum-dump or listed systematic overlaps an event. All control phases are null. - **Archives:** no VSX, ASAS-SN, EB-table, WDS or SB9 match; no TOI/CTOI; no literature found. - **Physics:** - f(M) = 0.0168, so m2 ≈ 0.43 Msun. At B the separation is 0.78a near periastron. - The 3.24 h duration and 1.30 h ingress (1.25 h predicted) match a partial occultation of a ~0.44 Rsun companion, whose TESS light is ~5–9 ppt. A transit at the same geometry would be ~80–100 ppt deep. - No transit is seen at A (< 0.15 ppt, 2σ). A Monte Carlo over the Gaia errors and stellar parameters gives 48% of plausible configurations with no detectable transit there, so its absence is consistent.

Caveats: - Only one eclipse kind is seen, so there is no e cos ω / e sin ω check from the light curve. - A wide-box dip at phase 0.385 (S/N 4.7) is not at Gaia's A (0.472 ± 0.015). It looks driven by a single s102 dip at BTJD 4132.90 (phase 0.41).

Next eclipses (BTJD): - Occultation: 4402.9219 ± 0.0014, 4410.7278, 4418.5337 (generally 3653.55840 + 7.805870·(E−262)). - A transit at A, if one exists, would fall near 4406.61 ± 0.12.

A122. TYC 3612-1132-1 (TIC 384525203)

status
: confirmed (per-object re-analysis 2026-10-18; grazing EB on an ellipsoidal evolved A star, depths 5.8/3.4 ppt, 35 + 39 events; beaming matches Gaia's K; the 3.0 ppt box depth sat on the ellipsoidal minimum at P/2)

P_days
: 4.055307

eP
: 0.000006

T0_primary_BTJD
: 2826.03683

depth_primary
: 0.58

depth_secondary
: 0.34

duration_h
: 3.22

source_path
: Fable 5\made\sb1-scale\pervet\vet384525203\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 384525203 (TYC 3612-1132-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.68, P 4.05515 d, 3.0 ppt, 76 events, dphi +0.014.

Verdict: CONFIRMED. A grazing detached eclipsing binary on an evolved A star (Teff 7709, R 4.1 R⊙) that is strongly ellipsoidal. Both eclipses are detected. Gaia SB1: P 4.055826 ± 0.000548, e 0.039 ± 0.046, K1 51.9 km/s.

1. Archives - cats.py at r = 60″: - Nothing in VSX at the target. Within 60″: a ROT star at 46″, ZTF EW P = 0.27767 d at 49″, an SR at 53″. None has the 4.055-d period. - ASAS-SN: none. Prša, IJspeert, Kostov t2–t4: none. - SIMBAD: TYC 3612-1132-1 [SB*] only. - Gaia DR3: no rows in vari_summary, vari_eclipsing_binary or vari_classifier. RUWE 1.12, duplicated_source true, ESP-HS type A. - SB9 and WDS: none. ExoFOP: no TOI or CTOI. - No separate web search was run; SIMBAD has nothing beyond the SB* entry. **New EB.**

2. Eclipses (raw SAP, QUALITY == 0; model is a per-orbit Legendre-2 plus 4 orbital harmonics fitted outside eclipse) - Primary: 5.76 ± 0.13 ppt, total duration 3.22 ± 0.06 h, ingress 1.29 h, flat bottom 0.6 h (nearly V-shaped). - Secondary: 3.39 ± 0.18 ppt, total duration 3.41 ± 0.09 h. - Background is flat through the events. Folded in-minus-out background is +2.0σ at primary and −2.8σ at secondary;

opposite signs can't produce two dips. - Momentum-dump (bit-32) cadences: only 3, in s056 secondary windows. - Depth by sector: primary 5.35–6.63 ppt, secondary 2.85–3.77 ppt. - Exception: s077 secondary 5.05 ppt (noisiest sector). - Exception: QLP s016 primary 11 ppt, on scattered-light ramps; TESS-SPOC s016 doesn't reproduce it. - Blind census (3-h box, at least 2 ppt): 73 dips. 68 are eclipses; the other 5 all sit on s016/s077 ramps. - The sector-systematic time s083 3564.5 overlaps one primary, which is a normal 5.0 ppt event.

3. Ephemeris

	Events	T0 (BTJD)	P (d)	O−C rms
Primary	35	2826.03683 ± 0.00076	4.055307 ± 0.000006	5 min
Secondary	39	2827.98974 ± 0.0014	4.055285 ± 0.000010	10 min

- Against Gaia's period: −0.95σ.
- Gaia conjunctions, using the ω−t_peri correlation (+0.999): conjunction A (primary eclipsed) is +0.031 ± 0.048 phase from the deep eclipse at the s016 epoch. The batch's dphi labels agree with this, because its t0 at 1741.251 is a secondary eclipse.
- e cos ω = −0.0289 ± 0.0006; Gaia gives −0.016 ± 0.055.
- e sin ω from the duration ratio comes out as +0.029. It isn't reliable for eclipses this grazing.

4. Independence - TESS-SPOC s016 SAP: 4.86 ± 0.15 / 2.85 ± 0.16 ppt. PDCSAP: 6.10 / 3.49 ppt. - Own TESScut s056 photometry, primary: 5.49 / 5.40 / 5.44 ppt at r = 1.0 / 1.6 / 2.5 px, so the depth does not change with aperture. - s056 difference image: both eclipse kinds peak on the target pixel; the deficit centroid is within 1.7σ of the target. - Per-sector centroids put the dimming source 1.4–6.7″ from the light centroid. - Neighbours and the dip each would need: G 13.96 at 28.9″ needs 11.9%; G 14.34 at 38.4″ needs 16.7%; G 14.63 at 23.7″ needs 21.9%; G 14.84 at 46.4″ needs 26.7%. Anything fainter than G 16.3 would need more than 100%, including G 16.52 at 10.9″. - None of the neighbours is a plausible source: centroids, aperture-invariance, and the eclipses landing on the SB1's own ellipsoidal minima all rule them out.

5. Physics - Mass function f(M) = 0.0586 M⊙. For M1 = 1.6–2.6, m2(i = 90°) = 0.67–0.90 M⊙ and a = 14–16 R⊙. - Eclipses need i above about 70–72°. A central eclipse would last about 10 h; the observed 3.2 h means a strongly grazing chord. - Ellipsoidal semi-amplitude: about 5.8 ppt undiluted, against a predicted 7–13 ppt × sin²i. That is consistent within the uncertainty. - Beaming: +0.394 ppt observed, 0.39 ppt predicted, with the right sign. - Which star is in front: at the deep eclipse (conjunction A) the companion crosses the A star's limb. - Depth ratio 0.59: higher than a naive surface-brightness ratio would give for this companion. It is explained by the grazing-limb geometry plus irradiation from the 53 L⊙ primary.

6. Next eclipses (BTJD) - Primary: 4334.6109, 4338.6662, 4342.7215, 4346.7768 (±3 min) - Secondary: 4332.5005, 4336.5558, 4340.6111, 4344.6664 (±6 min) - What would settle it further: a joint ellipsoidal, beaming and eclipse light-curve model, which would give i and the light ratio.

Paper row: P = 4.0553 d (not the 2.03-d half period), depths 5.8 / 3.4 ppt, ellipsoidal (~9.6 ppt peak-to-peak) and beaming present, e cos ω −0.029. The "peak_on_target false" receipt is a half-period-box artefact, not an offset flag.

A123. TYC 5962-1229-1 (TIC 78448518)

status
: confirmed (per-object re-analysis 2026-10-18; eccentric F + K dwarf, 7 + 7 events, deeper eclipse 3.9 ppt at conjunction B, secondary 1.8 ppt at phase 0.6436; e cos omega -0.227 vs Gaia -0.224)

P_days
: 11.874406

eP
: 0.000027

T0_primary_BTJD
: 1477.12450

depth_primary
: 0.39

depth_secondary
: 0.18

duration_h
: 2.95

source_path
: Fable 5\made\sb1-scale\pervet\vet78448518\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 78448518 (GSC 05962-01229 / TYC 5962-1229): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.92, P 11.87442 d, 3.3 ppt, 4 events, dphi +0.011.

Verdict: CONFIRMED. An eccentric, grazing F + K-dwarf eclipsing binary with both eclipses detected. Gaia SB1: P 11.872953 ± 0.000877, e 0.229 ± 0.008, ω 192.0 ± 2.6°, K1 37.6 km/s.

1. Archives - cats.py at r = 60″: VSX has only a ROT star (P = 2.218 d) at 56″. ASAS-SN and the EB tables: none. - SIMBAD: GSC 05962-01229 [SB*], plus TYC 5962-1229-1 and -2 at 0.3″. - WDS 06505-2223 = TDS4200: a 0.6″ Tycho pair (11.57 / 11.79), note "X" (dubious). Gaia shows a single source, with ipd_frac_multi_peak 0 and RUWE 1.38. - Gaia DR3 variability tables: none. SB9: none. ExoFOP: no TOI or CTOI. - No separate web search was run; SIMBAD has nothing beyond the SB* entry. **New EB.**

2. Primary eclipse (at conjunction B) - 7 covered events: 1489.00, 1500.88, 1512.75, 2213.33, 2225.21, 3673.88, 3685.77. The epoch at 1477.13 falls in a data gap. The batch's sapdepth check counted only 4. - Joint fit with exposure-time supersampling: depth 3.87 ± 0.13 ppt, duration 2.95 ± 0.14 h, ingress 1.18 h. - Per-event depths 3.69–4.25 ppt, all consistent. - Background and centroids through the events: all within 2.5σ. Flags: only one bit-32 cadence, outside the core. - No overlap with the sector-systematic times.

3. Ephemeris - T0 = BTJD 1477.12450 ± 0.00296, P = 11.874406 ± 0.000027 d. Errors are scaled by $\chi^2/\mathrm{dof} = 5.8$, since O–C scatter is about ±8 min. - Against Gaia's period: +1.66σ. - Jackknife: P stays within 11.874362–11.874427, at +1.60 to +1.68σ from Gaia for every dropped event. No event owns the result. - Gaia conjunction B is predicted −0.0106 ± 0.0075 phase from the eclipse at the s006/s007 epoch.

Secondary eclipse (at conjunction A) - Phase 0.6436 ± 0.0002, depth 1.84 ± 0.18 ppt (V-shaped), duration 1.79 ± 0.15 h, $\Delta\chi^2$ 177. - Present in all 7 covered epochs at 1.3–2.9 ppt each; dropping any one epoch leaves $\Delta\chi^2$ between 134 and 177. - Gaia predicts the separation between conjunctions A and B as 0.6414 ± 0.0051, so they agree to 0.4σ. - Exact Kepler solution: $e \cos \omega = -0.2271$; Gaia gives −0.2240.

4. Independence - TESS-SPOC primary / secondary: s006 3.64–3.88 / 2.08; s007 3.70–3.84 / 2.28; s033 3.60–3.78 / 1.45–1.62 ppt. - TGLC s006/s007: 3.2–4.9 / 1.6–2.7 ppt. Eleanor-lite was downloaded but not measured; the TASOC corrected flux was unusable. - Own TESScut s087 photometry, primary: 3.79 / 3.77 / 3.88 ppt at r = 1.0 / 1.6 / 2.5 px, so the depth does not change with aperture. - s087 primary difference image peaks on the target (–1.4″). The secondary difference image is too weak to place. - The batch's s033 difference image also had the peak on target, 1.6″ off. - Neighbours: G 16.51 at 57″ would need a 67% dip; G 16.92 at 32″ would need 97%. Nothing brighter lies within 78″. - Census (3-h box, at least 1.5 ppt): the 7 primaries plus one s087 single at 3683.53 (S/N 4.2, unexplained).

5. Physics - Mass function $f(M) = 0.0602\ M_\odot$. For M1 = 1.2–1.5, $m2(i = 90°) = 0.58–0.65\ M_\odot$ and a = 27 $R_\odot$. - Eclipses need i above about 85.5°. - Grazing model (`s2_geom.py`: two limb-darkened disks on Gaia's orbit) fits all four observables with $\chi^2 \approx 1$: $i \approx 85.9°$, R2 0.50–0.65 $R_\odot$, companion surface brightness ratio J ≈ 0.24, so T2 ≈ 4300 K, matching m2. - **Which star is in front:** - Deep eclipse at conjunction B: the F star occults the K dwarf. Projected separation there is 10% smaller, because $e \sin \omega = -0.048$. - Shallow eclipse at conjunction A: the K dwarf grazes the F star's limb. - The ω-flipped alternative fails ($\chi^2 \approx 1800$). - Ellipsoidal (about 0.1 ppt) and beaming (about 0.4 ppt) signals are too small to test against QLP systematics at this period.

6. Next eclipses (BTJD, ±10–12 min) - Deep (conjunction B): 4338.8563, 4350.7308, 4362.6052 - Shallow (conjunction A): 4334.6243, 4346.4987, 4358.3731 - What would settle it further: a light-curve fit to the 200-s data to measure i and R2, and RVs to check for the companion's lines.

Paper row: 7 + 7 events (not 4), add the secondary at phase 0.6436 (1.8 ppt), $e \cos \omega$ −0.227, P 11.874406 ± 0.000027. Nothing argues for striking or demoting it. Optional footnote: WDS TDS4200, a 0.6″ Tycho double flagged "X" (dubious), which Gaia does not see.

A124. TYC 3371-625-1 (TIC 310162414)

status
: confirmed (per-object re-analysis 2026-10-18; SB2 twin, q 0.97, equal grazing eclipses at both conjunctions, 4.1 ppt, 21 + 10 events; circular; TESS-SPOC confirms)

P_days
: 4.551256

eP
: 0.000007

T0_primary_BTJD
: 1819.32778

depth_primary
: 0.41

depth_secondary
: 0.41

duration_h
: 2.45

source_path
: Fable 5\made\sb1-scale\pervet\vet310162414\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 310162414 (TYC 3371-625-1, Gaia DR3 2619686483000747 52): per-object vet

Opus 5.5-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.75, P 4.55127 d, 3.5 ppt, 26 events, dphi +0.001, diffimg 2.34″.

Verdict: CONFIRMED (on-target grazing eclipses in a near-twin SB2)

1. Archives (`cats-out.txt`, `archives-out.txt`) - cats.py --r 60: - VSX: only Gaia DR3 2619687470820522 24 [ROT], P 0.596 d, at 49″ (a different star). - ASAS-SN, Prša+2022, IJspeert+2024, Kostov t2/t3/t4: none. - SIMBAD: TYC 3371-625-1 [SB*]. - Gaia DR3 has no vari_summary or vari_eclipsing_binary row. - nss_two_body_orbit SB2: - P 4.551136 ± 7.4e-5, e 0.0025 ± 0.0092, ω 35.9 ± 145.7 (undefined at e ≈ 0). - K1 87.77 ± 0.85, K2 90.51 ± 0.85, RUWE 1.79. - SB9, WDS and VizieR Gaia veb within 60″: none. ExoFOP: no TOI or CTOI. Web search on the TYC/TIC/Gaia names: no hits. - MAST has QLP s19/59/73/86, TESS-SPOC s19/59/73, TEQUILA s19. - Not previously catalogued as eclipsing.

2. Eclipses (`vet1.py`, `vet1-out.txt`, `events_310162414.png`; QLP SAP, q==0, untrimmed) - 31 QLP events fitted; TESS-SPOC adds the s19 odd eclipses QLP lacked core points for (1817.05, 1821.60, 1826.14, 1835.26, 1839.80). - Shape: median T14 2.45 h, T23 0.55 h, V-ish, consistent with grazing. - Background z-scores are within ±2 except s086 3639.806 (+5.2). That event is in the orbit-start scattered-light ramp and is compromised. - s086 3648.92 is 0.4 d from the listed 3649.3 systematic and has 10 flagged cadences, but it is on ephemeris. - No bit-32 momentum dumps within 0.5 d of any event. - Blind dip census: nothing else in s059 or s073. The s086 3641.3–3641.5 dips are inside a pre-gap ramp flagged 2^30 (`look_s086_3641.png`).

3. Ephemeris

Kind	N	P (d)	T0 (BTJD)	O−C rms
Even	21	4.551255 ± 2e-6	1819.32698	9.1 min
Odd	10	4.551246 ± 5e-6	1819.33291 (half-integer epochs)	5.2 min
Joint	31	4.551256 ± 2e-6 (×3.4 for red noise)	1819.32778	—

- Joint P vs Gaia: +1.6σ.
- Gaia's nominal conjunction B is at phase +0.007 of the even eclipses and A at −0.495. The Gaia period error alone grows to 0.009 phase by the TESS epochs.
- e cos ω = 0.002 ± 0.0014. The duration difference (2.35 vs 2.50 h) gives e sin ω ≈ 0.03 ± similar. Both are consistent with a circular orbit.
- SPOC minus QLP mid-times: −0.4 min median, 2.8 min rms.

4. Independence (`vet2-out.txt`)

TESS-SPOC (aperture 10–13 px, CROWDSAP 0.91–0.94):

	Even	Odd
SAP depth (ppt)	3.85	3.93
PDCSAP depth (ppt)	4.18	4.17

- Stacked centroid shift: −0.02 ± 0.03 and +0.01 ± 0.03 millipixel (30 events).
- Neighbours (G < 17): G 16.05 at 27″ needs 57%; G 13.90 at 41″ needs 7.8%; G 13.58 at 45″ needs 5.8%. All are excluded by the matching depths in two very different apertures plus the centroid null, and independently by the SB2 match.

5. Physics (`phys.py`) - M1 $\sin^3 i$ = 1.356 and M2 $\sin^3 i$ = 1.315 Msun; a sin i = 16.04 Rsun. - For ~1.5 Msun F stars: i ≈ 75°, a ≈ 16.6 Rsun, luminosity radius 2.16 Rsun each. - b/(R1+R2) ≈ 0.96 reproduces T14 = 2.45 h, and the crude overlap depth is 2–3 ppt against 4.1 observed. This is consistent. - Out-of-eclipse: ellipsoidal cos2φ = −1.14 ± 0.01 ppt (−1.13 / −1.16 / −1.08 per sector). No reflection or beaming asymmetry (< 0.15 ppt). - Mild tension: the predicted ellipsoidal amplitude is ~2.5 ppt. ~30% third light would explain it (RUWE 1.79 allows it). - The two kinds are interchangeable; each star is in front at one conjunction.

6. Next eclipses (BTJD, ±5 min) - Even: 4336.1725, 4340.7237, 4345.2750, 4349.8262 - Odd: 4338.4481, 4342.9993, 4347.5506, 4352.1019

Paper row: keep it. Note: "SB2 twin (q 0.97); equal grazing eclipses at both conjunctions; ellipsoidal 1.1 ppt; TESS-SPOC confirms. Trapezoid depth 4.1 ppt."

What would settle it further: a light-curve model (i, R1, R2, third light) fitted jointly with K1/K2.

A125. TYC 3634-2713-1 (TIC 431278074)

status
: confirmed (per-object re-analysis 2026-10-18; deeper eclipse 5.3 ppt is the companion's occultation at conjunction B; new 1.3 ppt eclipse at conjunction A, phase 0.553; e cos omega -0.0834 +- 0.0006 = Gaia's -0.0835)

P_days
: 6.386066

eP
: 0.000003

T0_primary_BTJD
: 1744.40878

depth_primary
: 0.53

depth_secondary
: 0.13

duration_h
: 3.6

source_path
: Fable 5\made\sb1-scale\pervet\vet431278074\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 431278074 (TYC 3634-2713-1, Gaia DR3 1990144878402555136): per-object vet

Opus 5.5-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.63, P 6.38604 d, 4.8 ppt, 13 events, dphi +0.010, diffimg 3.61″.

Verdict: CONFIRMED, with a second (shallow) eclipse found at the other conjunction

1. Archives - cats.py: VSX has only Gaia DR3 1990145084560963584 [DSCT|GDOR] at 38″ (G 14.57), a neighbour. ASAS-SN and the EB tables: none. SIMBAD: [SB*]. - Gaia DR3 has no vari row. nss SB1: - P 6.385793 ± 1.12e-4, e 0.0855 ± 0.0065, ω 192.47 ± 4.18. - t_peri 0.3475 ± 0.0729, K1 64.06 ± 0.45, RUWE 0.85. - SB9, WDS and veb: none. ExoFOP: no TOI or CTOI. Web search: no hits. - MAST has QLP and TESS-SPOC for s16/17/57/77/84. Not previously catalogued as eclipsing.

2. Eclipses (`vet1.py`, `foldfit.py`, `stack_AB_431278074.png`) - **Conjunction B** (ν = 270° − ω, primary in front, companion occulted): 14 QLP events. - Per-event trapezoid depth 5.78 ppt; stacked 5.25 ± 0.09 (QLP) and 5.42 ± 0.08 (SPOC PDCSAP). - T14 3.6 h, T23 1.1 h. - Per-event depth chi² 163/13, but SPOC

reproduces each event's depth. - **Conjunction A** (companion transits the primary): phase 0.5531 ± 0.0004 (QLP) / 0.5529 (SPOC). - Depth 1.20 ± 0.10 / 1.35 ± 0.06 ppt (jackknife by event), T14 2.2–2.4 h.

Sector	s16	s17	s57	s77	s84
QLP depth (ppt)	1.47	1.43	1.26	0.93	1.24

- The blind census picks these up independently: 1741.52, 1767.06, 1779.90, 1786.23, 2859.07, 2871.86, 2878.27, 3587.16, 3593.51, 3599.92.
- s077 3417.56 has background z −11.6 and three bit-32 cadences nearby (`look_s077_3417.png`). It is on ephemeris and SPOC agrees.
- SPOC SAP "11 ppt" at s084 3596.4 lies in a flagged pre-gap ramp and is a systematic (`look_s084_3596.png`).

3. Ephemeris - Conjunction B: P 6.386066 ± 3e-6 (formal), T0 1744.40878 ± 0.0006, O−C rms 3.1 min. - Gaia: Δ = +0.00027 d = +2.4σ. Gaia's nominal conjunction B falls at phase −0.001. - Spacing A – B = 0.55310 ± 0.00037, so **e cos ω = −0.0834 ± 0.0006** (A taken as the primary eclipse), against **Gaia −0.0835**. - Durations: A is shorter (2.2–2.4 h vs 3.6 h), consistent in sign with Gaia's e sin ω = −0.0185, which puts the stars 3.8% further apart at A. For grazing chords the duration ratio amplifies e sin ω, so it doesn't measure it directly. - SPOC minus QLP mid-times: 0.0 min median, 4.1 min rms.

4. Independence

TESS-SPOC (11–17 px, CROWDSAP 0.90–0.93):

	B (deep)	A (shallow)
SAP depth (ppt)	5.81	—
PDCSAP depth (ppt)	5.90	1.35

- Stacked centroid shift: +0.03 ± 0.04 and −0.08 ± 0.05 millipixel.
- Batch diffimg: peak on target, 3.61″.
- Neighbours: the nearest bright ones need 22% (G 14.57, the δ Sct/γ Dor at 38″) and 31% (G 14.92 at 39″). Everything else needs 44% to more than 100%. The SB1 phase match at both conjunctions rules out a blend independently.

5. Physics (`phys.py`) - f(M) = 0.172 Msun, so M2 ≥ 0.92–1.05 Msun for M1 1.2–1.4: a G dwarf giving ~4–6% of the TESS-band light. That is consistent with SB1. - Uniform-disk grazing solution: i ≈ 76°, b/(R1+R2) = 0.92–0.94 at B and 0.96–0.97 at A. - Predicted depths: 5.3 ppt at B; 1.9–3.3 ppt at A before limb darkening, which suppresses grazing transits of the primary's limb and brings it to the observed 1.2–1.35. Predicted T14 is 4.2–4.8 h at B and 2.8–3.6 h at A. - This explains why the occultation of the companion is deeper than the transit: the stars are 3.8% further apart at A and the geometry is grazing. - Ellipsoidal cos2φ = −2.7 ppt in every sector, against ~6–7 predicted. Same factor-2 offset as TIC 310162414, so the formula or catalogue radius is the likely cause.

6. Next eclipses (BTJD, ±6 min) - B (deep): 4337.1516, 4343.5376, 4349.9237 - A (shallow): 4340.6837, 4347.0698, 4353.4558

Paper row: - Keep it; add a note: "deep eclipse = occultation of the G-dwarf companion at Gaia conjunction B; shallow 1.3 ppt eclipse at conjunction A, phase 0.553; e cos ω −0.0834 ± 0.0006 matches Gaia." - Change the depth from 4.8 ppt (box) to 5.3 ppt (trapezoid). - Optionally note the +2.4σ period offset.

What would settle it further: an SB2 spectrum to see the companion, or a limb-darkened model fitting both depths and durations.

A126. CD-61 607 (TIC 279648140)

status
: confirmed (per-object re-analysis 2026-10-18; grazing EB + ellipsoidal + Doppler beaming at Gaia's K, M-dwarf companion; 6.0 ppt, secondary 0.71 ppt at phase 0.502; VSX type DSCT|GDOR|SXPHE is a type revision; unpromoted SPOC multi-sector TCE)

P_days
: 3.3502765

eP
: 0.0000006

T0_primary_BTJD
: 2829.361644

depth_primary
: 0.60

depth_secondary
: 0.071

duration_h
: 3.42

source_path
: Fable 5\made\sb1-scale\pervet\vet279648140\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 279648140 (CD-61 607): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.24, P 3.35025 d, 5.4 ppt, 53 events, dphi +0.021.

Verdict: CONFIRMED (grazing EB, plus ellipsoidal variation and Doppler beaming, M-dwarf companion)

Deciding numbers - **Period and conjunction:** P = 3.3502765 ± 0.0000006 d from 64 fitted events, against Gaia 3.3504026 ± 0.000058 (2.2σ). Mid-eclipse is −0.011 phase from Gaia conjunction A (companion in front) at the Gaia epoch, which is −1.9σ using the full Gaia covariance (corr(ω, t_p) = 0.9996). Conjunction B is 0.5 phase away, so it cannot be the match. - **Secondary eclipse:** phase 0.50186 ± 0.00040, depth 710 ± 38 ppm (SNR 26), present in every season. This gives e cos ω = +0.0029 ± 0.0006, against Gaia −0.009 ± 0.015. - **Light curve follows the orbit:** raw SAP shows an ellipsoidal wave of −6.1 ppt at P/2 in all 10 sectors. It also shows a Doppler-beaming term of +450 ppm at the correct phase and sign; Gaia's K = 41.9 km/s predicts about 560 ppm. A blend cannot do this.

Archives - VSX lists only Gaia's DSCT|GDOR|SXPHE at 0″. That is a type revision to eclipsing plus ellipsoidal, not prior art. - Kostov t2 (unvetted) is the only EB-table hit. SIMBAD has CD-61 607 [SB*] with 4 refs (RAVE ×3, Gaia multiplicity). No TOI or CTOI; web search found nothing. - **SPOC multi-**

sector DV (s56–s69) has TCE_1 at P 3.3502547, 6040 ppm, SNR 75, and TCE_2 at phase 0.501, 661 ppm, SNR 11. So a pipeline caught both eclipses before us, but nothing was published and it never became a TOI. Say so in the paper.

Eclipses and ephemeris - T0 = BTJD 2829.361644 ± 0.000167. O–C rms 1.9 min, jackknife max 0.34σ. T14 = 3.42 h (constant), T12 ≈ 1.3 h, so nearly V-shaped. - No momentum-dump cadences in any event core. Background and centroids are clean through every event, and no event falls near a listed sector-wide systematic. - A blind phase scan finds only the primary and the secondary. Odd/even differ by 2.0σ after per-sector normalisation, which is not significant.

Depth drop in the latest sectors - QLP depth falls from 6.0–6.35 ppt to 5.3–5.7 ppt in s95–97. - The ellipsoidal amplitude (6.1 → 5.2 ppt) and the secondary (≈760 → 658 ppm) drop by the same factor in the same sectors. - That points to extra dilution in that year's QLP processing, not a change in the eclipse.

Independence - TESS-SPOC PDCSAP depth is 5.91–6.04 ppt across s3, s28, s29, s68 and s69, with timing within 1 min. - TGLC PSF photometry in s2 gives 6.03 ± 0.24 ppt. - The batch difference image (s97) is on target. The only neighbour brighter than G 17 is G 16.64 at 1.45″, which would need a 197% dip, so no neighbour can host the signal.

Physics - f(m) = 0.0255 M⊙. With M1 = 1.19, m2 = 0.40–0.42 M⊙ (an early-M dwarf); a = 11.0 R⊙. - The secondary-to-primary depth ratio implies T2 ≈ 3600 K, which fits that M dwarf. - The geometry is grazing, i ≈ 70–76°. A central transit would last 9.8 h; the observed 3.4 h only fits a grazing chord. - **Open tension:** the ellipsoidal amplitude implies R1 ≈ 2.7 R⊙, but the TIC gives 3.71 R⊙, which would predict about 17 ppt instead of the observed 6.1. This does not affect the eclipse claim; a joint light-curve and RV model would resolve it.

Changes to the paper row - P should be **3.3502765(6)**. The row's 3.35025 is 39σ off; that only adds up to 19 min over the baseline, but the fitted value should be quoted. - Undiluted depth is **6.0 ppt** (the row's 5.4 is the folded box value). - Add the secondary: 0.71 ppt at phase 0.502. - Note the VSX type revision and the SPOC TCE.

Next eclipses (BTJD) - Primary (±0.5 min): 4333.6358, 4336.9861, 4340.3363, 4343.6866. - Secondary (±2 min): 4335.3172, 4338.6674, 4342.0177, 4345.3680.

A127. TYC 8737-1439-1 (TIC 74180670)

status
: confirmed (per-object re-analysis 2026-10-18; eccentric grazing EB with an M dwarf; transit 5.4 ppt QLP (6.2 corrected), occultation 3.3 ppt at phase 0.668 matching Gaia's e cos omega to 0.09 sigma; 6 + 5 events; jackknife clean)

P_days
: 18.4215088

eP
: 0.0000285

T0_primary_BTJD
: 3686.04707

depth_primary
: 0.54

depth_secondary
: 0.33

duration_h
: 3.0

source_path
: Fable 5\made\sb1-scale\pervet\vet74180670\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 74180670 (TYC 8737-1439-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.95, P 18.42155 d, 5.5 ppt, 5 events, dphi +0.028.

Verdict: CONFIRMED (eccentric, grazing EB with an M-dwarf companion; the verdict does not rest on any single event)

Deciding numbers - **Secondary matches Gaia's eccentric orbit:** it sits at phase 0.66815 ± 0.00011, and Gaia's e = 0.301, ω = 331° predict B at 0.6665 ± 0.0173 after A (+0.09σ). The implied e cos ω is 0.2646, against Gaia 0.2615 ± 0.029. A background EB would have to share both the period and this e cos ω. - **Period:** primary P = 18.4215088 ± 0.0000285 d, against Gaia 18.43027 ± 0.01423 (0.62σ). The secondaries give their own P = 18.4215431 ± 0.0000304, which agrees to 0.8σ. - **Geometry fits Gaia's orbit with an M dwarf:** m2 = 0.28 M⊙ at i ≈ 87.7° reproduces both the reversed durations and both depths:

	Predicted (i ≈ 87.7°)	Observed
T14 at A (companion in front)	3.0 h	2.9–3.0 h

	Predicted (i ≈ 87.7°)	Observed
T14 at B (companion behind)	4.0 h	3.8–4.0 h
Transit depth	~6.4 ppt	6.1–6.5 ppt (PDCSAP)
Occultation depth	~3.2 ppt (max 3.6 at T2 = 3300 K)	3.9–4.0 ppt (PDCSAP)

The occultation runs slightly deeper than predicted, which a T2 of about 3450 K would fix.

Tiny-n check - There are 5 QLP primaries; dropping any one moves P by at most 0.72σ. - 5 secondaries fix P independently (s12, s39, s100 and s104 individually at ≥12σ; s66 at 4.7σ), and s103 has a partial one. - The s12 primary that QLP lost to a gap is recovered by TARS (5.9 ppt) and eleanor-lite (5.4 ppt). That makes 6 primaries. - The s39 primary sits at the start of a background ramp (+1500 e^-). TESS-SPOC PDCSAP and TARS both recover it, and the result holds without it.

Archives - VSX only has an unrelated star at 48″. No EB-table hit, including Kostov t2. SIMBAD has TYC 8737-1439-1 [SB*], 1 ref (Gaia). No TOI or CTOI; web search found nothing. This is new.

Systematics and neighbours - The blind census finds nothing at other phases except orbit-edge ramps (s66, s103) and features right before the s104 data gap. - About 7% dilution: contratio 0.066, CROWDSAP 0.93. - 8 neighbours could in principle host the dip. The strongest is G 14.63 at 36.9″, which would need 16% primary and 10% secondary dips. - The diff image is on target (0.68″, but only 2 events in s104). The secondary-phase match and the consistent geometry rule out a blend much more strongly than the image does.

Changes to the paper row - P should be **18.4215088(285)**; T0 = BTJD 3686.04707(105). - QLP depth is 5.3–5.5 ppt; dilution-corrected about 6.2 ppt. - Add the secondary: 3.3 ppt in QLP (3.9 ppt in PDCSAP) at phase 0.668. - "5 events" should read 6 primaries plus 5 secondaries. - The c3 eyeread note "secondary ~0.68" is confirmed.

Next eclipses (BTJD) - Primary (±2 min): 4349.2214, 4367.6429, 4386.0644. - Secondary (±4 min): 4343.1105, 4361.5320, 4379.9536.

A128. TYC 1643-218-1 (TIC 219543346)

status
: confirmed (per-object re-analysis 2026-10-18; grazing occultation of the companion at conjunction B on a subgiant, 6.9 ppt, 7 events; conjunction A empty as predicted; TASOC independent; the automatic phase-0.22 'secondary' was an s081 ramp artefact)

P_days
: 19.205350

eP
: 0.000065

T0_primary_BTJD
: 2919.47988

depth_primary
: 0.69

duration_h
: 4.56

source_path
: Fable 5\made\sb1-scale\pervet\vet219543346\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 219543346 (TYC 1643-218-1, Gaia DR3 1829189638955084672): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.36, P 19.20532 d, 5.9 ppt, 7 events, dphi +0.005.

VERDICT: CONFIRMED. The single eclipse per orbit is at Gaia conj B. It is a grazing occultation of the companion by the subgiant primary; conj A misses as the geometry predicts.

Star and orbit - G 11.36, Tmag 10.96. TIC: Teff 6324 K, R 3.49 Rsun (a subgiant), M 1.25, d 833 pc. RUWE 1.52. - Gaia NSS SB1 (full row with corr_vec): P 19.20610 ± 0.00375 d, e 0.437 ± 0.018, ω 348.5 ± 5.1°, K1 54.2 ± 3.2 km/s, significance 17. - f(M) = 0.231 Msun; m2(i=90°) = 1.08 Msun — a ~1.06 Rsun, ~6000 K dwarf carrying about 7% of the TESS light.

1. Archives - `cats.py --r 60`: VSX has only two unrelated ROT stars (40", 42"). ASAS-SN: none. SIMBAD: [SB*] only. Prša, IJspeert and Kostov: none. - No SB9 or WDS entry. Gaia `vari_summary` and `vari_eclipsing_binary` empty. ExoFOP: no TOI/CTOI, no imaging or spectroscopy. - Web search on the TYC, TIC and Gaia ids: nothing. Not previously catalogued as an eclipsing binary.

2. Eclipses (QLP SAP, Q0, no trimming; `vet.py`, `sec.py`, `look2.py`) - **Primaries:** 7 measured (s014, s041, s054, s055×2, s082×2), each S/N 12.7–21.7. - Weighted depth 6.94 ± 0.14 ppt. Depth constancy χ^2 15.8/6 (range 6.4–7.8). Odd 7.07 ± 0.18 vs even 6.66 ± 0.25. - Global trapezoid: T14 4.56 h (1σ 4.50–4.62), T12 1.82 h, flat bottom 0.9 h. Close to V-shaped: the flat-bottomed T12/T14 = 0.1 is worse by $\Delta\chi^2$

= +300. - **Events seen but not counted:** s014 E-64 is fully flagged in QLP, though the dip is visible (TASOC has it). s081 E31 is fully flagged and sits on a huge background ramp, unusable. - **Systematics:** background flat through every clean event; background regression bounds a fake dip at ≤ 0.43 ppt vs 3.6–4.9 ppt measured; centroids dX −0.37 ± 0.18, dY +0.32 ± 0.24 mpix; one bit-32 cadence in one window; nothing near a listed systematic time. - **Controls:** all control phases within ±0.3 ppt of zero. - **Census:** all 7 primaries are S/N ≥ 4 peaks; the off-ephemeris peaks (s054 2777.6, 2791.1; s081 3531.4, 3532.5) are background features (`census_offeph.png`). - **Secondary:** `vet.py`'s phase-0.22 "secondary" (2.9 ppt, S/N 14.6) is the s081 orbit-end ramp plus flagged s014 data (`look_sec022.png`). With s081 excluded, the best dip in phase 0.206–0.266 is 0.02 ± 0.15 ppt (T14 scan 1–14 h; no epoch over S/N 2). 3σ upper limit ≈ 0.45 ppt at conj A. **Must not be reported as a secondary.**

3. Ephemeris and Gaia - P = 19.205350 ± 0.000065 d, T0 = 2919.47988 ± 0.00199 BTJD. χ^2 23.5/5 (errors scaled ×2.17). O−C rms 6.8 min. - vs Gaia P: −0.20σ. - Jackknife: P shifts ≤ 48 ppm-d (≤ 0.75σ); depth stays 6.77–7.09 ppt; leave-one-sector-out P runs 19.205302–19.205389. E39 (s041) is +18.5 ± 4.6 min off the rest; dropping it gives χ^2 5.9/4 with P and depth unchanged — no single event owns the verdict. - Gaia conjunctions (Monte Carlo with corr_vec): every eclipse on **conj B**, +0.6 to +2.4 h from prediction (+0.04 to +0.25σ). Conj A falls 0.236 ± 0.010 in phase after each eclipse, 7–14σ away, and empty. - One eclipse kind only, so no photometric e cos ω / e sin ω. Gaia: e cos ω +0.426 ± 0.018, e sin ω −0.087 ± 0.038; the geometry is consistent.

4. Independence - **TASOC s014 ensemble FLUX_CORR:** E-63 6.18 ± 0.42 ppt; E-64 6.78 ± 0.38 ppt (the event QLP flagged). TASOC FLUX_RAW too noisy (7.8 ppt point-to-point). MAST has no SPOC, TESS-SPOC, TGLC or ELEANOR products. - **Difference image** (TESScut s082, 2 events): PSF fit prefers the target, χ^2 4.7e4 vs ≥ 2.6e6 for any neighbour; implied dip 6.17 ppt. Batch diffimg (s055): 2.73" offset, peak on target. - **Neighbours** (dip each would need): G 13.81 @ 55" → 56 ppt; G 14.58 @ 57" → 115 ppt; G 15.08 @ 46" → 180 ppt; G 11.48 @ 89" → 7.8 ppt — the only physical mimic, and the difference image rejects it (χ^2 600× worse, implied dip wrong sign).

5. Physics (`phys_ecc.py`) - a = 40.0 Rsun. Separation 0.745a at conj B (ν = 281°), 0.886a at conj A (ν = 101°): projected separation 1.19× larger at A. - Central eclipse at B would last 13.8 h; observed 4.56 h means b = 0.94 (R1+R2), i ≈ 81.7° — grazing, matching the V shape. - Conj A: projected separation 5.11 Rsun vs R1+R2 = 4.55 — a miss by 0.55 Rsun, robust to R1 ±10% and M1 1.25–1.45. - Gaia convention: B = primary in front, so an occultation of the companion. Predicted uniform-disc depth 4.4–6.3 ppt vs 6.94 observed. Depth, duration, shape and the missing A eclipse fit one geometry.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: (1) conj B timing within 0.04–0.25σ and P at −0.20σ from Gaia; (2) TASOC detects it independently at 6.2–6.8 ppt; (3) difference image on target; (4) < 0.45 ppt at conj A matches the predicted miss. - **Paper row:** no change. Note it is an occultation of the companion, and drop the s081 "secondary". - **Next primaries** (conj B, ±8 min): BTJD 4417.497, 4436.703, 4455.908. - **Conj A** (no eclipse expected; ±0.19 d): 4422.03, 4441.24, 4460.44. - **To settle further:** a clean sector covering both conjunctions, or an RV epoch near conj B.

A129. TYC 7166-2609-1 (TIC 4965877)

status
: confirmed (per-object re-analysis 2026-10-18; grazing occultation at conjunction B near periastron, 6.8 ppt trapezoid (batch box 6.0), 11 events not 8; heartbeat excluded; three pipelines agree)

P_days
: 10.220874

eP
: 0.000008

T0_primary_BTJD
: 2901.40080

depth_primary
: 0.68

duration_h
: 2.58

source_path
: Fable 5\made\sb1-scale\pervet\vet4965877\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 4965877 (TYC 7166-2609-1, Gaia DR3 5436996115010241664): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.45, P 10.22082 d, 6.0 ppt, 8 events, dphi +0.011.

VERDICT: CONFIRMED. A grazing eclipse at Gaia conj B, 21° from periastron. Conj A, near apastron, misses.

Star and orbit - G 11.45, Tmag 11.07. TIC (spectroscopic): Teff 6120 K, R 1.98, M 1.16, d 468 pc. RUWE 1.11. - Gaia NSS SB1: P 10.220388 ± 0.000711 d, e 0.189 ± 0.009, ω 290.7 ± 3.4°, K1 50.30 ± 0.44 km/s, significance 115. - f(M) = 0.128 Msun; m2(i=90°) = 0.78 Msun (a K-dwarf companion, about 8% of the light).

1. Archives - `cats.py`: VSX and ASAS-SN none. SIMBAD: [SB*] only. Prša, IJspeert and Kostov: none. - No SB9 or WDS entry. Gaia vari tables empty. ExoFOP: no TOI or CTOI. - Web search on the TYC, TIC and Gaia ids: nothing. Not previously catalogued as an eclipsing binary.

2. Eclipses - **Primaries:** 11 clean (s008, s009×2, s035×2, s062×3, s089×2, s099), each S/N 9.6–22.7. - Weighted depth 6.79 ± 0.14 ppt. Depth constancy χ^2 27/10: s008 and s099 ~5.8–5.9, s062 7.7 — QLP aperture/dilution differences, since SPOC agrees within the same range. - Odd 6.54 ± 0.21 vs even 6.95 ± 0.18. - Global trapezoid: T14 2.58 h, T12 1.03 h, near-V (flat-bottomed T12/T14 = 0.1 worse by $\Delta\chi^2$ = +245). - **Systematics:** no bit-32 cadences in any event window, no event near a listed systematic time;

background regression bounds a fake dip at ≤ 0.27 ppt; centroids dX +0.21 ± 0.17, dY +0.42 ± 0.21 mpix; per-epoch plots (`look_pri.png`) show flat background (s062 E11's ramp comes after the dip). - **Controls:** all phases within ±0.4 ppt, nothing above S/N 3. - **Census:** all 11 eclipses are S/N ≥ 4 peaks; four off-ephemeris peaks, all 1–1.7 ppt — no other dip of half the eclipse depth or more. - **Secondary:** none. At conj A (phase ~0.443 ± 0.04): 0.22 ± 0.31 ppt. Full-phase scan max is S/N 3.5 at phase 0.66 (0.45 ppt), not significant.

3. Ephemeris and Gaia - P = 10.220874 ± 0.000008 d, T0 = 2901.40080 ± 0.00075 BTJD. χ^2 23.7/9 (errors scaled ×1.62). O−C rms 4.1 min. - vs Gaia P: +0.00049 d = +0.68σ. - Jackknife: P shifts ≤ 6 ppm-d; every dropped event's O−C within 2.4σ; depth stays 6.69–7.05 ppt. - Gaia conjunctions: eclipses on **conj B**, +2.5 h (s008) to +5.4 h (s099), +0.86 to +1.15σ; the growth over time reflects the small period difference. Conj A is 0.44 in phase away (17–48σ). Gaia's phase(B) − phase(A) = 0.543 ± 0.007. - One eclipse kind only: no photometric e cos ω / e sin ω. Gaia: e cos ω +0.067 ± 0.011, e sin ω −0.177 ± 0.010.

4. Independence - **TESS-SPOC PDCSAP:** s8 6.67 ± 0.33, s9 7.46 ± 0.32, s35 7.35 ± 0.24 ppt. - **TGLC PSF photometry:** s8 5.88 ± 0.34, s9 6.77 ± 0.46 ppt (deblends neighbours; still sees the dip). - **SPOC TPF difference image, s35:** target favoured, χ^2 1.7e4 vs ≥ 5.0e5 for any neighbour; implied dip 7.66 ppt. The G 12.94 at 50" would need 9 ppt of its own and fits 32× worse. s9 single-event image: peak loss on the central pixel. Batch diffimg (s062): 1.54" offset, peak on target. - **Neighbours** (dip each would need): G 12.94 @ 50" → 24 ppt; G 12.10 @ 89" → 11 ppt; G 15.30 @ 19" → 218 ppt.

5. Physics - a = 24.7 Rsun. Separation 0.820a at conj B (ν = 339°), 1.171a at conj A (ν = 159°). - Central eclipse at B would last 7.3 h; observed 2.58 h means b = 0.94, i ≈ 82.6° — a graze. - Conj A misses by 0.94 Rsun. - An occultation of the companion. Predicted uniform-disc depth 3.7–5.7 ppt (M1 1.16–1.32, R1 ±10%) vs 6.8 observed — factor ~1.3, within the uncertainty of the companion's radius and light ratio. - **Heartbeat check** (conj B near periastron): with e = 0.19 any tidal pulse would be broad (~a day) and well under 1 ppt; the observed event is a 2.6 h V shape reproduced by SPOC and TGLC — not a heartbeat.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: (1) three pipelines agree — QLP 6.8, SPOC 6.7–7.5, TGLC 5.9–6.8 ppt; (2) P +0.68σ from Gaia with eclipses on conj B at ~1σ; (3) jackknife stable to 6 ppm-d and the difference image on target. - **Paper row:** n_events should be 11, not 8. Nothing else to strike. - **Next primaries** (±2 min): BTJD 4403.869, 4414.090, 4424.311. - **Conj A** (no eclipse expected; ±0.25 d): 4408.40, 4418.62, 4428.84. - **To settle further:** an RV epoch near conj B, or 2-min SPOC data for a proper impact-parameter fit.

A130. TYC 3261-1420-1 (TIC 240680969)

status
: confirmed (per-object re-analysis 2026-10-18; grazing M-dwarf transit of a low-RGB giant at conjunction A, 5 events not 4; depth genuinely variable 6.7-8.5 ppt across four pipelines, spots on a grazing chord)

P_days
: 18.047704

eP
: 0.000085

T0_primary_BTJD
: 1778.6842

depth_primary
: 0.67-0.85

duration_h
: 7.2

source_path
: Fable 5\made\sb1-scale\pervet\vet240680969\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 240680969 (TYC 3261-1420-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.83, P 18.04742 d, 6.5 ppt, 4 events, dphi +0.018, diffimg offset 0.89".

Verdict: CONFIRMED. The M dwarf companion transits the low-RGB primary at Gaia conjunction A, on a grazing chord (b ≈ 0.8).

Deciding numbers 1. **Right time.** There are 5 events: the paper's 4 plus an s017 event at 1778.684 (30-min cadence). All 5 sit at Gaia conjunction A, +1.8 to +1.9σ late (+0.31 to +0.67 d); the lag grows with time because the TESS period is slightly longer than Gaia's. Conjunction B is excluded at 22–44σ. 2. **Right period.** TESS: P = 18.047704 ± 0.000085 d, T0 = 1778.6842 ± 0.0063 BTJD (errors scaled by √(χ²/dof) = 3.5). Gaia: 18.04419 ± 0.00199 d (+1.76σ). Jackknife: dropping any one event gives P between 18.047681 and 18.047730 — no single event owns the period or the verdict. The four 200-s events alone predict the independent s017 event to −2.9 min. 3. **Right star.** The TESS-SPOC s057 difference image (2 events) prefers the target's PSF over every neighbour by $\Delta\chi^2 > 3e5$, implying a 6.48

ppt dip on the target. The loss centroid is within 0.1 px of the target in s057 and s084, 0.3 px in s085. The only neighbour that could physically host the signal (G 14.22 at 52", needing 11–15%) is excluded by the difference image.

Measurements (raw QLP SAP, QUALITY==0, quadratic baseline over ±0.9 d, trapezoid) - Shape: T14 = 0.30 d (7.2 h), ingress ≈ 0.25·T14.

Sector	tc (BTJD)	Depth (ppt)	O−C (min)
s017	1778.6838	8.5	−0.6
s057	2861.5550	6.6	+12.3
s057	2879.5864	6.9	−11.1
s084	3601.4992	6.8	−4.4
s085	3619.5526	8.0	+3.8

- **O−C scatter ±12 min against ~2 min formal** — probably red noise from the giant's surface plus spots on a 1.8 h ingress.
- **Depth varies between events** (~6.7 ppt s057/s084 vs ~8.3 ppt s017/s085), present in QLP-SAP, SPOC-SAP, PDCSAP and TARS (s085: 7.9/8.5/9.3/8.8 ppt), robust to baseline window 0.6–1.3 d and poly degree 1–3. Real, not a QLP artefact; likely spots on an active grazing chord (VSX: ROT, 58.5 d, ~4% modulation). A change in impact parameter from a third body can't be excluded with 5 events.
- **Systematics:** no listed systematic time within 2.9 d of an event; SAP_BKG and centroids smooth through every event; the s057 momentum dump is 0.8 d before the 2861.55 event. Blind census (0.30 d box, S/N > 6): only the 5 eclipses plus gap/segment-edge artefacts. Control phases 0.25/0.40/0.75/0.85 null below 0.25 ppt.

Secondary eclipse / e cos ω - Gaia places conjunction B 10.599 ± 0.132 d after A. Six covered windows give −0.09 ± 0.06 ppt (2σ upper limit ~0.15 ppt). No occultation. - So no light-curve test of e cos ω or e sin ω; Gaia's e = 0.202 ± 0.011, ω = 47.6 ± 3.4° stand untested.

Physics - f(M) = 0.0104 Msun, so m2(i = 90°) = 0.25–0.34 Msun for M1 = 1.0–1.6 Msun. TIC: R1 = 2.87 Rsun, Teff 4771 K. - At conj A the separation is 0.834a; a central transit would last 10–12 h. The observed 7.2 h total / 3.6 h flat need b ≈ 0.76–0.89 and k = 0.07–0.14 → R2 ≈ 0.2–0.35 Rsun, consistent with the M dwarf and the depth. - At conj B the separation is 1.08a, b_B ≈ 1.0–1.15: occultation grazing or missed; a full occultation of a ~3400 K M dwarf would be ~2 ppt, so the null favours b_A ≳ 0.83 — consistent. - Not a heartbeat: periastron at phase 0.92, 1.5 d before the event; the event is a sharp 7 h dip.

Archives - cats.py: VSX ASASSN-V J003741.47+510219.2 is type ROT, P 58.46 d — a type revision, not prior art. SIMBAD: [SB*]. - Gaia DR3: no vari_summary, no vari_eclipsing_binary. SB9, WDS, J/AJ/160/111: none. ExoFOP: no TOI, CTOI, imaging or spectroscopy. Web search: nothing.

Next eclipses (BTJD) - Primary: 4341.458 ± 0.014, 4359.506, 4377.554, 4395.601 (T = 1778.6842 + 18.047704·n). - Secondary (probably grazing or absent): 4352.06 ± 0.13, 4370.11, 4388.15.

Changes to the paper row: keep it. Events: 5, not 4. Depth: 6.7–8.5 ppt and variable, not a single 6.5. Period: 18.04770 d.

What would settle it further: RVs, and more sectors to see whether the depth variation tracks the rotation.

A131. TYC 4030-58-1 (TIC 52855832)

status
: confirmed (per-object re-analysis 2026-10-18; gamma Dor pulsator + grazing EB, both eclipses 9.1/5.0 ppt, 43 + 41 events; the batch's 7.1 ppt blended both at the half period; depth ratio wants a ~5600 K companion against Gaia's f(M) 0.56 Msun, so Gaia's K is probably diluted, likely SB2)

P_days
: 2.4113702

eP
: 0.0000015

T0_primary_BTJD
: 2632.73471

depth_primary
: 0.91

depth_secondary
: 0.50

duration_h
: 1.92

source_path
: Fable 5\made\sb1-scale\pervet\vet52855832\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 52855832 (TYC 4030-58-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.46, P 2.41137 d, 7.1 ppt, 47 events, dphi +0.011, diffimg offset 2.26".

Verdict: CONFIRMED. A strongly grazing EB with both eclipses detected; the primary is a γ Dor pulsator.

Deciding numbers 1. **Both eclipse kinds present in all 5 sectors**, including the 2019–20 30-min sectors, measured after prewhitening the γ Dor pulsation (10 frequencies per sector plus 4 orbital harmonics). - Primary: 9.06 ± 0.12 ppt, 42 of 43 events at S/N > 3. - Secondary: 5.01 ± 0.12 ppt, 36 of 41 events at S/N > 3. - TESS-SPOC s024 independently gives SAP 8.6 / 4.7 ppt and PDCSAP 9.9 / 5.5 ppt (primary / secondary). 2. **Right time and period.** - Primary: P = 2.4113702 ± 0.0000015 d, T(n = 348) = 2632.73471 ± 0.00044 BTJD, O−C rms 4.3 min. Secondary: P = 2.4113694 ± 0.0000020 d. - Gaia: P = 2.411344 ± 0.000139 d (+0.19σ). - The primary falls at Gaia conjunction A, +0.034 ± 0.129 d (+0.3σ), and is the deeper eclipse, so the hotter primary is the star being eclipsed. B excluded at 9.5σ for

the primary. Secondary at phase 0.49973 ± 0.00031 vs Gaia's B prediction 0.505 ± 0.016. 3. **e cos ω and e sin ω agree with Gaia.** Light curve: e cos ω = −0.0004 ± 0.0005; durations (1.92 and 2.04 h) give e sin ω ≈ +0.03 ± 0.03. Gaia: +0.015 ± 0.026 and +0.020 ± 0.024.

Pulsation is not the source - Dominant mode f1 = 1.2515 c/d = 3.017–3.019 × f_orb in every 200-s sector — close to but not locked to 3·f_orb (beat ~133 d). - The eclipses stay at fixed orbital phase across 6 years while the pulsation phase drifts relative to them. - Within s058 the pulsation troughs happen to line up with the eclipses — worth knowing when reading the plots.

Systematics - SAP_BKG shows no excess at either eclipse (in−out ≤ 0.5σ). Centroid shifts ≤ 0.003 px. - One s085 event (3616.57) is near the listed 3616.5 systematic; irrelevant among 84 events.

Independence and blends - The s024 TPF difference image (30-min, pulsation-limited) is inconclusive: peak S/N 3.6; can't separate the target from the G 16.27 at 12.6" (0.6 px), which would need ~75%. - The batch c4 diffimg puts the peak on target, 2.26" offset, but psf_frac_resid 0.895 — weak. - Neighbours that could host by dilution alone: G 11.08 at 81" (~6 ppt), G 13.76 at 37" (~75 ppt), others. - **What settles attribution:** both eclipse kinds match the target's own Gaia SB1 period and conjunctions within 0.2–0.3σ. A neighbour coincidence at that level is implausible.

Physics: a real tension for the paper's prose - f(M) = 0.0433 Msun. For M1 = 1.3–1.6 Msun and i ≥ 72°, m2 = 0.52–0.62 Msun → a ~3900–4100 K companion. - But the secondary/primary depth ratio is 0.55 (0.48 from s024 alone); in a grazing circular orbit that is the surface-brightness ratio, needing ~5600 K. - Likely resolution: **Gaia's K is diluted by the companion's lines in the RVS spectrum.** A 0.95 Msun, 5600 K companion would give a true K1 ≈ 82 km/s and a blended ≈ 66 km/s, against Gaia's 55.8. If so, the system is a likely SB2 and m2 ≈ 0.9 Msun. - Central transit would last ~5.6 h vs observed 1.9–2.0 h: strongly grazing, b/(R1+R2) ≈ 0.94. - e ≈ 0, so no heartbeat concern.

Archives - VSX lists Gaia DR3 522478549617967616 as DSCT|GDOR|SXPHE — a type revision: γ Dor pulsator plus EB. SIMBAD: [gD*]. Gaia DR3 has a vari_classification entry but no EB entry. - ASAS-SN, SB9, WDS, J/AJ/160/111: none. ExoFOP: no TOI or CTOI. Web search: nothing.

Period stability: 2019–20 sectors give 2.411383 ± 0.000012 d; 2022–25 give 2.411353 ± 0.000003 d (2.4σ apart); dropping s085 shifts P by 4σ. Mild, probably pulsation-driven timing noise; apsidal or third-body effects not excluded.

Next eclipses (BTJD) - Primary: 4335.1620 ± 0.0018, 4337.5734, 4339.9848. - Secondary: 4336.3665 ± 0.0023, 4338.7779, 4341.1892.

Changes to the paper row - The 7.1 ppt depth is the average over both eclipse kinds from the half-period search: replace with **primary 9.1 ppt and secondary 5.0 ppt**. - Event count: **43 primaries plus 41 secondaries**, not 47 mixed. - Note "γ Dor primary". If the paper interprets the companion from Gaia's K, flag the K-dilution/SB2 caveat.

What would settle it further: high-resolution spectra for the secondary's lines (an SB2 detection fixes both masses), and a 120-s/20-s TPF difference image.

A132. TYC 2263-89-1 (TIC 283862049)

status
: confirmed (per-object re-analysis 2026-10-18; grazing F8 + M dwarf at conjunction A, 8.0 ppt, 6 events not 4; SPOC TPF difference image on target)

P_days
: 10.144315

eP
: 0.000004

T0_primary_BTJD
: 2997.87878

depth_primary
: 0.80

duration_h
: 1.92

source_path
: Fable 5\made\sb1-scale\pervet\vet283862049\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 283862049 (TYC 2263-89-1, Gaia DR3 2873905966582180864): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.09, P 10.14437 d, 7.9 ppt, 4 events, dphi +0.000.

Verdict: CONFIRMED. The F8 dwarf (M 1.15, R 1.16 Rsun) is grazed by its Gaia SB1 companion, an M dwarf of about 0.5 Msun.

Archives - `cats.py --r 60`: no VSX, ASAS-SN, Prša, IJspeert or Kostov entry. SIMBAD: SB* at 0". - Gaia: no `vari_summary` or `vari_eclipsing_binary` entry. - Gaia NSS SB1: P 10.144581 ± 0.000635 d, e 0.080 ± 0.013, ω 168.2 ± 7.4°, K1 35.68 ± 0.46 km/s, significance 78, RUWE 1.90, ipd_frac_multi_peak 0. - SB9, WDS and J/AJ/160/111: none. ExoFOP: no TOI or CTOI. Web search: no hits. - **New as an eclipsing binary.**

Light curve (QLP SAP, QUALITY==0, no trimming; `vet-out.txt`) - Trapezoid fit: depth 7.98 ppt, T14 1.92 h, ingress T12 0.77 h. A V shape is disfavoured by only $\Delta\chi^2$ 14 (near-V). - **6 events, not 4** — sector 17 (30-min cadence) holds two the batch missed. Per-event depths: 8.22, 7.64, 8.63, 8.18, 7.87, 7.83 ppt, S/N 17–26. Weighted depth 8.10 ± 0.14 ppt; depth-constancy χ^2 5.0/5 across s17, s57, s84. - Systematics: no bit-32 cadences in any window, none near the known systematic times. Control phases null, all within ±0.43 ppt. Blind dip census finds 6 dips, all on the ephemeris. Background regression could fake at most 0.03 ppt per sector.

Ephemeris - P = 10.144315 ± 0.000004 d; T0 = 2997.87878 ± 0.00029 BTJD; χ^2 0.4/4. - Jackknife: leaving out one event or one sector, the dropped events are predicted to ≤ 1.6 min. - Gaia: $\Delta P = -0.42\sigma$. Conjunction A matched at −0.09 to −0.19σ in each sector. - Gaia's e cos ω −0.078 ± 0.013 and e sin ω +0.016 ± 0.010 predict secondary phase 0.4503 ± 0.0083.

Secondary eclipse - Not detected. Best point in the Gaia window: 0.51 ± 0.15 ppt at phase 0.442 (7% false-alarm rate); at 0.4503 it is −0.37 ± 0.17 ppt. TESS cannot measure e cos ω here.

Independence

Pipeline	s17 (ppt)	s57 (ppt)	s84 (ppt)
TESS-SPOC PDCSAP	8.01 ± 0.23	8.27 ± 0.21	7.60 ± 0.22
TARS	9.17 ± 0.43	7.99 ± 0.40	8.39 ± 0.38

- SPOC TPF difference images (s57, s84): the loss's position minus the light's position is −0.006 and +0.021 px along the axis to the G 14.0 star at 78"; a neighbour host would give +3.56 px. A PSF fit puts the loss on the target at 7.95 and 7.75 ppt.
- No star brighter than G 17 within 60"; the brightest within 60" is G 19.2, which would need an impossible dip.

Physics - f(M) = 0.047 Msun, so m2 = 0.46–0.55 Msun. a = 23.3 Rsun, a/R1 = 20.2. - A central transit would last 5.6 h; the observed 1.9 h requires a grazing chord. - Limb-darkened two-sphere fit with k = 0.44 (R2 0.51 Rsun): b = 1.34, dilution scale 1.09 — fits depth, duration and shape with no third light (TIC contratio 0.001), χ^2 equal to the trapezoid's. - Predicted secondary: $b_B \approx 1.38$, depth 0.3–0.5 ppt, duration ≈ 1.6 h — matches the non-detection. - No heartbeat: e = 0.08 and a 1.9 h V-shaped event rule it out. - The companion is in front at conjunction A, where all six events fall.

Next predicted eclipses (BTJD) - Primary (±1 min): 4407.9386, 4418.0829, 4428.2272. - Secondary (Gaia phase, ±2.0 h): 4412.507, 4422.651, 4432.795.

Paper row changes: n_events 4 → 6; depth 8.1 ppt; mark as grazing. Nothing to strike or demote.

A133. TYC 8666-1738-1 (TIC 359299569)

status
: confirmed (per-object re-analysis 2026-10-18; flat-bottomed M-dwarf transit of an F subgiant, 7.9 ppt QLP, 9.3 undiluted, 17 events; the batch diffimg flag was a crowding artefact, own localisation on target in 5 sectors at 8-36 sigma)

P_days
: 7.149090

eP
: 0.000002

T0_primary_BTJD
: 3352.39715

depth_primary
: 0.79

duration_h
: 7.08

source_path
: Fable 5\made\sb1-scale\pervet\vet359299569\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 359299569 (TYC 8666-1738-1, Gaia DR3 6063854975652939136): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.09, P 7.14910 d, 8.2 ppt, 17 events, dphi +0.029; was on the batch "diffimg peak not on target" outlier list.

Verdict: CONFIRMED. The F subgiant (TIC: Teff 6939 K, R 2.91 Rsun, M 1.52) is transited by its Gaia SB1 companion, an M dwarf of about 0.22 Msun. **The "diffimg peak not on target" flag is an artefact.**

Archives - `cats.py --r 60`: no VSX, ASAS-SN, Prša, IJspeert or Kostov entry. SIMBAD: SB* at 0". - Gaia: no variability entries. - Gaia NSS SB1: P 7.149945 ± 0.000850 d, e 0.045 ± 0.036, ω 200 ± 34°, K1 17.16 ± 0.56 km/s, significance 31, RUWE 1.07. - SB9, WDS and J/AJ/160/111: none. ExoFOP: no TOI or CTOI. Web search: no hits. - **New as an eclipsing binary.** - Crowded field (galactic latitude 6.6°): 23 Gaia stars brighter than G 17 within 90".

Light curve (sectors 11, 38, 64, 99, 100, 102; `vet-out.txt`, `fold_oc.png`) - Flat-bottomed transit: depth 7.88 ppt (QLP, diluted), T14 7.08 h, T12 0.71 h; steep ingress decisively preferred over V (Δχ² 4260). - 17 events at S/N 25–66. Odd vs even depths: 7.88 vs 7.74 ppt. - QLP depth varies by sector (s38 8.82, s64 8.17, s100 7.27 ppt; χ² 210/16) — QLP aperture contamination; the pixel-level depths don't show it.

- Background regression could fake at most 0.10 ppt. Two windows contain bit-32 cadences; none near the known systematic times. - Blind census: no off-ephemeris dip deeper than 3.9 ppt. Real red noise present (S/N-scan rms 2.8; off-ephemeris S/N 4–8 dips are ~1 ppt).

Ephemeris - P = 7.149090 ± 0.000002 d; T0 = 3352.39715 ± 0.00031 BTJD; χ^2 27.9/15; O–C rms 2.3 min. - Jackknife: no event or sector moves P by more than 1e–6 d. - Gaia: $\Delta P = -1.0\sigma$. Conjunction A matched at –1.14 to –1.37σ.

Secondary eclipse (tentative) - 0.41 ± 0.05 ppt at phase 0.499 (Gaia predicts B at 0.4768 ± 0.017). - After red noise, effective significance ≈ 3σ, and sectors disagree (s100: −0.10 ± 0.10 ppt). If real, e cos ω ≈ 0.

On-target vs neighbour (`localise.py`, `localise-out.txt`, `localise_*.png`) - Method: model the out-of-eclipse scene from Gaia positions (restricted to ±4 px because of a bright star at the TESScut corner), fit a free Gaussian to the difference image, jackknife over events.

Sector	Pixel product	Diff-image offset from target	Nearest alternative
s11	SPOC target pixels	0.07 px	$\Delta\chi^2$ +442
s38	SPOC target pixels	0.07 px	$\Delta\chi^2$ +711 (G 15.87 at 27", 36σ)
s64	TESScut	0.05 px	$\Delta\chi^2$ +2189 (30σ)
s100	TESScut	0.05 px	$\Delta\chi^2$ +2735 (36σ)
s102	TESScut	0.40 ± 0.27 px	$\Delta\chi^2$ +1676 (8σ)

- Aperture-invariance: depth falls monotonically with aperture radius, as an on-target dip must (s64: 8.82 → 6.91 ppt from r = 0.8 to 4 px). Undiluted depth 8.9–10.3 ppt (scene model: ~9.0–9.7).
- Single-pixel apertures on neighbours show much smaller depths — the leakage expected from the target.
- Independent deblending: TGLC PSF photometry s11 gives 8.97 ± 0.18 ppt. TESS-SPOC PDCSAP: 8.77 (s11), 9.04 (s38) ppt.
- A neighbour host would need: G 15.87 at 27" → 63% dip; G 14.0 at 50" → ~11%. All excluded by position.
- **What tripped the batch flag:** the batch's own deficit was only 0.9" from the target; its *light* centroid is pulled by crowding, and a bright star at the TESScut corner breaks any unrestricted scene fit.

Physics - f(M) = 0.0037 Msun, so m2 = 0.20–0.25 Msun; a = 18.8 Rsun. - Central transit would last 9.2 h; observed 7.1 h gives b ≈ 0.6–0.7. - Limb-darkened fit to the 3.3-min sectors: shape alone is degenerate between companion size and dilution; with dilution set from the pixel-level depth and a/R1 from Kepler: k = 0.10 (R2 ≈ 0.29 Rsun), b = 0.55, a/R1 = 7 (within $\Delta\chi^2$ 7 of best). - Transit stellar density 0.12–0.15 solar vs 0.062 ± 0.014 from TIC — consistent if R1 ≈ 2.6 Rsun, which Gaia's logg 3.79 suggests. - Ellipsoidal: 0.47 ppt observed vs 0.68 predicted. Predicted secondary 0.5–0.7 ppt undiluted (~0.4–0.55 in QLP), consistent with the tentative 0.41. - No heartbeat: e = 0.045 and a flat-bottomed 7 h transit rule it out.

Next predicted eclipses (BTJD) - Primary (±1 min): 4403.3134, 4410.4624, 4417.6115. - Secondary (Gaia phase, ±2.8 h): 4406.722, 4413.871. On the tentative phase-0.499 detection: 4406.881, 4414.030.

Paper row changes: remove the "diffimg peak not on target" outlier flag; note the depth is diluted (8.2 ppt in QLP, ~9.3 ppt undiluted). Nothing to strike or demote.

A134. TYC 9000-1011-1 (TIC 329314665)

status
: confirmed (per-object re-analysis 2026-10-18; grazing F + K dwarf, 9.4 ppt, secondary 3.3 ppt at phase 0.5002; the 7.53 arcsec batch offset was a clipped-centroid artefact, 7-sector PSF localisation on target, 11 arcsec neighbour excluded at dchi2 >= 292)

P_days
: 3.7733403

eP
: 0.0000019

T0_primary_BTJD
: 3773.739862

depth_primary
: 0.94

depth_secondary
: 0.33

duration_h
: 2.30

source_path
: Fable 5\made\sb1-scale\pervet\vet329314665\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 329314665 (TYC 9000-1011-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.45, P 3.77334 d, 8.3 ppt, 34 events, dphi +0.012, diffimg offset 7.53″ (the outlier).

Verdict: CONFIRMED (grazing F + K-dwarf EB, the Gaia SB1 itself; the 7.53″ batch offset is a centroiding artefact in a crowded field)

Deciding numbers - **Period and conjunction:** P = 3.7733403 ± 0.0000019 d from 33 primaries, against Gaia 3.773505 ± 0.000119 (–1.4σ). Mid-eclipse is –0.012 phase from Gaia conjunction A (companion in front), –0.3σ using the full covariance (corr(ω, t_p) = 0.9998, σ = 0.14 d). Conjunction B is excluded at 13σ. - **Secondary eclipse:** phase 0.50016 ± 0.00024, depth 3.34 ± 0.14 ppt (SNR 25), present in all 7 sectors. - e cos ω = +0.0003 ± 0.0004; T14 ratio 0.99 ± 0.03 → e sin ω = –0.003 ± 0.017. - Gaia has e = 0.014 ± 0.030 (e cos ω +0.009, e sin ω –0.011) — consistent. - **Localisation:** own PSF-fit difference images in 7 sectors (s11, s38, s64, s65, s99, s100, s101; several camera/CCD orientations). - The primary deficit lies 1.2–6.8″ from the target (median ~2.6″, about 0.12 px). - A fixed-position fit at

the target is the best fixed position in 6 of 7 sectors for the primary and 7 of 7 for the secondary. - The only cheap host is the red G 14.19 (RP 12.94) star at 11.2″ W, which would need a 5.6% dip; it fits worse than the target by $\Delta\chi^2$ +292 to +1659 in every sector.

Archives (queried 2026-10-18) - cats.py r = 60″: no VSX, ASAS-SN, Prša, IJspeert or Kostov t2/t3/t4 rows. SIMBAD: TYC 9000-1011-1 [SB*] only. - *VizieR battery: nothing in WDS, SB9, Gaia DR3 vari_* (classifier, EB, summary), RAVE or GALAH. Gaia phot_variable_flag NOT_AVAILABLE. - ExoFOP: no TOI, no CTOI, no notes. Web search on the TYC/TIC names: nothing. - MAST: no SPOC or TESS-SPOC light curve or DV exists, so no SPOC TCE. Prior art: none.

Eclipses and ephemeris (QLP SAP, QUALITY==0, no trimming; events_329314665_pfix/sfix.txt) - **Primary:** T0 = BTJD 3773.739862 ± 0.000333; O−C rms 3.2 min against a 2.4-min median error (χ^2/dof 56.7/31; errors scaled). Folded depth 8.64 ± 0.12 ppt in s99–101 and 9.39 ± 0.14 ppt over all sectors. T14 = 2.30 h, T12 = 0.87 h: near-V, grazing. - **Per-sector depths:** 7.95–10.6 ppt, varying with QLP aperture dilution (TIC contratio 0.61). Odd/even: 8.60 ± 0.16 vs 8.71 ± 0.15 ppt. - **Secondary:** P = 3.7733276 ± 0.0000043 from 20 events, 2.7σ from the primary P (secondary timings noisier; not considered significant). - **Systematics:** background change through events negligible; centroid shifts ≤ 0.004 px; no event core contains a QUALITY bit-32 dump. The one event near a listed systematic (s65 3090.76, 1 d from 3089.8) agrees with the rest. - **Blind dip census:** nothing else deeper than 3 ppt at SNR > 5 once the eclipses are masked.

Independence - TGLC s11: aperture 9.48 ± 0.26 ppt; PSF 6.49 ± 0.23 ppt. eleanor-lite s11: 10.69 ± 0.39 ppt. TARS s38/64/65: 8.56, 8.24, 7.83 ppt. Timing agrees with QLP to ≤ 4 min, and every pipeline shows the secondary (2.7–3.6 ppt). - **Caveat on TGLC:** the TGLC files of the neighbours at 11.2″, 15.1″ and 23.0″ show the same dip with the same absolute flux loss (~37–56 e/s, equal to the target's) — the target's deficit leaking into overlapping 3×3 apertures and broad PSF fits 0.5–1.1 px away. TGLC confirms the signal but cannot localise it; localisation rests on the multi-sector PSF-fit difference images (loc_prim, *loc_sec*). - **Neighbour table** (neighbours_out.txt, G < 17.5 within 120″, b = −3°): many stars could host 9 ppt by flux alone. Within 10″ only G 16.72 at 4.3″ comes close (needs 70% in RP, 116% in G); the difference-image fits disfavour it at $\Delta\chi^2$ +119 to +635 in 5 of 7 sectors. - **Why the batch 7.53″ is an artefact:** the batch number is a clipped-positive 5×5 centroid, pulled by noise in a field where the light centroid itself sits ~12″ off the target.

Physics - f(m) = 0.0909 M⊙. With M1 = 1.28 (TIC; R1 2.02, Teff 6400), m2(i = 90°) = 0.71 M⊙ and a = 12.8 R⊙. - Central T14 would be 6.2 h; the observed 2.3 h needs a projected separation of 0.93 (R1 + R2), i ≈ 78.5°, so m2 ≈ 0.73 M⊙. - Depth ratio 0.386 implies T2 ≈ 4830 K, a K2–3 dwarf of ~0.75 M⊙ — matching the mass function independently. - A limb-darkened grazing model (R1 2.02, R2 0.74) gives 9.5 ppt at T14 2.6 h, consistent with 8.6–10.7 ppt and 2.3 h (a trapezoid underestimates T14 for a V shape). - The deeper eclipse is at conj A, so the K dwarf is in front at primary eclipse.

Changes to the paper row - Keep the row. P = **3.7733403(19)**. - Replace the 7.53″ diffimg offset with "on target: PSF-fit deficit 1–7″, median 2.6″, 7 sectors; 11″ neighbour excluded, $\Delta\chi^2 \geq 292$". - Add the secondary: 3.3 ppt at phase 0.5002. - Undiluted depth 8–11 ppt by pipeline (contratio 0.61); quote ~9 ppt.

Next eclipses (BTJD) - Primary (±0.6 min): 4252.9541, 4256.7274, 4260.5008, 4264.2741. - Secondary (±3 min): 4254.8414, 4258.6147, 4262.3880, 4266.1614.

What would settle it further: a spectrum showing the K-dwarf lines (SB2), or a seeing-limited ground-based light curve, which resolves 11″ easily.

A135. TYC 9347-571-1 (TIC 182517223)

status
: confirmed (per-object re-analysis 2026-10-18; grazing F + early M, spotted primary, 9.7 ppt, 56 events; secondary 1.2 ppt at phase 0.4994; unpromoted SPOC s56-s69 multi-sector TCE, SNR 87, caught both eclipses)

P_days
: 2.2076198

eP
: 0.0000003

T0_primary_BTJD
: 3115.008748

depth_primary
: 0.97

depth_secondary
: 0.12

duration_h
: 1.38

source_path
: Fable 5\made\sb1-scale\pervet\vet182517223\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 182517223 (TYC 9347-571-1 = Flo 460): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 10.74, P 2.20762 d, 8.6 ppt, 56 events, dphi +0.001, diffimg 3.23″.

Verdict: CONFIRMED (grazing F + early-M EB with a spotted, likely synchronised primary; the Gaia SB1 itself)

Deciding numbers - **Period and conjunction:** P = 2.2076198 ± 0.0000003 d from 51 primaries, against Gaia 2.2076462 ± 0.000023 (–1.15σ). Mid-eclipse is +0.0004 phase from Gaia conjunction A (0.02σ; σ = 0.037 d with corr(ω, t_p) = 0.9999). Conjunction B excluded at more than 27σ. - **Secondary eclipse:** 1.18 ± 0.08 ppt (14.6σ) at phase 0.49942 ± 0.00044, from fold fits on locally normalised windows. - T14 ratio 1.01 ± 0.05. Control windows at phases 0.25 and 0.75: −27 ± 58 and −106 ± 55 ppm (null). - e cos ω = −0.0009 ± 0.0007 and e sin ω = +0.005 ± 0.027. Gaia: e = 0.014 ± 0.009, ω = 29 ± 45° (e cos ω +0.012), consistent at ~1.4σ. - **Localisation:** own PSF-fit difference images. - Primary (s27, s67, s68, s94): free-fit offsets 0.9–2.1″, mean (+0.1, +0.7)″. Every capable neighbour fits worse by $\Delta\chi^2 > 2400$ in every sector: G 11.70 at 35.9″ (needs 1.9%), G 12.79 at 44.4″

(6.1%), G 15.26 at 48.8″ (35%). - Secondary (5 sectors): mean (−0.2 ± 0.7, +1.0 ± 0.8)″, χ^2 10.6 for 10 dof against zero offset; the target is the best fixed position in 5 of 5 sectors. - The batch "deficit–light 10.6″" is just the light centroid being pulled by the G 11.70 star.

Archives (queried 2026-10-18) - cats.py r = 60″: no VSX, ASAS-SN, Prša, IJspeert or Kostov rows. SIMBAD: Flo 460 [SB*] only. - *VizieR battery: no WDS, SB9, Gaia vari_*, RAVE or GALAH entries. ExoFOP: no TOI or CTOI. Web search: nothing. - **SPOC s56–s69 DV (TESS-SPOC dvt file):** TCE_1: P 2.2076405, 10.16 ppt, SNR 87, impact 0.99. TCE_2: phase 0.4995, 1.41 ppt, SNR 13. **The pipeline saw both eclipses but no TOI or CTOI was ever made — disclose this in the paper.**

Eclipses and ephemeris (QLP SAP, QUALITY==0) - **Primary:** T0 = BTJD 3115.008748 ± 0.000100; O−C rms 4.0 min (median error 0.9 min, χ^2/dof 57.8/49; s28 noisy at rms 5 ppt). Folded depth 9.68 ± 0.08 ppt, T14 1.376 ± 0.009 h, T12 0.53 h. - Odd/even: 9.49 ± 0.11 vs 9.57 ± 0.11 ppt. Per-sector depths 8.0–10.4 ppt. - **Systematics:** no QUALITY bit-32 in any event core; background changes negligible in s67–s95; nearest listed systematic (s094 3875) is 1.6 d from any event. - **Light curve:** ~1% spot modulation near P_orb, removed by a local quadratic per event. - **Blind census:** with a 0.3-day high-pass, no other dips > 3 ppt (earlier 1-day-median census hits were spot residuals at mask edges).

Independence - TESS-SPOC PDCSAP primary: s1 9.35 ± 0.22, s27 9.95 ± 0.14, s67 10.00 ± 0.11, s68 9.89 ± 0.11 ppt (CROWDSAP 0.86–0.89), timing within ≤ 1.6 min. Secondary: 1.23–1.55 ppt in all four sectors. - TARS s94: primary 9.44 ± 0.14, secondary 1.54 ± 0.16 ppt. TGLC s1 consistent but noisy (9.2 ± 6 ppt).

Physics - f(m) = 0.0599 $M_\odot$. With M1 = 1.21 (TIC; R1 1.33, Teff 6235), m2(i = 90°) = 0.58 $M_\odot$ and a = 8.66 $R_\odot$. - Central T14 would be 3.8 h; the observed 1.38 h needs i ≈ 78°. - A limb-darkened grazing model (R2 0.61) gives 9.2 ppt at T14 1.46 h, against 9.7 ppt and 1.38 h observed. - Depth ratio 0.122 implies T2 ≈ 3650 K (0.8 μm blackbody) — a little cooler than a 0.6 $M_\odot$ dwarf (~3900 K) but within this crude model. - Companion is in front at the primary eclipse (conj A).

Changes to the paper row - Keep the row. P = **2.2076198(3)**. - Depth ~9.7 ppt (QLP fold) or 9.9–10.0 ppt (TESS-SPOC PDCSAP); the row's 8.6 is the box value. - Add the secondary: 1.2 ppt at phase 0.4994. - Note the SPOC TCE_1/TCE_2 prior detection.

Next eclipses (BTJD) - Primary (±0.3 min): 4251.9329, 4254.1406, 4256.3482, 4258.5558. - Secondary (±2 min): 4253.0355, 4255.2431, 4257.4507, 4259.6583.

What would settle it further: a high-resolution spectrum to detect the M-dwarf lines and measure v sin i, testing the synchronised-rotation reading of the spot modulation.

A136. TYC 4004-778-1 (TIC 344947332)

status
: confirmed (per-object re-analysis 2026-10-18; flat-bottomed total transit of an F star by a 0.16 Msun M dwarf, 9.0 ppt, 9 events; new 0.55 ppt secondary at phase 0.4845, T2 ~3200 K)

P_days
: 10.451460

eP
: 0.000014

T0_primary_BTJD
: 2954.44026

depth_primary
: 0.90

depth_secondary
: 0.055

duration_h
: 7.54

source_path
: Fable 5\made\sb1-scale\pervet\vet344947332\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-18)

Archived source: sanitized report text.

TIC 344947332 (TYC 4004-778-1): per-object vet

Opus-model subagent for Fable 5, 2026-10-18 (saved by Fable from the agent's final message; the write tool refused in-agent). Batch row: G 11.46, P 10.45153 d, 9.0 ppt, 9 events, dphi +0.007.

Verdict: CONFIRMED. An M dwarf (about 0.16 Msun, 0.19 Rsun) transits an evolved F star (6200–6400 K, about 2 Rsun) every 10.451460 d. The transit is total and flat-bottomed, 9.0 ppt deep, and falls on Gaia's conjunction A. A **new 0.55 ppt secondary eclipse at phase 0.4845** is the M dwarf being hidden; its depth implies T2 ≈ 3200 K, right for a 0.16 Msun dwarf.

1. Archives - `cats.py --r 60` with APPOINTMENTS_DATA set: no VSX or ASAS-SN entry; nothing in Prša, IJspeert or Kostov t2/t3/t4. SIMBAD: TYC 4004-778-1 [SB*]. - Gaia DR3: no vari_summary or vari_eclipsing_binary row. NSS SB1: P 10.45395±0.00724 d, e 0.081±0.082, ω 351±57°, K1 12.41±1.00 km/s, RUWE 0.86. - SB9, WDS and J/AJ/160/111: nothing. ExoFOP: no TOI, CTOI, imaging or spectroscopy. Web search: nothing. Uncatalogued as a variable.

2. Eclipses (raw QLP SAP_FLUX, QUALITY==0, no trimming) - All 9 events measured, S/N 18–43. Weighted depth 8.99±0.08 ppt; even 9.03±0.10, odd 8.91±0.14. A 10th event (BTJD 1773.42, s17) falls entirely in flagged cadences. - Shape (2-min sectors): T14 7.54 h, ingress 54 min, flat bottom 5.73 h. V shape rejected at $\Delta\chi^2 > 700$. - Geometry from the shape: k 0.095, b ≈ 0.55, a/R1 ≈ 10.1. - Background: SAP_BKG regression accounts for ≤ 0.11 ppt in every sector; in-eclipse centroid shift < 1 mpix; no event

near a listed systematic time. - Blind dip census: the 6 off-ephemeris dips are ≤ 1.3 ppt or sit on s024's post-gap scattered-light ramps (`look_s024.png`). - **Secondary eclipse:** 0.550±0.083 ppt at phase 0.4845 (0.4840–0.4855) in s57/77/84, $\Delta\chi^2$ 43. All 6 individual events positive; control phases give at most |S/N| 2.8; the 30-min sectors (s17+s24) show it independently at 0.50±0.16 ppt.

3. Ephemeris and Gaia - P = 10.451460±0.000014 d, T0 = 2954.44026±0.00080 BTJD, O–C rms 3.2 min. Against Gaia's P: **−0.34σ**. - Jackknife: dropping any one event moves P by ≤ 1.0e−5 d and depth by ≤ 0.1 ppt; the conjunction-A offset stays at −0.39σ for every drop. Largest timing outlier: s77 E156 (+8.2 min), normal depth and shape. - Monte Carlo with Gaia's correlations: primary on conjunction A at −0.29 to −0.41σ in every sector; B as the primary's location disfavoured at 1.9–4.8σ. Gaia's A-to-B separation 0.536±0.043 vs TESS 0.4845 (−1.2σ). - e cos ω from the secondary's phase: −0.0122±0.0008 vs Gaia +0.057±0.068 (−1.0σ). e sin ω from durations too weak to constrain.

4. Independence - TARS s17/24/57/77/84: eclipses at 7.4–8.6 ppt. TESS-SPOC s24 PDCSAP: 9.53±0.18 ppt. - TESS-SPOC s24 difference image: loss on the target, $\Delta\chi^2 \approx 27{,}800$ better than the next-best star — agreeing with the batch diffimg receipt (s57, 3.4″). - Neighbour dips needed: G 13.21 @ 30″ → 45 ppt; G 14.52 @ 15.6″ → 152 ppt; G 15.72 @ 81″ → 460 ppt; G 16.06 @ 55″ → 625 ppt; G 16.49 @ 11.2″ → 93%; G 16.95 @ 19″ → impossible. All excluded by the image.

5. Physics - f(M) = 0.00205 Msun → m2 = 0.15–0.17 Msun for M1 1.1–1.4 at i ≈ 87°. - Separation ~22–23 Rsun; central transit 7.6–8.0 h, fitting the observed 7.54 h at b 0.55. - R2 = k·R1 = 0.19–0.21 Rsun — right for a 0.16 Msun dwarf. - Surface-brightness ratio J = 0.061 → T2 ≈ 3200–3260 K. - M dwarf in front at conjunction A (deep total transit), F star in front at B (shallow total occultation). One geometry fits everything.

6. Row notes (nothing to strike or demote) - P: 10.451460(14) d (the CSV's 10.45153 is 5σ off at the new precision). T0: 2954.44026(80) BTJD. - The 9.0 ppt depth is correct; describe it as a flat-bottomed total transit. - **Add the secondary: phase 0.4845, 0.55 ppt.** - An F star + M dwarf EB with a planet-like transit; Gaia's K1 rules out a planet. - "9 events" stands; a 10th exists in flagged s17 cadences.

Next predicted eclipses (BTJD) - Primary: 4407.1932, 4417.6447, 4428.0961 (±3 min). - Secondary: 4401.806, 4412.257, 4422.708 (±12 min).

What would settle it further: a 2-min SPOC light curve for a limb-darkened transit fit, and one RV epoch to pin down ω.

A137. TYC 6579-2222-1 (TIC 1260529)

status
: confirmed (per-object re-analysis 2026-10-19; grazing eclipse of a subgiant at conjunction A, 10.85 ppt, 8 events + 1 flagged s099 event at the predicted time; new 0.40 ppt secondary at phase 0.514, independently in SPOC DV at 0.63 ppt; prior art: SPOC TCE s0056-s0069, never a TOI; open 2.5-sigma e sin omega tension)

P_days
: 10.992454

eP
: 0.000008

T0_primary_BTJD
: 2566.15999

depth_primary
: 1.09

depth_secondary
: 0.040

duration_h
: 3.78

source_path
: Fable 5\made\sb1-scale\pervet\vet1260529\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 1260529 (TYC 6579-2222-1, Gaia DR3 5652319283872158336): per-object vet

Opus 5.5 subagent, wave 3, 2026-10-19. Resumed a dead session's folder. Every number below comes from a file in this folder, either produced by the dead session's scripts (`vet-out.txt`, `bkgreg-out.txt`, `cats-out.txt`, `morecats-out.txt`, `mastlc-out.txt`) or by this session (`indep-out.txt`, `tglc-out.txt`, `diff2-out.txt`, `sec-out.txt`, `sec2-out.txt`, `hbtest-out.txt`, `geom-out.txt`, `physics-out.txt`, `conjepoch-out.txt`, `flagged-out.txt`). Downloads are in `C:\Home\sky\data\qlp-sb-p20-60\pervet1260529\`.

Verdict: CONFIRMED (an eclipsing SB1, grazing primary eclipse at Gaia conjunction A, with a shallow secondary)

The deciding numbers: - **Timing locked to the RV orbit.** At the Gaia epoch (BTJD ~389), where Gaia's period error does not propagate, the TESS primary falls **−0.28 h (−0.29σ) from conjunction A** (predicted 389.6657 ± 0.0401 d, MC with corr_vec; ω–Tp corr 0.9994). At the TESS sectors the offset grows from +3.5 h to +11.7 h (all +1.7..+1.9σ). That drift is just the period difference: TESS P 10.992454(8) d vs Gaia 10.990961(780), +1.9σ. - **On target.** The out-minus-in difference image, fitted

with a free-position Gaussian in five TESS-SPOC TPF sectors, puts the source **1.0, 3.6, 2.5, 1.8 and 1.1″ from the target** (s8, 34, 35, 61, 62). If the Gaussian is pinned on the 9.4″ G 15.43 Gaia VARIABLE / VSX ROT neighbour instead, χ^2 is worse by 103, 136, 162, 278 and 164. That neighbour would need an impossible ~50% dip (10.85 ppt × 10^(0.4·4.17) = 0.51). QLP centroids during eclipse do not move: dX −0.02 ± 0.23, dY +0.02 ± 0.12 mpix. A 9.4″ source would shift them by ~5 mpix. SPOC DV gives a difference-image offset of 0.35 ± 2.54″. The batch diffimg (s34) reports peak_on_target. - **Not a heartbeat.** On the same harmonic base, the trapezoid has χ^2 757 (1500 bins). A Kumar heartbeat with Gaia's e/ω is worse by **+2959**; with e free within 3σ of Gaia, by +2953; even with e free up to 0.9 (best e 0.84, ruled out by the RV e = 0.034), by +100. The conjunction lies at ν = 75°, not near periastron. - **Repeats in other pipelines.** SPOC PDCSAP gives 11.17 / 10.99 / 10.91 / 11.49 / 10.48 ppt (s8/34/35/61/62). SPOC SAP gives 9.85–10.15 ppt. TGLC s8 aperture gives 15.8 ppt; TGLC is deeper on both stars of this batch, and its psf_flux is all NaN for this star. QLP SAP gives 10.85 ± 0.12 ppt. - **Prior detection (not a catalogue entry).** SPOC flagged it as a TCE in s61–62: multi-sector DV `s0056-s0069`, P 10.99301 ± 0.00055, epoch 2972.8802, depth 11224 ± 251 ppm, 3.85 h, b 0.97, Rp 36 R⊕. Its weak-secondary test found **0.63 ± 0.17 ppt at 5.663 d = phase 0.515**. ExoFOP lists no TOI or CTOI. There is no VSX/ASAS-SN entry at the target, no SB9, no WDS, nothing in Gaia vari_eclipsing_binary or vari_summary, and nothing in Prša+2022, IJspeert+2024 or Kostov+2025. SIMBAD has it as SB* only. A web search on the TYC and TIC names found nothing.

1. Archives

`cats.py --r 60` (APPOINTMENTS_DATA = $TEMP\hunt-sb1-scale): VSX has only the ROT star Gaia DR3 5652319180792945152 at 9″ (G 15.43). That is a neighbour, not the target, and is excluded above. ASAS-SN: none. SIMBAD: TYC 6579-2222-1 [SB*]. `morecats.py`: SB9 none, WDS none, J/AJ/160/111 none; ExoFOP has no TOIs, CTOIs, imaging, spectroscopy or companions. Gaia TAP (`gaia.json`) gives SB1 P 10.990961(780), e 0.0342(107), ω 14.7(21.7)°, Tp −1.5258(6560), K1 40.59(50) km/s, 15 RV epochs, significance 81, RUWE 1.06, no vari rows. TIC: Teff 5657 K, logg 3.76, R 2.19 R⊙, M 1.0 (likely low for this subgiant), contratio 0.107.

2. Eclipse measurements (QLP SAP, QUALITY==0, untrimmed; `vet-out.txt`)

Global trapezoid: depth 10.96 ppt, T14 3.78 h, T12 1.51 h, flat bottom 0.76 h. The shape is nearly V, as expected for a grazing eclipse. There are 8 primaries, all with S/N 23–37 and depths 10.3–11.5 ppt (constancy χ^2 8.8/7). Odd and even depths are 10.98 ± 0.18 and 10.74 ± 0.16. Background: the median in-minus-out is +1.0 ppt of target, and a fake eclipse would need +80 e^-/s. Regressing on background implies at most 0.06 ppt. None of the 8 events is within 0.5 d of a listed sector-wide systematic. Two windows contain a bit-32 cadence, and their depths are normal. The control phases are all consistent with zero, with no S/N > 3 in any of them. Blind census: 11 dips at S/N ≥ 4; 8 are the eclipses, and none of the off-ephemeris dips is ≥ half the eclipse depth. Events in flagged data only: - s099, E137 (all cadences carry QLP bit 29): **11.49 ± 0.60 ppt at 4072.1205 ± 0.0023**. The ephemeris predicted 4072.1262 (O−C −8 ± 3 min), 2.9 yr after the last Q0 eclipse. - s035 2269.36: 10.8 ± 4.5 ppt, too noisy to use.

3. Ephemeris, secondary, e cos ω / e sin ω

Primary: **P = 10.992454 ± 0.000008 d, T0 = 2566.15999 ± 0.00035 BTJD** (n 8, χ^2/dof 5.0/6, O−C rms 1.1 min). Versus Gaia P: +1.9σ. Secondary (`sec-out.txt`, `sec2-out.txt`; per-epoch local quadratic + trapezoid, inverse-variance stack): - **0.40 ± 0.09 ppt at phase 0.5140 (± 0.0007 by profile; ± 0.0015 across duration scalings)**, S/N 4.5. - Control phases have |S/N| ≤ 2.7. Every sector group is positive: s8 0.65 ± 0.21, s34/35 0.35 ± 0.18, s61/62 0.40 ± 0.14, s99 0.23 ± 0.22. - A shorter duration (T14 × 0.6) is mildly preferred: S/N 5.0, 0.55 ± 0.11 ppt. - SPOC DV independently places its weak secondary at phase 0.515. - **e cos ω ≈ (π/2)(Δφ − 0.5) = +0.022 ± 0.002**, against Gaia's +0.031 ± 0.010 (0.9σ). Gaia predicts Δφ = 0.5196 ± 0.0066; this is 0.8σ from the measurement. - e sin ω cannot be measured from the durations at this S/N. See §5 for what the depth ratio implies.

4. Independence

- Other pipelines: TESS-SPOC (PDCSAP and SAP) for all five SPOC sectors, and TGLC s8 (`indep-out.txt`, `tglc-out.txt`).
- Difference images: the free-position fit in five TPF sectors (`diff2-out.txt`). The older pinned-PSF script (`tpfdiff.py`) is unreliable for single-event sectors (zero scatter); for s61 it gives target χ^2 24903 vs 66361 for the 9.4″ star.
- Neighbours (Gaia G < 17 within 90″) and the dip each would need: 13.28 at 70″ → 69 ppt; 14.34 at 55″ → 184 ppt; 14.64 at 34″ → 243 ppt; 15.36 at 18.8″ → 0.47; 15.43 at 9.4″ (VARIABLE/ROT) → 0.51. Every one of them is 1–3.6″-incompatible with the difference-image source at ≥ 7″ separation.

5. Physics

- Mass function: f(M) = 0.0761 M⊙. With i ≈ 84–86° and M1 1.0–1.4, m2 = 0.57–0.70 M⊙. The companion is an early-M or late-K dwarf (R2 ≈ 0.5–0.6 R⊙, T2 ≈ 3700–4000 K); a = 24–27 R⊙, a/R1 ≈ 11–12.
- The central-chord duration would be 8.6–9.4 h; the observed 3.8 h together with the near-V shape requires a **grazing** chord (b ≈ 1.06–1.11 R1). SPOC DV independently fits b 0.97, k 0.15.
- Conjunction A has the RV primary behind, so the companion transits the primary: the deep eclipse is the primary eclipse. That is what is observed.
- **Tension, worth stating once.** For e sin ω ≈ 0 a grazing geometry gives a secondary of ~2.5–3.2 ppt for this companion, not 0.40 (`geom.py`: limb-darkened two-sphere orbit model). Matching 11 ppt / 3.8–4.2 h primary **and** a 0.4–0.7 ppt, ~2.7 h secondary needs the secondary chord to be more grazing, which means **e sin ω ≈ +0.04** (e ≈ 0.046, ω ≈ 60°). Gaia's e sin ω is +0.008 ± 0.013, so the tension is ~2.5σ (and e, ω are within ~1–2σ).
- The alternative, a much cooler companion (J = 0.037 → T2 ≈ 2800 K), contradicts the mass function. The tension does not bear on whether this is an eclipse (that is settled by timing, centroids and shape). It does bear on the companion's parameters, and an RV e sin ω or a higher-S/N secondary would settle it.

Row fixes for the paper table

field	ledger	fix
P (d)	10.99253	**10.992454 ± 0.000008**
T0 (BTJD)	—	**2566.15999 ± 0.00035**
depth (ppt)	9.3	**10.85 ± 0.12** (QLP SAP trapezoid; SPOC PDCSAP 10.5–11.5)

field	ledger	fix
T14 (h)	—	**3.78** (near-V, grazing)
n events	8	8 Q0 (+1 flagged s099 at the predicted time)
dphi	+0.013	+0.013 to +0.044 across the sectors (Gaia P drift, +1.9σ); **−0.001 (−0.29σ) at the Gaia epoch**
secondary	—	**0.40 ± 0.09 ppt at phase 0.514** (S/N 4.5; SPOC DV 0.63 ± 0.17 at 0.515)
prior art	—	SPOC TCE (DV s0056–s0069, s61–62 data); not a TOI; not in any EB catalogue

Next predicted eclipses (BTJD)

- Primary: **4401.8998, 4412.8923, 4423.8847** (± 2 min). Folding in the s099 flagged event's −8 min O−C would shift these by a few minutes.
- Secondary (phase 0.514, ~0.4 ppt): **4407.5499, 4418.5424, 4429.5348** (± 24 min).

Addendum 2026-10-21 (Fable, 03:30 session): the e sin ω tension re-priced (`esw.py` / `esw-out.txt`)

Two tests on the §5 tension, one negative and one positive: - **It is not a covariance artefact.** The 2.5σ used Gaia's marginal e sin ω error; re-done against Gaia's full (P, e, ω, Tp) covariance, conditioned on the TESS e cos ω (+0.0220 ± 0.0012, 8× tighter than Gaia's), the answer does not move: corr(e cos ω, e sin ω) in the Gaia solution is −0.015, conditional e sin ω +0.0090 ± 0.0126, and the geometry-required +0.044 ± 0.004 sits **+2.65σ** away either way. Joint 2D Mahalanobis of (e cos ω_TESS, e sin ω_needed) under Gaia: 2.76 (p = 0.022). - **The secondary's duration independently supports the grazing reading.** The depth-matching geometry (e sin ω ≈ +0.044) predicts the secondary lasts ~0.71× the primary's T14; Gaia's e sin ω (+0.008) predicts ~1.0×. A fixed-template scan (fT 0.4–1.3, phase profiled) peaks at fT 0.5 (S/N 5.02) and prices the two predictions at **Δ(S/N)² = +3.4 in favour of the short secondary** — depth and duration agree with each other and disagree with Gaia. - Net: the eclipse geometry is self-consistent at **e ≈ 0.049, ω ≈ 63°** (vs Gaia e 0.034 ± 0.011, ω 15 ± 22°: 1.4σ and 2.2σ separately, p = 0.022 jointly). The economical reading is that Gaia's ω is the off quantity, not the secondary. Context: wave 4's 288160281 and wave 3's 260789413 / 430987029 all show the opposite-direction version (Gaia e biased high for near-circular orbits, the Lucy–Sweeney direction); 1260529 is so far the only case where the eclipses want *more* eccentricity than Gaia — which is what makes it worth the DR4 epoch-RV check rather than a bias explanation.

What would settle it further

- An RV e sin ω (or a higher-S/N secondary with a measured duration) to resolve the depth-ratio tension.
- A reprocessing of s099 that unflags the cadences (or a second pipeline for s099), to confirm the 2.9-year forward prediction.
- A spectrum to fix M1 and R1.

A138. TYC 5361-543-1 (TIC 306477840)

status
: confirmed (per-object re-analysis 2026-10-19; flat-bottomed M-dwarf transit of an F star at conjunction A near periastron, 10.15 ppt, 13 events; heartbeat rejected dchi2 +5150; no secondary, geometrically expected to miss)

P_days
: 5.520586

eP
: 0.000003

T0_primary_BTJD
: 2474.27805

depth_primary
: 1.02

duration_h
: 3.72

source_path
: Fable 5\made\sb1-scale\pervet\vet306477840\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 306477840 (TYC 5361-543-1, Gaia DR3 2994311049695999616): per-object vet

Opus 5.5 subagent, wave 3, 2026-10-19. Resumed a dead session's folder. Every number below comes from a file in this folder. The dead session left `vet-out.txt`, `bkgreg-out.txt`, `cats-out.txt`, `morecats-out.txt` and `mastlc-out.txt`. This session added `indep-out.txt`, `tglc-out.txt`, `diff2-out.txt`, `sec-out.txt`, `hbtest-out.txt` (+ `hbtest_306477840.png`), `physics-out.txt` and `conjepoch-out.txt`. Downloads are in `C:\Home\sky\data\qlp-sb-p20-60\pervet306477840\`.

Verdict: CONFIRMED (eclipsing SB1; primary eclipse at Gaia conjunction A; no secondary, as the geometry predicts)

The deciding numbers: - **Heartbeat test (mandatory: conjunction A sits at ν ≈ −11°, near periastron).** Both models share the same const + 3-harmonic base, fitted to 1000 bins (7.9 min); the trapezoid gives χ² 1189. - A Kumar heartbeat with Gaia's e = 0.129 and ω = 101° fixed (periastron phase and i free) is worse by **Δχ² = +5905**. - With e free within 3σ of Gaia (0.02–0.24) and ω free: **+5150**. - Even with e free to 0.9 (best e 0.76, 18σ from the RV e): +227. - The event is **flat-bottomed**: T14 3.72–3.75 h, T12/T14 0.35, flat 1.1 h. A Gaia-geometry heartbeat is a ~1-day-wide hump or dip. - **Timing locked to the RV orbit.** At the Gaia epoch the TESS primary is **+0.28 h (+0.18σ) from conjunction A** (387.4849 ± 0.0661; MC with corr_vec, ω–Tp corr 0.9946). At the TESS sectors the

offset is +1.9 to +5.1 h (+0.6σ). TESS P 5.520586(3) vs Gaia 5.520242(657): +0.5σ. - **On target.** The free-position difference image puts the source **0.7″ (s6) and 0.9″ (s33)** from the target. Pinning it on the 20″ G 15.52 neighbour is worse by Δχ² +1230 / +1810. The batch diffimg (s87) gives 1.63″, peak_on_target. - **Repeats in other pipelines.** SPOC PDCSAP 10.06 ± 0.15 (s6) and 10.60 ± 0.15 (s33); SPOC SAP 9.64 / 10.04; TGLC s6 aperture 11.4 ± 0.2, psf 7.9–13.7 per event; QLP SAP 10.15 ± 0.10 ppt. - **No prior art.** No VSX, ASAS-SN, SB9 or WDS entry; nothing in J/AJ/160/111, Gaia vari_, *or Prša/IJspeert/Kostov; ExoFOP has no TOI or CTOI; no SPOC DV/TCE among its TESS-SPOC products; SIMBAD lists SB* only; a web search on the TYC and TIC names found nothing.

1. Archives

`cats.py --r 60`: VSX none, ASAS-SN none, SIMBAD TYC 5361-543-1 [SB*]. `morecats.py`: SB9, WDS, J/AJ/160/111 none; ExoFOP empty. Gaia TAP (`gaia.json`): SB1 P 5.520242(657), e 0.129(35), ω 100.8(18.9)°, Tp −1.3835(2837), K1 21.99(1.24) km/s, 22 RV epochs, significance 17.8, RUWE 1.39, no vari rows. TIC: Teff 6484 K, logg 3.89, R 2.17 R⊙, M 1.32 M⊙, contratio 0.033.

2. Eclipse measurements (QLP SAP, QUALITY==0, untrimmed; `vet-out.txt`)

Global trapezoid: depth 10.26 ppt, T14 3.72 h, T12 1.30 h, flat 1.12 h. A V shape is rejected (T12/T14 = 0.5 is worse by Δχ² +103). There are 13 primaries across s6/s33/s87, all with S/N 18–37 and depths 9.3–11.0 ppt. - Depth constancy is χ² 22.1/12, driven by a sector offset: s6 9.65 ± 0.17, s33 10.75 ± 0.19, s87 10.14 ± 0.17. SPOC shows the same sign (s6 10.06, s33 10.60), so this is per-sector aperture dilution, not astrophysics. - Odd and even depths are 10.11 ± 0.15 and 10.18 ± 0.14. - Background regression implies ≤ 0.05 ppt. Centroid shifts in eclipse are −0.6 / −0.7 mpix (a 20″ source would give ~10 mpix). - Two events need comment: - E400 (3677.77): 25 flagged cadences and a background excursion (+100 ppt of target), yet its depth (9.68) is normal. - E402 (3688.81): coincides with the listed s087 systematic at 3688.8, and its depth is normal (10.16). Every other event is clean. Two windows contain a bit-32 cadence. - Control phases have no S/N > 3. Blind census: 14 dips at S/N ≥ 4; 13 are the eclipses, and the one other (1.7 ppt) is under half the eclipse depth.

3. Ephemeris, secondary, e cos ω / e sin ω

Primary: **P = 5.520586 ± 0.000003 d, T0 = 2474.27805 ± 0.00051 BTJD** (n 13, χ²/dof 25.1/11, errors scaled × 1.51, O−C rms 2.6 min). **No secondary.** - Gaia predicts Δφ(B−A) = 0.486 ± 0.026. - The best peak anywhere in 0.40–0.60 is S/N 2.1, below the controls (`sec-out.txt`). At phase 0.500: +0.06 ± 0.09 ppt. - The 3σ limit is ≲ 0.3 ppt over the Gaia window. So e cos ω and e sin ω cannot be measured photometrically. Gaia's e sin ω = +0.120 ± 0.035 is what makes the secondary miss (§5).

4. Independence

- Other pipelines: TESS-SPOC PDCSAP/SAP (s6, s33) and TGLC (s6), all showing the eclipse at the QLP ephemeris.
- Difference images: two TPF sectors (`diff2-out.txt`); the pinned-PSF `tpfdiff.py` also prefers the target (χ² 1210 vs 6031 / 1130 vs 3746).
- Neighbours (G < 17 within 90″) and the dip each would need:
- 15.52 at 20″ → 0.44, and excluded by the difference image.

- 13.96 at 84″ → 103 ppt.
- All others need ≥ 0.6 and lie ≥ 32″ away.

5. Physics

- Mass function: f(M) = 0.00593 M⊙. For M1 = 1.32, m2(i ≈ 86°) ≈ 0.24 M⊙, so the companion is an M dwarf (R2 ≈ 0.25 R⊙, k ≈ 0.115, T2 ≈ 3300 K). a ≈ 15.2 R⊙, a/R1 ≈ 7.0.
- Conjunction A (primary behind) is where the dip is, so this is the companion transiting the F star.
- At conjunction A the separation is r = a(1−e²)/(1+e sin ω) ≈ 0.87a. Central duration ≈ 5.9 h, so the observed 3.72 h gives **b ≈ 0.87**. That predicts T23/T14 ≈ 0.26 (observed flat/T14 = 0.30) and a depth of ≈ k² × LD ≈ 11 ppt (observed 10.2–10.8 after dilution). One geometry fits depth, duration and shape.
- **Secondary.** At conjunction B, b_B = b_A (1+e sin ω)/(1−e sin ω) ≈ 0.87 × 1.27 ≈ 1.10, against 1+k = 1.115. The occultation is at most a sliver, and even a full occultation of a 3300 K dwarf behind a 6484 K star would be only k²J ≈ 0.8 ppt. The null is the expected outcome.

Row fixes for the paper table

field	ledger	fix
P (d)	5.52057	**5.520586 ± 0.000003**
T0 (BTJD)	—	**2474.27805 ± 0.00051**
depth (ppt)	9.4	**10.15 ± 0.10** (QLP SAP trapezoid; per-sector 9.65–10.75; SPOC PDCSAP 10.06–10.60)
T14 (h)	—	**3.72** (flat-bottomed, T12 1.30 h)
n events	13	13 (unchanged)
dphi	+0.014	+0.014 to +0.039 across the sectors (+0.6σ); **+0.002 (+0.18σ) at the Gaia epoch**
secondary	—	none; < ~0.3 ppt (3σ), geometrically expected to miss (b_B ≈ 1.10)
heartbeat	—	rejected, Δχ² +5150 (Gaia-consistent e)

Next predicted eclipses (BTJD)

- Primary: **4400.9626, 4406.4832, 4412.0037** (± 2 min).
- Secondary (Gaia Δφ 0.486 ± 0.026; expected undetectable or a sliver): **4403.6456, 4409.1662, 4414.6867** (± 3.5 h).

What would settle it further

- It is already settled as an eclipse.
- RV-quality e and ω (or a detected sliver secondary) would pin b_B and the companion temperature.
- A spectrum would refine M1, which fixes m2.

A139. BD-13 3646 (TIC 1695417)

status
: confirmed (per-object re-analysis 2026-10-19; transit of a ~0.14 Msun M dwarf at conjunction A, 9.3 ppt; 5 events not 3, two recovered from QLP-flagged cadences via SPOC/TESS-SPOC; jackknife clean)

P_days
: 17.797939

eP
: 0.000010

T0_primary_BTJD
: 2326.38398

depth_primary
: 0.93

duration_h
: 4.44

source_path
: Fable 5\made\sb1-scale\pervet\vet1695417\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 1695417 (BD-13 3646, TYC 5542-763-1, Gaia DR3 3524875747112977920): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. This finishes a vet whose first session died after the vet.py pass; everything below is backed by files in this folder. Batch row: G 10.07, P 17.79741 d, 9.5 ppt, 3 events, dphi +0.017.

VERDICT: CONFIRMED. The companion transits the primary at Gaia conj A. It is a ~0.14 Msun M dwarf, and the geometry checks out against Gaia's e and ω. There are five events, not three, and none of them carries the result.

Star and orbit - G 10.08, Tmag 9.64. TIC: Teff 5889 K, R 1.60 ± 0.08, M 1.07, d 193 pc. RUWE 1.14. - Gaia NSS SB1 (TAP, with corr_vec): P 17.792314 ± 0.007005 d, e 0.143 ± 0.026, ω 66.8 ± 11.7°, K1 9.89 ± 0.25 km/s, significance 40. - f(M) 0.0017 Msun → m2(i=90°) 0.136 Msun.

1. Archives (`cats-out.txt`, `morecats-out.txt`, `gaiatap-out.txt`, all on disk) - `cats.py --r 60`: VSX and ASAS-SN none. SIMBAD: BD-13 3646 [SB*] only. Prša, IJspeert and Kostov t2/t3/t4: none. - SB9, WDS and J/AJ/160/111: none. - Gaia vari_summary and vari_eclipsing_binary: empty. - ExoFOP: no TOI or CTOI, no imaging, no spectroscopy, no companions. - Web search on the BD, TYC and TIC names (redone this session): nothing. - Not previously catalogued as an eclipsing binary.

2. Eclipses (`vet-out.txt`, `pipes-out.txt`, `pipes.png`, `look.png`) - **Events:** the three QUALITY==0 events in QLP SAP are s037 2326.384, s046 2557.758 and s046 2575.555, at 9.37–9.78 ppt and S/N 61–88. Two more events sit in cadences QLP flags: - **s010 1578.8705**, seen in SPOC 2-min PDC (8.83 ± 0.13 ppt, ±0.7 min), TESS-SPOC (8.62), TARS (7.91), TGLC aperture (8.58) and ELEANOR-lite (8.66). QLP's own s010 flux there is flagged and unusable. - **s037 2308.586**, in the sector-start ramp: QLP flagged 9.36 ± 0.15, TESS-SPOC PDC 9.27 ± 0.13, TARS 10.48. - The s091 primary (3785.8) falls in the mid-sector gap. - **Shape:** T14 4.44 h, T12 1.11 h (T12/T14 0.25), flat for 2.2 h. V-shaped ($\Delta\chi^2$ +342) and boxier (+563) fits are both worse. - **Depth:** weighted 9.34 ± 0.06 ppt over the five events. QLP gives 9.4–9.8 and SPOC/TESS-SPOC PDC 8.6–9.7. - **Odd/even:** 9.54 ± 0.08 vs 9.47 ± 0.16. - **Systematics:** - No bit-32 cadences in any window, and no event within 0.5 d of a listed sector-wide systematic. - In-minus-out background is −1.2 ppt of the target — the wrong sign for a background fake, which would need +199 e-/s. - Centroids dX −0.30 ± 0.10, dY +0.37 ± 0.50 mpix. - **Controls:** all 7 phases within ±0.07 ppt, none at S/N > 3. - **Blind census:** every eclipse is a census peak, and no off-ephemeris dip is ≥ half the depth (largest 1.27 ppt). - **Secondary:** - The phase-0.185 scan peak (0.74 ppt, S/N 11.5) comes from one s091 window at 3789.11, at a gap edge. The other three windows give 0.00 ± 0.08 (`xcheck-0185.txt`), so it is a systematic. - At Gaia conj B (phase 0.5348): −0.22 ± 0.05 ppt, systematics-dominated (χ^2 27.7/3). SPOC 2-min gives +0.16 ± 0.13. No occultation.

3. Ephemeris, jackknife, Gaia (`jack-out.txt`) - **All 5 events:** P = 17.797939 ± 0.000010 d, T0 = 2326.38398 ± 0.0002 BTJD. χ^2 0.4/3; O−C within ±0.5 min. - **vs Gaia:** P is +0.0056 d = +0.80σ from Gaia. The eclipses sit +0.028 phase (+0.67σ) from Gaia's predicted conj A, and 0.494 phase (11.5σ) from conj B. - **Jackknife (drop each of 5):** P stays within 17.797933–17.797942, depth 9.22–9.49 ppt, conj A +0.026 to +0.030 phase (all within 0.68σ). - **No single event decides the verdict.** - **Ledger's 3 events only:** P 17.797933 ± 0.000030. Dropping s037 E(2326) leaves the s046 pair one cycle apart, and P loosens to 17.79746 ± 0.00081, still +0.73σ from Gaia. - So among the original three, the s037 event set the period's precision. The two recovered events remove that dependence. - **e cos ω:** only one eclipse kind, so not measurable. - **e sin ω:** the duration gives the speed factor at the conjunction. With R1 = TIC that requires e sin ω ≈ +0.15 ± 0.06, against Gaia's +0.130 ± 0.031 (−0.3σ) (`phys2-out.txt`).

4. Independence - **Five pipelines see the dip:** SPOC 2-min, TESS-SPOC (s10/s37/s46), TARS (s10/37/46), TGLC and ELEANOR-lite (`pipes-out.txt`). - **Difference images, PRF-free** (`tpfdiff2-out.txt`): - s37: difference and out-of-eclipse centroids agree to 1.3"; per-pixel depth 5.9–10.7 ppt across the 9 brightest pixels. - s46: 0.8"; per-pixel depth 7.7–9.9 ppt, uniform, which is what you see when the target is the source. - s10 (one event): centroids agree to 0.8–2.0". The core pixel is anomalously shallow (2.3 ppt vs 9–15 around it) in both the SPOC 2-min and the FFI TPF. That looks like image motion during the flagged s10 stretch, not an offset source. - The Gaussian-PSF fit (`tpfdiff.py`) is unstable with 1–2 events and is not used. - Batch diffimg s46: peak on target, deficit–light 0.65". - **Neighbours** (dip each would need): G 15.14 @ 48" → ~99%; G 16.49 @ 35" → 340%, both impossible; G 13.63 @ 80" → 25%, but excluded by the difference images. Nothing brighter than G 15 within 60".

5. Physics (`phys_ecc` runs, `phys2-out.txt`) - **Orbit:** a = 30.5 Rsun. Conj A is at ν = 23° (r = 0.866a). Central T14 there would be 6.9 h; the observed 4.44 h gives b ≈ 0.76–0.83 and i ≈ 87.1°. - **Radius test:** solving the transit shape (Seager & Mallén-Ornelas) and applying Gaia's e and ω gives R1 = 1.59 (1.54–1.64) Rsun against TIC 1.60 ± 0.08. 99.9% of the Gaia draws land within 2σ. - **Companion:** k ≈ 0.10, so R2 ≈ 0.15–0.17 Rsun, which matches m2 = 0.136 Msun. - **Which star is in front:** the companion transits the primary at conj A. The uniform-disc prediction is 10.8 ppt; limb darkening at b ~0.8 brings it near the observed 9.3. - **Conj B:** the projected separation is ≈ R1+R2 (graze or miss). A full M-dwarf occultation would only be ≲0.5 ppt anyway, so the nondetection is expected. - **Heartbeat check** (conj A is 23° from periastron) (`heartbeat-out.txt`, `heartbeat.png`): a Kumar-family pulse loses to the

trapezoid by $\Delta\chi^2$ +700 even at e free (best e 0.87, which Gaia excludes), and by +18,065 at e ≤ 0.22. With e = 0.14 a tidal pulse would be days wide; the event is a 4.4 h flat-bottomed transit. **Not a heartbeat.**

6. Verdict, row fixes, predictions - **CONFIRMED.** Deciding numbers: 1. Five events in five pipelines, and the jackknife moves P by ≤ 9e-6 d with conj A held at +0.67σ. 2. P +0.80σ from Gaia; the eclipse sits on Gaia conj A. 3. The photometric R1 with Gaia's e and ω is 1.59 vs TIC 1.60. 4. Difference images on target; heartbeat rejected ($\Delta\chi^2 \geq 700$). - **Paper-table row fixes:** - n_events 3 → **5** (2 in QLP-flagged data, recovered with SPOC/TESS-SPOC). - P → **17.797939 ± 0.000010**. - Depth 9.5 → **9.3 ppt**: QLP 9.4–9.8; PDC 8.6–9.7. - dphi: +0.017 at the batch epoch, +0.028 at T0 2326 (sign confirmed +). - Note: transit of an M dwarf (~0.14 Msun, ~0.16 Rsun) at conj A. - **Next primaries** (±2 min): BTJD **4408.7428, 4426.5407, 4444.3387**. - **Conj B** (no eclipse expected, at most a sub-ppt graze; primary + 0.5348 P ± 0.26 d): BTJD **4418.26, 4436.06, 4453.86**. - **To settle further:** RVs would pin m2 and i, and a high-cadence transit would give limb-darkened k and b. It already reads as a good low-mass-EB benchmark.

A140. TYC 2741-1694-1 (TIC 430987029)

status
: confirmed (per-object re-analysis 2026-10-19; flat-bottomed transit of a ~0.19 Msun M dwarf on a pulsating gamma Dor-like F star, 8.6 ppt after prewhitening, 5 events; eclipse duration disfavours Gaia's e sin omega at 3.6 sigma; only two sectors)

P_days
: 9.920766

eP
: 0.000016

T0_primary_BTJD
: 3364.52570

depth_primary
: 0.86

duration_h
: 6.96

source_path
: Fable 5\made\sb1-scale\pervet\vet430987029\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 430987029 (TYC 2741-1694-1, Gaia DR3 1905357825924892928): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. This finishes a vet whose first session died before the light-curve pass; everything below is backed by files in this folder. Batch row (c4b promotee): G 11.46, P 9.92091 d, 10.0 ppt, 5 events, dphi +0.012.

VERDICT: CONFIRMED. A flat-bottomed 8.6-ppt transit on target, at Gaia's period and Gaia conj A, on a pulsating F star. **Caveat:** the eclipse duration is in 3.6σ tension with Gaia's e sin ω. The orbit is probably closer to circular than Gaia's weak e = 0.30.

Star and orbit - G 11.46, Tmag 11.16. TIC (spectroscopic): Teff 7001 K, R 1.88 ± 0.09, M 1.54, d 594 pc. RUWE 1.06; ipd_frac_multi_peak 1. - Gaia NSS SB1 (TAP, with corr_vec): P 9.921763 ± 0.004026 d, e 0.302 ± 0.105, ω 80.3 ± 19.0°, K1 13.4 ± 1.6 km/s, significance 8.2 (weak). - f(M) 0.0021 Msun → m2(i=90°) 0.185 Msun.

1. Archives (`cats-out.txt`, `morecats-out.txt`, `gaiatap-out.txt`) - `cats.py --r 60`: VSX and ASAS-SN none. SIMBAD: [SB*] only. EB tables none (also batch `c4-ebcats-out.txt`). - SB9, WDS and J/AJ/160/111: none. - Gaia vari tables: empty. - ExoFOP: no TOI or CTOI. - Web search on the TYC, TIC and Gaia ids (this session): nothing. - Not previously catalogued.

2. Eclipses (`vet-out.txt`, `puls-out.txt`, `pipes-out.txt`, `pipes.png`, `census2-out.txt`) - **Pulsating base:** the base pulsates at 2.03–2.06 c/d (~11.7 h; 3.1 ppt in s016, 1.1 ppt in s083), plus 2.81 c/d. That is ≈20.2× the orbital frequency, so it nearly survives a fold. - These were prewhitened before measuring; out-of-eclipse rms dropped from 2.7 to 1.6 ppt (s016) and from 2.3 to 2.1 ppt (s083). - **Events:** 5 primaries, all QUALITY==0: s016 1747.445 and 1757.359 (30-min), and s083 3562.943, 3572.862, 3582.781 (200 s). - **Shape:** T14 6.96 h, T12 ≈ 0.7 h, flat bottom 5.6 h. V-shaped fits are worse by $\Delta\chi^2$ +385. - **Depth (prewhitened):** s016 8.81 ± 0.31, s083 8.35 ± 0.13 (8.12/7.39/9.48; χ^2 41.7/2 from residual pulsation, so realistic error ~0.5 ppt). Overall ≈ **8.6 ppt**. - The batch's 10.0 and the raw s016 10–15 ppt were inflated by the pulsation. - **Odd/even:** 8.20 vs 8.21. - **Systematics:** - In-minus-out background is negative (−0.4 to −22 ppt of target, the wrong sign for a fake). A fake would need +42 e-/s. - Centroids dX −0.6 ± 0.2, dY +0.5 ± 0.2 mpix. - One bit-32 cadence near s016 1757.3 is excluded by Q0. No event near a listed sector-wide systematic. - **Blind census, prewhitened:** all 5 eclipses are the top 5 peaks (S/N 18–42). No off-ephemeris dip ≥ half depth; the largest is 3.9 ppt, a pulsation residual. - Raw census had 11 such dips, all s016 pulsation. - **Secondary:** - At Gaia conj B (phase 0.529 ± 0.066): −0.27 ± 0.14 ppt (white). The scan's S/N spread is 4.7×, so the realistic error is ±0.6 ppt. Not detected; the expected M-dwarf occultation is ≲0.5 ppt. - A scan feature at phase 0.675 (1.6 ppt, 4 of 5 windows positive, χ^2 23/4) is ~2.9σ in red-noise units and is the scan maximum over ~900 trials. Not claimed. - It would also need L2 ~1.6 ppt, roughly 3× what a 0.19 Msun companion gives.

3. Ephemeris, jackknife, Gaia (`puls-out.txt`, `jack-out.txt`) - **Prewhitened:** P = 9.920766 ± 0.000016 d, T0 = 3364.52570 ± 0.0009 BTJD. χ^2 4.7/3; O−C +5.3, −4.1, +2.4, −0.1, −1.8 min. - Raw: P 9.920841 ± 0.000082 with χ^2 50/3, from pulsation-driven timing. - **vs Gaia:** −0.23σ. The eclipses sit at Gaia conj A, −0.029 phase (−0.24σ), and 0.44 phase (3.1σ) from conj B. - The batch dphi was −0.012 (`c4-marg-vetsum-out.txt`); the **ledger's +0.012 has the wrong sign.** - **Jackknife (raw):** P 9.920794–9.921042, conj A −0.028 to −0.030 phase, depth 8.27–9.21 ppt. **No event owns the verdict.** - **e cos ω:** one eclipse kind, so not measurable. - **e sin ω:** see Physics.

4. Independence - **TESS-SPOC s16 (the only other pipeline):** PDC 13.97 and 9.23 ppt on the two s016 events (raw, pulsation-inflated, as QLP's raw 14.9/11.0). - **Difference images:** - TESS-SPOC s16 TPF: Gaussian-PSF fit favours the target (χ^2 13,745 vs ≥ 89,317 for any neighbour). The PRF-free centroid offset is 0.12 px (2.6"), and core-pixel depths are 11–13 ppt. - Batch diffimg s83 (3 events): deficit 1.16" from target, peak on the target pixel; depth ap1 8.36 > ap3 7.64 > ap5 7.26 ppt, i.e. it concentrates on the target. - TESScut s83 (`cutcheck-out.txt`): the target's 3×3 dims 6.8 ppt at eclipse vs |≤1.25| at 19 control phases. The G 11.67 star 146" away shows +3.1, inside its own ±2.8 control scatter. - **Neighbours** (depth each would need, `diffimg/430987029/diff_s83.json`): - G 16.50 @ 11.6": 105% (impossible). - G 15.52 @ 30.5": 43%; G 15.00 @ 41.3": 27%; G 16.13 @ 50.3": 75%. All three sit 1.5–2.4 px off and are excluded by the difference images.

5. Physics (`phys2-out.txt`, `phys_ecc` runs) - **Orbit and companion:** a = 23.3 Rsun. k ≈ 0.092, so R2 ≈ 0.19–0.20 Rsun, consistent with m2 = 0.185 Msun. The companion transits the F primary at conj A. - **Duration tension:** Gaia puts conj A at ν ≈ 10° (near periastron), where the central T14 would be only 5.0 h (R1 1.88), shorter than the observed 6.96 h. - Solving the transit shape gives b ≈ 0.42 and a/R1 = 11.0 (circular), i.e. R1 = 2.12 Rsun. - With Gaia's e and ω the implied R1 is 2.85 (2.56–3.21) vs TIC 1.88 ± 0.09; 0.1% of Gaia draws land within 3σ. - Matching the TIC radius needs e sin ω ≈ −0.12 ± 0.04, against Gaia's +0.28 ± 0.10: **3.6σ** (4.8σ if T12 = 1 h). - **Reading:** Gaia's e (a 2.9σ e from a significance-8 solution) and/or ω are overestimated. A near-circular orbit fits the transit with R1 ≈ 2.0–2.1 (NSS parallax gives R1 ≈ 1.97). - **Why the attribution stands:** P agrees to 0.0009 d and the eclipse sits at conj A to 0.24σ, which ties it to the Gaia companion. A conj-B attribution would fix the duration but is 3.1σ off in timing. - **Heartbeat check** (conj A near periastron at Gaia's e) (`heartbeat-out.txt`,

`heartbeat-clean-out.txt`): the trapezoid wins by $\Delta\chi^2$ +200 (raw) / +265 (prewhitened) at e free (best e 0.73), and by +729 / +916 at e ≤ 0.62. A heartbeat cannot make a 5.6 h flat bottom with 0.7 h ingress. **Not a heartbeat.**

6. Verdict, row fixes, predictions - **CONFIRMED.** Deciding numbers: 1. Five flat-bottomed transits, the top 5 census peaks after prewhitening, with no other dip ≥ half depth. 2. On target in three independent image tests: TESS-SPOC s16 TPF, batch s83 diffimg, TESScut s83 controls. 3. P −0.23σ from Gaia, eclipse at conj A −0.24σ, jackknife stable. 4. Heartbeat rejected ($\Delta\chi^2 \geq 200$). - **Caveats for the table:** only two sectors; a second pipeline exists only for s016; the Gaia e/ω duration tension. - **Paper-table row fixes:** - Depth 10.0 → **8.6 ppt** (pulsation-cleaned; s083 8.35 ± 0.5). - P → **9.920766 ± 0.000016**. - dphi sign → **−0.012** (−0.029 at the s083 epoch). - Note: pulsating (γ Dor-like 2.05 c/d) F primary; transit of a ~0.19 Msun M dwarf; Gaia e sin ω disfavoured by the transit duration (3.6σ). - **Next primaries** (±5 min including pulsation jitter): BTJD **4406.2061, 4416.1269, 4426.0477**. - **Conj B** (no eclipse expected; occultation ≲0.5 ppt): - On Gaia's e and ω: primary + 0.529 P ± 0.65 d → BTJD **4411.45, 4421.38, 4431.30**. - If the orbit is near-circular, as the duration suggests: **4411.17, 4421.09, 4431.01**. - **To settle further:** an RV orbit with more epochs (to fix e and ω), and 2-min or 200-s data in a new sector for limb-darkened b and k plus a deeper conj-B limit.

A141. TYC 3093-1294-1 (TIC 116246364)

status
: confirmed (per-object re-analysis 2026-10-19; M dwarf grazing a G subgiant at conjunction A 13 deg before periastron, 11.4 ppt, 12 events not 9; heartbeat excluded dchi2 5391; conjunction B misses by 1.2 Rsun, no secondary; centroids exclude every neighbour)

P_days
: 14.748239

eP
: 0.000011

T0_primary_BTJD
: 2765.23338

depth_primary
: 1.14

duration_h
: 3.12

source_path
: Fable 5\made\sb1-scale\pervet\vet116246364\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 116246364 (TYC 3093-1294-1, Gaia DR3 1344576143791084416): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. A first subagent died after running vet.py; this one resumed from its files on disk. Batch row: G 11.39, P 14.74818 d, 10.4 ppt, 9 events, dphi +0.004.

VERDICT: CONFIRMED. The companion grazes across the primary at Gaia's conjunction A, which is 13° before periastron. A heartbeat is ruled out by $\Delta\chi^2 = 5391$. Conjunction B misses on geometry, and no secondary is seen there.

Star and orbit - G 11.38, Tmag 10.92. TIC: Teff 5559 K, R 1.96 ± 0.11, M 0.98, d 388 pc. Gaia GSP-Phot gives logg 4.08 (TIC 3.85). RUWE 1.15. - Gaia NSS SB1 (full archive row with corr_vec, `gaia.json`): P 14.748178 ± 0.002110 d, e 0.250 ± 0.023, ω 103.0 ± 3.5°, Tp 2.151 ± 0.125 (BTJD 391.151), K1 27.63 ± 0.50 km/s, significance 54.9.

1. Archives - `cats.py` (`cats-out.txt`): VSX, ASAS-SN, Prša, IJspeert and Kostov all none. SIMBAD lists TYC 3093-1294-1 as [SB*] only. - `morecats-out.txt`: no SB9, WDS or J/AJ/160/111 entry. ExoFOP: no TOI or CTOI, no imaging, no spectroscopy. - Gaia vari_summary and vari_eclipsing_binary are empty. - Web search on the TYC, TIC and Gaia ids: nothing. - Not previously catalogued as an eclipsing binary.

2. Eclipses (raw QLP SAP, QUALITY == 0, no trimming; `vet-out.txt`, `events.png`, `fold_oc.png`) - **Primaries:** 12 measured, each S/N 15–33: s025 ×1, s026 ×2, s040 ×2, s052 ×2, s053 ×2, s079 ×1, s080 ×2. - Weighted depth 11.39 ± 0.12 ppt. Constancy χ^2 12.8/11. Per sector 11.08–12.10 ppt. - Odd 11.34 ±

0.18 vs even 11.44 ± 0.17. - **Shape:** global trapezoid T14 3.12 h, T12 1.25 h (T12/T14 0.4), flat 0.62 h. Fully V (T12/T14 0.5) is worse by $\Delta\chi^2$ +34; flat-bottomed (0.1) is worse by +779. - **Systematics:** - No bit-32 cadence in any event window, and no event within 0.5 d of a listed sector-wide systematic. - Background regression (`bkgreg-out.txt`): the largest fake dip background could produce is 0.42 ppt (s052); every other sector ≤ 0.11 ppt. - QLP centroids in eclipse: dX −0.66 ± 0.36, dY −0.62 ± 0.37 mpix. - **Controls:** seven control phases all within ±0.25 ppt, none above S/N 3. - **Blind census:** 28 dips at S/N ≥ 4. 13 are on the primary ephemeris. The 15 off-ephemeris dips are all ≤ 4.7 ppt, so there is no other dip of half the eclipse depth or more. - **Secondary** (own fixed-shape test `sec2.py`; `sec2-out-0.463-*.txt`): - At Gaia's conj B phase, 0.4634 ± 0.0109: depth −0.03 ± 0.10 ppt with a 3.1 h template, +0.05 ± 0.09 ppt with 5.1 h (the expected B/A duration ratio is 1.65). - Best value anywhere in ±0.04 of that phase is S/N 2.1. A same-width scan reaches that S/N at 81–90% of control centres. - **No secondary.** 3σ limit ≈ 0.3 ppt. - `sec.py` crashed (singular matrix). Its earlier S/N 3.7 "secondary" at phase 0.42 is not significant against its own control rms of 2.6.

3. Ephemeris and Gaia - P = 14.748239 ± 0.000011 d, T0 = 2765.23338 ± 0.00040 BTJD (E = 52). χ^2 17.0/10 (errors scaled ×1.31). O−C rms 1.7 min. - vs Gaia P: +0.000061 d = +0.03σ. - Gaia conjunctions (Monte Carlo of 20k draws with the full covariance): the eclipses sit on **conj A**, 1.53–1.66 h early. That is −0.0045 in phase, or −0.14 to −0.29σ. - Conj B is predicted at phase 0.4634 ± 0.0109 from A, and the eclipse is 15–27σ away from it. - True anomaly at the eclipse is 343.9° (Gaia mean elements) against 347.0° expected for conj A. - Only one eclipse kind, so no photometric e cos ω or e sin ω. Gaia: e cos ω −0.056 ± 0.017, e sin ω +0.243 ± 0.022.

4. Independence - **TESS-SPOC PDCSAP**, all seven sectors (`indep-out.txt`): 11.50, 11.62, 11.25, 11.54, 10.93, 11.07, 11.78 ppt (±0.2–0.37). SPOC-SAP gives 10.60–11.36 ppt. - **TASOC s026 FLUX_RAW** (`tasoc-out.txt`): 10.88 ± 0.48 and 10.18 ± 0.64 ppt. - **TESS-SPOC TPF difference images**, s40/s53/s80 (`tpfdiff-out.txt`): the target is the best single-star fit, with χ^2 1.7–2.4× lower than any neighbour. Implied own dip 8.9–9.7 ppt. - **Own centroid-shift test** (`censhift.py`, all seven TPFs, `censhift-out.txt`): - Measured in-eclipse shifts are all ≤ 1.5 mpix (typical ±0.2–0.8). - Each neighbour would produce 20–50 mpix and is excluded at χ^2 840–50,000 per sector. - The target hypothesis gives χ^2 0.9–66. The residual comes from a 0.1–0.3 px WCS/aperture offset of c0, not from a neighbour. - **Batch diffimg** (s40): peak on target, offset 3.0". - **Neighbours** (fractional dip each would need = 11.4 ppt × 10^(0.4 Δm)): - G 14.78 @ 88": 260 ppt - G 14.81 @ 89": 265 ppt - G 16.23 @ 48": 990 ppt - G 16.36 @ 44": > 100% (impossible) - All excluded by the centroids anyway.

5. Physics (`phys-out.txt`, `transfit-out.txt`, `hbtest-out.txt`) - f(M) = 0.0293 Msun. With M1 0.98, m2(i = 90°) = 0.378 Msun (an M dwarf, TESS-band light ratio about 0.5%). a = 28.0 Rsun. - Separation at conj A (ν 347°) is 0.753a (central T14 would be 7.4 h). At conj B (ν 167°) it is 1.240a. - The observed 3.12 h chord implies i ≈ 84.1°. At B the projected separation is 3.55 Rsun against R1 + R2 = 2.38 Rsun, so **B misses by 1.2 Rsun**. That matches the absent secondary. - **Limb-darkened eclipse fit** of the stacked short-cadence primaries, with v_t fixed at 15.34 Rsun/d: - R1 = 1.50–1.75 Rsun (1σ). The TIC R1 of 1.96 is disfavoured by $\Delta\chi^2$ +62. - The k–b graze is degenerate: R2 0.18–0.67 Rsun at b 0.90–1.24. That range includes the R2 ≈ 0.38 an 0.38 Msun dwarf would have. - The smaller R1 agrees with Gaia GSP-Phot logg 4.08, which gives R ≈ 1.5 Rsun at 0.98 Msun. The tension is with the TIC luminosity radius. It is worth a footnote, not a strike. - **Heartbeat check** (conj A is 13° before periastron and e = 0.25, so this is required; `hbtest.py`, `hbtest.png`): - Kumar et al. tidal model with free e (±3σ), t_peri (±0.6 d), i and φ against the trapezoid, same per-event quadratic baselines, 13 windows of ±1.2 d. - Best heartbeat: χ^2 6734. Trapezoid: χ^2 1343. **$\Delta\chi^2$ = +5391 for the trapezoid.** - Heartbeat plus trapezoid gives depth 11.17 ppt and improves χ^2 by only 3.4. - Physical tide scale is $(m2/M1)(R1/r_p)^3$ = 0.31 ppt,

spread over the ~54 h periastron passage. The observed event is 11.4 ppt in 3.1 h. - **Not a heartbeat.** - **Which star is in front:** conj A, so the M dwarf grazes across the G subgiant. The companion's occultation at B does not happen.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. P +0.03σ from Gaia, with the eclipse on Gaia conj A at −0.14 to −0.29σ. 2. Heartbeat excluded by $\Delta\chi^2$ 5391. 3. Three pipelines agree: QLP 11.4, TESS-SPOC PDCSAP 10.9–11.8, TASOC 10.2–10.9 ppt. The centroids exclude every neighbour at $\chi^2 \geq 840$. - **Paper row fixes:** - n_events 9 → **12**. QLP s025/s026 add three; the batch sapdepth only used s040 onward. - depth 10.4 → **11.4 ppt** (trapezoid). - P → **14.748239(11) d**. - dphi sign is **−0.0045** (eclipse earlier than conj A), not +0.004. - Primary-only system; no secondary (< 0.3 ppt, 3σ), as the geometry predicts. - **Next primaries** (±2 min): BTJD **4402.288, 4417.036, 4431.784**. - **Conj B** (no eclipse expected; ±0.16 d): BTJD 4409.12, 4423.87, 4438.62. - **To settle further:** a spectroscopic radius or logg for the primary (TIC 1.96 vs eclipse 1.5–1.75 Rsun), and an RV epoch at conj A.

A142. TYC 2971-946-1 (TIC 98952099)

status
: confirmed (per-object re-analysis 2026-10-19; late-M dwarf transiting an F subgiant at conjunction A, 11.05 ppt, 11 events; new 0.65 ppt secondary at phase 0.5012, T2 ~3000 K; the G 14.23 neighbour 9.5 arcsec away is excluded by in-eclipse centroids and must not be catalogued as an eclipser itself)

P_days
: 5.955676

eP
: 0.000004

T0_primary_BTJD
: 2444.81721

depth_primary
: 1.11

depth_secondary
: 0.065

duration_h
: 3.72

source_path
: Fable 5\made\sb1-scale\pervet\vet98952099\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 98952099 (TYC 2971-946-1, Gaia DR3 9260254304019905 28): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. A first subagent died after running vet.py; this one resumed from its files on disk. Batch row: G 11.46, P 5.95570 d, 10.4 ppt, 11 events, dphi +0.023.

VERDICT: CONFIRMED. A late-M dwarf transits an F subgiant at Gaia conj A. NEW: its occultation is detected at phase 0.5012 (0.65 ppt, S/N 6.1). The G 14.23 neighbour 9.5" away is excluded by in-eclipse centroids in all three sectors.

Star and orbit - G 11.46, Tmag 11.06. TIC: Teff 6070 K, R 2.23 ± 0.12, M 1.13, d 543 pc, contratio 0.085. RUWE 1.08. GSP-Phot logg 3.83. - Gaia NSS SB1 (full archive row with corr_vec, `gaia.json`): P 5.956743 ± 0.000662 d, e 0.033 ± 0.044, ω 3 ± 58°, Tp −0.394 ± 0.947, K1 16.03 ± 0.61 km/s, significance 26.5.

1. Archives - `cats.py` (`cats-out.txt`): VSX, ASAS-SN, Prša, IJspeert and Kostov all none. SIMBAD lists TYC 2971-946-1 as [SB*] only. - `morecats-out.txt`: no SB9 or WDS entry. ExoFOP: no TOI or CTOI. - Gaia vari tables are empty. - Web search on the TYC, TIC and Gaia ids: nothing. - MAST CAOM lists two SDSS/eBOSS spectra at this position (plates 3676 and 8276, MJD 55186 and 57067). Those are possible extra RV epochs, not EB prior art. - Not previously catalogued as an eclipsing binary.

2. Eclipses (raw QLP SAP, QUALITY == 0; `vet-out.txt`, `events.png`, `fold_oc.png`) - **Primaries:** 11 measured (s020 ×4, s047 ×4, s060 ×3), each S/N 28–37. - Weighted depth 11.05 ± 0.10 ppt. Constancy χ^2 17.3/10. Per sector 10.73 / 10.99 / 11.49. - Odd 11.00 ± 0.15 vs even 11.08 ± 0.14. - **Shape:** trapezoid T14 3.72 h, T12 1.30 h (T12/T14 0.35), flat 1.12 h. Fully V is worse by $\Delta\chi^2$ +128, so the bottom is flat. - **Systematics:** - One event window (s047 E124) contains a bit-32 cadence; its depth of 10.38 ppt is in family. - No event near a listed sector-wide systematic. - Background regression: the largest fake dip background could produce is 0.05 ppt. - QLP centroids in eclipse: dX 0.00 ± 0.19, dY −0.14 ± 0.30 mpix. - **Controls:** off-eclipse phases within ±0.15 ppt. The one exception is phase 0.5, at +0.67 ± 0.10 ppt: the secondary. - **Blind census:** 14 dips at S/N ≥ 4. 11 are the primaries. The other three are ≤ 1.95 ppt. - **Jackknife** (`sec-out.txt`): - Dropping any one event moves P by ≤ 5.4 ppm-d, and each dropped event's O–C sits within 2σ of the rest. - Dropping a whole sector leaves P in 5.955673–5.955684. - **Secondary — NEW** (own fixed-shape test `sec2.py`; `sec2-out-0.500-3.7h.txt`, `sec2_0.500_3.7h.png`): - Depth 0.647 ± 0.105 ppt (S/N 6.1) at phase 0.5012 ± 0.0008, measured from a fine scan. - The look-elsewhere scan reaches S/N 6.1 at 0 of 39 control centres (95th percentile 1.98). Random-phase S/N rms is 1.02. - 9 of 10 events are positive. Per sector: s020 0.75 ± 0.17, s047 0.44 ± 0.19, s060 0.73 ± 0.19. Constancy χ^2 4.4/9. - `sec.py`'s free-shape global fit had wandered to a 12.8 h template and missed it. Do not cite that file's "no secondary".

3. Ephemeris and Gaia - P = 5.955676 ± 0.000004 d, T0 = 2444.81721 ± 0.00034 BTJD (E = 100). χ^2 10.6/9. O–C rms 1.6 min. - vs Gaia P: −0.001066 d = −1.61σ. - Gaia conjunctions (Monte Carlo with the full covariance): the eclipses sit on **conj A**, at −1.04σ (s020), −1.32σ (s047) and −1.39σ (s060). In phase that is −0.027, −0.051 and −0.063. The growing offset comes from the −1.6σ P difference. - Gaia predicts phase(B) − phase(A) = 0.516 ± 0.021. Observed: 0.5012 ± 0.0008. - Photometric **e cos ω = +0.0019 ± 0.0013**, against Gaia's +0.026 ± 0.033 (0.7σ). The orbit is circular to within ~0.002 in e cos ω. - e sin ω is not measurable from durations at this S/N.

4. Independence - **TESS-SPOC PDCSAP** (`indep-out.txt`): s20 10.47 ± 0.16, s47 10.31 ± 0.13, s60 9.82 ± 0.17 ppt. SPOC-SAP: 9.4–9.7 ppt. - **Ellipsoidal A2** is +0.16 to +0.31 ppt in SPOC. QLP c2 is −0.285 ppt (minima at the conjunctions). - **TESS-SPOC TPF difference images**, s20/s47/s60 (`tpfdiff-out.txt`): - The target is favoured over the G 14.23 neighbour at 9.5" by $\Delta\chi^2$ 1180 / 1030 / 1960. - The loss-weighted centroid lands on the target pixel position in s20 and s47, and within 0.13 px of it in s60. - The neighbour would need a 66–124 ppt dip of its own. - **Own centroid-shift test** (`censhift.py`, `censhift-out.txt`): this is the decisive test for the 9.5" neighbour. - Measured in-eclipse shifts: s20 (+0.40 ± 0.13, −0.03 ± 0.13), s47 (+0.45 ± 0.10, −0.16 ± 0.13), s60 (+0.34 ± 0.16, +0.08 ± 0.14) mpix. - Predicted if the target owns the dip: (+0.40, +0.44), (+0.40, −0.78), (+0.40, −0.14) mpix, giving χ^2 12, 22 and 2.5. - Predicted if the G 14.23 neighbour owns it: (−3.9, −0.7), (−3.8, −1.7), (−3.9, −1.3) mpix, giving **χ^2 1069, 1956 and 844**. - **Batch diffimg** (s020): peak on target. The offset of 2.6" is a fraction of the 9.5" separation. - **Neighbours** (fractional dip each would need = 11.05 ppt × 10^(0.4 Δm)): - G 14.23 @ 9.5": 141 ppt (excluded above) - G 15.73 @ 35": 565 ppt (centroid $\chi^2 \geq 11{,}700$) - G 16.69 / 16.70 @ 83" / 72": > 100% (impossible)

5. Physics (`phys-out.txt`, `transfit-out.txt`, `transfit-out-contam.txt`) - f(M) = 0.00254 Msun. With M1 1.13, m2(i = 90°) = 0.162 Msun; m2 = 0.164 at the fitted i ≈ 82°. a = 15.1 Rsun. - Circular orbit: central T14 would be 7.3 h; observed 3.72 h. - **Limb-darkened eclipse fit** (v_t 15.93 Rsun/d, LD 0.33/0.23), with no dilution correction: - Best R1 1.90, k 0.120 (R2 0.23 Rsun), b 0.90. The 2σ range is R1 1.90–2.20, R2 ≥ 0.23. - The TIC R1 of 2.23 is $\Delta\chi^2$ +8 worse. - With the TIC contratio of 0.085 applied, the grid runs to a k = 0.49 deep graze at the k edge. That solution is unphysical for f(M) (a ~1 Rsun companion would be about a solar mass and bright). It shows the k–b degeneracy, not a competing answer. - An R2 of about 0.23 Rsun for a 0.16–0.18 Msun dwarf means ~15% inflation, or R1 nearer 1.9: normal-sized tension. - **Secondary as a test of the companion:** - The depth ratio 0.65 /

11.05 corrected for limb darkening gives a TESS-band surface-brightness ratio of about 0.047. - That implies **T2 ≈ 3000 K** (Planck at 0.8 μm, T1 6070 K). This is exactly what a 0.16–0.18 Msun M dwarf should be. - The occulted fraction is ~0.9 of its disc, consistent with the same chord at B (circular orbit, same impact parameter). - **Ellipsoidal:** predicted 0.58 ppt at R1 2.23, or about 0.36 ppt at R1 1.9. Observed 0.16–0.31 ppt. Same star, same order. - **Which star is in front:** conj A, so the M dwarf transits the F subgiant. Conj B, at phase 0.5012, is the M dwarf's occultation. - **Heartbeat:** irrelevant. e ≈ 0, the dip is flat-bottomed and 3.7 h long, and SPOC reproduces it.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. The in-eclipse centroids exclude the 9.5" G 14.23 neighbour at χ^2 844–1956 in each of three sectors, and the difference images agree. 2. Secondary at phase 0.5012 ± 0.0008 (S/N 6.1, look-elsewhere false-alarm rate 0/39). It gives e cos ω +0.0019 ± 0.0013, matching Gaia's near-circular orbit, and its depth gives T2 ≈ 3000 K, matching f(M). 3. P within 1.6σ of Gaia, with the eclipses on conj A at about –1.2σ. QLP, TESS-SPOC PDCSAP and SPOC-SAP agree (9.4–11.5 ppt). - **Paper row fixes:** - depth 10.4 → **11.05 ppt** (QLP trapezoid; TESS-SPOC PDCSAP 9.8–10.5). - P → **5.955676(4) d**. - **Add the secondary: 0.65 ± 0.11 ppt at phase 0.5012.** - n stays 11 primaries; 10 secondary windows are covered. - The dphi sign is **negative**: the eclipse is 0.027–0.063 in phase earlier than Gaia's conj A, growing with time from the P offset. - Note the 9.5" neighbour as excluded, so it must not be catalogued as an eclipser itself. - **Next primaries** (±2 min): BTJD **4404.235, 4410.190, 4416.146**. - **Next secondaries** (±7 min): BTJD **4407.220, 4413.175, 4419.131**. - **To settle further:** RVs from the two archival eBOSS spectra against the Gaia orbit, and a spectroscopic R1 (TIC 2.23 vs eclipse ~1.9 Rsun).

A143. CD-24 8975 (TIC 192973862)

status
: confirmed (per-object re-analysis 2026-10-19; occultation of the M-dwarf companion at conjunction B near periastron, 10.9 ppt, 9 events not 8, i ~87.5 deg; heartbeat excluded dchi2 15029; conjunction A empty as predicted)

P_days
: 19.211924

eP
: 0.000008

T0_primary_BTJD
: 3144.30777

depth_primary
: 1.09

duration_h
: 3.48

source_path
: Fable 5\made\sb1-scale\pervet\vet192973862\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 192973862 (CD-24 8975, TYC 6623-355-1, Gaia DR3 5472790166292593664): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. This finishes the run a first session started; anything that session only stated in prose was redone. Batch row: G 10.80, P 19.21187 d, 10.6 ppt, 8 events, dphi +0.015.

VERDICT: CONFIRMED. There is one eclipse per orbit, at Gaia conjunction B, 13 h after periastron. It is a total-ish occultation of the M-dwarf companion by the G dwarf. Three things back it: - A tidal (heartbeat) model loses to a trapezoid by $\Delta\chi^2$ 15,029. - TESS-SPOC, TGLC and TARS all show the eclipse. - The SPOC difference image puts it on the target.

Conjunction A is empty (0.00 ± 0.07 ppt), which is what the orbit geometry predicts.

Star and orbit - G 10.80, Tmag 10.33, d 164 pc. TIC: Teff 5646 K, R 1.05 Rsun, M 1.00 Msun, a dwarf. RUWE 1.33. - Gaia NSS SB1, full row with corr_vec: P 19.20495 ± 0.00489 d, e 0.3448 ± 0.0102, ω 247.48 ± 2.94°, K1 28.27 ± 0.32 km/s, significance 87. - f(M) = 0.0372 Msun. With M1 = 1.0, m2(i = 90°) = 0.42 Msun, an M dwarf (R2 ~0.46 Rsun, Teff ~3600 K). a = 34.0 Rsun.

1. Archives - `cats.py --r 60`: VSX none, ASAS-SN none. SIMBAD has CD-24 8975 as [SB*] only. Prša, IJspeert and Kostov: none. Batch ebcats: none. - SB9, WDS and J/AJ/160/111: none. Gaia vari_summary and vari_eclipsing_binary: empty. - ExoFOP: no TOI or CTOI, no imaging, spectroscopy or companions. - Web search on CD-24 8975, TYC 6623-355-1 and TIC 192973862: nothing. **Not previously catalogued as an eclipsing binary.**

2. Eclipses (QLP SAP, QUALITY = 0, no trimming; `vet.py`, `sec.py`, `bkgreg.py`, `hbtest.py`) - **Events:** 9, not 8 (s009, s035 ×2, s036, s062 ×2, s089, s099 ×2), each with S/N 13.6–64. - **Shape:** global trapezoid depth 10.98 ppt, T14 3.48 h, T12 1.39 h, flat bottom 0.7 h. A V shape is worse by $\Delta\chi^2$ 2780. - **Depths:** weighted 10.93 ± 0.08 ppt. Depth constancy χ^2 169/8, mostly from s099 (9.6–10.1 ppt, on a steep background) against 11.5–12.0 elsewhere. Odd 11.06 ± 0.10, even 10.72 ± 0.12. - **Systematics:** - Background regression fakes ≤ 0.09 ppt, against a 5.8–7.2 ppt dip in the same windows. - Centroid shifts are –1.1 ± 0.6 and +1.9 ± 0.4 mpix. - 2 events have a bit-32 cadence in the window; none is near a listed systematic time. - **Controls:** all 7 control phases within ±0.2 ppt. - **Census:** all 9 eclipses are S/N ≥ 4 peaks of the census. There are no off-ephemeris dips ≥ 5.5 ppt. The –10 ppt feature at phase –0.17 in the raw fold is an s099 orbit-start ramp (`offeph.png`). - **Heartbeat test** (conj B is 22.5° of true anomaly after periastron; `hbtest.png`): - Trapezoid: χ^2 183 over 160 bins. - Kumar+95 tide + heating + beaming, with Gaia e, ω (±2σ), periastron phase and i free: χ^2 15,212. - Tide plus trapezoid: χ^2 161, with the trapezoid at 11.39 ppt. - The expected tidal amplitude, $(m2/M1)(R1/r_{peri})^3$, is ~0.03 ppt. **Not a heartbeat.** - **Secondary at conj A** (phase 0.5884, T14 6.7 h predicted from the velocity ratio): - 5 epochs give –0.16 ± 0.06 ppt. - The one outlier (s089, 3731.97, –0.96 ppt) sits on the listed 3732 systematic. Without it the result is 0.00 ± 0.07 ppt (3σ < 0.2 ppt). - `vet.py`'s scan best (phase 0.6575, 0.6 ppt, T14 ×2) is 7σ from Gaia's conj A and not confirmed by `sec.py`. **Do not report it.**

3. Ephemeris and Gaia - P = 19.211924 ± 0.000008 d, T0 = 3144.30777 ± 0.00027 BTJD. χ^2 19.8/7 (errors ×1.68), O–C rms 1.5 min. - Jackknife: dropping any event or sector moves P by ≤ 7 ppm-d. Depth stays 10.78–11.26. - vs Gaia P: +0.00698 d = +1.43σ. - Every eclipse is at **conj B**: +7 to +29 h from prediction, +1.0 to +1.3σ. The lag grows with time because Gaia's P is 1.4σ short. Conj A is 0.41 phase away (9–24σ). - One eclipse kind only, so no photometric e cos ω / e sin ω. Gaia gives e cos ω –0.132 ± 0.015 and e sin ω –0.318 ± 0.014.

4. Independence (`indep-out.txt`, `tpfdiff-s0062/0035-out.txt`) - TESS-SPOC PDCSAP: s009 11.95 ± 0.27, s035 11.22 ± 0.15, s036 12.07 ± 0.19, s062 11.89 ± 0.14 ppt. - TGLC PSF (deblended), s009: 11.82 ± 0.40 ppt. - TARS s009–s089: 10.0–14.5 ppt. - SPOC TPF difference images: the PSF fit puts the loss on the target in both sectors. - s062: target χ^2 1.23e5 vs 4.88e5 for either neighbour; max-loss pixel S/N 578. - s035: target χ^2 1.04e4 vs ≥ 2.9e4 for any neighbour. - Batch diffimg (s35): peak on target. - **Neighbours** (dip each would need): G 11.76 @ 62.5″ → 26.5 ppt; G 15.55 @ 85.6″ → 868 ppt (impossible). Nothing else with G < 17 within 90″. The G 11.76 star is rejected by both difference images.

5. Physics (`phys-out.txt`) - **Conjunctions:** - B: ν = 22.5°, r = 0.668a, central T14 4.66 h. - A: ν = 202.5°, r = 1.293a, central T14 9.0 h. - **Inclination:** the observed 3.48 h gives a chord fraction of 0.75 → i = 87.5°. - **Conj A:** the projected separation is 1.95 Rsun vs R1+R2 = 1.51 Rsun, so it misses by 0.43 Rsun. This holds for R1 ±10%, matching the empty conj A. - **Which star is in front:** under the Gaia convention, B means the primary is in front, so the companion is occulted. - Predicted uniform-disc occultation depth: 13–17 ppt, against 10.93 observed. The crude Planck light ratio puts the companion at ~3% of the TESS light; consistent within the model's slack. - The other reading, a transit of the G star by an object this size, would be 40–95 ppt. **Ruled out.** - Depth, duration, flat-ish bottom and the missing A eclipse all fit one geometry.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. On Gaia conj B (+1.0 to +1.3σ), with P +1.4σ from Gaia. 2. Heartbeat excluded, $\Delta\chi^2$ 15,029. 3. Independent SPOC, TGLC and TARS depths of 11–12 ppt. 4. SPOC difference image on target in two sectors. 5. Conj A < 0.2 ppt (3σ), as the geometry predicts. - **Row fixes for the paper table:** - P 19.211924(8) d (CSV 19.21187). - Depth 10.9 ppt (trapezoid; CSV 10.6). - **9 events, not 8.** - Note that it is an occultation of the M-dwarf companion at conj B, near periastron, with i ≈ 87.5°. - Batch dphi +0.015 grows to +0.063 by s100 because of Gaia's P. - **Next eclipses** (conj B, ±1 min): BTJD **4412.2948, 4431.5067, 4450.7186**. - **Conj A** (no eclipse expected; ±~0.01 d from Gaia's e, ω): BTJD 4404.387, 4423.599, 4442.811. - **Periastron:** ~0.56 d before each eclipse. - **To settle further:** an RV epoch to tie the conjunction labels, or a high-precision eclipse to test for a total (flat) occultation.

A144. TYC 3624-860-1 (TIC 200398115)

status
: confirmed (per-object re-analysis 2026-10-19; eclipsing heartbeat binary: ~18 ppt peak-to-peak tidal curve with periastron at phase ~0.25, a 12.0 ppt M-dwarf transit at conjunction A on top (tide alone loses by dchi2 11000+) and a new 2.1 ppt occultation at phase 0.3625, 0.26 sigma from Gaia's conjunction separation; 23 events; the VSX EB at 27 arcsec is an unrelated 0.38-d neighbour)

P_days
: 7.305664

eP
: 0.000049

T0_primary_BTJD
: 3170.06233

depth_primary
: 1.20

depth_secondary
: 0.21

duration_h
: 8.8

source_path
: Fable 5\made\sb1-scale\pervet\vet200398115\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 200398115 (TYC 3624-860-1, Gaia DR3 1986778586111123968): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19. This finishes the run a first session started; anything that session only stated in prose was redone. Batch row (c4b promotee): G 11.09, P 7.30556 d, 11.0 ppt, 22 events, dphi +0.007.

VERDICT: CONFIRMED, as an eclipsing heartbeat binary. The out-of-eclipse light curve is a large heartbeat: ~18 ppt peak-to-peak, with the maximum at periastron, phase ~0.22–0.26. The eclipses sit on top of it: - **Primary at Gaia conj A:** a full transit of the F subgiant by the M-dwarf companion, 12.0 ppt and 8.8 h. Tide alone cannot make it ($\Delta\chi^2$ 11,000–12,750). - **New secondary at conj B:** the companion's occultation, 2.1–2.2 ppt at phase 0.3625. It sits 0.26σ from Gaia's conjunction separation, and its depth matches the predicted 1.2–2.1 ppt.

Both eclipses are on the target, not on the brighter G 9.93 neighbour 38″ away.

Star and orbit - G 11.09, Tmag 10.77, d 1034 pc. TIC: Teff 6726 K, R 4.26 Rsun, logg 3.3, L 33 Lsun, an evolved F star (TIC mass 1.43, likely 1.4–1.8). RUWE 0.94. - Crowded field: TIC contratio 1.15, mainly from G 9.93 at 38.4″. - Gaia NSS SB1, full row with corr_vec: P 7.305697 ± 0.001008 d, e 0.218

± 0.045, ω 192.7 ± 13.4°, K1 31.96 ± 1.52 km/s, significance 21. - f(M) = 0.0230 Msun. m2(i = 90°) = 0.43 Msun for M1 1.43, 0.50 for M1 1.8. a = 19.5–20.9 Rsun.

1. Archives - `cats.py --r 60`: - VSX has one entry, Gaia DR3 1986778483031908992 [E], P 0.38264 d, at 27″. This is the G 15.91 VARIABLE neighbour, unrelated: 7.3056/0.38264 = 19.09 is not commensurate, and it would need an 881 ppt dip. - ASAS-SN none. SIMBAD has TYC 3624-860-1 as [SB*] only. Prša, IJspeert and Kostov: none. - c4 ebcats: none. - SB9, WDS and J/AJ/160/111: none. Gaia vari tables empty for the target. ExoFOP: no TOI or CTOI. - Web search on TYC 3624-860-1, TIC and Gaia ids: nothing. MAST has QLP only; no SPOC, TESS-SPOC, TGLC, TASOC or ELEANOR products. - **Not previously catalogued** as an eclipsing binary or heartbeat star.

2. Light curve (QLP SAP, QUALITY = 0, no trimming; `vet.py`, `pcfit.py`, `persec.py`, `offeph.py`) - **Phase curve** (`fold_oc.png`, `pcfit-k.png`): sharp dip at phase 0; brightening to +9 ppt at 0.22; broad tidal low at 0.30–0.38 ending in a sharp step at 0.38; plateau to phase 1. - **Per-sector folds** (`persec.png`): the step at 0.38 repeats in **7/7 sectors**, including the 2019 30-min s016, at +3.9 to +5.5 ppt. - **Primary, `vet.py` local-linear fits: 23 events, not 22** (3 in s016, 4 in s056, 4 in s057, 3 in s076, 3 in s077, 2 in s083, 4 in s084), each with S/N 6.5–27. - Weighted depth 10.37 ± 0.10 ppt, T14 7.14 h, T12/T14 0.25, flat 3.6 h. A V shape is worse by Δχ² 497. - Depth constancy χ² 96/22; odd and even agree. - Background: median in-minus-out −0.07 ppt of target; a fake would need +77 e-/s. - Centroids +1.1 ± 0.5 and +0.5 ± 0.2 mpix. - Control phases are ±2 ppt because the heartbeat curve is not removed by 3 harmonics. That is expected, not a problem. - **Heartbeat vs eclipse, global model** (`pcfit.py`, 1000 bins; Kumar+95 tide + r² heating + beaming; e, ω, periastron phase, i gridded; trapezoids at conj A and at a free conj-B centre): - Tide only: χ² 14,566 (Kumar basis) / 12,987 (extended r⁴, r⁵ basis). - Tide + eclipse A: 1813 / 1782. **The phase-0 dip is an eclipse**, Δχ² 12,752 / 11,205. - Eclipse A in that model: depth 12.05 ppt, T14 8.77 h, T12/T14 0.35. - Tide + A + B: 1528 / 1519, **Δχ² 286 / 263 for the conj-B eclipse**. B: centre phase 0.3625, depth 2.24 / 2.09 ppt, T14 ~10.5 h. Its ingress is hidden in the tidal decline; the egress is the sharp step. - Tidal-fit geometry: e 0.27–0.30, ω 207–212°, periastron at phase 0.25–0.26, vs Gaia 0.218 ± 0.045 and 192.7 ± 13.4° (≤ 1.8σ with a simplified model). - Tidal amplitude 3–6 ppt (Kumar S), against an expected (m2/M1)(R1/a)³ ≈ 3 ppt. - **Census:** - The 5 deep off-ephemeris dips are artefacts or known features (`offeph.png`): - 3417.14 (s077 sector start, flagged ramp); - 2838.32 (s056 gap with a 40 mpix centroid jump); - 2860.72 (s057 gap edge); - 1744.40 (s016, wing of the primary); - 2836.41 (the conj-B feature). - The many 2–5 ppt census peaks are QLP point noise. QLP rms is 4.3 ppt per cadence here, against 1.8 ppt in my own TESScut photometry. - 1 event is near a listed systematic (3564.5, s083 E249), and its depth (11.3) is normal. - **Bogus vet.py secondary:** the 6 ppt dip its scan found at phase 0.3375 is mostly tidal; the real occultation is 2.1 ppt at 0.3625. Do not report the scan value.

3. Ephemeris and Gaia - P = 7.305664 ± 0.000049 d, T0 = 3170.06233 ± 0.00263 BTJD. χ² 261/21 (errors ×3.53), O–C rms 19.6 min. - The large timing scatter comes from fitting local-linear baselines on a steep heartbeat curve. The heartbeat itself is not a timing signal. - vs Gaia P: −0.03σ. - Every primary is at **conj A**, +0.6 to +0.8 h from prediction (+0.05 to +0.16σ). Conj B is 5.6–11σ away from the primary. - Secondary – primary = 0.3625 phase (±~0.005, two bases agree) vs Gaia 0.3658 (MC median 0.369 ± 0.027): 0.26σ. - With Gaia's e sin ω (−0.046), this gives **e cos ω = −0.2175 ± 0.008**, against Gaia's −0.206 ± 0.042. The secondary's duration is too poorly defined (hidden ingress) for a photometric e sin ω.

4. Independence (TESScut s56 and s76; `tpfdiff-s56/s76-out.txt`, `ownphot-out.txt`) - **Difference images, primary:** the PSF fit prefers the target. - s56: χ² 40.8 vs ≥ 512 for any star; G 9.93 gets 606. - s76: χ² 9.6 vs ≥ 145; G 9.93 gets 173. - Implied own dip on the target: 8.7–8.9 ppt. - Batch diffimg-c4b (s56): peak on target, 2.51″. - **Own aperture photometry** (no other pipeline exists): - Target, 3 px:

primary 8.97 ± 0.16 (s56) and 7.74 ± 0.23 (s76) ppt. Conj-B step +3.75 ± 0.27 and +3.08 ± 0.21 ppt. - G 9.93 neighbour, 2–4 px: primary 0.68 ± 0.10 and 0.13 ± 0.13 ppt (leakage). Step −0.07 and +0.25 ppt. - Larger target apertures get shallower (6.7 / 5.5 ppt), as expected when they pull in neighbour light. - **Both eclipses and the heartbeat live on the target.** QLP's 10–12 ppt is not badly diluted. - **Neighbours** (dip each would need, for 10.4 ppt): - G 9.93 @ 38.4″ → 3.6 ppt: possible in principle, excluded by the images and apertures above. - G 13.82 @ 40″ → 129 ppt; G 14.15 @ 37″ → 174 ppt: excluded by the images. - G 15.22 @ 30″ → 467 ppt; G 15.91 @ 27″ → 881 ppt; fainter stars > 1000 ppt: all impossible.

5. Physics (`phys-out.txt`) - **Conjunctions:** - A: ν = 257°, r = 1.00a, central T14 13.1–13.9 h. - B: ν = 77°, r = 0.91a, central T14 11.9–12.6 h. - **Inclination:** T14(A) 8.77 h gives b = 0.74–0.78 (R1+R2), i ≈ 79–80°. - At A the companion sits fully on the disc: a full transit, matching the flat bottom. - At B the companion is totally occulted: projected separation 3.2–3.3 Rsun < R1−R2. - **Predicted depths:** - Transit at A: 12.05 ppt (M1 1.43), 9.9–15.5 ppt across R1 ±10% and M1 1.43–1.8, against **12.0 observed**. - Occultation at B: 1.15 ppt (m2 0.43, Teff2 ~3600 K) to 2.14 ppt (m2 0.50, ~3900 K), against **2.1–2.2 observed**. That favours M1 ~1.6–1.8. - Periastron falls 0.22–0.26 phase after the primary, where the heartbeat peaks. - One geometry fits both eclipses, their phase separation and the tidal curve.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. Primary at Gaia conj A (+0.05 to +0.16σ), P −0.03σ. 2. Eclipse vs tide-only: $\Delta\chi^2$ 11,000–12,750. 3. Secondary at conj B: $\Delta\chi^2$ 263–286, step in 7/7 sectors, phase separation 0.26σ from Gaia. 4. Difference images and own photometry put both on the target. 5. Depths match an M-dwarf transit and occultation. - **Row fixes for the paper table:** - P 7.305664(49) d (CSV 7.30556). - **23 events, not 22.** - Primary depth 12.0 ppt on the tidal baseline (10.4 ppt local-linear; CSV 11.0). T14 8.8 h. - **Add a secondary:** 2.1 ppt at phase 0.3625, e cos ω −0.2175(80). - Flag it as an **eclipsing heartbeat star** (~18 ppt peak-to-peak tidal curve, periastron at phase ~0.25). The batch eye-read "messy osc + dip / sec at φ 0.35" was this. - The VSX EB at 27″ is an unrelated neighbour, not prior art. - **Next primaries** (conj A, ±13 min): BTJD **4404.7195, 4412.0252, 4419.3309**. - **Next secondaries** (conj B, ±~0.04 d): BTJD **4407.368, 4414.674, 4421.979**. - **Heartbeat maxima** (periastron, ±~0.15 d): BTJD ~4406.6, 4413.9, 4421.2. - **To settle further:** a physical light-curve model (e.g. PHOEBE with tides) to pin i, M1 and the secondary's ingress. Also a TESS-SPOC or TGLC product if one is ever released; QLP is noisy here.

A145. TYC 3344-341-1 (TIC 260789413)

status
: confirmed (per-object re-analysis 2026-10-19; grazing EB, subgiant + 0.64 Msun companion, 14.3 ppt, 18 primaries + 14 secondaries; secondary at phase 0.49996 gives e cos omega 0.0000(3), so Gaia's e = 0.039 is 4.5 sigma off, orbit circular; SPOC TPF difference image on target in 3 sectors)

P_days
: 4.746808

eP
: 0.000003

T0_primary_BTJD
: 3126.22496

depth_primary
: 1.43

depth_secondary
: 0.49

duration_h
: 3.30

source_path
: Fable 5\made\sb1-scale\pervet\vet260789413\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 260789413 (TYC 3344-341-1, Gaia DR3 2066629041855927O4) — per-object vet

Opus 5.5 subagent, wave 3 relaunch, 2026-10-19. Ledger row: G 10.69 | P 4.74683 | 12.2 ppt | 16 events | dphi +0.009.

Verdict: CONFIRMED (eclipsing SB1, detached and grazing, with both eclipses detected; circular orbit)

Deciding numbers: - 18/18 primaries detected individually (S/N 29–50), depth 14.28 ± 0.08 ppt, T14 3.30 h, V/U-shaped (T12/T14 0.4). - Secondary eclipse detected: 4.90 ppt, S/N 39 on the fold, 14 individual events; same duration as the primary. - Photometric P = 4.746808 ± 0.000003 d vs Gaia 4.746792 ± 0.000199 (+0.09σ). - Primary is +1.1–1.3 h (+0.010 phase) from Gaia conjunction A (0.4–0.8σ); secondary –1.4 h from conjunction B (0.4–1.2σ). - Difference images (TESS-SPOC TPFs, s019/s059/s073) put the loss on the target. In s059 the target-PSF fit has χ^2 1082 and the G 10.97 neighbour at 38.5″ has χ^2 44311. The secondary-phase difference image (s059) is also on the target: χ^2

302 vs 2691. - Another pipeline (TESS-SPOC PDCSAP) sees the same eclipse: 14.4 / 14.0 / 13.5 ppt in s019 / s059 / s073. - A heartbeat is excluded. The most generous tidal (Kumar-family) superset fit is worse than the eclipse model by $\Delta\chi^2 = +6141$. With Gaia's e and ω fixed it is worse by +20939.

1. Archives

- `cats.py 260789413 --r 60` (APPOINTMENTS_DATA = $TEMP\hunt-sb1-scale): VSX none; ASAS-SN none; SIMBAD TYC 3344-341-1 [SB*] at 0.0″ only. Not in Prša+2022, IJspeert+2024, or Kostov t2/t3/t4 (batch ebcats: "none of the vetted/unvetted EB tables").
- VizieR: SB9 none; WDS none; J/AJ/160/111 none.
- Gaia DR3: vari_summary, vari_eclipsing_binary and vari_classifier_result are empty. RUWE 1.06; phot_variable_flag NOT_AVAILABLE.
- ExoFOP: no TOI, CTOI, imaging, spectroscopy or companions.
- Web search on "TYC 3344-341-1" / "TIC 260789413": nothing relevant.
- **New eclipsing binary; no prior art.**

Full Gaia NSS SB1 row (TAP, with corr_vec): P 4.746792(199), e 0.039(8), ω 346.4(18.2)°, Tp 1.711(236) d after J2016, K1 55.52(56) km/s, significance 99.5.

2. Eclipse measurements

Inputs: QLP SAP_FLUX with QUALITY==0, no trimming; sectors s019 (30 min), s059, s073 and s086 (200 s). Detrending: a 3-d moving median plus 3 orbital harmonics, with the eclipses masked.

- **Primary (global trapezoid):** depth 14.37 ppt, T14 3.30 h, T12 1.32 h, flat bottom 0.66 h.
- **Primary, per event:** 18 events with timing errors of 0.7–1.8 min. Depth constancy χ^2 61/17. Per-sector depths are 14.64 / 14.61 / 13.78 / 14.01 ppt; the spread is consistent with QLP aperture dilution changing between sectors. Odd 14.14 ± 0.10 vs even 14.44 ± 0.11 ppt.
- **Secondary:** at phase 0.49996 ± 0.00018, depth 4.90 ppt, T14 3.30 h [3.23–3.37]. 14 events, depths 3.7–6.1 ppt.
- **Background:**
- Median in−out SAP_BKG over events is −0.42 ppt-equivalent; a fake dip would need +175 e^-/s.
- bkgreg: the flux change the background could imprint is ≤0.04 ppt in every sector.
- Two s086 events (E386/387) have background jumps of +6 / +23 ppt-equivalent, but their depths are normal.
- **Centroids:** dX −0.5 ± 0.2 mpix, dY +1.9 ± 0.4 mpix.
- **Quality and systematics:** no QUALITY bit 32 in any event window; no event within 0.5 d of a listed sector-wide systematic.
- **Control phases** (0.125–0.875 excluding 0.5): all |depth| ≤ 0.15 ppt, n(S/N > 3) = 0.
- **Blind dip census:** 40 dips with S/N ≥ 4 — 18 are primaries and 13 are secondaries at phase ±0.500. The other 9 are low-S/N s086 features (S/N 4–9). Three are ≥ 7 ppt (3652.909, 3657.679 and 3641.469); the first two sit 0.2 d before the primary eclipses E386/387 in the high-background stretch (the +6 / +23 ppt jumps). No unexplained eclipse-like dips elsewhere.

3. Ephemerides and orbit shape

- Primary: P 4.746808 ± 0.000003 d, T0 = 3126.22496 ± 0.00027 BTJD; χ^2/dof 48/16; O−C rms 1.6 min.

- Secondary: P 4.746810 ± 0.000006, T0 = 2971.95351 ± 0.00081; O−C rms 5.5 min.
- Δφ(sec − pri) = 0.49996 ± 0.00018 → **e cos ω = −0.0001 ± 0.0003**. Gaia gives +0.036 ± 0.008, which predicts Δφ = 0.5231 ± 0.0051: a **4.5σ tension**.
- Duration ratio 1.00 [0.97–1.02] → **e sin ω = 0.000 [−0.015, +0.010]**. Gaia gives −0.009 ± 0.012, which is consistent.
- The orbit is photometrically circular, as expected for a 4.7-d binary with a subgiant primary. Gaia's e = 0.039 (5σ from zero) is very probably spurious (Lucy–Sweeney-type bias or spot RV jitter). Gaia's conjunction times are still met because the conjunctions do not depend strongly on e at this level.

4. Independence and dilution

- **TESS-SPOC (FFI, crowding-corrected PDCSAP):** s019 14.42 ± 0.13 ppt; s059 14.03 ± 0.09; s073 13.48 ± 0.13. SPOC-SAP (not deblended) is 10.9–12.4 ppt.
- **Difference images (TESS-SPOC TPFs, stacked in-eclipse vs local out-of-eclipse):** the max-loss pixel and the loss-weighted centroid coincide with the target in all three sectors. Single-star PSF-fit χ^2 (target vs G 10.97 neighbour): s019 1168 vs 5470; s059 1082 vs 44311; s073 3385 vs 12717. The secondary (s059) is also on target: 302 vs 2691.
- **Batch diffrun (s059):** deficit 2.5″ from the target. The "14″ from the light centroid" receipt is the light centroid being pulled by the neighbour of nearly equal brightness, not an off-target deficit.
- **Neighbour table** (Gaia DR3, within 60″, G < 15), with the fractional dip each would need = 14.3 ppt × 10^(0.4ΔG):

Gaia DR3	G	sep	needed dip	diff-image χ^2 (s059)
206662835466118400	10.97	38.5″	18.4 ppt	44311 (target 1082)
206662526228474880	14.25	49.6″	380 ppt	47241
206662904185592448	14.97	22.3″	740 ppt	36097

The G 10.97 neighbour is the only one that could physically host the signal. The pixels exclude it, and so does the period match: P agrees with Gaia's SB1 to 0.09σ and the conjunction phase agrees to 0.01.

5. Physics

- f(M) = 0.0842 M⊙. With M1 = 1.13 ± 0.15 (TIC): m2(i = 90°) = 0.64 (0.59–0.69) M⊙, a = 14.4 R⊙.
- Central-eclipse T14 would be 7.6 h; the observed 3.3 h means a grazing geometry (i ≈ 79°, b ≈ 1.1 R1 for R1 = 2.36).
- **Which star is in front:** at A (primary, the deeper eclipse) the cooler companion is in front of the F/G primary. At B the primary occults the companion.
- Depth ratio sec/pri = 0.34. Expected TESS-band surface-brightness ratio for a ~0.65 M⊙ dwarf (~4300 K) on a 6050 K primary is ≈ 0.29. Consistent.
- **Mild tensions, both pointing the same way (not a vet issue):**
- The limb-darkened grazing fit only gives a physical, s ≤ 1 solution for a/R1 ≳ 7, i.e. R1 ≲ 2.05 R⊙ vs TIC 2.36 ± 0.22.
- Observed ellipsoidal A2 is ≈ 1.35 ppt (sector range 0.5–2.35, spot-contaminated; PDCSAP 1.4–1.6) vs 3.1 ppt predicted for R1 = 2.36. A2 ∝ $R1^3$, so this also favours R1 ≈ 1.8–2.0 R⊙.
- The A1/A2 harmonics change between sectors, which suggests spots.
- Shape: the LD grazing model beats the trapezoid (χ^2 321 vs 350 on 360 2-min bins).
- **Heartbeat check:**

- The conjunctions are not near periastron (Gaia periastron at phase 0.715), and the photometric e is ≈ 0.
- Free Kumar-family superset (e, Tp grid; free C, $1/r^3$, $\cos 2\nu/r^3$, $\sin 2\nu/r^3$): χ^2 7238 vs the eclipse model's 1097 on 1367 5-min bins ($\Delta\chi^2$ +6141). With 3 harmonics added it is still +2475.
- A heartbeat cannot produce two flat-bottomed 3.3-h dips 0.5 apart in phase.

6. Row fixes for the paper table

- P: 4.74683 → **4.746808 ± 0.000003 d**
- depth: 12.2 → **14.3 ppt** (QLP SAP; PDCSAP 13.5–14.4)
- n events: 16 → **18 primaries + 14 secondaries**
- dphi: +0.009 → +0.0096–0.0111 (fine as is)
- offset: 2.51″ (s059 deficit–target), unchanged
- Add a secondary eclipse: 4.9 ppt at phase 0.49996.
- Note: e cos ω = 0.0000 ± 0.0003 (Gaia e = 0.039 is 4.5σ discrepant; the orbit is circular).

7. Next predicted eclipses (BTJD; linear ephemeris)

- **Primary** (T0 3126.22496 + 4.746808 E): 4403.1163, 4407.8631, 4412.6099 (± 1 min).
- **Secondary** (T0 2971.95351 + 4.746808 E): 4400.7427, 4405.4895, 4410.2363 (± 2 min).

What would settle it further

It is already settled as an EB on this star. Still useful: - A double-lined spectrum (K2) for masses. - A light-curve model with free R1 to resolve the radius tension. - An RV re-reduction to test Gaia's e.

Files: vet.py / vet-out.txt, sec.py / sec-out.txt / sec.png, heartbeat.py / heartbeat-out.txt / heartbeat.png, tpfdiff*.png, indep.py, physics-out.txt, grazing-out.txt, grazing-aR-scan.txt, secratio.py, bkgreg-out.txt, gaia.json, cats-out.txt, morecats-out.txt, nbvari-out.txt. Downloads are in C:\Home\sky\data\qlp-sb-p20-60\pervet260789413.

A146. TYC 4952-149-1 (TIC 53256660)

status
: confirmed (per-object re-analysis 2026-10-19; grazing G + M-dwarf EB, 12.9 ppt, 8 events; secondary only marginal, 0.9 ppt at 3.5 sigma, held out of the tables; VSX ROT entry at the target is a type revision)

P_days
: 5.160140

eP
: 0.000004

T0_primary_BTJD
: 3411.01065

depth_primary
: 1.29

duration_h
: 2.28

source_path
: Fable 5\made\sb1-scale\pervet\vet53256660\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 53256660 (TYC 4952-149-1, Gaia DR3 3682572652570849792) — per-object vet

Opus 5.5 subagent, wave 3 relaunch, 2026-10-19. Ledger row: G 11.15 | P 5.16016 | 12.2 ppt | 7 events | dphi +0.015.

Verdict: CONFIRMED (grazing eclipsing SB1; primary secure, secondary marginal)

Deciding numbers: - 8/8 primaries detected individually (S/N 22–35). Depth 12.87 ± 0.16 ppt, constant: χ^2 4.5/7; s046 12.73 vs s091 12.97 ppt. T14 2.23–2.28 h, V/U-shaped. - Photometric P = 5.160140 ± 0.000004 d vs Gaia 5.160018 ± 0.000234 (+0.54σ). - The primary is +1.9 h / +2.6 h (+0.015 / +0.021 phase) from Gaia conjunction A, i.e. 0.75σ / 0.69σ. - Difference image (TESS-SPOC TPF, s046): the loss is on the target pixel, centroid (5.18, 5.19) vs target (5.26, 5.03). The batch diffrun-c4 (s091) has the deficit 1.7″ from the target and 0.18″ from the light centroid. - No Gaia source with G < 17 within 90″. The brightest within 60″ is G 17.9 at 37.5″, which would need a dip larger than its whole flux (~6.5×). - Another pipeline (TESS-SPOC PDCSAP, s046) gives 12.43 ± 0.19 ppt. - A heartbeat is excluded: $\Delta\chi^2$ = +2371 for the most generous tidal superset.

1. Archives

- `cats.py 53256660 --r 60`: VSX has "Gaia DR3 3682572652570849792" type ROT at 0″. **This is a type revision (ROT → EA + spots), not prior art.** ASAS-SN none; SIMBAD TYC 4952-149-1 [SB*] only. Not in Prša, IJspeert or Kostov t2/t3/t4 (batch ebcats: none).
- VizieR: SB9 none; WDS none; J/AJ/160/111 none.
- Gaia DR3:
- vari_summary is present (G range 0.043 mag over 57 epochs, in_vari_eclipsing_binary false).
- vari_classifier_result: SOLAR_LIKE (score 0.07).
- vari_rotation_modulation: no row (re-queried with SELECT *).
- vari_eclipsing_binary: none. RUWE 1.10.
- ExoFOP: no TOI, CTOI, imaging, spectroscopy or companions.
- Web search on "TYC 4952-149-1" / "TIC 53256660": nothing relevant.
- **New eclipsing binary.**

Full Gaia NSS SB1 row (TAP, with corr_vec): P 5.160018(234), e 0.022(12), ω 292.9(36.6)°, Tp 1.067(529) d after J2016, K1 34.29(48) km/s, significance 70.8.

2. Eclipse measurements

Inputs: QLP SAP_FLUX with QUALITY==0, no trimming; s046 (10 min) and s091 (200 s). Detrending: a 3-d moving median plus 3 orbital harmonics, with the eclipses masked.

- **Primary (global trapezoid):** 13.01 ppt, T14 2.28 h [2.22–2.28], T12 0.91 h, flat bottom 0.46 h.
- **Primary, per event:** timing errors 0.7–1.4 min. Odd 12.85 ± 0.20 vs even 12.90 ± 0.26 ppt.
- **Background:**
- The per-event in–out SAP_BKG varies a lot: +28 / +16 / +3 / −10 ppt-equivalent in s046, and +210 at E239 in s091. The depths stay constant regardless.
- bkgreg: the background-induced flux change is −0.004 ppt (s046) and −0.12 ppt (s091).
- **Centroids:** dX 0.0 ± 0.5, dY −1.9 ± 1.7 mpix.
- **Quality and systematics:** one event window contains a QUALITY bit-32 cadence (E1, s046), and its depth is normal. No event near a listed sector-wide systematic.
- **Control phases:** |depth| ≤ 0.21 ppt except phase 0.5 (+0.92 ± 0.16, the secondary).
- **Blind dip census:** 8 dips with S/N ≥ 4, all 8 on the primary ephemeris; nothing off-ephemeris.
- **Secondary:** 0.85 ppt at phase 0.4986 ± 0.0007 (best-fit shape). With the primary shape fixed at phase 0.500 it is 0.92 ± 0.27 ppt (3.5σ). Per event it is only 1–3σ. Treat it as **marginal**: its timing and its duration ratio (0.69) are not reliable.

3. Ephemerides and orbit shape

- Primary: P 5.160140 ± 0.000004 d, T0 = 3411.01065 ± 0.00041 BTJD; χ^2/dof 35/6 (errors scaled); O–C rms 1.8 min.
- Secondary (marginal): $\Delta\varphi = 0.4986 \pm 0.0007$ (formal error, realistically $\gtrsim 2\times$ larger) → $e \cos\omega \approx -0.002 \pm 0.002$. Gaia gives $+0.007 \pm 0.013$ and predicts $\Delta\varphi = 0.5045 \pm 0.0085$; consistent ($-0.7\sigma$).
- $e \sin\omega$: not measurable (secondary S/N too low). Gaia gives -0.016 ± 0.011.
- The secondary ephemeris lands +1.0 h / +1.7 h from Gaia conjunction B (0.4σ / 0.5σ).

4. Independence and dilution

- **TESS-SPOC s046:** PDCSAP 12.43 ± 0.19 ppt, SPOC-SAP 12.36 ± 0.19 ppt (4 events).
- **SPOC TPF difference image (s046, 4 events):** the max-loss pixel is the target pixel (S/N 76). Target-PSF fit implied dip 10.4 ppt (crude G→T scaling).
- **Neighbour table:** no Gaia DR3 source with G < 17 within 90″. The batch run lists 3 sources within 60″, the brightest G 17.9 at 37.5″. Needed dip = 12.9 ppt × 10^(0.4 × 6.75) ≈ 6.5× its flux: impossible.
- TIC contratio is 0.002, so dilution is negligible.

5. Physics

- f(M) = 0.0216 M⊙. With M1 = 1.00 ± 0.12: m2(i = 90°) = 0.34 (0.31–0.36) M⊙, an M dwarf; a = 13.9 R⊙.
- Central-eclipse T14 would be 5.0 h; the observed 2.23 h means grazing (i ≈ 83.5°, b ≈ 1.13 R1 with R1 = 1.39). The V-like shape fits this.
- **Which star is in front:** at A (primary eclipse) the M dwarf is in front of the G primary. At B the primary occults the M dwarf.
- Expected secondary/primary ≈ the TESS-band surface-brightness ratio (3400 K / 5620 K) ≈ 0.12 → ~1.5 ppt. Observed 0.85–0.92 ppt; same order (grazing and limb-darkening details matter).
- The LD grazing fit is physical (s ≤ 1) only for a/R1 ≳ 11, i.e. R1 ≲ 1.26 vs TIC 1.39 ± 0.07. This is weak (56 bins) and not a vet issue.
- Ellipsoidal amplitude: predicted 0.43 ppt; not measurable, because the out-of-eclipse harmonics are spot-dominated and change between sectors (s046 A1c +3.9 ppt, s091 A2c −3.5 ppt). This is consistent with Gaia's 0.043-mag G range and the VSX ROT label.
- **Heartbeat check:**
- e ≈ 0.02 and Gaia periastron is at phase 0.56, not at the primary.
- Free Kumar-family superset: χ^2 4217 vs 1846 for the eclipse model on 1103 5-min bins ($\Delta\chi^2$ +2371). With harmonics added it is still +1895. Gaia-fixed: χ^2 8608.

6. Row fixes for the paper table

- P: 5.16016 → **5.160140 ± 0.000004 d**
- depth: 12.2 → **12.9 ppt** (PDCSAP 12.4)
- n events: 7 → **8 primaries** (s046: 4, s091: 4)
- dphi: +0.015 (s046) / +0.021 (s091), fine as is
- offset: 1.7″, unchanged
- Note: VSX ROT → EB type revision; marginal secondary of 0.9 ppt at phase 0.4986.

7. Next predicted eclipses (BTJD; linear ephemeris)

- **Primary** (T0 3411.01065 + 5.160140 E): 4401.7575, 4406.9176, 4412.0777 (± 1 min).
- **Secondary** (marginal; T0 3501.3058 + 5.160140 E): 4404.330, 4409.490, 4414.651. The formal error is ± 5 min; ± 15 min is realistic. The primary + 0.5 P gives 4404.338, 4409.498.

What would settle it further

The primary is settled. Still useful: - More TESS sectors at 200 s, to firm up the secondary and measure e sin ω. - Spectroscopy, to check the M-dwarf companion via K2 or IR lines.

Files: vet.py / vet-out.txt, sec.py / sec-out.txt / sec.png, heartbeat.py / heartbeat-out.txt / heartbeat.png, tpfdiff_46.png, indep.py, physics-out.txt, grazing-out.txt, secratio.py, bkgreg-out.txt, gaia.json, gaia-rot-out.txt, cats-out.txt, morecats-out.txt, nbvari-out.txt. Downloads are in C:\Home\sky\data\qlp-sb-p20-60\pervet53256660.

A147. TYC 2136-2756-1 (TIC 403936502)

status
: confirmed (per-object re-analysis 2026-10-19; grazing eclipse of the Gaia SB2 at conjunction B 0.27 d before periastron, 14.0 ppt, 9 events + 2 flagged; heartbeat loses by dchi2 22713; conjunction A geometrically excluded; Gaia's own P is 4.3 sigma off, the batch dphi was that error extrapolated; batch diffimg flag overturned, its own offset 2.1 arcsec)

P_days
: 14.804287

eP
: 0.000006

T0_primary_BTJD
: 3043.96544

depth_primary
: 1.40

duration_h
: 2.82

source_path
: Fable 5\made\sb1-scale\pervet\vet403936502\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 403936502: per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19 (wave 3b). Ledger row: G 10.26, P 14.80432 d, 12.7 ppt, 9 events, dphi +0.019. Star: TYC 2136-2756-1, Gaia DR3 2038397358512598400. Gaia NSS **SB2**: P 14.801518 ± 0.000644 d, e 0.330 ± 0.007, ω 283.5 ± 1.2°, K1 59.4 ± 0.7 km/s, K2 66.0 ± 0.7 km/s.

Verdict: CONFIRMED. A grazing eclipse at one conjunction of the Gaia SB2; no eclipse is possible at the other.

Deciding numbers 1. **The eclipse belongs to the Gaia binary.** - The eclipse falls on conjunction B, where the Gaia primary passes in front. Measured at Gaia's own epoch (Tp, BTJD 394), the TESS ephemeris lands **1.06 h (0.003 in phase, 2.4σ) after Gaia's predicted conjunction B**. - The ledger's +0.019, and the +5σ offsets at TESS epochs (+6.9 h in s014 growing to +15 h in s081), come entirely from Gaia's period being off by 0.0028 d (4.3σ_Gaia). At TESS epochs the error accumulates over about 88–212 cycles. - Conjunction A is 6.6 d away, so there is no ambiguity about which conjunction this is. 2. **It is an eclipse, not a heartbeat.** - Conjunction B is only 0.265 d before periastron, which is exactly the blind spot that caught 391918838. - Nine events, each in a ±1.5 d window with its own quadratic baseline and noise renormalised per event (N = 5563), were fitted with competing models:

Model	χ²
Free trapezoid	5765
Kumar heartbeat, Gaia e and ω fixed (inclination and Tp scanned)	28478 (Δχ² = **+22713**)
Heartbeat with e free up to 0.97	6626 (Δχ² = +861)

- Adding a heartbeat term to the trapezoid takes only +0.31 (unit shape) and leaves the depth unchanged at 13.96 ppt (`shape_models.png`, `hb-out.txt`).
- The dip is 2.8 h long and V-shaped. A Gaia-e tide would be about 1 d wide. 3. **On target, and in other pipelines.**
- TESS-SPOC pixel difference images in s040, s041 and s074 peak on the target pixel (5,5). A PSF at the target's position fits best in every sector; the next-best star is G 16.48 at 5.8″, which would need a 2500–3100 ppt dip.
- TESS-SPOC PDCSAP gives 14.2–14.4 ppt (s040/41/74/81), SPOC-SAP 13.3–13.4, TARS 14.0–14.5 (s040/41/74/80/81) and QLP SAP 14.0.
- The batch diffimg "peak not on target" flag is overturned: its own deficit offset was 2.1″ (0.1 px). 4. **One consistent geometry.**
- The SB2 gives M1 $\sin^3 i$ = 1.341 ± 0.033 M⊙, M2 $\sin^3 i$ = 1.206 ± 0.030 M⊙ and a sin i = 34.65 Rsun.
- At conjunction B the stars are 23.4 Rsun apart with a transverse speed of 20.6 Rsun/d. The 2.82 h duration then gives a half-chord of 1.21 Rsun.
- Splitting the TIC combined-light radius (2.18) by $L \propto M^4$ gives R1 ≈ 1.5 and R2 ≈ 1.25 Rsun at 0.9× scale. That predicts a QLP depth of **14.8 ppt (observed 14.0)** at i ≈ 83.9°, impact 0.90 (R1+R2): a grazing eclipse of the Gaia secondary. The masses are then 1.36 + 1.23 M⊙.
- At conjunction A the separation is 1.945× larger. An eclipse there would need R1 + R2 < 1.41 Rsun, which is impossible for these masses. So **no second eclipse is expected**.
- The conjunction-A limit is consistent: a 3–12 h box at the predicted phase 0.4484 ± 0.0047 gives 0.27–0.39 ± 0.05 ppt. That is a broad, edge-of-scan trend, not an eclipse. The free secondary search's best candidate (0.28 ppt, 14 h, phase 0.429) is not repeatable: χ² for constant depth across events is 101/9, the only S/N > 3 event (s080) comes with a 10 mpix centroid shift, and 78 of 301 control phases reach |S/N| > 4.

Measurements (QLP SAP_FLUX, QUALITY = 0, no trimming) - **Shape:** depth 13.97 ± 0.09 ppt, T14 2.82 h (1σ 2.76–2.82), T12 1.13 h (T12/T14 0.4, 0.56 h flat). A V shape (T12/T14 0.5) is worse by Δχ² 68; a box (T12/T14 0.1) is worse by 2020. - **Events:** nine Q0 events, all with S/N 16–67. Two more sit in flagged-only cadences at the predicted times: s054 2792.29 (QUALITY 4096/6144) and s080 3488.09 (bit 30), with crude dips of 10.4 and 8.9 ppt. s014 has no cadences at its 1696.78 epoch. - **Depth constancy:** χ² 29.6/8, driven by s054 (17.9 ± 1.1, the noisiest sector at 2.8 ppt/h) and s080 (12.8 ± 0.35). Odd 14.13 ± 0.16, even 13.89 ± 0.11. - **Ephemeris:** P = 14.804287 ± 0.000006 d, T0 = 3043.96544 ± 0.00021 BTJD. χ²/dof 26.6/7 (errors ×1.95), O–C rms 1.1 min. Jackknife: every leave-one-event-out and leave-one-sector-out P is within 6 ppm-d. - **e cos ω / e sin ω:** these cannot be derived from timing with only one eclipse kind. The single-eclipse geometry agrees with Gaia (e cos ω +0.077, e sin ω −0.321): r_A/r_B = 1.945 is what removes the second eclipse. - **Systematics:** none of the events lies within 0.5 d of the listed sector-wide systematics; one window contains a bit-32 cadence. Background regression (unit-free) allows at most 1.24 ppt of fake depth per event, typically under 0.3. Centroid in–out: dX +0.3 ± 0.5, dY +0.35 ± 0.26 mpix. Control phases are all consistent with 0 (|depth| ≤ 0.22 ppt). The blind census found 17 dips at S/N ≥ 4: 9 on the ephemeris, 0 off-ephemeris deeper than 7 ppt. - **Neighbours (fractional dip each would need):**

Neighbour	Needed dip
G 14.82 at 47″	93%

Neighbour	Needed dip
G 15.38 at 33″ and 35″	155%
G 16.48 at 5.8″	430%

None can host it. - **Archives:** cats.py (VSX, ASAS-SN, Prša+2022, IJspeert+2024, Kostov t2/t3/t4) found nothing. SIMBAD has TYC 2136-2756-1 only as [SB*]. SB9, WDS and J/AJ/160/111 have nothing. Gaia vari_summary and vari_eclipsing_binary are empty. ExoFOP has no TOI/CTOI, imaging or spectroscopy. A web search on the TYC, TIC and Gaia ids found nothing. **New as an eclipsing binary.**

Row fixes for the paper table - P: 14.80432 → **14.804287 ± 0.000006 d**; T0 **3043.96544 ± 0.00021 BTJD**. - Depth: 12.7 → **14.0 ppt** (QLP SAP; PDCSAP 14.3). - Duration: **2.82 h**, V-ish, grazing. - Events: 9 Q0 (+2 in flagged cadences). - dphi: report the conjunction offset at the Gaia epoch, **+0.003 (1.06 h, 2.4σ)**. The +0.019 figure is an extrapolation of Gaia's P error. - Notes: eclipse at conjunction B only; the other conjunction is geometrically excluded; Gaia P is 4.3σ off. - Diffimg: on target (TESS-SPOC s040/41/74).

Next predicted times (BTJD) - Eclipse (conjunction B): **4346.7427, 4361.5470, 4376.3512** (± 0.0006 d). - Conjunction A, no eclipse expected: **4353.381, 4368.185, 4382.990** (± 0.07 d).

Would settle it further: a joint light-curve and RV model with the Gaia SB2 constraint, giving i, the radii and the individual masses (a good benchmark: two F stars, grazing, e = 0.33). Ground-based photometry at a predicted conjunction-A time would confirm the absence of a second eclipse.

A148. TYC 2669-4733-1 (TIC 85944269)

status
: confirmed (per-object re-analysis 2026-10-19; grazing EB + ellipsoidal, both eclipses in every sector, 17.3/11.1 ppt, 48 + 47 events; the batch's 13.1 ppt was a both-kinds average at P/2; secondary at phase 0.5019; tension resolved as far as the archives go (2026-10-20 addendum): Gaia's K1 is confirmed by the raw epoch-RV spread, so M2 = 0.30 Msun; R1 ~ 3 Rsun from the ellipsoidal amplitude, vbroad and the Gaia catalogues (TIC's 4.84 is blend-inflated); the companion is probably a stripped post-mass-transfer star (R CMa / EL CVn-precursor family, T2 ~ 5650 K from the depth ratio); a 1.5-2x luminosity excess hints at unresolved third light; no LAMOST (through DR11) or APOGEE spectrum, Gaia DR4 epoch RVs decide)

P_days
: 3.021219

eP
: 0.000003

T0_primary_BTJD
: 2951.66557

depth_primary
: 1.73

depth_secondary
: 1.11

duration_h
: 4.6

source_path
: Fable 5\made\sb1-scale\pervet\vet85944269\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 85944269: per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19 (wave 3b). Ledger row: G 11.48, P 3.02120 d, 13.1 ppt, 90 events, dphi +0.026. Star: TYC 2669-4733-1, Gaia DR3 2033384753750667264, d ≈ 1.04 kpc, Teff 6470 K. The TIC radius is 4.84 Rsun, but see Physics below. TIC contratio 0.62, |b| = 1.4°. Gaia NSS **SB1**: P 3.021253 ± 0.000242 d, e 0.085 ± 0.052, ω 150 ± 43°, K1 31.7 ± 3.0 km/s.

Verdict: CONFIRMED. A grazing eclipsing binary (primary and secondary) with ellipsoidal variation, and it is the Gaia SB1 itself. The component properties do not fit a simple picture (see Physics).

Deciding numbers 1. **Same period as the Gaia orbit.** P = 3.021219 ± 0.000003 d (TESS) against Gaia's 3.021253 ± 0.000242: **−0.14σ**. 2. **On the right conjunction.** The deeper eclipse falls on Gaia conjunction A (companion in front, primary eclipsed) at **−0.023 to −0.029 in phase (−1.7 to −2.1**

h, −0.33 to −0.55σ) in every sector. - The secondary sits at phase 0.50185 ± 0.00020. Gaia predicts 0.464 ± 0.032 (1.2σ). 3. **Two eclipse kinds, both detected in every sector.**

Eclipse	Depth (QLP SAP)	T14	T12/T14	Events (Q0)	S/N per event
Primary	17.33 ± 0.10 ppt	4.62 h	0.4	48	8–35
Secondary	11.02–11.16 ± 0.09 ppt	4.44 h	0.35	47	9–23

- The secondary has its own ephemeris: P2 = 3.021213 ± 0.000002, −1.6σ from P. 4. **On target.**
- TESS-SPOC difference images in s014, s041, s054 and s055 all favour a PSF at the target over the next-best star, G 12.73 at 22.8″, by $\Delta\chi^2$ = 995 / 8965 / 5575 / 5580. The loss-weighted centroid is within 0.5–1 px of the target.
- The batch diffimg agrees (s075: on target, 4.6″).
- QLP centroid in−out: dX −0.4 ± 0.3, dY +0.4 ± 0.3 mpix.
- Neighbour dips needed:

Neighbour	Needed dip
G 12.73 at 22.8″	5.5%
G 14.17 at 38″	21%
G 15.7–15.9 at 28–49″	88–99%
VSX Gaia DR3 2033384753703991680 (G 18.45 [L] at 25″)	~800%

Only the first two are photometrically possible, and the pixels exclude both. 5. **Independent pipelines.** - TESS-SPOC PDCSAP (crowding-corrected) gives a primary depth of 19.6–20.3 ppt (s014/41/54/55); SPOC-SAP gives 12.5–15.7 ppt. - The ratio of eclipse depth to ellipsoidal amplitude is the same within scatter in QLP (3.24–3.69), SPOC-SAP (2.97–3.67) and PDCSAP (3.50–4.63). The eclipses and the ellipsoidal variation therefore come from the same diluted source.

Measurements (QLP SAP_FLUX, QUALITY = 0, no trimming) - **Primary depth:** χ^2 for constant depth 114/47, from sector-to-sector QLP dilution (s055 16.85 against 17.4–17.7 elsewhere). Odd 17.49 ± 0.14, even 17.17 ± 0.14. - **Flagged cadences:** about 14 more primaries in flagged-only cadences show 11–20 ppt dips at the predicted times. One s014 epoch inside a scattered-light (QUALITY 6144) region does not. - **Ephemeris:** T0 = 2951.66557 ± 0.00045 BTJD. χ^2 145.8/46 (errors ×1.78), O−C rms 5.7 min. - The s014 primaries (30-min cadence) run 6–17 min early; dropping s014 moves P to 3.021209. - The s014 secondaries are not correspondingly early (about −2 min). So this is most likely a cadence/template effect, not an ETV. Worth a look, not a claim. - **Orbit shape:** e cos ω = +0.0031 ± 0.0003 from the secondary phase (light-travel time about 30 s, negligible), consistent with circular; Gaia gives −0.07 ± 0.05. - The duration ratio (0.96) implies e sin ω of about −0.02, but grazing eclipses make that number unreliable. - Gaia's e = 0.085 ± 0.052 is 1.6σ from zero. - This is a 3-d near-circular orbit, so the heartbeat blind spot does not apply. - **Ellipsoidal (cos 2φ semi-amplitude):** 4.3–5.4 ppt in QLP, 4.3–5.7 in PDCSAP. - **Systematics:** no event lies within 0.5 d of a listed sector-wide systematic, and five windows contain a bit-32 cadence. Background regression allows at most 1.9 ppt of fake depth per event (typically under 0.7), against a 17 ppt eclipse. Control phases 0.125–0.875 (excluding 0.5) are all within ±0.43 ppt. - **Blind census:** every dip deeper than 0.5× the primary is a primary or a secondary. The remaining S/N 4–5.6 dips are 2.4–3.6 ppt features clustered at phase ±0.3, which is ellipsoidal residue. - **Archives:** cats.py found no VSX, ASAS-SN, Prša, IJspeert or Kostov entry at the target; its only VSX hit is the G 18.45 [L] star at 25″. SIMBAD has TYC 2669-4733-1 only as [SB*]. SB9, WDS and J/AJ/160/111 have nothing. Gaia vari tables are empty. ExoFOP has no TOI/CTOI. A web search on the TYC, TIC and Gaia ids found nothing. **New as an eclipsing binary.**

Physics: a real tension, recorded rather than resolved - **Mass function:** f(m) = 0.0099 M⊙ (±29%). For M1 = 1.32–1.6 M⊙ that gives m2 = 0.30–0.37 M⊙ at i = 65–90°, and a ≈ 10.3–11.0 Rsun. - **The eclipses demand a bright companion.** The secondary/primary depth ratio of 0.644 fixes the companion's TESS surface brightness at 0.64× the primary's, i.e. **Teff2 ≈ 5600–5700 K**. A 0.3 M⊙ main-sequence star (about 3400 K) would give a ratio near 0.1. - **But a bright companion breaks the RVs.** A main-sequence companion that hot (about 1 M⊙) would need K1 ≈ 84 sin i km/s, against Gaia's 31.7 ± 3.0. - **The ellipsoidal amplitude disagrees with the TIC radius.** About 5.4 ppt implies $q(R1/a)^3 \approx 4.5 \times 10^{-3}$. With Gaia's q (about 0.23) that needs **R1 ≈ 2.8–3.1 Rsun, not the TIC's 4.84**; with 4.84 Rsun the ellipsoidal would be 21–30 ppt. With q of about 0.76, R1 would be about 2.1 Rsun. - **Durations are consistent but uninformative.** The 4.6 h eclipses are grazing (half-chord about 2.1 Rsun against a central duration of order 8–12 h), so they say nothing more. - **Possible resolutions:** - Gaia's SB1 K1 is depressed by line blending in an unrecognised SB2. - The companion is a low-mass but hot stripped/pre-He-WD star (EL CVn-like, though cool for that class). - There is third light inside the Gaia source. - None of these moves the eclipses off the Gaia binary: the period, the conjunction, the pixels and the shared ellipsoidal all tie them to it.

Row fixes for the paper table - P: 3.02120 → **3.021219 ± 0.000003 d**; T0 **2951.66557 ± 0.00045 BTJD**. The search found P2 = 1.5106 (half the period), so the resolved period is right. - Depth: 13.1 → **primary 17.3 ppt, secondary 11.1 ppt** (QLP SAP; PDCSAP primary about 20 ppt crowding-corrected). The 13.1 was the both-kinds average at P/2. - Durations: **T14 4.6 / 4.4 h**, V-ish, grazing. - Events: **48 primaries + 47 secondaries** in Q0. - dphi: primary on conjunction A at −0.026 (−0.4σ); secondary at phase 0.5019. - Notes: ellipsoidal 4.3–5.7 ppt; e cos ω +0.003. Flag the K1 / depth-ratio / ellipsoidal tension.

Next predicted times (BTJD) - Primary: **4335.3839, 4338.4052, 4341.4264** (± 0.0013 d). - Secondary: **4336.9001, 4339.9214, 4342.9426** (± 0.0015 d).

Would settle it further: - A spectrum to test for SB2 lines (decides the K1 question). - A light-curve model with ellipsoidal, beaming and reflection terms to fit R1/a, q and the light ratio jointly. - 2-min photometry of s014-like epochs to test the 10-min O−C offset.

Addendum (Fable, 2026-10-20 21:30 session): the tension, pushed as far as the archives go

Receipts: `spec.py/spec-out.txt`, `vsweep.py/vsweep-out.txt`, `tension.py/tension-out.txt`, `lamost_meta.py`.

The spectrum gate is a real dead end in current archives. LAMOST LRS has no spectrum of this star in DR1, DR2, DR4, DR5, DR7 or **DR11 (V/162, Luo+ 2026)** — 8″ cones, catalogue ids validated against a dense test field first (several guessed ids in `spec.py`'s first pass were nonexistent tables, which VizieR reports identically to a miss; `vsweep.py` is the corrected instrument). APOGEE DR17 (III/286): nothing. GCVS: nothing. So the SB2-line test waits for new data — realistically Gaia DR4 epoch RVs.

Gaia DR3 RV diagnostics kill the "K1 is secretly ~84 km/s" escape. `rv_amplitude_robust = 67.0 km/s` over 25 transits — the raw epoch-RV spread, no orbit fit involved — against 2·K1 = 63.4 (ratio 1.06). A 1 M⊙ main-sequence companion needs a spread of ~168. For blending to compress epoch RVs 2.5× the companion would have to rival the primary in RVS flux, which the 0.64 depth ratio already forbids (it fixes the companion's *surface brightness*, and a near-equal-flux companion would give a near-unity depth ratio and double lines Gaia's pipeline flags as SB2). Gaia's K1 stands; **M2 = 0.30 ± 0.03 M⊙** (f(M) = 0.0099, M1 1.32–1.6, i ≳ 75°).

The ellipsoidal half of the tension dissolves: TIC's radius is the outlier. Four instruments independent of the TIC converge on R1 ≈ 3 R⊙: ellipsoidal q(R1/a)³ → 2.80 (q 0.227, a 10.35); **Gaia vbroad = 54.3 ± 18.7 km/s** → R_sync = 3.2 ± 1.1 if tidally synchronized (P 3.02 d; TIC's 4.84 predicts 81 km/s, −1.4σ); SHBoost (V/160, Gaia DR3 ML) logg 3.70 + M 1.78 → 3.1; GDR2AP (V/155) R0best 3.41. The TIC's 4.84 comes from blended photometry at |b| = 1.4° with contratio 0.62.

What genuinely remains, and the reading it points to. A 0.30 M⊙ companion with TESS-band surface brightness 0.64× a 6466 K primary (T2 ≈ 5600–5700 K) cannot be a main-sequence star (an M dwarf gives ~0.1). The self-consistent picture is a **stripped post-mass-transfer companion** — the R CMa / EL CVn-precursor family: M2 ≈ 0.3, T2 ≈ 5650, R2 ≤ its Roche lobe of 2.7 R⊙ (tension.py; a detached, shrinking proto-He-WD at R2 ~ 1–2 R⊙ contributes 6–20% of the light and keeps the donor's own distortion signal small, as observed). The primary at R1 ≈ 3, M1 ≈ 1.3–1.8 would be the spun-up accretor. One loose thread survives: at plx 0.935 mas the source wants L ≈ 22–41 L⊙ (A_G 0–0.66), i.e. R ≈ 3.8–5.1 at 6466 K — a factor ~1.5–2 more light than an R1 = 3 primary plus stripped companion supply. Either the extinction/BC budget at 1 kpc in the plane absorbs it, or there is third light inside the Gaia source (RUWE 0.86 and ipd_frac_multi_peak 0 exclude only *resolved* companions; a tertiary would also dilute the TESS depths — ratios, and hence T2, unaffected).

For the paper row: keep CONFIRMED; the note should say "companion is likely a stripped post-mass-transfer star (R CMa/EL CVn-like): M2 ≈ 0.3 M⊙ from Gaia K1 (raw epoch spread confirms), T2 ≈ 5650 K from the depth ratio, R1 ≈ 3 R⊙ from ellipsoidal + vbroad + two Gaia AP catalogues (TIC's 4.84 is blend-inflated); possible unresolved third light at the ~30–50% level." No LAMOST/APOGEE spectrum exists; Gaia DR4 epoch RVs are the next decisive instrument.

A149. BD-17 691 (TIC 93103821)

status
: confirmed (per-object re-analysis 2026-10-19; grazing near-circular F + K-dwarf EB, 15.5 ppt, 24 primaries + 20 secondaries not 13; new 5.8 ppt secondary at phase 0.50000(25), T2 ~4700 K; conjunction offset -0.022)

P_days
: 1.6350307

eP
: 0.0000004

T0_primary_BTJD
: 1868.90788

depth_primary
: 1.55

depth_secondary
: 0.58

duration_h
: 1.44

source_path
: Fable 5\made\sb1-scale\pervet\vet93103821\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 93103821 (BD-17 691, TYC 5873-244-1, Gaia DR3 5102568177264285824): per-object vet

Opus 5.5 subagent for Fable 5, wave 3b, 2026-10-19. Batch row: G 11.03, P 1.63502 d, 13.5 ppt, 13 events, dphi +0.022.

VERDICT: CONFIRMED. A grazing, near-circular F + K-dwarf eclipsing binary. Both eclipses are present: the primary eclipse (15.5 ppt) falls at Gaia conjunction A, and the secondary (5.8 ppt) falls at phase 0.50000 ± 0.00025.

Star and orbit - G 11.03, Tmag 10.60. TIC: Teff 6247 K, R 1.29, M 1.21, d 250 pc. Gaia GSP-Phot: 6056 K, logg 4.21. RUWE 1.12. - Gaia NSS SB1 (full row with corr_vec from the archive): P 1.6350635 ± 0.0000198 d, e 0.015 ± 0.019, ω 180 ± 49°, K1 88.28 ± 1.74 km/s, significance 50.6. - f(M) = 0.1165 Msun. m2(i=90°) = 0.71–0.77 Msun for M1 1.05–1.21; a = 7.1–7.3 Rsun.

1. Archives - `cats.py --r 60`: VSX and ASAS-SN none. SIMBAD: BD-17 691 [SB*] only. Prša, IJspeert and Kostov t2/t3/t4: none. - SB9 and WDS none (VizieR). Gaia vari_summary and vari_eclipsing_binary empty. ExoFOP: no TOI or CTOI, no imaging or spectroscopy. - Batch ebcats:

none. Web search on BD-17 691, TYC 5873-244-1 and TIC 93103821: nothing. Not previously catalogued as an eclipsing binary.

2. Eclipses (QLP SAP, QUALITY==0, no trimming; s004 has 30-min cadence, s031 10-min) - **Primaries:** 24 measured — 10 in s004 and 14 in s031; the row says 13. Every one has S/N 10–48. - Per-sector depths: s004 13.7 ± 0.13, s031 15.35 ± 0.11 ppt. - Supersampled trapezoid: s031 15.55 ppt, T14 1.44 h, T12/T14 0.4 (V-ish, so a graze). s004 still gives 13.85 even with 30-min smearing modelled. The s004/s031 difference is therefore aperture dilution between sectors; SPOC shows a smaller version of it (14.6 vs 15.3). - Odd 14.48 ± 0.11 vs even 14.90 ± 0.12 — this reflects the sector mix, not an odd/even effect. - **Secondary:** 20 events, all S/N > 3; phase-0.5 control gives 4.98 ± 0.08 ppt. Supersampled fit: 5.77 ppt, T14 1.39 h, at phase 0.50000 ± 0.00025. Depth ratio sec/pri 0.37. - **Systematics:** - No event within 0.5 d of a listed sector-wide systematic time. Three primary windows contain a bit-32 cadence; their depths match the rest. - In-out SAP_BKG change is −1.3 ppt-equivalent (the wrong sign to fake a dip). Background regression implies ≤ 0.007 ppt. - Centroids dX −0.04 ± 0.12, dY −0.10 ± 0.12 mpix. - **Controls:** phases 0.125/0.25/0.375/0.625/0.75/0.875 all within ±0.22 ppt. - **Census:** 45 dips at S/N ≥ 4 — 22 on the primary, 20 on the secondary, and 3 others of 0.9–7.3 ppt at single epochs. No off-ephemeris dip at or above half the primary depth except one 7.3 ppt S/N 4.2 blip (s031 2163.52, phase 0.185), a single cadence group. - **Out of eclipse** (both eclipses masked): ellipsoidal cos2φ −1.8 to −2.2 ppt (minima at conjunctions, the correct sign); reflection-like cos φ −1.1 / −3.2 ppt (brighter at the secondary); sin terms disagree between sectors (30-min s004 systematics).

3. Ephemeris and Gaia - P = 1.6350307 ± 0.0000004 d, T0 = 1868.90788 ± 0.00009 BTJD. χ^2 62/22 (errors scaled ×1.68). O−C rms 0.6 min. - vs Gaia P: −0.000033 d = −1.66σ. - Gaia conjunctions (MC 20k draws with corr_vec): the primary sits on **conj A** at −0.88 h (s004, −3.0σ) and −1.18 h (s031, −2.6σ), i.e. −0.023/−0.030 in phase. Conj B is 0.49 phase away (31–45σ). - The ~3σ timing offset is the only tension. With e consistent with zero, Gaia's ω (±49°) and t_peri are nearly degenerate, so the propagated conjunction error (0.29–0.46 h) is probably underestimated. It is not a sign of misattribution: the secondary sits exactly at 0.5, and the deeper eclipse is the F star eclipsed, as conj A requires. - Photometric e cos ω = (π/2)(φ_sec − 0.5) = +0.0000 ± 0.0004 vs Gaia −0.0136 ± 0.014 (1.0σ). e sin ω ≈ (T2 − T1)/(T2 + T1) = −0.009 ± 0.018 vs Gaia 0.000 ± 0.017. Circular, consistent with Gaia. - Batch dphi sign: the eclipse precedes Gaia conj A, so dphi = **−0.022** (c4-vetsum has −0.022; the ledger's +0.022 is the wrong sign).

4. Independence - **TESS-SPOC PDCSAP:** s4 14.58 ± 0.11, s31 15.29 ± 0.09 ppt. SPOC-SAP: 14.52 and 15.20. - **TARS:** s4 14.45, s31 15.42. **TGLC s4:** PSF 12.48 ± 0.28, aperture 16.83 ± 0.19. Every pipeline sees it. - **SPOC TPF difference image, s31 (14 events):** peak loss on pixel (4,5), S/N 48. Loss-weighted centroid (5.18, 4.64) vs target (5.02, 4.52). Single-star fit on the target implies a 12.4 ppt own dip. - Batch diffimg s31: deficit-light 0.22″, peak on target. - **Neighbours:** none in Gaia DR3 brighter than G 17 within 90″. The brightest within 60″ is G 19.0 at 21.9″, which would need a 15.5 × 10^(0.4 × 7.97) ≈ 24,000 ppt dip — impossible. Dilution is excluded.

5. Physics - With M1 1.05–1.21 and R1 1.15–1.29: central T14 3.3–3.5 h vs 1.42 h observed. Chord fraction 0.40–0.43, b ≈ 0.91 (R1+R2), i ≈ 75–76°. - e ≈ 0, so both conjunctions graze identically. The companion (K dwarf, R ~0.75–0.8) transits the F star at A (primary eclipse) and is occulted at B. - Uniform-disc predictions: transit 15–19 ppt (limb darkening lowers a graze) vs 15.5 observed; occultation 7.3–8.5 vs 5.8 observed. - The depth ratio 0.37 implies a TESS-band surface-brightness ratio for T2 ≈ 4700 K, which matches a 0.71–0.77 Msun K dwarf. - Ellipsoidal predicted 2.9–4.4 ppt (R1 1.05–1.29) vs 1.8–2.2 observed. That is a factor ≤ 2 and favours the smaller Gaia-like R1. - One consistent geometry. **Heartbeat check:** e = 0.015 ± 0.019, so there is no periastron tide to mimic; the events are 1.4-h eclipses at both conjunctions.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. Both eclipses are present — 24 primaries at 15.5 ppt and 20 secondaries at 5.8 ppt — with the secondary at phase 0.50000 ± 0.00025, matching Gaia's e ≈ 0. 2. P is –1.7σ from Gaia and the primary sits on conj A. 3. QLP, SPOC, TARS and TGLC agree, the SPOC difference image peaks on target, and there is no neighbour brighter than G 17 within 90″. - **Paper row fixes:** - n_events 24 primaries (+ 20 secondaries), not 13. - Depth 15.5 ppt (10-min s031; 13.7–13.9 in 30-min s004 QLP), not 13.5. - Add a secondary of 5.8 ppt at φ 0.5000. - P 1.6350307(4). - dphi −0.022 (sign). - **Next primaries** (±1 min): BTJD 4401.5705, 4403.2055, 4404.8405. - **Next secondaries** (±1 min): BTJD 4402.3880, 4404.0230, 4405.6580. - **To settle further:** a light-curve model with limb darkening (b, k, surface-brightness ratio) on 2-min or 200-s data, and an SB2 spectrum to see the K dwarf and resolve the ~1 h Gaia conjunction offset.

A150. CD-35 10394 (TIC 7795044)

status
: confirmed (per-object re-analysis 2026-10-19; grazing transit of a subgiant at conjunction A 18 deg past periastron, 15.0 ppt, 8 events; heartbeat alone loses by dchi2 >= 10569 but a real ~1 ppt tidal dip sits under the eclipse; conjunction B misses as predicted; QLP s103's 12.6 ppt was an aperture artefact; call the host a subgiant, Gaia 6610 K, logg 3.6)

P_days
: 9.4320556

eP
: 0.0000068

T0_primary_BTJD
: 3194.85288

depth_primary
: 1.50

duration_h
: 4.62

source_path
: Fable 5\made\sb1-scale\pervet\vet7795044\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 7795044 (CD-35 10394, TYC 7335-183-1, Gaia DR3 6013256545738149888): per-object vet

Opus 5.5 subagent for Fable 5, wave 3b, 2026-10-19. Batch row: G 10.17, P 9.43204 d, 13.9 ppt, 7 events, dphi +0.001.

VERDICT: CONFIRMED. A grazing transit of a subgiant by a ~0.9 Msun companion at Gaia conjunction A, which lies 18° of true anomaly past periastron. That is the heartbeat blind spot, and the test is decisive: a heartbeat model loses to the trapezoid by $\Delta\chi^2 \geq 10{,}569$. A weak (~1–2 ppt) broad tidal dip at periastron is also present on top of the eclipse. Conj B misses, as the geometry predicts.

Star and orbit - G 10.18, Tmag 9.55. TIC: Teff 5531 K, R 4.0, lumclass GIANT, d 343 pc, E(B−V) 0.23. Gaia GSP-Phot: 6610 K, logg 3.61 — a subgiant. RUWE 0.91. - Gaia NSS SB1 (full row with corr_vec): P 9.432060 ± 0.000360 d, e 0.187 ± 0.007, ω 71.6 ± 2.8°, K1 53.03 ± 0.56 km/s, significance 95. - f(M) = 0.138 Msun. m2(i=90°) = 0.87–0.97 Msun for M1 1.3–1.6; a = 24–26 Rsun.

1. Archives - `cats.py --r 60`: VSX and ASAS-SN none. SIMBAD: CD-35 10394 [SB*] only. Prša, IJspeert and Kostov t2/t3/t4: none. - SB9 and WDS none. Gaia vari tables empty. ExoFOP: no TOI or CTOI, no imaging or spectroscopy. - Batch ebcats: none. Web search on CD-35 10394, TYC 7335-183-1 and TIC 7795044: nothing. Not previously catalogued as an eclipsing binary.

2. Eclipses (QLP SAP, QUALITY==0; s038 10-min, s065 and s103 200-s) - **Primaries:** 7 in QLP Q0 data (s038 ×2, s065 ×3, s103 ×2), each S/N 78–114. - Trapezoid: 14.4–15.0 ppt, T14 4.5–4.6 h, T12/T14 0.4 (V-ish). A flat bottom (T12/T14 0.1) is worse by $\Delta\chi^2$ 7754. - A third s103 event (BTJD 4166.35) is missing from QLP Q0 but present in TESScut, so **8 eclipses** in all. - **Depth "variation" is a QLP artefact.** QLP gives s038 15.15, s065 15.41, s103 12.65 ppt (χ^2 584/6). My own aperture photometry on TESScut cutouts (r = 1–3.5 px) gives s065 15.0–15.3 and s103 14.8–14.95 ppt, with all three s103 events at 14.5–15.1. The depth is constant at ~15.0 ppt. - **Systematics:** - Two windows contain a bit-32 cadence (s038; depths unchanged). None near a listed systematic time. - Background regression implies ≤ 0.07 ppt. Centroids dX +0.33 ± 0.11, dY +0.06 ± 0.05 mpix. - **Controls:** all phases within ±0.12 ppt. - **Census:** 8 dips at S/N ≥ 4 — 7 are the eclipses. The other is 0.9 ppt (S/N 5.9, s103 4169.90, phase 0.376). None off-ephemeris at ≥ 50% of the eclipse depth. - **Heartbeat test** (`hb.py`, ±0.75 d windows, a 4-d detrend that preserves broad shapes): - Trapezoid: χ^2 2990, depth 15.0 ppt, T14 4.62 h. - Kumar heartbeat with Gaia's e and ω fixed: χ^2 27,957 ($\Delta\chi^2$ +24,967). - Fully free heartbeat (e 0.05–0.7, ω, periastron time, i): best χ^2 13,558 ($\Delta\chi^2$ +10,569). It cannot make a 4.6-h V; see `heartbeat_test.png`. - Trapezoid plus Gaia-geometry tide: $\Delta\chi^2$ −734 vs trapezoid alone, tide coefficient S 1.7 ppt, eclipse depth 14.2 ppt. So a weak broad (~1 day) periastron tidal dip of about 1 ppt sits under the eclipse. It is real at $\Delta\chi^2$ 734, though partly degenerate with the detrending. - **Secondary at Gaia conj B** (phase 0.538 ± 0.006): 0.007 ± 0.060 ppt (T14 ×1.4), 3σ < 0.19 ppt, over 5 covered conj-B passes. The window maximum, 0.32 ppt at phase 0.556 (S/N 6.3), and the vet.py scan peaks at 0.45/0.25 (S/N 8–10) are red-noise-level features at the scan's 99th percentile. Not claimed.

3. Ephemeris and Gaia - P = 9.4320556 ± 0.0000068 d, T0 = 3194.85288 ± 0.00053 BTJD. χ^2 141/5 (errors scaled ×5.3). O−C rms 2.0 min. - vs Gaia P: −0.000004 d = −0.01σ. - Gaia conjunctions (MC with corr_vec): the eclipse sits on **conj A** at +0.13/+0.12/+0.10 h (s038/s065/s103) = +0.07/+0.05/+0.03σ, +0.0005 in phase. Conj B is 0.46 phase away (28–47σ). - One eclipse kind only, so no photometric e cos ω / e sin ω. Gaia: e cos ω +0.059 ± 0.010, e sin ω +0.177 ± 0.007; conj-B phase offset 0.538 ± 0.006, where the data show nothing.

4. Independence - **TESS-SPOC s65:** PDCSAP 15.14 ± 0.08, SAP 14.56 ± 0.08 ppt. - **TARS:** s38 14.04 ± 0.16, s65 15.46 ± 0.12. My own TESScut photometry: s65 15.0, s103 14.85. - **SPOC TPF difference image, s65:** peak loss on pixel (5,5), S/N 201. Loss-weighted centroid (4.73, 4.84) vs target (4.69, 4.82). Target fit χ^2 18,236 vs ≥ 85,422 for every neighbour. Batch diffimg s65: peak on target, 1.17″. - **Neighbours** (dip each would need for 15 ppt): G 13.97 @ 40″ → ~490 ppt; G 15.50 @ 55″ → ~1.9e3; G 15.77 @ 40″ → ~2.5e3; fainter ones impossible. Dilution is excluded.

5. Physics - For M1 1.3–1.6 and R1 3.4–4.0: conj A at ν 18.4°, separation 0.82a; conj B at ν 198°, separation 1.17a. - Central T14 at A 10–12 h vs 4.6 h observed: chord fraction 0.37–0.44, b ≈ 0.90–0.93 (R1+R2), i ≈ 77–79°. - At conj B the projected separation exceeds R1+R2 by 1.25–1.6 Rsun, so **conj B misses**, consistent with the < 0.19 ppt limit. - At A the companion (~0.9 Msun, about 4–7% of the TESS light) transits the primary. An occultation could not give 15 ppt. - Uniform-disc graze depth from the straight-chord inversion is 6–12 ppt vs 15 observed — within the crudeness of that inversion; a slightly larger companion or smaller b closes the gap. - The tidal term at periastron (~1 ppt) is the expected order for R1/r_peri ≈ 0.17–0.2 and q ≈ 0.6.

6. Verdict and predictions - **CONFIRMED.** Deciding numbers: 1. A 4.6-h V-shaped 15.0 ppt eclipse beats any heartbeat model by $\Delta\chi^2$ ≥ 10,569. 2. P matches Gaia to −0.01σ and the eclipse is on conj A at +0.07σ. 3. QLP, SPOC, TARS and my TESScut photometry agree at 14–15.5 ppt, and the SPOC difference image is on target (neighbours ≥ 4.7× worse in χ^2). - **Paper row fixes:** - Depth 15.0 ppt, not 13.9; QLP s103's 12.6 is a QLP aperture/background artefact. - n_events 8 (QLP Q0 has 7; the third s103 event is in TESScut). - P 9.4320556(68). - dphi +0.0005. - Note a weak periastron tidal (heartbeat)

component of ~1 ppt under the eclipse — an EB row with a heartbeat note, not a heartbeat row. - TIC "GIANT, R 4.0, 5531 K" vs Gaia 6610 K / logg 3.6: call it a subgiant. - **Next primaries** (±2 min): BTJD 4402.1560, 4411.5880, 4421.0201. - **Conj B** (no eclipse expected; Gaia ±~0.15 d): BTJD 4407.23, 4416.66, 4426.10. - **To settle further:** a joint eclipse + tide model on the 200-s data to pin b and k and the tidal amplitude, and a spectrum to check for an SB2 signature of the ~0.9 Msun companion.

A151. TYC 8977-7461-1 (TIC 324273420)

status
: confirmed (per-object re-analysis 2026-10-19; eccentric grazing double-eclipser, both eclipses on Gaia's conjunctions, 14.7-16.3 ppt QLP, 18-21 undiluted, 33 primaries + 38 secondaries; secondary at phase 0.5593, e cos omega +0.0932 = Gaia's +0.098(17); common-mode +-4 min ETV in both eclipse kinds, SPOC-independent, likely a third body; forward test in TESS s112-114, 2027, pre-registered by Opus 5.5)

P_days
: 3.9158631

eP
: 0.0000015

T0_primary_BTJD
: 3200.00630

depth_primary
: 1.5-1.6

depth_secondary
: 1.5

duration_h
: 3.48

source_path
: Fable 5\made\sb1-scale\pervet\vet324273420\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis (2026-10-19)

Archived source: sanitized report text.

TIC 324273420 (TYC 8977-7461-1): per-object vet

Opus 5.5 subagent for Fable 5, 2026-10-19 (wave 3b). Batch row: G 11.02, P 3.91587 d, 14.7 ppt, 32 events, dphi +0.011, diffimg offset 3.59″ (peak on target).

Verdict: CONFIRMED. This is an eccentric, grazing, double-eclipsing binary, and it is the Gaia SB1 itself. Both eclipses fall at their Gaia conjunctions, the secondary sits at phase 0.559, and a heartbeat model is excluded.

Deciding numbers - **Both eclipse kinds land on Gaia's conjunctions.** The comparison is made at the Gaia epoch with the full covariance; corr(ω, t_p) = 0.994, so σ(conj) = 0.026 / 0.023 d. - Primary minus conj A: +0.009 d (+0.0023 phase, +0.35σ). - Secondary minus conj B: −0.003 d (−0.15σ). - Swapping the two attributions is excluded at 66–77σ. - Without the correlation the conjunction errors would be 0.146 / 0.132 d, about 6× larger. - **Secondary eclipse:** 14.7 ppt (QLP fold), against 16.3 ppt

for the primary. It sits at phase 0.55932 ± 0.00014 and is seen 38 times. - The exact Keplerian inversion gives e cos ω = +0.0932 ± 0.0002. Gaia gives +0.098 ± 0.017 (0.3σ), and the Gaia MC predicts a secondary phase of 0.5615 ± 0.011. - **Period:** P(joint, both kinds) = 3.9158631 ± 0.0000005 d (formal; ±0.0000015 with the ETV scatter folded in). Gaia has 3.9157408 ± 0.0001487 (+0.8σ). - **Heartbeat check:** periastron falls at phase 0.71, and the conjunctions are at ν = 117° (A) and 297° (B), so neither is near periastron. A Kumar-type heartbeat model fitted to the folded SAP, with e, periastron phase and i all free, runs off to e = 0.74 and still fits worse than two trapezoids by $\Delta\chi^2 = 5054$ over 500 bins. It cannot produce the secondary at all. - A heartbeat term at Gaia's e and periastron, added on top of the eclipses, only absorbs the ~2 ppt out-of-eclipse ellipsoidal/periastron modulation. That modulation is itself physical (heartbeat.png).

Archives (queried 2026-10-19) - cats.py r = 60″ (APPOINTMENTS_DATA set): - Nothing at the target in VSX, ASAS-SN, Prša, IJspeert or Kostov t2/t3/t4. - Nearby VSX entries are all faint field variables 17–53″ away: an RRC of G 19.7 at 17″, an L of G 18.6 at 19″, a VAR at 29″ and an L at 53″. - SIMBAD has only TYC 8977-7461-1 [SB*]. - VizieR battery: WDS, SB9, Gaia DR3 vari_classifier / veb / vartot / vst, Hipparcos, RAVE and GALAH all returned nothing. Gaia phot_variable_flag is NOT_AVAILABLE and RUWE is 0.92. - ExoFOP JSON: no TOI, no CTOI, no notes, no files. - Web search on the TYC/TIC names: nothing. - MAST has TESS-SPOC (s10, 11, 37, 38), TGLC (s10, 11), eleanor-lite, TARS and T16 light curves, but no SPOC 2-min data or DV. - Prior art: none.

Eclipses and ephemeris (QLP SAP, QUALITY==0, no trimming; events_324273420_p/s.txt) - **Primary:** 33 events fitted (the s11 E12 fit is degenerate and gets no weight). - T0 = BTJD 3200.00630 ± 0.00037, P = 3.9158610 ± 0.0000015. - Per-event depth 14.3–19.3 ppt. T14 = 3.3–3.9 h (fold 3.48 h), T12 ≈ 1.3 h: grazing, U/V-shaped. - Sector means range from 14.9 (s100) to 17.9 ppt (s64); this follows QLP aperture dilution (contratio 0.70). - **Secondary:** 38 events. T0 = BTJD 3315.75654 ± 0.00038, P = 3.9158664 ± 0.0000016. Depth 10.9–16.8 ppt, T14 3.4–4.4 h (fold 4.10 h), so it is longer than the primary. - **Systematics:** - Background change through the events is tens to a few hundred e/s, random in sign. - Centroid shifts are ≤ 0.008 px. - No event is near a listed sector-wide systematic. - Only one event core contains a bit-32 dump (s38 E195, one cadence), and that event agrees with the rest. - **Blind dip census** (eclipses masked ±0.12 d, 3.5-h box): - Nothing deeper than 6.1 ppt anywhere; nothing above 3 ppt in s99/s100. - Every 3–6 ppt hit is either within 0.33 d of an eclipse (mask-edge leakage of the 1-d median detrend) or a residual of the ~2 ppt out-of-eclipse modulation or an orbit edge. - **ETV (new finding):** the O−C is curved and the curve is common to both eclipse kinds. - Season means for the primary: Y2 −3.3 ± 0.9, Y4 +4.2 ± 0.7, Y5 +1.4 ± 0.7, Y8 −1.5 ± 0.4 min. - The secondary moves in step with it. - A joint quadratic improves χ^2 by 155 (68 → 67 dof; χ^2/dof is still 2.9). - TESS-SPOC SAP and PDCSAP independently show the same offsets: s10/11 −1.6 to −4.5 min, s37/38 +4.5 min. So this is not a QLP artefact. - Interpretation: likely light-travel time from a third body (a sin i ≳ 0.5 AU), or a period change. It is not resolvable with four seasons. - **Possible apsidal motion (not claimed):** the secondary phase drifts from 0.55888 ± 0.00027 (Y2) to 0.55960 ± 0.00011 (Y8). That is P_sec − P_prim = 0.46 ± 0.09 s raw, about 3σ after error scaling, or roughly 0.2°/yr in ω.

Independence | Pipeline / sectors | Primary depth (ppt) | Secondary depth (ppt) | |---|---|---| | TESS-SPOC SAP, s10/11/37/38 (smaller aperture) | 13.5–14.2 | 10.8–11.7 | | TESS-SPOC PDCSAP, de-crowded (CROWDSAP 0.64–0.71) | 19.9–21.2 | 16.3–18.0 | | TGLC s10 aperture | 18.5 | 15.2 | | TGLC s10 PSF | 17.6 | 14.7 | | TARS s64 | 16.2 | 13.8 |

- Timing in every pipeline matches QLP's to within a few minutes.
- **Undiluted depths:** about 18–21 ppt for the primary and 15–18 ppt for the secondary.
- **Localisation:**

- The batch difference image (s38, 6 events) puts the deficit 3.59″ from the target with the peak on target (diffimg/diffrun-c4.jsonl). I did not re-derive it; nothing here contradicts it.
- Neighbour table (neighbours_out.txt): Galactic plane (b = −0.7°), 257 Gaia sources within 60″. The cheapest hosts would need large dips: G 12.95 at 41″ needs 8.6%, G 14.62 at 22″ needs 40%, G 15.11 at 25″ needs 63%.
- The decisive point is that both eclipses sit at the SB1's own two conjunctions and reproduce its e cos ω, which a background EB cannot do.

Physics - f(m) = 0.0831 M⊙ (K = 59.3 km/s, e = 0.110). - With M1 = 1.19 M⊙ (TIC; Teff 6193, logg 3.68, R1 2.62 R⊙), m2 = 0.66–0.68 M⊙ (i ≈ 77–80°) and a = 12.9 R⊙. A central eclipse would last 7.8 h; the observed 3.5–4.1 h makes both eclipses grazing. - **Which star is in front:** the deeper eclipse is at conj A, so the companion is in front at primary eclipse. - **One-geometry fit** (geom2.py: numerical limb-darkened overlap, eccentric orbit, e cos ω fixed, fitting both T14 values and the undiluted depths). It fits at $\chi^2 \approx 0$ with: - R1 ≈ 1.9–2.0 R⊙, - a K-dwarf companion (R2 0.8–0.9 R⊙, L2/L1 0.03–0.06 in the TESS band), - e sin ω −0.04 to −0.12, against Gaia's −0.050 ± 0.017. - The "nearly equal" depths (ratio ~0.85) come from geometry: at B the separation is smaller, so b is smaller, and the primary covers 25–50% of the K dwarf, while at A only ~2% of the big star is covered. - **Tension (flagged, not decisive):** the fit's R1 is ~25% below the TIC's 2.62 R⊙. With R1 fixed at 2.62 the primary comes out too shallow (11 vs 18–21 ppt). Two systematics push in this direction: trapezoid T14 is biased low for V-shapes, and the TIC radius uses the DR2 distance (the DR3 parallax 2.20 mas makes it 5% smaller). A third light from the ETV body would also enter here.

Changes to the paper row - Keep the row. - Period: P = **3.9158631(15)** d (joint, both kinds; ETV scatter included). - Primary T0 = BTJD **3200.0063(4)**. - Add the secondary: 14.7 ppt QLP (15–18 ppt undiluted) at phase **0.5593**. e cos ω = +0.0932 agrees with Gaia. - Depth: QLP 14.7–16.3 ppt diluted; undiluted ~18–21 ppt (PDCSAP/TGLC-PSF). - Event count: 33 primaries + 38 secondaries measured in QLP (the ledger's 32 counts primaries only). - dphi: +0.0023 from conj A at +0.35σ with full covariance (the batch's +0.011 used the 3.4-yr-extrapolated phase). - Diffimg 3.59″: keep it, annotated "peak on target". - Add the note: "common-mode ETV ±4 min (third body?)".

Next eclipses (BTJD; ±6 min, dominated by the ETV, not by the period error) - Primary: 4335.6055, 4339.5214, 4343.4373, 4347.3531 - Secondary: 4337.7969, 4341.7127, 4345.6286, 4349.5444

What would settle it further: RVs that resolve the K-dwarf lines (SB2) would give the masses and fix R1. More TESS seasons, or ground-based minima timings, would characterise the ETV (an LTTE orbit) and the apsidal drift.

Files: overview.png, heartbeat.png, events_324273420_{p,s}.{txt,png}, ephem_324273420_{p,s}.png, conj.py, season.py, etv_out.txt, geom2_out.txt, physics_out.txt, cats_out.txt, archives_out.txt, mast_inventory.txt, neighbours_out.txt, census_out.txt, census_dist.txt, exofop.json, next_eclipses.txt. Second-pipeline FITS are in C:\Home\sky\data\qlp-sb-p20-60\pervet-324273420.

A152. TYC 7652-366-1 (TIC 123378551)

status
: confirmed (per-object re-analysis 2026-10-19; M dwarf grazing the F star at conjunction A, 22.9 ppt QLP, 22 +- 1.5 dilution-corrected, T14 2.6 h, 14 primaries not 7; the G 12.41 neighbour at 18.2 arcsec, probably physical, excluded at 17.6-27.4 sigma per sector in 6 own difference images; the 3.42 arcsec flag was crowding dragging the light centroid; tentative 1.0 +- 0.17 ppt secondary at phase 0.207 held out of the tables; the 3.25-d 2% quasi-sinusoid in the QLP flux is a contaminant, not the target)

P_days
: 11.010611

eP
: 0.000006

T0_primary_BTJD
: 2972.09636

depth_primary
: 2.3

duration_h
: 2.6

source_path
: Fable 5.1\made\pervet-deep\vet123378551\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 123378551

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent; the agent's own byline below says "Opus 5.5 subagent", meaning an Opus-model subagent, launched from Fable 5.1's session). Batch row: G 11.13, P 11.01057 d, 20.8 ppt, 7 events, dphi +0.027, diffimg offset 3.42" (peak not on target, s34; 54 neighbours in 60", 2 possible).

TIC 123378551: per-object attribution vet (pervet-deep). Opus 5.5 subagent, 2026-10-19.

The star is TYC 7652-366-1 = Gaia DR3 5535420719437355648: G 11.13, Teff about 6400–6500 K, R about 1.35 Rsun, parallax 3.49 mas, RUWE 1.95.

Gaia DR3 SB1 orbit:

P (d)	e	ω (deg)	K1 (km/s)	corr(ω, Tp)
11.00834 ± 0.00126	0.436 ± 0.017	181.8 ± 3.0	23.85 ± 0.60	+0.89

Verdict: CONFIRMED

The eclipse is on the target, at Gaia's conjunction A (the companion passes in front of the Gaia primary). The difference-image flag is a crowding artefact.

What decides it: - **Difference images, six sectors.** I made my own for s7, 8, 34, 35, 62 and 88 (two events each, noise from 150 null windows, Gaussian-PRF scene fitted to every Gaia DR3 source within 90″). - In every sector the fitted deficit sits 1.1–3.3″ (0.05–0.16 px) from the target. - The only neighbour bright enough to make the dip is the G 12.41 star at 18.2″. As the host it is rejected by $\Delta\chi^2$ = +1908, +930, +1245, +2142, +2621, +1615 (s7, 8, 34, 35, 62, 88). Scaled by the best fit's reduced χ^2, that is 18.3, 17.6, 19.1, 20.9, 27.4 and 18.8σ. - A model-free check agrees: the fractional loss on the target's pixel (14.7–17.9 ppt) is larger than on that neighbour's pixel (5.4–13.6 ppt) in all six sectors. - **Timing.** The TESS ephemeris carried back to Gaia's epoch lands 0.065 d (1.0σ) from Gaia's conjunction A. P_TESS − P_Gaia = +0.0023 d (1.8σ). - **Second pipelines.** TESS-SPOC SAP and PDCSAP, TGLC PSF-fit, TARS (including s61, which QLP lacks) and CDIPS all show every covered event, within 3 minutes of the QLP times. - **Physics.** The mass function f(m) = 0.0113 Msun gives m2 ≈ 0.29–0.34 Msun, an M dwarf. A 2.6-hour, 22-ppt, V-shaped grazing eclipse at i ≈ 86° fits within the uncertainty in R1 and limb darkening (the geometry predicts 13–22 ppt).

1. Archives and web

- `cats.py 123378551 --r 60`: no VSX, no ASAS-SN. SIMBAD has only TYC 7652-366-1 [SB*]. No hit in Prša, IJspeert or Kostov.
- Gaia DR3: no vari_summary or vari_eclipsing_binary row for the target, and no eclipsing-binary row for any neighbour within 90″. One classifier row (5535420268461954816, ELL, score 0.08) lies outside 60″.
- SB9, WDS and J/AJ/160/111: none. ExoFOP: no TOI or CTOI, no imaging, spectroscopy or companions.
- A web search on all three identifiers found nothing that mentions the star.
- **Not previously catalogued as an eclipser.**
- The G 12.41 neighbour (Gaia DR3 5535420715138207360, 18.2″) has parallax 3.551 mas against the target's 3.492 mas. It is probably a physical wide companion (about 5000 au), so it had to be excluded with pixels, not with distance. It is not a catalogued variable.

2. Every eclipse

Fits use raw QLP SAP_FLUX with a trapezoid on a local quadratic baseline, integrated over the exposure (`ecl.py`). QLP has 12 primaries: 10 on QUALITY==0 data and 2 that need flagged cadences. TARS s61 adds 2 more (BTJD 2972.096 and 2983.107).

sector	cycle	tc (BTJD)	depth (ppt)	T14 (h)	T23 (h)	note
7	−68	1496.6757 ± 0.0008	23.4 ± 0.9	2.50	0.62	
7	−67	1507.6843 ± 0.0010	20.7 ± 0.9	2.40	0.72	
8	−66	1518.6984 ± 0.0010	21.8 ± 0.8	2.59	0.65	3 flagged cadences in the event
8	−64	1540.7190 ± 0.0010	21.8 ± 0.8	2.59	0.65	
34	−1	2234.3834 ± 0.0010	22.4 ± 0.9	2.69	0.40	
34	0	2245.3985 ± 0.0008	24.1 ± 0.8	2.50	0.50	
35	1	2256.4051 ± 0.0008	25.0 ± 0.6	2.69	0.40	needs flagged cadences before the event
35	3	2278.4262 ± 0.0008	23.4 ± 0.9	2.50	0.62	5 flagged cadences in the event
62	68	2994.1173 ± 0.0003	21.8 ± 0.5	2.50	0.62	
62	69	3005.1273 ± 0.0003	24.5 ± 0.5	2.69	0.40	
88	132	3698.7972 ± 0.0003	21.7 ± 0.4	2.50	0.75	
88	133	3709.8083 ± 0.0005	24.1 ± 0.5	2.69	0.27	

- One cycle is lost: s35 cycle 2 (2267.42) falls in the data gap.
- **Background:** the unit-free background "fake depth" is at most 0.35 ppt for every event.
- **QLP centroids:** SAP_X and SAP_Y shift by +0.0015 to +0.004 px and −0.0007 to +0.002 px in eclipse. That is the size and sign expected when the target dims while the light centroid sits off it toward the neighbours.
- **Blind census** (`census.py`, 2.6-hour box, every sector): the primaries are the only dips deeper than 5 ppt.
- **Second-pipeline depths:**
- TESS-SPOC PDCSAP (crowding-corrected; CROWDSAP 0.71–0.75): 20.8–23.7 ppt
- TGLC PSF-fit: 19.5–23.0 ppt
- TESS-SPOC SAP (not corrected for crowding): 14.7–17.6 ppt
- **The ~3.25-day, ~2% peak-to-peak quasi-sinusoid in every QLP and SPOC sector is not the target's** (`varimg.py`). Its in-phase fractional amplitude is 0.3–5.5 ppt on the target pixel and 20–60 ppt on pixels 2–3 px toward +column, from a source roughly 60″ or more away. It is contamination in the apertures.

3. Ephemeris and orbit

- **Primary, 12 events:** P = 11.010611 ± 0.000006 d, T0 = 2972.09636 ± 0.00034 BTJD.
- χ^2/dof = 5.6, with O−C within ±4 minutes and no trend. The errors are already scaled by $\sqrt{\chi^2}$_red; the contaminating 3.25-day signal is what inflates the scatter.
- Against Gaia: P differs by +0.00227 ± 0.00126 d (1.8σ).
- **Conjunction at Gaia's epoch** (Monte Carlo through the full NSS covariance):
- Gaia conjunction A = BTJD 384.538 ± 0.065. The TESS primary projected back = 384.603. The difference is +0.006 in phase (1.0σ).
- At TESS epochs Gaia's window widens to ±0.2–0.45 d through the error on Gaia's P. The batch row's dphi of +0.027 sits inside that.
- **Secondary** (`sscan.py`, fixed shape, with look-elsewhere control):
- At Gaia's predicted phase 0.2322 ± 0.0097: 0.27 ± 0.18 ppt, i.e. nothing.
- **Tentative secondary at phase 0.2070 ± 0.001: 0.99 ± 0.17 ppt.**
  - z = 5.96, against a z spread of 1.24 at control phases, so about 4.8σ effective. The largest control peak away from the primary's wings is z = 3.98.
  - It holds for box durations from 2.0 to 3.5 h. Per-window depths scatter about 1.8× the noise.
- If it is real, $e \cos \omega = -0.479 \pm 0.002$ against Gaia's -0.435 ± 0.017 (2.6σ).
- $e \sin \omega$ can't be measured because the secondary is too weak to give a duration. Gaia gives −0.013, which predicts near-equal durations (2.7 h against 2.6 h).

4. Independence and attribution

- **Light curves:** see §2. Every pipeline shows every covered event.
- **Difference images:** see the Verdict for the per-sector $\Delta\chi^2$ and σ. The fitted deficit's sky offsets from the target (east, north) in arcsec:

s34	s7	s8	s35	s62	s88
(+3.3, +0.3)	(+1.9, +1.0)	(+1.6, +0.3)	(+1.1, 0.0)	(+1.9, +0.7)	(+2.1, +0.9)

Per-event positions agree to within about 1.5″. - There is a small, consistent eastward bias of about 2″ (0.1 px). It is best read as a systematic of the Gaussian PRF model, whose scene residual is 0.16–0.24. No Gaia source within 6″ could host the dip: the nearest, G 19.37 at 6.1″, would need a 4100% eclipse. - **Neighbour dilution table** (depth needed = 22 ppt × 10^(0.4 ΔG)):

Gaia DR3	G	separation	needs	result
5535420715138207360	12.41	18.2″	7.1%	excluded at 17.6–27.4σ per sector
5535420341480232448	13.03	35.3″	12.7%	excluded at 19.7–31.9σ

Every other source within 60″ is G ≥ 17.6 and would need more than 100%. A host of the 22-ppt dip would have to be brighter than G ≈ 15.3. - **Why the batch flagged it:** - The 7.25″ in the receipt is the deficit's distance from the *light* centroid, which is dragged 0.1–0.3 px toward the neighbours. - The peak-deficit pixel (8,8) is next to the target pixel (7,8): sub-pixel quantization. - psf_frac_resid = 0.68 comes from fitting D = a·I_out in a scene where 30–45% of I_out is other stars. - **This is a crowding artefact, like the five shallow cases.**

5. Physics

- **Companion:** f(m) = 0.01129 Msun gives m2 = 0.285, 0.315 or 0.343 Msun for M1 = 1.15, 1.35 or 1.55. That is an early-to-mid M dwarf, R2 ≈ 0.31 Rsun, T2 ≈ 3400 K.
- **Orbit size:** a = 24.7 Rsun. Separation is 20.3 Rsun at conjunction A and 19.7 Rsun at conjunction B.
- **Primary eclipse is grazing:**
- A central chord at conjunction A would last 5.2 h. The observed 2.6 h means b ≈ 1.05 R1 and i ≈ 86°, which fits the V/narrow-U shape.
- For k = 0.2–0.35 at that duration, the grazing depth is 13–15 ppt with R1 = 1.35 and 20–22 ppt with R1 = 1.2 (linear limb darkening u = 0.3–0.45).
- The observed ~22 ppt (dilution-corrected, from PDCSAP and TGLC) is in that range. Grazing depth is steep in R1 and b, so this is consistency, not a measurement.
- **Which star is in front:** at conjunction A the companion is in front, so the observed eclipse is the M dwarf transiting the F star.
- **Expected secondary:** occultation of the M dwarf predicted at 1.4–2.0 ppt, with T14 about 2.7 h (b_B ≈ 1.03 R1). The tentative 1.0 ± 0.17 ppt is the right size.
- **Ellipsoidal variation:** about 60–100 ppm, undetectable under the 20-ppt contaminating signal.
- **Not a heartbeat:** conjunction A is at true anomaly −92°, well away from periastron, and the predicted tidal amplitude is about 0.1 ppt.
- **RUWE:** 1.95 is larger than this orbit explains (a1 sin i ≈ 0.08 mas), so another close companion is possible. Nothing here needs it.

Next eclipses (BTJD)

- **Primary** (±0.0008 d formal; allow ±0.003 d for the O−C scatter): 4337.4121, 4348.4227, 4359.4333, 4370.4440, 4381.4546, 4392.4652, 4403.4758.
- **Tentative secondary** (phase 0.207, ±0.011 d): 4339.691, 4350.702, 4361.713, 4372.723, 4383.734, 4394.744, 4405.755.
- **Secondary in Gaia's window** (phase 0.2322 ± 0.0097, about ±0.11 d): 4339.97, 4350.98, 4361.99, 4373.00, 4384.01, 4395.02, 4406.03.

What would settle it further

- Two or three more 200-second sectors, or a ground-based light curve at 1-ppt precision at the tentative secondary times. That would confirm the ~1-ppt secondary and pin e cos ω, which now sits 2.6σ from Gaia.
- One or two RVs near conjunction A or B; the primary should be at γ − 10.4 km/s at both.
- High-resolution imaging, to explain the RUWE of 1.95.

Row fixes for the paper table

- **P:** 11.010611 ± 0.000006 d (row: 11.01057).
- **T0:** 2972.09636 ± 0.00034 BTJD; the primary is at Gaia's conjunction A (companion in front). The same ephemeris at cycle 0 gives 2245.3985.
- **Depth:**
- QLP SAP trapezoid: 20.7–25.0 ppt, mean about 22.9. The row's 20.8 is the box mean.
- Dilution-corrected: about 22 ± 1.5 ppt (PDCSAP 20.8–23.7, TGLC PSF 19.5–23.0).
- Secondary: tentative 1.0 ± 0.17 ppt at phase 0.207; nothing at Gaia's 0.232 (0.27 ± 0.18 ppt).
- **Duration:** T14 = 2.6 ± 0.1 h, T23 ≈ 0.3–0.75 h (grazing, V/U shape).
- **Event count:**
- 12 primaries in QLP (10 on QUALITY==0, 2 needing flagged cadences) across s7, 8, 34, 35, 62 and 88.
- Plus 2 in TARS s61, for 14 distinct primaries.
- The row's 7 is an undercount.
- **dphi:** +0.006 ± 0.006 in phase at Gaia's epoch (1.0σ). The row's +0.027 is the TESS-epoch value, inside Gaia's widened window.
- **Difference-image flag: crowding artefact, ON TARGET.**
- Own difference images in 6 sectors put the deficit 1.1–3.3″ from the target.
- The G 12.41 neighbour at 18.2″ (Gaia DR3 5535420715138207360, probably physical, needs 7.1%) is rejected at $\Delta\chi^2$ +930 to +2621 (17.6–27.4σ) per sector.
- No neighbour is a catalogued eclipser.
- The status can go from candidate to confirmed.

A153. TYC 8187-1030-1 (TIC 375435321)

status
: confirmed (per-object re-analysis 2026-10-19; flat-bottomed transit of a 0.24 Msun M dwarf at conjunction A, 21.5 ppt QLP, 23 +- 1 undiluted, T14 9.42 h, 8 + 8 events; new 1.32 ppt secondary at phase 0.3034 = Gaia's 0.305(21), e cos omega -0.308; on target in 6/6 difference images at 5.6-8.2 sigma, the 2.73 arcsec flag a pixel-boundary artefact; conjunction offset is -0.028, the batch's +0.030 a sign slip; T2 ~3300 K)

P_days
: 18.214018

eP
: 0.000011

T0_primary_BTJD
: 3498.99671

depth_primary
: 2.15

depth_secondary
: 0.13

duration_h
: 9.42

source_path
: Fable 5.1\made\pervet-deep\vet375435321\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 375435321

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.27, P 18.21422 d, 22.5 ppt, 7 events, dphi +0.030, diffimg offset 2.73" (peak not on target, s90).

Verdict: CONFIRMED. The eclipsing binary is the target, on Gaia's orbit.

TYC 8187-1030-1 = Gaia DR3 5407588611377633792. G 11.27, Teff 6306 K, R 1.89 Rsun, d 445 pc. The field is crowded: TIC contamination ratio 0.42, 88 Gaia sources within 60″. Gaia NSS SB1: P 18.22161 ± 0.00749 d, e 0.368 ± 0.031, ω 213.4 ± 7.6°, K1 15.75 km/s.

Three things decide it: 1. **The secondary is where Gaia's orbit puts it.** It sits at phase 0.3034 ± 0.0001; Gaia predicts 0.3052 ± 0.0210 (0.1σ). It is 1.32 ± 0.09 ppt deep (S/N 15) and shows in each of the 8 covered epochs (S/N 4.4–8.1). Four other pipelines recover it too. At Gaia's e sin ω, the eclipse timing gives e = 0.367 and ω = 213.0°, against Gaia's 0.368 and 213.4°. 2. **The difference images put the dip on the target** in 6 of 6 clean sector images. The fitted deficit is 1.5–2.3″ from the target. $\Delta\chi^2$ against the best neighbour is 268–826, which is 5.6–8.2σ per sector after scaling. 3. **The physics closes.** The mass function implies a 0.24 Msun M dwarf at i ≈ 88.5°. - The 23 ppt undiluted, U-shaped

transit gives R2 ≈ 0.27–0.29 Rsun. - The 1.35 ppt flat occultation implies T2 ≈ 3300 K. - One inclination fits both durations: 9.42 h at the wide conjunction, and 6.62 h predicted against 6.96 ± 0.30 h measured at the close one.

1. Archives and web - `cats.py --r 60`: no VSX entry at the target. Within 60″ there are two Gaia-classifier-only VSX objects: one ROT (G 12.9) at 51″ and one DSCT|GDOR (G 15.1) at 58″. ASAS-SN, Prša, IJspeert and Kostov t2/t3/t4: nothing. SIMBAD lists it as [SB*], with Gaia DR3 multiplicity as its only reference. - ExoFOP: no TOI or CTOI, no spectra or imaging. SB9, WDS, GCVS and Gaia vari_eclipsing_binary: nothing within 60″. A web search turned up nothing. - Neither the target nor any neighbour is a catalogued eclipsing binary.

2. Eclipses Method: SAP_FLUX with QUALITY == 0, no trimming. The primary shape is 21.5 ppt deep (QLP, diluted), T14 9.42 h, ingress 1.88 h, flat bottom 5.65 h. The batch's "6.3 h" was the box width, not the eclipse.

E	sector	tc (BTJD)	depth (ppt, QLP)	S/N
0	s9	1550.0973	21.8	82
1	s9	1568.3120	21.6	38 (new; not in sapdepth)
2	s10	1586.5267	22.3	65
41	s36	2296.8713	23.5	88
81	s63	3025.4276	23.3	113
121	s90	3753.9934	22.0	99
122	s90	3772.2061	22.4	105
138	s99	4063.6317	18.5	144

- The depth spread between sectors is aperture dilution. s99 also has a pre-eclipse baseline sag that pulls its depth down.
- No event lies near a listed sector-wide systematic. QLP centroid shifts are ≤ 7 mpix.
- Seven off-eclipse control phases are all consistent with zero.
- The off-ephemeris dips in the blind census are gap-edge or momentum-dump artefacts.
- Secondaries (QLP):

sector	tc (BTJD)	depth (ppt)	S/N
s10	1573.837	1.20	4.7
s10	1592.051	0.91	4.4
s36	2284.184	1.25	4.6
s36	2302.398	1.26	5.5
s63	3030.958	1.48	6.7
s90	3759.519	1.66	6.5
s99	4050.943	1.02	6.5
s99	4069.157	1.22	8.1

The depths are constant (χ^2 8.3/7). Across 390 control phases the S/N rms is 1.92, with a maximum of 7.4.

3. Ephemeris and orbit shape - P = 18.214018 ± 0.000011 d. T0 = 1550.0968 BTJD at E = 0, or 3498.99671 ± 0.00056 at E = 107. O–C rms is 2.7 min. - Against Gaia's P: –1.01σ. - Against Gaia's conjunctions: the primary is –0.028 phase (–1.05σ) from conjunction A, and the secondary about –0.030 from B. Both shift by the same amount, so this is Gaia's period/periastron-time error, not a misattribution. - e cos ω = –0.308 (at e sin ω = –0.20), against Gaia's –0.304 ± 0.035. The duration ratio gives e sin ω ≈ –0.15 ± 0.02 in the equal-chord approximation, against Gaia's –0.20 ± 0.045.

4. Independence, attribution, dilution - **Other pipelines** see the primary:

pipeline	depth (ppt)
TGLC s9, PSF photometry	20.9 ± 0.4
TGLC s9, decontaminated aperture	23.5 ± 0.3
ELEANOR-lite	17.3–17.6
T16	19.3–20.8
TARS (5 sectors)	17.1–20.0

They recover the secondary too: TGLC 1.37, ELEANOR 0.90, T16 1.20, TARS 1.21–1.66 ppt. - **Difference images** (TESScut, Gaussian scene model at Gaia positions):

sector (event)	offset from target	Δχ² vs best neighbour	scaled σ
s9	1.5″	308 (G 14.23 at 6.3″)	5.6
s10	2.3″	268 (G 14.23 at 6.3″)	7.1
s36	2.3″	361 (G 15.05 at 7.2″)	5.8
s63	1.5″	626 (G 15.05 at 7.2″)	7.6
s90 (3772.21)	1.5″	579 (G 15.05 at 7.2″)	6.9
s99	2.3″	826 (G 15.05 at 7.2″)	8.2

- **One event excluded.** On its own, the s90 event at 3753.99 localises 11″ off. Its ingress falls in a momentum-dump gap (3753.66–3753.90, QUALITY 32/40), so its before and during images straddle a pointing reset.
- **Why the batch flagged it.** The batch used only event 3772.22 and measured 2.73″ (0.13 px), which is consistent with the target. The "peak not on target" flag came from the target sitting at x = 7.49 px, on a pixel boundary (peak pixel [8,7] against target pixel [7,7]).
- **Neighbours as hosts:**

Gaia DR3	G	separation	fractional dip it would need
5407588508298418560	14.23	6.3″	290–320 ppt
5407588611377633920	15.05	7.2″	640–720 ppt
5407588611377634048	13.44	12.0″	110–155 ppt

The position fits disfavour all three, and none is a catalogued eclipser. - **Dilution.** Small apertures on the TESScut images give 21.5–23.4 ppt undiluted; TGLC's decontaminated aperture gives 23.5. Adopted: 23 ± 1 ppt primary, about 1.35 ppt secondary.

5. Physics - f(M) = 0.00593 Msun. With M1 = 1.30, m2 = 0.242 Msun (0.224–0.259 for M1 1.15–1.45) and a = 33.7 Rsun. - At conjunction A the Gaia star is behind. The deep eclipse is the M dwarf crossing the F star at r/a = 1.084; the shallow one is the M dwarf being hidden at r/a = 0.719. - i = 88.55° gives impact parameters 0.49 and 0.32. R1 ≈ 2.0 Rsun matches both durations exactly. - The occultation-to-transit surface-brightness ratio is 0.065, which fits T2 ≈ 3300 K. - Ellipsoidal (~33 ppm) and beaming (~210 ppm) signals are too small to see.

Next predicted eclipses (BTJD)

E	primary	secondary
157	4409.6976 (±1 min)	4415.2237 (±3 min)
158	4427.9116	4433.4378
159	4446.1256	4451.6518
160	4464.3397	4469.8658

What would settle it further: two RV epochs to tie Gaia's orbit to the eclipse epoch, and a near-IR spectrum to look for the M dwarf. A seeing-limited ground light curve of one primary would separate the 6.3″ and 12″ neighbours; a 23 ppt dip is easy from the ground.

Row fixes for the paper table - P = 18.214018 ± 0.000011 d (table: 18.21422). T0 = 1550.0968 BTJD (E = 0). - Depths: - Primary: 21.5 ppt QLP (sector range 18.5–23.5), 23 ± 1 ppt undiluted. - Secondary: 1.32 ± 0.09 ppt at phase 0.3034 (new). - Durations: primary 9.42 h (ingress 1.88 h, flat 5.65 h); secondary 6.96 h. The table's 6.3 h was the box. - Events: 8 primaries (table says 7) and 8 secondaries. - Conjunction offset: **−0.028 phase (eclipse early)**. The table's +0.030 is a sign slip. - Orbit from eclipses: $e \approx 0.367$, $\omega \approx 213°$, $e \cos \omega = -0.308$. - Difference-image flag: a pixel-boundary artefact plus neighbours pulling the light centroid. The dip is on the target in 6 of 6 clean images (5.6–8.2σ); s90 event 3753.99 is excluded for the momentum dump. - Status: candidate → **confirmed**.

A154. TYC 9075-1173-1 (TIC 375302184)

status
: confirmed (per-object re-analysis 2026-10-19; M dwarf transiting a +-3% gamma Dor-like pulsator at conjunction A, 28.5 ppt QLP, ~32 undiluted, T14 3.16 h, 74 events in 9 sectors; new 1.9 ppt secondary at phase 0.580 = Gaia's 0.563(31), e cos omega 0.126; on target in 7/8 own per-pixel difference images, the G 12.45 neighbour at 33 arcsec rejected every sector; the 9.7 arcsec flag came from box windows sitting inside the eclipse on a pulsator; VSX/ASAS-SN ROT 0.884 d at the target is the pulsation, a type revision)

P_days
: 2.5455862

eP
: 0.0000005

T0_primary_BTJD
: 3571.13574

depth_primary
: 2.85

depth_secondary
: 0.19

duration_h
: 3.16

source_path
: Fable 5.1\made\pervet-deep\vet375302184\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 375302184

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.18, P 2.54558 d, 27.4 ppt, 65 events, dphi +0.017, diffimg offset 9.7" (peak not on target, s93).

TYC 9075-1173-1, Gaia DR3 6445014184777521024, G 11.18, Teff 7060 K, R 1.50 R⊙ (TIC). Gaia SB1: P 2.545525 ± 0.000084 d, e 0.307 ± 0.047, ω 288.3 ± 8.0°, K1 33.3 ± 1.8 km/s.

Verdict: CONFIRMED. The target is an eclipsing binary on Gaia's orbit, and the difference-image flag is an artefact.

Deciding numbers: - **Attribution.** I made my own per-pixel difference images from 8 independent sector products: TESScut s27, s67, s93, s101 and s104, plus TESS-SPOC TPFs s13, s27 and s67. Each pixel's flux around each eclipse is fitted with the same cubic-baseline + trapezoid model as the light curve, so the 3% pulsation is modelled per pixel. A Gaussian PSF calibrated on the out-of-eclipse image was then fitted to the deficit. - The deficit sits **0.24–1.59″ from the target in 7 of 8 products**. s93

gives 3.45″, but that fit has a PSF σ of 0.98 px and χ²/dof 2.4, so the offset is a PSF-model systematic. Every offset is below 0.17 px. - Putting the eclipse on the only neighbour that could host it is rejected in every sector, by Δχ² = +124 (s101), +1318 (s93), +1834 (s104), +4560 (s27), +5567 (s67), and +3120 / +10524 / +17210 in the TPFs. That neighbour is Gaia DR3 6445014253499167744: G 12.45, 33.3″ away, needs an 11% eclipse. It is also ASAS-SN SR P = 330 d and SIMBAD LP* with RV +277 km/s, a halo giant. - Every fainter Gaia star is worse, and any G ≥ 15 star would need an eclipse deeper than 100%. - The deficit flux is 32.4 ppt (median) of the PSF-fitted *target* flux. - **Gaia orbit.** - P: TESS 2.5455862 ± 0.0000005 d vs Gaia 2.5455249 ± 0.0000843 d, **+0.73σ**. - Timing: the primary falls on Gaia conjunction A (companion in front) **+0.32 h = +0.20σ** from prediction at the Gaia epoch. - **Secondary eclipse detected:** 1.9 ± 0.24 ppt at phase **0.580 ± 0.003**, against the Gaia prediction of 0.563 ± 0.031 (+0.55σ). - It is 10.7σ above a look-elsewhere control: z std 1.26 over the 233 clear phases. - It is positive in all 9 sectors. - It gives e cos ω = 0.126 ± 0.006, against Gaia's 0.096 ± 0.046 (+0.7σ). TESS therefore confirms the eccentricity independently. - **Physics.** - f(m) = 0.0084 M⊙, so m2 ≈ 0.31 M⊙ for M1 = 1.56 (0.28–0.34 for M1 1.3–1.8): an M dwarf. - The trapezoid geometry gives k ≈ 0.18, b_A ≈ 0.76, i ≈ 85.5°, and R1 = 1.29 R⊙ (TIC 1.50) from T14 and the eccentric orbital speed at conjunction A. - That predicts a *total* occultation at B (b_B 0.42) lasting 2.1 h, observed 1.5–3 h. - The secondary depth implies T2 ≈ 3500 K, which is what a 0.3 M⊙ dwarf should be. - A 3% primary eclipse with K1 = 33 km/s is exactly the self-consistent case: a ~0.23 R⊙ dwarf transiting a 1.3–1.5 R⊙ F star.

What was done

1. **Archives** (cats.py --r 60 with APPOINTMENTS_DATA, Gaia TAP, SIMBAD TAP, ExoFOP JSON, a web search): see `cats-out.txt`, `gaiatap-out.txt`, `progress.md`. - VSX/ASAS-SN list the target as **ASASSN-V J191934.99-620008.1, type ROT, P = 0.884 d**. That is a type revision (the pulsation below), not prior art. - Neither star is in Prša+2022, IJspeert+2024, Kostov+2025 or Gaia vari_eclipsing_binary. - Gaia vari_summary flags the target as a short-timescale + MS oscillator (G std 0.034 mag, range 0.15 mag). - ExoFOP: no TOI or CTOI, no spectra; contratio 0.30. The web search found nothing on any of the star's names. - SIMBAD has the target as SB* from Gaia DR3; its other bibcodes are RAVE survey papers.
2. **Light curve** (`look.png`): the raw SAP flux carries a **±3% multiperiodic ~0.9-d beating pulsation** (γ Dor-like). It is the target's own, given Gaia's 0.034-mag scatter and that the neighbour is a 330-d SR. - The eclipses are superposed on it. I measured every covered event with a cubic baseline over ±0.25 d plus a trapezoid (`measure.py`, `events_W0.25_d3.json/png`). - That gives **74 primaries in 9 sectors** (13, 27, 67, 93, 94, 101–104). Global T14 = **3.16 h** (per-event 3.18 ± 0.15 h); ingress is 0.33 of T14, so the flat bottom is about 1.1 h. - Background, unit-free regression (`scan.py`): median fake signal +0.003 ppt. Only one event (s93 n861, already discrepant at 15.8 ppt with β = 3.5) has a large background term (13 ppt). - QLP centroid shifts in eclipse are a few mpx, as a 30%-contaminated aperture produces. TESS-SPOC MOM_CENTR shifts are ≤ 1.3 mpx. - Blind census: 71 of 95 distinct dips deeper than 8 ppt are the predicted primaries. The other 24 are 8–15 ppt at scattered phases during beat maxima (baseline failures). Nothing else of primary depth is present.
3. **Ephemeris** (`ephem.py`, `ephem-out.txt`, `oc.png`): 70 events give χ²/dof = 8, O–C rms 1.8 min. - The sector-mean O–C wanders between −2.4 and +2.1 min. The same offsets appear in TARS, so the structure is in the data. It is either pulsation-induced timing noise or a slow ETV and is not resolved here.
4. **Second pipelines** (`indep.py`, `indep-out.txt`): TESS-SPOC SAP / PDCSAP for s13, s27 and s67, and TARS for s13, s27, s67, s93 and s94. Every eclipse is recovered at the same times. - The crowding-

corrected PDCSAP depths are **31.6 / 32.2 / 33.0 ppt** (CROWDSAP 0.93 / 0.89 / 0.83). - Difference images: `pixfit.py`, `pixfit_*.png/json`, `pixfit-out.txt`, `pixfit-tpf-out.txt`.

5. **Physics:** `physics.py`, `physics-out.txt` (numbers above).

Why the batch flag fired (probable). The batch `diffstack` built its out-of-eclipse windows from the 1.9-h search box. The true T14 is 3.16 h, so those windows sat 1.4–3.3 h from mid-eclipse and partly inside the eclipse. Median windows do not cancel a 3% pulsation. And its 5×5 positive-clipped centroid has the G 12.45 neighbour 1.6 px away. Modelling the pulsation per pixel and fitting a PSF puts the deficit on the target.

Next eclipses (BTJD)

Formal error ±0.3 min; allow about ±5 min for the O–C wander. - Primary: 4334.8116, 4337.3572, 4339.9028, 4342.4483, 4344.9939, 4347.5395. - Secondary (phase 0.580 ± 0.003, ±9 min, ~2 ppt): 4333.7425, 4336.2880, 4338.8336, 4341.3792, 4343.9248, 4346.4704.

What would settle it further

- A few RVs at the quadratures, to tie K1 and γ to the TESS ephemeris.
- A joint pulsation + eclipse model with limb darkening, for k, b, R1 and e sin ω (the secondary's duration is poorly constrained at 1.5–3 h).
- An O–C analysis with the pulsations prewhitened, to tell whether the ±2-min sector-to-sector timing structure is a third body.

Row fixes for the paper table

field	batch row	per-object value
P (d)	2.54558	**2.5455862 ± 0.0000005**
T0 (BTJD)	1656.8543	**3571.13574 ± 0.00015** (1656.85633 in the batch-epoch global fit)
depth (ppt)	27.4 (box)	QLP trapezoid 28.5 median (25.3–31.7, aperture-dependent, declining with sector); **dilution-corrected ≈ 32** (PDCSAP 31.6–33.0; PSF deficit 32.4)
secondary	—	**1.9 ± 0.24 ppt at phase 0.580** (10.7σ vs control), T ≈ 1.5–3 h
duration	1.9 h (box)	**T14 3.16 h**, ingress/egress 1.03 h each, flat ~1.1 h
events	65	**74 covered primaries in 9 sectors** (70 in the ephemeris fit); secondary stacked over 72 cycles
conj. offset	+0.017	primary = conjunction A, +0.20σ at the Gaia epoch; secondary phase +0.55σ from Gaia conjunction B
diffimg flag	9.7″, peak not on target	**artefact: on target.** 0.24–1.59″ in 7 of 8 sector products (s93 3.45″, PSF-model systematic). G 12.45 neighbour rejected by $\Delta\chi^2 \geq 124$ in every sector.
note	—	Target pulsates at ±3% (γ Dor-like; ASAS-SN "ROT 0.884 d" is this). Companion is an M dwarf (m2 ≈ 0.31 $M_\odot$, T2 ≈ 3500 K). e cos ω = 0.126 from TESS.

A155. TYC 2327-1770-1 (TIC 285385215)

status
: confirmed (per-object re-analysis 2026-10-19; grazing transit of a 0.33 Msun M dwarf on a G8/K0 dwarf at conjunction A, 64-67 ppt QLP, 63.7 PDCSAP, T14 2.94 h, 7 primaries not 2 (the batch kept QUALITY==0 at 200 s) + 8 secondaries; new 5.85 ppt secondary at phase 0.50029, e cos omega +0.0005(4), circular; on target in 3/3 SPOC TPF difference images at 10-134 sigma, the 0.98 arcsec flag sub-pixel WCS scatter; the G 17.3 neighbour at 20 arcsec is a wide co-moving M dwarf; starspot modulation 17-32 ppt at 9.4-9.8 d)

P_days
: 8.2074497

eP
: 0.0000015

T0_primary_BTJD
: 1797.97260

depth_primary
: 6.5

depth_secondary
: 0.66

duration_h
: 2.94

source_path
: Fable 5.1\made\pervet-deep\vet285385215\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 285385215

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.11, P 8.20744 d, 61.7 ppt, 2 events, dphi +0.017, diffimg offset 0.98" (peak not on target, s58).

Verdict: CONFIRMED. TIC 285385215 is an eclipsing binary on Gaia's SB1 orbit, and the eclipses are on the target. It is TYC 2327-1770-1 (Gaia DR3 132815820610967680, G 11.11), a G8/K0 dwarf at 156 pc. The companion is an M dwarf of about 0.33 Msun in a grazing orbit (b ≈ 0.88).

The deciding numbers

Events. The primary eclipse is measured at 7 epochs, not 2:

Sector	Epochs measured
s18	E0, E1, E2
s58	E134, E135
s85	E221, E222

- E133 (2889.59) falls in a data gap.
- E223 (3628.23) sits in a scattered-light mess (background ×20). The dip is visible, but the measurement is rejected.
- The batch count of "2 events" came from keeping only QUALITY==0 at 200-s cadence.
- Of the 7, five are QUALITY==0 in at least one pipeline. The flagged ones carry bit 4096 (s18) or QLP bit 30 (s58/s85).
- Through every event, the background-implied fake depth is below 1.1 ppt, the centroid shift is below 0.002 px, and O−C is ≤ 1 min.

Secondary eclipse. Detected at 8 epochs at phase 0.50029 ± 0.00025. - Depth: 5.6–6.0 ppt in TESS-SPOC PDCSAP, 5.9–7.4 ppt in QLP SAP. - Strength: $\Delta\chi^2$ 207–567 per event.

Ephemerides. - Primary: P = 8.2074497 ± 0.0000015 d, T0 = BTJD 1797.97260 ± 0.00021. χ^2/dof 10.8/5; O−C ≤ 1.0 min. - Secondary: P = 8.2074441 ± 0.0000153 d, T0 = 1802.07870 ± 0.00208. O−C up to 10 min, from scattered light in s85. - Gaia P = 8.207075 ± 0.001056 d, so Δ/σ = +0.35.

Match to Gaia's conjunctions. Monte Carlo over Gaia's errors with corr_vec; corr(ω, t_peri) = +0.99986.

Eclipse	Gaia conjunction	Offset s18	Offset s58	Offset s85
Primary	A (companion in front)	+0.78σ	+0.70σ	+0.68σ
Secondary	B	+0.13σ	+0.32σ	+0.38σ

- The primary offsets are +3.7 to +6.6 h.
- Gaia predicts the secondary 0.515 ± 0.014 in phase after the primary; the measured 0.5003 is 1.1σ from that.

Eccentricity. - From the eclipse timing: e cos ω = +0.0005 ± 0.0004. - From the duration ratio (secondary/primary = 1.04): e sin ω ≈ +0.02 ± 0.03. This is coarse. - Gaia: e cos ω = +0.024 ± 0.022, e sin ω = +0.001 ± 0.028. - The two agree. The orbit is effectively circular.

Attribution. Own difference images from the TESS-SPOC target-pixel files in all three sectors: - In every primary composite, the deficit centre is 0.9–1.7″ (≤ 0.08 px) from the direct-image centre. - Placing the deficit on the target rather than on the G 17.32 neighbour at 20.3″ improves χ^2 by:

Composite	$\Delta\chi^2$	Significance
Primary s18	+99.5	10σ
Primary s58	+17931	134σ
Primary s85	+1249	35σ
Secondary s58	+273	16.5σ

- Every Gaia neighbour within 60″ would need a fractional dip above 1100% for the primary and above 200% for the secondary.

Physics.

Fit: two spheres with quadratic limb darkening, a circular orbit, and one light ratio shared by both eclipses. It is fitted jointly to the 200-s PDCSAP primary and secondary events.

Fitted parameter	Value
k = R2/R1	0.32 (2σ: 0.32–0.42)
a/R1	19.5
b	0.88 (1σ: 0.88–0.94)
L2/L1, TESS band	0.0087
Surface-brightness ratio J	0.085

With f(M) = 0.0251 Msun and M1 = 0.86–0.93 Msun, these imply:

Quantity	Implied	Independent value
m2 (at i = 87.4°)	0.33–0.345 Msun	
a	18.1–18.6 Rsun	
R1	0.93–0.95 Rsun	Gaia/TIC 1.01 ± 0.05
R2	0.30 Rsun	about 0.32 expected for a 0.33 Msun M dwarf
ρ1	1.07 solar	TIC 0.92 ± 0.22

- One geometry explains the V-shaped 64 ppt grazing primary, the 6 ppt secondary, the 2.9 h duration and K1 = 30.9 km/s. The eclipsed star has the target's density.
- The expected ellipsoidal amplitude is about 0.05 ppt, so its absence is expected.
- The 17–32 ppt peak-to-peak modulation is starspots. The Lomb–Scargle peak is at 9.4–9.8 d, near-synchronous with the orbit.

Blind dip census. In QUALITY==0 data, every dip deeper than 3 ppt in all three sectors is at a primary or secondary ephemeris time. Dips elsewhere appear only in flagged scattered-light stretches.

Archives

Source	What was queried	Result
`cats.py --r 60` (APPOINTMENTS_DATA set)	VSX, ASAS-SN, Prša+2022, IJspeert+2024, Kostov+2025 t2/t3/t4	none
`cats.py` SIMBAD	cone	TYC 2327-1770-1 [SB*] at 0.0″; its only reference is the Gaia DR3 multiplicity paper
VizieR	SB9 (30″), WDS (60″), Gaia vari classifier, VSX (120″)	none
Gaia DR3 archive	vari_summary, vari_eclipsing_binary	no rows
ExoFOP	TOI, CTOI, files, spectra, imaging	none; contamination ratio 0.0063
Web search	TYC and TIC names	nothing

- Gaia NSS row: P 8.20707, e 0.031 ± 0.027, ω 10.5 ± 53°, K1 30.92 ± 0.93 km/s, significance 33. RUWE 1.01, ipd_frac_multi_peak 0.
- **The eclipses are new.**

Neighbours (Gaia DR3, 60″; dip needed = 63.7 ppt × 10^(0.4ΔG))

Gaia DR3	sep	G	dip needed	note
132815820610967680	0	11.11	6.4%	target
132815820611210624	20.3″	17.32	~1960%	plx 6.83 vs 6.47: a wide M-dwarf companion (~3100 au projected); not variable in Gaia
132815816315381760	22.0″	20.05	>20000%	
132815820610967424	35.3″	18.83	~7800%	
132815781955415808	50.6″	19.63	>16000%	

Next predicted eclipses (BTJD)

Kind	Times	Uncertainty	Ephemeris
Primary	**4334.0746**, 4342.2820, 4350.4895, 4358.6969	±0.7 min	P 8.2074497

Kind	Times	Uncertainty	Ephemeris
Secondary	**4338.1789**, 4346.3864, 4354.5938, 4362.8013	±7.5 min	secondary's own

The primary ephemeris shifted by +0.50029 phase gives the secondary at 4338.1807 and so on.

What would settle it further

It is already settled as an eclipsing binary on the target. These would refine the parameters: - A few RVs to tighten K1 and e. - A spectroscopic Teff and log g for the primary. - A ground-based multi-band light curve of one primary eclipse (4334.07 or 4342.28) to pin down the grazing chord and the M dwarf's radius.

Row fixes for the paper table

Field	Value	Note
P	8.2074497 ± 0.0000015 d	batch: 8.20744
T0 (primary)	BTJD 1797.97260 ± 0.00021	
Primary depth	**63.7 ppt**	median of TESS-SPOC PDCSAP, crowding-corrected (CROWDSAP 0.992); range 62.0–65.1. QLP SAP: 64–66.6. The batch box depth (61.7) and search depth (51.3) are box-shape underestimates.
Secondary depth	5.85 ppt (PDCSAP, 5.6–6.0)	at phase 0.50029 ± 0.00025
Duration	primary T14 2.94 h, T12/T14 0.40 (near-V, grazing); secondary T14 3.06 h	the batch's "1.9 h" was the box width
Events	7 primaries measured (5 QUALITY==0 in at least one pipeline) in s18, s58, s85; 1 more visible but corrupted (E223); 1 in a gap (E133); 8 secondaries measured	
Conjunction offset	primary at Gaia A, +0.019 to +0.034 phase (≤ +0.78σ); secondary at Gaia B	

Difference-image flag: not an attribution problem. - The batch receipt's 0.98″ offset (0.05″ from the light centroid) is sub-pixel scatter in the WCS, not a sign of a different source. - Our own TESS-SPOC difference images put the deficit within 0.9–1.7″ of the target's direct-image centre. - The target beats the nearest neighbour by 10σ, 134σ and 35σ in s18, s58 and s85. - Clear the flag. Status: candidate → **confirmed**, with the secondary eclipse detected.

A156. TYC 7485-869-1 (TIC 159836610)

status
: confirmed (per-object re-analysis 2026-10-19; flat-bottomed transit of the F star by a 0.47 Msun M dwarf at conjunction A, 90.1 ppt QLP, 90 +- 2 undiluted, T14 4.74 h, 14 + 15 events; new 7.9 ppt secondary at phase 0.5128 = Gaia's 0.527(20), e cos omega +0.020; on target in 4/4 difference images incl. the SPOC target pixels at 47 sigma, the 2.36 arcsec flag a WCS zero-point/pixel-boundary artefact; Gaia RUWE 7.0 suggests a third body)

P_days
: 5.7022928

eP
: 0.0000004

T0_primary_BTJD
: 2618.89923

depth_primary
: 9.0

depth_secondary
: 0.79

duration_h
: 4.74

source_path
: Fable 5.1\made\pervet-deep\vet159836610\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 159836610

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.14, P 5.70229 d, 93.0 ppt, 14 events, dphi +0.010, diffimg offset 2.36" (peak not on target, s28).

Verdict: CONFIRMED. The eclipsing binary is the target, on Gaia's orbit.

TYC 7485-869-1 = Gaia DR3 6584193527718873856. G 11.14, RVS template Teff 6750 K, Gaia FLAME R 2.12 / M 1.35, parallax distance about 450 pc. Gaia RUWE is 7.0. Gaia NSS SB1: P 5.702265 ± 0.000573 d, e 0.107 ± 0.047, ω 63.5 ± 16.0°, K1 37.56 ± 1.19 km/s.

What decides it: - **Timing matches Gaia.** The period agrees to 0.05σ. Every sector's primary falls +0.2 to +0.6σ from Gaia's conjunction A. - **The secondary matches Gaia's orbit.** It is at phase 0.5128 ± 0.0001; Gaia predicts 0.5265 ± 0.0204 (0.7σ). It is 7.9 ppt deep, in 15 of 15 covered epochs, at S/N 31–51 each. - **The difference images put the dip on the target** in all four sectors, 0.0–1.1″ off. $\Delta\chi^2$ against the only plausible neighbour is 1,323–174,363 (6.6–21σ after scaling). The undiluted depth stays at 85–90 ppt for every aperture size. - **No Gaia neighbour can host it.** The brightest within 60″ (G 15.09 at 25.7″) would need 350% of its own flux. - **The physics closes.** An F star with a 0.47 Msun M dwarf: k ≈ 0.29, Teff2 ≈ 3600 K, and the durations ordered as e sin ω > 0 requires.

1. Archives and web - `cats.py --r 60`: VSX, ASAS-SN and every EB catalogue have nothing. SIMBAD lists it as [SB*] (RAVE DR4/5/6, APASS, Gaia DR3 multiplicity). - ExoFOP: no TOI or CTOI. SB9, WDS, GCVS and Gaia vari tables: nothing within 60″. A web search turned up nothing. - Not a catalogued eclipsing binary.

2. Eclipses Method: SAP_FLUX with QUALITY == 0, no trimming. The primary is 90.1 ppt deep (QLP), T14 4.74 h, ingress 1.18 h, flat bottom 2.37 h. A flat bottom is strongly preferred over a V shape ($\Delta\chi^2$ +16,000 for ingress/T14 = 0.5).

sector	E	tc (BTJD)	depth (ppt)
s1	0, 1, 2	1330.1813, 1335.8831, 1341.5854	91.5, 90.3, 90.7
s28	129–132	2065.7772, 2071.4790, 2077.1818, 2082.8836	94.8, 99.2, 94.9, 94.2
s68	320, 321, 323, 324	3154.9146, 3160.6169, 3172.0215, 3177.7243	95.3, 94.3, 94.4, 94.8
s95	448, 449, 451	3884.8082, 3890.5105, 3901.9151	83.1, 82.6, 82.3

- Each primary has S/N 93–355. Timing errors are about 0.4 min.
- Depth differences are between sectors (s95 is the lowest), which is dilution. Within a sector, odd and even events agree to ≤ 0.7%.
- E = 3 (1347.29) is missing from QLP's good-quality data but appears in ELEANOR-lite.
- Background shows nothing at the eclipse times; the per-event background column has broken units. Centroid shifts are about 2 mpix. Two windows contain a bit-32 cadence; no event lies near a listed systematic.
- Six control phases are zero. The seventh, phase 0.5, contains the secondary.
- The blind census finds only the primaries and secondaries, plus dips ≤ 3 ppt.
- Secondary: phase 0.5128, T14 5.28 h, ingress/T14 0.25, depth 7.905 ± 0.055 ppt (S/N 144). Per event it ranges 7.3–8.5 ppt. Across 149 control phases the S/N rms is 1.57, with a maximum of 2.83.

3. Ephemeris and orbit shape - P = 5.7022928 ± 0.0000004 d. T0 = 1330.18107 BTJD at E = 0, or 2618.89923 ± 0.00007 at E = 226. χ^2/dof 11.2/12, O–C rms 0.3 min. - Against Gaia's conjunction A: the eclipse is +0.009 to +0.0125 phase **late**. The table's +0.010 has the right sign. - e cos ω = +0.020 (range +0.016 to +0.020), against Gaia's +0.042 ± 0.032. - The duration ratio (secondary/primary) is 1.114 ± 0.029, giving e sin ω ≈ +0.054 ± 0.013 against Gaia's +0.092 ± 0.043. That puts e ≈ 0.06, against Gaia's 0.107 ± 0.047 (about 1σ).

4. Independence, attribution, dilution - **Other pipelines** see the primary:

pipeline	depth (ppt)
TESS-SPOC s68 PDCSAP (crowding-corrected)	89.6–90.1
TESS-SPOC s68 SAP	86.2–86.9
ELEANOR-lite s1	87–89
T16 s1	88–94
TASOC raw s1	85–86
TGLC s1	104–106

TASOC's corrected flux is broken, and TGLC's flux zero point looks high. The secondary appears in all of them at 7.2–10.3 ppt. - **Difference images:**

sector	source	offset from target	$\Delta\chi^2$ vs G 15.09 at 25.7″	other measure
s1	TESScut	1.0″	1,323	—
s28	TESScut	1.1″	9,074	—
s68	SPOC target-pixel file	0.0″ ± 0.6″	174,363	47σ (jackknife)
s95	TESScut	0.0″	83,114	50σ (jackknife)

The undiluted depth is aperture-invariant: 85–89 ppt (TESScut) and 89.3–90.3 ppt (target-pixel file) for radii 0.8–4 px. - **Why the batch flagged it.** The WCS puts the target at y = 7.43, but the light centroid is at y = 7.52, so the target is effectively on the row 7/8 boundary (peak pixel [7,8] against target pixel [7,7]). The batch's 2.36″ deficit-to-target offset is the WCS zero point: the light centroid is offset by the same amount, and there is no bright neighbour to pull it. - **Dilution.** Adopted undiluted primary 90 ± 2 ppt (QLP ranges 82.6–95 by sector); secondary 7.9 ppt. - **Caveat: Gaia RUWE is 7.0** (astrometric excess noise 1.24 mas, significance 1969). ipd_frac_multi_peak is 1% and the harmonic amplitude 0.016, so Gaia does not see a resolved double. The 5.7-day orbit moves the photocentre only about 44 μas, so a further, longer-period companion is likely. Its light would mean the true depth is somewhat larger. It does not move the attribution: the period, phases and e cos ω all tie the eclipse to the Gaia SB1.

5. Physics - f(M) = 0.0308 Msun. With M1 = 1.35, m2(i = 90°) = 0.466 Msun (0.435–0.497) and a = 16.4 Rsun. - Conjunction A is at r/a = 0.902 (faster), conjunction B at r/a = 1.093 (slower). So the primary eclipse should be the shorter one, and it is: 4.74 h against 5.28 h. - k = 0.28–0.30, so R2 ≈ 0.50–0.54 Rsun at R1 = 1.8 (a main-sequence 0.47 Msun dwarf is about 0.47 Rsun). - The flat bottom needs b ≈ 0.3–0.4, which favours R1 ≈ 1.6–1.8. At R1 = 1.8 the predicted secondary duration is 5.20 h against 5.28 h measured. FLAME's 2.12 assumes no extra light, and the RUWE suggests there is some. - The surface-brightness ratio is 0.097, which fits Teff2 ≈ 3600–3650 K. - The s68 phase curve shows terms of the expected size (0.3–0.6 ppt). But the second-harmonic sign is wrong for pure ellipsoidal variation, and one sector cannot separate spots or systematics, so I did not use it.

Next predicted eclipses (BTJD)

E	primary (±0.2 min)	secondary (±0.8 min)
539	4403.7169	4406.6410
540	4409.4192	4412.3433
541	4415.1215	4418.0456
542	4420.8238	4423.7479

What would settle it further: - RVs at quadrature to check K1 against the eclipse epoch. - High-resolution imaging or speckle to find the companion behind the RUWE of 7, and fix the third light and the true depth. - A near-IR spectrum to see the M dwarf, which would give a double-lined solution with masses and radii.

Row fixes for the paper table - P = 5.7022928 ± 0.0000004 d (table: 5.70229). T0 = 1330.18107 BTJD (E = 0). - Depths: - Primary: 90.1 ppt QLP trapezoid (sector range 82.6–95.0), 90 ± 2 ppt undiluted. The table's 93.0 is the sapdepth median. - Secondary: 7.9 ppt at phase 0.5128 (new). - Durations: primary 4.74 h (ingress 1.18 h, flat 2.37 h); secondary 5.28 h. - Events: 14 primaries in QLP (plus 1347.29 in the other pipelines) and 15 secondaries. - Conjunction offset: +0.009 to +0.0125 phase (eclipse late, +0.2 to +0.6σ). The table's +0.010 is fine. - Orbit from eclipses: e cos ω = +0.020, e sin ω ≈ +0.05. - Difference-image flag: a pixel-boundary and WCS zero-point artefact. The dip is on the target in 4 of 4 sectors (including the s68 target-pixel file, 47σ by jackknife), and no neighbour can host it. - Note: Gaia RUWE 7 suggests a third body. - Status: candidate → **confirmed**.

A157. TYC 1405-695-1 (TIC 86036265)

status
: confirmed (per-object re-analysis 2026-10-19; K dwarf transiting a spotted K giant at conjunction A with a flat-bottomed total occultation at B, ~100 ppt PDCSAP (QLP 103-110, 7% too deep), T14 10.7 h, 14 primaries + 15 secondaries incl. K2 C16 and TESS-SPOC s1751; new 86 ppt secondary at phase 0.50005, e cos omega +0.0001(1), circular; on target in 7/8 own difference images, every neighbour rejected at >= 30 sigma, the 2.85 arcsec flag a sub-pixel PSF offset; prior art: SPOC s42-s46 TCE at P/2 (SNR 542), never a TOI; VSX TAOS VAR 13.93 d and ASAS-SN ROT: at the target are the giant's 14-d rotation, a type revision; EPIC 251330444, BD+20 2296)

P_days
: 11.9972226

eP
: 0.0000022

T0_primary_BTJD
: 2507.95631

depth_primary
: 10.0

depth_secondary
: 8.6

duration_h
: 10.7

source_path
: Fable 5.1\made\pervet-deep\vet86036265\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 86036265

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.20, P 11.99731 d, 102.3 ppt, 14 events, dphi +0.022, diffimg offset 2.85" (peak not on target, s46).

Verdict: CONFIRMED. TIC 86036265 is a detached eclipsing binary on Gaia's SB1 orbit, and the eclipses are on the target. It is BD+20 2296 (EPIC 251330444, Gaia DR3 635898431051113216, G 11.18), a spotted K giant at about 355 pc (Teff 4930 K, R 2.67 Rsun). The companion is a K dwarf of about 0.8 Msun and 0.8 Rsun, with roughly the giant's surface brightness.

The two eclipse kinds look different: - **Primary (phase 0):** the dwarf passes in front of the giant. It is round-bottomed, about 100 ppt deep, and shows starspot-crossing bumps. - **Secondary (phase 0.5):** the giant fully hides the dwarf. It is flat-bottomed and 86 ppt deep.

The deciding numbers

Events. In TESS (QLP, s44–46 and s72): - 8 primaries: E0–E5 and E63–E64. - 7 usable secondaries, plus 2 partial ones at sector edges (2525.95 and 2537.95). - Background-implied fake depth is below 0.7 ppt and centroid shifts are below 0.004 px (QUALITY==0).

K2 C16 (BTJD 1095–1175; EVEREST, K2SFF and the Kepler pipeline light curve) adds 6 primaries and 7 secondaries. TESS-SPOC s1751 (BTJD 4055.8–4063.0, not in the QLP set) adds 1 secondary at 4061.597, 0.7 min from prediction. SPOC's own transit search had already flagged it: TCE_1 at P/2 = 5.99865 d, 99.4 ppt, SNR 542.

Ephemerides (TESS + K2). - Secondary, the clean flat-bottomed eclipse: P = 11.9972226 ± 0.0000022 d, T0(sec) = BTJD 2513.95553 ± 0.00020. 15 events, χ^2/dof 4.3/13, O–C rms 0.5 min. - Primary: P = 11.9972166 ± 0.0000091 d, T0 = 2507.95631 ± 0.00069. 14 events, O–C rms 3.3 min; spot crossings distort the transit mid-times. - The two periods differ by +0.5 ± 0.8 s. - Gaia P = 11.995577 ± 0.001562 d, so Δ = +0.00165 d = 1.05σ.

Match to Gaia's conjunctions. Monte Carlo with corr_vec; corr(ω, t_peri) = 0.99999.

Eclipse	Gaia conjunction	K2	s44–46	s72
Primary (transit, deeper)	A (companion in front)	+1.08σ	+1.33σ	+1.36σ
Secondary (occultation)	B	+1.11σ	+1.37σ	+1.39σ

The offset grows over time, from +0.009 to +0.040 in phase. That drift is Gaia's period being 1σ short, not a mismatch in which conjunction is which.

Eccentricity. - From the eclipse timing: secondary phase 0.50005 ± 0.00006, so e cos ω = +0.00008 ± 0.00009. - Both eclipses have T14 = 10.7 h, so e sin ω ≈ 0. - Gaia: e = 0.005 ± 0.014, e cos ω = +0.0001 ± 0.0117, Gaia-predicted phase separation 0.5001 ± 0.0075. - The orbit is circular.

Attribution. Own difference images from the SPOC 2-min target-pixel files (s44, s45, s46) and TESS-SPOC (s72), primary and secondary separately: - In 7 of 8 composites, the deficit centre is 1.1–5.2″ (≤ 0.25 px) from the direct-image centre. - The single-event s44 secondary composite is 9.7″ (0.46 px) off; it sits on a sector-start ramp. - These offsets come from fitting a Gaussian PSF at deficit S/N 38–148. The formal errors are tiny, so the offsets are model systematics, not evidence of another source. - Placing the deficit on the target rather than on any Gaia neighbour improves χ^2 by at least 930 (30σ) in every composite. Against the G 14.80 star at 69.9″ the improvement is 6145–145967. - Every star within 30″ is G 20.3–21.0 and would need a dip above 25000%. The G 14.80 star at 69.9″ would need 160–206%. - I also tried a K2 pixel difference image. It came out unusable (deficit 1–2 ppt of the direct image), most likely because Kp 11.2 is near saturation and I did not correct the roll motion. It is not counted either way.

Dilution-corrected depths.

Light curve	Primary (ppt)	Secondary (ppt)
TESS-SPOC / SPOC PDCSAP s44–46 (CROWDSAP 0.995–0.998)	97.2–101.1	83.9–87.5
TESS-SPOC s72	95.0, 95.9	85.7, 86.3
TESS-SPOC s1751	–	88.7
K2 EVEREST (Kepler band)	89.6–108.7	85.0–87.5
QLP SAP	102.7–109.9	91.0–95.2

QLP runs about 7% deeper than both SPOC and K2.

Physics.

Fit: two spheres with limb darkening, fitted jointly to 15 TESS-SPOC PDCSAP events. Starspot crossings make the formal errors meaningless; the values below are good to the grid step (k 0.01, a/R1 0.25).

Fitted parameter	Value
k = R2/R1	0.30
a/R1	10.5
b	≈ 0.02 (central)
L2/L1, TESS band	0.095
Surface-brightness ratio J	1.06

With f(M) = 0.1178 Msun and M1 = 1.0–1.4 Msun, these imply:

Quantity	Implied	Independent value
m2 (at i = 89.9°)	0.70–0.84 Msun	
a	26.3–28.9 Rsun	
R1	2.51–2.75 Rsun	TIC 2.67
R2	0.75–0.82 Rsun	

- J ≈ 1 means the companion's Teff is about the giant's, 4900–5000 K. That is a K1–K2 dwarf of about 0.8 Msun, which matches m2 from Gaia's mass function.
- The occultation depth is the same in the Kepler and TESS bands (86 ppt), as J ≈ 1 predicts.
- K1 = 45.6 km/s fits a 0.7–0.8 Msun companion; nothing about it is suspicious.
- The ellipsoidal signal is lost under 60–77 ppt peak-to-peak starspot modulation. Lomb–Scargle peaks at 14.5 d (s44–46), 14.2 d (s72) and 13.8 d (K2): sub-synchronous rotation of an RS CVn-like giant.

Blind dip census. In QUALITY==0 data, every dip deeper than 4 ppt is at an ephemeris eclipse. Dips elsewhere appear only in flagged scattered-light stretches of s72.

Archives

Source	What was queried	Result
`cats.py --r 60`	VSX	TAOS 061.00059 at 0″, type VAR, P = 13.93424 d
`cats.py --r 60`	ASAS-SN	variable at 1.0″, type ROT:, P = 415 d
`cats.py --r 60`	Prša, IJspeert, Kostov t2/t3/t4	none
`cats.py` SIMBAD	cone	BD+20 2296 [SB*]
SIMBAD bibliography	4 references	TAOS variability (2014), JSDC, mid-IR diameters, Gaia DR3 multiplicity; no eclipse paper
VizieR	SB9, WDS, Gaia vari classifier	none
Gaia DR3 archive	vari_summary, vari_eclipsing_binary	no rows
ExoFOP	TOI, CTOI	none; lists EPIC 251330444 with uncalibrated K2 photometry
Web search	BD+20 2296, EPIC 251330444, TIC 86036265	nothing

- Gaia NSS row: P 11.99558, e 0.005, K1 45.59 ± 0.65 km/s, significance 70. RUWE 1.17. No GSP-Phot or FLAME parameters.
- The TAOS 13.93 d period is the giant's rotation (we measure 13.8–14.5 d). That changes the variability type; it is not a prior eclipse detection.
- **No catalogue lists it as an eclipser.**

Neighbours (Gaia DR3; dip needed = 86 ppt × 10^(0.4ΔG), using the occultation depth)

Gaia DR3	sep	G	dip needed	note
6358984310511132116	0	11.18	8.6%	target
6358984267558828880	20.3″	20.97	~72000%	
6358984954753587520	26.2″	20.28	~38000%	
6358984654108509044	69.9″	14.80	~240%	plx 8.46; Gaia variable, VSX ROT; cannot host the dip, rejected at ≥ 78σ in the difference images

Next predicted eclipses (BTJD)

Kind	Times	Uncertainty
Secondary (occultation)	**4337.5334**, 4349.5306, 4361.5278, 4373.5250	±0.6 min
Primary (transit)	**4343.5304**, 4355.5277, 4367.5249, 4379.5221	±2.2 min

The secondary ephemeris shifted back by 0.50005 phase puts the primary at 4343.532, which is consistent.

What would settle it further

It is settled as an eclipsing binary on the target. These would yield masses and radii: - Double-lined spectroscopy to measure K2 for the dwarf, which contributes about 9% of the light; Gaia saw it as single-lined. - A spectroscopic Teff and log g for the giant. - A multi-band occultation light curve to pin down the dwarf's Teff. - Modelling the spot crossings during transit. This also gives a chance to measure the spin–orbit alignment.

Row fixes for the paper table

Field	Value	Note
P	11.9972226 ± 0.0000022 d	from secondaries, TESS + K2 + s1751; the batch's 11.99731 is 7.6 s too long
T0 (primary)	BTJD 2507.95631 ± 0.00069	the secondary ephemeris gives 2507.95692 for the same epoch
Primary depth	**~100 ppt** (PDCSAP s44–46: 97–101; s72: 95–96)	starspot-modulated. The batch's 102.3 and QLP's 103–110 are about 7% too deep.
Secondary depth	**86 ppt** (PDCSAP 84–89; K2 85–87.5)	flat-bottomed total occultation, so L2/(L1+L2) = 0.086
Duration	T14 = 10.7 h for both; T12/T14 0.25 (primary), 0.20 (secondary)	
Events	14 primaries (8 TESS, 6 K2); 15 secondaries (7 TESS, 7 K2, 1 s1751) plus 2 partial TESS	
Conjunction offset	primary at Gaia A, +0.009 (K2) to +0.040 (s72) phase, +1.1 to +1.4σ; secondary at Gaia B	the drift is Gaia's 1σ-short period

Difference-image flag: not an attribution problem. - The batch receipt (s46: 2.85″ from target, 1.6″ from the light centroid) is a sub-pixel offset from a crude PSF on a very bright eclipse. - Our own difference images in four sectors put the deficit within ≤ 0.25 px of the target's direct image in 7 of 8

composites, and the target beats every neighbour by ≥ 30σ. - No neighbour can physically host the dip. - Clear the flag. Status: candidate → **confirmed**.

Web sources consulted (no relevant hits for either star): - Seven new triply eclipsing triple star systems - Revising the properties of low mass eclipsing binary stars using TESS light curves - Discovery and characterisation of long-period eclipsing binary stars from K2 campaigns 1–3

A158. TYC 8142-243-1 (TIC 268342242)

status
: confirmed (per-object re-analysis 2026-10-19; K dwarf transiting an F star at conjunction A with a total occultation at B, 124 ppt QLP, ~125 PDCSAP, T14 9.14 h, 13 primaries + 14 secondaries; new 33 ppt secondary at phase 0.4924, e cos omega -0.012 = Gaia's; the 2.23 arcsec flag was 0.1 px of box-centroid systematics, no neighbour can host a 12% dip (>= 206% needed), SPOC DV offset 4.4 +- 3.8 arcsec; prior art: TESS-SPOC s56-s69 TCE at P/2; the VSX contact EB at 29 arcsec is unrelated)

P_days
: 14.6202005

eP
: 0.0000021

T0_primary_BTJD
: 2840.37583

depth_primary
: 12.4

depth_secondary
: 3.3

duration_h
: 9.14

source_path
: Fable 5.1\made\pervet-deep\vet268342242\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 268342242

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent). Batch row: G 11.08, P 14.62018 d, 121.7 ppt, 10 events, dphi +0.012, diffimg offset 2.23" (peak not on target, s7).

TYC 8142-243-1, Gaia DR3 5493180643792180352, G 11.08, Teff 6129 K (TIC; Gaia GSP-Phot 5827 K), R 1.92 R⊙, contratio 0.055. Gaia SB1: P 14.621257 ± 0.001851 d, e 0.2181 ± 0.0107, ω 264.45 ± 2.77°, K1 39.06 ± 0.47 km/s, significance 83.

Verdict: CONFIRMED. The target is a detached, eccentric EB with both eclipses, on Gaia's orbit, and the difference-image flag is an artefact.

Deciding numbers: - **Event count.** This is not a 2–3-event detection. The diffimg sector s7 alone holds 2 primaries and 1 secondary. In total there are **13 measured primaries** (9 fully covered and 4 partial) and **14 secondaries** (13 full and 1 partial), from 9 QLP sectors. Three more primary epochs and one

more secondary fall only in flagged cadences. - **Gaia orbit.** - P: TESS 14.6202005 ± 0.0000021 d vs Gaia 14.621257 ± 0.001851 d, −0.57σ. - Primary timing: at the Gaia epoch the primary is −2.10 h = −1.28σ from conjunction A. - Secondary timing: at the Gaia epoch the **secondary is +0.01 h = 0.0σ** from conjunction B (predicted 391.3802 ± 0.054, TESS 391.3807). - Phase separation 0.4924 vs Gaia 0.4863 ± 0.0069 (+0.9σ). That gives e cos ω = −0.0117 ± 0.001 (exact Sterne relation) vs Gaia −0.021 ± 0.011 (+0.9σ). - With Gaia's e and ω fixed, the four measured durations (T14 / T23 at A and B) are reproduced to $\chi^2 = 0.72$. Freeing e sin ω gives −0.225 vs Gaia's −0.217 ± 0.011 (0.7σ). - **Attribution.** - Flux budget: every Gaia neighbour within 90″ would need an eclipse of **≥ 206%** to make the 122-ppt primary (the brightest is G 14.15 at 56″). Only the target can host it. - The VSX/SIMBAD EB at 29″ (Gaia DR3 5493180536415078656, G 16.80, P 0.2452 d) would need more than 1000%. - **The SPOC DV report (TESS-SPOC s56–s69) independently puts the eclipse on target.** TCE 1 (P/2 = 7.31007 d fold, 33 ppt) has a difference-image offset from the TIC position of **4.4 ± 3.8″ (1.2σ)**. - The same DV report puts TCEs 2 and 3 (0.2453 d, 0.8 ppt) at 28.8 ± 2.8″, which is the 29″ contact EB. So that machinery does separate sources here. - My own difference images cover 16 products (TESScut s7/s88/s89; TESS-SPOC TPFs s7/s8/s9/s34/s35/s61/s62; primaries and secondaries separately). The deficit centroid lies **0.4–3.6″ from the Gaia target in 15 of 16**, and 5.1″ in the single-event tpf s9 primary. It is always closer to the target than the light centroid (1.8–7.4″ off, pulled by the box pedestal). The fractional loss at the target pixel is 95–124 ppt for the primary and 25–35 ppt for the secondary. - **Physics.** - f(m) = 0.0839 ± 0.0030 M⊙, so m2 ≈ 0.65 M⊙ for M1 = 1.16 (0.60–0.71 for M1 1.0–1.35); a = 30.7 R⊙. - The duration solution gives R1 1.76 R⊙ (TIC 1.92 ± 0.09), k 0.30 (R2 0.53 R⊙), i 88.3°, b_A 0.62 and b_B 0.40, so the secondary is a total occultation. - The depth ratio of 0.264 implies a surface-brightness ratio of about 0.3, so T2 ≈ 4300 K. That is what a 0.65 M⊙ K dwarf should be. - A 12.5% primary from a K1 = 39 km/s orbit is fully self-consistent.

What was done

1. **Archives** (`cats-out.txt`, `gaiatap-out.txt`): cats.py --r 60 found only the VSX "E" star at 29″ (above). - No ASAS-SN entry, and the target is not in Prša+2022, IJspeert+2024 or Kostov+2025. - SIMBAD has the target as SB*; its only bibcode is Gaia DR3 NSS. - Gaia vari tables: none. ExoFOP: no TOI or CTOI, no spectra or imaging. The web search found nothing. - The SPOC DV report (`dvr.xml` in `sky\data\qlp-sb-p20-60\pervet-deep-268342242\`) is a pipeline TCE, not a catalogue entry.
2. **Measurement** (`measure.py` with W 0.6 d, a quadratic baseline and a 7-point supersampled trapezoid for the 30/10-min FFIs; `events_pri/sec.json/png`; the per-epoch ledger `partial.py` → `partial-out.txt`). - Primary: T14 **9.14 h**, ingress 0.36 of T14, QLP depth 120–128 ppt (per-sector aperture). - Secondary: T14 **6.33 h**, ingress 0.26, 30–34 ppt. - The ledger gives, per epoch, the coverage fraction for QUALITY == 0 and for all cadences, the flag bits, the baselines on each side, and the depth with and without flagged cadences (changes ≤ 6 ppt). - Background gate (unit-free regression): at most 0.73 ppt. - Blind census: every dip deeper than 8 ppt at SNR > 7 is a primary, a secondary, or a baseline-window wing of one. There is no third family. - The broad 1–2-d dips in s61 (2963–2964, 2975–2976) are scattered-light systematics, not eclipse-shaped.
3. **Ephemerides** (`ephem.py`, `ephem-pri-out.txt`, `ephem-sec-out.txt`, `oc_pri.png`, `oc_sec.png`): - Primary: χ^2/dof 1.48, O−C rms 0.50 min. - Secondary: P 14.6202157 ± 0.0000041, T0 2730.61517 ± 0.00022, χ^2/dof 1.16, rms 1.17 min. - P_sec − P_pri = +1.3 s/cycle (3.3σ). This is possibly slow apsidal motion and is unconfirmed.
4. **Second pipeline** (`indep.py`, `indep-out.txt`): TESS-SPOC PDCSAP (crowding-corrected, CROWDSAP 0.93–0.94) gives primary depths **126.5 / 125.4 / 126.4 / 119.0 ppt** (s7/8/9/61) and secondary depths **33.1–33.3 ppt** (30.7 in s61). Times agree with QLP to under 1 min, and MOM_CENTR shifts are small. - Difference images: `pixfit2.py`, `pixfit2-out.txt`, `pixfit2-`

`summary.txt`. The Gaussian-PSF variant (`pixfit.py`, `pixfit-out.txt`) is unusable here: 2 events per product on a bright star.

5. **Physics:** `physics.py`, `physics-out.txt`.

Why the batch flag fired. 2.23″ is about 0.1 px of box-centroid systematics; my box centroids show the same 1–3″ scale. With n_possible = 0 by flux budget, and the SPOC DV offset at 1.2σ, it was a flag in name only.

Next eclipses (BTJD)

- Primary (~125 ppt, 9.1 h): 4346.2565, 4360.8767, 4375.4969, 4390.1171, 4404.7373 (±0.4 min).
- Secondary (~33 ppt, 6.3 h): 4338.8389, 4353.4591, 4368.0793, 4382.6995, 4397.3198 (±0.7 min).

What would settle it further

- An RV pair, to obtain K2 and absolute masses. This is a clean double-eclipse system with total secondaries, so it is a good benchmark candidate.
- A limb-darkened joint fit for k, i and R1.
- More sectors, to test the 3.3σ primary/secondary period difference (apsidal motion).

Row fixes for the paper table

field	batch row	per-object value
P (d)	14.62018	**14.6202005 ± 0.0000021** (secondary ephemeris 14.6202157 ± 0.0000041)
T0 (BTJD)	1495.3175	**2840.37583 ± 0.00012** (primary); secondary T0 2730.61517 ± 0.00022
depth (ppt)	121.7	QLP 120–128 (median ~124); **PDCSAP 119–126.5 (≈125 dilution-corrected)**
secondary	—	**33 ppt** (PDCSAP 30.7–33.3) at phase **0.4924**; total occultation
duration	6.3 h (box)	**primary T14 9.14 h** (flat 2.6 h); **secondary T14 6.33 h** (flat 3.0 h)
events	10 (2 in the diffimg sector)	**13 primaries (9 full + 4 partial) + 14 secondaries (13 full + 1 partial)**; 3 primary and 1 secondary epochs only in flagged cadences
conj. offset	+0.012	primary –1.28σ from conjunction A; secondary 0.0σ from conjunction B at the Gaia epoch
diffimg flag	2.23″, peak not on target	**artefact: on target.** No neighbour can make the dip (needs ≥ 206%). SPOC DV offset 4.4 ± 3.8″. Own deficit centroids 0.4–3.6″ (15/16).
note	—	K-dwarf companion: m2 ≈ 0.65 M⊙, R2 ≈ 0.53 R⊙, T2 ≈ 4300 K; i 88.3°. TESS e cos ω −0.012, e sin ω −0.225. SPOC TESS-SPOC TCE exists (s56–s69 DV, at P/2).

A159. TYC 6635-619-1 (TIC 437230270)

status
: confirmed (per-object re-analysis 2026-10-19; detached G + K-dwarf double eclipser, 137 ppt PDCSAP (QLP 132-140), new 12.0 ppt secondary at phase 0.62842, T14 4.1/3.35 h, 10 + 10 events; e cos omega 0.2005 vs Gaia's 0.252(13), 3.9 sigma with Gaia's covariance and 3.3 without: Table 2; on target in 5/5 primary and 4/4 secondary difference images, the 2.65 arcsec flag sub-pixel quantization, no neighbour within 60 arcsec brighter than G 16.6; conjunction offset is -0.023, the batch's +0.023 a sign slip; Kostov table 2 only)

P_days
: 11.870552

eP
: 0.000001

T0_primary_BTJD
: 2937.17038

depth_primary
: 13.7

depth_secondary
: 1.2

duration_h
: 4.1

source_path
: Fable 5.1\made\pervet-deep\vet437230270\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5.1 (2026-10-19)

Archived source: sanitized report text.

TIC 437230270

Opus-model subagent for Fable 5.1, 2026-10-19 (saved by Fable 5.1 from the agent's final message; the write tool refused in-agent; the agent's own byline below says "Opus 5.5 subagent", meaning an Opus-model subagent, launched from Fable 5.1's session). Batch row: G 11.49, P 11.87057 d, 124.7 ppt, 10 events, dphi +0.023, diffimg offset 2.65" (peak not on target, s9).

TIC 437230270: per-object attribution vet (pervet-deep). Opus 5.5 subagent, 2026-10-19.

The star is TYC 6635-619-1 = Gaia DR3 3537372345302366848: G 11.49, TIC Teff 5894 K (Gaia GSP-Phot 5661 K), TIC R 1.51 Rsun, TIC M 1.07 Msun, parallax 2.86 mas, RUWE 0.99.

Gaia DR3 SB1 orbit (goodness of fit 2.64):

P (d)	e	ω (deg)	K1 (km/s)	corr(ω, Tp)
11.874084 ± 0.001091	0.299 ± 0.014	32.6 ± 3.5	52.59 ± 0.81	+0.97

Verdict: CONFIRMED

This is a detached, double-eclipsing G-star + K-dwarf binary on the target. The primary is at Gaia's conjunction A, and there is a strong secondary. The difference-image flag is pixel quantization.

What decides it: - **Attribution.** Own difference images: - Primary, five sectors (s9, 36, 63, 90, 100): deficit 0.2–1.8″ from the target. - Secondary, four sectors (s36, 63, 90, 100): deficit 0.6–2.7″ from the target. - Only 4 Gaia sources lie within 60″, all G ≥ 16.64. To make the primary they would need eclipses of at least 1580% (the G 16.64 star would need 138% even for the secondary). They are also rejected directly by $\Delta\chi^2 \geq 39{,}000$ (15–38σ scaled) for the primary and +430 to +1216 (17–28σ) for the secondary. - **Both eclipses, consistently.** 10 primaries at 135–138 ppt (PDCSAP) and 10 secondaries at 11–13 ppt. Primary O−C is within 0.8 minutes. The secondary has the same period as the primary (difference +0.000004 ± 0.000007 d), so there is no apsidal motion over 7 years. - **Timing against Gaia.** At Gaia's epoch the primary lands 0.069 d (1.8σ) from conjunction A. - **Physics.** f(m) = 0.155 Msun gives m2 ≈ 0.82–0.86 Msun, a K dwarf. A two-eclipse geometry fits both depths and both durations at i ≈ 86.5–87° (§5).

1. Archives and web

- `cats.py 437230270 --r 60`: no VSX, no ASAS-SN. SIMBAD has TYC 6635-619-1 [SB*].
- Kostov+2025 table 2 (J/ApJS/279/50) has it at 0.001″. That is the unvetted ML list and doesn't count as prior art.
- Gaia DR3: no vari_summary or vari_eclipsing_binary row, and no variability rows for any neighbour.
- SB9, WDS and J/AJ/160/111: none. ExoFOP: no TOI or CTOI, no imaging or spectroscopy.
- A web search on all three identifiers found nothing.
- **Not previously catalogued as an eclipser.**
- MAST has QLP for s9, 36, 63, 90 and 100; TESS-SPOC for s9, 36 and 63; TGLC for s9; TARS for s9, 36, 63 and 90; ELEANOR-LITE and T16 for s9.

2. Every eclipse

Fits use raw QLP SAP_FLUX, QUALITY==0, with a trapezoid on a local quadratic baseline (`ecl.py`).

Primaries:

sector	tc (BTJD)	depth (ppt)	T14 (h)	T23 (h)
9	1548.3157 ± 0.0003	132.3	4.13	0.83
9	1560.1863 ± 0.0003	133.0	4.03	0.81
36	2284.2900	139.2	4.13	
36	2296.1605	137.8		
63	3020.2643	139.7		
63	3032.1348	138.3		
90	3756.2390	135.6		
90	3768.1091	135.7		
100	4076.7434	118.4		
100	4088.6135	118.7		

- All tc errors are about ±0.0003 d. Every primary has T14 of 4.03–4.13 h and T23 of about 0.8 h. Depth errors are 0.3–0.7 ppt.
- s100 cycle 215 (4100.49) has only flagged or partial data.

- The QLP depth in s100 is low (118 ppt) because of the QLP aperture or background in that sector. My own s100 difference image gives the same target-only depth as the other sectors (about 110–120 ppt in the core window).

Secondaries (phase 0.628): 1567.6441, 2291.7512, 2303.6183, 3015.8534, 3027.7255, 3039.5940 (flagged cadences, ±0.0033), 3751.8266, 3763.6987, 4084.2016, 4096.0746. Errors are ±0.0005–0.0013 d. Depths are 11.0–13.7 ppt, T14 3.17–3.65 h, T23 0.4–1.1 h.

- **Background and centroids:** the background fake depth is at most 0.46 ppt for every event. QLP centroid shifts are at most 0.009 px.
- **Blind census** (4-hour box, every sector): only the primaries and secondaries.
- **Second pipelines:**
- TESS-SPOC PDCSAP (s9, 36, 63; CROWDSAP 0.989–0.992): primary 136.2–138.0 ppt, secondary 11.6–13.0 ppt.
- TESS-SPOC SAP: 133.9–136.8 ppt and 11.3–12.8 ppt.
- TARS (s9–s90): 131–147 ppt and 10.6–16.1 ppt.
- TGLC s9: PSF 108.8 ppt / 8.8 ppt, aperture 151.6 ppt / 12.8 ppt (its calibration differs).
- Times agree to 1–2 minutes.

3. Ephemeris and orbit

- **Primary:** P = 11.870552 ± 0.000001 d, T0 = 2937.17038 ± 0.00009 BTJD (χ^2/dof = 0.76).
- **Secondary:** P = 11.870556 ± 0.000007 d, T0 = 3407.58163 ± 0.00044 BTJD (χ^2/dof = 4.4).
- **Phase separation:** 0.62842 ± 0.00004.
- **Against Gaia:**
- P_TESS − P_Gaia = −0.00353 d (3.2σ).
- At Gaia's epoch the primary is at 385.0016 against conjunction A at 384.933 ± 0.039 (+0.069 d, 1.8σ). The deep eclipse is conjunction A: the K dwarf in front of the G star.
- At the s9 epoch the primary is 0.28 d *before* Gaia's conjunction A (1548.316 against 1548.593 ± 0.132), so dphi = −0.023. That matches the c4 vetsum; **the paper table's +0.023 has the wrong sign.**
- **e cos ω** from the phase separation (exact Kepler): +0.2005 ± 0.0002. Gaia gives +0.2519 ± 0.0131 (phase separation 0.6603 ± 0.0081), a **3.9σ disagreement**. Gaia's goodness of fit of 2.64 suggests its errors are underestimated by roughly 1.5–2×. A companion supplying about 20–25% of the light could also bias an SB1 solution.
- **e sin ω** isn't free from the duration ratio alone (T14 of 4.10 and 3.35 h) because the impact parameters differ between the two conjunctions; see §5.

4. Independence and attribution

- **Light curves:** see §2; TESS-SPOC, TARS and TGLC all show both eclipses.
- **Difference images, primary.** Fitted deficit offsets from the target (east, north) in arcsec:

s9	s36	s63	s90 (1 event)	s100
(+0.2, +1.6)	(+1.1, +0.1)	(+1.4, +1.0)	(+1.6, −0.1)	(0.0, −0.2)

The fractional loss on the target pixel is 107–111 ppt, the largest in the scene. - **Difference images, secondary:** offsets are s36 (0.0, +1.4), s63 (0.0, +0.6), s90 (+0.4, +0.5) and s100 (+2.6, +0.2). - **Neighbour dilution table** (depth needed = 137 ppt × 10^(0.4 ΔG); every entry is impossible):

Gaia DR3	G	separation	needs
3537372345302783872	19.15	26.8″	15,873%
3537372418317693440	16.64	37.3″	1580% (138% even for the secondary)
3537372139144349696	18.54	41.4″	9037%
3537372139146119936	20.05	51.4″	36,476%

None is a catalogued variable. - **Why the batch flagged it:** the 2.65″ deficit-to-target offset with the peak on the pixel next to the target is sub-pixel quantization. Its psf_frac_resid (0.10) and deficit-to-light offset (0.9″) were already clean. **This is an artefact, ON TARGET.**

5. Physics

- **Companion:** f(m) = 0.1555 Msun gives m2 = 0.82–0.86 Msun for M1 = 1.07–1.15, a K dwarf (R2 ≈ 0.8 Rsun, T2 ≈ 5000 K). It supplies about 15–25% of the TESS light.
- **Gaia's orbit with the TIC radius fails:** with Gaia's (e, ω) and R1 = 1.51, a 4.1-hour primary can be at most about 65 ppt deep, and the secondary chord would miss (b_B ≈ 1.6 R1). That combination cannot produce the observed pair.
- **A free geometry fit does** (`geom.py`; e cos ω fixed at 0.2005, surface-brightness ratio J solved from the secondary, linear limb darkening, rectilinear chords; observables T14_A = 4.10 ± 0.10 h, T14_B = 3.35 ± 0.15 h, depth_A = 137 ± 4 ppt, depth_B = 12.0 ± 0.5 ppt):
- Solutions with $\chi^2 < 4$ need e sin ω = +0.10 to +0.29 (Gaia's +0.161 is inside), R1 = 1.0–1.35, k = 0.42–1.0, J = 0.23–1.5, i = 86.5–87.1°, e = 0.22–0.35.
- A Gaia-consistent example: R1 = 1.10, k = 0.74 (R2 = 0.81 Rsun, a main-sequence 0.82 Msun star), e sin ω = 0.16, J = 0.46, i = 86.8°, e = 0.257.
- This needs R1 smaller than the TIC's 1.51, which treats the binary as a single star. The remaining tension is between Gaia's (e, ω) and the photometric e cos ω (3.9σ).
- **Expected out-of-eclipse signal:** ellipsoidal plus reflection of about 0.2–0.4 ppt. The test was not attempted reliably because QLP and SPOC have 5–15 ppt of slow systematics.
- **Shape:** depths, durations, the 4-hour U/V primary and the K-dwarf secondary all fit one detached geometry. A 13.7% eclipse with K1 = 52.6 km/s and m2 ≈ 0.8 Msun is self-consistent, not the "deep eclipse with tiny K1" pattern.

Next eclipses (BTJD)

- **Primary** (±0.0002 d): 4337.8955, 4349.7661, 4361.6366, 4373.5072, 4385.3777, 4397.2483.
- **Secondary** (±0.0007 d): 4345.3556, 4357.2261, 4369.0967, 4380.9672, 4392.8378, 4404.7083.

What would settle it further

- High-resolution spectra to look for the K dwarf's lines (SB2). A full RV solution would test Gaia's ω and e against the photometric e cos ω = 0.2005.
- A joint light-curve model (limb darkening, third light about 0) with an SED fit to separate R1 from the companion's light.

Row fixes for the paper table

- **P:** 11.870552 ± 0.000001 d (row: 11.87057).
- **T0:** 2937.17038 ± 0.00009 BTJD (primary, Gaia's conjunction A).

- **Depth:**
- Primary 137 ± 1 ppt (PDCSAP, CROWDSAP 0.99; QLP SAP 132–140, s100 118). The row's 124.7 is the box median.
- Secondary 12.0 ± 0.5 ppt at phase 0.62842 ± 0.00004 (PDCSAP 11.6–13.0; QLP 11.0–13.7).
- **Duration:** primary T14 = 4.1 ± 0.1 h (T23 ≈ 0.8 h); secondary T14 = 3.35 ± 0.15 h.
- **Event count:** 10 primaries (s9, 36, 63, 90 and 100, two each, all QUALITY==0) and 10 secondaries (9 on QUALITY==0 plus s63 at 3039.594 using flagged cadences).
- **dphi:** −0.023 at the s9 epoch (row: +0.023; the sign is wrong). At Gaia's epoch: +0.069 d, 1.8σ.
- **e cos ω:** 0.2005 photometric against Gaia's 0.252 ± 0.013 (3.9σ).
- **Difference-image flag: pixel-quantization artefact, ON TARGET.**
- Primary: deficit 0.2–1.8″ from the target in 5 sectors.
- Secondary: deficit 0.6–2.7″ from the target in 4 sectors.
- No neighbour within 60″ can host either eclipse; the closest candidate would need 1580% (138% for the secondary).
- Not a catalogued eclipser, apart from the unvetted Kostov table 2 flag.
- The status can go from candidate to confirmed.

A160. TYC 7558-731-1 (TIC 142240367)

status
: confirmed (per-object re-analysis 2026-10-21; grazing G + 0.5-0.6 Msun companion at conjunction A, 19.1 ppt own-aperture (QLP 16.2 in s097/s104 is 200-s dilution), T14 3.42 h, 11 primaries + 9 secondaries; secondary 4.2-5.1 ppt at phase 0.50120, e cos omega +0.0019(1); TESS-SPOC s3 difference image on target; no neighbour brighter than G 14.4 within 74 arcsec)

P_days
: 7.541350

eP
: 0.000002

T0_primary_BTJD
: 3464.17253

depth_primary
: 1.91

depth_secondary
: 0.42

duration_h
: 3.42

source_path
: Fable 5\made\sb1-scale\pervet\vet142240367\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 142240367 (TYC 7558-731-1, Gaia DR3 4945899177740273920): per-object vet

Verdict: CONFIRMED. The eclipsing binary is the Gaia SB1 itself. It is grazing, with both eclipses detected.

Wave 4, 2026-10-20/21. A first Opus 5.5 subagent got as far as the secondary and ephemeris stage and was killed. A second one checked its receipts and re-measured the raw SAP independently (`check.py`), then did the physics (`check.py`, `esinw.py`) and this report. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Deciding numbers

- **Primary eclipse, 11 events (s003, s097, s104):**
- Every event has S/N 39–81.
- Own-aperture TESScut depth is **19.1 ± 0.1 ppt** in all three sectors (`cutphot-out.txt`).
- QLP SAP gives 18.5 ppt in s003 but 16.2 in s097/s104. That step is QLP dilution in the 200-s products, not a real change. TESS-SPOC s3 gives 19.0 (PDCSAP) and 18.8 (SAP).

- T14 is 3.42 h; the best T12/T14 is 0.4, so the shape is nearly a V.
- **Secondary eclipse, 9 events, S/N 15–26 each:**
- Phase **0.50120 ± 0.00005**.
- Depth 4.2 ppt in QLP, 4.7–5.1 ppt in the own aperture; T14 3.76 h.
- Fixed-shape test: S/N 26.9 at the measured phase, 16.9 at Gaia's predicted phase. The look-elsewhere controls (518 phases) have rms 1.5 and a maximum of 3.7.
- **Ephemerides:**
- Primary: P = 7.541350 ± 0.000002 d, T0 = 3464.17253 ± 0.00033 BTJD (E = −49), O−C rms 1.4 min.
- Secondary: P = 7.541351 ± 0.000008 d; P2 − P1 = +0.17σ.
- Against Gaia's P = 7.541328 ± 0.000329 d: **+0.07σ**.
- **Gaia conjunctions (Monte Carlo using `corr_vec`):**
- The primary sits on **conjunction A**: +0.60 h (+0.76σ) at the Gaia epoch, and +0.21 to +0.49σ in each sector.
- The secondary sits on conjunction B: −0.25 h (−0.43σ).
- **Orbit shape:**
- e cos ω = +0.0019 ± 0.0001, against Gaia's +0.0089 ± 0.0110 (−0.63σ).
- The duration ratio T_B/T_A is 1.10. The model lets the impact parameter scale with the separation at each conjunction, which matters for a grazing eclipse. It gives **e sin ω ≈ −0.016**, against Gaia's −0.022 ± 0.011 (about 0.5σ, same sign).
- **Which star is in front:** at conjunction A the primary is farthest away, so the companion is in front. The deeper eclipse is at A, as expected for a luminous primary.
- **Background and systematics:**
- The unit-free background regression gives |fake| median 0.02 ppt and maximum 0.39 ppt for the primaries; at most 0.10 ppt for the secondaries.
- Two primaries include bit-32 cadences (E −324 and E 16); their depths agree with the rest.
- No event falls within 0.5 d of a listed sector-wide systematic, and centroid shifts are under 1 mpix.
- **Blind dip census:** of 23 dips at S/N ≥ 4, 12 are primaries and 9 are secondaries. None of the off-ephemeris dips is deeper than half the eclipse; the largest is 1.2 ppt.
- **Raw re-check (`check-out.txt`):**
- Primaries: s003 18.9 ppt, s097 15.8, s104 15.9.
- Secondaries: 4.4, 4.0 and 4.0 ppt.
- The earlier cutout pass showed a 22–29 ppt "secondary" at s104 E = 47. It falls in a QLP data gap with no usable Q0 coverage, so it is an edge artefact, not an event.

Archives

- `cats.py --r 60`:
- VSX: none.
- ASAS-SN: none.
- Prša / IJspeert / Kostov: none.
- SIMBAD: TYC 7558-731-1, listed only as [SB*].
- SB9, WDS and J/AJ/160/111: none.
- ExoFOP: no TOI or CTOI; no imaging, spectroscopy or companions.

- Gaia DR3: vari_summary and vari_eclipsing_binary are empty; phot_variable_flag is NOT_AVAILABLE.
- A web search for "TYC 7558-731-1" (2026-10-21) found nothing about this star.
- **There is no prior eclipse record, so this is new.**

Independence

- **Other pipelines, s3, 3 events each:**
- TESS-SPOC PDCSAP: 18.98 ± 0.14 ppt.
- SPOC-SAP: 18.83 ppt.
- TGLC aperture: 22.5 ppt; TGLC PSF: 22.9 ppt.
- **Difference image (TESS-SPOC s3 TPF):**
- The flux loss peaks at S/N 32.8, and the loss-weighted centroid is on the target.
- The single-star fit gives χ^2 1803 for the target and 6487 for the neighbour.
- The batch diffimg for s104 puts the deficit 0.71″ from the light centroid, with the peak on target.
- **Neighbour dilution:**
- The only G < 15 star within 60″ is Gaia 4945899212100013056: G 14.38, 74″ away, Δm 3.31.
- To produce these eclipses it would need dips of about 350 ppt (primary) and 90 ppt (secondary), and the difference-image fit rejects it.
- TIC contratio is 0.0009.

Physics

- **Mass function and companion:**
- f(m) = 0.0643 $M_\odot$.
- With M1 = 1.07 ± 0.13 and R1 = 1.77 ± 0.08 (a slightly evolved G star, Teff 5898 K): m2(i = 90°) = 0.51–0.59 $M_\odot$ and a ≈ 19 $R_\odot$.
- **Grazing geometry:**
- A central chord would last about 7.2 h.
- The observed 3.42 h gives b ≈ 1.17 and i ≈ 84°.
- **Predicted against observed depths (uniform disks, main-sequence companion with R2 ≈ 0.59 $R_\odot$, Teff ≈ 3860 K):**
- Primary: 16–21 ppt predicted, 19 observed.
- Secondary: 3–4.6 ppt predicted, 4.3–4.9 observed.
- **Companion temperature and light:**
- The depth ratio of 0.256 implies Teff2 ≈ 4160 K.
- L2/L1 is about 2%, which fits a single-lined (SB1) spectrum.
- **Out-of-eclipse terms are small:** ellipsoidal about 0.04 ppt, c1 about −1.9 ppt.
- **No heartbeat:** e = 0.026, so heartbeat can be excluded.
- **One geometry fits everything:** a companion of about 0.55 $M_\odot$ grazing a 1.8 $R_\odot$ primary.

Next predicted eclipses (BTJD)

Errors are formal; the O–C rms is 1.4 min for the primary and 5 min for the secondary. - **Primary (conjunction A):** - 4338.9691, 4346.5105, 4354.0518 (±0.6 min). - 4406.8413, 4414.3826, 4421.9240. - **Secondary (conjunction B):** - 4335.2079, 4342.7492, 4350.2906 (±2.2 min). - 4403.0800, 4410.6214, 4418.1628.

What would settle it further

It is settled at the level the brief asks for. Two optional checks: - An RV phase check: the RV curve should cross γ, going from approaching to receding, at the primary eclipse. - Multi-colour photometry of a primary eclipse, to pin the grazing geometry and the companion's temperature.

A161. TYC 4564-1186-1 (TIC 288160281)

status
: confirmed (per-object re-analysis 2026-10-21; strongly grazing G8-K0 + 0.6-0.7 Msun companion, 16.2 ppt, T14 1.92 h, 41 primaries + 41 secondaries; secondary 6.71 ppt at phase 0.50726, e cos omega +0.0108(2); equal durations want e sin omega ~0, 2.5-3.3 sigma from Gaia's -0.0325(99), open tension; VSX ROT at the target is a type revision; Kostov table 2 only; on target in 7 TESS-SPOC difference images, the G 14.15 neighbour at 6.5 arcsec excluded)

P_days
: 7.063961

eP
: 0.000002

T0_primary_BTJD
: 2597.74250

depth_primary
: 1.62

depth_secondary
: 0.67

duration_h
: 1.92

source_path
: Fable 5\made\sb1-scale\pervet\vet288160281\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 288160281 (TYC 4564-1186-1, Gaia DR3 1715865311931719680): per-object vet

Verdict: CONFIRMED. The eclipsing binary is the Gaia SB1 itself, grazing, with both eclipses detected. One open tension: the eclipse durations want e sin ω ≈ 0, which is 2.5–3.3σ from Gaia's value.

Wave 4, 2026-10-20/21. The first subagent was killed; the second resumed, as for TIC 142240367. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Deciding numbers

- **Primary eclipse, 41 events in 13 sectors (s014–s075):**
- Every event has S/N 16–56.
- Weighted depth is 16.18 ± 0.07 ppt (QLP SAP). Depth constancy χ^2 is 70.8 for 40 degrees of freedom.
- Sector means run from 15.5 to 17.3 ppt; even and odd events give 16.19 and 16.20.
- T14 is 1.92 h, with best T12/T14 of 0.4.

- **Secondary eclipse, 41 events, S/N 4.8–21:**
- Phase **0.50726 ± 0.00005** (from the two ephemerides).
- Depth 6.71 ± 0.07 ppt; T14 1.92 h (1σ range 1.88–1.98).
- Fixed-shape test: S/N 32.1 at the measured phase. The look-elsewhere controls (518 phases) have rms 0.94 and a maximum of 2.8.
- At Gaia's predicted phase, 0.5154, the test finds nothing (−4.9σ, sitting on the wings), so the measured phase is where the secondary is.
- **Ephemerides:**
- Primary: P = 7.063961 ± 0.000002 d, T0 = 2597.74250 ± 0.00016 BTJD (E = 21), O−C rms 1.6 min.
- Secondary: P = 7.063962 ± 0.000004 d; P2 − P1 = +0.24σ.
- Against Gaia's P = 7.063673 ± 0.000331 d: **+0.87σ**.
- **Gaia conjunctions:**
- The primary sits on **conjunction A**: +1.06 h (+1.55σ) at the Gaia epoch.
- Within sectors the offset grows from +0.014 to +0.024 in phase (1.1–1.3σ). The growth comes from Gaia's P error, not from the eclipses (the wave-3 lesson).
- The secondary sits on conjunction B: −0.32 h (−0.39σ).
- **Orbit shape:**
- e cos ω = +0.0108 ± 0.0002, against Gaia's +0.0242 ± 0.0130 (−1.0σ).
- The durations are equal (ratio 1.00, range 0.98–1.03). In the grazing model that gives **e sin ω ≈ 0 ± about 0.005**.
- Gaia's e sin ω is −0.0325 ± 0.0099. At that value the model predicts T_B ≈ 2.6 h against the observed 1.92 h; a non-grazing chord would still predict a ratio of 0.94.
- So the tension is 2.5–3.3σ. The likely cause is Gaia's eccentricity being biased high for a nearly circular orbit (Lucy–Sweeney): if e sin ω ≈ 0, then e ≈ 0.011 against Gaia's 0.043 ± 0.010.
- This is a flagged open tension, like 1260529. It does not affect attribution: P, conjunction A timing and e cos ω all agree with Gaia.
- **Which star is in front:** the companion is in front at conjunction A, and that is where the deeper eclipse is.
- **Background and systematics:**
- The unit-free fake-ppt regression gives median 0.06 ppt and maximum 2.54 ppt for the primaries (E −8, s040), far below 16 ppt.
- One secondary, E 120 in s073, has fake 20 ppt and 15 flagged cadences. It is discounted; it has S/N 4.8 and is not needed.
- The `dBKG` columns in `vet.py` and `check.py` have broken units here, as BRIEF-ADAPT warns, and are not used.
- Five primaries include bit-32 cadences; their depths are normal.
- No event falls within 0.5 d of a sector-wide systematic, and centroid shifts are 0.1 mpix or less on average.
- **Blind dip census:**
- Of 96 dips at S/N ≥ 4, 40 are primaries.
- All 4 off-ephemeris dips deeper than half the primary depth are secondaries (phase −0.493).
- Everything else is 2.3 ppt or shallower, apart from three dips in noisy windows: 3300.53 (15.8 ppt, S/N 4.7 — the same flagged window as the discounted E 120 secondary) and 3346.39 (8.6 ppt, S/N 4.3, beside a primary).
- **Raw re-check (`check-out.txt`):**

- 41 primaries, all positive, with sector medians of 15.0–16.8 ppt (central-depth estimator).
- 37 secondaries, all positive, with a weighted depth of 6.31 ppt.

Archives

- **VSX:** Gaia DR3 1715865311931719680 is typed ROT at 0″, from the Gaia classifier. That is a type revision to EB, not prior art.
- **Kostov+2025 table 2:** flagged at 0.001″. This is an unvetted ML list and does not count as catalogued.
- SIMBAD lists TYC 4564-1186-1 as [SB*].
- ASAS-SN, SB9, WDS and J/AJ/160/111: none.
- ExoFOP: no TOI or CTOI; no imaging or spectroscopy.
- **Gaia DR3:**
- phot_variable_flag is VARIABLE, with G std 0.0072 mag and range 0.036 mag. That fits the eclipses.
- It is in the classification results but not in vari_eclipsing_binary, and vari_eb is empty.
- A web search for "TYC 4564-1186-1" (2026-10-21) found nothing.
- **There is no prior eclipse classification, so this is new.**

Independence

- **TESS-SPOC light curves, all 13 sectors (`indep-out.txt`):**
- PDCSAP: 15.1–16.2 ppt.
- SAP: 13.8–15.0 ppt.
- **Difference images, TESS-SPOC TPFs for 7 sectors (s040, 41, 48, 53, 60, 73, 74):**
- In every sector the loss peaks on the central pixel (S/N 19–44).
- The single-star fit always prefers the target over the 6.5″ neighbour: χ^2 536 vs 890, 510 vs 1366, 322 vs 1082, 355 vs 888, 194 vs 434, 495 vs 981, and 389 vs 718.
- The target's implied own dip is 12–14 ppt.
- The 30-min sectors s014–s026 crashed in `tpfdiff.py` (an empty slice) and were not redone; the 7 consistent sectors suffice.
- The batch diffimg (s040) puts the deficit 0.13″ from the light centroid, with the peak on target.
- **Neighbour dilution:**
- Gaia 1715865311931719552 is G 14.15, 6.5″ away (Δm 2.68).
- To produce these eclipses it would need dips of about **190 ppt** (primary) and **80 ppt** (secondary).
- It is unrelated to the target: parallax 1.12 vs 4.55 mas, proper motion (6.7, −7.8) vs (−31.2, 0.2) mas/yr.
- It is rejected by the difference-image χ^2 in all 7 sectors, by the centroid, and above all by the period: it would have to eclipse at Gaia's SB1 period to 0.9σ and at Gaia's conjunction A, with the secondary at Gaia's e cos ω.
- TIC contratio is 0.075.

Physics

- **Mass function and companion:**
- f(m) = 0.109 $M_\odot$.

- With M1 = 0.96 ± 0.12 and R1 = 1.11 ± 0.06 (a G8–K0 dwarf, Teff 5481 K): m2(i = 90°) = 0.61–0.70 M⊙ and a ≈ 18 R⊙.
- **Grazing geometry:**
- A central chord would last about 5.3 h.
- The observed 1.92 h gives b ≈ 1.5 and i ≈ 84.7°, so the eclipse is strongly grazing.
- **Predicted against observed depths (uniform disks, main-sequence companion):**
- Primary: 10–14 ppt predicted; 13.8–16.2 observed, depending on the pipeline's dilution correction.
- Secondary: 3–5.6 ppt predicted, 6.3–6.7 observed.
- That is agreement within a factor of about 1.3. A grazing depth is steeply sensitive to R2 and b, and the model ignores limb darkening, so this is consistent but not a precision check.
- **Companion temperature and light:**
- The depth ratio of 0.415 implies Teff2 ≈ 4380 K, against 4180 K for a main-sequence star of m2(i = 90°).
- L2/L1 is about 0.13–0.17, still compatible with Gaia treating it as SB1.
- **Out-of-eclipse terms are small:** ellipsoidal (−c2) about −0.34 ppt, c1 about −0.8 ppt.
- **No heartbeat:** e = 0.043 and the depth is 16 ppt, so the heartbeat route is excluded.

Next predicted eclipses (BTJD)

Errors are formal; the O–C rms is 1.6 min for the primary and 2.9 min for the secondary. - **Primary (conjunction A):** - 4335.4770, 4342.5409, 4349.6049 (±0.8 min). - 4406.1166, 4413.1805, 4420.2445. - **Secondary (conjunction B):** - 4339.0605, 4346.1244, 4353.1884 (±1.4 min). - 4402.6361, 4409.7001, 4416.7641.

What would settle it further

The eclipse attribution is settled. Two things would resolve the open questions: - An RV re-fit with e fixed near 0.011, or an independent RV orbit. That would test whether Gaia's e = 0.043 (e sin ω = −0.033) is the biased quantity. - A light-curve model with limb darkening. It would close the factor-1.3 depth gap and give i, R2/R1 and the companion's temperature.

A162. TYC 2205-1326-1 (TIC 258860596)

status
: confirmed (per-object re-analysis 2026-10-21; grazing F + ~0.45 Msun M dwarf at conjunction A, 18.1 ppt, T14 2.82 h, 10 + 10 events in s056/s083; new secondary 1.38 +- 0.13 ppt at phase 0.5014 (S/N 10.8, controls <= 1.9), e cos omega +0.0022(11); QLP-only star, own TESScut photometry reproduces both eclipses aperture-independently; the 4.71 arcsec batch offset resolves to 1.0 arcsec on target; no neighbour brighter than G 15.7 within 75 arcsec)

P_days
: 4.959311

eP
: 0.000003

T0_primary_BTJD
: 3056.15218

depth_primary
: 1.81

depth_secondary
: 0.14

duration_h
: 2.82

source_path
: Fable 5\made\sb1-scale\pervet\vet258860596\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 258860596 — per-object vet (sb1-scale, pervet)

Verdict: CONFIRMED — an eclipsing SB1 with a grazing primary eclipse at Gaia conjunction A and a shallow secondary at phase 0.501. P = 4.959311 d.

TYC 2205-1326-1, Gaia DR3 1783049316382818816, G 11.50, Tmag 11.19, TIC Teff 6900 K, R 1.69 Rsun, M 1.5 Msun. TESS data: QLP s056 and s083 (200 s). No other sectors, and no non-QLP light curves in MAST.

The work was done in two sessions (wave 4, 2026-10-20/21). The first was killed partway through. It left behind vet.py, sec.py, sec2.py, gaiaepoch.py, the archive queries, and their outputs, all in this folder. The second session checked the key numbers in those outputs, then added: - the full MAST CAOM listing (`caomall-out.txt`); - independent TESScut photometry and difference images (`cutphot-out.txt`, `cutdiff_s56/s83.png`, `cutfold_s56/s83.png`); - the physics and dilution check (`physics-out.txt`). (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

1. Archives (own pass)

- `cats.py --r 60`: VSX none; ASAS-SN none; SIMBAD has TYC 2205-1326-1 as `SB*` (Gaia SB1) only.
- SB9, WDS and J/AJ/160/111 return nothing.
- ExoFOP has no TOIs, CTOIs, imaging, spectroscopy or stellar companions.
- Gaia DR3: vari_summary and vari_eclipsing_binary are empty; phot_variable_flag is NOT_AVAILABLE; RUWE 0.86.
- Batch receipts: c4-ebcats has no hit in any vetted or unvetted EB table. c4-vetsum gives "dphi −0.007/−0.486 ATTR".
- Web searches for "TYC 2205-1326-1" and "TIC 258860596" find no page that mentions the star.
- **Not previously catalogued as an eclipsing binary.**

2. Eclipse measurements (raw QLP SAP_FLUX, QUALITY==0, no trimming)

Primary: - Trapezoid fit: depth 18.08 ppt, T14 2.82 h, T12 1.13 h. T12/T14 is 0.40, so the shape is close to a V. Flat bottom 0.56 h. - Ten individual eclipses give 18.13 ± 0.13 ppt. - The depth is constant (χ^2 9.9/9): even events 18.15 ± 0.20, odd 18.11 ± 0.18; s056 17.89 ± 0.18, s083 18.43 ± 0.20. - Every event has S/N 20–52. - None has a bit-32 cadence (momentum-dump flag) in its window. None falls within 0.5 d of a listed sector-wide systematic.

Controls: phases 0.125–0.875 (excluding 0.5) are all within ±0.13 ppt of zero. No control event reaches S/N > 3.

Background: - In-eclipse minus out-of-eclipse SAP_BKG is +0.46 ppt of the target's flux. - A fake dip of 18 ppt would need +91 e^-/s. - A unit-free regression gives a possible fake of at most 0.3 ppt per event. The one exception is the event at 2837.94, which has 14 flagged cadences, and even there the fake is only −2.2 ppt against an 11-ppt dip.

Centroids: dX −0.18 ± 0.27 mpix, dY +1.10 ± 0.93 mpix.

Blind dip census: - 14 dips at S/N ≥ 4. - 11 are on the primary ephemeris. The other 3 are 1.7–2.0 ppt dips at phase 0.5, which is the secondary. - There is no dip deeper than 9 ppt anywhere off the ephemeris.

Secondary: sec2 was used, not the shared sec.py. sec2 fixes the primary's shape at the predicted phase and compares against random phases. - At phase 0.5000: 1.38 ± 0.13 ppt, S/N 10.8. - 436 random phases with the same shape give a maximum S/N of 1.94 (rms 0.97). None reaches the secondary's S/N; it sits at 11.1× the control rms. - Changing the detrending (2 or 6 harmonics, or leaving the window unmasked) moves the depth by at most 0.02 ppt. - Per event (10 events): 1.48 ± 0.13 ppt, constant (χ^2 5.2/9). In raw SAP: 1.25 ± 0.15 ppt. - Clean subset (excluding the listed systematic at 3564.5): 1.47 ± 0.15 ppt. - Duration ratio T14(sec)/T14(pri) = 0.90 (1σ range 0.85–0.95).

3. Ephemeris and orbit

Primary ephemeris: - P = 4.959311 ± 0.000003 d. - T0 = 3056.15218 ± 0.00022 BTJD. - χ^2/dof 18.1/8 (errors scaled ×1.51); O−C rms 0.9 min. - Dropping any one event changes P by at most 2 ppm-d.

Comparison with Gaia: - Gaia P = 4.959237 ± 0.000701 d, which differs from the TESS period by +0.11σ. - The Monte Carlo used Gaia's correlation vector (the ω–Tp correlation is +1.00). At the Gaia epoch (BTJD 389) the primary lands on conjunction A at −0.64σ; at BTJD 3250 it lands at +0.03σ. -

The secondary lands on conjunction B at +0.17σ at both epochs.

Secondary timing: the per-event times put the secondary at phase 0.50139 ± 0.00067. A free secondary period gives 4.959290 ± 0.000029 d, −0.75σ from the primary period.

Orbit shape compared with Gaia:

quantity	TESS	Gaia	difference
e cos ω	+0.0022 ± 0.0011	−0.0097 ± 0.033	+0.36σ
e sin ω (from the duration ratio)	−0.053 ± 0.028	−0.002 ± 0.035	−1.1σ

Both agree with Gaia's e = 0.034 ± 0.037, which is consistent with a circular orbit.

4. Independence

MAST: CAOM searched within 10.8″ under all target names (so TGLC products filed under Gaia IDs would show up). There are only the QLP s56/s83 light curves, the SPOC and TICA full-frame images, and non-TESS imaging. No SPOC, TESS-SPOC, TGLC, TARS or eleanor light curve exists.

Own photometry from the TESScut cubes (SPOC-calibrated FFIs, own aperture and background, no QLP product). Depths in ppt:

aperture	s56 primary	s83 primary	s56 secondary	s83 secondary
r ≤ 1.0 px	18.14 ± 0.19	18.58 ± 0.23	1.30 ± 0.18	1.76 ± 0.25
r ≤ 1.5 px	18.12 ± 0.16	18.51 ± 0.20	1.25 ± 0.16	1.60 ± 0.20
r ≤ 2.0 px	18.49 ± 0.18	18.76 ± 0.22	1.03 ± 0.18	1.59 ± 0.21
r ≤ 3.0 px	18.61 ± 0.25	18.58 ± 0.33	0.68 ± 0.26	1.78 ± 0.28

- The primary depth does not change with aperture size. An off-target source would make it change.
- The control phases 0.25 and 0.75 are within ±0.6 ppt of zero.
- The s56 secondary falls as the aperture grows, but the two ends differ by only 2σ, and the s83 secondary is flat.

Difference images: - The centroid of the lost light is 1.0″ (s56) and 4.1″ (s83) from the centroid of the out-of-eclipse light. - The pixel with the most significant loss (S/N 21 in s56, 27 in s83) is the target pixel or the one next to it. - A PSF fit at the target gives χ^2 88 (s56) and 805 (s83). A fit at any neighbour gives $\chi^2 \geq 1194$ (s56) and ≥ 3895 (s83). - This agrees with the batch diffimg result for s56: peak on target, loss 1.09″ from the light centroid.

Dilution: - The brightest Gaia neighbour within 90″ is G 15.70 at 74.9″. It would need an 87% dip to produce the primary. - Every other neighbour would need more than 100%.

5. Physics

Companion mass and orbit size: - Mass function f(M) = 0.0245 ± 0.0031 Msun. - For a primary mass M1 of 1.35–1.65 Msun, the companion's minimum mass m2 (at i = 90°) is 0.43–0.48 Msun. - The separation is a ≈ 15.3 Rsun, or 9.1 primary radii.

Grazing geometry: - A main-sequence companion of R2 ≈ 0.49 Rsun gives a radius ratio k ≈ 0.29, so a central eclipse would last 5.4 h. - The observed 2.82 h implies an impact parameter b ≈ 1.10 primary radii (i ≈ 83°). That is grazing, since b < 1 + k. - Grazing explains both the near-V shape and a depth (18 ppt) far below k^2 (85 ppt).

Which star is in front: - At conjunction A (the primary eclipse), the M-dwarf companion passes in front of the F star. - At conjunction B, the F star passes in front of the companion.

Companion temperature: - The depth ratio gives a surface-brightness ratio J ≈ 0.076, so T2 ≈ 3500 K at 800 nm. - That is the right temperature for a 0.45 Msun main-sequence M dwarf. - The companion then supplies about 0.6% of the light in the TESS band.

The depths, both durations and the secondary depth all fit one grazing, circular F + early-M system. The eccentricity is low and the conjunction is not near periastron, so the heartbeat/tide confusion does not arise.

Verdict: CONFIRMED

The deciding numbers: - 10 of 10 primaries at S/N 20–52, with constant depth 18.13 ± 0.13 ppt. - P 4.959311 d, +0.11σ from Gaia's period. The primary falls on Gaia conjunction A (–0.6σ at the Gaia epoch) and the secondary on conjunction B (+0.2σ). - A secondary of 1.38 ± 0.13 ppt at phase 0.5014 with S/N 10.8, where no random phase gets above S/N 1.9. - Own TESScut photometry reproduces both eclipses: primary 18.1–18.8 ppt with no aperture dependence, secondary 1.3–1.6 ppt. - The difference image puts the loss on the target, and no neighbour can produce the dip. - The physics fits one geometry: a grazing eclipse by a ~0.45 Msun M dwarf.

Next predicted eclipses (BTJD; linear TESS ephemeris; secondary at phase 0.50139):

primary	secondary
4400.1256 ± 1.3 min	4402.6121 ± 5.0 min
4405.0849 ± 1.3 min	4407.5715 ± 5.0 min
4410.0442 ± 1.3 min	4412.5308 ± 5.0 min

What would settle it further: - Gaia DR4 epoch RVs, to check that the deep eclipse falls at the RV node Gaia predicts for conjunction A. - A third TESS sector, to tighten e sin ω. - Ground photometry: a 2.8-h, 18-ppt eclipse at Tmag 11.2 is easy to catch.

A163. TYC 1337-1803-1 (TIC 55114385)

status
: confirmed (per-object re-analysis 2026-10-21; grazing spotted G + 0.6-0.7 Msun companion at conjunction A, 17.8 ppt, T14 1.86 h, 23 primaries + 25 secondaries in 5 sectors, both eclipses in QLP/TESS-SPOC/SPOC/TARS; secondary 2.43 ppt at phase 0.50003, e cos omega 0.0000(2); Gaia's P is 2.24 sigma short, attribution holds at the Gaia epoch (-1.3 sigma), the 403936502 pattern; on target in 5 SPOC TPF difference images, the G 14.5 neighbour at 19.5 arcsec excluded)

P_days
: 4.268746

eP
: 0.000001

T0_primary_BTJD
: 2870.92006

depth_primary
: 1.78

depth_secondary
: 0.24

duration_h
: 1.86

source_path
: Fable 5\made\sb1-scale\pervet\vet55114385\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 55114385 — per-object vet (sb1-scale, pervet)

Verdict: CONFIRMED — an eclipsing SB1 with a grazing primary eclipse at Gaia conjunction A and a secondary at phase 0.5000. P = 4.268746 d.

TYC 1337-1803-1, Gaia DR3 3371848501900503296, G 10.59, Tmag 10.16, TIC Teff 5995 K, R 1.27 Rsun, M 1.105 Msun, d 201 pc. TESS data: QLP s043, s044, s045 (600 s) and s071, s072 (200 s). TESS-SPOC and TARS light curves and SPOC target pixel files exist for all five sectors.

Wave 4, 2026-10-20/21. The killed session had already run the archive checks, vet, sec, sec2, gaiaepoch, mastlc (with downloads), indep (primary and secondary) and tpfdiff on the five target pixel files. The second session re-checked those numbers and added the physics and dilution check (`physics-out.txt`). It also overrode `sec-out.txt`; see the secondary section below. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

1. Archives (own pass)

- `cats.py --r 60`: VSX none; ASAS-SN none; SIMBAD has TYC 1337-1803-1 as `SB*` only.
- SB9, WDS and J/AJ/160/111 return nothing.
- ExoFOP has no TOIs, CTOIs, imaging, spectroscopy or companions.
- Gaia DR3: vari_summary and vari_eclipsing_binary are empty; phot_variable_flag is NOT_AVAILABLE; RUWE 1.17.
- Batch receipts: c3-ebcats2 has no hit in any EB table. c3-vetsum gives "dphi +0.020/−0.474 ATTR".
- Web searches for "TYC 1337-1803-1" and "TIC 55114385" find no page that mentions the star.
- **Not previously catalogued as an eclipsing binary.**

2. Eclipse measurements (raw QLP SAP_FLUX, QUALITY==0)

Primary: - Trapezoid fit: depth 17.89 ppt, T14 1.86 h, T12 0.74 h. T12/T14 is 0.40, so the shape is close to a V. Flat bottom 0.37 h. - 23 individual eclipses, all at S/N 27–70, give 17.81 ± 0.07 ppt. - Even 17.80 ± 0.09, odd 17.81 ± 0.10. - By sector: 17.37, 17.89, 17.55, 18.31, 17.81 (each ±0.13–0.17). - The scatter is larger than the errors (χ^2 69/22). The amplitudes are about 0.5 ppt from sector to sector. That fits spot activity on a G star likely tidally synchronised to the 4.27-d orbit: the c1 harmonic at the orbital period is −2.1 ppt. - Three events have bit-32 cadences in their windows. Their depths (16.9, 18.4, 17.8 ppt) are normal. - No event falls within 0.5 d of a listed systematic.

Controls: phases 0.125–0.875 (excluding 0.5) are within ±0.11 ppt of zero.

Background: - In-eclipse minus out-of-eclipse SAP_BKG is −0.76 ppt of the target's flux. - A fake dip of 17.8 ppt would need +231 e^-/s.

Centroids: dX +0.32 ± 0.10 mpix, dY −0.05 ± 0.13 mpix.

Blind dip census: - 50 dips at S/N ≥ 4. - 22 are primaries. 25 of the off-ephemeris dips sit at phase ±0.50, which is the secondary. - There is no off-ephemeris dip of 8.9 ppt or more.

Secondary: - `sec-out.txt` is overridden. It was produced by the shared sec.py, which locked onto a 9.8-h feature at phase 0.47 with an inconsistent period (the known failure mode). sec2 was used instead. - At phase 0.5000: 2.43 ± 0.21 ppt, S/N 11.7. - 475 random phases give a maximum S/N of 2.70 (rms 0.73). The secondary's S/N is 16.0× the control rms. - The best phase is 0.4998 (1σ range 0.4993–0.4998). - Per event (25 events, all at S/N > 3): 2.46 ± 0.06 ppt. In raw SAP: 2.51 ± 0.07 ppt. - No event could be faked by the background: |fake| ≤ 0.29 ppt for every event. - The depth varies (χ^2 116/24; 1.98 in s072, 2.94 in s071). That is again consistent with spots. - Duration ratio T14(sec)/T14(pri) = 0.95 (1σ range 0.90–1.00).

3. Ephemeris and orbit

Primary ephemeris: - P = 4.268746 ± 0.000001 d. - T0 = 2870.92006 ± 0.00009 BTJD. - χ^2/dof 69/21 (errors scaled ×1.82); O−C rms 0.6 min. - Dropping any one sector leaves P within 1 ppm-d.

Comparison with Gaia: - Gaia P = 4.268512 ± 0.000105 d, +2.24σ from the TESS period. - Projecting from the Gaia epoch into the TESS sectors therefore drifts, giving +1.9σ to +2.0σ at conjunction A. - At the Gaia epoch itself, using Gaia's correlation vector (ω–Tp correlation +1.00): - the primary lands on conjunction A at −0.66 h (−1.33σ); - the secondary lands on conjunction B at −0.09 h

(−0.32σ). - Conjunction B is −27σ to −37σ from the primary, so B cannot be the primary. - So this is the 403936502 pattern: Gaia's own period is slightly off, and the attribution holds when compared at the Gaia epoch.

Secondary timing: the per-event times put the secondary at phase 0.50003 ± 0.00015.

Orbit shape compared with Gaia:

quantity	TESS	Gaia	difference
e cos ω	+0.00004 ± 0.00024	−0.0079 ± 0.0098	+0.81σ
e sin ω (from the duration ratio)	−0.026 ± 0.026	+0.019 ± 0.009	−1.6σ

The −1.6σ e sin ω difference is mild and uses a crude duration-ratio estimator. Gaia's e = 0.023 ± 0.009 is small. Overall the orbit is consistent with near-circular.

4. Independence

Other pipelines' light curves (`indep-out.txt`, `indep-sec-out.txt`). Depths in ppt:

pipeline	primary depth per sector	secondary depth per sector
TESS-SPOC PDCSAP	17.2–17.5	2.2–2.7
SPOC SAP	16.2–16.6	2.0–2.6
TARS	16.5–17.4	2.2–3.2
QLP SAP	17.4–18.2	2.2–2.8

- Every pipeline sees both eclipses in every sector.
- SPOC SAP is about 1 ppt shallower than PDCSAP. That is the expected crowding correction (TIC contamination ratio 0.05).

Difference images from the SPOC target pixel files (s43, s44, s45, s71, s72; `tpfdiff-out.txt`, `tpfdiff_*.png`): - In every sector the pixel with the most significant loss (S/N 34–49) is the target pixel. - A PSF fit at the target gives χ^2 766–1134, and implies the target's own dip is 12.9–14.1 ppt (with crude PSF normalisation). - The best neighbour fit gives χ^2 3730–6715. - The G 14.51 neighbour at 19.5″ would need a dip of 141–204 ppt in this test. - The batch diffimg result for s72 agrees: peak on target, loss 1.27″ from the light centroid.

Dilution: - The neighbour at G 14.51, 19.5″ would need a 66% primary dip and a 9% secondary. - The neighbour at G 14.76, 77″ would need 83%. - All others would need more than 100%. - Even the two that could in principle produce it are excluded by the difference images.

5. Physics

Companion mass and orbit size: - Mass function f(M) = 0.0891 ± 0.0018 Msun. - For a primary mass M1 of 0.99–1.22 Msun, the companion's minimum mass m2 (at i = 90°) is 0.61–0.69 Msun. - The separation is a ≈ 13.4 Rsun, or 10.5 primary radii.

Grazing geometry: - A main-sequence companion of R2 ≈ 0.68 Rsun gives k ≈ 0.53, so a central eclipse would last 4.8 h. - The observed 1.86 h implies an impact parameter b ≈ 1.41 primary radii (i ≈ 82°). That is grazing, since b < 1 + k = 1.53. - Grazing explains both the near-V shape and a depth (17.8 ppt) far below k^2 (≈ 290 ppt).

Which star is in front: - At conjunction A (the primary eclipse), the K/M-dwarf companion passes in front of the G star. - At conjunction B, the G star passes in front of the companion.

Companion temperature: - The depth ratio gives a surface-brightness ratio J ≈ 0.136, so T2 ≈ 3600 K at 800 nm. - A 0.65 Msun main-sequence star should be nearer 4000–4200 K. - That gap is within what the J estimate can resolve for a grazing chord, where limb darkening differs between the two eclipses and spots are present. - The companion supplies about 4% of the light in the TESS band.

Everything else fits one grazing, near-circular G + late-K/early-M system. The eccentricity is low and the conjunction is not near periastron, so the heartbeat/tide confusion does not arise.

Verdict: CONFIRMED

The deciding numbers: - 23 of 23 primaries at S/N 27–70, 17.81 ± 0.07 ppt, with even and odd depths equal. - The primary falls on Gaia conjunction A at the Gaia epoch (–1.3σ) and the secondary on conjunction B (–0.3σ). The 2.2σ difference between Gaia's period and the TESS period accounts for the drift into the TESS sectors (the 403936502 pattern). - A secondary of 2.43 ± 0.21 ppt at phase 0.5000 with S/N 11.7, where no random phase gets above S/N 2.7. - Both eclipses appear in TESS-SPOC PDCSAP, SPOC SAP, TARS and QLP in all five sectors. - The difference images from the SPOC target pixel files put the loss on the target in all five sectors, and the brightest neighbour would need a 66% dip. - The physics fits one geometry: a grazing eclipse by a ~0.65 Msun companion.

Next predicted eclipses (BTJD; linear TESS ephemeris; secondary at phase 0.50003):

primary	secondary
4403.4000 ± 0.6 min	4405.5345 ± 1.1 min
4407.6688 ± 0.6 min	4409.8033 ± 1.1 min
4411.9375 ± 0.6 min	4414.0720 ± 1.1 min

What would settle it further: - Gaia DR4 epoch RVs, to fix the conjunction-A attribution directly and resolve the 2.2σ period offset. - Spot-aware modelling, to explain the scatter in both depths. - Ground photometry of the 1.9-h, 18-ppt primary at Tmag 10.2 is easy.

A164. TYC 8062-84-1 (TIC 200745563)

status
: confirmed (per-object re-analysis 2026-10-21; single-eclipse eccentric system (e 0.23): occultation of a ~0.6 Msun K dwarf at conjunction B near periastron, 18.45 ppt, T14 5.10 h, flat-bottomed, 10 events; conjunction A empty (< 0.23 ppt) as geometry predicts (sky sep 1.09-1.21 (R1+R2)); occultation vs transit broken by R1 from the orbital speed; on target in 6 difference images, every neighbour would need > 100% of its light)

P_days
: 10.998961

eP
: 0.000002

T0_primary_BTJD
: 3412.11666

depth_primary
: 1.85

duration_h
: 5.10

source_path
: Fable 5\made\sb1-scale\pervet\vet200745563\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 200745563 (TYC 8062-84-1, Gaia DR3 4749248636894211840): per-object vet

Opus 5.5 subagent, 2026-10-21. This resumes a wave-4 session that was killed before it wrote a report. Its receipts were re-checked rather than copied: T0 − 184P reproduces the E=0 time to 1 min, and the QLP depths re-measured with indep.py match vet.py. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. One eclipse per orbit, at Gaia conjunction B. Only one eclipse kind is seen, and the geometry predicts the other one misses.

Deciding numbers: - **Eclipses:** 10 individual events in s003, s004, s030, s031 and s097, each at S/N 49–110. Weighted depth 18.45 ± 0.07 ppt (QLP SAP, Q==0). T14 5.10 h, T12 1.78 h. - **Period:** P = 10.998961 ± 0.000002 d against Gaia's 10.997818 ± 0.001256 d, a difference of **+0.91σ**. - **Timing:** every sector's eclipse lands on Gaia conjunction B, at +0.56σ to +0.78σ. Conjunction A is 11σ to 43σ away. - **On target:** - The eclipse appears in QLP, TESS-SPOC FFI, SPOC 2-min and TGLC at 17.5–22.5 ppt. - The target is the best single-star fit in all 6 difference images. - Every Gaia neighbour would need

a dip larger than 100% of its own light. - **Conjunction A is empty:** −0.04 ± 0.07 ppt, 95% upper limit 0.23 ppt. That is what the geometry predicts: - Conjunction B is near periastron (r = 0.785a), with impact parameter 0.72–0.80 (R1+R2). - Conjunction A is at r = 1.195a, so the impact parameter scales to 1.09–1.21 (R1+R2). It misses by 0.25–0.67 R⊙.

1. Archives (queried 2026-10-20/21)

- **cats.py --r 60** (re-run, same result): no hits in VSX, ASAS-SN, Prša+2022, IJspeert+2024 or Kostov+2025 t2/t3/t4. SIMBAD has only TYC 8062-84-1, typed [SB*].
- **Gaia DR3** vari_summary and vari_eclipsing_binary: empty.
- **SB9, WDS, J/AJ/160/111:** no entry.
- **ExoFOP:** no TOI, CTOI, imaging, spectroscopy or companions.
- **Extra VizieR cones at 30″** (ATLAS, ZTF and WISE variables, GCVS, VSX copy, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST DR7, APOGEE DR17, Prša+2022): no rows; each control cone returned rows, so the queries worked.
- J/A+A/674/A16 turned out not to be an EB table; it returns mixed sub-tables.
- Incidentally, one of its rows is a G 11.76 star 453″ away with the same proper motion and parallax (17.0/−10.6 mas/yr, ϖ 2.66 vs 2.68). It is a likely wide co-moving companion, far outside any aperture.
- **Web search** on "TYC 8062-84-1" and "TIC 200745563": no relevant hits.
- **Batch receipts:**
- c4-ebcats: none of the vetted or unvetted EB tables.
- c4-vetsum: dphi −0.428/+0.009, ATTR.
- diffrun-c4 s097: peak on target, deficit−target 4.16″.
- **No prior art.** This is a new eclipsing identification of a known Gaia SB1.

2. Eclipse measurements (vet-out.txt; raw QLP SAP_FLUX, Q==0, no trimming)

E	sector	tc (BTJD)	± (min)	depth (ppt)	S/N
0	s003	1388.3085	1.1	19.45 ± 0.39	50
1	s003	1399.3064	1.4	19.33 ± 0.40	49
4	s004	1432.3042	0.7	19.47 ± 0.23	85
67	s030	2125.2377	0.7	19.93 ± 0.23	88
69	s031	2147.2365	0.7	18.72 ± 0.22	85
71	s031	2169.2333	0.7	19.31 ± 0.23	86
232	s097	3940.0663	0.4	17.70 ± 0.16	110
233	s097	3951.0662	0.4	17.86 ± 0.18	100
235	s097	3973.0646	0.7	18.24 ± 0.21	86
236	s097	3984.0625	0.4	17.54 ± 0.17	100

- **Shape:** T14 5.10 h (1σ range 5.10–5.16 h), T12 1.78 h, flat bottom 1.53 h. A pure V shape is worse by $\Delta\chi^2$ +90.
- **Depth by sector:** s097 gives 17.8 ppt against 19.0–19.9 ppt in s003–s031. Crowding-corrected SPOC PDCSAP gives 18.3–19.1 ppt in the early sectors. Read as a QLP dilution difference between sectors.
- **Background:** the unit-free regression (bkgreg-out.txt) implies $|\Delta F| \leq 0.07$ ppt per sector, against a dip of 11–13 ppt in the same windows. Faking the eclipse would take +171 e^-/s of background; the

median in−out is −8 e^-/s.

- **Centroids:** dX −0.23 ± 0.29 mpix, dY +0.30 ± 0.10 mpix.
- **Quality flags:** one event (E=67) has a single bit-32 cadence, and its depth matches the others. No event is within 0.5 d of a listed sector-wide systematic.
- **Blind dip census:** 11 of the 25 dips at S/N ≥ 4 are on the ephemeris.
- **3967.406** (s097, 5.2 ppt): a pre-gap slide. Flux drops into flagged cadences (bits 32 and 30), X drifts, then a gap and a pointing jump (dip_3967.4.png). The same phase in other s097 cycles gives +0.18 ± 0.18 and −0.19 ± 0.19 ppt. It is a systematic.
- **3950.766:** a matched-filter edge artefact next to the 3951.07 primary and flagged bit-29 cadences.
- All other census dips are ≤ 1.8 ppt.
- **Control phases** 0.125–0.875: |depth| ≤ 0.16 ppt.

3. Ephemeris and orbit

- **Observed eclipse (conjunction B):** T0 = 3412.11666 ± 0.00017 BTJD (E = 184), P = 10.998961 ± 0.000002 d.
- χ^2/dof 16.6/8 (errors scaled ×1.44); O−C rms 0.8 min.
- **Against Gaia's B prediction** (Monte Carlo with corr_vec):

sector	measured − predicted
s003	+2.2 h (+0.56σ)
s004	+2.2 h (+0.57σ)
s030	+4.0 h (+0.68σ)
s031	+4.0 h (+0.68σ)
s097	+8.6 h (+0.78σ)

The drift is Gaia's P being 0.9σ short. - **Gaia orbit:** e cos ω = +0.097 ± 0.015, e sin ω = −0.207 ± 0.011. B − A in phase = 0.563 ± 0.010. - **e cos ω and e sin ω can't be measured from the photometry here**, because only one eclipse kind is visible. - **Duration test instead:** - At B (ν = 335°) a central chord would last 7.4–8.4 h for R1 = 2.0–2.5 R⊙. - The observed 5.1 h implies b ≈ 0.72–0.80 (R1+R2) and i ≈ 83°. - At A, with the orbital separation ×1.52 larger, the projected separation is 1.09–1.21 (R1+R2), so there is no eclipse. - **Fixed-shape test at the predicted A phase** (0.429 ± 0.010, scanned ±3σ, secA-out.txt):

duration	depth at predicted phase
T14 ×1	−0.04 ± 0.07 ppt
T14 ×1.5	−0.05 ± 0.07 ppt
T14 ×2	+0.07 ± 0.06 ppt

The best S/N inside the window is 1.5–2.6, below the median of the control windows (2.3–3.1). 95% upper limits: 0.23, 0.22 and 0.37 ppt.

4. Independence

- **Second pipelines** (indep-out.txt, same ephemeris), depth in ppt:

pipeline	s003	s004	s030	s031
TESS-SPOC PDCSAP	19.13 ± 0.18	19.10 ± 0.22	18.78 ± 0.15	18.43 ± 0.14
SPOC 2-min PDCSAP	—	—	18.70 ± 0.13	18.26 ± 0.13
TGLC aperture	21.6	22.5	—	—
QLP	19.4	19.0	19.4	19.0

- **Difference images** (6 TPFs: TESS-SPOC s003/4/30/31 and SPOC 2-min s030/31):
- The pixel with the largest flux loss is the target pixel every time, at S/N 30–52.
- The best single-star PSF fit is the target every time, with an implied own dip of 14.8–17.4 ppt.
- The 4.4″ neighbour (G 15.88) is next best by $\Delta\chi^2$ +77 to +600, but would need a 1100–1500 ppt dip.
- **Dilution table** (needed dip = 18.45 ppt × 10^(0.4ΔG)):

neighbour	ΔG	needed dip
G 15.88 at 4.4″	4.87	1630 ppt
G 15.37 at 28.7″	4.36	1020 ppt
G 15.92 at 34.8″	4.91	1690 ppt
G 16.43 at 44.7″	5.42	2700 ppt

None of them can produce the signal.

5. Physics (phys-out.txt, occfit-out.txt, occfit-vfix-out.txt)

- **Mass function** f(M) = 0.0804 M⊙. Taking M1 = 0.96–1.2 M⊙ (TIC: 0.96 ± 0.12 M⊙, R1 2.27 ± 0.12 R⊙, Teff 5500 K, a subgiant):
- m2 at i = 90° is 0.57–0.65 M⊙.
- a = 24.0–25.6 R⊙.
- Central durations: 11.2–12.8 h at A, 7.4–8.4 h at B.
- **Which star is in front.** In Gaia's convention the primary is the nearer star at B, so the observed event is the companion being occulted.
- Physical straight-chord fits to the 7 short-cadence events can't separate occultation from transit ($\Delta\chi^2$ 0.0–0.5).
- Fixing R1 from the orbital speed at B (v_t = 17.0 R⊙/d, R1 = 2.27 R⊙) breaks the tie:
  - **Occultation** needs k = 0.28 (R2 ≈ 0.64 R⊙), b = 0.93 R1, and a companion light fraction of 3.0%. A 0.57–0.62 M⊙ K dwarf is expected at R2 0.61–0.65 R⊙ and 2.7–3.5% of the TESS light. This is self-consistent with the Gaia SB1 companion.
  - **Transit** at the same R1 needs the primary to supply only 48% of the light. It becomes physical only for R1 ≳ 2.7 R⊙.
- With the occultation solution, the projected separation at A is 1.42 R1, beyond 1 + k = 1.28, so A misses. A transit at A would have been 25–75 ppt deep, which the data exclude.
- **Heartbeat check.** B lies 25° from periastron, with e = 0.23.
- A tidal signal would be about 1 ppt, spread over a periastron passage of roughly a day.
- The fitted orbital harmonics are all under 0.7 ppt.
- The observed event is 18 ppt, 5.1 h long and flat-bottomed. It is not a heartbeat.

6. Next predicted times (BTJD)

- **Observed eclipse** (companion occulted, conjunction B): **4336.0294, 4347.0283, 4358.0273** (± 0.0002 d), then every 10.998961 d.
- **Conjunction A** (no eclipse expected; upper limit 0.23 ppt): **4340.79, 4351.79, 4362.79** (± 0.13 d). This is phase 0.433 ± 0.012 after the observed eclipse.

What would settle it further

- A high-S/N spectrum might show the companion. A K dwarf at about 3% of the light could appear as a cross-correlation bump, which would show directly that B is the occultation.
- Multi-colour eclipse depths: an occulted companion should give a deeper eclipse in redder bands.
- More TESS coverage of conjunction A would tighten the null.

Files

- **Scripts:** vet.py, vetlib.py, indep.py (now also reads SPOC 2-min `*s_lc.fits`), tpfdiff.py, secA.py, occfit.py, phys_ecc.py, bkgreg.py, gaiatap.py, mastlc.py, morecats.py, extracats.py.
- **Outputs:** vet-out.txt, indep-out.txt, tpfdiff-out.txt, tpfdiff_*.png, secA-out.txt, secA-dips-out.txt, dip_3967.4.png, dip_3950.8.png, phys-out.txt, occfit-out.txt, occfit-vfix-out.txt, occfit_ph0.00.png, extracats-out.txt, cats-out2.txt.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet200745563\`.

A165. UCAC4 797-029409 (TIC 423927429)

status
: confirmed (per-object re-analysis 2026-10-21; circular G + 0.23 Msun M dwarf, 17.4 ppt, T14 2.04 h, 353 primaries in 23 sectors; new secondary 1.34 ppt at phase 0.5005 (S/N 24), e cos omega +0.0008(16); Doppler beaming +0.30 ppt matches K1's predicted +0.35 with the right sign; ellipsoidal 4.2 ppt; eclipse + ellipsoidal give R1 1.3-1.5 Rsun vs TIC 2.06 (open); O-C rms 1.5 min, possible ETV; Kostov table 2 only; on target in 200-s TPF difference images by dchi2 > 13,000)

P_days
: 1.4271182

eP
: 0.0000001

T0_primary_BTJD
: 2967.97498

depth_primary
: 1.74

depth_secondary
: 0.13

duration_h
: 2.04

source_path
: Fable 5\made\sb1-scale\pervet\vet423927429\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 423927429 (UCAC4 797-029409, Gaia DR3 2247293469576926336): per-object vet

Opus 5.5 subagent, 2026-10-21. This resumes a session that was killed with no progress.md; its vet.py, Gaia, catalogue and background outputs were reused and checked. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, circular, partially eclipsing SB1 with both eclipse kinds detected.

Deciding numbers: - **Primary eclipses:** 353 individual events in 23 sectors (s015–s086). Depth 17.41 ± 0.03 ppt, T14 2.04 h, T12/T14 0.35. - **Period:** P = 1.4271182 ± 0.0000001 d against Gaia's 1.427080 ± 0.000047 d, a difference of **+0.82σ**. - **Timing:** the primary falls on Gaia conjunction A (primary eclipsed) at +0.16σ to +0.46σ in every sector. Conjunction B is 6σ to 14σ away. - **Secondary eclipse:** 1.34 ± 0.05 ppt, S/N 24, at phase **0.50050 ± 0.00103**. That gives e cos ω = +0.0008 ± 0.0016, against

Gaia's −0.008 ± 0.039 (e = 0.025 ± 0.045). The secondary falls where Gaia's B − A = 0.495 ± 0.025 puts it. - **Doppler beaming matches Gaia's orbit:** sin φ = +0.30 ppt against about +0.35 ppt expected from K1 = 38.8 km/s. The sign independently confirms that the luminous star is the RV primary moving as Gaia says. - **On target:** TESS-SPOC PDCSAP gives 16.8–18.2 ppt in all 13 sectors checked. The target is the best fit in the difference images by $\Delta\chi^2 > 13{,}000$ over any neighbour.

1. Archives

- **cats.py --r 60** (2026-10-20; re-run 2026-10-21, identical result — `cats-out2.txt`):
- No hits in VSX, ASAS-SN, Prša+2022, IJspeert+2024, or Kostov t3/t4.
- The only hit is Kostov+2025 **table2**, an unvetted ML flag list (0.001″ match). It does not count as catalogued.
- SIMBAD has only UCAC4 797-029409, typed [SB*].
- **Gaia DR3** vari_summary and vari_eclipsing_binary: empty.
- **SB9, WDS, J/AJ/160/111:** no entry.
- **ExoFOP:** no TOI, CTOI, imaging or spectra.
- **Extra VizieR cones at 30″** (ATLAS, ZTF, WISE variables, GCVS, VSX copy, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST, APOGEE, Prša+2022): no rows; each control cone returned rows, so the queries worked. J/A+A/674/A16 is not an EB table.
- **Web search** on the TIC, UCAC4 and Gaia IDs: no relevant hits.
- **Batch receipts:**
- c4-vetsum: dphi +0.009/−0.477, ATTR.
- diffrun-c4 s056: peak on target, deficit–target 2.44″.
- c4-ebcats: Kostov table2 only.
- **No prior art** beyond the ML flag.

2. Measurements (vet-out.txt, sec2-out.txt; QLP SAP, Q==0, no trimming)

- **Primary depth per sector:** 16.2–18.3 ppt. χ^2/dof of depth constancy is 1267/352, a modest sector-to-sector scatter of about 0.5 ppt, typical of QLP dilution.
- **Odd vs even primaries:** 17.29 ± 0.04 vs 17.54 ± 0.04 ppt. The 4σ split is small and can't be a period-halving effect, because a real secondary is seen.
- **Fold fit with orbital harmonics:**
- Primary: T14 2.04 h, T12/T14 0.35, depth 17.6 ppt.
- Secondary: T14 1.84 h, T12/T14 0.30, depth 1.34 ppt.
- e sin ω from the duration ratio is about −0.05 ± 0.03. The 0.08 h duration grid limits this, and partial-occultation shape is degenerate with duration, so it is uninformative but consistent with 0.
- **Harmonics** (ppt, median over 23 sectors; sector scatter ~0.2):

term	amplitude	meaning
cos 2φ	−4.17	ellipsoidal
cos φ	−0.25	reflection, brightest at phase 0.5
sin φ	+0.30	beaming
other terms	≤ 0.14	—

TESS-SPOC independently gives an ellipsoidal amplitude of 3.5–4.4 ppt in every sector. - **Background:** the regression (bkgreg-out.txt) implies $|\Delta F| \leq 0.17$ ppt per sector, against a 13–15 ppt dip in the same windows. vet.py's dBKG column in s015 shows the known broken units and is ignored. - **Centroids:** dX −0.14 ± 0.10 mpix, dY −0.22 ± 0.12 mpix. - **Quality flags:** 10 of 353 events touch a bit-32 cadence and one is near a listed systematic. With 353 events none of them matter. - **Blind dip census:** 348 dips on the ephemeris. The five off-ephemeris dips at 9.7–13.3 ppt (2916.733, 2860.808, 2903.617, 3437.624, 3629.997) all sit about 3.2 h from a primary at the edge of a data gap. - Example: dip_2916.7.png shows a clean primary followed by a gap. The matched-filter baseline is corrupted by the eclipse. - None repeats at the same phase in neighbouring cycles. - They are artefacts. The many 2–4 ppt dips at phase ±0.5 are the secondary. - **Control phases:** 0.125, 0.25, 0.75 and 0.875 give $|\text{depth}| \leq 0.18$ ppt. Phases 0.375 and 0.625 give +0.25 ppt from ellipsoidal leakage in vet.py's estimator; sec2.py fits the harmonics properly.

3. Ephemeris

- **Primary (conjunction A):** T0 = 2967.97498 ± 0.00004 BTJD (E = 880), P = 1.4271182 ± 0.0000001 d.
- χ^2/dof 1694/351 (errors scaled ×2.2); O−C rms 1.5 min against a median timing error of 0.7 min.
- The excess is mostly from the noisy 30-min s016 events. Small eclipse-timing variations are not excluded, which is worth a look if anyone revisits.
- **Against Gaia's A prediction** (Monte Carlo with corr_vec): +0.17 h at s015 to +1.14 h at s086, i.e. +0.16σ to +0.46σ. The drift is Gaia's P being 0.8σ short.
- **Secondary − primary:** 0.50050 ± 0.00103 (jackknife over 23 sectors), against Gaia's 0.4950 ± 0.0247 (+0.2σ).

4. Independence

- **TESS-SPOC FFI** (13 sectors, indep-out.txt): PDCSAP 16.80–18.16 ppt, SPOC-SAP 16.25–17.14 ppt, ellipsoidal amplitude 3.46–4.43 ppt.
- **Difference images** (tpfdiff-out.txt, 200-s TPFs s059 and s079; the 30-min TPFs are too coarse for the in-window estimator):
- The pixel with the largest flux loss is the target pixel, at S/N 67 and 81.
- The best fit is the target, with implied own dips of 17.2 and 17.1 ppt.
- The next-best star (G 15.83 at 22″) is worse by $\Delta\chi^2$ +13,800 and would need a 190–340 ppt dip.
- **Dilution table** (needed dip = 17.4 ppt × 10^(0.4ΔG)):

neighbour	ΔG	needed dip
G 15.83 at 22.3″	4.33	940 ppt
G 16.92 at 43.8″	5.42	2540 ppt
G 15.35 at 67.3″	3.85	600 ppt (outside 60″)

None of them can produce the signal.

5. Physics (occfit-out.txt)

- **Mass function** f(M) = 0.0086 M⊙. With M1 ≈ 0.94 M⊙, m2 is about 0.23 M⊙, an M dwarf with R ≈ 0.24 R⊙. q ≈ 0.24, a ≈ 5.6 R⊙, v_t = 24.7 R⊙/d.

- **Transit at A** (companion in front, as Gaia's ω implies), from a physical fit to 207 events at 200-s cadence:
- k = 0.165, b = 0.895 R1, amplitude 0.93.
- That gives **R1 ≈ 1.50 R⊙ and R2 ≈ 0.25 R⊙**, exactly an M dwarf of the f(M) mass.
- **Occultation at A:** needs the companion to supply 3.8% of the light, impossible for a 0.23 M⊙ dwarf next to a G star (about 0.2% expected). This agrees with Gaia's convention.
- **Secondary:** at b ≈ 0.9 the M dwarf is about 70% occulted. With a TESS-band light ratio of about 2.4 ppt, the predicted depth is about 1.6 ppt; observed 1.34 ppt.
- **Ellipsoidal:** 1.3 × q × (R1/a)³ = 4.17 ppt gives R1/a ≈ 0.24, so R1 ≈ 1.33 R⊙. This agrees with the eclipse fit.
- **The radius tension** (the one open item):
- Eclipse and ellipsoidal fits independently give R1 ≈ 1.3–1.5 R⊙.
- TIC gives 2.06 ± 0.11 R⊙ (Teff 5400 K, L 3.26 L⊙). An own G-band estimate of L ≈ 3.1 L⊙ at about 5600 K gives about 1.9 R⊙.
- With R1 = 2.06, a 2.04 h transit of a 0.25 R⊙ companion would have to be a deep graze far shallower than 17 ppt, and the ellipsoidal amplitude would be about 15 ppt.
- Possible causes: a hotter, smaller primary (Gaia gspphot gives Teff 6712 K, logg 4.03, though bp_rp 0.98 argues against it), a modest third light, or the TIC luminosity.
- None of these affects whether the eclipses are real or belong to the binary.
- **Not a heartbeat or tidal artefact:** the orbit is circular, and both eclipses are short and sharp.

6. Next predicted times (BTJD)

- **Primary** (M dwarf transits the primary, conjunction A): **4335.1543, 4336.5814, 4338.0085** (± 0.0001 d; allow ±2 min for the χ^2 excess), then every 1.4271182 d.
- **Secondary** (M dwarf occulted, conjunction B): **4335.8685, 4337.2956, 4338.7228** (± 0.0015 d).

What would settle it further

- A spectroscopic Teff and log g for the primary would resolve the radius tension. So would high-resolution imaging for a possible third light.
- An eclipse-timing analysis restricted to the short-cadence sectors would test the 1.5-min O–C excess.
- RVs at quadrature would check K1 and allow a joint fit.

Files

- **Scripts:** vet.py, vetlib.py, indep.py, tpfdiff.py, secA.py, sec2.py, occfit.py, bkgreg.py, gaiatap.py, mastlc.py, morecats.py, extracats.py.
- **Outputs:** vet-out.txt, indep-out.txt, tpfdiff-out.txt, tpfdiff_59.png, tpfdiff_79.png, dips-out.txt, dip_*.png, sec2-out.txt, sec2.png, occfit-out.txt, occfit_ph0.00.png, extracats-out.txt, cats-out2.txt.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet423927429\`.

A166. CD-23 4400 (TIC 299650500)

status
: confirmed (per-object re-analysis 2026-10-21; grazing M-dwarf (~0.48 Msun) transit of a hybrid delta Sct/gamma Dor F star at conjunction A, 27 deg before periastron, 19.4 ppt, T14 4.5 h, 5 events over 2200 d; heartbeat rejected (dchi2 141-3500 per event); depth scatter 13-22 ppt is pulsation beating; no secondary, geometrically predicted to miss (sky sep 4.4 Rsun > R1+R2); on target in 3 SPOC TPF difference images; TARS, SPOC and TGLC agree)

P_days
: 18.337086

eP
: 0.000027

T0_primary_BTJD
: 3427.74767

depth_primary
: 1.94

duration_h
: 4.5

source_path
: Fable 5\made\sb1-scale\pervet\vet299650500\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 299650500 (CD-23 4400, Gaia DR3 2925231689064510464): per-object vet

Opus 5.5 subagent, 2026-10-21. This session resumed work from a session that was killed on 2026-10-20. The killed session's receipts (`vet-out.txt`, `indep-out.txt`, `neighbours-out.txt`, `gaiatap-out.txt`) were checked and hold up. New work this session: `extra.py` → `extra-out.txt`, and `tpfdiff.py` on the three SPOC TPFs → `tpfdiff-out.txt`, `tpfdiff_*.png`. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED: a grazing transit of an M-dwarf companion across the F primary, at Gaia conjunction A

The deciding numbers: - **Ephemeris:** P = 18.337086 ± 0.000027 d and T0 = 3427.74767 ± 0.00092 BTJD, from 5 primary eclipses across s006, s007, s033 and s087, a 2200-d baseline. The O−C rms is 4.7 min. In the jackknife, each dropped event is predicted to within ≤ 12 min. - **Agreement with Gaia:** Gaia's P is 18.33751 ± 0.01016, which is −0.04σ from ours. Every eclipse falls on the Gaia conjunction A predicted from the orbit (MC with corr_vec), at −0.6 to −1.9 h with σ = 15–44 h. The chance of a random 18-d event landing that close is about 0.5%. - **On target:** in the SPOC TPF difference images,

the target is the best single-star fit in all three sectors. $\Delta\chi^2$ to the next-best star is +1970 (s006), +224 (s007) and +1470 (s033). The batch TESScut s087 diff image puts the deficit 1.7″ from the target. No Gaia neighbour within 60″ can make the dip: the brightest possible contaminant (G 14.13 at 32″) would need a 45% eclipse, and its diff-image fit is much worse. - **Independent pipelines:** TARS (4 sectors), SPOC PDCSAP and SAP (3 sectors) and TGLC PSF and aperture (s006) all recover the event at 12–24 ppt. That includes the shallow s007 event, which every pipeline measures at 12–13 ppt. - **Not a heartbeat:** conjunction A falls 27° of true anomaly before periastron, the attribution blind spot, so this was tested explicitly: - The free Gaussian σ per event is 1.0–1.1 h, against a periastron timescale τ of 1.32 d. - Tide+eclipse beats tide-only by $\Delta\chi^2$ = 141 to 3500 per event. - The expected equilibrium-tide amplitude is about 0.3 ppt, against an observed 19 ppt. - **Shape and physics are consistent with one geometry:** - f(M) = 0.0259 $M_\odot$, so m2 sin i ≈ 0.48 $M_\odot$ (an M dwarf, R2 ≈ 0.47 $R_\odot$, k ≈ 0.20) for M1 = 1.57 $M_\odot$ and R1 = 2.37 $R_\odot$ (TIC values, Teff 7090 K). - At conjunction A the separation is 24.9 $R_\odot$ and the transverse speed 143 km/s. A central chord would last 7.6 h; the observed T14 of 4.5 h with a V shape (T12/T14 = 0.4) implies the sky separation is 0.97 R1, so the transit is grazing and i ≈ 84.7°. - About half the M-dwarf disc crosses the limb, which predicts roughly 14–20 ppt, consistent with the measured 19.4 ppt. - Conjunction A has ν+ω = 90°, so the primary is farthest and the companion is in front: A is the transit. - **No secondary, as predicted:** at conjunction B the separation is 47.3 $R_\odot$, so the sky separation is 4.36 $R_\odot$, larger than R1+R2 = 2.84 $R_\odot$. A full occultation would be ≤ 3.1 ppt in any case. Fixed-shape fit across all 5 B windows: - At Gaia-B nominal, the depth is −0.57 ± 0.28 ppt with T14 = 4.5 h, and −0.80 ± 0.22 ppt with T14 = 8.6 h (the duration scaled by the speed ratio). - The best peak in the ±3σ window has S/N 6.0, which sits below the median control peak of 9.7. Controls exceeded it 80% of the time. - Secondary: **not detected, consistent with the geometry.**

Caveats worth carrying

- **The depth is not constant:** events measure 21.9, 13.3, 18.8, 21.9 and 18.1 ppt (χ^2 154/4). The star is a hybrid pulsator. The residual periodogram shows 10.18 c/d at 1.15 ppt (δ Sct) and a group near 2.4–2.5 c/d at up to 1.7 ppt per sector (γ Dor), with 2.1–2.3 ppt rms. Over a 4.5-h window, beating at this level shifts single-event depths by several ppt. The low s007 event is real flux (every pipeline sees it), and the most likely cause is pulsation phase. A change in inclination over 2200 d cannot be excluded with 5 events.
- **e cos ω and e sin ω cannot be measured** from a single eclipse kind. The only orbital-shape test is that the eclipse lands on Gaia's conjunction A, with e = 0.349 ± 0.039 and ω = 117 ± 8°; the B phase falls 0.3945 ± 0.0255 after A.
- **Many off-ephemeris census features at 3–9 ppt** (white-noise S/N up to 18) are pulsation and red noise. None sits on the ephemeris, and none is as deep as the eclipses in the 10- and 3.3-min sectors.

Archives (step 1)

- `cats.py 299650500 --r 60` (re-run this session): VSX none, ASAS-SN none, SIMBAD CD-23 4400 [SB*] at 0.0″. The ebcats battery (Prša+2022, IJspeert+2024, Kostov+2025) found none.
- Gaia DR3 vari_summary and vari_eclipsing_binary: no rows.
- SB9, WDS and J/AJ/160/111: none.
- ExoFOP: no TOI, CTOI, imaging, spectroscopy or stellar companions.
- A web search on "CD-23 4400", "TYC 6521-2291-1" and "TIC 299650500" found no mention.
- **Not previously catalogued as eclipsing.**

Next predicted eclipses (BTJD)

Kind	Times
Primary (A, transit)	4344.6020, 4362.9391, 4381.2761, 4399.6132 (each ± 2–3 min); then +18.337086 d per cycle
B (Gaia phase; no eclipse expected)	4351.84, 4370.17, 4388.51, 4406.85 (± 0.47 d)

What would settle it further

- **RVs at the primary eclipse:** a Rossiter–McLaughlin signal or a K1 phase check would pin down which star is in front.
- **A clean high-cadence eclipse:** fitting it jointly with a pulsation model would give a real depth and k.
- **A Gaia DR4 epoch-RV refit:** e and ω can then be tested against the transit duration.

A167. TYC 8994-1423-1 (TIC 449475961)

status
: confirmed (per-object re-analysis 2026-10-21; the Gaia SB2 itself, near-twin grazing F/G EB (q 0.965), 19.2/19.3 ppt, T14 3.0 h, 30 + 30 events; secondary at phase 0.49985(12), e cos omega -0.0002(2) vs Gaia +0.0256(97), circular; Gaia P 8.0 sigma off but conjunctions match at the Gaia epoch (perturbed solution, gof 12.2, RUWE 2.70): Table 2; the 7.0 arcsec batch offset overturned, own 5-sector localisation 0.9-3.1 arcsec on target (box-centring bias in the batch centroid); common-mode +-3 min ETV, likely tertiary)

P_days
: 4.1731519

eP
: 0.0000014

T0_primary_BTJD
: 3037.02702

depth_primary
: 1.92

depth_secondary
: 1.93

duration_h
: 3.0

source_path
: Fable 5\made\sb1-scale\pervet\vet449475961\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 449475961 (TYC 8994-1423-1, Gaia DR3 5868525051360765952): per-object vet

Opus 5.5 subagent, 2026-10-21. This session resumed work from a session that was killed on 2026-10-20; its `vet-out.txt`, `indep-out.txt`, `neighbours-out.txt` and `gaiatap-out.txt` were checked and hold up. New work this session: - `locate.py` → `locate-out.txt` and `locate_s*.png`: own localization over 5 sectors. - `timing.py` → `timing-out.txt` and `timing-events.json`: both eclipse kinds, ETV, e cos ω, and the comparison with Gaia at its own epoch. - TESScut s100 and s101 cutouts, saved in `C:\Home\sky\data\qlp-sb-p20-60\vet449475961\`. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED: a detached, near-twin, grazing EB, and it is the Gaia SB2 itself. The eclipse is on target.

Localization (the special instruction: the batch reported a 7.0″ offset)

Own test. For each sector: 1. Stack both eclipse kinds (they are equal depth) into in-eclipse and out-of-eclipse images. 2. Calibrate the WCS by fitting the out-of-eclipse image with Gaussian PSFs at all Gaia stars with G < 14.6 within 5 px, with free fluxes, a common shift and a free width. 3. Fit the difference image with one PSF at a free position, and separately at each star's calibrated position.

Sector	Product	Eclipses	Free-fit deficit vs calibrated target	Δχ² of next-best star (G 14.49 at 17″)	Deficit / target flux
s011	SPOC TPF	7	1.3″ ± 1.3″	+3493	2.27%
s038	SPOC TPF	11	1.8″ ± 1.3″	+8600	2.06%
s064	TESScut	12	3.1″ ± 1.9″	+14045	2.05%
s100	TESScut	12	0.9″ ± 0.6″	+3027	1.94%
s101	TESScut	11	1.5″ (± 4″)	+37 (errors over-estimated, reduced χ² 0.03)	1.92%

- The halves-split is stable to ≤ 0.8″ in every sector.
- The missing flux is 1.9–2.3% of the target's own modelled flux, the same as the light-curve depth. So the deficit is the target's light, not a neighbour's.
- **Neighbour table:** no Gaia source of any listed magnitude lies within 17″.
- Nearest are G 14.49 at 17.0″ (would need a 28–44% dip, with twin-equal primary and secondary) and G 15.18 at 18.4″ (26–74%). Their fixed-position fits lose by Δχ² ≥ 3000 in every sector except s101 (+37, the sector with over-estimated errors).
- Brighter stars further out, such as G 12.27 at 72″ (Gaia VARIABLE, would need 9%), are 3.5 px away and would pull the deficit tens of arcseconds off, not 1–3″.
- The VSX RS (41″, G 17.2) and L (48″, G 18.9) neighbours would need far more than 100%.

Why the batch said 7.0″. `diffimg/diffstack.py` takes a moment centroid in a 5×5 box centred on the rounded target pixel (ix = 7) and compares it with the uncalibrated catalogue position (x = 7.28). In `diff_s64.json`, the light centroid (6.98, 7.93) and the deficit centroid (6.96, 8.00) agree to 0.07 px (1.5″). Both sit about 0.30 px from the catalogue position because the box is centred off the star. That box-centring bias, not a displaced deficit, is the 7.0″. **Worth flagging for other batch stars whose target sits near a pixel edge.**

Not separable: Gaia RUWE 2.70 and ipd_frac_multi_peak 2 hint at an unresolved companion within about 1″. TESS cannot separate that, so "on target" here means on the unresolved Gaia source.

Timing and the −8σ period tension

- **Ephemerides (raw QLP SAP, Q0, 30 events of each kind):**
- Primary: P = 4.1731519 ± 0.0000014 d, T = 3037.02702 ± 0.00033 BTJD.
- Secondary: P = 4.1731519 ± 0.0000016 d. P_sec − P_prim = 0.00 ± 0.18 s per cycle, so no apsidal motion.
- **Secondary phase:** 0.49985 ± 0.00012, which gives e cos ω = −0.0002 ± 0.0002. Gaia's +0.0256 ± 0.0097 is −2.7σ away. Gaia's e = 0.035 ± 0.011 is most likely a spurious-eccentricity artefact of a poor fit (gof 12.2). The orbit is circular, as expected for an F/G twin at 4.17 d.

- **Common-mode ETV:** per-sector O−C runs −1.0, −2.7, +2.8, +3.0, +0.6 and −2.0 min for primaries, and −1.6, −2.3, +2.6, +3.1, +1.3 and −1.6 min for secondaries, each ± 0.2–0.5 min.
- Primaries and secondaries move together, which suggests a light-travel-time or third-body effect rather than starspots or apsidal motion. It fits the RUWE hint.
- The s100→s101 swing of −2.6 min within one month would mean a short outer period. Alternatively it is a QLP time-stamp effect; the two are not separated here.
- **Gaia's period:** 4.173881 ± 0.000092, which is +8.0σ from TESS. Comparing at Gaia's own epoch (BTJD ~389, MC with corr_vec):
- Gaia conjunction A is 386.956 ± 0.012, and the TESS primary falls 2.9 h later.
- Gaia conjunction B is 389.111 ± 0.016, and the TESS secondary falls 1.2 h later.
- Near s011 the offsets are −2.3 h (A) and −4.0 h (B).
- An unrelated EB would land at a random phase. Instead the eclipses match Gaia's conjunctions to 0.01–0.04 in phase at epochs 1200 d apart. The leftover hours are the size Gaia's own period error implies (0.00073 d accumulates 4.4 h across its 1038-d baseline).
- The −8σ is therefore Gaia's error bar, understated in a perturbed solution (gof 12.2, RUWE 2.7, possibly the tertiary). It is not evidence of a different star.

Light curve, independence, physics

- **Depths:**
- Primary 19.16 ± 0.06 ppt, secondary 19.29 ± 0.09 ppt. Odd and even primaries are equal (19.15 vs 19.16).
- T14 is 3.0 h with a V shape (T12/T14 = 0.4).
- The control phases at 0.125–0.875 (excluding 0.5) are all 0 within ±0.16 ppt.
- Per-sector depths run from 17.2 (s101) to 21.4 ppt (s065) and **both kinds track each other**: for example s101 17.2/17.0 and s065 21.4/21.1. That is aperture or dilution in QLP. The adjacent-sector jump from s100 (19.2) to s101 (17.2) is too fast to be precession.
- **Background and systematics:** the background regression shows +0.27 ppt in eclipse, against +223 e^-/s needed to fake the dip. One event has a bit-32 cadence; none falls on a listed sector-wide systematic.
- **Blind census:** every off-ephemeris dip at S/N ≥ 4 is a secondary, or a matched-filter sidelobe 0.17–0.19 d from an eclipse.
- **Second pipelines:** all at the same ephemeris.
- TARS: 20.3–21.4 ppt in s011, s038, s064 and s065.
- SPOC SAP: 20.2 (s011) and 19.5 ppt (s038).
- PDCSAP: 23.7–24.6 ppt; crowding correction is applied there.
- **Physics:**
- The SB2 gives K1 82.6 and K2 85.7 km/s, so M1 $\sin^3 i$ = 1.05 $M_\odot$, M2 $\sin^3 i$ = 1.01 $M_\odot$ (q = 0.965), and a sin i = 13.9 $R_\odot$.
- Treat it as twin F/G stars (Gaia Teff 6356 K, logg 3.8), each about 1.5 $R_\odot$ since the TIC's 2.13 $R_\odot$ treats the pair as one star.
- A central eclipse would last about 6.9 h; the observed 3.0-h V shape implies grazing, at a sky separation of about 0.90 (R1+R2), i ≈ 79°, masses about 1.11 and 1.07 $M_\odot$.
- A two-limb grazing overlap gives roughly 1–2% per eclipse, equal for both kinds. That is consistent with the 19 ppt twin depths and with Gaia's near-equal K1 and K2. The SB2 needs sin i close to 1, so eclipses were expected a priori.

Archives (step 1)

- `cats.py 449475961 --r 60` (re-run this session):
- VSX has nothing at the target; Gaia DR3 5868524978316679552 [RS] at 41″ and 5868525051340908800 [L] at 48″ are both far too faint to matter.
- ASAS-SN: none. SIMBAD: TYC 8994-1423-1 [SB*].
- The ebcats battery (Prša, IJspeert, Kostov) found none.
- Gaia vari_summary and vari_eclipsing_binary: no rows.
- SB9, WDS and J/AJ/160/111: none.
- ExoFOP: no TOI, CTOI or follow-up.
- A web search on "TYC 8994-1423-1", "TIC 449475961" and "Gaia DR3 5868525051360765952" found no mention.
- **Not previously catalogued as eclipsing.**

Next predicted eclipses (BTJD; ± ~5 min, dominated by the ±3-min ETV)

Kind	Times
Primary	4334.8773, 4339.0504, 4343.2236, 4347.3968; then +4.1731519 d per cycle
Secondary	4336.9638, 4341.1370, 4345.3101, 4349.4833

What would settle it further

- **The ETV:** check whether the per-sector O−C pattern appears in SPOC or TARS times for s011, s038, s064 and s065. If it does, model it as LTTE to look for a tertiary, which RUWE 2.7 supports.
- **Gaia DR4 epoch RVs:** they should reproduce P = 4.17315 and a circular orbit.
- **High-resolution imaging:** this would resolve the ipd_frac_multi_peak companion.

A168. HD 259137 (TIC 234880209)

status
: confirmed (per-object re-analysis 2026-10-21; G2V (HD 259137) + 0.11 Msun M dwarf, flat-bottomed, 18.76 ppt, T14 2.22 h, 15 events; prior art: DTARPS-S planet candidate (Melton+ 2024, AJ 167, 203, Rp 13.4 Re, no TOI), reclassified stellar by the Gaia orbit, count as a reclassification; tentative 0.5-0.9 ppt secondary not robust, not claimed; on target in the TESS-SPOC s033 difference image, the G 11.65 star at 66 arcsec rejected)

P_days
: 4.270292

eP
: 0.000001

T0_primary_BTJD
: 2508.56208

depth_primary
: 1.88

duration_h
: 2.22

source_path
: Fable 5\made\sb1-scale\pervet\vet234880209\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 234880209 (HD 259137, G2V, Tmag 10.14): per-object vet

Verdict: CONFIRMED. HD 259137 is an eclipsing binary: a G dwarf eclipsed every 4.270292 d by a ≈0.11 Msun M dwarf.

Prior-art caveat: the transit signal is already published as a *planet* candidate. It appears in the DTARPS-S Galactic-plane list (Melton+ 2024, AJ 167, 203, table4): P 4.27030, depth 0.0188, Rp 13.4 Re, T0 1470.880, ProbRF 0.75, no TOI. What is new is the classification: the Gaia SB1 orbit shows the companion is a star.

Wave 4, 2026-10-20/21. Resumed from a killed session; its receipts (gaia.json, cats-out.txt, exofop.json) checked out against fresh runs. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

1. Archives

- `cats.py --r 60`: VSX none, ASAS-SN none, SIMBAD HD 259137 [SB*].
- Gaia DR3 NSS SB1: P 4.270326 ± 0.00049, e 0.113 ± 0.053, ω 86.5 ± 24°, K1 13.25 ± 0.81 km/s. vari_summary and vari_eclipsing_binary are empty. RUWE 1.19.

- VizieR cones found no entry in: SB9, WDS, GCVS, Prša+2022, IJspeert+2024, Gaia vEB, Kostov table2, Gaia-ASAS-SN, LAMOST.
- SIMBAD lists G2V, SB*, 7 references, including DTARPS II, which led to the table4 match above.
- ExoFOP: no TOI or CTOI. A web search found no papers about the star.
- MARVELS (Grieves+ 2018, J/MNRAS/481/3244): G2V dwarf, M 0.96, R 0.88, [Fe/H] −0.31 (implied density 1.41 solar; the eclipse duration gives 1.05, between MARVELS and TIC — fits b ≈ 0.75; m2 ≈ 0.106 M⊙ with M1 = 0.96, barely moved). Not in DIAmante (Montalto+ 2020), matching the DIAmante=false flag in its DTARPS row. Gaia DR3 multiplicity (J/A+A/674/A34): the same SB1, no variability class.

2. Eclipse measurements (QLP SAP, QUALITY==0, no trimming)

Outputs: vet-out.txt, events.png, fold_oc.png. - 15 primaries in s006, s033 and s087, each with S/N 31–85. - Shape: **depth 18.76 ± 0.07 ppt, T14 2.22 h, T12 0.55 h, flat bottom 1.11 h**. A V shape is rejected (Δχ² +623). - Depth by sector: 18.19 / 19.62 / 18.56 ppt. The same pattern shows up in other pipelines (SPOC PDCSAP 17.87 / 18.42, TARS 18.28 / 18.53 / 17.86), so it comes from the apertures, not the star. - Odd 18.80 ± 0.11 vs even 18.72 ± 0.10 ppt. - No momentum-dump (bit 32) cadences in any window and no event near a listed sector-wide systematic time. - Centroid shifts: dX +0.3 ± 0.9 mpix, dY −0.5 ± 0.2 mpix. - Control phases: all below 0.17 ppt except phase 0.5. - Blind dip census: the 6 dips off the ephemeris are all ≤ 1.5 ppt. None reaches 9.4 ppt (half the eclipse depth).

3. Ephemeris and Gaia

- **P = 4.270292 ± 0.000001 d, T0 = 2508.56208 ± 0.00018 BTJD**, O−C rms 1.0 min over 6 years.
- Gaia P differs by **−0.07σ**.
- The eclipse sits at Gaia conjunction A (companion in front) to within +0.18σ / +0.07σ / +0.00σ in the three sectors.
- Gaia predicts the other conjunction at phase 0.503 ± 0.031; Gaia e cos ω = 0.005 ± 0.049, e sin ω = 0.103 ± 0.049.
- Secondary eclipse (sec2.py): a tentative 0.5–0.9 ppt dip appears near phase 0.49–0.51, but it is **not robust**:
- its phase moves from 0.4925 to 0.5075 depending on the detrending mask;
- it is absent in s087 (−0.03 ± 0.14 ppt);
- individual events scatter from −0.6 to +3.4 ppt.
- So no secondary is claimed and no e cos ω / e sin ω is derived from the light curve. If the dip is real, |e cos ω| ≲ 0.02, which agrees with Gaia.

4. Independence

- Other pipelines see the same eclipse at the same times: TESS-SPOC PDCSAP 17.87 (s6) and 18.42 (s33), SPOC SAP 17.47 / 18.20, TARS 18.28 / 18.53 / 17.86.
- TGLC s6 aperture flux gives 26.0 ppt; its PSF flux is all NaN.
- Difference image from the TESS-SPOC s033 pixel file (5 events): the strongest loss is on the target pixel (S/N 41.8). In the single-star fit, the target wins with χ² 1173 against 9476 for the next star (G 11.65 at 65.8″); the target's implied dip is 16.3 ppt.

- Neighbours: G 11.65 at 65.8″ would need a 50 ppt dip and is rejected by the difference image. G 16.2 at 85″ and G 16.8 at 87″ would need more than 100%. TIC contamination ratio is 0.014.

5. Physics (TIC M1 1.04, R1 1.055)

- Mass function f(m) = 0.00101 Msun, so **m2(i = 90°) = 0.110 Msun** (0.099–0.120). a = 11.6 Rsun.
- k = 0.137, so R2 ≈ 0.145 Rsun.
- From the duration: b ≈ 0.75, a/R1 ≈ 11.25 (11.0 expected from TIC values), stellar density 1.05 solar (0.89 from TIC), i ≈ 86°.
- Expected occultation 0.8–1.2 ppt for a 2900–3100 K companion. Ellipsoidal and beaming signals are ≤ 0.16 ppt.
- The 3 ppt cos 2φ term appears only in s033, in every pipeline, so it is probably spot modulation on the G dwarf.
- Everything fits one geometry.

6. Verdict and predictions

CONFIRMED. G2V + ≈0.11 Msun M-dwarf eclipsing SB1. The deciding numbers: - P agrees with Gaia to −0.07σ, and the eclipse sits within 0.2σ of Gaia conjunction A in all 3 sectors. - 15/15 events give a constant 18.8 ppt depth in QLP, SPOC and TARS. - The difference image is on target ($\Delta\chi^2 \approx 8300$). - m2 = 0.11 Msun and R2 ≈ 0.145 Rsun are consistent.

It is prior art as a DTARPS planet candidate and should be reported as a stellar companion.

Next eclipses (BTJD): - Primary (±0.7 min): 4336.2471, 4340.5173, 4344.7876, 4349.0579 - Secondary (tentative, phase 0.50 ± 0.01, ±1 h): 4338.382, 4342.653, 4346.923, 4351.193

What would settle it further: the occultation in future 200-s sectors, or RVs to pin down e.

A169. TYC 4020-206-1 (TIC 284533163)

status
: confirmed (per-object re-analysis 2026-10-21; delta Sct/gamma Dor hybrid F + ~0.21 Msun M dwarf, flat-bottomed, 16.34 ppt cleaned (T14 4.98 h), 50 events; the batch relMAD 0.39 was ~80% pulsation beating + a uniform x0.63 dilution in QLP s017 only (SPOC s017 normal), cleaned relMAD 0.07; new secondary 1.15 ppt at phase 0.4995 in all 6 sectors; ellipsoidal 3.2-4.5 ppt vs 4.5 predicted from K1; e cos omega 0.000(5) vs Gaia 0.136(76), circular; on target in 3 TESS-SPOC difference images)

P_days
: 2.349378

eP
: 0.000002

T0_primary_BTJD
: 2995.69666

depth_primary
: 1.63

depth_secondary
: 0.12

duration_h
: 4.98

source_path
: Fable 5\made\sb1-scale\pervet\vet284533163\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 284533163 (TYC 4020-206-1, F star, Teff 6431 TIC / 7077 Gaia, R ≈ 2.2–2.5 Rsun): per-object vet

Verdict: CONFIRMED. A δ Scuti / γ Dor hybrid F star is eclipsed every 2.349378 d by a ≈0.21 Msun M dwarf. The batch depth scatter (relMAD 0.39) is about 80% the star's own pulsation and about 20% a QLP sector-17 dilution problem. Once both are handled, the eclipse depth is constant.

Wave 4, 2026-10-20/21. Resumed from a killed session; receipts checked against fresh runs. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

1. Archives

- `cats.py --r 60`: VSX shows only a ROT star (G 16.37) at 49″. ASAS-SN none. SIMBAD [SB*] with one reference (Gaia DR3 multiplicity).

- Gaia NSS SB1: P 2.349632 ± 0.000163, e 0.175 ± 0.094, ω 326 ± 19°, K1 25.9 ± 2.1 km/s. vari tables empty.
- Not in any EB table (batch c4-ebcats and own VizieR cones: SB9, WDS, GCVS, Prša, IJspeert, Gaia vEB, Kostov).
- J/A+A/689/A177 returns a wide comoving-pair row (G 19.1 at 184″), which is irrelevant.
- ExoFOP: no TOI or CTOI. A web search found nothing. DTARPS and DIAmante don't cover this northern field (Dec +63°).
- MARVELS and DIAmante: none.

2. Chasing the depth scatter (the special instruction: relMAD 0.39)

The star pulsates. A Lomb-Scargle periodogram of the out-of-eclipse light (puls.py) shows coherent modes in every sector from 2019 to 2024:

Frequency (c/d)	Semi-amplitude (ppt)
11.759	10.4
11.995	8.3
11.25	3.7
12.85	1.3
1.33	4.2
1.43	3.4

The ~2 h main modes beat across a 5 h eclipse. That is why the batch saw an out-of-eclipse scatter of about 10 ppt per event and a relMAD of 0.39.

After removing 20 sinusoids per sector (clean.py):

Sample	relMAD
All 50 events	0.124 (raw SAP, same estimator: 0.269)
Excluding s017	0.070
s058 onward	0.054

- Event-to-event rms is 1.45 ppt, the same as the same estimator at control phases (1.24–1.72 ppt). The remaining scatter is noise.
- Leftover pulsation imprint on the depths is ≤ 1 ppt.

QLP s017 is an instrumental outlier. Every variability amplitude in QLP s017 is ×0.63 of the other sectors (11.759 c/d mode 6.5 vs 10.4 ppt; 1.33 c/d mode 2.8 vs 4.2 ppt), and its eclipse depth is ×0.65 (10.8 vs 16.4 ppt). - The depth-to-pulsation ratio is the same in every product: 1.66 in QLP s17, against 1.51–1.67 elsewhere. So something dilutes all of the star's light uniformly. - The cleaned residual in s017 is 3× noisier (5.6 vs 1.6–1.8 ppt in s018/s024 at the same cadence). - QLP s017 has higher background (SAP_BKG 193 vs 109 e^-/s in s018/s024), and 420 of its 1130 cadences are flagged (bit 4096 ×294, 6144 ×126). - **TESS-SPOC s017 looks normal**: PDCSAP depth 17.6 ± 1.0 ppt, pulsation 10.85 ppt, rms 1.5 ppt (spoc-out.txt). The problem is QLP s017 only.

Background: a per-event regression of flux against SAP_BKG gives a median possible fake dip of 0.13 ppt (max 1.9), uncorrelated with depth (r = +0.06). **Other checks:** 2 events have bit-32 cadences; their depths (18.3 vs 16.3 ppt) show no systematic. No events at the sector-wide systematic times. Centroid shift +0.8 ± 0.2 mpix.

3. Eclipse measurements, ephemeris, Gaia

- Cleaned shape: **depth 16.34 ± 0.06 ppt (event-scatter error ≈ 0.2), T14 4.98 h, T12 0.75 h, flat 3.49 h**. A V shape is rejected (Δχ² +6142).
- Odd 16.35 vs even 16.33 ± 0.09 ppt.
- Depth by sector (cleaned QLP): s018 16.59, s024 16.57, s058 15.81, s078 16.33, s085 17.13 ppt.
- **P = 2.349378 ± 0.000002 d, T0 = 2995.69666 ± 0.00046 BTJD**. O−C rms 8.9 min, from pulsation leftovers.
- Gaia P is off by −1.58σ (0.00026 d). The eclipse sits at Gaia conjunction A within −0.17σ to −0.93σ; the offset grows across sectors because of Gaia's P (compare at the Gaia epoch).
- **Secondary eclipse at phase 0.4975–0.500, depth 1.15 ± 0.30–0.38 ppt** (event-scatter errors; deep.py, sec_robust.py):
- positive in all 6 sectors, both parities, and both detrendings (orbital harmonics removed, or kept with a local quadratic baseline);
- the strongest peak anywhere else in the phase scan reaches only 2.6σ.
- The secondary phase gives e cos ω ≈ 0.00 ± 0.005. Gaia predicted phase 0.587 ± 0.048 (e cos ω 0.136 ± 0.076), a 1.8σ difference. Gaia's e itself is only 1.9σ from zero, and a 2.35 d orbit is expected to be circular.
- e sin ω can't be measured: the secondary is too shallow to give a duration ratio.
- **Ellipsoidal variation:** a cos 2φ term of 3.2–4.5 ppt per sector (undiluted sectors) against 4.5 ppt predicted from K1. This independently ties the RV companion to the eclipse period.

4. Independence

- TESS-SPOC (crowding factors 0.84–0.89): PDCSAP depth 17.6 / 18.3 / 16.1 ppt and SAP 16.2 / 14.8 / 14.6 ppt in s17/18/24.
- Difference images from the TESS-SPOC s17/18/24 pixel files: **the target wins every time** (χ² 433 / 669 / 381 against the next star's 1252 / 1039 / 728); the target's implied dip is 20.4 / 15.7 / 15.3 ppt. The batch QLP difference image also put the deficit on target.
- Neighbours: G 14.72 at 21″ would need about a 33% eclipse and fits worse. G 16.26 at 20″ would need more than 100%. G 13.6–14.7 stars at 53–83″ would need 12–33% and are rejected by the fits.

5. Physics (TIC M1 1.3, R1 2.50)

- Mass function f(m) = 0.00404 Msun, so **m2 = 0.21 Msun** (0.19–0.23). a = 8.5 Rsun.
- From the duration: b ≈ 0.52, i ≈ 81°, a/R1 3.93 (so R1 ≈ 2.2 Rsun).
- k = 0.128, so R2 ≈ 0.28–0.32 Rsun: a mildly inflated or irradiated M dwarf, or the b–limb-darkening degeneracy.
- Expected occultation: 0.9–1.5 ppt (thermal plus 0.42 ppt reflection) against 1.15 measured. Expected ellipsoidal 4.5 ppt against 3.2–4.5 measured.
- The companion is in front at conjunction A. Everything fits one geometry.

6. Verdict and predictions

CONFIRMED. The deciding numbers: - Cleaned depth is constant (relMAD 0.07 excluding s017, 0.054 in s058+), with event scatter at the noise floor. - s017 is explained by a uniform ×0.63 QLP dilution, confirmed by the clean SPOC s017 light curve. - The difference image is on target in 3 sectors. -

The secondary (1.15 ppt at phase 0.50) and the ellipsoidal amplitude both match the physics of a ≈0.21 Msun companion. - P agrees with Gaia at –1.6σ, and the eclipse sits at conjunction A.

Not catalogued anywhere. The pulsating primary is a bonus target (δ Sct in an eclipsing binary).

Next eclipses (BTJD): - Primary (±2 min): 4337.1915, 4339.5409, 4341.8903, 4344.2396 - Secondary (phase 0.499 ± 0.003, ±10 min): 4338.363, 4340.713, 4343.062, 4345.412

What would settle it further: RVs to check whether e = 0, and a combined eclipse-plus-pulsation model to sharpen the depth and b.

A170. TYC 4114-604-1 (TIC 88457347)

status
: confirmed (per-object re-analysis 2026-10-21; eccentric (e 0.23) F + 0.29 Msun M dwarf, flat-bottomed, 20.6 ppt, T14 3.30 h, 14 events (s020 adds 4); periastron brightening 0.8 ppt at Gaia's predicted phase; occultation hint 0.33 ppt at Gaia's conj-B phase 0.361, not claimed; the 6.4 arcsec batch offset not real (uncalibrated WCS + PSF-wing pull toward the box centre), own fits 0.4-1.7 arcsec on target in 6 products; RUWE 9.08, unresolved ~1 arcsec companion; no G < 17 neighbour within 54 arcsec)

P_days
: 5.4689808

eP
: 0.0000012

T0_primary_BTJD
: 2730.21937

depth_primary
: 2.06

duration_h
: 3.30

source_path
: Fable 5\made\sb1-scale\pervet\vet88457347\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 88457347 (TYC 4114-604-1, Gaia DR3 1099411576514297728): per-object vet

Opus 5.5 subagent, wave 5, 2026-10-21. No earlier session had worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. An eccentric F + M-dwarf EB. The eclipse is on target, and the 6.4″ batch offset is a centroiding artefact.

The numbers that decide it: - **Period:** 14 primary eclipses (the batch had 10; s020 adds 4) give P = 5.4689808 ± 0.0000012 d. That is +0.04σ from Gaia's period. - **Gaia epoch:** the TESS primary, extrapolated back to BTJD ~389, lands **−0.10 h (−0.10σ) from Gaia's conjunction A**. - **Periastron brightening:** a ~0.8 ppt brightening appears in all four sectors at phase 0.15. Gaia puts periastron at 0.127 ± 0.008. - **Occultation hint:** 0.33 ± 0.08 ppt at Gaia's conjunction-B phase (0.362, well away from 0.5). - **Localization:** the PSF-fit deficit is 0.4–1.7″ from the target in 6 pixel products.

1. Archives

- cats.py (r = 60″): nothing in VSX, ASAS-SN, Prša, IJspeert or Kostov. SIMBAD lists it as SB*.
- Gaia vari_summary and vari_eclipsing_binary: no rows.
- SB9, WDS, J/AJ/160/111: nothing. ExoFOP: no TOI, CTOI, imaging or spectroscopy.
- Web search on all three ids found nothing.
- Gaia NSS: SB1, e 0.233 ± 0.021, ω 159.2 ± 3.7°, K1 26.41 km/s.
- **RUWE 9.08.**
- Not previously catalogued as eclipsing.

2. Light curve (raw QLP SAP, Q0, no trimming)

- **Primary:** 20.57 ± 0.07 ppt, T14 3.30 h, T12 1.15 h, flat bottom 1.0 h. A V shape is worse by $\Delta\chi^2$ +573.
- **Per-sector depths:** 20.8 / 20.1 / 20.7 / 20.6 ppt. Odd and even agree (20.44 vs 20.69).
- **Control phases:** all 0 within ±0.13 ppt.
- **Background:** the unit-free regression gives fake dips ≤ 0.7% of the depth in 13 events.
- E264 (BTJD 3288.06) sits on a +3475 e^-/s scattered-light ramp. It still fakes only 3%, and its depth is unchanged with background as a regressor.
- **Flags and systematics:** 3 events have a bit-32 cadence, with no depth effect. E201 is 0.3 d from the listed 2943.2 systematic, and its depth is normal.
- **Blind census:** every off-ephemeris dip is either the s060 sector-start scattered light (2939.2–2939.5) or a sidelobe.
- **TESS-SPOC (independent pipeline):** s060 PDCSAP 20.43 / SAP 20.35 ppt, and s073 20.62 / 20.33 ppt, at the same ephemeris.

3. Ephemeris, secondary, e cos ω

- **Ephemeris:** T0 = 2730.21937 ± 0.00008 BTJD. χ^2/dof 12.6/12, O−C rms 0.5 min.
- **Conjunction A per sector:** within ±0.02 h of every sector's eclipse.
- **Phase 0.5:** nothing (+0.12 ± 0.07 ppt), which is correct for e cos ω = −0.22.
- **Fixed-shape test at Gaia's B phase:** +0.33 ± 0.08 ppt (S/N 4.3).
- All four sectors are positive.
- The look-elsewhere controls have rms 1.25σ and a maximum of +2.44σ.
- The best phase in the scan is 0.3612.
- **e cos ω:** about −0.218, against Gaia's −0.2175 ± 0.022.
- **e sin ω:** not measurable from a 4σ hint. The preferred occultation (2.5 h, shorter than the transit) fits Gaia's more distant conjunction B (r_B/r_A = 1.18).

4. Localization

- **Batch hypothesis confirmed:**
- In diff_s47.json the light centroid (8.051, 7.267) and the deficit centroid (8.039, 7.319) agree to 1.1″.
- Both sit about 0.30 px from the uncalibrated catalogue x = 7.734, inside a box centred on the rounded pixel 8.

- **A correction to the wave-4 explanation:** boxbias.py shows that a pure box-centring bias on a Gaussian PSF is only 0.00–0.02 px.
- The remainder comes from the uncalibrated WCS (+0.16 px, 3.5″, in s047) plus the PSF wings or a residual pedestal pulling toward the box centre.
- So it is the same family of artefact as wave 4, but box-centring alone doesn't account for its size.
- **Own PSF fit at calibrated positions** puts the deficit this far from the target:

Product	Offset from target
TESScut s020	1.5″
TESScut s047	0.8″
TESScut s060	0.4″ (errors over-estimated, as in 449475961's s101)
TESScut s073	0.4″ (errors over-estimated)
SPOC TPF s060	1.7″
SPOC TPF s073	0.4″

- In every product the missing flux is 1.93–2.05% of the target's own flux.
- **Neighbours:** there is no G < 17 neighbour within 54″. The G 14.46 star at 54″ would need a 47% dip and loses its fixed-position fit everywhere.
- **Caveat:** RUWE 9 means an unresolved companion within about 1″, so "on target" means on the Gaia source.

5. Physics

- **Companion:** f(m) = 0.0096 M⊙. With M1 ≈ 1.3, m2 ≈ 0.29 M⊙ (an M dwarf, R about 0.31 R⊙), and a ≈ 15.3 R⊙.
- **Shape fit:** the limb-darkened model wants k = 0.16–0.18 at b ≈ 0.85, so R1 ≈ 1.7 R⊙. k ≥ 0.35 is excluded.
- **Duration:** a predicted T14 of about 3.1–3.4 h matches the 3.30 h observed.
- **Occultation:** the predicted occultation at B for Gaia's e sin ω is 0.24–0.47 ppt, which fits the 0.33 ppt hint.
- One geometry fits everything.

Next predicted eclipses (BTJD)

Kind	Times	Uncertainty
Primary	4403.7275, 4409.1965, 4414.6655, 4420.1345 (then +5.4689808 d)	±0.5 min
Secondary (~0.3 ppt, phase 0.3612)	4405.703, 4411.172, 4416.641, 4422.110	±~1.8 h

What would settle it further

More sectors stacked at phase 0.36 to turn the occultation into a detection; imaging or Gaia DR4 for the RUWE-9 companion; RVs for M1.

Row fixes: P 5.4689808(12); depth 20.6 ppt; 14 events; the 6.4″ offset is not real.

A171. TYC 3248-1904-1 (TIC 191146789)

status
: confirmed (per-object re-analysis 2026-10-21; F + companion of m2 >= 0.93 Msun (f(M) 0.178), 22.7 ppt, T14 3.42 h, 8 events (not 3: 5 recovered from flagged cadences), jackknife clean; physics tension: no occultation (<= 0.8 ppt) where a main-sequence companion that massive should show 10-18 ppt at Gaia's e sin omega, so e sin omega >= 0.06-0.08 (3-5 sigma, the more-eccentric direction) or a dark M+M pair companion; SB2 lines the cheap test; ellipsoidal 0.3-0.9 ppt; on target in 4 products by dchi2 >= 550)

P_days
: 7.7886371

eP
: 0.0000029

T0_primary_BTJD
: 2851.29421

depth_primary
: 2.27

duration_h
: 3.42

source_path
: Fable 5\made\sb1-scale\pervet\vet191146789\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 191146789 (TYC 3248-1904-1, Gaia DR3 3892247497093565 44): per-object vet

Opus 5.5 subagent, wave 5, 2026-10-21. No earlier session had worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED (the eclipse is real, on target, and tied to the Gaia orbit), with a physics tension flagged

The numbers that decide it: - **Events:** mining flagged cadences took the count from **3 to 8**. All are listed below. - **Period:** P = 7.7886371 ± 0.0000029 d, −0.38σ from Gaia. - **Gaia conjunction A:** +0.70, +0.44 and +0.26 h from the eclipse in s017, s057 and s084 (≤ 0.65σ), and +1.05 h (1.8σ) at the Gaia epoch. - **Localization:** the deficit is 0.0–3.1″ from the target in 4 products.

The 8 events (events.py):

Sector	Event	Type	Depth (ppt)
s017	E−139	Q0	23.7
s017	E−137	Q0	23.5

Sector	Event	Type	Depth (ppt)
s057	E1	Q0, pre-ingress flagged	22.1
s057	E2	every cadence flagged (bit 2^{30}), complete eclipse	21.9
s057	E3	ingress only	timing only
s084	E95	Q0	23.4
s084	E96	Q0	22.9
s084	E97	Q0	22.7

Jackknife: no single event controls the result. - Dropping any one event moves P by ≤ 1.7×10^{-6} d and the depth by ≤ 0.16 ppt. Each dropped event lands within 2.0σ of the rest. - Dropping a whole sector leaves P stable to ≤ 3×10^{-6} d. - The batch's three s084 events, used alone, predict all five earlier events to within 1.1–1.3σ, so those are independent recoveries. With Gaia's P instead, the earlier events land within 0.4–0.7σ.

Archives

Nothing eclipsing anywhere. Same battery as 88457347 (VSX, ASAS-SN, Prša, IJspeert, Kostov, SB9, WDS, J/AJ/160/111, ExoFOP, Gaia vari tables, web search); SIMBAD lists it as SB*. Gaia: e 0.035 ± 0.012, ω 37 ± 18°, K1 60.49 km/s, RUWE 1.33. Not previously catalogued as eclipsing.

Light curve

- **Primary:** 22.7 ± 0.2 ppt, T14 3.42 h, T12/T14 0.4.
- **Background:** fakes ≤ 1.8% of the depth. E235 sits on a +939 e^-/s ramp; that is the 1.8% case, and its depth holds.
- **Systematics:** no bit-32 cadences, and no event is near a listed systematic time.
- **Control phases:** all within ±0.28 ppt.
- **TESS-SPOC s017:** PDCSAP 22.5 / SAP 21.7 ppt at the same ephemeris.
- **Ellipsoidal variation:** present in every sector and pipeline, A2 +0.31 to +0.87 ppt with minima at the conjunctions.

Secondary

- **Gaia prediction:** phase(B) 0.5162 ± 0.0061.
- **Fixed test at that phase:** +0.39 ± 0.12 ppt (3.2σ).
- **Best point in the scan:** 0.79 ± 0.13 ppt (6.2σ) at phase 0.520 with a short 2.5 h duration. It is positive in all three sectors (s017 +0.31 ± 0.30, s057 +0.79 ± 0.19, s084 +1.05 ± 0.21 ppt).
- **Controls:** rms 1.33σ, maximum +3.67σ.
- **e cos ω:** about +0.03, against Gaia's +0.025 ± 0.0095.
- **Overall:** no secondary deeper than about 1 ppt at any phase.

Localization

- **Batch:** deficit–target 1.14″.
- **Own PSF fit:**

Product	Offset from target	Missing flux / target flux
TESScut s017	2.1″	2.25%

Product	Offset from target	Missing flux / target flux
TESScut s057	0.0″	2.01%
TESScut s084	0.9″	2.04%
SPOC TPF s017	3.1″ ± 1.9″	1.80%

- The next-best star's fit is worse by $\Delta\chi^2 \geq 550$ in every product.
- **Neighbours:** G 15.25 and 15.71 at 34″ would need 71% and 118% dips; G 14.74 at 83″ would need 49%.

Physics: the tension

- **Companion mass:** f(m) = 0.178 $M_\odot$, so $m_2 \geq 0.93\ M_\odot$. A main-sequence companion that massive (about 5540 K) would carry 10–20% of the light.
- **Secondary prediction:** with Gaia's e sin ω (+0.020 ± 0.013), physics.py predicts a 10–18 ppt secondary eclipse. Observed: ≤ 0.8 ppt.
- **Shape can't decide it:** k 0.16–0.50 all fit (k ≥ 0.25 grazing). A grazing k ≈ 0.43 is within $\Delta\chi^2$ 1.8 of the best fit.
- **To hide the occultation**, a normal main-sequence companion needs e sin ω ≳ 0.06–0.08 (R_1 1.6–2.5). That is 3–5σ from Gaia's e sin ω, and about 3σ more eccentric overall (e ≈ 0.07 vs 0.035 ± 0.012).
- It would make the conjunction-B chord ultra-grazing and short, which fits the 2.5-h hint at 0.52.
- It runs in the same direction as 1260529, against the usual small-e failures.
- **Alternative:** a dark but massive companion, for example a close M+M pair.
- A single compact object can't make a 22 ppt transit.
- The depth constancy, χ^2/dof 12.3/5, mildly allows varying transit shapes.
- **Duration:** under both hypotheses, T14 and Kepler's a give $R_1 \approx 1.6$–$1.8\ R_\odot$, against TIC's 2.2.
- None of this undermines the eclipse or its tie to the Gaia orbit. It concerns what the companion is.

Next predicted eclipses (BTJD)

Kind	Times	Uncertainty
Primary	4401.2330, 4409.0216, 4416.8103, 4424.5989 (then +7.7886371 d)	±1–2 min
Secondary (≤0.8 ppt, phase ~0.519)	4405.275, 4413.064, 4420.853, 4428.641	±~1.1 h

What would settle it further

- **RV spectroscopy:** look for SB2 lines; a G-type companion at 10–20% light should show them. This also tests e directly.
- **Occultation:** more sectors to confirm the phase-0.52 dip and its duration, which gives e sin ω.
- **Gaia DR4** epoch RVs.

Row fixes: P 7.7886371(29); depth 22.7 ppt; **8 events (not 3)**; offset confirmed on target; flag the companion-light / e sin ω tension.

A172. TYC 4282-552-1 (TIC 434263468)

status
: confirmed (per-object re-analysis 2026-10-21; circular F subgiant + ~0.3 Msun M dwarf, flat-bottomed transit, 23.85 ppt, T14 8.07 h, 23 events in 9 sectors; total occultation 1.4 ppt at phase 0.50002(34), S/N 17.7, e cos omega 0.0000(5); QLP depth scatter 17-28 ppt is QLP dilution, PDCSAP 23-25 constant; prior art: unpublished TESS-SPOC DV TCEs for both eclipses (s0056-s0069), never a TOI, SPOC's s57 secondary centroid toward the G 9.19 neighbour at 67 arcsec overruled by 9 own difference images (dchi2 ~97,000))

P_days
: 8.077758

eP
: 0.000004

T0_primary_BTJD
: 3199.37690

depth_primary
: 2.39

depth_secondary
: 0.14

duration_h
: 8.07

source_path
: Fable 5\made\sb1-scale\pervet\vet434263468\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

REPORT — TIC 434263468 (TYC 4282-552-1, Gaia DR3 2207241593875197952)

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder; no previous session worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, circular F-subgiant + M-dwarf SB1 with both eclipse kinds detected on target.

Deciding numbers: - **Primary:** 23 events in 9 QLP sectors (s017–s085), each with S/N 22–136. Flat-bottomed transit, T14 8.04–8.10 h, T12/T14 0.15. - Depth is 23.85 ± 0.05 ppt in QLP. QLP depths scatter 17–28 ppt (χ^2/dof 855/22). - TESS-SPOC PDCSAP gives 23.1–25.2 ppt in all 9 sectors, so the QLP scatter is a QLP dilution systematic. - **Period:** P = 8.077758 ± 0.000004 d against Gaia's 8.078179 ± 0.000548 d, a difference of **−0.77σ**. - **Timing:** the primary falls on Gaia conjunction A (companion in front) at −0.87σ to −0.99σ in every sector. Conjunction B is 17–35σ away. - **Secondary eclipse:**

1.34–1.43 ppt in PDCSAP, present in 24 of 26 individual events. - S/N is 17.7 with red-noise-scaled errors; the control windows reach at most 5.5. - Phase is **0.50002 ± 0.00034** (PDCSAP jackknife) and 0.50069 ± 0.00082 (QLP). - That gives e cos ω = +0.0000 ± 0.0005, against Gaia's +0.0105 ± 0.0147 (0.7σ). - Duration ratio D2/D1 is 0.98–1.00, so e sin ω ≈ −0.002 ± 0.026, against Gaia's +0.001 ± 0.018. - **On target:** both eclipse kinds are on target in the pixels. - Primary: the SPOC DV TIC offset is 0.65″ (s57) and 0.74″ (s58). - Own difference images over 9 sectors make the target the best fit. The G 9.19 neighbour at 67″ is worse by $\Sigma\chi^2 \approx 97{,}000$. - At the secondary, the target is best, or within $\Delta\chi^2 \leq 7$ of a G 15.6 star at 10.5″ that would need a 12–54 ppt own dip. The G 9.19 star is worse by about 500.

1. Archives

- **cats.py --r 60:**
- VSX has only a YSO 27″ away, fainter than G 17. It would need a dip of more than 9,000 ppt, so it cannot be the source.
- No hits in ASAS-SN, Prša+2022, IJspeert+2024, or Kostov t2/t3/t4.
- SIMBAD: TYC 4282-552-1 [SB*].
- **Gaia DR3:** vari_summary and vari_eclipsing_binary are empty. NSS is SB1, significance 44.6, flags 0, ruwe 1.02.
- **SB9, WDS, J/AJ/160/111:** none.
- **ExoFOP:** no TOI, CTOI, imaging, spectra or companions.
- **Extra VizieR cones at 30″** (ATLAS, ZTF, WISE, GCVS, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST, APOGEE, Prša+2022): no rows. Each control cone returned rows, so the queries worked.
- **Web search** on "TYC 4282-552-1" and "TIC 434263468": nothing relevant.
- **New find, a TESS-SPOC DV report for s0056–s0069 on MAST** (`dv-out.txt`):
- TCE 1 is the primary: P 8.077874 ± 0.000077 d, 25.4 ppt, 8.26 h, Rp/R* 0.149, a/R* 8.61.
- TCE 2 is the secondary: epoch 2856.065 (phase 0.4993), 1.56 ± 0.10 ppt, 7.6 h, MES 15.
- Neither became a TOI.
- SPOC's s57 difference image for TCE 2 is offset 70″ toward the G 9.19 neighbour. The per-sector difference images here do not reproduce that offset (§4), so it is treated as a low-S/N centroid from a single sector.
- **Batch receipts:**
- c3-vetsum: dphi −0.006/+0.483, ATTR.
- diffrun s058: peak on target, deficit–target 2.23″. The 21.5″ deficit–light offset is the light centroid being pulled by the G 9.19 star.
- ebcats: none.
- **No prior art** beyond the unpublished SPOC TCEs.

2. Measurements (vet-out.txt; QLP SAP, Q==0, no trimming)

- **Odd vs even:** 23.86 ± 0.08 vs 23.84 ± 0.06 ppt.
- **Background:** the unit-free regression (bkgreg-out.txt) implies a flux change of ≤ 0.11 ppt per sector. The exception is s077 at 1.4 ppt, against a 14 ppt dip in the same windows.
- **Quality flags:** 2 events touch a bit-32 cadence. None is within 0.5 d of a listed sector-wide systematic.
- **Centroids:** QLP dX is +4.8 ± 1.2 mpix and dY is −4.9 ± 1.3 mpix. That is the expected shift away from contaminating light (CROWDSAP 0.78–0.88).

- **Dip census:** the only off-ephemeris dip at half the eclipse depth or more is at 2867.637 (16 ppt).
- It is the matched filter's window running into the primary 0.55 d later, and 26 cadences there are flagged.
- Many 1–4 ppt red-noise dips are present.
- **Second eclipse kind:**
- vet.py's free scan peaked at phase 0.4875. That was red noise; secA's control windows reach S/N 24.6.
- With red-noise-scaled errors (secscan-out.txt), the scan peaks at exactly 0.5000 in both QLP (S/N 7.7) and PDCSAP (S/N 17.7).

3. Ephemeris

- **Primary:** T0 = 3199.37690 ± 0.00022 BTJD, P = 8.077758 ± 0.000004 d.
- χ^2/dof is 147/21 (errors scaled ×2.65); O−C rms 3.6 min.
- **Against Gaia's conjunction A:** −1.9 h at s017 to −4.8 h at s085. The drift is Gaia's P being 0.77σ long.
- **Phase separation:** Gaia's B−A is 0.5067 ± 0.0094; measured 0.50002 ± 0.00034.

4. Independence

- **TESS-SPOC light curves** (9 sectors): PDCSAP depth 23.1–25.2 ppt. SPOC-SAP, uncorrected for crowding, gives 19.3–20.7 ppt.
- **Difference images** (tpfdiff2.py on 11×11 TPFs):
- Primary: the target pixel loses 20–22 ppt. The G 9.19 star's pixel changes by −0.5 to +1.7 ppt, so that star could only be contributing ≤ 2 ppt of its own dip.
- Secondary: the target pixel loses 1.0–2.0 ppt (S/N up to 14). The neighbour's pixel averages about 0.
- **Neighbour table** (needed dip = 23.8 ppt × 10^(0.4ΔG)):

Neighbour	Needed dip	Verdict
G 9.19 at 66.9″	7.6 ppt	excluded by pixels ($\Sigma\chi^2$ +97,000)
G 15.63 at 10.5″	about 2,860 ppt	impossible
G 15.25 at 13.7″	about 2,000 ppt	impossible

5. Physics

- **Mass function:** f(M) = 0.0125 M⊙. With M1 = 1.30 M⊙, m2(i = 90°) is 0.29–0.35 M⊙ and a ≈ 19.9 R⊙.
- **Duration:** the central-transit T14 would be 8.95 h. The observed 8.04–8.10 h gives b ≈ 0.4–0.5 and i ≈ 86.4°.
- **Radii:** SPOC's k = 0.149 gives R2 = 0.34–0.38 R⊙ (using R1 = 2.31 R⊙ from SPOC's a/R*, or 2.53 R⊙ from TIC). That fits an M dwarf of the f(M) mass.
- **Which star is in front:**
- At conjunction A the M dwarf transits the F star.
- At conjunction B the M dwarf is totally occulted (flat bottom). The depth ratio 0.058 implies T2 ≈ 3200–3300 K.
- The predicted occultation depth is 1.4–1.9 ppt; observed 1.34–1.43 ppt.
- **Ellipsoidal:** predicted 0.64 ppt; PDCSAP gives A2 = 0.35–0.57 ppt.

- **Heartbeat:** not an issue, because the orbit is circular.

6. Next predicted times (BTJD)

- **Primary** (conjunction A): **4338.3408, 4346.4185, 4354.4963** (± 0.0006 d), then every 8.077758 d.
- **Secondary** (conjunction B): **4342.3798, 4350.4576, 4358.5353** (± 0.003 d).

What would settle it further

- RVs for a joint orbit fit.
- A spectroscopic log g to resolve the R1 question (TIC 2.53 R⊙ vs SPOC 2.31 R⊙).
- Reporting the SPOC TCE as a CTOI/EB.

Files

- **Scripts:** vet.py (TMAX/T14G overrides), vetlib, bkgreg, secA, secscan, secphase, tpfdiff2, indep, dvparse, phys_ecc, physics, gaiatap, morecats, extracats, mastlc.
- **Outputs:** the matching *-out.txt files, plus vet-results.json, fold_oc.png and events.png.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet434263468\`.

A173. BD-02 2880 (TIC 293844315)

status
: confirmed (per-object re-analysis 2026-10-21; eccentric (e 0.36) G + 0.22 Msun M dwarf transit at conjunction A, 22.5 ppt QLP (SPOC PDCSAP 23.0-23.3), T14 4.38 h, 7 events S/N 69-177; tentative grazing occultation 0.49 ppt at phase 0.3309 = Gaia's predicted B-A to -0.2 sigma (3 SPOC events, per-event depths inconsistent, kept tentative); SPOC DV TCE for the primary, never a TOI; LAMOST DR7 spectra exist; on target in 7 SPOC TPF difference images, SPOC DV offsets 0.2-0.5 arcsec)

P_days
: 14.774809

eP
: 0.000003

T0_primary_BTJD
: 2810.95109

depth_primary
: 2.25

duration_h
: 4.38

source_path
: Fable 5\made\sb1-scale\pervet\vet293844315\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

REPORT — TIC 293844315 (BD−02 2880, Gaia DR3 3837326955675043456)

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. An eccentric (e ≈ 0.36) G-dwarf + M-dwarf SB1; the primary eclipse is solidly attributed and on target, and the second eclipse is only tentatively detected.

Deciding numbers: - **Primary:** 7 events in 4 QLP sectors (s008, s035, s062, s099), each with S/N 69–177. Depth 22.5 ppt (QLP), T14 4.38 h, T12/T14 0.25. - QLP s099 is 21.2 ppt against about 24 ppt elsewhere. - SPOC 2-min PDCSAP gives 23.0–23.3 ppt in all 4 sectors, so the s099 value is a QLP dilution effect. - **Period:** P = 14.774809 ± 0.000003 d against Gaia's 14.773343 ± 0.001224 d, a difference of **+1.20σ**. - **Timing:** the eclipse falls on Gaia conjunction A at +1.34σ to +1.54σ in every sector. Conjunction B is 14–32σ away. - **On target:** - SPOC DV TIC offsets are 0.16–0.53″ (s8, s35, s62). - Own difference images over 7 TPFs make the target the best fit. Every neighbour is worse by Σχ²

of at least 60,000. - **Second eclipse at B (tentative):** - SPOC 2-min PDCSAP shows **0.49 ± 0.08 ppt at phase 0.3309 (0.3303–0.3356)**, from 3 events. This is the top of the whole-phase scan; control windows reach at most 2.4. - Gaia predicts B–A = 0.3333 ± 0.0105, so this is −0.2σ. - Converted to e cos ω (at Gaia's e sin ω) it gives −0.260 (−0.264 to −0.256), against Gaia's −0.256 ± 0.017. - A free fit gives a V-shaped graze, 0.5–0.77 ppt and 9.8 h (8.9–11.8 h). - Per-event depths of 0.33, 0.70 and 1.14 ppt are not mutually consistent, and QLP (2 events) does not see it. **Treat it as tentative.** The CONFIRMED rests on the primary.

1. Archives

- **cats.py --r 60:** no hits in VSX, ASAS-SN, Prša+2022, IJspeert+2024 or Kostov. SIMBAD: BD−02 2880 [SB*].
- **Gaia DR3:** vari tables empty. NSS is SB1, significance 37.3, flags 0, ruwe 1.05.
- **SB9, WDS, J/AJ/160/111:** none.
- **ExoFOP:** no TOI, CTOI, imaging or spectra.
- **Extra VizieR cones:** none in any variable or X-ray table. LAMOST DR7 LRS has spectra at 2.2″, which could supply RVs.
- **Web search** on "BD-02 2880" and "TIC 293844315": nothing relevant.
- **MAST:** SPOC DV reports for s1–s36 and s1–s65 contain one TCE, the primary only. No TOI was made.
- **Batch receipts:**
- c3-vetsum: dphi +0.011/−0.322, ATTR.
- diffrun s008: peak on target, 1.95″.
- ebcats: none.

2. Measurements (QLP SAP, Q==0, no trimming)

- **Odd vs even:** 22.61 ± 0.09 vs 22.29 ± 0.09 ppt. This tracks the s099 dilution, not a period problem.
- **Background:** the unit-free regression implies a flux change of ≤ 0.39 ppt per sector (s062), against a 16–18 ppt dip in the same windows.
- **Quality flags:** no event has a bit-32 cadence, and none is near a listed sector-wide systematic.
- **Centroids:** dX −0.3 mpix, dY −3.4 ± 1.0 mpix.
- **Control phases:** all |depth| ≤ 0.14 ppt.
- **Dip census:** no off-ephemeris dip of half the eclipse depth or more.
- A 9.3 ppt feature at 4067.13 sits 0.32 d after a primary, where the window runs into the eclipse.
- Phase 0.5 shows −0.03 ± 0.09 ppt; there is no circular-orbit secondary.
- **Second eclipse in QLP at B** (secA): 2 events, 0.0 ± 0.11 ppt; a weak S/N 3.1 peak at phase 0.329. Too thin to use on its own.

3. Ephemeris

- **Primary:** T0 = 2810.95109 ± 0.00021 BTJD, P = 14.774809 ± 0.000003 d.
- χ^2/dof is 22.5/5; O−C rms 0.6 min.
- **Against Gaia's conjunction A:** +3.7 h (s008) to +9.7 h (s099). The drift is Gaia's P being 1.2σ short.

4. Independence

- **SPOC 2-min and TESS-SPOC light curves:** PDCSAP depth 23.0–23.3 ppt in s008, s035, s062 and s099.
- **Difference images:**
- Primary: the target pixel loses 22.7–23.9 ppt. The G 10.91 star at 69″ shows ≤ 1.4 ppt at its pixel.
- Secondary candidate: the target pixel loses 1.78 ppt (S/N 7.9) in s062, but there is nothing in s099. Inconclusive.
- **Neighbour table** (needed dip = 23.8 ppt × 10^(0.4ΔG)):

Neighbour	Needed dip	Verdict
G 10.91 at 69″	42 ppt	excluded by pixels
G 15.07 at 42.5″	about 900 ppt	excluded
G 14.35 at 62″	about 1,700 ppt	excluded

5. Physics

- **Mass function:** f(M) = 0.0063 M⊙. With M1 = 1.10 M⊙, m2 ≈ 0.22 M⊙ (R2 ≈ 0.24 R⊙) and a = 27.8 R⊙.
- **Transit:** a 23 ppt transit across R1 = 1.55 R⊙ gives k ≈ 0.15, so R2 ≈ 0.23 R⊙. Consistent.
- **Conjunction A:** separation 0.695a, central T14 5.47 h. The observed 4.38 h gives a chord fraction of 0.80 and i ≈ 86.8°.
- **Conjunction B:** separation 1.164a, so the projected separation sits right at R1+R2 (marginal). That predicts at most a graze of an M dwarf whose total light is about 1.8 ppt, with a long chord (about 9.2 h central).
- This is consistent with the tentative 0.5–0.7 ppt, roughly 9.8 h V-shaped feature.
- It also explains why no clear secondary is seen.
- **Heartbeat:** the eclipse is near conjunction, not periastron, and is short and sharp. No heartbeat or tidal confusion.

6. Next predicted times (BTJD)

- **Primary** (M dwarf transits the G star, conjunction A): **4347.5312, 4362.3060, 4377.0808** (± 0.0004 d), then every 14.774809 d.
- **Secondary** (grazing occultation, conjunction B, tentative): **4352.4195, 4367.1943, 4381.9691** (± 0.04 d).

What would settle it further

- More 2-min sectors covering phase 0.331 to confirm the graze.
- LAMOST or new RVs to tighten e and ω.

Files

- **Scripts:** vet.py, vetlib, bkgreg, secA, secscan, secB, tpfdiff2, indep, dvparse, phys_ecc, gaiatap, morecats, extracats, mastlc.
- **Outputs:** the matching *-out.txt files, plus vet-results.json, fold_oc.png and events.png.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet293844315\`.

A174. TYC 5452-573-1 (TIC 348784186)

status
: confirmed (per-object re-analysis 2026-10-21; circular grazing F/G + K1-K2 dwarf (q 0.62, T2 4750-5000 K = the f(M) mass), 26.4 ppt, T14 3.00 h, 23 primaries + 23 secondaries; secondary 11.6 ppt at phase 0.50012(4), e cos omega +0.0001; the batch 'peak not on target' s088 flag overturned (P/2 stack mixing unequal eclipses), own TPF difference images on target by dchi2 > 9,000 in both kinds; R1 tension: eclipse + ellipsoidal 1.5 vs TIC 2.0 Rsun, companion light; the G 13.7 neighbour at 34 arcsec excluded by dchi2 +12,000 to +35,000)

P_days
: 4.0868675

eP
: 0.0000004

T0_primary_BTJD
: 2978.32495

depth_primary
: 2.64

depth_secondary
: 1.16

duration_h
: 3.00

source_path
: Fable 5\made\sb1-scale\pervet\vet348784186\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

REPORT — TIC 348784186 (TYC 5452-573-1, Gaia DR3 5736614545403168512)

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder; no earlier session worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, circular, grazing SB1 with both eclipse kinds strong and on target.

Deciding numbers: - **Primary eclipses:** 23 individual events in 5 sectors (s008, s035, s061, s088, s099), each at S/N 63–149. Depth 26.41 ± 0.05 ppt, T14 3.00 h, T12/T14 0.40 (V-ish, grazing). - **Secondary eclipses:** 23 individual events, each at S/N 28–73. Depth 11.61 ± 0.04 ppt, T14 2.94 h. - **Period:** 4.0868675 ± 0.0000004 d from the primaries and 4.0868686 ± 0.0000007 d from the secondaries. Gaia's is 4.086794 ± 0.000064 d, so the difference is **+1.14σ**. - **Which conjunction:** the deeper eclipse sits on Gaia conjunction A (primary eclipsed) at **+0.61σ to +1.01σ** in every sector. The

drift across sectors comes from Gaia's P being short. The shallow eclipse sits on conjunction B at +1.5σ to +1.8σ. - **Phase separation** 0.50012 ± 0.00004 gives e cos ω ≈ +0.0001 ± 0.0001, i.e. circular. - Gaia has e cos ω −0.0095 (e 0.0097 ± 0.0073), and Gaia's predicted B−A phase of 0.4953 ± 0.0038 is 1.2σ away. - This is the wave-4 small-e pattern, but mild here. - **Same source for both eclipse kinds:** the primary/secondary depth ratio is 2.23–2.36 in every sector. In s099 both depths drop together (26.9→23.8 and 11.9→10.5 ppt), which is a sector dilution change. - **On target:** TESS-SPOC difference images in s035 and s061 have their max-loss pixel on the target at S/N 65 and 109. The target fit beats every neighbour by $\Delta\chi^2 > 9{,}000$. The s061 secondary difference image is also on target (S/N 42).

1. Archives

- **cats.py --r 60:** no VSX or ASAS-SN entry, and no EB-table hits. SIMBAD has only TYC 5452-573-1, typed [SB*].
- **Gaia DR3:** vari_summary and vari_eclipsing_binary are empty; RUWE 1.10.
- **SB9, WDS, J/AJ/160/111:** none.
- **ExoFOP:** no TOI or CTOI, and no imaging, spectra or companions.
- **Extra VizieR cones at 30″** (ATLAS, ZTF, WISE variables, GCVS, VSX, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST, APOGEE, Prša+2022): no rows, and the control cones returned rows. J/A+A/674/A16 is not an EB table.
- **Web search** ("TYC 5452-573-1" OR "TIC 348784186" eclipsing binary): no page mentions the star.
- **Batch receipts:**
- c3-vetsum: dphi +0.002/−0.492, ATTR.
- c3-sapdepth: alternating ~25/~11 ppt.
- **diffrun s088: peak_on_target false, deficit−target 3.51″, n_possible 1 (the G 13.71 at 34.7″). The TPFs in §4 contradict this.** The likely cause is stacking at P/2 (mixing the 26- and 12-ppt eclipses) plus the wave-4 centroiding artefact; deficit−light is only 1.9″.
- **No prior art.** This is a new EB.

2. Measurements (QLP SAP, Q==0, no trimming)

- **Per-sector depths:** primary 27.02, 26.89, 27.16, 26.95 and 23.82 ppt. Secondary 11.44, 11.82, 12.17, 11.93 and 10.53 ppt.
- **Odd vs even primaries:** 26.35 ± 0.06 vs 26.46 ± 0.06 ppt.
- **Shape:** a pure V is worse by $\Delta\chi^2$ 700 and a box by $\Delta\chi^2$ 18,700.
- **Harmonics (ppt):** cos 2φ −1.10 (ellipsoidal, correct sign); cos φ −0.42 (reflection, brightest at phase 0.5); sin φ +0.56 ± 0.48 (beaming sign, weak).
- **Background:** the unit-free regression implies $|\Delta F| \leq 0.11$ ppt per sector, against 14–17 ppt (primary) and 6.5–7.4 ppt (secondary) in the same windows. vet.py's dBKG column has the known broken units and is ignored.
- **Centroids:** QLP centroid shifts are under 1 mpix.
- **Quality flags:**
- 4 primaries touch a bit-32 cadence, and none sits at a listed systematic time.
- Where flagged cadences cover an eclipse they show the same dip (27.4–28.3 ppt).
- **Control phases** 0.125–0.875 (excluding 0.5): |depth| ≤ 0.07 ppt, n 20–23 each.
- **Blind dip census:** 47 dips at S/N ≥ 4: 23 primaries, 22 secondaries, and two 1.5–1.8-ppt dips at S/N < 5. One of those two is at the s061 systematic (2981.9). Nothing off-ephemeris reaches half the

eclipse depth.

3. Ephemerides

- **Primary (conjunction A):** T0 = 2978.32495 ± 0.00007 BTJD, P = 4.0868675 ± 0.0000004 d. χ^2/dof 41/21, O−C rms 0.5 min.
- **Secondary (conjunction B):** T0 = 3143.84359 ± 0.00016 BTJD, P = 4.0868686 ± 0.0000007 d. χ^2/dof 138/21, O−C rms 1.0 min; the scatter is from the 30-min and 10-min cadence sectors.
- **P_sec − P_pri** = +0.10 ± 0.07 s: no apsidal motion.
- **Secondary offset from phase 0.5:** +0.7 ± 0.25 min. The Rømer delay accounts for about 16 s of it, leaving e cos ω ≈ +0.0001.
- **e sin ω** from the duration ratio is −0.010 ± 0.010; Gaia gives −0.002.
- **Against Gaia** (Monte Carlo with corr_vec): conjunction A − observed primary runs from +0.27 h (s008) to +1.36 h (s099).

4. Independence

- **Other pipelines:** TESS-SPOC exists for s008, s035 and s061. GSFC-ELEANOR-LITE, T16, TGLC and TARS also exist and were not used.
- **TESS-SPOC primary depths:** PDCSAP 26.2–26.8 ppt and SPOC-SAP 24.6–25.5 ppt. Ellipsoidal amplitude 1.1–1.7 ppt.
- **Difference images:**
- s061 primary: target χ^2 5,932 (implied own dip 20.4 ppt), against 29,175 for the G 15.91 star and 41,189 for the G 13.71 star.
- s035 primary: target χ^2 2,817, against 9,683 or more for every neighbour.
- s061 secondary: target χ^2 785, against 4,335.
- **Neighbour dilution table** (needed dip = depth × 10^(0.4ΔG)):

Neighbour	Primary needs	Secondary needs
G 13.71 at 34.4″	595 ppt	261 ppt
G 15.04 at 45.9″	≈2,000 ppt	—
G 15.91, 16.48, 16.53, 16.68 (20–49″)	>1,000 ppt	—
G 14.08 at 68″ (outside 60″)	835 ppt	—

Only the G 13.71 star is arithmetically possible, and the pixels exclude it by $\Delta\chi^2$ +35,000 (s061) and +12,000 (s035).

5. Physics

- **Mass function** f(M) = 0.1126 M⊙ (K1 64.31 ± 0.41 km/s). With M1 = 1.22 ± 0.19 M⊙, m2(90°) = 0.76 M⊙ (0.69–0.83), so a K dwarf. q ≈ 0.62, a ≈ 13.5 R⊙.
- **Which star is in front:** the deeper eclipse is at conjunction A, the F star eclipsed, as Gaia's convention requires.
- **Companion temperature:** the depth ratio 0.44 gives the surface-brightness ratio J. Two-sphere fits to both folds give J = 0.39–0.47, so T2 ≈ 4750–5000 K. That is a K1–K2 dwarf of exactly the f(M) mass.
- **Geometry:** grazing, b ≈ 1–1.6 R1, i ≈ 80–84°. With third light fixed at the TIC contamination ratio, R1/a = 0.112, so R1 ≈ 1.5 R⊙. The k–b–third-light degeneracy is strong; with free third light the fit

runs off to an unphysical k = 1.2.
- **Ellipsoidal:** predicted ≈ 1.1 ppt for R1 1.5 R⊙, observed 1.10–1.23 ppt. TIC's R1 of 2.0 R⊙ would give 2.5 ppt.
- **Open item, radius tension:** the eclipse and ellipsoidal fits want R1 ≈ 1.5 R⊙, against TIC 2.0 R⊙ and ≈ 1.8 R⊙ from Gaia logg. The companion's light inflates the single-star radius, which explains part of it. It does not affect the verdict.
- **Not a heartbeat:** the orbit is circular and both eclipses are short.

6. Next predicted eclipses (BTJD; formal errors 0.2–0.4 min, allow ±2 min)

- **Primary** (conjunction A, ~26 ppt): **4335.1650, 4339.2518, 4343.3387**, then every 4.0868675 d.
- **Secondary** (conjunction B, ~11.6 ppt): **4337.2092, 4341.2961, 4345.3830**, then every 4.0868686 d.

What would settle it further

- A spectrum to detect the K dwarf. As an SB2 (K2 ≈ 100 km/s) it would give masses.
- A spectroscopic Teff and log g, for R1.
- High-resolution imaging for third light.

Files in vet348784186

- **Scripts:** vet.py (patched: secondary mask, T14 ceiling, T14G mask width), vetlib.py, ecosw.py, ebfit.py, flagged.py, sec2.py, bkgreg.py, indep.py, tpfdiff.py, physics.py, gaiatap.py, mastlc.py, morecats.py, extracats.py.
- **Primary-eclipse outputs:** vet-out.txt, vet-results.json, fold_oc.png, events.png.
- **Secondary-eclipse outputs (secv):** vet-out.txt, vet-results.json, fold_oc.png, events.png, tpfdiff-sec-out.txt, bkgreg-out.txt.
- **Analysis outputs:** ecosw-out.txt, sec2-out.txt, ebfit-out.txt, ebfit-dfix-out.txt, ebfit*.png, bkgreg-out.txt, flagged-out.txt, flagged-sec-out.txt, indep-out.txt, tpfdiff-out.txt, tpfdiff_35.png, tpfdiff_61.png, physics-out.txt.
- **Archive outputs:** cats-out.txt, gaiatap-out.txt, morecats-out.txt, extracats-out.txt, mastlc-list.txt, mastlc-dl.txt.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet348784186\`.

A175. TYC 4674-364-1 (TIC 269504659)

status
: confirmed (per-object re-analysis 2026-10-21; G + 0.27 Msun M dwarf transit at conjunction A, 23.6 ppt, T14 4.44 h, 6 events (one from a bit-32 window); new secondary 1.45 ppt at phase 0.5357, e cos omega +0.0560(5), a real eccentricity; APOGEE DR17's three RVs phased on the eclipse ephemeris follow Gaia's orbit (rms 0.30 km/s), an independent spectroscopic tie; on target in 4 difference images; no Gaia source brighter than G 20.1 within 60 arcsec)

P_days
: 11.391807

eP
: 0.000015

T0_primary_BTJD
: 2800.01713

depth_primary
: 2.36

depth_secondary
: 0.15

duration_h
: 4.44

source_path
: Fable 5\made\sb1-scale\pervet\vet269504659\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

REPORT — TIC 269504659 (TYC 4674-364-1, 2MASS J00551372-0007541, Gaia DR3 2536832294627315072)

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder; no earlier session worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached SB1 with an M-dwarf companion; the M dwarf transits a G subgiant/dwarf. Secondary eclipse detected; APOGEE RVs independently phase the orbit to the eclipses.

Deciding numbers: - **Primary eclipses:** 6 individual events in s042, s043 and s070, each at S/N 61–117. This is one more than the batch's 5: the 3221.515 event has a bit-32 cadence in its window. Depth 23.57 ± 0.10 ppt, T14 4.44 h, T12/T14 0.35. - **Period:** P = 11.391807 ± 0.000015 d against Gaia's 11.390958 ± 0.001045 d, a difference of **+0.81σ**. - **Which conjunction:** the primary sits on Gaia

conjunction A at **+0.06σ to +0.26σ** (+0.3 h to +1.7 h). Conjunction B is 21σ to 27σ away. - **Secondary eclipse (new):** - In a fixed-shape test inside ±3σ of Gaia's predicted phase (0.5177 ± 0.0122), the best window is at phase 0.5361 with 1.47 ± 0.12 ppt, S/N 12.2. - In 35 look-elsewhere control windows the best S/N has median 2.0 and maximum 5.9; none reaches ours. - Per event: 4 events at S/N 5.4–8.4, depths 1.24, 1.41, 1.48 and 1.60 ppt (weighted 1.45 ± 0.11, χ^2 1.5/3). T14 4.92 h (range 4.68–5.22). - It is also present in TESS-SPOC and SPOC 2-min light curves at 1.35–1.89 ppt. - **Eccentricity:** phase separation 0.53566 ± 0.00031 gives **$e \cos \omega = +0.0560 \pm 0.0005$** (Rømer delay ~1 min, negligible). - Against Gaia's +0.028 ± 0.019 that is +1.4σ; against Gaia's predicted B–A phase it is +1.5σ. - The duration ratio gives $e \sin \omega \approx +0.05 \pm 0.03$, against Gaia's −0.015 ± 0.016. This is soft, per the wave-4 lesson. - **APOGEE DR17 (independent RVs):** three visits (BTJD 2079.94, 2080.92, 2083.89) at +2.51, −3.29 and −33.78 km/s. - Phased on the eclipse ephemeris alone, with Gaia's K and γ, the model matches with rms 0.30 km/s after a constant +1.76 km/s zero-point offset. - A free circular fit gives K = 20.9 km/s (Gaia 21.34 ± 0.57). The RV descending node falls within +0.004 phase of the deep eclipse. - So the luminous RV star is the one eclipsed at the deep eclipse. (With three points and three free parameters the free fit itself is exact, not a test.)

1. Archives

- **cats.py --r 60:** no VSX or ASAS-SN entry, and no EB-table hits. SIMBAD has only TYC 4674-364-1, typed [SB*].
- **Gaia DR3:** vari tables are empty; RUWE 1.29, ipd_frac_multi_peak 0.
- **SB9, WDS, J/AJ/160/111:** none.
- **ExoFOP:** no TOI or CTOI, and no imaging, spectra or companions. The SPOC 2-min light curve exists for s070.
- **Extra VizieR cones at 30″:** no variability or X-ray rows (ATLAS, ZTF, WISE, GCVS, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST, Prša+2022), and the control cones returned rows. APOGEE DR17 has the star (Teff 5770 K, logg 4.03), which is the source of the RVs above.
- **Web search** ("TYC 4674-364-1" OR "TIC 269504659" eclipsing binary): no page mentions the star.
- **Batch receipts:**
- c3-vetsum: dphi +0.001/+0.483, ATTR.
- c3-sapdepth: 5 events at 21.6–23.9 ppt.
- diffrun s043: peak on target, deficit–light 0.44″.
- **No prior art.** This is a new EB.

2. Measurements (QLP SAP, Q==0, no trimming)

- **Primary depth per sector:** s042 23.58, s043 24.39, s070 23.02 ppt. χ^2/dof of depth constancy is 46/5, consistent with sector-to-sector dilution.
- **Odd vs even:** 23.45 ± 0.14 vs 23.70 ± 0.15 ppt.
- **Shape:** a box is worse by $\Delta\chi^2$ 4,700 and a V by $\Delta\chi^2$ 490. There is a flat bottom of about 1.3 h.
- **Timing:** O–C is within ±2 min of the linear ephemeris. The formal 0.4-min errors sit at the fit's grid floor (0.72-min steps).
- **Background:**
- Primary: the regression implies $|\Delta F| \leq 0.06$ ppt per sector against 15–16 ppt dips in the same windows. The large dBKG on 3221.515 is the known broken-unit column.

- Secondary: s042 0.000 and s070 0.002 ppt. In s043 the background could account for up to 0.33 ppt of a 1.1-ppt dip, and the other sectors carry the detection anyway.
- **Centroids:** secondary dX +0.02 ± 0.19 mpix, dY −1.7 ± 1.1 mpix.
- **Quality flags:** 2 primaries and 2 secondaries touch a bit-32 cadence; none is near a listed systematic time.
- **Controls:** primary-run control phases give |depth| ≤ 0.27 ppt. Phase 0.500 gives +0.27 ± 0.10 ppt, which is wing leakage from the 0.536 secondary.
- **Blind dip census:** 14 dips at S/N ≥ 4.
- 6 are the primaries.
- 4 are the 1.2–1.6-ppt dips at phase −0.465 (= 0.535), i.e. the secondary.
- 3208.92 (1.7 ppt, phase 0.895) has a large s070 background excursion, and its next-cycle repeat measures 0.08 ± 0.22 ppt.
- 3221.94 sits on the post-eclipse shoulder.
- The rest are under 1 ppt.

3. Ephemerides

- **Primary (conjunction A):** T0 = 2800.01713 ± 0.00049 BTJD, P = 11.391807 ± 0.000015 d (χ^2 scaled ×4.5).
- **Secondary (conjunction B):** T0 = 2863.07833 ± 0.00347 BTJD, P = 11.391710 ± 0.000105 d, consistent with the primary's P. χ^2/dof 0.6/2.
- **Against Gaia** (Monte Carlo with corr_vec):
- Conjunction A – observed primary: +0.32 h (s042), +0.37 h (s043), +1.72 h (s070), i.e. 0.06σ–0.26σ.
- Conjunction B – observed secondary: +5.1 to +6.3 h, about +1.0σ.

4. Independence

- **Other pipelines** (TESS-SPOC s042, s043, s070 and SPOC 2-min s070): primary depth PDCSAP 22.2–22.9 ppt and SAP 22.9–23.1 ppt; secondary 1.35–1.89 ppt in all of them.
- **Difference images** (four TPFs: TESS-SPOC s042, s043, s070 and SPOC s070): the max-loss pixel is on or adjacent to the target at S/N 23–34, the loss-weighted centroid is within about 0.4 px of the target, and the implied own dip is 19.2–20.6 ppt.
- **Neighbours:** Gaia DR3 has no source brighter than G 20.14 within 60″; the six there are G 20.1–21.3 at 25–49″. The brightest would need a 116,000-ppt dip for the primary and 7,000 ppt for the secondary, so all are impossible.

5. Physics

- **Mass function** f(M) = 0.0114 M⊙. With M1 = 1.05 ± 0.14 M⊙, m2 = 0.27 M⊙ (0.25–0.29), an M dwarf. q ≈ 0.26, a ≈ 23.4 R⊙.
- **Transit geometry:** depth 23.6 ppt gives k ≈ 0.16, and T14 4.44 h with T12/T14 0.35 gives b ≈ 0.8. Then R1/a ≈ 0.061, so **R1 ≈ 1.4–1.6 R⊙** and R2 ≈ 0.23–0.26 R⊙. That is the size of an M dwarf with the f(M) mass.
- TIC gives R1 = 1.79 ± 0.09 R⊙.
- log g gives R1 ≈ 1.55 R⊙ (Gaia 4.08) and ≈ 1.64 R⊙ (APOGEE 4.03).
- So the mild tension is with TIC only.

- occfit.py's free fits are degenerate (transit R1 1.31 at amplitude 0.45; occultation R1 1.55) and are not used.
- **Secondary depth:** a total occultation of an M dwarf with $k^2 \approx 0.025$ needs $J \approx 0.06$, i.e. $T2 \approx$ 3100–3300 K for a G primary (TIC 5839 K, APOGEE 5770 K). That is the expected temperature for 0.27 $M_\odot$. Observed 1.45 ppt against roughly 1.5–2.2 ppt predicted.
- **Ellipsoidal and beaming:** predicted 0.15 and 0.28 ppt, below the noise (measured harmonics ≤ 0.17 ppt), and the measured cos 2φ term has the wrong sign. Not detected.
- **Not a heartbeat:** the orbit is low-e, and the eclipse is a flat-bottomed transit.

6. Next predicted eclipses (BTJD)

- **Primary** (M dwarf transits, conjunction A, ~23.6 ppt, T14 4.4 h): **4337.9111, 4349.3029, 4360.6947** (± 3 min), then every 11.391807 d.
- **Secondary** (M dwarf occulted, conjunction B, ~1.45 ppt, T14 ~4.9 h): **4344.0132, 4355.4050, 4366.7968** (± 6 min).

What would settle it further

- A few more RVs, ideally from Gaia DR4 epoch RVs or new APOGEE/HARPS visits, for a joint RV + eclipse fit of e and ω.
- More TESS sectors, to firm up the 1.45-ppt secondary's phase and duration (e sin ω).
- A spectroscopic R1.

Files in vet269504659

- **Scripts:** vet.py (patched as for 348784186), vetlib.py, secA.py, bkgreg.py, indep.py, tpfdiff.py, occfit.py, physics.py, apogee.py, nb60.py, gaiatap.py, mastlc.py, morecats.py, extracats.py.
- **Primary-eclipse outputs:** vet-out.txt, vet-results.json, fold_oc.png, events.png, oc-out.txt.
- **Secondary-eclipse outputs:** secA-out.txt; secv\vet-out.txt, vet-results.json, events.png, bkgreg-out.txt, indep-sec-out.txt.
- **Analysis outputs:** bkgreg-out.txt, indep-out.txt, tpfdiff-out.txt, tpfdiff_*.png, occfit-out.txt, physics-out.txt, apogee-out.txt.
- **Archive outputs:** nb60-out.txt, cats-out.txt, gaiatap-out.txt, morecats-out.txt, extracats-out.txt, mastlc-list.txt, mastlc-dl.txt.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet269504659\`.

A176. TYC 4104-606-1 (TIC 55214475)

status
: confirmed (per-object re-analysis 2026-10-21; grazing F/G + 0.44 Msun M dwarf, 25.97 ppt, T14 1.92 h, 17 events (not 12: 2 from flagged cadences); new secondary 1.54 ppt at phase 0.49681(17): e cos omega -0.00502(26), 19 sigma from zero, Gaia's e 0.019 (2.3 sigma) is real, the counter-example to the small-e-is-spurious pattern; VSX ROT (Gaia's 3.97-d spot signal) is a type revision; prior art: TESS-SPOC DV TCEs for both kinds (s060, s073), never a TOI; on target in 2 TESS-SPOC difference images; no neighbour brighter than G 17.2 within 60 arcsec)

P_days
: 4.8136773

eP
: 0.0000007

T0_primary_BTJD
: 2786.95100

depth_primary
: 2.60

depth_secondary
: 0.15

duration_h
: 1.92

source_path
: Fable 5\made\sb1-scale\pervet\vet55214475\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 55214475 (TYC 4104-606-1, Gaia DR3 1008430975045858048): per-object vet

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder; protocol VET-BRIEF + BRIEF-ADAPT (through wave 4). (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, grazing F/G + M-dwarf EB with both eclipses detected. The M dwarf transits the primary at Gaia conjunction A, and the orbit is slightly but measurably eccentric.

Deciding numbers: - **Primary eclipses:** 15 events with clean (Q0) cadences in s019/s059/s060/s073, plus 2 recovered from flagged cadences: 1829.029 (23.1 ± 0.6 ppt) and 3287.573 (26.4 ± 0.5 ppt). That makes **17, not the row's 12**. - Depth 25.97 ± 0.08 ppt, T14 1.92 h, T12/T14 0.40. - **Period:** P = 4.8136773 ± 0.0000007 d against Gaia's 4.813799 ± 0.000164 d (−0.74σ). - **Timing:** O−C rms 0.4 min; χ^2/dof 18/13. - **Attribution:** the primary sits on Gaia conjunction A at −0.94σ (s019, the sector nearest

the Gaia epoch) to −0.87σ (s073). Conjunction B is 25σ to 52σ away. - **NEW secondary eclipse:** 1.54 ± 0.07 ppt, S/N 19.9, at phase **0.49681 ± 0.00017**. - The four sectors agree (χ²/dof 2.4/3). - Look-elsewhere test: 55 random phases give a maximum S/N of 2.8; none reaches the secondary's S/N. - **Eccentricity: e cos ω = −0.00502 ± 0.00026**, against Gaia's −0.0108 ± 0.0075 (0.8σ). Gaia's predicted B − A separation is 0.4931 ± 0.0048 (0.8σ). - The light-travel-time delay is about 30 s, against the observed 22-min offset, so the eccentricity is real (e ≥ 0.005). - **On target:** - TESS-SPOC difference images (s060, s073) put the best fit on the target, with implied own dips of 21.3 and 18.5 ppt. The other star in the fit is worse by Δχ² 4,500–5,750. - No Gaia source within 60″ is brighter than G 17.21, so every neighbour would need a dip larger than 100% of its own flux.

1. Archives

- **cats.py --r 60:**
- VSX: "Gaia DR3 1008430975045858048", type **ROT**, P 3.97154 d, at 0″. This is the Gaia DR3 rotation-modulation classification, a non-eclipsing type, so it is a **type revision, not prior art**.
- ASAS-SN: none.
- SIMBAD: TYC 4104-606-1, type [SB*].
- Prša+2022, IJspeert+2024, Kostov tables 2–4: no hits.
- **Gaia DR3:**
- vari_eclipsing_binary: empty.
- vari_summary: in_vari_rotation_modulation = true; G range 0.029 mag; phot_variable_flag VARIABLE.
- NSS SB1 orbit: P 4.813799, e 0.019 ± 0.008, ω 129 ± 21°, K1 39.99 ± 0.33 km/s, significance 122. The full row with corr_vec is in gaia.json.
- RUWE 1.16; gspphot Teff 5909 K, logg 4.29.
- **SB9, WDS, J/AJ/160/111:** no entries.
- **ExoFOP:** no TOI, CTOI, imaging, spectra or companions.
- **VizieR cones at 30″** (extracats-out.txt): ATLAS, ZTF, WISE variables, GCVS, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST DR7, APOGEE DR17 and Prša all empty, and every control cone returned rows. The only hit is the VSX ROT row.
- **Web search** on the TYC, TIC and Gaia IDs: no relevant hits.
- **Prior art to disclose: SPOC DV from the TESS-SPOC full-frame-image pipeline produced detections (TCEs) in s060 and s073.**
- TCE_1: P 4.81379 d, 26.6 ppt, 2.01 h, MES 147.
- TCE_2: phase ~0.497, 1.75–1.80 ppt, MES 9–10.
- The pipeline saw both eclipses, but it never became a TOI or CTOI (dvcheck-out.txt).
- **Batch receipts:**
- c3-vetsum: dphi −0.010/+0.497, ATTR.
- diffrun s059: peak on target; the deficit is 0.58″ from the light centroid.

2. Measurements (vet-out.txt, sec2-out.txt, sectime-out.txt; QLP SAP_FLUX, QUALITY==0, no trimming)

- **Per-sector primary depth:** 25.4–26.5 ppt. The depth-constancy χ²/dof of 53/14 is mild.
- **Odd vs even:** 25.84 ± 0.11 vs 26.15 ± 0.13 ppt (1.8σ).

- **Background:** the unit-free regression (bkgreg-out.txt) implies at most 0.023 ppt of fake dip in any sector.
- **Centroids:** dX +0.15 ± 0.13 mpix, dY +0.20 ± 0.11 mpix.
- **Quality flags and known systematics:** no event touches a momentum-dump cadence (QUALITY bit 32), and none falls near a listed sector-wide systematic time.
- **Flagged cadences:** two more full-depth events, at 1829.029 (s019) and 3287.573 (s073, an orbit that QLP flagged with bit 30 but which is clean).
- **Blind dip census:**
- 33 dips at S/N ≥ 4: 14 are primaries and 12 are the secondary (phases 0.494–0.498).
- The remaining dips are 0.6–1.2 ppt with S/N ≤ 4.7.
- Nothing off-ephemeris is deeper than 2.1 ppt.
- **Control phases:** |depth| ≤ 0.11 ppt.
- **Eclipse shapes:**
- Primary: near-V graze. A flat-bottomed shape (T12/T14 0.1) is worse by $\Delta\chi^2$ 5,300, and 0.25 by $\Delta\chi^2$ 1,160.
- Secondary: T14 1.80 ± 0.18 h, giving e sin ω ≈ −0.03 ± 0.05. That is uninformative but consistent with Gaia.
- **Out-of-eclipse variability** (rotper-out.txt): spot modulation with a 4.6–5.0 d peak and 1.7–4.9 ppt semi-amplitude, i.e. near-synchronous rotation.
- This is Gaia's "3.97 d ROT" signal.
- The cos φ ≈ −2.6 ppt term at the orbital period, with sector scatter 1.4–1.9 ppt, is spots, not reflection.

3. Ephemeris

- **Primary:** P = 4.8136773 ± 0.0000007 d; T0 = 2786.95100 ± 0.00008 BTJD (E = 201).
- **Secondary − primary, per sector:** s019 0.49628, s059 0.49686, s060 0.49697, s073 0.49717; weighted mean 0.49681 ± 0.00017.
- **Versus Gaia:**
- The eclipse is 1.3 h before Gaia's conjunction A at s019 and 2.2 h before it at s073; this grows slowly because Gaia's P is 0.7σ long.
- e cos ω agrees at 0.8σ.

4. Independence

- **Other pipelines:**
- TARS: 25.5–26.7 ppt in all 4 sectors.
- TESS-SPOC PDCSAP and SAP: 24.5–26.3 ppt in s019, s060 and s073.
- **Difference images** (TESS-SPOC 200-s target pixel files):
- The pixel with the largest flux loss is the target pixel, at S/N 29.8 and 32.4.
- The loss-weighted centroid is on the target.
- s019 has only 30-min target pixel files, too coarse for the estimator.
- **Dilution table** (Gaia DR3 within 60″; needed dip = 26 ppt × 10^(0.4ΔG)): G 17.21 at 45.3″ → 10,250 ppt; G 18.93 at 8.9″ → 49,700 ppt; five stars at G 20.2–21.1 → over 160,000 ppt. None can produce the signal.

5. Physics (physics-out.txt, occfit-out.txt)

- **Companion:** f(M) = 0.0319 M⊙. With M1 = 1.20 ± 0.19 M⊙, m2(90°) = 0.44 M⊙ (range 0.40–0.48), an early-to-mid M dwarf.
- **Orbit scale:** a = 14.1 R⊙; tangential speed 18.7 R⊙/d at conjunction A.
- **Grazing geometry:** a central eclipse would last 3.9 h, so the observed 1.92 h needs i ≈ 84.6° and a mid-eclipse separation of 0.87 × (R1 + R2).
- **Which star is in front** (limb-darkened physical fit, 12 events at 200 s):
- Transit of the primary fits with k 0.285, b 1.02, and gives **R1 = 1.02 R⊙, matching TIC's 1.05 R⊙**.
- The occultation alternative needs the companion to supply 22% of the light, which is impossible here (expected light ratio L2/L1 ≈ 0.03).
- So the M dwarf is in front at conjunction A. Conjunction B is predicted to eclipse (margin 0.16 R⊙), and it is seen.
- **Secondary depth:** the secondary/primary ratio of 0.059 implies T2 ≈ 3200 K (blackbody). That is a little cool for 0.44 M⊙, but the treatment is crude, so there is no material tension.
- **Ellipsoidal and beaming:** predicted 0.19 and 0.53 ppt, buried under 2–5 ppt of spots.
- **Heartbeat:** not relevant here; the eclipses are short and sharp at both conjunctions.
- **Gaia eccentricity: a counter-example to the wave-4 small-e pattern.** Gaia's e of 0.019 is only 2.3σ, yet the eclipses independently require e cos ω = −0.0050, 19σ from zero. This is a real small eccentricity, not a Lucy–Sweeney artefact. Worth a sentence in the paper.

6. Next predicted times (BTJD; 2026-10-21 ≈ BTJD 4335)

- **Primary** (M dwarf transits the primary, conjunction A): **4336.9551, 4341.7688, 4346.5825**, ± 0.4 min formal (allow ~2 min); then every 4.8136773 d.
- **Secondary** (M dwarf occulted, conjunction B): **4339.3466, 4344.1603, 4348.9739**, ± 1.2 min.

What would settle it further

Reality and attribution are settled. What remains open: - Spectroscopic Teff for both stars. - Chromospheric activity, which the near-synchronous spots predict. - A joint RV + light-curve fit for i, R2 and e cos ω.

Row fixes

- Events: 17 (15 clean + 2 from flagged cadences).
- Depth: 25.97 ppt.
- P: 4.8136773(7) d.
- Secondary: 1.54 ppt at phase 0.4968.
- VSX ROT entry is a type revision.
- Disclose the SPOC TCE detection.

Files

- **Scripts:** vet.py, vetlib.py, sec2.py, sectime.py, bkgreg.py, rotper.py, missing.py, flagged.py, indep.py, tpfdiff.py, occfit.py, physics.py, phys_ecc.py, gaiatap.py, morecats.py, extracats.py, mastlc.py, dvcheck.py, simbad_lamost.py, nextecl.py.

- **Outputs:** the matching `*-out.txt` files, fold_oc.png, events.png, sec2.png, the tpfdiff images, gaia.json, nb60.json and vet-results.json.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet55214475\`.

A177. TYC 1429-565-1 (TIC 63858295)

status
: confirmed (per-object re-analysis 2026-10-21; grazing spotted G + 0.34 Msun M dwarf, near-synchronous rotation (spots 3.7 -> 10 ppt), 26.42 ppt, T14 1.86 h, 26 events (not 18: 3 from flagged cadences); new secondary 2.53 ppt at phase 0.50020(28), circular; prior art: SPOC DV TCEs for both kinds (2-min target s045/s046, s042-s046 multi-sector run), never a TOI; LAMOST G1 spectrum; on target in 3 difference images (S/N 60-75); only Gaia source within 60 arcsec is G 19.3; SIMBAD was unreachable on 2026-10-21, re-check)

P_days
: 3.4331770

eP
: 0.0000009

T0_primary_BTJD
: 2599.88038

depth_primary
: 2.64

depth_secondary
: 0.25

duration_h
: 1.86

source_path
: Fable 5\made\sb1-scale\pervet\vet63858295\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 63858295 (TYC 1429-565-1, Gaia DR3 3982492520952574336): per-object vet

Opus 5.5 subagent, 2026-10-21, wave 5. Fresh folder; protocol VET-BRIEF + BRIEF-ADAPT (through wave 4). (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, grazing G + M-dwarf EB on a spotted, near-synchronous primary, with both eclipses detected. The M dwarf transits the primary at Gaia conjunction A, and the orbit is circular.

Deciding numbers: - **Primary eclipses:** 23 events with clean cadences in s022/s045/s046/s072, plus 3 recovered from flagged cadences: 1899.512 (26.4 ± 0.7 ppt), 1916.678 (27.5 ± 1.1) and 3283.083 (27.0 ± 0.4). That makes **26, not the row's 18**. - Depth 26.42 ± 0.10 ppt, T14 1.86 h, T12/T14 0.40. - **Period:** P = 3.4331770 ± 0.0000009 d against Gaia's 3.433232 ± 0.000273 d (−0.20σ). - **Attribution:**

the primary sits on Gaia conjunction A at −0.02σ (s022) to −0.15σ (s072). Conjunction B is 7σ to 14σ away. - **NEW secondary eclipse:** 2.53 ± 0.10 ppt, S/N 17.6, at phase **0.50020 ± 0.00028**. - Look-elsewhere test: 55 random phases give a maximum S/N of 3.5; none reaches the secondary's S/N. - **Eccentricity: e cos ω = +0.0003 ± 0.0004**, against Gaia's +0.030 ± 0.027 (1.1σ). Gaia's predicted B − A separation is 0.519 ± 0.017 (1.1σ). - Gaia's e of 0.052 ± 0.039 is not significant, and the orbit is circular. This is the usual wave-4 small-e pattern and is harmless here. - **On target:** - Difference images from 120-s SPOC (s045, s046) and 200-s TESS-SPOC (s072): the max-loss pixel is the target pixel at S/N 60–75, and the target fit gives implied dips of 20.3–21.6 ppt. - The only Gaia source within 60″ is G 19.31 at 50.2″, which would need a 43,000 ppt dip.

1. Archives

- **cats.py --r 60:**
- VSX: none. ASAS-SN: none.
- Prša+2022, IJspeert+2024, Kostov tables: no hits.
- **SIMBAD was unreachable** (connection timeouts on 2026-10-21, also via SIMBAD TAP on both hosts). Re-check when it is back.
- **Gaia DR3:**
- vari_summary and vari_eclipsing_binary: both empty.
- NSS SB1 orbit: P 3.433232, e 0.052 ± 0.039, ω 322 ± 35°, K1 38.43 ± 0.87 km/s, significance 44.
- RUWE 1.12; gspphot Teff 5715 K, logg 4.32.
- **SB9, WDS, J/AJ/160/111:** no entries.
- **ExoFOP:** no TOI, CTOI, imaging, spectra or companions.
- **VizieR cones at 30″:** ATLAS, ZTF, WISE variables, GCVS, VSX, ROSAT, 2RXS, 4XMM, ASAS-SN, APOGEE and Prša all empty, and the controls returned rows.
- LAMOST DR7 LRS has two spectra. 2015-01-11: class STAR, subclass **G1**, z gives about +20 km/s at one epoch. 2017-02-14: S/N ≤ 2.6, unusable. No stellar parameters are in the catalogue.
- **Web search** on the TYC, TIC and Gaia IDs: no relevant hits.
- **Prior art to disclose:** the star was a 2-min SPOC target in s045/s046, and SPOC DV produced detections (TCEs) in:
- the s042–s046 multi-sector run: TCE_1 P 3.43325 d, 25.9 ppt, MES 158; TCE_2 at phase ~0.50, 2.24 ppt, MES 15;
- single-sector s045 and s046;
- TESS-SPOC s045, s046 and s072. The pipeline saw both eclipses repeatedly, but it never became a TOI or CTOI (dvcheck-out.txt).
- **Batch receipts:**
- c4-vetsum: dphi +0.004/+0.478, ATTR.
- diffrun-c4 s045: peak on target; deficit 0.41″ from the light centroid.

2. Measurements (QLP SAP_FLUX, QUALITY==0, no trimming)

- **Per-sector primary depth:** 26.2–26.6 ppt; depth-constancy χ^2/dof 33.6/22.
- **Odd vs even:** 26.24 ± 0.17 vs 26.52 ± 0.13 ppt (1.3σ).
- **Background:** the unit-free regression implies at most 0.08 ppt of fake dip in any sector.
- s072 has a −226-unit background change in eclipse, but the regression slope turns it into only −0.08 ppt.

- vet.py's dBKG column is ignored, per the wave-3 lesson.
- **Centroids:** dX +0.08 ± 0.10 mpix, dY +0.01 ± 0.10 mpix.
- **Momentum dumps:** 4 events have a bit-32 cadence in their window; with 26 events, none of them matters. None falls near a listed sector-wide systematic time.
- **Blind dip census:**
- 43 dips at S/N ≥ 4: 23 are primaries and 17 are the secondary (phases ±0.498–0.500).
- Two other dips are 1.4–1.9 ppt at S/N ~4.2–4.4.
- One 16.5 ppt "dip" at 3272.913 sits 0.13 d after the primary at 3272.783, inside the matched filter's ±0.15 d window. It is the primary leaking into the filter (S/N 6.3 against the primary's 54), the same artefact as in 423927429, and it does not repeat.
- **Control phases:** |depth| ≤ 0.21 ppt.
- **Eclipse shapes:**
- Primary: near-V graze. A flat-bottomed shape (T12/T14 0.1) is worse by $\Delta\chi^2$ 494.
- Secondary: T14 1.88 ± 0.07 h, giving e sin ω ≈ +0.011 ± 0.019, consistent with Gaia's −0.033 ± 0.036.
- **Timing scatter:** O−C rms 0.8 min, with χ^2/dof 120/21 (errors scaled ×2.4). This fits spot-crossing and spot-baseline distortion on an active star rather than ETVs; the s046 offset is +0.6 min.
- **Out-of-eclipse variability** (rotper-out.txt): strong spot modulation, peak 3.34–3.40 d against P_orb 3.433 (equal within the periodogram resolution, about 0.45 d), semi-amplitude 3.7 → 10.1 ppt from s022 to s072. A near-synchronous, increasingly active primary.

3. Ephemeris

- **Primary:** P = 3.4331770 ± 0.0000009 d; T0 = 2599.88038 ± 0.00012 BTJD (E = 203).
- **Secondary − primary, per sector:** s022 0.49962, s045 0.49980, s046 0.50076, s072 0.50023; weighted mean 0.50020 ± 0.00028 (χ^2/dof 4.9/3).
- **Versus Gaia:**
- The eclipse lies −0.06 h from Gaia's conjunction A at s022 and −0.84 h at s072 (σ about 3–5 h).
- The A–B separation agrees at 1.1σ.

4. Independence

- **Other pipelines:**
- SPOC 2-min PDCSAP / SAP: 25.8–25.9 ppt (s045, s046).
- TESS-SPOC PDCSAP: 24.9–26.1 ppt (4 sectors).
- TARS: 24.7–26.0 ppt (4 sectors).
- **Difference images:**
- SPOC 120-s s045: max-loss pixel S/N 69.5; target fit gives 20.6 ppt.
- SPOC 120-s s046: S/N 75.0; 20.3 ppt.
- TESS-SPOC 200-s s072: S/N 60.6; 21.6 ppt.
- In each case the loss-weighted centroid is on the target.
- **Dilution:** the only Gaia DR3 source within 60″ is G 19.31 at 50.2″ (needed dip 43,000 ppt). The TIC contamination ratio is 0.0003.

5. Physics (physics-out.txt, occfit-out.txt)

- **Companion:** f(M) = 0.0201 M⊙. With M1 = 1.07 ± 0.13 M⊙, m2(90°) = 0.34 M⊙ (range 0.32–0.37), an M dwarf.
- **Orbit scale:** a = 10.75 R⊙; tangential speed 19.1 R⊙/d at conjunction A.
- **Grazing geometry:** a central eclipse would last 3.7–3.8 h, so the observed 1.86 h needs i ≈ 83.1° and a mid-eclipse separation of 0.87 × (R1 + R2).
- **Which star is in front** (limb-darkened physical fit, 18 events):
- Transit of the primary fits with k 0.285, b 1.02, and gives **R1 = 1.00 R⊙** (TIC 1.13, LAMOST G1, consistent).
- The occultation alternative needs the companion to supply 9.4% of the light, against an expected light ratio of about 0.008. Impossible.
- So the M dwarf is in front at conjunction A. Conjunction B is predicted to eclipse (margin 0.27 R⊙), and it is seen.
- **Secondary depth:** the secondary/primary ratio of 0.095 implies T2 ≈ 3350 K for T1 ≈ 5900 K (blackbody), consistent with a 0.34 M⊙ dwarf.
- **Ellipsoidal and beaming:** predicted 0.47 / 0.51 ppt. The measured cos 2φ (+0.8 ppt) and sin φ terms are dominated by the spot modulation (sector scatter 1.5–4.8 ppt) and are unusable.
- **Heartbeat:** not relevant; the orbit is circular with sharp eclipses at both conjunctions.

6. Next predicted times (BTJD; 2026-10-21 ≈ BTJD 4335)

- **Primary** (M dwarf transits the primary, conjunction A): **4337.0679, 4340.5011, 4343.9343**, ± 0.7 min formal (allow ~2–3 min given the spot-driven timing excess); then every 3.4331770 d.
- **Secondary** (M dwarf occulted, conjunction B): **4335.3520, 4338.7852, 4342.2184**, ± 1.5 min.

What would settle it further

- A SIMBAD re-check once the service is back.
- A higher-S/N spectrum (the LAMOST one is low-resolution) for Teff and an RV point.
- A spot-corrected eclipse-timing analysis to confirm the 0.8-min O−C excess is activity, not a third body.

Row fixes

- Events: 26 (23 clean + 3 from flagged cadences).
- Depth: 26.42 ppt.
- P: 3.4331770(9) d.
- Secondary: 2.53 ppt at phase 0.5002.
- Disclose the SPOC TCE detection (2-min target, s042–s046 multi-sector run, both eclipses).

Files

- **Scripts:** the same set as 55214475.
- **Outputs:** the matching `*-out.txt` files, fold_oc.png, events.png, sec2.png, the tpfdiff images, gaia.json, nb60.json and vet-results.json.

- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet63858295\.`

A178. TYC 9406-845-1 (TIC 323007878)

status
: confirmed (per-object re-analysis 2026-10-21; circular grazing F + K dwarf (T2 ~4800 K = the f(M) mass 0.70-0.78 Msun), 25.5 ppt, T14 3.36 h, 52 primaries + 50 secondaries in 11 sectors, conjunction A at +0.10 sigma throughout; secondary 9.32 ppt at phase 0.50019(5), e cos omega +0.0002; QLP s094 dilution step scales both kinds x0.87 together; isolated field (nearest G < 17 at 59.6 arcsec); on target in 3 SPOC TPF difference images (S/N 35-116); no prior art of any kind; R1 1.8-1.95 vs TIC 2.08 Rsun, mild)

P_days
: 4.7034219

eP
: 0.0000005

T0_primary_BTJD
: 2822.87791

depth_primary
: 2.55

depth_secondary
: 0.93

duration_h
: 3.36

source_path
: Fable 5\made\sb1-scale\pervet\vet323007878\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 323007878 (TYC 9406-845-1, Gaia DR3 5196347929180662272): per-object vet

Opus 5.5 subagent, wave 5, 2026-10-21. Fresh folder; every number below comes from a file in it: - Outputs: `vet-out.txt`, `ecl2-out.txt`, `sec2-out.txt`, `bkgreg-out.txt`, `indep-out.txt`, `tpfdiff-out.txt`, `geom-out.txt`, `dips-out.txt`. - Archive and query outputs: `gaiatap-out.txt`, `cats-out*.txt`, `morecats-out.txt`, `extracats-out.txt`, `mastlc-*.txt`. - Downloads: `C:\Home\sky\data\qlp-sb-p20-60\pervet323007878\`. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A detached, circular, grazing eclipsing SB1 (F primary with a K-dwarf companion); both eclipse kinds detected.

Deciding numbers: - **Primary eclipse:** 52 events in 11 QLP sectors (s011–s094). - Each is S/N 13–87, apart from one event lost in the noisy QLP s027. - Depth 25.5 ± 0.07 ppt, T14 3.36 h, T12/T14 0.40. The shape is V-like, as a graze should be. - **Secondary eclipse:** 50 events, 9.32 ± 0.06 ppt, T14 3.40 h. It is seen individually at S/N 20–33 in every sector. - **Period:** P = 4.7034219 ± 0.0000005 d against Gaia's 4.703408 ± 0.000134 d, a difference of **+0.11σ**. - **Timing:** the primary falls **+0.10σ** from Gaia conjunction A in every sector. The offset runs from +0.09 h at s011 to +0.24 h at s094 (Monte Carlo with corr_vec). Conjunction B is 24–65σ away. - **Circular orbit:** the secondary falls at phase **0.50019 ± 0.00005**. - That gives e cos ω = +0.0003 ± 0.0001, of which about +0.00008 is the ~21 s light-travel delay. - Gaia gives e cos ω +0.000 ± 0.009 and e sin ω +0.001 ± 0.009. - The duration ratio gives e sin ω +0.006 ± 0.017. - **On target:** SPOC TPF difference images (s038, s065, s067) put the flux loss on the target pixel at S/N 35–116. - The target fits better than any neighbour by $\Delta\chi^2$ 4,600–24,000. - The field is isolated: only 3 Gaia sources with G < 17 lie within 90″, and the nearest is at 59.6″.

1. Archives

- **cats.py --r 60** (APPOINTMENTS_DATA set): no hits in VSX, ASAS-SN, Prša+2022, IJspeert+2024 or Kostov+2025 (any table). SIMBAD has only TYC 9406-845-1, typed [SB*] (first run timed out; re-run is `cats-out2.txt`).
- **Gaia DR3 TAP:** vari_summary and vari_eclipsing_binary empty; phot_variable_flag NOT_AVAILABLE. RUWE 0.84, ipd_frac_multi_peak 0. gspphot Teff 6347 K, logg 3.80.
- **SB9, WDS and J/AJ/160/111:** no entry.
- **ExoFOP:** no TOI, CTOI, imaging, spectroscopy or stellar companions.
- **Extra VizieR cones at 30″** (ATLAS, ZTF, WISE, GCVS, VSX, ROSAT, 2RXS, 4XMM, ASAS-SN, LAMOST, APOGEE, Prša): no rows. The ATLAS, ZTF, GCVS, ROSAT, LAMOST, APOGEE and Prša control cones were also empty — those catalogues simply don't cover dec −81° well.
- **Web search** on "TYC 9406-845-1" and "TIC 323007878": nothing relevant.
- **SPOC:** no 2-min target and no CAOM TESS-mission product, so no SPOC DV/TCE. TESS-SPOC FFI products don't run DV.
- **Gaia orbit:** SB1, P 4.703408(134) d, e 0.006(10), ω 84(107)°, K1 57.47(56) km/s, significance 102.
- **TIC:** Teff 6499 K, R 2.08 R⊙, M 1.33 M⊙, d 497 pc, contratio 0.012.
- **Batch receipts:** c4-vetsum dphi +0.000/+0.500 ATTR; diffrun-c4 s064 peak on target, 0.71″; c4-sapdepth 24–26.5 ppt per event.
- **Prior art:** none found.

2. Eclipse measurements (QLP SAP, QUALITY==0, untrimmed)

- **Global trapezoid** (`vet-out.txt`): 25.2 ppt, T14 3.36 h, T12 1.34 h, flat bottom 0.67 h. A box shape (T12/T14 0.1) is worse by $\Delta\chi^2$ +1899.
- **Individual primaries** (`ecl2-out.txt`, local quadratic baseline): weighted mean **25.51 ± 0.07 ppt**.
- Odd 25.94 ± 0.08 and even 25.70 ± 0.08 ppt.
- Per sector the depth is 24.8–27.4 ppt, except s094 at 23.1 ppt. In s094 the four primaries (22.8–23.6 ppt) and the secondaries (7.8–8.1 ppt) are all scaled by the same ×0.87. That is a QLP s094 dilution step, not the star.
- The depth-constancy χ^2/dof of 434/51 reflects this sector-level QLP scaling of about ±1 ppt.

- **Individual secondaries:** 9.32 ± 0.06 ppt, χ^2/dof 137/49.
- **Background** (unit-free regression): $|\Delta F| \leq 0.22$ ppt in 10 sectors and 0.91 ppt in s066, against 13–17 ppt dips in the same windows. The dBKG columns have the known broken units and are ignored.
- **Centroids:** dX −2.0 ± 0.4 mpix, dY −1.4 ± 0.5 mpix — tiny, consistent with an on-target dip in a nearly empty aperture.
- **Quality flags and systematics:** 2 primaries touch a bit-32 cadence, depths normal. No event within 0.5 d of a listed sector-wide systematic.
- **Control phases:** |depth| ≤ 0.13 ppt. Phase 0.5 holds the real secondary.
- **Blind dip census:** 132 peaks at S/N ≥ 4. 56 are primaries and ~46 are secondaries at phase ±0.500. Five off-ephemeris dips are ≥ 12.9 ppt, and none is real:
- Four are in QLP s027 (15.7 ppt rms, scattered-light dominated; e.g. 2047.98 is a −108 ppt SAP step with a 0.05 px centroid jump; SPOC's s027 is clean).
- The fifth, 3138.21, is a primary egress against a QLP bit-30-flagged gap; SAP is actually higher inside that window.
- None repeats at the same phase in neighbouring cycles (`dips-out.txt`).
- **Fold with harmonics** (`sec2-out.txt`): primary 24.9 ppt / 3.36 h; secondary 9.34 ± 0.17 ppt / 3.40 ± 0.11 h at phase 0.5000.

3. Ephemerides (`ec12-out.txt`; events with S/N > 3; the 5σ clip removed nothing)

kind	P (d)	T0 (BTJD)	n	χ^2/dof	O−C rms
primary (conj. A)	4.7034219 ± 0.0000005	2822.87791 ± 0.00009	52	76/50	2.1 min (median error 0.7 min)
secondary (conj. B)	4.7034239 ± 0.0000015	2834.63735 ± 0.00023	50	67/48	3.1 min (median error 1.9 min)

- **Period difference:** P(sec) − P(pri) = +0.17 s (+1.25σ). No apsidal motion.
- **Phase separation:** 0.50019 ± 0.00005. Gaia predicts B − A = 0.5000 ± 0.0058. (sec2's fold gives 0.4998 ± 0.0002, but its phase grid is 0.0005.)
- **e cos ω:** removing the light-travel delay (≈21 s) leaves about +0.0002. The orbit is circular.
- **Comparison with Gaia:** conjunction A matched at +0.10σ throughout, P at +0.11σ. Even with σω = 107°, using corr_vec gives a conjunction-time error of only 0.04–0.10 d — small enough to pin the attribution.

4. Independence

- **Other pipelines** (`indep-out.txt`): TESS-SPOC PDCSAP in s011, s013, s027, s038, s039, s065, s067 gives **25.3–26.1 ppt**; SPOC-SAP 25.1–25.7 ppt; GSFC-ELEANOR-LITE CORR 25.8 (s011) and 25.4 ppt (s013). SPOC's ellipsoidal amplitude A2 is 1.1–1.5 ppt in every sector.
- **Eleanor s012** (a sector QLP lacks) shows 8.2 ppt primaries and ~2 ppt secondaries at the right times. Its raw flux is one third of s011/s013 — a 4-pixel square aperture on another CCD with twice the background. A product problem; the timing still matches.
- **Difference images** (`tpfdiff-out.txt`):

sector	max-loss pixel	S/N	χ^2 target	χ^2 best neighbour
s038	target (5,5)	77	1007	18,055
s065	target (5,5)	116	1848	25,831
s067	adjacent (5,6)	35	651	5,230

The target's implied own dip is 21.6–21.9 ppt, and the loss-weighted centroid is within 0.1–0.3 px of the target. - **Neighbour dilution table** (needed dip = 25.5 ppt × 10^(0.4ΔG)): G 16.58 at 59.6″ → 2.85 (impossible); G 12.96 at 64.5″ → 101 ppt (excluded by the difference images); G 15.99 at 74.5″ → 1.64 (impossible).

5. Physics (`geom-out.txt`)

- **Mass function:** f(M) = 0.0925 M⊙. With M1 = 1.2–1.45 M⊙ and i ≈ 81–82°, m2 = 0.70–0.78 M⊙, a K dwarf. a = 14.6–15.4 R⊙.
- **Which star is in front:** at conjunction A the RV primary is behind, so the K dwarf transits the F star — the deep eclipse, as observed. At B the F star occults the K dwarf.
- **One geometry fits both eclipses:** two limb-darkened spheres on a circular orbit, m2 from f(M) at each i, reproduce both (primary 25.5 ppt / 3.36 h; secondary 9.2–9.5 ppt / 3.31–3.41 h; $\chi^2 < 3$ for 10 combinations). Fitted: R1 1.8–1.95 R⊙, R2 0.65–0.90 R⊙ (poorly constrained in a graze), **T2 ≈ 4800 K** (4600 and 5000 K both fail), L2/L1 4–8% in TESS. Geometry: i 81.2–82.0°, b 1.10–1.29 R1 — a graze; a central chord would last ~6.6 h.
- **Depth ratio:** 9.32/25.5 = 0.365 is the TESS surface-brightness ratio of a ~4820 K star to a 6500 K star — right for 0.72–0.78 M⊙.
- **Ellipsoidal:** ~1.3 ppt predicted, 1.1–1.5 ppt observed in SPOC.
- **Mild tension:** fitted R1 ≈ 1.8–1.95 R⊙ is 1.5–3σ below TIC's 2.08 ± 0.09 — same direction as 423927429; doesn't affect the verdict.
- **Not a heartbeat:** e ≈ 0, symmetric eclipses at both conjunctions.

Row fixes for the paper table

field	batch	fix
P (d)	4.70343	**4.7034219 ± 0.0000005**
T0 (BTJD)	1599.9843	**2822.87791 ± 0.00009**
depth (ppt)	24.6	**25.5 ± 0.1** (QLP SAP per event; SPOC PDCSAP 25.3–26.1)
T14 (h)	1.9 (search box)	**3.36** (grazing, T12/T14 0.40)
n events	43	52 Q0 primaries + 50 secondaries
dphi	+0.000	+0.0008 to +0.0021 across sectors; **+0.10σ** from conjunction A throughout
secondary	—	**9.32 ± 0.06 ppt at phase 0.50019 ± 0.00005** (e cos ω ≈ 0)
prior art	—	none (no EB catalogue, TOI, CTOI or SPOC TCE)

Next predicted eclipses (BTJD)

- **Primary** (K dwarf transits, conjunction A): **4403.2277, 4407.9311, 4412.6345**, then every 4.7034219 d. Formal ±0.5 min; allow ±2 min.
- **Secondary** (K dwarf occulted, conjunction B): **4400.8775, 4405.5809, 4410.2843**. Formal ±1 min; allow ±3 min.

What would settle it further

Already settled as an eclipsing SB1. For better parameters: - A spectrum (Teff, logg) to fix R1 and test the R1 tension. - An SB2 detection of the K dwarf (expected K2 ≈ 103 km/s, L2/L1 ≈ 6% in TESS and more in the red) would give model-independent masses.

A179. TYC 3616-2395-1 (TIC 384284045)

status
: confirmed (per-object re-analysis 2026-10-21; grazing near-twin G+G SB2 (M 1.07 + 1.15 Msun), both eclipses 28.89/25.58 ppt, T14 2.20/2.24 h, 34 + 37 events; the deeper eclipse is at conjunction B: Gaia's SB2 'primary' (K1 > K2) is the cooler, less massive star, a labelling note; phase separation 0.50002(2), circular; T0 is the deeper eclipse (conjunction B); spotted (RS CVn/BY Dra-like), 2RXS X-ray counterpart; prior art: SPOC 2-min s057 TCE at P/2 (odd/even statistic 246), never a TOI; on target in 3 difference images by dchi2 > 21,000; 3-sigma period split between the kinds = spot timing noise, flagged)

P_days
: 3.787011

eP
: 0.000002

T0_primary_BTJD
: 3277.42043

depth_primary
: 2.89

depth_secondary
: 2.56

duration_h
: 2.20

source_path
: Fable 5\made\sb1-scale\pervet\vet384284045\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet waves 4-5 (2026-10-21)

Archived source: sanitized report text.

TIC 384284045 (TYC 3616-2395-1, Gaia DR3 1980920834829359488): per-object vet

Opus 5.5 subagent, wave 5, 2026-10-21. Fresh folder; every number below comes from a file in it: - Outputs: `vet-out.txt`, `ecl2-out.txt`, `sec2-out.txt`, `bkgreg-out.txt`, `indep-out.txt`, `indep2-out.txt`, `tpfdiff-out.txt`, `geom-out.txt`, `dips-out.txt`. - Archive and query outputs: `gaiatap-out.txt`, `cats-out.txt`, `morecats-out.txt`, `extracats-out.txt`, `mastlc-*.txt`. - Downloads, including the SPOC DV xml: `C:\Home\sky\data\qlp-sb-p20-60\pervet384284045\`. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A grazing, circular, near-twin G-type eclipsing SB2; both eclipses detected and both match Gaia's conjunctions.

Deciding numbers: - **Two eclipses of similar depth, each fixed to one Gaia conjunction:** - The deeper one is **28.89 ± 0.06 ppt** (34 events) and falls at Gaia conjunction **B**, at −1.6σ in every sector. - The shallower one is **25.58 ± 0.05 ppt** (37 events) and falls at conjunction A. - Their phase separation is **0.50002 ± 0.00002**, against Gaia's predicted 0.5000 ± 0.0041. - The search's "P2" half-period alias (1.8935 d) came from these two near-equal eclipses. The batch period of 3.787 d is the right one. - **Period:** 3.7870110 ± 0.0000007 d (mean of the two kinds), against Gaia's 3.787087 ± 0.000049 d. That is −1.5σ, which is also why the conjunction offset is −1.6σ. - **Which star is which:** the RVs and the photometry agree independently. - Gaia's K1 (91.47) is larger than K2 (85.31), so Gaia's "primary" is the *less* massive star (q = 0.933). - The eclipse at A, where Gaia's primary is eclipsed, is the *shallower* one. Its depth ratio of 0.886 means that star has the lower surface brightness, about 200–250 K cooler. - So the more massive star is the hotter one, as it should be on the main sequence. Gaia's SB2 "primary" label went to the cooler star of a near-equal pair. That is a labelling note, not a conflict. - **On target:** difference images (2-min SPOC s57, TESS-SPOC s76 and s83) put the flux loss on or adjacent to the target pixel at S/N 87–112. - The target beats every neighbour by $\Delta\chi^2 > 21{,}000$. - The best neighbour, G 13.72, would need a 58% dip. - **Prior detection (not a catalogue entry):** SPOC 2-min **s57 TCE** (DV `s0057-s0057`). - Found at the half period: P 1.893481 d, epoch 2855.1687, depth 26,927 ± 141 ppm, 2.31 h, b 0.95, "Rp" 36 R⊕. - Its odd/even depth statistic is **246**: SPOC saw the alternating depths. - It is not a TOI or CTOI.

1. Archives

- **cats.py --r 60** (APPOINTMENTS_DATA set):
- VSX has only Gaia DR3 1980920078927243648 at 49.8″ (G 16.54, type S, 16.30–16.89 mag) — a neighbour, not the target; see §4.
- ASAS-SN: none. Prša+2022, IJspeert+2024 and Kostov+2025: none.
- SIMBAD has TYC 3616-2395-1, typed [SB*].
- **Gaia DR3 TAP:** vari_summary and vari_eclipsing_binary empty. RUWE 0.79. gspphot Teff 5744 K, logg 4.08. NSS flags 8388608.
- **SB9, WDS and J/AJ/160/111:** no entry.
- **ExoFOP:** no TOI, CTOI, imaging, spectroscopy or companions.
- **Extra VizieR cones at 30″:**
- **2RXS J215010.4+514807** lies 8.6″ away with a positional error of 9.9″, so it is consistent with the target. That fits an active, tidally synchronized binary (§5).
- The ATLAS row at 27″ is a G ≈ 17.6 "dubious" star, not the target.
- ZTF, WISE, GCVS, VSX, ROSAT bright, 4XMM, ASAS-SN, LAMOST, APOGEE and Prša return nothing.
- J/A+A/674/A16 is a Gaia source table, not an EB classification.
- **Web search** on "TYC 3616-2395-1" and "TIC 384284045": nothing relevant.
- **Gaia orbit:** SB2, P 3.787087(49) d, e 0.0025(81), ω 91(166)°, K1 91.47(91) km/s, K2 85.31 km/s, significance 100.
- **TIC** (treats the pair as one star): Teff 5776 K, R 1.47 R⊙, M 1.04 M⊙, L 2.18 L⊙, d 207 pc, contratio 0.092.
- **Batch receipts:** c3-vetsum P2-tagged, dphi +0.488/−0.012 ATTR; diffrun s083 peak on target, deficit–target 4.3″, deficit–light 1.57″; c3-sapdepth alternating ~24/~28 ppt.

2. Eclipse measurements (QLP SAP, QUALITY==0, untrimmed; 6 sectors s056–s084, 200 s cadence)

- **Global trapezoid** (deeper eclipse): 28.6 ppt, T14 2.22 h, T12 0.89 h, flat bottom 0.44 h. A box shape is worse by $\Delta\chi^2$ +1729 — V-like, grazing.
- **Deeper eclipse, conjunction B** (`ecl2-out.txt`): 28.89 ± 0.06 ppt, 34 events, χ^2/dof 181/33. Per-sector means 27.8–29.3 ppt (s056 lowest). Odd 28.93 ± 0.07 and even 28.74 ± 0.07 ppt.
- **Shallower eclipse, conjunction A:** 25.58 ± 0.05 ppt, 37 events, χ^2/dof 169/36.
- **Fold with harmonics** (`sec2-out.txt`): 28.9 ppt / 2.20 h and 25.5 ppt / 2.24 h. The equal durations again point to e sin ω ≈ 0.
- **Background:** |ΔF| ≤ 0.06 ppt in every sector, against a 17 ppt dip in the same windows.
- **Centroids:** dX +0.27 ± 0.08 mpix, dY −0.69 ± 0.05 mpix — negligible.
- **Quality flags and systematics:** one event has a bit-32 cadence, depth normal. No event near a listed sector-wide systematic; s083 does not touch 3564.5.
- **Control phases:** |depth| ≤ 0.17 ppt.
- **Blind dip census:** 87 dips at S/N ≥ 4; 36 on the deeper ephemeris and 36 are the other eclipse at phase ±0.500. The rest are not real:
- 2845.86 (15.9 ppt, phase 0.041) is the baseline just after an eclipse, next to a bit-32 cadence. SAP inside the window is −2 ppt, and the same phase in other cycles is *brighter*.
- The remaining dips are 1.5–2 ppt features in s084, the noisiest sector at 8.9 ppt rms.
- **Starspots:** the orbital-period cos φ and sin φ terms vary by ±6–15 ppt from sector to sector in both QLP and SPOC. This is spot modulation on synchronized G stars, consistent with the 2RXS detection. Spots also explain the depth-constancy χ^2 and s056's lower depth and A/B ratio.

3. Ephemerides (`ecl2-out.txt`; events with S/N > 3; the 5σ clip removed nothing)

kind	P (d)	T0 (BTJD)	n	χ^2/dof (scale)	O−C rms
deeper (conj. B)	3.7870109 ± 0.0000007	3277.42043 ± 0.00006	34	227/32 (×2.67)	0.59 min (median error 0.29 min)
shallower (conj. A)	3.7870141 ± 0.0000007	3290.67506 ± 0.00006	37	180/35 (×2.27)	0.46 min (median error 0.29 min)

- **Period difference:** P(A) − P(B) = +0.27 s per cycle, +3.0σ. With e ≈ 0 this is not apsidal motion; the most likely cause is spot-induced timing noise, anti-correlated between the two kinds, going with the χ^2 excess. Worth a look if anyone revisits.
- **Phase separation:** 0.50002 ± 0.00002, so e cos ω = +0.00004 ± 0.00004. For twins the light-travel delay is about 0.
- **Comparison with Gaia** (Monte Carlo with corr_vec): the deeper eclipse is −1.23 h (s056) to −1.63 h (s084) from conjunction B, all −1.58σ to −1.62σ. Conjunction A is 43–56σ away from it. The drift between sectors is Gaia's P being 1.5σ long.

4. Independence

- **Other pipelines:** TESS-SPOC PDCSAP at the deeper eclipse gives 27.8–29.6 ppt in all six sectors; SPOC-SAP 25.2–27.6 ppt. The **SPOC 2-min s57** light curve gives 29.03 / 25.14 ppt. With both kinds in PDCSAP (`indep2-out.txt`), the A/B depth ratio is **0.87–0.90** in s057, s076, s077, s083 and s084, and 0.96 in the spotty s056.
- **SPOC DV** (s57): difference-image centroid offset from the TIC position 1.72 ± 2.50″.

- **Difference images** (`tpfdiff-out.txt`):

sector	max-loss pixel	S/N	χ^2 target	χ^2 best neighbour
s057 (2-min)	(4,5), next to target	112	3975	31,584
s076	target (5,5)	87	3695	25,039
s083	(5,6), adjacent	105	8182	42,694

The target's implied own dip is 21.5–23.1 ppt. The s083 fit overrides the batch's 4.3″ deficit–target figure: the target beats every neighbour by $\Delta\chi^2$ 34,500 there. - **Neighbour dilution table** (needed dip = 28.9 ppt × 10^(0.4ΔG); 24 Gaia sources with G < 17 within 90″): G 13.72 at 45.3″ → 577 ppt; G 13.83 at 57.7″ → 640 ppt; G 14.31 at 43.4″ → 0.99 (impossible); G 15.84 at 33.1″ → >1 (impossible); the VSX "S" star G 16.54 at 49.8″ → 7.8 (impossible). The difference images exclude every one.

5. Physics (`geom-out.txt`)

- **Masses from the SB2 solution:** $M_A \sin^3 i = 1.046\ M_\odot$ (Gaia "primary") and $M_B \sin^3 i = 1.122\ M_\odot$. $a \sin i = 13.23\ R_\odot$. At the fitted $i \approx 82.6$–$83.0°$, **$M_A \approx 1.07$ and $M_B \approx 1.15\ M_\odot$** — two early-G or late-F dwarfs. Their summed luminosity of ~2 $L_\odot$ matches TIC's single-star 2.18 $L_\odot$, and the single-star-equivalent R of $1.47 \approx \sqrt{2} \times 1.04\ R_\odot$.
- **One geometry fits both eclipses:** two limb-darkened spheres, circular orbit, masses from the SB2 solution at each i → 28.9 ppt at B and 25.3 ppt at A, both ~2.26 h ($\chi^2 \approx 1.3$). Fitted: $R_A \approx R_B \approx 0.95$–$1.05\ R_\odot$, **$T_B - T_A \approx 200$–$250$ K**, i 82.6°, $b \approx 1.7\ R_\odot$ — a graze (a central chord would last ~4.8 h). SPOC DV independently gives b 0.95, radius ratio 0.23 in its planet model.
- **Which star is in front:** at conjunction A, star B (more massive, hotter) passes in front of star A, giving the shallower eclipse; at B the reverse. RVs and photometry agree.
- **Activity:** a 3.79-d synchronized pair of G dwarfs is expected to be spotted and X-ray bright. 2RXS and the ±10 ppt sector-to-sector rotational wave both bear this out (RS CVn / BY Dra-like).
- **Not a heartbeat:** $e \approx 0$, short sharp eclipses at both conjunctions.

Row fixes for the paper table

field	batch	fix
P (d)	3.787032 (P2-resolved)	**3.787011 ± 0.000002** (the two kinds give 3.7870109 and 3.7870141)
T0 (BTJD)	2826.7648	**3277.42043 ± 0.00006** (deeper eclipse, conjunction B)
depth (ppt)	25.2	**28.89 ± 0.06** (deeper) / **25.58 ± 0.05** (shallower); SPOC PDCSAP 27.8–29.6 / 25.1–26.5
T14 (h)	1.2 (search box)	**2.20 / 2.24** (grazing, T12/T14 0.40)
n events	71	34 deeper + 37 shallower (Q0, all fitted)
dphi	+0.488 / −0.012	the deeper eclipse is on **conjunction B** at −1.6σ; Gaia's "primary" is the cooler, less massive star
secondary	—	25.58 ppt at phase 0.50002 ± 0.00002 ($e \cos\omega \approx 0$)
prior art	—	**SPOC 2-min s57 TCE** at P/2 (26.9 ppt, odd/even statistic 246); not a TOI; 2RXS X-ray counterpart; not in any EB catalogue
note	—	spotted (RS CVn/BY Dra-like); spot timing noise gives an O−C χ^2 excess of about ×2.5

Next predicted eclipses (BTJD)

- **Deeper eclipse** (conjunction B, hotter and more massive star eclipsed): **4402.1627, 4405.9497, 4409.7367**, then every 3.787011 d.

- **Shallower eclipse** (conjunction A): **4400.2702, 4404.0572, 4407.8442**.
- Formal error ±0.3 min. Allow ±3 min, because of the +0.27 s per cycle difference between the two kinds and spot timing noise.

What would settle it further

Already settled as an eclipsing SB2. Two follow-ups would add value: - A ground RV and light-curve joint fit would give full model-independent masses and radii to ~1–2%. It is a bright (V 10.5) double-lined G+G graze. - A spot-modelled eclipse-timing analysis would test whether the 3σ period difference between the two kinds is spot noise.

A180. TYC 2609-823-1 (TIC 39307147)

status
: confirmed (adversarial re-analysis 2026-10-21; eccentric (e 0.36) G star 1.39 Rsun + 0.69 Msun dwarf, i 89.1 deg; A 24-26% full transit x6 + B 9.8-10.2% total occultation x5 (2020-2024), QLP and TESS-SPOC; Gaia P -1.5 sigma explains the 0.6-0.9 d early conjunctions; e, omega within 0.9 sigma of Gaia; difference image on target to 0.7 arcsec; Kostov t2 only)

P_days
: 23.421703

eP
: 0.000013

T0_primary_BTJD
: 1989.52348

depth_primary
: 24-26

depth_secondary
: 9.8-10.2

duration_h
: 6.9

source_path
: Fable 5.1\made\appointments\vet39307147\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 39307147 (TYC 2609-823-1, Gaia DR3 4600354696406809984): adversarial check

Checker: Opus 5.5, 2026-10-21. This resumes checker #1 (killed by a usage limit after steps 1-4). I reran or spot-checked steps 1-4 with new code (`verify.py`) and wrote step 5 (`physics2.py`).

Verdict: CONFIRMED

This is a detached, eccentric, double-eclipsing binary at P = 23.421703 d. Both eclipse kinds repeat across 2020, 2022 and 2024, and two pipelines recover them. The eclipses sit on the target pixel. One geometry fits both eclipses, and that geometry agrees with Gaia's SB1 orbit (e, omega, K1, P) and with the TIC primary. Prior art: Kostov+2025 table2 only (an unvetted ML flag list), so the star is uncatalogued as an EB.

Deciding numbers

- **Eclipses.** There are 6 A eclipses (s25, s26, s52, s53, s79, s80). Depth is 24.4-25.5% in QLP and 23.4-23.9% in TESS-SPOC; trapezoid T14 is 6.8-7.0 h. There are 5 usable B eclipses (s25, s26, s52, and

two in s80) at 9.8-10.2% and T14 7.8-8.0 h. A sixth B in s79 is recovered in all-flagged data at 3458.562 +- 9.5 min and is consistent with the ephemeris, but it was not used.

- **My own refit** (`verify.txt`, all cadences, raw SAP) matches checker #1 to within 0.6 min in time and 0.3% in depth.
- **Ephemeris A** (deep): T0 = BTJD 1989.52348 +- 0.00059, P = 23.421703 +- 0.000013 d, chi2 2.0/4, rms O-C 0.76 min.
- **Ephemeris B**: T0 = BTJD 2006.41760 +- 0.00116, P = 23.421703 +- 0.000025 d, chi2 12/3, rms 1.8 min. The s52 B eclipse is 2.9 min early, in a half-flagged segment. B-A phase is 0.72130.
- **Against Gaia.** Gaia gives P = 23.41591 +- 0.00396, which differs from TESS by -0.0058 d (-1.5 sigma). That difference is the whole reason Gaia's conjunctions run 0.56 to 0.93 d early, growing with epoch. Every measured eclipse lies within +1.5 sigma of its Gaia-predicted conjunction.
- Gaia's e and omega predict a B-A phase of 0.7274 +- 0.0073; I measure 0.72130 (-0.8 sigma).
- **Eclipse-model fit** (two LD spheres, Keplerian; chi2/N 1.00, rms 1.16 ppt): e = 0.3595, omega = 9.45 deg, i = 89.06 deg, R1/a = 0.0333, R2/a = 0.0168, J_TESS = 0.429.
- Compared with Gaia: e 0.368 +- 0.012 (-0.7 sigma) and omega 6.16 +- 3.63 (+0.9 sigma).
- Checker #1's ebfit.py prints "primary omega 189.45". That is a label slip: the model's Z>0 axis points toward the observer, which already absorbs the 180 deg. The value is 9.45 deg in Gaia's convention. e cos(omega) +0.3546 and the phase both have the same sign as Gaia's.
- **Pixels and neighbours.**
- TESScut s80 difference images: the deficit centroid is 0.3" (A) and 0.7" (B) from the light centroid, the peak deficit is on the target pixel, and the depth is the same in 1x1, 3x3 and 5x5 apertures.
- No Gaia source within 60" is brighter than G 17.4 (Delta G >= 6.2). Every neighbour would need a dip above 2900%. The nearest bright star (G 15.1, 65.7") shows only spill-over.
- **Census.** Nothing above 0.3% outside the eclipses in s25, s53, s79 or s80. The only features are short 0.3-0.7% wiggles in the noisy, partly flagged s26 and s52 segments.

Physics (step 5)

- **Masses.** f(M) = 0.107 [0.102, 0.112] Msun (K1 37.87 km/s). With the TIC M1 = 1.07 Msun and i = 89.06 deg, m2 = 0.69 Msun and a = 41.6 Rsun.
- **Primary.** R1 = 0.0333a = 1.39 Rsun. That gives logg1 = 4.18; Gaia GSP-Phot says 4.14, and TIC R is 1.50.
- **Companion.** R2 = 0.70 Rsun, as expected for a 0.69 Msun dwarf (about 0.66-0.68). The surface-brightness ratio gives T2 of about 4690 K (blackbody, T1 = 5888 K), slightly warmer than a 0.69 Msun dwarf (about 4400-4500 K). That is within the crudeness of the blackbody approach.
- **Light ratio.** L2/L1 = 0.11 in the TESS band. Gaia's SB1 classification is still plausible at that ratio, but light from the companion may slightly dilute K1.
- **A eclipse** (nu = 90 deg - omega): the companion passes in front of the primary. Separation 0.822a, b = 0.41, a full transit, model T14 7.61 h.
- **B eclipse:** the primary passes in front, and the companion is totally occulted (b = 0.46 < 1-k = 0.495). Separation 0.925a, model T14 8.48 h, flat 1.1 h. Depth is L2/(L1+L2) = 9.87%.
- **Consistency.** The B/A duration ratio follows from e sin(omega) > 0. Depth A is consistent with k^2 = 0.255 diluted by the companion's light, after limb darkening. Gaia's rv_amplitude_robust of 78.4 km/s is close to 2K1 = 75.7 km/s.
- **Trapezoid versus model.** The trapezoid T14 values are shorter than the model contact-to-contact values (A 6.9 vs 7.6 h, B 7.85 vs 8.5 h). This is the usual trapezoid bias on limb-darkened or

occultation ingress, not a tension in the data.

Next predicted eclipses (BTJD; linear ephemeris, propagated 1-sigma)

- A (deep, ~25%, about 7.6 h contact to contact): 4355.1155 +- 2.0 min, 4378.5372 +- 2.0 min, 4401.9589 +- 2.1 min, 4425.3806 +- 2.1 min
- B (~10%, about 8.5 h): 4348.5879 +- 4.0 min, 4372.0096 +- 4.0 min, 4395.4313 +- 4.0 min, 4418.8530 +- 4.1 min
- General formulas: A = 1989.52348 + 23.421703 E; B = 2006.41760 + 23.421703 E.

What would settle it further

- An SB2 spectrum, possibly in the red or IR, where L2/L1 rises. The companion's K2 of about 59 km/s would give true masses with i = 89.06 deg.
- Any further TESS sector, or one ground-based eclipse at the times above, to tighten P and check for apsidal motion. The B O-C chi2 of 12/3 hints at extra scatter, but it is driven by one half-flagged s52 point.
- Gaia DR4 epoch RVs, which would check the 0.0058 d period offset directly.

Archives queried (step 1)

- **cats.py --r 60** (rerun today): VSX none, ASAS-SN none, Prsa+2022 none, IJspeert+2024 none, Kostov+2025 t3/t4 none. The only hits are Kostov t2 at 0.001" and SIMBAD "TYC 2609-823-1" typed SB*.
- **Gaia DR3** vari_summary and vari_eclipsing_binary: 0 rows. phot_variable_flag is NOT_AVAILABLE.
- **VSX** 60", **SB9** 30" and **WDS** 30": 0.
- **ExoFOP:** no TOI or CTOI, contamination ratio 0.0037.
- **Web search** on "TYC 2609-823-1" / "TIC 39307147": nothing about this star.
- **MAST products:** QLP, TARS, TESS-SPOC (s25, s26, s52, s53, s79, s80) and TASOC (s26).

Files: verify.py/.txt, verify_spoc.txt, physics2.py/.txt (new). Checker #1's ecl*, ephem*, ecc*, secondpipe*, diff*, neighbours*, ebfit*, physics*, predict* are all re-verified.

A181. TYC 7081-350-1 (TIC 124856476)

status
: confirmed (adversarial re-analysis 2026-10-21; eccentric (e 0.464) G star 1.32 Rsun + 0.51 Msun M dwarf, i 88.4 deg; A 9.6-10.0% grazing transit (b 0.78) x6 + B 1.7-1.8% total occultation x6 (2018-2025), QLP/TESS-SPOC/TGLC; e, omega within 1.2 sigma of Gaia, every eclipse within 0.44 sigma of its predicted conjunction; RUWE 2.22, possible third light; on target in s87/s98 difference images; Kostov t2 only)

P_days
: 21.680622

eP
: 0.000006

T0_primary_BTJD
: 2210.25550

depth_primary
: 9.6-10.0

depth_secondary
: 1.7-1.8

duration_h
: 6.4

source_path
: Fable 5.1\made\appointments\vet124856476\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 124856476 (TYC 7081-350-1, Gaia DR3 2891700638862696576): adversarial check

Checker: Opus 5.5, 2026-10-21. This resumes checker #1 (killed by a usage limit after steps 1-4). I re-verified steps 1-4 with new code (`verify.py`) and added a TESScut s98 pixel check. Step 5 uses a reproduced ebfit, a bootstrap of its errors, and `physics2.py`.

Verdict: CONFIRMED (secure)

This is a detached, eccentric, double-eclipsing binary at P = 21.680622 d. - The deep eclipse is a grazing transit of the G-type primary by an M dwarf of about 0.5 Msun. The shallow eclipse is the total occultation of that M dwarf. - Five full A eclipses and six B eclipses span 2018-2025 and were measured in three pipelines (QLP, TESS-SPOC, TGLC) plus two TESScut sectors. They sit on the target pixel. - One geometry reproduces the depths, durations and phase, and it agrees with Gaia's SB1 orbit and the TIC primary. - Prior art: Kostov+2025 table2 only (unvetted ML flags), so the star is uncatalogued as an EB.

Deciding numbers

- **A eclipses:** s6 1473.1146, s7 1494.7947 (TESS-SPOC; QLP has only part of it), s33 2210.2552, s87 3684.5375, s98 4009.7477 and 4031.4276. Depth is 9.6-10.0% (s6, s33, s87). Trapezoid T14 is 6.4 h; it is V/U-shaped.
- **B eclipses:** s7 1499.7869, s87 3667.8488 and 3689.5279, s98 3993.0577 (flagged), 4014.7393, and 4036.4205 (flagged). Depth is 1.73-1.80% (s7, s87); T14 is 5.1-5.3 h, with a flat bottom of about 1.4 h in the model.
- **My own refit** (`verify.txt`, raw SAP, all cadences) agrees within 0.7 min. Two half- or all-flagged B eclipses differ by 2.2 min.
- **s98 depths.** QLP has A at 8.4% and B at 1.49% in s98. In the s98 TESScut pixels, A is 8.9-9.2% and B is 1.6-1.7%, the same in 1x1, 3x3 and 5x5 apertures, with the deficit on the target pixel (`diff_s98.txt`). The A/B ratio is 5.6 in both sectors. So the shallowing is mostly the QLP s98 aperture/background, not physics.
- **Ephemeris A:** T0 = BTJD 2210.25550 +- 0.00040, P = 21.680622 +- 0.000006 d, chi2 1.2/3, rms O-C 0.48 min.
- **Ephemeris B:** T0 = BTJD 2215.24730 +- 0.00060, P = 21.680621 +- 0.000008 d, chi2 6.0/4. B-A phase is 0.23024.
- **Against Gaia.**
- Period: Gaia's 21.68101 +- 0.00324 d differs from TESS by +0.12 sigma.
- Conjunction times: every measured eclipse lies within 0.44 sigma of Gaia's predicted conjunction. A runs about -0.15 d and B about +0.06 d.
- B-A phase: Gaia predicts 0.2202 +- 0.0066; I measure 0.23024 (+1.5 sigma).
- **Eclipse-model fit** (s87, two LD spheres, Keplerian; chi2/N 1.00, rms 0.73 ppt). I reproduced checker #1's fit exactly. It needs start values near e cos(omega) = -0.43; the default start finds a wrong minimum. The Hessian errors were NaN, so I replaced them with a 30-sample block bootstrap, which probably underestimates them.
- e = 0.4643 [0.4614, 0.4663]; omega = 201.6 [200.5, 202.4] deg; i = 88.35 deg.
- k = R2/R1 = 0.357; J = 0.139; L2/L1 (TESS) = 0.0178.
- Compared with Gaia: e 0.4764 +- 0.0103 (-1.2 sigma); omega 198.88 +- 2.66 (+1.0 sigma).
- ebfit.py prints "primary omega 21.7". That is a label slip, the same as for TIC 39307147: its Z>0 axis points toward the observer, and the correct value is 201.7 deg.
- **Pixels and neighbours.** In the s87 and s98 difference images, the deficit is 0.4-1.4" (A) and 2.5-3.0" (B) from the light centroid, and the peak is on the target pixel. Nothing within 60" is brighter than G 16.29 (Delta G 5.2), so a neighbour would need a dip above 200% for B and above 1100% for A.
- **Census.** Nothing above 0.3% outside the eclipses, except the s87 sector-start ramp at 3663.3 and an s98 pre-gap offset at 4038.4-4038.6. That offset is common-mode: 25 of 202 stars dip there.

Physics (step 5)

- **Masses.** f(M) = 0.0533 [0.0505, 0.0563] Msun (K1 32.15 km/s). With the TIC M1 = 1.06 and i = 88.35 deg, m2 = 0.51 Msun and a = 38.0 Rsun.
- **Primary.** R1 = 0.0347a = 1.32 Rsun (TIC 1.318). logg1 = 4.22 (Gaia GSP-Phot 4.18).
- **Companion.** R2 = 0.47 Rsun, as expected for a 0.5 Msun dwarf (about 0.47). The surface-brightness ratio gives T2 of about 3650 K (blackbody), close to a 0.5 Msun dwarf (about 3750 K).
- **Predicted occultation depth.** L2/(L1+L2) = 1.74%; observed 1.73-1.80%.

- **A eclipse** (nu = 90 deg - omega): the companion passes in front of the primary. Separation 0.947a, b_A = 0.78 > 1-k = 0.64, so the transit is grazing/partial. That explains why A (about 10%) is shallower than k^2 = 12.8% and why it is V/U-shaped. Model T14 is 6.81 h; trapezoid 6.4 h.
- **B eclipse:** the primary passes in front, and the companion is totally occulted (b_B = 0.55 < 0.64 in 100% of bootstrap resamples). Separation 0.669a, so the eclipse is faster. Model T14 5.38 h, T23 1.41 h; trapezoid T14 5.2 h, flat about 1.9 h.
- **Durations.** The B/A duration ratio of 0.81 is larger than Gaia's b=0 value of 0.73 (+2.5 sigma). The non-zero impact parameters explain the difference fully.
- **RV amplitude.** Gaia's rv_amplitude_robust of 67.3 km/s is close to 2K1 = 64.3 km/s.
- **Caveat: RUWE 2.22.** The inner orbit's photocentre wobble is only about 0.21 mas. TIC 39307147 has a similar wobble (about 0.18 mas) at a similar G and RUWE 1.02. So RUWE 2.2 hints at an unresolved further companion (ipd_frac_multi_peak = 0). Any third light would dilute both eclipses equally. It cannot be large, because the fitted R1 matches the TIC radius. It does not affect the verdict.

Next predicted eclipses (BTJD; linear ephemeris, propagated 1-sigma)

- A (about 10%, about 6.8 h contact to contact): 4334.9565 +- 1.1 min (2026-10-21, about 11 UT), 4356.6371 +- 1.1, 4378.3177 +- 1.1, 4399.9983 +- 1.1 min
- B (about 1.75%, about 5.4 h): 4339.9482 +- 1.5 min, 4361.6288 +- 1.5, 4383.3094 +- 1.5, 4404.9900 +- 1.5 min
- General formulas: A = 2210.25550 + 21.680622 E; B = 2215.24730 + 21.680621 E.

What would settle it further

- An SB2 or IR spectrum to measure K2 (expected about 67 km/s) and the true masses. A partial transit makes i well constrained.
- High-resolution imaging or Gaia DR4 astrometry to explain RUWE 2.2, which would test for third light.
- Further TESS sectors to test apsidal motion. With e = 0.46 and P = 21.7 d it should be slow.

Archives queried (step 1)

- **cats.py --r 60** (rerun today): VSX none, ASAS-SN none, Prsa+2022 none, IJspeert+2024 none, Kostov t3/t4 none. The only hits are Kostov t2 at 0.001" and SIMBAD "TYC 7081-350-1" typed SB*.
- **Gaia DR3** vari_summary and vari_eclipsing_binary: 0 rows (phot_variable_flag NOT_AVAILABLE).
- **VSX** 60", **SB9** 30" and **WDS** 30": 0.
- **ExoFOP:** no TOI or CTOI, contamination 0.0026.
- **Web search** on "TYC 7081-350-1" / "TIC 124856476": nothing about this star.
- **MAST:** QLP (s6, s7, s33, s87, s98), TESS-SPOC (s6, s7, s33), TGLC/T16/ELEANOR-LITE (s6, s7), TARS (s6, s7, s33, s87) and TASOC (s6).

Files: verify.py/.txt, verify_spoc.txt, diff_s98.txt + diff_*s98*.png/json, ebfit_s87_rerun.txt, ebfit_boot.py/.txt/.npy, physics2.py/.txt (new). ebfit.json was regenerated identically, and checker #1's copy is kept as ebfit_checker1.json. Checker #1's other files are re-verified.

A182. UCAC4 667-109998 (TIC 200222789)

status
: confirmed (adversarial re-analysis 2026-10-21; K giant 5.4 Rsun + ~6950 K, ~1.5 Msun companion, e 0.34; the deeper eclipse (B, T0) is the giant's total occultation of the companion, 13.5-14.6%, 24.2 h, flat 14.9 h, x3; A 6.1-7.0% rounded transit x2 (2018-2024); QLP-only, own TESScut photometry aperture-independent, difference image 0.8 arcsec on target; A-B separation +0.95 sigma, e -1.8 sigma, omega +1.6 sigma from Gaia; ATLAS 'dubious' entry is not an EB classification)

P_days
: 43.89933

eP
: 0.00007

T0_primary_BTJD
: 1741.61295

depth_primary
: 13.5-14.6

depth_secondary
: 6.1-7.0

duration_h
: 24.2

source_path
: Fable 5.1\made\appointments\vet200222789\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 200222789 (UCAC4 667-109998; Gaia DR3 1981914793343154944) -- adversarial check

Checkers: Opus 5.5 (steps 1-3 and the difference images, 2026-10-20; killed by a usage limit) and a second Opus 5.5 session that resumed on 2026-10-21. The second session re-checked steps 1-3 and did steps 4-6. Gaia SB1: P 43.90099 +- 0.00732 d, e 0.3505 +- 0.0085, w 332.23 +- 1.52 deg, K1 38.54 km/s. TIC: Tmag 10.86, Teff 5071, R 5.39 (giant), d 816 pc.

Verdict: CONFIRMED

This is a long-period eclipsing binary made of a K giant and a hotter, smaller companion. The deeper eclipse (B) is a total occultation of the companion behind the giant. The shallower eclipse (A) is the companion transiting the giant.

1. Archives (what was queried, what came back)

- cats.py --r 60 (run twice: 10-20 and again 10-21): VSX none; ASAS-SN none; Prsa+2022, IJspeert+2024 and Kostov+2025 t2/t3/t4 none within 5"; SIMBAD has UCAC4 667-109998 as [SB*], with one reference (the Gaia DR3 multiplicity paper).
- Gaia DR3 vari_summary 0 rows, vari_eclipsing_binary 0 rows, phot_variable_flag NOT_AVAILABLE. SB9 0; WDS 0; ExoFOP has no TOI or CTOI.
- VizieR within 30": ZTF Chen+2020 0; GCVS 0; WISE Chen+2018 0; ASAS-SN II/366 0; Gaia veb 0; APOGEE, GALAH, LAMOST and RAVE 0. ATLAS Heinze+2018 lists J337.8834+43.3708 with class "dubious" and long-fit period 79.9 d. That is not an eclipsing-binary classification and so not prior art.
- Web search on the UCAC4, TIC and Gaia DR3 ids: no page mentions the star.
- MAST CAOM (10"): QLP s16, s56, s83, s84 only. There are no SPOC, TESS-SPOC, TGLC, ELEANOR or other HLSP products.
- Gaia DR3 astrometry: RUWE 1.18, ipd_frac_multi_peak 0, rv_amplitude_robust 75.8 km/s, GSP-Phot Teff 5696 and logg 3.31. That Teff is a blend, pulled up by the hot companion.
- No prior art. The eclipses are new.

2. Eclipses (raw QLP SAP_FLUX, QUALITY==0, no trimming; flagged cadences inspected separately)

- B s16: 1741.61314 +- 0.4 min, depth 14.55%. Clean, flat bottom.
- B s56: 2839.09539 +- 0.5 min (fit error; the real error is larger), depth 14.29%. Ingress falls in the orbit gap, so the time rests on the egress plus late ingress. A flux-level step across the gap is handled by per-segment baselines.
- B s83: no eclipse; the sector ends 0.2 d before ingress.
- B s84: 3585.39085 +- 1.6 min, depth 13.52%. The bottom sits on a scattered-light swell up to about 1.7x the star's flux, giving a tilted floor.
- A s16: 1755.14825 +- 2.2 min, depth 6.22%. U-shaped.
- A s56: 2852.618, depth about 6.1%. Egress cut off by the sector end; one-sided, not used.
- A s84: 3598.9072 +- 7 min, depth 7.0%. The background swell peaks near mid-eclipse.
- Trapezoid fits: B has T14 24.2 h, ingress 4.7 h and a flat bottom of 14.9 h. A has T14 31.3 h and an ingress of 9.2 h (rounded, limb-darkened).
- Re-timing with my own code (mirror symmetry, block bootstrap): B s16 1741.61350 +- 8.5 min, B s84 3585.39700 +- 2.9 min, A s16 1755.14775 +- 8.2 min, A s84 3598.90350 +- 7.5 min. All agree with the list above to within 9 min.
- Blind dip census of all four sectors (1-h bins): nothing deeper than 0.5% outside the eclipses. The only dips deeper than 0.3% are background-swell residuals in s84.
- Centroids: s16 shows shifts of +18/+10 mpx during B. The difference image (step 4) resolves this as on-target.

3. Ephemeris

- Joint linear fit (common P; errors scaled x4.6, driven by s84): P = 43.89933 +- 0.00007 d; T0_B = BTJD 1741.61295 +- 1.8 min; T0_A = BTJD 1755.14708 +- 9.6 min. O-C: B +0.3, -1.3, +8.5 min; A +1.7, -17.1 min.

- Against Gaia: P differs by -0.0017 d, which is -0.23 sigma. The observed B time is 0.16 d earlier than Gaia's predicted conjunction (-0.5 sigma), and A is 0.06-0.12 d later (+0.2 to +0.7 sigma).
- The A-B separation is 13.534 d (phase 0.3083), against Gaia's prediction of 13.31 +- 0.23 d (+0.95 sigma).
- e cos w = +0.302 from the separation (Gaia +0.310). From the physical model (step 5): e 0.335, w 334.7, e sin w -0.143. Gaia has e 0.3505 and e sin w -0.163, so e is -1.8 sigma and w +1.6 sigma from Gaia.
- Possible small signal: in s84, B is 8 min late and A 17 min early, so A-B shrank by about 26 min over 42 cycles. Both s84 events sit on a large scattered-light swell, so I do not claim apsidal motion. Future eclipses would test it.

4. Independence

- No second pipeline exists for this star, so I did my own aperture photometry on TESScut 15x15 cutouts (1x1, 3x3 and 5x5 apertures).
- B: s16 1741.6137 (13.4%), s56 2839.0952 (13.5%), s84 3585.381 (13.4%).
- A: s16 1755.1490 (6.1%), s84 3598.9033 (5.6%).
- These times are within 1 min of QLP in s16 and s56, and within 6-14 min in s84 (the swell). The depth is the same in 1x1, 3x3 and 5x5 apertures to within about 1% absolute, so the signal is not diluted by anything off-target.
- Difference images (s16 A, s16 B, s56 B, s84 B):
- In s16 B, the centroid of the flux deficit lies 0.04 px (0.8") from the target.
- 84% of the deficit falls in the 3x3 box on the target, compared with 52-55% of the light.
- The two bright stars in the cutout, both more than 60" away, lose no flux.
- Neighbour dilution: within 60" the brightest Gaia neighbour is G 15.39 (42.8"). To fake A it would need a 238% dip, and B would need 532%; the others need 347-1600%. All impossible.

5. Physics (deeper-B question)

- Mass function f(m) = 0.214 Msun; a1 sin i = 31.3 Rsun. For M1 = 1.5 / 2.0 / 2.5 Msun: m2(i = 88 deg) = 1.14 / 1.34 / 1.51 Msun and a = 72 / 78 / 83 Rsun.
- Which star is in front: the SB1 w is the giant's, in the RV convention. So at B (nu = 270 - w = 297.8 deg) the giant is in front, and at A (nu = 117.8 deg) the companion is in front. The shapes match this: B is flat-bottomed (a total occultation of the small star), and A is rounded (a transit across a limb-darkened giant).
- One geometry fits both s16 eclipses. I fitted a two-sphere eccentric model with the giant's quadratic limb darkening fixed at 0.45/0.20:
- k = R2/R1 = 0.2545; companion light share in TESS L2 = 0.146; R1/a = 0.0764; i = 87.95 deg.
- Impact parameter b_B = 0.36, which is below 1 - k = 0.75, so the occultation is total, as observed. b_A = 0.48.
- Residual rms is 0.66 ppt for B, which is white. For A it is 1.07 ppt, with +-3 ppt structure across the bottom: spot crossings or a limb-darkening mismatch, not a geometry problem.
- The duration ratio comes out right without being forced. B happens near periastron (sky speed x1.24) and A near apastron (x0.89), and the larger A impact parameter partly offsets the difference.
- Why B is deeper: B depth equals L2 (14.6%). A depth is about k^2 (1 - L2) times limb darkening, about 6%. So per unit area the companion is 2.64x brighter than the giant in the TESS band. As

blackbodies, that means T2 is about 6950 K for T1 = 5071 K: an early-F or late-A dwarf.
- Sizes and masses: R2 = 1.41 / 1.52 / 1.62 Rsun and R1 = 5.5 / 6.0 / 6.4 Rsun for M1 = 1.5 / 2.0 / 2.5. A main-sequence star near 7000 K with R about 1.6 Rsun should have about 1.5 Msun, which needs M1 about 2.5 Msun. That gives logg1 = 3.23, against Gaia's 3.31, and R1 6.4 against TIC's 5.4. TIC assumed a single star; with 15% of the light from the companion and a blended Teff, its radius would come out low, so this is consistent.
- Depths, durations, shapes, the timing of both eclipse kinds, and the companion's light all fit one geometry.

6. Next predicted eclipses (BTJD; linear ephemeris; formal errors in brackets; allow +-30 min given the s84 O-C)

- A (transit, ~6%, 31 h): 4345.208 (11 min), 4389.107, 4433.006, 4476.906, 4520.805, 4564.704
- B (total occultation, ~14.5%, 24 h, flat 15 h): 4375.573 (5 min), 4419.472, 4463.372, 4507.271, 4551.170, 4595.070

What would settle it further

- Spectroscopy: a few spectra near quadrature should show a second set of lines from a ~7000 K star, with K2 of about 30-38 km/s. That would give an SB2 and dynamical masses. The prediction is M1 about 2.5 and M2 about 1.5 Msun.
- More eclipse timings (TESS, or ground-based for the 14.5% B eclipse at V about 11.5) to test the apparent shrinking of the A-B separation, which could be apsidal motion.
- A multi-band B eclipse would directly give the companion's colour and temperature.

Files: progress.md, verify.py/txt, zoom.py/zoom_B.png, photcut.*npz/png, cutfit.py/txt, diff_*.png/json, neighbours.txt, physics.py/txt, ebfit.py, ebfit_s16.json and ebfit_s16_fit.json/png, ephem_final.py/json/_out.json/txt.

A183. TYC 8201-1042-1 (TIC 146111348)

status
: confirmed (adversarial re-analysis 2026-10-21; K giant 5.6 Rsun + ~6000 K, ~1.35 Msun companion, e 0.17, i 87.4 deg; the deeper eclipse (B, T0) is a partial occultation of the companion, 12.4-14.2%, 25.3 h, x3; A 7.7-8.4% full transit x4 (2019-2025), QLP/eleanor/TGLC/T16; RAVE 2007 RV on the orbit; e -1.1 sigma, omega 0.0 sigma from Gaia; difference images 1.4-5.6 arcsec on target, the G 12.8 star at 55 arcsec loses 1.8% against 14% on the target)

P_days
: 49.24084

eP
: 0.00003

T0_primary_BTJD
: 1565.73335

depth_primary
: 12.4-14.2

depth_secondary
: 7.7-8.4

duration_h
: 25.3

source_path
: Fable 5.1\made\appointments\vet146111348\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 146111348 (TYC 8201-1042-1; Gaia DR3 5365525935061421952) -- adversarial check

Checkers: Opus 5.5 (steps 1-3 and the difference images, 2026-10-20; killed by a usage limit) and a second Opus 5.5 session that resumed on 2026-10-21. The second session re-checked steps 1-3 and did steps 4-6. Gaia SB1: P 49.23438 +- 0.01378 d, e 0.1854 +- 0.0119, w 136.30 +- 4.11 deg, K1 35.53 km/s, gamma 41.18 km/s. TIC: Tmag 10.54, Teff 4716, R 5.56 (giant), d 590 pc.

Verdict: CONFIRMED

This is a long-period eclipsing binary made of a K giant and a smaller, hotter companion of about 6000 K. The deeper eclipse (B) is a partial (grazing) occultation of the companion behind the giant. The shallower eclipse (A) is a full transit of the companion across the giant.

1. Archives (what was queried, what came back)

- cats.py --r 60: VSX none; ASAS-SN none; Prsa+2022, IJspeert+2024 and Kostov+2025 t2/t3/t4 none within 5"; SIMBAD has TYC 8201-1042-1 as [SB*] with 5 references (RAVE DR4/5/6, APASS-RAVE, Gaia DR3 multiplicity).
- Gaia DR3 vari_summary 0 rows, vari_eclipsing_binary 0 rows, phot_variable_flag NOT_AVAILABLE. SB9 0; WDS 0; ExoFOP has no TOI or CTOI. VizieR: ZTF 0 (the star is too far south), ATLAS 0, GCVS 0, WISE var 0, Gaia veb 0, Prsa 0.
- RAVE DR6 has one RV: 49.89 +- 0.63 km/s on 2007-03-28.
- Web search on the TYC, TIC and Gaia DR3 ids: no page mentions the star.
- MAST CAOM (10"): QLP s9, 10, 36, 63, 90, 99, 100; ELEANOR-LITE s9, 10; T16 s9, 10; TGLC s9, 10; TESS-SPOC s63 (TPF).
- Gaia: RUWE 1.15, ipd_frac_multi_peak 0, GSP-Phot Teff 4900 and logg 3.12.
- No prior art. The eclipses are new.

2. Eclipses (raw QLP SAP_FLUX, QUALITY==0, no trimming)

- B s9: 1565.73168 +- 1.4 min, depth 12.8%.
- B s36: 2304.34708 +- 0.9 min, depth 14.2%.
- B s100: 4077.01554 +- 0.8 min, depth 12.4%.
- A s9: 1545.049, depth 7.7%. Pre-ingress cadences flagged; one-sided, not used in the fit.
- A s10: 1594.29073 +- 0.9 min, depth 8.1%.
- A s36: 2283.66246 +- 0.7 min, depth 8.2%.
- A s63: 3022.27478 +- 0.7 min, depth 8.4%.
- A s90: 3760.895 +- 3.9 min. Egress only, on cadences flagged after the orbit gap.
- A s99: falls in the data gap.
- Shapes: B is rounded/V-like with T14 25.3 h. A has T14 21.8 h, ingress 7.8 h and a short, nearly flat bottom.
- Re-timing with my own code (mirror symmetry): B 1565.73400, 2304.34700, 4077.01525; A 1545.0505, 1594.29050, 2283.66300, 3022.27425. All agree with the list above to within 0.1-3.3 min.
- Census: nothing deeper than 0.5% outside the eclipses, apart from an orbit-start scattered-light ramp in s10 at 1585.1-1585.3. In-eclipse background dips of 1-3% of the star's flux are the star's own wings in the background annulus.
- Coverage: every predicted eclipse that falls inside the data is present, with the expected depth. None is missing.

3. Ephemeris

- Joint linear fit (common P; errors scaled x2.0): P = 49.24084 +- 0.00003 d; T0_B = BTJD 1565.73335 +- 1.7 min; T0_A = BTJD 1594.29078 +- 1.1 min. O-C: B -2.4, +1.7, -0.6 min; A -0.1, -0.0, -0.3, +11.4 min (s90 is egress-only on flagged cadences).
- Against Gaia: P differs by +0.0065 d, which is +0.47 sigma. The observed B time is 0.06-0.22 d later than Gaia's predicted conjunction (at most +0.54 sigma), and A is 0.05-0.21 d earlier (at most -1.0 sigma).
- The A-B separation is 28.557 d (phase 0.57995), against Gaia's prediction of 28.83 +- 0.39 d (-0.68 sigma).

- Physical model (step 5): e 0.1726, w 136.35 deg, so e cos w -0.125 and e sin w +0.119. Gaia has e cos w -0.134 and e sin w +0.128. e is -1.1 sigma from Gaia, and w agrees to 0.0 sigma.

4. Independence

- Three other pipelines show both eclipse kinds at the QLP times (s9/s10, timed with my mirror method):
- B 1565.73: ELEANOR-LITE raw 1565.73425 (12.8%); TGLC aperture 1565.73525 (16.0%); TGLC PSF 1565.73450 (14.9%); T16 1565.73400 (12.4%).
- A 1594.29: ELEANOR-LITE 1594.28975 (8.3%); TGLC 1594.29075 (aperture 9.7%, PSF 8.7%); T16 1594.29075 (7.2%).
- A 1545.05: ELEANOR-LITE 1545.05175; T16 1545.05000.
- All are within 0-5 min of QLP. TGLC's PSF photometry, which fits Gaia-positioned PSFs, puts the whole dip on the target.
- Difference images from TESScut (s9 B, s36 A, s36 B, s63 A, s100 B; I re-ran s36 B and s63 A and reproduced the earlier numbers exactly):
- The centroid of the flux deficit lies 1.4-5.6" from the target. The out-of-eclipse light centroid is 6-21" off, pulled by neighbours.
- 72-81% of the deficit falls in the 3x3 box on the target, compared with 23-37% of the light.
- Neighbours:
- Within 60", only the G 12.79 star at 55.5" is bright enough to host the dips in flux terms; it would need a 60% dip for B and 37% for A. Its pixel loses 1.8% (s36 B) and 0.4% (s63 A), against 14.0% and 8.2% in the target pixel.
- The G 13.32 star at 66" would need 98% and 60%; its pixel shows 0.7%.
- All other neighbours would need dips of more than 100%. Ruled out.

5. Physics (deeper-B question)

- Mass function f(m) = 0.217 Msun; a1 sin i = 34.0 Rsun.
- Which star is in front: the SB1 w is the giant's. A falls at nu = 90 - w = 313.7 deg, where the companion is in front, near periastron (r/a 0.856). B falls at nu = 133.7 deg, where the giant is in front, near apastron (r/a 1.107).
- One geometry fits both s36 eclipses. I fitted a two-sphere eccentric model with the giant's quadratic limb darkening fixed at 0.45/0.20:
- k = 0.312; L2 = 0.175; R1/a = 0.0615; i = 87.38 deg.
- Residuals are white, with rms 0.90 and 0.92 ppt.
- Impact parameter b_A = 0.64, which is below 1 - k = 0.69, so A is a full transit with a short, flattish, limb-darkened bottom.
- Impact parameter b_B = 0.82, which is above 1 - k, so B is a PARTIAL occultation. That gives the rounded, V-like shape, as observed. The larger separation at apastron is exactly what raises b at B.
- Why B is deeper: the giant covers part of a companion that supplies 17.5% of the TESS light, removing 13-14%. At A, the companion's disk (k^2 = 0.097) blocks about 8% of the cooler giant's light. Per unit area the companion is 2.18x brighter in TESS than the giant, which as blackbodies gives T2 about 5900-6150 K for T1 = 4716-4900 K.
- Sizes and masses: for M1 = 1.5 / 2.0 / 2.5 Msun, m2 = 1.16 / 1.35 / 1.53 Msun, R1 = 4.8 / 5.2 / 5.5 Rsun and R2 = 1.50 / 1.62 / 1.72 Rsun (logg1 3.25 / 3.31 / 3.35; TIC R1 5.56, Gaia logg 3.12). The

companion would be a slightly evolved F/G star of about 1.35 Msun around a giant of about 2 Msun. That is self-consistent.
- The single RAVE RV (49.89 km/s in 2007) matches the orbit's predicted 51.0-52.5 km/s (the orbit spans 0.9-71.9) to within 1-3 km/s.
- Depths, shapes, durations, both timings and the companion's light all fit one geometry.

6. Next predicted eclipses (BTJD; linear ephemeris; formal errors about 2 min, allow +-10 min)

- A (full transit, ~8.2%, 21.8 h): 4351.778, 4401.018, 4450.259, 4499.500, 4548.741, 4597.982
- B (partial occultation, ~13-14%, 25.3 h): 4372.461, 4421.702, 4470.943, 4520.183, 4569.424, 4618.665

What would settle it further

- Spectroscopy for an SB2: K2 should be about 32-40 km/s, with lines from a ~6000 K star. That would give dynamical masses.
- Multi-band photometry of a B eclipse: because B is a partial occultation, its depth should rise toward the blue, since the companion is hotter.
- Further TESS sectors to tighten e and w (the e of 0.173 vs Gaia's 0.185 is a 1.1 sigma difference).

Files: progress.md, verify.py/txt, zoom.py/zoom_all.png, pipe2.py/txt, diffimg_s36B.txt, diffimg_s63A.txt, diff_*.png/json, neighbours.txt, physics.py/txt, ebfit.py, ebfit_s36.json and ebfit_s36_fit.json/png, rave_check.py/txt, coverage.py/txt, ephem_final.py/json/_out.json/txt. Downloads are in C:\Home\sky\data\qlp-sb-p20-60\vet146111348.

A184. UCAC2 5492112 (TIC 220461013)

status
: confirmed (adversarial re-analysis 2026-10-21; spotted K giant 6.8 Rsun + 0.85-0.94 Msun G dwarf, e 0.30; the deeper eclipse (B, T0) is the dwarf's total occultation, 3.8-3.9% crowding-corrected, 53.4 h, flat 42.8 h, x23; A 1.2-2.3% transit, 31.7 h, with spot-crossing bumps x19, 42 sectors 2018-2025; A and B periods differ 3.0 sigma from spot jitter on A; e cos omega -0.181 vs Gaia -0.158(15); VSX/ASAS-SN ROT 54.6 d is the spot modulation: type revision to EA+ROT; TESS-SPOC TCE at 50.85 d never promoted; SPOC TPF pixels on target by chi2 188 vs 1536)

P_days
: 50.85317

eP
: 0.00006

T0_primary_BTJD
: 2760.1247

depth_primary
: 3.8-3.9

depth_secondary
: 1.2-2.3

duration_h
: 53.4

source_path
: Fable 5.1\made\appointments\vet220461013\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 220461013 (UCAC2 5492112, Gaia DR3 4762087153135575040) - adversarial check

VERDICT: CONFIRMED. It is a detached, eccentric eclipsing binary on the target: a spotted K giant (TIC R 6.84 Rsun, Teff 4550 K) with a G-dwarf companion. Each conjunction gives a different kind of eclipse. At A the companion transits the giant: rounded, 1.5-2.0% deep, 31.7 h long, with spot-crossing bumps. At B the giant totally occults the companion: flat-bottomed, 3.8-3.9% deep (crowding-corrected), 53.4 h long. Both repeat across 42 TESS sectors (2018-2025). The VSX/ASAS-SN classification "ROT P=54.57 d" describes the giant's spot modulation correctly, but the eclipses are missing from it. The type should be revised to EA+ROT (or EA/RS), with P = 50.8532 d. That is a type revision, not prior art.

1. ARCHIVES (queried 2026-10-20/21) - cats.py --r 60: VSX ASASSN-V J045855.35-594135.5 (0.1"), type ROT, P 54.57 d, amplitude 0.22 mag (g). ASAS-SN gives ROT with P 55.8 d. SIMBAD lists it as SB* only. It is not in Prsa+2022, IJspeert+2024, or Kostov+2025 tables 2/3/4. - Gaia DR3: vari_summary and vari_eclipsing_binary return 0 rows. NSS gives SB1 with P 50.8547 +- 0.0163 d, e 0.301 +- 0.011, w 121.7 +- 3.1 deg, K1 34.0 km/s. RUWE 1.03. - VizieR SB9 (30") and WDS (30") return 0 rows. ExoFOP (2026-10-20 run) has no TOI and no CTOI; it refused connections on 2026-10-21. - Web search on the TIC number, the ASASSN-V name, UCAC2 and 2MASS found nothing about this star. - MAST: the TESS-SPOC multi-sector DV report for s56-s69 contains TCE_4: P

50.8545 d, T0 2995.354, 16.9k ppm, 13.8 h, MES 7.9 (dvt_tces.txt). This is SPOC's automated pickup of the A eclipses (TCEs 1-3 are aliases). It was never made a TOI, and an automated TCE is not a catalogue classification. It is worth citing as an independent detection.

2. ECLIPSES (raw QLP SAP_FLUX, QUALITY==0, no trimming) The previous checker's fits were re-verified. A shape-fixed rerun gave identical numbers. A from-scratch free rerun reproduced every two-edge event to within 0.1 min. A separate model-free mirror-symmetry timing (mirror.py) agrees within 0.5-4 min for B and 1-9 min for clean A events (33 min for A s98, which has a spot bump). - A (19 events, 12 with both edges): T14 31.7 h, T12 4.7 h, QLP depth 1.2-2.3% (median 1.7%). The bottom is rounded, with ~0.5% bumps inside the eclipse: spot crossings on the giant. - B (23 events, 10 with both edges): T14 53.4 h, T12 5.3 h, flat bottom 42.8 h (total). QLP depth 3.1-4.4% (median 3.8%; 3.1-3.3% in s94-98, consistent with aperture/dilution differences). - Flagged cadences do not contradict anything: B s29 (2099.030) and B s97 (3980.604) fit; the flagged data in s68/s98 are junk. - Background and centroids (diag.txt): in clean events dBKG is within +-10 ppt of the star's flux (z<3), and the centroid shift is 0-2 mpix. Excursions occur only in scattered-light sectors, where the eclipse shape is unchanged. - Excluded events: B+31 s87 3675.5 (on the s087 ~3676 downturn, egress in the gap) and B+21 s68 (flagged). - Census of all 42 sectors (threshold 0.4%): nothing >= 0.9% deep outside the two eclipse families.

3. EPHEMERIDES (myephem.txt) - B: P = 50.85313 +- 0.00006 d, T0 = 2760.1247 +- 0.0011 BTJD (10 events, rms 4.9 min). - A: P = 50.85407 +- 0.00030 d, T0 = 2791.5637 +- 0.0049 BTJD (12 events, rms 22.7 min, 20.6 min spot jitter). The A/B periods differ at 3.0 sigma, most likely from spot-distorted A timings. - Joint P 50.85317 +- 0.00006 d. Gaia P 50.85471 +- 0.01629 d, so the difference is -0.09 sigma. - Phase of B after A: 0.38177 +- 0.00010. Gaia predicts 0.3965 +- 0.0096 (-1.5 sigma). T14B/T14A is 1.684; Gaia predicts 1.686 +- 0.045. - Eclipses give e cos w = -0.1807 and e sin w ~ +0.255 (e ~ 0.31, w ~ 125). Gaia gives -0.158 +- 0.015 and +0.256. - Against Gaia's conjunctions: A is +0.13 to +0.21 d late (+0.1 to +0.3 sigma); B is -0.54 to -0.61 d early (-0.4 to -0.6 sigma).

4. INDEPENDENCE - TESS-SPOC SAP gives the same mid-times as QLP: A s62 -2.4 min, s64 +7.5, s69 +6.4; B s64 +0.7, s66 +1.0 min. PDCSAP B depth is 3.86-3.92% (CROWDSAP 0.92). - SPOC TPF pixels: the deficit peaks on the target pixel, with equal depth in 1x1, 3x3 and 5x5 apertures. In single-source PSF fits the target alone is favoured over the best neighbour: chi2 188 vs 1536 (B s64), 1716 vs 4415 (B s66), 60 vs 459 (A s64), 30 vs 434 (A s62), 336 vs 393 (A s69). - Neighbours: G 14.61 at 10.7" would need a 33% A dip and a 74% B dip; G 14.89 at 26.2" would need 43% and 95%; all others more than 100%. Both are rejected by the pixels and by the Gaia orbit tie.

5. PHYSICS (physics.txt) - f(m) = 0.179 Msun. For M1 = 1.0-1.2, m2 = 0.85-0.94 Msun and a = 71-74 Rsun. Predicted b=0 T14 is A 31-32 h (observed 31.7) and B 52-54 h (observed 53.4). - At A the giant is behind and the companion transits it (rounded, spot bumps). At B the giant is in front and the companion is totally occulted (flat). - L2/L1 = 0.041 and k = 0.11-0.13 give a companion of ~0.75-0.87 Rsun at ~5900-6450 K: a G dwarf.

6. NEXT ECLIPSES (BTJD) - A: 4368.040, 4418.894, 4469.748, 4520.602 (formal +-0.26 h, realistic +-0.5 h). - B: 4336.572, 4387.425, 4438.278, 4489.131 (formal +-0.06 h, realistic +-0.15 h).

WHAT WOULD SETTLE IT FURTHER: a limb-darkened, spot-aware model fit (k, b, T2); spectroscopy for SB2 lines; more A timings to settle the A/B period difference; a VSX type revision to EA+ROT.

A185. TYC 4478-554-1 (TIC 427657329)

status
: confirmed (adversarial re-analysis 2026-10-21; spotted G/K giant 8.8 Rsun + 0.8-1.0 Msun dwarf, circular (e cos omega +0.0002(2); Gaia e 0.018(9)); the deeper eclipse (B, T0) is the dwarf's total occultation, 2.4-2.6%, 38.7 h, flat 31 h, x7; A 1.1-1.6% transit x4 with spot crossings on the floor only (2019-2025); Gaia P -2.1 sigma; VSX ROT 53.5 d is the spot modulation: type revision to EA+ROT; SPOC TPF pixels on target by chi2 304 vs 1716; B ingress wants R1 ~7.5 Rsun, 15% under TIC; Kostov t2 only)

P_days
: 47.52876

eP
: 0.00004

T0_primary_BTJD
: 2574.2519

depth_primary
: 2.4-2.6

depth_secondary
: 1.1-1.6

duration_h
: 38.7

source_path
: Fable 5.1\made\appointments\vet427657329\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 427657329 (TYC 4478-554-1, Gaia DR3 2214685184515156224) - adversarial check

VERDICT: CONFIRMED. It is a detached, near-circular eclipsing binary on the target: a spotted G/K giant (TIC R 8.8 Rsun, Teff 5120 K) with a hotter dwarf companion. The two eclipses are half an orbit apart. At B the giant totally occults the companion: flat-bottomed, 2.5% deep (QLP; PDCSAP 2.55-2.68%), 38.7 h long. At A the companion transits the giant: 1.3-1.6% deep, 39.3 h long. The "mid-eclipse bump" is spot crossings. Structured residuals of 1.5-3 ppt appear only in the A (transit) bottoms, identically in QLP and TESS-SPOC. The B (occultation) bottoms are as quiet as the out-of-eclipse light. The VSX "ROT P=53.5 d" (ASASSN-V J231724.84+682816.3, amplitude 0.11 mag g) is the giant's spot modulation. The eclipses are new: this is a type revision to EA+ROT with P = 47.5288 d, not prior art. Its only listing is Kostov+2025 table 2, the unvetted ML flag list.

1. ARCHIVES - cats.py --r 60: VSX ROT P 53.5 d (0.5"). ASAS-SN catalogue: none. Kostov+2025 table 2 (does not count). SIMBAD lists it as SB* only. It is not in Prsa+2022 or IJspeert+2024. - Gaia DR3: vari_summary and vari_eclipsing_binary return 0 rows. NSS gives SB1 with P 47.5479 +- 0.0091 d, e 0.018 +- 0.009, w 311 +- 24 deg, K1 28.4 km/s. RUWE 0.98. - SB9 and WDS return 0 rows. ExoFOP (2026-10-20) has no TOI and no CTOI; it refused connections on 2026-10-21. A web search on the TYC, ASASSN-V, TIC and 2MASS names found nothing. - SPOC DV (s56-69, s85): its TCEs are at 26.5, 11.7 and 9.0 d, detrending aliases. SPOC did not detect the eclipses at the true period.

2. ECLIPSES (raw QLP SAP, q==0, 10 sectors: s17/18/24/25/57/58/77/78/84/85) The previous checker's receipts were re-verified. The shape-fixed rerun is identical, and a from-scratch free rerun reproduces every event with two covered edges within 0.3 min. Model-free mirror timing agrees within 0.1-0.7 min for B and 4-10 min for A. - B (7 events: s17, s18, s24, s25, s57, s58, s85; three with both edges): T14 38.7 h, T12 3.7 h, flat bottom 31 h (total). Depth 2.44-2.58% in QLP; SPOC SAP 2.35-2.40%; PDCSAP (CROWDSAP 0.91) 2.55-2.68%. - A (4 events: s24, s58, s77, s84): T14 39.3 h, T12 5.9 h, depth 1.1-1.6% in QLP. The s84 event is cut by a flagged stretch in QLP, and SPOC SAP recovers it at 3596.117 +- 0.003. Another A event (1790.0) falls in the s17/s18 gap. - With flagged cadences included, B s17 (1766.261) and B s18 (1813.798) fit the ephemeris; the flagged s25 ingress is junk. - Background and centroids (diag.txt): clean in the s57/58/85 events (dBKG within 3 ppt, centroid shift 2 mpix). A s58 sits at the sector start, where the background ramps (z 57); the same event is clean in SPOC. - Census of 10 sectors (threshold 0.4%): nothing outside the two families, apart from a 0.5% wiggle at s17 1775.4.

3. EPHEMERIDES (myephem_loose.txt: events with at least one full edge; the strict two-edge fit gives the same P) - B: P = 47.52876 +- 0.00004 d, T0 = 2574.2519 +- 0.0005 BTJD (7 events, rms 2.0 min). - A: P = 47.52895 +- 0.00052 d, T0 = 2740.5971 +- 0.0069 BTJD (4 events, rms 17 min, 16 min spot jitter). A and B agree (+0.35 sigma). - Joint P 47.52876 +- 0.00004 d. Gaia P 47.5479 +- 0.0091 d, so the difference is -2.1 sigma, a bit large but acceptable. - Phase of B after A: 0.50012 +- 0.00014. Gaia predicts 0.5064 +- 0.0052 (-1.2 sigma). T14B/T14A 0.986; Gaia predicts 0.977 +- 0.016. - Eclipses give e cos w = +0.0002 +- 0.0002 and |e sin w| <~ 0.01, i.e. essentially circular. Gaia gives +0.012 and -0.014 (e 0.018 +- 0.009, so consistent with 0 at 2 sigma). - Gaia conjunctions (sigma 4.5-4.6 d, because t_periastron is poorly defined at e~0): A is -0.35 to -0.98 d off and B -0.59 to -1.33 d, all within 0.3 sigma. The drift follows the 2.1-sigma P difference.

4. INDEPENDENCE - TESS-SPOC FFI SAP gives the same times: B s57 +2.0 min, s58 +0.7, s85 -2.1; A s77 +1.5 min. A s84 is recovered (QLP partial). - SPOC TPF pixels (s57/58/85 B, s77 A): the deficit peaks on the target pixel, with equal fractional depth in 1x1 and 3x3 apertures (2.5%). The deficit is more concentrated than the light: 77-82% of the deficit sits in the 3x3, against 59-64% of the light. In single-source PSF fits the target alone is favoured: chi2 304 vs >= 1716 (B s58), 433 vs >= 1757 (B s85), 846 vs >= 1874 (B s57), 270 vs >= 894 (A s77). - Neighbours within 60": the only one that could do it is G 13.74 at 43.2", which would need a 15% A dip and a 26% B dip. Its pixel shows only 0.7-1.0% (target PSF wing) against 2.5% on the target, so it is excluded. G 15.52 at 7.2" would need 79% and 136%, so it is impossible.

5. PHYSICS - f(m) = 0.113 Msun. For M1 1.2-2.0, m2(i=90) = 0.76-1.0 Msun and a = 69-80 Rsun. - Geometry: A is at nu = 90 - w, so the giant is behind and the companion transits it (spot crossings, slightly rounded edges). At B the giant is in front and the companion is totally occulted (flat). The orbit is circular, so the durations are equal (observed ratio 0.986). - Durations: with TIC R1 8.8 and b=0, T14 would be 45-53 h, against 39 h observed. The B ingress (T12/T14 = 0.097) gives k ~ 0.107 at b ~ 0, which needs R1 ~ 7.2-7.8 Rsun (15% below TIC). Alternatively b ~ 0.6, but that would lengthen the B ingress to ~5.8 h. The likeliest reading is a TIC radius about 15% too large; a modest discrepancy, not a refutation. - Light: L2/L1 = 0.027 and k 0.106-0.117 give a companion of ~0.8-1.0 Rsun at ~6300-6700 K (blackbody estimate), which is a little hot for m2 0.8-1.0 Msun. The same mild tension appears in TIC 220461013, which suggests a systematic in the giant's TESS-band surface brightness from the crude blackbody.

6. NEXT ECLIPSES (BTJD) - A (transit, 39.3 h): 4356.581, 4404.110, 4451.639, 4499.168 (formal +-0.47 h, realistic +-0.6 h). - B (total occultation, 38.7 h): 4380.345, 4427.874, 4475.402, 4522.931 (formal +-0.04 h, realistic +-0.1 h).

WHAT WOULD SETTLE IT FURTHER: a physical light-curve model with limb darkening and spots (k, b, R1/a; it would test the TIC radius); spectroscopy for SB2 lines and the companion's temperature; a VSX type revision to EA+ROT.

A186. TYC 3444-504-1 (TIC 252806416)

status
: confirmed (adversarial re-analysis 2026-10-21; K giant 3.4-3.8 Rsun + 0.55-0.6 Msun dwarf, e 0.34, i 89-90 deg; A 2.9-3.0% transit, 13.9 h, x2 + B 0.75% total occultation, 27.7 h, flat 20 h, x2 (2019-2023); Gaia P +1.7 sigma was the classifier's 'A lag'; APOGEE DR17 RVs of 2015-18 fit the eclipse ephemeris at 0.34 km/s rms (5.9 with Gaia's P); the G 14.9 star at 3.7 arcsec excluded by centroid regression (chi2 894); TESS-SPOC s75 agrees; difference images on target)

P_days
: 42.44229

eP
: 0.00008

T0_primary_BTJD
: 2633.95593

depth_primary
: 2.9-3.0

depth_secondary
: 0.75

duration_h
: 13.9

source_path
: Fable 5.1\made\appointments\vet252806416\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 252806416 (TYC 3444-504-1, Gaia DR3 7833158992318530 56): adversarial check

Verdict: CONFIRMED. This is an eclipsing SB1. A K giant (Teff 4920 K, logg 3.4, R about 3.4-3.8 Rsun) is transited and occulted by a companion of about 0.55-0.6 Msun, on a 42.4423 d orbit with e about 0.35. Both eclipse kinds are on the target, both have the same period, and the timing matches the Gaia orbit and the 2015-2018 APOGEE RVs once Gaia's period is corrected. The "A lag" was a Gaia period error.

Checker: Opus 5.5 subagent, 2026-10-21. It resumed and verified the 2026-10-20 checker's step 1. All numbers come from scripts in this folder.

1. Archives

- `cats.py --r 60`: no entry in VSX, ASAS-SN, Prsa+2022, IJspeert+2024 or Kostov+2025 t2/t3/t4. SIMBAD lists it as SB* only.

- Gaia DR3 (rerun on 2026-10-21; it reproduces 10-20): vari_summary 0 rows, vari_eclipsing_binary 0 rows. NSS SB1 gives P 42.4205 +- 0.0125 d, e 0.374 +- 0.011, w 69.3 +- 2.9 deg, K1 25.81 +- 0.40 km/s, gamma -8.93.
- VizieR SB9, WDS and VSX (60") return nothing. ExoFOP has no TOI or CTOI. A web search on the TYC and TIC names finds nothing.
- The binarity was already known from APOGEE: Price-Whelan+2018 (J/AJ/156/18) lists it as a high-K binary, and SoS sets flagBinary. **The eclipses appear unreported.**
- MAST holds QLP s21/s48/s75 and TESS-SPOC s75.

2. Eclipses (raw QLP SAP_FLUX, untrimmed)

event	mid-time BTJD	depth	T14	T12	flat
B s21	1894.16853 +- 3.8 min	0.76%	27.65 h	3.69 h	20.3 h
B s48	2615.68798 +- 3.3 min	0.75%	27.70 h	3.69 h	20.3 h
A s48	2633.95598 +- 1.5 min	2.87%	13.99 h	2.78 h	(U-shaped)
A s75	3355.47474 +- 1.7 min	2.97%	13.9 h	2.6 h	(U-shaped)

- The s75 A egress runs into cadences flagged for scattered light (bit 2^30). A fixed-shape fit gives t0 3355.4735-3355.4756 depending on how many flagged cadences are included, about 3 min of spread.
- A in s21 (predicted 1869.995) ends before the sector starts at 1870.45, and the flat start of s21 is consistent with that.
- Background and centroids are smooth through every event.
- Blind census (q==0, 1-h bins, scatter 0.35-0.39 ppt): no other dip deeper than 0.15% in any sector.
- Common mode: the target is the deepest of 280-309 stars at each s21/s48 event. On its s75 CCD it shows 3.02% while 12 other stars show -0.10% to +0.02%.

3. Ephemeris

- A alone gives P 42.44228(9) and B alone gives 42.44232(21), so the two kinds agree.
- Joint fit: **P = 42.44229 +- 0.00008 d, T0_A = 2633.95593 +- 0.00099, T0_B = 2658.13002 +- 0.00188 BTJD**, chi2 0.03/1, all O-C under 0.5 min.
- P_TESS - P_Gaia = +0.0218 d (+1.74 sigma).
- Measured minus Gaia-predicted conjunction, unconditional: A +1.46 and +1.83 d (+1.75 sigma each); B +0.23 and +0.60 d. Conditioned on P = P_TESS: **A +0.02 d (+0.2 sigma), B -0.27 d (-0.9 sigma).** So the lag the classifier saw is entirely Gaia's period error.
- B-A phase is 0.56958(5), against Gaia's 0.5897 +- 0.0116 (-1.7 sigma).
- Duration ratio T_B/T_A is 1.99 +- 0.05, against 2.07 +- 0.06 for b=0 from Gaia (-1.1 sigma).
- Eclipse-derived e 0.344, w 72.5 deg: e cos w +0.103 (Gaia +0.133, -1.6 sigma). The b=0 e sin w of +0.328 is a lower limit, against Gaia's +0.350.

4. Independence

- **TESS-SPOC s75 SAP** (all cadences up to tc+0.45 d, because SPOC flags the egress): t0 3355.47575 (+1.4 min from QLP), depth 2.78%, T14 13.71 h. PDCSAP is NaN across the egress.
- **TESScut difference images** (s21 B, s48 A and B, s75 A): the deficit peaks on the target pixel every time. For s48 A the deficit centroid sits 0.4" from the light centroid, and the 1x1/3x3/5x5 depths are 2.9/2.8/2.8%. The G 15.02 star at 83" shows no deficit.

- **Neighbours.** The only Gaia source within 60" brighter than G 17 is a **G 14.89 star 3.7" away**, inside the same pixel. It would need a 73% A dip and a 19% B dip. Its parallax is 1.252 +- 0.030 mas against the target's 1.898 +- 0.025 and its proper motion differs, so it is unrelated.
- Centroid regression on the QLP SAP_X/Y: if the neighbour produced A, the centroid would move -4.9 mpx (s48) and -5.1 mpx (s75) in x. Measured: -0.06 +- 0.16 and +1.39 +- 0.44 mpx. **Ruled out at chi2 894 and 230.**
- For B, s48 favours the target (chi2 0.1 against 11.0). s21, at 30-min cadence with a trend, is ambiguous.
- **APOGEE DR17 visit RVs** (2015 and 2018, 61 orbits before s48), modelled on the eclipse ephemeris with Gaia's K and the eclipse e, w: rms 0.34 km/s, gamma -8.89, best shift of T_A +0.02 d. Using Gaia's P instead gives rms 5.9 km/s. This confirms the period and that A is the conjunction with the companion in front.

5. Physics

- f(M) = 0.0603 Msun, a1 sin i = 20.07 Rsun.
- With M1 about 1.2-1.4 (from logg 3.41 and R): m2(i=90) is 0.58-0.63 Msun and a is 62-65 Rsun.
- At A (r = 0.637a) the giant is behind and the companion transits it, which explains the limb-darkened U shape. At B (r = 1.323a) the giant is in front and the companion is totally occulted, which explains the flat bottom.
- One geometry fits T14_A, T14_B and T23_B (chi2 about 0.7): R1/a 0.052-0.058 (R1 3.3-3.7 Rsun), k = R2/R1 0.11-0.17 (R2 about 0.5 Rsun), i 89-90 deg, b_A <= 0.2.
- Depth A of 2.9% is about k^2 times limb darkening. `geom2.py` fits depth and durations together, with quadratic limb darkening (u1 0.5, u2 0.2) and Gaia's e sin w as a prior: chi2 0.66, k 0.152, b_A 0.12, b_B 0.24, e sin w 0.345 (Gaia 0.350).
- Depth B of 0.75% equals L2/(L1+L2). At k = 0.15 this needs a TESS-band surface-brightness ratio of 0.34, i.e. T2 about 3850 K, which fits a main-sequence companion of about 0.55-0.6 Msun.
- Everything is consistent with one system.

6. Next eclipses (BTJD; formal 1-sigma, plus about 3 min systematic from the s75 egress)

- A (2.9%, 13.9 h): **4374.0897 +- 4.3 min**, 4416.5320 +- 4.4, 4458.9743 +- 4.5, 4501.4166 +- 4.6.
- B (0.75%, 27.7 h, flat): **4355.8215 +- 6.4 min**, 4398.2638 +- 6.5, 4440.7061 +- 6.6, 4483.1484 +- 6.7.

What would settle it further

- A third epoch of either kind, to test linearity. There are only two of each now, so the joint fit has 1 dof.
- High-resolution spectroscopy or a combined RV + light-curve solution would give the masses and radii (an SB2 detection of the M dwarf is unlikely at a 0.8% flux ratio).
- Speckle or AO imaging is not needed for the 3.7" star, because the centroid test excludes it.

A187. UCAC4 240-011387 (TIC 170312515)

status
: confirmed (adversarial re-analysis 2026-10-21; G/K giant 4.5 Rsun + 0.6-0.7 Msun dwarf, e 0.33, i 88.5-89 deg; A 1.8-1.9% transit, 19.2 h, x4 + B 0.9% total occultation, 35 h, flat 26 h, x2 (2019-2025), TESS-SPOC/eleanor/TGLC agree; Gaia P -0.4 sigma, B-A phase +0.3 sigma; e sin omega 0.32 from durations vs Gaia 0.372(12), 2-2.5 sigma soft; difference images on target, nothing brighter than G 18.3 within 60 arcsec)

P_days
: 52.31098

eP
: 0.00005

T0_primary_BTJD
: 2232.33454

depth_primary
: 1.8-1.9

depth_secondary
: 0.9

duration_h
: 19.2

source_path
: Fable 5.1\made\appointments\vet170312515\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 170312515 (UCAC4 240-011387, Gaia DR3 5563236508191080576): adversarial check

Verdict: CONFIRMED. This is an eclipsing SB1 on a 52.3110 d orbit with e about 0.33-0.37. A G/K giant (R about 4.5 Rsun) is transited and occulted by a companion of about 0.6-0.7 Msun. There are four primaries and two flat-bottomed (total) secondaries, all on one ephemeris, which matches Gaia's period (-0.4 sigma) and Gaia's conjunction phase (+0.3 sigma). The signal sits on the target pixel and is reproduced by three other pipelines.

Checker: Opus 5.5 subagent, 2026-10-21. It resumed and verified the 2026-10-20 checker's step 1.

1. Archives

- `cats.py --r 60`: no entry in VSX, ASAS-SN, Prsa+2022, IJspeert+2024 or Kostov+2025. SIMBAD lists it as SB* only.

- Gaia DR3 (rerun on 2026-10-21): vari_summary 0, vari_eclipsing_binary 0. NSS SB1 gives P 52.3189 +- 0.0188 d, e 0.372 +- 0.012, w 91.4 +- 2.2 deg, K1 24.46 +- 0.37 km/s, gamma 15.03.
- SB9 and WDS return nothing. The nearest VSX entry is 101" away (ROT, G 15.3). ExoFOP has no TOI or CTOI.
- The VizieR all-catalogue 5" cone finds no variability, EB or orbit catalogue other than Gaia. Web searches on the UCAC4, TIC and 2MASS names find nothing.
- **The eclipses appear unreported.**

2. Eclipses (raw QLP SAP_FLUX, q==0, untrimmed; fixed-shape refits agree within 1.5 min)

event	mid-time BTJD	depth	T14	T12	flat
A s7	1499.98113 +- 2.8 min	1.79%	19.28 h	3.13 h	U-shaped
B s33	2205.97845 +- 3.2 min	0.90%	35.10 h	4.77 h	25.6 h
A s34	2232.33376 +- 2.4 min	1.87%	19.30 h	3.23 h	U
A s61	2964.68859 +- 2.8 min	1.84%	19.13 h	2.93 h	U (pre-ingress flagged for scattered light)
B s87	3670.68483 +- 3.0 min	0.89%	34.90 h	4.47 h	26.0 h
A s88	3697.04224 +- 2.0 min	1.83%	19.09 h	2.99 h	U (data gap and pointing jump in the ingress)

- Every conjunction that falls in the six sectors shows its eclipse, and no other conjunction is covered.
- Background is flat through every event. In s61 and s88 the centroids jump only at the data gaps.
- Blind census (q==0, 1-h bins, scatter 0.33-0.40 ppt): no other dip deeper than 0.15%, apart from a single 0.15% bin in s7.
- Common mode: same-CCD stars have a median dip of about 0 at every event. The few other stars dipping at those times are variables 2-50 deg away (e.g. TIC 174797643, an unrelated 24% EB, at s88).

3. Ephemeris

- A alone (4 events) gives P 52.31099(5), chi2 0.27/2. B alone (2 events) gives 52.31094(11).
- Joint fit: **P = 52.31098 +- 0.00005 d, T0_A = 2232.33454 +- 0.00097, T0_B = 2258.28885 +- 0.00165 BTJD**, chi2 0.44/3, all O-C under 1.2 min.
- P_TESS - P_Gaia = -0.0079 d (-0.42 sigma).
- Measured minus Gaia conjunction: A -0.21 to -0.55 d (-0.5 to -0.6 sigma), B -0.19 and -0.41 d (-0.3 sigma). Conditioned on P = P_TESS: A -0.07 d (-0.55 sigma), B +0.08 d (+0.18 sigma).
- B-A phase is 0.49615(3), against Gaia's 0.4935 +- 0.0100 (+0.27 sigma). That gives e cos w = -0.0058, against Gaia's -0.0094.
- Duration ratio T_B/T_A = 1.82, against 2.18 for b=0 from Gaia. The difference comes from the impact parameter: b_B = 2.2 b_A, so B is shortened more (see section 5).

4. Independence

- **TESS-SPOC s61 SAP**: A at 2964.68981 (+1.8 min from QLP), 1.72%, T14 19.11 h. PDCSAP gives +6.1 min and 1.69%.
- **GSFC-ELEANOR-LITE s7 RAW**: A at 1499.98008 (-1.5 min), 1.81%, 19.27 h.
- **TGLC s7**: aperture flux +0.0 min, 1.99%. The PSF flux, which fits the target's own Gaia PSF, gives +1.9 min, 1.64%, 19.27 h. TGLC's cal_* fluxes are detrended and flatten the eclipse, so they were not

used.

- **TESScut difference images** (s34 A, s61 A, s33 B, s87 B): the deficit peaks on the target pixel every time. The 3x3/5x5 depths are 1.8-1.9% (A) and 0.8-0.9% (B). The deficit is more concentrated on the target (74-82% inside the 3x3) than the light is (61-67%).
- **Neighbours**: nothing within 60" is brighter than G 18.35, and each such source would need a dip over 1000%. Further out:
- G 14.99 at 62" would need a 49% A dip; its pixel shows only +0.1 to +0.9% (the wing of the target's PSF).
- G 12.59 at 77" would need a 5.4% A dip; its pixel shows -0.5 to +0.3%.
- Both are excluded.

5. Physics

- f(M) = 0.0634 Msun, a1 sin i = 23.46 Rsun.
- For M1 1.2-1.6: m2(i=90) is 0.59-0.69 Msun and a is 71-78 Rsun.
- At A (r = 0.628a) the giant is behind and the companion transits it (U-shaped, limb-darkened). At B (r = 1.372a) the companion is totally occulted (flat bottom).
- One geometry fits all three durations and the A depth (quadratic limb darkening), chi2 0.13: R1/a 0.061, so R1 is 4.3-4.7 Rsun (TIC gives 4.45); k 0.122, so R2 is about 0.55 Rsun; b_A 0.22, b_B 0.44, i about 88.5-89 deg; e sin w 0.32 [0.295, 0.34].
- With Gaia's e sin w (0.372 +- 0.012) as a prior, chi2 rises to 7.0, which is about 2-2.5 sigma of tension. That is small enough to be the simplified trapezoid and limb-darkening model; it is not a contradiction.
- B depth 0.90% = L2/(L1+L2). This needs a surface-brightness ratio of 0.61, i.e. T2 about 4400 K, which fits a dwarf of about 0.7 Msun (m2 for M1 about 1.6).
- There are no APOGEE or LAMOST RVs, so no independent RV phase test was possible.

6. Next eclipses (BTJD, 1-sigma)

- A (1.8%, 19.2 h): **4377.0848 +- 2.4 min**, 4429.3958 +- 2.4, 4481.7067 +- 2.5, 4534.0177 +- 2.6.
- B (0.9%, 35 h, flat): **4350.7281 +- 2.8 min**, 4403.0391 +- 2.9, 4455.3501 +- 2.9, 4507.6611 +- 3.0.

What would settle it further

- A light-curve model with proper limb darkening, or a few epochs of ground or space RVs, to pin down e sin w (eclipse durations give 0.32, Gaia gives 0.372) and to get absolute masses and radii.
- A spectroscopic logg/R for the giant would fix M1.
- Nothing more is needed to establish that the target is the eclipser.

A188. UCAC4 275-003209 (TIC 176778312)

status
: confirmed (adversarial re-analysis 2026-10-21; eccentric (e ~0.55) F star + ~0.55 Msun (~3900 K) companion; A 9.3-9.7% curved transit, 7.6 h, x5 + B 1.2% flat total occultation, 8.1 h, x7 (2018-2025), SPOC/eleanor/TASOC/TGLC agree; B-A spacing 3.831 d vs Gaia 3.35(16): 2.9 sigma in e cos omega (0.544 vs 0.585(14)), a tension under the rule; B times wander ~50 min against A, non-monotonic, in every pipeline (not apsidal motion, not LTTE; tertiary or spots), B predictions +-40 min; difference images on target in SPOC TPFs and TESScut)

P_days
: 22.359363

eP
: 0.000023

T0_primary_BTJD
: 2575.79945

depth_primary
: 9.3-9.7

depth_secondary
: 1.2

duration_h
: 7.6

source_path
: Fable 5.1\made\appointments\vet176778312\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 176778312 (UCAC4 275-003209, Gaia DR3 5048026421933437696): adversarial check Checker: Opus 5.5, 2026-10-20/21. This run resumed after a usage-limit kill; every earlier receipt was re-run before use.

VERDICT: CONFIRMED. This is a new eccentric (e ~ 0.55), detached eclipsing binary at the Gaia SB1 period, and both conjunctions eclipse. A: a ~0.55 Msun companion transits the F star. The bottom is curved and the depth is 9.3%. B: the companion is totally occulted. The bottom is flat and the depth is 1.2%. The difference images put both kinds of eclipse on the target. I found no prior record of it as an eclipsing binary.

DECIDING NUMBERS

Measurements - 12 eclipses measured from raw SAP_FLUX with a trapezoid x quadratic-baseline model and prayer-bead errors. Every fit re-ran to better than 0.1 min. A x5: s4 1413.11 (missed by the classifier), s4 1435.47, s31 2150.97, s97 3939.72, s97 3984.44. B x7: s4 1431.62, s30 2124.79, s31 2147.15, s31 2169.52, s97 3935.88, s97 3958.24, s97 3980.61 (flagged cadences). - A shape: depth 9.32-9.70% (s4/s31). T14 7.63+-0.04 h, ingress 1.85 h, curved bottom. QLP s97 reads 7.1%, but the TESScut pixels give 9.5%. That is a QLP flux-scale artefact, and it scales both kinds by the same factor. - B shape: depth 1.20-1.25% (pixels at s97 give 1.1%). T14 8.07+-0.08 h, ingress 1.6-1.85 h, flat bottom 4.5-4.8 h.

Ephemeris - A-only fit: P = 22.359367 +- 0.000006 d, T0 = BTJD 2575.79943 +- 0.00033. O-C at most 1.6 min (chi2 20.7/3). - Joint fit (common P, errors scaled x10 because of the B anomaly below): P = 22.359363 +- 0.000023, T0_A = 2575.79945 +- 0.00127, T0_B = 2594.32784 +- 0.00456. - Against Gaia's period: Gaia P 22.35707 +- 0.00356, so the difference is +0.0023 d = +0.64 sigma. - Against Gaia's conjunction predictions: Observed A lands +0.29 to +0.57 d after the predicted A conjunction (+1.0 to +1.4 sigma). Observed B lands -0.20 to +0.09 d from the predicted B (within 0.9 sigma). So the A/B labelling agrees with Gaia's omega.

Orbit shape - Separation: B precedes A by 3.831 d. Gaia's e and omega predict 3.35 +- 0.16 d, a -2.9 sigma tension. Holding omega = 3 deg, the eclipses need e = 0.545 (e cos w 0.544), against Gaia's e 0.585 +- 0.014 (-2.8 sigma). This is a modest tension with the SB1 solution, not a contradiction. - Duration ratio: T14_B/T14_A = 1.063 +- 0.010, which gives e sin w = +0.030 +- 0.005. Gaia gives +0.013 +- 0.020 (omega 1.3 +- 2.0 deg), so the two are consistent. - Which star is in front: take z_primary = r sin(nu+w) sin i, with RV = dz/dt. At nu+w = 90 deg (A) the primary is behind, so the companion transits. That matches what is seen: A is the limb-darkened U and B is the flat total occultation.

Physics - f(M) = 0.0558 Msun. m2(i=90) = 0.50 / 0.545 / 0.58 Msun for M1 = 1.0 / 1.16 / 1.3. a = 38-41 Rsun. - Durations: the predicted central T14 is 8.5-9.1 h; the observed 7.6-8.1 h gives b ~ 0.4. - Radius ratio: the ingress/duration ratio of the total occultation gives k ~ 0.26-0.28, so R2 ~ 0.53 Rsun. That suits a 0.55 Msun dwarf. - Brightness: the surface-brightness ratio is about 0.15, so Teff2 is about 3900 K. That predicts L2/(L1+L2) = 0.013, against the observed 0.012. - One geometry accounts for all the depths, durations and shapes.

Independence - I refit SPOC SAP and PDCSAP (s4/s30/s31), eleanor, TASOC and TGLC (s4). Every clean eclipse reproduces within 0-2.2 min in t0. Depths are 9.0-10.2% (A) and 1.15-1.4% (B). TASOC is shallower (7.0/0.9%) because its aperture is diluted. - Difference images: SPOC TPF s4 (A x2), s31 (A), s30 (B), s31 (B), plus TESScut s97 (A, B). The deficit peaks on the target pixel, and its centroid is within 0.1-0.3 px of the target. The target pixel has the deepest fractional dip: A 9.4-9.6%, B 1.1-1.4%. - Neighbours within 60": the only one bright enough is G 13.19 at 58.4". It would need a 45% dip for A and 6% for B, and the difference images exclude it.

Census - With predicted eclipses masked, s4/s30/s31 have no other dip deeper than 0.15% (1-h scatter 0.3-0.4 ppt). - The one s97 census flag, at 3952.5, is not present in the raw cadences. - Every predicted eclipse with q==0 coverage was seen. - Three fall only on flagged cadences: A s30 2128.61, A s97 3962.08 and B s97 3980.61. I fit the last one anyway and it agrees.

ANOMALY WORTH FOLLOWING: THE B TIMES WANDER AGAINST A A is linear to within 1.6 min, but B is not. The interval from each B to the next A: s4 3.8507 d s30/31 3.8180, 3.8168, 3.8155 d s97 3.8394, 3.8406 d (3.8308 on flagged data) That is a 47-min drop over two years, then a 35-min rise. Each B time has a 1.4-1.7 min error, and a linear fit to B gives chi2 849/5. The wander is in the data, not the method: - SPOC, eleanor and TGLC reproduce the B times to within 2 min. - It is independent of fit window, baseline order and a fixed eclipse shape (variants B1-B3). Possible causes: - Apsidal motion: ruled out. It is monotonic, and at omega ~ 0 it barely moves the separation (6 min/deg). - Light-travel time from a third body: ruled out. It would shift A and B together. - Still open: A tertiary perturbing e dynamically. The separation changes by -12 d per unit e, so this needs delta-e ~ +-0.0015. Spot-distorted occultation profiles of the M-dwarf companion. This would need a large asymmetry. This does not affect the verdict, but B predictions should carry about +-40 min of systematic error.

PRIOR ART (queried 2026-10-20, re-run 2026-10-21) - cats.py --r 60: nothing in VSX, ASAS-SN, Prsa+2022, IJspeert+2024 or Kostov+2025 t2/t3/t4. SIMBAD lists UCAC4 275-003209 as [SB*], which comes from the Gaia SB1 alone. - Gaia DR3: vari_summary and vari_eclipsing_binary have no rows,

and phot_variable_flag is NOT_AVAILABLE. NSS SB1: P 22.357, e 0.585, K1 35.6. RUWE 1.15, rv_amplitude_robust 72 km/s. - SB9 and WDS: nothing. - ExoFOP: no TOI or CTOI. TIC contamination ratio 0.049. - Web search for "TIC 176778312", "UCAC4 275-003209" and "2MASS J03114308-3502457": no hits. => A new eclipsing binary.

EPHEMERIDES A (companion transits the primary): T0 = BTJD 2575.79945 +- 0.00127, P = 22.359363 +- 0.000023 d. The A-only fit gives 2575.79943 +- 0.00033 and P 22.359367 +- 0.000006. B (companion occulted): T0 = BTJD 2594.32784 +- 0.00456, same P. B precedes A by 3.831 d, and the interval wanders by up to about 0.02 d.

NEXT PREDICTED ECLIPSES (BTJD) A: 4342.189, 4364.549, 4386.908, 4409.267, each +-0.002. B: 4338.358, 4360.718, 4383.077, 4405.436, each +-0.005 formal, plus about +-0.03 d (+-40 min) systematic from the B wander.

WHAT WOULD SETTLE IT FURTHER 1. An RV pair near the two conjunctions would give absolute masses. Infrared spectroscopy is the best chance for an SB2, since lines from the 3900 K companion will be faint. 2. More TESS sectors with B coverage would map the B-A separation over time and test for a tertiary. Gaia DR4 epoch RVs and astrometry could show an outer orbit. 3. A proper eclipsing-binary light-curve model (limb darkening, b, k, J) should replace the trapezoids.

FILES (in vet176778312\; downloads in C:\Home\sky\data\qlp-sb-p20-60\vet176778312) - Fits: fits_176778312.*json, fit_*.png - Ephemeris and physics: ephem_176778312.json, betv.txt, physics.txt - Independence and difference images: indep.txt, diff.txt, diff_s97.txt, diff_*.png - Census and neighbours: census*.txt, neighbours.txt - Archive queries: archives.txt, arch2.txt, cats.txt

A189. TYC 7686-1190-1 (TIC 74965330)

status
: confirmed (adversarial re-analysis 2026-10-21; ~5200 K subgiant 3.6 Rsun + ~6100 K, ~1.05 Msun dwarf, e 0.34, i 88.2 deg; the deeper eclipse (B, T0) is the dwarf's total occultation, 10.4-11.1%, 19.0 h, flat 8.6 h, x2; A 6.6-7.0% transit, 17 h, x4 (2019-2025); labelling checked against Gaia's omega (a swap would be 11.5 d off); e, omega within 0.9 sigma; SPOC/eleanor/TGLC agree; difference images on target, four G 12-13 neighbours at 29-50 arcsec excluded; Kostov t2 only)

P_days
: 39.199094

eP
: 0.000029

T0_primary_BTJD
: 2671.06029

depth_primary
: 10.4-11.1

depth_secondary
: 6.6-7.0

duration_h
: 19.0

source_path
: Fable 5.1\made\appointments\vet74965330\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 74965330 (TYC 7686-1190-1, Gaia DR3 5427324157900849408): adversarial check Checker: Opus 5.5, 2026-10-20/21. This run resumed after a usage-limit kill; earlier receipts were spot-checked and the measurements were made fresh.

VERDICT: CONFIRMED. This is an eccentric (e ~ 0.34), detached, totally eclipsing binary at the Gaia SB1 period. Both conjunctions eclipse. The A/B labelling agrees with the Gaia omega: - Timing: both kinds land within 0.35 sigma of their predictions, and swapping the labels would put each eclipse about 11.5 d off. - Durations: the T14 ratio has the sign and size that e sin w predicts. B is deeper than A because the occulted companion, a ~1 Msun dwarf, has a higher TESS surface brightness than the cooler, larger RV star (R ~ 3.6 Rsun). That is physics, not a labelling error. Prior art is Kostov+2025 table2 only (an unvetted ML flag), so this is effectively new.

DECIDING NUMBERS

Measurements - 6 eclipses measured from raw SAP_FLUX (trapezoid x quadratic baseline, prayer-bead errors). A x4 (the companion transits the RV star): s9 1545.7711 +- 2.1 min (missed by the classifier) s36 2290.5566 +- 1.1 s62 2996.1396 +- 2.2 s89 3740.9213 +- 1.4 Depth 6.63-6.99%, T14 16.82-17.15 h, ingress 4.8-5.1 h, bottom 7.0-7.2 h. B x2 (the companion is occulted): s89 3729.4361 +- 1.2 min. Depth 10.41%, T14 18.99 h, ingress 5.20 h, flat bottom 8.60 h. s35 2279.0682 +- 2.5 min. The egress is on

q=4096 stray-light cadences, so the shape was fixed to s89. Including the flagged egress and freeing the shape gives 2279.0695, and the depth there is 11.1%. - The predicted B in s8 (1534.6) falls in the orbit gap.

Ephemeris - A-only fit: P = 39.199084 +- 0.000036, O-C at most 2.9 min (chi2 3.5/2). - B-only fit: P = 39.199131 +- 0.000052 (2 points). - Joint fit: P = 39.199094 +- 0.000029 d, T0_A = BTJD 2643.34748 +- 0.00062, T0_B = BTJD 2671.06029 +- 0.00105 (chi2 4.1/3). - Against the Gaia period: Gaia P 39.19755 +- 0.0105, a difference of +0.15 sigma.

A/B labelling against omega (the check that was asked for) - Conjunctions: observed A lands +0.15 to +0.23 d from the Gaia A (within 0.25 sigma), and B lands -0.23/-0.29 d from the Gaia B (within 0.35 sigma). - Separation: B precedes A by 11.486 d. Gaia predicts A->B 28.175 +- 0.558 d, observed 27.713 (-0.8 sigma). - Duration ratio: T14_B/T14_A = 1.122 +- 0.010, giving e sin w = +0.058 +- 0.004 (Gaia +0.068 +- 0.024). The eclipse solution is e = 0.336, w = 10.2 deg (e cos w 0.331), against Gaia e 0.357 +- 0.024, w 11.0 +- 3.9 (-0.9 and -0.2 sigma). - Geometry: with z = r sin(nu+w) sin i and RV = dz/dt, the RV star is behind at A (nu+w = 90) and in front at B. The flat bottom of B is a total occultation of the smaller companion by the larger RV star. B is longer because B is the slower conjunction (1 + e cos nu smaller), and that matches the e sin w > 0 that Gaia gives. - If the labels were swapped (omega + 180), every eclipse would sit about 11.5 d from its prediction and the duration ratio would invert. Neither happens.

Physics - f(M) = 0.177 Msun. m2(i=90) = 0.93 / 1.05 / 1.23 Msun for M1 = 1.2 / 1.5 / 2.0. a = 62-72 Rsun. - Contact-duration geometry at both conjunctions, with e and w from the eclipses: Durations alone allow k = 0.27-0.44, R1/a = 0.044-0.056, i > 88 deg. The A depth (k^2 L1/Ltot x limb darkening = 0.068) narrows this to k = 0.25-0.28, giving: R1/a ~ 0.054, i ~ 88.2 deg, b_A ~ 0.47, b_B ~ 0.53 R1 ~ 3.6 Rsun (TIC 3.47) and R2 ~ 0.96 Rsun - Surface brightness: J = (L2/L1)/k^2 = 1.5-1.9 in the TESS band. That puts the companion about 0.6-1 kK hotter than the RV star, e.g. a ~5200 K subgiant plus a ~6100 K, ~1.05 Msun dwarf (TIC composite Teff 5588). - The companion supplies ~11% of the light. A ~1 Msun dwarf beside a 3.6 Rsun subgiant/giant fits that. - Depths, durations and shapes fit one geometry. The trapezoid underestimates the T23 of A (7.1 h vs a model 8.0 h), which is the expected limb-darkening bias.

Independence - Same model, other pipelines: A s9: eleanor RAW/CORR -1.7/-0.9 min from QLP; TGLC aperture/PSF +1.8/+3.5 min. A s62: SPOC SAP/PDCSAP -2.6/-1.0 min. Depths range 4.2-7.8% with aperture dilution; the crowding-corrected PDCSAP gives 7.18%. - Difference images: SPOC TPF s62 (A); TESScut s89 (A and B) and s35 (B). The brightest deficit pixel is the target pixel every time, and the deficit centroid sits 0.09-0.21 px (<= 4.4 arcsec) from the target. The target pixel shows A 6.1-6.4% and B 9.2-9.4%. - Neighbours within 60 arcsec: four could in principle make the dips: G 12.17 at 29.2 arcsec (needs A 14% / B 22%) G 12.30 at 49.5 arcsec G 13.34 at 40.1 arcsec G 13.39 at 38.2 arcsec The difference images exclude all four. Their pixels show only 0.7-5.4%, the wings of the target, and the deficit centroid is more than 1 px from each.

Census - Outside the predicted eclipses there is no dip deeper than 0.35% in any sector (1-h scatter 0.63-0.74 ppt). - The s062 blips at 3006.9-3007.2 are the known s062 3007 common mode. - Every predicted eclipse inside the data was seen.

PRIOR ART (queried 2026-10-20, spot-checked 2026-10-21) - cats.py --r 60: VSX none, ASAS-SN none, Prsa+2022 none, IJspeert+2024 none. Kostov+2025 table2 (J/ApJS/279/50) lists TIC 74965330 at 0.001 arcsec. That is an unvetted ML flag list and does not count as catalogued; t3/t4 have nothing. SIMBAD lists TYC 7686-1190-1 as [SB*]. - Gaia DR3: vari_summary and vari_eclipsing_binary have no rows, and phot_variable_flag is NOT_AVAILABLE. NSS SB1: P 39.198, e 0.357, K1 37.7. RUWE 0.95,

rv_amplitude_robust 64 km/s. - SB9 and WDS: nothing. - ExoFOP: no TOI or CTOI. - Web search on TIC/TYC/2MASS names: no hits. => Effectively new. The first characterisation is here, including both eclipse kinds, the ephemeris and the orbit check.

EPHEMERIDES A (companion transits the RV star): T0 = BTJD 2643.34748 +- 0.00062, P = 39.199094 +- 0.000029 d. B (companion totally occulted): T0 = BTJD 2671.06029 +- 0.00105, same P. B precedes A by 11.486 d.

NEXT PREDICTED ECLIPSES (BTJD) A: 4368.108, 4407.307, 4446.506, 4485.705, each +-0.0015 (about 2 min). B: 4356.621, 4395.820, 4435.020, 4474.219, each +-0.0017 (about 2.4 min). Duration ~17 h (A) and ~19 h (B), so a ground campaign needs long nights or several sites.

WHAT WOULD SETTLE IT FURTHER 1. A spectrum (SB2 cross-correlation) would confirm a hotter ~6000 K dwarf beside a cooler subgiant, and with the Gaia K1 it would give K2 and absolute masses. 2. A full eclipse model with limb darkening would pin k, b and J. Even now, the flat bottom of the B occultation directly measures L2/Ltot = 0.104-0.111. 3. A second B with full coverage would firm up the B period on its own; the s35 egress is on flagged cadences.

FILES (in vet74965330\; downloads in C:\Home\sky\data\qlp-sb-p20-60\vet74965330) - Fits: fits_74965330.*json, fits_use.json, fit_*.png - Ephemeris, physics, census: ephem.txt, physics.txt, census.txt - Independence and difference images: indep.txt, diff.txt, diff_*.png - Neighbours and archive queries: neighbours.txt, archives.txt, arch2.txt, cats.txt

A190. TYC 9193-870-1 (TIC 307361219)

status
: confirmed (adversarial re-analysis 2026-10-21; K4-5 dwarf 0.70 Rsun + 0.5 Msun early-M dwarf, e 0.415, i 88.5 deg; 23 flat-bottomed 22% primaries (3.3 h, rms O-C 0.36 min) + 20 grazing V-shaped 0.43% secondaries at phase 0.7136 (2018-2025); e cos omega 0.33 vs Gaia 0.367 (~2 sigma); e sin omega +0.25 branch chosen on physics (the -0.20 branch needs R1 0.46 Rsun); TESS-SPOC TCEs at 38.1 and 27.2 d aliases, never promoted; difference images 1-4 arcsec on target, nothing brighter than G 15.9 within 60 arcsec)

P_days
: 23.828151

eP
: 0.0000013

T0_primary_BTJD
: 2089.30214

depth_primary
: 22

depth_secondary
: 0.43

duration_h
: 3.3

source_path
: Fable 5.1\made\appointments\vet307361219\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 307361219 (TYC 9193-870-1, Gaia DR3 5270203254792431744): adversarial check

Verdict: CONFIRMED. It is an eccentric detached eclipsing binary on target, with P = 23.828151 d. It shows a 22% primary eclipse (the Gaia RV primary, a K4-5 dwarf, is eclipsed at Gaia conjunction A) and a 0.43% grazing secondary at phase 0.7136 (conjunction B). The companion is an early-M dwarf of about 0.5 Msun.

Deciding numbers

- **Period.** 23 primaries over s2-s98 (2018-2025) give a linear ephemeris with rms O-C of 0.36 min. The secondary period agrees with the primary period to +0.007 +- 0.028 min/cycle. P_Gaia - P_TESS = +0.0009 d (0.2 sigma Gaia).
- **Timing against Gaia.** Gaia conjunction A, predicted at the deep eclipse: obs - pred = +0.18 +- 0.39 d. Conjunction B at the secondary: -0.33 +- 0.46 d. Swapping the two assignments puts them 14-60 sigma off.

- **On target.** The TESScut difference images (s63, s64) put the deficit centroid 1.0-1.5" from the light centroid for the primary and 1.7-4.0" for the secondary (one pixel is 21"). There is no Gaia neighbour brighter than G 15.89 within 60". A neighbour would need a primary dip above 1000%, so it cannot host the primary.
- **Independent pipeline.** TESS-SPOC raised two TCEs from s56-s69 (dvr.xml):
- TCE1 epoch 2994.77167 (ours 2994.77174), depth 22.8%, 3.62 h, SNR 932.
- TCE2 epoch 3011.7748 (ours 3011.7752), depth 0.40%, 2.97 h, SNR 19.
- Both had alias periods (38.125 d and 27.23 d) because of the gappy coverage, and neither became a TOI.
- Our fits to TESS-SPOC SAP s62/65/69 match QLP to within 0.4 min.
- **Physics.** A two-sphere model, with e cos w locked by the phase and e sin w fixed at +0.25 (Gaia +0.243), fits both eclipse shapes with these parameters:
- e 0.415, w 37.0 deg, i 88.54 deg
- R1/a 0.0187, R2/a 0.0131, L2/L1 (TESS) 0.074
- model secondary at phase 0.71359, depth 0.40%
- With f(M) = 0.0812 Msun, M1 0.76 gives m2 = 0.51 Msun, a = 37.7 Rsun, R1 = 0.70 Rsun and R2 = 0.49 Rsun: a K dwarf with an early-M dwarf, consistent with the Gaia luminosity (M_G 6.08).
- Impact parameters b_A 0.53 and b_B 0.89, which is why the secondary is grazing and V-shaped.

Archives (step 1)

- **Catalogues.** VSX, ASAS-SN, SB9, WDS, Prsa+2022, IJspeert+2024 and Kostov+2025 have nothing within 60".
- **SIMBAD** lists the star as SB* only (from Gaia NSS). Gaia vari_summary and vari_eclipsing_binary have no rows.
- **ExoFOP** has no TOI or CTOI (rechecked 2026-10-21).
- **Web search** on the names (TYC, TIC, 2MASS J08210101-6909391) found nothing.
- **Prior detection.** The only prior detection is the unpromoted SPOC TCE pair above. **The star is not catalogued as an eclipsing binary.**
- **Gaia NSS SB1:** P 23.8291 +- 0.0046, e 0.4405 +- 0.0141, w 33.5 +- 2.8, K1 35.68 +- 0.78 km/s.

Eclipse measurements (step 2; raw QLP SAP, quality==0, untrimmed trapezoid with quadratic baseline)

- **Primary.** 23 fitted.
- Depth 19.6-23.1% in QLP; the variation is sector dilution (s98 17.6-18.2%; TESS-SPOC 21.5-21.8%).
- T14 3.27-3.31 h in the 200-s sectors, T12 1.34 h, flat bottom 0.58 h.
- t0 errors 0.1-0.3 min.
- **Secondary.** 20 fitted.
- Depth 0.30-0.57% (typically 0.43%).
- T14 2.3-3.0 h (mean 2.7 h), nearly V-shaped.
- t0 errors 2-5 min.
- **Background and centroids.**
- Background drops during the primary (the star's wings fall in the annulus) and never rises. It is flat at the secondaries.

- Centroid shifts are 3-20 mpx in the primary and under 2 mpx in the secondary.
- **Blind census** (>0.2%, 1-h runs): nothing except sector-start and gap-edge features (s62 2989.0, s63 3014.4, s87 3665.4/3676.0, s98 4002.8-4003.2/4009.4). There is no third dip kind.

Ephemerides (step 3)

- **Primary (A; K dwarf eclipsed):** BTJD 2089.30214 +- 0.00005 + E x 23.828151 +- 0.0000013 d (chi2 90/21; errors scaled).
- **Secondary (B):** BTJD 2106.30502 +- 0.00091 + E x 23.828156 +- 0.000020 d (chi2 47/17; rms 4.6 min).
- **Secondary phase:** 0.71356 +- 0.00004.
- **e cos w** from the phase is 0.33 at e ~0.42 (Gaia 0.367).
- **Tension with Gaia's w.** At Gaia's e, the phase needs w = 42.0 deg against Gaia's 33.5 +- 2.8 (3.0 sigma on w alone). With e also free, the LC solution e 0.415 / w 37.0 is about 1.8 sigma in e and 1.2 sigma in w. Gaia predicts B - A = 0.7352; the observed value is 0.7136.
- **e sin w.** The light curve alone is bimodal in e sin w:
- The -0.20 branch has an SSR equal to +0.25 within dchi2 ~1, but needs R1 = 0.46 Rsun and L2/L1 = 0.005. That is unphysical for a 4730 K star with M_G 6.08 and a 0.5 Msun companion.
- e sin w = 0 is worse by dchi2 ~33.
- The +0.25 branch agrees with Gaia (e sin w +0.243).
- The grid run was stopped early. Only -0.20, -0.10, 0.00 and +0.25 were completed (ebgrid.txt, ebgrid_+0.25.png).

Independence (step 4)

TESS-SPOC SAP, the SPOC DV TCEs and the difference images are listed above. The previous checker's free ebfit.py result (e 0.573, L2/L1 2.6) is void: the optimizer lost the secondary and put it at phase 0.745.

Next predicted eclipses (BTJD)

Kind	Epoch	BTJD	Error	UTC
Primary	E95	4352.9765	0.2 min	2026-11-08 11:26
Primary	E96	4376.8046	0.2 min	
Primary	E97	4400.6328	0.2 min	
Primary	E98	4424.4609	0.2 min	
Secondary	E94	4346.1517	3.0 min	2026-11-01 15:38
Secondary	E95	4369.9798	3.0 min	
Secondary	E96	4393.8080	3.1 min	
Secondary	E97	4417.6362	3.1 min	

What would settle it further

- Two or three RV epochs (ground or archival spectra) to tie e and w down. The ~3-sigma tension in w between Gaia SB1 and the eclipse phase is the only loose end.
- A finished e sin w grid with a physically priored L2/L1 (rerun `ebgrid.py ... 0.1,0.15,0.2,0.3,0.35`).
- A spectrum to see whether the M companion is visible (SB2 at K2 ~ 53 km/s).

A191. TYC 3519-1236-1 (TIC 356308234)

status
: confirmed (adversarial re-analysis 2026-10-21; G subgiant 1.47 Rsun + 0.73 Msun K dwarf 0.78 Rsun, i ~90 deg, e 0.092; A 26-27% annular transit x8 (10.2 h) + B 15.2% flat total occultation x6 (11.3 h), 2019-2025, TESS-SPOC/TASOC agree; every element <= 1.1 sigma from Gaia; prior art: WASP0-TR-06 (Kane+ 2005, MNRAS 364, 1091): one partial event in 2000, spurious P 6.03 d, rejected as a planet; never catalogued as an EB; difference images 0.1-0.2 px on target, nothing brighter than G 16.2 within 60 arcsec)

P_days
: 42.860164

eP
: 0.000019

T0_primary_BTJD
: 1689.14215

depth_primary
: 26-27

depth_secondary
: 15.2

duration_h
: 10.2

source_path
: Fable 5.1\made\appointments\vet356308234\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 356308234 (TYC 3519-1236-1, Gaia DR3 1364083026255543936) - adversarial check, 2026-10-21

VERDICT: CONFIRMED. A detached, slightly eccentric EB, P = 42.86016 d: G subgiant (~1.0 Msun, 1.47 Rsun) + K dwarf (~0.73 Msun, 0.78 Rsun) at i ~ 90 deg. The eclipses are on the target, and the orbit agrees with the Gaia SB1 orbit to <=1.1 sigma in every element. Not catalogued as an EB, but not quite unknown: it is WASP0-TR-06 in Kane et al. 2005 (MNRAS 364, 1091). There it is a single partial "transit" from the 2000 Draco-field run: depth 13.2%, egress and bottom missed, spurious P = 6.03 d, rejected as a planet (companion 3.9 R_J). Nothing since identifies it as an EB or gives the real period.

1 ARCHIVES (2026-10-21) - cats.py (60"): nothing in VSX, ASAS-SN, Prsa22, IJspeert24 or Kostov25 t2/t3/t4. SIMBAD has SB* only. - VizieR cones (60"): 0 rows in ZTF-periodic, ATLAS, ASAS-SN II/366, WISE vars, VSX, SB9, WDS, ACVS, KELT, Justesen21 and the TESS OBA EBs. The J/A+A/674/A16 cone hit an unrelated table; the direct Gaia query below is the one that counts. - Gaia DR3 vari_classifier/summary/eclipsing_binary/rotation/LPV: all empty. The NSS solution is SB1: P 42.874 +- 0.013, e 0.091 +- 0.012, w 23.2 +- 9.6, K1 30.94 +- 0.40 km/s. - ExoFOP: no TOI or CTOI. ZTF light curve: 0 rows. - SIMBAD ids include WASP0-TR 06; refs are 2005MNRAS.364.1091K and Gaia DR3. No epoch was published for the WASP0 event, so it cannot be phased. - Web search on the TYC and TIC names: nothing.

2 ECLIPSES (QLP SAP_FLUX, q==0, untrimmed) 14 events. Trapezoid (ecl.py) and mirror-symmetry (ecl2.py) mid-times agree to <=0.1 min (1 min in s14). sec kind t0 BTJD +-min depth T14trap W50 14 P 1689.14172 0.4 25.8% 10.14h 6.54h 26 S 2012.72988 0.3 15.3% 11.26 7.76 26 P 2032.02385 0.3 26.5% 10.14 6.53 40 S 2398.47104 0.3 15.2% 11.39 7.79 40 P 2417.76502 0.2 26.5% 10.21 6.51 51 S 2698.49240 0.3 15.3% 11.33 7.82 (21/72 flagged within 6 h; the allq fit is the same) 53 P 2760.64591 0.2 26.6% 10.15 6.54 54 S 2784.21249 0.3 15.2% 11.27 7.74 74 P 3317.82795 0.3 26.9% 10.16 6.50 78 P 3446.40862 0.3 26.4% 10.16 6.48 79 S 3469.97518 0.4 15.1% 11.34 7.84 80 P 3489.27065 0.3 26.9% 10.15 6.54 (background bump on ingress; +2.3 min O-C) 81 S 3512.83488 0.4 15.2% 11.32 7.69 81 P 3532.12854 0.3 26.7% 10.16 6.48 - Shapes: P has a rounded (limb-darkened) bottom, an annular transit. S is flat-bottomed (~4.3 h), a total occultation. Ingress is 3.5 h in both. - Background: within +-2 sigma everywhere except s40 S, where it is 5 sigma lower (a trend; would brighten, not dim). Centroid shifts are 5-30 mpx with no consistent direction. - Census: unmasked at 3%, it finds exactly these 14 events. Masked at 0.3%, only orbit-edge features remain (s14 1695.9 and the end of the sector).

3 EPHEMERIDES AND ORBIT - P: T0 = 1689.14215 +- 0.00057, P = 42.860164 +- 0.000019 d. chi2 90/6, errors scaled. O-C rms 1.0 min, driven by s80. - S: T0 = 1712.70878 +- 0.00020, P = 42.860150 +- 0.000008 d, chi2 2.6/4. - Against Gaia: 42.87429 +- 0.01282 (-1.1 sigma). P falls at Gaia conjunction A (companion in front) and S at B. With the full covariance every event is -0.7 to -1.1 sigma early (the period mismatch accumulating). S phase is 0.54985 against Gaia's 0.5524 +- 0.0079. - EB model (ebfit2; the 200-s and 600/1800-s data give the same answer): ecosw 0.0783, esinw 0.048 (+-~0.005 syst), e 0.092, w 31.5 deg, i 89.9-90, R1/a 0.0237, R2/a 0.0126, J 0.63, L2/L1 0.18. - Against Gaia: ecosw 0.082 +- 0.012 (0.3 sig), esinw 0.035 +- 0.015 (0.9 sig), w 23.2 +- 9.6 (0.9 sig).

4 INDEPENDENCE - TESS-SPOC SAP/PDCSAP (s14, s26, s40, s81) and TASOC (s14, s26): same times within 0.3 min (1-2 min in s14) and same depths. The SPOC s81 primary is flagged from 3532.3 on (end of sector), but the raw minimum is 0.743. - TESScut difference images (s40, s81, both kinds): deficit centroid within 0.1-0.2 px of the target. Depth 26%/15% on the target pixel, <=0.7% at the G 12.5 neighbour (82"). - No Gaia star within 60" is brighter than G 16.2; a neighbour would need a depth of >2700%. - Conclusion: contamination is excluded.

5 PHYSICS - f(m) = 0.130 +- 0.005 Msun. M1 = 1.0 gives m2 = 0.73, a = 61.9 Rsun, R1 = 1.47, R2 = 0.78, K2 = 42 km/s. M1 = 1.2 gives m2 = 0.81. - J = 0.63 means T2 ~ 4700 K, a K dwarf. - At A the K dwarf transits the G star (27%, annular). At B the G star occults the K dwarf (flat; depth = L2 fraction 0.152, which matches the measured 15.2%). - Predicted central T14 with Gaia e, w: A 11.4 h, B 12.3 h. The model gives 11.3 and 12.4 h. - Everything fits one geometry.

6 NEXT ECLIPSES (BTJD) - P (A, 27%): 4346.4723, 4389.3325, 4432.1926, 4475.0528 (+-1.9 min) - S (B, 15%): 4370.0381, 4412.8982, 4455.7584, 4498.6185 (+-0.8 min)

FURTHER - Settled as an EB on the target. What would add most: SB2 spectroscopy for absolute masses (a totally eclipsing G+K benchmark), fits with free limb darkening and third light, a check of the s80 timing and any ETVs, and the WASP0/SuperWASP archive epochs to extend the baseline.

A192. TYC 3619-1706-1 (TIC 197484019)

status
: confirmed (adversarial re-analysis 2026-10-21; giant 12-16 Rsun (3-4 Msun) + hot ~14,000 K late-B companion 2-3 Msun giving 22-35% of the light, e ~0.01, i 80.6 deg, both eclipses partial; the deeper eclipse (B, T0) is the giant hiding the companion, 7.1-7.3%, 30.0 h, x4; A 0.3-0.5% at phase 0.5001 x4 (Gaia 0.512(8), -1.6 sigma), missed by fits at Gaia's phase; QLP-only, own TESScut photometry and difference images 1.5-5.9 arcsec on target, the G 12.0 star at 37 arcsec loses 2% against 7%; VSX/ASAS-SN ROT 49.65 d: type revision to EA; companion hot for its mass, UV would settle it)

P_days
: 51.88235

eP
: 0.00021

T0_primary_BTJD
: 2835.1830

depth_primary
: 7.1-7.3

depth_secondary
: 0.3-0.5

duration_h
: 30.0

source_path
: Fable 5.1\made\appointments\vet197484019\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 197484019 (TYC 3619-1706-1): adversarial check

Checker: Opus 5.5 subagent, 2026-10-21, resuming the 2026-10-20 checker killed by a usage limit. The file tool refused REPORT.md; this is the checker's full final-message text, saved by Fable 5.1. `report.txt` is the checker's own short version; `progress.md` has every step checkpointed.

Verdict: CONFIRMED. It is a two-eclipse binary, the eclipses are on this star, and they fall at Gaia's period and conjunctions. The VSX type should change from ROT to EA.

What changed from the claim - **Deep eclipse at Gaia conjunction B.** Four events, 7.1–7.3% deep, lasting 30.0 h in total. The 17.5 h in the claim was the width at half depth. The bottom is rounded, so the eclipse is partial. - **The secondary is real.** A 0.3–0.5% eclipse sits at phase 0.5001 (Gaia conjunction A), seen in s17, s57, s76 and s84. The suspected s076 dip near 3380 is one of these, at 3379.98. - **Why it was missed before:** the earlier fits looked at Gaia's predicted phase 0.512, which is 0.6 d away from where the eclipse actually is.

Archives - VSX lists ASASSN-V J222216.80+514740.9 as ROT, P 49.65 d, at 0". ASAS-SN also gives ROT 49.655 d. - SIMBAD lists it only as a spectroscopic binary (SB*). - Gaia: no eclipsing-binary entry. The variability summary says VARIABLE, range 0.150 mag, skewed toward faint (as eclipses would make it), with no class assigned. - Not in Prša, IJspeert or Kostov, SB9, WDS, or ExoFOP (no TOI or CTOI). A web search on its names found nothing. - MAST has only QLP light curves (s16, 17, 56, 57, 76, 77, 83, 84). - Rerunning `cats.py` gave byte-identical output to the earlier checker's.

Deep eclipses (trapezoid fits, stable across window 1.2–2.5 d, baseline order, and flagged or unflagged data)

Sector	Mid-time (BTJD)	Error (d)	Depth	Note
s16	1745.662	±0.007	~7.0%	Egress is flagged during a background excursion, so all cadences were used
s56	2835.1828	±0.001	7.14%	T14 30.1 h, ingress 11.5 h
s77	3405.8875	±0.001	7.3%	
s83	3561.5374	±0.001	7.23%	

Shallow eclipses (shape fixed to the deep one)

Sector	Mid-time (BTJD)	Error (d)	Depth
s17	1771.659	±0.036	0.49 ± 0.07%
s57	2861.050	±0.038	0.46 ± 0.08%
s76	3379.981	±0.014	0.46 ± 0.04% (baseline on one side only, before a data gap)
s84	3587.409	±0.023	0.31 ± 0.05%

- A blind scan over all phases has its strongest signal at phase 0.5015. Three sectors each peak on their own at 0.498–0.501.
- Stacks at four control phases give 0.00–0.04 ± 0.01%.
- Other stars on the same detectors show no matching dip at any of the event times (median −0.02% to +0.002% across 374–465 stars).
- A blind search for other dips found nothing above 1% apart from the four deep eclipses.

Ephemeris

	T0 (BTJD)	P (d)	Fit
Deep alone	2835.1829 ± 0.0015	51.88236 ± 0.00015	χ^2 5.5/2; O−C +12.5 (s16), −0.2, −2.1, +2.0 min
Joint, deep (B)	2835.1830 ± 0.0021	51.88235 ± 0.00021	
Joint, shallow (A)	2861.1286 ± 0.025	same P	phase 0.50008 ± 0.00048 after B

- P differs from Gaia's by −0.36σ.
- Gaia's predicted times: B 2835.17 ± 0.85 (observed 2835.18); A 2861.77 ± 0.85 (observed 2861.05–2861.13, −0.8σ).
- e cos ω from timing is −0.0001 ± 0.0008 (Gaia −0.019 ± 0.012, 1.6σ).
- From my eclipsing-binary model, for the giant's ω: e sin ω = −0.0077 ± 0.0014 (Gaia −0.012 ± 0.012). So e ≈ 0.008 against Gaia's 0.026 ± 0.013, all within 2σ.

Independent checks - No other pipeline exists in MAST, so I did my own photometry on TESScut cutouts. - Deep eclipses: s56 at 2835.1832 and s83 at 3561.535–3561.537, depth 6.1–6.7% (more dilution in my aperture), T14 30.0 h. - Shallow eclipses: s57 at 2861.006 (0.345%) and s84 at 3587.402 (0.243%). - Difference images for s56, s77 and s83: - The lost light is centred 0.07–0.28 px (1.5–5.9") from the target. - The bright neighbour's pixel dims only 1.8–2.2%, against 6.2–7.1% on the target. - Neighbours:

Neighbour	Separation	Deep eclipse would need	Shallow would need	Verdict
G 11.98	36.6"	11.6%	0.69%	Excluded by the difference images

Neighbour	Separation	Deep eclipse would need	Shallow would need	Verdict
G 14.17 (VSX δ Sct)	25.3"	87%	—	Excluded by the difference images
G 15.03	—	191%	—	Impossible

Physics - Mass function f(M) = 0.4595 M⊙, using Gaia K1 44.05 km/s. - My two-sphere model gives: - inclination 80.6° and giant radius 0.149 a; - companion radius 0.031 a (range 0.021–0.038) and surface-brightness ratio 0.083; - the companion supplies 22–35% of the light in the TESS band; - both eclipses are partial, and a total-occultation fit is worse by $\Delta\chi^2 \approx 140$. - Companion mass is 1.40 / 1.96 / 2.41 / 2.81 M⊙ for a giant of 1 / 2 / 3 / 4 M⊙. That gives a separation of 78–111 R⊙, giant radius 11.7–16.5 R⊙ and companion radius 2.5–3.5 R⊙. - **Which star is in front:** - At B (deep), the giant, which is the star Gaia measured, partly hides the companion. - At A (shallow), the companion crosses the giant's edge. - Equal durations and a depth ratio equal to the brightness ratio mean one geometry explains both. - **Temperature:** the companion is about 12× (8–20×) brighter per unit area, so about 14,000 K for a 5383 K giant (11,000–19,000 K across plausible giant temperatures). - A late-B/B9 companion of 2–3 M⊙ with a 3–4 M⊙ giant fits roughly. - There is mild tension, because the companion is hot for the mass allowed. Not a refutation. - The star is reddened (galactic latitude −4.6°), so the TIC temperature is uncertain. - VSX's 49.65 d "rotation" is probably an alias of the eclipses plus slow variability in the light curve. That is not settled.

Next eclipses (BTJD; today is about 4334.5)

Kind	E29	E30	E31	E32	E33
Deep B (±0.005 d)	4339.771	4391.654	4443.536	4495.418	4547.301
Shallow A (±0.026 d formal)	4365.717	4417.599	4469.481	4521.364	4573.246

The shallow eclipses' real timing scatter is about 1.5 h, and each eclipse lasts about 30 h.

What would settle it further - A UV measurement or spectrum (GALEX NUV, Swift/UVOT, Gaia XP) to see the hot companion and pin its temperature. - Photometry of the 4339.77 and 4391.65 deep eclipses, and of the 4365.72 shallow one. - Gaia DR4 epoch radial velocities and photometry. - A VSX revision: ROT → EA, P 51.88235 d, epoch BTJD 2835.1830, depths about 0.08 and 0.005 mag.

A193. TYC 5939-78-1 (TIC 160300523)

status
: confirmed (adversarial re-analysis 2026-10-21; 1.12 Msun F/G + 0.69 Msun K dwarf, e 0.457, i 88.3 deg, both grazing; A 11.1-11.6% transit, 5.15 h, x4 + B 0.9-1.1% occultation, 4.9 h, x3 (2018-2025), six pipelines within 2 min; e cos omega 0.443 vs Gaia 0.4426(138), e sin omega fits; the earlier 9-sigma e sin omega tension was a uniform-disk artefact on a grazing trapezoid; WDS ARA 337 (G 12.5 at 12.9 arcsec) is a background star (plx 1.30 vs 3.52 mas), excluded by the difference image and TGLC PSF depth)

P_days
: 37.312955

eP
: 0.000007

T0_primary_BTJD
: 1473.96743

depth_primary
: 11.1-11.6

depth_secondary
: 0.9-1.1

duration_h
: 5.15

source_path
: Fable 5.1\made\appointments\vet160300523\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 160300523 (TYC 5939-78-1) - adversarial check - VERDICT: CONFIRMED Checked by Opus 5.5, 2026-10-20/21. I resumed a run that a usage limit had cut short, re-ran or spot-checked every earlier output, and finished the rest. Scripts and outputs are in this folder; downloads are in sky\data\qlp-sb-p20-60\vet160300523.

WHAT IT IS A detached, eccentric, grazing eclipsing binary on the target: the Gaia SB1 orbit seen edge-on. A 0.69 Msun K dwarf orbits a 1.12 Msun F/G star at i = 88.3 deg, and both eclipses are partial.

DECIDING NUMBERS - Primary (deep) eclipse = Gaia conjunction A. Depth 11.1-11.6% in QLP SAP (9.4% in s98, where the aperture dilution differs) and 11.3-11.4% in TESS-SPOC PDCSAP. T14 5.15 h, T12 2.05 h, V/U-shaped. Seen at all 4 epochs. - Secondary (shallow) eclipse = conjunction B. Depth 0.94-1.14%, T14 4.75-5.02 h. 3 clean events, plus s98 E67, which falls across a data gap and is consistent with the ephemeris. - Period: 37.312955 +- 0.000007 d from the eclipses, against Gaia 37.3106 +- 0.0092 d (-0.26 sigma). - Secondary phase: 0.774030 +- 0.000036, against Gaia's B-A of 0.7741 +- 0.0077 (0.0 sigma). - Every eclipse falls +0.08 to +0.19 d after the Gaia-predicted conjunction (0.14-0.22 sigma). - Difference images put the deficit on the target to within 0.6" in all 4 (P and S, in s33 and s87). - The 12.9" neighbour (G 12.50) is 0.6 px from the deficit. It is a background star (parallax 1.30 vs

3.52 mas), so WDS ARA 337 is an optical pair. - Five other pipelines time the eclipses within 2 min of QLP. TGLC's PSF photometry, which deblends the neighbour, gives the deepest eclipse (15.3%), which points to the target.

1. ARCHIVES (re-run 2026-10-21) - cats.py: VSX none; ASAS-SN none; no hits in Prsa+2022, IJspeert+2024 or Kostov+2025 t2/t3/t4; SIMBAD lists the star as SB*. - Gaia DR3 vari_summary and vari_eclipsing_binary: 0 rows. The NSS solution is SB1: P 37.31057 +- 0.00921, e 0.4584 +- 0.0121, omega 15.0 +- 1.8 deg, K1 33.14 +- 0.53 km/s, RUWE 1.91. - SB9: 0 rows. WDS ARA 337 is the 12.9" field star. - ExoFOP: yesterday's run found no TOI or CTOI and a TIC contamination ratio of 0.276. Today the connection was refused twice, so that result is unverified. - Web search for "TYC 5939-78-1" OR "TIC 160300523": nothing on the star. - MAST light curves: QLP 6/33/87/98, TARS 6/33/87, TESS-SPOC 6/33, TGLC 6, eleanor-lite 6, TASOC 6. - Status: not catalogued as an eclipsing binary anywhere I checked.

2. ECLIPSES (raw QLP SAP, q==0, no trimming; trapezoid x quadratic baseline over +-0.6 d; the all-cadence fit is identical) sec kind E t0 BTJD +-min depth T14h T12h flat h 6 P 0 1473.96702 0.3 11.07% 5.16 2.07 1.02 33 P 20 2220.22666 0.2 11.60% 5.14 2.05 1.04 87 P 59 3675.43198 0.1 11.40% 5.14 2.05 1.05 98 P 68 4011.24813 0.1 9.37% 5.16 2.06 1.04 33 S 19 2211.79445 1.3 1.09% 4.77 1.84 1.10 87 S 58 3666.99942 1.4 1.14% 4.75 1.89 0.96 98 S 68 4040.12790 1.4 0.94% 5.02 2.14 0.73 98 S 67 4002.8131 3.0 (shape fixed, ingress only; gap 4002.797-4003.010 and a post-gap ramp) O-C +1.6 min; left out of the ephemeris - Common mode (about 400 stars per sector, at three secondaries and one primary): median dip 0.01% or less. The target ranks 1st or 2nd and is the only star above 5 sigma in 3 of the 4 tests. - SAP_BKG excursions in eclipse are small and mostly negative, which would raise the flux rather than fake a dip. - The x-centroid shifts 7-17 mpx in the primaries and about 1 mpx in the secondaries, in proportion to depth, as a fixed blend does when the target dims. - Blind census (0.2% threshold, 1-h bins): s6, s33 and s87 are clean. s98 has only dips 0.1-0.2 d before data gaps (3995.5 at 0.45%; 4009.7, 4024.2 and 4038.5 at 0.2-0.33%). At 3995.5, 27 unrelated stars also dip at more than 5 sigma, so this is a sector systematic.

3. EPHEMERIDES - Primary: BTJD 1473.96743(35) + 37.312955(7) E. O-C -0.59, +0.19, +0.32, -0.32 min. chi2 15.4/2, so the errors are scaled. A quadratic term gives chi2 1.8, but pipelines differ by about 1 min at 30-min cadence, so this is not significant. - Secondary: BTJD 1502.84878(131) + 37.312934(26) E. O-C -0.11, +0.69, -0.55 min; chi2 0.4/1. - Observed minus Gaia-predicted conjunction: A: s33 +0.08 d (0.18 sigma), s87 +0.17 d (0.22 sigma), s98 +0.19 d (0.22 sigma) B: s33 +0.08 d (0.14 sigma), s87 +0.17 d (0.19 sigma), s98 +0.19 d (0.20 sigma) - e cos w from the timing is 0.4425-0.4456, against Gaia's 0.4426 +- 0.0138 (0.0-0.2 sigma). - e sin w from my own limb-darkened two-sphere Keplerian fit to both s87 eclipses (ebfit.py) is 0.115, against Gaia's 0.1187 +- 0.0132. The profile is flat: dchi2 is 0.0 at -0.05, 4.4 at 0, 0.5 at 0.05, 0.2 at 0.1187 and 1.2 at 0.17, so the shapes barely constrain e sin w and Gaia's value fits. - Best fit: e 0.457, w 14.6 deg, i 88.26 deg, R1/a 0.0201, R2/a 0.0123, J 0.30, L2/L1 0.114, rms 0.92 ppt. - A uniform-disk model that reads the trapezoid flat as T23 (geom.py) gave e sin w -0.015, 9 sigma from Gaia. That is an artefact: both eclipses are grazing and limb-darkened, so there is no true T23.

4. INDEPENDENCE - Other pipelines, mid-time minus QLP: TESS-SPOC SAP/PDCSAP (s6, s33) -0.1 to -0.3 min; TGLC aperture/PSF (s6) -0.8/-1.1 min; TASOC (s6) +0.1 min; eleanor-lite (s6) -0.1 min; TARS (s6, s33, s87) -1.3 to +2.1 min. PDCSAP depths are 11.3-11.4% (P) and 1.14% (S). - Difference images (TESScut 15x15): the deficit centroid sits within 0.01-0.03 px of the target; the out-of-eclipse light centroid is pulled about 0.2 px toward the neighbour. The brightest deficit pixel is the target

pixel in 3 of 4 images, and the pixel next to it in S s33. - Neighbour dilution (depth needed = 11.4% x 10^(0.4 dm)): G 12.50 at 12.9" -> 35.6%, excluded by the difference image and the TGLC PSF depth. G 15.16 at 43.2" -> 417%, impossible. Everything else within 60" is fainter than G 17.

5. PHYSICS - f(M) = 0.0987 [0.0941, 0.1035] Msun. With M1 = 1.118 Msun (TIC), m2 = 0.685 Msun at i = 88.3 deg, and a = 57.2 Rsun. - The model gives R1 = 1.15 Rsun (TIC 1.24) and R2 = 0.70 Rsun (main sequence about 0.66). - J = 0.30, against a blackbody value of 0.33 for T2 of about 4400 K. - At A (separation 0.706 a) the K dwarf is in front of the primary; at B (0.896 a) the primary is in front. - Central T14 would be 7.45 h (A) and 9.46 h (B). The observed 5.15 and 4.85 h follow from b_A 0.66 and b_B 0.84, whose ratio is fixed by the orbit. - Both eclipses are partial. A total occultation would be about 8% deep; B is 1.1%. - All of this is one consistent geometry.
6. NEXT ECLIPSES (BTJD) Primary (A, about 11%): 4347.0649, 4384.3779, 4421.6909, 4459.0038, 4496.3168 (+-0.9 min each) Secondary (B, about 1.1%): 4338.6318, 4375.9447, 4413.2576, 4450.5706, 4487.8835 (+-3.4 min each)

WHAT WOULD SETTLE IT FURTHER - Near-IR SB2 RVs or multi-colour primary-eclipse photometry, to pin e sin w and the radii. - More TESS sectors, to tell whether the primary-timing chi2 of 15/2 is systematics or a real ETV. - It is ready for a VSX submission as a new EA.

A194. TYC 8925-649-1 (TIC 309861809)

status
: confirmed (adversarial re-analysis 2026-10-21; subgiant 3.5-4.1 Rsun + 0.8-1.0 Msun companion, e 0.43 adopted from Gaia; seven grazing 1.0-1.1% A transits (9.1 h) + one ingress, 2019-2025, TESS-SPOC/TARS and own TESScut photometry agree; no B (< 0.03%) as predicted (separation at B 1.19x that at A); Gaia P -1.2 sigma; VSX ROT 72.8 d probably real rotation: type revision to ROT+EA; difference images 0.2-0.4 px on target, the G 14.7 star at 36 arcsec loses 0.4% against 1.2%)

P_days
: 38.386132

eP
: 0.000039

T0_primary_BTJD
: 3769.65585

depth_primary
: 1.0-1.1

duration_h
: 9.1

source_path
: Fable 5.1\made\appointments\vet309861809\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 309861809 (TYC 8925-649-1): adversarial check

Checker: Opus 5.5 subagent, 2026-10-21, resuming the 2026-10-20 checker killed by a usage limit. The file tool refused REPORT.md; this is the checker's full final-message text, saved by Fable 5.1. `report.txt` is the checker's own short version; `progress.md` has every number.

Verdict: CONFIRMED. Eclipses of one kind only (the companion transiting the star Gaia measured, at conjunction A), on this star. Seven full events plus one ingress, 1.0–1.1% deep, lasting 9.1 h, grazing. The missing secondary is expected from the orbit. VSX should add EA to the existing ROT rather than replace it.

Archives - VSX lists ASASSN-V J082704.63-613929.2 as ROT, P 72.82 d (amplitude 0.09 in g). - SIMBAD lists it only as a spectroscopic binary (SB*). - Gaia: no variability or eclipsing-binary entry. - Not in SB9, WDS, or ExoFOP (no TOI or CTOI), nor in the eclipsing-binary catalogues. A web search found nothing. - Rerunning `cats.py` gave byte-identical output.

A-eclipse times (BTJD)

Sector	Time	Error (d)	Note
s35	2272.5966	±0.0013	
s37	2310.9820	±0.0021	QLP flags the whole event
s62	3001.9347	±0.0048	QLP flags the whole event
s63	3040.3213	±0.0038	QLP flags the whole event
s68	3155.4813	±0.0030	QLP background problem; SPOC is clean here
s89	3731.2698	±0.0013	
s90	3769.6549	±0.0016	

- In every sector: T14 8.9–9.6 h, ingress 3.3–3.9 h, depth 0.97–1.12%.
- The s10 ingress lands exactly on the predicted 1581.6464 ($\Delta\chi^2$ 2277).
- Other stars on the same detectors show nothing (median −0.06% to +0.07%).
- The blind dip search is clean apart from sector-edge ramps. A deep "dip" at s61 2974.87 is flagged scattered light that also appears in 93 other stars.
- Several events sit next to data gaps, but s90 is mid-orbit and clean. All seven times fit one linear ephemeris to 2.2 min rms over 1500 days, which rules out gap artefacts.

Ephemeris - **T0 = 3769.65585 ± 0.00099 BTJD, P = 38.386132 ± 0.000039 d** (χ^2 2.29/5). - P differs from Gaia's by −0.0066 d (−1.17σ). That accounts for the claimed −1.1σ offsets; at Gaia's own epoch the eclipse ephemeris matches Gaia's A to −0.09σ.

Secondary search at B (Gaia phase 0.7626 ± 0.006; searched phases 0.70–0.83 with durations of 6, 9 and 12 h) - Strongest signal S/N 2.4, i.e. nothing; 3σ upper limit 0.024% (worst case 0.05%). - A blind scan over all phases peaks at S/N 3.9. - With only one kind of eclipse, e cos ω and e sin ω cannot be measured, so Gaia's e = 0.433 and ω = 11.35° are adopted.

Independent checks - TESS-SPOC SAP: s62 3001.9298, s63 3040.3257 (unflagged cadences), s68 3155.4778 (unflagged, 0.97%). - TARS: s35 2272.5969, s37 2310.9828, s62 3001.9393, s63 3040.3188, s68 3155.4826, s89 3731.2692, depths 0.88–1.20%. - My own TESScut photometry: s90 3769.6546 (1.10%) and s89 3731.2696 (1.05%). - Difference images: the lost light is centred 0.22 px (s90) and 0.39 px (s89) from the target. - Neighbours:

Neighbour	Separation	Would need	Verdict
G 14.68	36.3"	31%	Its pixel dims only 0.4–0.5%, against 1.2% on the target
G 16.72	8.3"	203%	Impossible

Physics - Mass function f(M) = 0.1524 M⊙. Companion mass at i = 85° is 0.79 / 0.87 / 0.95 / 1.02 M⊙ for a primary of 1.0 / 1.2 / 1.4 / 1.6 M⊙; separation 58–66 R⊙. - My model, with Gaia's e and ω fixed, gives inclination 84.4–86.0° and primary radius 0.058–0.065 a. That is 3.5–4.1 R⊙, which agrees with the TIC's 3.77 R⊙. - The companion-to-primary size ratio is not pinned down (0.11–0.28). The best fit, 0.275, means a companion of about 1.1 R⊙, which matches a ~0.9 M⊙ dwarf. - **Why no secondary:** the two stars are 1.186× farther apart at B than at A. Measured against the sum of the radii, the sky separation at B is 0.97–1.05 across the model family: - the physically matching solutions (size ratio 0.15 or more) miss entirely at B; - the small-companion solutions graze at most. - So a secondary below 0.03% is predicted. - At A the companion passes in front of the subgiant. - A 1% eclipse cannot produce VSX's 72.8 d, 0.09-mag variation, so that is probably real rotation of the subgiant. Hence ROT+EA.

Next predicted times (BTJD) - **A eclipses:** 4345.4478, 4383.8340, 4422.2201, 4460.6062, 4498.9924 (±0.0016) - **B, where no eclipse is expected:** 4336.33, 4374.72, 4413.10, 4451.49, 4489.88 (±0.23)

What would settle it further - High-cadence photometry of the 4345.45 or 4383.83 eclipse to break the size/inclination degeneracy. - Deep photometry at B (e.g. 4374.72) to test a graze of 0.03% or less. - Gaia DR4 epoch RVs to refine e and ω. - An ASAS-SN/VSX revision to ROT+EA with this ephemeris.

A195. TYC 3481-179-1 (TIC 233640767)

status
: confirmed (adversarial re-analysis 2026-10-21; 1.2 Msun, 1.6 Rsun F star + 0.73 Msun K dwarf, e 0.489, i 89.1 deg; A 15.0-15.9% transit near apastron, 13.1 h, x4 + B 5.5-5.8% total occultation near periastron, 5.8 h, flat 2.5 h, x5 (2019-2024), TESS-SPOC/TARS agree; the classifier's five +1.8 sigma B offsets are one e cos omega offset, +1.55 sigma, joint p 0.19 against Gaia's covariance; no apsidal motion (P_B - P_A 0.04(3) min/cycle); isolated, nothing brighter than G 19 within 60 arcsec; difference images 0.3-0.8 arcsec on target)

P_days
: 30.305947

eP
: 0.000008

T0_primary_BTJD
: 1752.76490

depth_primary
: 15.0-15.9

depth_secondary
: 5.5-5.8

duration_h
: 13.1

source_path
: Fable 5.1\made\appointments\vet233640767\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 233640767 (TYC 3481-179-1) - adversarial check - VERDICT: CONFIRMED (the B offset is not significant) Checked by Opus 5.5, 2026-10-20/21. I resumed a run that a usage limit had cut short, re-ran or spot-checked every earlier output, and finished the rest. Scripts and outputs are in this folder; downloads are in sky\data\qlp-sb-p20-60\vet233640767.

WHAT IT IS A detached, eccentric eclipsing binary on the target: the Gaia SB1 orbit seen edge-on. - Primary eclipse (A): a 0.73 Msun K dwarf crosses a 1.2 Msun, R 1.6 Rsun F star near apastron (separation 1.32 a). Depth 15%, T14 about 13-14 h. - Secondary eclipse (B): a total occultation of the K dwarf near periastron (separation 0.53 a). Depth 5.5%, T14 about 5.8 h, flat for 2.5 h.

DECIDING NUMBERS - The B offset is one phase offset, not five independent ones. The secondary sits at phase 0.34680, against Gaia's B-A of 0.3256 +- 0.0137 (+1.55 sigma). The five B events at +1.75 to +1.87 sigma all come from that single e cos w offset (0.2184 eclipses vs 0.2511 +- 0.0209 Gaia, +1.56 sigma) and the period. - Against Gaia's full covariance (gaiacomp.py): P +1.58 sigma, tA -0.18, phase +1.55, e sin w -0.67, e cos w +1.56. Joint (P, tA, phase): d2 = 4.82 for 3 dof, p = 0.19. Adding e sin w: d2 = 5.93 for 4 dof, p = 0.20. So the Gaia orbit agrees with the eclipses at about 1.3 sigma overall. - Eclipse-derived elements: e = 0.489, w = 243.5 deg, against Gaia's e 0.4958 +- 0.0142 and w 239.5 +- 2.6 deg (-0.45 and +1.5 sigma). - The star is isolated. The brightest Gaia source within 60" is G 19.36, which

would need a depth of about 22000%. The difference images put the deficit on the light centroid to within 0.8". - TESS-SPOC and TARS reproduce the timing to within about 1 min, except at the gap-affected events.

1. ARCHIVES (re-run 2026-10-21) - cats.py --r 60: VSX none; ASAS-SN none; no hits in Prsa+2022, IJspeert+2024 or Kostov+2025 t2/t3/t4; SIMBAD lists the star as SB*. - Gaia DR3 vari_summary and vari_eclipsing_binary: 0 rows. The NSS solution is SB1: P 30.29790 +- 0.00510, e 0.4958 +- 0.0142, w 239.54 +- 2.60 deg, K1 36.80 +- 0.62 km/s, RUWE 1.18, Teff_gspphot 6089 K, logg 4.09. - SB9: 0 rows. WDS: 0 rows. - ExoFOP: yesterday's run found no TOI or CTOI and a contamination ratio of 0.00055. Today the connection was refused, so that result is unverified. - Web search for "TYC 3481-179-1" OR "TIC 233640767": nothing on the star. - MAST light curves: QLP, TARS and TESS-SPOC each have s16/23/24/50/51/77. - Status: not catalogued as an eclipsing binary anywhere I checked.

2. ECLIPSES (raw QLP SAP, no trimming; trapezoid x quadratic baseline; P window +-1.0 d, S window +-0.5 d) sec kind E t0 BTJD +-min depth T14h T12h flat h note 16 P 0 1752.76493 0.7 15.08% 13.04 5.08 2.88 23 P 6 1934.60028 0.6 15.90% 13.21 5.28 2.66 all-cadence fit; 19/27 in-eclipse cadences carry QUALITY 4096, but the flux is smooth 24 P 7 1964.90697 0.7 15.24% 13.11 5.23 2.65 77 P 55 3419.59200 0.4 15.05% 13.11 5.17 2.76 23 S 6 1945.11126 0.3 5.79% 5.79 1.63 2.52 24 S 7 1975.41581 0.3 5.53% 5.97 1.73 2.52 50 S 30 2672.4562 0.7 5.6% (shape fixed) ingress lost in gap 2672.249-2672.430; egress-only fit, error inflated to 2 min 51 S 31 2702.76057 0.3 5.61% 5.67 1.54 2.58 77 S 54 3399.79717 0.4 5.48% 5.65 1.50 2.65 Unusable: s16 S (1763.27; all flagged, at sector end) and s50/51 P (2692.25; between sectors). - Common mode (156-425 stars per sector, at 4 S and 2 P events): median dip under 0.02%. The target ranks 1st or 2nd among the q==0 stars every time. - SAP_BKG drops during the eclipses (strongest in the s77 primary), which is consistent with the target's own light reaching the background annulus. A drop cannot fake a dip. - Centroid shifts are 0-2 mpx, as expected for an unblended star. The exception is s50 S, which is post-gap. - Blind census (1-h bins, eclipses masked): nothing deeper than 0.4% in any sector. At the 0.2% level there are only short 0.2-0.35% noise excursions, mostly in the noisier s50 and s51; none is eclipse-like.

3. EPHEMERIDES - Primary: BTJD 1752.76490(29) + 30.305947(8) E. O-C +0.04, -0.43, +0.64, -0.00 min; chi2 1.3/2. - Secondary: BTJD 1763.27493(56) + 30.305976(21) E. O-C +0.69, -1.36, +2.87, +0.55, -0.67 min; chi2 30.4/3. The 30-min-cadence errors are underestimated: including flagged cadences moves s24 S by 1 min, and SPOC PDCSAP puts it +1.2 min later. The errors are scaled accordingly. - P_S - P_P = 0.041 +- 0.032 min/cycle, so no apsidal motion is detected. - Against Gaia: P(eclipses) - P(Gaia) = +0.0081 d (+1.58 sigma). There are no aliases (P/2 and 2P are each more than 2900 sigma away). - Observed minus Gaia-predicted conjunction: A: s16 -0.12 d (-0.37 sigma), s23 -0.07 (-0.21), s24 -0.07 (-0.19), s77 +0.32 (+0.67) B: s23 +0.57 (+1.87), s24 +0.58 (+1.86), s50 +0.77 (+1.80), s51 +0.78 (+1.80), s77 +0.96 (+1.75) - Geometry from my own limb-darkened two-sphere Keplerian model, fitted to both s77 eclipses at 200-s cadence (ebfit.py): e cos w -0.2184, e sin w -0.4379 (e 0.489, w 243.5 deg), i 89.11 deg, R1/a 0.0314, R2/a 0.0130, J 0.34, L2/L1 0.059, rms 1.29 ppt. - The shapes do pin e sin w here: dchi2 is 25.7 at Gaia's -0.4269, about +-0.002 formally (red noise not included). - Profile dchi2: 453 at -0.39, 104 at -0.46, 867 at -0.50. - Model T14: 14.06 h (A) and 6.12 h (B). The trapezoid underestimates A's because of the limb-darkened ingress.

4. INDEPENDENCE - Other pipelines, mid-time minus QLP: TESS-SPOC SAP/PDCSAP: s16 +0.3; s24 -0.3 to +1.2; s51 -0.4 to -0.5; s77 +0.1 min. PDCSAP depths are 14.8-15.0% (P) and 5.35-5.5% (S). TARS: s16 +0.9; s23 -0.2 (P) and +0.8 (S); s24 0.0/-0.7; s51 -0.5; s77 +2.2 (P) and +0.6 (S) min. - Exceptions: SPOC s23 P: -16 min. SPOC flags most of the in-eclipse cadences there; TARS agrees

with QLP. s50 S: +7 to +19 min in every pipeline, because the ingress is in the data gap. Hence the 2-min error. - Difference images (TESScut 15x15): P s77: 103 in-eclipse cadences; deficit centroid (7.69, 7.97) = light centroid (7.69, 7.97); depth 14.3%, 14.1% and 14.0% in 1x1, 3x3 and 5x5 apertures. S s77: offset 0.8"; depth 5.4-5.5%, flat across apertures. P s24: offset 0.3"; 14.4/14.3/14.3%. S s24: offset 0.7". The deficit share inside 3x3 matches the light share (0.82 vs 0.81). - Neighbour dilution (depth needed = 15% x 10^(0.4 dm)): G 19.36 at 49.5" -> about 22000%. G 20.07 at 40.5" -> about 43000%. Both impossible; nothing within 40".

5. PHYSICS - f(M) = 0.1024 [0.0974, 0.1076] Msun; a1 sin i = 19.1 Rsun. - For M1 = 1.22 Msun (TIC; Teff 6256 K, R 1.7 Rsun), m2 = 0.73 Msun at i = 89.1 deg, and a = 51.1 Rsun. For M1 from 1.1 to 1.35, m2 runs 0.69-0.77. - The model gives R1 = 1.60 Rsun and R2 = 0.66 Rsun (main sequence 0.70). rho1 is about 0.30 solar, which fits a slightly evolved F star and Gaia's logg 4.09. - J = 0.34, against a blackbody value of 0.32 for T2 of about 4500 K. L2/L1 = 0.059. - A (separation 1.317 a, b = 0.46 in units of R1+R2): the K dwarf transits the F star, giving 15%. Central T14 would be about 15.6 h; with b it is 13.8-14.1 h. Observed: 13.1 h by trapezoid, 14.06 h by the model. - B (separation 0.528 a, b = 0.18): the F star totally occults the K dwarf, since 0.0082 a < R1-R2 = 0.0184 a. The predicted depth is L2/(L1+L2) = 5.6%; observed 5.5%. The flat bottom of 2.5-2.65 h is real totality. - Everything fits one geometry, with Gaia's orbit agreeing at the 1.3-1.6 sigma level.
6. NEXT ECLIPSES (BTJD) Primary (A, 15%, about 14 h long): 4359.0764, 4389.3823, 4419.6883, 4449.9942, 4480.3002 (+-1.0-1.1 min) Secondary (B, 5.5%, total, about 6 h): 4339.2829, 4369.5889, 4399.8948, 4430.2008, 4460.5068 (+-2.6-2.8 min) Ephemerides: P 1752.76490(29) + 30.305947(8) E; S 1763.27493(56) + 30.305976(21) E.

WHAT WOULD SETTLE IT FURTHER - Gaia DR4 epoch RVs (or new RVs) fitted with e and w fixed to the eclipse values. This would test whether the 1.5-sigma phase and period offsets are just Gaia DR3 noise; I expect they are. - More 200-s TESS sectors, to tighten the secondary timing and test for apsidal motion. - It is ready for a VSX submission as a new EA (P 30.30595 d, secondary at phase 0.3468).

A196. TYC 4216-238-1 (TIC 233606058)

status
: confirmed (adversarial re-analysis 2026-10-21; G subgiant 2.2-2.4 Rsun (1.1-1.5 Msun) + 0.86-1.02 Msun, ~1.0 Rsun companion of the same surface brightness (J 1.07, L2/L1 0.2), e 0.098, i 89.0 deg; 36 A 17.7% rounded transits (10.9 h) + 44 B 17.1% flat total occultations (12.1 h) over 39 sectors s14-s86, rms O-C 1.0 min, no apsidal motion; TESS-SPOC/TASOC agree; e cos omega -0.0845 vs Gaia -0.059(11), 2.3 sigma, APOGEE 2018 RVs prefer the eclipse e, omega; Kostov t2 flags the 37-arcsec neighbour TIC 233606060, not this star; difference images 0.02-0.15 px on target; ExoFOP unreachable at the check)

P_days
: 21.737275

eP
: 0.000004

T0_primary_BTJD
: 1691.30913

depth_primary
: 17.7

depth_secondary
: 17.1

duration_h
: 10.9

source_path
: Fable 5.1\made\appointments\vet233606058\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

Adversarial check: TIC 233606058 (TYC 4216-238-1, Gaia DR3 2156722539268511360) Second vet run, resumed 2026-10-21 from the run killed on 2026-10-20. I re-verified the inherited Gaia, MAST and cats queries. Scripts and outputs are in this folder; downloads are in sky\data\qlp-sb-p20-60\vet233606058.

VERDICT: CONFIRMED This is a detached, slightly eccentric eclipsing binary on the target, at the Gaia SB1 period. It shows a transit (A, 17.7%) and a total occultation (B, 17.1%) at every covered conjunction from 2019 to 2024. I found it in no EB catalogue I could query; ExoFOP could not be checked.

1. ARCHIVES (what was queried and what came back) - cats.py --r 60 (rerun): no VSX, ASAS-SN, Prsa+2022, IJspeert+2024, or Kostov+2025 t2/t3/t4 entry. SIMBAD has only "SB*". - Gaia DR3 TAP: vari_eclipsing_binary, vari_summary and vari_classifier_result return 0 rows; phot_variable_flag is NOT_AVAILABLE. The NSS SB1 solution: P 21.73575(313) d, e 0.0826(116), w 135.7(6.3) deg, K1 42.02(44) km/s, corr(w, Tp) = 0.995. - VizieR 60" cones (archives2.txt) on about 28 catalogues returned 0 rows in each of these: Variable-star catalogues: VSX, GCVS, ASAS-SN II/366, Hipparcos/Tycho var, NSVS, ZTF (Chen+2020), ATLAS (Heinze+2018), WISE vars Eclipsing-binary catalogues: Prsa+2022, IJspeert+2021, IJspeert+2024, Justesen+2021, Rowan+2022 Other TESS / survey variability: Fetherolf+2023 Gaia DR3 variability: I/358 Binary /

multiple-star catalogues: SB9, WDS Spectroscopic surveys: RAVE, LAMOST, GALAH Exact-TIC lookups in Prsa+2022 and IJspeert+2024 also returned 0. - Non-empty but not prior art: - Kostov+2025 table2, the unvetted ML list, contains the 37" neighbour TIC 233606060 (Tmag 12.8), not the target. The likely cause is our eclipses leaking into that star's aperture. - Green+2023 lists the target with score 0.0075, a non-detection. - KELT (Oelkers+2018) lists only rms statistics. - The cone on "J/A+A/674/A16" returned a non-EB table and is not the Mowlavi EB catalogue; the Gaia TAP result above is the authoritative one. - APOGEE DR17 has 2 visits, used in section 4. - ExoFOP: connection refused, both from Python and from WebFetch. NOT CHECKED. - Web search on "TYC 4216-238-1", "TIC 233606058", the Gaia id and 2MASS J18512734+6120136 found nothing relevant.

2. ECLIPSE MEASUREMENTS Method: raw QLP SAP_FLUX, q==0, no trimming; a trapezoid times a quadratic baseline over +-1 d. Flagged cadences were fitted too (ecl_allq). I fitted 84 events in 39 sectors (s14-s86) and dropped four: - s55 A E52 (scattered light) - s59 A E57 (partial) - s84 A E88 (no ingress) - s57 B E54 (no ingress) I added s17 A E4 and s21 A E9, which can only be measured with the flagged cadences. That leaves 80 events: 36 A and 44 B.

Kind Depth T14 T12 Flat Shape A 17.66% (sd 0.30) 10.85 h (sd 0.05) 3.57 h 3.72 h rounded bottom (transit) B 17.12% (sd 0.28) 12.06 h (sd 0.07) 3.60 h 4.87 h flat bottom (total occultation)

- Background: the in-eclipse change is -0.3% (A) and -1.5% (B) of the target flux, against -0.04% and -0.15% at control phases. The background is not the cause; the small drop fits target light falling in the background annulus.
- Centroids: SAP_X/Y move by 15-20 millipixels in eclipse and about 0 at control phases. The difference image explains this: the light centroid is pulled toward the G 13.5 neighbour when the target dims.
- Blind census of all sectors (outside +-0.4 d of both eclipses, threshold 0.3%): nothing, apart from scattered-light noise in s54/s55 (maximum 4.4%) and one 0.33% wiggle in s22. There is no third family of dips.

3. EPHEMERIDES AND ORBIT Kind T0 (BTJD) P (d) chi2/dof (1-min floor) rms A 1691.30913 +- 0.00020 21.737275 +- 0.000004 32.3/34 0.98 min B 1701.00795 +- 0.00020 21.737274 +- 0.000004 42.8/42 1.00 min

- P_B - P_A = -0.002 +- 0.007 min per cycle, so no apsidal motion is detected over 6 years. A quadratic term is not needed.
- The TESS period minus Gaia's is +0.0015 d, which is +0.49 sigma.
- QLP s80 runs +3 min late for all three of its events. TESS-SPOC s80 is on the ephemeris, so this is a QLP timing artefact.
- The "+0.3 d A offset":
- Measured minus Gaia-predicted conjunction is A +0.24 to +0.38 d and B -0.11 to +0.03 d. With independent errors that is at most 0.54 sigma; with corr(w, Tp) = 0.995 it is 0.8-1.2 sigma.
- The cause is the B-A phase: measured 0.44618, Gaia-predicted 0.4627 +- 0.0067 (-2.5 sigma). That gives e cos w = -0.0845 against Gaia's -0.059 +- 0.011 (-2.4 sigma).
- The duration ratio T_B/T_A is 1.1117 +- 0.0013, against Gaia's 1.121 +- 0.023 (-0.4 sigma).
- My own two-sphere EB fit (ebfit.txt: s56/58/80/81, rms 0.91 ppt, chi2/N 1.00) gives: e cos w -0.08448 e sin w +0.0488 +- 0.0004 e 0.0975 (Gaia 0.083 +- 0.012) w 150.0 deg (Gaia 135.7 +- 6.3) i 88.99 deg R1/a 0.0535 R2/a 0.0234 J 1.07 L2/L1 0.205
- The APOGEE RVs prefer the eclipse values (section 4). Gaia's e and w are about 2 sigma off, most likely from blending with the companion's lines (L2/L1 = 0.2 at nearly equal Teff). This is a refinement, not a problem.

4. INDEPENDENCE - Other pipelines (secondpipe.txt): - TESS-SPOC SAP (s14/26/80/84/85) and TASOC (s14/26) reproduce both kinds with the same T14 and T12. Mid-times agree with QLP to within 0.2-4 min. - TESS-SPOC SAP depths are 15.3-16.4%. That is QLP's depth times CROWDSAP (0.90-0.95), i.e. 17.1-17.5% undiluted. - Difference images (TESScut s80 at 200 s, and s14; diff.txt and diff_*.png): - The flux deficit centres on the target: 0.02 px off in s80, 0.15 px in s14. The brightest deficit pixel is the target pixel, at 17.0% (A) and 16.6% (B). - The G 13.52 neighbour's pixel loses only 4-7%, which is the target's wings. - Neighbours (neighbours.txt): all 15 Gaia sources within 60" would need dips above 100%. The brightest, G 13.52 at 36.5", would need 128% (A) and 124% (B). Unresolved companions are unlikely: RUWE 0.83, ipd_frac_multi_peak 0. - Spectroscopy (apogee_rv.txt): APOGEE DR17 has 2 visits in 2018, 1.5 years before TESS: RV -71.25 and -58.28 km/s, Teff 5783, logg 4.00, vsini 8.7. - Phased on the TESS A ephemeris, the SB1 orbit predicts them to within -1.1 / +0.7 km/s with the eclipse e and w, and +1.2 / +3.5 km/s with Gaia's. - Swapping A and B gives +14 / +20 km/s, which is ruled out. - This independently ties the eclipses to the RV primary's orbit and confirms that A is the companion passing in front of the primary.
5. PHYSICS - f(M) = 0.1654 +- 0.005 Msun; a1 sin i = 17.99 Rsun; i = 89.0 deg from the light curve. - For M1 = 1.1-1.5 Msun: m2 0.86-1.02 Msun a 41-45 Rsun R1 2.19-2.38 Rsun R2 0.96-1.04 Rsun logg1 3.80-3.86 - The TIC radius of 2.53 Rsun, de-blended by sqrt(1 + L2/L1) = sqrt(1.205), gives 2.30 Rsun, which matches. The picture is a G subgiant (about 1.3 Msun, 2.3 Rsun, 5800 K) with a roughly 1 Msun G dwarf of the same surface brightness (J = 1.07). - Which star is in front: - At A (nu = 90 - w), the companion transits the primary. That gives the deeper, limb-darkened, rounded eclipse. - At B, the primary covers the companion completely. That gives the flat bottom, with depth = L2/(L1+L2) = 17.0%. - Separations are 0.939a at A and 1.054a at B, so B should be longer by a factor of 1.12; 1.11 is observed. The b=0 durations from the TIC radius are 12.6-14.6 h; the observed ones are shorter, consistent with b of about 0.4 (in units of R1+R2). - Depths, durations, the flat-bottomed B, the near-equal depths (J about 1) and the light ratio needed for an SB1 (a companion about 1.7 mag fainter) all fit one geometry.
6. NEXT ECLIPSES (BTJD; 1-sigma from the ephemeris is about 0.7 min, allow +-3 min) - A (transit, 17.7%, T14 10.9 h): 4343.2567, 4364.9940, 4386.7312, 4408.4685 - B (total occultation, 17.1%, T14 12.1 h, flat): 4352.9553, 4374.6926, 4396.4299, 4418.1672 - General: T_A = 1691.30913 + 21.737275 E; T_B = 1701.00795 + 21.737274 E.

WHAT WOULD SETTLE IT FURTHER 1. ExoFOP (TOI/CTOI or notes), checked from a machine that can reach it. 2. Spectra at quadrature to detect the secondary's lines (expected K2 of about 50-60 km/s, L2/L1 about 0.2) and get absolute masses. With i = 89 deg, this is a well-detached G subgiant + G dwarf benchmark. 3. A proper EB model with fitted limb darkening, to tighten J and k and set the w difference from Gaia. 4. Report it as new: it is not in VSX or the TESS EB catalogues.

A197. TYC 7612-1193-1 (TIC 300202855)

status
: confirmed (adversarial re-analysis 2026-10-21; F star 1.69 Rsun, 1.3 Msun + 0.57 Msun M dwarf, e 0.36, i 87.8 deg; A 6.4-6.9% flat-bottomed transit, 5.9 h, x4 at rms 0.10 min + B 0.7-0.8% total occultation x5 at phase 0.2792 (2019-2025); Gaia P +0.79 sigma, e, omega within 0.6 sigma; SPOC 2-min and TESS-SPOC reproduce both; prior flag: ExoFOP CTOI 300202855.01 (TSTPC single transit, PC, 2024-08-28, 7.85 ppt) is the s32 secondary eclipse, not a planet; difference image 0.4-2.2 arcsec on target)

P_days
: 25.762668

eP
: 0.0000034

T0_primary_BTJD
: 2211.64639

depth_primary
: 6.4-6.9

depth_secondary
: 0.7-0.8

duration_h
: 5.9

source_path
: Fable 5.1\made\appointments\vet300202855\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 300202855 (TYC 7612-1193-1, Gaia DR3 2884556351606337920): adversarial check

Verdict: CONFIRMED. It is an eccentric detached eclipsing binary on target, with P = 25.762668 d. The F-type Gaia RV primary is eclipsed at conjunction A, giving a 6.9% primary with a flat bottom, 5.9 h long. The secondary is a 0.75% flat-bottomed occultation of an early-M companion (about 0.57 Msun) at phase 0.2792.

On the period. The period difference from Gaia is not real tension: P_TESS - P_Gaia = +0.0155 d is 0.79 sigma of Gaia's 0.0197 d error.

On the CTOI. ExoFOP CTOI TIC 300202855.01 (TSTPC single transit, PC, 2024-08-28; epoch BTJD 2193.078 +- 0.12, 7.85 ppt, 5.3 h) is this binary's secondary eclipse in s32 (ours: 2193.0730). It is not a planet. The CTOI should be re-dispositioned as an EB, P 25.7627 d.

Deciding numbers

- **Period.** Two consecutive s98 primaries give P = 25.76247 directly. With s33 that fixes the cycle count (N = 71). The neighbouring counts would need 26.131 d or 25.405 d, which is impossible.
- **Ephemeris fit.** Four primaries fit a linear ephemeris with rms 0.10 min. Five secondaries (s7 to s98, 2019 to 2025) give P_S - P_P = -0.010 +- 0.049 min/cycle.
- **Gaia timing.** Conjunction A, predicted at the deep eclipse: obs - pred = -0.24 +- 0.52 d (TESS P) or +0.86 +- 1.49 d (Gaia P). Conjunction B: +0.04 +- 0.57 d or +1.14 +- 1.51 d. The swapped assignment is 13-23 sigma off.
- **Geometry against Gaia.** The constrained model at Gaia-like e sin w gives e 0.359 and w 170.4. Gaia SB1 has e 0.378 +- 0.032 and w 171.2 +- 4.7: agreement within 0.6 sigma. e cos w from the eclipse phase is -0.354 (Gaia -0.373).
- **On target.** In the TESScut s87 difference image the deficit centroid is 0.4" from the light centroid for the primary and 2.2" for the secondary (pixel 21"). Deficit share in the 3x3 is 0.76/0.83, against a light share of 0.75/0.74.
- **Neighbours.** The only Gaia neighbours within 60" brighter than G 18.5:
- G 16.07 at 34.4" (x68): primary would need 470%, impossible; secondary 51%.
- G 17.15 at 21.2" (x181): impossible for both.
- **Independent pipeline.** SPOC 2-min (s87, s98) and TESS-SPOC (s32, s33) SAP:
- Primaries at QLP times within -0.5..-0.1 min, depth 6.39-6.47% in every sector. The QLP s98 depth of 5.3% is QLP dilution.
- Secondaries within -1.7..+2.0 min, depth 0.67-0.78%.
- SPOC recovers a secondary flagged in QLP: s98 E70 at 4022.2272 +- 3.0 min, O-C +1.9 min.

Archives (step 1, previous checker; ExoFOP rechecked)

- VSX, ASAS-SN, SB9, WDS and the EB catalogues (Prsa, IJspeert, Kostov) have nothing.
- SIMBAD lists TYC 7612-1193-1 as SB* only. Gaia vari tables have no rows.
- Web search: nothing.
- **ExoFOP has the CTOI above**, the only prior flag. It was raised as a single transit and has no period.
- **Gaia:** G 11.50, BP-RP 0.69, plx 2.295 mas, Teff 6160 K (gspphot; TIC 6433 K, R 1.65 Rsun, M 1.3 Msun), RUWE 0.98.
- **Gaia NSS SB1:** P 25.7471 +- 0.0197, e 0.378 +- 0.032, w 171.2 +- 4.7, K1 28.91 km/s.

Eclipses (step 2; raw QLP SAP, quality==0, untrimmed trapezoid with quadratic baseline)

- **Primary.** Measured at s33 2211.64639, s87 3680.11844, s98 4015.03324 and 4040.79571 (each +- 0.3 min).
- T14 5.91-5.93 h, T12 2.24-2.31 h, flat 1.31-1.44 h.
- Depth 6.9% (5.3-5.4% in QLP s98).
- **Secondary.** Measured at s7 1497.486 (+- 4.6 min), s32 2193.0730 (3.3), s33 2218.8423 (3.1), s87 3687.3127 (2.6) and s98 3996.4625 (2.2).
- Depth 0.64-0.79% (0.30% in 30-min s7).
- T14 5.6-6.1 h, flat 0.6-2.3 h.

- **Flagged-only windows** also dip on time: s32 P 2185.88 (~6%, scattered light), s98 P 3989.27 (~4.5%), s98 S 4022.21, s6 S 1471.72 and s7 P 1516.06 (ingress).
- **Background and centroids.** No background rise that could make the dips; centroids move under 1 mpx except in the noisy s87.
- **Census.** Nothing else, apart from the s7 sector-start ramp (1492.45-1492.87, 0.5%, not at any conjunction) and the s98 4003.0 gap edge (also present in TIC 307361219: common mode).

Ephemerides (step 3)

- **Primary (A, F star eclipsed):** BTJD 2211.64639 +- 0.00020 + E x 25.762668 +- 0.0000034 d.
- **Secondary (B):** BTJD 2218.83963 +- 0.00172 + E x 25.762661 +- 0.000034 d.
- **Secondary phase:** 0.27921 +- 0.00007. Gaia predicts 0.2677.
- **e sin w.** The duration ratio is T_A/T_B = 1.00. e sin w is not separable from the radius ratio in the light curve alone: e sin w 0 and +0.06 give the same chi2. Gaia's e sin w (+0.058) selects the physical branch.

Physics (step 5)

- **Mass function.** f(M) = 0.0525 Msun. M1 1.2-1.4 gives m2 = 0.54-0.59 Msun and a = 44-46 Rsun.
- **Model at e sin w +0.06.** R1/a 0.0374, R2/a 0.0110, giving R1 = 1.69 Rsun (TIC 1.65) and R2 = 0.50 Rsun. i 87.75 deg, L2/L1 (TESS) 0.014.
- **Companion light.** The expected L2/L1 for a 0.57 Msun M0V against a 6200-6400 K F star is about 0.015-0.02. The flat-bottomed secondary is a total occultation whose depth equals the companion's light share.
- **The other branch is excluded.** At e sin w 0 the model needs a 1.3 Rsun companion with L2/L1 0.10, which an SB1 with m2 about 0.57 cannot be.
- Everything (depths, durations, shapes, the companion's light, the Gaia orbit) fits one geometry.

Next predicted eclipses (BTJD)

Kind	Epoch	BTJD	Error	UTC
Primary	E83	4349.9478	0.5 min	2026-11-05 10:44
Primary	E84	4375.7105	0.5 min	
Primary	E85	4401.4732	0.5 min	
Primary	E86	4427.2358	0.5 min	
Secondary	E83	4357.1405	4.8 min	2026-11-12 15:22
Secondary	E84	4382.9032	4.8 min	
Secondary	E85	4408.6658	4.8 min	
Secondary	E86	4434.4285	4.9 min	

What would settle it further

- An RV spectrum to detect the M companion (K2 ~ 65 km/s): that gives masses directly.
- Reporting to ExoFOP that CTOI 300202855.01 is the secondary eclipse of a 25.763-d EB.
- A light-curve fit with a mass-radius prior to pin k and i.

A198. TYC 8787-921-1 (TIC 320805993)

status
: confirmed (adversarial re-analysis 2026-10-21; F8/G0 1.1 Msun, 1.4 Rsun + 0.66 Msun, 0.65 Rsun K5 dwarf (5.7% of the light), e 0.314, i 89.5 deg; A 22.5-22.7% full transit near periastron, 7.55 h, x3 + B 5.3-5.4% near-total occultation near apastron, 13.4 h, x3 (2019-2025), O-C <= 0.6 min; e cos omega, e sin omega, e, omega all within 0.8 sigma of Gaia; depths from own TESScut photometry, QLP diluting s101/s103 by 15-35% in this field; TESS-SPOC agrees in s13/s27; difference images 0.1 px on target; nothing brighter than G 19.1 within 60 arcsec)

P_days
: 56.173887

eP
: 0.000011

T0_primary_BTJD
: 1676.76580

depth_primary
: 22.5-22.7

depth_secondary
: 5.3-5.4

duration_h
: 7.55

source_path
: Fable 5.1\made\appointments\vet320805993\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

Archived source: sanitized report text.

TIC 320805993 (TYC 8787-921-1, Gaia DR3 6447155071357750400) - adversarial check, 2026-10-21

VERDICT: CONFIRMED. A detached eccentric EB, P = 56.17388 d, e = 0.31, w = 109 deg, i = 89.5 deg. It is an F8/G0 primary (~1.1 Msun, ~1.4 Rsun, Teff 5900) and a K-dwarf companion (~0.66 Msun, ~0.65 Rsun, 5.7% of TESS light). - The eclipses are on the target, and the orbit agrees with Gaia SB1 to within 0.8 sigma in e cos w, e sin w, e and w. - No prior art found: not in any EB or variable catalogue, and the Gaia DR3 vari tables are empty. - One new finding: QLP under-reports the B depth in s101/s103 (4.6/3.5%). Own TESScut photometry gives a constant 5.3-5.4% in all three B events, and the B depth does not vary.

1 ARCHIVES (2026-10-21) - cats.py (60"): nothing in VSX, ASAS-SN, Prsa22, IJspeert24 or Kostov25 t2/t3/t4. SIMBAD has SB* only; its only reference is Gaia DR3 2023A&A...674A..34G. - VizieR cones (60"): 0 rows in ZTF-periodic, ATLAS, ASAS-SN II/366, WISE vars, VSX, SB9, WDS, ASAS ACVS, KELT, Justesen21 and the TESS OBA EBs. The J/A+A/674/A16 cone hit an unrelated X-ray table. - Gaia DR3 vari_classifier/summary/eclipsing_binary/rotation/LPV: all empty. phot_variable_flag NOT_AVAILABLE. - NSS SB1: P 56.168 +- 0.029, e 0.343 +- 0.038, w 111.3 +- 3.8, K1 26.78 +- 0.77 km/s, significance 35. - RUWE 1.52, ipd_frac_multi_peak 1. Only mildly elevated; the photocentre orbit could cause it. - ExoFOP: no TOI or CTOI. The TIC contamination ratio is 0.006. - No ZTF (dec -58.7). - Web search on the TYC, TIC and 2MASS names: nothing.

2 ECLIPSES Six events: P (deeper) x3 at A, S x3 at B. - QLP SAP q==0 (trapezoid ecl.py, mirror ecl2.py): s13 P 1676.76567 21.8%; s67 P 3137.28746 22.6%; s94 P (egress flagged in QLP, so own photometry is used); s27 S 2038.14965 5.27%; s101 S 4116.58093 4.56%; s103 S 4172.75443 3.54%. - Own TESScut photometry (r=1.5 px aperture, local background, q==0; cutfit.py), used for the ephemeris: sec kind t0 BTJD +-min depth T14trap T12 flat 13 P 1676.76578 0.3 22.70% 7.53h 2.57h 2.39h 67 P 3137.28727 0.8 22.45% 7.61 2.62 2.36 (allq: same t0) 94 P 3867.54726 0.6 22.65% 7.55 2.59 2.38 27 S 2038.15009 0.4 5.37% 13.44 5.08 3.28 101 S 4116.58283 0.3 5.36% 13.42 5.11 3.20 103 S 4172.75761 0.6 5.31% 13.49 5.13 3.24 (allq: same) - Depth check, 1x1 / 3x3 / 5x5 apertures on the cutouts: P 22.7-23.0% and S 5.2-5.5% in every sector. The QLP s101/s103 S depths (and the s94 P depth of ~20%) are diluted in QLP: a pipeline background/aperture issue, not astrophysical. TESS-SPOC SAP/PDCSAP: s13 P 22.3-22.6%, s27 S 5.24-5.31%. - Timing across pipelines: S mid-times differ between pipelines by up to 2-5 min (shallow 13-h events). Treat the S timing error as ~2 min. - Background and centroids: SAP_BKG is smooth through all six events (within +-1.9 sigma). The s67 eclipse sits ~0.3 d before a scattered-light ramp, not on a known common-mode time. - Census: unmasked at 2%, it finds exactly the six events (plus a flagged spike at 3869.62). Masked at 0.3%, every hit is at a segment start right after heavily flagged scattered-light/momentum blocks (s94 3869.7, s101 4113-4114, s103 4153) or at a sector start. No other dips. - Every covered A and B window contains its eclipse. There are no empty covered windows.

3 EPHEMERIDES AND ORBIT (TESScut timings) - P(A): T0 = 1676.76580 +- 0.00021, P = 56.173887 +- 0.000011 d, chi2 0.6/1, O-C <= 0.6 min. - S(B): T0 = 1701.10690 +- 0.00064, P = 56.173863 +- 0.000018 d, chi2 3.9/1 (errors scaled). - With QLP S timings: P = 56.173816 +- 0.000013. The P-S difference is 1-2 sigma, i.e. the S timing systematics; there is no evidence of apsidal motion. - Against Gaia: 56.16849 +- 0.02853, -0.2 sigma. - Against the Gaia conjunctions (full covariance): A events -0.34/-0.20/-0.13 d (-0.33/-0.11/-0.06 sigma); B events +0.68/+0.85/+0.86 d (+1.06/+0.58/+0.57 sigma). S phase is 0.43331, against Gaia 0.4168 +- 0.0203 (+0.8 sigma). - EB model (ebfit3 on own photometry; two LD spheres, Kepler orbit, u=0.45 fixed): ecosw -0.1002, esinw 0.297 (formal +-0.002; realistic +-~0.01), e 0.314, w 108.6 deg, i 89.50, R1/a 0.01855, R2/a 0.00869 (k 0.47), J 0.27, L2/L1 0.060. Model T14: A 8.3 h, B 14.4 h. - Against Gaia: ecosw -0.124 +- 0.030 (+0.8 sigma), esinw 0.319 +- 0.032 (-0.7 sigma), e 0.343 +- 0.038 (-0.8 sigma), w 111.3 +- 3.8 (-0.7 sigma). - The duration ratio B/A (T14 13.4/7.55 = 1.78) confirms the large positive e sin w: B occurs near apastron.

4 INDEPENDENCE - TESS-SPOC SAP/PDCSAP: s13 P at 1676.76564-6 (same as QLP to 0.0 min); s27 S at 2038.1501 (+0.6 min against QLP). - SPOC s67 flags the whole eclipse (scattered light), but the raw SAP shows it at 0.79. - Own TESScut photometry of all six events (above). - Difference images (s67 P, s94 P, s27 S, s101 S): the deficit peaks on the target pixel, centroid within 0.1 px (0.5-2"). Depth on the target pixel is 22.4-22.9% (P) and 5.3-5.4% (S). 81-85% of the deficit lies in the 3x3, versus 75-77% of the light. The neighbour pixels show only the wings of the target PSF. - Neighbour dilution: there is nothing brighter than G 19.1 within 60". Beyond that:

neighbour sep dG depth it would need for the 23% P G 13.74 71" 2.7 266% G 15.37 65.5" 4.3 1200% G 15.74 62" 4.7 1700% G 19.11 19" 8.0 >30000%

Contamination is excluded.

5 PHYSICS - f(m) = 0.093 +- 0.008 Msun.

M1 m2(i=89.5) a R1 R2 K2 1.0 0.63 72.6 Rs 1.35 0.63 1.1 0.66 74.5 Rs 1.38 0.65 45 km/s 1.3 0.72 78.1 Rs 1.45 0.68

- The primary has logg 4.2 (Gaia GSP-Phot 4.0, Teff 5940); the TIC gives 1.58 Rsun from blended light.

- J = 0.27 means T2 ~ 4100-4300 K: a K5 dwarf of ~0.65 Msun and R ~0.64 Rsun, consistent with the dynamical m2.
- At A (P, near periastron, r = 0.70 a) the K dwarf transits the primary, impact b ~ 0.33 R1: a full transit, 23%.
- At B (S, near apastron, r = 1.28 a) the primary occults the K dwarf with b ~ 0.60 R1, just short of total (1 - k = 0.53). The depth of 5.4-5.5% is close to the companion light fraction of 5.7%.
- Durations: the B/A ratio follows e sin w, and the model matches both shapes (rms 1.2 ppt, chi2/N 1.0).

6 NEXT ECLIPSES (BTJD; today ~ 4334.5) - P (A, 23%, 7.6 h): 4373.1124, 4429.2863, 4485.4602, 4541.6341 (+-1 min) - S (B, 5.4%, 13.4 h): 4341.2785, 4397.4523, 4453.6262, 4509.8001 (+-2 min realistic; formal 1.5)

FURTHER - Already settled as an EB on the target. - Double-lined spectroscopy, to get K2 and absolute masses. - A fit with free LD and third light, to refine i and e sin w and decide whether B is total. - The next S events at 4341.28 and 4397.45 would pin down the P(S) vs P(P) timing systematics. - A note for Fable: QLP s101/s103 depths for this field are diluted by ~15-35%. Depth-based comparisons on those sectors need a check against another pipeline.

A199. TYC 9380-1063-1 (TIC 391925531)

status
: confirmed (adversarial re-analysis 2026-10-21; F/G subgiant 2.7 Rsun, 1.3-1.6 Msun + 0.8-0.9 Msun K/G dwarf, e 0.38, i ~86.4 deg, southern CVZ; 23 grazing V-shaped 2.0% occultations of the companion at Gaia's B (T0), 6.7 h, s1-s98, rms 0.85 min; no A (< 110 ppm at 3 sigma) as the geometry requires (b_A 1.86 b_B); Gaia P -0.97 sigma; the single APOGEE RV phases as B (O-C +1.5 vs -39.7 km/s as A); prior flags: SPOC 2-min TCE (s1-s96 DV: 43.977843 d) and ExoFOP CTOI 391925531.01 (PC, Taaki et al. in prep., 2022) are this eclipse, a stellar companion, not a planet; the classifier's phase-0.225 dip is s98 scattered light; difference images on target, the G 14.6 star at 49 arcsec loses 0.3-1.1%)

P_days
: 43.977834

eP
: 0.000008

T0_primary_BTJD
: 2079.03934

depth_primary
: 2.0

duration_h
: 6.7

source_path
: Fable 5.1\made\appointments\vet391925531\REPORT.md

found_by
: Fable 5.1 (appointments search, G 11-11.5 subsample, 2026-10-20); confirmed by an Opus 5.5 adversarial re-analysis to VET-BRIEF (2026-10-21)

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Adversarial check: TIC 391925531 (TYC 9380-1063-1, Gaia DR3 5212819506717573504) Second vet run, resumed 2026-10-21. I read and re-used the inherited Gaia, MAST, archives2 and ExoFOP results; dvlist and dvparse were run here. Scripts and outputs are in this folder; downloads are in sky\data\qlp-sb-p20-60\vet391925531. Labels: the observed eclipse is at Gaia conjunction B (nu = 270 - w, the RV primary in front). In ecl/ephem files it is kind "P"; see progress.md.

VERDICT: CONFIRMED (eclipsing binary on the target) - It is a grazing occultation of the companion by the RV primary at conjunction B: 2.02% deep, about 6.7 h long, V-shaped, seen 23 times from s1 to s98. - There is no eclipse at conjunction A (< about 110 ppm), which the geometry requires. - "Really uncatalogued?" Not in any EB or variable-star catalogue, but the signal itself is NOT unknown: - SPOC 2-min TCE since the s1-s36 multi-sector run (2021); - ExoFOP CTOI TIC 391925531.01, disposition PC (Taaki et al., in prep., created 2022-12-09, P 43.95(5) d, epoch BTJD 2078.97(5), duration 6(2) h). Both treat it as a planet candidate. The new result is the identification: the Gaia SB1 orbit (f(M) = 0.12 Msun) puts the eclipse at the stellar companion's conjunction, so it is an EB and the CTOI should be re-dispositioned. - The phase-0.225 dip is an artefact (s98 scattered light at a data gap).

1. ARCHIVES (what was queried and what came back) - cats.py: VSX, ASAS-SN, Prsa+2022, IJspeert+2024 and Kostov+2025 t2/t3/t4 have nothing within 5". SIMBAD has only "SB*". - Gaia DR3: vari_eclipsing_binary, vari_summary and vari_classifier_result return 0 rows. NSS SB1: P

43.99434(1699) d, e 0.3818(257), w 232.14(3.27) deg, K1 32.16(1.63) km/s, gamma 12.34(55), corr(w, Tp) 0.902. GSP-Phot: Teff 6338, logg 3.82, R 2.66 Rsun. FLAME: M 1.63 Msun. RUWE 1.01. - VizieR 60" cones (archives2.txt) returned 0 rows in VSX, ASAS-SN, Prsa, IJspeert, GCVS, SB9, WDS, ZTF, ATLAS, KELT, Justesen, Rowan, Fetherolf, IJspeert+2021 and RAVE/LAMOST/GALAH. - Non-empty but not prior art: - Kostov+2025 table2 lists TIC 391925540 (Tmag 14.06, 48.7"), not the target. - Green+2023 gives the target a score of 0.000. - The cone on "J/A+A/674/A16" returned a different table; Gaia TAP is the authoritative source here. - APOGEE DR17: 1 visit (LCO, JD 2458146.644), vhelio -15.04 km/s, Teff 5981, logg 3.66, [M/H] -0.10, vsini 11.6. - ExoFOP (exofop.json): no TOI. CTOI TIC 391925531.01 as above. No imaging or spectroscopy entries. TIC contamination ratio 0.028. - SPOC DV (dv.txt), s1-s96 run: - P 43.977843(13), epoch 2079.03902(29), depth 20301(87) ppm, duration 6.98 h, b 0.945(19); - odd/even significance 0.22; suspectedEclipsingBinary false; - weak secondary maximum MES 3.5 at +22.48 d (phase 0.511), 167(45) ppm, which is not significant. The same TCE appears in the s1-s36 and s1-s39 runs. The TESS-SPOC s56-s69 TCE (29.3 d, 309 ppm) is unrelated noise. - Web search on the TIC/TYC names and the Taaki method found nothing. Tarrants & Mendes 2023 (arXiv 2310.12380, Gaia NSS vs planet candidates) uses TOIs and K2 only; this star is not in it.

2. ECLIPSE MEASUREMENTS Method: raw QLP SAP, q==0, no trimming; a trapezoid times a quadratic baseline. - Events: 22 q==0 fits, plus s39 from the flagged-cadence (all-q) fit, gives 23. No events were dropped. - Shape: depth 2.02% (median, sd 0.10), T14 6.70 h (sd 0.08), T12 2.59 h, flat 1.51 h. V/U-shaped, with ingress about 0.39 of the total, i.e. grazing. - Depth stability: QLP s96, s97 and s98 give 1.91, 1.73 and 1.72%. This is a QLP artefact: - SPOC 2-min s96 gives 2.09% (SAP) and 2.01% (PDCSAP); - my TESScut photometry gives 1.9% (1x1 aperture) in both s93 and s98. - Background: the in-eclipse change is -0.41% of the target flux, against +0.02% and +0.22% at control phases. The sign is the wrong way for a background-made dip. - Centroids: dx/dy = +0.3/+0.5 mpx (event sd 1.5/3.2), with controls at about 0. No motion. - A-conjunction slot (phase +0.6546 after B): - all 24 per-cycle fits give <= 0.04%; - a fold of 15 cycles over +-3 d (which covers +-4.5 sigma of Gaia's A-timing error of 0.67 d) has a deepest 6-h box of 33 +- 35 ppm. So depth_A < about 110 ppm (3 sigma). - A fold at phase 0.5 shows nothing (the SPOC weak secondary is not confirmed). - Phase-0.225 dip (phasedip_0225.txt, look_s98_0225.png): the classifier's fold zero, 1341.32, is B + 0.225 P. - In the 25 covered cycles, the q==0 depth at that phase is -0.05 to +0.06% (+-0.01%). - The 0.5% feature is s98 4023.8-4024.3: the q==0 tail of a mostly flagged low segment, under strong, falling scattered light (SAP_BKG 50000 -> 15000 e/s). It ends at a data gap, with a centroid jump across the gap. - Verdict: artefact. - Blind census (thr 0.3%): nothing outside the eclipses, except the known s94 3875 common-mode dip and s98 scattered light.

3. EPHEMERIS AND ORBIT - B: T0 = BTJD 2079.03934 +- 0.00018, P = 43.977834 +- 0.000008 d. chi2 14.4/21 (1-min floor); rms 0.85 min; no quadratic term. - SPOC DV agrees: 43.977843(13), 2079.03902(29). - The TESS period minus Gaia's is -0.0165 d = -0.97 sigma. - Measured minus Gaia-predicted B conjunction: -0.20 d (s1) drifting to -1.21 d (s98) because of the period difference; that is -0.39 to -0.86 sigma. - Only one eclipse kind is seen, so e cos w and e sin w cannot be measured from timing. Gaia predicts A - B = 0.6544 +- 0.0153 in phase. - Instead, Gaia's e and w were tested by the forward model (section 5): r_B = 0.656a near periastron and r_A = 1.223a. So b_A = 1.86 b_B, and a grazing B with no A is exactly what Gaia's e and w predict. - APOGEE: the 2018 RV, phased on the TESS ephemeris with the eclipse as B, matches the Gaia orbit (O-C +1.5 km/s). If the eclipse were A, O-C would be -39.7 km/s. This independently confirms the conjunction identification.

4. INDEPENDENCE - Other pipelines (secondpipe.txt): - SPOC 2-min SAP/PDCSAP s28, s93, s96; TESS-SPOC s1, s28; GSFC-ELEANOR-lite s1; TGLC s1 (aperture and PSF). - All recover the eclipse. Mid-times agree with QLP to within 0.1-2.3 min; T14 6.6-7.0 h; depth 1.92-2.39%. - Difference images (TESScut s93 and s98; diff.txt): - The deficit is centred on the target pixel; depth is 1.9 / 1.8 / 1.7% in 1x1 / 3x3 / 5x5 apertures, in both sectors. - The pixels of the G 14.56 (48.8") and G 15.85 (47.3") neighbours show only 0.3-1.1%, which is the target's wings. - Neighbours (neighbours.txt): within 60", the only source brighter than G 17 besides the target is G 14.56 at 48.8". It would need a 49% eclipse, which is excluded by the difference image and by the absence of centroid motion (expected about 40 mpx, observed 0.3 +- 0.3). G 15.85 would need 162%, which is impossible. Gaia RUWE is 1.01 and ipd_frac_multi_peak is 0.
5. PHYSICS - f(M) = 0.1196 [0.102, 0.139] Msun; a1 sin i = 25.8 Rsun. - For M1 = 1.25-1.65 Msun: m2(i about 86 deg) = 0.79-0.93 Msun (a K/G dwarf, R2 0.75-0.89 Rsun, Teff 5000-5500 K); a = 66-72 Rsun. - Which star is in front: - At B (nu + w = 270), the RV primary (F/G subgiant, R about 2.7-3.0) passes in front of the companion. This is the observed eclipse: a partial occultation of the companion. - At A, the companion would transit, but only at 1.86 times the projected separation. - b=0 durations would be 12.6-13.3 h at B and 23-25 h at A; the observed 6.7 h at B means a high impact parameter. - Forward model (graze.txt; MS companion, limb darkening, blackbody J): - i = 86.2-86.6 deg reproduces depth_B = 2.02% with b_B = 0.72-0.82 (in units of R1+R2), T14 6.9-9.2 h (first-to-last contact), and depth_A = 0.000%. - The Gaia-FLAME-like case (M1 1.65, R1 2.7) gives T14 6.85 h, against 6.70 observed. - Depth, duration, the V shape, the missing A eclipse, the RV phase and the SB1 nature (L2/L1 about 0.05) all fit one geometry. - A planet is excluded by K1 = 32 km/s at this period.
6. NEXT EVENTS (BTJD) - B eclipse (2.0%, about 6.7 h, grazing occultation of the companion): 4365.8867, 4409.8646, 4453.8424, 4497.8202 (+-0.7 min formal; allow +-3 min). - General: T_B = 2079.03934 + 43.977834 E. - A conjunction (no eclipse expected; < 110 ppm observed): 4350.69, 4394.67, 4438.64, 4482.62 (+-0.67 d).

WHAT WOULD SETTLE IT FURTHER 1. Tell ExoFOP: CTOI 391925531.01 is an EB (a stellar companion with f(M) = 0.12 Msun), via a CTOI comment or a disposition update. 2. A few high-resolution RVs to refine K1, e and w and search for the secondary's lines (L2/L1 about 0.05 in TESS, more in the red). Combined with the eclipse they would fix i, about 86.4 deg. 3. A proper grazing-EB fit (with ld and J free) to the 2-min SPOC data, to bound R2/R1 and L2/L1. 4. Optional: report it to VSX as an EA with the ephemeris above, noting that only one eclipse per orbit exists.

A200. TYC 3697-940-1 (TIC 374560569)

status
: confirmed (per-object re-analysis 2026-10-21; grazing F star + 0.50-0.57 Msun M dwarf (b ~0.95, i ~86 deg), 27.9 ppt trapezoid (box 25.3), T14 5.57 h, 3 primaries + 4 secondaries in s058/s085 only; new secondary 4.17 +- 0.31 ppt at phase 0.49948(31), e cos omega -0.0008(5) vs Gaia +0.011(11); jackknife clean, the secondaries recover the s085 primary alone; conjunction A at the Gaia epoch +0.29 sigma; the 2.13 arcsec batch offset was the uncalibrated WCS, own PSF localisation 0.4-0.6 +- 0.9 arcsec on target, dchi2 > 1500 to every neighbour; a/R1 wants R1 2.05-2.2 vs TIC 2.37 Rsun (1.5-2.5 sigma, open); ExoFOP 503 at the check, batch sweep had no TOI)

P_days
: 13.893272

eP
: 0.000012

T0_primary_BTJD
: 2892.25831

depth_primary
: 2.79

depth_secondary
: 0.42

duration_h
: 5.57

source_path
: Fable 5\made\sb1-scale\pervet\vet374560569\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet wave 6 (2026-10-21)

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TIC 374560569: per-object vet (sb1-scale wave 6)

Opus subagent, 2026-10-21. Standard: `Fable 5.1\made\appointments\VET-BRIEF.md + pervet\BRIEF-ADAPT.md`. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.) Star: TYC 3697-940-1 = Gaia DR3 505449450965010176. G 11.32, Tmag 10.87, Teff 6102 K, R 2.37 ± 0.11 R⊙, M 1.15 M⊙ (TIC), d 503 pc, contratio 0.081. Gaia SB1: P 13.894558 ± 0.001888 d, e 0.017 ± 0.011, ω 31 ± 49°, K1 31.60 ± 0.48 km/s, RUWE 1.39. Batch row: 25.3 ppt, P 13.89323 d, n = 3, diffimg offset 2.13″.

Verdict: CONFIRMED

The 28 ppt primary eclipse is real, sits on the target, and belongs to the Gaia SB1 orbit. The worry was n = 3. It goes away for a reason the batch did not see: **a secondary eclipse at phase 0.4995 shows up on its own in all four secondary windows covered** (4.17 ± 0.31 ppt, S/N 14 folded, S/N 6.3–6.8 in each window). That makes 7 eclipse events, not 3. No single event owns the verdict or the period.

What decides it: - **Timing.** The TESS primary, extrapolated back to the Gaia epoch, lands 0.37 h from Gaia conjunction A: +0.29σ, using this star's own corr_vec, with ω–T_p correlation +1.000. At the TESS epochs the offset is −0.64σ and −0.65σ. P_TESS − P_Gaia = −0.68σ. The secondary phase of 0.49948 ± 0.00031 matches Gaia's predicted B − A phase of 0.5070 ± 0.0070 (–1.1σ). - **Localization.** Difference image fitted with a WCS-calibrated PSF. The flux deficit lies 0.6″ from the target in s058 (2 eclipses) and 0.4″ in s085 (1 eclipse), with a 1σ radius of 0.9″. A fit fixed on the target beats every neighbour by $\Delta\chi^2 > 1500$. Deficit/target is 2.6%. The batch figures (2.13″ deficit to target, 6.42″ deficit to light) belong to the wave-4 class of centroiding artefacts. - **Second pipeline.** Own TESScut aperture photometry recovers E0, E1 and E52 at 27.5–28.9 ppt. The times agree with QLP to 0–1.4 min. - **Physics.** f(m) = 0.0454 M⊙, so m2 = 0.50–0.57 M⊙ for M1 = 1.15–1.45. The depth ratio gives a TESS surface-brightness ratio J = 0.133, which implies T_eff2 ≈ 3720 K (blackbody): an early-M dwarf of about 0.5 M⊙, as the mass function requires. The joint limb-darkened fit allows R2 ≈ 0.46–0.50 R⊙ (k 0.22–0.24, b 0.93–0.96), a main-sequence 0.5 M⊙ star.

Confidence in the event count: 3 primaries in raw Q0 data and 4 secondaries. All seven come from only **two sectors** (s058, s085), which TESScut confirms are the only sectors that observed the star. Within that limit the solution is overdetermined: leaving out each event in turn moves nothing (below).

Archives and web (step 1)

- `cats.py --r 60`: no VSX, ASAS-SN, Prša+2022, IJspeert+2024, Kostov+2025 t2/t3/t4 entries (the silent tables returned no rows and no errors). SIMBAD has TYC 3697-940-1 [SB*], whose only variability measure is the Gaia DR3 SB1 period of 13.894558 d. SIMBAD's parent is Cl Stock 2 with membership 0 (Cantat-Gaudin+2018), so it is a non-member: Stock 2 is at about 370–400 pc and the star at about 515 pc.
- Gaia DR3 vari_summary and vari_eclipsing_binary: no rows. SB9, WDS, J/AJ/160/111: none.
- ExoFOP returned HTTP 503 twice, so the TOI/CTOI check is **unverified**. The batch-wise review has no TOI here.
- Web search on TIC, TYC and Gaia ids: nothing about this star.
- MAST CAOM: only the QLP s058/s085 light curves, plus SPOC and TICA FFIs. There are no SPOC, TESS-SPOC, TGLC or eleanor light curves.
- **Prior art: none found.** It is a known Gaia SB1, not a known eclipser.

Eclipse measurements (step 2): raw SAP_FLUX, QUALITY = 0, no trimming

The shared primary trapezoid is T14 5.57 h, T12 2.12 h. Point σ is 1.8 ppt per 200-s cadence. Red-noise β at T14 binning is 2.12.

event	BTJD mid	± (min)	depth (ppt)	notes
pri E0 s058	2892.25800	0.7	28.09 ± 0.30	bkg +0.26 sd; y centroid −0.006 px
pri E1 s058	2906.15203	0.9	27.76 ± 0.33	bkg +0.10 sd; y centroid −0.006 px
pri E52 s085	3614.70846	0.7	27.94 ± 0.31	bkg −0.34 sd; y centroid −0.006 px
pri E53 s085	(3628.602)	–	**unusable**	QUALITY 2048 throughout
sec s058	2885.3087	12	4.08 ± 0.62	
sec s058	2899.1994	13	4.30 ± 0.65	
sec s085	3621.6451	12	4.07 ± 0.61	
sec s085	3635.5384	12	4.23 ± 0.63	

- Flagged-cadence mining: the only extra epoch is E53 (3628.60). QLP flags all of it with 2048 (scattered light). SAP scatter there is 1.25 (relative), and TESScut plane-background photometry gives 57 ppt point-to-point scatter, against 1.7 ppt at E52. It cannot be recovered and **is not counted**. A nominally negative core there (−44 to −83 ppt) is not evidence. No other primary or secondary epoch was missed.
- Background through each event is flat in its own units (sd). Every event falls outside the sector-systematic times: E52 at 3614.71 is 1.8 d before the s085 feature at 3616.5. No QUALITY-32 slide falls inside any eclipse.
- The y centroid shift of −0.006 px is identical in all three events. That is what an on-target 28 ppt dip looks like given about 8% contaminating light.
- Blind census of both sectors (box width T14/2, > 4 ppt or S/N > 6): it finds exactly the 3 primaries and the 4 secondaries and nothing else. Outside the Gaia window, the strongest control phase in the secondary scan reaches S/N 1.9.
- Depths: primary 27.9 ± 0.2 ppt (trapezoid maximum; box depth 25.3, matching the batch). Secondary 4.17 ± 0.31 ppt. Both are constant across events (secondary χ^2 0.10/3).

Ephemeris, O−C, jackknife (step 3)

- 3 primaries: **P = 13.893272 ± 0.000012 d**, T0 = 2892.25831 ± 0.00039 BTJD (corr −0.62), χ^2 0.95/1. O−C is −0.4, +0.7 and −0.0 min. Gaia P differs by −0.68σ.
- 4 secondaries alone: P 13.893162 ± 0.000161, χ^2 0.04/2. Their phase relative to the primary is **0.49948 ± 0.00031** (−10.5 ± 6.2 min), so **e cos ω = −0.0008 ± 0.0005**. Gaia gives +0.011 ± 0.011, a −1.1σ difference. Light-travel time is about 1 min, negligible.
- Jackknife over all 7 events (primaries and secondaries with a free secondary offset): joint P 13.893272 ± 0.000012, χ^2 1.45/4. Leaving out E0 gives 13.893262, E1 gives 13.893277, any one secondary gives 13.893272. Leaving out **E52**, the only long-baseline primary, gives P 13.893197 ± 0.000158 from the s058 pair plus the 4 secondaries. That predicts E52 at 3614.7046, against 3614.7085 measured (about 1σ). The secondaries recover the s085 primary independently.
- Cycle count: the s058 pair alone gives P 13.89403 ± 0.00078, which predicts E52 to ±0.04 d. The 51- or 53-cycle aliases sit ±13.9 d away, so the count is unambiguous.
- e sin ω: the trapezoid duration ratio T14s/T14p is 1.013 (0.970–1.078), giving roughly +0.006 (−0.015 to +0.037). Both eclipses are partial (grazing), so a duration ratio is a soft constraint here. It agrees with Gaia's +0.006 ± 0.012. Per the wave-4 lesson, Gaia's small e is a soft prior. The timing (P, conjunction A at the Gaia epoch, e cos ω) is the attribution evidence, and it is clean.
- Gaia conjunctions (own corr_vec MC, ω−T_p corr +1.000):

epoch	Gaia conj A	TESS primary	Δ
near BTJD 389 (Gaia)	391.4538 ± 0.0538	391.4693 ± 0.0025	+0.37 h, +0.29σ
near 2900	2906.3711 ± 0.343	2906.1516	−0.64σ
near 3620	3614.9957 ± 0.439	3614.7085	−0.65σ

Conjunction A (ν + ω = 90°, primary farthest from us) is the companion in front of the F star. That is the deep eclipse, as expected for an M-dwarf companion.

Independence (step 4)

- Second pipeline (`phot.py`, own TESScut aperture photometry with r = 1.5 and 2.5 px and an annulus background): E0 28.35/27.48, E1 28.62/27.84, E52 28.82/28.85 ppt. Mid-times agree with QLP to

within 0–1.4 min.

- Difference image (`locate.py`, TESScut s058/s085, PSF model with WCS calibrated on all Gaia G < 17 stars; calibration shift 0.8″ and 1.2″): the free single-PSF deficit is 0.6″ and 0.4″ from the target, 1σ radius 0.9″. Fixed-position fits: the target has χ^2 822 (s058) and 88 (s085). The next best is G 16.04 at 26″, at $\Delta\chi^2$ 8332 and 1523, and it would need a 31–33% dip. The light-centroid to catalogue mismatch is 1.6–2.2″ with the uncalibrated WCS, which is where the batch's 2.13″ / 6.42″ come from.
- Neighbour dilution table, as the depth each neighbour would need (25.3 ppt × 10^(0.4Δm)):

neighbour	sep	needs
G 13.48	53.6″	185 ppt
G 14.11	81″	330 ppt
G 14.21	68″	360 ppt
G 14.99	64″	740 ppt
G 15.96	38″	1.8 (impossible)
G 16.04	26″	1.9 (impossible)

The three brightest are excluded by the PSF localization ($\Delta\chi^2 > 1500$, and the deficit is within 0.6″ of the target). A blend inside 1″ is not excluded by imaging, but it would have to copy the Gaia SB1 period and conjunction phase to 0.3σ. Not credible.

Physics (step 5)

- f(m) = 0.0454 $M_\odot$. For M1 = 1.15, 1.30, 1.45 $M_\odot$, m2(i ≈ 86°) = 0.498, 0.535, 0.571 $M_\odot$ and a = 28.7, 29.8, 30.8 $R_\odot$.
- Joint limb-darkened two-sphere fit (`physics.py`, `kprof.py`), circular orbit, quadratic LD (0.32, 0.22 for the primary; 0.35, 0.30 for the companion). a/R1 = 14.0–14.5, robust. i ≈ 85.6–86°. Both eclipses are partial (b > 1 − k). The usual grazing k–b degeneracy leaves χ^2 flat for k 0.20–0.38. The best coarse point is k 0.24, b 0.96, giving R2 ≈ 0.50 $R_\odot$ for M1 1.3, which suits a 0.5 $M_\odot$ dwarf. k ≤ 0.16 is excluded. Third light of 0.08 (TIC contratio) changes almost nothing (a/R1 14.3, J 0.132).
- Surface-brightness ratio J = 0.133, so T_{eff2} ≈ 3720 K. L2/L1 (TESS band) ≈ 0.015. A 0.50–0.57 $M_\odot$ M dwarf fits mass, temperature and radius together. Its light is too small to matter for Gaia's SB1 classification.
- Mild tension: a/R1 ≈ 14.1 implies R1 = 2.05–2.20 $R_\odot$ ($\rho_1 \approx 0.19\ \rho_\odot$), against TIC R 2.37 ± 0.11 (ρ 0.086). That is about 1.5–2.5σ depending on M1 and LD. It is not decisive in a grazing, LD-sensitive fit, and it does not touch attribution. Spectroscopic R1 or log g would settle it.
- Predicted central T14 for TIC R1 would be 11–11.6 h. The observed 5.6 h is consistent with the high impact parameter.

Row fixes

- **P** = 13.893272 ± 0.000012 d (batch 13.89323). T0 = 2892.25831 ± 0.00039 BTJD (batch 2892.2614 was 4.5 min late).
- **Depth**: primary 27.9 ± 0.2 ppt (trapezoid maximum; the box depth of 25.3 ppt is a box statistic, not a fix). **Secondary 4.17 ± 0.31 ppt at phase 0.49948 ± 0.00031**, new.
- **Event count**: 3 → **7** (3 primaries + 4 secondaries). E53 (3628.60) was observed but is unusable (scattered light).

- **dphi**: primary −0.014 phase from Gaia conj A at the TESS epoch (−0.64σ). The well-measured comparison is the Gaia epoch: +0.0011 phase, +0.29σ. Conj B at +0.477 in the batch convention; the measured secondary is at 0.4995 against Gaia 0.507 ± 0.007.
- **Diffimg offset**: 2.13″ → **0.4–0.6″ ± 0.9″** after WCS calibration (artefact class).

Next predicted eclipses (BTJD)

E	primary	± (min)	secondary	± (min)
105	4351.0519	1.6	4357.9913	6.4
106	4364.9452	1.6	4371.8845	6.4
107	4378.8384	1.6	4385.7778	6.4
108	4392.7317	1.6	4399.6711	6.4
109	4406.6250	1.6	4413.5643	6.4
110	4420.5183	1.7	4427.4576	6.4

What would settle it further

1. One more TESS sector or a ground-based primary. At V 11.4 a 28 ppt, 5.6 h eclipse is easy.
2. A spectroscopic log g or R1, to resolve the R1 2.1 vs 2.37 tension.
3. An ExoFOP TOI/CTOI re-check when the site is back (503 tonight).
4. RVs or a double-lined spectrum in the red, to detect the M dwarf (expected K2 ≈ 70 km/s).

Files (all in this folder)

- gaiatap.py → gaia.json, gaiatap-out.txt
- cats-out.txt, morecats.py → morecats-out.txt, mastlc.py → mastlc-out.txt
- look.py
- vet.py → vet-out.txt, vet-results.json, events.png
- locate.py → locate-out.txt, locate_s058/s085.png
- phot.py → phot-out.txt, phot.png
- e53.py → e53-out.txt, e53.png
- conj.py → conj-out.txt, ephem.json
- sec.py → sec-out.txt, sec.png
- sec2.py → sec2-out.txt, sec2.json, sec2.png
- physics.py → physics-out.txt, physics.json, physics.png
- kprof.py → kprof-out.txt
- next.py → next-out.txt
- TESScut cubes: `C:\Home\sky\data\qlp-sb-p20-60\vet374560569\`

A201. TYC 1365-707-1 (TIC 70465753)

status
: confirmed (per-object re-analysis 2026-10-21; grazing (b 1.1-1.2) spotted G dwarf + 0.39 Msun M dwarf, circular, 26.6 ppt (TESS-SPOC PDCSAP 26.2-26.8), T14 2.16 h, 18 primaries + 11 occultations in 5 sectors; new occultation 2.3 ppt at phase 0.5001(2), S/N 16, e cos omega +0.0003(3) vs Gaia -0.028(19), 1.5 sigma circular-ward; VSX ROT (Gaia classifier) at the target: type revision; the 3.87 arcsec batch offset was the uncalibrated WCS (light and deficit centroids agree to 0.19 arcsec), own localisation 0.4-1.2 arcsec on target, dchi2 >= 1949 to every neighbour; ExoFOP 503 at the check)

P_days
: 6.4475682

eP
: 0.0000015

T0_primary_BTJD
: 2831.15209

depth_primary
: 2.66

depth_secondary
: 0.23

duration_h
: 2.16

source_path
: Fable 5\made\sb1-scale\pervet\vet70465753\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet wave 6 (2026-10-21)

Archived source: sanitized report text.

TIC 70465753 (TYC 1365-707-1, Gaia DR3 6715344619026787 84): per-object vet

Opus subagent, sb1-scale pervet wave 6, 2026-10-21. No earlier session had worked this star. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. A grazing eclipsing G dwarf + M dwarf on a circular 6.4476-d orbit. Both eclipses are detected, both are on target, and the 3.87″ batch diffimg offset is a WCS artefact.

The numbers that decide it: - **Primary eclipse:** 18 events, 26.56 ± 0.08 ppt deep, all at S/N 56–86. - **Occultation:** 2.35 ± 0.14 ppt (S/N 16) at phase 0.5001 ± 0.0002. It is seen in every sector that covers it. - **Period:** P = 6.4475682 ± 0.0000015 d, which is +1.7σ from Gaia's period. - **Gaia conjunctions:** the primary lands 1.1–2.7 h after Gaia's conjunction A in every sector (0.4–0.7σ). At the

Gaia epoch it is −2.2σ from A, and the secondary is −1.0σ from B. - **Localization:** the PSF-fit deficit is 0.4–1.2″ from the WCS-calibrated target in three TESScut sectors. A fixed-position fit on the target beats every neighbour by $\Delta\chi^2 > 1900$. - **Independent pipeline:** TESS-SPOC PDCSAP gives 26.2–26.8 ppt in all 5 sectors. - **Physics:** the mass function gives m2 ≈ 0.39 M⊙. A grazing chord (b ≈ 1.1–1.2) predicts T14 = 2.1–2.5 h against 2.16 h observed. The secondary/primary depth ratio of 0.081 implies a T_eff2 of about 3300 K (blackbody).

1. Archives

- **cats.py (r = 60″, APPOINTMENTS_DATA set):**
- VSX has Gaia DR3 671534461902678784 as **ROT** with no period. That is a type revision to EA/EB, not prior art.
- ASAS-SN: none. SIMBAD: TYC 1365-707-1 as **SB***.
- **SIMBAD bibliography:** the only bibcode is 2023A&A...674A..34G, the Gaia DR3 NSS paper.
- **morecats.py:** nothing in B/sb9, B/wds or J/AJ/160/111.
- **ExoFOP:** returned HTTP 503 for both the JSON and the page (Retry-After 3600), so I could not check it this session. The batch sweep (Opus 5.5's TOI/CTOI review) had no entry for it.
- **Batch ebcats (c3/c4):** none of the vetted or unvetted EB tables.
- **Web search** for "TYC 1365-707-1", "TIC 70465753" and the Gaia id: no relevant hits.
- **Gaia DR3:**
- vari_eclipsing_binary has no row. vari_summary has a row (in_vari_classification_result = true, but no EB and no rotation-modulation class). G range is 0.031 mag over 23 epochs, phot_variable_flag VARIABLE.
- NSS SB1: P 6.446975 ± 0.000342, e 0.040 ± 0.027, ω 219 ± 23°, K1 34.84 ± 0.51 km/s, significance 68. RUWE 1.11, ipd_frac_multi_peak 0.
- TIC: T_eff 5810 K, R 1.00 R⊙, M 1.05 M⊙, d 174 pc.
- **Conclusion:** not previously catalogued as eclipsing.

2. Light curve (raw QLP SAP, QUALITY == 0, no trimming)

Files: `vet-out.txt`, `events.png`, `fold_oc.png`, `bkgfake-out.txt`, `sec2-out.txt`. - **Primary shape:** depth 26.56 ± 0.08 ppt, T14 2.16 h, T12 0.86 h, flat part 0.43 h — nearly V-shaped. A flat-bottomed trapezoid (T12/T14 = 0.1) is worse by $\Delta\chi^2$ +7123, and a pure V by +201. - **Per-sector depths:** 27.16 / 26.24 / 26.50 / 27.15 / 25.98 ppt. The constancy χ^2 is 44/17, a sector-to-sector scatter of ±0.5 ppt that is plausibly aperture/dilution. Odd and even agree (26.62 vs 26.48). - **Control phases (0.125–0.875 except 0.5):** all 0 within ±0.08 ppt. Phase 0.5 is the occultation. - **Background gate (unit-free regression):** the fake dip is ≤ 1.8% of the depth in all 18 events. - The worst case is E6 (s045, BTJD 2541.01), which sits on a +2788 e^-/s ramp. Its depth is unchanged with background as a regressor. - **Flags and systematics:** no bit-32 cadence inside any eclipse window, and no event within 0.5 d of a listed sector-wide systematic time. Centroid shifts in eclipse are dX +0.07 ± 0.11 mpix and dY −0.25 ± 0.15 mpix. - **Blind census:** 30 dips at S/N ≥ 4. Eighteen are the primaries. Eleven are occultations at phase ±0.500, each 1.9–3.1 ppt. - The one "15-ppt" off-ephemeris peak (BTJD 3266.38, S/N 4.1) is an occultation that falls inside a 0.22-d momentum-dump data gap in s072. It is a fit to the gap edges, not an event. - **Occultation:** 2.16–2.35 ppt depending on the shape, at phase 0.50000 (±1.4 min on the fold). - Per sector: s044 2.50 ± 0.33, s045 2.36 ± 0.25, s046 2.21 ± 0.19, s072 2.01 ± 0.27 ppt. s071 has

no coverage of it. - It leads the vet.py phase scan at S/N 16.4; the scan's 99th percentile is 5.2. - **Rotational modulation:** about 1 ppt. The first harmonic amplitude is about 1.2 ppt, consistent with the ROT classification (a spotted, probably tidally synchronised G dwarf).

3. Ephemeris and orbit

Files: `timing-out.txt`, `gaiaepoch-out.txt`. - **Primary ephemeris:** P = 6.4475682 ± 0.0000015 d, T0 = 2831.15209 ± 0.00009 BTJD. χ^2/dof 38/15; O−C rms 0.5 min; per-sector mean O−C within ±0.33 min. - **Secondary:** 11 events, P_sec = 6.4475952 ± 0.0000340. P_sec − P_prim = 2.3 ± 2.9 s per cycle, so no apsidal motion is detected. - **Secondary phase:** 0.50020 ± 0.00018 from per-event timing, and 0.50000 from the fold. So **e cos ω = +0.0003 ± 0.0003**. Gaia's −0.028 ± 0.019 is 1.5σ away, in the circular direction — the Lucy–Sweeney sign again, as in the wave-4 list. - **e sin ω:** - A trapezoid duration ratio of 0.78 would imply −0.12. But both eclipses are grazing, and the occultation's contacts sink into the noise, so that number means nothing. - The limb-darkened two-sphere joint fit (`twosphere.py`, one geometry for both eclipses) leaves e sin ω **unconstrained over −0.10…+0.10** ($\Delta\chi^2 < 1$ everywhere, because of the k–b degeneracy at this occultation S/N). - So: no tension with Gaia's −0.024 ± 0.022, and no measurement either. - **Comparison with Gaia** (MC with this star's own corr_vec; ω–T_p correlation +0.996): - **Per TESS sector:** the primary is +1.08, +1.13, +1.19, +2.68 and +2.74 h from conjunction A (+0.37 to +0.73σ). - **At the Gaia epoch (BTJD ~389):** - The primary is −3.56 h from A (−2.2σ). - The secondary is −0.68 h from B (−1.0σ). - Gaia's A→B separation is 0.482 ± 0.012 in phase, against 0.5001 measured. Most of the −2.2σ is Gaia's small spurious e pulling A and B apart. - The deep eclipse is at A (companion in front), as it should be for an M dwarf transiting the G star.

4. Localization: the 3.87″ batch offset is an artefact

Files: `locate-s044-out.txt`, `locate-s046-out.txt`, `locate-s072-out.txt`, `locate_s0*.png`. - **Batch receipt, diff_s44.json:** - The light centroid (7.876, 7.362) and the deficit centroid (7.879, 7.371) agree to **0.19″** (0.003, 0.008 px). - Both sit (+0.156, +0.098) px from the uncalibrated catalogue spot. That is exactly the wave-4/5 artefact family. - The peak-deficit pixel is the target pixel. - **Own pass** (21×21 TESScut cubes; WCS calibrated by a Gaussian-PSF model of all G < 16.5 stars on the out-of-eclipse image; free single-PSF fit to the difference image):

Sector	WCS shift needed	Deficit vs calibrated target	1σ radius	Target fixed-fit $\Delta\chi^2$ to next star	Deficit / target flux
s044	3.0″	1.2″	1.2″	9226	2.36%
s046	1.9″	0.4″	0.8″	1949	2.45%
s072	2.5″	0.6″	0.6″	3110	2.41%

- **Stability:** half-sample fits are 0.4–2.4″ from the target.
- **Mechanism:** the uncalibrated WCS alone is off by 1.9–3.0″. That, plus the moment-centroid box pull, accounts for the batch's 3.87″.
- **s044 caveat:** it has reduced χ^2 13.5. The Gaussian PSF is imperfect, but no neighbour comes close.
- **Neighbours** (`neighbours-out.txt`; T from Gaia colours): the dip each would need is
- G 14.75 at 41.7″: 94%
- G 13.66 at 64.8″: 35%
- G 15.15 at 71.1″: 149%

All of them lose the fixed-position fit by $\Delta\chi^2 \geq 1949$ in every sector. - **TESS-SPOC (independent pipeline):** at the same ephemeris PDCSAP gives s044 26.42, s045 26.33, s046 26.16, s071 26.78, s072 26.43 ppt. SPOC-SAP gives 25.45–26.44 ppt (`indep-out.txt`). - **Caveat:** RUWE 1.11 and ipd_frac_multi_peak 0, so no sign of an unresolved third light.

5. Physics

File: `physics-out.txt`. - **Companion:** f(m) = 0.0283 M⊙. With M1 = 1.045, m2 = 0.39 M⊙ (i ≈ 86°) — an M dwarf with R2 ≈ 0.40 R⊙. a = 16.4 R⊙ (16.4 R1). - **Chord:** a central transit would be about 180 ppt deep. The observed 26.6 ppt needs a grazing chord, b_A ≈ 1.13–1.22 R1 (for R1 0.9–1.1). The predicted T14 is 2.07–2.50 h against 2.16 h observed, and the V shape fits a grazing chord. - **Occultation depth:** - The ms() relation (T2 3634 K, blackbody J 0.146) predicts about 4.3 ppt at e sin ω = 0. - The observed ratio 0.081 means a blackbody J of about 0.08, i.e. T_eff2 ≈ 3300 K. That is cooler than the crude relation but within normal scatter for a 0.35–0.4 M⊙ M dwarf, whose TESS-band flux a blackbody overestimates. - Alternatively, a small |e sin ω| ≈ 0.03 at the relation's temperature. The light curve cannot distinguish the two. - **Companion light:** about 1.5–2.5%, negligible for the depth. - One geometry (circular, grazing, k ≈ 0.35–0.45) accounts for both eclipses, the duration, the depth ratio and the RV mass function.

Row fixes

- **P:** 6.4475682 ± 0.0000015 d (the batch's 6.44759 is fine to 5 decimals).
- **T0:** 2831.15209 ± 0.00009 BTJD (primary).
- **Depth:** 26.6 ppt (trapezoid fit; the batch's 25.5 is a median of core points in sapdepth; TESS-SPOC PDCSAP gives 26.4). Occultation 2.3 ppt at phase 0.5001.
- **Events:** 18 primaries (unchanged) plus 11 occultations.
- **dphi:** +0.007 to +0.018 vs Gaia A per sector, as the batch had. At the Gaia epoch it is −0.023 (−2.2σ), driven by Gaia's spurious e.
- **Offset:** the 3.87″ is not real; the deficit is 0.4–1.2″ from the target.

Next predicted eclipses (BTJD)

Kind	Times	Uncertainty
Primary (~26.6 ppt)	4404.3587, 4410.8063, 4417.2539, 4423.7014 (then +6.4475682 d)	±0.5 min
Secondary (~2.3 ppt, phase 0.5001)	4407.5832, 4414.0307, 4420.4783, 4426.9259	±2 min

What would settle it further

- **e sin ω:** more 200-s sectors through the occultation, with ingress and egress resolved.
- **Masses and radii:** one RV epoch plus a spectroscopic T_eff for star 1 would turn the system into a mass–radius point for the M dwarf. The grazing geometry limits R2 precision.
- **Archive:** re-check ExoFOP when it is back up (it was 503 tonight).

A202. TYC 5995-1398-1 (TIC 348536911)

status
: confirmed (per-object re-analysis 2026-10-21; grazing (b 1.04, i ~76 deg) G subgiant 2.12 Rsun + ~0.5 Msun M dwarf, circular, RS CVn-like spots (+-10 ppt humps), 28.2 ppt, T14 3.36 h, 36 primaries + 36 secondaries in 4 sectors (s007 added); secondary 3.84 ppt at phase 0.49993(45), S/N 58, e cos omega -0.0001(7) vs Gaia +0.051(15): 3.5 sigma, and Gaia P 3.4 sigma long (2.5540664(374) vs 2.553938(5)): Table 2; circular-conditioned conjunctions still match at the Gaia epoch (A -2.0, B -1.6 sigma); the G 11.44 star at 9.6 arcsec (WDS ARA 52, optical: plx 0.72 vs 3.19 mas) excluded by the pixel-gradient sign test in both kinds x3 sectors; spot-induced O-C +-6 min; ExoFOP 503 at the check)

P_days
: 2.553938

eP
: 0.000005

T0_primary_BTJD
: 2998.52091

depth_primary
: 2.82

depth_secondary
: 0.38

duration_h
: 3.36

source_path
: Fable 5\made\sb1-scale\pervet\vet348536911\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet wave 6 (2026-10-21)

Archived source: sanitized report text.

TIC 348536911 (TYC 5995-1398-1, Gaia DR3 5721091468448223488): per-object vet

Opus subagent, 2026-10-21, following VET-BRIEF and BRIEF-ADAPT. The data is raw QLP SAP_FLUX, QUALITY==0, no trimming, unless stated otherwise. The parent saved this file because the write tool refused it for the subagent.

Verdict: CONFIRMED. The target eclipses: a grazing, circular eclipsing SB1 (a G subgiant with an early M dwarf), with both eclipse kinds seen.

Deciding numbers: - **Primary eclipses:** 36 events in 4 sectors (s007, s034, s061, s088). - QLP depth 28.2 ppt, T14 3.36 h, T12/T14 0.40. - Odd and even depths are equal: 28.24 ± 0.07 vs 28.15 ± 0.07 ppt. - **Secondary eclipse:** 3.84 ± 0.07 ppt (S/N 58) at phase **0.49993 ± 0.00045**. - Twelve control phases give ≤ 0.57 ppt. - It is seen in every pipeline and every sector. - **Period:** P = 2.553938 ± 0.000005 d. Gaia gives 2.554066 ± 0.000037 d, so TESS is −3.4σ shorter. - **Timing against Gaia:** the primary falls on Gaia conjunction A, meaning the RV primary is the star eclipsed. - At the TESS sectors the offset is +0.1σ to +2.4σ; at the Gaia epoch it is −1.28 h (−3.9σ). - All of that offset comes from Gaia's e cos ω. Restricted to circular draws, A lands −0.30 h (−2.0σ) from the TESS primary and B lands −0.24 h (−1.6σ) from the TESS secondary. - TESS gives e cos ω = −0.0001 ± 0.0007, against Gaia's +0.051 ± 0.015 (3.5σ). This is a seventh small-e Gaia orbit failing in the circular direction. - **On target, despite a G 11.44 star 9.6″ away.** That star is WDS ARA 52 and is not bound to the target. - Test: in the TESS-SPOC pixel data, the fractional flux loss per pixel was measured along the line from the target to the neighbour. - It falls toward the neighbour in all 3 sectors, for both eclipse kinds. - Primary: −2.9 ± 0.2, −5.9 ± 0.6 and −4.1 ± 0.7 ppt/px. Secondary: −0.6 ± 0.3, −1.4 ± 0.4 and −1.3 ± 0.5 ppt/px. - A target eclipse predicts about −4.3 ppt/px; a neighbour eclipse predicts about +12. - **Physics:** a limb-darkened two-sphere fit gives R1 = 2.12 R⊙ (TIC 2.07 ± 0.10), R2 ≈ 0.63 R⊙, b ≈ 1.04, i ≈ 76°. - Gaia's f(M) = 0.0513 M⊙ then gives m2 ≈ 0.51–0.53 M⊙. - The predicted ellipsoidal amplitude is 6.9 ppt; the observed 3.8–6.3 ppt is contaminated by starspots. - The secondary depth fits an M dwarf at T2 ≈ 3400–3500 K.

Row fixes:

	batch row	corrected
P (d)	2.55395	**2.553938 ± 0.000005**
depth (ppt)	26.2	**28.2**, fixed-shape fit; sector means 28.6 / 29.0 / 28.3 / 27.7
events	27	**36**, adding s007 (30-min cadence), which the batch lacked
T14 (h)	1.9	**3.36**
dphi		secondary at phase **0.49993 ± 0.00045**; Gaia A – TESS primary = −0.021 ± 0.006 phase at the Gaia epoch (−0.005 if circular)
diffimg	2.75″	stands: its deficit–light 2.62″ matches the 2.47″ a target eclipse predicts (a neighbour eclipse predicts 7.1″)

1. Archives

- **cats.py --r 60** (`cats-out.txt`, with APPOINTMENTS_DATA set):
- No hits in VSX, ASAS-SN, Prša+2022, IJspeert+2024, or Kostov t2/t3/t4. There is not even a Kostov table2 flag.
- SIMBAD has TYC 5995-1398-1, typed [SB*].
- **Gaia DR3** (`gaiatap-out.txt`, `gaia.json`):
- vari_summary and vari_eclipsing_binary are empty.
- NSS solution: SB1, significance 46, flags 0. P 2.5540664 ± 0.0000374, e 0.054 ± 0.015, ω 15.4 ± 12.9°, Tp −1.146 ± 0.092, K1 57.88 ± 1.25 km/s.
- RUWE 1.03.
- **SB9 and J/AJ/160/111:** no entry.

- **WDS** (`morecats-out.txt`): 08066–1645 is ARA 52, a 9.6–9.8″ pair first measured in 1903, PA 264° to 261°, mags 10.79 and 11.90.
- The companion is Gaia 5721091365369010176 = TIC 348536909.
- It is an optical pair (`pair-out.txt`): parallax 0.717 against 3.191 mas, different proper motion, RV +47 against −9 km/s, and Teff 7640 K.
- **ExoFOP:** returned HTTP 503 and was not checked. The batch sweep found no TOI or CTOI.
- **Extra VizieR cones at 30″** (`extracats-out.txt`; control cones returned rows):
- The only hit is an ATLAS "dubious" variable at 23.5″ with G about 17. It would need a dip deeper than 100%, so it is irrelevant.
- J/A+A/674/A16 is not an EB table.
- **Web search:** "TIC 348536911" and "TYC 5995-1398-1" returned nothing relevant.
- **Batch receipts:**
- c3-vetsum: dphi −0.002/+0.465, ATTR.
- c3-sapdepth: 27 events, median 26.19 ppt. Its box estimator underestimates the depth.
- diffrun s088: peak on target, deficit–target 2.75″, deficit–light 2.62″.
- **No prior art as an eclipsing binary.** This is a new eclipsing binary in a known SB1.

2. Measurements (`vet-out.txt`, `sec2-out.txt`, `census-out.txt`)

- **Primary shape:** the fold of the 200-s sectors gives 27.9 ppt, T14 3.36 h, T12/T14 0.40.
- Each event was fitted with that fixed shape on a quadratic baseline within ±0.4 d, supersampled for the 30-min sector.
- Depth constancy χ^2 is 118/35. The ~1 ppt sector-to-sector steps come from QLP's reference flux, dilution and spots.
- **Background:** the unit-free regression gives ≤ 0.32 ppt per event.
- **Centroids:** in-eclipse shifts are 0–2 mpix.
- **Quality flags:** no event touches a bit-32 cadence, and none sits at a listed sector-wide systematic time.
- Three events have no Q==0 points in eclipse (2242.55, 2247.66, 2975.53) and were skipped.
- **Out-of-eclipse light** (`timeseries.png`, `look.png`): a ±10 ppt modulation, combining the ellipsoidal effect with large spot humps.
- The humps migrate in phase from sector to sector, like a synchronised active star of the RS CVn type.
- A beaming tie is therefore not possible: the expected ~0.6 ppt is swamped.
- `sec.py`'s global harmonic model fails on this light curve. `sec2.py` supersedes it.
- **Secondary:**
- Local fixed-shape fits give 3.84 ± 0.07 ppt at phase 0.4990 (fold) and 0.49993 ± 0.00045 (sector-weighted).
- Controls at phases 0.15–0.85 are ≤ 0.57 ppt; the largest, at 0.40 and 0.60, is leakage from the V-shaped minimum.
- The fitted secondary T14 of 1.44 h is not physical. The shallow eclipse sits in a V-shaped ellipsoidal minimum, and the quadratic baseline absorbs its wings.
- A grazing circular orbit has equal chords at both eclipses, so **no e sin ω is derived from durations**.
- **Eclipse timing variations (ETV):** sector-mean primary O−C is −6.5, +4.2, +1.3 and −1.3 min (χ^2/dof 1549/34).

- TESS-SPOC, TGLC and TARS reproduce it (−6.3, +4.0, +1.2, −1.2 min), so it is astrophysical, not a QLP artefact.
- O−C correlates with the local baseline slope at r = −0.84. After correcting for slope the sector means are −0.4, +2.3, +2.3 and +1.8 min, so the shifts are spot-induced.
- Secondary minus primary per sector is −3.7, −2.9, +4.7 and +1.0 min (± 2–3). That is not significantly anti-correlated, so there is no apsidal-motion or light-travel-time claim.
- The period error is inflated to 5×10^{-6} d because of this.
- **Blind census:** every dip over 6 ppt is a primary or a secondary, except four of 6–10 ppt.
- Those four sit at phase 0.89–0.93 or 0.11, about 0.27 d from a primary. They are artefacts of the fitting window's edge.
- No unexplained dips remain.

3. Ephemeris and Gaia (`conj-out.txt`, `conjcirc-out.txt`)

The Monte Carlo uses this star's own corr_vec (ω–Tp correlation 0.999). - **Primary ephemeris:** T0 = 2998.52091 ± 0.00035 BTJD, P = 2.5539380 ± 0.0000050 d (formal error 0.0000012). - **Secondary phase:** 0.49993 ± 0.00045, against Gaia's (B–A)/P = 0.532 ± 0.009 (−3.5σ). That gives e cos ω = −0.0001 ± 0.0007. - **Gaia A minus the TESS primary:**

epoch	Gaia A − TESS primary
Gaia epoch	−1.28 h (−3.9σ)
s007	+0.06 h
s034	+0.94 h
s061	+1.83 h
s088	+2.70 h (+2.4σ)

The drift across the sectors is Gaia's period running 3.4σ long. - **Gaia B minus the TESS secondary:** +0.70 h at the Gaia epoch, and +2.1 to +4.7 h in the sectors. - **Conditioned on e cos ω ≈ 0:** A is −0.30 h and B is −0.24 h from the TESS events at the Gaia epoch. - **Attribution:** the primary is conjunction A, unambiguously; the other assignment would be off by P/2.

4. Independence (`indep-out.txt`, `tpf2-out.txt`)

- **Second pipelines (primary depth):**

pipeline	sectors	primary depth (ppt)
TESS-SPOC PDCSAP	s007 / s034 / s061	28.4 / 28.3 / 27.8
TESS-SPOC SAP (diluted; crowding correction ~0.72)	same	20.2–20.4
TGLC PSF	s007	28.5
TARS	all four	20.8–24.3
eleanor-lite	s007	18.2

The secondary appears in all of them at 2.6–4.9 ppt. - **The neighbour's own light curves** (TIC 348536909): QLP shows 81–83 ppt and TGLC PSF shows 64 ppt. This is **not** evidence against the target. - QLP measures flux on difference images and adds back each star's catalogue flux. - So a target eclipse seen in the overlapping neighbour aperture gets divided by a reference flux 0.347 times smaller. - The observed ratio 83/28 = 2.96 matches 1/0.347 = 2.88, and either hypothesis predicts that. - Likewise, TGLC's PSF fit cannot split a pair 0.46 px apart. - **TESS-SPOC TPF difference images**

(s007, s034, s061; both eclipse kinds): - The PSF-free sign test (D/O gradient along target→neighbour) is negative in all 6 cases, matching a target eclipse in sign and size. - Deficit-minus-light offset along target→neighbour, where the target predicts −0.12 px and the neighbour predicts +0.34 px:

```
| eclipse | s007 | s034 | s061 |
|---|---|---|---|
| primary (px) | +0.13 | −0.02 | −0.04 |
| secondary (px) | −0.14 | −0.10 | −0.21 |

Five of six favour the target. s007 is ambiguous at 30-min cadence.
```

- Two-Gaussian fits are degenerate at this separation (negative flux ratios) and are disregarded.
- **Dilution:** if the G 11.44 neighbour were the eclipser, it would need an ~81 ppt dip of its own. It would also have to share the target's Gaia SB1 period and conjunction phase by chance.
- The next-brightest star within 60″ is G 14.36 at 54″. It would need about 1000 ppt.

5. Physics (`physics-out.txt`)

- **Mass function and orbit:** f(M) = 0.0513 M⊙. With M1 = 1.09 M⊙: m2(90°) = 0.51 M⊙, q = 0.47, a = 9.20 R⊙.
- **Primary eclipse fit:** quadratic limb darkening and no dilution, since QLP's reference flux leaves the depth essentially undiluted.
- Result: R1/a = 0.23, k = 0.30, b = 1.04.
- So R1 = 2.12 R⊙, which matches TIC (2.07, log g 3.84, a subgiant).
- R2 ≈ 0.63 R⊙ and i ≈ 76°, giving m2 ≈ 0.53 M⊙.
- R2 runs about 20% large for that mass. The grazing geometry, the coarse grid and spots allow that; the grid was resolution-limited, so no formal ranges are given.
- **Eclipse geometry:** at A the M dwarf transits the G star. At B about 39% of the M dwarf is occulted.
- **Secondary:** 3.84 ppt means L2/L1 ≥ 9.8 ppt and a surface-brightness ratio of 0.108, i.e. T2 ≈ 3450 K. That is slightly cool for 0.53 M⊙ but within the approximations.
- **Ellipsoidal:** 6.9 ppt predicted against 3.8–6.3 ppt observed. Consistent.
- **Not a heartbeat:** the orbit is circular, and both eclipses are sharp and at fixed phases.

6. Next predicted eclipse times (BTJD)

- **Primary** (M dwarf transits, conjunction A): **4336.7844, 4339.3384, 4341.8923**, then every 2.553938 d.
- The ephemeris gives ±4 min; spot jitter adds up to ±6 min.
- **Secondary** (M dwarf occulted, conjunction B): **4338.0612, 4340.6152, 4343.1691** (± 5 min).

What would settle it further

- **Radial velocities**, ideally red or infrared so the secondary's lines show. These would give the true K1 and e and resolve the two Gaia tensions (P at 3.4σ, e cos ω at 3.5σ).
- **A spot-aware timing fit** would turn the O−C into a real ETV test.
- **ExoFOP**, once the service is back.
- **Third light:** the 9.6″ pair is already resolved, so imaging would only help to bound unresolved light closer in.

Files

- **Scripts:** vet.py, vetlib.py (copied from vet423927429), sec.py (superseded), sec2.py, look.py, conj.py, conjcirc.py, gaiatap.py, morecats.py, extracats.py, mastlc.py, indep.py, tpf2.py, physics.py, census.py.
- **Outputs:** vet-out.txt, vet-results.json, sec2-out.txt, sec2-results.json, conj-out.txt, conjcirc-out.txt, indep-out.txt, tpf2-out.txt, physics-out.txt, census-out.txt, cats-out.txt, gaiatap-out.txt, morecats-out.txt, extracats-out.txt, pair-out.txt, mastlc-*.txt.
- **Figures:** fold_oc.png, sec2.png, look.png, timeseries.png.
- **Data:** `C:\Home\sky\data\qlp-sb-p20-60\pervet348536911\`, with the neighbour's files in `nbr\`.

A203. TYC 8972-1194-1 (TIC 315849332)

status
: confirmed (per-object re-analysis 2026-10-21; grazing (b ~1.05) subgiant 1.86 Rsun + 0.56-0.62 Msun K7/M0 dwarf, 29.2 ppt QLP (37-40 undiluted, CROWDSAP 0.53-0.62, b +0.7 deg), T14 3.74 h, 21 primaries + 21 secondaries in 6 sectors; new secondary 7.6 ppt at phase 0.50020(19), e cos omega +0.0003(3) vs Gaia +0.006(18); conjunctions A +0.9 and B +0.6 sigma at the Gaia epoch; e sin omega ~+0.007 from durations vs Gaia -0.036(13), ~3 sigma circular-ward, soft (grazing trapezoid bias); the 3.13 arcsec batch offset is a WCS/crowding artefact, TGLC PSF at the Gaia position gives the full depth, the only neighbour under 100% (G 11.8 at 41 arcsec) never ranks in the localisation; ExoFOP 503 at the check)

P_days
: 5.4921244

eP
: 0.0000015

T0_primary_BTJD
: 2753.70951

depth_primary
: 2.92

depth_secondary
: 0.76

duration_h
: 3.74

source_path
: Fable 5\made\sb1-scale\pervet\vet315849332\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet wave 6 (2026-10-21)

Archived source: sanitized report text.

TIC 315849332: per-object vet (sb1-scale, G 11-11.5 subsample)

Opus subagent, sb1-scale pervet wave 6, 2026-10-21. (Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED. This is a grazing, double-eclipsing binary on the target, and it is the Gaia SB1. Gaia's orbit predicts both conjunctions, and both eclipses fall on them.

Batch row: 27.5 ppt, P 5.49211 d, n = 20, diffimg offset 3.13″. Star: TYC 8972-1194-1 = Gaia DR3 5336005937706167808. G 11.18, Tmag 10.68, Teff 5499 K, R 1.86 R⊙, logg 3.88 (a subgiant). It sits at b = +0.7°, and the TIC contamination ratio is 1.10.

Deciding numbers

- **Primary eclipses.** 21 primaries, all individually detected at S/N 14–115. The batch said 20; the extra one is E5, s011 1600.363. All six QLP sectors (s010/011/037/064/099/100) have them. None falls on a listed sector-wide systematic time. Odd and even depths agree (29.12 vs 29.24 ppt). The in–out background is −0.6 ppt-equivalent (234 e^-/s would be needed to fake the eclipse). Centroid shifts are about 1 mpix.
- **Secondary eclipse.** It is found at the ephemeris, not at a free-scan peak. The fixed-shape control at phase 0.5 gives +6.8 ppt with 24/24 events at S/N > 3. All other control phases are within ±0.4 ppt. Individual secondaries show up as census peaks at S/N 6–26. The fold gives 7.6 ± 0.16 ppt at phase 0.50020 ± 0.00019.
- **Ephemeris (timing.py, raw SAP).**
- Primary: P = 5.4921244 ± 0.0000015 d, T0 = 2753.70951 ± 0.00034 BTJD; O−C rms 2.3 min.
- Secondary: P = 5.4921159 ± 0.0000048 d (−0.74 ± 0.43 s per cycle relative to the primary, not significant).
- e cos ω = +0.0003 ± 0.0003.
- **Gaia attribution (MC 20k draws with this star's own corr_vec).**
- Gaia P = 5.492272 ± 0.000409 d, which is Δ −0.36σ from TESS.
- At the Gaia epoch: conjunction A vs the TESS primary is +0.80 h (+0.9σ); conjunction B vs the TESS secondary is +0.45 h (+0.6σ).
- In TESS sectors, conjunction A matches the primary to between −0.01 and −0.29σ in every sector.
- Gaia's predicted phase(B) − phase(A) is 0.5036 ± 0.011; measured 0.5002. Gaia e cos ω is +0.006 ± 0.018 (−0.3σ from the TESS value).
- **Independence: both eclipses appear in every other pipeline.**

Pipeline	Sectors	Primary (ppt)	Secondary (ppt)
TARS	s10/11/37/64	26.7–28.3	6.5–7.4
TESS-SPOC SAP	s10/11/37	20.5–22.6	5.2–5.4
TESS-SPOC PDCSAP (CROWDSAP 0.53–0.62)	s10/11/37	36.9–40.2	9.3–9.7
TGLC PSF photometry at the Gaia position	s10/11	32.1–34.2	7.5–8.6

The dilution-free secondary/primary depth ratio is 0.23–0.26 in every pipeline and sector. Sector-to-sector depth changes in QLP (s010 30.9 → s064 33.6 → s100 27.2 ppt, χ^2 424/20) are aperture dilution in a crowded field. - **Neighbours (neighbours-out.txt).** The only neighbour bright enough to matter is G 11.80 (BP−RP 2.48, T ≈ 10.72) at 41″, which would need a 3.0% dip. The 66.7″ and 85″ stars at G 12.2–12.5 would need 6.2%. Every other star within 30″ would need ≥ 17.9%, mostly > 100%. The VSX EA star at 27″ (P 0.915 d) is G 16.63 and would need > 100%. - **Localization.** - Batch s064 diffimg: deficit–target 3.13″, deficit–light 5.17″, peak_on_target true. - Own locate.py on the TESS-SPOC TPFs: - s010: the target is the best fixed-position source by $\Delta\chi^2$ 3254. It implies a 5.8% dip of the target model, which is crowding-diluted. - s011: the WCS calibration failed. The calibration shift was 10.8″ and the target model flux only 138 e^-/s, so this sector is not usable. - s037: the best fixed position is the G 13.39 star at 11.7″, ahead of the target by $\Delta\chi^2$ 388. That star would need a 157% dip, which is impossible. - The 41″ star (2 px away, inside the fit radius) is not in the top six positions in any sector. - Wave-4/5 lesson: the PSF-model WCS shifts are 6–11″ and unstable (reduced χ^2 ≈ 20, 71 model stars in an 11×11 stamp). Moment light–target offsets (4.9–7.8″) are as large as deficit–target offsets (4.6–8.3″). So the few-arcsecond offsets are the calibration/crowding artefact, not a displaced source. - Taken together: TGLC's PSF photometry at the Gaia position recovers the full depth, no catalogued neighbour can make the dip except at > 100% or at 2 px, and the Gaia orbit times both conjunctions. The signal is on target.

Physics

- Mass function f(m) = 0.0735 M⊙. That gives m2 = 0.557 M⊙ for M1 = 0.96 (TIC) and 0.618 M⊙ for M1 = 1.15, with i ≈ 82.5°.
- Separation a = 15.05 R⊙ = 8.09 R1.
- A 0.56 M⊙ dwarf (R2 ≈ 0.56, k ≈ 0.30) would make a central transit of about 100 ppt. The observed eclipse is grazing, with b_A ≈ 1.04–1.09 R1. The predicted grazing T14 is 3.85–4.3 h against a measured 3.70–3.78 h. The trapezoid flat (0.76 h) is not totality.
- Conjunction A has the companion in front, so it is the deeper eclipse of the G subgiant. Conjunction B is the occultation of the companion.
- The depth ratio of 0.258, in a limb-darkened two-sphere model (twosphere.py), gives J(TESS) ≈ 0.25, so T2 ≈ 3900–3930 K. That is a K7/M0 dwarf of about 0.55–0.6 M⊙, matching the mass function.
- physics.py's built-in main-sequence relation gives T2 4300 K and overpredicts the secondary (13 ppt at e sin ω = 0). That relation runs too hot; it is not a tension in the data.
- e sin ω from the two-sphere duration solve is +0.007 (1σ +0.001..+0.013); the T14 ratio sec/prim is 0.981 ± 0.017. Gaia gives e sin ω = −0.036 ± 0.0125 and e = 0.041 ± 0.012. That is about 3σ in the circular (Lucy–Sweeney) direction, the seventh case of the wave-4 pattern. It is soft, because trapezoid T14s of grazing V-shapes are biased.
- The out-of-eclipse fold shows about ±3 ppt orbital-phase modulation (asymmetric hump, s1 −1.8 ppt, c2 −2.0 ppt). This looks like ellipsoidal variation plus spots or reflection. It was not modelled further.
- One loose end in the blind census: a 17.9-ppt, S/N 8.9 peak at 4088.051 (s100) lies 0.245 d before the 4088.296 primary, at the window edge. It is the matched filter catching that primary's ingress, not a separate event. No other off-ephemeris dip reaches half the eclipse depth.

Prior art

- cats.py (r 60″): no VSX entry at the target. VSX has neighbours only: an EA star with P 0.91475 d at 27″ (G 16.63) and another at 55″ (G 16.76). ASAS-SN: none. SIMBAD lists it as SB* with one reference, Gaia DR3 2023A&A...674A..34G.
- SB9, WDS and J/AJ/160/111: none.
- Gaia DR3: vari_summary and vari_eclipsing_binary have no rows; phot_variable_flag is NOT_AVAILABLE.
- ExoFOP was down tonight: HTTP 503, maintenance page via both urllib and curl. So TOI/CTOI was not re-checked here; the batch sweep covered it.
- A web search on the TIC, TYC and Gaia DR3 ids found nothing.
- **No eclipsing prior art found. The eclipses are new; the SB1 is Gaia's.**

Row fixes

- P = **5.4921244 ± 0.0000015 d** (batch 5.49211). T0 = 2753.70951 ± 0.00034 BTJD (primary).
- Depth: QLP-SAP weighted **29.2 ± 0.1 ppt** (sector range 27.2–33.6; batch 27.5 was the median of the batch estimator). Crowding-corrected PDCSAP is 37–40 ppt.
- Secondary: **7.6 ± 0.2 ppt** (QLP) at phase **0.50020 ± 0.00019**.
- Event count: **21** primaries (batch 20) and 21 secondaries.

- dphi: the primary sits on conjunction A, +0.006 phase at the Gaia epoch (+0.9σ) and −0.0001..−0.013 in TESS sectors. The secondary sits on conjunction B, +0.003 phase at the Gaia epoch (+0.6σ). Equivalently B − A = 0.5002 vs Gaia's 0.5036 ± 0.011. The batch's +0.002/+0.498 is confirmed, and the 0.498 is now an eclipse.
- Diffimg 3.13″: a calibration/crowding artefact, not an offset source.
- T14 3.70–3.78 h, grazing.

Next predicted eclipses (BTJD, linear ephemerides)

- Primary: 4335.4413, 4340.9335, 4346.4256, 4351.9177 (±1 min).
- Secondary: 4338.1861, 4343.6782, 4349.1703, 4354.6624 (±2.5 min).

What would settle it further

- An RV spectrum to check for SB2 lines of a 3900 K companion. In the red, the K7/M0 is about 3–5% of the light.
- A proper limb-darkened light-curve fit with a dilution prior, to get e sin ω and i without trapezoid bias.
- A WCS-calibrated PRF (not Gaussian) difference image in s064/s099, to drive the residual offset down. This is a nicety, not a doubt.

Files

- Output: vet-out.txt, fold_oc.png, events.png, timing-out.txt, shapes-out.txt, twosphere-out.txt, physics-out.txt, neighbours-out.txt, indep-out.txt, locate-s00{10,11,37}-out.txt, gaiatap-out.txt, cats-out.txt, morecats-out.txt, mastlc-out.txt.
- Downloaded products are in C:\Home\sky\data\qlp-sb-p20-60\vet315849332.

A204. TYC 8675-1668-1 (TIC 453601725)

status
: confirmed (per-object re-analysis 2026-10-21; the Gaia SB2 itself, near-twin solar-type pair (1.05 + 1.08 Msun, i 84-85 deg), both eclipses partial, 30.4/29.4 ppt QLP (~36/35 undiluted, TESS-SPOC PDCSAP), T14 2.58 h, 25 primaries + 27 secondaries in 7 sectors (the batch counted one kind); secondary at phase 0.51577(7), e cos omega +0.0248 vs Gaia +0.0085(83), +1.9 sigma in the more-eccentric direction; e sin omega +0.005 to +0.010 from geometry (Gaia +0.0071(87)); VSX ROT (Gaia classifier) at the target: type revision; open ~6-min common-mode ETV across s099-s102 (QLP-only sectors): LTTE or QLP systematic; on target in TESS-SPOC TPFs by dchi2 ~4700-5150; CDIPS and TARS agree; ExoFOP 503 at the check)

P_days
: 5.5666840

eP
: 0.0000021

T0_primary_BTJD
: 3765.53158

depth_primary
: 3.04

depth_secondary
: 2.94

duration_h
: 2.58

source_path
: Fable 5\made\sb1-scale\pervet\vet453601725\REPORT.md

found_by
: Fable 5 (G 10-11.5 batch fold of Gaia orbits, 2026-10-14 to 17); confirmed by an Opus-model per-object re-analysis for Fable 5, pervet wave 6 (2026-10-21)

Archived source: sanitized report text.

TIC 453601725 — per-object vet (pervet wave 6, Opus subagent, 2026-10-21)

(Saved by the parent session; the file-write tool refused REPORT.md for the subagent.)

Verdict: CONFIRMED — a detached, partially eclipsing near-twin solar-type SB2 (Gaia DR3 5871568499564025472, TYC 8675-1668-1), P = 5.566684 d. Both eclipses are present, and each one lands on its own Gaia RV conjunction. The source is on target in two independent pipelines and in the TESS-SPOC pixels.

Batch row: 28.1 ppt, G 10.88, P 5.56669 d, 22 events, dphi −0.007, diffimg 0.72″.

Row fixes

field	batch	this vet
P	5.56669	**5.5666840 ± 0.0000021 d** (pri). Sec: 5.5666831 ± 0.0000022. Gaia: 5.566740 ± 0.000155 (−0.37σ)
T0 (pri)	1605.6571	**3765.53158 ± 0.00026 BTJD**. Sec T0 3790.66947 ± 0.00028
depth	28.1 ppt	QLP SAP: **pri 30.4 ± 0.1, sec 29.4 ± 0.1 ppt** (sector-dependent dilution, 27.4–34.6). Undiluted (TESS-SPOC PDCSAP s11+s38): **pri ~36.3, sec ~35.2 ppt**
event count	22	**25 pri + 27 sec = 52**. s011 (30-min) adds 3 pri; the batch counted only one kind
dphi	−0.007	**pri −0.0075 to −0.013 phase vs Gaia conj A** (−0.74 to −1.22σ). **Sec −0.001 phase vs conj B** (s065, ~−0.1σ). The sign is correct
sec phase	—	0.51577 ± 0.00007 → e cos ω = +0.0248

1. Archives and web

- `cats.py --r 60` (`APPOINTMENTS_DATA` set):
- VSX lists the target itself as **ROT**, with Gaia DR3 as the source. That is a type revision, not prior art: Gaia's classifier called the eclipses rotation. There is also a ROT neighbour at 57″, G 16.20.
- ASAS-SN: none.
- SIMBAD: TYC 8675-1668-1 [SB*] at 0.0″.
- Batch sweeps (c3/c4 ebcats): "none of the vetted/unvetted EB tables". That covers Prša+2022, IJspeert+2024 and Kostov.
- Gaia DR3 TAP:
- NSS **SB2**: P 5.566740(155), e 0.0136(97), ω 39(36)°, K1 77.87(87), K2 76.03, γ −17.0, gof 1.57, significance 89.9.
- vari_summary row exists (in_vari_classification_result true), but in_vari_eclipsing_binary is false and vari_eclipsing_binary is empty.
- RUWE 0.91, ipd_frac_multi_peak 0.
- SB9, WDS, J/AJ/160/111 (VizieR): none.
- ExoFOP json: HTTP 503 twice, so not checked tonight. The batch TOI/CTOI sweep found nothing.
- Web search on "TYC 8675-1668-1" / "TIC 453601725": no paper names the star.
- **No prior eclipse claim found.** The only catalogue entry is Gaia's ROT classification.

2. Eclipse measurements (QLP SAP_FLUX, Q==0, no trimming)

Scripts: `vet.py`, `ecl2.py`, `persector.py`. - **Global trapezoid:** depth 31.1 ppt, T14 2.58 h, T12 1.03 h. A V shape fits badly: $\Delta\chi^2$ +11317 at T12/T14 = 0.1. - **Per event:** all 25 primaries and 27 secondaries have S/N 40–136. - Odd/even primary: 30.59 / 30.31 ppt. - Only 2 events have any bit-32 cadence in the window. - None is within 0.5 d of a listed sector-wide systematic. - **Background:** the in−out excursion is −2 e^-/s (−0.2 ppt equivalent), where a fake eclipse would need +306 e^-/s. The ecl2 dBKG(rel) column has the known broken units and is not used. - **Centroids:** dX +0.17 ± 0.12 mpix, dY +0.10 ± 0.17 mpix, both null. - **Control phases** 0.125–0.875 are null (|d| < 0.11 ppt). The exception is 0.500, at −9 ppt, which is the wing of the secondary at 0.516. - **Blind dip census:** 51 dips with S/N ≥ 4. 25 are the primaries and the other 26 are all the secondary (phase −0.484). Nothing else is in the data. - **Depth vs sector:** the QLP depth drifts from 34.6 (s065) to 27.4 (s102), with χ^2/dof 1355/24. **T14 stays**

constant at 2.50–2.64 h in free per-sector fits, for both kinds. - The sec/pri ratio is also stable: 0.93–0.99. - So the drift is aperture dilution, not a change in inclination. QLP has no crowding correction, while TESS-SPOC PDCSAP gives 36–37 ppt.

3. Ephemeris, O−C and comparison with Gaia

- **Per kind:**
- pri: P 5.5666840(21), O−C rms 1.80 min.
- sec: P 5.5666831(22), O−C rms 1.89 min.
- P(sec) − P(pri) = −0.08 s (−0.3σ), so no apsidal motion is visible.
- **Gaia conjunctions:** Monte Carlo with this star's own corr_vec, through `vetlib.gaia_draws`.
- The primary sits at **conj A**, −0.74 to −1.22σ across the seven sectors.
- The secondary sits at **conj B**, ~−0.1σ.
- Gaia predicts phase(B) − phase(A) = 0.5054 ± 0.0053; measured 0.51577 ± 0.00007, a 1.96σ difference.
- **e cos ω:** the eclipse value is +0.0248. Gaia gives +0.0085 ± 0.0083, a 1.9σ difference in the *more*-eccentric direction (like 1260529 and 191146789).
- **e sin ω:** `geom.py` (two limb-darkened spheres, numerical overlap, e cos ω fixed at +0.0248) gives **+0.005 to +0.010** as the only acceptable range.
- Best χ^2 by e sin ω: 0.5 at +0.010, 5.1 at 0, 77 at +0.015.
- Gaia: +0.0071 ± 0.0087, so it agrees.
- Combined: eclipse e ≈ 0.026, ω ≈ 17°, against Gaia e 0.0136(97) (1.2σ) and ω 39(36)° (0.6σ). Consistent.
- **Common-mode ETV (open):**
- Both kinds show the *same* O−C pattern inside s099–s102: about +2.5…+3 min at s099, falling to −3…−3.7 min at s102, a swing of ~6 min in ~100 d.
- s011–s065 sit near 0. TARS and TESS-SPOC reproduce s011/s038/s065, but s099–s102 are QLP-only.
- Apsidal motion and spots would anti-correlate the two kinds; this does not. It could be light-travel time from a third body or a QLP timing or systematic effect in those sectors.
- It does not affect the verdict. Flagged for follow-up, in the same family as 324273420's common-mode ETV.

4. Independence

- **Second pipelines** (`persector.py`; both kinds in every file):
- TESS-SPOC PDCSAP: s11 pri 37.3 / sec 34.4, s38 35.7 / 35.7 ppt.
- CDIPS IRM1: s11 43.3 / 40.9, s38 37.5 / 37.2.
- TARS: s11 33.9 / 31.3, s38 30.8 / 31.8, s65 33.2 / 31.9.
- In every pipeline T14 is 2.50–2.64 h and the timing agrees within 1.4 min.
- Also at MAST but not downloaded: GSFC-ELEANOR-LITE, PATHOS and TGLC (s11).
- **TESS-SPOC difference images** (`tpfdiff.py`):
- s38: 5 events, peak loss in the target pixel at S/N 40.6. The target PSF fits with χ^2 1593, against 6319 for the next-best star (G 13.99 at 27.7″).
- s11: 1 event, S/N 60.9, χ^2 1429 vs 6583.
- The batch diffimg for s65 has deficit–light 0.26″ and deficit–target 0.72″, peak on target. Clean.

- **Neighbour dilution:** dip each neighbour would need to make the 30 ppt seen in QLP.
- G 13.99 at 27.7″: ×17.6 → **53%** (and the tpfdiff fit wants 106–120 ppt of it).
- G 14.49 at 47.5″: ×27.6 → 84%.
- G 15.19 at 26.3″: ×53 → impossible.
- G 15.70 at 13.8″: ×85 → impossible.
- Every fainter star is impossible. None survives the pixel test.

5. Physics

- **Masses:** Gaia's K1/K2 give M_A sin³i 1.039 and M_B sin³i 1.064 M⊙, with q = M_A/M_B = 0.976 and a sin i 16.93 R⊙.
- **Geometry** (`geom.py`, best solutions):
- i ≈ 84.0–85.0°, so M_A ≈ 1.05 and M_B ≈ 1.08 M⊙, a ≈ 17.0 R⊙.
- R ≈ 0.9–1.05 R⊙ each, which fits the TIC combined radius of 1.40 ≈ √2 × 1.0. Teff 5900–6000 K each (TIC combined 6093).
- Both eclipses are **partial**: a cos i ≈ 1.5–1.8 R⊙ against R_A + R_B ≈ 2. The trapezoid's 0.52-h "flat" is not totality.
- **Which star is in front:**
- Gaia's primary (star A, higher K, slightly lower mass) is eclipsed at conj A, the deeper eclipse.
- Star B is eclipsed at conj B.
- The depth difference (B/A = 0.966) and duration ratio (0.982) are explained by e sin ω ≈ +0.0075 (a smaller impact parameter at A) with T_A ≤ T_B, rather than by a hotter low-mass star.
- **Consistency:** with ~50/50 light, the undiluted depths of ~36/35 ppt and the 2.6-h durations all fit one geometry. Nothing needs third light beyond what PDCSAP already removes.

6. Verdict

CONFIRMED. The deciding numbers: - 52 eclipses of both kinds, each at S/N 40–136, with zero events unexplained in the blind census. - P agrees with Gaia's SB2 RV period to −0.37σ. - The primary is at Gaia conj A (≤1.2σ) and the secondary at conj B (~0.1σ). - e sin ω from geometry (+0.005…+0.010) matches Gaia's +0.0071. e cos ω is in 1.9σ tension in the eccentric direction, which is soft below e ~ 0.05. - The difference image is on target ($\Delta\chi^2$ ≈ 4700–5150 against the best neighbour). The only plausible neighbour would need a 53% dip. - Depth and timing are reproduced by TESS-SPOC, CDIPS and TARS.

Next predicted eclipses (BTJD; formal ±0.5 min, but allow ±5 min given the ETV): - Primary (conj A): **4400.1336, 4405.7002, 4411.2669** - Secondary (conj B): **4403.0046, 4408.5713, 4414.1380**

What would settle it further: - Fresh TESS sectors, or ground timings, to test whether the ~6-min common-mode ETV is real (a third body via LTTE) or a QLP systematic. - An independent light curve for s099–s102: TESScut photometry or TGLC. - A joint RV + light-curve fit to sharpen e and ω.

Files

All in `C:\Home\Fable 5\made\sb1-scale\pervet\vet453601725\`: - Outputs: `vet-out.txt`, `vet-results.json`, `ecl2-out.txt`, `persector-out.txt/.json`, `geom-out.txt`, `gaiatap-out.txt`, `gaia.json`, `cats-out.txt`, `morecats-out.txt`, `mastlc-list.txt`, `mastlc-dl.txt`. - Figures: `events.png`, `fold_oc.png`, `tpfdiff_11.png`,

tpfdiff_38.png. - Downloads: C:\Home\sky\data\qlp-sb-p20-60\vet453601725\.

R evidence: individual adjudicated reports

R001. TIC 373138363

physical_verdict: CONFIRMED

novelty_status: limited: fresh nine-catalog IDIA cones: no target prior EB match; nearby unvetted flags separated

completion: all required gates attempted/adjudicated; limitations retained

next_check: Obtain stellar-radius constraints and joint RV/eclipse characterization.

Archived source: sanitized report text.

TIC 373138363: full adversarial review, 2026-10-30

Physical verdict: CONFIRMED. Repeated primary and secondary eclipses, matching SB1 period/conjunction, independent reduction, target-centered prior image and no capable resolved G<15 neighbor establish an eclipsing target. Stellar dimensions remain model dependent.

Review completion: all required gates were attempted and adjudicated below; unavailable services and inconclusive models are explicit limitations. This is not automatic confirmation of novelty. No catalogue/paper edits or commits.

Deciding measurements

- Raw QLP SAP, QUALITY=0, no gap trimming: 9 primary fits. Median trapezoid depth 29.71 ppt (event range 25.47 to 30.13; median formal/red-noise error 0.29 ppt). Sector medians: {5: 29.85, 32: 30.07, 98: 25.83}.
- Primary T14 4.19 h, conditional profile/grid range 4.10 to 4.28 h; T12 1.47 h; T23 1.26 h. These are trapezoid shape summaries, not a claim of true totality in grazing events.
- P = 7.240208264 +/- 0.000001939 d; Tref = 2671.293626 +/- 0.000328 BTJD at cycle 170. T0 = 1440.458221 BTJD. Chi2/dof 0.55/7; O-C rms 0.62 min. Per-event timing/errors/O-C are in fit-results.json.
- Conditional secondary: 9 fitted windows, 9 individual >3-sigma fits; depth 2.92 +/- 0.07 ppt after between-event scatter; phase 0.500031 +/- 0.000145. T14 4.19 h (range 3.93 to 4.28), T12 1.47 h, T23 1.26 h. Weak/modulated shape uncertainty can exceed these conditional errors.
- Gaia SB1 P offset -0.73 sigma. At BTJD389, primary minus conjunction A = +0.21 h versus Gaia sigma 0.89 h. Full per-event conjunction comparison uses this source's covariance. Bit index127 and 15 correlations asserted against ESA's SB1 ordering; omega-Tp correlation +0.999976.
- First-order e cos(omega) = +0.0000 +/- 0.0002; for eccentric systems compare exact Gaia conjunction separation rather than treating this expansion as exact. Gaia e=0.0104 +/- 0.0142, omega=292.96 +/- 90.22 degrees.

Gate adjudication

Gate	Status	Evidence and limits
Own archives/prior art	completed; no target prior EB match	ExoFOP HTTP200, TOI=0/CTOI=0; SIMBAD CFA exact Gaia ID is SB*. Shared ARI Gaia variability cache has no EB rows among70 sources. CDS and Harvard timed out, then IDIA succeeded: fresh 60arcsec cones in VSX, ASAS-SN, SB9, WDS, Kostov table2/3/4, IJspeert and Prsa. All nine table sections returned; no target EB prior-art match. Neighbor ROT/unvetted table2 matches are distinct sources. Raw responses and hashes in *_idia.tsv/idia-receipts.json. Identifier web searches have no pertinent hit, not proof of novelty.
Individual events/ephemerides	measured	fit-results.json, secondary-fit.json: integrated-exposure trapezoids, local quadratic baselines, fixed shared shape per kind, profile timing error with cadence/grid floors; primary residual bin-beta range 1.00 to 1.22; ephemeris errors inflated when chi2/dof>1. Secondary ephemeris stored when >=3 strong events; two-event cases use primary P.
Flags/background/centroids	tested; conditional	Each fit records flagged core counts, OR bits, template centroids/errors, background regression and refitted depth. Maximum predicted fake-background primary depth 0.22 ppt. Native background units never treated as flux. Sparse flags/gaps remain exclusions, not clean nulls.
Blind census	measured; coverage limited	19 candidate dip groups at max(4 robust per-cadence sigma,3ppt), including 1 away from measured eclipses. Thresholded rolling median can detect intrinsic modulation/gap edges. For straightforward objects the lone off-phase group lacks sufficient clean baseline for event fitting (offphase-checks.json); no claim it is an eclipse.
Other eclipse/controls	measured; conditional	Gaia-fixed-phase tests and off-phase controls in orbit-gates.json; optimized secondary timing/shape separately in secondary-fit.json. Any quoted null upper limit is primary-shape/Gaia-phase conditional, not general exclusion.
Independent pipeline	passed for aperture events	TESS-SPOC SAP/PDCSAP retrieval and QUALITY=0 core estimates in indep_measure.txt; different reduction of same TESS data, not a separate observatory. Amplitude differences include dilution/sampling.
Neighbors/localization	supports target	Fresh ARI G<15 cone within60arcsec; required fractional dip estimates in neighbors_summary.csv are G-band proxies without aperture throughput. Historical difference-image deficit-target offset 0.39arcsec, peak on target. See star-specific verdict and limitations.
Physics/e sin(omega)	modeled; conditional	RV mass function 0.03001 Msun; minimum m2 0.436 to 0.436 Msun for listed M1 hypotheses. Predicted central primary T14 7.85 to 7.85 h versus observed 4.19 h, compatible with a noncentral chord. Numerical limb-darkened two-disk fits in geometry.json explore both shapes with fixed TIC dimensions, free depth scales and physical scale<=1; degeneracies prevent a robust independent e sin(omega) measurement. Do not interpret the coarse-grid profile interval as a calibrated orbital confidence interval.

Next predictions (BTJD, from 2026-10-30 noon): primary [np.float64(4351.02194), np.float64(4358.26215)]; secondary [np.float64(4347.40207), np.float64(4354.64227)]. Timing extrapolations inherit ephemeris and secondary-shape systematics; no precise future uncertainty for the modulated system.

Next deciding check: Obtain stellar-radius constraints and joint RV/eclipse characterization.

Provenance: ../downloads.json and ../receipts.json; shared-full-review Gaia receipts; independent CAOM responses and indep_sha256.txt. Scripts fit_batch.py/orbit_gates.py/secondary_fit.py/geometry.py reproduce numeric results. Optional Terra personal note was not supplied; its scientific contribution is preserved verbatim in INDEPENDENT-NOTE.md. Lead: Sol batch A.

R002. TIC 270181578

physical_verdict: CONFIRMED

novelty_status: limited: fresh nine-catalog IDIA cones: no target prior EB match; nearby unvetted flags separated

completion: all required gates attempted/adjudicated; limitations retained

next_check: Characterize high-RUWE multiplicity with high-resolution imaging and better primary mass/radius; then joint light/RV fit.

Archived source: sanitized report text.

TIC 270181578: full adversarial review, 2026-10-30

Physical verdict: CONFIRMED. Repeated primary and secondary eclipses match the SB1 ephemeris, an independent pipeline recovers the deficits, the historical deficit is centered on target and no capable resolved G<15 neighbor exists. RUWE 7.688 and conditional dilution limit stellar characterization, not the supported eclipsing-target classification.

Review completion: all required gates were attempted and adjudicated below; unavailable services and inconclusive models are explicit limitations. This is not automatic confirmation of novelty. No catalogue/paper edits or commits.

Deciding measurements

- Raw QLP SAP, QUALITY=0, no gap trimming: 8 primary fits. Median trapezoid depth 27.56 ppt (event range 25.78 to 28.16; median formal/red-noise error 0.30 ppt). Sector medians: {55: 27.66, 82: 27.55}.
- Primary T14 5.76 h, conditional profile/grid range 5.63 to 5.90 h; T12 1.44 h; T23 2.88 h. These are trapezoid shape summaries, not a claim of true totality in grazing events.
- P = 5.573958606 +/- 0.000005421 d; Tref = 3179.434865 +/- 0.000356 BTJD at cycle 68. T0 = 2800.405680 BTJD. Chi2/dof 11.70/6; O-C rms 1.26 min. Per-event timing/errors/O-C are in fit-results.json.
- Conditional secondary: 6 fitted windows, 6 individual >3-sigma fits; depth 2.56 +/- 0.08 ppt after between-event scatter; phase 0.500343 +/- 0.000461. T14 6.03 h (range 5.63 to 6.16), T12 1.51 h, T23 3.01 h. Weak/modulated shape uncertainty can exceed these conditional errors.
- Gaia SB1 P offset +0.82 sigma. At BTJD389, primary minus conjunction A = +0.43 h versus Gaia sigma 0.64 h. Full per-event conjunction comparison uses this source's covariance. Bit index127 and 15 correlations asserted against ESA's SB1 ordering; omega-Tp correlation +0.999825.
- First-order e cos(omega) = +0.0005 +/- 0.0007; for eccentric systems compare exact Gaia conjunction separation rather than treating this expansion as exact. Gaia e=0.0218 +/- 0.0127, omega=341.15 +/- 45.66 degrees.

Gate adjudication

Gate	Status	Evidence and limits
Own archives/prior art	completed; no target prior EB match	ExoFOP HTTP200, TOI=0/CTOI=0; SIMBAD CFA exact Gaia ID is SB*. Shared ARI Gaia variability cache has no EB rows among70 sources. CDS and Harvard timed out, then IDIA succeeded: fresh 60arcsec cones in VSX, ASAS-SN, SB9, WDS, Kostov table2/3/4, IJspeert and Prsa. All nine table sections returned; no target EB prior-art match. Neighbor ROT/unvetted table2 matches are distinct sources. Raw responses and hashes in *_idia.tsv/idia-receipts.json. Identifier web searches have no pertinent hit, not proof of novelty.
Individual events/ephemerides	measured	fit-results.json, secondary-fit.json: integrated-exposure trapezoids, local quadratic baselines, fixed shared shape per kind, profile timing error with cadence/grid floors; primary residual bin-beta range 1.25 to 1.78; ephemeris errors inflated when chi2/dof>1. Secondary ephemeris stored when >=3 strong events; two-event cases use primary P.
Flags/background/centroids	tested; conditional	Each fit records flagged core counts, OR bits, template centroids/errors, background regression and refitted depth. Maximum predicted fake-background primary depth 1.53 ppt. Native background units never treated as flux. Sparse flags/gaps remain exclusions, not clean nulls.
Blind census	measured; coverage limited	17 candidate dip groups at max(4 robust per-cadence sigma,3ppt), including 1 away from measured eclipses. Thresholded rolling median can detect intrinsic modulation/gap edges. For straightforward objects the lone off-phase group lacks sufficient clean baseline for event fitting (offphase-checks.json); no claim it is an eclipse.
Other eclipse/controls	measured; conditional	Gaia-fixed-phase tests and off-phase controls in orbit-gates.json; optimized secondary timing/shape separately in secondary-fit.json. Any quoted null upper limit is primary-shape/Gaia-phase conditional, not general exclusion.
Independent pipeline	passed for aperture events	TESS-SPOC SAP/PDCSAP retrieval and QUALITY=0 core estimates in indep_measure.txt; different reduction of same TESS data, not a separate observatory. Amplitude differences include dilution/sampling.
Neighbors/localization	supports target	Fresh ARI G<15 cone within60arcsec; required fractional dip estimates in neighbors_summary.csv are G-band proxies without aperture throughput. Historical difference-image deficit-target offset 1.47arcsec, peak on target. See star-specific verdict and limitations.
Physics/e sin(omega)	modeled; conditional	RV mass function 0.02443 Msun; minimum m2 0.334 to 0.472 Msun for listed M1 hypotheses. Predicted central primary T14 6.96 to 7.88 h versus observed 5.76 h, compatible with a noncentral chord. Numerical limb-darkened two-disk fits in geometry.json explore both shapes with fixed TIC dimensions, free depth scales and physical scale<=1; degeneracies prevent a robust independent e sin(omega) measurement. Do not interpret the coarse-grid profile interval as a calibrated orbital confidence interval.

Next predictions (BTJD, from 2026-10-30 noon): primary [np.float64(4344.39221), np.float64(4349.96617)]; secondary [np.float64(4347.18111), np.float64(4352.75506)]. Timing extrapolations inherit ephemeris and secondary-shape systematics; no precise future uncertainty for the modulated system.

Next deciding check: Characterize high-RUWE multiplicity with high-resolution imaging and better primary mass/radius; then joint light/RV fit.

Provenance: ../downloads.json and ../receipts.json; shared-full-review Gaia receipts; independent CAOM responses and indep_sha256.txt. Scripts fit_batch.py/orbit_gates.py/secondary_fit.py/geometry.py reproduce numeric results. Optional Terra personal note was not supplied; its scientific contribution is preserved verbatim in INDEPENDENT-NOTE.md. Lead: Sol batch A.

Specific caution: no other G<15 resolved neighbor within60arcsec, but this does not exclude unresolved light. RUWE7.688 and conditional geometry primary scale~0.45 motivate multiplicity/dilution follow-up rather than a unique binary geometry.

R003. TIC 298645930

physical_verdict: PLAUSIBLE

novelty_status: limited: fresh nine-catalog IDIA cones: no target prior EB match; nearby unvetted flags separated

completion: all required gates attempted/adjudicated; limitations retained

next_check: Calibrate a realistic PRF/multi-source pixel model and separate periodic modulation before assessing secondary/heartbeat alternatives; do not classify neighbor ROT as target prior art.

Archived source: sanitized report text.

TIC 298645930: full adversarial review, 2026-10-30

Physical verdict: PLAUSIBLE. The target aperture contains repeated sharp primary deficits on strong variable modulation. Secondary amplitude changes materially with template; timing and blind census are contaminated. A real 24.78 arcsec neighbor and inadequate calibrated PSF fit leave target attribution/model unresolved.

Review completion: all required gates were attempted and adjudicated below; unavailable services and inconclusive models are explicit limitations. This is not automatic confirmation of novelty. No catalogue/paper edits or commits.

Deciding measurements

- Raw QLP SAP, QUALITY=0, no gap trimming: 117 primary fits. Median trapezoid depth 30.05 ppt (event range 27.23 to 32.95; median formal/red-noise error 0.77 ppt). Sector medians: {14: 29.09, 25: 29.52, 26: 30.43, 40: 30.34, 41: 29.09, 52: 29.57, 54: 30.05, 58: 30.11, 74: 30.0, 79: 30.16, 80: 31.14, 81: 31.37, 85: 30.08}.
- Primary T14 3.50 h, conditional profile/grid range 3.41 to 3.58 h; T12 1.22 h; T23 1.05 h. These are trapezoid shape summaries, not a claim of true totality in grazing events.
- P = 2.518978031 +/- 0.000001199 d; Tref = 2820.951170 +/- 0.000214 BTJD at cycle 451. T0 = 1684.892078 BTJD. Chi2/dof 1543.62/115; O-C rms 4.66 min. Per-event timing/errors/O-C are in fit-results.json.
- Conditional secondary: 116 fitted windows, 98 individual >3-sigma fits; depth 6.64 +/- 0.20 ppt after between-event scatter; phase 0.500532 +/- 0.000685. T14 5.41 h (range 4.63 to 5.84), T12 2.43 h, T23 0.54 h. Weak/modulated shape uncertainty can exceed these conditional errors.
- Gaia SB1 P offset -0.37 sigma. At BTJD389, primary minus conjunction A = +0.68 h versus Gaia sigma 1.00 h. Full per-event conjunction comparison uses this source's covariance. Bit index127 and 15 correlations asserted against ESA's SB1 ordering; omega-Tp correlation +0.995618.
- First-order e cos(omega) = +0.0008 +/- 0.0011; for eccentric systems compare exact Gaia conjunction separation rather than treating this expansion as exact. Gaia e=0.0830 +/- 0.0564, omega=50.26 +/- 34.46 degrees.

Gate adjudication

Gate	Status	Evidence and limits
Own archives/prior art	completed; no target prior EB match	ExoFOP HTTP200, TOI=0/CTOI=0; SIMBAD CFA exact Gaia ID is SB*. Shared ARI Gaia variability cache has no EB rows among70 sources. CDS and Harvard timed out, then IDIA succeeded: fresh 60arcsec cones in VSX, ASAS-SN, SB9, WDS, Kostov table2/3/4, IJspeert and Prsa. All nine table sections returned; no target EB prior-art match. Neighbor ROT/unvetted table2 matches are distinct sources. Raw responses and hashes in *_idia.tsv/idia-receipts.json. Identifier web searches have no pertinent hit, not proof of novelty.
Individual events/ephemerides	measured	fit-results.json, secondary-fit.json: integrated-exposure trapezoids, local quadratic baselines, fixed shared shape per kind, profile timing error with cadence/grid floors; primary residual bin-beta range 1.00 to 2.08; ephemeris errors inflated when chi2/dof>1. Secondary ephemeris stored when >=3 strong events; two-event cases use primary P.
Flags/background/centroids	tested; conditional	Each fit records flagged core counts, OR bits, template centroids/errors, background regression and refitted depth. Maximum predicted fake-background primary depth 6.10 ppt. Native background units never treated as flux. Sparse flags/gaps remain exclusions, not clean nulls.
Blind census	measured; coverage limited	333 candidate dip groups at max(4 robust per-cadence sigma,3ppt), including 92 away from measured eclipses. Thresholded rolling median can detect intrinsic modulation/gap edges. For straightforward objects the lone off-phase group lacks sufficient clean baseline for event fitting (offphase-checks.json); no claim it is an eclipse.
Other eclipse/controls	measured; conditional	Gaia-fixed-phase tests and off-phase controls in orbit-gates.json; optimized secondary timing/shape separately in secondary-fit.json. Any quoted null upper limit is primary-shape/Gaia-phase conditional, not general exclusion.
Independent pipeline	passed for aperture events	TESS-SPOC SAP/PDCSAP retrieval and QUALITY=0 core estimates in indep_measure.txt; different reduction of same TESS data, not a separate observatory. Amplitude differences include dilution/sampling.
Neighbors/localization	inconclusive	Fresh ARI G<15 cone within60arcsec; required fractional dip estimates in neighbors_summary.csv are G-band proxies without aperture throughput. Historical difference-image deficit-target offset 2.68arcsec, peak on target. See star-specific verdict and limitations.
Physics/e sin(omega)	modeled; conditional	RV mass function 0.01252 Msun; minimum m2 0.393 to 0.393 Msun for listed M1 hypotheses. Predicted central primary T14 4.65 to 4.65 h versus observed 3.50 h, compatible with a noncentral chord. Numerical limb-darkened two-disk fits in geometry.json explore both shapes with fixed TIC dimensions, free depth scales and physical scale<=1; degeneracies prevent a robust independent e sin(omega) measurement. Do not interpret the coarse-grid profile interval as a calibrated orbital confidence interval.

Next predictions (BTJD, from 2026-10-30 noon): primary [np.float64(4344.93288), np.float64(4347.45186)]; secondary [np.float64(4346.19371), np.float64(4348.71268)]. Timing extrapolations inherit ephemeris and secondary-shape systematics; no precise future uncertainty for the modulated system.

Next deciding check: Calibrate a realistic PRF/multi-source pixel model and separate periodic modulation before assessing secondary/heartbeat alternatives; do not classify neighbor ROT as target prior art.

Provenance: ../downloads.json and ../receipts.json; shared-full-review Gaia receipts; independent CAOM responses and indep_sha256.txt. Scripts fit_batch.py/orbit_gates.py/secondary_fit.py/geometry.py reproduce numeric results. Optional Terra personal note was not supplied; its scientific contribution is preserved verbatim in INDEPENDENT-NOTE.md. Lead: Sol batch A.

Specific caution: source2123135139098776320 is a real G13.62 neighbor24.78arcsec away; its ROT classification is not the target2123135139098776448. Pixel298.py reused a hashed existing s41 cube: target beats neighbor by delta chi2 about22825, but free centroid~10arcsec and WCS shift at grid boundary make precise localization unreliable. Baseline sensitivity, broad secondary shape and large off-phase census preclude confirmation. Tide298.py compares equilibrium-tide plus five orbital

harmonics per sector against the same plus a narrow primary: delta BIC17274 favors the eclipse component, with primary depths28.4 to30.8ppt. Correlations prohibit formal FAP; secondary and calibrated localization remain unresolved.

R004. TIC 362610608

physical_verdict: CONFIRMED

novelty_status: limited: fresh nine-catalog IDIA cones: no target prior EB match; nearby unvetted flags separated

completion: all required gates attempted/adjudicated; limitations retained

next_check: Compare per-sector apertures to explain s99 depth decrease; improve PRF and stellar characterization.

Archived source: sanitized report text.

TIC 362610608: full adversarial review, 2026-10-30

Physical verdict: CONFIRMED. Repeated primary and secondary eclipses agree with the Gaia orbit and an independent pipeline. Own s63 pixel-template fitting places the calibrated deficit 0.23 arcsec from target and beats the capable 13.11 arcsec neighbor by delta chi2 about14709. This resolves the historical crowding concern for the eclipse source; precise astrometry still has PSF systematics.

Review completion: all required gates were attempted and adjudicated below; unavailable services and inconclusive models are explicit limitations. This is not automatic confirmation of novelty. No catalogue/paper edits or commits.

Deciding measurements

- Raw QLP SAP, QUALITY=0, no gap trimming: 18 primary fits. Median trapezoid depth 30.92 ppt (event range 25.43 to 32.05; median formal/red-noise error 0.32 ppt). Sector medians: {9: 30.08, 10: 30.13, 36: 31.67, 62: 31.51, 63: 31.72, 89: 30.69, 90: 31.23, 99: 25.68}.
- Primary T14 5.24 h, conditional profile/grid range 5.11 to 5.37 h; T12 1.84 h; T23 1.57 h. These are trapezoid shape summaries, not a claim of true totality in grazing events.
- P = 9.494540661 +/- 0.000002536 d; Tref = 2930.088859 +/- 0.000194 BTJD at cycle 145. T0 = 1553.380464 BTJD. Chi2/dof 7.88/16; O-C rms 0.86 min. Per-event timing/errors/O-C are in fit-results.json.
- Conditional secondary: 17 fitted windows, 17 individual >3-sigma fits; depth 1.79 +/- 0.08 ppt after between-event scatter; phase 0.533366 +/- 0.000182. T14 4.72 h (range 4.59 to 4.85), T12 0.71 h, T23 3.31 h. Weak/modulated shape uncertainty can exceed these conditional errors.
- Gaia SB1 P offset -0.09 sigma. At BTJD389, primary minus conjunction A = +0.70 h versus Gaia sigma 2.35 h. Full per-event conjunction comparison uses this source's covariance. Bit index127 and 15 correlations asserted against ESA's SB1 ordering; omega-Tp correlation +0.996035.
- First-order e cos(omega) = +0.0524 +/- 0.0003; for eccentric systems compare exact Gaia conjunction separation rather than treating this expansion as exact. Gaia e=0.0795 +/- 0.0242, omega=322.13 +/- 12.83 degrees.

Gate adjudication

Gate	Status	Evidence and limits
Own archives/prior art	completed; no target prior EB match	ExoFOP HTTP200, TOI=0/CTOI=0; SIMBAD CFA exact Gaia ID is SB*. Shared ARI Gaia variability cache has no EB rows among70 sources. CDS and Harvard timed out, then IDIA succeeded: fresh 60arcsec cones in VSX, ASAS-SN, SB9, WDS, Kostov table2/3/4, IJspeert and Prsa. All nine table sections returned; no target EB prior-art match. Neighbor ROT/unvetted table2 matches are distinct sources. Raw responses and hashes in *_idia.tsv/idia-receipts.json. Identifier web searches have no pertinent hit, not proof of novelty.
Individual events/ephemerides	measured	fit-results.json, secondary-fit.json: integrated-exposure trapezoids, local quadratic baselines, fixed shared shape per kind, profile timing error with cadence/grid floors; primary residual bin-beta range 1.00 to 1.52; ephemeris errors inflated when chi2/dof>1. Secondary ephemeris stored when >=3 strong events; two-event cases use primary P.
Flags/background/centroids	tested; conditional	Each fit records flagged core counts, OR bits, template centroids/errors, background regression and refitted depth. Maximum predicted fake-background primary depth 0.28 ppt. Native background units never treated as flux. Sparse flags/gaps remain exclusions, not clean nulls.
Blind census	measured; coverage limited	20 candidate dip groups at max(4 robust per-cadence sigma,3ppt), including 1 away from measured eclipses. Thresholded rolling median can detect intrinsic modulation/gap edges. For straightforward objects the lone off-phase group lacks sufficient clean baseline for event fitting (offphase-checks.json); no claim it is an eclipse.
Other eclipse/controls	measured; conditional	Gaia-fixed-phase tests and off-phase controls in orbit-gates.json; optimized secondary timing/shape separately in secondary-fit.json. Any quoted null upper limit is primary-shape/Gaia-phase conditional, not general exclusion.
Independent pipeline	passed for aperture events	TESS-SPOC SAP/PDCSAP retrieval and QUALITY=0 core estimates in indep_measure.txt; different reduction of same TESS data, not a separate observatory. Amplitude differences include dilution/sampling.
Neighbors/localization	supports target	Fresh ARI G<15 cone within60arcsec; required fractional dip estimates in neighbors_summary.csv are G-band proxies without aperture throughput. Historical difference-image deficit-target offset 3.92arcsec, peak on target. See star-specific verdict and limitations.
Physics/e sin(omega)	modeled; conditional	RV mass function 0.01122 Msun; minimum m2 0.260 to 0.260 Msun for listed M1 hypotheses. Predicted central primary T14 8.14 to 8.14 h versus observed 5.24 h, compatible with a noncentral chord. Numerical limb-darkened two-disk fits in geometry.json explore both shapes with fixed TIC dimensions, free depth scales and physical scale<=1; degeneracies prevent a robust independent e sin(omega) measurement. Do not interpret the coarse-grid profile interval as a calibrated orbital confidence interval.

Next predictions (BTJD, from 2026-10-30 noon): primary [np.float64(4344.77542), np.float64(4354.26996)]; secondary [np.float64(4349.83949), np.float64(4359.33403)]. Timing extrapolations inherit ephemeris and secondary-shape systematics; no precise future uncertainty for the modulated system.

Next deciding check: Compare per-sector apertures to explain s99 depth decrease; improve PRF and stellar characterization.

Provenance: ../downloads.json and ../receipts.json; shared-full-review Gaia receipts; independent CAOM responses and indep_sha256.txt. Scripts fit_batch.py/orbit_gates.py/secondary_fit.py/geometry.py reproduce numeric results. Optional Terra personal note was not supplied; its scientific contribution is preserved verbatim in INDEPENDENT-NOTE.md. Lead: Sol batch A.

Own pixel362.py uses the existing hashed s63 TESScut and three eclipse templates. Target fixed-source chi2=4556 versus nearest capable neighbor19265, free best0.23arcsec from calibrated target. Gaussian PSF shape is approximate and formal chi2 ignores PSF covariance; relative source discrimination is strong, precise centroid uncertainty is not claimed.

R005. TIC 399501645

physical_verdict: CONFIRMED

novelty_status: limited: fresh nine-catalog IDIA cones: no target prior EB match; nearby unvetted flags separated

completion: all required gates attempted/adjudicated; limitations retained

next_check: Improve weak secondary shape and stellar-radius/dilution characterization.

Archived source: sanitized report text.

TIC 399501645: full adversarial review, 2026-10-30

Physical verdict: CONFIRMED. Three primaries and two repeated eccentric-phase secondary events, Gaia timing, an independent reduction and target-centered historical deficit support an eclipsing target. The sole capable resolved neighbor is 51.87 arcsec away; existing image has peak on target and 3.62 arcsec deficit-target offset.

Review completion: all required gates were attempted and adjudicated below; unavailable services and inconclusive models are explicit limitations. This is not automatic confirmation of novelty. No catalogue/paper edits or commits.

Deciding measurements

- Raw QLP SAP, QUALITY=0, no gap trimming: 3 primary fits. Median trapezoid depth 28.89 ppt (event range 28.61 to 28.94; median formal/red-noise error 0.28 ppt). Sector medians: {54: 28.78, 81: 28.89}.
- Primary T14 4.54 h, conditional profile/grid range 4.45 to 4.63 h; T12 1.13 h; T23 2.27 h. These are trapezoid shape summaries, not a claim of true totality in grazing events.
- P = 11.985795081 +/- 0.000014084 d; Tref = 3024.962624 +/- 0.000415 BTJD at cycle 21. T0 = 2773.260927 BTJD. Chi2/dof 1.61/1; O-C rms 0.60 min. Per-event timing/errors/O-C are in fit-results.json.
- Conditional secondary: 2 fitted windows, 2 individual >3-sigma fits; depth 1.50 +/- 0.18 ppt after between-event scatter; phase 0.430268 +/- 0.000344. T14 3.67 h (range 3.06 to 4.80), T12 0.55 h, T23 2.57 h. Weak/modulated shape uncertainty can exceed these conditional errors.
- Gaia SB1 P offset +0.08 sigma. At BTJD389, primary minus conjunction A = -0.91 h versus Gaia sigma 3.77 h. Full per-event conjunction comparison uses this source's covariance. Bit index127 and 15 correlations asserted against ESA's SB1 ordering; omega-Tp correlation +0.985543.
- First-order e cos(omega) = -0.1095 +/- 0.0005; for eccentric systems compare exact Gaia conjunction separation rather than treating this expansion as exact. Gaia e=0.2151 +/- 0.0284, omega=242.11 +/- 9.69 degrees.

Gate adjudication

Gate	Status	Evidence and limits
Own archives/prior art	completed; no target prior EB match	ExoFOP HTTP200, TOI=0/CTOI=0; SIMBAD CFA exact Gaia ID is SB*. Shared ARI Gaia variability cache has no EB rows among70 sources. CDS and Harvard timed out, then IDIA succeeded: fresh 60arcsec cones in VSX, ASAS-SN, SB9, WDS, Kostov table2/3/4, IJspeert and Prsa. All nine table sections returned; no target EB prior-art match. Neighbor ROT/unvetted table2 matches are distinct sources. Raw responses and hashes in *_idia.tsv/idia-receipts.json. Identifier web searches have no pertinent hit, not proof of novelty.
Individual events/ephemerides	measured	fit-results.json, secondary-fit.json: integrated-exposure trapezoids, local quadratic baselines, fixed shared shape per kind, profile timing error with cadence/grid floors; primary residual bin-beta range 1.02 to 1.46; ephemeris errors inflated when chi2/dof>1. Secondary ephemeris stored when >=3 strong events; two-event cases use primary P.
Flags/background/centroids	tested; conditional	Each fit records flagged core counts, OR bits, template centroids/errors, background regression and refitted depth. Maximum predicted fake-background primary depth 0.19 ppt. Native background units never treated as flux. Sparse flags/gaps remain exclusions, not clean nulls.
Blind census	measured; coverage limited	4 candidate dip groups at max(4 robust per-cadence sigma,3ppt), including 1 away from measured eclipses. Thresholded rolling median can detect intrinsic modulation/gap edges. For straightforward objects the lone off-phase group lacks sufficient clean baseline for event fitting (offphase-checks.json); no claim it is an eclipse.
Other eclipse/controls	measured; conditional	Gaia-fixed-phase tests and off-phase controls in orbit-gates.json; optimized secondary timing/shape separately in secondary-fit.json. Any quoted null upper limit is primary-shape/Gaia-phase conditional, not general exclusion.
Independent pipeline	passed for aperture events	TESS-SPOC SAP/PDCSAP retrieval and QUALITY=0 core estimates in indep_measure.txt; different reduction of same TESS data, not a separate observatory. Amplitude differences include dilution/sampling.
Neighbors/localization	supports target	Fresh ARI G<15 cone within60arcsec; required fractional dip estimates in neighbors_summary.csv are G-band proxies without aperture throughput. Historical difference-image deficit-target offset 3.62arcsec, peak on target. See star-specific verdict and limitations.
Physics/e sin(omega)	modeled; conditional	RV mass function 0.00459 Msun; minimum m2 0.211 to 0.211 Msun for listed M1 hypotheses. Predicted central primary T14 6.62 to 6.62 h versus observed 4.54 h, compatible with a noncentral chord. Numerical limb-darkened two-disk fits in geometry.json explore both shapes with fixed TIC dimensions, free depth scales and physical scale<=1; degeneracies prevent a robust independent e sin(omega) measurement. Do not interpret the coarse-grid profile interval as a calibrated orbital confidence interval.

Next predictions (BTJD, from 2026-10-30 noon): primary [np.float64(4355.38588), np.float64(4367.37167)]; secondary [np.float64(4348.55719), np.float64(4360.54299)]. Timing extrapolations inherit ephemeris and secondary-shape systematics; no precise future uncertainty for the modulated system.

Next deciding check: Improve weak secondary shape and stellar-radius/dilution characterization.

Provenance: ../downloads.json and ../receipts.json; shared-full-review Gaia receipts; independent CAOM responses and indep_sha256.txt. Scripts fit_batch.py/orbit_gates.py/secondary_fit.py/geometry.py reproduce numeric results. Optional Terra personal note was not supplied; its scientific contribution is preserved verbatim in INDEPENDENT-NOTE.md. Lead: Sol batch A.

R006. TIC 295945076

physical_verdict: CONFIRMED

novelty_status: no prior target eclipsing classification in successfully queried vetted sources; unvetted flags separate

completion: full_attempted_and_adjudicated

next_check: Refine secondary timing and Gaia small-e orbit; multi-sector calibrated pixels

Archived source: sanitized report text.

TIC 295945076 (TYC 5969-485-1): full attempted and adjudicated review

Verdict: CONFIRMED physical eclipsing target/system associated with the Gaia SB1. Repeated primary and secondary eclipses, orbital timing, independent photometry and localization support this classification. Exact component properties remain uncertain; this is not a catalogue edit or publication action.

Deciding light-curve evidence

13 primary event fits pass the stated basic fit/coverage screen; the stricter cadence/gap screen retains 10. Median fitted primary depth is 32.32 ppt, with median conditional per-event depth error 0.51 ppt. Median T14/T12/T23 are 3.12/1.25/0.65 hours. These are fitted trapezoid descriptors; partial eclipses do not imply physical totality. Individual profile ranges, counts and gaps are in 295945076_measurements.json. Shape grids and baseline systematics limit errors, especially at profile boundaries.

16 secondary timing fits give median depth 21.64 ppt and phase 0.521923 +/- 0.00250 cycles. The latter is a conservative heuristic envelope from phase-grid resolution, width alternatives and leave-one-sector scatter, not a calibrated confidence interval. Fixed-phase/alternate-width controls and covered windows are preserved in secondary JSONs. Approximate small-e e*cos(omega) = +0.0344 +/- 0.0039; no trapezoid duration-ratio estimate of e*sin(omega) is claimed.

Timing and Gaia

Primary T0 = 2217.12888360 +/- 0.00030392 BTJD, P = 5.908926218 +/- 0.000002164 d. Primary O-C rms 1.37 min. Covariance is scaled by max(1, chi-square/dof); stricter-mask P = 5.908927003 d. Secondary P = 5.908926897 +/- 0.000002068 d, O-C rms 1.65 min; primary-secondary difference +0.06 +/- 0.26 seconds/cycle. Covariances and all O-C values remain in JSON.

The Gaia comparison uses this source's full SB1 row, bit_index 127, its own 15-element correlation vector and the documented six-parameter order. The primary is +1.41 sigma from conjunction A near the Gaia epoch; photometric-Gaia period difference is -0.35 sigma. Per-sector A/B timing comparisons are saved. Physical-e rejection fraction 0.0% means the Gaussian orbit approximation is conditional and small-e priors remain soft.

Independence, systematics and attribution

TESS-SPOC sector 34 independently recovers time-aligned dips: SAP 29.1 to 31.1 ppt and PDCSAP 33.5 to 35.3 ppt. Their difference is a dilution/detrending uncertainty, not evidence of absent eclipses. Raw QLP SAP with QUALITY=0 is the primary source; no post-gap trim was used. Flags, raw centroids, native-unit background excursions and gap coverage were examined. Background regression changes primary depths by at most 1.2%; this is a unit-free sensitivity check and does not convert SAP_BKG to fractional stellar flux. The secondary is at phase 0.52192, not the Gaia median 0.53683. The difference is about 1.7 sigma after the heuristic timing envelope; the Gaia orbit is not contradicted. Sector 87 has baseline/depth shifts. The grazing solution has no securely measured totality.

The blind coarse-box census retains 29 peaks, 3 gap-flagged, and 0 classified off the two eclipse ephemerides. Its max(3 ppt, three flank-MAD) threshold is not a significance calibration or exhaustive shallow-event census. Fresh TESScut template fitting in sector 34 uses two events. The target is preferred to all listed neighbours; free and split-event PSF centroids lie within about 0.07 pixel of the catalogue position. These are unweighted Gaussian-PSF/WCS diagnostics, not calibrated localization sigma. Fresh 60-arcsec Gaia neighbours and G-proxy dilution requirements are in neighbours-dilution.csv; aperture throughput, colour and unresolved companions remain limitations.

Physics and archives

RV mass function 0.1309 Msun; using catalog M1=1.188, edge-on M2min=0.804 Msun and a=17.31 Rsun. A shared limb-darkened two-disc model was actually tested against both eclipse shapes, with finite-exposure sampling, radius/impact/e*sin(omega)/light-ratio freedom. Illustrative best k=0.390, b_A=1.12, R1 scale=0.85, R2=0.602 Rsun; diagnostic score 98.4 for 120 bins. Correlated bins, fixed limb darkening, local trends, no third light and coarse bounds prevent precision physical inference. This is a plausible geometry family, not independently measured masses/radii.

Fresh SIMBAD identifies the target SB*. ExoFOP has no TOI/CTOI. Successful IDIA cones find no target entry in ASAS-SN, SB9, WDS, Prsa, IJspeert or vetted Kostov table3/4. Kostov table2 is unvetted and is not prior confirmation; 372376841 appears there. Root's successful 60-arcsec VSX sweep returns 0 rows here: no eclipsing entry at this target. The 372 cone's known EB is the different 58-arcsec neighbour. Gaia vari_eclipsing_binary has no target row in the successful shared query. Exact TIC/main-name web searches found no relevant prior eclipsing record; 416's KIC11418931 is absent from the Kepler EB catalogue. This supports novelty only within the queried sources. Initial failed CDS queries were superseded by working mirrors, not treated as null results.

Next eclipses and follow-up

After 2026-10-30: primary BTJD 4344.34232, 4350.25125, 4356.16017; secondary 4347.42653, 4353.33546, 4359.24439. Linear-covariance prediction errors can be optimistic; secondary scheduling allowance is at least about 21 minutes from the heuristic phase envelope, plus accumulated ephemeris uncertainty. Primary covariance propagation and sampling sensitivity are retained in the ephemeris files. Follow-up priorities are robust component radii/light ratios, baseline-sensitive events and calibrated multi-sector localization; they do not negate the repeated attributed eclipses.

Provenance: parent acquisition.csv/selected.json/prior-receipts.txt; full-vet scripts and per-target JSONs; successful archive response hashes; shared Gaia/VSX receipts. Sol lead reviewed code and compact evidence plus exactly two new pixel maps; Terra performed bulk extraction/model attempts. Original catalogue and work directories remain unchanged.

R007. TIC 372376841

physical_verdict: CONFIRMED

novelty_status: no prior target eclipsing classification in successfully queried vetted sources; unvetted flags separate

completion: full_attempted_and_adjudicated

next_check: Resolve SAP/PDCSAP dilution difference and component radii

Archived source: sanitized report text.

TIC 372376841 (TYC 9210-30-1): full attempted and adjudicated review

Verdict: CONFIRMED physical eclipsing target/system associated with the Gaia SB1. Repeated primary and secondary eclipses, orbital timing, independent photometry and localization support this classification. Exact component properties remain uncertain; this is not a catalogue edit or publication action.

Deciding light-curve evidence

19 primary event fits pass the stated basic fit/coverage screen; the stricter cadence/gap screen retains 19. Median fitted primary depth is 31.58 ppt, with median conditional per-event depth error 0.22 ppt. Median T14/T12/T23 are 9.96/1.99/5.98 hours. These are fitted trapezoid descriptors; partial eclipses do not imply physical totality. Individual profile ranges, counts and gaps are in 372376841_measurements.json. Shape grids and baseline systematics limit errors, especially at profile boundaries.

18 secondary timing fits give median depth 4.64 ppt and phase 0.502467 +/- 0.00253 cycles. The latter is a conservative heuristic envelope from phase-grid resolution, width alternatives and leave-one-sector scatter, not a calibrated confidence interval. Fixed-phase/alternate-width controls and covered windows are preserved in secondary JSONs. Approximate small-e e*cos(omega) = +0.0039 +/- 0.0040; no trapezoid duration-ratio estimate of e*sin(omega) is claimed.

Timing and Gaia

Primary T0 = 2356.43238487 +/- 0.00087188 BTJD, P = 8.419131165 +/- 0.000008952 d. Primary O-C rms 4.28 min. Covariance is scaled by max(1, chi-square/dof); stricter-mask P = 8.419131165 d. Secondary P = 8.419142052 +/- 0.000021879 d, O-C rms 12.85 min; primary-secondary difference +0.94 +/- 2.04 seconds/cycle. Covariances and all O-C values remain in JSON.

The Gaia comparison uses this source's full SB1 row, bit_index 127, its own 15-element correlation vector and the documented six-parameter order. The primary is -0.14 sigma from conjunction A near the Gaia epoch; photometric-Gaia period difference is -0.40 sigma. Per-sector A/B timing comparisons are saved. Physical-e rejection fraction 15.1% means the Gaussian orbit approximation is conditional and small-e priors remain soft.

Independence, systematics and attribution

TESS-SPOC sector 10 independently recovers time-aligned dips: SAP 21.3 to 23.6 ppt and PDCSAP 28.9 to 32.4 ppt. Their difference is a dilution/detrending uncertainty, not evidence of absent eclipses. Raw QLP SAP with QUALITY=0 is the primary source; no post-gap trim was used. Flags, raw centroids, native-unit background excursions and gap coverage were examined. Background regression changes primary depths by at most 3.2%; this is a unit-free sensitivity check and does not convert SAP_BKG to fractional stellar flux. The independent SAP/PDCSAP depth difference is substantial. The 16-arcsec neighbour could mimic the depth in a large aperture, but the new pixel test selects the target. A separate known EB 58 arcsec away has period 1.78258 d and is too faint to explain the depth under the G-band proxy.

The blind coarse-box census retains 35 peaks, 2 gap-flagged, and 0 classified off the two eclipse ephemerides. Its max(3 ppt, three flank-MAD) threshold is not a significance calibration or exhaustive shallow-event census. Fresh TESScut template fitting in sector 10 uses two events. The target is preferred to all listed neighbours; free and split-event PSF centroids lie within about 0.07 pixel of the catalogue position. These are unweighted Gaussian-PSF/WCS diagnostics, not calibrated localization sigma. Fresh 60-arcsec Gaia neighbours and G-proxy dilution requirements are in neighbours-dilution.csv; aperture throughput, colour and unresolved companions remain limitations.

Physics and archives

RV mass function 0.0344 Msun; using catalog M1=1.070, edge-on M2min=0.425 Msun and a=19.91 Rsun. A shared limb-darkened two-disc model was actually tested against both eclipse shapes, with finite-exposure sampling, radius/impact/e*sin(omega)/light-ratio freedom. Illustrative best k=0.176, b_A=0.56, R1 scale=1.00, R2=0.546 Rsun; diagnostic score 193.9 for 120 bins. Correlated bins, fixed limb darkening, local trends, no third light and coarse bounds prevent precision physical inference. This is a plausible geometry family, not independently measured masses/radii.

Fresh SIMBAD identifies the target SB*. ExoFOP has no TOI/CTOI. Successful IDIA cones find no target entry in ASAS-SN, SB9, WDS, Prsa, IJspeert or vetted Kostov table3/4. Kostov table2 is unvetted and is not prior confirmation; 372376841 appears there. Root's successful 60-arcsec VSX sweep returns 1 rows here: no eclipsing entry at this target. The 372 cone's known EB is the different 58-arcsec neighbour. Gaia vari_eclipsing_binary has no target row in the successful shared query. Exact TIC/main-name web searches found no relevant prior eclipsing record; 416's KIC11418931 is absent from the Kepler EB catalogue. This supports novelty only within the queried sources. Initial failed CDS queries were superseded by working mirrors, not treated as null results.

Next eclipses and follow-up

After 2026-10-30: primary BTJD 4351.76647, 4360.18560, 4368.60473; secondary 4347.57965, 4355.99879, 4364.41793. Linear-covariance prediction errors can be optimistic; secondary scheduling allowance is at least about 31 minutes from the heuristic phase envelope, plus accumulated ephemeris uncertainty. Primary covariance propagation and sampling sensitivity are retained in the ephemeris files. Follow-up priorities are robust component radii/light ratios, baseline-sensitive events and calibrated multi-sector localization; they do not negate the repeated attributed eclipses.

Provenance: parent acquisition.csv/selected.json/prior-receipts.txt; full-vet scripts and per-target JSONs; successful archive response hashes; shared Gaia/VSX receipts. Sol lead reviewed code and compact evidence plus exactly two new pixel maps; Terra performed bulk extraction/model attempts.

Original catalogue and work directories remain unchanged.

R008. TIC 416525506

physical_verdict: CONFIRMED

novelty_status: no prior target eclipsing classification in successfully queried vetted sources; unvetted flags separate

completion: full_attempted_and_adjudicated

next_check: Inspect S55 cycle232 background-sensitive event; refine component geometry

Archived source: sanitized report text.

TIC 416525506 (TYC 3566-1556-1): full attempted and adjudicated review

Verdict: CONFIRMED physical eclipsing target/system associated with the Gaia SB1. Repeated primary and secondary eclipses, orbital timing, independent photometry and localization support this classification. Exact component properties remain uncertain; this is not a catalogue edit or publication action.

Deciding light-curve evidence

51 primary event fits pass the stated basic fit/coverage screen; the stricter cadence/gap screen retains 49. Median fitted primary depth is 33.54 ppt, with median conditional per-event depth error 0.60 ppt. Median T14/T12/T23 are 3.00/1.05/0.83 hours. These are fitted trapezoid descriptors; partial eclipses do not imply physical totality. Individual profile ranges, counts and gaps are in 416525506_measurements.json. Shape grids and baseline systematics limit errors, especially at profile boundaries.

49 secondary timing fits give median depth 2.52 ppt and phase 0.499824 +/- 0.00250 cycles. The latter is a conservative heuristic envelope from phase-grid resolution, width alternatives and leave-one-sector scatter, not a calibrated confidence interval. Fixed-phase/alternate-width controls and covered windows are preserved in secondary JSONs. Approximate small-e e*cos(omega) = -0.0003 +/- 0.0039; no trapezoid duration-ratio estimate of e*sin(omega) is claimed.

Timing and Gaia

Primary T0 = 2840.07978816 +/- 0.00013162 BTJD, P = 4.895374455 +/- 0.000001051 d. Primary O-C rms 1.35 min. Covariance is scaled by max(1, chi-square/dof); stricter-mask P = 4.895374539 d. Secondary P = 4.895396424 +/- 0.000011483 d, O-C rms 16.93 min; primary-secondary difference +1.90 +/- 1.00 seconds/cycle. Covariances and all O-C values remain in JSON.

The Gaia comparison uses this source's full SB1 row, bit_index 127, its own 15-element correlation vector and the documented six-parameter order. The primary is +0.25 sigma from conjunction A near the Gaia epoch; photometric-Gaia period difference is -0.19 sigma. Per-sector A/B timing comparisons are saved. Physical-e rejection fraction 33.9% means the Gaussian orbit approximation is conditional and small-e priors remain soft.

Independence, systematics and attribution

TESS-SPOC sector 15 independently recovers time-aligned dips: SAP 31.9 to 37.0 ppt and PDCSAP 32.9 to 39.3 ppt. Their difference is a dilution/detrending uncertainty, not evidence of absent eclipses. Raw QLP SAP with QUALITY=0 is the primary source; no post-gap trim was used. Flags, raw centroids, native-unit background excursions and gap coverage were examined. Background regression changes primary depths by at most 14.4%; this is a unit-free sensitivity check and does not convert SAP_BKG to fractional stellar flux. S55 cycle 232 is background-sensitive: raw 32.55 versus background-regressed 37.23 ppt. Excluding it changes P by only 0.0048 seconds. It is retained as an affected event, not pristine evidence.

The blind coarse-box census retains 45 peaks, 0 gap-flagged, and 0 classified off the two eclipse ephemerides. Its max(3 ppt, three flank-MAD) threshold is not a significance calibration or exhaustive shallow-event census. The valid prior difference-image receipt is reused, with on-target peak and small deficit-to-target offset (0.87 arcsec for 416525506, 1.63 for 451493289, 2.50 for 348737250). No new pixel cube was needed; this is explicitly prior attribution evidence. Fresh 60-arcsec Gaia neighbours and G-proxy dilution requirements are in neighbours-dilution.csv; aperture throughput, colour and unresolved companions remain limitations.

Physics and archives

RV mass function 0.0091 Msun; using catalog M1=1.117, edge-on M2min=0.258 Msun and a=13.49 Rsun. A shared limb-darkened two-disc model was actually tested against both eclipse shapes, with finite-exposure sampling, radius/impact/e*sin(omega)/light-ratio freedom. Illustrative best k=0.214, b_A=0.88, R1 scale=1.00, R2=0.289 Rsun; diagnostic score 50.0 for 120 bins. Correlated bins, fixed limb darkening, local trends, no third light and coarse bounds prevent precision physical inference. This is a plausible geometry family, not independently measured masses/radii.

Fresh SIMBAD identifies the target SB*. ExoFOP has no TOI/CTOI. Successful IDIA cones find no target entry in ASAS-SN, SB9, WDS, Prsa, IJspeert or vetted Kostov table3/4. Kostov table2 is unvetted and is not prior confirmation; 372376841 appears there. Root's successful 60-arcsec VSX sweep returns 0 rows here: no eclipsing entry at this target. The 372 cone's known EB is the different 58-arcsec neighbour. Gaia vari_eclipsing_binary has no target row in the successful shared query. Exact TIC/main-name web searches found no relevant prior eclipsing record; 416's KIC11418931 is absent from the Kepler EB catalogue. This supports novelty only within the queried sources. Initial failed CDS queries were superseded by working mirrors, not treated as null results.

Next eclipses and follow-up

After 2026-10-30: primary BTJD 4347.85512, 4352.75049, 4357.64587; secondary 4345.41192, 4350.30732, 4355.20272. Linear-covariance prediction errors can be optimistic; secondary scheduling allowance is at least about 18 minutes from the heuristic phase envelope, plus accumulated ephemeris uncertainty. Primary covariance propagation and sampling sensitivity are retained in the ephemeris files. Follow-up priorities are robust component radii/light ratios, baseline-sensitive events and calibrated multi-sector localization; they do not negate the repeated attributed eclipses.

Provenance: parent acquisition.csv/selected.json/prior-receipts.txt; full-vet scripts and per-target JSONs; successful archive response hashes; shared Gaia/VSX receipts. Sol lead reviewed code and compact evidence plus exactly two new pixel maps; Terra performed bulk extraction/model attempts. Original catalogue and work directories remain unchanged.

R009. TIC 451493289

physical_verdict: CONFIRMED

novelty_status: no prior target eclipsing classification in successfully queried vetted sources; unvetted flags separate

completion: full_attempted_and_adjudicated

next_check: Resolve catalog radius versus joint eclipse geometry; sector baseline variation

Archived source: sanitized report text.

TIC 451493289 (HD 142624): full attempted and adjudicated review

Verdict: CONFIRMED physical eclipsing target/system associated with the Gaia SB1. Repeated primary and secondary eclipses, orbital timing, independent photometry and localization support this classification. Exact component properties remain uncertain; this is not a catalogue edit or publication action.

Deciding light-curve evidence

23 primary event fits pass the stated basic fit/coverage screen; the stricter cadence/gap screen retains 20. Median fitted primary depth is 33.18 ppt, with median conditional per-event depth error 0.16 ppt. Median T14/T12/T23 are 8.76/1.75/5.26 hours. These are fitted trapezoid descriptors; partial eclipses do not imply physical totality. Individual profile ranges, counts and gaps are in 451493289_measurements.json. Shape grids and baseline systematics limit errors, especially at profile boundaries.

25 secondary timing fits give median depth 3.16 ppt and phase 0.500165 +/- 0.00250 cycles. The latter is a conservative heuristic envelope from phase-grid resolution, width alternatives and leave-one-sector scatter, not a calibrated confidence interval. Fixed-phase/alternate-width controls and covered windows are preserved in secondary JSONs. Approximate small-e e*cos(omega) = +0.0003 +/- 0.0039; no trapezoid duration-ratio estimate of e*sin(omega) is claimed.

Timing and Gaia

Primary T0 = 3106.41985304 +/- 0.00023937 BTJD, P = 6.940427624 +/- 0.000002516 d. Primary O-C rms 2.52 min. Covariance is scaled by max(1, chi-square/dof); stricter-mask P = 6.940428228 d. Secondary P = 6.940422652 +/- 0.000010554 d, O-C rms 8.17 min; primary-secondary difference -0.43 +/- 0.94 seconds/cycle. Covariances and all O-C values remain in JSON.

The Gaia comparison uses this source's full SB1 row, bit_index 127, its own 15-element correlation vector and the documented six-parameter order. The primary is +0.17 sigma from conjunction A near the Gaia epoch; photometric-Gaia period difference is -0.11 sigma. Per-sector A/B timing comparisons are saved. Physical-e rejection fraction 28.1% means the Gaussian orbit approximation is conditional and small-e priors remain soft.

Independence, systematics and attribution

TESS-SPOC sector 12 independently recovers time-aligned dips: SAP 32.4 to 32.8 ppt and PDCSAP 35.1 to 35.5 ppt. Their difference is a dilution/detrending uncertainty, not evidence of absent eclipses. Raw QLP SAP with QUALITY=0 is the primary source; no post-gap trim was used. Flags, raw centroids, native-unit background excursions and gap coverage were examined. Background regression changes primary depths by at most 1.4%; this is a unit-free sensitivity check and does not convert SAP_BKG to fractional stellar flux. The joint model fits poorly near the catalog dimensions. A refined illustrative fit needs R1 about 2.52 Rsun, 0.75 times TIC 3.355 +/- 0.144 Rsun. This is a large unresolved stellar-dimension/model tension, not a measured correction to the catalogue. S93/S101 have extreme flagged excursions and S100/S102 depth/baseline changes.

The blind coarse-box census retains 28 peaks, 5 gap-flagged, and 0 classified off the two eclipse ephemerides. Its max(3 ppt, three flank-MAD) threshold is not a significance calibration or exhaustive shallow-event census. The valid prior difference-image receipt is reused, with on-target peak and small deficit-to-target offset (0.87 arcsec for 416525506, 1.63 for 451493289, 2.50 for 348737250). No new pixel cube was needed; this is explicitly prior attribution evidence. Fresh 60-arcsec Gaia neighbours and G-proxy dilution requirements are in neighbours-dilution.csv; aperture throughput, colour and unresolved companions remain limitations.

Physics and archives

RV mass function 0.0240 Msun; using catalog M1=1.240, edge-on M2min=0.401 Msun and a=18.06 Rsun. A shared limb-darkened two-disc model was actually tested against both eclipse shapes, with finite-exposure sampling, radius/impact/e*sin(omega)/light-ratio freedom. Illustrative best k=0.170, b_A=0.16, R1 scale=0.75, R2=0.428 Rsun; diagnostic score 84.5 for 120 bins. Correlated bins, fixed limb darkening, local trends, no third light and coarse bounds prevent precision physical inference. This is a plausible geometry family, not independently measured masses/radii.

Fresh SIMBAD identifies the target SB*. ExoFOP has no TOI/CTOI. Successful IDIA cones find no target entry in ASAS-SN, SB9, WDS, Prsa, IJspeert or vetted Kostov table3/4. Kostov table2 is unvetted and is not prior confirmation; 372376841 appears there. Root's successful 60-arcsec VSX sweep returns 0 rows here: no eclipsing entry at this target. The 372 cone's known EB is the different 58-arcsec neighbour. Gaia vari_eclipsing_binary has no target row in the successful shared query. Exact TIC/main-name web searches found no relevant prior eclipsing record; 416's KIC11418931 is absent from the Kepler EB catalogue. This supports novelty only within the queried sources. Initial failed CDS queries were superseded by working mirrors, not treated as null results.

Next eclipses and follow-up

After 2026-10-30: primary BTJD 4348.75640, 4355.69683, 4362.63725; secondary 4345.28679, 4352.22722, 4359.16764. Linear-covariance prediction errors can be optimistic; secondary scheduling allowance is at least about 25 minutes from the heuristic phase envelope, plus accumulated ephemeris uncertainty. Primary covariance propagation and sampling sensitivity are retained in the ephemeris files. Follow-up priorities are robust component radii/light ratios, baseline-sensitive events and calibrated multi-sector localization; they do not negate the repeated attributed eclipses.

Provenance: parent acquisition.csv/selected.json/prior-receipts.txt; full-vet scripts and per-target JSONs; successful archive response hashes; shared Gaia/VSX receipts. Sol lead reviewed code and compact evidence plus exactly two new pixel maps; Terra performed bulk extraction/model attempts.

Original catalogue and work directories remain unchanged.

R010. TIC 348737250

physical_verdict: CONFIRMED

novelty_status: no prior target eclipsing classification in successfully queried vetted sources; unvetted flags separate

completion: full_attempted_and_adjudicated

next_check: Refine component radius/light ratio and calibrated localization

Archived source: sanitized report text.

TIC 348737250 (TYC 6539-657-1): full attempted and adjudicated review

Verdict: CONFIRMED physical eclipsing target/system associated with the Gaia SB1. Repeated primary and secondary eclipses, orbital timing, independent photometry and localization support this classification. Exact component properties remain uncertain; this is not a catalogue edit or publication action.

Deciding light-curve evidence

14 primary event fits pass the stated basic fit/coverage screen; the stricter cadence/gap screen retains 14. Median fitted primary depth is 34.66 ppt, with median conditional per-event depth error 0.21 ppt. Median T14/T12/T23 are 8.04/1.61/4.82 hours. These are fitted trapezoid descriptors; partial eclipses do not imply physical totality. Individual profile ranges, counts and gaps are in 348737250_measurements.json. Shape grids and baseline systematics limit errors, especially at profile boundaries.

17 secondary timing fits give median depth 3.52 ppt and phase 0.499842 +/- 0.00250 cycles. The latter is a conservative heuristic envelope from phase-grid resolution, width alternatives and leave-one-sector scatter, not a calibrated confidence interval. Fixed-phase/alternate-width controls and covered windows are preserved in secondary JSONs. Approximate small-e e*cos(omega) = -0.0002 +/- 0.0039; no trapezoid duration-ratio estimate of e*sin(omega) is claimed.

Timing and Gaia

Primary T0 = 2968.49132442 +/- 0.00011718 BTJD, P = 6.101247833 +/- 0.000001073 d. Primary O-C rms 0.40 min. Covariance is scaled by max(1, chi-square/dof); stricter-mask P = 6.101247833 d. Secondary P = 6.101247451 +/- 0.000013112 d, O-C rms 8.25 min; primary-secondary difference -0.03 +/- 1.14 seconds/cycle. Covariances and all O-C values remain in JSON.

The Gaia comparison uses this source's full SB1 row, bit_index 127, its own 15-element correlation vector and the documented six-parameter order. The primary is -0.78 sigma from conjunction A near the Gaia epoch; photometric-Gaia period difference is +0.15 sigma. Per-sector A/B timing comparisons are saved. Physical-e rejection fraction 32.8% means the Gaussian orbit approximation is conditional and small-e priors remain soft.

Independence, systematics and attribution

TESS-SPOC sector 34 independently recovers time-aligned dips: SAP 32.2 to 32.9 ppt and PDCSAP 34.9 to 35.3 ppt. Their difference is a dilution/detrending uncertainty, not evidence of absent eclipses. Raw QLP SAP with QUALITY=0 is the primary source; no post-gap trim was used. Flags, raw centroids, native-unit background excursions and gap coverage were examined. Background regression changes primary depths by at most 0.9%; this is a unit-free sensitivity check and does not convert SAP_BKG to fractional stellar flux. At the catalog radius the simple R2=M2min relation underpredicts depth. Allowing radius/model freedom provides an illustrative binary geometry. The nearby 33-arcsec source would need about a 96% G-band dip under equal-throughput assumptions; prior target-centered difference imaging supports attribution.

The blind coarse-box census retains 17 peaks, 0 gap-flagged, and 0 classified off the two eclipse ephemerides. Its max(3 ppt, three flank-MAD) threshold is not a significance calibration or exhaustive shallow-event census. The valid prior difference-image receipt is reused, with on-target peak and small deficit-to-target offset (0.87 arcsec for 416525506, 1.63 for 451493289, 2.50 for 348737250). No new pixel cube was needed; this is explicitly prior attribution evidence. Fresh 60-arcsec Gaia neighbours and G-proxy dilution requirements are in neighbours-dilution.csv; aperture throughput, colour and unresolved companions remain limitations.

Physics and archives

RV mass function 0.0276 Msun; using catalog M1=1.290, edge-on M2min=0.434 Msun and a=16.85 Rsun. A shared limb-darkened two-disc model was actually tested against both eclipse shapes, with finite-exposure sampling, radius/impact/e*sin(omega)/light-ratio freedom. Illustrative best k=0.184, b_A=0.56, R1 scale=1.00, R2=0.537 Rsun; diagnostic score 219.7 for 120 bins. Correlated bins, fixed limb darkening, local trends, no third light and coarse bounds prevent precision physical inference. This is a plausible geometry family, not independently measured masses/radii.

Fresh SIMBAD identifies the target SB*. ExoFOP has no TOI/CTOI. Successful IDIA cones find no target entry in ASAS-SN, SB9, WDS, Prsa, IJspeert or vetted Kostov table3/4. Kostov table2 is unvetted and is not prior confirmation; 372376841 appears there. Root's successful 60-arcsec VSX sweep returns 0 rows here: no eclipsing entry at this target. The 372 cone's known EB is the different 58-arcsec neighbour. Gaia vari_eclipsing_binary has no target row in the successful shared query. Exact TIC/main-name web searches found no relevant prior eclipsing record; 416's KIC11418931 is absent from the Kepler EB catalogue. This supports novelty only within the queried sources. Initial failed CDS queries were superseded by working mirrors, not treated as null results.

Next eclipses and follow-up

After 2026-10-30: primary BTJD 4347.37333, 4353.47458, 4359.57583; secondary 4344.32165, 4350.42290, 4356.52415. Linear-covariance prediction errors can be optimistic; secondary scheduling allowance is at least about 22 minutes from the heuristic phase envelope, plus accumulated ephemeris uncertainty. Primary covariance propagation and sampling sensitivity are retained in the ephemeris files. Follow-up priorities are robust component radii/light ratios, baseline-sensitive events and calibrated multi-sector localization; they do not negate the repeated attributed eclipses.

Provenance: parent acquisition.csv/selected.json/prior-receipts.txt; full-vet scripts and per-target JSONs; successful archive response hashes; shared Gaia/VSX receipts. Sol lead reviewed code and compact evidence plus exactly two new pixel maps; Terra performed bulk extraction/model attempts.

Original catalogue and work directories remain unchanged.

R011. TIC 158545440

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 158545440 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Six primary and seven repeat secondary events across sectors, a matching spectroscopic period, no capable catalogued neighbour, and on-target prior difference imaging establish the target EB. Moderate eccentricity and the long primary are compatible with the tested orbital geometry.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 6 accepted primary events in 4 sectors; median depth 33.531ppt (range31.279–35.827). T14=10.6272h, profile range10.4382–10.8162h; ingress T12=1.5941h, nominal trapezoid T23=7.4390h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=17.4542044155±0.0000100957d, Tref=2890.7786891±0.0006466BTJD (reference cycle77); T0=1546.8049491. Chi2/dof=5.62/4; O-C RMS=1.852min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found17 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.1075ppt; including native-unit background regression changes an individual depth by at most0.52%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=2; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
9	1546.8045136	0.0013797	33.572	0.302	-0.627	2
9	1564.2588041	0.0013284	33.491	0.295	-0.503	0
36	2297.3363488	0.0013284	35.827	0.297	0.878	0
90	3763.4914382	0.0013284	34.985	0.312	3.641	0
100	4077.6633539	0.0013284	31.564	0.247	-1.779	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
100	4095.1176444	0.0013284	31.279	0.293	-1.655	0

Gaia orbit and secondary gate

Exact source5394156908605834624, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=17.4546757450±0.0044738376d; photometric difference=-0.105sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=-0.130238d with Gaia-only sigma0.166589d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.168044±0.018986, omega=237.9841deg. Gaia relative other-conjunction phase=0.4427770; measured optimized phase=0.4523714. Status: supported repeat secondary. Optimized depth=3.9810±0.1486ppt; 7 individual >3sigma events among7 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=17.4542108868±0.0000117053d, Tref=2706.6756088±0.0007771, O-C RMS1.011min, chi2/dof0.61/5.

First-order timing proxy e cos(omega)˜-0.07481 versus Gaia-0.08909; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=-0.1000, k=0.200, impact proxy=0.235; chi2/dof=160.28/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector9; header target=158545440, sector=9 checked. SHA256=1e98356da57b33f7eea3133f33c7eb98fbdf603beac935d8c0500432a30ab521. At actual fitted QLP times, the independent product gives depth range33.533–35.792ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has0 resolved neighbours, 0 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip

Prior difference image sector9, 2 events: peak on target=True; deficit/target separation4.24arcsec; deficit/light separation3.43arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.024829Msun; assumed TIC M1=1.170Msun gives edge-on minimum M2=0.3929Msun, a=32.862Rsun. A main-sequence-radius approximation gives central T14(A/B)=12.245/9.190h compared with measured10.627h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 7724-540-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4356.931860BTJD, subsequent 4374.386064. Other-kind nominal prediction: 4347.371432BTJD, subsequent4364.825643. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.11ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 10 predicted windows, status counts {'fit': 7, 'insufficient': 3}; conditional depth 1.4579ppt, control-inflated 3sigma upper value 4.2784ppt. Optimized secondary windows status counts{'fit': 7, 'insufficient': 3}; fitted secondary T14=8.7372h, T12=1.3106h, trapezoid T23=6.1160h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R012. TIC 339787086

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Jointly refit RV and eclipse timings; assess sector-dependent baseline effects and the separate 281-day astrometric solution.

Archived source: sanitized report text.

TIC 339787086 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Repeated ~36/40 ppt alternating eclipses in four sectors, independent reduction and absence of a sufficiently bright resolved neighbour establish the target EB. The Gaia SB2 association is supported by a nearly equal spectroscopic/photometric period and Gaia-epoch conjunction within about2 Gaia-only sigma, but the published Gaia orbit does not predict later eclipse times accurately. This is a confirmed target EB with an imperfect orbital solution, not a precision ephemeris or a validated association with the separate 281-day astrometric orbit.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 22 accepted primary events in 4 sectors; median depth 36.214ppt (range35.212–37.136). T14=3.6695h, profile range3.5826–3.7564h; ingress T12=1.2843h, nominal trapezoid T23=1.1009h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=3.7794134843±0.0000086571d, Tref=2701.1571438±0.0012138BTJD (reference cycle269); T0=1684.4949165. Chi2/dof=1796.62/20; O-C RMS=6.510min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found50 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.5694ppt; including native-unit background regression changes an individual depth by at most1.45%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=8; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
14	1684.4916695	0.0022731	36.859	0.479	-4.676	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
14	1688.2729397	0.0022731	37.136	0.422	-2.002	1
14	1695.8318106	0.0022731	36.652	0.579	-1.939	0
14	1699.6103286	0.0021263	36.311	0.420	-3.228	0
40	2395.0181926	0.0004862	36.377	0.402	-9.301	5
40	2406.3574163	0.0005176	35.861	0.538	-7.885	7
40	2410.1359343	0.0005358	36.320	0.396	-9.175	8
40	2413.9172045	0.0005012	36.745	0.387	-6.501	6
40	2417.6957226	0.0005012	36.440	0.480	-7.790	6
54	2772.9727448	0.0004599	36.042	0.342	9.713	3
54	2776.7512629	0.0004862	36.559	0.416	8.423	5
54	2780.5306983	0.0005012	36.133	0.437	8.455	6
54	2784.3101337	0.0005012	35.212	0.446	8.486	6
54	2788.0895692	0.0005012	36.295	0.452	8.518	6
54	2791.8680872	0.0005012	36.377	0.452	7.229	6
54	2795.6475227	0.0004725	36.067	0.377	7.260	4
81	3509.9479769	0.0004587	35.822	0.274	-5.260	0
81	3517.5068478	0.0004587	35.493	0.306	-5.196	0
81	3521.2872006	0.0004587	35.712	0.366	-3.844	0
81	3525.0675534	0.0004587	35.382	0.354	-2.491	0
81	3528.8479062	0.0004587	35.773	0.311	-1.139	0
81	3532.6282590	0.0004587	35.411	0.305	0.214	0

Gaia orbit and secondary gate

Exact source4322674031257647744, solutionSB2, bit_index255, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=3.7789813432±0.0000964621d; photometric difference=4.462sigma. Primary association is conjunctionB; extrapolated Gaia-epoch offset=-0.033647d with Gaia-only sigma0.016760d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.044911±0.012884, omega=356.0555deg. Gaia relative other-conjunction phase=0.4714858; measured optimized phase=0.4981861. Status: supported repeat secondary. Optimized depth=40.0133±0.1599ppt; 15 individual >3sigma events among15 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=3.7793989754±0.0000098278d, Tref=2714.3818786±0.0015154, O-C RMS7.598min, chi2/dof861.66/13.

First-order timing proxy e cos(omega)˜0.00285 versus Gaia0.04480; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0750,

k=0.250, impact proxy=0.935; chi2/dof=3255.67/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector54; header target=339787086, sector=54 checked. SHA256=09afd87c6b08a025369eaa253ebe6586a187a910b0b115262463f65a05e2d3bf. At actual fitted QLP times, the independent product gives depth range34.251–36.984ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has0 resolved neighbours, 0 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip

Prior difference image sector40, 8 events: peak on target=True; deficit/target separation5.1arcsec; deficit/light separation3.57arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

SB2 dynamical q=0.9262, M1 sin³i=1.5107Msun, M2 sin³i=1.3992Msun and a sin i=14.580Rsun. These supersede the SB1/TIC-M1 surrogate in orbit-gates physics. The TIC assumed M1 is below this SB2 minimum; it is not a valid measured primary mass. Geometry chi2 is poor and must not be promoted to component characterization. Strong paired eclipses identify the target EB despite that model failure. The separate Gaia Orbital solution P280.989±2.275d is retained as possible multiplicity context; no causal explanation of eclipse timing changes is established. Linear timing residuals are sector-coherent (about-9 to+10min); published Gaia conjunction offsets accumulate across sectors.

SIMBAD exact Gaia-ID name: TYC 1604-1130-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4345.202009BTJD, subsequent 4348.981423. Other-kind nominal prediction: 4347.082236BTJD, subsequent4350.861635. These extrapolations are conditional. Linear timing extrapolation needs the systematic margin specified below. Jointly refit RV and eclipse timings; assess sector-dependent baseline effects and the separate 281-day astrometric solution.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 2.17ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

The photometric versus Gaia SB2 period discrepancy is4.46sigma under the quoted uncertainties. Primary chi2/dof1796.6/20 and sector-coherent O-C of roughly-9 to+10min demonstrate limitations of a single linear ephemeris and its formal timing errors. At2026-10-30 the propagated formal linear timing scale alone is about5.7min; it is not a complete predictive error. Use at least +/-0.05d (72min) around a nominal prediction as a provisional observing search window, not a calibrated confidence interval or guarantee; further timing evolution may require more. The target is confirmed as an EB; the exact connection of all eclipses to the published SB2 orbital parameters remains imperfect, and the281-day astrometric-orbit association remains unestablished.

Fixed Gaia-phase other-conjunction coverage: 28 predicted windows, status counts {'fit': 16, 'insufficient': 12}; conditional depth -9.4273ppt, control-inflated 3sigma upper value 8.4461ppt. Optimized secondary windows status counts{'fit': 15, 'insufficient': 13}; fitted secondary T14=3.6695h, T12=1.2843h, trapezoid T23=1.1009h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R013. TIC 341432079

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 341432079 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Repeated primaries and secondaries, independent reduction and new per-pixel localization favour the target over both capable neighbours. Mild Gaia eccentric-orbit/secondary-phase tension is a characterization issue.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 17 accepted primary events in 5 sectors; median depth 36.134ppt (range31.254–36.730). T14=4.9829h, profile range4.8526–5.1132h; ingress T12=1.2457h, nominal trapezoid T23=2.4914h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=5.1087338002±0.0000034021d, Tref=2703.1501461±0.0003281BTJD (reference cycle199); T0=1686.5121199. Chi2/dof=48.41/15; O-C RMS=4.197min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found36 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 3.6506ppt; including native-unit background regression changes an individual depth by at most11.11%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=12; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
14	1686.5053543	0.0019019	36.439	0.255	-9.742	0
14	1701.8328828	0.0019019	36.730	0.339	-7.831	0
40	2391.5093783	0.0080972	31.254	4.694	-11.529	12
40	2406.8468725	0.0018686	36.077	1.407	4.733	10
41	2422.1694181	0.0006229	36.252	0.304	-0.531	0
41	2427.2781790	0.0006229	36.551	0.267	-0.492	0
41	2437.4969466	0.0006229	35.799	0.280	1.380	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
41	2442.6044618	0.0006229	36.134	0.281	-0.375	0
54	2774.6726759	0.0006229	35.378	0.301	0.369	8
54	2779.7814368	0.0006229	36.195	0.410	0.408	8
54	2784.8914434	0.0006229	36.406	0.397	2.241	8
54	2790.0002043	0.0006229	35.203	0.299	2.280	12
54	2795.1064738	0.0006229	35.115	0.333	-1.268	5
81	3510.3292654	0.0006229	36.198	0.316	-1.183	0
81	3520.5467872	0.0006229	36.103	0.260	-1.104	0
81	3525.6567938	0.0006229	36.124	0.254	0.728	0
81	3530.7643090	0.0006229	36.333	0.308	-1.026	5

Gaia orbit and secondary gate

Exact source1823986543821182080, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=5.1086672779±0.0004283853d; photometric difference=0.155sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=-0.018922d with Gaia-only sigma0.021890d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.028067±0.013215, omega=159.2772deg. Gaia relative other-conjunction phase=0.4832894; measured optimized phase=0.5003486. Status: supported repeat secondary. Optimized depth=2.6987±0.0754ppt; 20 individual >3sigma events among20 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=5.1087474561±0.0000088999d, Tref=2659.7262720±0.0010573, O-C RMS6.685min, chi2/dof14.36/18.

First-order timing proxy e cos(omega)˜0.00055 versus Gaia-0.02625; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0500, k=0.250, impact proxy=0.515; chi2/dof=201.03/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector54; header target=341432079, sector=54 checked. SHA256=8b4646a460f2ad8470e901d17a81dcb2e0f11498297c3717695e53fff5bf411f. At actual fitted QLP times, the independent product gives depth range31.798–38.504ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has5 resolved neighbours, 2 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip
1823986642546802944	41.07	13.815	0.720
1823986578180902784	47.44	13.626	0.605
1823986711266237568	47.63	14.969	2.082
1823986642546807808	54.74	14.479	1.326
1823986646900389632	56.74	14.337	1.163

Prior difference image sector40, 4 events: peak on target=True; deficit/target separation6.92arcsec; deficit/light separation8.03arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

A new per-pixel eclipse-template regression on hashed existing TESScut cubes tests the ambiguity directly. OOT multi-source Gaussian calibration: sigma0.700pixel, dx-0.080,dy0.080. Free deficit center is1.878arcsec from target; fixed target chi2=3719.06, best alternative-source chi2=40034.52, delta=36315.45. This strongly distinguishes target from resolved neighbours; PSF residual covariance prevents interpreting these formal chi2 values as calibrated significance or precise astrometry. See pixel-localization.json and pixel_selective.py.

Physical interpretation, archives and next check

RV mass function f(M)=0.021608Msun; assumed TIC M1=1.499Msun gives edge-on minimum M2=0.4319Msun, a=15.544Rsun. A main-sequence-radius approximation gives central T14(A/B)=6.160/6.283h compared with measured4.983h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: HD 350336, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4348.162430BTJD, subsequent 4353.271164. Other-kind nominal prediction: 4345.612933BTJD, subsequent4350.721680. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.30ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple

sectors.

Fixed Gaia-phase other-conjunction coverage: 26 predicted windows, status counts {'insufficient': 7, 'fit': 19}; conditional depth 0.8746ppt, control-inflated 3sigma upper value 9.6162ppt. Optimized secondary windows status counts{'fit': 20, 'insufficient': 6}; fitted secondary T14=5.5043h, T12=1.3761h, trapezoid T23=2.7521h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R014. TIC 144749615

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 144749615 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Forty-three primary and37 secondary detections over multiple sectors, an independent reduction, on-target prior deficit and a capable neighbour almost50arcsec away establish the target EB. A grazing/contact-shape degeneracy limits radii and eccentricity inference.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 43 accepted primary events in 6 sectors; median depth 36.665ppt (range34.601–38.636). T14=1.9289h, profile range1.8919–1.9658h; ingress T12=0.6751h, nominal trapezoid T23=0.5787h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=2.7920366462±0.0000003597d, Tref=2501.1001829±0.0001170BTJD (reference cycle361); T0=1493.1749536. Chi2/dof=55.39/41; O-C RMS=0.536min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found93 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 1.1885ppt; including native-unit background regression changes an individual depth by at most3.33%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=5; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
7	1495.9673345	0.0030071	37.561	0.491	0.496	0
7	1498.7588800	0.0030071	37.090	0.503	-0.211	0
7	1501.5509078	0.0030071	37.491	0.490	-0.224	0
7	1507.1344812	0.0034723	38.049	0.450	-0.944	0
7	1509.9265089	0.0034723	37.636	0.526	-0.957	0
7	1512.7195011	0.0034723	37.836	0.501	0.419	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
7	1515.5110467	0.0034723	38.636	0.470	-0.288	0
8	1518.3030745	0.0034722	36.582	0.459	-0.301	0
8	1521.0951022	0.0034722	37.522	0.470	-0.314	0
8	1523.8871300	0.0034722	35.970	0.406	-0.326	0
8	1526.6791578	0.0034722	36.133	0.454	-0.339	0
8	1537.8467866	0.0030070	35.584	0.416	-1.085	0
8	1540.6397788	0.0030070	35.036	0.496	0.291	0
34	2230.2725675	0.0005787	36.913	0.554	-0.087	0
34	2233.0641130	0.0005787	36.321	0.433	-0.795	0
34	2235.8571052	0.0005787	35.742	0.439	0.581	0
34	2238.6486508	0.0005787	37.154	0.548	-0.126	0
34	2244.2327063	0.0005787	37.718	0.532	-0.151	0
34	2247.0252163	0.0006682	35.600	0.621	0.530	3
34	2249.8172441	0.0005787	37.720	0.465	0.517	0
34	2252.6083074	0.0006044	37.014	0.531	-0.884	1
35	2258.1928452	0.0007577	37.205	1.365	-0.215	4
35	2263.7769007	0.0006682	34.601	1.052	-0.241	2
35	2274.9450118	0.0006044	36.003	0.509	-0.292	1
35	2277.7370396	0.0006339	37.060	0.504	-0.305	1
61	2964.5778005	0.0002411	36.624	0.508	-0.671	0
61	2967.3703104	0.0002411	36.872	0.527	0.011	0
61	2970.1623382	0.0002411	37.024	0.620	-0.002	0
61	2972.9543660	0.0002411	36.785	0.368	-0.015	0
61	2978.5384215	0.0002411	36.501	0.395	-0.040	0
61	2981.3309315	0.0002411	35.455	0.580	0.641	5
61	2984.1224771	0.0002411	37.717	0.691	-0.066	0
61	2986.9149871	0.0002411	36.665	0.505	0.616	0
88	3690.5083968	0.0002411	36.064	0.504	0.868	0
88	3693.3004245	0.0002411	37.459	0.772	0.855	0
88	3696.0924523	0.0002411	36.006	0.539	0.842	0
88	3698.8835157	0.0002411	36.942	0.493	-0.559	0
88	3701.6760256	0.0002411	36.597	0.471	0.122	0
88	3704.4680534	0.0002411	36.173	0.402	0.109	0
88	3707.2595990	0.0002411	36.074	0.478	-0.598	0
88	3710.0511445	0.0002411	36.096	0.426	-1.305	0
88	3712.8436545	0.0002411	35.346	0.714	-0.623	0
88	3715.6361645	0.0002411	35.649	0.481	0.058	0

Gaia orbit and secondary gate

Exact source5547705769022471552, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=2.7919428822±0.0000425200d; photometric difference=2.205sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=-0.028526d with Gaia-only sigma0.009918d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.033481±0.011326, omega=173.4671deg. Gaia relative other-conjunction phase=0.4788273; measured optimized phase=0.5004199. Status: supported repeat secondary. Optimized depth=5.3267±0.0838ppt; 37 individual >3sigma events among38 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=2.7920357079±0.0000013896d, Tref=2502.4970829±0.0003789, O-C RMS3.034min, chi2/dof45.57/35.

First-order timing proxy e cos(omega)˜0.00066 versus Gaia-0.03326; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0500, k=0.250, impact proxy=0.900; chi2/dof=283.65/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector34; header target=144749615, sector=34 checked. SHA256=5b5a6d339dd2ce89feb528970adc7771329bb610f3506a3f925005f5b1916ec6. At actual fitted QLP times, the independent product gives depth range32.123–40.323ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has1 resolved neighbours, 1 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip
5547699962226697216	49.62	13.202	0.292

Prior difference image sector88, 10 events: peak on target=True; deficit/target separation5.31arcsec; deficit/light separation2.53arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.059742Msun; assumed TIC M1=1.060Msun gives edge-on minimum M2=0.5333Msun, a=9.746Rsun. A main-sequence-radius approximation gives central T14(A/B)=3.798/3.827h compared with measured1.929h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 7125-1627-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4343.844369BTJD, subsequent 4346.636406. Other-kind nominal prediction: 4345.240650BTJD, subsequent4348.032686. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 2.83ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 56 predicted windows, status counts {'insufficient': 15, 'fit': 41}; conditional depth -0.7991ppt, control-inflated 3sigma upper value 6.7308ppt. Optimized secondary windows status counts{'insufficient': 18, 'fit': 38}; fitted secondary T14=1.8550h, T12=0.6492h, trapezoid T23=0.5565h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R015. TIC 304033253

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited; VSX ROT type revision

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 304033253 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Thirty-five primary and36 secondary events, independent reduction, no capable resolved neighbour and deficit/light agreement0.36arcsec establish an EB on the Gaia spectroscopic target. Existing VSX ROT classification becomes a type-revision caveat; it is not prior target EB art.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 35 accepted primary events in 9 sectors; median depth 39.851ppt (range34.115–42.058). T14=2.2423h, profile range2.1836–2.3010h; ingress T12=0.7848h, nominal trapezoid T23=0.6727h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=5.5172481514±0.0000005943d, Tref=2777.1869001±0.0000886BTJD (reference cycle263); T0=1326.1506363. Chi2/dof=32.81/33; O-C RMS=0.653min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found80 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 2.3021ppt; including native-unit background regression changes an individual depth by at most5.80%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=12; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
1	1326.1514240	0.0030070	39.147	0.397	1.134	0
1	1331.6686796	0.0030070	39.851	0.413	1.145	0
1	1337.1859351	0.0026895	39.441	0.473	1.156	0
1	1342.7037512	0.0026895	40.084	0.431	1.973	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
13	1662.7028909	0.0026896	38.108	0.313	0.169	0
13	1673.7379626	0.0030071	38.150	0.351	0.998	0
13	1679.2546576	0.0030071	37.944	0.327	0.201	0
27	2037.8751466	0.0005560	41.150	0.386	-0.722	0
27	2043.3929627	0.0005358	40.223	0.432	0.096	0
27	2054.4280344	0.0005560	40.797	0.437	0.925	1
27	2059.9447294	0.0005560	40.046	0.518	0.128	0
28	2065.4619849	0.0005358	40.311	0.358	0.139	0
28	2070.9798010	0.0005560	40.089	0.436	0.957	0
28	2076.4970566	0.0005358	39.794	0.438	0.967	0
28	2082.0143121	0.0005560	40.373	0.459	0.978	0
67	3130.2906228	0.0002803	41.311	0.425	-0.229	0
67	3135.8078783	0.0002803	42.058	0.497	-0.218	0
67	3141.3251338	0.0002803	41.075	0.482	-0.208	0
67	3146.8418288	0.0002803	40.684	0.551	-1.004	12
67	3152.3596449	0.0005606	41.328	1.031	-0.186	2
68	3157.8763399	0.0002803	40.828	0.492	-0.983	0
68	3163.3941560	0.0002803	40.400	0.355	-0.165	0
68	3168.9114115	0.0002803	40.728	0.400	-0.155	0
68	3174.4286671	0.0002803	39.709	0.409	-0.144	0
68	3179.9464832	0.0002803	39.957	0.505	0.674	6
93	3830.9815160	0.0002803	39.699	0.441	0.315	0
93	3847.5332826	0.0002803	38.458	0.361	0.347	0
93	3853.0499776	0.0002803	38.763	0.408	-0.449	0
94	3858.5677937	0.0002803	34.352	0.342	0.369	0
94	3864.0850492	0.0002803	34.115	0.308	0.379	0
94	3880.6368159	0.0002803	34.405	0.371	0.411	2
95	3886.1540714	0.0002803	35.693	0.341	0.422	0
95	3891.6707664	0.0002803	35.360	0.333	-0.375	0
95	3897.1885825	0.0002803	35.711	0.397	0.443	0
95	3902.7052775	0.0002803	35.332	0.314	-0.354	0

Gaia orbit and secondary gate

Exact source4636522684725084800, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=5.5164605934±0.0003372100d; photometric difference=2.336sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=0.015376d with Gaia-only sigma0.032704d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.067909±0.013169, omega=126.4955deg. Gaia relative other-conjunction phase=0.4742558; measured optimized phase=0.4639456. Status: supported repeat secondary. Optimized depth=4.1880±0.0833ppt; 36 individual >3sigma events among36 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=5.5172492207±0.0000022602d, Tref=2680.4365304±0.0003712, O-C RMS3.451min, chi2/dof39.34/34.

First-order timing proxy e cos(omega)˜-0.05663 versus Gaia-0.04039; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0750, k=0.375, impact proxy=1.005; chi2/dof=96.50/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product SPOC, sector28; header target=304033253, sector=28 checked. SHA256=b713ead4f671b944fd3698729c70352238dad41f3c065fea69c7d9fe70e03cf6. At actual fitted QLP times, the independent product gives depth range38.350–39.100ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has0 resolved neighbours, 0 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip

Prior difference image sector67, 5 events: peak on target=True; deficit/target separation3.29arcsec; deficit/light separation0.36arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.075990Msun; assumed TIC M1=1.100Msun gives edge-on minimum M2=0.6044Msun, a=15.694Rsun. A main-sequence-radius approximation gives central T14(A/B)=4.387/4.894h compared with measured2.242h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 9352-146-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited; VSX ROT type revision. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4344.085375BTJD, subsequent 4349.602623. Other-kind nominal prediction: 4346.645795BTJD, subsequent4352.163044. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No

unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.61ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 44 predicted windows, status counts {'fit': 36, 'insufficient': 8}; conditional depth -0.5004ppt, control-inflated 3sigma upper value 0.4948ppt. Optimized secondary windows status counts{'fit': 36, 'insufficient': 8}; fitted secondary T14=2.0077h, T12=0.7027h, trapezoid T23=0.6023h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R016. TIC 117919258

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Obtain a second sector to verify the secondary ephemeris and reduce extrapolation uncertainty.

Archived source: sanitized report text.

TIC 117919258 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Four deep primary events, two supporting clean secondary events, an independent reduction and precise prior localization establish the target EB. All data are one sector; the local orbital period is useful but extrapolation to Gaia epoch or future years has appreciable uncertainty.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 4 accepted primary events in 1 sectors; median depth 37.161ppt (range36.577–37.897). T14=4.7222h, profile range4.5919–4.8526h; ingress T12=1.1806h, nominal trapezoid T23=2.3611h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=7.2428348001±0.0005008657d, Tref=2597.7537849±0.0006134BTJD (reference cycle2); T0=2583.2681153. Chi2/dof=7.20/2; O-C RMS=1.140min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found6 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.1384ppt; including native-unit background regression changes an individual depth by at most0.35%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=4; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
47	2583.2677611	0.0005903	37.435	0.366	-0.510	4
47	2590.5108320	0.0005903	36.886	0.376	-0.170	1
47	2597.7550835	0.0005903	36.577	0.427	1.870	2
47	2604.9957933	0.0005903	37.897	0.435	-1.190	4

Gaia orbit and secondary gate

Exact source917229955854793216, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=7.2428161380±0.0012557446d; photometric difference=0.014sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=-0.057061d with Gaia-only sigma0.079628d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.022529±0.030761, omega=284.7313deg. Gaia relative other-conjunction phase=0.5036479; measured optimized phase=0.5036479. Status: supported repeat secondary. Optimized depth=2.6775±0.1556ppt; 2 individual >3sigma events among2 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Only two secondary events: no independent secondary period is claimed.

First-order timing proxy e cos(omega)˜0.00573 versus Gaia0.00573; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.1250, k=0.225, impact proxy=0.655; chi2/dof=532.74/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product SPOC, sector47; header target=117919258, sector=47 checked. SHA256=1bb2f08e7ee6062adac0e944aeaa8bbe5eb44f52b2d113fea099be7c3cd62198. At actual fitted QLP times, the independent product gives depth range34.654–35.580ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has0 resolved neighbours, 0 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip

Prior difference image sector47, 4 events: peak on target=True; deficit/target separation1.21arcsec; deficit/light separation0.13arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.019734Msun; assumed TIC M1=1.140Msun gives edge-on minimum M2=0.3530Msun, a=18.005Rsun. A main-sequence-radius approximation gives central T14(A/B)=7.079/6.777h compared with measured4.722h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 2988-1320-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4350.519806BTJD, subsequent 4357.762641. Other-kind nominal prediction: 4346.924810BTJD, subsequent4354.167645. These extrapolations are conditional. The secondary prediction uses the primary period; only two secondary events are available. Obtain a second sector to verify the secondary ephemeris and reduce extrapolation uncertainty.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.23ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The0.23ppt control is well below the2.68ppt secondary; both individual events support it, but their two-event span cannot provide an independent secondary period.

Fixed Gaia-phase other-conjunction coverage: 3 predicted windows, status counts {'fit': 2, 'insufficient': 1}; conditional depth 2.6772ppt, control-inflated 3sigma upper value 7.7450ppt. Optimized secondary windows status counts{'fit': 2, 'insufficient': 1}; fitted secondary T14=4.7222h, T12=1.1806h, trapezoid T23=2.3611h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R017. TIC 97509220

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Obtain another eclipse sector to improve local period and test the weak secondary against phase controls.

Archived source: sanitized report text.

TIC 97509220 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Four ~37 ppt primary eclipses are repeated in rawSAP and TARS, and targeted pixels locate the disappearing light on the target rather than either capable neighbour. The local period agrees with the Gaia SB1 period to0.48sigma and RV requires a stellar companion. The optimized ~1.8ppt secondary is not secure because an off-phase control is deeper; that ambiguity does not remove the attributed primary EB.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 4 accepted primary events in 1 sectors; median depth 37.089ppt (range37.042–37.450). T14=4.0171h, profile range3.9302–4.1040h; ingress T12=0.6026h, nominal trapezoid T23=2.8120h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=5.2102346549±0.0001587900d, Tref=3815.9763862±0.0002511BTJD (reference cycle2); T0=3805.5559169. Chi2/dof=1.00/2; O-C RMS=0.362min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found4 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.0809ppt; including native-unit background regression changes an individual depth by at most0.21%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=0; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
92	3805.5556658	0.0005021	37.450	0.433	-0.362	0
92	3810.7664026	0.0005021	37.130	0.385	0.362	0
92	3821.1868719	0.0005021	37.042	0.359	0.362	0

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
92	3826.3966044	0.0005021	37.048	0.360	-0.362	0

Gaia orbit and secondary gate

Exact source6767648451564895360, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=5.2106396818±0.0008243086d; photometric difference=-0.482sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=0.131501d with Gaia-only sigma0.096159d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.046223±0.041905, omega=300.1219deg. Gaia relative other-conjunction phase=0.5147779; measured optimized phase=0.5007016. Status: ambiguous secondary; no secure detection. Optimized depth=1.7957±0.2166ppt; 4 individual >3sigma events among4 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=5.2119796745±0.0020251515d, Tref=3818.5851514±0.0026340, O-C RMS7.729min, chi2/dof5.49/2.

First-order timing proxy e cos(omega)˜0.00110 versus Gaia0.02320; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0250, k=0.200, impact proxy=0.305; chi2/dof=312.21/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TARS, sector92; header target=97509220, sector=92 checked. SHA256=a18537b7a081d89fb267694c539dff8c73c303f605092066637bc10542aae323. At actual fitted QLP times, the independent product gives depth range36.826–39.345ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has2 resolved neighbours, 2 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip
6767648451564893568	33.62	13.634	0.478
6767648245406462080	48.16	11.694	0.080

Prior difference image sector92, 4 events: peak on target=True; deficit/target separation1.19arcsec; deficit/light separation14.94arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

A new per-pixel eclipse-template regression on hashed existing TESScut cubes tests the ambiguity directly. OOT multi-source Gaussian calibration: sigma0.550pixel, dx-0.000,dy-0.000. Free deficit center is0.840arcsec from target; fixed target chi2=11700.55, best alternative-source chi2=86931.84, delta=75231.28. This strongly distinguishes target from resolved neighbours; PSF residual covariance prevents interpreting these formal chi2 values as calibrated significance or precise astrometry. See pixel-localization.json and pixel_selective.py.

Physical interpretation, archives and next check

RV mass function f(M)=0.004633Msun; assumed TIC M1=1.178Msun gives edge-on minimum M2=0.2072Msun, a=14.099Rsun. A main-sequence-radius approximation gives central T14(A/B)=4.888/4.512h compared with measured4.017h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 6894-2722-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4347.420321BTJD, subsequent 4352.630556. Other-kind nominal prediction: 4344.995098BTJD, subsequent4350.207078. These extrapolations are conditional. The secondary prediction is tentative because its detection is ambiguous. Obtain another eclipse sector to improve local period and test the weak secondary against phase controls.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 3.43ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The3.43ppt control exceeds the candidate1.80ppt secondary, so no secure secondary or geometry derived from that candidate is claimed. Primary-only EB classification rests on the repeatability, pixels, independent TARS reduction and RV companion.

Fixed Gaia-phase other-conjunction coverage: 6 predicted windows, status counts {'insufficient': 2, 'fit': 4}; conditional depth 0.5904ppt, control-inflated 3sigma upper value 9.6586ppt. Optimized secondary windows status counts{'insufficient': 2, 'fit': 4}; fitted secondary T14=4.1909h, T12=0.6286h, trapezoid T23=2.9336h. Uncovered windows are not counted as nulls. The optimized secondary remains ambiguous. Full event tables and all exclusions are in the cited JSON files.

R018. TIC 148618714

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited; existing unvetted Kostov table2 flag

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 148618714 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Twenty-six primary and29 secondary events, independent reduction, prior on-target difference imaging and a sole capable neighbour51arcsec away establish the target EB. Kostov table2 is an existing unvetted flag; the Gaia eccentric orbit and detailed component geometry need revision.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 26 accepted primary events in 5 sectors; median depth 40.759ppt (range37.608–42.679). T14=5.2436h, profile range5.1132–5.3739h; ingress T12=1.8353h, nominal trapezoid T23=1.5731h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=3.4810873424±0.0000015148d, Tref=2754.3623775±0.0001896BTJD (reference cycle261); T0=1845.7985811. Chi2/dof=28.92/24; O-C RMS=1.411min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found77 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.8216ppt; including native-unit background regression changes an individual depth by at most1.76%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=6; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
20	1845.7991436	0.0019018	42.365	0.420	0.810	0
20	1849.2802551	0.0019018	42.679	0.451	0.845	0
20	1852.7600556	0.0019018	42.618	0.745	-1.008	0
20	1859.7222786	0.0021263	40.955	0.453	-0.939	2

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
20	1863.2047010	0.0018133	40.742	0.359	0.984	0
20	1866.6871233	0.0018133	41.164	0.373	2.906	0
47	2583.7882215	0.0006554	42.064	0.930	-1.262	1
47	2587.2693330	0.0006554	41.321	0.705	-1.227	4
47	2590.7517553	0.0006554	40.952	0.602	0.695	2
47	2597.7139783	0.0006554	40.704	0.383	0.765	3
47	2601.1950898	0.0006554	41.830	0.408	0.800	3
47	2604.6762012	0.0006554	40.657	0.284	0.834	3
60	2942.3400815	0.0006554	40.553	0.898	-1.458	2
60	2945.8225039	0.0006554	40.776	0.570	0.464	0
60	2949.3023044	0.0006554	40.175	0.324	-1.389	0
60	2959.7443280	0.0013109	40.455	1.058	-3.172	0
73	3290.4499180	0.0006554	39.570	0.296	0.129	0
73	3293.9310294	0.0006554	40.329	0.580	0.164	0
73	3297.4121409	0.0006554	39.623	0.519	0.199	0
73	3307.8554753	0.0006554	37.608	0.624	0.303	0
73	3311.3365868	0.0006554	38.056	0.815	0.338	0
74	3318.3001206	0.0013109	40.529	1.288	2.295	6
74	3321.7812321	0.0013109	40.943	1.086	2.330	0
74	3325.2623436	0.0006554	41.431	0.726	2.364	0
74	3335.7030562	0.0006554	40.918	0.655	-1.307	0
74	3339.1841677	0.0006554	39.987	0.280	-1.272	0

Gaia orbit and secondary gate

Exact source1087362780203935872, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=3.4810325585±0.0000584772d; photometric difference=0.937sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=0.062686d with Gaia-only sigma0.022793d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.045542±0.014508, omega=350.5302deg. Gaia relative other-conjunction phase=0.5285888; measured optimized phase=0.4997888. Status: supported repeat secondary. Optimized depth=12.0918±0.1237ppt; 29 individual >3sigma events among29 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=3.4810870188±0.0000022446d, Tref=2724.7733307±0.0003246, O-C RMS2.092min, chi2/dof57.43/27.

First-order timing proxy e cos(omega)˜-0.00033 versus Gaia0.04492; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0250,

k=0.225, impact proxy=0.795; chi2/dof=138.12/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product SPOC, sector47; header target=148618714, sector=47 checked. SHA256=c9ef3a370b27831388390290627ad4b3aabf5ea88ce86038dfcffc04dc1160d1. At actual fitted QLP times, the independent product gives depth range40.309–43.390ppt. All per-event depths/errors, columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has1 resolved neighbours, 1 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip
1087362715779956992	51.07	13.403	0.761

Prior difference image sector20, 6 events: peak on target=True; deficit/target separation2.13arcsec; deficit/light separation1.05arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.059058Msun; assumed TIC M1=1.073Msun gives edge-on minimum M2=0.5344Msun, a=11.323Rsun. A main-sequence-radius approximation gives central T14(A/B)=7.621/7.508h compared with measured5.244h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: TYC 4111-343-1, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited; existing unvetted Kostov table2 flag. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4345.219293BTJD, subsequent 4348.700380. Other-kind nominal prediction: 4346.959881BTJD, subsequent4350.440968. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 1.60ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 39 predicted windows, status counts {'fit': 28, 'insufficient': 11}; conditional depth 1.5778ppt, control-inflated 3sigma upper value 13.2808ppt. Optimized secondary windows status counts{'fit': 29, 'insufficient': 10}; fitted secondary T14=5.2436h, T12=1.8353h, trapezoid T23=1.5731h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R019. TIC 266540667

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 266540667 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Twenty-one primary and19 secondary events, independent reduction, no capable resolved neighbour and deficit/light agreement0.52arcsec establish the target EB.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 21 accepted primary events in 4 sectors; median depth 38.463ppt (range33.423–39.154). T14=4.9829h, profile range4.8526–5.1132h; ingress T12=0.7474h, nominal trapezoid T23=3.4880h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=3.6993685260±0.0000007262d, Tref=2558.9080632±0.0001783BTJD (reference cycle333); T0=1327.0183441. Chi2/dof=5.76/19; O-C RMS=0.639min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found42 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 0.2472ppt; including native-unit background regression changes an individual depth by at most0.68%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=8; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
1	1327.0180543	0.0020047	38.739	0.356	-0.417	1
1	1330.7174177	0.0019018	38.607	0.343	-0.425	0
1	1334.4167811	0.0019018	38.463	0.354	-0.432	0
1	1338.1161446	0.0019018	38.530	0.361	-0.439	0
1	1341.8167537	0.0024552	38.477	0.455	1.347	4
1	1345.5161171	0.0020047	38.795	0.316	1.340	0
28	2063.1926218	0.0006229	38.522	0.279	-0.085	7

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
28	2066.8919852	0.0006229	38.934	0.459	-0.092	3
28	2070.5913486	0.0006229	38.501	0.348	-0.100	3
28	2077.9900755	0.0006229	39.154	0.280	-0.114	8
28	2081.6894389	0.0006229	38.692	0.333	-0.122	6
68	3158.2054416	0.0006229	37.996	0.333	-0.465	0
68	3165.6054142	0.0006229	38.426	0.427	1.314	0
68	3169.3035319	0.0006229	38.272	0.326	-0.487	0
68	3173.0041410	0.0006229	38.166	0.354	1.300	0
68	3176.7022587	0.0006229	38.012	0.345	-0.502	0
95	3886.9812822	0.0006229	33.489	0.275	-0.118	0
95	3890.6806457	0.0006229	33.550	0.288	-0.125	0
95	3898.0793725	0.0006229	33.435	0.286	-0.140	0
95	3901.7787359	0.0006229	33.541	0.297	-0.147	0
95	3905.4780994	0.0006229	33.423	0.294	-0.155	0

Gaia orbit and secondary gate

Exact source6776367269533991424, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=3.6995626746±0.0001569517d; photometric difference=-1.237sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=0.019372d with Gaia-only sigma0.021131d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.043473±0.018095, omega=273.3458deg. Gaia relative other-conjunction phase=0.5016168; measured optimized phase=0.4920431. Status: supported repeat secondary. Optimized depth=1.9621±0.0483ppt; 19 individual >3sigma events among19 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=3.6993717105±0.0000031356d, Tref=2645.8156813±0.0007784, O-C RMS5.972min, chi2/dof35.81/17.

First-order timing proxy e cos(omega)˜-0.01250 versus Gaia0.00254; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=-0.0250, k=0.200, impact proxy=0.235; chi2/dof=83.11/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector1; header target=266540667, sector=1 checked. SHA256=90b127fb1e1083944f96bbb64f8c4bdf1388a2bfa77912cedaf076500102a584. At actual fitted QLP times, the independent product gives depth range37.125–37.598ppt. All per-event depths/errors,

columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has0 resolved neighbours, 0 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip

Prior difference image sector1, 6 events: peak on target=True; deficit/target separation4.06arcsec; deficit/light separation0.52arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.012880Msun; assumed TIC M1=1.320Msun gives edge-on minimum M2=0.3269Msun, a=11.888Rsun. A main-sequence-radius approximation gives central T14(A/B)=5.624/5.157h compared with measured4.983h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: CD-36 14708, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4345.703061BTJD, subsequent 4349.402430. Other-kind nominal prediction: 4343.827296BTJD, subsequent4347.526668. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.64ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 27 predicted windows, status counts {'fit': 19, 'insufficient': 8}; conditional depth 1.5605ppt, control-inflated 3sigma upper value 4.3987ppt. Optimized secondary windows status counts{'fit': 19, 'insufficient': 8}; fitted secondary T14=4.7222h, T12=0.7083h, trapezoid T23=3.3055h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R020. TIC 285145433

physical_verdict: CONFIRMED

novelty_status: No vetted target EB found in queried catalogues; bibliography limited

completion: COMPLETE

next_check: Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Archived source: sanitized report text.

TIC 285145433 — CONFIRMED target eclipsing binary

Full-review completion: COMPLETE. Physical verdict and novelty are separate. Nineteen primary and20 secondary events in eight sectors, independent reduction and a deficit only1.07arcsec from target (0.35arcsec from its light centroid) establish the target EB. The known EB49arcsec away is a different Gaia/TIC source and is inconsistent with the location of the disappearing light.

Raw photometry and timing

All available QLP products in qlp-manifest.json were examined from SAP_FLUX with QUALITY==0, no gap trimming; flagged core counts remain in every event row. Exposure-integrated trapezoids with local polynomial baselines, red-noise inflation and grid-error floors were fitted. 19 accepted primary events in 8 sectors; median depth 41.599ppt (range40.391–42.630). T14=5.5043h, profile range5.3739–5.6346h; ingress T12=1.9265h, nominal trapezoid T23=1.6513h. Trapezoid flat duration is not physical totality for grazing events.

Primary linear ephemeris: P=8.2345804211±0.0000022000d, Tref=2867.9040354±0.0001792BTJD (reference cycle167); T0=1492.7291051. Chi2/dof=14.89/17; O-C RMS=0.943min. Errors are conditional on this baseline/shape and include fit red-noise/chi2 scaling, not all model systematics. The original resolved CSV and P/P2 were checked; all use the resolved full orbital period. Individual event values and exclusions are retained in fit-results.json.

The blind rolling-median dip census found42 threshold groups; there are zero groups deeper than half the median primary depth outside ±1.5T14 around either fitted eclipse phase. It is a detection census, not an astrophysical classification of small fluctuations. No isolated sector-wide systematic is used as deciding evidence. Maximum fitted background-correlated fake depth is 1.2162ppt; including native-unit background regression changes an individual depth by at most2.03%. Background units were not divided into normalized flux. Maximum flagged core count among accepted windows=7; rejected/partial windows are not treated as nulls.

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
7	1500.9640387	0.0019018	41.859	0.292	0.508	1
7	1509.1972295	0.0018133	42.630	0.360	-1.493	0
8	1525.6677392	0.0018133	40.689	0.288	0.450	1
34	2233.8404896	0.0006880	41.599	0.319	-1.229	0
34	2250.3109993	0.0006880	40.997	0.383	0.713	0
35	2258.5455662	0.0006880	42.461	0.441	0.694	7

Sector	Mid BTJD	Timing error d	Depth ppt	Error ppt	O-C min	Core flagged
35	2275.0146999	0.0006880	41.368	0.311	0.655	6
61	2966.7183165	0.0006880	41.371	0.304	-0.985	0
61	2983.1888263	0.0006880	41.368	0.280	0.958	0
62	2991.4220171	0.0006880	42.587	0.343	-1.044	0
62	2999.6579600	0.0006880	42.001	0.316	0.919	0
62	3007.8911508	0.0006880	42.415	0.397	-1.083	0
88	3691.3615766	0.0006880	40.812	0.338	-0.721	0
88	3699.5961435	0.0006880	40.391	0.323	-0.741	0
88	3707.8307103	0.0006880	40.893	0.326	-0.760	0
88	3716.0652772	0.0006880	40.574	0.313	-0.780	3
89	3724.3012201	0.0006880	42.207	0.340	1.182	0
89	3732.5357870	0.0006880	41.821	0.296	1.163	5
89	3740.7703539	0.0006880	41.709	0.301	1.143	0

Gaia orbit and secondary gate

Exact source5533012033062952064, solutionSB1, bit_index127, own correlation vector used. SB1 covariance order P,gamma,K1,e,omega,Tp (15 lower-triangle correlations); SB2 order P,gamma,K1,K2,e,omega,Tp (21). Gaia P=8.2341002650±0.0006526900d; photometric difference=0.736sigma. Primary association is conjunctionA; extrapolated Gaia-epoch offset=-0.032863d with Gaia-only sigma0.038259d. Individual epochs, covariance-propagated conjunction errors and timing residuals are in orbit-gates.json; the Gaia-only sigma does not include extrapolated photometric uncertainty.

Gaia e=0.087537±0.015648, omega=178.1652deg. Gaia relative other-conjunction phase=0.4443719; measured optimized phase=0.4504993. Status: supported repeat secondary. Optimized depth=5.4519±0.0655ppt; 20 individual >3sigma events among20 fitted windows. Phase was optimized, so the formal S/N is conditional and is not a blind false-alarm probability. Fixed Gaia-phase tests, every covered/partial/flagged other-conjunction window and off-phase controls are retained in orbit-gates.json; optimized individual measurements, T14/ingress/flat durations and uncertainty are in secondary-fit.json/secondary-shape.json. The fixed-primary-shape upper limit applies only at its tested phase and shape; it is not a null across the Gaia uncertainty envelope.

Secondary ephemeris P=8.2345779563±0.0000067427d, Tref=2764.5618612±0.0006654, O-C RMS4.831min, chi2/dof23.63/18.

First-order timing proxy e cos(omega)˜-0.07776 versus Gaia-0.08749; this small-e proxy is approximate and uncalibrated, especially when secondary is weak. Gaia and photometric phase disagreements are retained, not converted into a false precise e claim. Numerical limb-darkened two-disk geometry was tested instead of identifying grazing trapezoid T23 with totality. Coarse best e sin(omega)=0.0250, k=0.250, impact proxy=0.760; chi2/dof=178.39/121. Fixed TIC dimensions and freely fitted flux scales make this a feasibility diagnostic, not a posterior, measured radius, or a guarantee of a unique stellar solution.

Independent reduction and attribution

MAST product TESS-SPOC, sector35; header target=285145433, sector=35 checked. SHA256=149b42bcde95cef5d22f278ea294071832e632ca9a13bc6682b097aa5ca4d462. At actual fitted QLP times, the independent product gives depth range36.478–42.298ppt. All per-event depths/errors,

columns, URLs and target metadata are retained in independent-manifest-results.json. This is an independent reduction of TESS pixels, not an independent observatory.

Fresh ARI Gaia60arcsec cone to G<15 has3 resolved neighbours, 3 capable of this depth under a G-band flux-ratio approximation. Fainter sources cannot fully explain a ~35–40ppt dip individually. Required fractional dips are indicative because G and TESS bandpasses/aperture throughput differ.

Gaia source	Separation arcsec	G	Required fractional dip
5533011929983740800	36.27	14.274	0.759
5533011277148716288	42.88	14.305	0.781
5533011998703204352	48.81	13.099	0.257

Prior difference image sector88, 4 events: peak on target=True; deficit/target separation1.07arcsec; deficit/light separation0.35arcsec. Duplicate old/new-schema rows are one measurement, not independent repetitions. Exact records are in parent diffrun JSONL receipts.

Physical interpretation, archives and next check

RV mass function f(M)=0.031270Msun; assumed TIC M1=1.490Msun gives edge-on minimum M2=0.4981Msun, a=21.577Rsun. A main-sequence-radius approximation gives central T14(A/B)=7.728/7.771h compared with measured5.504h; inclination/grazing/dilution and stellar evolution remain degenerate. These approximations do not veto an attributed EB. In Gaia primary-omega convention, conjunctionA puts the spectroscopic primary behind its companion, while B puts the primary in front (positive RV is receding); flux labels alone do not uniquely assign components in a blended/grazing system.

SIMBAD exact Gaia-ID name: UCAC4 233-021434, typeSB*. Fresh IDIA nine-catalogue cones found no target vetted EB classification; shared current Gaia vari_eclipsing_binary and ExoFOP TOI/CTOI checks are negative. Catalogue details, nearby variables, query limitations and weak web-name search are in CATALOGUE-NOTE.md. Novelty status: No vetted target EB found in queried catalogues; bibliography limited. This is not a complete publication-discovery claim.

Next nominal primary after2026-10-30: 4350.128511BTJD, subsequent 4358.363092. Other-kind nominal prediction: 4345.600829BTJD, subsequent4353.835407. These extrapolations are conditional. Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Review gates are completed with their stated limitations: archives/current prior-art pass; raw primary and secondary/coverage analysis; individual timing/error and O-C; own-source Gaia covariance; independent reduction; neighbours and localization; physical feasibility; supported physical verdict. No unavailable measurement has been silently counted as a detection. Source catalogues were not edited.

Final adversarial review details

The largest positive off-phase fixed-template control depth was 0.18ppt. Controls calibrate low-frequency/modulated residuals; they are not an exact multiple-trial probability for the optimized secondary. The repeat secondary signal is stronger than the off-phase controls and recurs in multiple sectors.

Fixed Gaia-phase other-conjunction coverage: 26 predicted windows, status counts {'fit': 19, 'insufficient': 7}; conditional depth 4.0538ppt, control-inflated 3sigma upper value 9.1666ppt. Optimized secondary windows status counts{'fit': 20, 'insufficient': 6}; fitted secondary T14=5.5043h,

T12=1.9265h, trapezoid T23=1.6513h. Uncovered windows are not counted as nulls. Full event tables and all exclusions are in the cited JSON files.

R021. TIC 240360239

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 240360239: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Both eclipse kinds repeat over many cycles and agree with the SB1 ephemeris. Own sector56 pixel fits resolve the capable neighbor44.8arcsec away. Absolute depth depends on baseline: independent aperture quadratic43.5–47.4ppt versus linear54.8–61.1ppt; use the timing and source attribution more strongly than exact radius/depth inference.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 8 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 42.258ppt; fitted total duration 6.026h, grid/profile interval [5.89531034 6.156]h; ingress 2.109h and flat 1.808h. The fitted flat in a partial eclipse does not imply physical totality.

P=3.0634160272±1.8e-06d; Tref=2933.2458724±0.00038BTJD at cycle398; T0=1714.0062935. 45timings; O-C RMS3.89min; chi2/dof=109.88/43. Sector depths: {"15": {"n": 5, "median_depth": 0.043195034056900984, "min_depth": 0.040556928514787094, "max_depth": 0.04565328730818138}, "16": {"n": 6, "median_depth": 0.042427105312243876, "min_depth": 0.04178605015058184, "max_depth": 0.04327859451866343}, "56": {"n": 9, "median_depth": 0.04207188261487375, "min_depth": 0.04064162754013664, "max_depth": 0.044091585335620964}, "76": {"n": 9, "median_depth": 0.04391299182566162, "min_depth": 0.04141500648272432, "max_depth": 0.04588257778916765}, "82": {"n": 8, "median_depth": 0.03601949845980092, "min_depth": 0.0303410111633611, "max_depth": 0.03727172875301374}, "83": {"n": 8, "median_depth": 0.04211502655510492, "min_depth": 0.04032057980415939, "max_depth": 0.04271652298911235}}.

Secondary conditional fit: phase0.5012908; depth14.250ppt; formal error0.209ppt; empirical error0.274ppt; 43covered fitted windows, 43individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions

remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 3.0634265233182356, "P_error": 6.474212840507014e-06, "Tref": 2922.5280385751366, "Tref_error": 0.0014387327676760179, "reference_cycle": 394, "T0": 1715.5379883877517, "chi2": 76.92372608844698, "dof": 41, "n": 43, "oc_rms_minutes": 14.201485305447916}.

Largest background-projected fake primary depth1.798ppt, median depth change on background regression0.003ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has106runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source1967872449309986688; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-0.192sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.00689/0.01999d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.5096908389353895, "phase_error": 0.01140439890040454, "gaia_ecos": 0.015222417215959298, "gaia_ecos_error": 0.017914642178757847, "gaia_esin": -0.008145004382872047, "gaia_esin_error": 0.019542132519259948, "phase_difference_sigma": -0.73018729454949, "measured_phase": 0.5013559384253966, "measured_phase_error": 0.0004858495198859212}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.5013559384253966, "phase_error_conservative": 0.0004858495198859212, "ecos_first_order": 0.0021299030979730475, "ecos_first_order_error": 0.0007631706412118691, "gaia_ecos": 0.015222417215959298, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 0.9, "m2_min": 0.4132320769176533, "f_mass": 0.040916559974095955, "a_rsun": 9.720561574385453, "r1_rsun": 2.95293, "r2_approx_rsun": 0.4931229564807353, "central_T14_hours": {"A": 8.363512338965787, "B": 8.228371373408397}, "observed_T14_hours": 6.025655172413792}, {"m1": 1.2, "m2_min": 0.48864288688961, "f_mass": 0.040916559974095955, "a_rsun": 10.570368632161358, "r1_rsun": 2.95293, "r2_approx_rsun": 0.5638885354661474, "central_T14_hours": {"A": 7.849065206317354, "B": 7.7222368825564525}, "observed_T14_hours": 6.025655172413792}, {"m1": 1.6, "m2_min": 0.5792041248427657, "f_mass": 0.040916559974095955, "a_rsun": 11.508273263350315, "r1_rsun": 2.95293, "r2_approx_rsun": 0.646048789149516, "central_T14_hours": {"A": 7.377806012694136, "B": 7.258592482798882}, "observed_T14_hours": 6.025655172413792}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 169.57645268095058, "dof": 121, "k": 0.27499999999999997, "b_A": 0.7250000000000001, "e_sin_omega": 0.0749999999999999, "primary_scale": 0.5336523507597534, "secondary_scale": 0.014044171154945294}, "profile_interval_approx_95": {"k": [0.24999999999999997, 0.29999999999999993], "b_A": [0.6900000000000001, 0.76], "e_sin_omega": [-5.551115123125783e-17, 0.12499999999999994]}, "error_inflation": 1.4014582866194263, "assumptions": "Circular projected chord with eccentric

speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 56, "provenance": "TESScut independently constructed aperture", "source_receipt": "240360239/pixel-localization.json", "events": 9, "quadratic_depth_ppt_range": [43.47681781916156, 47.39516354559122], "linear_depth_ppt_range": [54.76635391134967, 61.133725702283414], "note": "No exact-target SPOC/TESS-SPOC/TARS product in shared MAST inventory. Same TESS pixels; 3-pixel-diameter aperture plus outer-pixel median background. Strong baseline sensitivity; conditional fixed-shape fit, formal errors in pixel-localization.json; no quality issues admitted in TESScut."}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 2non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "1967872277518445824", "ra": "322.76452393303657", "dec": "43.32961378581258", "phot_g_mean_mag": "13.389073", "bp_rp": "1.6941042", "sep": "44.756191799968654", "required_fraction_G_approx": 0.27392900231140316, "capable": true}]. Previously verified difference-image receipt: {"tic": "240360239", "sector": 56, "n_events": 8, "deficit_minus_target_arcsec": 5.11, "deficit_minus_light_arcsec": 4.39, "peak_on_target": true, "depth_ap3": 0.03991, "psf_frac_resid": 0.415, "n_neighbours_60": 91, "n_possible": 1, "brightest_nb": {"G": 13.389073, "sep": 44.8}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 240360239_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 1967872449309986688", "main_id": "TYC 3195-1090-1", "otype": "SB*", "ra": "322.77762787244", "dec": "43.32165883249"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0,

"II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4345.480661BTJD; next opposite-kind(B) center 4347.021372BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0349d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R022. TIC 341735257

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 341735257: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Seven primary and eight secondary measurements, independent reduction and own sector79 localization support target over the capable neighbor18.53arcsec away. Primary geometry touches the low-impact-parameter grid boundary; dimensions are not uniquely measured.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 3 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 40.409ppt; fitted total duration 6.547h, grid/profile interval [6.41668966 6.67737931]h; ingress 1.637h and flat 3.274h. The fitted flat in a partial eclipse does not imply physical totality.

P=4.5489678768±3.4e-06d; Tref=2835.0205600±0.00039BTJD at cycle187; T0=1984.3635671. 7timings; O-C RMS0.26min; chi2/dof=0.27/5. Sector depths: {"25": {"n": 2, "median_depth": 0.040678420421151454, "min_depth": 0.040576112269576727, "max_depth": 0.040780728572726174}, "52": {"n": 2, "median_depth": 0.040162344763977814, "min_depth": 0.0399156458786883, "max_depth": 0.04040904364926733}, "79": {"n": 3, "median_depth": 0.04022093992581822, "min_depth": 0.04015077549582523, "max_depth": 0.04068302914841864}}.

Secondary conditional fit: phase0.4993437; depth2.314ppt; formal error0.103ppt; empirical error0.111ppt; 8covered fitted windows, 8individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 4.548977969353895, "P_error": 2.8050966799525166e-05, "Tref": 2828.1988997016065, "Tref_error": 0.0027014280340816148, "reference_cycle": 185, "T0": 1986.6379753711358, "chi2": 16.09662652231133, "dof": 6, "n": 8, "oc_rms_minutes": 33.021762282463854}.

Largest background-projected fake primary depth0.747ppt, median depth change on background regression0.218ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has25runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source4563408868034620288; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=0.044sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: -0.05082/0.02838d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.48455837581019257, "phase_error": 0.010446737911137425, "gaia_ecos": -0.02423401823610156, "gaia_ecos_error": 0.01639803613220015, "gaia_esin": 0.04447895390727373, "gaia_esin_error": 0.016006364986290534, "phase_difference_sigma": 1.513543999441726, "measured_phase": 0.5003960390329338, "measured_phase_error": 0.0006000974671189269}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.5003960390329338, "phase_error_conservative": 0.0006000974671189269, "ecos_first_order": 0.0006220966581998742, "ecos_first_order_error": 0.0009426308970693317, "gaia_ecos": -0.02423401823610156, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.08, "m2_min": 0.3378847893842394, "f_mass": 0.0191877493837463, "a_rsun": 12.979543691883121, "r1_rsun": 2.15694, "r2_approx_rsun": 0.41977339831150834, "central_T14_hours": {"A": 6.5966365431944, "B": 7.210775800404036}, "observed_T14_hours": 6.547034482758621}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 254.25894830508668, "dof": 121, "k": 0.19999999999999998, "b_A": 0.2, "e_sin_omega": 0.0749999999999999, "primary_scale": 0.8833628196290009, "secondary_scale": 0.00211399794871453}, "profile_interval_approx_95": {"k": [0.19999999999999998, 0.19999999999999998], "b_A": [0.2, 0.30500000000000005], "e_sin_omega": [0.04999999999999993, 0.0749999999999999]}, "error_inflation": 2.101313622356088, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 79, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 3453.6804555770764, "qlp": {"1": {"depth": 0.04154721195312732, "error": 0.0006831337543429956, "red_noise_beta": 2.6452485442695957, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.040746566188523584, "background_template_native_sigma": 0.9130817293434513}, "2": {"depth": 0.04029554204542685, "error": 0.0006077143561812174, "red_noise_beta": 2.265624686163599, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.0405005743330241, "background_template_native_sigma": 0.28085114652932536}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.03515732104864991, "error": 0.0004940334428576085, "red_noise_beta": 2.535973332316316, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.03434905244548717, "background_template_native_sigma": 0.9120065735775219}, "2": {"depth": 0.034521563693009925, "error": 0.0004983731479080002, "red_noise_beta": 2.363820912135039, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.03444928000856417, "background_template_native_sigma": 0.29778889771188044}}, "PDCSAP_FLUX": {"1": {"depth": 0.03874095181697651, "error": 0.0005084010272828205, "red_noise_beta": 2.4032473419467872, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.03854277972962165, "background_template_native_sigma": 0.9120065735775219}, "2": {"depth": 0.03854468628984889, "error": 0.0005623837969571777, "red_noise_beta": 2.3856467653036404, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.03861164200618334, "background_template_native_sigma": 0.29778889771188044}}}}, {"tc": 3467.3270235219047, "qlp": {"1": {"depth": 0.042654832777548356, "error": 0.0008503791798639908, "red_noise_beta": 2.9905007725934962, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.04218865234045374, "background_template_native_sigma": 0.6318575031102031}, "2": {"depth": 0.040330427671139396, "error": 0.000460453641797862, "red_noise_beta": 1.8491113861639312, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.04009064456142027, "background_template_native_sigma": -0.20916644412509425}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.03662909077773672, "error": 0.0006668421973075178, "red_noise_beta": 2.9818137916393344, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.035865916918287916, "background_template_native_sigma": 0.6755875664626914}, "2": {"depth": 0.034859916631961745, "error": 0.0003937891403538127, "red_noise_beta": 1.9719735861493928, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.03468826065624026, "background_template_native_sigma": -0.24323186936709043}}, "PDCSAP_FLUX": {"1": {"depth": 0.040647525545603504, "error": 0.0006340866554601208, "red_noise_beta": 2.7293163341218527, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.04013355880822512, "background_template_native_sigma": 0.6755875664626914}, "2": {"depth": 0.03915509362800975, "error": 0.0004635982262483793 7, "red_noise_beta": 2.046433460294119, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.038855750208330944, "background_template_native_sigma": -0.24323186936709043}}}}, {"tc": 3471.8758795035137, "qlp": {"1": {"depth": 0.044123787860138854, "error": 0.0011961456776006768, "red_noise_beta": 3.3925809113814713, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.043697552090016356, "background_template_native_sigma": 0.2779296198242311}, "2": {"depth": 0.04072349384057578, "error": 0.0005553612295250114, "red_noise_beta": 2.071366984924221, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.04063757964897949, "background_template_native_sigma": -0.07277560206368713}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.03781462582899058, "error": 0.0010100934355657984, "red_noise_beta": 3.495679354431587, "n_core": 118, "baseline_degree": 1, "background_regressed_depth":

0.03725635683168862, "background_template_native_sigma": 0.2867731869555697}, "2": {"depth": 0.03495388430855177, "error": 0.0004845877222452042, "red_noise_beta": 2.2774867046281835, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.03491331038864946, "background_template_native_sigma": -0.0963253912147757}}, "PDCSAP_FLUX": {"1": {"depth": 0.04316449805328493, "error": 0.0012384212755345277, "red_noise_beta": 3.6059632418181544, "n_core": 118, "baseline_degree": 1, "background_regressed_depth": 0.04244502900025052, "background_template_native_sigma": 0.28677318695556997}, "2": {"depth": 0.03955503031251004, "error": 0.0005109601263709818, "red_noise_beta": 2.211379694222449, "n_core": 118, "baseline_degree": 2, "background_regressed_depth": 0.03951107864467825, "background_template_native_sigma": -0.0963253912147757}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [40.295542045426856, 40.723493840575784], "error_ppt_range": [0.460453641797862, 0.6077143561812174], "linear_minus_quadratic_ppt_range": [1.2516699077004674, 3.4002940195630718], "background_regressed_minus_original_ppt_range": [-0.23978310971912303, 0.20503228759724546]}, "SAP_FLUX": {"depth_ppt_range": [34.521563693009924, 34.95388430855177], "error_ppt_range": [0.3937891403538127, 0.4983731479080002], "linear_minus_quadratic_ppt_range": [0.6357573556399834, 2.8607415204388102], "background_regressed_minus_original_ppt_range": [-0.1716559757214825, -0.040573919902309186]}, "PDCSAP_FLUX": {"depth_ppt_range": [38.54468628984889, 39.555030312510034], "error_ppt_range": [0.4635982262483794, 0.5623837969571778], "linear_minus_quadratic_ppt_range": [0.19626552712762035, 3.6094677407748907], "background_regressed_minus_original_ppt_range": [-0.2993434196788061, 0.06695571633445058]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 1non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "4563408906690778496", "ra": "248.83228595764535", "dec": "19.28120702495508", "phot_g_mean_mag": "13.7653885", "bp_rp": "1.2936888", "sep": "18.531505916080608", "required_fraction_G_approx": 0.4092380176028601, "capable": true}]. Previously verified difference-image receipt: {"tic": "341735257", "sector": 79, "n_events": 3, "deficit_minus_target_arcsec": 3.67, "deficit_minus_light_arcsec": 4.85, "peak_on_target": true, "depth_ap3": 0.03517, "psf_frac_resid": 0.439, "n_neighbours_60": 9, "n_possible": 1, "brightest_nb": {"G": 13.7653885, "sep": 18.5}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 341735257_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0,

"J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 4563408868034620288", "main_id": "TYC 1527-427-1", "otype": "SB*", "ra": "248.83529478723", "dec": "19.27697559619"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4345.277895BTJD; next opposite-kind(B) center 4347.557541BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0475d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R023. TIC 306553279

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Characterize temperatures/component assignment and test the covered null secondary with resolved multiband photometry.

Archived source: sanitized report text.

TIC 306553279: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target eclipsing system with one observed eclipse kind. It matches Gaia B, not A; the resolved11.9604d period is used. The opposite conjunction is genuinely covered and null under the tested widths/phases. Eccentric geometry can hide that eclipse. Narrow events survive the smooth tide+harmonics adversary. Which luminous component is being eclipsed remains a characterization issue, particularly because the deeper event is at B; no unique temperature/radius solution is claimed.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 1 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 42.822ppt; fitted total duration 4.365h, grid/profile interval [4.2777931 4.45158621]h; ingress 1.528h and flat 1.309h. The fitted flat in a partial eclipse does not imply physical totality.

P=11.9603919006±6.5e-06d; Tref=2893.1955235±0.00025BTJD at cycle92; T0=1792.8394687. 10timings; O-C RMS0.82min; chi2/dof=10.21/8. Sector depths: {"18": {"n": 2, "median_depth": 0.042418834983267484, "min_depth": 0.04222764615530861, "max_depth": 0.042610023811226354}, "57": {"n": 3, "median_depth": 0.043240664256883596, "min_depth": 0.04280206104869187, "max_depth": 0.043272315703038404}, "58": {"n": 2, "median_depth": 0.04253810691493681, "min_depth": 0.042213574152250605, "max_depth": 0.04286263967762302}, "85": {"n": 3, "median_depth": 0.04284163095684671, "min_depth": 0.04237920443287885, "max_depth": 0.043501176504966134}}.

Secondary conditional fit: phase0.4749771; depth0.019ppt; formal error0.157ppt; empirical error0.278ppt; 7covered fitted windows, 0individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions

remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: `null`.

Largest background-projected fake primary depth0.596ppt, median depth change on background regression0.028ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has12runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source4166128628047258888; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesB; delta P=0.006sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.01804/0.02085d.

Exact Gaia relative-phase and e-components: `{"predicted_phase": 0.4755289524547869, "phase_error": 0.008709757073548352, "gaia_ecos": 0.03510262211776811, "gaia_ecos_error": 0.012384711466424706, "gaia_esin": -0.40800987239781494, "gaia_esin_error": 0.014807112656206215}`. Photometric phase diagnostic: `null`. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: `[{"m1": 1.23, "m2_min": 0.9308356483556182, "f_mass": 0.17273309293776085, "a_rsun": 28.453895991508386, "r1_rsun": 2.74955, "r2_approx_rsun": 0.9442748130024795, "central_T14_hours": {"A": 18.279546071223077, "B": 7.685536176521885}, "observed_T14_hours": 4.364689655172413}]`. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: `{"status": "insufficient secondary shape"}`. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Covered null opposite-kind test:45conditional phase/shape tests across5durations and±2sigma Gaia phase covariance; phase sigma0.008653. Pointwise3sigma upper limits range[0.33148917 1.06195296]ppt. Each test preserves its individual windows and exclusions in null-secondary.json; no assertion that uncovered/flagged windows are null. Feasible eccentric missing-eclipse geometry: `[{"radius_ratio_assumption": 0.34342885672291085, "b_primary_duration_approx": 1.1057641274758563, "b_other": 2.6299877910727765, "contact": 1.343428856722911, "other_misses": true, "es_gaia": -0.40800987239781494, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}]`. Smooth equilibrium tide+five harmonics versus same+narrow primary yields deltaBIC=43938.8, RSS[0.618845085018788, 0.14725190329299398]; sector narrow depths[{'sector': 18, 'eclipse_depth': 0.04411493585234402}, {'sector': 57, 'eclipse_depth': 0.0429810612361468}, {'sector': 58, 'eclipse_depth': 0.04231467093096759}, {'sector': 85, 'eclipse_depth': 0.043089020203868074}]. This is an adversarial smooth-variability test, not calibrated false-alarm probability.

Independence, dilution and source

Own alternate photometry: {"sector": 18, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 1792.8390735172413, "qlp": {"1": {"depth": 0.04310015712170297, "error": 0.0005316092710387254, "red_noise_beta": 1.1552673008289465, "n_core": 8, "baseline_degree": 1, "background_regressed_depth": 0.04369198624437644, "background_template_native_sigma": 1.4997204732770812}, "2": {"depth": 0.042193239450043205, "error": 0.0004504773219951539, "red_noise_beta": 1.0552554521727737, "n_core": 8, "baseline_degree": 2, "background_regressed_depth": 0.04264944513982229, "background_template_native_sigma": 2.105259035298262}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.03852937096213136, "error": 0.0004998412319961181, "red_noise_beta": 1.1252456278684426, "n_core": 8, "baseline_degree": 1, "background_regressed_depth": 0.03906887067404397, "background_template_native_sigma": 1.7935428893642182}, "2": {"depth": 0.03754100413774975, "error": 0.00037534985856355513, "red_noise_beta": 1, "n_core": 8, "baseline_degree": 2, "background_regressed_depth": 0.038010959774478055, "background_template_native_sigma": 1.8954520204192342}}, "PDCSAP_FLUX": {"1": {"depth": 0.04349944061598075, "error": 0.0005190741123263998, "red_noise_beta": 1, "n_core": 8, "baseline_degree": 1, "background_regressed_depth": 0.04422618384617931, "background_template_native_sigma": 1.7935428893642182}, "2": {"depth": 0.04308954678016501, "error": 0.0005663439803490912, "red_noise_beta": 1, "n_core": 8, "baseline_degree": 2, "background_regressed_depth": 0.04381901961655223, "background_template_native_sigma": 1.8954520204192342}}}}, {"tc": 1804.7995059395432, "qlp": {"1": {"depth": 0.04330670660542389, "error": 0.000413853704870886, "red_noise_beta": 1.0697231731482282, "n_core": 9, "baseline_degree": 1, "background_regressed_depth": 0.04338345949988512, "background_template_native_sigma": 0.6610066341824832}, "2": {"depth": 0.042672738553423385, "error": 0.00037967220013164054, "red_noise_beta": 1, "n_core": 9, "baseline_degree": 2, "background_regressed_depth": 0.04274712199088542, "background_template_native_sigma": 0.46215103139075386}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.03797306268979021, "error": 0.00033888021595696787, "red_noise_beta": 1.0138367421779433, "n_core": 9, "baseline_degree": 1, "background_regressed_depth": 0.03806362627465568, "background_template_native_sigma": 2.1394587363999578}, "2": {"depth": 0.03735888255635299, "error": 0.000314796763521595, "red_noise_beta": 1, "n_core": 9, "baseline_degree": 2, "background_regressed_depth": 0.037653451381339374, "background_template_native_sigma": 1.8044282371446339}}, "PDCSAP_FLUX": {"1": {"depth": 0.04221740298426695, "error": 0.00037639437023055216, "red_noise_beta": 1.0330408444217472, "n_core": 9, "baseline_degree": 1, "background_regressed_depth": 0.04235083401832769, "background_template_native_sigma": 2.1394587363999578}, "2": {"depth": 0.041816834281484055, "error": 0.0003865870562820381, "red_noise_beta": 1, "n_core": 9, "baseline_degree": 2, "background_regressed_depth": 0.04207571597642326, "background_template_native_sigma": 1.8044282371446339}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [42.1932394500432, 42.672738553423386], "error_ppt_range": [0.37967220013164055, 0.45047732199515395], "linear_minus_quadratic_ppt_range": [0.6339680520005067, 0.9069176716597627], "background_regressed_minus_original_ppt_range": [0.07438343746203613, 0.45620568977908227]}, "SAP_FLUX": {"depth_ppt_range": [37.35888255635299, 37.54100413774975], "error_ppt_range": [0.3147967635215 95, 0.3753498585635551], "linear_minus_quadratic_ppt_range": [0.6141801334372254, 0.988366824381609], "background_regressed_minus_original_ppt_range": [0.29456882498638676,

0.4699556367283031]}, "PDCSAP_FLUX": {"depth_ppt_range": [41.816834281484056, 43.08954678016501], "error_ppt_range": [0.3865870562820381, 0.5663439803490912], "linear_minus_quadratic_ppt_range": [0.40056870278289136, 0.40989383581574446], "background_regressed_minus_original_ppt_range": [0.2588816949392059, 0.7294728363872183]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 1non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "416612897164458496", "ra": "13.453173778266867", "dec": "51.84858884601202", "phot_g_mean_mag": "14.515168", "bp_rp": "0.8689308", "sep": "28.73801697711168", "required_fraction_G_approx": 0.7831010122064801, "capable": true}]. Previously verified difference-image receipt: {"tic": "306553279", "sector": 57, "n_events": 3, "deficit_minus_target_arcsec": 1.47, "deficit_minus_light_arcsec": 1.6, "peak_on_target": true, "depth_ap3": 0.04065, "psf_frac_resid": 0.206, "n_neighbours_60": 34, "n_possible": 1, "brightest_nb": {"G": 14.515168, "sep": 28.7}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 306553279_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 2, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 416612862804725888", "main_id": "TYC 3274-477-1", "otype": "SB*", "ra": "13.459995485580002", "dec": "51.84182051089001"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 2, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(B) center 4352.363335BTJD; next opposite-kind(A) center 4346.090456BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.1042d before future-period

uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Characterize temperatures/component assignment and test the covered null secondary with resolved multiband photometry.

R024. TIC 56534111

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 56534111: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Both kinds repeat, independent TARS confirms the narrow primary and own sector88 per-pixel regression favors target over all six capable neighbors. Nearby VSX pulsator and IJspeert entries belong to different Gaia/TIC sources. Difference-light centroid bias and aperture dilution explain why the original offset alone was insufficient. TARS depths35.6–38.5ppt versus QLP40.8–44.5ppt are a dilution/baseline caveat.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 10 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 44.400ppt; fitted total duration 2.360h, grid/profile interval [2.30095862 2.41826897]h; ingress 0.826h and flat 0.708h. The fitted flat in a partial eclipse does not imply physical totality.

P=3.1852584072±5.5e-07d; Tref=2913.6633407±9.9e-05BTJD at cycle438; T0=1518.5201583. 49timings; O-C RMS1.23min; chi2/dof=104.95/47. Sector depths: {"8": {"n": 6, "median_depth": 0.04272071270976918, "min_depth": 0.04133628702933415, "max_depth": 0.045030990659239406}, "34": {"n": 5, "median_depth": 0.04433780777525329, "min_depth": 0.043239986374396924, "max_depth": 0.0448362733807555}, "35": {"n": 5, "median_depth": 0.04502494212361518, "min_depth": 0.04407310169887147, "max_depth": 0.0453396756280471}, "61": {"n": 8, "median_depth": 0.0467154334313336, "min_depth": 0.04476680995143728, "max_depth": 0.047129694713508}, "62": {"n": 8, "median_depth": 0.046710496067889754, "min_depth": 0.04495666346885815, "max_depth": 0.048357550889201045}, "88": {"n": 9, "median_depth": 0.04237118390209334, "min_depth": 0.04098563989628061, "max_depth": 0.04462175804540915}, "89": {"n": 8, "median_depth": 0.04280706743603792, "min_depth": 0.041469899307077646, "max_depth": 0.04440024595521745}}.

Secondary conditional fit: phase0.5000476; depth5.113ppt; formal error0.155ppt; empirical error0.181ppt; 38covered fitted windows, 31individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 3.185262011045394, "P_error": 4.4377180438844585e-06, "Tref": 2988.516106056921, "Tref_error": 0.0008161545094866513, "reference_cycle": 461, "T0": 1520.1103189649946, "chi2": 31.436986329112724, "dof": 29, "n": 31, "oc_rms_minutes": 6.727508116128878}.

Largest background-projected fake primary depth0.941ppt, median depth change on background regression0.025ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has121runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source5543595481010515712; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-1.098sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.01807/0.01847d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.49529662888990356, "phase_error": 0.011891014369430467, "gaia_ecos": -0.0073865096677807425, "gaia_ecos_error": 0.018677844839175926, "gaia_esin": 0.02078203866651639, "gaia_esin_error": 0.016074572920876034, "phase_difference_sigma": 0.3719517587336258, "measured_phase": 0.49972057058776703, "measured_phase_error": 0.00026010451960406777}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.49972057058776703, "phase_error_conservative": 0.00026010451960406777, "ecos_first_order": -0.0004389266943340026, "ecos_first_order_error": 0.0004085712239768208, "gaia_ecos": -0.0073865096677807425, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.14, "m2_min": 0.442410103145765, "f_mass": 0.034580957529360634, "a_rsun": 10.616582762877737, "r1_rsun": 1.18767, "r2_approx_rsun": 0.5207868731738282, "central_T14_hours": {"A": 3.8353317048057627, "B": 3.998126945366193}, "observed_T14_hours": 2.359613793103448}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 144.27865301853979, "dof": 121, "k": 0.42499999999999993, "b_A": 1.0050000000000001, "e_sin_omega": 0.04999999999999993, "primary_scale": 0.6273324883905627, "secondary_scale": 0.017233737723469346}, "profile_interval_approx_95": {"k": [0.27499999999999997, 0.44999999999999996], "b_A": [0.76, 1.145], "e_sin_omega": [-0.07500000000000004, 0.1999999999999999]}, "error_inflation": 1.1923855621366923, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",

"primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 88, "provenance": "TARS", "fits_identity_validated": true, "quality_column_present": false, "events": [{"tc": 3690.8671268381054, "qlp": {"1": {"depth": 0.04408024261013339, "error": 0.0014149976939896345, "red_noise_beta": 2.093619472003843, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.04325960906731251, "background_template_native_sigma": -0.7167947346298095}, "2": {"depth": 0.04155047710338953, "error": 0.0008870008758414057, "red_noise_beta": 1.4232873819589316, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.04107719771621186, "background_template_native_sigma": -0.6042656716131537}}, "independent": {"FLUX": {"1": {"depth": 0.03923292834813722, "error": 0.0008509990529493386, "red_noise_beta": 1.435172527878349, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03796481914901471, "error": 0.0007228384650110012, "red_noise_beta": 1.1577192362103812, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3694.0506115229423, "qlp": {"1": {"depth": 0.04465452119550402, "error": 0.0017684580176556696, "red_noise_beta": 2.629325592501437, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.0447582892052417, "background_template_native_sigma": 0.06628185059541677}, "2": {"depth": 0.04412448728328514, "error": 0.0018947033015640948, "red_noise_beta": 2.5773936396644226, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.04446946187812505, "background_template_native_sigma": 0.23123677135180132}}, "independent": {"FLUX": {"1": {"depth": 0.03887364766605507, "error": 0.0009332104588892305, "red_noise_beta": 1.5869887125585032, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03847105480013184, "error": 0.0009810454916249329, "red_noise_beta": 1.5242988832710023, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3697.235865918124, "qlp": {"1": {"depth": 0.04248208594366458, "error": 0.0008337612607625911, "red_noise_beta": 1.6223599362122998, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.04258162364680554, "background_template_native_sigma": 0.3358257170607588}, "2": {"depth": 0.043081547239634636, "error": 0.0008585214248458793, "red_noise_beta": 1.5359474746478543, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.043074203891932394, "background_template_native_sigma": -0.032997004913553174}}, "independent": {"FLUX": {"1": {"depth": 0.03572330989510094, "error": 0.000633586174881051, "red_noise_beta": 1.1951683531484152, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.035611633484967535, "error": 0.0006939782683666367, "red_noise_beta": 1.1878347125028312, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3700.421120313306, "qlp": {"1": {"depth": 0.043394561823376976, "error": 0.0016158125773002448, "red_noise_beta": 2.444284334501767, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.043312936960738846, "background_template_native_sigma": -0.1771106839904188}, "2": {"depth": 0.04399203644011536, "error": 0.0017698990911620758,

"red_noise_beta": 2.4539959451150066, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.0439108317181234, "background_template_native_sigma": -0.1761186644102641}}, "independent": {"FLUX": {"1": {"depth": 0.03592274302352161, "error": 0.0008117771943432669, "red_noise_beta": 1.550966514330854, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.035670626197132245, "error": 0.000871319554969494, "red_noise_beta": 1.5180584774591046, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3703.6075545153844, "qlp": {"1": {"depth": 0.04414892429566677, "error": 0.0010236163620008132, "red_noise_beta": 1.6585945088037883, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.04409821209851075, "background_template_native_sigma": -0.13006013800701796}, "2": {"depth": 0.04248320402343571, "error": 0.000719774854514858, "red_noise_beta": 1.1443531643988207, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.042476919330018285, "background_template_native_sigma": -0.01960270920266226}}, "independent": {"FLUX": {"1": {"depth": 0.03656990250889208, "error": 0.0007088223689413678, "red_noise_beta": 1.2697789449489134, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03613204947298118, "error": 0.0007381621573109285, "red_noise_beta": 1.2057131078832717, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3706.7928089105662, "qlp": {"1": {"depth": 0.03649274027380918, "error": 0.0025400666966279095, "red_noise_beta": 3.1774981633044836, "n_core": 42, "baseline_degree": 1, "background_regressed_depth": 0.03611645534363707, "background_template_native_sigma": 0.583219142521697}, "2": {"depth": 0.04078295045867178, "error": 0.000864113346891053, "red_noise_beta": 1.4303316545021403, "n_core": 42, "baseline_degree": 2, "background_regressed_depth": 0.041137247619937575, "background_template_native_sigma": 1.038244753748405}}, "independent": {"FLUX": {"1": {"depth": 0.03581047749438897, "error": 0.0009456378712081822, "red_noise_beta": 1.7024922527092243, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.037279937291952524, "error": 0.0005705165102781809, "red_noise_beta": 1, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3709.978063305748, "qlp": {"1": {"depth": 0.04132361082408482, "error": 0.000661720802673406, "red_noise_beta": 1.3248232112588272, "n_core": 41, "baseline_degree": 1, "background_regressed_depth": 0.04134639538438145, "background_template_native_sigma": 0.19960573087278644}, "2": {"depth": 0.04181414911641789, "error": 0.0006787705398819675, "red_noise_beta": 1.2498828231616566, "n_core": 41, "baseline_degree": 2, "background_regressed_depth": 0.041842962048208455, "background_template_native_sigma": 0.2380453788731843}}, "independent": {"FLUX": {"1": {"depth": 0.035415029464492466, "error": 0.0009205361074474441, "red_noise_beta": 1.66898003243022, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03631713887915443, "error": 0.0007709618052762999, "red_noise_beta": 1.3030299899919067, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3713.1627277974812, "qlp": {"1": {"depth": 0.04811116412121642, "error": 0.002562085613323461, "red_noise_beta": 3.221468899835536, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": 0.04812000952693092, "background_template_native_sigma": 0.04316831279983507}, "2": {"depth": 0.04451833753865746, "error": 0.0016016015126500017, "red_noise_beta": 2.304482662043859, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": 0.044522265690455885, "background_template_native_sigma": 0.016436367429436504}}, "independent": {"FLUX": {"1":

{"depth": 0.03806439797471662, "error": 0.000956208212597945, "red_noise_beta": 1.7090916566988394, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03699999012371252, "error": 0.0008156254081674137, "red_noise_beta": 1.3723268690278292, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3716.347982192663, "qlp": {"1": {"depth": 0.04013837301719542, "error": 0.0011770201967862813, "red_noise_beta": 1.9820450805504948, "n_core": 40, "baseline_degree": 1, "background_regressed_depth": 0.04045362121726251, "background_template_native_sigma": 0.44902052158125483}, "2": {"depth": 0.04203472509076876, "error": 0.0006539643334526413, "red_noise_beta": 1.1140084962557333, "n_core": 40, "baseline_degree": 2, "background_regressed_depth": 0.04232037749043913, "background_template_native_sigma": 0.42923540883654115}}, "independent": {"FLUX": {"1": {"depth": 0.0362224246815385, "error": 0.0004880221869977586, "red_noise_beta": 1, "n_core": 43, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.036580130488234684, "error": 0.0005336661186966717, "red_noise_beta": 1, "n_core": 43, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [40.78295045867178, 44.518337538657455], "error_ppt_range": [0.6539643334526413, 1.8947033015640948], "linear_minus_quadratic_ppt_range": [-4.2902101848626, 3.59282658255896], "background_regressed_minus_original_ppt_range": [-0.473279387177665, 0.3542971612657958]}, "FLUX": {"depth_ppt_range": [35.611633484967534, 38.47105480013184], "error_ppt_range": [0.5336661186966717, 0.9810454916249328], "linear_minus_quadratic_ppt_range": [-1.4694597975635562, 1.2681091991225135], "background_regressed_minus_original_ppt_range": null}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 6non-target sources, 6capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "5543595412291039488", "ra": "127.16510961978122", "dec": "-34.134958655038005", "phot_g_mean_mag": "12.165077", "bp_rp": "0.6110878", "sep": "27.51793915294223", "required_fraction_G_approx": 0.09883140647940428, "capable": true}, {"source_id": "5543595485318597376", "ra": "127.15587181399464", "dec": "-34.12116864359313", "phot_g_mean_mag": "11.442989", "bp_rp": "0.64363194", "sep": "33.83884364036981", "required_fraction_G_approx": 0.050822938997607064, "capable": true}, {"source_id": "5543618884301814528", "ra": "127.17958885704554", "dec": "-34.13107352387466", "phot_g_mean_mag": "13.332588", "bp_rp": "0.819376", "sep": "47.1016158988344", "required_fraction_G_approx": 0.2896672363814341, "capable": true}, {"source_id": "5543595446650777728", "ra": "127.14941135718036", "dec": "-34.13225152527005", "phot_g_mean_mag": "11.848819", "bp_rp": "0.57396126", "sep": "48.161198137967396", "required_fraction_G_approx": 0.07385701919069311, "capable": true}, {"source_id": "5543595416599119488", "ra": "127.17377910512144", "dec": "-34.139414627633855", "phot_g_mean_mag": "13.937604", "bp_rp": "0.8576994", "sep": "51.62644290288045", "required_fraction_G_approx": 0.505714930685685, "capable": true}, {"source_id": "5543595450958866944", "ra": "127.14868408591232", "dec": "-34.134199502517895", "phot_g_mean_mag": "13.564283", "bp_rp": "1.9017773", "sep": "53.06339279835212",

"required_fraction_G_approx": 0.358572827564861, "capable": true}]. Previously verified difference-image receipt: {"tic": "56534111", "sector": 88, "n_events": 9, "deficit_minus_target_arcsec": 4.13, "deficit_minus_light_arcsec": 9.34, "peak_on_target": true, "depth_ap3": 0.02535, "psf_frac_resid": 0.676, "n_neighbours_60": 77, "n_possible": 6, "brightest_nb": {"G": 11.442989, "sep": 33.8}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 56534111_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 1, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 3, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 5543595481010515712", "main_id": "TYC 7143-211-1", "otype": "SB*", "ra": "127.16457016887", "dec": "-34.12761335487"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 1, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 3, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4347.029624BTJD; next opposite-kind(B) center 4345.437723BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0379d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R025. TIC 31357336

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Refit the Gaia small-e orbit jointly with both eclipse ephemerides.

Archived source: sanitized report text.

TIC 31357336: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Primary/secondary timing is essentially circular; Gaia relative-phase tension is2.87sigma and primary epoch389 timing differs2.9sigma, suggesting refinement of its low-e RV orbit. It does not erase the repeatable target-localized eclipses. The capable neighbor48.6arcsec away is a VSX pulsator, distinct from target; original deficit/light separation1.16arcsec strongly favors target. Grazing geometry means trapezoid flat duration cannot measure e sin omega.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 1 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 42.697ppt; fitted total duration 4.712h, grid/profile interval [4.62537931 4.79917241]h; ingress 1.649h and flat 1.414h. The fitted flat in a partial eclipse does not imply physical totality.

P=6.4429678687±1.3e-06d; Tref=2337.8270301±0.00024BTJD at cycle139; T0=1442.2544964. 15timings; O-C RMS0.87min; chi2/dof=4.46/13. Sector depths: {"5": {"n": 4, "median_depth": 0.04239539670642163, "min_depth": 0.04192931417272335, "max_depth": 0.04282549234676403}, "6": {"n": 3, "median_depth": 0.04403142026932625, "min_depth": 0.04393396029810865, "max_depth": 0.04416734660083965}, "32": {"n": 4, "median_depth": 0.04300713355395081, "min_depth": 0.042696564069592075, "max_depth": 0.04310583490597633}, "98": {"n": 4, "median_depth": 0.03591196578385179, "min_depth": 0.035556586122775166, "max_depth": 0.03631464487644695}}.

Secondary conditional fit: phase0.4974591; depth5.849ppt; formal error0.073ppt; empirical error0.208ppt; 13covered fitted windows, 13individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 6.4429628106235235,

"P_error": 3.864682991386542e-06, "Tref": 2611.6526121278744, "Tref_error": 0.0007001810392303333, "reference_cycle": 181, "T0": 1445.4763434050167, "chi2": 18.98616801345126, "dof": 11, "n": 13, "oc_rms_minutes": 3.7526442759968877}.

Largest background-projected fake primary depth0.519ppt, median depth change on background regression0.215ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has36runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source2908637241099225600; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=1.816sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: -0.10815/0.03742d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.4669848133363157, "phase_error": 0.011490094003234412, "gaia_ecos": -0.05188157560063782, "gaia_ecos_error": 0.018070038786678667, "gaia_esin": -0.008433597846669192, "gaia_esin_error": 0.01460927245812901, "phase_difference_sigma": 2.868618447137942, "measured_phase": 0.4999472355617157, "measured_phase_error": 0.00011760966257992077}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.4999472355617157, "phase_error_conservative": 0.00011760966257992077, "ecos_first_order": -8.28821858423441e-05, "ecos_first_order_error": 0.00018474082597612674, "gaia_ecos": -0.05188157560063782, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.05, "m2_min": 0.455379064635996, "f_mass": 0.04167037781703822, "a_rsun": 16.698685601318314, "r1_rsun": 2.06675, "r2_approx_rsun": 0.5329647040907759, "central_T14_hours": {"A": 7.71660131049374, "B": 7.587532401376177}, "observed_T14_hours": 4.7122758620689655}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 125.92123075171807, "dof": 121, "k": 0.3749999999999999, "b_A": 1.0750000000000002, "e_sin_omega": -5.551115123125783e-17, "primary_scale": 0.997895384280781, "secondary_scale": 0.01920862987355482}, "profile_interval_approx_95": {"k": [0.27499999999999997, 0.3999999999999999], "b_A": [0.935, 1.11], "e_sin_omega": [-5.551115123125783e-17, 0.02499999999999994]}, "error_inflation": 1.0406713285265956, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 32, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 2176.7529528790196, "qlp": {"1": {"depth": 0.04349596171714658, "error": 0.0003818413413561788, "red_noise_beta": 1.253220515558488, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.04374123454865687, "background_template_native_sigma": 0.7218897649592133}, "2": {"depth": 0.0431761319015988, "error": 0.00041009353467573564, "red_noise_beta": 1.2300585425116068, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.0432969766558066, "background_template_native_sigma": 0.2890454550882381}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04412266541475429, "error": 0.0003436615596789273, "red_noise_beta": 1.1649196753554516, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.044792388799806994, "background_template_native_sigma": 1.541571726682434}, "2": {"depth": 0.04388252290472125, "error": 0.0003727409756198782, "red_noise_beta": 1.1478413553839895, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.044487434999424744, "background_template_native_sigma": 1.262301904808965}}, "PDCSAP_FLUX": {"1": {"depth": 0.045339728940815914, "error": 0.00034699104656767886, "red_noise_beta": 1.1644313606074843, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.04601777398945456, "background_template_native_sigma": 1.541571726682434}, "2": {"depth": 0.04512634322810551, "error": 0.00038153116586194934, "red_noise_beta": 1.1608049669957272, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.04573412175213035, "background_template_native_sigma": 1.262301904808965}}}}, {"tc": 2183.1959274118362, "qlp": {"1": {"depth": 0.04368457064287349, "error": 0.0003446217981902 23, "red_noise_beta": 1.1848596642019404, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.043991391614654186, "background_template_native_sigma": 0.4765236436080497}, "2": {"depth": 0.043333196469138066, "error": 0.0003621693315853639, "red_noise_beta": 1.1427408791708005, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.043692062541799755, "background_template_native_sigma": 0.19779354597066934}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.044870085283080945, "error": 0.00036350251006509116, "red_noise_beta": 1.2572729998196677, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.04514325973030176, "background_template_native_sigma": 0.5132000142980161}, "2": {"depth": 0.04445890554662474, "error": 0.00038347046818945824, "red_noise_beta": 1.2266563165716429, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.0448883363395479, "background_template_native_sigma": 0.249629065395671}}, "PDCSAP_FLUX": {"1": {"depth": 0.046002224378565035, "error": 0.0003820498968089407, "red_noise_beta": 1.28543457894062, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.04624306512747405, "background_template_native_sigma": 0.5132000142980161}, "2": {"depth": 0.04552079634862043, "error": 0.0003963678047117229, "red_noise_beta": 1.2438723508503262, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.04596334532757356, "background_template_native_sigma": 0.249629065395671}}}}, {"tc": 2189.638901944653, "qlp": {"1": {"depth": 0.043834544737298815, "error": 0.00039841272710066755, "red_noise_beta": 1.3475281380226285, "n_core": 29, "baseline_degree": 1, "background_regressed_depth": 0.043806420487506686, "background_template_native_sigma": -0.08949081260734482}, "2": {"depth": 0.042972316829791454, "error": 0.00034159492253602915, "red_noise_beta": 1.1759665673608062, "n_core": 29, "baseline_degree": 2, "background_regressed_depth": 0.04299059793874859, "background_template_native_sigma": 0.07070757314141057}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04476728009796988, "error": 0.00041792160405121933, "red_noise_beta": 1.3874678590557232, "n_core": 29, "baseline_degree": 1, "background_regressed_depth":

0.044846904660017575, "background_template_native_sigma": 0.2728826140861077}, "2": {"depth": 0.04394750776889422, "error": 0.000386282572994905, "red_noise_beta": 1.286773817786107, "n_core": 29, "baseline_degree": 2, "background_regressed_depth": 0.0440362499341757, "background_template_native_sigma": 0.32802538679505666}}, "PDCSAP_FLUX": {"1": {"depth": 0.045681498060910666, "error": 0.00041621195267789067, "red_noise_beta": 1.3801824036272237, "n_core": 29, "baseline_degree": 1, "background_regressed_depth": 0.04576604572672005, "background_template_native_sigma": 0.2728826140861077}, "2": {"depth": 0.04488352115574264, "error": 0.0003884280154276107, "red_noise_beta": 1.2838916527501831, "n_core": 29, "baseline_degree": 2, "background_regressed_depth": 0.04497837898362352, "background_template_native_sigma": 0.32802538679505666}}}}, {"tc": 2196.0818764774695, "qlp": {"1": {"depth": 0.044113736776166056, "error": 0.0004107762368388225, "red_noise_beta": 1.329814913300526, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.044479349886195355, "background_template_native_sigma": 0.8318635519993202}, "2": {"depth": 0.043287661238771015, "error": 0.0003509604314618985, "red_noise_beta": 1.1364104736343246, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.04364182962306282, "background_template_native_sigma": 0.7677184657459285}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04533803056558976, "error": 0.00035376180299351485, "red_noise_beta": 1.2305522091052905, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.045436313731026204, "background_template_native_sigma": 1.3618104726763471}, "2": {"depth": 0.04462631546833843, "error": 0.00033424783085534766, "red_noise_beta": 1.1439464159742512, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.04482461785016168, "background_template_native_sigma": 0.9223368610014978}}, "PDCSAP_FLUX": {"1": {"depth": 0.046315688752757316, "error": 0.00035920249795570 89, "red_noise_beta": 1.233812354206725, "n_core": 28, "baseline_degree": 1, "background_regressed_depth": 0.04640963393784804, "background_template_native_sigma": 1.3618104726763471}, "2": {"depth": 0.04560380966065024, "error": 0.00034226271011527645, "red_noise_beta": 1.1539563934660868, "n_core": 28, "baseline_degree": 2, "background_regressed_depth": 0.0457989549771077, "background_template_native_sigma": 0.9223368610014978}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [42.97231682979145, 43.333196469138066], "error_ppt_range": [0.34159492253602913, 0.41009353467573567], "linear_minus_quadratic_ppt_range": [0.3198298155477758, 0.8622279075073611], "background_regressed_minus_original_ppt_range": [0.018281108957136916, 0.3588660726616885]}, "SAP_FLUX": {"depth_ppt_range": [43.88252290472125, 44.62631546833843], "error_ppt_range": [0.33424783085534765, 0.386282572994905], "linear_minus_quadratic_ppt_range": [0.24014251003304232, 0.8197723290756589], "background_regressed_minus_original_ppt_range": [0.08874216528147877, 0.6049120947034942]}, "PDCSAP_FLUX": {"depth_ppt_range": [44.883521155742635, 45.603809660650235], "error_ppt_range": [0.3422627101152764, 0.3963678047117229], "linear_minus_quadratic_ppt_range": [0.21338571271040108, 0.7979769051680297], "background_regressed_minus_original_ppt_range": [0.09485782788088565, 0.6077785240248337]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 1non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "2908634664118846592", "ra":

"83.19265742168255", "dec": "-26.902305997849947", "phot_g_mean_mag": "14.7080145", "bp_rp": "0.63252735", "sep": "48.60909111672991", "required_fraction_G_approx": 0.8596703087407617, "capable": true}]. Previously verified difference-image receipt: {"tic": "31357336", "sector": 5, "n_events": 4, "deficit_minus_target_arcsec": 3.01, "deficit_minus_light_arcsec": 1.16, "peak_on_target": true, "depth_ap3": 0.04518, "psf_frac_resid": 0.47, "n_neighbours_60": 4, "n_possible": 1, "brightest_nb": {"G": 14.7080145, "sep": 48.6}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 31357336_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 1, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 2908637241099225600", "main_id": "UCAC4 316-007455", "otype": "SB*", "ra": "83.17754781003626", "dec": "-26.90335445260722"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 1, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4348.033005BTJD; next opposite-kind(B) center 4344.809608BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0740d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Refit the Gaia small-e orbit jointly with both eclipse ephemerides.

R026. TIC 342454977

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 342454977: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Over100 independently timed events per kind, alternate TARS reduction, and original near-target difference image support target. Simple geometry has chi2/dof~7.1, indicating stellar dimensions, dilution or baseline/model mismatch; its grid intervals are not calibrated orbital constraints. TARS baseline changes depth by2.3–7.3ppt. Physical confirmation rests on the attributed repeat eclipses and SB1 match.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 23 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 45.403ppt; fitted total duration 3.496h, grid/profile interval [3.40882759 3.58262069]h; ingress 1.224h and flat 1.049h. The fitted flat in a partial eclipse does not imply physical totality.

P=2.3572171939±2e-07d; Tref=2784.0675749±5.6e-05BTJD at cycle536; T0=1520.5991590. 112timings; O-C RMS0.81min; chi2/dof=129.10/110. Sector depths: {"8": {"n": 5, "median_depth": 0.04265878860464408, "min_depth": 0.04235867825650347, "max_depth": 0.043146357981555764}, "9": {"n": 9, "median_depth": 0.04338067549429067, "min_depth": 0.04215297677087241, "max_depth": 0.04370723712773967}, "10": {"n": 8, "median_depth": 0.04461168427396722, "min_depth": 0.043601493295798856, "max_depth": 0.04497154955055468}, "34": {"n": 9, "median_depth": 0.04491190703239951, "min_depth": 0.04411077033724689, "max_depth": 0.04568264805250375}, "35": {"n": 8, "median_depth": 0.04538060895209929, "min_depth": 0.04416918830228644, "max_depth": 0.046547748271491544}, "36": {"n": 8, "median_depth": 0.04620445584335496, "min_depth": 0.04583501749409942, "max_depth": 0.047123635258291545}, "61": {"n": 10, "median_depth": 0.045599711019941674, "min_depth": 0.04433805863347272, "max_depth": 0.04667282329068989}, "62":

{"n": 9, "median_depth": 0.04663566281122505, "min_depth": 0.04573787419753326, "max_depth": 0.04734955447048573}, "63": {"n": 11, "median_depth": 0.047528597138123116, "min_depth": 0.046567592860200235, "max_depth": 0.04835219969719257}, "88": {"n": 12, "median_depth": 0.045637082459720424, "min_depth": 0.04497379980886952, "max_depth": 0.046571280768612985}, "89": {"n": 11, "median_depth": 0.045821793646669405, "min_depth": 0.045083479587563584, "max_depth": 0.046186663169335285}, "90": {"n": 12, "median_depth": 0.04491036985569993, "min_depth": 0.04401951695585111, "max_depth": 0.04608036534132991}}.

Secondary conditional fit: phase0.5000576; depth6.555ppt; formal error0.038ppt; empirical error0.050ppt; 103covered fitted windows, 103individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 2.357217047784891, "P_error": 7.28088663561703e-07, "Tref": 2740.4590401590995, "Tref_error": 0.00024303855533469766, "reference_cycle": 517, "T0": 1521.7778264543108, "chi2": 146.82615245307835, "dof": 101, "n": 103, "oc_rms_minutes": 3.878624305802949}.

Largest background-projected fake primary depth0.635ppt, median depth change on background regression0.089ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has318runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source5318912277038701696; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=0.662sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: -0.02606/0.01461d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.5084575651493011, "phase_error": 0.009137030450295191, "gaia_ecos": 0.013278897115884345, "gaia_ecos_error": 0.014344838684912325, "gaia_esin": -0.03153123328240862, "gaia_esin_error": 0.015806456516436328, "phase_difference_sigma": -0.9264757846618187, "measured_phase": 0.4999917582609509, "measured_phase_error": 0.00010598121671172912}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.4999917582609509, "phase_error_conservative": 0.00010598121671172912, "ecos_first_order": -1.2946093424759168e-05, "ecos_first_order_error": 0.000166474905920038, "gaia_ecos": 0.013278897115884345, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.15, "m2_min": 0.5066919817034703, "f_mass": 0.04739674907404368, "a_rsun": 8.819559604800181, "r1_rsun": 2.05657, "r2_approx_rsun": 0.5804906452647437, "central_T14_hours": {"A": 5.556322809857988, "B": 5.216638067298117}, "observed_T14_hours": 3.4957241379310346}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 857.1050959725244, "dof": 121, "k":

0.42499999999999993, "b_A": 1.11, "e_sin_omega": -0.02500000000000005, "primary_scale": 1.0, "secondary_scale": 0.01989605725963778}, "profile_interval_approx_95": {"k": [0.27499999999999997, 0.42499999999999993], "b_A": [0.7950000000000002, 1.11], "e_sin_omega": [-0.02500000000000005, 0.1499999999999999]}, "error_inflation": 7.083513189855574, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 88, "provenance": "TARS", "fits_identity_validated": true, "quality_column_present": false, "events": [{"tc": 3691.5964166928575, "qlp": {"1": {"depth": 0.053057867664972454, "error": 0.0028661028798635966, "red_noise_beta": 3.678433649645796, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.04902652149437008, "background_template_native_sigma": -1.113156105259998}, "2": {"depth": 0.046824552754260416, "error": 0.0006328341886968832, "red_noise_beta": 1.4836870200602097, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.046204875319868624, "background_template_native_sigma": -0.8261536184700238}}, "independent": {"FLUX": {"1": {"depth": 0.051812117758248434, "error": 0.00346927751304823, "red_noise_beta": 3.717825398861288, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.044509082568451847, "error": 0.0012967247560315412, "red_noise_beta": 2.3821942686916153, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3693.9536285788913, "qlp": {"1": {"depth": 0.05263944123881273, "error": 0.0028133200775625187, "red_noise_beta": 3.8059672675459644, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.047133902208476096, "background_template_native_sigma": -0.36960530127872276}, "2": {"depth": 0.04641878910697994, "error": 0.0005018745729580749, "red_noise_beta": 1.5130292836497743, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.046088801956043576, "background_template_native_sigma": -0.07344186597926652}}, "independent": {"FLUX": {"1": {"depth": 0.04837722672471509, "error": 0.0029282385584506456, "red_noise_beta": 3.61829712787916, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.0419195680580539, "error": 0.0005999948103778922, "red_noise_beta": 1.3421263765394702, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3696.310840464926, "qlp": {"1": {"depth": 0.052064175894060906, "error": 0.002799920495393122, "red_noise_beta": 3.8039499773392165, "n_core": 61, "baseline_degree": 1, "background_regressed_depth": 0.0520957868971344, "background_template_native_sigma": 0.1565328006282593}, "2": {"depth": 0.046082955462845916, "error": 0.0006387012741713888, "red_noise_beta": 1.7528146396699593, "n_core": 61, "baseline_degree": 2, "background_regressed_depth": 0.046068104469216505, "background_template_native_sigma": -0.02542385826331871}}, "independent": {"FLUX": {"1": {"depth": 0.049656633623084985, "error": 0.003103858974232075, "red_noise_beta": 3.6610195637690133, "n_core": 62, "baseline_degree": 1, "background_regressed_depth": null,

"background_template_native_sigma": null}, "2": {"depth": 0.04323189004753591, "error": 0.0012102150358622734, "red_noise_beta": 2.2768403632789305, "n_core": 62, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3698.668052350961, "qlp": {"1": {"depth": 0.05118902388609341, "error": 0.0024031244060208467, "red_noise_beta": 3.7150554947932157, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.05127219122632117, "background_template_native_sigma": 0.4815496096975802}, "2": {"depth": 0.04600638697020434, "error": 0.00069098748109665, "red_noise_beta": 1.958223716997477, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04613112018100137, "background_template_native_sigma": 0.4082300007916667}}, "independent": {"FLUX": {"1": {"depth": 0.049509348959921734, "error": 0.0025589286077717915, "red_noise_beta": 3.5348883633730046, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.0442140842774186, "error": 0.0009858052058933637, "red_noise_beta": 2.024115153097746, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3701.02439030596, "qlp": {"1": {"depth": 0.05026939201903794, "error": 0.0020416276882426546, "red_noise_beta": 3.6185589196199652, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.05037122660621079, "background_template_native_sigma": 0.3290184905492699}, "2": {"depth": 0.04597529993118368, "error": 0.0006298042118784593, "red_noise_beta": 1.7885825444903953, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04608230191554124, "background_template_native_sigma": 0.2511463717568079}}, "independent": {"FLUX": {"1": {"depth": 0.04828918594367681, "error": 0.002186040164637249, "red_noise_beta": 3.300256111241164, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.04368718769403389, "error": 0.0006879332206274425, "red_noise_beta": 1.4309808866918783, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3703.3824761230294, "qlp": {"1": {"depth": 0.04919894969833795, "error": 0.0016647368391630858, "red_noise_beta": 3.3513557050473386, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.04945161278470409, "background_template_native_sigma": 0.4946196030378441}, "2": {"depth": 0.045656572380176695, "error": 0.0005543682046665922, "red_noise_beta": 1.593740852578105, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04588139153472318, "background_template_native_sigma": 0.6128273411871191}}, "independent": {"FLUX": {"1": {"depth": 0.04822680832561588, "error": 0.002042485998189533, "red_noise_beta": 3.2646786783835546, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.04397421643415254, "error": 0.000744315843819268, "red_noise_beta": 1.5926823759532307, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3705.739688009064, "qlp": {"1": {"depth": 0.049289675273803055, "error": 0.0015014108500914993, "red_noise_beta": 3.1482563163834305, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.046791459881384063, "background_template_native_sigma": -1.0013891110460367}, "2": {"depth": 0.046104288360825835, "error": 0.0005461174272644372, "red_noise_beta": 1.5002350978260939, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04674747686465119, "background_template_native_sigma": 0.26902332529343204}}, "independent": {"FLUX": {"1": {"depth": 0.048371942081232015, "error": 0.0017523727377597052, "red_noise_beta": 3.0022273018896355, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.044713652138202414, "error": 0.0007116568124396 36, "red_noise_beta": 1.5007809507605077, "n_core": 63, "baseline_degree": 2,

"background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3708.096899895098, "qlp": {"1": {"depth": 0.047730787460284876, "error": 0.0012008128553900104, "red_noise_beta": 2.8984812279995524, "n_core": 62, "baseline_degree": 1, "background_regressed_depth": 0.04531781080465804, "background_template_native_sigma": -0.5321550949989861}, "2": {"depth": 0.04511583919916738, "error": 0.0003849009303804497, "red_noise_beta": 1.1514736080035248, "n_core": 62, "baseline_degree": 2, "background_regressed_depth": 0.04460935402178746, "background_template_native_sigma": -0.22383071416968564}}, "independent": {"FLUX": {"1": {"depth": 0.04589928476708102, "error": 0.0012077695214905899, "red_noise_beta": 2.4947262313903518, "n_core": 62, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.043316327904607016, "error": 0.0004952779497581231, "red_noise_beta": 1.1324636692040486, "n_core": 62, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3710.4541117811327, "qlp": {"1": {"depth": 0.04756507377571423, "error": 0.0008825168229093555, "red_noise_beta": 2.5661928638687823, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.04755446842271857, "background_template_native_sigma": -0.03179805015615619}, "2": {"depth": 0.04568387854748933, "error": 0.0003663347898231283, "red_noise_beta": 1.1941091087199147, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.045571037769271606, "background_template_native_sigma": -0.2385804144590324}}, "independent": {"FLUX": {"1": {"depth": 0.046349445880677805, "error": 0.0011727180250927511, "red_noise_beta": 2.3727450920427593, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.043875970951691844, "error": 0.00048717131015149607, "red_noise_beta": 1.0594069137842184, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3712.8113236671666, "qlp": {"1": {"depth": 0.04689980489627076, "error": 0.0008496017802694106, "red_noise_beta": 2.360109467514791, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.047067419007480495, "background_template_native_sigma": 0.25348793244976203}, "2": {"depth": 0.0451456386124973, "error": 0.0004327295732786163, "red_noise_beta": 1.277505896170106, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04536171197258537, "background_template_native_sigma": 0.3607140263745439}}, "independent": {"FLUX": {"1": {"depth": 0.044056411211568655, "error": 0.0010871378508544292, "red_noise_beta": 2.2990975423709625, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.041760526769424, "error": 0.0005057924424889109, "red_noise_beta": 1.1351873326032362, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3715.1685355532013, "qlp": {"1": {"depth": 0.04713363395509599, "error": 0.0009727537645329972, "red_noise_beta": 2.584506214885352, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": 0.04719234430282982, "background_template_native_sigma": 0.1334946567860193}, "2": {"depth": 0.04506237398490665, "error": 0.0004093172644383012, "red_noise_beta": 1.2215372755230194, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": 0.04513079733897752, "background_template_native_sigma": 0.16854626094162922}}, "independent": {"FLUX": {"1": {"depth": 0.04526842023036884, "error": 0.001217601837552703, "red_noise_beta": 2.4384632925394683, "n_core": 63, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.042719173174056414, "error": 0.0005532859010514126, "red_noise_beta": 1.2012097379692013, "n_core": 63, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 3717.5248735082014, "qlp": {"1": {"depth": 0.04739241263488989, "error": 0.0009771509769938118,

"red_noise_beta": 2.2894995289154103, "n_core": 57, "baseline_degree": 1, "background_regressed_depth": 0.04777765731106114, "background_template_native_sigma": 0.4760690489703298}, "2": {"depth": 0.04537751259482565, "error": 0.0004621617141373611, "red_noise_beta": 1.211937828194203, "n_core": 57, "baseline_degree": 2, "background_regressed_depth": 0.04581390709543914, "background_template_native_sigma": 0.7134467750479223}}, "independent": {"FLUX": {"1": {"depth": 0.04625564091570088, "error": 0.0017508773595083177, "red_noise_beta": 2.6865347918706237, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.04276095491954395, "error": 0.0006789507556918725, "red_noise_beta": 1.2232335835732566, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [45.06237398490665, 46.82455275426042], "error_ppt_range": [0.3663347898231283, 0.6909874810966501], "linear_minus_quadratic_ppt_range": [1.7541662837734644, 6.233314910712037], "background_regressed_minus_original_ppt_range": [-0.6196774343917923, 0.6431885038253524]}, "FLUX": {"depth_ppt_range": [41.760526769424004, 44.71365213820241], "error_ppt_range": [0.48717131015149606, 1.2967247560315411], "linear_minus_quadratic_ppt_range": [2.2958844421446525, 7.303035189796588], "background_regressed_minus_original_ppt_range": null}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 2non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "5318912277038701952", "ra": "124.07045043479013", "dec": "-56.27531039067747", "phot_g_mean_mag": "13.533747", "bp_rp": "1.6203833", "sep": "32.66136349363487", "required_fraction_G_approx": 0.38952253593477015, "capable": true}]. Previously verified difference-image receipt: {"tic": "342454977", "sector": 88, "n_events": 12, "deficit_minus_target_arcsec": 0.87, "deficit_minus_light_arcsec": 2.86, "peak_on_target": true, "depth_ap3": 0.03834, "psf_frac_resid": 0.21, "n_neighbours_60": 42, "n_possible": 1, "brightest_nb": {"G": 13.533747, "sep": 32.7}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 342454977_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 5318912277038701696", "main_id": "TYC 8575-1796-1", "otype": "SB*", "ra": "124.06308225819", "dec": "-56.28336922564"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0,

"II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4344.545357BTJD; next opposite-kind(B) center 4345.723850BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0215d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R027. TIC 455214882

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Constrain grazing geometry and component light with resolved multiband photometry.

Archived source: sanitized report text.

TIC 455214882: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target eclipsing system with one observed kind at Gaia A. Six repeat events, independent TESS-SPOC, no capable Gaia neighbor and original deficit-light separation0.28arcsec establish attribution. Covered opposite windows remain null; eccentric grazing geometry provides a feasible missed B eclipse. A narrow eclipse is required beyond smooth tide+five harmonics. Component light/radii remain uncertain.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 2 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 44.937ppt; fitted total duration 6.026h, grid/profile interval [5.89531034 6.156]h; ingress 2.109h and flat 1.808h. The fitted flat in a partial eclipse does not imply physical totality.

P=12.8713558881±7.7e-06d; Tref=2728.3114746±0.00046BTJD at cycle96; T0=1492.6613094. 6timings; O-C RMS0.91min; chi2/dof=1.29/4. Sector depths: {"7": {"n": 2, "median_depth": 0.04494180958609065, "min_depth": 0.04485255459788128, "max_depth": 0.04503106457430002}, "61": {"n": 2, "median_depth": 0.04510688998449283, "min_depth": 0.04502229681003766, "max_depth": 0.045191483158948}, "88": {"n": 2, "median_depth": 0.04468874702633373, "min_depth": 0.04459049192064623, "max_depth": 0.044787002132021234}}.

Secondary conditional fit: phase0.7034939; depth-1.646ppt; formal error0.377ppt; empirical error0.377ppt; 5covered fitted windows, 0individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: `null`.

Largest background-projected fake primary depth0.820ppt, median depth change on background regression0.036ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has8runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source3090276562005552640; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-0.355sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.07488/0.03416d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.7104263954971599, "phase_error": 0.008672709642367452, "gaia_ecos": 0.33373646276258834, "gaia_ecos_error": 0.014599900803751014, "gaia_esin": 0.13947929487728022, "gaia_esin_error": 0.017506443153724648}. Photometric phase diagnostic: null. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.02, "m2_min": 0.701082833765827, "f_mass": 0.11633340890182671, "a_rsun": 27.698913505652023, "r1_rsun": 2.77355, "r2_approx_rsun": 0.7526888224232011, "central_T14_hours": {"A": 10.24216436528471, "B": 13.562409549817229}, "observed_T14_hours": 6.025655172413792}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "insufficient secondary shape"}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Covered null opposite-kind test:45conditional phase/shape tests across5durations and±2sigma Gaia phase covariance; phase sigma0.008626. Pointwise3sigma upper limits range[0.5391442 3.30094099]ppt. Each test preserves its individual windows and exclusions in null-secondary.json; no assertion that uncovered/flagged windows are null. Feasible eccentric missing-eclipse geometry: [{"radius_ratio_assumption": 0.27138101798172054, "b_primary_duration_approx": 1.028075905037991, "b_other": 1.361351563511708, "contact": 1.2713810179817204, "other_misses": true, "es_gaia": 0.13947929487728022, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}]. Smooth equilibrium tide+five harmonics versus same+narrow primary yields deltaBIC=22190.5, RSS[0.5855337356032324, 0.19491540527228002]; sector narrow depths[{'sector': 7, 'eclipse_depth': 0.044767262380909445}, {'sector': 61, 'eclipse_depth': 0.045349840219698866}, {'sector': 88, 'eclipse_depth': 0.044012060953399}]. This is an adversarial smooth-variability test, not calibrated false-alarm probability.

Independence, dilution and source

Own alternate photometry: {"sector": 61, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 2972.867587270667, "qlp": {"1": {"depth": 0.04472275758534232, "error": 0.00033786786941631563, "red_noise_beta": 1.6348762656534035, "n_core": 109, "baseline_degree": 1, "background_regressed_depth": 0.04475955171986736, "background_template_native_sigma": 0.08136082835331833}, "2": {"depth": 0.0449300824174668, "error": 0.00036759776888433906, "red_noise_beta": 1.6107040648000923, "n_core": 109, "baseline_degree": 2, "background_regressed_depth": 0.04492869850017607, "background_template_native_sigma": -0.0030936236788745505}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.043347778825227064, "error": 0.0003115246413444248, "red_noise_beta": 1.7068622536979108, "n_core": 109, "baseline_degree": 1, "background_regressed_depth": 0.043460673623943374, "background_template_native_sigma": 0.37871061111215193}, "2": {"depth": 0.043513587403610124, "error": 0.000340406970122398, "red_noise_beta": 1.687871034353712, "n_core": 109, "baseline_degree": 2, "background_regressed_depth": 0.04362570548260003, "background_template_native_sigma": 0.37627992205288013}}, "PDCSAP_FLUX": {"1": {"depth": 0.04311679384347829, "error": 0.0003482099122235503, "red_noise_beta": 1.8364099060017234, "n_core": 109, "baseline_degree": 1, "background_regressed_depth": 0.04323017105728225, "background_template_native_sigma": 0.37871061111215193}, "2": {"depth": 0.04366438401734131, "error": 0.00034190211280192274, "red_noise_beta": 1.6755861294114183, "n_core": 109, "baseline_degree": 2, "background_regressed_depth": 0.04377686087186763, "background_template_native_sigma": 0.37627992205288013}}}}, {"tc": 2985.7389593338908, "qlp": {"1": {"depth": 0.0453855351961962, "error": 0.0003415336150174341, "red_noise_beta": 1.7547695106389694, "n_core": 108, "baseline_degree": 1, "background_regressed_depth": 0.04542275827117351, "background_template_native_sigma": 0.0973340581209229}, "2": {"depth": 0.04532058553022763, "error": 0.00037780289477398104, "red_noise_beta": 1.7520272953659024, "n_core": 108, "baseline_degree": 2, "background_regressed_depth": 0.045393434533534324, "background_template_native_sigma": 0.19097349122069027}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04399103950034038, "error": 0.00033120003413974537, "red_noise_beta": 1.9192328983876772, "n_core": 108, "baseline_degree": 1, "background_regressed_depth": 0.044122819196991604, "background_template_native_sigma": 0.3054139114420517}, "2": {"depth": 0.043971140305980594, "error": 0.0003669899159486866, "red_noise_beta": 1.9188235217243472, "n_core": 108, "baseline_degree": 2, "background_regressed_depth": 0.0440939699156985, "background_template_native_sigma": 0.2844770963454244}}, "PDCSAP_FLUX": {"1": {"depth": 0.044220961999119054, "error": 0.00035588654914322615, "red_noise_beta": 2.014383927067976, "n_core": 108, "baseline_degree": 1, "background_regressed_depth": 0.04435039051495063, "background_template_native_sigma": 0.3054139114420517}, "2": {"depth": 0.0441628687641613, "error": 0.00039385925146419955, "red_noise_beta": 2.012129207942644, "n_core": 108, "baseline_degree": 2, "background_regressed_depth": 0.04428361840941898, "background_template_native_sigma": 0.2844770963454244}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [44.9300824174668, 45.32058553022763], "error_ppt_range": [0.36759776888433904, 0.37780289477398105], "linear_minus_quadratic_ppt_range": [-0.20732483212448222, 0.0649496659685736], "background_regressed_minus_original_ppt_range": [-0.001383917290730119, 0.07284900330669503]}, "SAP_FLUX": {"depth_ppt_range": [43.513587403610124, 43.9711403059806], "error_ppt_range": [0.340406970122398, 0.3669899159486866], "linear_minus_quadratic_ppt_range":

[-0.16580857838306012, 0.019899194359788985], "background_regressed_minus_original_ppt_range": [0.11211807898990622, 0.12282960971790602]}, "PDCSAP_FLUX": {"depth_ppt_range": [43.66438401734131, 44.162868764161296], "error_ppt_range": [0.34190211280192273, 0.39385925146419953], "linear_minus_quadratic_ppt_range": [-0.5475901738630218, 0.058093234957756756], "background_regressed_minus_original_ppt_range": [0.112476854526318, 0.12074964525768395]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 0non-target sources, 0capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: `[]`. Previously verified difference-image receipt: {"tic": "455214882", "sector": 7, "n_events": 2, "deficit_minus_target_arcsec": 2.16, "deficit_minus_light_arcsec": 0.28, "peak_on_target": true, "depth_ap3": 0.04301, "psf_frac_resid": 0.06, "n_neighbours_60": 9, "n_possible": 0, "brightest_nb": {"G": 16.120275, "sep": 51.3}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 455214882_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 3090276562005552640", "main_id": "TYC 200-238-1", "otype": "SB*", "ra": "125.53991436827998", "dec": "2.57824779448"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4350.102317BTJD; next opposite-kind(B) center 4346.375112BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.1116d before future-period

uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Constrain grazing geometry and component light with resolved multiband photometry.

R028. TIC 140477577

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 140477577: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB. Nine primary and seven secondary timings, independent TESS-SPOC, no capable neighbor and original deficit-target0.13arcsec support attribution. Both kinds agree with a shared period and Gaia conjunctions; radius ratio/impact parameter remain degenerate.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 4 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 44.301ppt; fitted total duration 4.365h, grid/profile interval [4.2777931 4.45158621]h; ingress 1.528h and flat 1.309h. The fitted flat in a partial eclipse does not imply physical totality.

P=11.0432740534±2.9e-06d; Tref=3014.2181020±0.00026BTJD at cycle147; T0=1390.8568162. 9timings; O-C RMS0.93min; chi2/dof=9.46/7. Sector depths: {"3": {"n": 2, "median_depth": 0.04878636791549222, "min_depth": 0.04869701249331665, "max_depth": 0.048875723337667785}, "30": {"n": 2, "median_depth": 0.04947297304562644, "min_depth": 0.04916642816382513, "max_depth": 0.04977951792742774}, "97": {"n": 4, "median_depth": 0.043785832520039744, "min_depth": 0.04284172132286076, "max_depth": 0.04430110046488444}, "104": {"n": 1, "median_depth": 0.04327859843446422, "min_depth": 0.04327859843446422, "max_depth": 0.04327859843446422}}.

Secondary conditional fit: phase0.5031951; depth10.356ppt; formal error0.091ppt; empirical error0.264ppt; 7covered fitted windows, 7individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 11.043276710835402,

"P_error": 1.4709084993852531e-05, "Tref": 3395.2474023085065, "Tref_error": 0.0009906694640639511, "reference_cycle": 181, "T0": 1396.4143176472987, "chi2": 40.6874773007216, "dof": 5, "n": 7, "oc_rms_minutes": 2.0875026333590907}.

Largest background-projected fake primary depth0.195ppt, median depth change on background regression0.013ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has22runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source5016430031163524480; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-0.774sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: -0.00366/0.04855d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.505934892982184, "phase_error": 0.00929364563421681, "gaia_ecos": 0.00931909650661067, "gaia_ecos_error": 0.014596851983612378, "gaia_esin": 0.027581142884742033, "gaia_esin_error": 0.01044839560262718, "phase_difference_sigma": -0.2853524759591847, "measured_phase": 0.5032827626267415, "measured_phase_error": 0.00010384980672758288}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.5032827626267415, "phase_error_conservative": 0.00010384980672758288, "ecos_first_order": 0.00515655147582518, "ecos_first_order_error": 0.00016312689494604713, "gaia_ecos": 0.00931909650661067, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.06, "m2_min": 0.577714335572155, "f_mass": 0.07188911413177981, "a_rsun": 24.599752243640452, "r1_rsun": 1.49259, "r2_approx_rsun": 0.6447190688249793, "central_T14_hours": {"A": 7.130469160119484, "B": 7.534958413493129}, "observed_T14_hours": 4.364689655172413}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 169.22970074359574, "dof": 121, "k": 0.35, "b_A": 0.97, "e_sin_omega": 0.02499999999999994, "primary_scale": 0.8280405272155779, "secondary_scale": 0.023093972364664844}, "profile_interval_approx_95": {"k": [0.2999999999999993, 0.3999999999999999], "b_A": [0.7950000000000002, 1.04], "e_sin_omega": [0.02499999999999994, 0.1499999999999999]}, "error_inflation": 1.3985925681288904, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 30, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 2119.712954084695, "qlp": {"1": {"depth": 0.04917593353118339, "error": 0.0004608191122473068, "red_noise_beta": 1.4624381360662972, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.0492194825418715, "background_template_native_sigma": 0.28074630154737057}, "2": {"depth": 0.04922726425236863, "error": 0.0005114626811656413, "red_noise_beta": 1.4628317296916038, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04932826529720101, "background_template_native_sigma": 0.6045086394104411}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04844354088907827, "error": 0.0003715987053600328, "red_noise_beta": 1.2915018664175486, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.04856001412705258, "background_template_native_sigma": 0.5126688911432278}, "2": {"depth": 0.048436755170796486, "error": 0.00041101145706423246, "red_noise_beta": 1.2912541167676508, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.0486111026482837, "background_template_native_sigma": 0.7534597128189063}}, "PDCSAP_FLUX": {"1": {"depth": 0.04855170140906241, "error": 0.00039063885413288455, "red_noise_beta": 1.3282880791131129, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.04867416410656726, "background_template_native_sigma": 0.5126688911432278}, "2": {"depth": 0.04853314070073548, "error": 0.0004319221885275096, "red_noise_beta": 1.3276441273743138, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04871569811484246, "background_template_native_sigma": 0.7534597128189063}}}}, {"tc": 2130.7562123461666, "qlp": {"1": {"depth": 0.0493239118199997, "error": 0.0004487916796732662, "red_noise_beta": 1.4409399791922257, "n_core": 23, "baseline_degree": 1, "background_regressed_depth": 0.049773195183342146, "background_template_native_sigma": 0.5458639017392374}, "2": {"depth": 0.04961712651041612, "error": 0.00048724187144648894, "red_noise_beta": 1.4217207113569177, "n_core": 23, "baseline_degree": 2, "background_regressed_depth": 0.0497128772880779, "background_template_native_sigma": 0.10090394617865256}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04860752010047281, "error": 0.00040051191980332903, "red_noise_beta": 1.4470057598318817, "n_core": 26, "baseline_degree": 1, "background_regressed_depth": 0.04892737148723741, "background_template_native_sigma": 0.7473364442759319}, "2": {"depth": 0.04883077878324373, "error": 0.000435571000123666, "red_noise_beta": 1.429589587074336, "n_core": 26, "baseline_degree": 2, "background_regressed_depth": 0.048938504178808934, "background_template_native_sigma": 0.28168240148515766}}, "PDCSAP_FLUX": {"1": {"depth": 0.04679182126464343, "error": 0.0010207882738537076, "red_noise_beta": 2.0914332239671185, "n_core": 26, "baseline_degree": 1, "background_regressed_depth": 0.048643251507153866, "background_template_native_sigma": 0.7473364442759319}, "2": {"depth": 0.04891849181615109, "error": 0.0006023772996370941, "red_noise_beta": 1.6503011229834124, "n_core": 26, "baseline_degree": 2, "background_regressed_depth": 0.049095298785625005, "background_template_native_sigma": 0.28168240148515766}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [49.227264252368634, 49.617126510416114], "error_ppt_range": [0.48724187144648895, 0.5114626811656413], "linear_minus_quadratic_ppt_range": [-0.29321469041641846, -0.051330721185245676], "background_regressed_minus_original_ppt_range": [0.0957507776617808, 0.10100104483237726]}, "SAP_FLUX": {"depth_ppt_range": [48.43675517079649, 48.83077878324373], "error_ppt_range": [0.4110114570642325, 0.435571000123666], "linear_minus_quadratic_ppt_range": [-0.22325868277092015, 0.006785718281783482],

"background_regressed_minus_original_ppt_range": [0.10772539556520161, 0.17434747748721474]}, "PDCSAP_FLUX": {"depth_ppt_range": [48.53314070073548, 48.91849181615109], "error_ppt_range": [0.4319221885275096, 0.6023772996370941], "linear_minus_quadratic_ppt_range": [-2.126670551507663, 0.01856070832693263], "background_regressed_minus_original_ppt_range": [0.17680696947391245, 0.18255741410697773]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 0non-target sources, 0capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: []. Previously verified difference-image receipt: {"tic": "140477577", "sector": 97, "n_events": 4, "deficit_minus_target_arcsec": 0.13, "deficit_minus_light_arcsec": 0.5, "peak_on_target": true, "depth_ap3": 0.04912, "psf_frac_resid": 0.024, "n_neighbours_60": 2, "n_possible": 0, "brightest_nb": {"G": 16.800104, "sep": 45.5}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 140477577_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 5016430031163524480", "main_id": "TYC 7002-124-1", "otype": "SB*", "ra": "20.969605739340004", "dec": "-31.84203159695"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4350.454262BTJD; next opposite-kind(B) center 4344.969199BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.1026d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R029. TIC 190972398

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Obtain joint RV/eclipse and stellar-radius constraints for characterization.

Archived source: sanitized report text.

TIC 190972398: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED eccentric target EB. Five primary and four secondary timings establish the strongly non-half-cycle secondary, phase0.71817, agreeing with Gaia exact eccentric timing. A first-order e cos omega approximation is inadequate at e~0.35. Original deficit-light0.33arcsec and no capable neighbors support source attribution. The simplified chord geometry has residual tension and is only a feasibility exploration.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 1 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 44.664ppt; fitted total duration 7.068h, grid/profile interval [6.93806897 7.19875862]h; ingress 2.474h and flat 2.121h. The fitted flat in a partial eclipse does not imply physical totality.

P=13.1298707269±1.5e-05d; Tref=3156.5989263±0.0004BTJD at cycle105; T0=1777.9625000. 5timings; O-C RMS0.00min; chi2/dof=0.00/3. Sector depths: {"57": {"n": 3, "median_depth": 0.044664036712894865, "min_depth": 0.04463544722800895, "max_depth": 0.04474620553805678}, "84": {"n": 2, "median_depth": 0.04435486921943038, "min_depth": 0.044026010481784464, "max_depth": 0.04468372795707631}}.

Secondary conditional fit: phase0.7181502; depth13.416ppt; formal error0.078ppt; empirical error0.099ppt; 4covered fitted windows, 4individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 13.129889431857293,

"P_error": 1.1512647120611785e-05, "Tref": 2969.0801138641796, "Tref_error": 0.0005304224112872449, "reference_cycle": 90, "T0": 1787.3900649970233, "chi2": 1.9790564052829005, "dof": 2, "n": 4, "oc_rms_minutes": 0.897054488254963}.

Largest background-projected fake primary depth0.067ppt, median depth change on background regression0.019ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has10runs and0deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source3889299431538245l2; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-1.023sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.00386/0.02304d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.7185988332582892, "phase_error": 0.0031203412919599714, "gaia_ecos": 0.35045629080467156, "gaia_ecos_error": 0.0052647447820774705, "gaia_esin": -0.03742570786405258, "gaia_esin_error": 0.008208742369371705, "phase_difference_sigma": -0.136654603545318, "measured_phase": 0.7181723653906893, "measured_phase_error": 5.185003203201545e-05}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.7181723653906893, "phase_error_conservative": 5.185003203201545e-05, "ecos_first_order": 0.34270435016384876, "ecos_first_order_error": 8.14458398600876e-05, "gaia_ecos": 0.35045629080467156, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.18, "m2_min": 0.5788852178840147, "f_mass": 0.0627050302021314, "a_rsun": 28.27319889373608, "r1_rsun": 2.70215, "r2_approx_rsun": 0.6457642044346009, "central_T14_hours": {"A": 11.548111296335994, "B": 10.714902254990603}, "observed_T14_hours": 7.068413793103448}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 445.75325340387667, "dof": 121, "k": 0.27499999999999997, "b_A": 0.9000000000000001, "e_sin_omega": 0.04999999999999993, "primary_scale": 0.960517424620003, "secondary_scale": 0.029968610781151688}, "profile_interval_approx_95": {"k": [0.24999999999999997, 0.27499999999999997], "b_A": [0.865, 0.9000000000000001], "e_sin_omega": [0.04999999999999993, 0.04999999999999993]}, "error_inflation": 3.6839111851560054, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 57, "provenance": "TESS-SPOC", "fits_identity_validated": true, "quality_column_present": true, "events": [{"tc": 2854.611899602175, "qlp": {"1": {"depth": 0.04472477157482349, "error": 0.00024470914021991095, "red_noise_beta": 1.9566636201212837, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.0447475194989279, "background_template_native_sigma": 0.2464968389225974}, "2": {"depth": 0.04468215753285637, "error": 0.000270868012296406, "red_noise_beta": 1.955172055115573, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04470094709163349, "background_template_native_sigma": 0.20204455023202858}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04406585330801195, "error": 0.00023293675375396118, "red_noise_beta": 1.948213535717528, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.04426997449571573, "background_template_native_sigma": 1.1949054927083167}, "2": {"depth": 0.04398752801881483, "error": 0.0002569685844799612, "red_noise_beta": 1.9419396388952073, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.044144502789760065, "background_template_native_sigma": 0.8127760664176484}}, "PDCSAP_FLUX": {"1": {"depth": 0.04408061291597245, "error": 0.00024059600816555684, "red_noise_beta": 2.008507988848922, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.04429021952861417, "background_template_native_sigma": 1.1949054927083167}, "2": {"depth": 0.0441796506176252, "error": 0.0002641285923689293, "red_noise_beta": 1.9937925285119626, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04431818422901884, "background_template_native_sigma": 0.8127760664176484}}}}, {"tc": 2867.7417703290303, "qlp": {"1": {"depth": 0.04476317614582649, "error": 0.00037604538268158085, "red_noise_beta": 2.1727075869554846, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.044757354658069744, "background_template_native_sigma": -0.02964061075471291}, "2": {"depth": 0.04476137568145427, "error": 0.00037659247897467267, "red_noise_beta": 2.1721699106416494, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04475516399567403, "background_template_native_sigma": -0.03158866721973025}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04383619038397499, "error": 0.00027738167212346533, "red_noise_beta": 2.037169008263014, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.04384911046883224, "background_template_native_sigma": 0.6562158513973241}, "2": {"depth": 0.04406930207355989, "error": 0.0003062994503812836, "red_noise_beta": 1.997190708046558, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.044092809781801384, "background_template_native_sigma": -0.17321244855431847}}, "PDCSAP_FLUX": {"1": {"depth": 0.04221037031990928, "error": 0.0006510548264222183, "red_noise_beta": 3.334655831848257, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.04265160459288104, "background_template_native_sigma": 0.6562158513973241}, "2": {"depth": 0.04221259584320054, "error": 0.0007390016313661323, "red_noise_beta": 3.334548932694045, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04202715498298544, "background_template_native_sigma": -0.17321244855431847}}}}, {"tc": 2880.8716410558864, "qlp": {"1": {"depth": 0.04488701549058649, "error": 0.00023924939229221968, "red_noise_beta": 1.943168662074237, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.0447612555691215, "background_template_native_sigma": -0.6719751561259436}, "2": {"depth": 0.04467819674843978, "error": 0.0002556857742454408, "red_noise_beta": 1.8899095082080999, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.044681547889662714, "background_template_native_sigma": 0.021225732001359573}}, "independent": {"SAP_FLUX": {"1": {"depth": 0.04423978719211993, "error": 0.00022800094951337668, "red_noise_beta": 1.9225213239571137, "n_core": 127, "baseline_degree": 1, "background_regressed_depth":

0.04404559491448965, "background_template_native_sigma": -0.8224734938043332}, "2": {"depth": 0.04406464036651982, "error": 0.0002458789197571634, "red_noise_beta": 1.8829534980062153, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04407397278159663, "background_template_native_sigma": 0.030968469128463346}}, "PDCSAP_FLUX": {"1": {"depth": 0.044220926132216086, "error": 0.00023733216942312468, "red_noise_beta": 2.004633531778997, "n_core": 127, "baseline_degree": 1, "background_regressed_depth": 0.043983855404954135, "background_template_native_sigma": -0.8224734938043332}, "2": {"depth": 0.04402532535076174, "error": 0.0002552084460963235, "red_noise_beta": 1.9609192643678304, "n_core": 127, "baseline_degree": 2, "background_regressed_depth": 0.04403811506207405, "background_template_native_sigma": 0.030968469128463346}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [44.67819674843978, 44.76137568145427], "error_ppt_range": [0.25568577424544076, 0.3765924789746727], "linear_minus_quadratic_ppt_range": [0.0018004643722244196, 0.20881874214671164], "background_regressed_minus_original_ppt_range": [-0.006211685780237175, 0.01878955877712002]}, "SAP_FLUX": {"depth_ppt_range": [43.98752801881483, 44.06930207355989], "error_ppt_range": [0.2458789197571634, 0.3062994503812836], "linear_minus_quadratic_ppt_range": [-0.2331116895849042, 0.17514682560011013], "background_regressed_minus_original_ppt_range": [0.009332415076813982, 0.15697477094523543]}, "PDCSAP_FLUX": {"depth_ppt_range": [42.212595843200546, 44.1796506176252], "error_ppt_range": [0.2552084460963235, 0.7390016313661323], "linear_minus_quadratic_ppt_range": [-0.099037701652753, 0.19560078145434284], "background_regressed_minus_original_ppt_range": [-0.18544086021510253, 0.1385336113936403]}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 0non-target sources, 0capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: `[]`. Previously verified difference-image receipt: {"tic": "190972398", "sector": 57, "n_events": 3, "deficit_minus_target_arcsec": 0.81, "deficit_minus_light_arcsec": 0.33, "peak_on_target": true, "depth_ap3": 0.04432, "psf_frac_resid": 0.021, "n_neighbours_60": 13, "n_possible": 0, "brightest_nb": {"G": 16.021408, "sep": 48.6}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 190972398_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 3889299431538245 12", "main_id": "TYC 3248-1414-1", "otype": "SB*", "ra": "6.689702405700001", "dec": "45.550125854140006"}], "gaia_vari_summary": [{"source_id":

"388929943153824512", "abbe_mag_bp": "0.6782814", "abbe_mag_g_fov": "0.7166718", "abbe_mag_rp": "0.64446557", "in_vari_agn": "0", "in_vari_cepheid": "0", "in_vari_classification_result": "0", "in_vari_compact_companion": "0", "in_vari_eclipsing_binary": "0", "in_vari_long_period_variable": "0", "in_vari_microlensing": "0", "in_vari_ms_oscillator": "0", "in_vari_planetary_transit": "0", "in_vari_rotation_modulation": "0", "in_vari_rrlyrae": "0", "in_vari_short_timescale": "0", "iqr_mag_bp": "0.0024472817", "iqr_mag_g_fov": "0.0021513263", "iqr_mag_rp": "0.001970497", "kurtosis_mag_bp": "41.715927", "kurtosis_mag_g_fov": "59.799152", "kurtosis_mag_rp": "47.664467", "mad_mag_bp": "0.0018564602", "mad_mag_g_fov": "0.0015762588", "mad_mag_rp": "0.001563861", "max_mag_bp": "10.631521", "max_mag_g_fov": "10.353991", "max_mag_rp": "9.907961", "mean_mag_bp": "10.58335", "mean_mag_g_fov": "10.30274", "mean_mag_rp": "9.856969", "mean_obs_time_bp": "2256.4103581373292", "mean_obs_time_g_fov": "2247.7512437548044", "mean_obs_time_rp": "2240.420671553533", "median_mag_bp": "10.582317", "median_mag_g_fov": "10.302015", "median_mag_rp": "9.8559885", "min_mag_bp": "10.577114", "min_mag_g_fov": "10.297248", "min_mag_rp": "9.852898", "num_selected_bp": "86", "num_selected_g_fov": "83", "num_selected_rp": "76", "outlier_median_bp": "54.560253", "outlier_median_g_fov": "110.87519", "outlier_median_rp": "61.692944", "range_mag_bp": "0.054407105", "range_mag_g_fov": "0.056742553", "range_mag_rp": "0.05506278", "skewness_mag_bp": "5.866806", "skewness_mag_g_fov": "7.269978", "skewness_mag_rp": "6.5774217", "solution_id": "375316653866487564", "std_dev_mag_bp": "0.0062958035", "std_dev_mag_g_fov": "0.0061687077", "std_dev_mag_rp": "0.006683394", "std_dev_over_rms_err_mag_bp": "5.515204", "std_dev_over_rms_err_mag_g_fov": "5.494393", "std_dev_over_rms_err_mag_rp": "8.2557535", "stetson_mag_bp": "2.2895546", "stetson_mag_g_fov": "2.7428415", "stetson_mag_rp": "2.4086356", "time_duration_bp": "999.04816", "time_duration_g_fov": "999.04816", "time_duration_rp": "998.872", "trimmed_range_mag_bp": "0.010094259", "trimmed_range_mag_g_fov": "0.006335222", "trimmed_range_mag_rp": "0.006643199"}], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4351.417162BTJD; next opposite-kind(B) center 4347.718504BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0410d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Obtain joint RV/eclipse and stellar-radius constraints for characterization.

R030. TIC 349273287

physical_verdict: CONFIRMED

novelty_status: limited: no target prior EB match in fresh nine-catalog cones; not exhaustive bibliography

completion: all applicable gates attempted/adjudicated; limitations retained

next_check: Calibrate a multi-source empirical PSF and obtain less diluted eclipse photometry.

Archived source: sanitized report text.

TIC 349273287: CONFIRMED

Full review completed; all applicable gates attempted/adjudicated, with limitations below. Reviewed2026-10-30 by Sol lead with fresh Terra independent photometry/localization helper. Original candidate status and absence of original vet directory checked before work; original files unchanged.

CONFIRMED target EB.147primary and127 individually significant secondary timings plus TARS establish recurrence. Own sector97 pixel regression favors target over the flagged27.85arcsec neighbor after including a bright source outside the60arcsec Gaia cone. Moving that empirical nuisance source±0.2pixel keeps target preferred; exact centroid errors are not calibrated. Blind-census extra features in sectors62/65/69 are broad quality-contaminated ramps, not repeat eclipse-shaped events. Sector65 feature coincides with the known3082.8 common-mode interval and its fitted depth drops35.6 to16.5ppt on background regression. These features limit baseline purity, not the well-localized5.54727d eclipses.

Raw SAP measurements and timing

QUALITY==0 raw SAP, no post-gap trimming or detrended QLP flux. All flagged cadences inspected numerically; 37 fitted event cores contain some flagged points, excluded from fits. Exposure-integrated trapezoids with local quadratic baselines, residual-bin red-noise inflation and grid floors; timing errors and chi2-scaled ephemeris retained. No primary timing scan reaches an edge. Individual mid-times, errors, depths/errors, total/ingress/flat durations, O-C, quality, background and centroid statistics are in event-measurements.csv, including explicit insufficient-coverage windows. Shared-shape duration uncertainties are conditional; individual CSV durations are not separately fitted shape posteriors.

Primary median depth 44.772ppt; fitted total duration 4.191h, grid/profile interval [4.104 4.2777931]h; ingress 1.048h and flat 2.095h. The fitted flat in a partial eclipse does not imply physical totality.

P=5.5472713261±5.9e-07d; Tref=2595.8318055±8e-05BTJD at cycle228; T0=1331.0539431. 147timings; O-C RMS1.41min; chi2/dof=311.79/145. Sector depths: {"1": {"n": 2, "median_depth": 0.045201415837557844, "min_depth": 0.044525502363387884, "max_depth": 0.045877329311727805}, "2": {"n": 3, "median_depth": 0.04432193533672404, "min_depth": 0.04363387262618118, "max_depth": 0.04436718211394969}, "4": {"n": 2, "median_depth": 0.04423164754326235, "min_depth": 0.04246336667018296, "max_depth": 0.04599992841634175}, "5": {"n": 4, "median_depth": 0.044120763290096995, "min_depth": 0.042840963804182605, "max_depth": 0.04511042672757787}, "6": {"n": 3, "median_depth": 0.04390775126834872, "min_depth": 0.04375740547621887, "max_depth": 0.044067958941599705}, "7": {"n": 3, "median_depth": 0.046293527108933516, "min_depth":

0.04477193674383739, "max_depth": 0.04672566885586524}, "8": {"n": 1, "median_depth": 0.04404677757273511, "min_depth": 0.04404677757273511, "max_depth": 0.04404677757273511}, "9": {"n": 3, "median_depth": 0.044433751030575216, "min_depth": 0.04281272505234333, "max_depth": 0.04518850594515574}, "10": {"n": 4, "median_depth": 0.04457531884055077, "min_depth": 0.04354492585971543, "max_depth": 0.04515885336175857}, "11": {"n": 2, "median_depth": 0.04341466804471329, "min_depth": 0.043383706074067274, "max_depth": 0.04344563001535931}, "12": {"n": 4, "median_depth": 0.04475160679635512, "min_depth": 0.04206076599855942, "max_depth": 0.046156055576636726}, "13": {"n": 5, "median_depth": 0.043821464589111556, "min_depth": 0.04165391975472044, "max_depth": 0.04495609211099319}, "27": {"n": 4, "median_depth": 0.04551174523151919, "min_depth": 0.04361466393814126, "max_depth": 0.0472952035986212}, "28": {"n": 3, "median_depth": 0.044878841819729905, "min_depth": 0.044126638011195755, "max_depth": 0.04670582449621045}, "30": {"n": 3, "median_depth": 0.04675282673138345, "min_depth": 0.04653472894465007, "max_depth": 0.04783016043190327}, "31": {"n": 4, "median_depth": 0.04598843782669591, "min_depth": 0.04465262053245084, "max_depth": 0.04715691298768504}, "32": {"n": 3, "median_depth": 0.04362480777084899, "min_depth": 0.04283734825318054, "max_depth": 0.04497865952041755}, "33": {"n": 5, "median_depth": 0.04675781026596509, "min_depth": 0.045619121852144354, "max_depth": 0.047367781581457075}, "34": {"n": 4, "median_depth": 0.0446757474976507, "min_depth": 0.04432242860662969, "max_depth": 0.046371567195609295}, "35": {"n": 4, "median_depth": 0.04450669938523276, "min_depth": 0.04369048752986498, "max_depth": 0.047440066838419816}, "36": {"n": 4, "median_depth": 0.046378872988327324, "min_depth": 0.04407195152343948, "max_depth": 0.047804191210053575}, "37": {"n": 4, "median_depth": 0.04548166014639017, "min_depth": 0.04449993968848613, "max_depth": 0.04930836608028599}, "38": {"n": 3, "median_depth": 0.04473292900854669, "min_depth": 0.04407233423084851, "max_depth": 0.04609206926913425}, "39": {"n": 4, "median_depth": 0.04557678114326344, "min_depth": 0.04340968246494515, "max_depth": 0.046715712310497336}, "61": {"n": 4, "median_depth": 0.04603507786041587, "min_depth": 0.04471929715311263, "max_depth": 0.0463153213216558}, "62": {"n": 4, "median_depth": 0.04452554791465504, "min_depth": 0.04203483169576238, "max_depth": 0.04555293248263755}, "63": {"n": 5, "median_depth": 0.0450711073958525, "min_depth": 0.04295811866767346, "max_depth": 0.04656510046695577}, "64": {"n": 5, "median_depth": 0.044638545362496417, "min_depth": 0.043636217833534455, "max_depth": 0.04555752722193714}, "65": {"n": 5, "median_depth": 0.04659558200823676, "min_depth": 0.04498023188566051, "max_depth": 0.047715428345687}, "67": {"n": 4, "median_depth": 0.04545685939756251, "min_depth": 0.044317754118721145, "max_depth": 0.04771194812613899}, "68": {"n": 4, "median_depth": 0.04677635209992348, "min_depth": 0.044838095447029026, "max_depth": 0.047484790372071116}, "69": {"n": 3, "median_depth": 0.047507321843370136, "min_depth": 0.04705011000786664, "max_depth": 0.04819852901228976}, "87": {"n": 5, "median_depth": 0.043618694801225835, "min_depth": 0.04281226431914129, "max_depth": 0.04469167662977288}, "88": {"n": 5, "median_depth": 0.0456914918731509, "min_depth": 0.0452232412672358, "max_depth": 0.04683201195864464}, "89": {"n": 3, "median_depth": 0.04649716305981128, "min_depth": 0.043756353326188935, "max_depth": 0.046685567983043}, "90": {"n": 4, "median_depth": 0.04508337228311389, "min_depth": 0.04429389281767151, "max_depth": 0.04545869857587415}, "94": {"n": 4, "median_depth": 0.04056112685780164, "min_depth": 0.039805469542809745, "max_depth": 0.04234056333244035}, "95": {"n": 2, "median_depth": 0.040296791714182024, "min_depth": 0.040234783330780594, "max_depth": 0.040358800097583455}, "96": {"n": 3, "median_depth": 0.03863211616420122, "min_depth": 0.0385379032763762, "max_depth": 0.0393360539936664}, "97": {"n": 6, "median_depth": 0.03901370030004328, "min_depth": 0.03831466475820937, "max_depth": 0.04007699034920062}}.

Secondary conditional fit: phase0.4998008; depth2.427ppt; formal error0.036ppt; empirical error0.080ppt; 155covered fitted windows, 127individual >3sigma detections. Phase optimization is not a blind significance claim; repeat event detections, fixed Gaia/control tests and full coverage exclusions remain in secondary-fit.json/orbit-gates.json. Secondary ephemeris: {"P": 5.547266676984206, "P_error": 5.2504965575836e-06, "Tref": 2620.794451612628, "Tref_error": 0.0008839801786537521, "reference_cycle": 232, "T0": 1333.8285825522923, "chi2": 382.237416641922, "dof": 125, "n": 127, "oc_rms_minutes": 18.495997890214607}.

Largest background-projected fake primary depth1.186ppt, median depth change on background regression0.064ppt. Native SAP_BKG is never divided by normalized flux. Blind rolling-median census has233runs and4deep off-eclipse-phase runs; flags/gaps and broad ramps are retained rather than counted as astrophysical eclipses. See review-diagnostics.json and any offphase-controls/context products.

Gaia covariance, geometry and physics

Exact source5285787668183123968; SB1, bit_index127,15lower-triangle correlations verified independently for this source. Covariance order P,gamma,K1,e,omega,Tp, with angular units converted. Own20000/40000draw checks; no copied correlation. Primary matchesA; delta P=-0.335sigma. Conjunction comparisons at every event and Gaia epoch389 are in orbit-gates.json. Epoch389(primary) offset/error: 0.02986/0.04763d.

Exact Gaia relative-phase and e-components: {"predicted_phase": 0.5191116735433868, "phase_error": 0.01584202633564283, "gaia_ecos": 0.030015284699584602, "gaia_ecos_error": 0.02488537103741595, "gaia_esin": -0.025513461422614292, "gaia_esin_error": 0.0259211186496339, "phase_difference_sigma": -1.2069690053729387, "measured_phase": 0.49998986301270915, "measured_phase_error": 0.00016004777087373577}. Photometric phase diagnostic: {"phase_at_primary_Tref": 0.49998986301270915, "phase_error_conservative": 0.00016004777087373577, "ecos_first_order": -1.5923142401240176e-05, "ecos_first_order_error": 0.00025140245060017535, "gaia_ecos": 0.030015284699584602, "warning": "First order relation inaccurate at large e; phase itself is measured and compared with exact Gaia prediction. Per-kind ephemeris covariance absent, so phase uncertainty conservatively adds period extrapolation in quadrature."}. When e is small, timing can refine a biased small-e RV solution; phase matching and source localization establish system identity. The first-order e cos omega diagnostic is explicitly not used for precise high-e inference.

RV mass function and conditional dimensions/central durations: [{"m1": 1.11, "m2_min": 0.24313196265853793, "f_mass": 0.007849569918631536, "a_rsun": 14.586308377798185, "r1_rsun": 1.26441, "r2_approx_rsun": 0.32260689462371345, "central_T14_hours": {"A": 4.727982048124764, "B": 4.4927297727927415}, "observed_T14_hours": 4.190896551724138}]. Minimum companion masses are stellar under the listed M1 assumptions. Companion radius uses an explicit main-sequence approximation, not a measurement. Limb-darkened two-disk shape exploration: {"status": "conditional shape feasibility", "best": {"chi2": 166.1077614355591, "dof": 121, "k": 0.22499999999999998, "b_A": 0.375, "e_sin_omega": -5.551115123125783e-17, "primary_scale": 0.7977060624718236, "secondary_scale": 0.002790645856685999}, "profile_interval_approx_95": {"k": [0.22499999999999998, 0.22499999999999998], "b_A": [0.30500000000000005, 0.375], "e_sin_omega": [-5.551115123125783e-17, 0.02499999999999994]}, "error_inflation": 1.3727914168228024, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64,

"secondary_profile_n": 64}. Its coarse fixed-TIC dimensions, depth/dilution freedom and simplified chord speeds limit parameter inference; profile ranges are not calibrated uncertainties. In the standard primary-RV convention A has the companion in front and B has the primary in front; observed depth alone cannot identify which component is hotter, and unresolved multiplicity/stellar dimensions remain characterization caveats.

Independence, dilution and source

Own alternate photometry: {"sector": 33, "provenance": "TARS", "fits_identity_validated": true, "quality_column_present": false, "events": [{"tc": 2201.975998077601, "qlp": {"1": {"depth": 0.04729013193696316, "error": 0.0008621890274496433, "red_noise_beta": 1.4127299474073003, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.04740964897773502, "background_template_native_sigma": 0.13666307075962703}, "2": {"depth": 0.04706641570428088, "error": 0.000814702988477017, "red_noise_beta": 1.3517420129567677, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04718830360607497, "background_template_native_sigma": 0.14103248458707343}}, "independent": {"FLUX": {"1": {"depth": 0.043741551330053544, "error": 0.0008351183395324834, "red_noise_beta": 1.4724786908978447, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.044346571446342624, "error": 0.0007595667763018409, "red_noise_beta": 1.3298048655138357, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 2207.52326558128, "qlp": {"1": {"depth": 0.0465072297818301, "error": 0.0007629092181100671, "red_noise_beta": 1.4946481140024308, "n_core": 24, "baseline_degree": 1, "background_regressed_depth": 0.04649931209698839, "background_template_native_sigma": -0.014135901213862288}, "2": {"depth": 0.047503838982918656, "error": 0.0007427179482921763, "red_noise_beta": 1.3810287523590508, "n_core": 24, "baseline_degree": 2, "background_regressed_depth": 0.0476655549542058, "background_template_native_sigma": 0.22404713252691588}}, "independent": {"FLUX": {"1": {"depth": 0.041756947546444476, "error": 0.0007939868425022602, "red_noise_beta": 1.5041948806891439, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.04231666601406693, "error": 0.0008531525993219918, "red_noise_beta": 1.4602211443598176, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 2213.070533084959, "qlp": {"1": {"depth": 0.046386648626237946, "error": 0.0006615760388371808, "red_noise_beta": 1.3299096957330547, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.04717734235696923, "background_template_native_sigma": 0.2708129218935127}, "2": {"depth": 0.04661605183286252, "error": 0.0007314859262809935, "red_noise_beta": 1.3164114238018094, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04690865433563481, "background_template_native_sigma": 0.07088164512201353}}, "independent": {"FLUX": {"1": {"depth": 0.03922640512874606, "error": 0.0006684108189041511, "red_noise_beta": 1.3144493188409476, "n_core": 26, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03909524046672762, "error": 0.0007495979540979332, "red_noise_beta": 1.3157823799186406, "n_core": 26, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 2218.6167528645, "qlp": {"1": {"depth": 0.045224819594130446, "error": 0.0005346695973587831, "red_noise_beta": 1.2110535114747183, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.045222197149061595, "background_template_native_sigma": -0.004970110450619194}, "2": {"depth": 0.045736153575350325, "error": 0.0005774900812944937,

"red_noise_beta": 1.1896846247508113, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04577499596668371, "background_template_native_sigma": 0.07122377378233516}}, "independent": {"FLUX": {"1": {"depth": 0.04019599874330395, "error": 0.0006829672189346332, "red_noise_beta": 1.293831000902702, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03984814749446423, "error": 0.0007500853575825128, "red_noise_beta": 1.2760376683516323, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}, {"tc": 2224.165068092317, "qlp": {"1": {"depth": 0.04620877482008761, "error": 0.000564908494629953, "red_noise_beta": 1.296460788263264, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": 0.04637724573815028, "background_template_native_sigma": 0.4366870850847495}, "2": {"depth": 0.045861382979112054, "error": 0.0006207897230102404, "red_noise_beta": 1.2836859940728105, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": 0.04608749814129999, "background_template_native_sigma": 0.6218088041589283}}, "independent": {"FLUX": {"1": {"depth": 0.03956006431870327, "error": 0.0005583898802662342, "red_noise_beta": 1.2772817755753143, "n_core": 25, "baseline_degree": 1, "background_regressed_depth": null, "background_template_native_sigma": null}, "2": {"depth": 0.03946276400587444, "error": 0.0006256405684335679, "red_noise_beta": 1.2783453589545202, "n_core": 25, "baseline_degree": 2, "background_regressed_depth": null, "background_template_native_sigma": null}}}}], "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified.", "measurement_ranges": {"qlp": {"depth_ppt_range": [45.73615357535032, 47.50383898291866], "error_ppt_range": [0.5774900812944936, 0.8147029884770169], "linear_minus_quadratic_ppt_range": [-0.996609201088558, 0.34739184097555553], "background_regressed_minus_original_ppt_range": [0.03884239133338657, 0.292602502772292]}, "FLUX": {"depth_ppt_range": [39.09524046672762, 44.34657144634262], "error_ppt_range": [0.6256405684335679, 0.8531525993219918], "linear_minus_quadratic_ppt_range": [-0.6050201162890803, 0.34785124883972285], "background_regressed_minus_original_ppt_range": null}}}. Detailed individual baseline/error/quality/product identity and hashes are in independent-summary.json and independent-manifest.json. These are independently reduced TESS pixels, not independent observatory data. Independent products use their own available quality cuts; TARS has no quality column.

GaiaG<15 neighbor cone: 1non-target sources, 1capable by the G-band total-flux upper-bound approximation depth×10^(0.4deltaG). Exact neighbor table in neighbors-summary.json; aperture throughput/color can change dilution. Capable rows: [{"source_id": "5285787663885673216", "ra": "109.12614638104118", "dec": "-63.20485948937643", "phot_g_mean_mag": "11.81346", "bp_rp": "0.7271261", "sep": "27.847305224395946", "required_fraction_G_approx": 0.06255100350905454, "capable": true}]. Previously verified difference-image receipt: {"tic": "349273287", "sector": 97, "n_events": 8, "deficit_minus_target_arcsec": 3.45, "deficit_minus_light_arcsec": 32.61, "peak_on_target": true, "depth_ap3": 0.02523, "psf_frac_resid": 0.931, "n_neighbours_60": 17, "n_possible": 1, "brightest_nb": {"G": 11.81346, "sep": 27.8}}. Duplicate schemas are one measurement. New per-pixel localization where required is in this folder's pixel-localization.json and nuisance variants; model residuals and event consistency matter more than uncalibrated formal chi2. No unresolved source ambiguity remains in the tested target-versus-neighbor comparisons.

Archives and novelty

Fresh2026-10-30 VizieRIDIA60arcsec cones returned all9catalog sections: VSX, ASAS-SNII/366, SB9, WDS, Kostovtable2/3/4, IJspeert and Prsa. Raw response 349273287_idia.tsv; URLs/response hashes in idia-receipts.json. Parsed exact rows in catalogue-rows.json. Target-vs-neighbor IDs checked; Kostovtable2 is unvetted and not published EB classification. Cone row counts: {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 1, "J_ApJS_258_16_tess-ebs": 0}. Current shared Gaia variability exact-source and ExoFOP checks: {"simbad": [{"id": "Gaia DR3 5285787668183123968", "main_id": "TYC 8913-1676-1", "otype": "SB*", "ra": "109.13565811467", "dec": "-63.198414717809996"}], "gaia_vari_summary": [], "gaia_eb": [], "exofop": {"toi_count": 0, "ctoi_count": 0}, "catalog_counts": {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 1, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 1, "J_ApJS_258_16_tess-ebs": 0}}. TIC349273287 ExoFOP was retried successfully; its receipt is retained.

No target prior EB entry established by these catalogs. SIMBAD exact Gaia match is SB*, already known spectroscopic binarity, not a complete bibliography. Name web searches returned no reliable target-specific EB result; search non-detection is weak and not an exhaustive literature clearance. Nearby VSX/IJspeert flags do not establish target novelty or prior EB status. Novelty status is limited prior-art coverage/no target EB match, separate from physical CONFIRMED.

Prediction and next check

Using epochs after2026-10-30T20UTC (BTJD4344.33333), next primary-kind(A) center 4348.769545BTJD; next opposite-kind(B) center 4345.994388BTJD. Repeated secondary uses its own fitted ephemeris; null opposite kind uses Gaia phase with sigma0.0879d before future-period uncertainty, so it may geometrically miss. Ephemeris errors are conditional and future checks should include O-C scatter.

Calibrate a multi-source empirical PSF and obtain less diluted eclipse photometry.

R031. TIC 273724854

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 273724854: CONFIRMED attributed eclipsing system

Six primary timings span multiple sectors; seven secondaries repeat near phase0.695. The moderate eccentric orbit agrees in period and primary epoch. Approximate geometry permits both eclipses. Pixel regression favors the target over the53arcsec capable neighbor in each event; its free-centroid offset4.7arcsec is small against that separation, with PSF mismatch retained.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2853d, profile range[0.277425, 0.293175], ingress fraction0.25 (T23 conditional, not proof of totality). Median primary depth44.763ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=9.041956626 ±9.678556676e-06d, Tref=2927.950691 ±0.0004053232998 BTJD, cycle reference122. T0=1824.831983. chi2/dof=0.029854297050686464/4; O-C RMS=0.09218086295min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
19	1824.831917	0.00160734773	44.93632343	0.5075781034	0.2853	0.071325	0.14265	-0.09449746962	1.161871849	0	-0.1028859096	45.046177295
59	2918.908811	0.0008559	43.872969971	0.3661178898	0.2853	0.071325	0.14265	0.1092632175	1.468745512	0	0.01533949834	43.856609 29
59	2927.950684	0.0008559	43.8440073	0.4220158322	0.2853	0.071325	0.14265	-0.01128381824	1.573964585	0	-0.03751571804	43.883734 24
73	3289.629024	0.0008559	46.230831150552	0.9611180552	0.2853	0.071325	0.14265	0.09681875585	2.442799146	0	0.2406042462	45.887337 45
73	3298.670897	0.0008559	44.589473356995	0.5015926995	0.2853	0.071325	0.14265	-0.02372827927	1.292258957	0	-0.393731809	44.984320 87
73	3307.71277	0.0008559	45.6165876	0.4307219484	0.2853	0.071325	0.14265	-0.144275315	1.230426303	2	-0.09083485453	45.71441418

Own Gaia source1929602018676005l2, SB1, bit_index127, w–Tp correlation0.96854. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=9.041166927771348±7.057007E-4d, e=0.3093853258047935±0.011946757, omega=12.56309934848314±4.2138906deg. Photometric P difference=1.118922352sigma. Primary attributed toA; propagated epoch389 offset=-0.03517±0.04258d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.6896982; optimized diagnostic phase=0.6953041. Fit statistic: {"n": 7, "depth": 0.007425178157771617, "error": 0.00012300165827370543, "snr": 60.36648824074373, "scatter": 0.00042515022967767164, "chi2": 3.917500169582251, "dof": 6, "empirical_error": 0.00016069168250987308, "n_individual_gt3sigma": 7}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: {"P": 9.04193618014397, "P_error": 2.6687011700870094e-05, "Tref": 2509.2647986106467, "Tref_error": 0.0016056105913202592, "reference_cycle": 75, "T0": 1831.1195850998488, "chi2": 11.685381665561682, "dof": 5, "n": 7, "oc_rms_minutes": 9.046030913720168}.

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
19	1822.075035	0.002851 2	7.470328588	0.2863836604	0.31688	0.110888	0.095004	-3.764311664
19	1831.122694	0.002851 2	7.007721607	0.3221512083	0.31688	0.110888	0.095004	4.476586827
19	1840.177956	0.001900 8	6.420023059	1.459748 4829	0.31688	0.110888	0.095004	23.66609332

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
59	2916.1574 89	0.00285 12	7.625 5097 29	0.29485 2510 4	0.3168	0.11088	0.09504	8.00 9685 595
59	2925 .1899 42	0.00190 08	7.473 0814 04	0.31738 9820 2	0.3168	0.11088	0.09504	-5.6466 3191 6
59	2934 .235 7	0.00190 08	7.306 7447 48	0.26274 1299 8	0.3168	0.11088	0.09504	-0.1428 8542 65
73	3295 .9158 66	0.00285 12	7.790 4256 35	0.35615 5737 6	0.3168	0.11088	0.09504	3.771 9661 63

Measured fitted secondary phase at its reference epoch=0.6952111; first-order e cos(omega) in primary-to-secondary convention=0.30664 +/- 0.00043 (conservative sum of reference-time and period errors; no full ephemeris cross-covariance), Gaia same-convention=0.30198. This first-order estimator is approximate for eccentric or grazing systems; exact Gaia conjunction-phase comparison above is preferred. Duration ratio=1.110; grazing/partial chords prohibit direct trapezoid-duration conversion into e sin(omega). Own limb-darkened two-sphere feasibility result: {"status": "conditional shape feasibility", "best": {"chi2": 225.13029886865672, "dof": 121, "k": 0.22499999999999998, "b_A": 0.655, "e_sin_omega": 0.1499999999999999, "primary_scale": 0.9000537804575524, "secondary_scale": 0.009449668592239915}, "profile_interval_approx_95": {"k": [0.22499999999999998, 0.22499999999999998], "b_A": [0.655, 0.6900000000000001], "e_sin_omega": [0.09999999999999992, 0.1499999999999999]}, "error_inflation": 1.8605809823855928, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. These coarse fixed-TIC profiles are not calibrated orbital uncertainties.

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector19; 1 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At1824.831917: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04355195395010385, "error": 0.0003911369403096833, "red_noise_beta": 1.2014324951173394, "n_core": 14, "baseline_degree": 1, "background_regressed_depth": 0.04306274611625808, "background_template_native_sigma": 0.7960818938787566}, "2": {"depth": 0.043140735238071486, "error": 0.00041692528481073734, "red_noise_beta": 1.1877397565403531, "n_core": 14, "baseline_degree": 2, "background_regressed_depth": 0.04327897683182085, "background_template_native_sigma": 0.5675679276572787}}, "PDCSAP_FLUX": {"1": {"depth": 0.045979381519573176, "error": 0.00043126088496552516, "red_noise_beta": 1.2230350620560142, "n_core": 14, "baseline_degree": 1, "background_regressed_depth": 0.045104820897872544, "background_template_native_sigma": 0.7960818938787566}, "2": {"depth": 0.04531476956867521, "error": 0.0004262180948746483, "red_noise_beta": 1.187117823672853, "n_core": 14, "baseline_degree": 2, "background_regressed_depth": 0.045435061704022915, "background_template_native_sigma": 0.5675679276572787}}}. Own hashed pixel regression sector59, 2 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset4.70arcsec; target residual fraction0.257. Every event favors['192960201867600512', '192960201867600512']. Fixed source comparisons: [{"source_id": "192960201867600512", "sep_arcsec": 0.0, "G": 11.472519, "chi2": 11677.044015504745, "amplitude": 65.44914280829774, "fractional_residual": 0.25727901268918874}, {"source_id": "192959965645830912", "sep_arcsec": 53.16328012646885, "G": 14.687721, "chi2": 90245.0047334416, "amplitude": 4.210440195577153, "fractional_residual": 0.9906768093907801}]. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are retained. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "273724854", "sector": 59, "n_events": 2, "deficit_minus_target_arcsec": 2.6, "deficit_minus_light_arcsec": 2.1, "peak_on_target": true, "depth_ap3": 0.04359, "psf_frac_resid": 0.617, "n_neighbours_60": 35, "n_possible": 1, "brightest_nb": {"G": 14.687721, "sep": 53.2}}]. Own Gaia<15 cone neighbor table: [{"source_id": "192959965645830912", "ra": "86.13734320473145", "dec": "42.03834407565086", "phot_g_mean_mag": "14.687721", "bp_rp": "1.0360403", "sep": "53.16328012646885", "required_fraction_G_approx": 0.8649659256549902, "capable": true}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 192960201867600512", "main_id": "TYC 2919-1976-1", "otype": "SB*", "ra": "86.13983267498", "dec": "42.05302697271001"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.259,

"m2_min": 0.5358755879215997, "f_mass": 0.04776647768394166, "a_rsun": 22.195530540925894, "r1_rsun": 2.77935, "r2_approx_rsun": 0.6070873428823376, "central_T14_hours": {"A": 9.389234528416505, "B": 10.74412607707296}, "observed_T14_hours": 6.8472}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains15 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.394ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Six primary timings span multiple sectors; seven secondaries repeat near phase0.695. The moderate eccentric orbit agrees in period and primary epoch. Approximate geometry permits both eclipses. Pixel regression favors the target over the53arcsec capable neighbor in each event; its free-centroid offset4.7arcsec is small against that separation, with PSF mismatch retained.

Next nominal eclipses after2026-10-30T00UTC: primary4347.53788BTJD; other4344.78283BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R032. TIC 365181558

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 365181558: CONFIRMED attributed eclipsing system

Only one QLP event has sufficient clean bilateral baseline for timing, so the reported photometric period is held at search value and has no fitted error. A second event2cycles later is independently recovered from the pixel aperture:42.80±0.99ppt linear /44.11±0.97ppt quadratic, versus43.68±0.44 /43.63±0.50ppt first event. This repeat, the exact-target alternate reduction, own on-target deficit and Gaia orbit establish an EB, but cannot support precise new photometric ephemeris. The single shallow secondary is conditional and does not independently establish a period.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2184827586d, profile range[0.21305172413793103, 0.22391379310344825], ingress fraction0.25 (T23 conditional, not proof of totality). Median primary depth46.434ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=8.838698731 ±unconstrainedd, Tref=1932.567311 ±0.001901841125 BTJD, cycle reference0. T0=1932.567311. chi2/dof=None/0; O-C RMS=unconstrainedmin. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
23	1932.567311	0.001901841125	46.43383169	0.4815481542	0.2184827586	0.0546206896	0.1092413793	0	1	0	-0.1686590157	46.52181031

Own Gaia source3714465189925105792, SB1, bit_index127, w–Tp correlation0.96102. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=8.83733575878259±7.336397E-4d, e=0.28180411463678257±0.025335394, omega=105.57813498611576±5.4078736deg. Photometric P difference=unconstrainedsigma. Primary attributed toA; propagated epoch389 offset=-0.04313±0.03763d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.4499932; optimized diagnostic phase=0.4382651. Fit statistic: {"n": 1, "depth":

0.002006875263963851, "error": 0.00019373368403424667, "snr": 10.358938219587525, "scatter": 0.0, "chi2": 0.0, "dof": 0, "empirical_error": 0.00019373368403424667, "n_individual_gt3sigma": 1}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `null`.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
23	1954.147863	0.004910275862	2.00687526 4	0.19373368 4	0.27279310 34	0.02182344828	0.22914620 69	0

No robust independent secondary period/e cos omega is claimed; geometry follows the null test or conditional model, not a noisy optimized phase.

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector23; 1 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At1932.567311: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.0439359300522893, "error": 0.00039186978682026995, "red_noise_beta": 1, "n_core": 10, "baseline_degree": 1, "background_regressed_depth": 0.0439690666770447, "background_template_native_sigma": 0.958719077294971}, "2": {"depth": 0.04382475515672699, "error": 0.0004405057758257776, "red_noise_beta": 1, "n_core": 10, "baseline_degree": 2, "background_regressed_depth": 0.0438931963547015, "background_template_native_sigma": 0.39212301483688283}}, "PDCSAP_FLUX": {"1": {"depth": 0.04374777368829497, "error": 0.00037863900564150615, "red_noise_beta": 1, "n_core": 10, "baseline_degree": 1, "background_regressed_depth": 0.04386794700104291, "background_template_native_sigma": 0.958719077294971}, "2": {"depth": 0.043820481872623315, "error": 0.00042625633457306263, "red_noise_beta": 1, "n_core": 10, "baseline_degree": 2, "background_regressed_depth": 0.043868925635098054, "background_template_native_sigma": 0.39212301483688283}}}. Own hashed pixel regression sector23, 2 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset0.00arcsec; target residual fraction0.156. Every event favors['3714465189925105792', '3714465189925105792']. Fixed source comparisons: [{"source_id": "3714465189925105792", "sep_arcsec": 0.0, "G": 11.202918, "chi2": 1924.3702171375119, "amplitude": 101.79929331577931, "fractional_residual": 0.15606556387939782}]. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are retained. Original batch deficit receipts (duplicate schema rows are not independent observations): `[]`. Own Gaia<15 cone neighbor table: `[]`. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-

source SIMBAD [{"id": "Gaia DR3 3714465189925105792", "main_id": "TYC 310-219-1", "otype": "SB*", "ra": "204.086302344", "dec": "4.534086278249999"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.24, "m2_min": 0.3699497801409628, "f_mass": 0.01953456130077691, "a_rsun": 21.082532012175843, "r1_rsun": 1.82988, "r2_approx_rsun": 0.4513508023720313, "central_T14_hours": {"A": 5.5126633985307265, "B": 9.620626275350599}, "observed_T14_hours": 5.243586206896551}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains3 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.169ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Only one QLP event has sufficient clean bilateral baseline for timing, so the reported photometric period is held at search value and has no fitted error. A second event2cycles later is independently recovered from the pixel aperture:42.80±0.99ppt linear /44.11±0.97ppt quadratic, versus43.68±0.44 /43.63±0.50ppt first event. This repeat, the exact-target alternate reduction, own on-target deficit and Gaia orbit establish an EB, but cannot support precise new photometric ephemeris. The single shallow secondary is conditional and does not independently establish a period.

Next nominal eclipses after2026-10-30T00UTC: primary4345.53206BTJD; other4349.40576BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R033. TIC 370251176

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 370251176: CONFIRMED attributed eclipsing system

Seven strong primary timings, independent reduction and clean target deficit establish an attributed EB. No reliable secondary is detected. Own covariance-phase and width tests provide conditional limits; eccentric geometry can miss the other eclipse. Component luminosity and radius remain unmeasured.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.160137931d, profile range[0.15651724137931033, 0.1637586206896552], ingress fraction0.35 (T23 conditional, not proof of totality). Median primary depth47.864ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=16.68750584 ±6.064876975e-06d, Tref=3541.374033 ±0.0002644314182 BTJD, cycle reference132. T0=1338.623261. chi2/dof=10.603727402630478/5; O-C RMS=0.8514481716min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
28	2072.873382	0.0004804137931	56.67842747	0.3551705184	0.1601379310	0.05604827586	0.04804137931	-0.1969748136	1.202103739	0	-0.1303640807	56.795775
68	3157.561519	0.0004804137931	55.83302512	0.327259399	0.1601379310	0.05604827586	0.04804137931	0.1737287825	1.625522559	0	-0.02216365957	55.876133 52
68	3174.249044	0.0004804137931	55.91655697	0.3635827527	0.1601379310	0.05604827586	0.04804137931	0.2007179417	1.646404418	7	-0.01871978994	55.954973 41
95	3891.81164	0.0004804137931	47.72575671	0.2979031399	0.1601379310	0.05604827586	0.04804137931	-0.02233995452	1.721304652	0	0.0006034118982	47.724404 1
102	4142.12547	0.0004804137931	47.86374535	0.3066580344	0.1601379310	0.05604827586	0.04804137931	1.766089151	1.649183525	0	-0.007873378962	47.838 71444
103	4158.811073	0.0004804137931	45.00448871	0.5946171864	0.1601379310	0.05604827586	0.04804137931	-0.9741051376	2.208469801	6	-0.07426521099	45.087 71453
103	4175.498597	0.0004804137931	46.75816532	0.2926274996	0.1601379310	0.05604827586	0.04804137931	-0.9471159804	1.466077853	1	-0.01433086532	46.783 46025

Own Gaia source6508371858702158336, SB1, bit_index127, w–Tp correlation0.91995. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=16.688483193177326±0.0018087598d, e=0.2917985375138582±0.012118004, omega=86.36949240225397±2.117758deg. Photometric P difference=-0.5403395005sigma. Primary attributed toA; propagated epoch389 offset=-0.00475±0.06332d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.5122951; optimized diagnostic phase=0.5093181. Fit statistic: `{"n": 9, "depth": 7.105353131474385e-05, "error": 4.886080158948841e-05, "snr": 1.4542031445106263, "scatter": 0.00036219784569752674, "chi2": 27.16099765344126, "dof": 8, "empirical_error": 0.00012073261523250891, "n_individual_gt3sigma": 0}`. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `null`.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
1	1330.418444	0.00828	0.06181560255	0.14174556661	0.2776	0.138	0	0
28	2064.605773	0.00579 6	-0.2544582029	0.14390107762	0.2776	0.138	0	0
28	2081.382703	0.00513 36	-0.8968435825	0.27323226291	0.2776	0.138	0	0
95	3883.706198	0.00828	0.2003832489	0.2038532874	0.2776	0.138	0	0
102	4133.929361	0.00828	-0.0998290814	0.12818913228	0.2776	0.138	0	0
102	4150.623491	0.0207	0.3251151746	0.15760395228	0.2776	0.138	0	0
103	4167.312653	0.02898	0.2897826125	0.13641970376	0.2776	0.138	0	0

secondary sector	mid BTJD	erord	depth ppt	error ppt	T14d	ingress d	T23d	O-Cmin
104	4183.915703	0.00828	0.10858035552	0.13648420688	0.276	0.138	0	0
104	4200.7688809	0.011592	0.23725997288	0.11561210616	0.276	0.138	0	0
No robust independent secondary period/e cos omega is claimed; geometry follows the null test or conditional model, not a noisy optimized phase.								
Secondary null tests: own covariance relative-phase sigma=0.007249, 5widths×9phases spanning±2sigma, pointwise conditional3sigma upper ranges[0.118, 0.73]ppt. Full coverage/excluded windows in null-secondary.json; upper limits are phase/shape/baseline conditional and not a simultaneous bound. Feasibility: [{"radius_ratio_assumption": 0.3116410122377534, "b_primary_duration_approx": 0.9968735760746208, "b_other": 1.8160264784053006, "contact": 1.3116410122377533, "other_misses": true, "es_gaia": 0.2912129426803021, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}].								

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector68; 2 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At3157.561519: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.05539586049102551, "error": 0.0003247941042219259, "red_noise_beta": 1.8379276719229136, "n_core": 69, "baseline_degree": 1, "background_regressed_depth": 0.055422971295843, "background_template_native_sigma": 0.155586954810369}, "2": {"depth": 0.0552443898359438, "error": 0.000354026609454435, "red_noise_beta": 1.814068179354614, "n_core": 69, "baseline_degree": 2, "background_regressed_depth": 0.055266994380274694, "background_template_native_sigma": 0.12859751084153398}}, "PDCSAP_FLUX": {"1": {"depth": 0.0555625278213786, "error": 0.0003273361064099826, "red_noise_beta": 1.8451467588566968, "n_core": 69, "baseline_degree": 1, "background_regressed_depth": 0.055589348749431014, "background_template_native_sigma": 0.155586954810369}, "2": {"depth": 0.05539539377774374, "error": 0.0003559667067749508, "red_noise_beta": 1.8184571382595889, "n_core": 69, "baseline_degree": 2, "background_regressed_depth": 0.055417778385624346, "background_template_native_sigma": 0.12859751084153398}}}. At3174.249044: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.05560344923086007, "error": 0.00031345334651038554, "red_noise_beta": 1.533009053136357, "n_core": 64, "baseline_degree": 1, "background_regressed_depth": 0.0558024241682578, "background_template_native_sigma": 0.8423766830927337}, "2": {"depth": 0.0553089115619223, "error": 0.000336287204297118, "red_noise_beta": 1.5089598177370114, "n_core": 64, "baseline_degree": 2, "background_regressed_depth": 0.055464239661047667, "background_template_native_sigma": 0.4699459530715501}}, "PDCSAP_FLUX": {"1": {"depth": 0.05315927509702478, "error": 0.0003272021501984101, "red_noise_beta": 2.9856865274542783, "n_core": 64, "baseline_degree": 1, "background_regressed_depth": 0.05352864961197137,

"background_template_native_sigma": 0.8423766830927337}, "2": {"depth": 0.05421390534209818, "error": 0.0006049788434068311, "red_noise_beta": 2.546696117871127, "n_core": 64, "baseline_degree": 2, "background_regressed_depth": 0.05431668891137594, "background_template_native_sigma": 0.4699459530715501}}}. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "370251176", "sector": 68, "n_events": 2, "deficit_minus_light_arcsec": 0.18, "peak_on_target": true, "depth_ap3": 0.05344, "psf_frac_resid": 0.015, "n_neighbours_60": 6, "n_possible": 0, "brightest_nb": {"G": 18.884909, "sep": 44.6}}, {"tic": "370251176", "sector": 68, "n_events": 2, "deficit_minus_target_arcsec": 1.86, "deficit_minus_light_arcsec": 0.18, "peak_on_target": true, "depth_ap3": 0.05344, "psf_frac_resid": 0.015, "n_neighbours_60": 6, "n_possible": 0, "brightest_nb": {"G": 18.884909, "sep": 44.6}}]. Own Gaia<15 cone neighbor table: []. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 2, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 6508371858702158336", "main_id": "CD-55 9079", "otype": "SB*", "ra": "335.9971527988", "dec": "-54.94798882242001"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.2, "m2_min": 0.3974178041670223, "f_mass": 0.02459828999077711, "a_rsun": 32.125890323744336, "r1_rsun": 1.53371, "r2_approx_rsun": 0.4779669368791648, "central_T14_hours": {"A": 5.9136992294338455, "B": 10.773115712691927}, "observed_T14_hours": 3.8433103448275863}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains14 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.130ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Seven strong primary timings, independent reduction and clean target deficit establish an attributed EB. No reliable secondary is detected. Own covariance-phase and width tests provide conditional limits; eccentric geometry can miss the other eclipse. Component luminosity and radius remain unmeasured.

Next nominal eclipses after2026-10-30T00UTC: primary4359.06182BTJD; other4350.87356BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R034. TIC 119301005

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 119301005: CONFIRMED attributed eclipsing system

The original primary-duration grid ended at0.6003d; widening gave interior0.73876d and improved timing chi2 from179/6 to29.3/6, though6.25min timing scatter remains. Pixel aperture primaries repeat59.9–67.2ppt across degree1/2 and2–4duration windows; the broad baseline is a material uncertainty. A secondary at1483.652 is44.54±1.43ppt linear /38.97±0.97ppt quadratic. Known neighboring EA TIC119301028 at58.5arcsec has a distinct2.44349d period; the target8.296d deficit beats both capable neighbors in every tested primary event. Missing TIC radius/mass precludes a unique geometry; the long eclipses permit evolved components, and no main-sequence radius veto is justified.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.7387627119d, profile range[0.7311317796610168, 0.7463936440677965], ingress fraction0.25 (T23 conditional, not proof of totality). Median primary depth63.837ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=8.29617034 ±1.958589559e-05d, Tref=2309.117459 ±0.001849475658 BTJD, cycle reference101. T0=1471.204255. chi2/dof=29.26632800787357/6; O-C RMS=6.251462933min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
6	1471.199386	0.00221628813 6	43.765310 87	0.7767706286	0.7387627119	0.184690678	0.3693813559	-7.011802609	1.353926442	6	1.165152419	43.144764 02
6	1479.508349	0.00221628813 6	44.24473261	0.8539835709	0.7387627119	0.184690678	0.3693813559	11.41008402	1.367791751	1	0.178900345 3	44.08117482
6	1487.799582	0.00221628813 6	40.75635217	0.8982101571	0.7387627119	0.184690678	0.3693813559	4.300331318	1.38543625	4	1.894915903	39.06589132
33	2209.562381	0.00221628813 6	65.11342478	1.4813484477	0.7387627119	0.184690678	0.3693813559	-1.488809625	2.195151166	1	1.257909972	63.999889 63

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
33	2217.853614	0.002216288136	72.38673049	1.513894442	0.7387627119	0.1846090678	0.3693813559	-8.598562322	2.162709148	0	1.297706186	70.899730 41
33	2226.153712	0.002216288136	72.16905208	1.102724906	0.7387627119	0.1846090678	0.3693813559	-2.942495359	2.158482067	1	0.7376477599	71.198692 34
87	3669.692609	0.004432576271	62.560 5158	3.031279156	0.7387627119	0.1846090678	0.3693813559	4.627952249	4.147386841	3	0.1931844929	62.414870 03
87	3686.28394	0.002216288136	69.729 94747	1.853374316	0.7387627119	0.1846090678	0.3693813559	3.174266516	4.117693897	1	0.0992274034 8	69.653865 26

Own Gaia source2954081396710806400, SB1, bit_index127, w–Tp correlation0.99985. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=8.29593229280462±0.0014021182d, e=0.04126349783700163±0.014851228, omega=274.4277617950491±33.09536deg. Photometric P difference=0.1697600605sigma. Primary attributed toB; propagated epoch389 offset=0.06218±0.05025d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.4979703; optimized diagnostic phase=0.5003703. Fit statistic: `{"n": 7, "depth": 0.0314983098438132, "error": 0.0005098671768247851, "snr": 61.77748102941237, "scatter": 0.00745345888905685, "chi2": 227.63404782787072, "dof": 6, "empirical_error": 0.0028171426610983127, "n_individual_gt3sigma": 7}`. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `{"P": 8.296251575217076, "P_error": 3.329466439670281e-05, "Tref": 2628.520306670817, "Tref_error": 0.003604945862852637, "reference_cycle": 139, "T0": 1475.3413377156432, "chi2": 17.925646333635807, "dof": 5, "n": 7, "oc_rms_minutes": 10.319697307160647}`.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
6	1475.348208	0.003601 8	24.4048 7801	0.9539 7433 4	0.6003	0.0900 045	0.4202 1	9.893 3944 87
6	1483.626369	0.003601 8	26.4151 5282	0.9177 0710 51	0.6003	0.0900 045	0.4202 1	-16.15 6544 66
33	2205.4148	0.003601 8	42.8408 9691	1.3490 2602 4	0.6003	0.0900 045	0.4202 1	4.785 821 307
33	2222.000 7141	0.003601 8	36.5594 9023	1.4845 4052 1	0.6003	0.0900 045	0.4202 1	4.551 8630 07
87	3665.544381	0.007203 6	43.4908 4316	2.3172 0241 8	0.6003	0.0900 045	0.4202 1	-10.61 5917 05

secondary sector	mid BTJD	erro rd	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
87	3673.851357	0.000547	43.5913669	2.174240011	0.60 03	0.09 0045	0.42 021	4.826879798
87	3682.133 12	0.000900 45	34.6924384938	2.504242449 38	0.60 03	0.09 0045	0.42 021	-16.036467 35
Measured fitted secondary phase at its reference epoch=0.5000349; first-order e cos(omega) in primary-to-secondary convention=0.00005 +/- 0.00118 (conservative sum of reference-time and period errors; no full ephemeris cross-covariance), Gaia same-convention=-0.00319. This first-order estimator is approximate for eccentric or grazing systems; exact Gaia conjunction-phase comparison above is preferred. Duration ratio=0.813; grazing/partial chords prohibit direct trapezoid-duration conversion into e sin(omega). Own limb-darkened two-sphere feasibility result: `{"status": "unavailable: TIC radius missing"}`. These coarse fixed-TIC profiles are not calibrated orbital uncertainties.								

Independent reduction and source attribution

No exact-target alternate-pipeline light curve covering an event in the shared MAST inventory. An independently extracted pixel aperture supplies the reduction check; both kinds were checked in targeted-aperture.json. This is the same observatory and pixels. Own hashed pixel regression sector6, 3 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset6.30arcsec; target residual fraction0.296. Every event favors['2954081396710806400', '2954081396710806400', '2954081396710806400']. Fixed source comparisons: `[{"source_id": "2954081396710806400", "sep_arcsec": 0.0, "G": 10.43796, "chi2": 135558.11381377012, "amplitude": 367.0172898748855, "fractional_residual": 0.2960384237301189}, {"source_id": "2954081461126345728", "sep_arcsec": 32.759954520234935, "G": 13.096158, "chi2": 210242.42749909993, "amplitude": 75.78600903502438, "fractional_residual": 0.9098780754576322}, {"source_id": "2954081431070539776", "sep_arcsec": 58.53715653230721, "G": 13.167544, "chi2": 276867.9188322554, "amplitude": 17.250458161574098, "fractional_residual": 0.9916968688903649}]`. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are retained. Original batch deficit receipts (duplicate schema rows are not independent observations): `[{"tic": "119301005", "sector": 6, "n_events": 3, "deficit_minus_light_arcsec": 2.02, "peak_on_target": true, "depth_ap3": 0.03097, "psf_frac_resid": 0.131, "n_neighbours_60": 44, "n_possible": 2, "brightest_nb": {"G": 13.096158, "sep": 32.8}}, {"tic": "119301005", "sector": 6, "n_events": 3, "deficit_minus_target_arcsec": 4.06, "deficit_minus_light_arcsec": 2.02, "peak_on_target": true, "depth_ap3": 0.03097, "psf_frac_resid": 0.131, "n_neighbours_60": 44, "n_possible": 2, "brightest_nb": {"G": 13.096158, "sep": 32.8}}]`. Own Gaia<15 cone neighbor table: `[{"source_id": "2954081461126345728", "ra": "100.49172358394289", "dec": "-11.326737972557588", "phot_g_mean_mag": "13.096158", "bp_rp": "1.7692165", "sep": "32.759954520234935", "required_fraction_G_approx": 0.7385016451876112, "capable": true}, {"source_id": "2954081431070539776", "ra": "100.51291798873905", "dec": "-11.321800440121843", "phot_g_mean_mag": "13.167544", "bp_rp": "0.81274414", "sep": "58.53715653230721",`

"required_fraction_G_approx": 0.7886891515744043, "capable": true}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 1, "II_366_catv2021": 2, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 4, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 1, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 2954081396710806400", "main_id": "UCAC4 394-016428", "otype": "SB*", "ra": "100.49962458556752", "dec": "-11.331528719709725"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 0.9, "m2_min": 0.7150798524560977, "f_mass": 0.1401766337877521, "a_rsun": 20.233882366321183, "r1_rsun": null, "r2_approx_rsun": 0.7646868704569845, "central_T14_hours": null, "observed_T14_hours": 17.73030508474576}, {"m1": 1.2, "m2_min": 0.8338937443946468, "f_mass": 0.1401766337877521, "a_rsun": 21.85029515558335, "r1_rsun": null, "r2_approx_rsun": 0.8647460217533551, "central_T14_hours": null, "observed_T14_hours": 17.73030508474576}, {"m1": 1.6, "m2_min": 0.9762184148676554, "f_mass": 0.1401766337877521, "a_rsun": 23.641522843484328, "r1_rsun": null, "r2_approx_rsun": 0.9809290506739875, "central_T14_hours": null, "observed_T14_hours": 17.73030508474576}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindB, other kindA. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains47 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=1.895ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: The original primary-duration grid ended at0.6003d; widening gave interior0.73876d and improved timing chi2 from179/6 to29.3/6, though6.25min timing scatter remains. Pixel aperture primaries repeat59.9–67.2ppt across degree1/2 and2–4duration windows; the broad baseline is a material uncertainty. A secondary at1483.652 is44.54±1.43ppt linear /38.97±0.97ppt quadratic. Known neighboring EA TIC119301028 at58.5arcsec has a distinct2.44349d period; the target8.296d deficit beats both capable neighbors in every tested primary event. Missing TIC radius/mass precludes a unique geometry; the long eclipses permit evolved components, and no main-sequence radius veto is justified.

Next nominal eclipses after2026-10-30T00UTC: primary4349.97536BTJD; other4345.83035BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R035. TIC 377656542

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 377656542: CONFIRMED attributed eclipsing system

Forty primary and39secondary fits establish two equal-depth kinds at full3.77777d; stronger half-period search is the expected ambiguity. Pixel apertures recover seven secondary events at44.8–45.7ppt linear or41.4–42.4ppt quadratic. The22.3arcsec capable neighbor is strongly disfavored for every primary. Gaia small eccentricity gives a3.3sigma primary-epoch tension, while period agrees; observed near-half-cycle separation favors a more circular orbit. Coarse fixed-TIC geometry is a poor characterization fit (chi2~4207/121, scale at boundary), not a calibrated radius/eccentricity posterior.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2108275862d, profile range[0.2072068965517241, 0.21444827586206897], ingress fraction0.35 (T23 conditional, not proof of totality). Median primary depth50.341ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=3.777765336 ±7.410532372e-07d, Tref=2715.627326 ±0.00011442131 BTJD, cycle reference302. T0=1574.742195. chi2/dof=41.58326142767368/38; O-C RMS=1.172873389min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
10	1574.74032	0.001901826711	47.18576452	0.5364969127	0.2108275862	0.07378965517	0.06324827586	-2.699280001	1.015931919	0	-0.1676803262	47.36638136
10	1578.519325	0.00181332062	47.13354981	0.5173698788	0.2108275862	0.07378965517	0.06324827586	-0.9147859498	1.023236128	0	-0.2586769905	47.28084592
10	1586.074804	0.002273117695	47.96112353	0.5533905947	0.2108275862	0.07378965517	0.06324827586	-0.9888985363	1.072661346	4	0.307120349	47.68722114
10	1589.851279	0.00181332062	47.42509619	0.5454327171	0.2108275862	0.07378965517	0.06324827586	-2.847505175	1.181645912	0	-0.07120891183	47.45285734
10	1593.629018	0.001901826711	47.2833353	0.5564478953	0.2108275862	0.07378965517	0.06324827586	-2.884561468	1	0	-0.2790769457	47.55881677

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
36	2284.962956	0.0006324827586	50.43653298	0.5251855868	0.2108275862	0.07378965517	0.06324827586	1.263438921	1.349282182	1	-0.38947471	50.82425274
36	2288.73943	0.0006324827586	50.57654205	0.4264338054	0.2108275862	0.07378965517	0.06324827586	-0.5951677177	1.200704437	0	-0.1239125959	50.66969446
36	2296.294909	0.0006324827586	50.22595585	0.535246677	0.2108275862	0.07378965517	0.06324827586	-0.6692803038	1.253358907	0	-0.1140871638	50.38810216
36	2300.072649	0.0006324827586	50.36747586	0.4892555177	0.2108275862	0.07378965517	0.06324827586	-0.7063365976	1.202382125	0	-0.2497916789	50.58379265
36	2303.850389	0.0006324827586	50.82596804	0.5016455951	0.2108275862	0.07378965517	0.06324827586	-0.7433928907	1.247063269	0	-0.2326274054	51.0540931
37	2311.405868	0.0006324827586	50.65876443	0.5382102682	0.2108275862	0.07378965517	0.06324827586	-0.8175054775	1.321780866	0	-0.0101233304	50.67110475
37	2315.184872	0.0006324827586	50.59847406	0.4792004426	0.2108275862	0.07378965517	0.06324827586	0.9669885744	1.250910773	0	-0.1983453956	50.81028759
37	2322.739087	0.0006324827586	50.78656138	0.4501770328	0.2108275862	0.07378965517	0.06324827586	-0.9286743573	1.069676395	0	0.1509513789	50.62639217
37	2326.516826	0.0006324827586	50.26480706	0.5074362288	0.2108275862	0.07378965517	0.06324827586	-0.9657306511	1.182076913	1	-0.0629007462	50.33775347
37	2330.294566	0.0006324827586	50.29838386	0.4665042041	0.2108275862	0.07378965517	0.06324827586	-1.002786944	1.082086512	0	-0.1432478167	50.45568318
38	2337.850045	0.0006324827586	50.31473347	0.4217722626	0.2108275862	0.07378965517	0.06324827586	-1.07689953	1.044065674	0	0.0998080916	50.19604568
38	2341.62905	0.0006324827586	50.78202575	0.4885333832	0.2108275862	0.07378965517	0.06324827586	0.7075945215	1.268809007	0	0.0231567847	50.75318358
38	2349.184529	0.0006324827586	51.135009195	0.6247525237	0.2108275862	0.07378965517	0.06324827586	0.6334819347	1.295929038	0	-9.121293133	52.01568253
38	2352.962268	0.0006324827586	51.241322211	0.4744114781	0.2108275862	0.07378965517	0.06324827586	0.5964256416	1.082516417	0	0.1380861035	51.10988311
38	2356.740008	0.0006324827586	50.68657564	0.5204284483	0.2108275862	0.07378965517	0.06324827586	0.5593693479	1.229343988	0	0.3618709851	50.34991323
63	3017.848233	0.0006324827586	50.30221263	0.4845209575	0.2108275862	0.07378965517	0.06324827586	-0.4608309523	1.291469324	0	0.0897691049	50.21330803
63	3021.627238	0.0006324827586	50.77203116	0.659804372	0.2108275862	0.07378965517	0.06324827586	1.323663099	1.764691145	0	0.1789286872	50.58566558
63	3025.404977	0.0006324827586	50.19641918	0.5060293007	0.2108275862	0.07378965517	0.06324827586	1.286606806	1.370681553	0	0.0664763403	50.12993281
63	3029.182717	0.0006324827586	50.23936352	0.4240063376	0.2108275862	0.07378965517	0.06324827586	1.249550513	1.076654874	0	-0.1165252549	50.35244753
63	3032.960457	0.0006324827586	50.63055619	0.5750227895	0.2108275862	0.07378965517	0.06324827586	1.212494219	1.440834713	0	0.0639035110	50.56024329
63	3036.736931	0.0006324827586	49.88553344	0.4635207345	0.2108275862	0.07378965517	0.06324827586	-0.646112419	1.286982693	0	-0.0667979978	49.95569643
64	3044.293675	0.0006324827586	51.079558636	0.5234889602	0.2108275862	0.07378965517	0.06324827586	1.101325339	1.431876653	0	-0.1410642736	51.19763503
64	3048.071415	0.0006324827586	51.345933488	0.4638683943	0.2108275862	0.07378965517	0.06324827586	1.064269046	1.211379655	0	-0.1110789712	51.45191445
64	3051.849155	0.0006324827586	51.177900721	0.5197199576	0.2108275862	0.07378965517	0.06324827586	1.027212752	1.416239062	0	-0.1547510187	51.34910225
64	3055.626894	0.0006324827586	51.470526668	0.5221010771	0.2108275862	0.07378965517	0.06324827586	0.9901564592	1.313960182	0	-0.0493078309	51.52452204
64	3059.404634	0.0006324827586	51.118356624	0.4426543747	0.2108275862	0.07378965517	0.06324827586	0.9531001655	1.144737551	0	0.1215030379	50.99383117
64	3063.182374	0.0006324827586	51.92200509	0.4856001958	0.2108275862	0.07378965517	0.06324827586	0.9160438724	1.319468593	0	0.1343472826	51.77052682
64	3066.958848	0.0006324827586	52.03660377	0.4412689574	0.2108275862	0.07378965517	0.06324827586	-0.9425627656	1.096356054	2	0.5126086023	51.54605413

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
90	3750.734776	0.0006324827586	50.02723337	0.3905087984	0.2108275862	0.07378965517	0.06324827586	-0.3635504793	1.185570364	0	-0.06396321381	50.09603742
90	3754.512516	0.0006324827586	49.22105464	0.5056507363	0.2108275862	0.07378965517	0.06324827586	-0.400606773	1.483975889	0	-0.05974799255	49.27149368
90	3758.290255	0.0006324827586	50.07252409	0.4773755969	0.2108275862	0.07378965517	0.06324827586	-0.4376630661	1.304977961	0	0.07534969005	50.00178923
90	3762.067995	0.0006324827586	49.14441384	0.5063265981	0.2108275862	0.07378965517	0.06324827586	-0.4747193592	1.368609657	0	-0.04719183461	49.18872274
90	3765.845734	0.0006324827586	49.35746825	0.4392556192	0.2108275862	0.07378965517	0.06324827586	-0.5117756529	1.24056535	0	-0.02718891912	49.38554549
90	3769.622209	0.0006324827586	49.4416009	0.4839788654	0.2108275862	0.07378965517	0.06324827586	-2.370382292	1.343488913	0	-0.04905625181	49.48433148
90	3773.401214	0.0006324827586	50.2533654	0.4543188452	0.2108275862	0.07378965517	0.06324827586	-0.5858882397	1.072003909	6	-0.2189737154	50.46218142

Own Gaia source5245490498321111552, SB1, bit_index127, w–Tp correlation0.99881. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=3.777662244369182±8.284E-5d, e=0.03871030140712028±0.008295542, omega=34.75465399247972±15.501584deg. Photometric P difference=1.244419165sigma. Primary attributed toB; propagated epoch389 offset=-0.04829±0.01464d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.4797512; optimized diagnostic phase=0.4998419. Fit statistic: {"n": 39, "depth": 0.04383457656950189, "error": 7.536767133112232e-05, "snr": 581.6098042477378, "scatter": 0.0012645633910706485, "chi2": 283.08263111721055, "dof": 38, "empirical_error": 0.00020249220118164336, "n_individual_gt3sigma": 39}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: {"P": 3.7777651924574553, "P_error": 7.882696620477997e-07, "Tref": 2706.1827965107354, "Tref_error": 0.00011763574129888012, "reference_cycle": 299, "T0": 1576.6310039659563, "chi2": 41.25538689656929, "dof": 37, "n": 39, "oc_rms_minutes": 1.0424982821383069}.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
10	1572.853398	0.0021263069 04	41.860557 51	0.6291436058	0.2108275862	0.0737896551 7	0.0632482758 6	1.0670964 86

secondary sector	midBTJD	erorrd	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
10	1576.630 48	0.00 1901 8267 11	41.6 8693 212	0.41 2839 1932	0.21 0821 7586 2	0.0 0737 8965 17	0.0 6324 8275 86	-0.75 4246 9026
10	1580.4 0824 6	0.00 1901 8267 11	41.0 4674 665	0.66 3488 6656	0.21 0821 7586 2	0.0 0737 8965 17	0.0 6324 8275 86	-0.75 4039 9459
10	1587.9 6251 1	0.00 1901 8267 11	40.5 7138 534	0.41 5252 5577	0.21 0821 7586 2	0.0 0737 8965 17	0.0 6324 8275 86	-2.57 5176 377
10	1591.74 1542	0.00 1813 3206 2	40.6 9541 768	0.43 2797 3443	0.21 0821 7586 2	0.0 0737 8965 17	0.0 6324 8275 86	-0.75 3419 0754
36	2283.0 7386 3	0.00 0632 4827 586	44.3 5536 2231	0.45 7498 6487	0.21 0821 7586 2	0.0 0737 8965 17	0.0 6324 8275 86	1.1 0600 4359

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
36	2290.628129	0.000632486275866	44.8486124 5	0.463372199 8	0.210825862	0.073789651 7	0.063248275 86	-0.715132 0724
36	2298.183659	0.000632486275866	44.3499595 51	0.508225438 4	0.210825862	0.073789651 7	0.063248275 86	-0.714718 1589
36	2301.961425	0.000632486275866	44.1276751 51	0.507556916	0.210825862	0.073789651 7	0.063248275 86	-0.714511 2019
37	2313.294721	0.000632486275866	43.7947531 3	0.537484362	0.210825862	0.073789651 7	0.063248275 86	-0.713890 332
37	2317.072486	0.000632486275866	44.6985433 2	0.541061989	0.210825862	0.073789651 7	0.063248275 86	-0.713683 375

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
37	2320.850251	0.000632482758	43.725884666	0.5691529325	0.210827586	0.00737896551	0.0632482758	-0.7134764188
37	2324.628017	0.000632482758	44.208524005	0.4596451345	0.210827586	0.00737896551	0.0632482758	-0.7132694616
37	2328.405782	0.000632482758	44.717016992	0.4093745801	0.210827586	0.00737896551	0.0632482758	-0.7130625045
38	2339.740343	0.000632482758	44.910207655	0.4144047284	0.210827586	0.00737896551	0.0632482758	1.1091087811
38	2343.516843	0.000632482758	44.766772625	0.4364824164	0.210827586	0.00737896551	0.0632482758	-0.7122346776
38	2351.073639	0.000632482758	44.243717737	0.4376197801	0.210827586	0.00737896551	0.0632482758	1.1097295981

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
38	2354.851404	0.000632482758 6	44.708905 05	0.4415473769	0.210825758 62	0.0737 8965517	0.0632 4827586	1.109936537
38	2358.627905	0.000632482758 6	44.223164 01	0.4538853934	0.210825758 62	0.0737 8965517	0.0632 4827586	-0.711406850 8
63	3015.960338	0.000632482758 6	43.819787 73	0.4922357234	0.210825758 62	0.0737 8965517	0.0632 4827586	1.146153973
63	3019.736839	0.000632482758 6	44.039975 09	0.4662670067	0.210825758 62	0.0737 8965517	0.0632 4827586	-0.675189414 6
63	3023.514604	0.000632482758 6	43.821253 06	0.5209007079	0.210825758 62	0.0737 8965517	0.0632 4827586	-0.674982458 2

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
63	3027.293634	0.000632482758	44.96171215	0.641620722	0.21082758622	0.00737896517	0.00632482758	1.14677484843
63	3031.070135	0.000632482758	43.42340129	0.494356097	0.21082758622	0.00737896517	0.00632482758	-0.674568545454
63	3034.849165	0.000632482758	43.70987387	0.469717936	0.21082758622	0.00737896517	0.00632482758	1.14718875757
63	3038.626293	0.000632482758	44.21713437	0.430204078	0.21082758622	0.00737896517	0.00632482758	1.14739575714
64	3042.403431	0.000632482758	44.97887777	0.443525288	0.21082758622	0.00737896517	0.00632482758	-0.673947674343

secondary sector	mid BTJD	error d	depth ppt	errorppt	T14d	ingress d	T23d	O-C min
64	3046.1824161	0.000632482758586	45.53049735	0.4429002393	0.2108258762	0.007378965517	0.006324827586	1.147809627
64	3049.960226	0.000632482758586	45.71799423	0.4961384247	0.2108258762	0.007378965517	0.006324827586	1.148016583
64	3053.736727	0.000632482758586	44.58678443	0.4737424443	0.2108258762	0.007378965517	0.006324827586	-0.6733268038
64	3057.515757	0.000632482758586	44.48737759	0.5025725942	0.2108258762	0.007378965517	0.006324827586	1.148434098
64	3061.293522	0.000632482758586	45.01640829	0.5382573775	0.2108258762	0.007378965517	0.006324827586	1.148637455

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
64	3065.071288	0.000632482758 6	45.735830833	0.507356416 6	0.210825758 62	0.007378965517	0.006324827586	1.148844411
90	3748.844284	0.000632482758 6	43.063913333	0.382153072 9	0.210825758 62	0.007378965517	0.006324827586	-2.456797102
90	3752.623144	0.000632482758 6	43.786491074	0.426524406 6	0.210825758 62	0.007378965517	0.006324827586	-0.635039805
90	3756.401079	0.000632482758 6	42.513857074	0.429351561 69	0.210825758 62	0.007378965517	0.006324827586	-0.634832847
90	3763.957875	0.000632482758 6	43.323017366	0.413767679 76	0.210825758 62	0.007378965517	0.006324827586	1.187131414

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
90	3767.734375	0.0006324827586	42.523883723	0.541560059	0.210820758622	0.007378965517	0.0632482758	-0.6342119736
90	3771.512141	0.0006324827586	43.605270234	0.486454982	0.210820758622	0.007378965517	0.0632482758	-0.6340050165

Measured fitted secondary phase at its reference epoch=0.4999691; first-order e cos(omega) in primary-to-secondary convention=-0.00005 +/- 0.00010 (conservative sum of reference-time and period errors; no full ephemeris cross-covariance), Gaia same-convention=-0.03180. This first-order estimator is approximate for eccentric or grazing systems; exact Gaia conjunction-phase comparison above is preferred. Duration ratio=1.000; grazing/partial chords prohibit direct trapezoid-duration conversion into e sin(omega). Own limb-darkened two-sphere feasibility result: {"status": "conditional shape feasibility", "best": {"chi2": 4207.017547844852, "dof": 121, "k": 0.29999999999999993, "b_A": 0.935, "e_sin_omega": 0.0749999999999999, "primary_scale": 1.0, "secondary_scale": 0.16860668693300973}, "profile_interval_approx_95": {"k": [0.29999999999999993, 0.32499999999999996], "b_A": [0.935, 0.97], "e_sin_omega": [0.0749999999999999, 0.0749999999999999]}, "error_inflation": 34.76874006483349, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. These coarse fixed-TIC profiles are not calibrated orbital uncertainties.

Independent reduction and source attribution

No exact-target alternate-pipeline light curve covering an event in the shared MAST inventory. An independently extracted pixel aperture supplies the reduction check; both kinds were checked in targeted-aperture.json. This is the same observatory and pixels. Own hashed pixel regression sector64, 7 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset0.00arcsec; target residual fraction0.144. Every event favors['5245490498321111552', '5245490498321111552', '5245490498321111552', '5245490498321111552', '5245490498321111552', '5245490498321111552', '5245490498321111552']. Fixed source comparisons: [{"source_id": "5245490498321111552", "sep_arcsec": 0.0, "G": 11.395722, "chi2": 14621.215927788628, "amplitude": 146.5515354451254, "fractional_residual": 0.14359548353666207}, {"source_id": "5245490498321099648", "sep_arcsec": 22.341921808683605, "G": 14.277299, "chi2": 127533.11723794657, "amplitude": 43.345997175377335, "fractional_residual": 0.8692859797680765}]. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are

retained. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "377656542", "sector": 64, "n_events": 13, "deficit_minus_target_arcsec": 1.98, "deficit_minus_light_arcsec": 0.48, "peak_on_target": true, "depth_ap3": 0.04155, "psf_frac_resid": 0.499, "n_neighbours_60": 138, "n_possible": 1, "brightest_nb": {"G": 14.277299, "sep": 22.3}}]. Own Gaia<15 cone neighbor table: [{"source_id": "5245490498321099648", "ra": "154.58268915356837", "dec": "-65.55639206555605", "phot_g_mean_mag": "14.277299", "bp_rp": "1.4797668", "sep": "22.341921808683605", "required_fraction_G_approx": 0.7154075864736388, "capable": true}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 6, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 5245490498321111552", "main_id": "TYC 8964-629-1", "otype": "SB*", "ra": "154.59674877562", "dec": "-65.55409741854"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.19, "m2_min": 0.9803942689762861, "f_mass": 0.20004376678090383, "a_rsun": 13.21598032597276, "r1_rsun": 3.22697, "r2_approx_rsun": 0.984284420615829, "central_T14_hours": {"A": 8.990657390389924, "B": 9.396410892943438}, "observed_T14_hours": 5.059862068965517}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindB, other kindA. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains104 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=9.121ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Forty primary and39secondary fits establish two equal-depth kinds at full3.77777d; stronger half-period search is the expected ambiguity. Pixel apertures recover seven secondary events at44.8–45.7ppt linear or41.4–42.4ppt quadratic. The22.3arcsec capable neighbor is strongly disfavored for every primary. Gaia small eccentricity gives a3.3sigma primary-epoch tension, while period agrees; observed near-half-cycle separation favors a more circular orbit. Coarse fixed-TIC geometry is a poor characterization fit (chi2~4207/121, scale at boundary), not a calibrated radius/eccentricity posterior.

Next nominal eclipses after2026-10-30T00UTC: primary4343.84419BTJD; other4345.73247BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R036. TIC 194609413

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 194609413: CONFIRMED attributed eclipsing system

Four primary timings, independent reduction and target-localized pixel deficit reject the48.7arcsec capable neighbor. No reliable secondary; phase/width limits and a feasible grazing eccentric geometry allow it to miss. The faint-companion main-sequence radius used in the feasibility test is an assumption.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2293448276d, profile range[0.22391379310344828, 0.2347758620689655], ingress fraction0.35 (T23 conditional, not proof of totality). Median primary depth49.663ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=15.33077163 ±1.422462442e-05d, Tref=2801.221223 ±0.0005419706051 BTJD, cycle reference65. T0=1804.721067. chi2/dof=0.7606137083833822/2; O-C RMS=0.7535575288min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
18	1804.72112	0.001813326333	50.32674284	0.9793934833	0.2293448276	0.08027068966	0.06880344828	0.0756225847	1	0	0.07480195953	50.35776859
58	2893.20477	0.001376068966	49.82140248	1.716360903	0.2293448276	0.08027068966	0.06880344828	-1.560034177	2.661764321	0	-0.01079955124	49.83162346
58	2908.536882	0.0006880344828	49.50444892	1.007114004	0.2293448276	0.08027068966	0.06880344828	0.3705587103	2.208867982	0	0.01719676331	49.49397313
85	3613.752126	0.0006880344828	48.97967369	1.123209391	0.2293448276	0.08027068966	0.06880344828	0.008562537114	2.186355664	0	-0.1534619076	49.11212174

Own Gaia source2361113646290323 20, SB1, bit_index127, w–Tp correlation0.99326. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=15.330110533129895±0.0014221929d, e=0.2834685700782087±0.02075948, omega=140.95550528175605±6.9377356deg. Photometric P difference=0.4648210472sigma. Primary

attributed toA; propagated epoch389 offset=-0.17262±0.09646d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.3587515; optimized diagnostic phase=0.3500174. Fit statistic: {"n": 5, "depth": 0.0001359262967476486, "error": 0.000517680569349946, "snr": 0.2625678937850264, "scatter": 0.0012451265499708918, "chi2": 5.030586216345322, "dof": 4, "empirical_error": 0.0005568375212649405, "n_individual_gt3sigma": 0}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `null`.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
18	1794.746 93	0.042 734 482 76	1.38 036 34 21	1.88 731 04 99	0.142 448 259	0.011 395 862 07	0.119 656 517	0
18	1810.046 078	0.042 734 482 76	0.65 659 335 73	0.85 852 136 64	0.142 448 259	0.011 395 862 07	0.119 656 517	0
58	2898.612 914	0.042 734 482 76	-1.25 246 181 8	1.63 785 761 3	0.142 448 259	0.011 395 862 07	0.119 656 517	0
85	3619.094 225	0.035 042 275 86	1.29 413 506	1.10 840 543 9	0.142 448 259	0.011 395 862 07	0.119 656 517	0
85	3634.449 782	0.009 401 586 207	-1.49 649 981 6	1.04 898 933 25	0.142 448 259	0.011 395 862 07	0.119 656 517	0
No robust independent secondary period/e cos omega is claimed; geometry follows the null test or conditional model, not a noisy optimized phase.								
Secondary null tests: own covariance relative-phase sigma=0.019043, 5widths×9phases spanning±2sigma, pointwise conditional3sigma upper ranges[0.895, 2.461]ppt. Full coverage/excluded windows in null-secondary.json; upper limits are phase/shape/baseline conditional and not a simultaneous bound. Feasibility: [{"radius_ratio_assumption": 0.29521287584129763, "b_primary_duration_approx": 1.0416648030839886, "b_other": 1.49453829597801, "contact": 1.2952128758412975, "other_misses": true, "es_gaia": 0.17856357523635008, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}].								

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector58; 2 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At2893.204770: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04568931360906524, "error": 0.0012451126115605173, "red_noise_beta": 2.4626294421320774, "n_core": 99, "baseline_degree": 1, "background_regressed_depth": 0.04611601682053894, "background_template_native_sigma": 0.19228093896855375}, "2": {"depth": 0.046244665469498604, "error": 0.0013683519292022835, "red_noise_beta": 2.4528454438616336, "n_core": 99, "baseline_degree": 2, "background_regressed_depth": 0.04624962893105362, "background_template_native_sigma": 0.01351184524165255}}, "PDCSAP_FLUX": {"1": {"depth": 0.04677161877102125, "error": 0.0013704418129170915, "red_noise_beta": 2.546771604586282, "n_core": 99, "baseline_degree": 1, "background_regressed_depth": 0.04740609924199172, "background_template_native_sigma": 0.19228093896855375}, "2": {"depth": 0.0477878122140029, "error": 0.0014611676412593882, "red_noise_beta": 2.483193680236721, "n_core": 99, "baseline_degree": 2, "background_regressed_depth": 0.047765157194760745, "background_template_native_sigma": 0.01351184524165255}}}. At2908.536882: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.046127367246634146, "error": 0.000937783117793986, "red_noise_beta": 2.4720743223286785, "n_core": 99, "baseline_degree": 1, "background_regressed_depth": 0.046140170525830325, "background_template_native_sigma": 0.6633361209376647}, "2": {"depth": 0.04603289034911997, "error": 0.0010386912660850778, "red_noise_beta": 2.4714448702861675, "n_core": 99, "baseline_degree": 2, "background_regressed_depth": 0.04604501404987372, "background_template_native_sigma": 0.7285965201619152}}, "PDCSAP_FLUX": {"1": {"depth": 0.04840961543714131, "error": 0.0009815500437451972, "red_noise_beta": 2.474793485423876, "n_core": 99, "baseline_degree": 1, "background_regressed_depth": 0.04841638770535483, "background_template_native_sigma": 0.6633361209376647}, "2": {"depth": 0.048267479068228585, "error": 0.0010866411916904892, "red_noise_beta": 2.473579724908912, "n_core": 99, "baseline_degree": 2, "background_regressed_depth": 0.04827197461493002, "background_template_native_sigma": 0.7285965201619152}}}. Own hashed pixel regression sector58, 2 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset2.10arcsec; target residual fraction0.157. Every event favors['236111364629032320', '236111364629032320']. Fixed source comparisons: [{"source_id": "236111364629032320", "sep_arcsec": 0.0, "G": 10.448637, "chi2": 14479.26317431367, "amplitude": 217.837406238467, "fractional_residual": 0.15748920206625192}, {"source_id": "236111433348507520", "sep_arcsec": 48.704068653209035, "G": 13.402758, "chi2": 81717.09846081014, "amplitude": 8.487074886500297, "fractional_residual": 0.9951774868907829}, {"source_id": "236111467708246272", "sep_arcsec": 53.82557815159796, "G": 14.619842, "chi2": 82978.76912931936, "amplitude": 3.972009677453152, "fractional_residual": 0.9965273587480183}, {"source_id": "236108409689820416", "sep_arcsec": 59.30724295823588, "G": 14.427484, "chi2": 83253.42133109302, "amplitude": 2.6034868259217827, "fractional_residual": 0.9968653621759151}]. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are retained. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "194609413", "sector": 58, "n_events": 2, "deficit_minus_light_arcsec": 1.61, "peak_on_target": true, "depth_ap3": 0.04459, "psf_frac_resid": 0.208, "n_neighbours_60": 20, "n_possible": 1, "brightest_nb": {"G": 13.402758, "sep": 48.7}}, {"tic": "194609413", "sector": 58, "n_events": 2, "deficit_minus_target_arcsec": 3.27, "deficit_minus_light_arcsec": 1.61, "peak_on_target": true, "depth_ap3": 0.04459,

"psf_frac_resid": 0.208, "n_neighbours_60": 20, "n_possible": 1, "brightest_nb": {"G": 13.402758, "sep": 48.7}}]. Own Gaia<15 cone neighbor table: [{"source_id": "236111433348507520", "ra": "50.80123041950704", "dec": "40.26433237171349", "phot_g_mean_mag": "13.402758", "bp_rp": "0.94339085", "sep": "48.704068653209035", "required_fraction_G_approx": 0.7545372775140392, "capable": true}, {"source_id": "236111467708246272", "ra": "50.82301039580128", "dec": "40.25542018390916", "phot_g_mean_mag": "14.619842", "bp_rp": "0.78073883", "sep": "53.82557815159796", "required_fraction_G_approx": 2.3148044081402293, "capable": false}, {"source_id": "236108409689820416", "ra": "50.825121710525146", "dec": "40.24668675783259", "phot_g_mean_mag": "14.427484", "bp_rp": "0.8015518", "sep": "59.30724295823588", "required_fraction_G_approx": 1.9389700493839084, "capable": false}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 236111364629032320", "main_id": "HD 275280", "otype": "SB*", "ra": "50.80433170988999", "dec": "40.25103565251"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.57, "m2_min": 0.7057784868363401, "f_mass": 0.06788053765359348, "a_rsun": 34.15948189937553, "r1_rsun": 2.5633, "r2_approx_rsun": 0.7567191646439982, "central_T14_hours": {"A": 9.261753064290565, "B": 13.288386630221835}, "observed_T14_hours": 5.504275862068965}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains251 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.153ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Four primary timings, independent reduction and target-localized pixel deficit reject the48.7arcsec capable neighbor. No reliable secondary; phase/width limits and a feasible grazing eccentric geometry allow it to miss. The faint-companion main-sequence radius used in the feasibility test is an assumption.

Next nominal eclipses after2026-10-30T00UTC: primary4349.62916BTJD; other4354.99519BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R037. TIC 323434013

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 323434013: CONFIRMED attributed eclipsing system

Gaia is SB2, not SB1: own bit255/21correlation ordering was verified against ESA documentation. Sixty-five fits of each kind, equal depths, alternate reduction and tight target deficit establish the EB at full3.071615d. SB2 amplitudes independently yield q0.9046, M1sin3i1.19244solar, M2sin3i1.07870solar and a sini1.6887Rsun. Fixed-TIC geometry fits poorly (chi2~1314/121); precise component radii and e sin omega are not inferred.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.1276448276d, profile range[0.12520086206896552, 0.13008879310344829], ingress fraction0.35 (T23 conditional, not proof of totality). Median primary depth50.345ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=3.071615251 ±2.854267142e-07d, Tref=2792.753784 ±6.330471416e-05 BTJD, cycle reference379. T0=1628.611603. chi2/dof=45.176890449595874/63; O-C RMS=0.5203017362min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
12	1634.754493	0.002455258272	48.90347802	0.4033689232	0.1276448276	0.04467568966	0.03829344828	-0.4908703201	1	0	-0.002303446448	49.39268934
12	1643.969334	0.002455258272	49.36732699	0.3866860205	0.1276448276	0.04467568966	0.03829344828	-0.498049519	1	0	-0.2382290344	49.77753914
12	1647.041713	0.002455258272	49.57163897	0.3195615959	0.1276448276	0.04467568966	0.03829344828	0.6024087253	1	0	-0.175941349	49.73872848
12	1650.112561	0.002455258272	50.06142233	0.4078008299	0.1276448276	0.04467568966	0.03829344828	-0.5028356512	1	0	-0.2007415937	50.01238822
13	1656.255788	0.002455218738	50.24030903	0.4941838232	0.1276448276	0.04467568966	0.03829344828	-0.5076217837	1.05006011	0	-0.4888417516	50.6146822
13	1659.327402	0.002273090857	50.07507201	0.5619332747	0.1276448276	0.04467568966	0.03829344828	-0.5100148498	1.02693219	0	0.03028817763	50.04854141

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
13	1662.399015	0.002455218738	49.69261274	0.3791001568	0.1276448276	0.04467568966	0.03829344828	-0.5124079162	1.079422296	0	0.1355789242	49.533136 25
13	1665.470629	0.002273090857	49.76454082	0.4653586631	0.1276448276	0.04467568966	0.03829344828	-0.514800982	1	0	0.0447824241	49.73246953
13	1671.613856	0.002455218738	49.39567692	0.3783303859	0.1276448276	0.04467568966	0.03829344828	-0.5195871145	1.050729374	0	-0.3572832038	49.43214177
13	1674.686236	0.002455218738	49.66022826	0.3185304599	0.1276448276	0.04467568966	0.03829344828	0.5808711295	1	0	-0.3187589 42	49.87250048
13	1677.757083	0.002455218738	49.70979739	0.3516534365	0.1276448276	0.04467568966	0.03829344828	-0.5243732471	1.007911462	0	-0.02870551791	49.78071879
13	1680.829463	0.002455218738	49.70040685	0.3797023794	0.1276448276	0.04467568966	0.03829344828	0.5760849973	1	0	-0.1608434416	49.79583729
27	2037.136639	0.0004725037233	50.24741415	0.3543557534	0.1276448276	0.04467568966	0.03829344828	0.2984893185	1.152078057	0	-0.1085633902	50.2274636
27	2040.208253	0.0004599013529	50.28542349	0.3772267222	0.1276448276	0.04467568966	0.03829344828	0.2960962521	1.254778592	0	0.3634682133	49.89353956
27	2043.279866	0.0004725037233	50.32224931	0.3139909596	0.1276448276	0.04467568966	0.03829344828	0.293703186	1.042063861	0	0.0481616413 5	50.26785117
27	2046.35148	0.0004725037233	50.6088402	0.4628057262	0.1276448276	0.04467568966	0.03829344828	0.2913101196	1.439421774	0	-0.2335705547	50.67479261
27	2049.423094	0.0004599013529	50.58311388	0.3737845039	0.1276448276	0.04467568966	0.03829344828	0.2889170535	1.18951195	0	-0.1932304952	50.73789981
27	2052.494707	0.0004725037233	50.42620442	0.3669049199	0.1276448276	0.04467568966	0.03829344828	0.2865239874	1.224424433	0	-0.1718485643	50.39240491
27	2055.566321	0.0004725037233	50.74435296	0.3136158189	0.1276448276	0.04467568966	0.03829344828	0.2841309206	1.065343908	0	0.0041524196 38	50.73917741
27	2058.637934	0.0004599013529	51.107121 38	0.3570093942	0.1276448276	0.04467568966	0.03829344828	0.2817378552	1.194016121	0	0.6627314764	50.69428907
39	2362.772768	0.0004725139862	50.46688516	0.3580738535	0.1276448276	0.04467568966	0.03829344828	0.04482430108	1.191073156	0	-0.0307470552 8	50.49051212
39	2365.799293	0.0004599113421	50.51596545	0.3806000295	0.1276448276	0.04467568966	0.03829344828	0.04243123498	1.375103959	0	-0.1599380247	50.53650459
39	2368.870907	0.0004725139862	50.66202839	0.4484193794	0.1276448276	0.04467568966	0.03829344828	0.04003816823	1.447804429	0	0.0270157762 1	50.63142377
39	2371.94252	0.0004725139862	50.57247825	0.3928893446	0.1276448276	0.04467568966	0.03829344828	0.03764510278	1.433277413	0	0.0010781236 51	50.57342262
39	2378.085748	0.0004725139862	50.69979388	0.462820932	0.1276448276	0.04467568966	0.03829344828	0.03285897059	1.423019803	0	0.0892787161 3	50.60305134
39	2381.156595	0.0004862128806	50.34467489	0.3996417894	0.1276448276	0.04467568966	0.03829344828	-1.072385406	1.249723638	1	0.0238350952 1	50.27493053
39	2384.228975	0.0004599113421	50.42410654	0.4138708757	0.1276448276	0.04467568966	0.03829344828	0.02807283774	1.374201715	0	0.1679391219	50.17642929
65	3069.199571	0.0003829344828	49.19814262	0.428048371	0.1276448276	0.04467568966	0.03829344828	0.5972703862	1.889906689	0	-0.1255686835	49.33144018
65	3072.271185	0.0003829344828	49.52029055	0.4865837588	0.1276448276	0.04467568966	0.03829344828	0.5948773201	1.957704078	0	-0.0951156839	49.66655409
65	3075.342798	0.0003829344828	50.45501433	0.3927125673	0.1276448276	0.04467568966	0.03829344828	0.5924842534	1.756725803	0	-0.1646365202	50.62072397
65	3078.413646	0.0003829344828	50.81547162	0.4280452227	0.1276448276	0.04467568966	0.03829344828	-0.5127601232	1.829391501	0	0.0423469025 5	50.75654772
65	3081.486025	0.0003829344828	50.39089083	0.3703884968	0.1276448276	0.04467568966	0.03829344828	0.5876981212	1.606229969	0	-0.5369061551	50.8242649
65	3084.557639	0.0003829344828	50.89368209	0.3962900907	0.1276448276	0.04467568966	0.03829344828	0.5853050544	1.900887478	0	-0.1775778349	50.99301453
65	3087.629253	0.0003829344828	50.70013276	0.4329839128	0.1276448276	0.04467568966	0.03829344828	0.5829119896	1.908148978	0	-0.0704376193	50.75888263

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
65	3090.7001	0.0003829344828	50.42170929	0.3476312443	0.1276448276	0.04467568966	0.03829344828	-0.5223323876	1.490173554	0	-0.07025032228	50.45846902
65	3093.77248	0.0003829344828	50.17944968	0.427861856	0.1276448276	0.04467568966	0.03829344828	0.5781258568	1.880668328	0	-0.2463107374	50.41574633
66	3099.915707	0.0003829344828	50.95497526	0.3682203549	0.1276448276	0.04467568966	0.03829344828	0.5733397239	1.557266996	0	-0.07700584132	51.0271977
66	3102.986555	0.0003829344828	51.38885522	0.3675827145	0.1276448276	0.04467568966	0.03829344828	-0.5319046526	1.598414547	0	0.06473322125	51.32628919
66	3106.058168	0.0003829344828	51.4904485	0.3387057665	0.1276448276	0.04467568966	0.03829344828	-0.5342977187	1.485498605	0	-0.0674407519	51.27509103
66	3109.129782	0.0003829344828	51.5107062	0.368883015	0.1276448276	0.04467568966	0.03829344828	-0.5366907854	1.45995641	0	-0.09059294216	51.32479646
66	3115.273009	0.0003829344828	51.22042453	0.3938354027	0.1276448276	0.04467568966	0.03829344828	-0.5414769176	1.629970968	0	-0.01053838484	51.23421247
66	3118.344623	0.0003829344828	51.219669655	0.477692306	0.1276448276	0.04467568966	0.03829344828	-0.5438699837	1.834310296	0	-0.0122836196	51.23354203
66	3121.416236	0.0003829344828	50.97408171	0.3384386454	0.1276448276	0.04467568966	0.03829344828	-0.5462630492	1.430012093	0	0.0528457154	50.819228826
67	3127.559463	0.0003829344828	51.635445016	0.3833933556	0.1276448276	0.04467568966	0.03829344828	-0.551049182	1.702086266	0	-0.03647502482	51.676823333
67	3130.631077	0.0003829344828	51.156122103	0.5115466773	0.1276448276	0.04467568966	0.03829344828	-0.5534422481	2.11631649	0	-0.04366814507	51.2031952
67	3133.703457	0.0003829344828	51.4575868	0.4240370848	0.1276448276	0.04467568966	0.03829344828	0.5470159955	1.919794177	0	-0.055961327	51.5118166
67	3136.774304	0.0003829344828	51.40388538	0.5040732512	0.1276448276	0.04467568966	0.03829344828	-0.5582283803	2.074211166	0	0.2052307441	51.187115135
67	3142.917531	0.0003829344828	50.95223061	0.3662407151	0.1276448276	0.04467568966	0.03829344828	-0.5630145132	1.615485442	0	-0.01705534147	50.9745709
67	3145.989145	0.0003829344828	50.88166683	0.4086896691	0.1276448276	0.04467568966	0.03829344828	-0.5654075793	1.540710082	0	-0.03434854166	50.914888015
67	3149.060759	0.0003829344828	51.170007689	0.384769506	0.1276448276	0.04467568966	0.03829344828	-0.5678006454	1.705060834	0	0.0401692821414	51.120916821
67	3152.132372	0.0003829344828	50.41233849	0.4146106534	0.1276448276	0.04467568966	0.03829344828	-0.5701937121	1.694049928	4	-0.03402972965	50.32246517
93	3830.960507	0.0003829344828	48.20473248	0.4546503619	0.1276448276	0.04467568966	0.03829344828	1.10664128	1.937136925	0	0.07341934709	48.135477781
93	3834.031355	0.0003829344828	48.48611462	0.37879211	0.1276448276	0.04467568966	0.03829344828	0.001396901789	1.6006838	0	-0.02030347402	48.514377746
93	3837.103734	0.0003829344828	48.65020194	0.3630749711	0.1276448276	0.04467568966	0.03829344828	1.101855147	1.527231963	0	-0.02051758334	48.67379704
93	3840.175348	0.0003829344828	49.55015503	0.3721186158	0.1276448276	0.04467568966	0.03829344828	1.09946208	1.569235281	0	-0.01119011309	49.5333137
93	3846.317809	0.0003829344828	49.63619922	0.3577743839	0.1276448276	0.04467568966	0.03829344828	-0.008175361945	1.555046619	0	-0.04664942482	49.70040675
93	3849.389423	0.0003829344828	49.8922918	0.3845613939	0.1276448276	0.04467568966	0.03829344828	-0.0105684287	1.653593016	1	-0.01895527156	49.93277006
93	3852.461036	0.0003829344828	50.32628986	0.3888436252	0.1276448276	0.04467568966	0.03829344828	-0.0129614949414	1.606541817	0	-0.06839641248	50.43553244
94	3858.604264	0.0003829344828	44.76068108	0.3073557008	0.1276448276	0.04467568966	0.03829344828	-0.01774762764	1.566292387	0	0.0003533966864	44.75942872
94	3861.675877	0.0003829344828	44.96222888	0.3023833755	0.1276448276	0.04467568966	0.03829344828	-0.02014069309	1.507133071	0	-0.004158534555	44.974891
94	3864.747491	0.0003829344828	45.111660925	0.3186851691	0.1276448276	0.04467568966	0.03829344828	-0.02253375918	1.634774949	0	0.006293494489	45.10319119
94	3870.890718	0.0003829344828	45.09383553	0.361779957	0.1276448276	0.04467568966	0.03829344828	-0.02731989138	1.668443445	0	0.0157174390606	45.07841004

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
94	3873.962332	0.0003829344828	45.03023413	0.3415725424	0.1276448276	0.04467568966	0.03829344828	-0.02971295879	1.50297962	19	0.0418507208	44.9978885
94	3877.033945	0.0003829344828	44.88553533	0.2836105785	0.1276448276	0.04467568966	0.03829344828	-0.03210602357	1.478052557	0	-0.01227000429	44.88971971
94	3880.104793	0.0003829344828	44.7548868	0.3684044711	0.1276448276	0.04467568966	0.03829344828	-1.137350401	1.761035966	2	-0.002389322133	44.75524598

Own Gaia source4614246265612443136, SB2, bit_index255, w–Tp correlation0.99968. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=3.071649485331793±5.8143956E-5d, e=0.02040543100811207±0.01071757, omega=26.77193835734906±18.379349deg. Photometric P difference=-0.5887709616sigma. Primary attributed toA; propagated epoch389 offset=0.01213±0.00864d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.5115979; optimized diagnostic phase=0.4999621. Fit statistic: {"n": 65, "depth": 0.04669163530518724, "error": 4.460431787412611e-05, "snr": 1046.7963087553892, "scatter": 0.0014487196095831502, "chi2": 1195.9576818993694, "dof": 64, "empirical_error": 0.00017969155226882775, "n_individual_gt3sigma": 65}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: {"P": 3.071615025371487, "P_error": 2.9312477119348145e-07, "Tref": 2735.9289403519183, "Tref_error": 6.524732211279344e-05, "reference_cycle": 360, "T0": 1630.1475312181828, "chi2": 26.462864400374787, "dof": 63, "n": 65, "oc_rms_minutes": 0.5100512728148995}.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
12	1630.147295	0.002455258272	45.93088051	0.2247194208276	0.1276448276	0.04467568966	0.03829344828	-0.3406703406
12	1633.218144	0.002455258272	45.89623391	0.3518901	0.1276448276	0.04467568966	0.03829344828	-1.443196103

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
12	1636.290525	0.000307064977	44.6880390404	0.28993049054	0.12764482766	0.004467568966	0.0038293448288	-0.3400192448
12	1642.433756	0.000268960068	46.1107028485	0.37842905022	0.12764482766	0.004467568966	0.0038293448288	-0.3393681487
12	1645.505371	0.000245582572	46.1470782828	0.29253823288	0.12764482766	0.004467568966	0.0038293448288	-0.3390426012
12	1648.576986	0.000245582572	46.4119862525	0.41872178833	0.12764482766	0.004467568966	0.0038293448288	-0.3387170533
12	1651.646801	0.000245582572	46.5878011414	0.33312934566	0.12764482766	0.004467568966	0.0038293448288	-0.3383915051

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
13	1663.935062	0.000245521 87838	47.7378204 9	0.4265508519	0.1276448276	0.0044675689 66	0.0382934 4828	-0.3370893136
13	1667.005912	0.000245521 87838	47.9236546 6	0.4189800736	0.1276448276	0.0044675689 66	0.0382934 4828	-1.4396150 76
13	1670.078293	0.000245521 87838	47.4248650 9	0.3923339901 3	0.1276448276	0.0044675689 66	0.0382934 4828	-0.3364382175
13	1673.149142	0.000268955 7373	46.3161779 9	0.3226374231	0.1276448276	0.0044675689 66	0.0382934 4828	-1.4389639 8
13	1676.221523	0.000227309 0857	47.1175289 6	0.4211340349	0.1276448276	0.0044675689 66	0.0382934 4828	-0.3357871221

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
13	1679.292373	0.000245521 8738	46.4524970 4	0.3189366 82	0.1276448 2766	0.0446756 8966	0.0382934 828	-1.4383 12884
27	2038.672889	0.000459 9013529	47.8018173 5	0.3669573 2766	0.1276448 2766	0.0446756 8966	0.0382934 828	0.8054 7884 25
27	2041.744504	0.000472 5037233	48.0974 5031	0.3930683 027	0.1276448 2766	0.0446756 8966	0.0382934 828	0.8058 0439 04
27	2044.815354	0.000472 5037233	48.3668 1338	0.3409627 185	0.1276448 2766	0.0446756 8966	0.0382934 828	-0.2967 2137 21
27	2047.886969	0.000459 9013529	48.5860 3366	0.3632989 603	0.1276448 2766	0.0446756 8966	0.0382934 828	-0.2963 9582 39

secondary sector	midBTJD	error d	depth ppt	errorppt	T14d	ingress d	T23d	O-Cmin
27	2050.958584	0.000472503723	47.976445222	0.427823	0.127644827	0.044675689664	0.038293448	-0.2960702764
27	2054.030199	0.000472503723	48.142555563	0.321772	0.127644827	0.044675689664	0.038293448	-0.2957447285
27	2057.101815	0.000459901352	48.237073152	0.424522	0.127644827	0.044675689664	0.038293448	-0.2954191806
27	2060.173433	0.000486202301	46.937682644	0.376365	0.127644827	0.044675689664	0.038293448	-0.2950936327
39	2364.263434	0.000472513986	47.525124333	0.341969	0.127644827	0.044675689664	0.038293448	-0.2628643886

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
39	2367.334955	0.000472513982	47.963239599	0.372555613688	0.127644827676	0.044675689664	0.038293448288	-0.262538840808
39	2370.407336	0.000472513982	48.356459884	0.405053523176	0.127644827676	0.044675689664	0.038293448288	0.840638017575
39	2373.478185	0.000459913421	48.535730340	0.425705980807	0.127644827676	0.044675689664	0.038293448288	-0.261887745
39	2376.549801	0.000472513982	48.333585853	0.430761512	0.127644827676	0.044675689664	0.038293448288	-0.261562619771
39	2379.621416	0.000472513982	48.056997676	0.366293000996	0.127644827676	0.044675689664	0.038293448288	-0.261236364699

secondary sector

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
39	2385.764646	0.000472513986 2	47.250647 47	0.3431059709	0.1276448276	0.0446756896 6	0.0382934482 8	-0.2605855535
39	2388.836262	0.000486212880 6	46.613004 43	0.4160001411	0.1276448276	0.0446756896 6	0.0382934482 8	-0.2602600056
65	3070.734848	0.000382934482 8	47.332316 57	0.4479107866	0.1276448276	0.0446756896 6	0.0382934482 8	-0.1879883676
65	3073.806463	0.000382934482 8	46.902218 2	0.3867914036	0.1276448276	0.0446756896 6	0.0382934482 8	-0.1876628204
65	3076.878078	0.000382934482 8	46.925473 28	0.3949768584	0.1276448276	0.0446756896 6	0.0382934482 8	-0.1873372725

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
65	3079.94969 3	0.00038293448288	46.975735999	0.43241373948	0.12764482766	0.04467568966	0.03829344828	-0.1870117246
65	3086.09292 24	0.00038293448288	45.967407210	0.43255044676	0.12764482766	0.04467568966	0.03829344828	-0.1863606289
65	3089.16453 9	0.00038293448288	45.829749191	0.39530607388	0.12764482766	0.04467568966	0.03829344828	-0.1860350801
65	3092.23615 4	0.00038293448288	45.772513	0.42039688607	0.12764482766	0.04467568966	0.03829344828	-0.1857095325
65	3095.30777	0.00038293448288	46.001291515	0.37574334451	0.12764482766	0.04467568966	0.03829344828	-0.1853839852

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
66	3098.379385	0.000382934482	46.8917121	0.38406168	0.12764482	0.04467568	0.03829344	-0.18505843
66	3101.451	0.000382934482	47.0775976	0.39890136	0.12764482	0.04467568	0.03829344	-0.18473288
66	3104.523381	0.000382934482	46.7042867	0.41611697	0.12764482	0.04467568	0.03829344	0.9184439688
66	3107.594231	0.000382934482	47.3719037	0.31038680	0.12764482	0.04467568	0.03829344	-0.18408179
66	3110.666612	0.000382934482	47.3804429	0.47341341	0.12764482	0.04467568	0.03829344	0.9191909

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
66	3113.737461	0.000382934482	47.2930942	0.331953367	0.1276448276	0.044675689664	0.038293448	-0.183430697
66	3116.808076	0.000382934482	47.2876157	0.403616090	0.1276448276	0.044675689664	0.038293448	-0.183105149
66	3119.880692	0.000382934482	47.137447	0.330464710	0.1276448276	0.044675689664	0.038293448	-0.182779601
66	3122.953073	0.000382934482	47.2271520	0.393895648	0.1276448276	0.044675689664	0.038293448	0.9203972561
67	3129.096303	0.000382934482	47.514355	0.423843647	0.1276448276	0.044675689664	0.038293448	0.9210483525

secondary sector

secondary sector	mid BTJD	error d	depth ppt	err or ppt	T1 4d	ingress d	T2 3d	O-C min
67	3132.1679918	0.000382934482828	47.765608633	0.371733864	0.127644827676	0.044675689660	0.038293448280	0.921373901 1
67	3135.2395334	0.000382934482828	47.394431 5	0.388897721 3	0.127644827676	0.044675689660	0.038293448280	0.921699483
67	3141.3819198	0.000382934482828	47.063148 48	0.334706399 4	0.127644827676	0.044675689660	0.038293448280	-0.180500 7664
67	3144.4536144	0.000382934482828	47.446949 76	0.401974158 58	0.127644827676	0.044675689660	0.038293448280	-0.180175 2185
67	3147.5252299	0.000382934482828	48.006178 7	0.581709641 41	0.127644827676	0.044675689660	0.038293448280	-0.179849 6699

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
67	3150.596844	0.000382934482	48.142491916	0.2731500245	0.1276448276	0.0446756896	0.0382934482	-0.17952412 27
93	3832.49543	0.000382934482	47.476175 8	0.3143658521	0.1276448276	0.0446756896	0.0382934482	-0.10725248 61
93	3835.567045	0.000382934482	47.412351 6	0.3260179526	0.1276448276	0.0446756896	0.0382934482	-0.10692693 69
93	3838.638666	0.000382934482	47.562739	0.3331249889	0.1276448276	0.0446756896	0.0382934482	-0.10660138 96
93	3844.781891	0.000382934482	46.957306 92	0.3553295594	0.1276448276	0.0446756896	0.0382934482	-0.10595029 32

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
93	3847.853506	0.000382934482	47.065527	0.327965650	0.127644827	0.044675689	0.038293448	-0.10562474
93	3850.925121	0.000382934482	47.042906	0.318496267	0.127644827	0.044675689	0.038293448	-0.10529919
93	3853.996737	0.000382934482	47.030997	0.362601274	0.127644827	0.044675689	0.038293448	-0.10497364
94	3860.139967	0.000382934482	42.371461	0.331462411	0.127644827	0.044675689	0.038293448	-0.10432255
94	3863.211582	0.000382934482	42.381061	0.346626598	0.127644827	0.044675689	0.038293448	-0.10399700

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
94	3866.283198	0.000382934828	42.5510782	0.33409584276	0.12764482766	0.044675689664	0.038293448288	-0.10367145810
94	3872.426428	0.000382934828	42.6623741	0.37769068644	0.12764482766	0.044675689664	0.038293448288	-0.10302036230
94	3878.569659	0.000382934828	42.975818	0.26222704584	0.12764482766	0.044675689664	0.038293448288	-0.10236926650

Measured fitted secondary phase at its reference epoch=0.5000126; first-order e cos(omega) in primary-to-secondary convention=0.00002 +/- 0.00007 (conservative sum of reference-time and period errors; no full ephemeris cross-covariance), Gaia same-convention=0.01822. This first-order estimator is approximate for eccentric or grazing systems; exact Gaia conjunction-phase comparison above is preferred. Duration ratio=1.000; grazing/partial chords prohibit direct trapezoid-duration conversion into e sin(omega). Own limb-darkened two-sphere feasibility result: {"status": "conditional shape feasibility", "best": {"chi2": 1314.4326882021792, "dof": 121, "k": 0.32499999999999996, "b_A": 0.97, "e_sin_omega": 0.04999999999999993, "primary_scale": 1.0, "secondary_scale": 0.15705399450566007}, "profile_interval_approx_95": {"k": [0.24999999999999997, 0.35], "b_A": [0.8300000000000001, 1.0050000000000001], "e_sin_omega": [0.04999999999999993, 0.09999999999999992]}, "error_inflation": 10.863080067786605, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. These coarse fixed-TIC profiles are not calibrated orbital uncertainties.

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector65; 9 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At3069.199571: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.047836890364784834, "error":

0.0005449104110206311, "red_noise_beta": 2.4465155932471534, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04751631333030296, "background_template_native_sigma": 0.3492614112735265}, "2": {"depth": 0.046861035449981145, "error": 0.0004070310126127413, "red_noise_beta": 1.914089121838743, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.046930233894490525, "background_template_native_sigma": 0.16515772666207684}}, "PDCSAP_FLUX": {"1": {"depth": 0.04932839405816213, "error": 0.0005666474910696327, "red_noise_beta": 2.4663377551040053, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.048982922034064964, "background_template_native_sigma": 0.3492614112735265}, "2": {"depth": 0.04830345414591452, "error": 0.00041690418948283654, "red_noise_beta": 1.912889176513531, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04837072102378518, "background_template_native_sigma": 0.16515772666207684}}}. At3072.271185: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.047969618939596496, "error": 0.000510168500629105, "red_noise_beta": 2.293372430977631, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04806772034864459, "background_template_native_sigma": 0.37332868446289796}, "2": {"depth": 0.047243459138779836, "error": 0.0004677489246836202, "red_noise_beta": 2.0489210946322, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.047300633528912166, "background_template_native_sigma": 1.0875456838364936}}, "PDCSAP_FLUX": {"1": {"depth": 0.04942613195774554, "error": 0.0005266990820771379, "red_noise_beta": 2.303092717107297, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.049529403825348345, "background_template_native_sigma": 0.37332868446289796}, "2": {"depth": 0.04868223811491831, "error": 0.00048341541619212846, "red_noise_beta": 2.058565269780652, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.048751145200185884, "background_template_native_sigma": 1.0875456838364936}}}. At3075.342798: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04831721496916941, "error": 0.000377477117704088, "red_noise_beta": 1.9717066900691864, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.048358750022202836, "background_template_native_sigma": 0.2513799979770865}, "2": {"depth": 0.04800478308917369, "error": 0.00041183559908773, "red_noise_beta": 1.9755570074125208, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04807636953134417, "background_template_native_sigma": 0.034481852874115995}}, "PDCSAP_FLUX": {"1": {"depth": 0.04978405389924306, "error": 0.0003917877854730695, "red_noise_beta": 1.9942706482853083, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.049841664650528174, "background_template_native_sigma": 0.2513799979770865}, "2": {"depth": 0.049474454488779, "error": 0.00042844275269566915, "red_noise_beta": 2.000298426084776, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04955033433887244, "background_template_native_sigma": 0.034481852874115995}}}. At3078.413646: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04874900038225 16, "error": 0.0004127705701956199, "red_noise_beta": 2.026637707251751, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04822983841117942, "background_template_native_sigma": 1.0957933176554728}, "2": {"depth": 0.048249285854390554, "error": 0.0003989021570800136, "red_noise_beta": 1.8424011058099723, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04818754561916363, "background_template_native_sigma": 0.22121593022488228}}, "PDCSAP_FLUX": {"1": {"depth": 0.05021857314125755, "error": 0.00042202042948425628 04, "red_noise_beta": 2.029676965995674, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04970197346675168, "background_template_native_sigma": 1.0957933176554728}, "2": {"depth": 0.049712229599089906, "error": 0.00040961879769029977, "red_noise_beta": 1.8518838130601574, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04965538298292361,

"background_template_native_sigma": 0.22121593022488228}}}. At3081.486025: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04810517860741927, "error": 0.00036467591238765627, "red_noise_beta": 1.8528739188420318, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04844648359422767, "background_template_native_sigma": 0.71726810861929}, "2": {"depth": 0.04778466447056494, "error": 0.00039277463330454734, "red_noise_beta": 1.8325712972002173, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.048114653140111416, "background_template_native_sigma": 0.7682219312989591}}, "PDCSAP_FLUX": {"1": {"depth": 0.049560461799830466, "error": 0.000375798159475723, "red_noise_beta": 1.8492703617391621, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.049935623495036115, "background_template_native_sigma": 0.71726810861929}, "2": {"depth": 0.04922799460909148, "error": 0.000404685401384818, "red_noise_beta": 1.829013037207618, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04959309905267898, "background_template_native_sigma": 0.7682219312989591}}}. At3084.557639: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04858507184297725, "error": 0.00038393713378192976, "red_noise_beta": 2.120227193674034, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04874475972504164, "background_template_native_sigma": 0.9092719322614469}, "2": {"depth": 0.04823983233264624, "error": 0.00040960277452482845, "red_noise_beta": 2.091547112263129, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.0483984759616134, "background_template_native_sigma": 0.8822590299430806}}, "PDCSAP_FLUX": {"1": {"depth": 0.05001822870854587, "error": 0.00039303540402414 58, "red_noise_beta": 2.109594682993706, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.05018089917253465, "background_template_native_sigma": 0.9092719322614469}, "2": {"depth": 0.04964792766195014, "error": 0.00041660282223297 88, "red_noise_beta": 2.072267538537739, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04980973113644217, "background_template_native_sigma": 0.8822590299430806}}}. At3087.629253: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04799222765772045, "error": 0.0003552275224497715, "red_noise_beta": 1.8498029170513859, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.04807466198488935, "background_template_native_sigma": 0.5429307657631269}, "2": {"depth": 0.048034304703546755, "error": 0.0003901420521640965, "red_noise_beta": 1.8306904759450664, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.048142135954438975, "background_template_native_sigma": 0.6942069648661181}}, "PDCSAP_FLUX": {"1": {"depth": 0.04952903590488413, "error": 0.00035751041332454675, "red_noise_beta": 1.8169166947055242, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.049616562182304216, "background_template_native_sigma": 0.5429307657631269}, "2": {"depth": 0.04952623781377637, "error": 0.00039711557552706933, "red_noise_beta": 1.8180138152424952, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.049638957429507205, "background_template_native_sigma": 0.6942069648661181}}}. At3090.700100: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04769907666814212, "error": 0.0003079854584698072, "red_noise_beta": 1.5504263720199325, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.047653779540458414, "background_template_native_sigma": -0.28213195696667664}, "2": {"depth": 0.04767905957464956, "error": 0.00033647538526884773, "red_noise_beta": 1.5502550628196363, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.0477368636970146, "background_template_native_sigma": 0.12692385577852916}}, "PDCSAP_FLUX": {"1": {"depth": 0.0491891905844186, "error": 0.0003159981653356597, "red_noise_beta": 1.556728357203403, "n_core": 55, "baseline_degree": 1, "background_regressed_depth": 0.049114027721330264, "background_template_native_sigma": -0.28213195696667664}, "2": {"depth": 0.0491151360356513,

"error": 0.00034431182536632366, "red_noise_beta": 1.5543530917808732, "n_core": 55, "baseline_degree": 2, "background_regressed_depth": 0.04917757570870915, "background_template_native_sigma": 0.12692385577852916}}}. At3093.772480: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.047027962027332455, "error": 0.00047355581845187107, "red_noise_beta": 2.384089322913508, "n_core": 56, "baseline_degree": 1, "background_regressed_depth": 0.04703122209443175, "background_template_native_sigma": 0.05254209069249003}, "2": {"depth": 0.0475817541243957, "error": 0.0004156411020240634, "red_noise_beta": 1.9933672272158298, "n_core": 56, "baseline_degree": 2, "background_regressed_depth": 0.04782817521255266, "background_template_native_sigma": 0.21396785044592323}}, "PDCSAP_FLUX": {"1": {"depth": 0.04840225782579816, "error": 0.00047926148492953856, "red_noise_beta": 2.370934287926704, "n_core": 56, "baseline_degree": 1, "background_regressed_depth": 0.04840496208785293, "background_template_native_sigma": 0.05254209069249003}, "2": {"depth": 0.048959211204510376, "error": 0.00042272967240425434, "red_noise_beta": 1.9894585641570925, "n_core": 56, "baseline_degree": 2, "background_regressed_depth": 0.04920397000176509, "background_template_native_sigma": 0.21396785044592323}}}. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "323434013", "sector": 65, "n_events": 18, "deficit_minus_light_arcsec": 0.18, "peak_on_target": true, "depth_ap3": 0.04649, "psf_frac_resid": 0.026, "n_neighbours_60": 10, "n_possible": 0, "brightest_nb": {"G": 14.487941, "sep": 6.3}}, {"tic": "323434013", "sector": 65, "n_events": 18, "deficit_minus_target_arcsec": 1.14, "deficit_minus_light_arcsec": 0.18, "peak_on_target": true, "depth_ap3": 0.04649, "psf_frac_resid": 0.026, "n_neighbours_60": 10, "n_possible": 0, "brightest_nb": {"G": 14.487941, "sep": 6.3}}]. Own Gaia<15 cone neighbor table: [{"source_id": "4614246265614433920", "ra": "69.81669734810306", "dec": "-85.91178666984327", "phot_g_mean_mag": "14.487941", "bp_rp": "1.2466431", "sep": "6.342704277472143", "required_fraction_G_approx": 2.5656224115005863, "capable": false}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 4614246265612443136", "main_id": "CPD-86 55", "otype": "SB*", "ra": "69.805040901", "dec": "-85.9135537774"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.09, "m2_min": 1.030439716797158, "f_mass": 0.24334134739252788, "a_rsun": 11.42424484004741, "r1_rsun": 1.99168, "r2_approx_rsun": 1.0172387043893005, "central_T14_hours": {"A": 6.122826992497429, "B": 6.236426190721182}, "observed_T14_hours": 3.063475862068966}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. SB2 physical dimensions from both measured amplitudes: { "q_M2_M1": 0.9046159660720485, "M1_sin3i_solar":

1.1924440308717033, "M2_sin3i_solar": 1.0787039089738535, "a_sini_Rsun": 11.688731391988755, "source": "own Gaia SB2 K1,K2; no adopted TIC mass needed", "covariance_source": "https://gea.esac.esa.int/archive/documentation/GDR3/Gaia_archive/chap_datamodel/sec_dm_non-single_stars_tables/ssec_dm_nss_two_body_orbit.html", "ordering_verified": "SB2 bit255: P,gamma,K1,K2,e,omega,Tp;21 correlations. ESA upper-column-major equivalent to symmetric lower-row-major." }. Blind raw-SAP rolling-median census contains171 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.663ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Gaia is SB2, not SB1: own bit255/21correlation ordering was verified against ESA documentation. Sixty-five fits of each kind, equal depths, alternate reduction and tight target deficit establish the EB at full3.071615d. SB2 amplitudes independently yield q0.9046, M1sin3i1.19244solar, M2sin3i1.07870solar and a sini11.6887Rsun. Fixed-TIC geometry fits poorly (chi2~1314/121); precise component radii and e sin omega are not inferred.

Next nominal eclipses after2026-10-30T00UTC: primary4343.91949BTJD; other4345.45518BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R038. TIC 327209313

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 327209313: CONFIRMED attributed eclipsing system

Fourteen primary and18secondary fits repeat with independent reduction and an attributed deficit. Own source fit beats the42.7arcsec capable neighbor in all four pixel events. Secondary depths show sector scatter beyond formal errors; empirical errors and O-C are retained. Conditional two-disk geometry is feasible, not a parameter posterior.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2293448276d, profile range[0.22391379310344828, 0.2347758620689655], ingress fraction0.35 (T23 conditional, not proof of totality). Median primary depth51.888ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=5.931777383 ±2.79291006e-06d, Tref=2951.383903 ±0.000373023731 BTJD, cycle reference217. T0=1664.188211. chi2/dof=34.580653964741636/12; O-C RMS=1.737593268min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
13	1664.186128	0.002126260161	54.54481429	1.011802952	0.2293448276	0.08027068966	0.06880344828	-2.999211132	1	3	0.6678468792	53.87908691
13	1670.119302	0.001813280758	51.77098845	1.08143224	0.2293448276	0.08027068966	0.06880344828	-0.9883728564	1.025634149	0	-0.6328942863	52.35994086
13	1676.0511	0.001813280758	52.09332809	0.8699536889	0.2293448276	0.08027068966	0.06880344828	-0.9590738906	1.078185342	0	0.4859291692	51.61706385
39	2364.136884	0.0006880344828	49.86678946	0.8989525177	0.2293448276	0.08027068966	0.06880344828	-1.523472485	1.214220783	6	-0.4665083969	50.32883806
39	2370.070058	0.0006880344828	50.00155619	0.5909289421	0.2293448276	0.08027068966	0.06880344828	0.4873657913	1.082108657	8	0.05598244625	49.94979354
39	2381.933653	0.0006880344828	51.15387181	0.7747817412	0.2293448276	0.08027068966	0.06880344828	0.5459637229	1.086987982	6	-0.7175235973	51.81822746
66	3099.678427	0.0006880344828	52.72904189	0.6394921272	0.2293448276	0.08027068966	0.06880344828	0.1280599569	1.202382662	0	-0.4261618846	53.15218312

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
66	3105.610224	0.0006880344828	53.5384232	0.6925165598	0.2293448276	0.08027068966	0.06880344828	0.157358923	1.196105821	0	-0.3404083244	53.87569733
66	3117.47382	0.0006880344828	53.5829019	0.6078514358	0.2293448276	0.08027068966	0.06880344828	0.2159568546	1.071767675	0	-0.04227789448	53.6241747
93	3835.219969	0.0006880344828	52.00551895	0.6063600799	0.2293448276	0.08027068966	0.06880344828	1.779592399	1.124482394	0	0.3038684626	51.69542107
93	3847.083564	0.0006880344828	51.678131154	0.6889712252	0.2293448276	0.08027068966	0.06880344828	1.83819033	1.257227924	0	0.1935885356	51.47026966
93	3853.015362	0.0006880344828	53.40728629	0.8799702835	0.2293448276	0.08027068966	0.06880344828	1.867489297	1.2447811	1	0.3252565899	53.07896605
104	4179.261485	0.0006880344828	45.62233586	0.749484361	0.2293448276	0.08027068966	0.06880344828	-0.4841462067	2.430283279	0	-1.337448003	46.856 8882
104	4185.190531	0.0006880344828	46.93213141	0.4568274202	0.2293448276	0.08027068966	0.06880344828	-4.417925863	1.647595473	17	-0.07212658431	47.00305299

Own Gaia source4037063482176211072, SB1, bit_index127, w–Tp correlation0.98859. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=5.931859063076344±7.4450846E-4d, e=0.07234927626403258±0.024204032, omega=233.989694091892±11.78907deg. Photometric P difference=-0.1097096665sigma. Primary attributed toA; propagated epoch389 offset=0.03635±0.05462d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.4728822; optimized diagnostic phase=0.4704525. Fit statistic: {"n": 18, "depth": 0.016725396266449415, "error": 9.986514936606058e-05, "snr": 167.47981025033727, "scatter": 0.0015923081304868773, "chi2": 267.8705166544092, "dof": 17, "empirical_error": 0.000375310625601915, "n_individual_gt3sigma": 18}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: {"P": 5.931769659411875, "P_error": 4.070797529508901e-06, "Tref": 2989.7647452089705, "Tref_error": 0.0007197506858108495, "reference_cycle": 223, "T0": 1666.9801111601223, "chi2": 54.396146716012346, "dof": 16, "n": 18, "oc_rms_minutes": 4.379471201175646}.

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
13	1661.0499936	0.0021618620069	19.217250999	0.7411262633	0.240206896	0.0840724137 9	0.0720620689 7	2.295742861

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
13	1666.975948	0.0036031034448	16.635863637	1.102102548	0.240206896	0.0840724137	0.0720620689	-5.9946851
13	1672.9106008	0.0021618620 69	18.074159833	0.748303983 5	0.240206896	0.0840724137	0.0720620689	-1.8327891 64
13	1678.8380622	0.0036031034448	18.3187091 2	1.299964115	0.240206896	0.0840724137	0.0720620689	-8.0478303 5
39	2366.9285622	0.0014412413 79	19.550589 31	0.539395487 3	0.240206896	0.0840724137	0.0720620689	-0.5315650 663
39	2372.8574577	0.0021618620 69	19.913454 1	0.695936685	0.240206896	0.0840724137	0.0720620689	-4.6712186 65

secondary sector	midBTJD	erord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
39	2378.790675	0.0021618620 69	19.290582 58	0.9292961 4161 4	0.2402068966	0.0840724137 9	0.0720620689 7	-2.5847090505
39	2384.723894	0.0014412413 79	19.230075 56	0.6608961 2559	0.2402068966	0.0840724137 9	0.0720620689 7	-0.4982003461
66	3102.468957	0.0014412413 79	19.461916 9	0.5924075 931	0.2402068966	0.0840724137 9	0.0720620689 7	0.8475100466
66	3114.333953	0.0014412413 79	19.442056 16	0.5922243 0222	0.2402068966	0.0840724137 9	0.0720620689 7	2.9451407 8
66	3120.26717 2	0.0014412413 79	19.383486 98	0.5972433 702	0.2402068966	0.0840724137 9	0.0720620689 7	5.0316499 39

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14d	ingress d	T23d	O-C min
93	3832.080458	0.0021618620 69	18.305269 92	0.801489 2393	0.240206 896 6	0.008 4072 4137 9	0.007 2062 0689 7	6.366 2387 59
93	3838.015118	0.0014 4124 1379	19.771 1892 6	0.647 1826 2778	0.240 2068 966	0.008 4072 4137 9	0.007 2062 0689 7	10.52 8135 5
93	3843.936806	0.0014 4124 1379	20.33 6497 09	0.782 9083 024	0.240 2068 966	0.008 4072 4137 9	0.007 2062 0689 7	-3.98 8456 026
104	4182.049558	0.0007 2062 0689 7	15.55 7953 5	0.207 9817 593	0.240 2068 966	0.008 4072 4137 9	0.007 2062 0689 7	-1.27 9138 751
104	4187.982777	0.0007 2062 0689 7	15.79 8180 54	0.233 6947 313	0.240 2068 966	0.008 4072 4137 9	0.007 2062 0689 7	0.807 3704 089

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
104	4193.9131313	0.000720620689	15.92153372	0.2769721287	0.2402006896	0.0840724379	0.0720620689	-1.2568956065
104	4199.8448911	0.000720620689	15.76900037	0.2373127377	0.2402006896	0.0840724379	0.0720620689	-1.2457740311

Measured fitted secondary phase at its reference epoch=0.4703780; first-order e cos(omega) in primary-to-secondary convention=-0.04653 +/- 0.00029 (conservative sum of reference-time and period errors; no full ephemeris cross-covariance), Gaia same-convention=-0.04254. This first-order estimator is approximate for eccentric or grazing systems; exact Gaia conjunction-phase comparison above is preferred. Duration ratio=1.047; grazing/partial chords prohibit direct trapezoid-duration conversion into e sin(omega). Own limb-darkened two-sphere feasibility result: {"status": "conditional shape feasibility", "best": {"chi2": 208.69729665570085, "dof": 121, "k": 0.27499999999999997, "b_A": 0.865, "e_sin_omega": 0.0749999999999999, "primary_scale": 0.9615635319997947, "secondary_scale": 0.04238972733794568}, "profile_interval_approx_95": {"k": [0.27499999999999997, 0.29999999999999993], "b_A": [0.865, 0.9000000000000001], "e_sin_omega": [0.0749999999999999, 0.0749999999999999]}, "error_inflation": 1.7247710467413293, "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.", "primary_profile_n": 64, "secondary_profile_n": 64}. These coarse fixed-TIC profiles are not calibrated orbital uncertainties.

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector39; 3 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At2364.136884: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04402252811053591, "error": 0.0003431105218521327, "red_noise_beta": 1.392754605608463, "n_core": 33, "baseline_degree": 1, "background_regressed_depth": 0.044071598192687925, "background_template_native_sigma": 0.09652473872594597}, "2": {"depth": 0.04381819488236002, "error": 0.000374227692138964, "red_noise_beta": 1.378768682757763, "n_core": 33, "baseline_degree": 2, "background_regressed_depth": 0.043854345860215214, "background_template_native_sigma": 0.06446493783153504}}, "PDCSAP_FLUX": {"1": {"depth": 0.06548517788795354, "error": 0.0005051235018217839, "red_noise_beta": 1.364868967335991, "n_core": 33, "baseline_degree": 1, "background_regressed_depth": 0.06554365922861356, "background_template_native_sigma":

0.09652473872594597}, "2": {"depth": 0.0651927139258962, "error": 0.0005523680823856802, "red_noise_beta": 1.353694070617889, "n_core": 33, "baseline_degree": 2, "background_regressed_depth": 0.06523654793599544, "background_template_native_sigma": 0.06446493783153504}}}. At2370.070058: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04390460381822652, "error": 0.00030218841836947947, "red_noise_beta": 1.234647049847842, "n_core": 33, "baseline_degree": 1, "background_regressed_depth": 0.04390141285373685, "background_template_native_sigma": -0.022696246866369164}, "2": {"depth": 0.04370349085125302, "error": 0.000330600643832611, "red_noise_beta": 1.2258173652184106, "n_core": 33, "baseline_degree": 2, "background_regressed_depth": 0.04377445460007015, "background_template_native_sigma": 0.6334124982596071}}, "PDCSAP_FLUX": {"1": {"depth": 0.06543817907679668, "error": 0.0004488268158317776, "red_noise_beta": 1.2389470037454036, "n_core": 33, "baseline_degree": 1, "background_regressed_depth": 0.06543397016344693, "background_template_native_sigma": -0.022696246866369164}, "2": {"depth": 0.06527974441323128, "error": 0.0004963478725028533, "red_noise_beta": 1.2365641087679609, "n_core": 33, "baseline_degree": 2, "background_regressed_depth": 0.0653910990391185, "background_template_native_sigma": 0.6334124982596071}}}. At2381.933653: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04532110795926395, "error": 0.0007348238771637359, "red_noise_beta": 1.4394695230792578, "n_core": 32, "baseline_degree": 1, "background_regressed_depth": 0.04461883784198507, "background_template_native_sigma": -0.9947422708412029}, "2": {"depth": 0.04402679029618366, "error": 0.0006632944767068587, "red_noise_beta": 1.274161275122413, "n_core": 32, "baseline_degree": 2, "background_regressed_depth": 0.0440088462524523, "background_template_native_sigma": -0.04835428032752621}}, "PDCSAP_FLUX": {"1": {"depth": 0.06535898572411436, "error": 0.0005943325474500386, "red_noise_beta": 1.5790381212321762, "n_core": 32, "baseline_degree": 1, "background_regressed_depth": 0.06525357504310472, "background_template_native_sigma": -0.9947422708412029}, "2": {"depth": 0.06510695936451924, "error": 0.0006519597384916586, "red_noise_beta": 1.5727676399920278, "n_core": 32, "baseline_degree": 2, "background_regressed_depth": 0.0651057593708182, "background_template_native_sigma": -0.04835428032752621}}}. Own hashed pixel regression sector104, 4 events. Approximate Gaussian PSF calibrated on out-of-eclipse Gaia sources. Free centroid offset2.10arcsec; target residual fraction0.318. Every event favors['4037063482176211072', '4037063482176211072', '4037063482176211072', '4037063482176211072']. Fixed source comparisons: [{"source_id": "4037063482176211072", "sep_arcsec": 0.0, "G": 11.171228, "chi2": 11196.376585746093, "amplitude": 169.5043576437327, "fractional_residual": 0.31815525883941165}, {"source_id": "4037063409055337088", "sep_arcsec": 42.73174482873894, "G": 13.585817, "chi2": 33287.99439129555, "amplitude": 24.062265928953924, "fractional_residual": 0.9782580552888445}, {"source_id": "4037063409056618624", "sep_arcsec": 48.12721386852933, "G": 14.581224, "chi2": 35295.69771746952, "amplitude": 17.537111117813414, "fractional_residual": 0.9839463961681812}, {"source_id": "4037063516535929472", "sep_arcsec": 57.34676062061514, "G": 14.412503, "chi2": 37104.32966333872, "amplitude": 3.340433726729843, "fractional_residual": 0.9867878807213636}]. Formal chi2 does not include PSF mismatch/correlated pixels; compare source separation and repeatability, not nominal sigma. Pixel source path/hash and independent-aperture depths are retained. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "327209313", "sector": 104, "n_events": 3, "deficit_minus_target_arcsec": 3.94, "deficit_minus_light_arcsec": 1.5, "peak_on_target": true, "depth_ap3": 0.05002, "psf_frac_resid": 0.596, "n_neighbours_60": 830, "n_possible": 1, "brightest_nb": {"G": 13.585817, "sep": 42.7}}]. Own Gaia<15 cone neighbor table: [{"source_id": "4037063409055337088", "ra": "270.6867718863953", "dec": "-37.596656938428666", "phot_g_mean_mag": "13.585817", "bp_rp": "2.293107", "sep": "42.73174482873894",

"required_fraction_G_approx": 0.47962813925844194, "capable": true}, {"source_id": "4037063409056618624", "ra": "270.6887671858549", "dec": "-37.59758620423919", "phot_g_mean_mag": "14.581224", "bp_rp": "0.9892988", "sep": "48.12721386852933", "required_fraction_G_approx": 1.199685624230758, "capable": false}, {"source_id": "4037063516535929472", "ra": "270.7001622472945", "dec": "-37.58038584830463", "phot_g_mean_mag": "14.412503", "bp_rp": "1.7172813", "sep": "57.34676062061514", "required_fraction_G_approx": 1.0270203634830204, "capable": false}]. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 12, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 4037063482176211072", "main_id": "TYC 7899-2345-1", "otype": "SB*", "ra": "270.68126766041", "dec": "-37.58564696309"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.14, "m2_min": 0.5746779425281592, "f_mass": 0.06455181567490499, "a_rsun": 16.505391297538058, "r1_rsun": 2.61761, "r2_approx_rsun": 0.6420067919156252, "central_T14_hours": {"A": 9.480780740508429, "B": 8.432425671243626}, "observed_T14_hours": 5.504275862068965}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains37 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=1.337ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Fourteen primary and18secondary fits repeat with independent reduction and an attributed deficit. Own source fit beats the42.7arcsec capable neighbor in all four pixel events. Secondary depths show sector scatter beyond formal errors; empirical errors and O-C are retained. Conditional two-disk geometry is feasible, not a parameter posterior.

Next nominal eclipses after2026-10-30T00UTC: primary4345.35159BTJD; other4348.14221BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R039. TIC 330659284

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 330659284: CONFIRMED attributed eclipsing system

Four primary timings and six faint repeatable secondary windows establish the pair; five secondaries individually exceed3sigma. Independent reduction and isolated on-target pixel deficit support attribution. Secondary timing is appreciably offset from Gaia nominal phase, within the relative-phase uncertainty tested, and is measured rather than forced to that nominal phase.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.1891034483d, profile range[0.18548275862068964, 0.1927241379310345], ingress fraction0.25 (T23 conditional, not proof of totality). Median primary depth48.886ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=15.63231187 ±1.643576137e-05d, Tref=2728.71382 ±0.0003330016671 BTJD, cycle reference14. T0=2509.861454. chi2/dof=2.7563946392018903/2; O-C RMS=0.6781470825min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
44	2509.861835	0.0005673103448	49.35875502	0.4168355581	0.1891034483	0.04727586207	0.09455172414	0.5477688876	1.122550866	0	0.0057696177778	49.353138 62
46	2556.758749	0.0005673103448	47.66270038	0.3462800021	0.1891034483	0.04727586207	0.09455172414	0.5179177217	1.079843175	3	0.0061488585 81	47.663711 24
46	2572.38992	0.0005673103448	48.41258294	0.4268629953	0.1891034483	0.04727586207	0.09455172414	-1.125886459	1.169412963	3	-0.06255037317	48.47986638
72	3275.844777	0.0005673103448	49.91188314	0.4504214978	0.1891034483	0.04727586207	0.09455172414	0.06019985012	1.042909312	0	1.335072452	48.49815488

Own Gaia source607391083840036480, SB1, bit_index127, w–Tp correlation0.99388. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=15.631403214205429±0.0048393435d, e=0.10387992143456555±0.044260405, omega=318.21617054349275±18.211214deg. Photometric P difference=0.1877621988sigma. Primary

attributed toA; propagated epoch389 offset=0.21701±0.26170d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.5493811; optimized diagnostic phase=0.5560834. Fit statistic: {"n": 6, "depth": -0.0006461810301584904, "error": 0.00020234712989151906, "snr": -3.193428196905568, "scatter": 0.0006915700133664855, "chi2": 10.669614818237166, "dof": 5, "empirical_error": 0.00028233227569293863, "n_individual_gt3sigma": 0}. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `null`.

secondary sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min
44	2502.930393	0.03247944828	0.7966606581	0.5436161476	0.17462068897	0.02619310345	0.12223448278	0
44	2518.520796	0.05238620699	-0.50546834023	0.5313559903	0.17462068897	0.02619310345	0.12223448278	0
45	2534.205494	0.02619310345	-1.000784957	0.5064683	0.17462068897	0.02619310345	0.12223448278	0
45	2549.810565	0.015192	-0.9977459638	0.48481857833	0.17462068897	0.02619310345	0.12223448278	0
72	3268.914722	0.01571586207	-1.357117129	0.460533883	0.17462068897	0.02619310345	0.12223448278	0
72	3284.589991	0.05238620699	-0.46272564049	0.464749139 4	0.17462068897	0.02619310345	0.12223448278	0

No robust independent secondary period/e cos omega is claimed; geometry follows the null test or conditional model, not a noisy optimized phase.

Independent reduction and source attribution

Exact target/sector identity validated: SPOC sector46; 2 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At2556.758749: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.045862287886657416, "error": 0.0003785350575869863, "red_noise_beta": 1.7163199350247476, "n_core": 136, "baseline_degree": 1, "background_regressed_depth": 0.04617791666450382, "background_template_native_sigma": 0.3596880488917207}, "2": {"depth": 0.045566903700372356, "error": 0.0004102734734889531, "red_noise_beta": 1.6725596693750855, "n_core": 136, "baseline_degree": 2, "background_regressed_depth": 0.04587292106342136, "background_template_native_sigma": 0.34768688895348093}}, "PDCSAP_FLUX": {"1": {"depth": 0.04579836443593918, "error": 0.0003603285589746838, "red_noise_beta": 1.6595421953217813, "n_core": 136, "baseline_degree": 1, "background_regressed_depth": 0.04611973264651487, "background_template_native_sigma": 0.3596880488917207}, "2": {"depth": 0.04558936430674134, "error": 0.00039712743910533406, "red_noise_beta": 1.6409254826942246, "n_core": 136, "baseline_degree": 2, "background_regressed_depth": 0.045900670479078406, "background_template_native_sigma": 0.34768688895348093}}}. At2572.389920: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04609442894991948, "error": 0.0003297090051258604, "red_noise_beta": 1.4310945441250797, "n_core": 129, "baseline_degree": 1, "background_regressed_depth": 0.04657587901472007, "background_template_native_sigma": 0.4352266118672036}, "2": {"depth": 0.04609605630632885,

"error": 0.00036857716417711665, "red_noise_beta": 1.4310090353920781, "n_core": 129, "baseline_degree": 2, "background_regressed_depth": 0.046432958967183476, "background_template_native_sigma": 0.29762845251364173}}, "PDCSAP_FLUX": {"1": {"depth": 0.04688677558640622, "error": 0.0003673767419943409, "red_noise_beta": 1.585213742327168, "n_core": 129, "baseline_degree": 1, "background_regressed_depth": 0.04736375820146355, "background_template_native_sigma": 0.4352266118672036}, "2": {"depth": 0.04699045013923904, "error": 0.0004060322901082836, "red_noise_beta": 1.5679424852655868, "n_core": 129, "baseline_degree": 2, "background_regressed_depth": 0.04731903187111674, "background_template_native_sigma": 0.29762845251364173}}}. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "330659284", "sector": 46, "n_events": 2, "deficit_minus_target_arcsec": 1.66, "deficit_minus_light_arcsec": 2.26, "peak_on_target": true, "depth_ap3": 0.04584, "psf_frac_resid": 0.13, "n_neighbours_60": 2, "n_possible": 0, "brightest_nb": {"G": 21.110783, "sep": 46.4}}]. Own Gaia<15 cone neighbor table: []. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 0, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 607391083840036480", "main_id": "TYC 1401-1146-1", "otype": "SB*", "ra": "136.5339856446", "dec": "15.280782218589998"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 1.09, "m2_min": 0.20721399359928933, "f_mass": 0.005287300485962327, "a_rsun": 28.692844672266908, "r1_rsun": 1.02992, "r2_approx_rsun": 0.28388030601269715, "central_T14_hours": {"A": 5.842663705669587, "B": 5.086195583881944}, "observed_T14_hours": 4.53848275862069}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains5 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=1.335ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Four primary timings and six faint repeatable secondary windows establish the pair; five secondaries individually exceed3sigma. Independent reduction and isolated on-target pixel deficit support attribution. Secondary timing is appreciably offset from Gaia

nominal phase, within the relative-phase uncertainty tested, and is measured rather than forced to that nominal phase.

Next nominal eclipses after2026-10-30T00UTC: primary4354.47425BTJD; other4347.53481BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R040. TIC 82620363

physical_verdict: CONFIRMED

novelty_status: no_target_EB_match_in_queried_archives; limited_prior_art_coverage

completion: complete

next_check: Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Archived source: sanitized report text.

TIC 82620363: CONFIRMED attributed eclipsing system

Eight primary timings and the alternate reduction establish repeating eclipses with Gaia attribution and no G<15capable neighbor. Existing deficit peaks on target, although its PSF residual is poor, so do not quote a precision centroid. No reliable secondary; conditional width/phase limits vary strongly for broad templates, up to59.6ppt, so no single uniform strong exclusion is claimed. Approximate eccentric geometry permits a miss.

Physical verdict: **CONFIRMED**. Full review: **complete**. Novelty: no target EB match in the current specified archives; limited prior-art coverage, not a claim of discovery priority. Physical EB identification is distinct from component characterization.

Measurements and timing

Raw QLP SAP_FLUX, clean QUALITY==0 cadences, untrimmed; flagged cores are counted. Shared exposure-integrated trapezoid T14=0.2076206897d, profile range[0.20218965517241377, 0.213051724137931], ingress fraction0.25 (T23 conditional, not proof of totality). Median primary depth49.932ppt. Shared-shape durations have profile/grid uncertainty; individual event depths and timings are free. Errors include residual binning beta and a grid/cadence floor. Primary ephemeris: P=11.16810796 ±6.114836697e-06d, Tref=2556.609709 ±0.0004437977045 BTJD, cycle reference68. T0=1797.178367. chi2/dof=0.6820944163579461/6; O-C RMS=0.6437982038min. Individual fitted events below; all excluded/uncovered windows remain in fit-results.json.

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
18	1797.178174	0.001901848856	51.168319 5	0.4061431753	0.2076206897	0.0519051724 1	0.1038103448	-0.2778518369	1.019524855	0	-0.2885351376	51.415706 42
18	1808.346382	0.002004724717	52.331684 58	0.4385426946	0.2076206897	0.0519051724 1	0.1038103448	-0.1349846973	1.086448839	1	-0.2432691749	52.304637 41
19	1819.515834	0.001901783188	49.237 38653	0.5892531918	0.2076206897	0.0519051724 1	0.1038103448	1.801725201	1.010447551	0	-0.1880011535	49.335687 23
19	1830.682796	0.002273065675	49.540 30262	0.8417337888	0.2076206897	0.0519051724 1	0.1038103448	0.1507495828	1.025850095	2	-0.6836910764	50.306 99203
59	2913.988926	0.000622862069	49.917 90691	0.5316408231	0.2076206897	0.0519051724 1	0.1038103448	-0.341879916	1.958569488	0	-0.1757833808	50.1192 638
59	2936.325341	0.000622862069	49.945 21076	0.3786036579	0.2076206897	0.0519051724 1	0.1038103448	-0.05614563619	1.436467366	0	-0.0039818477997	49.949 07062

sector	mid BTJD	error d	depth ppt	error ppt	T14 d	ingress d	T23 d	O-C min	beta	flagged core	background fake ppt	BG refit ppt
86	3639.916164	0.000622862069	51.647186609	0.7008666118	0.2076206897	0.05190517241	0.1038103448	-0.02472961482	1.717262272	1	-0.2897789315	51.971030 86
86	3662.252579	0.000622862069	49.537176887	0.4082770 74	0.2076206897	0.05190517241	0.1038103448	0.261004665	1.484634284	1	-0.1410168291	49.673484 64

Own Gaia source462316159559941760, SB1, bit_index127, w–Tp correlation0.92187. Strict covariance layout asserted (SB1six parameters/15correlations; SB2seven/21). ESA layout. Gaia P=11.168589202440996±7.7344035E-4d, e=0.3554047188113604±0.016045228, omega=112.42186148744912±2.1248167deg. Photometric P difference=-0.6221842839sigma. Primary attributed toA; propagated epoch389 offset=0.01340±0.02050d (Gaia-only uncertainty; photometric extrapolation may add). Every observed epoch comparison is preserved in orbit-gates.json. Secondary Gaia relative phase=0.4089429; optimized diagnostic phase=0.4370604. Fit statistic: `{"n": 5, "depth": -0.0006511999075934945, "error": 0.00013699441530911195, "snr": -4.75347776859471, "scatter": 0.0002670127161872876, "chi2": 3.1665954159531915, "dof": 4, "empirical_error": 0.00013699441530911195, "n_individual_gt3sigma": 0}`. Phase optimization is conditional; weak scan peaks are not detections. Fixed Gaia-phase controls, every secondary coverage exclusion and individual errors are preserved in orbit-gates.json/secondary-fit.json. Secondary ephemeris: `null`.

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-C min
18	1813.1294 81	0.0961 6841 379	-0.5273 6254 62	0.4715 0246 13	0.3271 0344 83	0.0261 6827 586	0.2747 6689 66	0
19	1824.4251 6	0.0504 6827 59	-0.6970 0407 81	0.3713 3732 3	0.3271 0344 83	0.0261 6827 586	0.2747 6689 66	0
59	2918.9684 32	0.0049 0655 1724	-1.2183 0619 8	0.3571 3259 06	0.3271 0344 83	0.0261 6827 586	0.2747 6689 66	0

secondary sector	midBTJD	errord	depthppt	errorppt	T14d	ingressd	T23d	O-Cmin
59	2929.942241	0.009813103448	-0.4989384373	0.2237342959	0.3271034483	0.0261682758	0.2747668966	0
86	3644.699188	0.009813103448	-0.5672783893	0.2678641411	0.3271034483	0.0261682758	0.2747668966	0
No robust independent secondary period/e cos omega is claimed; geometry follows the null test or conditional model, not a noisy optimized phase.								
Secondary null tests: own covariance relative-phase sigma=0.009315, 5widths×9phases spanning±2sigma, pointwise conditional3sigma upper ranges[0.538, 59.588]ppt. Full coverage/excluded windows in null-secondary.json; upper limits are phase/shape/baseline conditional and not a simultaneous bound. Feasibility: [{"radius_ratio_assumption": 0.2011205446250908, "b_primary_duration_approx": 0.8091782936077454, "b_other": 1.6010140132665311, "contact": 1.2011205446250908, "other_misses": true, "es_gaia": 0.32853632359556373, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}, {"radius_ratio_assumption": 0.23064807956874134, "b_primary_duration_approx": 0.761191960699198, "b_other": 1.5060698062372957, "contact": 1.2306480795687413, "other_misses": true, "es_gaia": 0.32853632359556373, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}, {"radius_ratio_assumption": 0.26493932851567836, "b_primary_duration_approx": 0.6986387178258419, "b_other": 1.3823039820590513, "contact": 1.2649393285156783, "other_misses": true, "es_gaia": 0.32853632359556373, "note": "Central-duration/chord approximation with main-sequence companion radius, not measured component dimensions; establishes feasible missing eclipse, not unique geometry."}].								

Independent reduction and source attribution

Exact target/sector identity validated: TESS-SPOC sector59; 2 events measured with linear/quadratic baselines and red-noise inflation. Same TESS pixels, independent pipeline, not independent observatory. See independent-summary.json and hashed independent-manifest.json. At2913.988926: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04917940200628943, "error": 0.00038630420402259354, "red_noise_beta": 2.0439165957726915, "n_core": 89, "baseline_degree": 1, "background_regressed_depth": 0.049086476547536555, "background_template_native_sigma": 0.2873529301128743}, "2": {"depth": 0.04898543039672694, "error": 0.00042658754847358575, "red_noise_beta": 2.0280236073010167, "n_core": 89, "baseline_degree": 2, "background_regressed_depth": 0.048955673105880275, "background_template_native_sigma":

-0.061833286147886114}}, "PDCSAP_FLUX": {"1": {"depth": 0.05290840373719531, "error": 0.00041558968868925136, "red_noise_beta": 2.0631139304849047, "n_core": 89, "baseline_degree": 1, "background_regressed_depth": 0.05305254456941335, "background_template_native_sigma": 0.2873529301128743}, "2": {"depth": 0.05304825975862393, "error": 0.0004586349554625894, "red_noise_beta": 2.041313397961749, "n_core": 89, "baseline_degree": 2, "background_regressed_depth": 0.052995573211275994, "background_template_native_sigma": -0.061833286147886114}}}. At2936.325341: independent raw fits {"SAP_FLUX": {"1": {"depth": 0.04874348578329389, "error": 0.00032948073109963824, "red_noise_beta": 1.7341158006496311, "n_core": 90, "baseline_degree": 1, "background_regressed_depth": 0.048996214949135766, "background_template_native_sigma": 0.7051795888083273}, "2": {"depth": 0.048794108885490566, "error": 0.0003592677488114731, "red_noise_beta": 1.7242885746782806, "n_core": 90, "baseline_degree": 2, "background_regressed_depth": 0.04903518988090448, "background_template_native_sigma": 0.6740216321115888}}, "PDCSAP_FLUX": {"1": {"depth": 0.05336150633310747, "error": 0.00036396176993721386, "red_noise_beta": 1.7227088886719593, "n_core": 90, "baseline_degree": 1, "background_regressed_depth": 0.05360692387600596, "background_template_native_sigma": 0.7051795888083273}, "2": {"depth": 0.05322518965898409, "error": 0.00040758283469048814, "red_noise_beta": 1.7627842597240522, "n_core": 90, "baseline_degree": 2, "background_regressed_depth": 0.053461559329093744, "background_template_native_sigma": 0.6740216321115888}}}. Original batch deficit receipts (duplicate schema rows are not independent observations): [{"tic": "82620363", "sector": 18, "n_events": 2, "deficit_minus_target_arcsec": 0.87, "deficit_minus_light_arcsec": 2.36, "peak_on_target": true, "depth_ap3": 0.04947, "psf_frac_resid": 0.971, "n_neighbours_60": 29, "n_possible": 0, "brightest_nb": {"G": 15.619723, "sep": 56.9}}]. Own Gaia<15 cone neighbor table: []. Required fraction uses G contrast and observed depth, an approximate bandpass/dilution test; original deeper Gaia cone receipts above test fainter neighbors.

Archives, physical plausibility, and verdict

Fresh IDIA60arcsec queries returned all nine table sections: VSX, ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert, Prsa. Counts {"B_vsx_vsx": 0, "II_366_catv2021": 0, "B_sb9_main": 0, "B_wds_wds": 0, "J_ApJS_279_50_table2": 3, "J_ApJS_279_50_table3": 0, "J_ApJS_279_50_table4": 0, "J_A_A_691_A242_obaf-eb1": 0, "J_ApJS_258_16_tess-ebs": 0}. Cone matches were checked by exact TIC/Gaia/coordinates; table2 is unvetted and nearby sources do not establish target prior art. Exact-source SIMBAD [{"id": "Gaia DR3 462316159559941760", "main_id": "TYC 4062-394-1", "otype": "SB*", "ra": "51.827049658750006", "dec": "60.18350188889"}]. Shared successfully queried Gaia variability/EB rows [] / []; ExoFOP {"toi_count": 0, "ctoi_count": 0}. Exact-TIC/name web searches returned no results. Current archive responses, URLs and hashes retained; bibliography/name search is not exhaustive. RV mass function/minimum masses and central-duration assumptions: [{"m1": 0.9, "m2_min": 0.3433992546283835, "f_mass": 0.02619249315593742, "a_rsun": 22.61032651826692, "r1_rsun": 2.11438, "r2_approx_rsun": 0.42524525714439954, "central_T14_hours": {"A": 6.7425956237939175, "B": 13.340681732024553}, "observed_T14_hours": 4.982896551724137}, {"m1": 1.2, "m2_min": 0.40753611994164674, "f_mass": 0.02619249315593742, "a_rsun": 24.63146330557811, "r1_rsun": 2.11438, "r2_approx_rsun": 0.48767768647855536, "central_T14_hours": {"A": 6.341485709994628, "B": 12.547058623340307}, "observed_T14_hours": 4.982896551724137}, {"m1": 1.6, "m2_min": 0.484631726339817, "f_mass": 0.02619249315593742, "a_rsun": 26.860437422609678, "r1_rsun": 2.11438, "r2_approx_rsun": 0.56018241742698, "central_T14_hours": {"A": 5.977285054157102, "B": 11.826462979286369}, "observed_T14_hours": 4.982896551724137}]. Radius estimates here are explicit approximate main-sequence hypotheses, not measured constraints; physical

EB confirmation does not require precise component masses/radii. At A the primary is behind its companion in the adopted Gaia-primary omega convention; at B the primary is in front. Primary eclipse kindA, other kindB. Unknown flux ratios/multiplicity can alter component assignment. Blind raw-SAP rolling-median census contains15 runs;0 off-primary/off-secondary runs deeper than half median primary after duration allowances. This census is a diagnostic, not proof that every small run is stellar. Sector-wide dip list and quality/background gates are addressed by repeated events, independent reduction and localization; known systematic times alone cannot explain these orbital event sequences. Backgrounds were regressed in their native units and projected through the same template into fake normalized depth; raw SAP_BKG values are never divided by normalized SAP flux. Maximum absolute fake primary depth=0.684ppt; event-level BG refits, flags, centroids and scan-edge status remain in JSON. Shape/baseline limitations above bound precision; they do not erase the repeated source-localized eclipses.

CONFIRMED target eclipsing system: Eight primary timings and the alternate reduction establish repeating eclipses with Gaia attribution and no G<15capable neighbor. Existing deficit peaks on target, although its PSF residual is poor, so do not quote a precision centroid. No reliable secondary; conditional width/phase limits vary strongly for broad templates, up to59.6ppt, so no single uniform strong exclusion is claimed. Approximate eccentric geometry permits a miss.

Next nominal eclipses after2026-10-30T00UTC: primary4343.50698BTJD; other4348.38812BTJD. Other phase is photometric conditional where detected, Gaia/diagnostic where absent. These are planning predictions, not certified precise schedules; extrapolate covariance/model scatter, especially held-period365181558. RV epoch conventions and component identities remain explicit above. Next check: obtain targeted multi-band resolved eclipse photometry and joint RV/light-curve modeling for component dimensions; preserve baseline/systematic and secondary-coverage limits.

Reproducibility: Python/numpy/astropy scripts within this folder; fit_batch.py -> orbit_gates.py -> secondary_fit.py -> shape_secondary.py -> secondary_fit.py -> geometry.py; targeted aperture and null tests as applicable. Downloads and shared archive inputs have SHA256 manifests. Original live candidate status/vet directory were checked before work (source-audit.json); source paper unchanged, no commits.

R041. TIC 30311062

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 30311062: CONFIRMED

Repeated primary and secondary eclipses, concordant independent reduction and own pixel regression support an eclipsing target. The capable 25.8 arcsec neighbour is strongly disfavoured in every pixel event.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 7688-1322-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

22 accepted primary fits across 5 sectors; median depth 53.465 ppt. Shared exposure-integrated trapezoid T14=2.5942 h, profile range [2.53557931 2.65288966] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=3.8955409024 ± 6.08e-07 d; Tref=2476.7759143 ± 0.000168 BTJD, reference cycle 245; χ^2/dof=13.75/20, O−C RMS=0.511 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 23 usable windows, 23 individual >3σ; combined depth 6.4164 ± 0.1063 ppt (event-scatter error). Fitted phase=0.4999561, Gaia nominal=0.4722082. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=3.8955419398 ± 2.52e-06 d; Tref=2404.7079841 ± 0.00062 BTJD, reference cycle 226; χ^2/dof=41.86/21, O−C RMS=3.592 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	8	1522.3683029 ± 0.00269	53.181 ± 0.560	2.594/0.908/0.778	1	-0.130025
P	8	1526.2638395 ± 0.00269	52.748 ± 0.640	2.594/0.908/0.778	1	-0.136153
P	8	1537.9504495 ± 0.00269	53.435 ± 0.566	2.594/0.908/0.778	1	-0.154534
P	9	1545.7415228 ± 0.00269	54.128 ± 0.630	2.594/0.908/0.778	1.07634	-0.166789
P	9	1549.6370594 ± 0.00269	52.964 ± 0.628	2.594/0.908/0.778	1	-0.172916
P	9	1553.5319475 ± 0.00269	53.443 ± 0.611	2.594/0.908/0.778	1	-1.11297
P	9	1561.3243179 ± 0.00269	53.364 ± 0.641	2.594/0.908/0.778	1	0.742627
P	9	1565.2192060 ± 0.00269	53.106 ± 0.554	2.594/0.908/0.778	1	-0.197425
P	35	2258.6253778 ± 0.000556	54.759 ± 0.536	2.594/0.908/0.778	1.01045	-0.354146

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	35	2262.5209144 ± 0.000536	54.320 ± 0.649	2.594/0.908/0.778	1.05479	-0.360273
P	35	2278.1030610 ± 0.000579	54.247 ± 0.632	2.594/0.908/0.778	1	-0.384782
P	36	2285.8947829 ± 0.000556	54.177 ± 0.589	2.594/0.908/0.778	1.03355	0.536888
P	36	2289.7903195 ± 0.000518	53.920 ± 0.657	2.594/0.908/0.778	1.24742	0.53076
P	36	2297.5813928 ± 0.000518	54.881 ± 0.747	2.594/0.908/0.778	1.27875	0.518506
P	36	2301.4762809 ± 0.000501	54.413 ± 0.594	2.594/0.908/0.778	1.31815	-0.421546
P	89	3719.4535662 ± 0.000324	53.487 ± 0.515	2.594/0.908/0.778	1.12975	0.149918
P	89	3723.3484543 ± 0.000324	53.415 ± 0.549	2.594/0.908/0.778	1.30092	-0.790133
P	89	3727.2446395 ± 0.000324	53.521 ± 0.543	2.594/0.908/0.778	1.18982	0.137664
P	89	3731.1401762 ± 0.000324	53.144 ± 0.617	2.594/0.908/0.778	1.25237	0.131537
P	89	3735.0350643 ± 0.000324	53.385 ± 0.581	2.594/0.908/0.778	1.23284	-0.808515
P	89	3742.8274347 ± 0.000324	53.173 ± 0.573	2.594/0.908/0.778	1.19895	1.04708
P	89	3746.7223228 ± 0.000324	53.846 ± 0.662	2.594/0.908/0.778	1.07113	0.107028
S	8	1524.3172896 ± 0.00246	5.984 ± 0.353	2.594/0.908/0.778	1.08619	2.56884
S	8	1528.2095877 ± 0.00246	6.843 ± 0.419	2.594/0.908/0.778	1	-2.10228
S	8	1539.8968590 ± 0.00301	6.304 ± 0.378	2.594/0.908/0.778	1	-1.17284
S	9	1547.6885893 ± 0.00246	6.103 ± 0.463	2.594/0.908/0.778	1	-0.241899
S	9	1551.5821846 ± 0.00269	6.915 ± 0.447	2.594/0.908/0.778	1	-3.04517
S	9	1559.3732664 ± 0.00269	6.372 ± 0.438	2.594/0.908/0.778	1	-3.04815
S	9	1563.2726986 ± 0.00301	6.162 ± 0.479	2.594/0.908/0.778	1.06847	2.5539
S	9	1567.1688881 ± 0.00269	6.457 ± 0.503	2.594/0.908/0.778	1.08676	3.48633
S	35	2256.6763850 ± 0.00162	7.115 ± 0.451	2.594/0.908/0.778	1	-1.44771
S	35	2260.5719259 ± 0.00195	7.461 ± 0.640	2.594/0.908/0.778	1.24434	-1.4492
S	35	2264.4694125 ± 0.00292	5.867 ± 0.623	2.594/0.908/0.778	1.21943	1.35108
S	35	2272.2546573 ± 0.00227	6.930 ± 0.541	2.594/0.908/0.778	1.05792	-7.05723
S	35	2276.1579809 ± 0.00389	5.491 ± 0.533	2.594/0.908/0.778	1.13964	4.14837
S	36	2283.9484141 ± 0.00162	6.373 ± 0.491	2.594/0.908/0.778	1.10519	3.21146
S	36	2287.8426579 ± 0.00162	6.981 ± 0.447	2.594/0.908/0.778	1	1.34212
S	36	2291.7375502 ± 0.00195	5.562 ± 0.465	2.594/0.908/0.778	1.21341	0.406697
S	36	2303.4248215 ± 0.00195	6.642 ± 0.447	2.594/0.908/0.778	1.00254	1.33614
S	89	3721.3952244 ± 0.00162	5.689 ± 0.440	2.594/0.908/0.778	1	-8.54687
S	89	3725.2978995 ± 0.00162	6.279 ± 0.439	2.594/0.908/0.778	1.00749	1.7248
S	89	3729.1940889 ± 0.0013	6.363 ± 0.424	2.594/0.908/0.778	1.0649	2.65723
S	89	3736.9832250 ± 0.0013	7.050 ± 0.433	2.594/0.908/0.778	1	-0.147527
S	89	3740.8755232 ± 0.00162	6.712 ± 0.470	2.594/0.908/0.778	1.0687	-4.81864
S	89	3744.7794953 ± 0.00162	6.217 ± 0.478	2.594/0.908/0.778	1.04952	7.32088

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 22 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9937; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 50 triggers; 1 outside the two eclipse phase windows. Largest off-phase dip=4.015 ppt, versus primary median 53.465 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 5331998419615383296; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=0.736σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.05773932081262956,
      "epoch389_gaia_error": 0.0333814287225213
    },
    {
      "kind": "B",
      "epoch389_offset": -1.9017475239081705,
      "epoch389_gaia_error": 0.02602670286563926
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.47220815810774097,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.45725756759080327,
      0.47363311595792956,
      0.48741877166881104
    ],
    "gaia_relative_phase_sigma": 0.014681907149251458,
    "gaia_e_cos_omega_quantiles": [
      -0.06714117234692414,
      -0.04140133817282157,
      -0.019757666776569708
    ],
    "gaia_e_sin_omega_quantiles": [
      -0.04411436145591112,
      -0.025652195219393575,
      -0.01049851667360484
    ],
    "photometric_phase": 0.49995606427953015,
    "photometric_e_cos_omega_first_order": -6.901406832912049e-05,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      -5.551115123125783e-17,
      0.1499999999999999
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 1.26,
      "m2_min": 0.6217224924042701,
      "f_mass": 0.06787011513003473,
      "a_rsun": 12.862494442295503,
      "r1_rsun": 1.248,
      "r2_approx_rsun": 0.6837183934664118,
      "central_T14_hours": {
        "A": 4.59984449468575,
```

```
        "B": 4.334056685019584
      },
      "observed_T14_hours": 2.594234482758621
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 131.23966706443295,
      "dof": 121,
      "k": 0.44999999999999996,
      "b_A": 1.0750000000000002,
      "e_sin_omega": -5.551115123125783e-17,
      "primary_scale": 0.8772294505408279,
      "secondary_scale": 0.019771395014470575
    },
    "profile_interval_approx_95": {
      "k": [
        0.32499999999999996,
        0.44999999999999996
      ],
      "b_A": [
        0.8300000000000001,
        1.0750000000000002
      ],
      "e_sin_omega": [
        -5.551115123125783e-17,
        0.1499999999999999
      ]
    },
    "error_inflation": 1.0846253476399417,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TARS, sector 89: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
3719.45357	FLUX	41.8469	0.692875	42.7122	47	n/a
3723.34845	FLUX	42.2192	0.48355	43.0431	46	n/a
3727.24464	FLUX	43.2549	0.775645	44.523	47	n/a
3731.14018	FLUX	43.1723	0.633141	44.5356	47	n/a

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
3735.03506	FLUX	45.4171	0.637084	46.6709	47	n/a
3742.82743	FLUX	48.7159	0.958973	49.6814	46	n/a
3746.72232	FLUX	48.7572	1.0262	48.3701	46	n/a

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable
5331998423911867520	25.7815	14.190861	0.673572	True

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "30311062",
    "sector": 89,
    "n_events": 8,
    "deficit_minus_target_arcsec": 3.67,
    "deficit_minus_light_arcsec": 4.15,
    "peak_on_target": true,
    "depth_ap3": 0.04381,
    "psf_frac_resid": 0.373,
    "n_neighbours_60": 28,
    "n_possible": 1,
    "brightest_nb": {
      "G": 14.190861,
      "sep": 25.8
    }
  }
]
```


pixel-localization.json:

Ancillary log block 3: 104 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 6, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4346.63555; next other-kind approximately 4344.68712 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Repeated primary and secondary eclipses, concordant independent reduction and own pixel regression support an eclipsing target. The capable 25.8 arcsec neighbour is strongly disfavoured in every pixel event.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R042. TIC 290592570

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 290592570: CONFIRMED

Two eclipse kinds repeat at the resolved orbital period and match the RV system. Own calibrated pixel fits favour the target over its 9.1 arcsec bright neighbour in both covered events. Depths depend on aperture/dilution; this does not remove the eclipses.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 7467-193-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

8 accepted primary fits across 3 sectors; median depth 81.856 ppt. Shared exposure-integrated trapezoid T14=4.1909 h, profile range [4.104 4.2777931] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=8.8871978965 ± 2.08e-06 d; Tref=3005.4048697 ± 0.000185 BTJD, reference cycle 108; χ^2/dof=3.97/6, O–C RMS=0.531 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 7 usable windows, 7 individual >3σ; combined depth 34.6232 ± 0.9374 ppt (event-scatter error). Fitted phase=0.4970615, Gaia nominal=0.5005982. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=8.8872001208 ± 2.29e-06 d; Tref=2876.5145337 ± 0.000198 BTJD, reference cycle 93; χ^2/dof=2.48/5, O–C RMS=0.449 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O–C min
P	27	2036.7007263 ± 0.000524	82.945 ± 0.623	4.191/1.467/1.257	1.30221	0.615282
P	27	2045.5868954 ± 0.000524	82.665 ± 0.562	4.191/1.467/1.257	1.42372	-0.866118
P	27	2054.4751600 ± 0.000524	82.428 ± 0.653	4.191/1.467/1.257	1.18092	0.669927
P	67	3129.8252582 ± 0.000524	82.125 ± 0.640	4.191/1.467/1.257	2.01262	-0.550187
P	67	3147.5996920 ± 0.000524	81.588 ± 1.024	4.191/1.467/1.257	2.16595	-0.495542
P	94	3858.5759939 ± 0.000524	69.843 ± 0.550	4.191/1.467/1.257	2.03132	0.181556
P	94	3867.4632108 ± 0.000524	69.440 ± 0.719	4.191/1.467/1.257	2.03712	0.208879
P	94	3876.3504276 ± 0.000524	69.841 ± 0.583	4.191/1.467/1.257	2.07721	0.236202
S	27	2041.1177827 ± 0.000524	37.003 ± 0.473	4.191/1.467/1.257	1.48798	0.0869243

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
S	27	2050.0049806 ± 0.000524	37.035 ± 0.315	4.191/1.467/1.257	1.1644	0.0837213
S	27	2058.8921785 ± 0.000524	36.954 ± 0.363	4.191/1.467/1.257	1.03461	0.0805183
S	67	3134.2431240 ± 0.000524	35.657 ± 0.333	4.191/1.467/1.257	1.37515	-0.307047
S	67	3143.1303219 ± 0.000524	35.918 ± 0.334	4.191/1.467/1.257	1.51749	-0.31025
S	94	3862.9933515 ± 0.000524	31.221 ± 0.337	4.191/1.467/1.257	1.55127	-0.569694
S	94	3871.8815971 ± 0.000524	31.058 ± 0.286	4.191/1.467/1.257	1.31994	0.935826

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 20 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9993; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 19 triggers; 2 outside the two eclipse phase windows. Largest off-phase dip=6.451 ppt, versus primary median 81.856 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 6695056769751681664; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=2.098σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": 0.014802134114063392,
      "epoch389_gaia_error": 0.020711153170966536
    },
    {
      "kind": "B",
      "epoch389_offset": -4.432148571263724,
      "epoch389_gaia_error": 0.022799711586115157
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.5005982136030157,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.4966178325324745,
      0.5004188405102833,
      0.5041068990386895
    ],
    "gaia_relative_phase_sigma": 0.004174853945575952,
    "gaia_e_cos_omega_quantiles": [
      -0.005311852857204527,
      0.0006578823338586681,
      0.006450399662696864
    ],
    "gaia_e_sin_omega_quantiles": [
      0.009480644197146965,
```

```
      0.017069386767230507,
      0.02496020609739401
    ],
    "photometric_phase": 0.49706147181960564,
    "photometric_e_cos_omega_first_order": -0.004615829271946755,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      0.0749999999999999,
      0.09999999999999992
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 1.07,
      "m2_min": 0.7405549838443879,
      "f_mass": 0.12389349106526702,
      "a_rsun": 22.005243695569593,
      "r1_rsun": 1.53232,
      "r2_approx_rsun": 0.7864042415243565,
      "central_T14_hours": {
        "A": 7.025347153451335,
        "B": 7.283732188639722
      },
      "observed_T14_hours": 4.190896551724138
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 150.22054515725134,
      "dof": 121,
      "k": 0.3749999999999999,
      "b_A": 0.9000000000000001,
      "e_sin_omega": 0.0749999999999999,
      "primary_scale": 0.987098050162664,
      "secondary_scale": 0.09794134347107186
    },
    "profile_interval_approx_95": {
      "k": [
        0.35,
        0.3749999999999999
      ],
      "b_A": [
        0.865,
        0.9000000000000001
      ],
      "e_sin_omega": [
        0.0749999999999999,
        0.09999999999999992
      ]
    },
    "error_inflation": 1.2414921087376143,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```

Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TESS-SPOC, sector 27: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2036.70073	SAP_FLUX	47.4536	0.373625	47.625	25	47.5145
2036.70073	PDCSAP_FLUX	80.3903	0.621327	80.7928	25	80.4691
2045.5869	SAP_FLUX	46.9238	0.338916	46.907	22	46.93
2045.5869	PDCSAP_FLUX	79.4024	0.570665	79.3774	22	79.4059
2054.47516	SAP_FLUX	46.6413	0.339209	46.5793	22	46.7156
2054.47516	PDCSAP_FLUX	78.8202	0.616742	78.2573	22	78.0879

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable
6695056838471158912	9.06145	11.102676	0.122916	True

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "290592570",
    "sector": 27,
    "n_events": 5,
    "deficit_minus_light_arcsec": 3.57,
    "peak_on_target": true,
    "depth_ap3": 0.03128,
    "psf_frac_resid": 0.178,
    "n_neighbours_60": 12,
    "n_possible": 1,
    "brightest_nb": {
      "G": 11.102676,
      "sep": 9.1
    }
  },
  {
    "tic": "290592570",
    "sector": 27,
    "n_events": 5,
    "deficit_minus_target_arcsec": 2.84,
    "deficit_minus_light_arcsec": 3.57,
    "peak_on_target": true,
    "depth_ap3": 0.03128,
    "psf_frac_resid": 0.178,
    "n_neighbours_60": 12,
    "n_possible": 1,
```

```
    "brightest_nb": {
      "G": 11.102676,
      "sep": 9.1
    }
  }
]
```


pixel-localization.json:


```
{
  "source_path": "C:\\Home\\sky\\data\\sb1-scale-c3\\vet290592570\\tesscut_s027.fits",
  "sha256": "ff435b27c5815b8d9838fd593e3296d14c4371688601054cbda4c08da484d289",
  "sector": 27,
  "calibration": {
    "sigma_pixels": 0.55,
    "dx_pixels": -0.10000000000000009,
    "dy_pixels": -0.20000000000000007
  },
  "fixed_fits": [
    {
      "source_id": "6695056769751681664",
      "sep_arcsec": 0.0,
      "G": 10.661288,
      "chi2": 2979.049171055395,
      "amplitude": 436.41668389677454,
      "fractional_residual": 0.1317029655590075
    },
    {
      "source_id": "6695056838471158912",
      "sep_arcsec": 9.061447109205899,
      "G": 11.102676,
      "chi2": 7676.342720513076,
      "amplitude": 410.7548080270191,
      "fractional_residual": 0.33015318181326486
    }
  ],
  "free_best": {
    "chi2": 2741.3711180634955,
    "pixel_x": 8.072174859875762,
    "pixel_y": 7.504140334399938,
    "distance_from_target_arcsec": 2.969848480983278,
    "fractional_residual": 0.11570610833772482
  },
  "event_source_comparisons": [
    {
      "tc": 2045.5874969191968,
      "best_source": "6695056769751681664",
      "delta_chi2_neighbor_minus_target": [
        0.0,
        1677.9542048550059
      ]
    },
    {
      "tc": 2054.4746948156812,
      "best_source": "6695056769751681664",
      "delta_chi2_neighbor_minus_target": [
        0.0,
        3008.219703521313
      ]
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal
```

```
significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional
only; degree comparison probes baseline sensitivity."
}
```

Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 1, 'J_ApJS_279_50_table2': 2, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4347.37175; next other-kind approximately 4351.78918 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Two eclipse kinds repeat at the resolved orbital period and match the RV system. Own calibrated pixel fits favour the target over its 9.1 arcsec bright neighbour in both covered events. Depths depend on aperture/dilution; this does not remove the eclipses.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R043. TIC 257800191

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 257800191: CONFIRMED

Five primary and three secondary fits, independent SPOC recovery, Gaia timing and existing on-target difference imaging establish a target EB. The short baseline and only three secondaries limit precise characterization.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 25-254-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

5 accepted primary fits across 2 sectors; median depth 55.095 ppt. Shared exposure-integrated trapezoid T14=3.6695 h, profile range [3.58262069 3.75641379] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=9.3696078307 ± 0.000176 d; Tref=2478.7629717 ± 0.000308 BTJD, reference cycle 3; χ^2/dof=6.16/3, O−C RMS=0.767 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 3 usable windows, 3 individual >3σ; combined depth 4.0265 ± 0.2471 ppt (event-scatter error). Fitted phase=0.5014626, Gaia nominal=0.4890536. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=9.3708618782 ± 0.00167 d; Tref=2474.0926564 ± 0.00276 BTJD, reference cycle 2; χ^2/dof=0.72/1, O−C RMS=2.543 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	42	2450.6537174 ± 0.000459	53.501 ± 0.373	3.670/1.284/1.101	1	-0.620355
P	42	2469.3938828 ± 0.000473	53.245 ± 0.408	3.670/1.284/1.101	1.03938	0.747227
P	43	2478.7630481 ± 0.000486	56.048 ± 0.423	3.670/1.284/1.101	1.09176	0.109992
P	43	2488.1331307 ± 0.000459	55.095 ± 0.492	3.670/1.284/1.101	1.07246	0.793783
P	43	2497.5013787 ± 0.000501	55.807 ± 0.497	3.670/1.284/1.101	1.03818	-1.16448
S	42	2455.3493345 ± 0.00332	4.526 ± 0.456	3.322/0.830/1.661	1.19648	-2.30123
S	42	2464.7239253 ± 0.00332	3.961 ± 0.353	3.322/0.830/1.661	1	3.06831
S	43	2492.8327488 ± 0.00581	3.497 ± 0.530	3.322/0.830/1.661	1.29076	-2.34917

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 18 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9990; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 7 triggers; 0 outside the two eclipse phase windows. Largest off-phase dip=0.000 ppt, versus primary median 55.095 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 2576503365647719296; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=0.263σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.013880566533089223,
      "epoch389_gaia_error": 0.13795174384628345
    },
    {
      "kind": "B",
      "epoch389_offset": -4.672073138101268,
      "epoch389_gaia_error": 0.17519736659561588
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.48905362894084214,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.472916065677294,
      0.49806626367464596,
      0.5283207143284699
    ],
    "gaia_relative_phase_sigma": 0.030157576131930944,
    "gaia_e_cos_omega_quantiles": [
      -0.042515154829713186,
      -0.0030362296051438696,
      0.044419166831589224
    ],
    "gaia_e_sin_omega_quantiles": [
      -0.02225486360812121,
      0.01171071885411363,
      0.05637572795125282
    ],
    "photometric_phase": 0.5014626420295186,
    "photometric_e_cos_omega_first_order": 0.002297512727383687,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      -5.551115123125783e-17,
```

```
      0.12499999999999994
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 1.08,
      "m2_min": 0.29714436243856335,
      "f_mass": 0.013833865467616557,
      "a_rsun": 20.807084636958,
      "r1_rsun": 1.16399,
      "r2_approx_rsun": 0.3787686238202824,
      "central_T14_hours": {
        "A": 5.18081862507593,
        "B": 5.433884258577503
      },
      "observed_T14_hours": 3.66951724137931
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 315.87272175582666,
      "dof": 121,
      "k": 0.24999999999999997,
      "b_A": 0.76,
      "e_sin_omega": 0.02499999999999994,
      "primary_scale": 0.9401934685971229,
      "secondary_scale": 0.0045837524255479215
    },
    "profile_interval_approx_95": {
      "k": [
        0.24999999999999997,
        0.29999999999999993
      ],
      "b_A": [
        0.6900000000000001,
        0.7950000000000002
      ],
      "e_sin_omega": [
        -5.551115123125783e-17,
        0.12499999999999994
      ]
    },
    "error_inflation": 2.6105183616184022,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

SPOC, sector 43: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2478.76305	SAP_FLUX	52.0055	0.324193	52.2631	110	52.1665
2478.76305	PDCSAP_FLUX	52.0933	0.332694	52.2622	110	52.2587
2488.13313	SAP_FLUX	52.0191	0.410616	52.0258	110	52.2409
2488.13313	PDCSAP_FLUX	52.0407	0.421528	52.1537	110	52.277
2497.50138	SAP_FLUX	51.8506	0.372642	51.9421	110	52.0324
2497.50138	PDCSAP_FLUX	51.8485	0.403447	51.8173	110	52.0233

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "257800191",
    "sector": 43,
    "n_events": 3,
    "deficit_minus_target_arcsec": 1.64,
    "deficit_minus_light_arcsec": 0.34,
    "peak_on_target": true,
    "depth_ap3": 0.05056,
    "psf_frac_resid": 0.289,
    "n_neighbours_60": 3,
    "n_possible": 0,
    "brightest_nb": {
      "G": 18.742105,
      "sep": 59.3
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 0, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4352.68454; next other-kind approximately 4348.01171 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Five primary and three secondary fits, independent SPOC recovery, Gaia timing and existing on-target difference imaging establish a target EB. The short baseline and only three secondaries limit precise characterization.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R044. TIC 6946549

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 6946549: CONFIRMED

Repeated primary/secondary eclipses survive independent SPOC photometry, background regression and on-target difference imaging; no capable 60 arcsec neighbour. Small Gaia-period tension and scatter are characterization caveats.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies HD 176388 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

8 accepted primary fits across 3 sectors; median depth 55.972 ppt. Shared exposure-integrated trapezoid T14=5.2436 h, profile range [5.11324138 5.37393103] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=8.0519229126 ± 9.94e-06 d; Tref=3849.0884984 ± 0.00053 BTJD, reference cycle 89; χ^2/dof=31.36/6, O−C RMS=1.869 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 6 usable windows, 6 individual >3σ; combined depth 25.0713 ± 0.8874 ppt (event-scatter error). Fitted phase=0.4875889, Gaia nominal=0.4865036. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=8.0519333073 ± 1.23e-05 d; Tref=3732.2376496 ± 0.000675 BTJD, reference cycle 74; χ^2/dof=25.46/4, O−C RMS=1.944 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	67	3132.4660327 ± 0.000655	66.100 ± 0.457	5.244/1.835/1.573	2.21905	-1.91011
P	67	3148.5712635 ± 0.000655	65.822 ± 0.621	5.244/1.835/1.573	2.99759	0.0842835
P	94	3865.1943893 ± 0.000655	55.815 ± 0.412	5.244/1.835/1.573	2.3215	2.94491
P	94	3873.2463492 ± 0.000655	54.038 ± 0.405	5.244/1.835/1.573	2.09984	2.99826
P	104	4179.2168951 ± 0.000655	56.165 ± 0.567	5.244/1.835/1.573	2.8705	-0.637441
P	104	4187.2688551 ± 0.000655	56.017 ± 0.492	5.244/1.835/1.573	2.71761	-0.584089
P	104	4195.3208151 ± 0.000655	55.927 ± 0.430	5.244/1.835/1.573	2.35325	-0.530737
P	104	4203.3714641 ± 0.000655	55.803 ± 0.423	5.244/1.835/1.573	2.37893	-2.36508
S	67	3128.3414648 ± 0.000655	28.463 ± 0.236	5.244/1.835/1.573	1.59084	-1.70891

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
S	67	3144.4466215 ± 0.000655	28.285 ± 0.221	5.244/1.835/1.573	1.52571	0.148846
S	94	3861.0703825 ± 0.000655	23.603 ± 0.223	5.244/1.835/1.573	1.67309	2.59204
S	94	3877.1742284 ± 0.000655	23.237 ± 0.195	5.244/1.835/1.573	1.52335	2.56211
S	104	4183.1446773 ± 0.000655	24.186 ± 0.214	5.244/1.835/1.573	1.71009	-1.78207
S	104	4199.2485231 ± 0.000655	24.267 ± 0.164	5.244/1.835/1.573	1.45878	-1.81201

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 11 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9995; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 30 triggers; 13 outside the two eclipse phase windows. Largest off-phase dip=8.657 ppt, versus primary median 55.972 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 6657873519885638144; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=-2.164σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": 0.00874866586434564,
      "epoch389_gaia_error": 0.0426999455321721
    },
    {
      "kind": "B",
      "epoch389_offset": -3.9175385711929493,
      "epoch389_gaia_error": 0.01897283970891443
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.4865035711117025,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.4810694076491121,
      0.48750646816807414,
      0.49439481584739775
    ],
    "gaia_relative_phase_sigma": 0.0067444774266091365,
    "gaia_e_cos_omega_quantiles": [
      -0.029736052483049096,
      -0.01962106588925358,
      -0.008800758145478952
    ],
    "gaia_e_sin_omega_quantiles": [
      0.010963023803343356,
      0.02161645960729645,
```

```
      0.03170726388345989
    ],
    "photometric_phase": 0.48758894046256385,
    "photometric_e_cos_omega_first_order": -0.01949524673303748,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      0.1499999999999999,
      0.1499999999999999
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 1.65,
      "m2_min": 0.9907179702334377,
      "f_mass": 0.13944600606556237,
      "a_rsun": 23.36851181084498,
      "r1_rsun": 2.40266,
      "r2_approx_rsun": 0.992567457978695,
      "central_T14_hours": {
        "A": 8.73067717476338,
        "B": 9.14561808925202
      },
      "observed_T14_hours": 5.243586206896551
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 98.42753795213136,
      "dof": 121,
      "k": 0.27499999999999997,
      "b_A": 0.76,
      "e_sin_omega": 0.1499999999999999,
      "primary_scale": 0.834241799100291,
      "secondary_scale": 0.06758187226133039
    },
    "profile_interval_approx_95": {
      "k": [
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      ],
      "b_A": [
        0.76,
        0.76
      ],
      "e_sin_omega": [
        0.1499999999999999,
        0.1499999999999999
      ]
    },
    "error_inflation": 1,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```

Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

SPOC, sector 104: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
4187.26886	SAP_FLUX	62.9947	0.868542	63.0532	157	63.1334
4187.26886	PDCSAP_FLUX	64.1356	0.867554	64.0838	157	64.2649
4195.32082	SAP_FLUX	63.009	0.850754	63.0117	157	63.2245
4195.32082	PDCSAP_FLUX	63.8489	0.877183	63.919	157	64.0534
4203.37146	SAP_FLUX	62.7654	1.03422	62.8591	157	62.8934
4203.37146	PDCSAP_FLUX	63.9878	1.06392	64.1162	157	64.1165

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "6946549",
    "sector": 104,
    "n_events": 4,
    "deficit_minus_light_arcsec": 0.43,
    "peak_on_target": true,
    "depth_ap3": 0.06137,
    "psf_frac_resid": 0.02,
    "n_neighbours_60": 41,
    "n_possible": 0,
    "brightest_nb": {
      "G": 15.628898,
      "sep": 48.3
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  },
  {
    "tic": "6946549",
    "sector": 104,
    "n_events": 4,
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    "deficit_minus_light_arcsec": 0.43,
    "peak_on_target": true,
    "depth_ap3": 0.06137,
    "psf_frac_resid": 0.02,
    "n_neighbours_60": 41,
    "n_possible": 0,
```

```
    "brightest_nb": {
      "G": 15.628898,
      "sep": 48.3
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 0, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4348.30772; next other-kind approximately 4352.23494 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Repeated primary/secondary eclipses survive independent SPOC photometry, background regression and on-target difference imaging; no capable 60 arcsec neighbour. Small Gaia-period tension and scatter are characterization caveats.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R045. TIC 149021625

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 149021625: CONFIRMED

Thirty-three primary and thirty secondary events, independent reduction and clean target-localized deficit establish a target EB. Gaia small-e geometry and simplified shape model are not precise component constraints.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 6990-1025-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

33 accepted primary fits across 4 sectors; median depth 53.599 ppt. Shared exposure-integrated trapezoid T14=3.1808 h, profile range [3.12213103 3.23944138] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=2.3565823031 ± 2.95e-07 d; Tref=2586.3495796 ± 0.000114 BTJD, reference cycle 522; χ^2/dof=19.28/31, O−C RMS=0.541 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 30 usable windows, 30 individual >3σ; combined depth 13.9046 ± 0.1891 ppt (event-scatter error). Fitted phase=0.4997485, Gaia nominal=0.4889485. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=2.3565807011 ± 7.4e-07 d; Tref=2573.3888172 ± 0.000291 BTJD, reference cycle 516; χ^2/dof=121.26/28, O−C RMS=1.692 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	2	1358.5697063 ± 0.00246	55.656 ± 0.410	3.181/1.113/0.954	1.0061	-0.710425
P	2	1360.9262886 ± 0.00227	55.653 ± 0.481	3.181/1.113/0.954	1	-0.710452
P	2	1363.2836661 ± 0.00227	55.453 ± 0.431	3.181/1.113/0.954	1	0.434604
P	2	1365.6402484 ± 0.00227	55.217 ± 0.529	3.181/1.113/0.954	1	0.434577
P	2	1370.3534129 ± 0.00246	55.415 ± 0.566	3.181/1.113/0.954	1	0.434523
P	2	1372.7099952 ± 0.00246	55.243 ± 0.491	3.181/1.113/0.954	1	0.434496
P	2	1375.0665775 ± 0.00246	55.417 ± 0.383	3.181/1.113/0.954	1.02682	0.434469
P	2	1377.4231598 ± 0.00246	55.631 ± 0.425	3.181/1.113/0.954	1.01346	0.434442
P	2	1379.7797421 ± 0.00246	56.155 ± 0.620	3.181/1.113/0.954	1.02544	0.434415

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	29	2089.1110097 ± 0.000518	53.947 ± 0.476	3.181/1.113/0.954	1.22591	0.426309
P	29	2091.4675920 ± 0.000473	54.341 ± 0.426	3.181/1.113/0.954	1.13444	0.426282
P	29	2093.8233790 ± 0.000536	53.064 ± 0.447	3.181/1.113/0.954	1.12772	-0.718828
P	29	2096.1807565 ± 0.000486	53.621 ± 0.528	3.181/1.113/0.954	1.22618	0.426228
P	29	2098.5373388 ± 0.000536	54.808 ± 0.800	3.181/1.113/0.954	1.35129	0.426201
P	29	2103.2505034 ± 0.000501	54.623 ± 0.566	3.181/1.113/0.954	1.32356	0.426148
P	29	2105.6062905 ± 0.000579	53.695 ± 0.523	3.181/1.113/0.954	1.18807	-0.718962
P	29	2110.3194550 ± 0.000579	53.807 ± 0.534	3.181/1.113/0.954	1.10435	-0.719016
P	69	3184.9217719 ± 0.000398	53.385 ± 0.567	3.181/1.113/0.954	2.04255	0.413786
P	69	3187.2775590 ± 0.000398	53.599 ± 0.556	3.181/1.113/0.954	1.98625	-0.731324
P	69	3189.6349365 ± 0.000398	53.551 ± 0.531	3.181/1.113/0.954	1.85915	0.413732
P	69	3191.9915188 ± 0.000398	53.534 ± 0.448	3.181/1.113/0.954	1.54211	0.413705
P	69	3196.7046833 ± 0.000398	53.542 ± 0.567	3.181/1.113/0.954	1.98439	0.413652
P	69	3199.0604704 ± 0.000398	53.479 ± 0.506	3.181/1.113/0.954	1.77829	-0.731458
P	69	3201.4178479 ± 0.000398	52.786 ± 0.495	3.181/1.113/0.954	1.44392	0.413598
P	69	3203.7736350 ± 0.000398	53.177 ± 0.474	3.181/1.113/0.954	1.81008	-0.731512
P	69	3206.1302173 ± 0.000398	52.490 ± 0.431	3.181/1.113/0.954	1.45592	-0.731539
P	96	3910.7483203 ± 0.000398	47.091 ± 0.406	3.181/1.113/0.954	1.80161	-0.739592
P	96	3913.1056978 ± 0.000398	45.907 ± 0.395	3.181/1.113/0.954	1.6089	0.405464
P	96	3915.4622801 ± 0.000398	45.551 ± 0.392	3.181/1.113/0.954	1.79652	0.405438
P	96	3917.8188624 ± 0.000398	45.620 ± 0.396	3.181/1.113/0.954	1.67908	0.405411
P	96	3924.8886092 ± 0.000398	46.001 ± 0.533	3.181/1.113/0.954	1.82935	0.40533
P	96	3929.6017738 ± 0.000398	45.433 ± 0.467	3.181/1.113/0.954	1.96203	0.405276
P	96	3931.9575609 ± 0.000398	45.337 ± 0.493	3.181/1.113/0.954	1.84315	-0.739834
S	2	1355.0355874 ± 0.00227	15.094 ± 0.349	3.415/1.537/0.342	1	-1.45066
S	2	1357.3913158 ± 0.00227	14.531 ± 0.388	3.415/1.537/0.342	1	-2.6779
S	2	1359.7487520 ± 0.00227	14.241 ± 0.436	3.415/1.537/0.342	1.09823	-1.44604
S	2	1362.1070420 ± 0.00227	14.533 ± 0.437	3.415/1.537/0.342	1.08271	1.01536
S	2	1364.4619166 ± 0.00246	14.525 ± 0.571	3.415/1.537/0.342	1.1582	-1.44143
S	2	1369.1767889 ± 0.00227	14.845 ± 0.584	3.415/1.537/0.342	1.17713	1.02228
S	2	1371.5333712 ± 0.00227	14.358 ± 0.408	3.415/1.537/0.342	1.0614	1.02458
S	2	1373.8890996 ± 0.00227	14.500 ± 0.460	3.415/1.537/0.342	1.10247	-0.202655
S	2	1376.2448281 ± 0.00246	14.600 ± 0.370	3.415/1.537/0.342	1	-1.42989
S	29	2090.2892659 ± 0.000536	14.596 ± 0.330	3.415/1.537/0.342	1.12361	-0.730886
S	29	2092.6458482 ± 0.000518	14.893 ± 0.362	3.415/1.537/0.342	1.2572	-0.728579
S	29	2104.4296136 ± 0.000518	14.933 ± 0.409	3.415/1.537/0.342	1.23963	0.512503
S	29	2106.7879036 ± 0.000536	14.971 ± 0.351	3.415/1.537/0.342	1.12585	2.9739
S	29	2109.1444859 ± 0.000501	14.839 ± 0.469	3.415/1.537/0.342	1.24764	2.97621
S	29	2111.4985067 ± 0.000854	15.214 ± 0.482	3.415/1.537/0.342	1.10033	-0.710123
S	69	3186.0991830 ± 0.000854	14.670 ± 0.392	3.415/1.537/0.342	1.41362	-0.887695
S	69	3188.4549115 ± 0.000427	15.851 ± 0.402	3.415/1.537/0.342	1.43641	-2.11493
S	69	3190.8123476 ± 0.000854	14.716 ± 0.540	3.415/1.537/0.342	1.97763	-0.883081
S	69	3193.1680761 ± 0.000854	13.550 ± 0.438	3.415/1.537/0.342	1.34797	-2.11032
S	69	3195.5238046 ± 0.000854	14.627 ± 0.456	3.415/1.537/0.342	1.55709	-3.33756
S	69	3197.8812407 ± 0.000427	14.280 ± 0.292	3.415/1.537/0.342	1.23569	-2.10571
S	69	3200.2378230 ± 0.000427	15.262 ± 0.332	3.415/1.537/0.342	1.28247	-2.1034
S	69	3202.5952592 ± 0.000427	15.009 ± 0.240	3.415/1.537/0.342	1	-0.871546
S	69	3204.9535492 ± 0.000427	15.134 ± 0.438	3.415/1.537/0.342	1.58258	1.58985
S	96	3911.9256786 ± 0.000427	12.441 ± 0.236	3.415/1.537/0.342	1.07323	-1.4067
S	96	3916.6405509 ± 0.000427	12.296 ± 0.224	3.415/1.537/0.342	1.10342	1.05701
S	96	3918.9971332 ± 0.000427	12.393 ± 0.243	3.415/1.537/0.342	1.12446	1.05932

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
S	96	3926.0677339 ± 0.00128	12.124 ± 0.574	3.415/1.537/0.342	1.12624	2.29578
S	96	3928.4234624 ± 0.000427	12.375 ± 0.272	3.415/1.537/0.342	1.23781	1.06854
S	96	3930.7808985 ± 0.000427	12.424 ± 0.236	3.415/1.537/0.342	1.15559	2.3004

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 37 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9914; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 110 triggers; 28 outside the two eclipse phase windows. Largest off-phase dip=8.641 ppt, versus primary median 53.599 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 2320146732073841408; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=0.000σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.016199541222931657,
      "epoch389_gaia_error": 0.01642193785552541
    },
    {
      "kind": "B",
      "epoch389_offset": -1.1695032553310512,
      "epoch389_gaia_error": 0.009850584901260957
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.4889484978109181,
    "gaia_relative_phase_quantiles_16_50_84": [
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      0.4894701463956819,
      0.4990415276347008
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    "gaia_relative_phase_sigma": 0.009713624032738821,
    "gaia_e_cos_omega_quantiles": [
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      -0.016513726794503268,
      -0.0015021293599789447
    ],
    "gaia_e_sin_omega_quantiles": [
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      -0.057491540080314305,
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    ],
    "photometric_phase": 0.49974849781091807,
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    "photometric_e_cos_omega_first_order": -0.0003950587147907733,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
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      0.09999999999999992
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
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      "m2_min": 0.5952686102564131,
      "f_mass": 0.06401137408774017,
      "a_rsun": 9.09089119754108,
      "r1_rsun": 2.14238,
      "r2_approx_rsun": 0.660344227175269,
      "central_T14_hours": {
        "A": 5.889593925393931,
        "B": 5.229010420662292
      },
      "observed_T14_hours": 3.180786206896552
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 1028.4800915447115,
      "dof": 121,
      "k": 0.24999999999999997,
      "b_A": 0.8300000000000001,
      "e_sin_omega": 0.09999999999999992,
      "primary_scale": 1.0,
      "secondary_scale": 0.03158288534300896
    },
    "profile_interval_approx_95": {
      "k": [
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      ],
      "b_A": [
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        0.935
      ],
      "e_sin_omega": [
        0.0749999999999999,
        0.09999999999999992
      ]
    },
    "error_inflation": 8.499835467311666,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180°

change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TESS-SPOC, sector 2: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
1358.56971	SAP_FLUX	52.2193	0.358926	55.5245	6	52.2426
1358.56971	PDCSAP_FLUX	52.3859	0.372415	55.7678	6	52.3274
1360.92629	SAP_FLUX	52.4785	0.461041	55.5936	6	52.6734
1360.92629	PDCSAP_FLUX	52.5531	0.460458	55.63	6	52.7678
1363.28367	SAP_FLUX	52.3568	0.408617	55.415	7	52.0293
1363.28367	PDCSAP_FLUX	52.3625	0.411738	55.4904	7	52.09
1365.64025	SAP_FLUX	52.1133	0.471168	55.2168	7	52.0159
1365.64025	PDCSAP_FLUX	52.1506	0.482132	55.3331	7	52.0035
1370.35341	SAP_FLUX	52.4701	0.513911	55.2895	6	52.4411
1370.35341	PDCSAP_FLUX	52.4428	0.51376	55.3337	6	52.4377
1372.71	SAP_FLUX	52.0073	0.420846	55.7467	6	51.9978
1372.71	PDCSAP_FLUX	52.1425	0.416406	55.8787	6	52.0829
1375.06658	SAP_FLUX	52.2928	0.31855	55.416	6	52.6607
1375.06658	PDCSAP_FLUX	52.4702	0.322058	55.588	6	52.8435
1377.42316	SAP_FLUX	52.4963	0.437528	55.6725	6	52.794
1377.42316	PDCSAP_FLUX	52.7184	0.432692	55.9125	6	53.0365
1379.77974	SAP_FLUX	52.951	0.60816	56.5663	6	52.3906
1379.77974	PDCSAP_FLUX	52.9949	0.530457	56.4432	6	52.2268

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


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[
  {
    "tic": "149021625",
    "sector": 29,
    "n_events": 9,
    "deficit_minus_light_arcsec": 0.19,
    "peak_on_target": true,
    "depth_ap3": 0.05279,
    "psf_frac_resid": 0.034,
    "n_neighbours_60": 4,
    "n_possible": 0,
    "brightest_nb": {
      "G": 18.08154,
      "sep": 48.6
    }
  },
  {
```

```
    "tic": "149021625",
    "sector": 29,
    "n_events": 9,
    "deficit_minus_target_arcsec": 4.12,
    "deficit_minus_light_arcsec": 0.19,
    "peak_on_target": true,
    "depth_ap3": 0.05279,
    "psf_frac_resid": 0.034,
    "n_neighbours_60": 4,
    "n_possible": 0,
    "brightest_nb": {
      "G": 18.08154,
      "sep": 48.6
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 0, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

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Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4344.35998; next other-kind approximately 4345.53954 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Thirty-three primary and thirty secondary events, independent reduction and clean target-localized deficit establish a target EB. Gaia small-e geometry and simplified shape model are not precise component constraints.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R046. TIC 362082706

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 362082706: CONFIRMED

Both kinds repeat, independent TESS-SPOC confirms them and own four-event pixel regression favours the target over the 4.3 arcsec companion each time. Aperture dilution and a close visual companion limit component characterization.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 9501-275-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

28 accepted primary fits across 8 sectors; median depth 55.655 ppt. Shared exposure-integrated trapezoid T14=7.9812 h, profile range [7.7922 8.1702] h; fractional ingress=0.250. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=7.3636980984 ± 2.13e-06 d; Tref=2583.9981175 ± 0.000209 BTJD, reference cycle 129; χ^2/dof=15.52/26, O−C RMS=1.180 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 23 usable windows, 23 individual >3σ; combined depth 7.9893 ± 0.0882 ppt (event-scatter error). Fitted phase=0.5003570, Gaia nominal=0.4804912. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=7.3637003388 ± 6.27e-06 d; Tref=2396.2255355 ± 0.00052 BTJD, reference cycle 103; χ^2/dof=90.80/21, O−C RMS=3.468 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	12	1634.0827000 ± 0.0015	53.205 ± 0.334	7.981/1.995/3.991	1.02909	2.35751
P	12	1641.4444025 ± 0.0019	53.250 ± 0.536	7.981/1.995/3.991	1.17634	-0.51619
P	12	1648.8081002 ± 0.0015	53.803 ± 0.320	7.981/1.995/3.991	1.12029	-0.516661
P	13	1656.1717980 ± 0.0015	54.954 ± 0.359	7.981/1.995/3.991	1.14115	-0.517131
P	13	1663.5374911 ± 0.0015	54.471 ± 0.333	7.981/1.995/3.991	1.15353	2.35563
P	13	1670.8991936 ± 0.0015	54.670 ± 0.332	7.981/1.995/3.991	1.13786	-0.518071
P	13	1678.2628913 ± 0.0015	54.325 ± 0.350	7.981/1.995/3.991	1.13818	-0.518541
P	27	2039.0840822 ± 0.000998	56.845 ± 0.333	7.981/1.995/3.991	1.42856	-0.541582
P	27	2046.4497752 ± 0.000998	58.926 ± 0.458	7.981/1.995/3.991	1.3228	2.33118

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	27	2053.8114777 ± 0.000998	57.017 ± 0.341	7.981/1.995/3.991	1.39732	-0.542523
P	38	2340.9956908 ± 0.000998	55.585 ± 0.355	7.981/1.995/3.991	1.54415	-0.560861
P	38	2355.7230863 ± 0.000998	55.629 ± 0.308	7.981/1.995/3.991	1.30591	-0.561802
P	39	2363.0867841 ± 0.000998	56.315 ± 0.323	7.981/1.995/3.991	1.44511	-0.562272
P	39	2370.4504819 ± 0.000998	56.792 ± 0.381	7.981/1.995/3.991	1.52542	-0.562742
P	39	2377.8141797 ± 0.000998	56.333 ± 0.334	7.981/1.995/3.991	1.45542	-0.563212
P	39	2385.1798727 ± 0.000998	56.469 ± 0.336	7.981/1.995/3.991	1.3947	2.30955
P	65	3070.0017702 ± 0.000998	55.999 ± 0.329	7.981/1.995/3.991	1.63426	-0.607413
P	65	3077.3654680 ± 0.000998	55.999 ± 0.301	7.981/1.995/3.991	1.58674	-0.607883
P	65	3084.7291658 ± 0.000998	55.681 ± 0.291	7.981/1.995/3.991	1.52804	-0.608354
P	65	3092.0928635 ± 0.000998	55.950 ± 0.315	7.981/1.995/3.991	1.58668	-0.608824
P	67	3128.9113524 ± 0.000998	55.968 ± 0.340	7.981/1.995/3.991	1.71839	-0.611175
P	67	3136.2750502 ± 0.000998	55.379 ± 0.358	7.981/1.995/3.991	1.61195	-0.611645
P	67	3143.6387479 ± 0.000998	56.252 ± 0.311	7.981/1.995/3.991	1.5822	-0.612115
P	67	3151.0024457 ± 0.000998	56.494 ± 0.349	7.981/1.995/3.991	1.65251	-0.612586
P	94	3857.9174318 ± 0.000998	48.246 ± 0.285	7.981/1.995/3.991	1.72412	-0.657727
P	94	3865.2811296 ± 0.000998	48.074 ± 0.313	7.981/1.995/3.991	1.97872	-0.658197
P	94	3872.6468226 ± 0.000998	48.129 ± 0.276	7.981/1.995/3.991	1.7676	2.21456
P	94	3880.0105204 ± 0.000998	48.191 ± 0.310	7.981/1.995/3.991	1.77394	2.21409
S	12	1630.4018429 ± 0.00155	8.129 ± 0.252	8.359/2.090/4.180	1.05541	1.64535
S	12	1645.1292391 ± 0.0015	7.862 ± 0.182	8.359/2.090/4.180	1.01787	1.6389
S	13	1659.8587251 ± 0.0015	7.863 ± 0.191	8.359/2.090/4.180	1	4.64175
S	13	1667.2224232 ± 0.00146	8.066 ± 0.187	8.359/2.090/4.180	1	4.63853
S	13	1674.5819417 ± 0.0015	7.799 ± 0.206	8.359/2.090/4.180	1	-1.38332
S	13	1681.9435500 ± 0.00146	7.750 ± 0.178	8.359/2.090/4.180	1	-4.39586
S	27	2042.7668466 ± 0.00104	8.476 ± 0.221	8.359/2.090/4.180	1.18317	-1.54463
S	27	2050.1326345 ± 0.00104	8.464 ± 0.201	8.359/2.090/4.180	1.07283	1.46146
S	27	2057.4900632 ± 0.00104	8.562 ± 0.225	8.359/2.090/4.180	1.0871	-7.5697
S	38	2337.3147705 ± 0.00209	7.878 ± 0.229	8.359/2.090/4.180	1.18018	-1.67367
S	38	2344.6826482 ± 0.00104	7.775 ± 0.196	8.359/2.090/4.180	1.0364	4.34172
S	38	2352.0421667 ± 0.00209	8.221 ± 0.207	8.359/2.090/4.180	1.11942	-1.68013
S	38	2359.4037750 ± 0.00104	7.755 ± 0.195	8.359/2.090/4.180	1.11589	-4.69266
S	39	2366.7695629 ± 0.00104	8.025 ± 0.194	8.359/2.090/4.180	1.05742	-1.68658
S	39	2374.1353508 ± 0.00104	7.698 ± 0.233	8.359/2.090/4.180	1.02606	1.31951
S	39	2381.5011387 ± 0.00104	8.115 ± 0.220	8.359/2.090/4.180	1.00434	4.32559
S	39	2388.8627470 ± 0.00104	8.590 ± 0.241	8.359/2.090/4.180	1.08761	1.31305
S	65	3073.6866702 ± 0.00104	8.115 ± 0.170	8.359/2.090/4.180	1.00298	1.01302
S	65	3081.0482785 ± 0.00104	7.730 ± 0.177	8.359/2.090/4.180	1	-1.99951
S	65	3088.4119766 ± 0.00104	8.292 ± 0.186	8.359/2.090/4.180	1.07718	-2.00274
S	67	3132.5983447 ± 0.00104	8.313 ± 0.204	8.359/2.090/4.180	1.08124	3.99653
S	94	3861.6002769 ± 0.00104	7.325 ± 0.186	8.359/2.090/4.180	1.31009	-2.34149
S	94	3876.3339425 ± 0.00209	6.638 ± 0.342	8.359/2.090/4.180	2.00373	6.68

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 10 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9953; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 59 triggers; 4 outside the two eclipse phase windows. Largest off-phase dip=9.801 ppt, versus primary median 55.655 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 5189686537983293056; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=0.021σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.04814634115541594,
      "epoch389_gaia_error": 0.04458956229643425
    },
    {
      "kind": "B",
      "epoch389_offset": -3.614083248113502,
      "epoch389_gaia_error": 0.04964988648834859
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.4804912000615088,
    "gaia_relative_phase_quantiles_16_50_84": [
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      0.484777206805325,
      0.49496896342594154
    ],
    "gaia_relative_phase_sigma": 0.01105024237931731,
    "gaia_e_cos_omega_quantiles": [
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      -0.02390119917460249,
      -0.007902367656297383
    ],
    "gaia_e_sin_omega_quantiles": [
      -0.02234867706356318,
      -0.0005833491213913074,
      0.0179731026393232
    ],
    "photometric_phase": 0.500357033511672,
    "photometric_e_cos_omega_first_order": 0.0005608269286771306,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
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      0.0749999999999999
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
```

```
      "m1": 1.14,
      "m2_min": 0.5082905613213733,
      "f_mass": 0.04833576450987126,
      "a_rsun": 18.8153585704395,
      "r1_rsun": 2.69573,
      "r2_approx_rsun": 0.5819553112169709,
      "central_T14_hours": {
        "A": 9.801786251408714,
        "B": 9.788346337757467
      },
      "observed_T14_hours": 7.981199999999999
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 739.6320192502759,
      "dof": 121,
      "k": 0.22499999999999998,
      "b_A": 0.55,
      "e_sin_omega": 0.0749999999999999,
      "primary_scale": 1.0,
      "secondary_scale": 0.007973886206696083
    },
    "profile_interval_approx_95": {
      "k": [
        0.22499999999999998,
        0.22499999999999998
      ],
      "b_A": [
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        0.55
      ],
      "e_sin_omega": [
        0.0749999999999999,
        0.0749999999999999
      ]
    },
    "error_inflation": 6.112661316117983,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TESS-SPOC, sector 39: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale

differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2363.08678	SAP_FLUX	40.5266	0.199111	40.7336	48	40.6089
2363.08678	PDCSAP_FLUX	54.8768	0.269137	55.2004	48	54.9502
2370.45048	SAP_FLUX	40.5244	0.210739	40.8042	48	40.5827
2370.45048	PDCSAP_FLUX	54.8639	0.28941	55.2489	48	54.9202
2377.81418	SAP_FLUX	40.481	0.214093	40.8304	48	40.5728
2377.81418	PDCSAP_FLUX	54.8757	0.295696	55.3864	48	55.0335
2385.17987	SAP_FLUX	40.5443	0.233002	40.6927	48	40.5494
2385.17987	PDCSAP_FLUX	54.9127	0.31128	55.2023	48	54.9195

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable
5189686533688447360	4.31219	12.431944	0.168632	True

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "362082706",
    "sector": 13,
    "n_events": 4,
    "deficit_minus_target_arcsec": 1.73,
    "deficit_minus_light_arcsec": 1.95,
    "peak_on_target": true,
    "depth_ap3": 0.04019,
    "psf_frac_resid": 0.16,
    "n_neighbours_60": 14,
    "n_possible": 1,
    "brightest_nb": {
      "G": 12.431944,
      "sep": 4.3
    }
  }
]
```


pixel-localization.json:


```
{
  "source_path": "C:\\Home\\sky\\data\\sb1-scale-c3\\vet362082706\\tesscut_s013.fits",
  "sha256": "f46a40c6a1e525c196d32d5ca4443f69f0dbebc488aeb635f2a7332e40b3ce76",
  "sector": 13,
  "calibration": {
    "sigma_pixels": 0.6500000000000001,
    "dx_pixels": -1.1102230246251565e-16,
    "dy_pixels": -1.1102230246251565e-16
  },
  "fixed_fits": [
    {
      "source_id": "5189686537983293056",
      "sep_arcsec": 0.0,
      "G": 11.22835,
      "chi2": 1996.701481197056,
      "amplitude": 140.9404784614059,
      "fractional_residual": 0.12190014470004588
    },
    {
```

```
      "source_id": "5189686533688447360",
      "sep_arcsec": 4.312192960707238,
      "G": 12.431944,
      "chi2": 7522.548143135841,
      "amplitude": 132.96840530086183,
      "fractional_residual": 0.27448080029943295
    }
  ],
  "free_best": {
    "chi2": 1619.1616896457203,
    "pixel_x": 7.518108243307379,
    "pixel_y": 7.239774828355809,
    "distance_from_target_arcsec": 2.0999999999998433,
    "fractional_residual": 0.11897284471581845
  },
  "event_source_comparisons": [
    {
      "tc": 1656.1721571341586,
      "best_source": "5189686537983293056",
      "delta_chi2_neighbor_minus_target": [
        0.0,
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      ]
    },
    {
      "tc": 1663.5358552325536,
      "best_source": "5189686537983293056",
      "delta_chi2_neighbor_minus_target": [
        0.0,
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      ]
    },
    {
      "tc": 1670.8995533309487,
      "best_source": "5189686537983293056",
      "delta_chi2_neighbor_minus_target": [
        0.0,
        1290.9261603347416
      ]
    },
    {
      "tc": 1678.2632514293439,
      "best_source": "5189686537983293056",
      "delta_chi2_neighbor_minus_target": [
        0.0,
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      ]
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 1, 'J_ApJS_279_50_table2': 1, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI=

{'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4351.28566; next other-kind approximately 4347.60530 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Both kinds repeat, independent TESS-SPOC confirms them and own four-event pixel regression favours the target over the 4.3 arcsec companion each time. Aperture dilution and a close visual companion limit component characterization.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R047. TIC 436727157

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Constrain SB2 flux ratio/component radii and test the missing conjunction with resolved photometry.

Archived source: sanitized report text.

TIC 436727157: CONFIRMED

Fifteen repeated sharp primary eclipses, independent photometry, own SB2 covariance timing and target-localized deficit establish an eclipsing target. The undetected other eclipse is a physical geometry/brightness constraint, not a veto; a conditional chord calculation allows it to miss.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies HD 286162 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

15 accepted primary fits across 6 sectors; median depth 59.423 ppt. Shared exposure-integrated trapezoid T14=3.8433 h, profile range [3.75641379 4.2777931] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=8.5313599670 ± 1.65e-06 d; Tref=2722.8717159 ± 0.000137 BTJD, reference cycle 150; χ^2/dof=12.38/13, O−C RMS=0.695 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 18 usable windows, 0 individual >3σ; combined depth 0.0223 ± 0.0611 ppt (event-scatter error). Fitted phase=0.6019792, Gaia nominal=0.5997267. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

No reliable independent ephemeris.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	5	1443.1670825 ± 0.00227	57.250 ± 0.467	3.843/1.345/1.153	1	-0.919266
P	5	1460.2298450 ± 0.00246	57.722 ± 0.547	3.843/1.345/1.153	1.19038	-0.857935
P	32	2176.8649102 ± 0.00048	58.099 ± 0.624	3.843/1.345/1.153	1.66841	0.334359
P	32	2193.9276727 ± 0.00048	57.999 ± 0.549	3.843/1.345/1.153	1.60647	0.39569
P	43	2475.4622936 ± 0.00048	63.349 ± 0.540	3.843/1.345/1.153	1.40777	0.0240531
P	43	2483.9936748 ± 0.00048	63.691 ± 0.427	3.843/1.345/1.153	1.18384	0.0547184
P	43	2492.5250561 ± 0.00048	63.242 ± 0.508	3.843/1.345/1.153	1.45442	0.0853837
P	44	2501.0564374 ± 0.00048	62.149 ± 0.561	3.843/1.345/1.153	1.58572	0.116049
P	44	2509.5878186 ± 0.00048	61.595 ± 0.464	3.843/1.345/1.153	1.29597	0.146714
P	44	2518.1182391 ± 0.00048	61.780 ± 0.417	3.843/1.345/1.153	1.23816	-1.20621
P	71	3243.2846855 ± 0.00048	59.425 ± 0.418	3.843/1.345/1.153	1.62367	0.0167476

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	71	3251.8160668 ± 0.00048	59.423 ± 0.459	3.843/1.345/1.153	1.67078	0.0474129
P	98	4002.5747354 ± 0.00048	49.742 ± 0.914	3.843/1.345/1.153	2.45161	-1.40481
P	98	4011.1070775 ± 0.00048	49.931 ± 0.429	3.843/1.345/1.153	2.06607	0.00944207
P	98	4028.1708008 ± 0.00048	49.717 ± 0.381	3.843/1.345/1.153	1.6424	1.45436
S	5	1439.8201032 ± 0.00769	0.254 ± 0.183	3.843/0.307/3.228	1	0
S	5	1448.2649887 ± 0.011	0.215 ± 0.126	3.843/0.307/3.228	1	0
S	5	1456.8328601 ± 0.0293	0.277 ± 0.189	3.843/0.307/3.228	1	0
S	32	2181.9802015 ± 0.0423	-0.303 ± 0.179	3.843/0.307/3.228	1.14894	0
S	32	2190.5797803 ± 0.048	0.211 ± 0.175	3.843/0.307/3.228	1.07743	0
S	32	2199.0150575 ± 0.00192	0.219 ± 0.141	3.843/0.307/3.228	1	0
S	43	2480.6460192 ± 0.048	0.162 ± 0.180	3.843/0.307/3.228	1	0
S	43	2489.1168470 ± 0.0207	-0.350 ± 0.160	3.843/0.307/3.228	1	0
S	43	2497.6568544 ± 0.00817	0.380 ± 0.199	3.843/0.307/3.228	1.08751	0
S	44	2506.2045484 ± 0.0163	-0.231 ± 0.168	3.843/0.307/3.228	1.04827	0
S	44	2514.7166918 ± 0.0187	0.337 ± 0.181	3.843/0.307/3.228	1.00742	0
S	44	2523.2326786 ± 0.0298	-0.388 ± 0.187	3.843/0.307/3.228	1.17561	0
S	71	3239.9120747 ± 0.0283	-0.205 ± 0.157	3.843/0.307/3.228	1.02941	0
S	71	3256.9699905 ± 0.048	-0.145 ± 0.147	3.843/0.307/3.228	1	0
S	98	3999.2252108 ± 0.048	0.116 ± 0.120	3.843/0.307/3.228	1	0
S	98	4016.2446935 ± 0.0115	0.342 ± 0.167	3.843/0.307/3.228	1.10401	0
S	98	4024.7260904 ± 0.048	-0.075 ± 0.128	3.843/0.307/3.228	1.06793	0
S	98	4041.8512641 ± 0.011	-0.221 ± 0.128	3.843/0.307/3.228	1	0

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 10 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9913; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 20 triggers; 3 outside the two eclipse phase windows. Largest off-phase dip=17.025 ppt, versus primary median 59.423 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 3405360860398901376; covariance bit_index=255; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=1.518σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
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    {
      "kind": "B",
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```
    }
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with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "insufficient secondary shape",
    "geometry_e_sin_omega_conditional_range": null,
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
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  },
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      "K2_kms": 69.19587339273912,
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      "m1_sin3i": 1.0769416917838774,
      "m2_sin3i": 1.0650010251126478,
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      "r1_rsun": 1.93785,
      "r2_approx_rsun": 1.0365482307936966,
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        "B": 9.858371026837387
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    }
  ],
  "shape_feasibility": {
    "status": "insufficient secondary shape"
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

The other-conjunction covariance/width-tested pointwise limits are retained below. Every individual clean/flagged/insufficient window is retained in null-secondary.json. These are conditional pointwise limits, not a simultaneous confidence envelope; fitted coverage uses actual clean cadences.

Ancillary log block 2: 384 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independence and attribution

TESS-SPOC, sector 43: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2475.46229	SAP_FLUX	61.1033	0.486189	61.6726	23	61.3665
2475.46229	PDCSAP_FLUX	62.1273	0.496598	62.7091	23	62.3781
2483.99367	SAP_FLUX	61.3789	0.545969	61.6627	22	61.3614
2483.99367	PDCSAP_FLUX	62.4115	0.562225	62.6926	22	62.3972
2492.52506	SAP_FLUX	61.6574	0.517553	61.6879	22	61.7504
2492.52506	PDCSAP_FLUX	62.5431	0.529098	62.557	22	62.6386

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
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  {
    "tic": "436727157",
    "sector": 98,
    "n_events": 5,
    "deficit_minus_light_arcsec": 1.0,
    "peak_on_target": true,
    "depth_ap3": 0.05935,
    "psf_frac_resid": 0.041,
    "n_neighbours_60": 16,
    "n_possible": 0,
    "brightest_nb": {
      "G": 16.155735,
      "sep": 47.6
    }
  },
  {
    "tic": "436727157",
    "sector": 98,
    "n_events": 5,
    "deficit_minus_target_arcsec": 5.06,
    "deficit_minus_light_arcsec": 1.0,
    "peak_on_target": true,
    "depth_ap3": 0.05935,
    "psf_frac_resid": 0.041,
    "n_neighbours_60": 16,
    "n_possible": 0,
    "brightest_nb": {
```

```
      "G": 16.155735,
      "sep": 47.6
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 0, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4352.36147; next other-kind approximately 4357.47795 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Constrain SB2 flux ratio/component radii and test the missing conjunction with resolved photometry.

Adjudication: **CONFIRMED**. Fifteen repeated sharp primary eclipses, independent photometry, own SB2 covariance timing and target-localized deficit establish an eclipsing target. The undetected other eclipse is a physical geometry/brightness constraint, not a veto; a conditional chord calculation allows it to miss.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R048. TIC 279166593

physical_verdict: PLAUSIBLE

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Calibrated TESS PRF/WCS or spatially resolved eclipse photometry to distinguish target and 11.2 arcsec neighbour.

Archived source: sanitized report text.

TIC 279166593: PLAUSIBLE

The recurring two-kind eclipse signal is secure, but its source is unresolved. Circular and expanded elliptical crowded-field calibrations reverse target versus 11.2 arcsec neighbour preference and fail their calibration boundaries. This is an adjudicated plausible host, not an unattempted review.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 3179-181-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

26 accepted primary fits across 8 sectors; median depth 58.852 ppt. Shared exposure-integrated trapezoid T14=3.6695 h, profile range [3.58262069 3.75641379] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=7.1665201901 ± 3.14e-06 d; Tref=2992.7709899 ± 0.000175 BTJD, reference cycle 178; χ^2/dof=54.30/24, O–C RMS=1.743 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 23 usable windows, 23 individual >3σ; combined depth 6.8618 ± 0.1520 ppt (event-scatter error). Fitted phase=0.5610385, Gaia nominal=0.5494189. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=7.1665383675 ± 9.88e-06 d; Tref=3046.9579591 ± 0.000845 BTJD, reference cycle 185; χ^2/dof=16.59/21, O–C RMS=5.236 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O–C min
P	15	1717.1259436 ± 0.00227	59.493 ± 0.856	3.670/1.284/1.101	1	-6.41157
P	15	1731.4607993 ± 0.00227	60.272 ± 0.903	3.670/1.284/1.101	1	-3.79753
P	16	1752.9640002 ± 0.00227	59.490 ± 0.850	3.670/1.284/1.101	1	1.44457
P	16	1760.1305107 ± 0.00227	59.180 ± 1.075	3.670/1.284/1.101	1	1.43056
P	55	2799.2745281 ± 0.000459	59.480 ± 0.754	3.670/1.284/1.101	1	-0.600032
P	55	2806.4419559 ± 0.000459	59.937 ± 0.861	3.670/1.284/1.101	1.03586	0.70699
P	55	2813.6075490 ± 0.000459	59.268 ± 0.790	3.670/1.284/1.101	1.12497	-0.62804
P	55	2820.7749768 ± 0.000459	58.670 ± 0.736	3.670/1.284/1.101	1	0.678982
P	56	2827.9414873 ± 0.000459	58.821 ± 0.888	3.670/1.284/1.101	1.34812	0.664978

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	56	2835.1070804 ± 0.000459	60.224 ± 0.721	3.670/1.284/1.101	1.06407	-0.670052
P	56	2842.2745082 ± 0.000459	58.047 ± 0.638	3.670/1.284/1.101	1	0.63697
P	56	2849.4401013 ± 0.000459	58.442 ± 0.787	3.670/1.284/1.101	1.1636	-0.698061
P	75	3343.9311582 ± 0.000459	58.883 ± 0.849	3.670/1.284/1.101	1	0.977709
P	75	3351.0967513 ± 0.000459	57.281 ± 0.955	3.670/1.284/1.101	1.28548	-0.357322
P	75	3358.2650965 ± 0.000459	57.991 ± 0.857	3.670/1.284/1.101	1.04299	2.27073
P	75	3365.4288548 ± 0.000459	60.130 ± 0.834	3.670/1.284/1.101	1.24018	-1.70636
P	76	3372.5972000 ± 0.000459	59.347 ± 0.745	3.670/1.284/1.101	1	0.921692
P	76	3379.7627931 ± 0.000459	57.843 ± 0.743	3.670/1.284/1.101	1.09271	-0.413338
P	76	3386.9302210 ± 0.000459	58.554 ± 0.730	3.670/1.284/1.101	1.01038	0.893684
P	76	3394.0958140 ± 0.000459	58.553 ± 0.727	3.670/1.284/1.101	1.09345	-0.441346
P	82	3537.4260233 ± 0.000459	58.698 ± 0.653	3.670/1.284/1.101	1	-0.721428
P	82	3544.5934512 ± 0.000459	59.678 ± 0.718	3.670/1.284/1.101	1.0783	0.585594
P	82	3551.7599617 ± 0.000459	58.793 ± 0.700	3.670/1.284/1.101	1.16176	0.57159
P	83	3566.0920652 ± 0.000459	57.964 ± 0.801	3.670/1.284/1.101	1.26395	-0.777445
P	83	3573.2594931 ± 0.000459	58.421 ± 0.715	3.670/1.284/1.101	1.12176	0.529577
P	83	3580.4241688 ± 0.000459	60.160 ± 0.822	3.670/1.284/1.101	1.3236	-2.12648
S	15	1713.9816868 ± 0.00528	7.270 ± 1.030	3.843/1.345/1.153	1.0474	-0.195826
S	15	1728.3118447 ± 0.00432	6.043 ± 0.744	3.843/1.345/1.153	1	-4.39895
S	16	1742.6525717 ± 0.00528	6.677 ± 0.956	3.843/1.345/1.153	1.13374	6.61743
S	16	1756.9836904 ± 0.00384	7.034 ± 0.707	3.843/1.345/1.153	1	3.7979
S	55	2817.6306002 ± 0.00673	5.304 ± 0.621	3.843/1.345/1.153	1	2.69109
S	56	2831.9588364 ± 0.00673	8.067 ± 0.766	3.843/1.345/1.153	1	-4.27922
S	56	2839.1272783 ± 0.00384	8.123 ± 0.659	3.843/1.345/1.153	1	-1.53821
S	56	2846.2947593 ± 0.00336	7.443 ± 0.737	3.843/1.345/1.153	1.20325	-0.180794
S	75	3340.7827307 ± 0.00336	6.278 ± 0.783	3.843/1.345/1.153	1.10183	-4.75409
S	75	3347.9444468 ± 0.0048	7.700 ± 0.744	3.843/1.345/1.153	1	-11.6982
S	75	3355.1148103 ± 0.00384	5.821 ± 0.785	3.843/1.345/1.153	1.08767	-6.19004
S	75	3362.2842130 ± 0.00432	6.715 ± 0.748	3.843/1.345/1.153	1	-2.06544
S	76	3369.4449682 ± 0.00432	6.553 ± 0.925	3.843/1.345/1.153	1	-10.3932
S	76	3376.6201358 ± 0.00432	6.984 ± 0.688	3.843/1.345/1.153	1.02506	2.03299
S	76	3383.7866560 ± 0.0048	7.284 ± 0.853	3.843/1.345/1.153	1	2.00681
S	76	3390.9531762 ± 0.00625	6.240 ± 0.985	3.843/1.345/1.153	1.40247	1.98064
S	82	3534.2874233 ± 0.00288	7.488 ± 0.667	3.843/1.345/1.153	1.0648	6.99149
S	82	3541.4568260 ± 0.00528	6.902 ± 0.707	3.843/1.345/1.153	1.13634	11.1161
S	82	3548.6175812 ± 0.00288	6.836 ± 0.582	3.843/1.345/1.153	1.03234	2.78836
S	82	3555.7821798 ± 0.00336	6.971 ± 0.669	3.843/1.345/1.153	1.07311	-0.00499427
S	83	3562.9458175 ± 0.00432	6.236 ± 0.724	3.843/1.345/1.153	1.20668	-4.18195
S	83	3570.1171418 ± 0.00288	6.855 ± 0.627	3.843/1.345/1.153	1	2.70984
S	83	3577.2817403 ± 0.00336	6.895 ± 0.623	3.843/1.345/1.153	1	-0.083521

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 6 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9644; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 35 triggers; 0 outside the two eclipse phase windows. Largest off-phase dip=0.000 ppt, versus primary median 58.852 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow

isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 2163072940491937792; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=1.196σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.07060489699193795,
      "epoch389_gaia_error": 0.05479794425110205
    },
    {
      "kind": "B",
      "epoch389_offset": 3.17172064857607,
      "epoch389_gaia_error": 0.043040953130135604
    }
  ],
  "eccentricity_comparison": {
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with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
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    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
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        "B": 5.898571139192928
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      "observed_T14_hours": 3.66951724137931
    }
  ],
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    "best": {
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      "k": 0.32499999999999996,
      "b_A": 0.865,
      "e_sin_omega": 0.04999999999999993,
      "primary_scale": 0.8618357995958955,
      "secondary_scale": 0.01314651994533462
    },
    "profile_interval_approx_95": {
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      ],
      "b_A": [
        0.8300000000000001,
        1.04
      ],
      "e_sin_omega": [
        0.02499999999999994,
        0.09999999999999992
      ]
    },
    "error_inflation": 1.1850731473368827,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TARS, sector 55: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2799.27453	FLUX	49.1299	0.929347	50.2586	22	n/a

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2806.44196	FLUX	48.2587	0.607853	49.2018	22	n/a
2813.60755	FLUX	45.6596	0.677751	46.1439	22	n/a
2820.77498	FLUX	46.7241	0.75298	47.0785	22	n/a

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable
2163072940491938176	11.1971	14.208706	0.733051	True
2163073009211419776	34.6857	14.790378	1.25257	False
2163072974851680128	49.152	12.565834	0.16143	True
2163072837412723584	58.1347	14.521498	0.977803	True

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "279166593",
    "sector": 55,
    "n_events": 4,
    "deficit_minus_target_arcsec": 5.09,
    "deficit_minus_light_arcsec": 2.57,
    "peak_on_target": true,
    "depth_ap3": 0.04222,
    "psf_frac_resid": 0.861,
    "n_neighbours_60": 160,
    "n_possible": 3,
    "brightest_nb": {
      "G": 12.565834,
      "sep": 49.2
    }
  }
]
```


pixel-localization.json:

Ancillary log block 3: 108 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

pixel-localization-elliptical.json:

Ancillary log block 4: 111 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 7, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4347.24331; next other-kind approximately 4351.26127 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Calibrated TESS PRF/WCS or spatially resolved eclipse photometry to distinguish target and 11.2 arcsec neighbour.

Adjudication: **PLAUSIBLE**. The recurring two-kind eclipse signal is secure, but its source is unresolved. Circular and expanded elliptical crowded-field calibrations reverse target versus 11.2 arcsec neighbour preference and fail their calibration boundaries. This is an adjudicated plausible host, not an unattempted review.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R049. TIC 67651413

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 67651413: CONFIRMED

Both eclipse kinds repeat, independent TESS-SPOC and on-target difference imaging agree, and no capable neighbour is present. The Gaia-period discrepancy and poor simplified two-disk shape fit preclude precise orbital/stellar claims but do not invalidate the target EB.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 2996-1253-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

7 accepted primary fits across 2 sectors; median depth 65.013 ppt. Shared exposure-integrated trapezoid T14=3.6695 h, profile range [3.58262069 3.75641379] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=6.8329952426 ± 1.23e-05 d; Tref=2305.9547009 ± 0.000581 BTJD, reference cycle 63; χ^2/dof=2.27/5, O−C RMS=0.403 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 6 usable windows, 6 individual >3σ; combined depth 25.7249 ± 0.5592 ppt (event-scatter error). Fitted phase=0.5002217, Gaia nominal=0.5042494. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=6.8329911329 ± 1.3e-05 d; Tref=2247.8747538 ± 0.000705 BTJD, reference cycle 54; χ^2/dof=3.10/4, O−C RMS=0.995 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	21	1875.4758521 ± 0.00227	66.160 ± 0.524	3.670/1.284/1.101	1	-0.213773
P	21	1889.1419252 ± 0.00227	66.500 ± 0.593	3.670/1.284/1.101	1	-0.0948858
P	21	1895.9749617 ± 0.00227	66.869 ± 0.577	3.670/1.284/1.101	1.00068	-0.0354423
P	48	2613.4401271 ± 0.000486	65.013 ± 0.482	3.670/1.284/1.101	1.20366	0.922018
P	48	2620.2722462 ± 0.000473	63.634 ± 0.451	3.670/1.284/1.101	1.03414	-0.339565
P	48	2627.1052827 ± 0.00046	63.436 ± 0.481	3.670/1.284/1.101	1.19358	-0.280121
P	48	2633.9383193 ± 0.000473	63.751 ± 0.463	3.670/1.284/1.101	1.14724	-0.220678
S	21	1872.0610179 ± 0.00227	23.628 ± 0.343	3.670/1.284/1.101	1	1.11804
S	21	1878.8921784 ± 0.00227	25.426 ± 0.284	3.670/1.284/1.101	1.05253	-1.51809

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
S	21	1892.5600036 ± 0.00269	25.783 ± 0.325	3.670/1.284/1.101	1	1.1358
S	48	2610.0226693 ± 0.000459	28.137 ± 0.324	3.670/1.284/1.101	1.07217	-0.88487
S	48	2616.8565820 ± 0.000473	26.216 ± 0.311	3.670/1.284/1.101	1.23265	0.442074
S	48	2630.5225724 ± 0.00046	24.873 ± 0.345	3.670/1.284/1.101	1.26416	0.45391

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 15 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9967; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 15 triggers; 2 outside the two eclipse phase windows. Largest off-phase dip=4.541 ppt, versus primary median 65.013 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 806524493741663232; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=-3.531σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": 0.00544786216897819,
      "epoch389_gaia_error": 0.015698303660971
    },
    {
      "kind": "B",
      "epoch389_offset": 3.397196715908194,
      "epoch389_gaia_error": 0.010279629624072086
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.5042494264253734,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.5008141581445732,
      0.5034658602629234,
      0.5070693921197647
    ],
    "gaia_relative_phase_sigma": 0.0031395388254875423,
    "gaia_e_cos_omega_quantiles": [
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      0.005443977775436568,
      0.011104741264862305
    ],
    "gaia_e_sin_omega_quantiles": [
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      0.0009789453178621137,
      0.006187132831248789
    ],
```

```
    "photometric_phase": 0.5002217082237235,
    "photometric_e_cos_omega_first_order": 0.0003482584634450499,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      0.09999999999999992,
      0.1499999999999999
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 0.96,
      "m2_min": 0.7294188010569504,
      "f_mass": 0.13597410649170086,
      "a_rsun": 18.048945139917684,
      "r1_rsun": 1.61617,
      "r2_approx_rsun": 0.7769294117537343,
      "central_T14_hours": {
        "A": 6.90911196099168,
        "B": 6.934319908175338
      },
      "observed_T14_hours": 3.66951724137931
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 1316.6690679652888,
      "dof": 121,
      "k": 0.29999999999999993,
      "b_A": 0.8300000000000001,
      "e_sin_omega": 0.1499999999999999,
      "primary_scale": 0.9497979712015495,
      "secondary_scale": 0.14998433377066753
    },
    "profile_interval_approx_95": {
      "k": [
        0.27499999999999997,
        0.35
      ],
      "b_A": [
        0.7950000000000002,
        0.935
      ],
      "e_sin_omega": [
        0.09999999999999992,
        0.1499999999999999
      ]
    },
    "error_inflation": 10.881562545167677,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```

Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TESS-SPOC, sector 48: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2613.44013	SAP_FLUX	62.854	0.688027	62.1941	22	63.0521
2613.44013	PDCSAP_FLUX	63.1019	0.639768	62.306	22	63.3559
2620.27225	SAP_FLUX	61.8112	0.562852	61.3112	22	61.9741
2620.27225	PDCSAP_FLUX	62.3953	0.573241	61.8308	22	62.5857
2627.10528	SAP_FLUX	61.4131	0.526416	60.7633	22	61.4082
2627.10528	PDCSAP_FLUX	61.9325	0.529296	61.2836	22	61.9282
2633.93832	SAP_FLUX	61.1641	0.531187	60.7391	21	61.3927
2633.93832	PDCSAP_FLUX	61.69	0.551966	61.2047	21	61.9085

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "67651413",
    "sector": 48,
    "n_events": 7,
    "deficit_minus_light_arcsec": 0.14,
    "peak_on_target": true,
    "depth_ap3": 0.04483,
    "psf_frac_resid": 0.013,
    "n_neighbours_60": 4,
    "n_possible": 0,
    "brightest_nb": {
      "G": 16.661469,
      "sep": 36.3
    }
  },
  {
    "tic": "67651413",
    "sector": 48,
    "n_events": 7,
    "deficit_minus_target_arcsec": 0.52,
    "deficit_minus_light_arcsec": 0.14,
    "peak_on_target": true,
    "depth_ap3": 0.04483,
    "psf_frac_resid": 0.013,
```

```
    "n_neighbours_60": 4,
    "n_possible": 0,
    "brightest_nb": {
      "G": 16.661469,
      "sep": 36.3
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 0, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4349.02028; next other-kind approximately 4345.60452 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Both eclipse kinds repeat, independent TESS-SPOC and on-target difference imaging agree, and no capable neighbour is present. The Gaia-period discrepancy and poor simplified two-disk shape fit preclude precise orbital/stellar claims but do not invalidate the target EB.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R050. TIC 244352837

physical_verdict: CONFIRMED

novelty_status: No prior target EB found; limited prior-art coverage

completion: COMPLETE (adjudicated)

next_check: Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Archived source: sanitized report text.

TIC 244352837: CONFIRMED

Repeated primary/secondary eclipses, independent TESS-SPOC, Gaia conjunction timing and existing localized difference imaging establish a target EB. Sector depth variation and the nonzero PSF residual limit exact dimensions.

Full-review completion: COMPLETE (adjudicated). Novelty: no prior target EB classification found in successful queried catalogues; limited prior-art coverage, not a guaranteed discovery claim. Current SIMBAD identifies TYC 2145-1191-1 as SB*; known spectroscopic binarity is distinct from prior eclipse classification.

Raw SAP, individual events and ephemerides

23 accepted primary fits across 5 sectors; median depth 58.857 ppt. Shared exposure-integrated trapezoid T14=2.5942 h, profile range [2.53557931 2.65288966] h; fractional ingress=0.350. T12 and T23 below are trapezoid coordinates; a grazing flat segment is not physical totality.

P=4.2052503571 ± 1.28e-06 d; Tref=2605.2806012 ± 0.000167 BTJD, reference cycle 219; χ^2/dof=33.07/21, O−C RMS=3.427 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

Secondary: 20 usable windows, 20 individual >3σ; combined depth 15.3442 ± 0.2003 ppt (event-scatter error). Fitted phase=0.4876446, Gaia nominal=0.4803570. Phase was optimized; repeatability and controls matter more than its single-trial S/N.

P=4.2052477430 ± 3.36e-06 d; Tref=2737.6941275 ± 0.000464 BTJD, reference cycle 250; χ^2/dof=97.65/18, O−C RMS=3.821 min. Errors include grid/red-noise and ephemeris scatter scaling, conditional on shape/baseline.

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	14	1684.3244630 ± 0.00246	59.499 ± 0.633	2.594/0.908/0.778	1	-9.08644
P	14	1688.5303637 ± 0.00269	59.963 ± 0.935	2.594/0.908/0.778	1	-8.14992
P	14	1701.1454716 ± 0.00269	59.066 ± 0.577	2.594/0.908/0.778	1	-9.07605
P	14	1709.5559759 ± 0.00269	60.821 ± 0.649	2.594/0.908/0.778	1	-9.07085
P	41	2420.2500769 ± 0.000518	59.694 ± 0.562	2.594/0.908/0.778	1.13382	0.707676
P	41	2424.4553291 ± 0.000501	59.641 ± 0.576	2.594/0.908/0.778	1.02566	0.710275
P	41	2428.6599327 ± 0.000518	58.585 ± 0.668	2.594/0.908/0.778	1.09011	-0.22105
P	41	2437.0704370 ± 0.000518	59.456 ± 0.580	2.594/0.908/0.778	1.10819	-0.215851
P	41	2441.2763377 ± 0.000501	60.173 ± 0.563	2.594/0.908/0.778	1.01182	0.720672

kind	sector	mid BTJD ± d	depth ± ppt	T14/T12/T23 h	β	O−C min
P	41	2445.4809413 ± 0.000518	59.427 ± 0.559	2.594/0.908/0.778	1.02208	-0.210653
P	54	2773.4906100 ± 0.000579	58.379 ± 0.773	2.594/0.908/0.778	1.07421	-0.00790891
P	54	2781.9011143 ± 0.000579	59.487 ± 0.679	2.594/0.908/0.778	1	-0.00271035
P	54	2786.1063665 ± 0.000579	59.222 ± 0.737	2.594/0.908/0.778	1.13409	-0.000111065
P	55	2798.7227715 ± 0.000518	55.862 ± 0.650	2.594/0.908/0.778	1.17359	0.941611
P	55	2802.9280237 ± 0.000501	55.572 ± 0.578	2.594/0.908/0.778	1.04418	0.94421
P	55	2807.1319787 ± 0.000518	55.412 ± 0.602	2.594/0.908/0.778	1.05424	-0.921039
P	55	2811.3378795 ± 0.000501	54.892 ± 0.650	2.594/0.908/0.778	1.2061	0.0154846
P	55	2815.5431316 ± 0.000518	55.621 ± 0.518	2.594/0.908/0.778	1.02968	0.0180839
P	55	2819.7483838 ± 0.000501	55.137 ± 0.504	2.594/0.908/0.778	1	0.0206832
P	55	2823.9536359 ± 0.000501	55.764 ± 0.564	2.594/0.908/0.778	1.053	0.0232825
P	81	3509.4090898 ± 0.000324	58.631 ± 0.590	2.594/0.908/0.778	1.22198	-0.486959
P	81	3517.8195941 ± 0.000324	57.421 ± 1.079	2.594/0.908/0.778	1	-0.481761
P	81	3522.0254949 ± 0.000324	58.857 ± 0.586	2.594/0.908/0.778	1.30996	0.454763
S	14	1686.3743684 ± 0.00269	16.469 ± 0.626	2.829/1.273/0.283	1.06547	-11.2657
S	14	1698.9901194 ± 0.00246	15.585 ± 0.574	2.829/1.273/0.283	1	-11.2544
S	41	2422.2995458 ± 0.00106	16.052 ± 0.601	2.829/1.273/0.283	1.00201	-1.44147
S	41	2430.7114609 ± 0.000707	13.790 ± 0.533	2.829/1.273/0.283	1.04975	0.602833
S	41	2434.9181257 ± 0.000707	15.039 ± 0.542	2.829/1.273/0.283	1	2.64337
S	41	2439.1205472 ± 0.000707	16.438 ± 0.534	2.829/1.273/0.283	1.03776	-1.42641
S	41	2443.3279192 ± 0.00106	15.911 ± 0.667	2.829/1.273/0.283	1.03876	1.63251
S	54	2771.3353254 ± 0.00141	15.685 ± 0.808	2.829/1.273/0.283	1.22017	-1.12903
S	54	2775.5391613 ± 0.00177	15.284 ± 1.038	2.829/1.273/0.283	1.10532	-3.16204
S	54	2783.9517837 ± 0.00106	16.031 ± 0.782	2.829/1.273/0.283	1	-0.0993509
S	54	2788.1563268 ± 0.00141	14.687 ± 0.840	2.829/1.273/0.283	1	-1.11397
S	55	2800.7720779 ± 0.000707	14.716 ± 0.459	2.829/1.273/0.283	1.01692	-1.10268
S	55	2804.9787427 ± 0.000707	14.830 ± 0.503	2.829/1.273/0.283	1	0.937859
S	55	2809.1839931 ± 0.000707	13.728 ± 0.531	2.829/1.273/0.283	1	0.941623
S	55	2813.3885362 ± 0.000707	14.800 ± 0.500	2.829/1.273/0.283	1	-0.0730006
S	55	2817.5966154 ± 0.00106	13.661 ± 0.521	2.829/1.273/0.283	1.0789	4.00432
S	81	3507.2562595 ± 0.000354	15.683 ± 0.438	2.829/1.273/0.283	1	2.58489
S	81	3511.4579738 ± 0.000707	15.679 ± 0.552	2.829/1.273/0.283	1.13138	-2.50328
S	81	3524.0737249 ± 0.000354	16.710 ± 0.412	2.829/1.273/0.283	1	-2.49199
S	81	3528.2810969 ± 0.000707	16.172 ± 0.495	2.829/1.273/0.283	1	0.566936

All primary timing scans interior: True. QUALITY=0 fits retain all untrimmed cadences; 26 flagged core cadences inspected separately in event records. Minimum background-regressed/ordinary primary depth ratio=0.9876; background is standardized in native units, never divided by normalized SAP. Centroid template shifts/errors are retained per event in fit-results.json and are not by themselves source localization.

Blind rolling-median census: 51 triggers; 0 outside the two eclipse phase windows. Largest off-phase dip=0.000 ppt, versus primary median 58.857 ppt. No off-phase event of comparable primary depth. Off-phase s094 triggers cluster near the known 3875 BTJD sector systematic where applicable; shallow isolated census triggers are not additional classified eclipses. Full trigger/threshold/duration list and every rejected/insufficient event window remain in fit-results.json; coverage failures are not nulls.

Own Gaia covariance and physical interpretation

Source 1834764536559315712; covariance bit_index=127; source and SB1/SB2 layout asserted before reconstruction. Photometric period minus Gaia period=0.894σ. Primary matches conjunction A. Per-event Gaia predictions, correlated timing errors and residuals are in orbit-gates.json; no correlation copied from another source.


```
{
  "gaia_primary_epoch_comparison": [
    {
      "kind": "A",
      "epoch389_offset": -0.019972265849446558,
      "epoch389_gaia_error": 0.010230450460690065
    },
    {
      "kind": "B",
      "epoch389_offset": -2.041841994234005,
      "epoch389_gaia_error": 0.010618020625431284
    }
  ],
  "eccentricity_comparison": {
    "bit_index": 127,
    "gaia_primary": "A",
    "gaia_relative_phase_nominal": 0.4803570279169995,
    "gaia_relative_phase_quantiles_16_50_84": [
      0.4763248969635718,
      0.4807985378881131,
      0.48521288122537637
    ],
    "gaia_relative_phase_sigma": 0.004422658325468349,
    "gaia_e_cos_omega_quantiles": [
      -0.03719420060183956,
      -0.030161348436015,
      -0.023225349223494646
    ],
    "gaia_e_sin_omega_quantiles": [
      -0.02366481549601782,
      -0.01679556287941748,
      -0.009991619811236379
    ],
    "photometric_phase": 0.4876445681184507,
    "photometric_e_cos_omega_first_order": -0.019407867015502206,
    "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
    "geometry_status": "conditional shape feasibility",
    "geometry_e_sin_omega_conditional_range": [
      -5.551115123125783e-17,
      0.1999999999999999
    ],
    "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
  },
  "physics": [
    {
      "m1": 1.285,
      "m2_min": 0.6781209924604608,
      "f_mass": 0.0809147007924894,
      "a_rsun": 13.728070440775658,
      "r1_rsun": 1.27387,
      "r2_approx_rsun": 0.7329017373012959,
      "central_T14_hours": {
        "A": 4.776029269323283,
```

```
        "B": 4.613038549043756
      },
      "observed_T14_hours": 2.594234482758621
    }
  ],
  "shape_feasibility": {
    "status": "conditional shape feasibility",
    "best": {
      "chi2": 152.12805813941532,
      "dof": 121,
      "k": 0.3749999999999999,
      "b_A": 0.865,
      "e_sin_omega": 0.09999999999999992,
      "primary_scale": 0.6611285098582685,
      "secondary_scale": 0.0402512923258836
    },
    "profile_interval_approx_95": {
      "k": [
        0.35,
        0.44999999999999996
      ],
      "b_A": [
        0.7950000000000002,
        1.0750000000000002
      ],
      "e_sin_omega": [
        -5.551115123125783e-17,
        0.1999999999999999
      ]
    },
    "error_inflation": 1.2572566788381432,
    "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
    "primary_profile_n": 64,
    "secondary_profile_n": 64
  }
}
```


Physical quantities using TIC primary properties and approximate companion radii are conditional feasibility checks. The mass function and minimum mass establish stellar-scale companions; main-sequence radii are not measured constraints. A/B conjunction uses Gaia primary ω without a 180° change. Two-disk models use limb darkening and partial overlap; no e sinω is inferred by treating a grazing trapezoid T23 as totality. Free depth scales, grid boundaries, χ^2 and dilution prevent precise radii/mass-ratio claims.

Independence and attribution

TESS-SPOC, sector 55: FITS target+sector identity validated against the exact MAST row. Same TESS pixels, independent pipeline; not a second observatory. Linear/quadratic baseline sensitivity and background regressions tested. Aperture and PDCSAP dilution corrections explain depth scale differences; eclipse recurrence survives.

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2798.72277	SAP_FLUX	54.6136	0.512337	54.718	13	54.8337
2798.72277	PDCSAP_FLUX	61.8513	0.587804	61.9782	13	62.0213
2802.92802	SAP_FLUX	54.182	0.452245	54.5761	14	54.1532
2802.92802	PDCSAP_FLUX	61.2786	0.519447	61.5981	14	61.1971

BTJD	column	quadratic ppt	β error ppt	linear ppt	core n	background ppt
2807.13198	SAP_FLUX	54.8504	0.563556	55.1295	14	54.9284
2807.13198	PDCSAP_FLUX	62.188	0.629562	62.5067	14	62.2648
2811.33788	SAP_FLUX	54.4703	0.448777	54.8125	13	55.1102
2811.33788	PDCSAP_FLUX	62.0685	0.516515	62.3566	13	62.7069
2815.54313	SAP_FLUX	54.7285	0.463232	55.0122	12	54.6177
2815.54313	PDCSAP_FLUX	62.0795	0.54683	62.4566	12	62.0415
2819.74838	SAP_FLUX	54.7983	0.476243	55.3693	15	54.769
2819.74838	PDCSAP_FLUX	62.4694	0.540061	63.1362	15	62.4217
2823.95364	SAP_FLUX	54.5109	0.459084	54.6535	14	55.4083
2823.95364	PDCSAP_FLUX	62.0313	0.551003	62.2142	14	63.1379

Exact URI/URL, header, SHA256 and retrieval UTC are in independent-manifest.json. Neighbour required fractional dips use G as an approximate passband, not a calibrated TESS dilution correction.

Gaia source	sep arcsec	G	required fractional dip	capable

Original batch difference-image evidence (old/new schemas represent the same events, not independent repeats):


```
[
  {
    "tic": "244352837",
    "sector": 55,
    "n_events": 7,
    "deficit_minus_target_arcsec": 4.83,
    "deficit_minus_light_arcsec": 0.92,
    "peak_on_target": true,
    "depth_ap3": 0.05487,
    "psf_frac_resid": 0.312,
    "n_neighbours_60": 89,
    "n_possible": 0,
    "brightest_nb": {
      "G": 15.90056,
      "sep": 24.3
    }
  }
]
```


Archives and novelty

Fresh 60 arcsec IDIA cones returned all nine requested table sections. Counts: {'B_vsx_vsx': 0, 'II_366_catv2021': 0, 'B_sb9_main': 0, 'B_wds_wds': 0, 'J_ApJS_279_50_table2': 3, 'J_ApJS_279_50_table3': 0, 'J_ApJS_279_50_table4': 0, 'J_A_A_691_A242_obaf-eb1': 0, 'J_ApJS_258_16_tess-ebs': 0}. Gaia EB matches=0; vari_summary rows=0; ExoFOP TOI/CTOI= {'toi_count': 0, 'ctoi_count': 0}. SIMBAD exact Gaia identifier response is retained in shared cache. WDS records at 290592570/362082706 are visual companions, not an eclipse classification. Kostov table2 rows are unvetted flags and cone neighbours; they do not establish prior published EB classification. There are no target matches in VSX, ASAS-SN, SB9, vetted Kostov3/4, IJspeert or Prša in this fresh pass.

Exact catalogue rows including positions/identifiers are in catalogue-rows.json. General web search on all ten SIMBAD names plus eclipsing returned no relevant exact-target publication (an unrelated aggregated exoplanet-catalogue result was discarded); this is weaker than a complete bibliography

search. Novelty remains limited to the successful named checks. Shared/fresh archive query receipts, response hashes and timestamps are indexed in archive-provenance.json and idia-receipts.json.

Predictions and next check

After evaluation epoch BTJD 4344.32544: next primary approximately 4346.25425; next other-kind approximately 4348.30567 BTJD. These are extrapolations using photometric primary P and the observed secondary epoch (or Gaia phase for a null); formal small timing errors do not include future orbital changes or model systematics. For the three-event secondary at257800191 use these only as scheduling estimates.

Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.

Adjudication: **CONFIRMED**. Repeated primary/secondary eclipses, independent TESS-SPOC, Gaia conjunction timing and existing localized difference imaging establish a target EB. Sector depth variation and the nonzero PSF residual limit exact dimensions.

Reproducibility: source-audit.json verifies original candidate status and absent original vet folder before claim. fit_batch.py → orbit_gates.py → secondary_fit.py → shape_secondary.py → secondary_fit.py → geometry.py; null_secondary.py and covariance_geometry.py adapt SB2 covariance as needed. Independent and pixel scripts preserve separate provenance. No original catalogue/paper edits or commits.

R051. TIC 93213423

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 93213423: CONFIRMED

Repeated primary and secondary eclipses, a coherent multi-sector ephemeris, independent TESS-SPOC recovery and target-centred batch deficits establish the target EB. The small secondary phase displacement from the Gaia median is a soft-orbit characterization caveat.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=3.340044799 +/- 0.000000539 d; Tref=2606.2852128 +/- 0.0001484 BTJD at cycle340; T0=1470.6699813. 27 fitted primary events, O-C RMS=0.460 min; chi2/dof=3.71/25. Shape T14=5.504h, profile range=[0.22391379310344828, 0.2347758620689655], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
6	1470.6705479	0.001813	56.048	0.814	0.816	0
6	1474.0105773	0.001813	56.131	0.663	0.794	0
6	1480.6906362	0.001813	54.522	0.485	0.750	0
6	1484.0306656	0.001813	54.309	0.469	0.727	0
6	1487.3706950	0.001813	54.645	0.445	0.705	0
32	2175.4195113	0.000688	57.696	0.467	0.112	0
32	2178.7595407	0.000688	57.371	0.444	0.089	1
32	2182.0995701	0.000688	57.505	0.396	0.067	1
32	2188.7796290	0.000688	57.791	0.433	0.023	0
32	2192.1196585	0.000688	57.561	0.532	0.001	0
32	2195.4596879	0.000688	57.538	0.407	-0.021	0
32	2198.7997173	0.000688	57.402	0.473	-0.043	0
33	2202.1397468	0.000688	59.572	0.592	-0.065	0
33	2205.4797762	0.000688	59.704	0.506	-0.088	0
33	2208.8198056	0.000688	59.961	0.522	-0.110	0
33	2212.1598351	0.000688	59.952	0.527	-0.132	0
33	2218.8398939	0.000688	59.522	0.529	-0.176	0
33	2222.1799234	0.000688	59.672	0.527	-0.198	0
33	2225.5199528	0.000688	59.956	0.502	-0.220	0
98	3999.0838404	0.000688	48.482	0.405	-0.077	0

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
98	4002.4238699	0.000688	48.876	0.413	-0.099	0
98	4012.4439582	0.000688	48.430	0.426	-0.166	0
98	4015.7839876	0.000688	48.601	0.424	-0.188	0
98	4025.8040759	0.000688	48.946	0.403	-0.254	0
98	4029.1441054	0.000688	48.607	0.435	-0.276	0
98	4039.1655697	0.000688	48.691	0.468	1.639	0
98	4042.5042231	0.000688	48.621	0.395	-0.365	0

Primary median depth=57.371ppt; maximum absolute background-regression fake depth=0.327ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9999462. Primary assigned conjunctionA; period discrepancy=-0.142sigma. ConjunctionA at Gaia epoch: offset=0.011636d, Gaia-only sigma=0.009232d; including photometric extrapolation sigma=0.009240d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=1.660093d, Gaia-only sigma=0.009006d; including photometric extrapolation sigma=0.009015d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.32,
    "m2_min": 0.44235364823060586,
    "f_mass": 0.027869085399606808,
    "a_rsun": 11.358066803045336,
    "r1_rsun": 2.04811,
    "r2_approx_rsun": 0.5207337073826673,
    "central_T14_hours": {
      "A": 5.770122758161012,
      "B": 5.771039096345279
    },
    "observed_T14_hours": 5.504275862068965
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
```

```
  "best": {
    "chi2": 80.61193521929933,
    "dof": 121,
    "k": 0.22499999999999998,
    "b_A": 0.34,
    "e_sin_omega": -5.551115123125783e-17,
    "primary_scale": 0.9861342005182465,
    "secondary_scale": 0.004594321661181541
  },
  "profile_interval_approx_95": {
    "k": [
      0.22499999999999998,
      0.22499999999999998
    ],
    "b_A": [
      0.27,
      0.34
    ],
    "e_sin_omega": [
      -5.551115123125783e-17,
      0.02499999999999994
    ]
  },
  "error_inflation": 1,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.498901; pooled depth=4.680ppt, empirical error=0.081ppt; 27/27 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.5073008654398913,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.5020331733019257,
    0.5063513152865134,
    0.5119820789147937
  ],
  "gaia_relative_phase_sigma": 0.004909515560665673,
  "gaia_e_cos_omega_quantiles": [
    0.0031936989053881286,
    0.009976516022701644,
    0.018821554711331803
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.009466876184681375,
    -0.0002099496349373057,
    0.006244346914788412
  ],
  "photometric_phase": 0.49890086543989126,
  "photometric_e_cos_omega_first_order": -0.0017265165296721312,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
```

with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    -5.551115123125783e-17,
    0.02499999999999994
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}

sector	secondary mid BTJD	error d	depth ppt	error ppt
6	1469.0133294	0.007865	5.941	1.041
6	1475.6711337	0.001966	5.018	0.270
6	1479.0124894	0.001902	4.784	0.128
6	1482.3551560	0.001902	4.541	0.147
6	1485.6938899	0.001902	4.690	0.142
6	1489.0365565	0.001813	4.500	0.132
32	2177.0831633	0.000655	4.991	0.123
32	2180.4245189	0.000655	5.037	0.138
32	2183.7671855	0.000655	4.894	0.123
32	2190.4472751	0.000655	4.777	0.130
32	2193.7873199	0.000655	5.171	0.139
32	2197.1234320	0.000655	5.078	0.132
33	2203.8035216	0.001311	5.038	0.163
33	2207.1448773	0.000655	5.194	0.153
33	2210.4862330	0.000655	5.088	0.144
33	2217.1650117	0.001311	4.999	0.208
33	2220.5050565	0.000655	5.300	0.127
33	2223.8451013	0.000655	5.264	0.127
98	3997.4102004	0.000655	4.219	0.114
98	4000.7502452	0.000655	4.449	0.098
98	4010.7716904	0.000655	4.175	0.101
98	4014.1091135	0.000655	4.403	0.092
98	4017.4504691	0.000655	4.077	0.181
98	4027.4719144	0.000655	4.337	0.130
98	4030.8106483	0.001311	4.638	0.170
98	4040.8294718	0.000655	4.278	0.097
98	4044.1695166	0.000655	4.436	0.122

Secondary ephemeris: {"P": 3.340046052276308, "P_error": 9.909578681513317e-07, "Tref": 2648.031457938721, "Tref_error": 0.0002897530745744818, "reference_cycle": 352, "T0": 1472.3352475374606, "chi2": 85.3655446193316, "dof": 25, "n": 27, "oc_rms_minutes": 5.458855997680531}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TESS-SPOC sector33, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).

```
{
  "tc": 2202.139746762542,
  "qlp": {
    "1": {
      "depth": 0.0600649047073018,
      "error": 0.0005902903378002644,
      "red_noise_beta": 1.5811983523825757,
      "n_core": 33,
      "baseline_degree": 1,
      "background_regressed_depth": 0.060151204618899966,
      "background_template_native_sigma": 0.5000839610876912
    },
    "2": {
      "depth": 0.05962707018516755,
      "error": 0.0006012767025867044,
      "red_noise_beta": 1.559193837824628,
      "n_core": 33,
      "baseline_degree": 2,
      "background_regressed_depth": 0.05964945955946948,
      "background_template_native_sigma": 0.6620531657047218
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05740428175535831,
        "error": 0.00043156696433219557,
        "red_noise_beta": 1.284745677565976,
        "n_core": 33,
        "baseline_degree": 1,
        "background_regressed_depth": 0.057213331642321764,
        "background_template_native_sigma": 1.1231207066052524
      },
      "2": {
        "depth": 0.056855633306232604,
        "error": 0.00044838305617960346,
        "red_noise_beta": 1.2757936995653905,
        "n_core": 33,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05667849634483026,
        "background_template_native_sigma": 1.109443988241273
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.058212223050603945,
        "error": 0.0004355133693376432,
        "red_noise_beta": 1.2840275303078628,
        "n_core": 33,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05801671202053907,
        "background_template_native_sigma": 1.1231207066052524
      },
      "2": {
        "depth": 0.057703867463138006,
        "error": 0.0004565326123140108,
        "red_noise_beta": 1.276652213318582,
        "n_core": 33,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05752137669444432,
        "background_template_native_sigma": 1.109443988241273
      }
    }
```

```
  }
}

{
  "tc": 2205.479776199814,
  "qlp": {
    "1": {
      "depth": 0.061172558795048446,
      "error": 0.0006109750857147577,
      "red_noise_beta": 1.6973857650165645,
      "n_core": 33,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06125148818183048,
      "background_template_native_sigma": 1.1309348083095763
    },
    "2": {
      "depth": 0.05974662708111812,
      "error": 0.0005119434854652825,
      "red_noise_beta": 1.5494500953389037,
      "n_core": 33,
      "baseline_degree": 2,
      "background_regressed_depth": 0.05975227212915644,
      "background_template_native_sigma": 1.2171791958528626
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05821271710872041,
        "error": 0.0005703117033196052,
        "red_noise_beta": 1.6870399422866855,
        "n_core": 33,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05811556961266888,
        "background_template_native_sigma": 1.6328257877445025
      },
      "2": {
        "depth": 0.05691064534929745,
        "error": 0.00048651460593924,
        "red_noise_beta": 1.5474727254184741,
        "n_core": 33,
        "baseline_degree": 2,
        "background_regressed_depth": 0.0567831369692732,
        "background_template_native_sigma": 1.6546444740377138
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.05912179813603917,
        "error": 0.000581607436835985,
        "red_noise_beta": 1.6949468012764894,
        "n_core": 33,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05901393162486257,
        "background_template_native_sigma": 1.6328257877445025
      },
      "2": {
        "depth": 0.057795444401436406,
        "error": 0.0004967494620723318,
        "red_noise_beta": 1.5590792940173666,
        "n_core": 33,
        "baseline_degree": 2,
        "background_regressed_depth": 0.0576565291359601,
```

```
        "background_template_native_sigma": 1.6546444740377138
      }
    }
  }
}
```


Batch difference image: sector98, deficit-minus-light=1.92arcsec; deficit-minus-WCS=3.97arcsec; PSF fractional residual=0.256. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```


Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 2966750897393787264", "main_id": "TYC 5925-1148-1", "otype": "SB*", "ra": "85.72547642917", "dec": "-20.61338249434"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4346.44855, conditional linear-ephemeris1sigma=0.00032d. Next secondary template phase: BTJD4344.77486; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R052. TIC 369984032

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 369984032: CONFIRMED

Eleven sharp primary eclipses and nine individually detected secondary eclipses recur with the RV period. Independent SPOC and a 0.22 arcsec deficit-minus-light offset support target attribution. Secondary phase and duration favor an orbit adjustment; the Gaia small-e solution is not exact geometry.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=6.191975075 +/- 0.000001084 d; Tref=3003.9768643 +/- 0.0001350 BTJD at cycle156; T0=2038.0287527. 11 fitted primary events, O-C RMS=0.566 min; chi2/dof=8.49/9. Shape T14=3.496h, profile range=[0.14203448275862068, 0.14927586206896554], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
27	2038.0283739	0.000518	55.950	0.617	-0.545	6
27	2044.2212284	0.000448	56.319	0.511	0.721	0
27	2050.4123349	0.000460	55.779	0.588	-0.530	2
27	2056.6051893	0.000448	57.150	0.540	0.736	1
67	3127.8160664	0.000437	59.011	0.516	-0.431	0
67	3134.0080469	0.000437	58.751	0.481	-0.423	0
67	3146.3920079	0.000437	58.655	0.555	-0.408	6
94	3858.4688904	0.000437	49.831	0.546	-0.769	0
94	3864.6617448	0.000437	48.728	0.589	0.497	0
94	3870.8537253	0.000437	49.724	0.524	0.505	0
94	3877.0457058	0.000437	50.179	0.465	0.513	0

Primary median depth=55.950ppt; maximum absolute background-regression fake depth=0.533ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.99170786. Primary assigned conjunctionA; period discrepancy=0.284sigma. ConjunctionA at Gaia epoch: offset=0.032229d, Gaia-only sigma=0.025339d; including photometric extrapolation sigma=0.025343d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=2.940142d, Gaia-only sigma=0.025386d; including photometric extrapolation sigma=0.025390d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 0.93,
    "m2_min": 0.21696372442651174,
    "f_mass": 0.007763579726418871,
    "a_rsun": 14.85374538209675,
    "r1_rsun": 1.0942,
    "r2_approx_rsun": 0.2945165482799229,
    "central_T14_hours": {
      "A": 4.938026569015326,
      "B": 3.951303423643637
    },
    "observed_T14_hours": 3.4957241379310346
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 162.08754592945033,
    "dof": 121,
    "k": 0.24999999999999997,
    "b_A": 0.655,
    "e_sin_omega": -0.07500000000000004,
    "primary_scale": 0.8806811510750339,
    "secondary_scale": 0.0035002100947217076
  },
  "profile_interval_approx_95": {
    "k": [
      0.24999999999999997,
      0.27499999999999997
    ],
    "b_A": [
      0.6200000000000001,
      0.6900000000000001
```

```
    ],
    "e_sin_omega": [
      -0.10000000000000003,
      -0.02500000000000005
    ]
  },
  "error_inflation": 1.3395664952847135,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.517995; pooled depth=3.577ppt, empirical error=0.125ppt; 9/9 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.5309603441520124,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.523410681306229,
    0.530381493633695,
    0.5376378235853407
  ],
  "gaia_relative_phase_sigma": 0.007180840272767503,
  "gaia_e_cos_omega_quantiles": [
    0.0365610236474968,
    0.04745252182036956,
    0.058769833387361164
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.1276125928027583,
    -0.11058125096800794,
    -0.09365925554177115
  ],
  "photometric_phase": 0.5179951756826143,
  "photometric_e_cos_omega_first_order": 0.028266755862279433,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    -0.10000000000000003,
    -0.02500000000000005
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
27	2041.2353354	0.001661	3.954	0.243
27	2047.4273105	0.001246	4.168	0.185
27	2053.6176246	0.003322	3.468	0.298
27	2059.8095997	0.002907	3.057	0.281
67	3131.0246095	0.001661	3.707	0.282

sector	secondary mid BTJD	error d	depth ppt	error ppt
67	3143.4118816	0.001661	3.920	0.222
67	3149.5972128	0.001661	3.380	0.213
94	3861.6760074	0.001661	3.051	0.186
94	3880.2544240	0.001661	3.361	0.210

Secondary ephemeris: {"P": 6.191981387275571, "P_error": 5.8812551748908645e-06, "Tref": 2821.424918507847, "Tref_error": 0.0006987163113386135, "reference_cycle": 126, "T0": 2041.235263711125, "chi2": 10.290291936118514, "dof": 7, "n": 9, "oc_rms_minutes": 2.691674808189136}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

SPOC sector27, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 2038.0283739310344,
  "qlp": {
    "1": {
      "depth": 0.05544831850915039,
      "error": 0.0006075101485218567,
      "red_noise_beta": 1.3236280356675048,
      "n_core": 15,
      "baseline_degree": 1,
      "background_regressed_depth": 0.05653920131256707,
      "background_template_native_sigma": 1.2174354190663599
    },
    "2": {
      "depth": 0.05596608929110563,
      "error": 0.0006257338524523678,
      "red_noise_beta": 1.2825801495513485,
      "n_core": 15,
      "baseline_degree": 2,
      "background_regressed_depth": 0.05660670638044414,
      "background_template_native_sigma": 0.7787862757175406
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05472051701584885,
        "error": 0.0004828891538746284 4,
        "red_noise_beta": 2.1544797878489383,
        "n_core": 105,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05478499728377427,
        "background_template_native_sigma": 0.1675896167866095
      },
      "2": {
        "depth": 0.05511328707288404,
        "error": 0.0005090369367455475,
        "red_noise_beta": 2.053337170541624,
        "n_core": 105,
```

```
        "baseline_degree": 2,
        "background_regressed_depth": 0.05530003127314893,
        "background_template_native_sigma": 0.3244485539512207
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.05410160430239408,
        "error": 0.0004880302626318257,
        "red_noise_beta": 2.2066033053810554,
        "n_core": 105,
        "baseline_degree": 1,
        "background_regressed_depth": 0.054209931996161725,
        "background_template_native_sigma": 0.1675896167866095
      },
      "2": {
        "depth": 0.05442645334230298,
        "error": 0.0005082273121034446,
        "red_noise_beta": 2.071297645568529,
        "n_core": 105,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05469441196226321,
        "background_template_native_sigma": 0.3244485539512207
      }
    }
  }
}

{
  "tc": 2044.2212283528704,
  "qlp": {
    "1": {
      "depth": 0.05632601748977561,
      "error": 0.0004621771123033072,
      "red_noise_beta": 1.197839540481482,
      "n_core": 21,
      "baseline_degree": 1,
      "background_regressed_depth": 0.05644642341109225,
      "background_template_native_sigma": 0.11452728992981114
    },
    "2": {
      "depth": 0.056108383435219476,
      "error": 0.0005202292628429232,
      "red_noise_beta": 1.1929381188187107,
      "n_core": 21,
      "baseline_degree": 2,
      "background_regressed_depth": 0.05635310647812987,
      "background_template_native_sigma": 0.2383412903894266
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.055148430260156735,
        "error": 0.000512677310253226,
        "red_noise_beta": 2.006954158980317,
        "n_core": 105,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05483521388852523,
        "background_template_native_sigma": 1.1801403525645067
      },
      "2": {
        "depth": 0.0549988023939722,
```

```
        "error": 0.0005701450022847196,
        "red_noise_beta": 2.002637131581066,
        "n_core": 105,
        "baseline_degree": 2,
        "background_regressed_depth": 0.0548298182938013,
        "background_template_native_sigma": 0.658022601034829
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.05560006412547815,
        "error": 0.0006305896351484395,
        "red_noise_beta": 2.376643410936905,
        "n_core": 105,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05580307854607954,
        "background_template_native_sigma": 1.1801403525645067
      },
      "2": {
        "depth": 0.05556710140951724,
        "error": 0.0007050438801499339,
        "red_noise_beta": 2.3812322121367,
        "n_core": 105,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05574315152395262,
        "background_template_native_sigma": 0.658022601034829
      }
    }
  }
}
```


Batch difference image: sector27, deficit-minus-light=0.22arcsec; deficit-minus-WCS=3.43arcsec; PSF fractional residual=0.016. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```


Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 6669738865254236928", "main_id": "TYC 8405-863-1", "otype": "SB*", "ra": "306.35616748992", "dec": "-45.82921107506"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4347.63546, conditional linear-ephemeris1sigma=0.00027d. Next secondary template phase: BTJD4344.65089; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R053. TIC 345589621

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 345589621: CONFIRMED

Two well separated primary eclipses, matching the 20-day RV clock, are independently recovered in TESS-SPOC sector24. The difference image is target centred with no capable catalogued contaminant. This is sufficient for the physical target EB; the period fit has zero residual degrees of freedom and secondary coverage is inadequate.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=20.000360036 +/- 0.000024718 d; Tref=2694.5261566 +/- 0.0009071 BTJD at cycle36; T0=1974.5131953. 2 fitted primary events, O-C RMS=0.000 min; chi2/dof=0.00/0. Shape T14=7.981h, profile range=[0.32467499999999994, 0.340425], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
24	1974.5131953	0.001504	58.078	0.448	0.000	0
78	3434.5394779	0.000998	58.594	0.523	0.000	0

Primary median depth=58.336ppt; maximum absolute background-regression fake depth=0.663ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9603577. Primary assigned conjunctionA; period discrepancy=0.171sigma. ConjunctionA at Gaia epoch: offset=-0.096245d, Gaia-only sigma=0.071477d; including photometric

extrapolation sigma=0.071539d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=-9.680029d, Gaia-only sigma=0.115127d; including photometric extrapolation sigma=0.115166d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 0.964,
    "m2_min": 0.5356365955277238,
    "f_mass": 0.06833428634959808,
    "a_rsun": 35.49109899088749,
    "r1_rsun": 2.60194,
    "r2_approx_rsun": 0.6068707318143161,
    "central_T14_hours": {
      "A": 10.818166216716904,
      "B": 17.620147959364633
    },
    "observed_T14_hours": 7.981199999999999
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "insufficient secondary shape"
}
```


Secondary evidence

Optimized secondary phase=0.485432; pooled depth=nanppt, empirical error=0.000ppt; 0/0 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.479113476458235,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.47101748869567495,
    0.4792035671685599,
    0.4872518580388723
  ],
  "gaia_relative_phase_sigma": 0.008147970110627355,
  "gaia_e_cos_omega_quantiles": [
    -0.044203753297525974,
    -0.031738770087817794,
    -0.019447853478028132
  ],
  "gaia_e_sin_omega_quantiles": [
    0.22586490644842322,
    0.23881331097301925,
    0.25183462087252556
```

```
  ],
  "photometric_phase": null,
  "photometric_e_cos_omega_first_order": null,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "insufficient secondary shape",
  "geometry_e_sin_omega_conditional_range": null,
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


No clean fixed-width Gaia-median secondary windows exist. The automatic4.79ppt upper limit in orbit-gates.json has n=0 and is INVALID. null-secondary.json records conditional windows for alternate phases/widths; no uniform secondary upper bound is claimed. Two primary timing points provide no O-C degrees of freedom.

sector	secondary mid BTJD	error d	depth ppt	error ppt

Secondary ephemeris: null. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TESS-SPOC sector24, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 1974.5131953,
  "qlp": {
    "1": {
      "depth": 0.05760282411800896,
      "error": 0.0004293355177955828,
      "red_noise_beta": 1.3001115208932328,
      "n_core": 16,
      "baseline_degree": 1,
      "background_regressed_depth": 0.057734593767114865,
      "background_template_native_sigma": 1.3866321541103368
    },
    "2": {
      "depth": 0.05779512532460251,
      "error": 0.0004791571248199714,
      "red_noise_beta": 1.296421533877869,
      "n_core": 16,
      "baseline_degree": 2,
      "background_regressed_depth": 0.05781159320892652,
      "background_template_native_sigma": 0.5649673766552274
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.0564430791418671,
        "error": 0.00041332557127438655,
        "red_noise_beta": 1.2489254034679051,
        "n_core": 15,
        "baseline_degree": 1,
```

```
        "background_regressed_depth": 0.05599696702075248,
        "background_template_native_sigma": 0.7433486427510385
      },
      "2": {
        "depth": 0.0567041447186976,
        "error": 0.0004521947381604087,
        "red_noise_beta": 1.2409271046956984,
        "n_core": 15,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05630031441653478,
        "background_template_native_sigma": 0.7709207962880049
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.05569747082662542,
        "error": 0.0005086515257390553,
        "red_noise_beta": 1.330766705361284,
        "n_core": 15,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05539697835201073,
        "background_template_native_sigma": 0.7433486427510385
      },
      "2": {
        "depth": 0.05669194312899613,
        "error": 0.0004516346060164972,
        "red_noise_beta": 1.2790062994037348,
        "n_core": 15,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05661688429194981,
        "background_template_native_sigma": 0.7709207962880049
      }
    }
  }
}
```


Batch difference image: sector24, deficit-minus-light=0.27arcsec; deficit-minus-WCS=3.24arcsec; PSF fractional residual=0.026. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```


Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 1211102922603965056", "main_id": "TYC 1500-1410-1", "otype": "SB*", "ra": "234.42013392665", "dec": "20.78540394511"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4354.55604, conditional linear-ephemeris1sigma=0.00224d. Next secondary template phase: BTJD4344.26449; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R054. TIC 135251275

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 135251275: CONFIRMED

Eighteen primaries and fifteen individually detected secondaries, independent TESS-SPOC recovery and an isolated target-centred deficit establish an EB. The target VSX ROT classification at half the EB period is a type revision, not prior published eclipse classification.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=6.105646911 +/- 0.000001827 d; Tref=3009.8346314 +/- 0.0002533 BTJD at cycle235; T0=1575.0076073. 18 fitted primary events, O-C RMS=1.494 min; chi2/dof=39.42/16. Shape T14=4.365h, profile range=[0.17824137931034478, 0.18548275862068964], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
10	1575.0059912	0.002126	60.481	0.281	-2.327	0
10	1581.1116453	0.002273	59.900	0.299	-2.317	1
10	1587.2172995	0.002126	59.773	0.337	-2.306	0
10	1593.3229537	0.002126	58.902	0.431	-2.296	1
37	2313.7901448	0.000546	59.268	0.334	-1.064	0
37	2326.0025443	0.000546	58.838	0.349	0.528	6
37	2332.1071073	0.000546	60.286	0.370	-1.032	1
64	3046.4697355	0.000546	60.975	0.434	1.761	0
64	3052.5742985	0.000546	60.800	0.355	0.200	0
64	3058.6810438	0.000546	60.822	0.340	1.781	0
64	3064.7856068	0.000546	60.979	0.417	0.221	0
100	4078.3241978	0.000546	48.615	0.263	1.954	0
100	4084.4287608	0.000546	47.946	0.241	0.393	0
100	4090.5344149	0.000546	47.779	0.222	0.404	0
100	4096.6400691	0.000546	47.552	0.228	0.414	0
101	4102.7446321	0.000546	54.509	0.223	-1.147	1
101	4108.8502862	0.000546	54.290	0.279	-1.136	0
101	4121.0605034	0.000546	54.244	0.321	-2.687	0

Primary median depth=59.085ppt; maximum absolute background-regression fake depth=0.328ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.99819833. Primary assigned conjunctionA; period discrepancy=-1.416sigma. ConjunctionA at Gaia epoch: offset=0.005504d, Gaia-only sigma=0.017923d; including photometric extrapolation sigma=0.017942d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=2.977768d, Gaia-only sigma=0.023676d; including photometric extrapolation sigma=0.023690d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.12,
    "m2_min": 0.3217649084560672,
    "f_mass": 0.01602607361553534,
    "a_rsun": 15.882203192374531,
    "r1_rsun": 1.33258,
    "r2_approx_rsun": 0.40367415299918485,
    "central_T14_hours": {
      "A": 4.941302533533435,
      "B": 5.26023326018618
    },
    "observed_T14_hours": 4.364689655172413
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 86.13030993913411,
    "dof": 121,
    "k": 0.24999999999999997,
    "b_A": 0.41000000000000003,
    "e_sin_omega": 0.02499999999999994,
    "primary_scale": 0.8250323133496351,
    "secondary_scale": 0.003482843666689779
  },
  "profile_interval_approx_95": {
```

```
    "k": [
      0.24999999999999997,
      0.24999999999999997
    ],
    "b_A": [
      0.41000000000000003,
      0.41000000000000003
    ],
    "e_sin_omega": [
      0.02499999999999994,
      0.02499999999999994
    ]
  },
  "error_inflation": 1,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.515004; pooled depth=3.400ppt, empirical error=0.080ppt; 15/15 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.513859540553136,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.5079552045608652,
    0.5130932989253477,
    0.5190140803521471
  ],
  "gaia_relative_phase_sigma": 0.00555616551437239,
  "gaia_e_cos_omega_quantiles": [
    0.012490613215469208,
    0.0205566180379574,
    0.029856015974012353
  ],
  "gaia_e_sin_omega_quantiles": [
    0.020471947144851853,
    0.030324597736285895,
    0.04097332049909722
  ],
  "photometric_phase": 0.515003535233445,
  "photometric_e_cos_omega_first_order": 0.023567498033633213,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    0.02499999999999994,
    0.02499999999999994
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```

sector	secondary mid BTJD	error d	depth ppt	error ppt
10	1572.0453424	0.002126	3.588	0.197
10	1590.3633308	0.002273	3.702	0.237
37	2310.8307141	0.002619	3.202	0.224
37	2316.9374087	0.001572	3.887	0.212
37	2323.0367693	0.001048	3.524	0.196
37	2329.1476548	0.001572	3.568	0.193
64	3043.5104389	0.001572	3.492	0.189
64	3049.6139903	0.001572	3.614	0.161
64	3055.7196372	0.001048	3.208	0.168
64	3061.8315705	0.002095	3.946	0.205
100	4075.3626714	0.001572	3.227	0.133
100	4081.4672706	0.002619	2.858	0.178
101	4105.8856673	0.001048	3.008	0.172
101	4118.1011520	0.002619	3.057	0.227
101	4124.2036558	0.002095	3.578	0.187

Secondary ephemeris: {"P": 6.1056433427028995, "P_error": 6.218189820321831e-06, "Tref": 3012.978681641863, "Tref_error": 0.0008183373070155994, "reference_cycle": 235, "T0": 1578.1524961066818, "chi2": 52.71900553666333, "dof": 13, "n": 15, "oc_rms_minutes": 4.007131258451903}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TESS-SPOC sector10, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 1575.0059911724136,
  "qlp": {
    "1": {
      "depth": 0.05965499822983591,
      "error": 0.00042901637908482457,
      "red_noise_beta": 1.1366491018251947,
      "n_core": 8,
      "baseline_degree": 1,
      "background_regressed_depth": 0.05967028615817798,
      "background_template_native_sigma": 1.6716532186911472
    },
    "2": {
      "depth": 0.06065744004512028,
      "error": 0.00028166543957104185,
      "red_noise_beta": 1,
      "n_core": 8,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06112471155755894,
      "background_template_native_sigma": 2.0616206282132246
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
```

```
        "depth": 0.06002157711757433,
        "error": 0.00044828664334098866,
        "red_noise_beta": 1.1933449313036724,
        "n_core": 8,
        "baseline_degree": 1,
        "background_regressed_depth": 0.06026111479697441,
        "background_template_native_sigma": 2.1524564404429145
      },
      "2": {
        "depth": 0.06107365170873486,
        "error": 0.00025913440751378856,
        "red_noise_beta": 1,
        "n_core": 8,
        "baseline_degree": 2,
        "background_regressed_depth": 0.06125642250580076,
        "background_template_native_sigma": 2.1256577745566707
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.060717061777675205,
        "error": 0.0004507160029220888,
        "red_noise_beta": 1.2018698058442923,
        "n_core": 8,
        "baseline_degree": 1,
        "background_regressed_depth": 0.060930353022331296,
        "background_template_native_sigma": 2.1524564404429145
      },
      "2": {
        "depth": 0.06176664085901266,
        "error": 0.00025897966700204924,
        "red_noise_beta": 1,
        "n_core": 8,
        "baseline_degree": 2,
        "background_regressed_depth": 0.06192360318264485,
        "background_template_native_sigma": 2.1256577745566707
      }
    }
  }
}

{
  "tc": 1581.1116453348473,
  "qlp": {
    "1": {
      "depth": 0.05926165105752432,
      "error": 0.00042494643156807283,
      "red_noise_beta": 1.1158975186885691,
      "n_core": 7,
      "baseline_degree": 1,
      "background_regressed_depth": 0.059096139390387256,
      "background_template_native_sigma": 1.1019720056735456
    },
    "2": {
      "depth": 0.05999576838956972,
      "error": 0.0002995811624470777,
      "red_noise_beta": 1,
      "n_core": 7,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06015068629321189,
      "background_template_native_sigma": 1.6476478054575852
    }
  },
```

```
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.059231306653297594,
        "error": 0.0006417750737190021,
        "red_noise_beta": 1.2499770628637563,
        "n_core": 8,
        "baseline_degree": 1,
        "background_regressed_depth": 0.060051117949736346,
        "background_template_native_sigma": 2.1098677309473217
      },
      "2": {
        "depth": 0.06078744494269334,
        "error": 0.00029848054886784945,
        "red_noise_beta": 1,
        "n_core": 8,
        "baseline_degree": 2,
        "background_regressed_depth": 0.060948020906227626,
        "background_template_native_sigma": 1.9497087119639738
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.06008921840643354,
        "error": 0.0004888177379467468,
        "red_noise_beta": 1.1737301310916575,
        "n_core": 8,
        "baseline_degree": 1,
        "background_regressed_depth": 0.060802543654708816,
        "background_template_native_sigma": 2.1098677309473217
      },
      "2": {
        "depth": 0.061285445392131876,
        "error": 0.0002743681036614407,
        "red_noise_beta": 1,
        "n_core": 8,
        "baseline_degree": 2,
        "background_regressed_depth": 0.06148821819333466,
        "background_template_native_sigma": 1.9497087119639738
      }
    }
  }
}
```


Batch difference image: sector64, deficit-minus-light=0.28arcsec; deficit-minus-WCS=2.71arcsec; PSF fractional residual=0.048. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```


Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 6149942887797746560", "main_id": "TYC 7761-128-1", "otype": "SB*", "ra": "186.24282481663997", "dec": "-39.65764672227"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[{"source_id": "6149942887797746560", "abbe_mag_bp": "0.7316816", "abbe_mag_g_fov": "0.83544135", "abbe_mag_rp": "0.8087098", "in_vari_agn": "0", "in_vari_cepheid": "0", "in_vari_classification_result": "0", "in_vari_compact_companion": "0",

"in_vari_eclipsing_binary": "0", "in_vari_long_period_variable": "0", "in_vari_microlensing": "0", "in_vari_ms_oscillator": "0", "in_vari_planetary_transit": "0", "in_vari_rotation_modulation": "1", "in_vari_rrlyrae": "0", "in_vari_short_timescale": "0", "iqr_mag_bp": "0.012998429", "iqr_mag_g_fov": "0.012881311", "iqr_mag_rp": "0.009710297", "kurtosis_mag_bp": "10.108827", "kurtosis_mag_g_fov": "12.088275", "kurtosis_mag_rp": "14.100673", "mad_mag_bp": "0.009011926", "mad_mag_g_fov": "0.009010335", "mad_mag_rp": "0.0063914703", "max_mag_bp": "11.343665", "max_mag_g_fov": "11.032574", "max_mag_rp": "10.548094", "mean_mag_bp": "11.268182", "mean_mag_g_fov": "10.958851", "mean_mag_rp": "10.479972", "mean_obs_time_bp": "2330.844023750595", "mean_obs_time_g_fov": "2317.9323218075233", "mean_obs_time_rp": "2336.4123146163847", "median_mag_bp": "11.267736", "median_mag_g_fov": "10.957714", "median_mag_rp": "10.4789505", "min_mag_bp": "11.243395", "min_mag_g_fov": "10.938451", "min_mag_rp": "10.461985", "num_selected_bp": "65", "num_selected_g_fov": "68", "num_selected_rp": "66", "outlier_median_bp": "83.29495", "outlier_median_g_fov": "239.64076", "outlier_median_rp": "83.274185", "range_mag_bp": "0.100269906", "range_mag_g_fov": "0.09412307", "range_mag_rp": "0.08610976", "skewness_mag_bp": "2.4383008", "skewness_mag_g_fov": "2.8286264", "skewness_mag_rp": "3.2817688", "solution_id": "375316653866487564", "std_dev_mag_bp": "0.015749298", "std_dev_mag_g_fov": "0.014831815", "std_dev_mag_rp": "0.013656788", "std_dev_over_rms_err_mag_bp": "18.177307", "std_dev_over_rms_err_mag_g_fov": "4.1553025", "std_dev_over_rms_err_mag_rp": "17.317661", "stetson_mag_bp": "10.758741", "stetson_mag_g_fov": "12.363876", "stetson_mag_rp": "8.684481", "time_duration_bp": "1001.1367", "time_duration_g_fov": "1001.2107", "time_duration_rp": "1001.1367", "trimmed_range_mag_bp": "0.04053065", "trimmed_range_mag_g_fov": "0.035522208", "trimmed_range_mag_rp": "0.027912533"}]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": [{"_r": "0.04", "Name": "Gaia DR3 6149942887797746560", "Type": "ROT", "Period": "3.036790000000", "RAJ2000": "186.24281", "DEJ2000": "-39.65765"}]}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4346.97130, conditional linear-ephemeris1sigma=0.00047d. Next secondary template phase: BTJD4344.01009; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R055. TIC 435190459

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 435190459: CONFIRMED

Eight deep repeated primary eclipses at the RV clock survive independent TESS-SPOC reduction and target-centred localization. Secondary non-detection does not veto a target EB. Its pointwise limits are conditional on phase and width; radius/light-ratio assumptions are not measurements.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=6.638903127 +/- 0.000005260 d; Tref=3202.1243494 +/- 0.0002894 BTJD at cycle56; T0=2830.3457742. 8 fitted primary events, O-C RMS=0.995 min; chi2/dof=5.70/6. Shape T14=6.547h, profile range=[0.26736206896551723, 0.27822413793103445], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
56	2830.3454897	0.000818	58.570	0.617	-0.410	0
56	2836.9859477	0.000818	58.471	0.593	1.829	1
56	2843.6231322	0.000818	59.030	0.458	-0.646	0
56	2850.2619535	0.000818	59.401	0.564	-0.763	0
83	3560.6240116	0.000818	60.439	0.478	-1.594	6
83	3567.2644696	0.000818	59.945	0.492	0.646	2
83	3573.9032909	0.000818	59.343	0.542	0.528	0
83	3580.5421121	0.000818	59.051	0.484	0.410	0

Primary median depth=59.197ppt; maximum absolute background-regression fake depth=1.438ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9946027. Primary assigned conjunctionA; period discrepancy=-0.119sigma. ConjunctionA at Gaia epoch: offset=0.157191d, Gaia-only sigma=0.090549d; including photometric extrapolation sigma=0.090577d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=-3.001407d, Gaia-only sigma=0.071989d; including photometric extrapolation sigma=0.072024d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.04,
    "m2_min": 0.3096944634559846,
    "f_mass": 0.016305322433816067,
    "a_rsun": 16.42797739824315,
    "r1_rsun": 1.59862,
    "r2_approx_rsun": 0.3915135101641129,
    "central_T14_hours": {
      "A": 7.434438724894848,
      "B": 5.069938347239261
    },
    "observed_T14_hours": 6.547034482758621
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "insufficient secondary shape"
}
```


Secondary evidence

Optimized secondary phase=0.446979; pooled depth=3.931ppt, empirical error=0.117ppt; 0/8 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.47498018379158913,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.45396318485306825,
    0.47535980638116565,
    0.5001577700582943
  ],
  "gaia_relative_phase_sigma": 0.02341750040402469,
  "gaia_e_cos_omega_quantiles": [
    -0.07126936460425878,
    -0.03802967769626962,
```

```
    0.00024259385537845393
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.21455364141512775,
    -0.18643453512332384,
    -0.15686729176544462
  ],
  "photometric_phase": null,
  "photometric_e_cos_omega_first_order": null,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "insufficient secondary shape",
  "geometry_e_sin_omega_conditional_range": null,
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


No individual secondary exceeds3sigma. null-secondary.json tests covariance phases and widths: pointwise limits span0.58–4.81ppt and are not a uniform global upper bound. A faint companion and uncertain chord/radius/light ratio can account for a missing secondary; the assumed main-sequence radius is not a measured exclusion.

sector	secondary mid BTJD	error d	depth ppt	error ppt
56	2826.6185530	0.000558	3.846	0.292
56	2833.2574562	0.000558	3.475	0.340
56	2839.8963593	0.000558	3.777	0.276
56	2846.5352624	0.000558	3.717	0.299
83	3563.5368001	0.000558	3.828	0.294
83	3570.1757032	0.000558	4.091	0.318
83	3576.8146064	0.000558	4.060	0.267
83	3583.4535095	0.000558	4.667	0.320

Secondary ephemeris: null. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TESS-SPOC sector83, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 3560.624011591666,
  "qlp": {
    "1": {
      "depth": 0.06012117241347394,
      "error": 0.0004531036642680692,
      "red_noise_beta": 1.8548437651913054,
      "n_core": 112,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06011361050771707,
      "background_template_native_sigma": -0.006016107745134231
    },
    "2": {
      "depth": 0.06069633534156806,
      "error": 0.00047602965132843394,
```

```
      "red_noise_beta": 1.7826363128129086,
      "n_core": 112,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06057489869015148,
      "background_template_native_sigma": -0.11219312738923046
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.056157528859484794,
        "error": 0.00042751522208951524,
        "red_noise_beta": 2.0022585199597454,
        "n_core": 111,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05618919598670773,
        "background_template_native_sigma": 0.10016136529383748
      },
      "2": {
        "depth": 0.0566511291081388,
        "error": 0.0004515883115895924,
        "red_noise_beta": 1.9351016888237207,
        "n_core": 111,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05665526765619395,
        "background_template_native_sigma": -0.022070966496170075
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.056322813326895094,
        "error": 0.00040148148018327224,
        "red_noise_beta": 1.9482825628797609,
        "n_core": 111,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05636947658600153,
        "background_template_native_sigma": 0.10016136529383748
      },
      "2": {
        "depth": 0.056598059559810905,
        "error": 0.00043586077020426093,
        "red_noise_beta": 1.9136993168726697,
        "n_core": 111,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05659311114950755,
        "background_template_native_sigma": -0.022070966496170075
      }
    }
  }
}

{
  "tc": 3567.264469605509,
  "qlp": {
    "1": {
      "depth": 0.05954630061516232,
      "error": 0.0004781889263476012,
      "red_noise_beta": 1.8900433266968115,
      "n_core": 116,
      "baseline_degree": 1,
      "background_regressed_depth": 0.05984374132301949,
      "background_template_native_sigma": 0.4703473954106571
    },
```

```
    "2": {
      "depth": 0.06002995156665453,
      "error": 0.000497742952706455,
      "red_noise_beta": 1.780700550382042,
      "n_core": 116,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06018356727163635,
      "background_template_native_sigma": 0.28823204221457865
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05566563057540342,
        "error": 0.00043158129527178776,
        "red_noise_beta": 2.047793012616552,
        "n_core": 116,
        "baseline_degree": 1,
        "background_regressed_depth": 0.0557783919125477,
        "background_template_native_sigma": 0.38792767774618
      },
      "2": {
        "depth": 0.056082314483056735,
        "error": 0.0004533653596821236,
        "red_noise_beta": 1.9485561834976115,
        "n_core": 116,
        "baseline_degree": 2,
        "background_regressed_depth": 0.056121316435812654,
        "background_template_native_sigma": 0.27252607792339284
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.056072852932239,
        "error": 0.00041384096265338757,
        "red_noise_beta": 1.9727597301765256,
        "n_core": 116,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05614432380168172,
        "background_template_native_sigma": 0.38792767774618
      },
      "2": {
        "depth": 0.05631132951261328,
        "error": 0.0004532596613317053,
        "red_noise_beta": 1.942131211104606,
        "n_core": 116,
        "baseline_degree": 2,
        "background_regressed_depth": 0.056338739826469336,
        "background_template_native_sigma": 0.27252607792339284
      }
    }
  }
}
```


Batch difference image: sector56, deficit-minus-light=0.12arcsec; deficit-minus-WCS=0.8arcsec; PSF fractional residual=0.283. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 2812131799856915968", "main_id": "UCAC4 517-141331", "otype": "SB*", "ra": "349.76188432019666", "dec": "13.285064206886112"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4344.01569, conditional linear-ephemeris1sigma=0.00095d. Next secondary template phase: BTJD4346.98314; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R056. TIC 150696021

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 150696021: CONFIRMED

Seven primaries and five independently detected secondaries recur with the RV period. The G13.86 neighbor19.3 arcsec away is capable in a dilution-only test, but per-pixel regression in four events favors the target every time, with the free centroid on target. Physical EB is confirmed; geometry and component luminosities remain conditional.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=7.760015435 +/- 0.000012413 d; Tref=3154.8847230 +/- 0.0006042 BTJD at cycle42; T0=2828.9640747. 7 fitted primary events, O-C RMS=3.173 min; chi2/dof=17.66/5. Shape T14=5.765h, profile range=[0.2347758620689655, 0.24563793103448273], ingress fraction=0.35; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
56	2828.9633588	0.000721	62.666	0.443	-1.031	0
56	2836.7219254	0.001441	61.393	1.591	-3.117	0
56	2844.4848157	0.000721	62.243	0.498	1.023	0
56	2852.2491472	0.002162	59.027	2.534	7.238	0
83	3566.1655448	0.000721	61.404	0.443	0.005	0
83	3573.9269939	0.000721	62.146	0.524	2.070	0
83	3581.6841192	0.000721	62.052	0.436	-2.092	0

Primary median depth=62.052ppt; maximum absolute background-regression fake depth=0.459ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9998698. Primary assigned conjunctionA; period discrepancy=1.585sigma. ConjunctionA at Gaia epoch: offset=-0.027202d, Gaia-only sigma=0.045108d; including photometric extrapolation sigma=0.045329d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=-3.836545d, Gaia-only sigma=0.032742d; including photometric extrapolation sigma=0.033046d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.04,
    "m2_min": 0.5761781061580149,
    "f_mass": 0.07323046921808257,
    "a_rsun": 19.35622502108516,
    "r1_rsun": 2.21953,
    "r2_approx_rsun": 0.6433471796556982,
    "central_T14_hours": {
      "A": 8.690817982022358,
      "B": 8.841584787207033
    },
    "observed_T14_hours": 5.764965517241379
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 159.6936691260499,
    "dof": 121,
    "k": 0.27499999999999997,
    "b_A": 0.7950000000000002,
    "e_sin_omega": 0.09999999999999992,
    "primary_scale": 0.965047812455021,
    "secondary_scale": 0.03576951779831132
  },
  "profile_interval_approx_95": {
    "k": [
      0.27499999999999997,
      0.27499999999999997
    ],
    "b_A": [
      0.7950000000000002,
      0.7950000000000002
    ],
    "e_sin_omega": [
      0.09999999999999992,
      0.09999999999999992
    ]
  },
  "error_inflation": 1.3197823894714868,
```

```
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.500334; pooled depth=18.164ppt, empirical error=0.291ppt; 5/5 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.48795244650292313,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.48208999389226315,
    0.49135494381916045,
    0.49861539131737936
  ],
  "gaia_relative_phase_sigma": 0.008786686515241409,
  "gaia_e_cos_omega_quantiles": [
    -0.028132585905592058,
    -0.013577905964244942,
    -0.0021748370861388523
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.006122724569791174,
    0.0064428506808386935,
    0.023589563952668976
  ],
  "photometric_phase": 0.500334220666431,
  "photometric_e_cos_omega_first_order": 0.0005249925951686977,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    0.09999999999999992,
    0.09999999999999992
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
56	2832.8452347	0.000721	17.713	0.357
56	2840.6052502	0.000721	18.322	0.267
56	2848.3638244	0.000721	19.407	0.312
83	3562.2881269	0.000721	18.111	0.284
83	3570.0452598	0.000721	17.567	0.235

Secondary ephemeris: {"P": 7.760020297125379, "P_error": 1.3299521850682019e-05, "Tref": 3127.725531071351, "Tref_error": 0.000609286499375676, "reference_cycle": 38, "T0": 2832.844759780586, "chi2": 10.722296342587057, "dof": 3, "n": 5, "oc_rms_minutes":

1.519595923750579}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TESS-SPOC sector56, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 2828.963358758621,
  "qlp": {
    "1": {
      "depth": 0.06318850663584949,
      "error": 0.00041747144977343105,
      "red_noise_beta": 1.7296763722544908,
      "n_core": 104,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06324672190506934,
      "background_template_native_sigma": 0.13955170518400567
    },
    "2": {
      "depth": 0.06292773318870623,
      "error": 0.0004512306029557722,
      "red_noise_beta": 1.6938440822198069,
      "n_core": 104,
      "baseline_degree": 2,
      "background_regressed_depth": 0.0629704148190884,
      "background_template_native_sigma": 0.10126210262284988
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05669834430782072,
        "error": 0.00037498421791336116,
        "red_noise_beta": 1.8061769624469866,
        "n_core": 104,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05672563580678699,
        "background_template_native_sigma": 0.34741947025150316
      },
      "2": {
        "depth": 0.05656965951709012,
        "error": 0.0004130892078825761,
        "red_noise_beta": 1.7976618721637025,
        "n_core": 104,
        "baseline_degree": 2,
        "background_regressed_depth": 0.056591328924534445,
        "background_template_native_sigma": 0.2502854431842343
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.0631983945458318,
        "error": 0.0007024024432912891,
        "red_noise_beta": 2.6691119853672465,
        "n_core": 104,
```

```
        "baseline_degree": 1,
        "background_regressed_depth": 0.06303200093126982,
        "background_template_native_sigma": 0.34741947025150316
      },
      "2": {
        "depth": 0.06258498316350793,
        "error": 0.0007396150306909728,
        "red_noise_beta": 2.586373262416255,
        "n_core": 104,
        "baseline_degree": 2,
        "background_regressed_depth": 0.06247346276822013,
        "background_template_native_sigma": 0.2502854431842343
      }
    }
  }
}

{
  "tc": 2836.7219253572894,
  "qlp": {
    "1": {
      "depth": 0.05779548979003091,
      "error": 0.0016914676725034198,
      "red_noise_beta": 3.260992815015399,
      "n_core": 104,
      "baseline_degree": 1,
      "background_regressed_depth": 0.059677131334476795,
      "background_template_native_sigma": -1.1402130119008185
    },
    "2": {
      "depth": 0.06048425963833052,
      "error": 0.0015860016817420789,
      "red_noise_beta": 3.068950849767196,
      "n_core": 104,
      "baseline_degree": 2,
      "background_regressed_depth": 0.060855541728110155,
      "background_template_native_sigma": -0.33848661278126757
    }
  },
  "independent": {
    "SAP_FLUX": {
      "1": {
        "depth": 0.05629363327148408,
        "error": 0.0004109561110479489,
        "red_noise_beta": 1.9079512047866956,
        "n_core": 104,
        "baseline_degree": 1,
        "background_regressed_depth": 0.05612608951901542,
        "background_template_native_sigma": -0.9355152621273938
      },
      "2": {
        "depth": 0.05639880386723798,
        "error": 0.00045394519266711894,
        "red_noise_beta": 1.903115475189127,
        "n_core": 104,
        "baseline_degree": 2,
        "background_regressed_depth": 0.05631892450001077,
        "background_template_native_sigma": -0.25821173226487065
      }
    },
    "PDCSAP_FLUX": {
      "1": {
        "depth": 0.0573290979749512,
```

```
        "error": 0.0014905494147139015,
        "red_noise_beta": 3.6557481348981473,
        "n_core": 104,
        "baseline_degree": 1,
        "background_regressed_depth": 0.06025255682689771,
        "background_template_native_sigma": -0.9355152621273938
      },
      "2": {
        "depth": 0.059568478937806174,
        "error": 0.0014216459275203017,
        "red_noise_beta": 3.5510603280767636,
        "n_core": 104,
        "baseline_degree": 2,
        "background_regressed_depth": 0.06035714104162983,
        "background_template_native_sigma": -0.25821173226487065
      }
    }
  }
}
```


Batch difference image: sector56, deficit-minus-light=2.17arcsec; deficit-minus-WCS=3.82arcsec; PSF fractional residual=0.168. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[
  {
    "source_id": "1881109883677908224",
    "ra": "338.4257912309341",
    "dec": "26.974684521364487",
    "phot_g_mean_mag": "13.864994",
    "bp_rp": "1.2449017",
    "sep": "19.30089117189746",
    "required_fraction_G_approx": 0.6781701809487413,
    "capable": true
  }
]
```


pixel-localization.json:


```
{
  "calibration": {
    "sigma_pixels": 0.6500000000000001,
    "dx_pixels": 0.09999999999999987,
    "dy_pixels": -1.1102230246251565e-16
  },
  "free_best": {
    "chi2": 9653.358143825943,
    "pixel_x": 7.099169734214097,
    "pixel_y": 7.023534238890427,
    "distance_from_target_arcsec": 2.1102042644709938e-13,
    "fractional_residual": 0.11270363370544775
  },
  "fixed_fits": [
    {
      "source_id": "1881109883677907584",
      "sep_arcsec": 0.0,
      "G": 11.268545,
      "chi2": 9653.358143825833,
      "amplitude": 145.262575953447,
      "fractional_residual": 0.11270363370544757
    },
```

```
    {
      "source_id": "1881109883677908224",
      "sep_arcsec": 19.30089117189746,
      "G": 13.864994,
      "chi2": 100245.0204413415,
      "amplitude": 67.76468725916556,
      "fractional_residual": 0.7806075339091613
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 1881109883677907584", "main_id": "TYC 2227-1936-1", "otype": "SB*", "ra": "338.41975979931", "dec": "26.97456707901"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": [{"_r": "19.62", "Name": "Gaia DR3 1881109883677908224", "Type": "ROT", "Period": "", "RAJ2000": "338.42587", "DEJ2000": "26.97467"}]}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4349.92710, conditional linear-ephemeris1sigma=0.00200d. Next secondary template phase: BTJD4346.04969; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R057. TIC 345050437

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 345050437: CONFIRMED

Thirty-three primaries and repeated secondaries establish a coherent EB. The old high PSF residual is not accepted uncritically: fresh template regression favors the target in every usable sector24 event and its free centroid is only2.97 arcsec away. A known E-type variable44.25 arcsec away has period2.5355d, inconsistent with the target4.48083d clock. Elevated geometry chi-square precludes precise radii/orbit claims.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=4.480834621 +/- 0.000001500 d; Tref=2910.6551981 +/- 0.0001744 BTJD at cycle255; T0=1768.0423698. 33 fitted primary events, O-C RMS=1.518 min; chi2/dof=47.90/31. Shape T14=3.670h, profile range=[0.14927586206896548, 0.15651724137931033], ingress fraction=0.35; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
17	1768.0405000	0.002690	63.258	0.862	-2.693	2
17	1781.4830276	0.002126	64.164	1.133	-2.658	0
17	1785.9638701	0.002126	61.973	0.785	-2.647	0
24	1956.2368033	0.002273	66.176	0.811	-0.894	0
24	1960.7158110	0.002273	69.063	0.854	-3.524	0
24	1965.1984883	0.002455	67.042	1.182	-0.871	1
24	1969.6774961	0.002273	64.284	2.107	-3.502	0
24	1974.1601733	0.002273	67.753	1.005	-0.848	0
24	1978.6410159	0.002126	66.248	0.847	-0.837	0
57	2856.8879848	0.000917	65.935	1.804	4.035	0
57	2861.3660751	0.000459	64.653	0.540	0.083	0
57	2870.3277602	0.000459	64.925	0.531	0.106	0
57	2879.2894452	0.000459	65.006	0.658	0.129	0
58	2883.7702878	0.000459	62.257	1.001	0.140	0
58	2888.2502129	0.000459	64.256	0.789	-1.169	1
58	2892.7328902	0.000917	60.170	1.978	1.484	0
58	2901.6936578	0.000459	61.906	0.973	0.186	0
58	2906.1745004	0.000459	62.911	0.832	0.197	0

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
77	3399.0662603	0.000459	65.241	0.836	0.128	0
77	3403.5471028	0.000459	65.296	0.869	0.139	1
77	3421.4713903	0.000459	64.815	0.757	1.505	1
78	3439.3929256	0.000459	66.309	0.572	-1.091	0
78	3443.8756029	0.000459	64.791	0.797	1.562	0
78	3448.3555280	0.000459	65.014	0.651	0.253	0
84	3587.2607288	0.000459	73.145	1.467	-0.716	0
84	3591.7424887	0.000459	64.233	0.583	0.617	0
84	3600.7041737	0.000459	65.542	0.639	0.639	0
84	3605.1840989	0.000917	60.120	2.160	-0.670	0
84	3609.6649414	0.000459	64.949	1.022	-0.659	0
85	3614.1467013	0.000459	63.183	0.608	0.674	0
85	3618.6266265	0.000459	63.938	0.714	-0.636	0
85	3623.1074690	0.000459	62.384	1.369	-0.625	5
85	3632.0682366	0.000459	62.772	0.552	-1.923	0

Primary median depth=64.791ppt; maximum absolute background-regression fake depth=1.579ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9999283. Primary assigned conjunctionA; period discrepancy=-1.904sigma. ConjunctionA at Gaia epoch: offset=-0.003781d, Gaia-only sigma=0.022320d; including photometric extrapolation sigma=0.022336d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=2.220511d, Gaia-only sigma=0.020657d; including photometric extrapolation sigma=0.020675d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 0.9,
    "m2_min": 0.4565141915123945,
    "f_mass": 0.05170281215759653,
    "a_rsun": 12.662074986282596,
    "r1_rsun": 1.08915,
    "r2_approx_rsun": 0.5340272595790694,
    "central_T14_hours": {
      "A": 4.352033694299332,
      "B": 4.424834246325007
    },
    "observed_T14_hours": 3.66951724137931
  },
  {
    "m1": 1.2,
```

```
    "m2_min": 0.5386596407959161,
    "f_mass": 0.05170281215759653,
    "a_rsun": 13.754188064991082,
    "r1_rsun": 1.08915,
    "r2_approx_rsun": 0.6096092521064337,
    "central_T14_hours": {
      "A": 4.193030846537644,
      "B": 4.263171608703266
    },
    "observed_T14_hours": 3.66951724137931
  },
  {
    "m1": 1.6,
    "m2_min": 0.6372582512480045,
    "f_mass": 0.05170281215759653,
    "a_rsun": 14.960133650254367,
    "r1_rsun": 1.08915,
    "r2_approx_rsun": 0.6973525189018747,
    "central_T14_hours": {
      "A": 4.0541455829552095,
      "B": 4.121963080971872
    },
    "observed_T14_hours": 3.66951724137931
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 350.28861292667847,
    "dof": 121,
    "k": 0.44999999999999996,
    "b_A": 0.5150000000000001,
    "e_sin_omega": -0.10000000000000003,
    "primary_scale": 0.3141153485690935,
    "secondary_scale": 0.008500752548683271
  },
  "profile_interval_approx_95": {
    "k": [
      0.44999999999999996,
      0.44999999999999996
    ],
    "b_A": [
      0.5150000000000001,
      0.585
    ],
    "e_sin_omega": [
      -0.12500000000000003,
      -0.10000000000000003
    ]
  },
  "error_inflation": 2.89494721427007,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
```

```
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.500743; pooled depth=8.053ppt, empirical error=0.310ppt; 34/37 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.5055202121952085,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.49673731135095534,
    0.503280255570373,
    0.511822094639491
  ],
  "gaia_relative_phase_sigma": 0.00864590819504258,
  "gaia_e_cos_omega_quantiles": [
    -0.005123750498972375,
    0.005151058698107992,
    0.018568974484470233
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.003993631637848673,
    0.005677233822333128,
    0.019493104206848237
  ],
  "photometric_phase": 0.5007430849291679,
  "photometric_e_cos_omega_first_order": 0.0011672350772335184,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    -0.12500000000000003,
    -0.10000000000000003
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
17	1770.2907036	0.005046	6.153	0.798
17	1779.2661336	0.009632	10.538	1.871
17	1783.7249511	0.003670	6.752	0.594
24	1958.4784187	0.005504	7.445	0.936
24	1962.9620054	0.004128	7.363	0.781
24	1967.4382532	0.007339	8.291	1.333
24	1971.9502787	0.013761	9.407	2.354
24	1976.3999224	0.003211	8.136	0.645
24	1980.8853439	0.002752	9.720	0.625
57	2854.6380038	0.016972	4.268	2.182
57	2859.1215906	0.003670	9.650	0.992
57	2868.0878467	0.002752	7.532	0.710
57	2872.5714334	0.005046	8.591	1.044

sector	secondary mid BTJD	error d	depth ppt	error ppt
57	2877.0513507	0.024311	3.271	2.366
57	2881.5285158	0.001376	8.604	0.464
58	2886.0157721	0.004587	8.398	0.962
58	2890.4938546	0.003211	9.052	0.752
58	2894.9856977	0.006880	13.571	1.824
58	2899.4536890	0.002293	6.826	0.567
58	2903.9354410	0.003211	8.270	0.750
58	2908.4116888	0.002752	9.000	0.588
77	3396.8235798	0.018806	3.773	2.249
77	3401.3044144	0.002752	7.833	0.532
77	3405.7916707	0.003670	7.165	0.747
77	3419.2341746	0.002752	8.263	0.607
78	3437.1520088	0.002752	8.748	0.782
78	3441.6310086	0.005046	6.482	0.841
78	3446.1145954	0.002293	8.642	0.489
84	3585.0259729	0.005046	7.311	1.105
84	3589.4967164	0.011467	5.665	1.712
84	3593.9839726	0.001835	8.085	0.453
84	3598.4693941	0.005504	6.148	0.996
84	3602.9456419	0.004128	9.436	1.087
84	3607.4558326	0.010091	10.111	2.060
85	3616.3927326	0.002752	9.401	0.658
85	3620.8662282	0.001835	7.851	0.391
85	3634.3023104	0.004587	7.351	0.768

Secondary ephemeris: {"P": 4.4808319786037885, "P_error": 5.409492670163906e-06, "Tref": 2912.896438265481, "Tref_error": 0.0006805155783497134, "reference_cycle": 255, "T0": 1770.2842837215148, "chi2": 47.73252386567204, "dof": 32, "n": 34, "oc_rms_minutes": 12.164741998817926}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TARS sector24, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 1956.236803265395,
  "qlp": {
    "1": {
      "depth": 0.06645511892419378,
      "error": 0.000755561014613154,
      "red_noise_beta": 1.1136289997349256,
      "n_core": 7,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06618119316628734,
      "background_template_native_sigma": -0.8281131610642404
    },
    "2": {
      "depth": 0.06628522694414007,
```

```
      "error": 0.0008369618879326378,
      "red_noise_beta": 1.0970671419745532,
      "n_core": 7,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06616293083817126,
      "background_template_native_sigma": -0.1728613681302516
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.061070363448494584,
        "error": 0.0017553930763530643,
        "red_noise_beta": 1.2571522617855257,
        "n_core": 7,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.06363394266684382,
        "error": 0.0016147786880699835,
        "red_noise_beta": 1.2797488779993245,
        "n_core": 7,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}

{
  "tc": 1960.7158110278717,
  "qlp": {
    "1": {
      "depth": 0.06900039186511787,
      "error": 0.0008746875033901591,
      "red_noise_beta": 1.2104800142767422,
      "n_core": 7,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06946905757910311,
      "background_template_native_sigma": 0.8038477578531368
    },
    "2": {
      "depth": 0.06926760681753494,
      "error": 0.0009814149536555125,
      "red_noise_beta": 1.2209212300146388,
      "n_core": 7,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06942456198162125,
      "background_template_native_sigma": 0.26466868079975486
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.06049487917754053,
        "error": 0.0009592494724983237,
        "red_noise_beta": 1.043133779087984,
        "n_core": 7,
        "baseline_degree": 1,
        "background_regressed_depth": null,
```

```
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.06084796925770194,
        "error": 0.0010703109094375098,
        "red_noise_beta": 1.046870845821653,
        "n_core": 7,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}
```


Batch difference image: sector24, deficit-minus-light=2.7arcsec; deficit-minus-WCS=2.81arcsec; PSF fractional residual=0.848. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[
  {
    "source_id": "2011135380006552832",
    "ra": "355.3461946027936",
    "dec": "59.525463809467325",
    "phot_g_mean_mag": "14.965737",
    "bp_rp": "2.291256",
    "sep": "23.196200041677034",
    "required_fraction_G_approx": 1.6076962265747736,
    "capable": false
  }
]
```


pixel-localization.json:


```
{
  "calibration": {
    "sigma_pixels": 0.45,
    "dx_pixels": -0.10000000000000009,
    "dy_pixels": -0.10000000000000009
  },
  "free_best": {
    "chi2": 1770.9734435675846,
    "pixel_x": 7.739989671941279,
    "pixel_y": 7.834463584517652,
    "distance_from_target_arcsec": 2.969848480983278,
    "fractional_residual": 0.22305336449585392
  },
  "fixed_fits": [
    {
      "source_id": "2011135311287080704",
      "sep_arcsec": 0.0,
      "G": 11.479009,
      "chi2": 1981.1935672920263,
      "amplitude": 231.18128046755353,
      "fractional_residual": 0.24112415322488306
    },
    {
      "source_id": "2011135380006552832",
      "sep_arcsec": 23.196200041677034,
      "G": 14.965737,
      "chi2": 12840.70754461235,
```

```
      "amplitude": 95.78201985484202,
      "fractional_residual": 0.8014067555286477
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 2011135311287080704", "main_id": "TYC 4012-825-1", "otype": "SB*", "ra": "355.34922032052", "dec": "59.51927035952001"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": [{"_r": "44.25", "Name": "Gaia DR3 2011136788755832704", "Type": "E", "Period": "2.535500000000", "RAJ2000": "355.37332", "DEJ2000": "59.52063"}]}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4344.52228, conditional linear-ephemeris1sigma=0.00051d. Next secondary template phase: BTJD4346.76602; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R058. TIC 236531158

physical_verdict: PLAUSIBLE

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Obtain a calibrated empirical TESS PRF with color-dependent neighbor fluxes or ground-based resolved eclipse photometry of the target and12arcsec neighbor.

Archived source: sanitized report text.

TIC 236531158: PLAUSIBLE

The eclipsing signal is secure, with27 primaries and28 individually detected secondary events, but the host remains materially ambiguous. Free-amplitude calibrated circular/elliptical models favor the12arcsec G13.84 neighbor in13/13 events; enforcing Gaia-G flux ratios flips the result to target in13/13 events. PSF/color/dilution calibration is not controlled sufficiently to resolve this model dependence. PLAUSIBLE refers to the assigned target EB; it is not a claim that the eclipse signal itself is doubtful.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=2.020569926 +/- 0.000000533 d; Tref=2853.3540841 +/- 0.0001656 BTJD at cycle564; T0=1713.7526460. 27 fitted primary events, O-C RMS=2.545 min; chi2/dof=27.03/25. Shape T14=2.477h, profile range=[0.10076120689655173, 0.10564913793103449], ingress fraction=0.35; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
15	1711.7274469	0.002690	66.887	0.730	-6.666	0
15	1713.7486308	0.002690	67.981	0.512	-5.782	0
15	1715.7691954	0.002690	67.163	0.576	-5.790	0
15	1717.7897600	0.002690	67.274	0.681	-5.797	0
15	1727.8932024	0.002690	65.241	0.641	-4.944	0
15	1729.9137670	0.002690	66.162	0.604	-4.951	0
15	1731.9343317	0.002690	66.778	0.627	-4.959	0
55	2798.7992675	0.000518	64.193	0.572	0.823	0
55	2800.8192129	0.000518	64.166	0.513	-0.077	0
55	2802.8403968	0.000536	64.836	0.947	0.808	1
55	2804.8603422	0.000556	64.057	0.666	-0.092	2
55	2812.9432199	0.000518	64.862	0.650	0.769	0
55	2814.9637846	0.000518	64.734	0.508	0.762	0
55	2816.9843492	0.000518	64.410	0.543	0.754	0
55	2819.0042946	0.000518	65.454	0.630	-0.145	0
82	3534.2860311	0.000310	64.909	0.505	-0.170	0

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
82	3536.3065957	0.000310	64.952	0.596	-0.178	0
82	3538.3277796	0.000310	65.241	0.611	0.707	6
82	3540.3477250	0.000310	64.945	0.665	-0.193	0
82	3542.3682896	0.000310	64.824	0.611	-0.200	0
82	3544.3888543	0.000310	64.184	0.614	-0.208	0
82	3546.4094189	0.000310	64.609	0.602	-0.216	0
82	3548.4299835	0.000310	64.464	0.497	-0.223	0
82	3550.4505481	0.000310	65.080	0.615	-0.231	0
82	3552.4717320	0.000310	64.418	0.616	0.653	0
82	3554.4916774	0.000310	64.810	0.588	-0.246	0
82	3556.5122420	0.000310	64.634	0.653	-0.254	0

Primary median depth=64.862ppt; maximum absolute background-regression fake depth=0.728ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.99966836. Primary assigned conjunctionA; period discrepancy=-0.532sigma. ConjunctionA at Gaia epoch: offset=0.012333d, Gaia-only sigma=0.015010d; including photometric extrapolation sigma=0.015025d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=0.996579d, Gaia-only sigma=0.008721d; including photometric extrapolation sigma=0.008747d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.225,
    "m2_min": 0.38348180443751545,
    "f_mass": 0.021797263881794683,
    "a_rsun": 7.880635700035171,
    "r1_rsun": 1.46735,
    "r2_approx_rsun": 0.46451079345321333,
    "central_T14_hours": {
      "A": 3.724241547042809,
      "B": 3.8426873934471626
    },
    "observed_T14_hours": 2.4769241379310345
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV

primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 406.6147734396419,
    "dof": 121,
    "k": 0.35,
    "b_A": 0.76,
    "e_sin_omega": 0.17499999999999988,
    "primary_scale": 0.6531649686670947,
    "secondary_scale": 0.018009764761063012
  },
  "profile_interval_approx_95": {
    "k": [
      0.29999999999999993,
      0.3999999999999999
    ],
    "b_A": [
      0.7250000000000001,
      0.865
    ],
    "e_sin_omega": [
      0.04999999999999993,
      0.1999999999999999
    ]
  },
  "error_inflation": 3.360452673054892,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.500215; pooled depth=5.555ppt, empirical error=0.101ppt; 28/28 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.514752958945282,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.5028053286584397,
    0.5126002222947604,
    0.5232478994997485
  ],
  "gaia_relative_phase_sigma": 0.009847679506689013,
  "gaia_e_cos_omega_quantiles": [
    0.004405492321579887,
    0.01978956972136035,
    0.03651711001651194
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.0028630625834132183,
```

```
    0.011006944769087767,
    0.025523306015155546
  ],
  "photometric_phase": 0.5002149473714801,
  "photometric_e_cos_omega_first_order": 0.00033763854157514315,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape,
with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    0.04999999999999993,
    0.1999999999999999
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a
calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
15	1712.7407617	0.002712	6.334	0.578
15	1716.7724112	0.002690	6.725	0.554
15	1718.8072167	0.003728	4.746	0.620
15	1726.8847512	0.003728	5.914	0.679
15	1728.8985423	0.002712	5.699	0.574
15	1730.9177564	0.003050	5.338	0.572
15	1732.9457831	0.003728	5.488	0.635
55	2797.7915570	0.002373	6.054	0.473
55	2799.8073818	0.002373	5.698	0.492
55	2801.8313411	0.002712	5.446	0.524
55	2803.8525889	0.003050	4.946	0.509
55	2805.8745146	0.003050	4.932	0.598
55	2811.9335129	0.002373	4.739	0.452
55	2813.9547607	0.002373	6.407	0.523
55	2815.9705854	0.002373	5.149	0.532
55	2817.9945448	0.002034	5.922	0.470
55	2820.0185041	0.003050	4.335	0.522
55	2824.0535430	0.002373	5.459	0.521
82	3535.2968684	0.002034	6.232	0.457
82	3537.3221835	0.002034	5.192	0.467
82	3541.3599339	0.001695	5.544	0.452
82	3543.3791481	0.002712	5.376	0.626
82	3547.4189322	0.002034	6.003	0.457
82	3549.4374684	0.002373	5.687	0.463
82	3551.4573605	0.002712	5.370	0.464
82	3553.4799641	0.001695	5.643	0.464
82	3555.5018897	0.001695	5.459	0.477
82	3557.5211039	0.003389	5.416	0.695

Secondary ephemeris: {"P": 2.020574908592863, "P_error": 1.9481989092571177e-06, "Tref": 2801.8285105165855, "Tref_error": 0.000642512030683303, "reference_cycle": 538, "T0": 1714.7592096936253, "chi2": 47.113966495630685, "dof": 26, "n": 28, "oc_rms_minutes": 5.0529832295951245}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TARS sector82, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 3534.286031114435,
  "qlp": {
    "1": {
      "depth": 0.06632714692680686,
      "error": 0.0007571202785204066,
      "red_noise_beta": 1.6894819126436342,
      "n_core": 45,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06606653037798706,
      "background_template_native_sigma": -0.6750000621966564
    },
    "2": {
      "depth": 0.06508582233867669,
      "error": 0.0005167555412750362,
      "red_noise_beta": 1.1187856010307209,
      "n_core": 45,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06479885502601164,
      "background_template_native_sigma": -0.7118204707611383
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.06070331304734206,
        "error": 0.0008661550419790073,
        "red_noise_beta": 1.695868593231188,
        "n_core": 45,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.059587036892043595,
        "error": 0.0007103550833787949,
        "red_noise_beta": 1.312385491712957,
        "n_core": 45,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}

{
  "tc": 3536.306595743863,
  "qlp": {
    "1": {
      "depth": 0.06574393277427075,
      "error": 0.0007212640191589774,
      "red_noise_beta": 1.6424688802003156,
      "n_core": 45,
      "baseline_degree": 1,
```

```
      "background_regressed_depth": 0.06562384288193808,
      "background_template_native_sigma": -0.33686693829468967
    },
    "2": {
      "depth": 0.06490440449452724,
      "error": 0.0006155739692819678,
      "red_noise_beta": 1.3099347620520052,
      "n_core": 45,
      "baseline_degree": 2,
      "background_regressed_depth": 0.0647660719648585,
      "background_template_native_sigma": -0.37573087353344803
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.06181594136039163,
        "error": 0.000670178974250891,
        "red_noise_beta": 1.2978879222773796,
        "n_core": 45,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.06103010705403677,
        "error": 0.000595145186187066,
        "red_noise_beta": 1.0643042051224818,
        "n_core": 45,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}
```


Batch difference image: sector82, deficit-minus-light=6.95arcsec; deficit-minus-WCS=2.58arcsec; PSF fractional residual=0.348. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[
  {
    "source_id": "1866269770941896704",
    "ra": "314.55606991437077",
    "dec": "33.49908260749376",
    "phot_g_mean_mag": "13.844145",
    "bp_rp": "1.5537663",
    "sep": "11.982427639472563",
    "required_fraction_G_approx": 0.7205101623427412,
    "capable": true
  },
  {
    "source_id": "1866269740889217792",
    "ra": "314.5529659749947",
    "dec": "33.48117007892408",
    "phot_g_mean_mag": "13.662389",
    "bp_rp": "0.76188564",
    "sep": "58.220486692174504",
    "required_fraction_G_approx": 0.6094494797392379,
    "capable": true
```

```
  }
]
```


pixel-localization.json:


```
{
  "calibration": {
    "sigma_pixels": 0.55,
    "dx_pixels": 0.4999999999999998,
    "dy_pixels": -1.1102230246251565e-16
  },
  "free_best": {
    "chi2": 47315.681498489655,
    "pixel_x": 7.174194900952417,
    "pixel_y": 7.3476791902740874,
    "distance_from_target_arcsec": 10.707940978544986,
    "fractional_residual": 0.1715812765169125
  },
  "fixed_fits": [
    {
      "source_id": "1866269775241800320",
      "sep_arcsec": 0.0,
      "G": 11.230027,
      "chi2": 165759.92220863828,
      "amplitude": 102.62168405392656,
      "fractional_residual": 0.5725200452773824
    },
    {
      "source_id": "1866269770941896704",
      "sep_arcsec": 11.982427639472563,
      "G": 13.844145,
      "chi2": 46144.81675599125,
      "amplitude": 193.11354889769873,
      "fractional_residual": 0.17752679146606454
    },
    {
      "source_id": "1866269740889217792",
      "sep_arcsec": 58.220486692174504,
      "G": 13.662389,
      "chi2": 304064.04866461636,
      "amplitude": 1.3928228462623233,
      "fractional_residual": 0.9842874784762555
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

pixel-localization-elliptical.json:


```
{
  "calibration": {
    "sigma_pixels": [
      0.7,
      0.7
    ],
    "dx_pixels": 0.5999999999999996,
    "dy_pixels": -2.220446049250313e-16
  },
```

```
  "free_best": {
    "chi2": 26273.92467241758,
    "pixel_x": 6.974194900952417,
    "pixel_y": 7.447679190274087,
    "distance_from_target_arcsec": 14.700000000000097,
    "fractional_residual": 0.19155541550787739
  },
  "fixed_fits": [
    {
      "source_id": "1866269775241800320",
      "sep_arcsec": 0.0,
      "G": 11.230027,
      "chi2": 186693.57804486455,
      "amplitude": 56.19139872628159,
      "fractional_residual": 0.6853134991794976
    },
    {
      "source_id": "1866269770941896704",
      "sep_arcsec": 11.982427639472563,
      "G": 13.844145,
      "chi2": 36479.33196592168,
      "amplitude": 123.29976294843169,
      "fractional_residual": 0.16788037749808551
    },
    {
      "source_id": "1866269740889217792",
      "sep_arcsec": 58.220486692174504,
      "G": 13.662389,
      "chi2": 304002.94721909065,
      "amplitude": 1.3085051972427295,
      "fractional_residual": 0.984317723110925
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate axis-aligned elliptical Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization-elliptical.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

pixel-localization-positive.json:


```
{
  "calibration": {
    "sigma_pixels": [
      0.7,
      0.7
    ],
    "dx_pixels": 0.3999999999999997,
    "dy_pixels": -2.220446049250313e-16
  },
  "free_best": {
    "chi2": 26273.924672417565,
    "pixel_x": 7.174194900952417,
    "pixel_y": 7.447679190274087,
    "distance_from_target_arcsec": 10.500000000000112,
    "fractional_residual": 0.19155541550787775
  },
  "fixed_fits": [
    {
      "source_id": "1866269775241800320",
```

```
      "sep_arcsec": 0.0,
      "G": 11.230027,
      "chi2": 133137.82774795697,
      "amplitude": 80.14175383076574,
      "fractional_residual": 0.5085662599239503
    },
    {
      "source_id": "1866269770941896704",
      "sep_arcsec": 11.982427639472563,
      "G": 13.844145,
      "chi2": 26903.62283595638,
      "amplitude": 128.05529719714332,
      "fractional_residual": 0.22544469831837582
    },
    {
      "source_id": "1866269740889217792",
      "sep_arcsec": 58.220486692174504,
      "G": 13.662389,
      "chi2": 303719.1677772248,
      "amplitude": 2.1538533515932112,
      "fractional_residual": 0.9843339355969052
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate axis-aligned elliptical Gaussian PSF
calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level
systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are
white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization-positive.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

pixel-localization-fluxratio.json:


```
{
  "calibration": {
    "sigma_pixels": [
      0.7,
      0.7
    ],
    "dx_pixels": -2.220446049250313e-16,
    "dy_pixels": -0.20000000000000018
  },
  "free_best": {
    "chi2": 26273.924672417525,
    "pixel_x": 7.574194900952415,
    "pixel_y": 7.647679190274086,
    "distance_from_target_arcsec": 4.695742752749475,
    "fractional_residual": 0.1915554155078779
  },
  "fixed_fits": [
    {
      "source_id": "1866269775241800320",
      "sep_arcsec": 0.0,
      "G": 11.230027,
      "chi2": 42437.805542130925,
      "amplitude": 119.57535410526069,
      "fractional_residual": 0.21969839118602286
    },
    {
      "source_id": "1866269770941896704",
      "sep_arcsec": 11.982427639472563,
      "G": 13.844145,
```

```
      "chi2": 112342.19144314181,
      "amplitude": 89.33836039100534,
      "fractional_residual": 0.5790939719413017
    },
    {
      "source_id": "1866269740889217792",
      "sep_arcsec": 58.220486692174504,
      "G": 13.662389,
      "chi2": 301198.37961262884,
      "amplitude": 5.549090714620524,
      "fractional_residual": 0.9837094551282867
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate axis-aligned elliptical Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization-fluxratio.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 1866269775241800320", "main_id": "TYC 2692-150-1", "otype": "SB*", "ra": "314.55257938221996", "dec": "33.4973524698"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Obtain a calibrated empirical TESS PRF with color-dependent neighbor fluxes or ground-based resolved eclipse photometry of the target and12arcsec neighbor. Next fitted primary after2026-10-30: BTJD4344.53469, conditional linear-ephemeris1sigma=0.00043d. Next secondary template phase: BTJD4343.52484; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R059. TIC 415751695

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 415751695: CONFIRMED

Twenty-one primaries and twenty secondary events plus independent TARS recovery establish a repeated EB. The old PSF residual0.931 is addressed by fresh regression: free deficit centroid2.97arcsec from target with fractional residual0.177 and no G<15 capable neighbor. No precise parameter inference follows from this approximate PSF model.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=4.995996617 +/- 0.000000890 d; Tref=3038.5788104 +/- 0.0001250 BTJD at cycle298; T0=1549.7718185. 21 fitted primary events, O-C RMS=0.662 min; chi2/dof=17.58/19. Shape T14=4.017h, profile range=[0.16375862068965513, 0.17099999999999999], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
9	1549.7717914	0.002273	59.973	0.402	-0.039	1
9	1554.7677764	0.002690	59.475	0.451	-0.056	3
9	1564.7597464	0.002126	60.189	0.432	-0.089	0
36	2284.1835911	0.000502	64.402	0.415	0.389	0
36	2289.1785718	0.000502	63.866	0.381	-1.074	1
36	2299.1715460	0.000502	64.795	0.417	0.338	3
36	2304.1675310	0.000502	63.828	0.445	0.322	2
63	3018.5953908	0.000502	62.993	0.395	0.816	0
63	3023.5903715	0.000502	63.181	0.357	-0.647	0
63	3028.5863565	0.000502	63.239	0.326	-0.663	0
63	3033.5833457	0.000502	62.276	0.478	0.766	2
63	3038.5783264	0.000502	62.947	0.387	-0.697	1
90	3748.0102012	0.000502	61.679	0.375	-0.186	0
90	3753.0061862	0.000502	61.590	0.362	-0.202	0
90	3758.0031754	0.000502	61.390	0.341	1.227	0
90	3762.9991604	0.000502	61.640	0.384	1.210	0
90	3767.9941411	0.000502	60.656	0.593	-0.253	2
90	3772.9901261	0.000502	61.139	0.486	-0.269	2
100	4077.7462138	0.000502	55.706	0.320	0.154	0

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
100	4082.7421987	0.000502	56.014	0.346	0.137	0
100	4097.7291494	0.000502	55.940	0.296	-1.359	0

Primary median depth=61.640ppt; maximum absolute background-regression fake depth=0.744ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9995584. Primary assigned conjunctionA; period discrepancy=0.544sigma. ConjunctionA at Gaia epoch: offset=-0.050309d, Gaia-only sigma=0.050469d; including photometric extrapolation sigma=0.050471d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=-2.485075d, Gaia-only sigma=0.043945d; including photometric extrapolation sigma=0.043948d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.24,
    "m2_min": 0.3253200269770704,
    "f_mass": 0.014051610905027786,
    "a_rsun": 14.279420960348848,
    "r1_rsun": 1.28734,
    "r2_approx_rsun": 0.4072383223929631,
    "central_T14_hours": {
      "A": 4.534153546644391,
      "B": 4.5223599294225565
    },
    "observed_T14_hours": 4.017103448275861
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 112.7818313174603,
    "dof": 121,
    "k": 0.27499999999999997,
    "b_A": 0.375,
    "e_sin_omega": -5.551115123125783e-17,
```

```
    "primary_scale": 0.7495084330789565,
    "secondary_scale": 0.003393019985787365
  },
  "profile_interval_approx_95": {
    "k": [
      0.27499999999999997,
      0.27499999999999997
    ],
    "b_A": [
      0.375,
      0.375
    ],
    "e_sin_omega": [
      -5.551115123125783e-17,
      -5.551115123125783e-17
    ]
  },
  "error_inflation": 1,
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.500364; pooled depth=3.447ppt, empirical error=0.063ppt; 20/20 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.48708226485372597,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.4716521118126703,
    0.48872067074725734,
    0.4992057440662677
  ],
  "gaia_relative_phase_sigma": 0.015905871809338275,
  "gaia_e_cos_omega_quantiles": [
    -0.04452333067583703,
    -0.01771165573260186,
    -0.0012476113066753707
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.01826477622258167,
    0.0016564655976043854,
    0.034809391413405985
  ],
  "photometric_phase": 0.5003640716992194,
  "photometric_e_cos_omega_first_order": 0.0005718824878237635,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    -5.551115123125783e-17,
    -5.551115123125783e-17
  ],
```

```
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
9	1552.2674449	0.004191	3.171	0.317
9	1562.2646767	0.003143	3.410	0.255
9	1567.2606733	0.003667	2.799	0.295
36	2286.6820907	0.003143	3.400	0.251
36	2291.6833260	0.003667	3.933	0.339
36	2296.6782749	0.002619	3.924	0.261
36	2301.6679851	0.003667	3.407	0.305
63	3016.0934059	0.003667	3.583	0.334
63	3021.0946412	0.002619	3.899	0.312
63	3026.0822560	0.002095	3.444	0.270
63	3031.0866344	0.002095	3.370	0.274
63	3036.0847265	0.003143	3.535	0.301
90	3750.5101473	0.003667	3.214	0.292
90	3755.5103348	0.002619	3.066	0.285
90	3765.5002326	0.003667	3.178	0.291
90	3770.4878474	0.002619	3.644	0.263
100	4075.2499274	0.002095	3.476	0.216
100	4080.2480194	0.002095	3.337	0.232
100	4090.2326786	0.001572	3.733	0.225
100	4095.2307707	0.002095	3.378	0.202

Secondary ephemeris: {"P": 4.995992343898419, "P_error": 5.284286784174196e-06, "Tref": 3021.093453886058, "Tref_error": 0.0009118139755815085, "reference_cycle": 294, "T0": 1552.2717047799226, "chi2": 38.79656292702458, "dof": 18, "n": 20, "oc_rms_minutes": 5.09305636646804}. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TARS sector90, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 3748.0102012187626,
  "qlp": {
    "1": {
      "depth": 0.06172720560114846,
      "error": 0.0003529641837227917,
      "red_noise_beta": 1.2464413978554645,
      "n_core": 72,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06220986786933617,
      "background_template_native_sigma": 0.5815925001153252
    },
    "2": {
      "depth": 0.06180258851366541,
      "error": 0.0003938001801668584 6,
```

```
      "red_noise_beta": 1.2447585271714376,
      "n_core": 72,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06207641879125174,
      "background_template_native_sigma": 0.3187601725086757
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.0635751270335014,
        "error": 0.0006117803328877753,
        "red_noise_beta": 1.4043809507897536,
        "n_core": 72,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.06385354232253854,
        "error": 0.0006685582578828625,
        "red_noise_beta": 1.375909782910015,
        "n_core": 72,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}

{
  "tc": 3753.006186192379,
  "qlp": {
    "1": {
      "depth": 0.061444738267793585,
      "error": 0.00034333138225790095,
      "red_noise_beta": 1.2469308018122431,
      "n_core": 72,
      "baseline_degree": 1,
      "background_regressed_depth": 0.061669624999183166,
      "background_template_native_sigma": 0.4121679953241458
    },
    "2": {
      "depth": 0.06159852707260317,
      "error": 0.000378557209084772,
      "red_noise_beta": 1.2320311726047433,
      "n_core": 72,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06180240659779209,
      "background_template_native_sigma": 0.3750345034275165
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.05993002954599291,
        "error": 0.0007463446653183674,
        "red_noise_beta": 1.4695507929668603,
        "n_core": 72,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
```

```
      },
      "2": {
        "depth": 0.06078209752046338,
        "error": 0.0007718792333783774,
        "red_noise_beta": 1.378342899171527,
        "n_core": 72,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}
```


Batch difference image: sector90, deficit-minus-light=3.54arcsec; deficit-minus-WCS=1.47arcsec; PSF fractional residual=0.931. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[]
```


pixel-localization.json:


```
{
  "calibration": {
    "sigma_pixels": 0.6500000000000001,
    "dx_pixels": -0.10000000000000009,
    "dy_pixels": 0.09999999999999987
  },
  "free_best": {
    "chi2": 25838.78000245633,
    "pixel_x": 7.241803277952407,
    "pixel_y": 7.167631316672429,
    "distance_from_target_arcsec": 2.9698484809834893,
    "fractional_residual": 0.17674795147329955
  },
  "fixed_fits": [
    {
      "source_id": "3549629597848975872",
      "sep_arcsec": 0.0,
      "G": 11.403031,
      "chi2": 28800.380434301325,
      "amplitude": 115.46216951474243,
      "fractional_residual": 0.19390534966841602
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 3549629597848975872", "main_id": "TYC 6082-305-1", "otype": "SB*", "ra": "164.63570637693", "dec": "-22.10958257191"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[];

current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4347.52992, conditional linear-ephemeris1sigma=0.00026d. Next secondary template phase: BTJD4345.03374; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R060. TIC 335862015

physical_verdict: CONFIRMED

novelty_status: LIMITED_PRIOR_ART_COVERAGE

completion: COMPLETED_WITH_ARCHIVE_LIMITATION

next_check: Retry unavailable catalogs; refine joint orbit and stellar characterization.

Archived source: sanitized report text.

TIC 335862015: CONFIRMED

Three deep primary and two secondary events in sector91, recovered by TARS and localized to target against a capable49.5arcsec neighbor in all three events, establish a physical EB. The short baseline, poor fixed-TIC geometry and discrepant Gaia-median secondary phase require future orbit/stellar characterization; they do not erase the localized eclipses.

Full review: COMPLETED_WITH_ARCHIVE_LIMITATION. Physical verdict is separate from novelty. Novelty: LIMITED_PRIOR_ART_COVERAGE; no prior target EB was established by the available current sources. New nine-catalog passes to both IDIA and Harvard timed out and are not negative matches.

Raw SAP and event measurements

Primary P=6.577232276 +/- 0.000439950 d; Tref=3790.2187995 +/- 0.0005680 BTJD at cycle2; T0=3777.0643350. 3 fitted primary events, O-C RMS=0.422 min; chi2/dof=0.29/1. Shape T14=7.603h, profile range=[0.30892499999999995, 0.324675], ingress fraction=0.25; T12=fr*T14 and nominal T23=T14-2*T12. Trapezoid flat time is not totality for grazing systems.

sector	mid BTJD	error d	depth ppt	error ppt	O-C min	flagged core
91	3777.0641992	0.000950	64.194	0.609	-0.196	0
91	3790.2192068	0.000950	64.158	0.556	0.587	0
91	3796.7957603	0.000950	64.514	0.565	-0.391	0

Primary median depth=64.194ppt; maximum absolute background-regression fake depth=0.017ppt. Background native units were not treated as normalized flux. 0 event timing fits hit a scan edge. Quality-zero fits use exposure integration, local quadratic baselines, red-noise beta and grid floors; flagged-core counts and centroid/background regressions are retained in fit-results.json. Source variability and shape errors may exceed these conditional errors.

Blind rolling-median census covers all available sectors without the old post-gap trim. No off-eclipse census dip exceeds30%of the primary depth outside0.08phase of either fitted eclipse kind. Smaller candidates, gaps, sector edges and all threshold details remain in fit-results.json; this is a diagnostic census, not a calibrated false-positive probability.

Gaia and physical feasibility

Own-source SB1 covariance verified: bit_index=127,15 correlations in P,gamma,K1,e,omega,Tp order; omega-Tp correlation=0.9971951. Primary assigned conjunctionA; period discrepancy=-0.676sigma. ConjunctionA at Gaia epoch: offset=0.403832d, Gaia-only sigma=0.067549d; including photometric extrapolation sigma=0.237324d. Full epoch-by-epoch comparisons are retained in orbit-gates.json. ConjunctionB at Gaia epoch: offset=-3.120511d, Gaia-only sigma=0.101962d; including photometric extrapolation sigma=0.249311d. Full epoch-by-epoch comparisons are retained in orbit-gates.json.

Mass-function/assumed-stellar-radius checks (not measured binary parameters):


```
[
  {
    "m1": 1.03,
    "m2_min": 0.5133488316115586,
    "f_mass": 0.05679492608907814,
    "a_rsun": 17.073422623562458,
    "r1_rsun": 2.27928,
    "r2_approx_rsun": 0.5865837968306765,
    "central_T14_hours": {
      "A": 8.28802641296841,
      "B": 8.557267064253015
    },
    "observed_T14_hours": 7.603199999999999
  }
]
```


Limb-darkened two-sphere feasibility permits free primary/secondary flux scales and fixed TIC stellar dimensions. It does not identify the foreground component from eclipse depth alone. Using Gaia primary spectroscopic omega and positive RV away from the observer, at A (nu+omega=90deg) the RV primary is behind and the companion is foreground; at B (270deg) the primary is foreground. The deeper eclipse is assigned A here; luminosity and component labeling remain conditional. Minimum RV companion mass and eclipse luminosity/shape jointly allow a stellar companion.


```
{
  "status": "conditional shape feasibility",
  "best": {
    "chi2": 956.7807932435624,
    "dof": 121,
    "k": 0.24999999999999997,
    "b_A": 0.23500000000000001,
    "e_sin_omega": 0.04999999999999993,
    "primary_scale": 0.9129607255190911,
    "secondary_scale": 0.008347226128724176
  },
  "profile_interval_approx_95": {
    "k": [
      0.24999999999999997,
      0.24999999999999997
    ],
    "b_A": [
      0.2,
      0.23500000000000001
    ],
    "e_sin_omega": [
      0.04999999999999993,
      0.04999999999999993
    ]
  },
  "error_inflation": 7.907279283004647,
```

```
  "assumptions": "Circular projected chord with eccentric speed factors; fixed TIC R1/M1, u1=.4 u2=.6, free depth scales per kind, event-scatter bin errors and 0.15ppt floor. Contact shape and dilution degeneracies persist. No claim of precise e sin omega.",
  "primary_profile_n": 64,
  "secondary_profile_n": 64
}
```


Secondary evidence

Optimized secondary phase=0.500277; pooled depth=8.230ppt, empirical error=0.157ppt; 2/2 individually>3sigma. This scan optimizes phase and does not provide a calibrated phase error or global significance. Repeat events, independent reduction, and off-phase controls are more informative than its raw peak S/N.


```
{
  "bit_index": 127,
  "gaia_primary": "A",
  "gaia_relative_phase_nominal": 0.53597893199964,
  "gaia_relative_phase_quantiles_16_50_84": [
    0.5136469640922227,
    0.5354260178365298,
    0.562769152758785
  ],
  "gaia_relative_phase_sigma": 0.023913737021386976,
  "gaia_e_cos_omega_quantiles": [
    0.021436106625961644,
    0.05563908995520599,
    0.09866611237134963
  ],
  "gaia_e_sin_omega_quantiles": [
    -0.008708233389776276,
    0.01219100172992036,
    0.04755889384437401
  ],
  "photometric_phase": null,
  "photometric_e_cos_omega_first_order": null,
  "photometric_note": "First-order small-e approximation; fitted profile phase is optimized and conditional on eclipse shape, with no calibrated uncertainty. Omitted without three individual secondary detections.",
  "geometry_status": "conditional shape feasibility",
  "geometry_e_sin_omega_conditional_range": [
    0.04999999999999993,
    0.04999999999999993
  ],
  "geometry_note": "Coarse conditional feasibility range, fixed TIC properties and simplified limb-darkened shape; not a calibrated posterior. Check chi2 and grid boundaries in geometry.json."
}
```


sector	secondary mid BTJD	error d	depth ppt	error ppt
91	3780.3529676	0.000903	8.407	0.241
91	3793.5074322	0.000903	8.098	0.208

Secondary ephemeris: null. Two-event periods and optimized shapes are not precise independent orbit constraints. Covered/uncovered/flagged secondary windows are enumerated individually in orbit-gates.json secondary_windows, and null-secondary.json when applicable.

Independence and source attribution

TARS sector91, exact FITS identity validated=True. Same TESS pixels through a different reduction, not another observatory. Exact product/time coverage/URLs/SHA256 are in independent manifests; every measured event and degree1/2 background/baseline comparison is in independent-summary.json (alternate summary for345589621).


```
{
  "tc": 3777.0641992,
  "qlp": {
    "1": {
      "depth": 0.06546150571224707,
      "error": 0.0006033190373236466,
      "red_noise_beta": 2.432778713279399,
      "n_core": 137,
      "baseline_degree": 1,
      "background_regressed_depth": 0.06547532501264686,
      "background_template_native_sigma": 0.3822669075696377
    },
    "2": {
      "depth": 0.06462614598179718,
      "error": 0.0006250413303067231,
      "red_noise_beta": 2.322741905576321,
      "n_core": 137,
      "baseline_degree": 2,
      "background_regressed_depth": 0.06461390899878607,
      "background_template_native_sigma": -0.06196644739084536
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.07345961536855511,
        "error": 0.0006693451269136685,
        "red_noise_beta": 1.600139362680759,
        "n_core": 137,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.07350565707457783,
        "error": 0.0007473369010430314,
        "red_noise_beta": 1.5994856115880274,
        "n_core": 137,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}

{
  "tc": 3790.21920683697,
  "qlp": {
    "1": {
      "depth": 0.06477067923744757,
      "error": 0.0005382538618033496,
      "red_noise_beta": 2.193099247993015,
      "n_core": 136,
      "baseline_degree": 1,
```

```
      "background_regressed_depth": 0.06476844559500948,
      "background_template_native_sigma": -0.02349672625315945
    },
    "2": {
      "depth": 0.06409101255483318,
      "error": 0.0005622363902128375,
      "red_noise_beta": 2.0834206013445185,
      "n_core": 136,
      "baseline_degree": 2,
      "background_regressed_depth": 0.064092635290639,
      "background_template_native_sigma": 0.01983459784663881
    }
  },
  "independent": {
    "FLUX": {
      "1": {
        "depth": 0.06733148627972271,
        "error": 0.0007295288358141654,
        "red_noise_beta": 1.7134792410314077,
        "n_core": 136,
        "baseline_degree": 1,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      },
      "2": {
        "depth": 0.06696502653416919,
        "error": 0.0008114734167832886,
        "red_noise_beta": 1.6993420484511348,
        "n_core": 136,
        "baseline_degree": 2,
        "background_regressed_depth": null,
        "background_template_native_sigma": null
      }
    }
  }
}
```


Batch difference image: sector91, deficit-minus-light=5.5arcsec; deficit-minus-WCS=3.09arcsec; PSF fractional residual=0.308. Duplicate legacy/new-schema rows are a single observation. Relative light/deficit offsets receive more weight than uncalibrated WCS offsets.

Neighbor dilution within60arcsec, G<15 (Gaia G approximates TESS; not a color-corrected flux ratio):


```
[
  {
    "source_id": "6310152241747719552",
    "ra": "224.22186487079426",
    "dec": "-15.146044956835752",
    "phot_g_mean_mag": "12.536925",
    "bp_rp": "0.82973576",
    "sep": "49.53926082922907",
    "required_fraction_G_approx": 0.18294415583788728,
    "capable": true
  }
]
```


pixel-localization.json:


```
{
  "calibration": {
    "sigma_pixels": 0.7500000000000001,
    "dx_pixels": -0.10000000000000009,
    "dy_pixels": -1.1102230246251565e-16
```

```
  },
  "free_best": {
    "chi2": 16231.41189742049,
    "pixel_x": 7.205778292784393,
    "pixel_y": 7.461464210057608,
    "distance_from_target_arcsec": 2.1102042644709938e-13,
    "fractional_residual": 0.12311846200098397
  },
  "fixed_fits": [
    {
      "source_id": "6310152211684175104",
      "sep_arcsec": 0.0,
      "G": 11.399861,
      "chi2": 16231.411897420277,
      "amplitude": 107.5616393356673,
      "fractional_residual": 0.12311846200098274
    },
    {
      "source_id": "6310152241747719552",
      "sep_arcsec": 49.53926082922907,
      "G": 12.536925,
      "chi2": 260189.39519202718,
      "amplitude": 5.9570387695799445,
      "fractional_residual": 0.9881512892252862
    }
  ],
  "caution": "Same TESS pixels as QLP, independent aperture reduction. Approximate circular Gaussian PSF calibrated to multi-star out-of-eclipse image; formal chi2 excludes PSF mismatch, correlated pixels/time and sector-level systematics. No formal significance from delta-chi2; compare event repeatability and residuals. Aperture depth errors are white-noise conditional only; degree comparison probes baseline sensitivity."
}
```


Individual event source comparisons are in pixel-localization.json; approximate PSF chi-square excludes model/correlated-noise errors and is not formal localization significance.

Current archives and novelty

SIMBAD exact Gaia-ID: [{"id": "Gaia DR3 6310152211684175104", "main_id": "TYC 6168-1001-1", "otype": "SB*", "ra": "224.21401891473", "dec": "-15.157451048150003"}]. *This SB* classification establishes spectroscopy, not a prior eclipse discovery. Gaia vari_eclipsing_binary rows=0; vari_summary=[]; current ExoFOP={"toi_count": 0, "ctoi_count": 0}. Fresh shared VSX60arcsec result: {"B_vsx_vsx": []}. ROT is not prior eclipse classification. A cone neighbor must not be assigned to target.

Nine-catalog IDIA and Harvard cone requests each timed out (idia-receipts.json and idia-mirror-receipts.json). Consequently ASAS-SN, SB9, WDS, Kostov2/3/4, IJspeert and Prsa current per-object coverage is unavailable, not empty. Batched exact-name web queries on2026-10-30 returned no relevant source and do not prove novelty. Shared source receipts preserve current Gaia/VSX/SIMBAD/ExoFOP query provenance. The original paper/source records were checked live and remain unchanged.

Follow-up and predictions

Refine stellar radii, light ratio and RV/eclipsing joint orbit with resolved photometry or spectroscopy; retry unavailable catalogs before any novelty claim. Next fitted primary after2026-10-30: BTJD4349.28354, conditional linear-ephemeris1sigma=0.03740d. Next secondary template phase:

BTJD4345.99675; phase/systematic uncertainty is uncalibrated, so do not use it as a precise scheduling window. For undetected secondary kinds use the Gaia covariance windows, not this optimized scan prediction.

Reviewed by fresh Sol lead G with fresh GPT-6.1 Terra numerical helper; measurements and script assumptions were independently inspected.

R061. TIC 211231319

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 211231319: CONFIRMED

Repeated primary and secondary eclipses, independent SPOC recovery, own Gaia timing and on-target deficit establish a target EB; no bright capable neighbor.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 11 accepted primary fits; median depth62.881ppt; T14=5.0599h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 634 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 240 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 764 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 206 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "211231319",
    "sector": 92,
    "n_events": 4,
    "deficit_minus_target_arcsec": 1.61,
    "deficit_minus_light_arcsec": 0.53,
    "peak_on_target": true,
    "depth_ap3": 0.06209,
    "psf_frac_resid": 0.15,
    "n_neighbours_60": 19,
    "n_possible": 0,
    "brightest_nb": {
      "G": 15.675792,
      "sep": 43.3
    }
  }
]
```


Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 6749770323777264128",
      "main_id": "TYC 7440-1461-1",
      "otype": "SB*",
      "ra": "299.51092397216",
      "dec": "-31.327449139220004"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
```

```
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\211231319.tsv",
      "sha256": "561d9171d5a6318a11db31808ba3c1bba347627871e28b5e0c40e8618984d349"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\211231319.json",
      "sha256": "55e14f91c83a41dc54554154f4152af65ccf87d950343308a49dded6f20bd21c"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4348.51433; other conjunction~4345.60159. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Repeated primary and secondary eclipses, independent SPOC recovery, own Gaia timing and on-target deficit establish a target EB; no bright capable neighbor.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 62.88077035150803, "offphase_census_triggers": 5, "max_offphase_census_ppt": 8.896080455776945, "minimum_bg_regressed_depth_ratio": 0.9997180259837454}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R062. TIC 120522352

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 120522352: CONFIRMED

Eleven primary and thirteen secondary events, independent TARS recovery and on-target deficit establish a target EB.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 11 accepted primary fits; median depth63.583ppt; T14=3.6695h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 734 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 375 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 829 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 146 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "120522352",
    "sector": 96,
    "n_events": 3,
    "deficit_minus_target_arcsec": 2.18,
    "deficit_minus_light_arcsec": 0.29,
    "peak_on_target": true,
    "depth_ap3": 0.07204,
    "psf_frac_resid": 0.033,
    "n_neighbours_60": 2,
    "n_possible": 0,
    "brightest_nb": {
      "G": 20.015728,
      "sep": 28.3
    }
  }
]
```


Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 4996621435834932864",
      "main_id": "TYC 7523-58-1",
      "otype": "SB*",
      "ra": "2.69744682385",
      "dec": "-39.91993531301"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
```

```
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\120522352.tsv",
      "sha256": "e00930082a6aa6d763cf4e20cd59824376c8b1b88b6c9f2097aa3298f88fee39"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\120522352.json",
      "sha256": "e8c7349b5705a82298cfed6377cc45639951e17985e667773fb10ef501636a68"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4349.67772; other conjunction~4345.49504. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Eleven primary and thirteen secondary events, independent TARS recovery and on-target deficit establish a target EB.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 63.582568688096515, "offphase_census_triggers": 13, "max_offphase_census_ppt": 8.898341450854996, "minimum_bg_regressed_depth_ratio": 0.9928611228064197}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R063. TIC 192198579

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 192198579: CONFIRMED

Four primary and six secondary events plus independent TESS-SPOC support an EB. Own pixel fits favor target over the13.26arcsec neighbor in both events (target residual0.146 vs neighbor0.582). High geometry residual limits dimensions, not eclipse reality.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 4 accepted primary fits; median depth63.689ppt; T14=6.0257h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 363 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 123 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 528 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 141 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "977665638971641088",
    "ra": "105.67151458471844",
    "dec": "46.82832953686493",
    "phot_g_mean_mag": "12.131536",
    "bp_rp": "0.8744564",
    "sep": "13.263741925334552",
    "required_fraction_G_approx": 0.3883618714215827,
    "capable": true
  },
  {
    "source_id": "977667086376402688",
    "ra": "105.69205506549349",
    "dec": "46.83663159994198",
    "phot_g_mean_mag": "13.530179",
    "bp_rp": "0.71845627",
    "sep": "47.00996217611928",
    "required_fraction_G_approx": 1.4082954793008624,
    "capable": false
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "192198579",
    "sector": 47,
    "n_events": 2,
    "deficit_minus_light_arcsec": 1.13,
    "peak_on_target": true,
    "depth_ap3": 0.05272,
    "psf_frac_resid": 0.112,
    "n_neighbours_60": 14,
    "n_possible": 1,
    "brightest_nb": {
      "G": 12.131536,
      "sep": 13.3
    }
  },
  {
    "tic": "192198579",
    "sector": 47,
    "n_events": 2,
    "deficit_minus_target_arcsec": 4.37,
    "deficit_minus_light_arcsec": 1.13,
    "peak_on_target": true,
    "depth_ap3": 0.05272,
```

```
    "psf_frac_resid": 0.112,
    "n_neighbours_60": 14,
    "n_possible": 1,
    "brightest_nb": {
      "G": 12.131536,
      "sep": 13.3
    }
  }
]
```


pixel-localization.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 7: 131 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 977665643267391744",
      "main_id": "TYC 3391-623-1",
      "otype": "SB*",
      "ra": "105.67464026268",
      "dec": "46.83149278762"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\192198579.tsv",
      "sha256": "3b27bdfd3c2ffa34985ec536ad36957b97b0b7c78f7f1516e59408d1a928404e"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\192198579.json",
      "sha256": "6298d6fb593ac5f23f86a048c73a1e335aa59943af92a099a90bdfc627a46ef7"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4351.96211; other conjunction~4346.39040. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Four primary and six secondary events plus independent TESS-SPOC support an EB. Own pixel fits favor target over the13.26arcsec neighbor in both events (target residual0.146 vs neighbor0.582). High geometry residual limits dimensions, not eclipse reality.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 63.688964104405805, "offphase_census_triggers": 0, "max_offphase_census_ppt": 0, "minimum_bg_regressed_depth_ratio": 1.0012584955795731}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R064. TIC 1989681813

physical_verdict: PLAUSIBLE

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Obtain spatially resolved time-series photometry across a later narrow deficit and both Gaia conjunctions; use empirical calibrated TESS PRFs in a second sector to distinguish astrophysical deficit from nonstationary spatial contamination.

Archived source: sanitized report text.

TIC 1989681813: PLAUSIBLE

Later sectors retain a narrow recurring deficit after smooth tidal/harmonic removal, but broad/nonstationary earlier waveforms and complex deficit image (target residual0.991, free location22arcsec away) prevent secure eclipse/source interpretation. PLAUSIBLE after targeted broad-shape and tide tests.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 9 accepted primary fits; median depth71.178ppt; T14=14.4072h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 483 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 285 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 640 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Other-conjunction windows and width/phase-grid limits follow. These are pointwise conditional template limits, not a uniform simultaneous confidence bound; coverage varies.

Ancillary log block 4: 10348 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Broad waveform follow-up: sector-specific tide+harmonic baseline versus added broad/narrow trapezoid. RSS gains are descriptive; no calibrated model-selection probability. This supersedes interpreting the common grid-edge shape as a physical duration.


```
[
  {
    "sector": 27,
    "n": 3292,
    "baseline_rss": 55.3656882876341,
    "best_rss": 54.61816948258974,
    "T14": 1.5,
    "shift": 0.050000000000000044,
    "depth": 0.17987944391917976,
    "fractional_rss_improvement": 0.013501481299408242,
    "note": "sector-level 3 harmonics plus Gaia-eccentric heartbeat basis, quadratic drift; positive eclipse scanned in width
and shift; no calibrated significance; full sector, nonstationary modulation possible"
  },
  {
    "sector": 67,
    "n": 9306,
    "baseline_rss": 777.2255326552257,
    "best_rss": 773.389123794061,
    "T14": 0.8750000000000001,
    "shift": 0.0,
    "depth": 0.1804311176060867,
    "fractional_rss_improvement": 0.0049360304055612435,
    "note": "sector-level 3 harmonics plus Gaia-eccentric heartbeat basis, quadratic drift; positive eclipse scanned in width
and shift; no calibrated significance; full sector, nonstationary modulation possible"
  },
  {
    "sector": 94,
    "n": 8948,
    "baseline_rss": 0.65471512261252,
    "best_rss": 0.21796896153854636,
    "T14": 0.42500000000000004,
    "shift": -0.024999999999999994,
    "depth": 0.07316820724512163,
    "fractional_rss_improvement": 0.667078162684281,
    "note": "sector-level 3 harmonics plus Gaia-eccentric heartbeat basis, quadratic drift; positive eclipse scanned in width
and shift; no calibrated significance; full sector, nonstationary modulation possible"
  },
  {
    "sector": 104,
    "n": 8635,
    "baseline_rss": 0.7084769219871602,
    "best_rss": 0.21679224612175857,
    "T14": 0.42500000000000004,
    "shift": -0.024999999999999994,
    "depth": 0.08255713898910502,
    "fractional_rss_improvement": 0.6940023882306677,
    "note": "sector-level 3 harmonics plus Gaia-eccentric heartbeat basis, quadratic drift; positive eclipse scanned in width
and shift; no calibrated significance; full sector, nonstationary modulation possible"
```

```
  }
]
```


Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 6: 174 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "1989681813",
    "sector": 67,
    "n_events": 2,
    "deficit_minus_light_arcsec": 2.27,
    "peak_on_target": false,
    "depth_ap3": 0.07241,
    "psf_frac_resid": 0.987,
    "n_neighbours_60": 13,
    "n_possible": 0,
    "brightest_nb": {
      "G": 16.130404,
      "sep": 55.6
    }
  },
  {
    "tic": "1989681813",
    "sector": 67,
    "n_events": 2,
    "deficit_minus_target_arcsec": 3.12,
    "deficit_minus_light_arcsec": 2.27,
    "peak_on_target": false,
    "depth_ap3": 0.07241,
    "psf_frac_resid": 0.987,
    "n_neighbours_60": 13,
    "n_possible": 0,
    "brightest_nb": {
      "G": 16.130404,
      "sep": 55.6
    }
  }
]
```


pixel-localization.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 9: 127 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 6477619446187396352",
      "main_id": "TYC 8420-1363-1",
      "otype": "SB*",
      "ra": "312.87305676785",
      "dec": "-51.57112232347"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\1989681813.tsv",
      "sha256": "381e4d567fcbfd92a5c1cd666656949ec77158c5c9ca84aab86ed58735dc488c"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\1989681813.json",
      "sha256": "31dfc5aa72383e791be4545bf45d6f6f839631fa7b38c0cec92c4e064f320399"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4350.48570; other conjunction~4355.24293. Photometric extrapolations are scheduling estimates, conditional on stationary orbit; for1989681813 neither is a reliable physical eclipse ephemeris. For null secondary use Gaia phase uncertainty, not the single date.

Obtain spatially resolved time-series photometry across a later narrow deficit and both Gaia conjunctions; use empirical calibrated TESS PRFs in a second sector to distinguish astrophysical deficit from nonstationary spatial contamination.

Final adjudication: **PLAUSIBLE**. Later sectors retain a narrow recurring deficit after smooth tidal/harmonic removal, but broad/nonstationary earlier waveforms and complex deficit image (target residual0.991, free location22arcsec away) prevent secure eclipse/source interpretation. PLAUSIBLE after targeted broad-shape and tide tests.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 71.17807741090103, "offphase_census_triggers": 21, "max_offphase_census_ppt": 665.7661206505501, "minimum_bg_regressed_depth_ratio": 0.8106702617464214}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.1989681813 has large off-phase structure and remains plausible.

R065. TIC 250742811

physical_verdict: PLAUSIBLE

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Resolve the3.70arcsec pair through an eclipse using ground-based photometry, or fit empirical PRFs with externally calibrated WCS across multiple sectors; establish which member loses flux.

Archived source: sanitized report text.

TIC 250742811: PLAUSIBLE

The two-kind EB signal is secure, but target versus nearly equally bright3.70arcsec companion remains ambiguous. Circular PSF favors target; expanded calibration reverses preference and event rankings. RV agreement supports target but cannot resolve the contradictory localization.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 7 accepted primary fits; median depth64.993ppt; T14=6.5470h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 481 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 177 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 631 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 146 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "358654370790026496",
    "ra": "33.42414429486065",
    "dec": "50.42428787660143",
    "phot_g_mean_mag": "11.393224",
    "bp_rp": "0.82036114",
    "sep": "3.698370894837177",
    "required_fraction_G_approx": 0.0740187404938573,
    "capable": true
  },
  {
    "source_id": "358654198991334784",
    "ra": "33.44908630009805",
    "dec": "50.42142424462216",
    "phot_g_mean_mag": "11.251107",
    "bp_rp": "1.3166361",
    "sep": "54.51572619155012",
    "required_fraction_G_approx": 0.06493739921285417,
    "capable": true
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "250742811",
    "sector": 18,
    "n_events": 3,
    "deficit_minus_target_arcsec": 3.52,
    "deficit_minus_light_arcsec": 13.96,
    "peak_on_target": true,
    "depth_ap3": 0.03181,
    "psf_frac_resid": 0.556,
    "n_neighbours_60": 25,
    "n_possible": 2,
    "brightest_nb": {
      "G": 11.251107,
      "sep": 54.5
    }
  }
]
```

pixel-localization.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 7: 157 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

pixel-localization-elliptical.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 8: 160 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 358654375087124736",
      "main_id": "BD+49   589A",
      "otype": "SB*",
      "ra": "33.42547742392",
      "dec": "50.423916165079994"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\250742811.tsv",
      "sha256": "d8e07b8f2ee95adf57cfc3632e20bdf6b6367b77111db502ab6792b4988b58e5"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\250742811.json",
      "sha256": "f9ffa3988b02a9a268210ff1e080e88d719bb2bf9eee141e7079f50052145ed7"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4347.44576; other conjunction~4351.57159. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Resolve the3.70arcsec pair through an eclipse using ground-based photometry, or fit empirical PRFs with externally calibrated WCS across multiple sectors; establish which member loses flux.

Final adjudication: **PLAUSIBLE**. The two-kind EB signal is secure, but target versus nearly equally bright3.70arcsec companion remains ambiguous. Circular PSF favors target; expanded calibration reverses preference and event rankings. RV agreement supports target but cannot resolve the contradictory localization.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 64.99268676413543, "offphase_census_triggers": 2, "max_offphase_census_ppt": 9.971371674406647, "minimum_bg_regressed_depth_ratio": 0.9998030263893818}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R066. TIC 157000246

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 157000246: CONFIRMED

Eight primary and nine secondary events, independent SPOC and target-localized deficit establish an EB. The real secondary atphase0.3408 is recovered repeatedly despite the Gaia-median fixed-phase test missing it.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 8 accepted primary fits; median depth65.721ppt; T14=6.8472h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 531 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 168 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 647 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 141 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "5564099590459295744",
    "ra": "104.37601604838547",
    "dec": "-39.562843739485814",
    "phot_g_mean_mag": "14.366881",
    "bp_rp": "0.8590479",
    "sep": "41.78957198455989",
    "required_fraction_G_approx": 2.652230986085243,
    "capable": false
  },
  {
    "source_id": "5564099590459295616",
    "ra": "104.37530631567265",
    "dec": "-39.55917340625633",
    "phot_g_mean_mag": "13.891327",
    "bp_rp": "0.7629719",
    "sep": "47.89826746462356",
    "required_fraction_G_approx": 1.7115506061108694,
    "capable": false
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "157000246",
    "sector": 7,
    "n_events": 2,
    "deficit_minus_light_arcsec": 1.47,
    "peak_on_target": true,
    "depth_ap3": 0.06381,
    "psf_frac_resid": 0.093,
    "n_neighbours_60": 17,
    "n_possible": 0,
    "brightest_nb": {
      "G": 13.891327,
      "sep": 47.9
    }
  },
  {
    "tic": "157000246",
    "sector": 7,
    "n_events": 2,
    "deficit_minus_target_arcsec": 2.06,
    "deficit_minus_light_arcsec": 1.47,
    "peak_on_target": true,
    "depth_ap3": 0.06381,
```

```
    "psf_frac_resid": 0.093,
    "n_neighbours_60": 17,
    "n_possible": 0,
    "brightest_nb": {
      "G": 13.891327,
      "sep": 47.9
    }
  }
]
```


Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 5564099620520383360",
      "main_id": "TYC 7632-2311-1",
      "otype": "SB*",
      "ra": "104.39080983295",
      "dec": "-39.56517217139"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\157000246.tsv",
      "sha256": "227b8e76cc9b36fe080db52a4aa9d954dbb011a02bb11d823565fcee4216ce3a"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\157000246.json",
      "sha256": "507af03bc9c1c9308575f9d1d42e37d19633c8729c6b106d5c08829e4cd9876a"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4351.63910; other conjunction~4357.64009. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Eight primary and nine secondary events, independent SPOC and target-localized deficit establish an EB. The real secondary atphase0.3408 is recovered repeatedly despite the Gaia-median fixed-phase test missing it.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 65.72142094457375, "offphase_census_triggers": 1, "max_offphase_census_ppt": 3.3503941987336194, "minimum_bg_regressed_depth_ratio": 0.9965604850086657}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog. VSX neighbor Gaia DR3 5564099590459295616 at48.12arcsec is DSCT|GDOR|SXPHE; it is not target prior EB classification.

R067. TIC 351541670

physical_verdict: PLAUSIBLE

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Spatially resolved eclipse photometry must distinguish the21.77arcsec neighbor and the target/1.23arcsec pair; empirical PRF/WCS calibration in a second sector is the immediate archival deciding test.

Archived source: sanitized report text.

TIC 351541670: PLAUSIBLE

The two-kind EB is secure but host unresolved: circular calibration favors target while expanded calibration moves to21.77arcsec neighbor and a large WCS shift. A capable1.23arcsec companion also remains unresolved. Ranking depends on calibration; no unique host is claimed.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 8 accepted primary fits; median depth65.265ppt; T14=4.1909h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 635 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 213 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 703 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 271 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "4559553159927952384",
    "ra": "253.5961350234577",
    "dec": "18.154378972658453",
    "phot_g_mean_mag": "13.602669",
    "bp_rp": "",
    "sep": "1.2303465376990557",
    "required_fraction_G_approx": 0.6810318632699872,
    "capable": true
  },
  {
    "source_id": "4559553159931324288",
    "ra": "253.59831598437444",
    "dec": "18.148779591793883",
    "phot_g_mean_mag": "13.260843",
    "bp_rp": "0.8950949",
    "sep": "21.774090665368163",
    "required_fraction_G_approx": 0.497092295058387,
    "capable": true
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "351541670",
    "sector": 79,
    "n_events": 4,
    "deficit_minus_target_arcsec": 2.88,
    "deficit_minus_light_arcsec": 2.3,
    "peak_on_target": true,
    "depth_ap3": 0.05211,
    "psf_frac_resid": 0.109,
    "n_neighbours_60": 8,
    "n_possible": 2,
    "brightest_nb": {
      "G": 13.260843,
      "sep": 21.8
    }
  }
]
```

pixel-localization.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 7: 183 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

pixel-localization-elliptical.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 8: 186 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 4559553159927952128",
      "main_id": "Gaia DR2 4559553159927952128",
      "otype": "SB*",
      "ra": "253.59644221341",
      "dec": "18.15454619226"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\351541670.tsv",
      "sha256": "09d47c2c55a89ce8d562cdb86910d725fa2ed2cf0138d4ca3fa715ea676dea81"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\351541670.json",
      "sha256": "e967ccec9f3e00896896769839l9d580dd28dd762b381161fa1335f562f90f3f"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4347.28711; other conjunction~4345.20557. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Spatially resolved eclipse photometry must distinguish the21.77arcsec neighbor and the target/1.23arcsec pair; empirical PRF/WCS calibration in a second sector is the immediate archival deciding test.

Final adjudication: **PLAUSIBLE**. The two-kind EB is secure but host unresolved: circular calibration favors target while expanded calibration moves to21.77arcsec neighbor and a large WCS shift. A capable1.23arcsec companion also remains unresolved. Ranking depends on calibration; no unique host is claimed.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 65.26456258195196, "offphase_census_triggers": 6, "max_offphase_census_ppt": 18.88450986639667, "minimum_bg_regressed_depth_ratio": 0.969727431716339}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R068. TIC 253923749

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 253923749: CONFIRMED

Sixty-three primary timings and65/66 significant secondary windows, independent TESS-SPOC and clean on-target deficit establish an EB. High two-disk residual and radius-ratio grid boundary limit characterization.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 63 accepted primary fits; median depth77.503ppt; T14=2.0766h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 3701 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 1995 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 3101 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 921 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "6602147173915176192",
    "ra": "337.6715451866464",
    "dec": "-31.066619924662362",
    "phot_g_mean_mag": "14.648426",
    "bp_rp": "1.9394684",
    "sep": "7.319972990866362",
    "required_fraction_G_approx": 2.1426549445469885,
    "capable": false
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "253923749",
    "sector": 68,
    "n_events": 15,
    "deficit_minus_target_arcsec": 1.88,
    "deficit_minus_light_arcsec": 0.42,
    "peak_on_target": true,
    "depth_ap3": 0.07215,
    "psf_frac_resid": 0.057,
    "n_neighbours_60": 5,
    "n_possible": 0,
    "brightest_nb": {
      "G": 14.648426,
      "sep": 7.3
    }
  }
]
```


Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 6602147242633661952",
      "main_id": "TYC 7497-704-1",
      "otype": "SB*",
```

```
      "ra": "337.66895844721",
      "dec": "-31.06566951679"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\253923749.tsv",
      "sha256": "b029648a5e74c877283fc4d2f1afb1286e06e0bc085becf706cead9f43d33b4d"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\253923749.json",
      "sha256": "e8fca19eef68f6e8c1127f64169ce6f733d0bf5abaea71faba29f1fb082ab4d6"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4345.59446; other conjunction~4344.88694. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Sixty-three primary timings and65/66 significant secondary windows, independent TESS-SPOC and clean on-target deficit establish an EB. High two-disk residual and radius-ratio grid boundary limit characterization.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 77.50287933494327, "offphase_census_triggers": 91, "max_offphase_census_ppt": 32.78237955279006, "minimum_bg_regressed_depth_ratio": 0.950560065460254}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R069. TIC 389527961

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 389527961: CONFIRMED

Fourteen primary and eleven secondary events atresolved4.80078d, independent TESS-SPOC and on-target deficit establish an EB. Old high PSF residual is not used alone; deficit/light agree0.77arcsec and no capable bright neighbor exists.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 14 accepted primary fits; median depth78.903ppt; T14=5.0599h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 759 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.

Ancillary log block 2: 312 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 843 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.

Ancillary log block 4: 401 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "389527961",
    "sector": 1,
    "n_events": 8,
    "deficit_minus_light_arcsec": 0.77,
    "peak_on_target": true,
    "depth_ap3": 0.07165,
    "psf_frac_resid": 0.853,
    "n_neighbours_60": 11,
    "n_possible": 0,
    "brightest_nb": {
      "G": 17.093588,
      "sep": 29.1
    }
  },
  {
    "tic": "389527961",
    "sector": 1,
    "n_events": 8,
    "deficit_minus_target_arcsec": 0.8,
    "deficit_minus_light_arcsec": 0.77,
    "peak_on_target": true,
    "depth_ap3": 0.07165,
    "psf_frac_resid": 0.853,
    "n_neighbours_60": 11,
    "n_possible": 0,
    "brightest_nb": {
      "G": 17.093588,
      "sep": 29.1
    }
  }
]
```


Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
```

```
    {
      "id": "Gaia DR3 6774739854885748480",
      "main_id": "TYC 7964-1312-1",
      "otype": "SB*",
      "ra": "314.47436225951",
      "dec": "-38.66589876121"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\389527961.tsv",
      "sha256": "2b00dc8813ee695c696a13369bc0f0e23a92e2db078c837a356090a69957ecce"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\389527961.json",
      "sha256": "995756c63260f74b37062de395efec9cdecb99e00d81a84194ce457387a9f830"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4346.20144; other conjunction~4348.60262. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Fourteen primary and eleven secondary events atresolved4.80078d, independent TESS-SPOC and on-target deficit establish an EB. Old high PSF residual is not used alone; deficit/light agree0.77arcsec and no capable bright neighbor exists.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 78.90325488328814, "offphase_census_triggers": 6, "max_offphase_census_ppt": 5.197015273224914, "minimum_bg_regressed_depth_ratio": 0.9977356261112604}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog.

R070. TIC 196426780

physical_verdict: CONFIRMED

novelty_status: LIMITED: successful checks negative; multi-catalog service unavailable

completion: COMPLETE (adjudicated)

next_check: Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Archived source: sanitized report text.

TIC 196426780: CONFIRMED

Six sharp primary events, independent TESS-SPOC, own Gaia covariance timing and target-centered pixel deficit establish an eclipsing target. The46.89arcsec neighbor is disfavored (residual0.961 vs target0.123). Missing other eclipse is geometrically feasible; limits are conditional.

Full review: COMPLETE (adjudicated). Novelty: LIMITED; successful checks found no target EB classification, but several catalogues were unavailable.

Raw untrimmed QUALITY=0 SAP: 6 accepted primary fits; median depth71.389ppt; T14=5.5043h. Primary and secondary individual timing/depth errors, duration/ingress/flat coordinates, O-C and rejected windows are tabulated below. Errors are conditional on shape/local baseline with grid and red-noise inflation. Trapezoid flat time is not measured totality.

Ancillary log block 1: 396 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Raw-event controls and coverage

Background regressions use standardized native background units, not normalized flux ratios. Event records retain flagged-core counts, background-regressed depth and centroid template shifts. Centroid shifts are diagnostics, not source attribution. The blind rolling-median census below is diagnostic and uncalibrated; off-phase triggers, known common-mode sector times and shallow systematics must not be promoted to extra eclipses. Missing/flagged windows are not nulls.


```
{
  "blind_census": [
    {
      "sec": 8,
      "tc": 1524.3118556059544,
      "depth": 0.07270626387052537,
      "n": 11,
      "phase": 1.7553506930817075e-06,
      "threshold": 0.003,
      "duration": 0.22916702798170263
    },
    {
      "sec": 35,
      "tc": 2261.4584270791797,
      "depth": 0.07634606786916065,
      "n": 28,
      "phase": 6.122119216911415e-06,
      "threshold": 0.004934848082767148,
```

```
      "duration": 0.22222078302252157
    },
    {
      "sec": 61,
      "tc": 2979.70336295333,
      "depth": 0.0775130498297879,
      "n": 92,
      "phase": 0.9999870207873016,
      "threshold": 0.007303466133941194,
      "duration": 0.21759490788372204
    },
    {
      "sec": 62,
      "tc": 2998.6049595779386,
      "depth": 0.07714581203561033,
      "n": 92,
      "phase": 8.426876718203857e-06,
      "threshold": 0.007418212573200877,
      "duration": 0.21295933823512314
    },
    {
      "sec": 88,
      "tc": 3697.948596706823,
      "depth": 0.07589728845105992,
      "n": 92,
      "phase": 0.999983679273214,
      "threshold": 0.007751043592023509,
      "duration": 0.21528309664381595
    },
    {
      "sec": 88,
      "tc": 3716.847257249487,
      "depth": 0.07740754555178353,
      "n": 79,
      "phase": 0.9998497469088079,
      "threshold": 0.007751043592023509,
      "duration": 0.21296298692732307
    }
  ]
}
```


Own Gaia covariance and physical feasibility

Source identity, solution layout and bit order were asserted before covariance sampling. No borrowed correlations. Per-event conjunction predictions and residuals, period tension, mass function, companion minimum mass, orbital scale and central duration estimates follow. TIC properties and assumed companion radii are feasibility inputs, not measured dimensions. Two-disk limb-darkened models account for partial overlap; poor fits/grid edges prevent calibrated e sin omega or component posterior claims.

Ancillary log block 3: 568 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Other-conjunction windows and width/phase-grid limits follow. These are pointwise conditional template limits, not a uniform simultaneous confidence bound; coverage varies.

Ancillary log block 4: 8958 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Independent reduction and source attribution

Exact target and sector FITS identities were validated. This is a different pipeline using the same TESS pixels, not another observatory. Linear/quadratic baseline and background-sensitive depths are retained; aperture/PDCSAP depth differences reflect dilution and processing. TARS has no quality column.


```
{
  "sector": 35,
  "provenance": "TESS-SPOC",
  "fits_identity_validated": true,
  "quality_column_present": true,
  "events": [
    {
      "tc": 2261.458311841027,
      "qlp": {
        "1": {
          "depth": 0.07276783026155831,
          "error": 0.0004677990185897638,
          "red_noise_beta": 1.0405493481264427,
          "n_core": 29,
          "baseline_degree": 1,
          "background_regressed_depth": 0.07307085700416903,
          "background_template_native_sigma": 0.26287063635107766
        },
        "2": {
          "depth": 0.07294062565516207,
          "error": 0.000504908618790929,
          "red_noise_beta": 1.0366697305452923,
          "n_core": 29,
          "baseline_degree": 2,
          "background_regressed_depth": 0.07286838354547077,
          "background_template_native_sigma": -0.06061850622825145
        }
      },
      "independent": {
        "SAP_FLUX": {
          "1": {
            "depth": 0.06712873516304589,
            "error": 0.0003249189835140943,
            "red_noise_beta": 1.2684125194437679,
            "n_core": 33,
            "baseline_degree": 1,
            "background_regressed_depth": 0.06790784815945924,
            "background_template_native_sigma": 2.47164117603384
          },
          "2": {
            "depth": 0.06675005734711953,
            "error": 0.00033752823734742126,
            "red_noise_beta": 1.21830920294554,
            "n_core": 33,
            "baseline_degree": 2,
            "background_regressed_depth": 0.06760692822738709,
            "background_template_native_sigma": 2.2646372262770473
          }
        },
        "PDCSAP_FLUX": {
          "1": {
            "depth": 0.0729047017200105,
            "error": 0.00035251087571294904,
            "red_noise_beta": 1.2717484150516365,
            "n_core": 33,
```

```
            "baseline_degree": 1,
            "background_regressed_depth": 0.0739102344369366,
            "background_template_native_sigma": 2.47164117603384
          },
          "2": {
            "depth": 0.07326694153773082,
            "error": 0.00037186491100733924,
            "red_noise_beta": 1.2332673611526177,
            "n_core": 33,
            "baseline_degree": 2,
            "background_regressed_depth": 0.07409869489435418,
            "background_template_native_sigma": 2.2646372262770473
          }
        }
      }
    }
  ],
  "note": "Same TESS pixels; independent reduction only. Fixed QLP eclipse template and timing; uncertainties conditional and beta-inflated. TARS has no quality flags; its FLUX background provenance unspecified."
}
```


Gaia G<15 neighbors within60arcsec and required intrinsic fractional dips (G approximate passband; not calibrated TESS dilution):


```
[
  {
    "source_id": "5647004382107259264",
    "ra": "129.57950677048598",
    "dec": "-26.720640857429082",
    "phot_g_mean_mag": "13.9903",
    "bp_rp": "0.9273634",
    "sep": "46.88924914371803",
    "required_fraction_G_approx": 0.7238548263822135,
    "capable": true
  }
]
```


Original batch difference-image receipts; old/new schema duplicates are the same events:


```
[
  {
    "tic": "196426780",
    "sector": 88,
    "n_events": 2,
    "deficit_minus_target_arcsec": 5.42,
    "deficit_minus_light_arcsec": 4.26,
    "peak_on_target": true,
    "depth_ap3": 0.06425,
    "psf_frac_resid": 0.366,
    "n_neighbours_60": 48,
    "n_possible": 1,
    "brightest_nb": {
      "G": 13.9903,
      "sep": 46.9
    }
  }
]
```


pixel-localization.json: approximate Gaussian PRF/WCS sensitivity test; formal chi-square excludes PSF mismatch and correlated pixels. Ranking reversals explicitly prevent unique attribution.

Ancillary log block 8: 117 lines are preserved in linked text evidence and in the full archived report. Scientific prose and tables continue below.

Archives and novelty

Successful current shared60arcsec VSX, Gaia vari_summary/vari_eclipsing_binary, SIMBAD exact Gaia-ID and ExoFOP TOI/CTOI responses were rechecked. Exact paths/hashes below and shared receipts preserve timestamps. No Gaia EB/vari_summary entries and no TOI/CTOI entries for these targets. SIMBAD lists known spectroscopic binaries, not prior eclipse classifications.


```
{
  "simbad": [
    {
      "id": "Gaia DR3 5647003935424232576",
      "main_id": "TYC 6578-1113-1",
      "otype": "SB*",
      "ra": "129.58229701199",
      "dec": "-26.733459357909997"
    }
  ],
  "gaia_vari_summary": [],
  "gaia_eb": [],
  "exofop": {
    "toi_count": 0,
    "ctoi_count": 0
  },
  "shared_receipts": [
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\vsx\\196426780.tsv",
      "sha256": "89fde1594d2081006415ba5ae7d9b941047446b568eae04f3c0ccc4f266b917a"
    },
    {
      "path": "C:\\Home\\Astra\\made\\binary-handoff-2026-10-30\\shared-full-review\\exofop\\196426780.json",
      "sha256": "7c8a74206769dfa42ce127d2dd476eca820b38b44cdc2710687997f99b7033ae"
    }
  ]
}
```


One bounded Harvard multi-catalogue mirror probe timed out after batchG reported all IDIA9 requests timing out. ASAS-SN, SB9, WDS, Prsa, IJspeert and Kostov tables therefore remain unchecked in this fresh pass; failed service is not a negative match. VSX raw table sections were inspected; no target EB rows found. General web searches on the ten exact SIMBAD names plus eclipsing/binary found only unrelated results, not exact-target prior publications. This does not establish novelty or a complete bibliography.

Predictions and deciding next check

AtBTJD4344.32880: next primary~4359.49063; other conjunction~4372.26912. Photometric extrapolations are scheduling estimates, conditional on stationary orbit. For null secondary use Gaia phase uncertainty, not the single date.

Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.

Final adjudication: **CONFIRMED**. Six sharp primary events, independent TESS-SPOC, own Gaia covariance timing and target-centered pixel deficit establish an eclipsing target. The46.89arcsec neighbor is disfavored (residual0.961 vs target0.123). Missing other eclipse is geometrically feasible; limits are conditional.

Reproducibility: scripts and manifests retained in this batch. Original source audit completed before work; no source/paper edits or commits.

Lead control audit: {"median_primary_ppt": 71.38866728804093, "offphase_census_triggers": 0, "max_offphase_census_ppt": 0, "minimum_bg_regressed_depth_ratio": 0.9988502115682464}. Census trigger centers/durations can merge broad eclipse wings, especially253923749; they are not a calibrated blind-search catalog. VSX neighbor Gaia DR3 5647004382107259264 at47.04arcsec is ROT; it is not target prior EB classification.

Returned objects: reconciled adjudications

TIC 195865529 - CONFIRMED

Unique count addition: no.

Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.

Catalogue scope: **already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX

Full source report

TIC 468181994 - CONFIRMED (weaker)

Unique count addition: yes.

Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.

Catalogue scope: not in VSX/ASAS-SN/Gaia-EB (per report)

Full source report

TIC 470235665 - CONFIRMED

Unique count addition: yes.

QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 379251709 - CONFIRMED

Unique count addition: yes.

Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/ASAS-SN within 60"; SIMBAD SB*

Full source report

TIC 70665829 - CONFIRMED

Unique count addition: no.

Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"

Full source report; existing A4 evidence

TIC 331293467 - CONFIRMED

Unique count addition: yes.

Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.

Catalogue scope: not in VSX/PrÅia/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)

Full source report

TIC 283532878 - CONFIRMED

Unique count addition: yes.

Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add â‰¤3 ppt dips

Full source report

TIC 407877153 - CONFIRMED

Unique count addition: yes.

Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 15855794 - CONFIRMED

Unique count addition: yes.

Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert within 60"

Full source report

TIC 92881948 - CONFIRMED

Unique count addition: yes.

VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.

Catalogue scope: VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue

Full source report

TIC 468827621 - CONFIRMED

Unique count addition: yes.

Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 416285595 - CONFIRMED, with caveats

Unique count addition: yes.

Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.

Catalogue scope: not in VSX/Gaia-EB/PrÅia/IJspeert/SB9 within 60"; SB*

Full source report

TIC 372683900 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 13106408 - ELLIPSOIDAL

Unique count addition: no.

Smooth double wave; nearby 364-day SR unrelated.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 372683622 - ELLIPSOIDAL

Unique count addition: no.

Double wave plus variable spots; no eclipse.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 319185391 - PLAUSIBLE

Unique count addition: no.

Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 62328710 - NEIGHBOUR_SIGNAL

Unique count addition: no.

Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.

Catalogue scope: Enumerated catalogue cones only; SIMBAD SB* for first four; no exhaustive literature novelty claim. QLP only, one pixel sector per star.

Full source report

TIC 300903537 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 23112764 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 373062090 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465073693 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 316405449 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 465263741 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

TIC 441741446 - UNVETTED

Unique count addition: no.

Screen-positive only; no confirmation inference.

Catalogue scope: Not adjudicated in this returned vet batch.

Full source report

Returned full reports

vet01 full report

Archived source: sanitized report text.

vet01: adversarial check of TIC 195865529 (ATO J310.7264+43.3062) and TIC 468181994 (UCAC4 023-009594)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A first agent ran `s1–s8a` and was killed by a container restart. This run kept its downloads and outputs, checked its logs and continued with `s8–s17`. Everything is in `/home/user/eclipses-paper-review/work/c9a-vets/vet01/`. Downloads are in the scratchpad at `data/vet01/<tic>/`; the TESScut cubes were deleted after use. Depths are in ppt, times in BTJD (BJD_TDB − 2457000), durations in hours. A = conjunction at ν = 90° − ω, when the companion is in front of Gaia's RV star. B = ν = 270° − ω.

TIC 195865529: CONFIRMED. A detached F-star + K-dwarf eclipsing binary at the Gaia period. It is already a catalogued EB in ASAS-SN (II/366, EA, P = 3.8310 d) and ATLAS (class DBH at P/2), though not in VSX. - (1) Individual events: 48 primaries and 49 secondaries were re-measured one by one from raw QLP SAP (QUALITY==0). Another 3 primaries were fitted but excluded as partial. - (2) Period:

P_ecl = 3.8310456 ± 0.0000009 d, which is −0.15σ from Gaia's. - (3) Pixels: the deficit is on the target in all 8 sectors, across 4 camera/CCD orientations. Every Gaia neighbour within 90″ is excluded by the pixels. The dilution arithmetic alone excludes all but one: the G 13.46 star at 36″ would need ≈ 100%. - (4) Independence: QLP SAP, my TESScut aperture and TASOC agree after correcting for each aperture's dilution: undiluted primary 183 / 192 / 196 ppt, secondary 48 / 51 / 47 ppt. - (5) Geometry: the primary is conjunction A (A within 0.18σ, B excluded at 3.3–3.9σ). The secondary sits at phase 0.49999 ± 0.00007; Gaia's e and ω predict 0.479 ± 0.024 (+0.87σ). - The screen was right about depths (≈ 185 / 48 ppt), phases and period. Those depths are on QLP's apparently deblended scale, though. Raw apertures see the same eclipse diluted: 127–144 ppt in my r = 1.5 px aperture and 71 ppt in TASOC, whose header gives 64% contamination.

TIC 468181994: CONFIRMED. This is the weaker of the two; see the caveats. A late-F/G star with an M-dwarf-mass companion (m2 sin i ≈ 0.21–0.24 M⊙). It is not in VSX, Gaia-EB, Prša or IJspeert within 60″. - (1) Individual events: 37 primaries were re-measured individually in QLP SAP (one more was excluded as partial), plus 7 in TESScut sector 66, which QLP does not cover. The 1.3 ppt secondary was measured at fixed shape in 39 windows: 37 are positive and 28 are above 2σ. - (2) Period: P_ecl = 3.9969190 ± 0.0000006 d, which is **−2.31σ** from Gaia's 3.999219 ± 0.000996 d. This is the weakest link. - (3) Pixels: the 2.85″ neighbour (G 16.75) cannot be excluded by arithmetic (it would need 73% primary and 4.1% secondary depths). The difference images reject it: the mean centroid over 8 sectors lies 0.51″ from the target and 3.2″ from the neighbour (≈ 16σ), and the PSF χ² prefers the target in every sector. Every other G < 18 neighbour is excluded by arithmetic. - (4) Independence: QLP SAP, my TESScut aperture and ELEANOR-lite agree, at 21–24 ppt per sector. - (5) Geometry: the secondary sits at phase 0.4956 ± 0.0012, and Gaia predicts 0.534 ± 0.040 (A) or 0.466 ± 0.040 (B), both within 1σ. Propagated forward with Gaia's P, the primary's absolute phase does not decide between A and B (1.2σ vs 0.7σ). Propagated back with the eclipse P, A is preferred (0.01σ vs 2.2σ). - The screen was right (≈ 23 ppt, ≈ 1.5 ppt near 0.5, P 3.9969 d). The secondary is measurably *before* phase 0.5, at 0.4956. Its depth is 1.24–1.41 ppt. - Gaia shows RUWE 12.6 and ipd_frac_multi_peak = 39%: probably an unresolved sub-arcsecond companion, which TESS pixels cannot test.

1. Archives and web

- **Cones** (`s2_archives.out`, `s2b_archives.out`; the VizieR mirror was sanity-checked on Algol, which returned β Per):
- 195865529:
  - VSX (60″): only `Gaia DR3 2066587835904958592`, type YSO, 15.34–15.39 mag, at **35.76″**. That is the G 15.36 neighbour, not the target.
  - Not in Prša+2022, IJspeert+2021, Gaia DR3 VEB (I/358/veb), Gaia varisum (10″) or SB9.
  - **ASAS-SN II/366 at 0.16″: ASASSN-V J204254.34+431822.5, Type EA, Per 3.8310 d, Amp 0.12 mag, Prob 1.000.**
  - **ATLAS J/AJ/156/241 at 0.09″: class DBH (P(DBH) = 0.98), period 1.915571 d (= P/2).**
  - **So it is already a catalogued eclipsing binary**, though not in VSX.
- 468181994: no rows in VSX, J/ApJS/258/16, J/A+A/652/A120, I/358/veb, I/357/varisum, B/sb9, ATLAS or ASAS-SN II/366 (radii 10–60″, as listed in the scripts).
- The all-catalogue 5″ query returned only astrometric/photometric tables (I/...) for both stars.
- **SIMBAD (harvard mirror, 10″ cone):** both are type SB*, with 1 bibcode each: ATO J310.7264+43.3062 and UCAC4 023-009594. "TIC " is not a SIMBAD identifier for either star

(`s2c_simbad_raw.out`).

- **ExoFOP:** tois = [], ctois = [] for both.
- **WebSearch** on all names: nothing mentions either star (`s15_websearch.out`).
- **Gaia DR3 extras** (`s10_gaia_extra.out`):
- 195865529: RUWE 0.94, ipd_frac_multi_peak 0, rv_amplitude_robust 147 km/s (2K1 = 132), Teff_gspphot 6608 K, FLAME R = 2.03 R⊙, M = 1.50 M⊙.
- 468181994: **RUWE 12.57, astrometric_excess_noise 1.53 mas (sig 2803), ipd_frac_multi_peak 39**, rv_amplitude_robust 52.9 km/s (2K1 = 44.7), Teff_gspphot 5775 K, logg 4.08, FLAME R = 2.91 R⊙ (FLAME uses the gspphot distance of 608 pc; the parallax distance is 1044 pc).
- The 2.85″ neighbour 6342991943947141504: G 16.75, BP−RP 1.51, parallax 1.717 ± 0.048 mas, RUWE 1.10.
- The NSS orbit rows match targets.csv. conf_spectro_period = 1.0 for both.

2. Independence, systematics, pixels

- **Pipelines available.** Neither star has TESS-SPOC FFI light curves or 2-min SPOC data (`s1_download_lc.out`).
- 195865529: QLP s15/41/55/56/75/76/82/83, plus TASOC for s15.
- 468181994: QLP s12/13/27/39/67/93/94, plus GSFC-ELEANOR-LITE for s12/13. TESScut covers s66, where QLP has nothing.
- My own photometry: an 11×11 TESScut cube per sector. Background = per-cadence median of the faintest 20% of pixels; aperture r = 1.5 px (6–8 px); QUALITY == 0 (`s8_pixels.py`).
- **Depth method.** One method for every pipeline (`s11_depths.out`): ephemeris and shape fixed, a quadratic baseline per event, and an exposure-supersampled trapezoid amplitude.

TIC 195865529 primary / secondary (ppt)	s15	s41	s55	s56	s75	s76	s82	s83
QLP SAP	182.0 / 49.2	186.2 / 48.2	182.5 / 46.7	182.2 / 46.9	184.3 / 49.2	183.8 / 48.3	182.4 / 48.4	181.9 / 47.6
TESScut ap., raw	141.2 / 38.6	134.7 / 36.0	143.1 / 37.4	127.4 / 33.5	140.9 / 38.1	143.0 / 38.8	129.6 / 34.3	143.5 / 38.0
TESScut ÷ target fraction	183.3 / 50.2	188.1 / 50.3	194.4 / 50.8	197.4 / 51.9	187.2 / 50.6	186.0 / 50.5	204.2 / 54.1	197.5 / 52.3
TASOC raw (AP_CONT 0.637)	71.3 / 16.9 → undiluted 196.2 / 46.5							

- Typical errors are 0.3–0.9 ppt.
- In `s12_dilution.out`, the target's flux fraction in my aperture is 0.635–0.770 from TIC Tmag and the fitted PSF width, and 0.585–0.803 from the fitted scene.
- Undiluted TESScut depths: mean primary 192.3 (sd 7.2), secondary 51.3 (sd 1.4), S/P 0.267. QLP: 183.2 (sd 1.5) / 48.1 (sd 0.9), S/P 0.262.
- So QLP SAP behaves as if it were deblended with the TIC contamination ratio (0.367). The TESScut and TASOC numbers agree with it to ≈ 5%, which is the accuracy of my Gaussian-PSF dilution model. Depth *ratios* agree to 2%.
- Note: the raw aperture depths of this star are not comparable across pipelines without that correction.
- **TIC 468181994 primary (ppt):**
- QLP SAP: 21.0, 22.3, 23.7, 23.2, —, 23.0, 23.3, 21.2.

- TESScut: 22.3, 21.8, 21.9, 22.3, 22.3 (s66), 22.4, 22.8, 22.0.
- ELEANOR-lite raw: 24.3 (s12), 22.7 (s13). ELEANOR-lite corr: 23.4, 22.4.
- These are sectors 12, 13, 27, 39, 66, 67, 93, 94. The target fraction in my aperture is 0.952–0.960.
- Secondary per sector: QLP 1.07–1.55, TESScut 0.96–2.28 (s67 ± 0.98), ELEANOR-lite 0.94–1.61, all with errors of 0.16–0.46.
- All pipelines agree within about 2σ.
- **Difference images** (`s8_pixels_*.out`, `s9_pixsummary.out`). Each is the median out-of-eclipse frame minus the median in-eclipse frame, primaries only. 50 primaries per star went into the stacks, covering all 8 sectors of each. A free-position Gaussian PSF was fitted, and the zero point is a scene model of the out-of-eclipse image (TIC stars with T < 17). Errors come from bootstrapping over events. Two single-event difference images per sector are also listed.
- **195865529:**
  - Per-sector offsets from the target: 1.2–4.3″. Single events: 1.0–4.3″. Sector mean (+0.20 ± 0.57, +1.37 ± 0.91)″.
  - The nearest G < 18 neighbour is 18.9″ away. The difference centroid lies 17–23″ from it (≥ 101σ in every sector) and 32–40″ from the 33–36″ group.
  - PSF hypothesis χ^2: the target is best in all 8 sectors. The next best, the 18.8″ star, is worse by $\Delta\chi^2 \geq 16{,}096$.
  - Arithmetic, as the depth each star would need: the G 13.46 star at 36.0″ has a maximum flux fraction of 18.0%, so it would need 101.6% (primary) and 26.5% (secondary). It is excluded only marginally by arithmetic, but cleanly by pixels. Every other G < 18 neighbour within 90″ would need ≥ 393%.
  - The TIC's T 18.3 star at 10.7″ (G 19.5) contributes about 0.1% and is irrelevant.
- **468181994:**
  - Per-sector offsets: 0.41–1.53″. Mean (+0.50 ± 0.20, −0.12 ± 0.19)″, error from the scatter between sectors.
  - The 2.85″ neighbour is at (–1.67, +2.22)″. Per sector, the difference centroid is 2.3–3.8″ from it (7–24σ); the mean is ≈ 3.2″, or ≈ 16σ.
  - PSF hypothesis $\Delta\chi^2$ (neighbour minus target): +119, +190, +184, +230, +58, +14, +44, +256, so the target is preferred in all 8 sectors.
  - Arithmetic: the neighbour's flux fraction is 3.0–3.1%, so it would need **73.3% / 4.1%**, which arithmetic cannot exclude. All others (44–86″) would need ≥ 293% for the primary.
  - Figures: `fig_468181994.png` and `fig_195865529.png` (out-of-eclipse image, difference image, raw fold). In both, the difference image peaks on the target pixel.
- **Common mode:** not done with other QLP stars. A dip that repeats strictly on one ephemeris across 8 sectors over 6 years, and whose per-pixel difference image follows the target PSF only, is not a sector-wide systematic.
- **Blind period search** on raw SAP (`s4_period_search.out`, BLS 0.5–20 d): the best peaks are 3.83107 d and 3.99694 d. The other peaks are aliases (P/2, 2P, 3P, 4P).

3. TIC 195865529 (P 3.831 d)

Raw QLP SAP, QUALITY==0. Each event is fitted with a trapezoid times a quadratic baseline, with red-noise β-scaled errors (`s6_events.out`). Summary per sector (`s17_tables.out`); every event is listed in `events_table_195865529.md`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s15 ×4 (lost: 1721.87, 1733.36, 1737.19 flagged)	1714.199–1729.524	182.0 / 0.6	5.19 / 2.18	QLP, TESScut, TASOC
P s41 ×6 (lost 2442.10)	2422.946–2445.932	186.4 / 1.1	5.23 / 2.15	QLP, TESScut
P s55 ×6 (lost 2809.88)	2798.388–2821.374	182.5 / 0.9	5.23 / 2.16	QLP, TESScut
P s56 ×7 (lost 2825.21)	2829.037–2852.023	181.7 / 1.0	5.23 / 2.18	QLP, TESScut
P s75 ×6 (partial 3346.23)	3342.396–3365.382	184.1 / 2.4	5.20 / 2.16	QLP, TESScut
P s76 ×6 (partial 3388.37)	3369.213–3392.199	183.2 / 1.5	5.22 / 2.17	QLP, TESScut
P s82 ×6 (partial 3545.44)	3533.948–3556.934	182.4 / 0.7	5.24 / 2.15	QLP, TESScut
P s83 ×7	3560.766–3583.752	181.8 / 1.0	5.20 / 2.15	QLP, TESScut
S s15 ×4 (3 flagged)	1712.285–1731.440	48.9 / 1.2	5.34 / 2.53	QLP, TESScut, TASOC
S s41 ×6, s55 ×7, s56 ×6	2421.033–2850.108	48.0, 46.4, 46.8	5.23–5.29 / 2.44–2.53	QLP, TESScut
S s75 ×8, s76 ×7, s82 ×6, s83 ×5	3340.481–3581.835	49.4, 48.0, 48.4, 47.3	5.22–5.29 / 2.42–2.53	QLP, TESScut

- **Ephemeris and O−C** (`s7_ephem.out`): 48 primaries give P = 3.8310456 ± 0.0000009 d and T0 = 3346.226971 ± 0.000140. Red. χ^2 is 9.43 (errors scaled) and the O−C rms is 1.31 min; sector means lie between −3.4 and +1.8 min.
- **Secondaries:** P = 3.8310460 ± 0.0000013 (P_sec − P_prim = +0.0000004 ± 0.0000015). Phase **0.49999 ± 0.00007**, so e cos ω = −0.00002 ± 0.00010. The durations are equal (T14 5.22 vs 5.25 h), so e sin ω ≈ 0 as well. The orbit is circular.
- **Comparison with Gaia:** P_ecl − P_Gaia = −0.000058 d = −0.15σ.
- **Odd/even primaries:**
- Raw: 182.72 ± 0.21 vs 183.04 ± 0.24 ppt. Sector-normalised: 0.9985 ± 0.0011 vs 1.0021 ± 0.0013 (−2.0σ, driven by the depth differences between sectors).
- Odd/even secondaries: 47.81 ± 0.19 vs 47.60 ± 0.18.
- A distinct 48 ppt secondary at phase 0.5 rules out 2P and P/2.

4. TIC 468181994 (P 3.997 d)

Same method (`s6_events.out`). Every primary is listed in `events_table_468181994.md`; the individual secondaries at fixed phase and shape are in `s7b_sec468.out`.

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s12 ×4 (lost 1624.89 gap, 1628.89 and 1640.88 flagged)	1632.882–1648.870	21.1 / 0.5	3.73 / 2.40	QLP, TESScut, ELEANOR-lite
P s13 ×5 (1660.86, 1664.86 partial)	1656.865–1680.846	22.1 / 1.6	3.68 / 2.32	QLP, TESScut, ELEANOR-lite
P s27 ×5 (2048.56 gap, 2060.55 partial)	2036.572–2056.556	24.0 / 1.4	3.66 / 2.39	QLP, TESScut
P s39 ×6 (2368.32 flagged in QLP; clean in TESScut)	2364.320–2388.300	23.2 / 1.8	3.66 / 2.42	QLP, TESScut
P s66 ×7 (TESScut only)	3099.752–3123.734 (predicted)	21.5–23.2 each, 22.27 ± 0.22	shape fixed	TESScut

Event	Mid BTJD (range)	Depth (ppt, mean / sd)	T14 / T23 (h)	pipelines
P s67 ×5 (3139.72 flagged, 3147.72 partial)	3127.731–3151.712	23.3 / 1.0	3.60 / 2.54	QLP, TESScut
P s93 ×6 (3843.18 flagged)	3831.188–3855.169	23.2 / 1.1	3.70 / 2.40	QLP, TESScut
P s94 ×6	3859.167–3879.151	20.8 / 1.0	3.73 / 2.43	QLP, TESScut
S, 39 windows s12–s94, phase fixed at 0.4956	1630.867–3877.136	weighted mean 1.24 ± 0.07 (χ^2r 2.54)	primary shape	QLP (TESScut and ELEANOR per sector, §2)

- **Ephemeris** (`s7_ephem.out`): 37 primaries give P = 3.9969190 ± 0.0000006 d and T0 = 2384.303949 ± 0.000146. Red. χ^2 0.75, O–C rms 1.07 min.
- **Comparison with Gaia:** P_ecl – P_Gaia = −0.002300 d = **−2.31σ**.
- **Odd/even:** raw 22.18 ± 0.16 vs 22.55 ± 0.17 ppt; sector-normalised −1.79σ.
- **Secondary:**
- Fold fit: phase 0.4953 ± 0.0006, depth 1.41 ± 0.09 ppt. With the shape free: 0.4952 ± 0.0005, 1.39 ± 0.09 ppt, T14 3.57 ± 0.21 h (`s7_ephem.out`).
- **Null boxes** at phases 0.25 and 0.75: −0.25 ± 0.09 and −0.01 ± 0.09 ppt. The formal errors therefore understate the systematics by up to ≈ 3×. Even so, the secondary stands at ≈ 5σ against the worst null.
- It is present in both halves: s12–39 phase 0.4951, 1.27 ± 0.11 ppt; s67–94 phase 0.4956, 1.25 ± 0.11 ppt.
- Per-sector phases run from 0.4881 to 0.4970: mean 0.4941, standard error 0.0012 (`s13_geometry.out`).
- **The by-eye screen did not resolve** that the secondary is ≈ 25 min early, nor that it is ≈ 1.3 ppt rather than 1.5.

Geometry vs Gaia

(`s13_geometry.out`: 20,000 draws of P, Tp, e, ω; e truncated at 0.) - **195865529** (Gaia e = 0.074 ± 0.040, ω = 235 ± 33°, K1 = 66.18): - Primary minus conjunction A: −0.029 ± 0.161 phase with Gaia's P propagated forward over 771.6 cycles; −0.016 ± 0.141 with the eclipse P propagated back. - Primary minus B: +0.496 ± 0.149 and −0.490 ± 0.127, i.e. 3.3σ and 3.9σ. - **So the primary is conjunction A:** the hotter RV star is eclipsed by the companion. - Predicted secondary phase 0.479 ± 0.024, observed 0.49999 (+0.87σ). Gaia's e cos ω = −0.033 ± 0.040 vs 0.0000 ± 0.0001 observed. Gaia's e is a 1.8σ detection, and the eclipses show the orbit is circular. - **Physics:** - f(m) = 0.1141 M⊙, so m2 = 0.84–0.86 M⊙ for M1 = 1.43–1.50, and a = 13.5–13.7 R⊙. - The flat-bottom trapezoid solution (no limb darkening) gives R1 ≈ 1.72 R⊙, R2 ≈ 0.71 R⊙, k = 0.41, ρ1 ≈ 0.46 ρ⊙. That agrees with FLAME's 2.03 R⊙; TIC's 2.83 R⊙ looks too large. - The depth ratio S/P = 0.267 equals a Planck (800 nm) surface-brightness ratio for T2 ≈ 4500 K against 6600 K. A 0.85 M⊙ dwarf would be about 5000–5200 K (ratio 0.40–0.46). This crude check ignores limb darkening and the partial-eclipse geometry; it is roughly consistent, not exact. - The light-time shift of the secondary is 16–17 s. - **468181994** (Gaia e = 0.085 ± 0.084, ω = 20 ± 49°, K1 = 22.35): - Gaia's P error alone amounts to 0.497 d over the 498.9 cycles to TESS. - Primary minus A: forward −0.286 ± 0.233 (1.23σ), back −0.001 ± 0.192 (0.01σ). Primary minus B: forward +0.163 ± 0.240 (0.68σ), back +0.447 ± 0.202 (2.2σ). - **Undecided with Gaia's own P. With the eclipse period, A is preferred at about 2σ.** - Predicted secondary phase 0.534 ± 0.040 if the primary is A, 0.466 ± 0.040 if B; observed 0.4956 ± 0.0012 (−0.96σ or +0.74σ). - Observed e cos ω ≈ −0.0069 ± 0.0019. Gaia gives +0.053 ± 0.066. - The light-time (Rømer) delay is only 39–43 s (0.0001 phase), so the ≈ 25 min offset points to a small real eccentricity. Template or systematic error at the 1.3

ppt level cannot be excluded. - **Physics:** - f(m) = 0.0046 M⊙, so m2 = 0.21–0.24 M⊙ and a = 11.9–12.7 R⊙. - The trapezoid solution gives R1 ≈ 1.47 R⊙, R2 ≈ 0.22 R⊙, k = 0.152, b ≈ 0.50, ρ1 ≈ 0.54 ρ⊙. R2 suits a 0.2 M⊙ M dwarf. - S/P = 0.062 vs a Planck ratio of 0.077 for 3200 K against 5800 K, consistent with occultation of an M dwarf. - Tension: Gaia's photometric size for the primary (FLAME 2.91 R⊙; logg 4.08 implies ≈ 1.8 R⊙) is larger than the eclipse-implied 1.47 R⊙. Light from the suspected sub-arcsecond companion would inflate the photometric radius and dilute the depths.

Ephemerides and next eclipses

(`s14_ephem_next.out`.) BJD_TDB is converted to geocentric UTC with the barycentric light-time correction **removed** (astropy, builtin ephemeris); the uncorrected times differ by ≤ 2.5 min. - **TIC 195865529:** - P = 3.8310456 ± 0.0000009 d; T0 = BTJD 3346.22697 ± 0.00014 = BJD_TDB 2460346.22697. - Next primaries (geocentric UTC): **2026-11-01 15:04, 2026-11-05 11:00, 2026-11-09 06:57** (BTJD 4346.1299, 4349.9609, 4353.7920; ± 0.4 min formal). T14 ≈ 5.2 h, depth ≈ 18% (on QLP's scale). - Secondaries: 2026-11-03 13:02, 2026-11-07 08:59, 2026-11-11 04:56. - **TIC 468181994:** - P = 3.9969190 ± 0.0000006 d; T0 = BTJD 2384.30395 ± 0.00015 = BJD_TDB 2459384.30395. - Next primaries (geocentric UTC): **2026-11-02 07:00, 2026-11-06 06:55, 2026-11-10 06:51** (BTJD 4346.7912, 4350.7881, 4354.7850; ± 0.5 min). T14 ≈ 3.7 h, depth ≈ 2.3%. - Secondaries (phase 0.4956, ± 7 min): 2026-11-04 06:32, 2026-11-08 06:28, 2026-11-12 06:24.

Caveats

- **468181994:**
- RUWE 12.6 and ipd_frac_multi_peak 39% point to a companion within about 1″. The 2.85″ neighbour is probably not the cause, since Gaia resolves it as a separate source; this is my inference, not a measurement. TESS pixels cannot separate a sub-arcsecond companion. The case for the eclipses being on the RV star is that the eclipse period equals the SB1 period to 0.06%.
- That period agreement is only at 2.3σ given Gaia's quoted error. With a 3σ threshold this is a pass; with a 2σ threshold it would make the star PROBABLE.
- Speckle or AO imaging and a few RVs at quadrature (K1 ≈ 22 km/s) would settle it.
- **195865529:** depth comparisons across pipelines depend on my Gaussian-PSF dilution model (≈ 5%). The 36″ G 13.46 star is excluded by the pixels, not by arithmetic.
- **Not done:** the common-mode test with other stars; any light-curve model with limb darkening (the physics above is trapezoid-level). No TESS-SPOC or 2-min data exist for either star.
- **The `s8` pixel analysis cannot be re-run** without downloading the cubes again (`s3_tesscut_dl.py`), because they were deleted per the brief. Its outputs are saved in `pix/`.
- **Inherited fits:** `s6` marked four primaries and one secondary "ok" although their covariance had failed (partial coverage). `s7` and `s17` exclude them from timing; they are labelled partial above.

Files

- **Scripts and logs:**
- `s1_download_lc`, `s1b_download_other`, `s2_archives`, `s2b_archives`, `s3_tesscut_dl`, `s4_period_search`, `s5_fold_scan`, `s6_events`, `s7_ephem`, `s7b_sec468`, `s8a_tic_neighbours`: from the first agent, checked and reused.
- `s8_pixels`, `s9_pixsummary`, `s10_gaia_extra`, `s11_depths`, `s12_dilution`, `s13_geometry`, `s14_ephem_next`, `s15_websearch` (log only), `s16_plots`, `s17_tables`: this run. Each `.py` has a matching `.out`; s8 has

`s8_pixels_<tic>.out`.

- **Data products:** `lcio.py`, `events_qlp.csv`, `events_table_<tic>.md`, `ephem.json`, `s11_depths.json`, `fold_<tic>.txt`, `tic_neighbours_<tic>.csv`, `pix/` (aperture light curves, difference and scene images, per-sector JSON), `fig_<tic>.png`.

TIC 195865529: CONFIRMED; P_ecl=3.8310456±0.0000009 d; depth1=183 ppt (QLP, ≈deblended; 192±7 TESScut undiluted); depth2=48 ppt at phase 0.49999±0.00007; pixel: on target; pipelines: 3 (QLP, own TESScut, TASOC); Gaia geometry: pass (primary = conjunction A at 0.1–0.2σ, B excluded at 3.3–3.9σ; circular orbit) TIC 468181994: CONFIRMED (weaker: P_ecl vs Gaia at −2.3σ; suspected sub-arcsec companion); P_ecl=3.9969190±0.0000006 d; depth1=22.5 ppt (QLP) / 23.2 (TESScut); depth2=1.24–1.41 ppt at phase 0.4956±0.0012; pixel: on target (2.85″ G16.75 neighbour rejected at ≈16σ); pipelines: 3 (QLP, own TESScut, ELEANOR-lite); Gaia geometry: pass (secondary phase within 1σ; A vs B undecided with Gaia P, A preferred at ~2σ with eclipse P)

vet02 full report

Archived source: sanitized report text.

vet02: adversarial check of TIC 470235665 (UCAC4 859-013225) and TIC 379251709 (UCAC4 788-006141)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. A container restart killed the first agent part-way through. This run picked up its outputs (`s1`–`s8`: archives, downloads, raw-SAP event fits, ephemerides, the 379251709 secondary search, Gaia geometry), checked them, and added the missing parts: `s1c` (SIMBAD bibcodes, the 5″ all-catalogue list for 379251709 that `s1b` crashed on), `s9`/`s10` (TESScut pixels and own photometry), `s11` (independence), `s12` (physics, ephemerides), `s13` (secondary-phase drift), `s14` (fold plots) and `s15` (event tables). Scripts and outputs are in this folder. Light curves are in the scratchpad `data/vet02/<tic>/`. Each TESScut cube was deleted after use. Depths are in ppt and times in BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star (true anomaly 90° − ω). B = the companion is behind it.

TIC 470235665: CONFIRMED. Every check passed: 1. 25 primaries and 25 secondaries are each re-measured from raw SAP in QLP and TESS-SPOC, sectors 18–85. 2. The eclipse period is 10.326183 ± 0.000002 d, +0.82σ from Gaia's 10.3239 ± 0.0028 d. 3. The difference images in s19 (camera 3) and s79 (camera 4) put the deficit on the target. Its centre is 0.29–0.74″ from the out-of-eclipse centroid. Every neighbour would need an intrinsic depth above 100% even if all its flux fell in the aperture. 4. QLP SAP, TESS-SPOC SAP and my own TESScut aperture all see every event they cover. SPOC and TESScut depths agree to about 1%; QLP is 4% deeper. No 2-min data exist. 5. The primary falls on Gaia conjunction A, −0.07 ± 0.11 d away; B is excluded at 19σ. The secondary phase (0.52275) is +0.7σ from Gaia's prediction. The B/A duration ratio gives e sin ω ≥ 0.17, against Gaia's 0.21 ± 0.05.

The screen got this one right (P 10.3262 d, secondary at phase 0.52). Its depths (128/84 ppt) match QLP. TESS-SPOC SAP and TESScut give 122/81 ppt.

TIC 379251709: CONFIRMED. Every check passed: 1. 18 primaries are re-measured individually. The two in s18 that TESS-SPOC flags are recovered in QLP and in my TESScut photometry. 2. The eclipse period is 12.323912 ± 0.000007 d, −0.53σ from Gaia's 12.3261 ± 0.0042 d. 3. The primary deficit

is on the target in s18 (camera 2) and s52 (camera 4), 0.10″ and 1.86 ± 0.65″ from the out-of-eclipse centroid. All neighbours are excluded for the primary even in the worst case. 4. QLP, TESS-SPOC and TESScut depths agree (34–38 ppt). 5. The primary is at conjunction A (−0.17 ± 0.22 d); B is excluded at 28σ with Gaia's correlations, 4.2σ without. The secondary phase (0.4960 ± 0.0010) is +0.8σ from the 0.471 ± 0.031 that Gaia's e and ω predict.

Weak points: - The 2.5 ppt secondary is detected only in folds: QLP S/N 8.5, TESS-SPOC S/N 3.8. It is too shallow for a pixel test or for the neighbour arithmetic. - The short, V-shaped primary implies a primary radius of about 1.3 R⊙, against the TIC's 1.88 R⊙ (see Caveats).

The screen's 31 ppt primary depth is low by about 15%: the raw-SAP depth is 34–36 ppt (TESScut 36.5–38.1). Its period and secondary phase are right.

1. Archives and web

- **VizieR cones, 60″ (`s1_archives.out`):** VSX (B/vsx/vsx), Prša+2022 (J/ApJS/258/16), IJspeert+2021 (J/A+A/652/A120), Gaia DR3 EBs (I/358/veb), Gaia DR3 variability summary (I/357/varisum), Gaia vclassre, SB9 and ASAS-SN II/366 return **0 rows for both stars**. The same query returns 1 row for Algol, so the cones work. Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.
- **All-catalogue 5″ list (`s1b`, `s1c`):** only astrometric and photometric I/ catalogues come back (47 and 51 tables). The mirror's list stops alphabetically inside I/, so this query is not a completeness check; the targeted cones above are.
- **SIMBAD (`s1_archives.out`, `s1c_archives.out`):**
- 470235665 = UCAC4 859-013225, type **SB***, 1 bibcode (Gaia DR3), RV −66.6 ± 9.2 km/s.
- 379251709 = UCAC4 788-006141, type **SB***, 1 bibcode, RV −34.0 ± 4.0 km/s.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Web search:** the 2MASS, UCAC4 and TIC names return nothing relevant for either star.
- **Gaia DR3 orbits (`gaia_nss_tap.csv`):**
- 470235665: SB1, P 10.32387 ± 0.00280 d, e 0.210 ± 0.051, ω 90.3 ± 13.7°, K1 47.5 ± 3.8 km/s, flags 8192.
- 379251709: SB1, P 12.32612 ± 0.00417 d, e 0.097 ± 0.049, ω 122.6 ± 26.8°, K1 23.0 ± 1.1 km/s.
- **TIC stellar parameters (`s1c_archives.out`):**
- 470235665: contamination ratio 0.059; Teff 5353 K, R 2.30 R⊙, log g 3.68.
- 379251709: contamination ratio 0.041; Teff 5279 K, R 1.88 R⊙, log g 3.85.

2. Independence, systematics, pixels

- **Pipelines (`s11_independence.out`, raw SAP, QUALITY==0, trapezoid on a quadratic baseline):**
- 470235665 primary: QLP 127.4 ppt (sd 2.6, N = 23); TESS-SPOC 122.4 (sd 2.0, N = 15); TESScut r ≤ 1.5 px aperture 122.1 ± 0.3 (s19) and 123.3 ± 1.0 (s79).
- 470235665 secondary: QLP 84.0 (sd 1.6); TESS-SPOC 80.9 (sd 1.5); TESScut 80.8 ± 0.3 and 80.9 ± 0.7.
- QLP/SPOC depth ratio over the shared sectors: 1.040 ± 0.007 (primary) and 1.038 ± 0.006 (secondary). This is an aperture/background systematic, much larger than the 0.5 ppt formal errors. TESS-SPOC and TESScut agree within 1%.

- 379251709 primary: QLP 34.2 (sd 3.1, N = 17, including the noisy s24/s25); TESS-SPOC 36.3 (sd 1.4, N = 4); TESScut 38.1 ± 0.8 (s18) and 36.5 ± 0.1 (s52).
- T14 agrees between pipelines: 6.44 h (QLP) vs 6.45 h (SPOC) for the 470235665 primary, 9.02 vs 9.01 h for its secondary, and 4.05 vs 4.22 h for the 379251709 primary.
- Neither star has 2-min SPOC data.
- **Common mode (`s10_offtarget.out`):** this uses off-target pixels inside each cutout instead of other QLP stars.
- Summed over pixels more than 3 px from the target, the stacked primary deficit is –1.1 to +3.2 ppt, against 120 or 33–38 ppt on target.
- The brightest off-target pixels are stars beyond 90″ at the cutout edges, some brighter than the targets. They show –4.2 to +2.2 ppt.
- The eclipses also repeat on a strict period over 6 years, in three cameras and two cadence regimes. A sector-wide systematic cannot do that.
- **Census of non-eclipse dips (`s4_census.out`):**
- Apart from the eclipses, the 4–9 ppt features in raw SAP sit within about 0.3 d of gaps.
- TESS-SPOC s18 starts with a ramp (27–54 ppt, 1790.7–1791.5) **in both stars**, so it is a systematic.
- 379251709 QLP s24 is dominated by scattered light (σ30 = 8.1 ppt). Its events are kept but carry large errors.
- **Difference images (`s9_tesscut_<tic>_s<sec>.txt`; median in-eclipse minus median local out-of-eclipse, per event, averaged; Gaussian-fit centres; 21″/px):**

Star / sector (camera)	kind	N ev	deficit – OOT centroid	deficit – target WCS	frac. depth on target pixel
470235665 s19 (cam 3)	P / S	3 / 2	0.29″ (±0.37) / 0.28″ (±0.14)	0.96″ / 0.96″	122.9 / 76.6 ppt
470235665 s79 (cam 4)	P / S	3 / 3	0.33″ (±0.22) / 0.74″ (±0.26)	1.43″ / 1.58″	120.8 / 77.7 ppt
379251709 s18 (cam 2)	P	2	1.86″ (±0.65)	1.30″	33.2 ppt
379251709 s52 (cam 4)	P	2	0.10″ (±0.46)	1.32″	31.3 ppt

The 379251709 secondary difference images (s18: 2 events; s52: 1 event at the sector start) are noise. No pixel location is claimed for the secondary. - **Dilution arithmetic** (aperture r ≤ 1.5 px; fluxes from Gaia G). The "worst case" puts 100% of the neighbour's flux in the aperture. - **470235665:** the nearest neighbour is G 16.39 at 5.5″, on the target pixel. It would need an intrinsic depth of 320–356% for the primary and 210–234% for the secondary (PRF and worst-case values, both sectors). G 16.19 at 57″ needs at least 269% / 176%; G 15.51 at 69″ needs at least 149% / 98%. That last 98% (s79 secondary) is the only worst case below 100%. That star lies 70″ from the deficit centre, the fractional depth at its pixel is 2.8–8.6 ppt, and it is excluded for the primary. All neighbours are **excluded**. - **379251709 primary:** the brightest neighbour is G 16.57 at 67″ and needs at least 109–128% in the worst case. The nearest is G 17.33 at 30″ and needs at least 216–253%. All neighbours are **excluded**. - **379251709 secondary:** with a 2.5 ppt depth, any neighbour with a worst-case share above about 2.5% could produce it arithmetically. This is not excluded by pixels. It rests on timing: the dip sits exactly at the occultation phase the primary's e cos ω demands. - ZTF was not needed: for the primaries, no neighbour is arithmetically viable.

3. TIC 470235665 (P 10.3262 d)

Raw SAP, QUALITY==0, supersampled trapezoid on a local quadratic baseline, errors max(covariance, prayer-bead) (`events_470235665.csv`, `s5_events.py`). "TESScut" is my own aperture photometry (`s9`) with the shape fixed.

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E0	18	1795.7429 ± 0.0003	131.2 / 124.2 / –	6.44 / 2.51	QLP+SPOC
P E1	18	1806.0694 ± 0.0003	128.0 / 121.8 / –	6.45 / 2.60	QLP+SPOC
P E2	19	1816.3945 (TESScut)	– / – / 122.5	(fixed)	TESScut (QLP lost, SPOC partial)
P E3	19	1826.7204 ± 0.0003	130.4 / 124.3 / 122.5	6.45 / 2.44	QLP+SPOC+TESScut
P E4	19	1837.0465 ± 0.0003	130.1 / 123.2 / 121.5	6.45 / 2.46	QLP+SPOC+TESScut
P E16	24	1960.9600 ± 0.0004	123.9 / 123.0 / –	6.42 / 2.46	QLP+SPOC
P E17	24	1971.2876 ± 0.0004	126.1 / 125.2 / –	6.38 / 2.43	QLP+SPOC
P E18	24	1981.6139 ± 0.0003	124.7 / 123.7 / –	6.46 / 2.40	QLP+SPOC
P E19	25	1991.9398 ± 0.0004	– / 121.7 / –	6.48 / 2.49	SPOC (QLP partial)
P E20	25	2002.2652 ± 0.0004	– / 125.9 / –	6.49 / 2.33	SPOC (QLP partial)
P E21	26	2012.5925 ± 0.0004	126.5 / 120.5 / –	6.47 / 2.44	QLP+SPOC
P E23	26	2033.2442 ± 0.0003	126.5 / 120.0 / –	6.40 / 2.48	QLP+SPOC
P E90	52	2725.0982 ± 0.0003	128.1 / 122.5 / –	6.44 / 2.40	QLP+SPOC
P E91	52	2735.4253 ± 0.0003	122.1 / 119.4 / –	6.48 / 2.52	QLP+SPOC
P E92	53	2745.7508 ± 0.0003	124.8 / 120.3 / –	6.49 / 2.38	QLP+SPOC
P E94	53	2766.4038 ± 0.0002	125.6 / 120.6 / –	6.42 / 2.52	QLP+SPOC
P E106	58	2890.3180 ± 0.0003	130.6 / – / –	6.43 / 2.48	QLP
P E107	58	2900.6438 ± 0.0003	131.6 / – / –	6.46 / 2.43	QLP
P E145	73	3293.0389 ± 0.0003	129.6 / – / –	6.44 / 2.43	QLP
P E146	73	3303.3666 ± 0.0004	126.2 / – / –	6.44 / 2.42	QLP
P E160	78	3447.9313 ± 0.0003	124.3 / – / –	6.43 / 2.43	QLP
P E161	79	3458.2570 ± 0.0004	126.7 / – / 121.5	6.46 / 2.44	QLP+TESScut
P E162	79	3468.5833 ± 0.0003	129.9 / – / 124.8	6.44 / 2.40	QLP+TESScut
P E163	79	3478.9099 ± 0.0003	128.3 / – / 123.7	6.43 / 2.51	QLP+TESScut
P E176	85	3613.1506 ± 0.0003	129.7 / – / –	6.48 / 2.43	QLP
P E178	85	3633.8032 ± 0.0003	125.7 / – / –	6.45 / 2.50	QLP
S E1	18	1811.4664 ± 0.0005	85.7 / 81.7 / –	8.98 / 2.79	QLP+SPOC
S E2	19	1821.7930 ± 0.0006	84.4 / 81.7 / 80.5	9.06 / 2.77	QLP+SPOC+TESScut
S E3	19	1832.1197 ± 0.0005	86.7 / 82.3 / 81.0	8.95 / 2.79	QLP+SPOC+TESScut
S E16	24	1966.3596 ± 0.0008	82.5 / 81.8 / –	9.09 / 2.78	QLP+SPOC
S E17	24	1976.6851 ± 0.0005	83.2 / 82.6 / –	8.97 / 2.87	QLP+SPOC
S E18	25	1987.0117 ± 0.0006	86.9 / 82.8 / –	8.98 / 2.91	QLP+SPOC
S E19	25	1997.3387 ± 0.0008	85.7 / 81.8 / –	9.03 / 2.64	QLP+SPOC
S E20	25	2007.6649 ± 0.0006	– / 82.3 / –	8.97 / 2.69	SPOC (QLP partial)
S E21	26	2017.9920 ± 0.0006	82.4 / 78.8 / –	8.96 / 2.88	QLP+SPOC
S E22	26	2028.3179 ± 0.0005	82.8 / 79.1 / –	9.01 / 2.83	QLP+SPOC
S E89	52	2720.1718 ± 0.0007	– / 79.8 / –	9.02 / 2.78	SPOC (QLP lost)
S E91	52	2740.8227 ± 0.0005	83.7 / 80.3 / –	9.06 / 2.82	QLP+SPOC
S E92	53	2751.1493 ± 0.0009	81.6 / 79.2 / –	9.13 / 2.68	QLP+SPOC
S E93	53	2761.4760 ± 0.0006	81.1 / 79.1 / –	8.99 / 2.76	QLP+SPOC
S E105	58	2885.3895 ± 0.0005	84.5 / – / –	9.04 / 2.72	QLP
S E106	58	2895.7163 ± 0.0005	86.8 / – / –	9.16 / 2.83	QLP
S E107	58	2906.0419 ± 0.0005	84.9 / – / –	9.06 / 2.77	QLP
S E145	73	3298.4358 ± 0.0005	84.0 / – / –	8.97 / 2.96	QLP
S E146	73	3308.7628 ± 0.0005	83.7 / – / –	8.94 / 2.79	QLP
S E159	78	3443.0031 ± 0.0006	82.0 / – / –	9.01 / 2.88	QLP
S E160	79	3453.3286 ± 0.0005	82.6 / – / 79.7	8.95 / 2.84	QLP+TESScut
S E161	79	3463.6543 ± 0.0006	83.5 / – / 80.7	9.09 / 2.82	QLP+TESScut
S E162	79	3473.9798 ± 0.0006	84.7 / – / 82.3	9.02 / 2.77	QLP+TESScut
S E176	85	3618.5472 ± 0.0005	84.5 / – / –	8.94 / 2.88	QLP
S E177	85	3628.8739 ± 0.0008	84.6 / – / –	9.01 / 2.72	QLP

Not measured (gap / flagged / too little coverage): pri E22 (s26, pred 2022.918): QLP lost, SPOC lost; pri E93 (s53, pred 2756.079): QLP lost, SPOC lost; pri E159 (s78, pred 3437.608): QLP partial; pri E177 (s85, pred 3623.479): QLP lost; sec E-1 (s18, pred 1790.822): SPOC partial; sec E0 (s18, pred 1801.148): QLP lost, SPOC lost; sec E15 (s24, pred 1956.041): QLP partial, SPOC partial; sec E90 (s52, pred 2730.506): QLP lost, SPOC lost; sec E144 (s73, pred 3288.121): QLP partial

Period (`s6_period.out`): - Primaries, one value per epoch: N = 25, **P = 10.326183 ± 0.000002 d**, T0 = 1795.74233 ± 0.00018, O–C rms 0.9 min (reduced χ^2 3.5; errors scaled). - Fitting QLP alone and TESS-SPOC alone gives the same P (10.326182). - Secondaries: N = 25, P = 10.326173 ± 0.000002 d. - Joint fit: P = 10.326180 ± 0.000001 d, secondary phase **0.52275 ± 0.00002**. - Gaia P differs by **+0.82σ**.

Odd/even: primary QLP 127.66 ± 0.75 vs 128.24 ± 0.85 ppt (−0.5σ); primary SPOC +1.0σ; secondary +0.02σ / +0.28σ. The two eclipse kinds differ in depth (127 vs 84 ppt) and duration (6.44 vs 9.02 h), so the true period is not twice this one, and the odd/even test rules out a stacked P2.

Shape (`s14_fold_470235665.png`): both eclipses have rounded bottoms. The trapezoid "T23" (2.45 / 2.81 h) is not a resolved flat bottom, so totality is not established.

Secondary-phase drift (`s13_secphase_470.out`): P_sec − P_pri = (−10.0 ± 2.8) × 10^{-6} d (−3.5σ), which would be −2.6 min over 178 cycles. In QLP the secondary phase goes 0.52284 → 0.52277 → 0.52269 across 2019–20, 2022 and 2022–24. TESS-SPOC shows no change between 2019–20 and 2022 (0.52281 vs 0.52280). This hints at apsidal motion or a cadence effect. It is not claimed.

4. TIC 379251709 (P 12.3239 d)

Event	Sector	Mid BTJD	Depth QLP / SPOC / TESScut (ppt)	T14 / T23 (h)	pipelines
P E-13	18	1801.0439 ± 0.0014	35.1 / – / 39.0	4.24 / 1.16	QLP+TESScut (SPOC lost)
P E-12	18	1813.3689 ± 0.0011	35.2 / – / 37.3	4.13 / 0.89	QLP+TESScut (SPOC lost)
P E0	24	1961.2533 ± 0.0009	– / 36.7 / –	4.20 / 1.04	SPOC (QLP noisy, e_tc>0.005 d)
P E1	24	1973.5750 ± 0.0010	35.8 / 35.2 / –	4.24 / 1.06	QLP+SPOC
P E2	25	1985.9080 ± 0.0028	29.7 / – / –	4.31 / 1.53	QLP
P E3	25	1998.2305 ± 0.0043	27.7 / – / –	2.91 / 0.58	QLP (SPOC lost)
P E62	52	2725.3357 ± 0.0008	33.2 / 35.2 / 36.6	4.17 / 1.00	QLP+SPOC+TESScut
P E63	52	2737.6604 ± 0.0008	36.8 / 38.2 / 36.4	4.27 / 0.67	QLP+SPOC+TESScut
P E75	58	2885.5465 ± 0.0008	34.5 / – / –	4.04 / 1.03	QLP
P E76	58	2897.8705 ± 0.0009	34.7 / – / –	4.10 / 0.87	QLP
P E78	59	2922.5198 ± 0.0009	33.7 / – / –	4.13 / 1.07	QLP
P E79	59	2934.8432 ± 0.0009	34.2 / – / –	4.13 / 0.85	QLP
P E120	78	3440.1238 ± 0.0009	36.1 / – / –	4.09 / 0.99	QLP
P E122	79	3464.7714 ± 0.0009	39.9 / – / –	4.28 / 0.32	QLP
P E123	79	3477.0951 ± 0.0013	34.1 / – / –	4.20 / 1.07	QLP
P E134	85	3612.6582 ± 0.0010	38.4 / – / –	4.22 / 0.86	QLP
P E136	86	3637.3031 ± 0.0011	32.1 / – / –	3.89 / 1.15	QLP
P E138	86	3661.9554 ± 0.0010	30.2 / – / –	3.82 / 1.28	QLP

Not measured (gap / flagged / too little coverage): pri E77 (s58, pred 2910.177): QLP lost; pri E121 (s79, pred 3452.429): QLP lost; pri E135 (s85, pred 3624.963): QLP lost; pri E137 (s86, pred 3649.611): QLP lost

Period (`s6_period.out`): - One value per epoch: N = 18, **P = 12.323912 ± 0.000007 d**, T0 = 1961.25375 ± 0.00060, O–C rms 3.1 min. The largest residuals are +9.2 and +7.2 min, in the noisy QLP s25. - QLP alone: 12.323905 ± 0.000009 d. TESS-SPOC alone (N = 4): 12.323931 ± 0.000018 d. - Gaia P

differs by **−0.53σ**.

Odd/even: QLP 33.87 ± 1.04 vs 34.73 ± 1.27 ppt (−0.5σ); SPOC −0.55σ.

Shape: V-shaped, T14 4.1–4.2 h, trapezoid T23 about 1.0 h.

Secondary (`s7_secondary_379.out`; primaries masked, 1.5-d running median, box scan over phase): - QLP, excluding s24/s25: the strongest peak in the scan is at phase 0.4955. A trapezoid fit gives **phase 0.49597 ± 0.00035, depth 2.47 ± 0.29 ppt (S/N 8.5), T14 4.50 ± 0.43 h**. - TESS-SPOC: 0.49741 ± 0.00040, 2.59 ± 0.68 ppt (S/N 3.8), 2.6σ apart in phase from QLP. The SPOC scan's top peak (phase 0.2235) is the s18 start-of-sector ramp (`s7b_otherpeaks_379.out`). - Individual events (excluding the noisy QLP s24): 1.1–4.1 ppt at S/N 1.6–7.3. Only s59 E78 (4.12 ± 0.56 ppt) is individually significant.

Geometry vs Gaia

`s8_geometry.out`: Monte Carlo with 20000 draws using Gaia's corr_vec. Parameter order is assumed to be P, Vcm, K1, e, ω, Tp. The column-major upper triangle gives a positive-definite matrix with ω–Tp = 0.98–0.99; the row-major one does not. Uncorrelated draws are shown in brackets. `s12_physics_ephem.out` repeats the calculation with nominal values as an independent check.

- **470235665:**
- Taking the TESS ephemeris back to the Gaia epoch, the primary is −0.068 ± 0.108 d from **conjunction A** [−0.080 ± 0.427] and +5.09 d from B (+19σ [+7.5σ]). Taking Gaia's ephemeris forward to TESS gives +0.245 ± 0.405 d from A. The nominal check in `s12` gives −0.079 d.
- Because ω ≈ 90°, conjunction A falls at periastron. The primary is the companion transiting the RV star.
- Gaia predicts the secondary at phase 0.4994 ± 0.0329; measured 0.52275, **+0.71σ**. TESS-only e cos ω ≈ +0.036 vs Gaia −0.001 ± 0.051.
- T14(B)/T14(A) = 1.397 gives e sin ω ≥ +0.166, against Gaia +0.205 ± 0.050. The eclipse at periastron (A) is the shorter one, as the orbit requires.
- **379251709:**
- Backward: the primary is −0.170 ± 0.224 d from **conjunction A** [−0.189 ± 1.218] and −5.96 d from B (−28σ [−4.2σ]). Forward: −0.473 ± 0.418 d from A.
- Gaia predicts the secondary at 0.4711 ± 0.0314; measured 0.4960 ± 0.0010, **+0.79σ**. TESS e cos ω ≈ −0.006 vs Gaia −0.045 ± 0.049.
- Duration ratio 1.066 gives e sin ω ≥ 0.03 vs Gaia +0.068 ± 0.041. This test is weak.
- **Physics (`s12_physics_ephem.out`; rough two-sphere model without limb darkening):**
- **470235665:**
  - f(m) = 0.107 M⊙, so m2 = 0.64–0.77 M⊙ for M1 = 0.93–1.3 M⊙ and a = 23.2–25.4 R⊙.
  - The depths imply k ≈ 0.37 and a companion light share ≥ 8%. The two T14 values with Gaia's e and ω give R1/a = 0.079 and b_A ≈ 0.45, so R1 = 1.8–2.0 R⊙ (TIC 2.30), R2 = 0.67–0.73 R⊙ and i ≈ 87°.
  - J ≈ 0.66, so T2 ≈ 4800 K.
  - A subgiant with a K-dwarf companion: radius, temperature and mass agree with the mass function to about 15–20%. b_B ≈ 0.70 > 1 − k says the B eclipse is partial, which fits the rounded secondary.
- **379251709:**
  - f(m) = 0.0153 M⊙, so m2 = 0.28–0.35 M⊙ (an M dwarf, R2 ≈ 0.3–0.4 R⊙).

- Read as a non-grazing transit (T23/T14 = 0.22), the shape needs b ≈ 0.77 and **R1 ≈ 1.2–1.5 R⊙, R2 ≈ 0.25–0.30 R⊙**, which is consistent with m2. With the TIC's R1 = 1.88 R⊙ the 4.2 h eclipse would have to be grazing (b ≈ 0.99), and that is hard to square with a 36 ppt depth.
- A 3200–3500 K companion would give a full-occultation depth of 4–8 ppt; the measured secondary is about 2.5 ppt. Consistent.

Ephemerides and next eclipses

From `s12_physics_ephem.out`. The barycentric light-time correction (+2.8 to +5.2 min) was removed to give UTC at the geocentre. Errors are formal, from T0 and P. - **TIC 470235665 primary:** T0 = BTJD 1795.74233 ± 0.00018 (BJD_TDB 2458795.74233), **P = 10.326183 ± 0.000002 d**, T14 6.45 h. - Next primaries: BTJD 4346.3095 = **2026-11-01 19:21 UTC**; 4356.6357 = **2026-11-12 03:11**; 4366.9619 = **2026-11-22 11:00** (±0.8 min). - Secondary: T0 = 1801.14120 ± 0.00023, P = 10.326173 ± 0.000002 d, T14 9.0 h. Next: 2026-11-07 04:52, 2026-11-17 12:42, 2026-11-27 20:31 UTC. - **TIC 379251709 primary:** T0 = BTJD 1961.25375 ± 0.00060 (BJD_TDB 2458961.25375), **P = 12.323912 ± 0.000007 d**, T14 4.2 h; secondary at phase 0.4960 ± 0.0010. - Next primaries: BTJD 4352.0927 = **2026-11-07 14:07 UTC**; 4364.4166 = **2026-11-19 21:53**; 4376.7405 = **2026-12-02 05:40** (±2.1 min).

Caveats

- **SAP depths depend on the pipeline at the 4% level** (QLP deeper) and none are corrected for crowding (TIC contamination 4–6%). "Agreement" here means within about 5%, not within the formal errors.
- **The 379251709 secondary is a fold-level detection** (QLP S/N 8.5, SPOC S/N 3.8, phases 2.6σ apart). It has not been located in pixels or excluded from neighbours arithmetically. Its exact match to the e cos ω phase is what ties it to the target.
- **379251709 radius tension:** the eclipse shape prefers R1 ≈ 1.3 R⊙, about 30% below the TIC's 1.88 R⊙, unless the eclipse is grazing. This is not a failed check, but the TIC Teff/radius deserves a look. A spectrum, or a proper fit with limb darkening on the 200-s data, would settle it.
- **470235665:** the secondary-phase drift (−2.6 min over 5 years, 3.5σ, seen in QLP only) is unconfirmed. The rough physical model is crude and does not decide whether either eclipse is total.
- **Gaia geometry:** the forward predictions (±0.4 d) are limited by Gaia's P error. The backward test (±0.1–0.2 d) assumes the corr_vec ordering stated above; without correlations the errors grow to ±0.4–1.2 d and B is still rejected at 4.2σ and 7.5σ.
- **TESScut times** are not barycentred to the target, which puts them up to about ±3 min off (O−C −3.1 to +3.0 min). They are not used in any ephemeris.
- **Not done:** the QLP-neighbour common-mode test; TESScut-only sectors 60 and 86 of 470235665, which no pipeline covers; ZTF.

Files

- Scripts and outputs are in this folder. **From the first agent** (checked here, not re-run):
- `s1_archives.py/.out`, `s1b_archives.py/.out` (crashed for 379251709)
- `s2_download.py/.out`, `s3_load.py/.out` (per-star `lc_sap_q0.npz` in scratch)
- `s4_census.py/.out`, `s5_events.py` → `events_<tic>.csv`
- `s6_period.py/.out`
- `s7_secondary_379.py/.out/.txt`, `s7b_otherpeaks_379.py/.out`

- `s8a_gaia_tap.py/.out` → `gaia_nss_tap.csv`, `s8_geometry.py/.out/.txt`
- **Added in this run:**
- `s1c_archives.py/.out`
- `s9_tesscut.py` → `s9_tesscut_<tic>_s<sec>.txt/.npz/.png` (470235665 s19, s79; 379251709 s18, s52)
- `s10_offtarget.py/.out`
- `s11_independence.py/.out`
- `s12_physics_ephem.py/.out`
- `s13_secphase_470.py/.out`
- `s14_foldplot.py` → `s14_fold_<tic>.png`
- `s15_event_table.py/.out`
- **Data:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet02/<tic>/` (QLP and TESS-SPOC FITS; TESScut cubes deleted).

Summary - TIC 470235665: CONFIRMED; P_ecl=10.326183 d; depth1=122–127 ppt; depth2=81–84 ppt at phase 0.52275; pixel: on target; pipelines: 3; Gaia geometry: pass - TIC 379251709: CONFIRMED; P_ecl=12.323912 d; depth1=34–38 ppt; depth2=2.5 ppt at phase 0.4960; pixel: on target (primary; secondary untested); pipelines: 3; Gaia geometry: pass

vet03 full report

Archived source: sanitized report text.

vet03: adversarial check of TIC 70665829 (UCAC4 537-050325) and TIC 331293467 (UCAC4 150-142995)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart. The predecessor wrote `s1`–`s7` and their outputs. Its pixel run for 331293467 stopped part-way through s100 (`s7_pixels_331293467.out`). I checked those outputs, re-ran the pixel analysis for 331293467 in all four sectors (`s7b`), and wrote `s9`–`s14` plus `s2c`, `s10a`, `s12b`. All scripts, `.out`/`.json` outputs and the `pix_*.png` plots are in this folder. Downloads are under `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet03/<tic>/`; the TESScut cubes have been deleted. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly ν = 90° − ω). **B** = the RV star passes in front of the companion. Depths are in ppt of the raw (undetrended) flux unless marked "undiluted".

TIC 70665829: CONFIRMED. A K dwarf with an early-M companion, P = 3.4657654 d. - **(1) Events, pass:** 24 primaries and 28 secondaries (SPOC), and 22 and 28 (QLP), measured one by one from SAP QUALITY==0 flux. - **(2) Period, pass:** the eclipse period matches Gaia's at −0.43σ. - **(3) Pixels, pass:** the difference images in s46, s48 and s72 (three cameras) sit 0.5–1.2″ from the target. The only neighbour (G 14.44 at 64″) would need an intrinsic depth above 1500%. - **(4) Independence, pass:** QLP, TESS-SPOC, my TESScut aperture photometry and ZTF g/r all see both eclipses. Absolute depths differ by up to ~10% between apertures. The secondary/primary depth ratio is 0.216–0.233 in every pipeline, so the differences are dilution or background, not the signal. - **(5) Geometry, pass:** the secondary falls at phase 0.46048, which is e cos ω = −0.0621. Gaia predicts 0.4547 +0.024/−0.024 if the deep eclipse is A (+0.24σ), and needs 3.6–5.6σ if it is B. The deep eclipse

coincides with conjunction A (+0.4 ± 0.5 h when Gaia's orbit is conditioned on the TESS period). So the deep event is Gaia's RV star (the K dwarf) being eclipsed. ZTF colours agree: the deep eclipse is deeper in g, the shallow one deeper in r. - **Screen:** correct on all points (deep ~300–330 ppt at phase 0, ~65–72 ppt secondary at 0.4605, flat baseline, P 3.4658 d).

TIC 331293467: CONFIRMED. An A/F star with a K-dwarf companion, P = 5.2568199 d. The light is heavily blended, so the undiluted depths are uncertain. - **(1) Events, pass:** 21 primaries and 24 secondaries measured individually from QLP SAP QUALITY==0. Every secondary is above 3σ, and null phases 0.25/0.75 give −0.11 ± 0.16 and −0.09 ± 0.15 ppt. - **(2) Period, pass:** +0.51σ from Gaia. - **(3) Pixels and dilution, pass:** the diff-image source sits 0.3–3.6″ from the target in s38, s99, s100 and s101 (cameras 2, 4, 3, 2). - The G 17.07 star at 8″ needs 305–458%, the G 16.07 at 24″ needs 143–287%, and the G 16.14 at 14″ needs 91–115% (over 100% in s99). All are arithmetically excluded or effectively so. - The only neighbour that passes the arithmetic, G 13.78 at 30.5″ (needs 25–84%), is excluded by position. The fit is 10–19× worse at its position, and the measured offset points 0.00–0.11 of the way toward it. - ZTF has no data at Dec −60. No higher-resolution light curve was available. - **(4) Independence, pass in dilution-free form:** no TESS-SPOC or SPOC light curve exists. My TESScut 3×3 photometry sees every eclipse with the same T14 (4.05–4.08 h against QLP 4.07 h) and a secondary/primary ratio of 0.168–0.202 against QLP 0.160–0.176. Raw depths differ by aperture (QLP 61–73, TESScut 47–56 ppt). The undiluted primary depth is model-dependent: 80–89 ppt from PSF amplitudes, 80–116 ppt from aperture shares. - **(5) Geometry, pass:** the secondary is at phase 0.50013 ± 0.00023 (Gaia predicts 0.5015, so this cannot tell A from B). The deep eclipse sits on Gaia's conjunction A: +0.1 ± 1.0 h when conditioned on the TESS period, and +0.49σ on Gaia alone with correlations. B is excluded at 6.7σ (Gaia alone) or ≫10σ (conditioned). This test needs Gaia's ω–T_p correlation (0.9997); with independent errors it does not decide. Landing on the target's own spectroscopic conjunction is further evidence that the signal is on the target. - **Screen:** the ~66 ppt and ~11 ppt figures are right for QLP SAP. The undiluted target depths are about 8–11% and 1.3–2%.

1. Archives and web

- **Catalogue cones (`s2_archives.py`, 60″, VizieR mirror):**
- 70665829: nothing in VSX, Prsa+2022, IJspeert+2021, Gaia DR3 VEB, Gaia varisum or SB9.
- 331293467: the same, except one VSX entry at 33.7″: Gaia DR3 5866907631080924672, type L, G 18.4–18.63. It is not our star and is fainter than the 90″ neighbour list (G < 18).
- Neither star is a catalogued eclipsing binary within 60″ in these catalogues.
- **Sanity checks:** the VSX cone was checked on Algol (1 row at 0.11″, `s2_archives.out`). `s2c` showed the mirror serves Prsa (nearest EB 1621″ from 331293467) and IJspeert (nearest 2860″ from 70665829). The mirror then began timing out, so Gaia-VEB, varisum and SB9 were not sanity-checked (`s2c_catalogue_sanity.out`). The all-catalogue 5″ query (`s2b`) returned only I/ catalogues, even for Algol, so it is truncated and was not relied on.
- **SIMBAD (10″):**
- 70665829 = UCAC4 537-050325, type SB*, 3 bibcodes (2023A&A...674A..34G, 2022AJ....163..152S, 2022ApJS..259...35A).
- 331293467 = UCAC4 150-142995, type SB*, 1 bibcode (2023A&A...674A..34G).
- **ExoFOP:** no TOIs or CTOIs for either star (`s2b_allcat.out`).
- **Gaia NSS rows (`s10a_gaia_nss.csv`):** both are SB1 with flags 0. corr(ω, T_p) is 0.9984 for 70665829 and 0.9997 for 331293467.

2. Independence, systematics, pixels

- **Pipelines (`s1_download_lc.out`):**
- 70665829: QLP s45, s46, s48, s72; TESS-SPOC s45, s46, s48, s72 and "s1751". MAST labels that last product sector 1751 (camera 1, BTJD 4055.76–4063.00, 1480/3131 good cadences); it is used as given. No 2-min SPOC.
- 331293467: QLP only (s11, s38, s99–s102). No TESS-SPOC and no 2-min data.
- **Per-sector depths, primary / secondary in ppt (`s9_indep.out`):**

TIC / sector	QLP SAP	TESS-SPOC SAP	TESScut 3×3 raw	TESScut / target share	PSF amp / target flux	d2/d1 (QLP, SPOC, TESScut)
70665829 s45	325.3 / 72.3	300.3 / 67.3	—	—	—	0.222, 0.224, —
70665829 s46	306.6 / 66.1	296.2 / 66.6	297.6 / 67.2	325.3 / 73.4 (0.915)	287.5 / 65.8	0.216, 0.225, 0.226
70665829 s48	312.2 / 70.6	282.7 / 64.1	288.1 / 67.1	285.2 / 66.4 (1.010)	264.6 / 60.4	0.226, 0.227, 0.233
70665829 s72	329.8 / 72.2	299.0 / 67.9	300.6 / 66.2	340.8 / 75.0 (0.882)	290.9 / 64.1	0.219, 0.227, 0.220
331293467 s11	61.4 / 9.8	—	—	—	—	0.160
331293467 s38	64.7 / 10.6	—	46.7 / 9.4	80.5 / 16.3 (0.580)	79.6 / 14.2	0.163, —, 0.202
331293467 s99	72.6 / 12.0	—	50.4 / 9.2	115.8 / 21.1 (0.435)	89.3 / 14.6	0.165, —, 0.182
331293467 s100	70.1 / 11.7	—	51.3 / 8.6	99.2 / 16.6 (0.517)	87.5 / 12.7	0.166, —, 0.168
331293467 s101	61.4 / 10.7	—	55.5 / 9.6	106.7 / 18.5 (0.520)	81.5 / 12.1	0.174, —, 0.173
331293467 s102	70.5 / 12.4	—	—	—	—	0.176

- For 70665829, QLP SAP is 4–10% deeper than TESS-SPOC SAP in every sector, but the d2/d1 ratios match. This is a multiplicative aperture or background-scale difference, not a different signal. TESS-SPOC CROWDSAP is 0.966 (s46) and 0.969 (s1751).
- For 331293467, raw depths depend on the aperture (TIC contamination ratio 0.70), while d2/d1 and T14 agree.
- **ZTF (`s13_ztf.out`, 2″ cone):**
- 70665829: 1789 rows from 2018 to 2025 (HJD−2457000 1202.8–3969.0), folded on the TESS ephemeris.
  - Primary core: zg 361 and 316 ppt, zr 325 and 315 ppt (4–8 points per field).
  - Secondary core: zg 26 and −9 ppt, zr 60 and 33 ppt.
  - So the eclipses are seen by an independent instrument, the ephemeris holds back to 2018, and the secondary is redder.
- 331293467: 0 rows (Dec −60, outside ZTF).
- **Common mode:** I did not fetch other QLP stars. Both signals repeat at a fixed period across 3 cameras (70665829) and 3 cameras over 7 years (331293467), with secondaries at fixed phase, and the difference images show no off-target deficit. A sector-wide systematic cannot do that.
- **Blind dip census (`s3_events_*.out`, 5×MAD below a running median; `s11_census.out`):** every dip is a primary or secondary except:
- TESS-SPOC s1751 at 4059.25–4059.7: a ramp from 0.935 to 0.98 right after a data gap (phase 0.77–0.90), background flat.

- QLP s48 at 2621.1–2621.5: QLP background −110 to −210, many cadences flagged 2048/4096.
- Both are systematics.
- **Pixels (`s7_pixels_70665829.out`, `s7b_pixels_331293467.out`):** median in-eclipse minus out-of-eclipse images from the TESScut 11×11 cubes, a PSF-fitted source position, and the dilution needed for every Gaia neighbour within 90″ with G < 18, in a 3×3 aperture (scene model with Gaussian PSF and TIC-style T from G and BP−RP).
- **70665829, primary:** offset 1.0 ± 0.2″ (s46), 1.2 ± 0.6″ (s48), 0.5 ± 0.1″ (s72).
- **70665829, secondary:** 2.3 ± 0.3″, 1.1 ± 0.8″, 1.8 ± 0.1″.
- **70665829, neighbour:** the only one, G 14.44 at 64.3″, needs 7253–47260% (primary) and 1573–10587% (secondary), with a χ^2 ratio of 41–78. Excluded.
- **331293467, primary offsets:** 0.3 ± 0.2″ (s38), 2.0 ± 0.2″ (s99), 1.2 ± 0.3″ (s100), 3.6 ± 0.5″ (s101). Bootstrap errors are surely underestimates of the systematics, but every offset is far smaller than any separation that matters.
- **331293467, neighbour by neighbour:**
  - G 17.07 at 8.0″: needs 377/458/334/305%. χ^2 ratio 3.6–4.6. Offset projection onto it is 0.03/0.23/0.00/−0.18 of the way.
  - G 16.14 at 14.0″: needs 91/115/99/97%. χ^2 ratio 3.8–10.5. Projection −0.01/0.01/−0.08/0.24.
  - G 16.07 at 24.4″: needs 143–287%.
  - G 13.78 at 30.5″: needs 84/45/28/25%, so not excluded by arithmetic. χ^2 ratio 19.1/10.1/19.1/16.0 and projection 0.00/−0.00/0.04/−0.11, so excluded by position.
  - G 14.73 at 47.3″: needs 96–5754%, χ^2 ratio 10–20.
  - All other neighbours need more than 150%.
- **331293467, secondary:** the fit gives 0.2 ± 0.7″, 1.9 ± 1.0″, 3.2 ± 0.6″, 3.2 ± 1.3″. The χ^2 cannot separate the target from the 8″ and 14″ stars here (ratio ~1.0–1.1), and a G 12.85 VARIABLE star at 89.4″ on the cutout edge fits the secondary diff image better (ratio 0.14–0.36). That star contributes 0.0000 of the aperture flux, so it is noise in a 4–5 event average, not a source. The secondary is tied to the target because it keeps a fixed phase (0.5002) and a fixed d2/d1 with the on-target primary.

3. TIC 70665829

Measurements (raw SAP QUALITY==0; trapezoid with quadratic baseline; `s3_events_70665829.out` lists every event). Sector summaries:

Event	Mid BTJD (first … last)	Depth (SPOC / QLP / TESScut)	T14 / T23	pipelines
Primary s45 ×6	2528.21356 … 2549.00771	300.3 / 325.3 / —	2.83 / 0.53 h (all SPOC primaries; range 2.80–2.90)	SPOC, QLP
Primary s46 ×6	2555.93930 … 2576.73431	296.2 / 306.6 / 297.6	"	SPOC, QLP, TESScut
Primary s48 ×6	2611.39079 … 2635.65071 (2614.85610 SPOC only)	282.7 / 312.2 / 288.1	"	SPOC, QLP, TESScut
Primary s72 ×5	3262.95482 … 3283.74972	299.0 / 329.8 / 300.6	"	SPOC, QLP, TESScut
Primary s1751 ×1	4060.08215 ± 0.00010	299.86	2.84 / 0.53 h	SPOC
Secondary s45 ×7	2526.34374 … 2547.13826	67.3 / 72.3 / —	2.77 / 0.54 h (all SPOC secondaries; range 2.57–2.88)	SPOC, QLP
Secondary s46 ×8	2554.07007 … 2578.32973	66.6 / 66.1 / 67.2	"	SPOC, QLP, TESScut

Event	Mid BTJD (first … last)	Depth (SPOC / QLP / TESScut)	T14 / T23	pipelines
Secondary s48 ×6	2612.98690 … 2633.78190	64.1 / 70.6 / 67.1	"	SPOC, QLP, TESScut
Secondary s72 ×6	3264.54980 … 3285.34512	67.9 / 72.2 / 66.2	"	SPOC, QLP, TESScut
Secondary s1751 ×1	4061.67233 (partial, egress lost)	54.78	—	SPOC

- **Lost events:** in TESS-SPOC, s45/46/48/72 lose 1/2/5/4 events to gaps (`s11_census.out`). QLP flags more of s48: it has 1 full and 11 partial events there.
- **Period (`s12_ephem.out`, full-coverage events, joint fit):**
- SPOC: T0 = 2791.61111 ± 0.00009 BTJD, P = 3.4657654 ± 0.0000007 d. QLP: 3.4657623 ± 0.0000006 d.
- The 3σ difference comes from the single s1751 primary. Without it, SPOC primaries give 3.4657637 ± 0.0000012, and the s1751 eclipse arrives +2.30 min late (`s12b_no1751.out`).
- Primary O−C has χ^2_r of 24–30 against the formal errors: sector means range from −1.9 to +1.3 min (s48 is early in both pipelines while its secondaries are not). This was not pursued, but the realistic timing error is about ±2 min.
- Gaia P = 3.4658878 ± 0.0002846, so $(P - P_{Gaia})/\sigma = -0.43$ for both pipelines.
- **Odd/even (`s4_period_70665829.out`):** primary +0.62 ± 0.82 ppt (SPOC), −0.87 ± 1.24 (QLP). Secondary +0.17 ± 0.54, −1.06 ± 0.74. No difference.
- **Which eclipse is primary:** the primary (300 ppt) and secondary (67 ppt) differ by a factor of 4.5 and sit at an off-centre phase. So P is neither half nor double the true period, and the deeper event is unambiguous.

4. TIC 331293467

Measurements (QLP SAP QUALITY==0. Primaries have a free trapezoid fit (`s3_events_331293467.out`). Secondaries use the primary's shape with free t0 and depth (`s5_secondary_331293467.out`)):

Event	Mid BTJD (first … last)	Depth QLP (TESScut raw)	T14 / T23	pipelines
Primary s11 ×4	1601.23407 … 1622.26045	61.4	4.07 / 0.83 h (all primaries; range 3.93–4.23)	QLP
Primary s38 ×5	2337.18980 … 2358.21445	64.7 (46.7)	"	QLP, TESScut
Primary s99 ×2	4050.91270, 4066.68276	72.6 (50.4)	"	QLP, TESScut
Primary s100 ×3	4077.19684 … 4098.22307	70.1 (51.3)	"	QLP, TESScut
Primary s101 ×4	4103.47964 … 4124.50456	61.4 (55.5)	"	QLP, TESScut
Primary s102 ×3	4129.76349 … 4150.78766 (last one partial)	70.5	"	QLP
Secondary ×24 (3/4/4/5/4/4 per sector)	1603.8613 … 4148.1608	11.42 ± 0.15 weighted mean (8.6–13.4 each)	primary shape (TESScut free fit 3.5–4.2 h)	QLP, TESScut

- **Lost events:** 14 events fall in gaps across the six sectors (`s11_census.out`).
- **Period (`s12_ephem.out`, full-coverage, joint):** T0 = 3467.40446 ± 0.00029 BTJD, P = 5.2568199 ± 0.0000020 d.
- All primaries: 5.2568187 ± 0.0000017. Secondaries only: 5.2568291 ± 0.0000074 (1.4σ).

- Gaia P = 5.2565467 ± 0.0005365, so +0.51σ.
- **Odd/even, primary:** −1.19 ± 0.49 ppt (formal 2.4σ, 1.8% of the depth). It is driven by sectors with 1–2 events and a partial s102 event, and the sector-to-sector depth scatter (61–73 ppt) is much larger. At 2P both halves are the same eclipse type, and the 11 ppt secondary at phase 0.5002 rules out P/2. Not a sign of a wrong period.

Geometry vs Gaia

All from `s10_geometry.out`: 20000 draws each with independent errors, with Gaia's correlation matrix, and with that matrix conditioned on the TESS period.

70665829. Gaia: e = 0.0769 ± 0.0397, ω = 175.4 ± 22.2°, T_p = 390.5724 ± 0.2071 BTJD. - **Secondary phase:** measured shallow-after-deep phase is 0.46048 ± 0.00008. The implied e cos ω = −0.0621, the same exact or to first order. - **Gaia prediction:** (t_B − t_A)/P = 0.4547 (+0.0239/−0.0238). That is below 0.5 in 100% of draws, with e cos ω = −0.0711 ± 0.0362. - Deep = A matches at +0.24σ. Deep = B would need 0.5395, at +3.55σ. - Conditioned on the TESS period: 0.4630 ± 0.0136, giving A −0.19σ and B +5.64σ. - **Absolute timing (630 cycles from T_p):** - Independent errors: deep minus A = −0.0215 ± 0.0999 P (−0.22σ); minus B = −4.71σ. - Gaia correlated: deep minus A = −0.0204 ± 0.0585 P (−1.7 h, −0.35σ); minus B = −11.7σ. - Conditioned on the TESS period: deep minus A = +0.0051 ± 0.0064 P (+0.4 h, +0.80σ); minus B = −56σ. - **Conclusion: deep eclipse = conjunction A.** Gaia's ω decides this through two independent routes (secondary phase and absolute timing). - **Durations:** T14 is 2.77 h for the secondary and 2.83 h for the primary (SPOC). This is consistent with Gaia's small e sin ω = +0.0032 ± 0.0310. - **Physics (`s14_physics.out`, M1 = 0.85 M⊙):** f(m) = 0.0807 M⊙, giving m2 ≈ 0.54 M⊙ (≈3760 K, ≈0.52 R⊙). a = 10.75 R⊙ (a/R1 = 14.2). A central eclipse would last 3.15 h against 2.83 h observed, so b > 0. The expected d2/d1 ≈ J(800 nm) ≈ 0.28 against 0.22 observed. - A cooler M-dwarf companion makes the K dwarf's eclipse the deep one, as the timing says. ZTF agrees: the deep eclipse is deepest in g, the shallow one deeper in r.

331293467. Gaia: e = 0.0215 ± 0.0250, ω = 287.0 ± 69.8°, T_p = 389.8053 ± 1.0004 BTJD. - **Secondary phase:** 0.50013 ± 0.00023 (joint fit) and 0.5002 ± 0.0001 (`s5`). That gives e cos ω = +0.0003 to +0.0004 ± 0.0002, against Gaia's +0.0024 ± 0.0227. Consistent, but it cannot separate A from B. - **Absolute timing (699 cycles from T_p):** - Independent errors: deep minus A = +0.14σ, minus B = −1.90σ. Not decisive. - Gaia correlated: deep minus A = +0.0321 ± 0.0649 P (+4.1 h, +0.49σ); minus B = −6.66σ. - Conditioned on the TESS period: deep minus A = +0.0009 ± 0.0077 P (+0.1 h, +0.12σ); minus B = +81σ. Conditioning shifts T_p by +0.29 d and ω by +20.7°, because corr(P, T_p) = 0.57. - **Conclusion: deep eclipse = conjunction A,** the K dwarf in front of the A/F star. This rests on Gaia's corr(ω, T_p) = 0.9997, which is physically expected at e ≈ 0. - **Physics (M1 = 1.69 M⊙):** f(m) = 0.0630 M⊙, giving m2 ≈ 0.71 M⊙ (≈4580 K, ≈0.70 R⊙, k ≈ 0.41). a = 17.04 R⊙. A central eclipse would last 5.76 h against 4.07 h observed (b > 0). The expected d2/d1 ≈ 0.21 against 0.16 observed. The undiluted primary depth of 8–11% is below $k^2 \approx 0.17$, as a partial or high-b transit should be.

Ephemerides and next eclipses

From `s12_ephem.out`. Times are BJD_TDB / BTJD, converted to UTC calendar dates without the barycentric-to-Earth light-time correction (up to ±8.3 min). - **TIC 70665829:** T0 = 2791.61111 ± 0.00009 BTJD (BJD_TDB 2459791.61111), P = 3.4657654 ± 0.0000007 d (SPOC). The secondary follows at +1.59591 ± 0.00029 d (phase 0.46048). Realistic timing error is ±2 min (see §3). - Next primaries: 4344.2740 (2026-10-30 18:33 UTC), 4347.7398 (2026-11-03 05:44), 4351.2055 (2026-11-06 16:54). The QLP ephemeris is 1–2 min earlier. - Next secondaries: 4345.8699 (2026-11-01 08:51),

4349.3357 (2026-11-04 20:02), 4352.8014 (2026-11-08 07:12). - **TIC 331293467:** T0 = 3467.40446 ± 0.00029 BTJD (BJD_TDB 2460467.40446), P = 5.2568199 ± 0.0000020 d. The secondary follows at +2.62912 ± 0.00123 d (phase 0.50013). - Next primaries: 4345.2934 ± 0.0004 (2026-10-31 19:01 UTC), 4350.5502 (2026-11-06 01:11), 4355.8070 (2026-11-11 07:20). - Next secondaries: 4347.9225 (2026-11-03 10:07), 4353.1793 (2026-11-08 16:17), 4358.4361 (2026-11-13 22:26).

Caveats

- **331293467 has a single pipeline (QLP) plus my own TESScut photometry.** Its undiluted depths rest on a Gaussian-PSF scene model: primary 80–116 ppt (79.6–89.3 ppt from PSF amplitude), secondary ~13–21 ppt. Only the fact that the signal is on the target, the timing and the ratios are robust.
- **Attribution for 331293467 is pixel-level, not resolved.** ZTF has no data at Dec −60. The G 13.78 star at 30.5″ is excluded only by position (and by the timing match to Gaia's conjunction), not by arithmetic. Bootstrap centroid errors (0.2–0.5″) understate the systematics, but the closest call is still 0.24 of the way toward any neighbour.
- **Gaia geometry for 331293467 is decisive only with the corr_vec decoding** (order P, Vcm, K, e, ω, T_p, column-major upper triangle; the 0.9997 ω–T_p term confirms it).
- **70665829 timing:** the O−C scatter between sectors (±1–2 min) and the late s1751 primary (+2.3 min) are unexplained. Candidates are timestamps in the unusual "sector 1751" product, or real timing variations. This does not affect the classification.
- **Archives:** the cones for Gaia-VEB, varisum and SB9 were not checked on a known object, because the mirror failed during `s2c`. Their empty results stand from `s2` only.
- **Common mode:** not done with other stars (see §2).

Files

- **Light curves:** `lcio.py`, `s1_download_lc.py` / `.out`.
- **Archives:** `s2_archives.py` / `.out`, `s2b_allcat.py` / `.out`, `s2c_catalogue_sanity.py` / `.out`.
- **Events:** `s3_events.py`, `s3_events_70665829.out` / `.json`, `s3_events_331293467.out` / `.json`.
- **Periods:** `s4_period.py`, `s4_period_*.out`, `s4_ephem_*.json`; `s5_secondary_331.py`, `s5_secondary_331293467.out` / `.json`.
- **Pixels:** `s6_tesscut.py` / `.out`; `s7_pixels.py`, `s7_pixels_70665829.out`, `s7_pixels_331293467.out` (interrupted; superseded by s7b); `s7b_pixels.py`, `s7b_pixels_331293467.out`; `pix_70665829_s46/s48/s72.png`, `pix_331293467_s38/s99/s100/s101.png`; `tesscut_ap3x3_*.txt` (own photometry).
- **Independence:** `s9_indep.py` / `.out` / `.json`.
- **Geometry:** `s10a_gaia_query.py` / `.out`, `s10a_gaia_nss.csv`, `s10_geometry.py` / `.out` / `.json`.
- **Census:** `s11_census.py` / `.out`.
- **Ephemerides:** `s12_ephem.py` / `.out` / `.json`, `s12b_no1751.py` / `.out`.
- **ZTF:** `s13_ztf.py` / `.out` (raw CSVs in the scratchpad `ztf/` folder).
- **Physics:** `s14_physics.py` / `.out`.

TIC 70665829: CONFIRMED; P_ecl=3.4657654±0.0000007 d; depth1=283–330 ppt (pipeline-dependent; SPOC ~300); depth2=64–72 ppt at phase 0.46048; pixel: on target; pipelines: 4 (QLP, TESS-SPOC, TESScut, ZTF); Gaia geometry: pass (deep = conjunction A, Gaia RV star eclipsed) TIC 331293467: CONFIRMED; P_ecl=5.2568199±0.0000020 d; depth1=61–73 ppt QLP raw (≈80–116 ppt

undiluted); depth2=11.4 ppt QLP (≈13–21 ppt undiluted) at phase 0.50013; pixel: on target (30.5″ G13.78 excluded by position only); pipelines: 2 (QLP, TESScut); Gaia geometry: pass (deep = conjunction A; needs Gaia ω–T_p correlation)

vet04 full report

Archived source: sanitized report text.

vet04: adversarial check of TIC 283532878 (UCAC4 740-075601) and TIC 407877153 (UCAC4 054-021425)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed the work of an earlier agent that a container restart killed part-way through. I re-read its scripts and outputs `s01`–`s11` and kept what they had measured. I re-ran the archive step it had not finished (`s12`), and found that the first VizieR pass gave false empty results (see §1). New steps: a difference-image test with real noise weighting (`s13`, `s13b`), because the `s11` "chi2" values were unweighted sums of squares. Also new: ZTF (`s14`), plots (`s15`), a common-mode test (`s16`), extra geometry, physics and ephemeris (`s17`), and a merged event table (`s18`). All of these are in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`. The downloads are in the scratchpad at `.../scratchpad/data/vet04/<tic>/`. I deleted the TESScut cubes after `s13`. Notation: **A** = the companion passes in front of Gaia's RV star (true anomaly ν = 90° − ω); **B** = the RV star passes in front of the companion. Depths are in ppt, times in BTJD = BJD_TDB − 2457000, durations in hours.

TIC 283532878: CONFIRMED. - (1) Each event fitted on its own in raw QLP SAP flux, QUALITY==0: 11 primaries and 9 full secondaries, plus 1 partial secondary. One more primary (s16, 1744.53) is visible only in TESScut. - (2) The eclipses give P = 17.446019 ± 0.000019 d. Gaia's 17.4298 ± 0.0089 d is +1.83σ away. - (3) Difference images in all 8 sectors (cameras 2, 3 and 4) put both the primary and the secondary on the target. Primary offsets are 0.4–1.5″. Secondary offsets are 1.0–5.3″, each within ~2σ. Every Gaia neighbour is rejected by $\Delta\chi^2 \geq 1119$ for the primary and $\Delta\chi^2 \geq 41.7$ for the secondary in every sector. The dilution arithmetic alone excludes every neighbour for the primary (each would need ≥ 218%). For the 10 ppt secondary it does **not** exclude three neighbours (G 14.50 at 37″, G 14.93 at 28″, G 15.74 at 29.5″, each needing 30–47%); the pixels do. - (4) QLP SAP and my own TESScut photometry agree on the secondary (9–10 ppt in both). For the primary, QLP gives 70–79 ppt and the TESScut aperture 65–72 ppt, with 13–20% of that aperture's light from neighbours. The PSF-deblended depth is 71–77 ppt. No TESS-SPOC or 2-min data exist. - (5) The secondary is at phase 0.67812 ± 0.00030; Gaia's e and ω predict 0.712 ± 0.024 (–1.4σ). The primary falls 0.20 d from Gaia's conjunction A (0.4–0.7σ); B is excluded at > 11σ. - **Screen check:** the screen was right. The "~9 ppt dip near 0.675" is the eccentric secondary. It is not a one-sector glitch: it appears in all 8 sectors and 4 seasons at constant phase (0.6775–0.6795). Its depth is 9–10 ppt; 10–11 ppt deblended.

TIC 407877153: CONFIRMED. - (1) Each event fitted on its own: 9 full primaries and 9 secondaries in raw SAP, plus extra partial events (two primaries, two secondaries). - (2) The eclipses give P = 17.089294 ± 0.000008 d. Gaia's 17.0902 ± 0.0063 d is −0.14σ away. - (3) Difference images put the primary on the target in 6 sectors ($\Delta\chi^2 \geq 337$ against every neighbour). The secondary also fits best on the target in 6 sectors ($\Delta\chi^2 \geq 7$ per sector; ≥ 166 summed over sectors). The arithmetic needs ≥ 441% (primary) and ≥ 51% (secondary) on any neighbour. - (4) Three pipelines: TESS-SPOC SAP (28.6–29.4 ppt), my own TESScut aperture (29.4–31.0 ppt) and QLP SAP. QLP gives 27–29.6 ppt in s12/s39 but

only 24–27 ppt in s93–s102, a QLP dilution issue. The secondary is 2.6–4.2 ppt in all three. - (5) The secondary is at phase 0.54690 ± 0.00017; the prediction is 0.559 ± 0.044 (−0.28σ). The primary is at conjunction A (0.16–0.64σ); B is excluded at ≥ 5σ. - **Screen check:** the secondary is **3.0 ppt, not ~2 ppt** (stack 2.98 ± 0.13). The primary is ~29 ppt in SPOC and TESScut, not ~27 (27 is QLP's diluted value). The phase of 0.545 was right.

1. Archives and web

- **VizieR cones** (`s12d_recheck.out`, mirror vizier.idia.ac.za, run immediately after an Algol VSX control returned bet Per, EA/SD):
- **TIC 283532878** is not in VSX, Gaia DR3 EB (I/358/veb), Prsa+2022, IJspeert+2021, Gaia varisum or SB9 within 60″ of the target. Both VSX and Gaia-EB list one row at **29.6″: Gaia DR3 2202301140233408128**, G 17.66 (the "VARIABLE" neighbour in `neighbours_90as.csv`). It is VSX type E, P = 1.79747 d, G 17.61–18.17; Gaia EB eclipse depths are 0.39 and 0.60 (`s12e_neighbour_eb.out`). It is excluded as the source below.
- **TIC 407877153:** nothing in any of those six catalogues within 60″.
- **First-pass warning:** the first pass (`s12_archives.out`) returned 0 rows everywhere, including the EB above, and its Algol control failed later (`s12c_controls.out`: VSX 0 rows for Algol). The mirror was intermittently returning empty tables. The all-catalogue 5″ query is likewise unreliable here: its Algol control gave 0 tables (`s12b_archives.out`), so I do not use it.
- **SIMBAD** (sim-coo 10″, `s12b_archives.out`): both stars are type **SB*** with 1 bibcode (2023A&A...674A..34G, the Gaia DR3 NSS paper). TIC 283532878 = UCAC4 740-075601 = 2MASS J21512860+5759057. TIC 407877153 = UCAC4 054-021425 = 2MASS J18230801-7921092.
- **ExoFOP:** no TOI. "CTOIs 0" on both pages.
- **Conclusion:** neither star is a catalogued eclipsing binary within 60″. Both are known only as Gaia SB1s.

2. Independence, systematics, pixels

- **Pipelines:**
- 283532878: QLP SAP (8 sectors) plus my own TESScut aperture photometry (8 sectors). The two agree on the secondary. The primary is shallower in my aperture because it holds 13–20% neighbour light (`s11_pixels_283532878.out`: target fraction 0.796–0.873).
- 407877153: QLP, TESS-SPOC (s12, s13, s39) and TESScut. SPOC and TESScut agree on the primary to about 1 ppt; QLP s93–s102 are ~10–15% shallower. No 2-min SPOC data exist for either star.
- **Difference images, noise-weighted (`s13_pixnoise_*.out`, `s13b_summary.out`):** in each sector I averaged the per-event (median out − median in) images. The per-pixel noise comes from 300 pseudo-events at random out-of-eclipse times with the same windows. I then fitted Gaussian-PSF single-source models for the target and for every Gaia neighbour with G < 18 within 90″. Results:
- 283532878 P: S/N 40–154 per sector; least-rejected neighbour $\Delta\chi^2$ = 73,292 summed (≥ 1119 in every sector); PSF-amplitude depth on target 70.6–77.3 ppt.
- 283532878 S: S/N 8.6–14.9; $\Delta\chi^2 \geq 41.7$ in every sector, ≥ 977 summed; depth 9.8–12.9 ppt; free-fit offsets 1.0–5.3″ (1σ 1.3–2.9″).
- 407877153 P: S/N 18.6–84; $\Delta\chi^2 \geq 337$; depth 30.0–35.3 ppt.
- 407877153 S: S/N 2.8–8.7; $\Delta\chi^2 \geq 7.1$ per sector, ≥ 166 summed; offsets 3.6–9.9″ with 1σ 1.9–12″.
- I looked at two of the images (`s15_diff_283532878_s57_S.png`, `s15_diff_407877153_s13_S.png`). In both, the deficit sits on the target pixel, away from the bright neighbours in the direct image.

- **Dilution arithmetic** (per-sector aperture flux fractions, `s11_pixels_*.out`):
- **283532878 primary:** every neighbour needs ≥ 218% intrinsic depth, so all are excluded.
- **283532878 secondary (10.2 ppt):** most neighbours are excluded arithmetically (≥ 82% needed; the G 17.66 Gaia EB at 29.7″ needs 181–240%). Three are **not**: G 14.50 at 37.0″ needs 29.6–100%, G 14.93 at 28.3″ needs 30–47%, and G 15.74 at 29.5″ needs 31.8–39%. The difference images reject them instead ($\Delta\chi^2 \geq 41.7$ each sector).
- **407877153:** the primary needs ≥ 441% on any neighbour; the secondary needs ≥ 51.5% (G 14.68 at 60″, s13) and 61–135% for the G 17.23 star at 46″.
- **Same period and phase drift:** both secondaries share the primary's period (283532878: P_S – P_P = −0.00030 ± 0.00010 d; 407877153: +0.000013 ± 0.000059 d, `s05_ephem.out`) and sit at fixed phase over 4–5 seasons. A neighbour's EB would have to share the period exactly. The catalogued neighbour EB (P = 1.7975 d; 17.446/1.7975 = 9.71, not commensurate) would drift in phase.
- **ZTF (`s14_ztf.out`):** not informative. IRSA returns only 54–94 catflags==0 points per object for the target and the five closest bright neighbours, with 0–1 points inside any eclipse window.
- **Common mode (`s16_commonmode.out`):** not done. The MAST product queries kept failing (RemoteServiceError and proxy resets, `s16_commonmode.out`), and the coordinator then dropped this optional step. The pixel and two-pipeline evidence above stands without it.
- **Phase 0.5 of 283532878** (`s17_geometry2.out`, from 10 fixed-time fits in `s08`): +0.35 ± 0.30 ppt. There is no eclipse at phase 0.5. $\chi^2 = 45/9$ shows red noise at the ~2–3 ppt level.

3. TIC 283532878 (P = 17.446 d, e = 0.35)

Measurements (`s18_event_table.out`, from `s04`/`s08`/`s11`; trapezoid with quadratic baseline, raw SAP QUALITY==0; "fixed" means the shape is fixed from the 200-s stack). There is no TESS-SPOC column because this star has no TESS-SPOC data.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
P n=−1 s16	1744.5307 ± 0.0011 (TESScut)	QLP: 13 flagged cadences	67.6 ± 1.0	–	TESScut
S n=−1 s16	1756.3642 ± 0.0079	7.5 ± 1.2 / 7.6 ± 1.1	10.1 ± 0.8	5.53 / 3.45	QLP, TESScut
P n=0 s16	1761.9800 ± 0.0015	70.8 ± 1.7 / 70.4 ± 1.4	66.9 ± 1.1	7.01 / 2.00	QLP, TESScut
S n=0 s17	1773.8350 ± 0.0048	9.6 ± 0.7 / 9.8 ± 0.7	9.2 ± 0.8	8.21 / 3.43	QLP, TESScut
P n=1 s17	1779.4207 ± 0.0007	73.6 ± 0.8 / 73.9 ± 0.9	69.6 ± 0.8	7.29 / 2.12	QLP, TESScut
P n=61 s56	2826.1851 ± 0.0009	74.1 ± 1.1 / 74.1 ± 0.9	66.8 ± 0.6	7.20 / 2.05	QLP, TESScut
S n=61 s56	2838.0248 ± 0.0037 (TESScut)	12.8 ± 2.0 (egress in gap)	10.7 ± 0.6	–	QLP (partial), TESScut
P n=62 s56	2843.6288 ± 0.0010	75.5 ± 1.3 / 75.2 ± 1.0	66.8 ± 0.5	7.04 / 2.02	QLP, TESScut
S n=62 s57	2855.4637 ± 0.0090	9.2 ± 1.5 / 8.6 ± 1.3	9.3 ± 0.5	6.08 / 2.19	QLP, TESScut
P n=63 s57	2861.0760 ± 0.0010	73.5 ± 1.3 / 73.3 ± 1.2	68.4 ± 0.6	6.95 / 2.05	QLP, TESScut
S n=63 s57	2872.9161 ± 0.0070	9.0 ± 1.0 / 9.2 ± 1.0	9.2 ± 0.5	6.87 / 3.03	QLP, TESScut
P n=64 s57	2878.5193 ± 0.0005	79.2 ± 0.7 / 78.6 ± 0.6	69.6 ± 0.5	7.11 / 1.97	QLP, TESScut
S n=92 s76	3378.8457 ± 0.0067	8.9 ± 1.1 / 9.4 ± 1.1	9.8 ± 0.5	6.34 / 3.85	QLP, TESScut
P n=93 s76	3384.4553 ± 0.0007	76.4 ± 0.9 / 77.3 ± 0.8	68.8 ± 0.6	6.76 / 2.26	QLP, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth own TESScut ap. (ppt)	T14 / T23 (h)	Pipelines
S n=93 s77	3396.2636 ± 0.0050	10.4 ± 0.7 / 10.1 ± 0.8	9.4 ± 0.5	8.89 / 2.96	QLP, TESScut
P n=94 s77	3401.9005 ± 0.0008	75.4 ± 1.0 / 75.5 ± 0.8	71.8 ± 0.7	6.98 / 2.10	QLP, TESScut
S n=94 s77	(3413.73)	in the 3407.0–3417.2 data gap	–	–	none
P n=95 s77	3419.3449 ± 0.0008	74.9 ± 1.0 / 74.1 ± 0.9	70.9 ± 0.5	7.22 / 1.89	QLP, TESScut
S n=103 s83	3570.7490 ± 0.0055	9.6 ± 1.0 / 9.9 ± 0.9	9.4 ± 0.7	6.33 / 3.35	QLP, TESScut
P n=104 s83	3576.3613 ± 0.0011	74.6 ± 1.3 / 74.4 ± 1.1	68.3 ± 0.7	7.03 / 2.05	QLP, TESScut
S n=104 s84	3588.1988 ± 0.0068	9.8 ± 1.3 / 9.9 ± 1.3	9.1 ± 0.7	5.67 / 3.36	QLP, TESScut
P n=105 s84	3593.8100 ± 0.0005	76.4 ± 0.7 / 76.6 ± 0.6	65.5 ± 0.5	7.07 / 2.09	QLP, TESScut
S n=105 s84	3605.6254 ± 0.0035	12.8 ± 0.7 / 12.6 ± 0.6	10.2 ± 0.4	7.32 / 2.48	QLP, TESScut

- **Stacked shape** (200-s QLP, `s08_shape_oddeven.out`):
- Primary: 75.41 ± 0.39 ppt, T14 = 7.028 ± 0.034 h, T23 = 2.08 ± 0.05 h. This is a near-V shape: a grazing or partial eclipse.
- Secondary: 10.24 ± 0.40 ppt, T14 = 6.66 ± 0.25 h, T23 = 2.78 ± 0.36 h.
- **Period** (`s05_ephem.out`):
- 11 primaries alone: P = 17.446025 ± 0.000017 d (χ^2 = 56/9; O–C within ±7 min). 9 secondaries alone: 17.445724 ± 0.000099 d.
- The joint fit gives P = 17.446019 ± 0.000019 d, +0.0162 d from Gaia's period = **+1.83σ_Gaia**.
- The cycle count is unambiguous: consecutive primaries within s56, s57 and s77 are 17.44 d apart.
- **Odd/even** (QLP fixed shape): 76.25 ± 0.38 vs 75.36 ± 0.33 ppt. The difference is 1.8σ, smaller than the 1.5–2.6 ppt scatter between sectors. A secondary of different depth exists at phase 0.678, so 2P is ruled out, and phase 0.5 is empty, so P/2 is ruled out.
- **Secondary phase by season** (`s17_geometry2.out`): 0.67954 ± 0.00026 (s16–17), 0.67854 ± 0.00033 (s57), 0.67750 ± 0.00025 (s76–77), 0.67775 ± 0.00016 (s83–84).
- **Physics:** f(m) = 0.0334 $M_\odot$. With the TIC M1 = 1.35 $M_\odot$, m2 ≥ 0.48 $M_\odot$ and a = 34.6 $R_\odot$. The depth ratio S/P = 0.136 implies a TESS-band surface-brightness ratio ≈ T2 3830 K, consistent with an early-M companion with R2 ≳ 0.5 $R_\odot$. A central eclipse at conjunction A would last ~9.5 h; the observed 7.0 h and V shape mean a high impact parameter. The duration ratio S/P = 0.948 ± 0.036 gives e sin ω ≈ −0.03 ± 0.02, against Gaia −0.086 ± 0.044. The equal-b assumption behind that number is weak for grazing eclipses.

4. TIC 407877153 (P = 17.089 d, e = 0.13)

Measurements (`s18_event_table.out`). TESS-SPOC depths are fixed-shape fits where available.

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
S n=−1 s12	1626.5135 ± 0.0171 (TESScut)	QLP: 17 flagged	–	2.9 ± 0.7	–	TESScut
P n=0 s12	1634.2635 ± 0.0017	27.5 ± 0.5 / 27.4 ± 0.6	29.2 ± 0.5	30.1 ± 0.7	9.44 / 6.10	QLP, SPOC, TESScut
S n=0 s12	1643.5993 ± 0.0077	3.2 ± 0.6 / 2.9 ± 0.5	3.6 ± 0.5	2.9 ± 0.6	7.62 / 7.29	QLP, SPOC, TESScut

Event	Mid BTJD	Depth QLP SAP free / fixed (ppt)	Depth TESS-SPOC SAP (ppt)	Depth own TESScut (ppt)	T14 / T23 (h)	Pipelines
P n=1 s12	1651.3509 ± 0.0012	27.5 ± 0.6 (partial)	29.4 ± 0.5	29.9 ± 0.6	9.72 / 5.81	QLP (partial), SPOC, TESScut
S n=1 s13	1660.7055 ± 0.0069	3.0 ± 0.5	2.9 ± 0.4	3.1 ± 0.4	7.46 / 6.61	QLP, SPOC, TESScut
P n=2 s13	(1668.43)	in the 1667.6–1668.7 gap	gap	–	–	none
S n=2 s13	1677.7897 ± 0.0079	2.9 ± 0.5 / 2.6 ± 0.4	3.0 ± 0.3	2.9 ± 0.4	7.81 / 7.10	QLP, SPOC, TESScut
P n=43 s39	2369.1015 ± 0.0009	29.5 ± 0.4 / 29.0 ± 0.4	28.6 ± 0.4	29.5 ± 0.4	9.53 / 6.10	QLP, SPOC, TESScut
S n=43 s39	2378.4374 ± 0.0064	3.8 ± 0.7 / 3.0 ± 0.4	2.9 ± 0.3	3.2 ± 0.3	8.62 / 6.29	QLP, SPOC, TESScut
P n=44 s39	2386.1901 ± 0.0010	29.6 ± 0.5 / 29.3 ± 0.4	28.7 ± 0.4	29.4 ± 0.4	9.65 / 6.05	QLP, SPOC, TESScut
S n=128 s93	3831.0358 ± 0.0104	4.2 ± 0.6 / 3.5 ± 0.4	–	3.0 ± 0.4	9.27 / 2.83	QLP, TESScut
P n=129 s93	3838.7816 ± 0.0010	27.1 ± 0.4 / 27.3 ± 0.4	–	29.5 ± 0.4	9.38 / 6.55	QLP, TESScut
S n=129 s93	3848.1399 ± 0.0053	2.9 ± 0.3 / 2.9 ± 0.4	–	3.7 ± 0.4	7.66 / 6.63	QLP, TESScut
S n=130 s94	3865.2086 ± 0.0068	3.2 ± 0.3 / 3.2 ± 0.3	–	4.2 ± 0.4	9.56 / 5.54	QLP, TESScut
P n=131 s94	3872.9595 ± 0.0013	24.7 ± 0.4 / 24.3 ± 0.4	–	29.6 ± 0.5	9.85 / 5.89	QLP, TESScut
P n=143 s100	4078.0293 ± 0.0011	26.8 ± 0.4 / 26.7 ± 0.4	–	31.0 ± 0.5	9.45 / 6.20	QLP, TESScut
S n=143 s100	(4087.36)	64 flagged cadences	–	–	–	none
P n=144 s100	4095.1193 ± 0.0010	24.7 ± 0.4 / 24.8 ± 0.3	–	29.4 ± 0.4	9.51 / 6.45	QLP, TESScut
S n=144 s101	4104.4747 ± 0.0102	2.9 ± 0.3 / 2.6 ± 0.2	–	no cube	12.06 / 3.76	QLP
P n=145 s101	(4112.23)	156 flagged cadences	–	no cube	–	none
S n=145 s101	4121.5442 ± 0.0073	3.0 ± 0.3 / 2.8 ± 0.3	–	no cube	8.90 / 4.73	QLP
P n=146 s102	4129.2998 ± 0.0009	25.0 ± 0.3 / 24.9 ± 0.4	–	29.5 ± 0.4	9.37 / 6.28	QLP, TESScut
S n=146 s102	4138.6654 ± 0.0176 (TESScut)	QLP: 155 flagged	–	4.3 ± 1.1	–	TESScut
P n=147 s102	4146.3904 ± 0.0024	23.7 ± 0.7 (ingress lost)	–	25.0 ± 0.7	9.11 / 6.53	QLP (partial), TESScut

- **Stacked shape** (200-s QLP):
- Primary: 25.53 ± 0.16 ppt (QLP-diluted), T14 = 9.459 ± 0.051 h, T23 = 6.38 ± 0.08 h. **Flat-bottomed**, a total transit.
- Secondary: 2.98 ± 0.13 ppt, T14 = 8.46 ± 0.31 h, T23 = 5.71 ± 0.50 h. **Flat-bottomed**, an occultation.
- **Period:**
- 9 primaries: P = 17.089294 ± 0.000006 d (χ^2 = 6.0/7; O–C within ±2.3 min). 9 secondaries: 17.089307 ± 0.000059 d.
- Joint fit: P = 17.089294 ± 0.000008 d, **−0.14σ_Gaia**.
- Secondary phase by season: 0.54630–0.54713, steady.

- **Odd/even:** TESS-SPOC 28.91 ± 0.31 vs 28.91 ± 0.30 ppt. The QLP odd/even difference (26.10 vs 26.91) comes from which sectors fall in each group: QLP s93+ are diluted. 2P and P/2 are excluded as for the other star.
- **Physics:**
- f(m) = 0.0216 M⊙. With the TIC M1 = 0.87 M⊙, m2 ≥ 0.31 M⊙ and a = 29.5 R⊙.
- The flat-bottom depth gives k ≈ 0.16, so R2 ≈ 0.31 R⊙ with the TIC R1 = 1.91 R⊙. That matches a 0.31 M⊙ M dwarf.
- The occultation/transit ratio of 0.117 gives T2 ≈ 3200 K (TESS-band Planck).
- A central T14 at conjunction A would be 10.6 h; observed 9.46 h, so b is moderate.
- The duration ratio S/P = 0.894 ± 0.033 gives e sin ω = −0.056 ± 0.018, against Gaia −0.078 ± 0.066.
- Everything hangs together as a subgiant + M-dwarf SB1.

Geometry vs Gaia

- **Method** (`s07_geometry.out`, `s17_geometry2.out`; 20,000 Monte Carlo draws): I propagated P, T_p, e and ω; conjunction A is at ν = 90° − ω and B at ν = 270° − ω. The ω–T_p correlation was taken either as 0 or as the last `corr_vec` element (0.927 and 0.985). That second case assumes the SB1 parameter order ends with (arg_periastron, t_periastron), which I have not verified; results are given both ways.
- **283532878:**
- Predicted B–A separation: 0.7122 ± 0.0237. Measured secondary phase: 0.67812 ± 0.00030, **−1.44σ**. The other assignment (primary = B) is off by 16σ.
- The measured phase implies e cos ω = 0.2823 ± 0.0012 (exact Kepler, Gaia e sin ω draws), against Gaia 0.338 ± 0.039 (–1.42σ).
- Absolute timing, propagated with the TESS period: the primary is +0.0112 phase (+0.20 d) from conjunction A (0.37σ uncorrelated, 0.71σ correlated) and 0.30 phase from B (11.9σ / 23.7σ). Propagated with Gaia's period, A is still within 1.7–2.0σ.
- **The deep eclipse is A:** the F-type RV star is eclipsed by its companion. **Pass.**
- **407877153:**
- Predicted separation 0.5589 ± 0.0435; measured 0.54690 ± 0.00017, **−0.28σ**. The other assignment is off by 2.4σ.
- Implied e cos ω = 0.0733 ± 0.0006, against Gaia 0.093 ± 0.068.
- The primary is +0.0167 phase (+0.29 d) from A (0.16σ / 0.64σ) and 5.1σ / 18.9σ from B.
- **The deep eclipse is A**, the subgiant RV star transited by an M dwarf. **Pass.**

Ephemerides and next eclipses

From `s05_ephem_*.json` and `s17_geometry2.out`. UTC times are geocentric: I removed the barycentric light-time with astropy (offset given in brackets). - **TIC 283532878:** T0 = BJD_TDB 2458761.97572 ± 0.00150 (BTJD 1761.97572), **P = 17.446019 ± 0.000019 d**. Secondary T0 = 1773.80624 ± 0.00535 (phase 0.67812). - Next primaries: BTJD 4343.9865 = 2026-10-30 11:36 UTC [+3.4 min]; 4361.4325 = 2026-11-16 22:18; 4378.8785 = 2026-12-04 09:02. Each ±2.3 min formal; T14 = 7.0 h. - Next secondaries: BTJD 4355.817, 4373.263, 4390.709. The secondary phase has moved by ~0.002 between seasons, so allow ±1 h. - **TIC 407877153:** T0 = BJD_TDB 2458634.26184 ± 0.00084 (BTJD 1634.26184), **P = 17.089294 ± 0.000008 d**. Secondary T0 = 1643.60789 ± 0.00296 (phase 0.54690).

- Next primaries: BTJD 4351.4596 = 2026-11-06 23:03 UTC [−3.1 min]; 4368.5489 = 2026-11-24 01:13; 4385.6382 = 2026-12-11 03:22. Each ±1.0 min; T14 = 9.5 h. - Next secondaries: BTJD 4343.716, 4360.806, 4377.895.

Caveats

- **Secondary-phase drift in 283532878:** the phase falls 0.0018 between 2019 and 2024 (≈ 45 min, ~4σ formal), and the joint timing χ^2 is 132/17. GR plus tidal apsidal motion at a/R1 ≈ 18 would give far less (~2.5×10^{-4} rad over 105 cycles, against the ~0.03 rad needed). I therefore treat this as underestimated timing errors for the 10 ppt, red-noise-affected secondaries, not as physics. It does not affect the Gaia test, whose tolerance is ±0.024.
- **Catalogued EB 29.7″ from 283532878:** this is G 17.66, P = 1.7975 d, eclipses 40–60% deep in G. With 0.4–0.6% of the light in my apertures it can add dips of ≤ 3 ppt at unrelated phases. That may explain some event-to-event scatter in the secondary depths (for example S n=105, 12.6 ppt). It cannot produce the secondary: the period is wrong, the depth needed is 181–240%, and the pixels reject it.
- **Gaussian-PSF limits:** the pixel model uses a Gaussian PSF. Formal centroid offsets for the primaries (up to 1.5″ for 283532878; 5.1″ and 7.6″ in s93 and s102 for 407877153, the s102 stack including a partial event) exceed their noise-only errors. That is model systematics; every neighbour is ≥ 26″ away and rejected by $\Delta\chi^2$ in the hundreds or more.
- **Deblended depths are model-dependent:** for 283532878, about 71–77 ppt (PSF amplitude) up to ~80 ppt (aperture depth ÷ target fraction). The TIC contamination ratio is 0.32. QLP's SAP depths for 407877153 in s93–s102 are diluted by ~10–15%.
- **Not used:** ZTF (too few epochs), 2-min SPOC (none exists), TESScut for s101 of 407877153 (not downloaded; QLP covers its two secondaries).
- **Correlation assumption:** the `corr_vec` ordering behind the correlated Monte Carlo case is assumed, not verified. The secondary-phase test does not use T_p at all.

Files

All in `/home/user/eclipses-paper-review/work/c9a-vets/vet04/`: - **Shared code:** `common.py` (loader, Gaia constants), `fitlib.py` (trapezoid fitter). - **Earlier agent, read and kept:** - `s01_download` (QLP/TESS-SPOC light curves); `s02_inventory` (sectors, gaps, flags); `s03_blind` (blind dip census); `s04_events` (+ `.json`, per-event free fits); `s05_ephem` (+ `s05_ephem_<tic>.json`); `s06_gaia_query` (+ `s06_gaia_nss.csv`, `s06_gaia_source.csv`); `s07_geometry` (Monte Carlo conjunctions); `s08_shape_oddeven` (+ `.json`, `s08_stack_*.npz`); `s10_tesscut_dl`; `s11_pixels` (+ `_<tic>.out/.json`, `s11_diff_*.npy`; aperture photometry, dilution). - **This run:** - `s12_archives.out` (first pass, false negatives), `s12b_archives` (SIMBAD by coordinates, ExoFOP), `s12c_controls` (failed Algol control), `s12d_recheck` (cones with passing control), `s12e_neighbour_eb` (neighbour EB rows). - `s13_pixnoise` (+ `_<tic>.out/.json`, `s13_diff_*.npz`), `s13b_summary`. - `s14_ztf`, `s15_plot_diff` (+ `s15_diff_*.png`), `s16_commonmode`, `s17_geometry2`, `s18_event_table`. - **Downloads:** `/tmp/claude-0/-home-user-eclipses-paper-review/55c92ff4-798a-544f-be03-9888c6804905/scratchpad/data/vet04/` (TESScut cubes deleted).

TIC 283532878: CONFIRMED; P_ecl=17.446019±0.000019 d; depth1=75 ppt (QLP SAP; 71–77 deblended); depth2=10 ppt at phase 0.6781; pixel: on target; pipelines: 2 (QLP, own TESScut); Gaia geometry: pass TIC 407877153: CONFIRMED; P_ecl=17.089294±0.000008 d; depth1=29 ppt (TESS-SPOC/TESScut; 25–27 QLP s93+); depth2=3.0 ppt at phase 0.5469; pixel: on target; pipelines: 3 (QLP, TESS-SPOC, own TESScut); Gaia geometry: pass

vet05 full report

Archived source: sanitized report text.

vet05: adversarial check of TIC 15855794 (UCAC4 564-009141) and TIC 92881948 (WISE J053256.98-225927.8)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This run resumed from a predecessor that was killed by a container restart; that run had finished s1–s9 and part of s10. This run added s10b/c/d (archives), s11 (own TESScut photometry, difference images, dilution), s12 (master event table, ephemerides), s13 (Gaia geometry MC) and s14 (physics). The coordinator cut the budget, so the report is short. Scripts and outputs are in this folder. Downloads are in the scratchpad `data/vet05/<tic>/`; all TESScut cubes have been deleted. Depths are in ppt, times in BTJD = BJD_TDB − 2457000. "C" = own TESScut photometry with a 3×3 aperture, "Q" = QLP SAP, "S" = TESS-SPOC SAP (QUALITY==0 throughout).

TIC 15855794: CONFIRMED. (1) Repeating events: I measured 12 primaries and 11 secondaries one by one from raw flux. (2) Period: the eclipse period is 5.8911458 ± 0.0000038 d, +0.54σ from Gaia. (3) Pixels: in s44, s70 and s71 the difference-image deficit lies 0.4–0.9″ from the target, and every neighbour is excluded. (4) Pipelines: Q, S and C agree on depth to within 4%. (5) Geometry: the primary falls at Gaia conjunction A (+0.03 ± 0.08 d at the Gaia epoch; conjunction B is excluded at ~40σ), and the secondary phase of 0.50182 matches Gaia's prediction of 0.502 ± 0.021. Corrections to the screen: the primary is 35.3 ppt after dilution correction, not "probably larger", and it lasts 3.25 h. The secondary is 5.0–5.4 ppt.

TIC 92881948: CONFIRMED. (1) Repeating events: 24 primaries and 22 secondaries measured one by one. (2) Period: 8.8049754 ± 0.0000013 d, +0.43σ from Gaia. (3) Pixels: in all 7 sectors the deficit lies 0.3–1.6″ from the target, and the depth scales with the target's flux fraction in the aperture. (4) Pipelines: Q, S and C agree once dilution is accounted for. (5) Geometry: the primary falls at conjunction A (+0.06 ± 0.09 d; B is excluded at ~44σ). The secondary phase is 0.48334 ± 0.00008, so e cos ω = −0.026, which is 1.0σ from Gaia's prediction of 0.502 ± 0.018. Corrections to the screen: the primary depth is 16.4 ppt (Q) / 18.3 ppt (C) / 19.0 ppt dilution-corrected, not ~11 ppt. It lasts 1.69 h, not the 1.2 h that BLS gave. The secondary is 5.0–5.3 ppt at phase 0.4833.

Not tested for either star: the common-mode test (brief item 8, outside the five protocol checks) and next-eclipse UTC predictions (budget).

1. Archives and web (s10, s10b, s10c, s10d)

- **15855794:** no rows within 60″ in VSX, Prsa+2022 (J/ApJS/258/16), IJspeert+2021, Gaia DR3 vari_eclipsing_binary / I/358/veb, or SB9. It is not in Gaia vari_summary. SIMBAD: SB*, 1 bibcode (2023A&A...674A..34G). ExoFOP: TOIs 0, CTOIs 0.
- **92881948:** VSX has one entry at 1.35″: "Gaia DR3 2964506347485091072", type **ROT**, G 12.76–12.80. Nothing in Prsa, IJspeert, Gaia-EB or SB9. Gaia vari_summary: in_vari_eclipsing_binary = 0; the classifier's best class is SOLAR_LIKE (0.63). SIMBAD: SB*, 1 bibcode. ExoFOP: TOIs 0, CTOIs 0.
- **Neither star is a catalogued eclipsing binary.**

- **Positive controls:** each catalogue cone found a known row (Prsa, I/358/veb, IJspeert and SB9 via control rows; VSX via the 92881948 row itself).
- **Brief error:** the brief lists "I/357/varisum", but Gaia DR3 variability is I/358 (I/357 is the NSS part), so Gaia variability was queried through TAP.

2. Independence, systematics, pixels (s11_*.out, s12_compile.out, s14_physics.out)

- **Depths, 15855794 (all events):**
- Primary: Q 35.84 ± 0.27 (n=11), S 34.56 ± 0.32 (n=4), C 34.90 ± 0.23 (n=12), C dilution-corrected 35.34 ± 0.23.
- Secondary: Q 5.00 ± 0.30, S 4.91 ± 0.43, C 5.36 ± 0.24.
- The C primary χ^2 of 34/11 comes from E124 (27.6 ppt, at the end of the s70 orbit, a cadence QLP flags).
- **Depths, 92881948:**
- Primary: Q 16.43 ± 0.24, S 18.20 ± 0.24, C 18.33 ± 0.18, C r2.5 15.90 ± 0.23. Divided by the modelled target fraction, 3×3 gives 18.99 ± 0.19 and r2.5 gives 19.17 ± 0.28: they agree.
- QLP is shallower because its larger aperture takes in a G 13.67 star (60″) and a G 14.56 star (53″). TESS-SPOC CROWDSAP is 0.90–0.94.
- Secondary: Q 5.02 ± 0.21, S 5.33 ± 0.24, C 5.08 ± 0.18.
- **Difference images** (median of the eclipse core minus median of the flanks, per sector; pixel-integrated Gaussian fit; errors from 200 fake-event stacks):
- 15855794, primary: offsets from the target 0.4″ (s44, 1.1σ), 0.5″ (s70 without E124, 0.8σ; the run including E124 is distorted by that event) and 0.9″ (s71, 2.1σ). Every neighbour is >45″ from the centroid at >100σ. Secondary: 1.2–3.5″, ≤1.0σ.
- 92881948, primary: offsets 1.3″ (s5), 1.6″ (s6), 1.4″ (s32), 0.9″ (s98), 0.3″ (s107), 1.5″ (s108), 0.9″ (s109), all ≤2.8σ, with errors of 0.02–0.06 px. The nearest neighbour is at 29″. In every sector the pixel with the deepest fractional dip is the target pixel or one next to it.
- **Dilution arithmetic** (Gaussian PSF, and a PSF 1.5× broader):
- 15855794: in the 3×3 aperture the brightest neighbour (G 16.03, 45″) contributes ≤1.9% of the flux. It would need an intrinsic depth of ≥180% to make the primary: excluded. All other neighbours contribute ≤0.6%: excluded.
- 92881948: in 3×3, G 13.67 (60″) contributes 0.11–1.44% and would need ≥127%: excluded. G 14.56 contributes ≤3.95% and would need ≥46%; it is not excluded by arithmetic alone, but it is excluded by the centroid (53″ away) and because the depth grows in the smaller aperture. G 16.46 (29″) contributes ≤2.4% and would need ≥75%; it is excluded by the centroid.

3. TIC 15855794 (joint shape fit, s6: P 35.25 ± 0.31 ppt, T14 3.255 ± 0.025 h, T23 0.70 ± 0.03 h; S 4.89 ± 0.38 ppt, T14 3.14 ± 0.19 h)

There is no 2-min SPOC data. The s43 TESScut cut-out is empty (RAW_CNTS = –1, s11_15855794_s43_check.out).

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E–1	2500.18	lost (gap/flags)	–
P E0	2503.12275 ± 0.00077	34.25 / 34.40 / 34.20	S Q C
S E0	2506.0791 ± 0.0049	5.10 / 4.79 / 4.99	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
P E1	2509.01328 ± 0.00066	34.46 / 35.24 / 36.59	S Q C
S E1	2511.97	lost (gap)	–
P E2	2514.90506 ± 0.00059	34.45 / 36.19 / 34.79	S Q C
S E2	2517.8619 ± 0.0062	4.54 / 5.00 / 4.47	S Q C
P E3	2520.79709 ± 0.00074	35.07 / 36.27 / 35.79	S Q C
S E3	2523.7436 ± 0.0048	4.99 / 5.99 / 5.26	S Q C
P E120	3210.06068 ± 0.00106	– / 37.03 / 35.22	Q C
S E120	3213.0275 ± 0.0078	– / 4.10 / 5.43	Q C
P E121	3215.95111 ± 0.00085	– / 36.48 / 35.55	Q C
S E121	3218.9089 ± 0.0063	– / 4.91 / 5.50	Q C
P E122	3221.84	lost (flags, all pipelines)	–
S E122	3224.8002 ± 0.0061	– / 5.38 / 5.60	Q C
P E123	3227.73365 ± 0.00081	– / 37.28 / 34.94	Q C
S E123	3230.6864 ± 0.0057	– / 5.35 / 5.74	Q C
P E124	3233.6255 ± 0.0019	– / flagged / 27.64 (end of orbit, suspect)	C
S E124	3236.5893 ± 0.0065	– / 4.79 / 5.79	Q C
P E125	3239.51638 ± 0.00085	– / 36.17 / 35.01	Q C
S E125	3242.4652 ± 0.0066	– / 4.37 / 5.01	Q C
P E126	3245.40761 ± 0.00090	– / 35.35 / 34.55	Q C
S E126	3248.36	lost (flags)	–
P E127	3251.29757 ± 0.00090	– / 34.85 / 33.77	Q C
S E127	3254.2567 ± 0.0054	– / 5.55 / 5.62	Q C
P E128	3257.18858 ± 0.00085	– / 35.70 / 35.34	Q C
P E264	4058.39	lost (gap; segment "s1751")	–
S E264	4061.3461 ± 0.0060	– / – / 4.92	C

- **Period** (s12): primaries only, P = 5.8911449 ± 0.0000038 (χ^2 4.8/10). Secondaries only, P = 5.8911779 ± 0.0000229. TESScut alone, P = 5.8911437 ± 0.0000042. All are +0.53 to +0.56σ from Gaia's 5.890538 ± 0.001133.
- **Odd/even primaries:** Δ = +0.6σ (S), +0.7σ (Q), +1.8σ (C). There is no hint of 2P.
- **Primary O−C** scatter is ≤1.2 min.

4. TIC 92881948 (s6: P 16.63 ± 0.45 ppt [Q/S fast cadence], T14 1.692 ± 0.030 h, T23 0.20 ± 0.05 h (V-shaped); S 5.00 ± 0.35 ppt, T14 1.76 ± 0.12 h)

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E−1	1440.6495 ± 0.0027	6.09 / 5.99 / 4.97	S Q C
P E0	1445.2014 ± 0.0011	16.40 / 16.19 / 18.08	S Q C
S E0	1449.4544 ± 0.0028	4.41 / 4.01 / 5.19	S Q C
P E1	1454.0066 ± 0.0010	17.78 / 16.75 / 18.38	S Q C
S E1	1458.2549 ± 0.0035	5.06 / 5.73 / 6.65	S Q C
P E2	1462.8129 ± 0.0010	18.96 / 16.96 / 18.83	S Q C
P E3	1471.6159 ± 0.0006	19.98 / flagged / 19.95	S C
S E3	1475.8763 ± 0.0021	6.41 / 4.72 / 5.95	S Q C
P E4	1480.4223 ± 0.0006	18.24 / 17.06 / 17.14	S Q C
S E4	1484.6802 ± 0.0035	4.51 / 3.93 / 3.63	S Q C
P E5	1489.2272 ± 0.0011	17.53 / 15.78 / 18.18	S Q C
P E83	2176.0155 ± 0.0007	17.85 / 17.50 / 18.44	S Q C

Event	Mid BTJD	Depth ppt (S / Q / C)	pipelines
S E83	2180.2716 ± 0.0022	5.09 / 4.98 / 5.29	S Q C
P E84	2184.8210 ± 0.0007	17.48 / 14.82 (partial) / 18.06	S Q C
S E84	2189.0738 ± 0.0033	4.85 / 4.97 / 5.33	S Q C
P E85	2193.6244 ± 0.0008	18.44 / 17.15 (partial) / 17.92	S Q C
S E85	2197.8764 ± 0.0020	5.93 / 5.49 / 5.81	S Q C
P E289	3989.8400 ± 0.0014	– / flagged / 17.88	C
S E289	3994.0970 ± 0.0035	– / flagged / 4.38	C
P E290	3998.6438 ± 0.0007	– / 15.62 / 17.53	Q C
S E290	4002.90	lost (gap)	–
P E291	4007.4468 ± 0.0012	– / flagged / 19.97	C
S E291	4011.7102 ± 0.0022	– / 4.56 / 5.24	Q C
P E292	4016.2547 ± 0.0007	– / 16.51 / 18.36	Q C
S E292	4020.5174 ± 0.0073	– / flagged / 3.79	C
P E293	4025.0597 ± 0.0008	– / 16.72 / 18.94	Q C
S E293	4029.3183 ± 0.0022	– / 5.16 / 5.29	Q C
P E294	4033.8637 ± 0.0015	– / flagged / 21.07	C
S E294	4038.1176 ± 0.0031	– / flagged / 5.58	C
P E295	4042.6692 ± 0.0007	– / 15.56 / 17.12	Q C
P E320 / E321 / E322 / E323 (s107)	4262.7954 ± 0.0013 / 4271.5989 ± 0.0008 / 4280.4041 ± 0.0008 / 4289.2104 ± 0.0008	C: 19.30 / 18.47 / 18.99 / 17.11	C
S E320 / E321 / E322 (s107), E323 (s108)	4267.0482 ± 0.0031 / 4275.8536 ± 0.0029 / 4284.6636 ± 0.0036 / 4293.4656 ± 0.0032	C: 4.83 / 4.72 / 3.76 / 4.79	C
P E324 (s108)	4298.01	partial (no pre-ingress baseline)	–
S E324 / E325 (s108), E326 (s109)	4302.2707 ± 0.0029 / 4311.0736 ± 0.0024 / 4319.8790 ± 0.0039	C: 4.47 / 5.87 / 6.60	C
P E325 / E326 (s108)	4306.8209 ± 0.0011 / 4315.6232 ± 0.0008	C: 17.93 / 18.20	C
P E327 / E328 (s109)	4324.4292 ± 0.0009 / 4333.2316 ± 0.0021	C: 18.87 / 18.88	C
S E327 / E328 (s109)	4328.6871 ± 0.0024 / 4337.4867 ± 0.0031	C: 5.75 / 5.60	C

- **Sectors without other pipelines:** sectors 107–109 exist only in TESScut (no QLP or SPOC light curves). There is no 2-min SPOC data.
- **Period:** primaries only, P = 8.8049749 ± 0.0000013 (χ^2 26.8/22). Secondaries only, P = 8.8049820 ± 0.0000049. TESScut alone, P = 8.8049753 ± 0.0000014. All are +0.43σ from Gaia's 8.804360 ± 0.001441.
- **Odd/even primaries:** +1.6σ (S), +0.2σ (Q), +0.7σ (C).
- **Secondaries** are not at phase 0.5 and differ from the primaries in depth, so this is not an equal-pair P/2 system.

Geometry vs Gaia (s13_geometry.out; 20000 draws, NSS correlation matrix)

- **How the test works:** with ω and T_peri correlated at +0.9996, conjunction times are well defined. Without the correlations the prediction is uninformative (σ ≈ 2 d). The order of the first four corr_vec parameters is uncertain, so all 24 orders were run; they give the same answer.
- **15855794:**
- Observed primary minus conjunction A, at the Gaia epoch: +0.030 d (robust σ 0.080; over the 24 orders +0.23 to +0.46σ). Minus conjunction B: −2.93 d (−39σ).

- Carried forward with Gaia's P: +0.65σ.
- Secondary phase: predicted 0.5019 ± 0.0209, observed 0.50182 ± 0.00030; e cos ω = +0.0029 ± 0.0005, against Gaia's +0.003 ± 0.033.
- **92881948:**
- Primary minus conjunction A: +0.060 d (σ 0.086; +0.35 to +0.69σ). Minus conjunction B: −43σ.
- Forward with Gaia's P: +0.52σ.
- Secondary phase: predicted 0.5016 ± 0.0181, observed 0.48334, a 1.0σ difference (32% of draws lie as far from 0.5). e cos ω = −0.0262, against Gaia's +0.003 ± 0.028.
- **Which conjunction:** in both stars the primary is conjunction A, the companion passing in front of the RV star.
- **Physics (s14, uniform discs):**
- 15855794: f(m) = 0.0437 M⊙, so m2 ≥ 0.49 M⊙ (M dwarf, about 3630 K). The observed T14 needs b ≈ 1.07 (grazing). Predicted depths are 32.2 / 4.2 ppt; observed 35.3 / 5.4 ppt. The depth ratio implies a companion at 3760 K.
- 92881948: f(m) = 0.0144 M⊙, so m2 ≥ 0.22 M⊙. Grazing, b ≈ 1.22. Predicted 14.6 / 4.0 ppt; observed 19.0 / 5.3 ppt. The depth ratio of 0.277 matches the Planck ratio (0.275) for a 3180 K companion.
- Both are physically sane.

Ephemerides and next eclipses (s12_compile.out)

- **15855794:** T_ref = BTJD 2886.04720 ± 0.00024 (BJD_TDB 2459886.04720), P = 5.8911458 ± 0.0000038 d. Secondary at +2.95631 ± 0.00179 d.
- **92881948:** T_ref = BTJD 3047.70744 ± 0.00019 (BJD_TDB 2460047.70744), P = 8.8049754 ± 0.0000013 d. Secondary at +4.25578 ± 0.00070 d.
- **Next three eclipses in UTC:** not computed (budget cut). Use T_ref + nP. Those times are BJD_TDB, so no barycentric correction has been applied for an observer.

Caveats

- **Common mode:** not tested with other stars. Partial substitutes: the residual sky and the neighbour pixels show no dip at the target's fractional depth, and the depth scales with the target fraction.
- **TESScut times:** taken from the FFI time stamps; the barycentric correction is for the FFI, not the target (errors under a minute).
- **15855794 E124:** shallow (27.6 ppt), at the end of an orbit, and flagged by QLP. Its primary-stack difference image was redone without it (s11_15855794_s70_exclE124).
- **PSF model:** a Gaussian with a 1.5× variant, not the TESS PRF. Neighbour exclusions rely on the 3×3 arithmetic plus the centroids.

Files

`s1–s9` (predecessor run); `s10b_archives_redo`, `s10c_vizier_check`, `s10d_controls_gaiavar` (.py/.out); `s11_tesscut.py` with `s11_<tic>_s<sec>.{out,json}`, `_lc.npz`, `_diff.npz`, `s11b_redownload_s70`, `s11_15855794_s43_check.out`; `s12_compile.{py,out,json}`; `s13_geometry.{py,out,json}`; `s14_physics.{py,out,json}`; plus `common.py` and `fitlib.py`.

vet06 full report

Archived source: sanitized report text.

vet06: adversarial check of TIC 468827621 (UCAC4 796-015685) and TIC 416285595 (UCAC4 571-080398)

Opus 5.5 subagent, 2026-10-30, for Fable 5.1. This is a reduced-depth run: the coordinator cut the budget mid-task. It covers the five protocol checks only; physics, next-eclipse lists, the common-mode test and the literature search were skipped.

Light curves were fetched from MAST's `Download/file` endpoint (`s2`), because `lightkurve.search_lightcurve` hung at MAST and archive.stsci.edu serves an incomplete TLS chain. Data are in `scratchpad/data/vet06/<tic>/`. Each TESScut cube was deleted after use.

Depths are in ppt. Times are BTJD (BJD_TDB − 2457000). A = the companion passes in front of Gaia's RV star ($\nu = 90° - \omega$); B = the companion is behind it.

TIC 468827621: CONFIRMED. All five checks pass: 1. **Individual events.** 10 primary and 9 secondary epochs are fitted individually from raw SAP (QUALITY==0) in QLP s19, 20, 59, 60, 73 and TESS-SPOC s19, 20. 2. **Period.** The eclipse period is 11.041067 ± 0.000001 d (primaries, O−C rms 0.5 min). It is −0.16σ from Gaia's 11.04138 ± 0.00196 d. The secondaries alone give 11.041068 ± 0.000005 d. 3. **Pixels.** The primary deficit is 1.42″ (s19) and 1.50″ (s59) from the target's WCS position. The only neighbour bright enough in principle is G 13.40 at 70.6″. With its PRF share of my aperture it would need a 4800–7800% intrinsic eclipse, and the fractional deficit at its pixel is only 1–4 ppt. Both other neighbours need more than 100% even in the worst case. 4. **Pipelines.** QLP, TESS-SPOC and my TESScut aperture all see the primary: 192, 184 and 192 ppt. 5. **Geometry.** The primary is at conjunction A (−0.14 ± 0.09 d); B is excluded at 31σ (20σ without Gaia's correlations). The secondary phase, **0.72649 ± 0.00003**, is +0.56σ from Gaia's predicted 0.718 ± 0.015 (from e = 0.355, ω = 350°). The eclipse at A, nearer periastron, is the longer one: 6.0 h against 5.05 h. That is the direction $e \sin\omega = -0.06$ requires.

The screen was right on P, on both depths (QLP) and on the secondary phase: 0.7265 measured against its by-eye 0.72. TESS-SPOC SAP depths are about 4% shallower than QLP (184/29.6 against 192/31.5 ppt).

TIC 416285595: CONFIRMED. All five checks pass, with weak points below: 1. **Individual events.** 21 primary and 19 secondary epochs are fitted individually: QLP s14, 40, 54, 80, 81 and TESS-SPOC s40. 2. **Period.** P = 4.615357 ± 0.000003 d (O−C rms 2.4 min), −0.37σ from Gaia's 4.615565 ± 0.000559 d. 3. **Pixels.** The primary deficit is 2.16″ (s40, camera 1) and 0.58″ (s80, camera 2) from the target. Every one of the 43 neighbours needs more than 100% with its PRF share; the minimum is 126%, for G 14.75 at 37.9″ in s40. The neighbours that pass only in the unphysical worst case (100% of their flux in my aperture) all lie 37″ or more away, at least 1.8 px from a deficit centre that sits within 0.1 px of the target. So the primary is on the target, and the screen's contamination concern (ratio 0.27) does not change the attribution of the primary. 4. **Pipelines.** QLP and TESS-SPOC see every event at the same times and durations. Depth agreement is the weakest part (see the weak points). 5. **Geometry.** The primary is at conjunction A (−0.04 ± 0.08 d); B is excluded at 29σ with correlations, but only 3.3σ without. The secondary phase, 0.50011 ± 0.00027, is +0.28σ from Gaia's 0.49 ± 0.03.

Weak points: - **Depths disagree between pipelines.** QLP is deeper than TESS-SPOC in s40: 61.7 against 54.7 ppt, about 12% and about 4σ apart. My TESScut aperture gives 70 (s40) and 89 ppt (s80), but its background (median of pixels ≥ 4 px away) is unreliable in this crowded field. I attribute the spread to crowding and background, since none of these is a crowding-corrected depth. Strict agreement within errors fails. - **The secondary is not tested in pixels.** At 12.9 ppt, G 14.75 at 37.9″ could produce it arithmetically (it would need about 26%). Only the exact phase 0.5001 and the matching period tie it to the target. - **The geometry test is weak.** e ≈ 0.08, so phase 0.5 is the generic expectation, and the A/B decision rests on Gaia's ω–Tp correlation of 0.998.

The screen was right on P and on the secondary phase. Its 59 ppt primary depth matches QLP but not TESS-SPOC (54.7). Its 11 ppt secondary is low: QLP and TESS-SPOC both give 12.9 ppt. The primary is near-V: T14 3.6 h, trapezoid T23 about 0.7 h.

1. Archives and web

- **VizieR mirror cones (`s1_archives.out`, `s1b_archives.out`):** both stars return 0 rows within 60″ in VSX, Prša+2022, IJspeert+2021, Gaia DR3 veb / vclassre / vari (10″), SB9, ASAS-SN II/366 and J/AJ/162/170. Algol returns 1 VSX row, so the cones work.
- Neither star is a catalogued eclipsing binary: none is listed in VSX, Gaia-EB, Prša or IJspeert within 60″.
- 468827621 is in the Gaia DR3 short-timescale table I/358/vst (amplitude 0.044 mag) and is flagged VARIABLE in gaia_source.
- **SIMBAD:** both are SB*, with 2 bibcodes (468827621) and 1 bibcode (416285595), all Gaia DR3. The SIMBAD lookup by TIC ID fails; the 10″ coordinate cone finds both.
- **ExoFOP:** 0 TOIs and 0 CTOIs for both.
- **Gaia SB1 rows (`gaia_tbosb1.tsv`, with corvec):**
- 468827621: e 0.3555 ± 0.0239, ω 350.13 ± 3.87°, K1 42.03 km/s.
- 416285595: e 0.078 ± 0.044, ω 104 ± 36°, K1 46.4 km/s.

2. Independence, systematics, pixels

- **Pipeline depths (`s3_events.out`, per-event means):**
- 468827621 primary: QLP 191.9 (sd 3.6, N = 10); TESS-SPOC 184.5 (sd 5.0, N = 4); TESScut 191.9 (s19) and 191.7 (s59).
- 468827621 secondary: QLP 31.5 (sd 1.0); TESS-SPOC 29.6 (sd 1.2).
- 416285595 primary: QLP 58.6 (sd 3.3, N = 20); TESS-SPOC 54.7 (sd 1.3, N = 6); TESScut 69.8 (s40) and 89.4 (s80), background-limited.
- 416285595 secondary: QLP 12.9 (sd 2.5); TESS-SPOC 12.9 (sd 1.4).
- Neither star has 2-min SPOC data.
- **Odd/even (primaries):** 468827621 +0.38σ (QLP) and −0.09σ (SPOC); 416285595 −1.2σ (QLP) and +1.1σ (SPOC). The two eclipse kinds differ in depth (and, for 468827621, in duration), so there is no hidden P/2 or 2P.
- **Difference images (`s4_tesscut_<tic>_s<sec>.txt/.png`; mean over primaries of median in-eclipse minus local out-of-eclipse; Gaussian-fit centre):**

Star / sector (cam)	N ev	deficit − target WCS	frac. depth, target pixel	aperture share of target
468827621 s19 (cam 2)	2	1.42″ (deficit − OOT 1.17 ± 0.33″)	158 ppt	0.962
468827621 s59 (cam 2)	3	1.50″ (deficit − OOT 1.62 ± 0.35″)	117 ppt	0.957

Star / sector (cam)	N ev	deficit − target WCS	frac. depth, target pixel	aperture share of target
416285595 s40 (cam 1)	4	2.16″	34 ppt	0.826
416285595 s80 (cam 2)	5	0.58″	48 ppt	0.854

- For 416285595 the out-of-eclipse Gaussian locked onto a brighter star at the cutout edge, so only the WCS offset is quoted.
- Both 468827621 sectors are in camera 2. That is the only camera available.

3. TIC 468827621 (P 11.041067 d)

`events_468827621.csv`: trapezoid on a quadratic baseline; errors are the covariance scaled to the rms, inflated by a 10-shift prayer bead.

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s19	1819.85379 ± 0.00025	196.9 / 189.9	6.00 / 2.33	QLP+SPOC+TESScut
P E1 s19	1830.89477 ± 0.00025	196.6 / 188.9	5.99 / 2.34	QLP+SPOC+TESScut
P E3 s20	1852.97691 ± 0.00024	186.4 / 179.4	6.02 / 2.33	QLP+SPOC
P E4 s20	1864.01807 ± 0.00027	187.5 / 179.6	6.00 / 2.36	QLP+SPOC
P E99 s59	2912.91944 ± 0.00020	195.2 / –	6.03 / 2.31	QLP+TESScut
P E101 s59	2935.00116 ± 0.00019	194.2 / –	5.99 / 2.39	QLP+TESScut
P E102 s60	2946.04227 ± 0.00038	187.8 / –	6.03 / 2.33	QLP
P E103 s60	2957.08343 ± 0.00018	190.3 / –	6.02 / 2.31	QLP
P E133 s73	3288.31654 ± 0.00045	191.6 / –	5.96 / 2.35	QLP
P E135 s73	3310.39800 ± 0.00018	192.1 / –	5.99 / 2.36	QLP
S E−1 s19	1816.83466 ± 0.00143	– / 30.8	5.04 / 2.27	SPOC
S E1 s19	1838.91713 ± 0.00110	31.9 / 30.3	5.14 / 2.17	QLP+SPOC
S E2 s20	1849.95666 ± 0.00095	29.1 / 27.8	5.01 / 2.51	QLP+SPOC
S E3 s20	1860.99765 ± 0.00098	31.3 / 29.6	4.99 / 2.31	QLP+SPOC
S E99 s59	2920.94045 ± 0.00089	32.4 / –	5.12 / 2.21	QLP
S E100 s59	2931.98226 ± 0.00096	32.2 / –	5.04 / 2.47	QLP
S E101 s60	2943.02108 ± 0.00108	31.2 / –	4.87 / 2.49	QLP
S E133 s73	3296.33725 ± 0.00087	32.2 / –	5.07 / 2.29	QLP
S E134 s73	3307.37832 ± 0.00080	31.4 / –	5.00 / 2.48	QLP

- **Lost events:** P E100 (s59), P E134 (s73), S E0 (s19) and S E102 (s60) fall in gaps or flagged data.
- **Ephemerides:**
- Primaries: P = 11.041067 ± 0.000001 d, O−C within ±1.3 min.
- Secondaries: P = 11.041068 ± 0.000005 d, O−C rms 1.0 min.
- **SPOC secondary odd/even:** +2.5σ, from only 4 events. The s20 SPOC depths are uniformly low, so this is a sector systematic, not two eclipse kinds.

4. TIC 416285595 (P 4.615357 d)

`events_416285595.csv` has all 40 fitted events plus the lost ones. A representative subset:

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E0 s14	1687.52122 ± 0.00101	52.9 / –	3.46 / 0.91	QLP
P E153 s40	2393.67735 ± 0.00069	56.4 / 54.6	3.64 / 0.66	QLP+SPOC+TESScut
P E154 s40	2398.29120 ± 0.00056	60.4 / 53.1	3.51 / 0.90	QLP+SPOC
P E155 s40	2402.90615 ± 0.00051	64.2 / 56.8	3.74 / 0.60	QLP+SPOC+TESScut
P E156 s40	2407.52291 ± 0.00075	63.5 / 54.2	3.55 / 0.90	QLP+SPOC+TESScut
P E158 s40	2416.75340 ± 0.00049	64.0 / 55.8	3.70 / 0.70	QLP+SPOC+TESScut

Event	Mid BTJD	Depth QLP / SPOC (ppt)	T14 / T23 (h)	pipelines
P E236 s54	2776.75002 ± 0.00096	59.9 / –	3.41 / 0.89	QLP
P E392 s80	3496.74568 ± 0.00070	57.9 / –	3.58 / 0.82	QLP+TESScut
P E398 s81	3524.43758 ± 0.00072	55.4 / –	3.62 / 0.77	QLP
S E152 s40	2391.37357 ± 0.00232	– / 10.7	2.98 / 1.42	SPOC (QLP fit degenerate)
S E156 s40	2409.82774 ± 0.00208	9.9 / 14.0	3.15 / 0.30	QLP+SPOC
S E157 s40	2414.44236 ± 0.00167	13.6 / 14.7	3.50 / 0.20	QLP+SPOC
S E395 s81	3512.89836 ± 0.00325	11.8 / –	3.22 / 1.18	QLP
S E397 s81	3522.13865 ± 0.00319	12.3 / –	3.39 / 0.95	QLP

- **Primary ephemeris:** N = 21 epochs, O–C rms 2.4 min. The two s14 events (30-min cadence) are at −7 min.
- **Secondaries:** P = 4.615370 ± 0.000009 d, O–C rms 8.7 min. Some single-event depth errors are degenerate (T23 → 0); see the csv.

Geometry vs Gaia

`s0_gaia_predict.out`, `s5_geometry.out` and `s5b_unwrap.out`. The Monte Carlo uses 20000 draws with Gaia's corvec, ordered P, Vcm, K1, e, ω, Tp (column-major upper triangle, positive-definite for both stars). Uncorrelated values are in brackets. "Backward" means the TESS primary ephemeris taken back to the Gaia epoch. - **468827621:** - Backward: primary − A = −0.142 ± 0.087 d (−1.6σ) [−0.142 ± 0.170]; primary − B = +2.97 ± 0.10 d (+31σ) [+20σ]. - Forward (Gaia ephemeris taken to TESS): −0.18 ± 0.28 d from A. - The deep eclipse is conjunction A. - Secondary phase: observed 0.72649 against predicted 0.7182 ± 0.0149, **+0.56σ**. If the deep eclipse were B, the shallow one would be predicted at 0.282. - **416285595:** - Backward: primary − A = −0.043 ± 0.075 d [−0.05 ± 0.62]; primary − B = ±2.31 ± 0.08 d (29σ) [±0.70, 3.3σ]. - Secondary phase: observed 0.50011 against predicted 0.4915 ± 0.0304, +0.28σ. - Because e is small, the secondary-phase test is weak.

Ephemerides

Errors are formal (`s3_events.out`); TESS-pipeline times are already barycentric. - **TIC 468827621 primary:** T0 = BTJD 1819.85371 ± 0.00010 (BJD_TDB 2458819.85371), P = 11.041067 ± 0.000001 d, T14 6.0 h. Secondary at phase 0.72649 ± 0.00003, T14 5.05 h. - **TIC 416285595 primary:** T0 = BTJD 1687.52620 ± 0.00067 (BJD_TDB 2458687.52620), P = 4.615357 ± 0.000003 d, T14 3.6 h. Secondary at phase 0.50011 ± 0.00027.

Caveats

- **SAP depths are not crowding-corrected.** The QLP/SPOC offsets are 4% (468827621) and about 12% (416285595).
- **TESScut photometry** uses a crude background: the median of pixels ≥ 4 px away. In the crowded 416285595 field it gives depths of 70–89 ppt, so it is used for location, not for depth.
- **Secondaries were not pixel-tested** for either star. For 468827621, the G 16.17 star at 24.6″ could arithmetically produce the 31 ppt secondary in the worst case, and for 416285595 the G 14.75 star at 37.9″ could produce the 12.9 ppt one. For both stars only timing ties the secondary to the target.
- **Dilution uses a Gaussian PRF** (σ 0.75 px) with G-band fluxes, not the TESS PRF.

- **The 468827621 difference images** are both from camera 2; no other orientation was available in the sectors used.
- **Not done:** common-mode test, ZTF, physics sanity check, next-eclipse lists.

Files

`s0_gaia_predict.py/.out`, `s1_archives.py/.out`, `s1b_archives.py/.out`, `gaia_tbosb1.tsv`, `s2_download.py/.out`, `s3_events.py/.out`, `events_468827621.csv`, `events_416285595.csv`, `s3_ephem_<tic>.npy`, `s4_tesscut.py`, `s4_tesscut_<tic>_s<sec>.txt/.npz/.png` (468827621 s19, s59; 416285595 s40, s80), `s5_geometry.py/.out`, `s5b_unwrap.py/.out`, `REPORT.md`.

Summary - TIC 468827621: CONFIRMED; P_ecl=11.041067 d; depth1=184–192 ppt; depth2=29.6–31.5 ppt at phase 0.72649; pixel: on target (primary); pipelines: 3; Gaia geometry: pass - TIC 416285595: CONFIRMED; P_ecl=4.615357 d; depth1=55–59 ppt (QLP s40 62); depth2=12.9 ppt at phase 0.50011; pixel: on target (primary; secondary untested); pipelines: 3 (depths agree only to ~12%); Gaia geometry: pass (weak, e≈0)

vet11 full report

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vet11: Gaia-period folds of five sinusoidal c9a candidates

Independent check (Opus 5.5 subagent, 2026-10-30, for Fable 5.1). I resumed a predecessor that was killed by a container restart. Its downloads, s3 light-curve fits, s4 pixel maps and the first half of the catalogue cones were reused. I added: s2b (the rest of the catalogue cones and the Gaia DR3 variability tables), s5b (the Roche toy model, crash fixed), s6 (a refold at P_phot, which removes the phase-drift artefacts of the Gaia-P fold), s8 (a narrow-dip test at min I), s7 (TIC 62328710 frequencies and ZTF), s9/s9b (Gaia covariance and conjunctions) and s10 (ExoFOP). Data are QLP SAP only, QUALITY==0, 10-min bins, with a per-segment cubic baseline. No TESS-SPOC light curves exist for any of the five. Depths are in ppt, measured from the mean of the two maxima, with per-sector SEM errors. Phases are at P_phot with 0 = min I (the deeper minimum). "Conj A" = the primary at superior conjunction (companion in front). Conjunction errors come from 40000 draws with the full Gaia corr_vec. The earlier independent-draw MC ignored corr(omega,Tp) of about 0.99 and overstated the errors. Coverage was cut short by a budget limit: one pixel sector per star, no difference images of the min-I dip, and no TESS-SPOC or TESScut-aperture depth comparison.

TIC 372683900 (Gaia P 1.476502 +- 0.000081 d, e 0.30)

quantity	value
LS top peak / P_phot (chi2 scan)	0.73825 d (=P/2); 1.476507 +- 0.000001 d, ratio 1.000003 (+0.05 sigma)
min I / min II depth	31.07 +- 0.31 at 0 / 28.18 +- 0.19 at 0.5011 +- 0.0008; diff 2.88 +- 0.35 (8.3 sigma), 11/12 sectors
harmonics K=6	A1 2.35, A2 14.79, A3 0.67, A4 0.27 (+-0.05); min widths 0.271 / 0.244 (cos2phi = 0.250)
narrow dip at min I	0.63 ppt, empirical 1.7 sigma (none)
conjunctions	min I - conj B +0.087 +- 0.117; min II - conj A -0.042 +- 0.127; min I at conj A excluded at 3.6 sigma
minima separation	0.5007 +- 0.0018 vs Gaia-predicted 0.372 +- 0.039 (3.3 sigma): photometry says e cos w ~ 0, Gaia says -0.21

quantity	value
pixel (s57)	amplitude centroid 0.9" from target. A G=15.23 star at 1.8" is unresolved, but it would need 128 ppt intrinsic and would have to match Gaia P to 1e-4 d

ELLIPSOIDAL: a smooth double wave at the Gaia period with no eclipse narrowing. The unequal minima are deeper at conj B, which matches the Morris A1 sign reversal at low i. The s5b toy model at i ~ 20 deg gives A1 1.74, A2 14.87, deeper at B by 2.69 ppt. Caveat: with the TIC R1 that needs a dark M2 ~ 4 Msun, so R1 is probably overestimated. Gaia's e = 0.30 conflicts with the minima separation.

TIC 13106408 (Gaia P 2.637505 +- 0.000401 d)

quantity	value
LS top / P_phot	1.31884 d (=P/2); 2.637721 +- 0.000002, ratio 1.000082 (+0.54 sigma)
min I / min II depth	46.85 +- 0.33 at 0 / 40.63 +- 0.26 at 0.5009 +- 0.0010; diff 6.22 +- 0.39 (15.8 sigma), 7/7 sectors
harmonics K=6	A1 2.06, A2 21.77, A3 1.40, A4 0.72 (+-0.03); widths 0.252 / 0.272 (min II flattened)
narrow dip at min I	0.36 ppt, 2.3 sigma (not significant)
conjunctions	min I - conj A +0.079 +- 0.166; min I - conj B -0.427 +- 0.167 (2.6 sigma)
pixel (s58)	on target (1.9"). The best neighbour (G 14.4, 15") would need 91 ppt

ELLIPSOIDAL: a large-amplitude, near-Roche-filling double wave at the Gaia period with unequal minima and a flattened min II. The K=6 residuals are white and there are no eclipse features. The conjunction test only mildly prefers min I at conj A.

TIC 372683622 (Gaia P 3.809006 +- 0.000179 d)

quantity	value
LS top / P_phot	1.90468 d (=P/2); 3.809282 +- 0.000014, ratio 1.000073 (+1.5 sigma)
min I / min II depth	7.73 +- 0.16 at 0 / 6.45 +- 0.19 at 0.4943 +- 0.0061; diff 1.28 +- 0.30 (4.2 sigma) but variable, 0.3-3.2 ppt between sectors
harmonics K=6	A1 0.78, A2 3.23, A3 0.13, A4 0.12; per-sector A3/A4 ~0.5 and incoherent (spots)
narrow dip at min I	0.24 ppt, 2.1 sigma (none)
conjunctions	min I - conj B +0.075 +- 0.040; min II - conj A +0.044 +- 0.036; min I at conj A excluded at 12.5 sigma
pixel (s56)	on target (2.6")

ELLIPSOIDAL: a double wave locked to the Gaia period with minima at the two conjunctions. The deeper minimum is at conj B, as for low-i ellipsoidals; the toy model at i ~ 20 deg gives A2 3.2 and deeper at B by 1.7 ppt. On top of this there is variable spot modulation, and no eclipse.

TIC 319185391 (Gaia P 2.677901 +- 0.000187 d)

quantity	value
LS top / P_phot	1.33902 d (=P/2); 2.678053 +- 0.000009, ratio 1.000057 (+0.8 sigma)
min I / min II depth	8.46 +- 0.30 at 0 / 6.57 +- 0.14 at 0.5102 +- 0.0040; diff 1.89 +- 0.23 (8.3 sigma), 6/6 sectors
harmonics K=6	A1 0.70, A2 3.64, A3 0.23, A4 0.04
narrow dip at min I	box 1.15 +- 0.17 ppt, centre +0.007, core 0.012 phase (0.8 h), profile about 0.03 phase (about 1.9 h). Empirical S/N 5.3; 0/400 random phases as deep; per sector 0.46/0.84/2.01/0.85/0.96/1.48 (6/6 > 0, weighted 1.04 +- 0.17); even 1.26 +- 0.25, odd 1.04 +- 0.29; at min II -0.03 +- 0.21. The same test on the other three stars gives 1.7-2.3 sigma

quantity	value
conjunctions	min I - conj A +0.057 +- 0.084 (conj B excluded at 6.0 sigma); separation 0.511 +- 0.003 vs Gaia 0.529 +- 0.028
pixel (s85)	the ellipsoidal signal is on target (5.2"). The dip itself has no pixel test
other	Gaia DR3 also has an astrometric orbit for this source, P = 434.5 +- 8.2 d, e = 0.70 (likely a triple)

DETACHED-EB (grazing, fold-only): an ellipsoidal double wave plus a phase-locked 1 ppt, about 2 h dip at min I, which sits at Gaia conj A (companion in front). This is the eclipse that the flattened P/2 fold hid. Individual events are about 1 sigma, so they cannot be seen singly, and the dip has no difference image. That puts it at probable-level evidence.

TIC 62328710 (Gaia P 0.938088 +- 0.000042 d)

quantity	value
dominant signal	f1 = 6.390610 +- 0.000005 c/d (0.156480 d), 3.09 ppt; f1/f_orb = 5.9950, not an orbital harmonic (19 sigma in P). Second frequency f = 5.806788 c/d, 1.26 ppt. Both are stable in all 7 sectors
Gaia-P signal	after prewhitening, A2 = 0.058 +- 0.046 ppt (< 0.15 ppt, 2 sigma). Minima: none
pixel (s56)	amplitude centroid 45" from target, on Gaia DR3 2165710355333626240 (G 13.38, 44.3"; next source worse by dchi2 642). f = 5.807 c/d also peaks there
ZTF	neighbour: top peak 6.39060 c/d, 9.6 (g) / 8.9 (r) mmag, noise 99th percentile 2.3/4.6; 5.807 c/d 4.2/5.1 mmag. Target: 0.16/0.70 mmag at f1 (noise 1.5/1.7)
catalogue	the neighbour is Gaia VarFlag VARIABLE (vari_short_timescale). Target not in VSX/Gaia-EB within 60"

OTHER: the "P/3" wave is a multiperiodic delta Scuti-type pulsator, the neighbour 44" away, diluted into the QLP aperture. The target is flat at the Gaia period. With any R1 >= 1.8 Rsun, ellipsoidal A2 < 0.15 ppt would require i < 5 deg, so the Gaia 0.938 d SB1 orbit is suspect.

Summary

TIC	class	P_phot / P_Gaia	min1 (ppt, phase)	min2 (ppt, phase)	pixel	catalogues
372683900	ELLIPSOIDAL	1.000003	31.07 at conj B	28.18 at conj A	on target (1.8" nbr unresolved)	none; SIMBAD SB*
13106408	ELLIPSOIDAL	1.000082	46.85	40.63	on target	VSX only an SR at 29"; SB*
372683622	ELLIPSOIDAL	1.000073	7.73 at conj B	6.45 at conj A	on target	none; SB*
319185391	DETACHED-EB	1.000057	8.46 + 1.0 ppt dip at conj A	6.57 at conj B	on target (dip untested)	none; SB*; Gaia astrometric 434 d
62328710	OTHER	f1 = 5.995 f_orb	none (A2 < 0.15)	none	neighbour 44"	nbr Gaia VARIABLE

TIC 372683900: ELLIPSOIDAL; P_phot=1.476507 d (1.000003); min1=31.07+-0.31 ppt at phase 0 (conj B +0.09+-0.12); min2=28.18+-0.19 at phase 0.5011 (conj A -0.04+-0.13); pixel: on target; catalogue: not in VSX/Gaia-EB/Prsa/IJspeert/SB9, SIMBAD SB* TIC 13106408: ELLIPSOIDAL; P_phot=2.637721 d (1.000082); min1=46.85+-0.33 ppt at phase 0 (conj A +0.08+-0.17); min2=40.63+-0.26 at phase 0.5009; pixel: on target; catalogue: none (VSX SR at 29" unrelated), SIMBAD SB* TIC 372683622: ELLIPSOIDAL; P_phot=3.809282 d (1.000073); min1=7.73+-0.16 ppt at phase 0 (conj B +0.075+-0.040); min2=6.45+-0.19 at phase 0.4943 (conj A +0.044+-0.036); pixel: on target; catalogue: none, SIMBAD SB* TIC 319185391: DETACHED-EB (grazing, fold-only); P_phot=2.678053 d (1.000057); min1=8.46+-0.30 ppt at phase 0 (conj A +0.06+-0.08) with a

1.04+-0.17 ppt, about 2 h dip; min2=6.57+-0.14 at phase 0.5102; pixel: on target (ellipsoidal signal; dip untested); catalogue: none, SIMBAD SB*, Gaia astrometric 434 d orbit TIC 62328710: OTHER (delta Sct neighbour); P_phot=0.156480 d (1/5.995 of Gaia P); min1=none (Gaia-P A2<0.15 ppt); min2=none; pixel: neighbour Gaia DR3 2165710355333626240 at 44" (ZTF confirms 6.3906 c/d there, not on target); catalogue: neighbour Gaia VARIABLE

Caveats

Only QLP is available, so there is no second pipeline. The Gaia-P fold in s3 has phase drift: its residual "dips" flip sign between sectors, so use the s6/s8 numbers, which are at P_phot. The s5b Roche model is a toy (blackbody at 800 nm, no reflection) and uses TIC radii, which look overestimated (13106408 and 62328710 overfill their Roche lobes). ExoFOP: no TOI or CTOI for any of the five (s10). The ms_oscillator TAP query failed.

Files

s1_download.py/.log, s2_catalogs.py/.txt (+s2_err.log), s2b_catalogs.py/.txt, s3_lc.py + s3_.txt/.json + fold_.png, s4_pixels.py + s4_*s.txt/.npz + pixs.png, s5b_roche.py/.txt (s5_roche.* crashed), s6_pphot.py* + *s6*.txt/.json, s7_62328710.py/.txt/.json, s8_dip319.py + s8__b10.txt/.json + dip_319185391.png, s9_gaia_corr.py + s9_gaia_nss.csv, s9b_conj_corr.py/.txt, s10_exofop.sh/.txt, common.py.

Returned batch summary

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Chunk-9 (c9a) candidates: screening and vets, 2026-10-30

Cloud session (Fable 5.1 lead, Opus 5.5 helpers) on the 24-star bundle Astra packaged from Fable 5's `sb1-scale` chunk 9 (montage sheets c9a_00 and c9a_01; roster order = best z descending). Everything here is on the branch `claude/zealous-pasteur-8mmme2`.

What was done

1. **Screen** (`SCREEN_RESULTS.csv`, `SCREEN_REPORT.md`): first-pass visual triage of the two montage sheets, three Opus helpers at eight panels each plus a second look at the borderline calls. 19 eclipse-like, 5 non-eclipse-like, 0 ambiguous, 0 unreadable.
2. **Shared inputs** (`targets.csv`, `neighbours_90as.csv`, `prep.py`): TIC rows, Gaia DR3 NSS SB1 orbits (VizieR I/357 via the idia mirror; the CDS and ESA hosts were unreachable), Gaia neighbours within 90", QLP / TESS-SPOC / 2-min / TESScut sector lists.
3. **Vets** (`vet01`–`vet06`, `vet11`; protocol in `VET_BRIEF.md`): independent adversarial checks to the paper's Section 2.5 tier definition (events from raw SAP, period from the eclipses, difference images and dilution, independent pipeline, Gaia geometry). `VERDICTS.md` is the one-line-per-star table; each `vetNN/REPORT.md` holds the evidence, scripts and outputs.

Results

	n	TICs
vetted CONFIRMED	1 2	195865529*, 468181994, 470235665, 379251709, 70665829 (= paper's A4, blind control, recovered), 331293467, 283532878, 407877153, 15855794, 92881948, 468827621, 416285595 (with caveats: pipeline depths differ ~12%)
of which already a catalogued eclipsing binary	1	195865529 (ASAS-SN EA, same period; not in VSX)
sinusoids refolded at Gaia P: ellipsoidal	3	372683900, 13106408, 372683622
sinusoids: grazing detached EB, fold-level only (PLAUSIBLE at best)	1	319185391 (also a Gaia 434 d astrometric orbit: likely triple)
sinusoids: signal from a neighbour (δ Scuti at 44"); target flat at the Gaia period	1	62328710 (Gaia's SB1 orbit for it is suspect)
eclipse-like, **not yet vetted** (credits ran out)	7	300903537, 23112764, 373062090, 465073693, 316405449, 465263741, 441741446

Side findings worth a catalogue note: a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d) sits 29.7" from 283532878 and may leak ≤3 ppt dips into its light curve; 372683900's equal, half-period-spaced minima conflict with Gaia's e = 0.30 at 3.3σ; several secondaries are off phase 0.5 and match Gaia's e cos ω (470235665, 283532878, 407877153, 92881948, 468827621).

Cost note, for the next batch

Each full vet cost ~240k Opus tokens, about two stars per agent. Most of that was execution depth the tier does not need (every individual event fitted, own TESScut photometry on top of two pipelines, dilution for every neighbour within 90", Monte Carlo geometry, physics checks, ephemerides, literature search) plus each agent rebuilding its tooling from scratch, and a container restart that cost wave 1 roughly 1.7×. For the remaining seven: reuse the scripts in `vet04/` or `vet05/` (`common.py`, `fitlib.py`), a handful of events per star, one difference image per sector in two sectors, only the neighbours that matter, quoted pipeline depths, simple error propagation, two-page report. `VET_BRIEF.md` should be trimmed accordingly before reuse.

Data-access notes (this machine, 2026-10-30)

MAST light curves and TESScut work (pass coordinates to TESScut, not names; `lightkurve.search_lightcurve` sometimes hangs, in which case fetch `mast:HLSP/...` URIs directly). Gaia: ESA TAP times out, Heidelberg TAP is slow on joins; VizieR mirror `vizier.idia.ac.za` works but is intermittent, so run a known-positive (Algol) check before trusting an empty cone. The Gaia DR3 variability tables are I/358, not I/357/varisum as the brief says. AAVSO VSX API is behind Cloudflare; use VizieR B/vsx.

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c9a vets: running verdict table

Tiers per the paper's Conventions (Confirmed / Probable / Plausible / Refuted / Open). One line per star, pasted from each vet's summary; see vetNN/REPORT.md for the evidence.

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet01	1958652529	CONFIRMED	3.83104 56(9)	183 ppt QLP; 192±7 TESScut undiluted	48 ppt @ 0.49999	on target, all 8 sectors	3 (QLP, own TESScut, TASOC)	pass; primary = conj A	**already catalogued**: ASAS-SN EA P 3.8310 d at 0.16"; ATLAS DBH at P/2; not in VSX	recovery, not new
vet01	468181994	CONFIRMED (weaker)	3.99691 90(6)	22.5 ppt QLP / 23.2 TESScut	1.24–1.41 ppt @ 0.4956±0.0012	on target; 2.85" G16.75 neighbour rejected ~16σ	3 (QLP, own TESScut, ELEANOR-lite)	pass; secondary within 1σ; A vs B undecided	not in VSX/ASAS-SN/Gaia-EB (per report)	P_ecl −2.3σ from Gaia; RUWE 12.6 suggests sub-arcsec companion; secondary 25 min early (e cos ω ≈ −0.007)
vet02	47023566 5	CONFIRMED	10.3261 83(2)	122–127 ppt	81–84 ppt @ 0.52275	on target, s19 & s79 (2 cameras)	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded 19σ; secondary +0.7σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 859-013225; subgiant + K dwarf; possible secondary-phase drift (3.5σ, QLP only, not claimed)
vet02	37925170 9	CONFIRMED	12.3239 12(7)	34–38 ppt (screen's 31 was 15% low)	2.5 ppt @ 0.4960±0.0010 (fold-level, S/N 8.5 QLP)	on target (primary), s18 & s52	3 (QLP, TESS-SPOC, own TESScut)	pass; primary = conj A, B excluded; secondary +0.8σ	not in VSX/Gaia-EB/Prša/IJspeert/ASAS-SN within 60"; SIMBAD SB*	UCAC4 788-006141; M-dwarf companion; eclipse shape prefers R1≈1.3 R⊙ vs TIC 1.88
vet03	70665829	CONFIRMED	3.46576 54(7)	283–330 ppt (pipeline-dependent; SPOC ~300)	64–72 ppt @ 0.46048	on target, s46/s48/s72 (3 cameras); only neighbour needs >1500%	4 (QLP, TESS-SPOC, own TESScut, ZTF g/r)	pass; deep = conj A (RV star eclipsed) +0.24σ; reverse 3.6–5.6σ off	not in VSX/Prša/IJspeert within 60"	**BLIND CONTROL = paper's A4**: recovered independently (paper: P 3.4657540±0.0000093, secondary at 0.461, deep eclipse on the K dwarf)
vet03	33129346 7	CONFIRMED	5.25681 99(20)	61–73 ppt QLP raw (≈80–116 undiluted)	11.4 ppt QLP (≈13–21 undiluted) @ 0.50013	on target, 4 sectors / 3 cameras; G13.78 at 30.5" passes dilution arithmetic (needs 25–84%) but excluded by position (fit 10–19× worse there)	2 (QLP, own TESScut; no SPOC)	pass; deep = conj A +0.49σ, other excluded ≥6.7σ (relies on Gaia ω–Tp correlation)	not in VSX/Prša/IJspeert within 60"; Gaia-EB/varisum/SB9 cones from predecessor run only (mirror timed out)	A/F star + K dwarf; heavy blend (contamination 0.70) so undiluted depths uncertain; Dec −60, no ZTF
vet04	28353287 8	CONFIRMED	17.4460 19(19) (+1.8σ vs Gaia)	75 ppt QLP (71–77 undiluted)	10 ppt @ 0.6781 (Gaia predicts 0.712±0.024, −1.4σ)	on target, 8 sectors; 3 neighbours at 28–37" pass dilution arithmetic (need 30–47%) but rejected by noise-weighted difference images ($\Delta\chi^2 \geq 41.7$ each sector)	2 (QLP, own TESScut; no SPOC)	pass; primary = conj A, B excluded >11σ	not in VSX/Gaia-EB/Prša/IJspeert on target; **a catalogued EB (Gaia DR3 2202301140233408128, P 1.7975 d, G 17.66) sits 29.7" away**, excluded as source but may add ≤3 ppt dips	eccentric secondary confirmed in all 8 sectors over 4 seasons; secondary phase drifts ~45 min 2019–2024 (~4σ formal, likely underestimated errors)

vet	TIC	tier	P_ecl (d)	depth1	depth2 @ phase	pixel	pipelines	Gaia geometry	catalogue status	notes
vet04	407877153	CONFIRMED	17.089294(8) (−0.14σ)	29 ppt (SPOC, TESScut; QLP 24–27 diluted)	3.0 ppt flat-bottomed @ 0.5469 (Gaia 0.559±0.044, −0.3σ)	on target; secondary pixel S/N weak (Δχ²≥7 per sector, ≥166 summed)	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's depths low (27→29, 2→3 ppt)
vet05	158557994	CONFIRMED	5.891145 8(38) (+0.54σ)	35.3 ppt dilution-corrected (QLP 35.8, SPOC 34.6, TESScut 34.9); T14 3.25 h	5.0–5.4 ppt @ 0.50182±0.00030 (Gaia 0.502±0.021)	on target, s44/s70/s71 (0.4–0.9"); all neighbours excluded	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.03±0.08 d, B excluded ~40σ	not in VSX/Gaia-EB/Prša/IJspeert within 60"	screen's 35 ppt was right
vet05	928819488	CONFIRMED	8.8049754(13) (+0.43σ)	16.4 QLP / 18.2 SPOC / 19.0 ppt dilution-corrected; T14 1.69 h V-shaped	5.0–5.3 ppt @ 0.48334±0.00008 (e cos ω −0.026; Gaia 0.502±0.018, 1.0σ)	on target, all 7 sectors (0.3–1.6"); G14.56 at 53" excluded by centroid and aperture dependence	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A +0.06±0.09 d, B excluded ~43σ	VSX lists it as ROT (Gaia DR3 variable) only; not in any EB catalogue	screen's 11 ppt was low (16–19 ppt); QLP ~10% shallower than SPOC (dilution)
vet06	468827621	CONFIRMED	11.041067(1) (−0.16σ)	184–192 ppt (QLP 192, SPOC 184, TESScut 192)	29.6–31.5 ppt @ 0.72649±0.00003 (Gaia 0.718±0.015, +0.56σ)	on target (primary), s19 & s59 but both camera 2; secondary not pixel-tested (G16.17 at 24.6" could supply it in the worst case)	3 (QLP, TESS-SPOC, own TESS cut)	pass; primary = conj A −0.14±0.09 d, B excluded 31σ; longer eclipse nearer periastron as e sin ω requires	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; screen right on everything
vet06	416285595	CONFIRMED, with caveats	4.615357(3) (−0.37σ)	55–59 ppt; QLP s40 61.7 vs SPOC 54.7 (~12%, ~4σ apart; TESScut 70–89 unreliable in crowded field)	12.9 ppt @ 0.50011±0.00027	on target (primary), s40 cam 1 & s80 cam 2; all 43 neighbours need >100% (min 126%); secondary not pixel-tested (G14.75 at 37.9" could supply ~28%)	3, but depths agree only to ~12%	pass but weak (e≈0.08 so phase 0.5 expected anyway; A/B rests on Gaia ω–Tp correlation)	not in VSX/Gaia-EB/Prša/IJspeert/SB9 within 60"; SB*	reduced-depth run; if depth agreement within errors is required this drops to PROBABLE; two worst-case secondary figures are the agent's hand estimates, not in an output file

vet11: the five sinusoidal (P2) candidates, refolded at the Gaia period (QLP only; no second pipeline exists)

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
372683900	ELLIPSOIDAL	1.476507 (1.000003)	31.1 ppt @ 0 (conj B)	28.2 ppt @ 0.5011 (conj A)	on target (G15.2 at 1.8" unresolved)	none; SIMBAD SB*	minima 0.5007 apart vs Gaia e=0.30 predicting 0.372±0.039 (3.3σ conflict); TIC radius probably overestimated

TIC	class	P_phot (d) / Gaia ratio	min1	min2	pixel	catalogue	notes
131064008	ELLIPSOIDAL	2.637721 (1.000082)	46.9 ppt @ 0 (conj A)	40.6 ppt @ 0.5009, flattened	on target	none (unrelated 364 d SR at 29"); SB*	
372683622	ELLIPSOIDAL	3.809282 (1.000073)	7.7 ppt @ 0 (conj B; A excluded 12.5σ)	6.5 ppt @ 0.4943	on target	none; SB*	spot modulation varying sector to sector
319185391	DETACHED-EB (grazing; fold-level only)	2.678053 (1.000057)	8.5 ppt @ 0 (conj A; B excluded 6σ) + 1.04±0.17 ppt ~2 h dip	6.6 ppt @ 0.5102	on target (ellipsoidal); dip itself untested	none; SB*; Gaia also has a 434 d astrometric orbit (likely triple)	dip in all 6 sectors, odd and even; single events ~1σ; same test gives 1.7–2.3σ on the other three → PLAUSIBLE at best
62328710	OTHER (δ Scuti neighbour)	0.156480 (1/5.995 of Gaia P, not a harmonic)	none (<0.15 ppt at Gaia P)	none	neighbour Gaia DR3 2165710355333626240 (G13.38, 44"); ZTF: 6.3906 c/d at ~9 mmag there, flat on target	neighbour flagged VARIABLE by Gaia	target flat at 0.938 d: Gaia's SB1 orbit for it is suspect